

Team Caplet

insight_v1 research record: strategy for Laura Gao

Wharton Global High School Investment Competition 2026-27

Research run 27-28 September 2026 · about 90 AI agent runs, every step audited

Scope: the Trading Notes Analysis (23 Oct) and the Investment Policy Statement (6 Nov)

AI-use statement. Everything in this record was produced by an AI assistant (Claude Code) as research and brainstorming for the six students of Team Caplet. None of it is text to submit: the team decides and writes every word, and records its AI use in the Works Cited pages as Wharton's AI policy requires. Laura Gao's public professional record only; the team's own client documents are not included.

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Executive summary

Team Caplet · Wharton Global High School Investment Competition 2026-27 · client: Laura Gao
Research run 27-28 September 2026. **Serves:** the Trading Notes Analysis (due 5 p.m. ET Fri 23 Oct = 8 a.m. AEDT Sat 24 Oct) and the Investment Policy Statement (due 5 p.m. ET Fri 6 Nov = 9 a.m. AEDT Sat 7 Nov; trading ends and the strategy freezes). Nothing here is for the Final Report or the finale.

AI-use statement. Everything in this record was produced by an AI assistant (Claude Code) as research and brainstorming for the six students. None of it is text to submit. The team decides and writes every word. Wharton's AI policy requires any AI use to be recorded in the Works Cited pages.

1. The answer in one page

The recommended strategy (pending the team's vote; full specification in Part 2):

1. **January 2027:** the first \$300,000 buys ten U.S. Treasury bonds, one maturing just before each \$50,000 payment (2033-2042). This happens before any money takes stock-market risk. If falling rates make the ladder cost more than \$300,000, the latest payments are bought first, and the 2028 deposit completes the rest.
2. **January 2028:** the \$150,000 deposit first finishes any unbought payment. It then buys a Treasury that repays \$150,000 just before 2033. That is the bought floor of the range Laura can tell co-sponsors. The rest (about \$40,000) goes into one world stock fund, untouched until 2033.
3. **2031:** nothing is traded. Laura tells co-sponsors a range: the floor she already owns, up to the floor plus half of what the stock fund is worth that day.
4. **2033:** the ten bonds become the operating reserve. The facility gets the floor plus half the stock fund, never more than the announced top. The rest stays with Laura as flexibility.

Why it is better than the earlier plan (lock the growth money in 2031): about the same middle outcome (\$207k vs \$210k total facility-plus-flexibility money), better bad cases (\$182k vs \$168k at the 5th percentile), and across every six-year window of U.S. market history since 1928, a worst case of \$169k vs \$120k (model and history estimates; `scripts/E6_final_checks.py`). The price: less stock (about 9% of her money in 2028-2032) and a smaller median gift.

Why Laura would choose us (blind contest, a simulation): three independent simulated-Laura evaluators, grounded only in the case and her verified public words, each read six anonymised strategies (ours and five rival approaches) in a different order. **All three ranked ours first** (Borda score 18 of 18; next best 15). They cited that the 2031 bottom is money she already owns, so she cannot overpromise; that the payments do not depend on markets (rival plans miss a payment in up to 3.2% of model runs, 41-46% if the 2028 deposit never arrives, and that deposit comes from income she herself called "unstable"); and that the plan names its own costs. **All three asked for one fix:** show "appropriate balance" more visibly (the plan reads as timid), say where the leap is (the residency itself), and price the certainty in one line.

Before the first WInS order: three recorded team votes (adopt lock-early; recommended design or the 2031-lock alternative; which book WInS shows), a screenshot of WInS Session Rules (position limit), a zero-risk test of the note-box length, and every ticker checked as listed in WInS. Target fills by Fri 2 Oct (U.S. time); hard stop 9 Oct.

2. Discoveries that changed the plan

- **This year's WInS rules are public** (SMAApply Trading Details and FAQ): \$300,000 starting cash (exactly Laura's 2027 deposit; the \$150,000 is not added), 200-trade cap, any ETF or U.S. Treasury bond listed in WInS, no sector minimum, no day trading, trade notes cannot be edited. A per-security position limit exists but is shown only when logged in.
- **The \$300,000 fit is knife-edge.** The ten-payment ladder cost more than \$300,000 on 173 of 185 trading days of 2026. The real ladder (Treasury STRIPS maturing each 15 November) costs \$294,387, leaving only \$5,613 (0.19 percentage points of rate) of headroom. In about 1 in 3 modelled rate paths it costs more by January 2027, so the rule for that case matters.

- **Two official 2026-27 documents were missing** from the repo (the Investment Competition Guide and Infographic). They are now filed. The Guide says judges want "the reasoning, assumptions, and tradeoffs behind those decisions."
- **Laura's words, verified.** 91 of her public quotes were confirmed word for word and in context; 7 were excluded for privacy. The case's pull quote ("The only person who needs to believe in something is yourself.") cannot be verified as hers, so cite it as the case's words. A popular reading (she "only leapt once a contract secured a floor") comes from a reporter, not from her, and was dropped. No Laura quotes go into the Trading Notes or the IPS.
- **The real field is smaller than 6,000.** In 2025-26, 6,300+ teams registered but about 2,300 submitted final reports.

3. How the run worked (about 90 agent runs, every step audited)

Phase	What happened	Output in this record
A. Foundations	Official documents catalogued (324 entries), every number traced to a source, stakeholders mapped, WInS rules found	Part 3, Phase A
WInS week 1	Fund research, allocation, red team, revision; later v1 with both book options	Part 3, WInS
B. Questions	20 agents, 10 lenses (incl. client synthesis and a contrarian lens): 472 questions, 271 quotes	Part 3, Phase B
C. Filter	Merged to 248; three skeptics per question; 48 survived (so-what, anchor, 2-of-3, simplicity)	Part 3, Phase C; Appendix B
D. Research	12 specialists (with Python models), 5 audits, quote verification (91 Laura quotes verified)	Part 3, Phase D
E. Strategy	Architect, 4 red teams (judge, Laura, rival, pre-mortem), chief strategist	Part 2; Part 3, Phase E
Contest	Blind Laura's Choice: 6 anonymised strategies, 3 evaluators	Part 2
Outputs	3 writers, 1 verifier (19 problems found, 18 fixed)	Part 1; Part 3, verification

4. How to read this record

- **Part 1** holds the six working documents for the team, and is all most students need. Start with the *Trading Notes pack* (due first), then the *IPS specification*.
- **Part 2** holds the final strategy specification and the blind contest (the reasoning behind Part 1).
- **Part 3** is the full research record, phase by phase, for anyone who wants to check a claim or learn the method. Every claim carries a status label: VERIFIED-PRIMARY (checked on the original page), VERIFIED-REPO-FILE (from an official file in the repo), SNIPPET-UNVERIFIED, ASSUMPTION or MODEL (a model estimate, not a forecast), PARAPHRASE-UNVERIFIED.
- **Appendices:** the brief every agent worked from, the 48 surviving questions, and the index of all Python scripts (every model number can be re-run from the repo).
- Later files override earlier ones. Where a specialist file ends with "Audit corrections", the corrections win.

What this teaches

A long research effort is only useful if its conclusions fit on one page and every number can be traced back. The work behind this plan was large; the plan itself is simple: buy the promises when the money arrives, let the rest grow, and never tell anyone a number you do not already own.

Part 1 - The deliverables for the team

Start here. Six working documents for the Trading Notes (Oct 23) and the IPS (Nov 6).

1. Trading Notes pack
2. IPS specification
3. Securities and allocation
4. Why Laura would choose us
5. Strategy changes and top insights
6. Open questions and team decisions

Trading Notes pack: trades, gates, note elements and checks (insight_v1, writer W1)

Summary (5 lines)

1. Before any order: three recorded team votes (lock-early; REC or ALT; WInS book (ii)R), each after every student writes 3-5 own-words lines, then the Session Rules screenshot, listing checks and a zero-risk note-box test (E6 s5.4, INT).
2. Recommended book (ii)R at \$300,000: IEF + TLH (the operations hedge, about 66%), the building minimum (2032 Treasury note, else IBTM, else VGIT/SPTI, about 24%), VT (about 9%), cash float about 1%; all orders in one session, hedge first (E6 s5.2, INT).
3. The three notes are TLH (supported and tested), the building minimum (refined only if the change was logged before its order, else supported) and VT (supported, carrying the growth-first story) (E6 s5.5, INT).
4. Students draft every note offline from element lists (300 characters until the box test), a second student copy-checks, and two outside readers test the drafts; reflections are student-written without AI (E6 s5.4, s5.7).
5. Submit on Thu Oct 22 AEDT; the deadline is Fri Oct 23, 5:00 p.m. ET = Sat Oct 24, 08:00 AEDT (TN guide; E6 s7).

Serves: Trading Notes (TN). Nothing here is note or reflection text. It is AI-generated research (Wharton AI policy: log it; students write every word). Authority: `research/insight_v1/phase_E/E6_final_spec.md` (sections 0, 2, 5, 7-9); details copied from `wins_now/securities_and_allocation_v1.md` ("ticket v1") and `phase_D/trading_now_brief.md` only where E6 points to them. **Status: PROVISIONAL until the team votes.**

Status labels: VP = verified on the primary page (by the named agent and date); VRF = verified in an official repo file; SNIP = snippet, unverified; ASM = assumption; MODEL = model output, never a forecast; DER = W1's arithmetic from labelled inputs; INT = judgement; UNVERIFIED = not yet checked.

Terms (first use):

Term	Meaning
ETF	Exchange-traded fund: a fund bought and sold like a share
Treasury	A bond issued by the U.S. government
STRIPS / zero-coupon	A Treasury that pays one amount on one date and nothing before
Rung	One bond in a ladder, maturing before one payment
Operations hedge	IEF + TLH in WInS: funds that stand in for the ten-payment ladder
Building minimum	The Treasury bought with the 2028 deposit that repays its full dollar amount before 2033
Stock fund	One broad world stock fund holding everything else
Duration	Rate sensitivity: about the % a bond fund falls if rates rise 1 percentage point
bp	Basis point: 0.01 percentage point
Position limit (cap)	The most WInS lets one security be of the portfolio (shown in Session Rules)
Scaling marker	The few words inside a note that say WInS shows Laura's plan "after both deposits" / "after her 2028 deposit"
Day trading	Buying and selling the same security on one U.S. trading day (not permitted, User Guide, VP via A4)
Ex-dividend date	The day a fund's price drops by the distribution it is about to pay
REC / ALT	E6's recommended design (minimum bought in 2028) / the alternative (E1's 2031 lock)

1. Gate: tick every box before the first order (E6 s5.4; E5 calendar)

#	Check	Owner (role)	Done
a1	Every student writes 3-5 lines in their own words, no AI file open (R-W51; PM-01)	each student	[]

#	Check	Owner (role)	Done
a2	Vote 1: adopt lock-early, recorded with first names, date, reason (E6 D1)	team leader	[]
a3	Vote 2: REC or ALT, using E6 s7 D2 test (values statement + two-minute explain-back of the four dates), recorded before any building or stock order (this dated entry is what makes a "refined" note B possible)	team leader	[]
a4	Vote 3: WInS book (ii)R or (iii) (E6 D3); the two-reader check (row e3) is its tie-breaker	team leader	[]
b1	Session Rules screenshot (Portfolio Summary > Session Rules): single-security limit, any bond or type limit, day-trading tooltip, trades allowed, commissions	trader	[]
b2	Read the logged-in Trading Details page. Activity minimum or whole-type cap shown: stop, ask Wharton (Contact Us); never add trades to meet a count (PM-15). "First trade by Oct 10" stays UNVERIFIED until read	trader	[]
b3	Pick the position-limit branch (section 3) before ordering	trader + rates analyst	[]
c1	Search IEF, TLH, VT, IBTM, SPTL, VGIT: each shows as an ETF with the issuer's full name	trader	[]
c2	Bonds drop-down: every U.S. Treasury maturing Jul-Dec 2032 (coupon, price, accrued interest); search by maturity, not a built-up symbol (AX1b item 10)	trader	[]
d1	Note-box test at zero risk: type about 450 characters on the order-review screen, see if they are cut, clear them, do not press Confirm ; record the result. Until a limit is seen or ruled out, draft to 300 characters (UNVERIFIED for WInS; a sister product caps at 300, VP; Wharton's own example is 413 characters, VRF)	trader	[]
e1	Notes drafted offline by students from section 6, no research file open (PM-04)	note writers	[]
e2	Second student restates each note's role in one sentence, then runs the five-word copy check (section 8)	second reader	[]
e3	Two outside readers pass the "what does Laura's real money hold in January 2027?" check (section 8)	second reader	[]
f1	Roles: one trader + backup (only they place orders); note writers; second reader; log keeper; data owner; every student drafts at least one note or reflection and one IPS block (PM-05; R-W38, VP)	team leader	[]
g1	Cash: at least \$1,000 left at the next open's prices; share counts recomputed from the last close (T2 finding 1); no sells on a day anything is bought	trader	[]
h1	Team names its own labels for the three parts in a meeting; logs that the proposals came from an AI source (E2 F10; T2 finding 8)	log keeper	[]
i1	Decision log and AI-use log started (section 9)	log keeper	[]

Never: an order before the gate; the advisor placing a trade ("Students should place trades on WInS, NOT the advisor", VP via E6 s5.9).

2. Orders

All dollar figures: illustration at the 2026-09-25 closes (VP via ticket v1 s12; SPTL/SPTI as of 9/24), \$300,000 WInS cash (VP via brief s14). Shares rounded down (DER). **Recompute every share count from the last close before entering orders** (T2 finding 1). Orders placed while the market is closed fill at the next open (R-W62, VP). Commissions: \$25 per ETF trade, \$10 per bond trade (VP via brief s14).

2.1 Recommended book (ii)R: Laura's plan on 2 January 2028, scaled to \$300,000 (E6 s5.1-5.2, INT)

Entry order	Holding	Weight	\$ at \$300k	~Shares	Job / Guide role word	Primary, then alternates	Status	2025-26 list (history only, VRF)
1	IEF (7-10y Treasuries)	23.5%	\$70,500	~783	Operations: future funding	IEF; VGIT, then SPTI	PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK	IEF yes; VGIT no; SPTI no

Entry order	Holding	Weight	\$ at \$300k	~Shares	Job / Guide role word	Primary, then alternates	Status	2025-26 list (history only, VRF)
1	TLH (10-20y Treasuries)	42.5%	\$127,500	~1,365	Operations: future funding (one duration-weighted decision with IEF)	TLH; if missing, IEF 40.9% + TLT 25.1%	PENDING WlnS AVAILABILITY + POSITION-LIMIT CHECK	TLH no; TLT yes
2	Building minimum: 4.125% Treasury note 15-Nov-2032 (CUSIP 91282CFV8, VP per S1 via D1), else IBTM	24.3%	\$72,900	note ≈ \$75k face (≈ \$72,570 at the model price 95.27 + 1.49 accrued, ASM); IBTM ~3,336	Building minimum: "risk management" or "future funding" (pick one word, record it)	1) 2032 note (\$10); 2) IBTM (\$25); 3) VGIT (~1,281), then SPTI (~2,669), noted as "moves like, not dated"	PENDING WlnS AVAILABILITY + POSITION-LIMIT CHECK	IBTM no; VGIT no
3	VT (world stocks)	8.7%	\$26,100	~163	Stock fund: growth	VT; VTI + VXUS (62/38: ~\$16,180 / ~\$9,920, ~42 / ~114 sh, DER); then ITOT + IXUS	PENDING WlnS AVAILABILITY + POSITION-LIMIT CHECK	VT, VTI, VXUS yes; ITOT, IXUS not found (W1 grep of competition/historical/2025_26/, 2026-09-28)
-	Cash float	~1%	~\$3,090 before, ~\$2,990 after 4 × \$25 (IBTM route, DER)	-	Liquidity	Keep at least \$1,000	-	-

- Hedge mix IEF 35.6% / TLH 64.4% of the hedge gives "about 10 years" of rate sensitivity, like the payments' (issuer durations IEF 6.86y, TLH 11.58y on 9/25, VP via ticket v1; 9.90y liability, VRF). Formula: IEF share of hedge = (D_TLH – 9.90) / (D_TLH – D_IEF); re-read durations on the trade day (ticket v1 s10).
- Weights trace to the plan on 2 Jan 2028: ladder 65.9%, minimum 25.3%, stock fund 8.7% of about \$464k (MODEL, E6 [7]); the 1% float comes out of the minimum's share (25.3 – 1.0 = 24.3, DER).
- Entry: IEF and TLH first, then the minimum, then VT, **all in one session**, so the time stamps show promise-first (D7, INT). Target fills by Fri Oct 2 ET; hard stop Fri Oct 9 ET (ASM team rule).
- Volume rule (no trade above 2× daily volume, VP): every order is tiny against volume (IBTM ~3,336 vs ~216k/day; TLH ~1,365 vs ~1.97m/day; VP via ticket v1/E4). Read Vanguard volumes in WlnS.

2.2 Fallbacks (decided in advance; E6 s5.10)

Trigger	Book	Orders at \$300k (illustration, DER)	Notes change
No dated 2032 instrument listed (no Jul-Dec 2032 note, no IBTM)	(ii)R with a stand-in	IEF \$70,500 / TLH \$127,500 / VGIT \$72,900 (~1,281; else SPTI ~2,669; about 4.8-4.9y) / VT \$26,100 / float	Note B says it moves like a Treasury maturing in late 2032; never claims a date. Real plan unchanged
Two-reader check fails twice after redrafting	(iii): Laura's literal January-2027 book	Re-price the ladder on the trade date (A2_curve_recheck.py + about \$2.1k for Nov-15 dates; hand check about \$294,400 + \$292 per 1bp fall since Sep 25, ASM). Sep-25 illustration: IEF 34.9% \$104,700 (~1,163) / TLH 63.1% \$189,300 (~2,027); then, only after both show Filled , SGOV (else BIL, then SHV) with the leftover if ≥ \$1,000 (~\$4,500, ~45 SGOV); cash ~\$1,400 after 3 × \$25. Leftover under \$1,000: a listed dated payment rung; otherwise revert to (ii)R. Never a token sliver	One reflection answers "why so little stock" with forced-versus-chosen content. Without a rung, IEF and TLH are one decision: accept that in the vote and log it, or choose (ii)R (T2 finding 4)

Trigger	Book	Orders at \$300k (illustration, DER)	Notes change
Team votes ALT	Ticket v1 (ii) at 50/50	IEF 23.5% \$70,500 / TLH 42.5% \$127,500 / VT 17.0% \$51,000 (~318) / VGSH 16.0% \$48,000 (~833) / cash ~1% (~\$3,100 after 4 × \$25, DER)	Note C becomes the VGSH note: never "the floor" or "already owned". Record the growth-money bond choice (short or dated, AX1 C18) before the VGSH order

Every ticker in this table is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK** (SGOV, BIL, SHV, VGSH, VGIT, SPTI included).

3. Position-limit branches (read both Session Rules lines first; E6 s5.3, ticket v1 s3)

Single-security limit shown	Book (ii)R
None, or 43% or more	As in 2.1
25-42%	IEF 23.5 / TLH at the cap minus 1 / SPTL the rest of the hedge. Minimum held one point under the cap if the cap is under 25.3%. At 25%: IEF 23.5 / TLH 24 / SPTL 18.5 / IBTM 24 / VT 8.7 / cash about 1.3%; at \$300k: \$70,500 / \$72,000 (~771) / \$55,500 (~2,286) / \$72,000 (~3,295) / \$26,100 (~163) / ~\$3,850 after 5 × \$25 (E6 [7]; \$ and shares DER)
Under 25%	Hedge per ticket v1 s3: IEF, TLH, SPTL at the cap minus 1; VGIT, then VGLT, take the rest (at a 20% cap: IEF 19 / TLH 19 / SPTL 19 / VGIT 9, about 9.9y, ticket v1). Minimum: IBTM at the cap minus 1, the rest in VGIT. VGIT bought for both jobs counts as one holding against the cap: add the two before ordering (at 20%: 9 + 5.3 = 14.3%, DER). If 5 or more hedge funds would be needed (a cap of about 17% or less), stop and ask Wharton (Contact Us) before trading
A separate, higher bond limit is shown	The 2032 note may carry the whole minimum
IEF not listed	VGIT (else SPTI) re-solved: VGIT 16.6 / TLH 49.4 (SPTI 16.3 / 49.7); TLH rises, so re-check the cap row (ticket v1 s12, DER)

- Never add to a holding within one point of the cap. A 20-24% stock fall, or a 10-12% fall with a 50bp rally, can lift TLH above a 25% cap (T2 [5]; ASM that WInS re-checks the limit).
- A pre-chosen branch (for example the SPTL split) is **not** a "refined" decision (AY2 D7 C1). An order rejected by a limit nobody could see is a genuine rule-driven trade.
- SPTL's alternate is VGLT. Describe every branch as "about 10 years"; never two decimals. Every ticker in this section is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK**.

4. Trading calendar (ET for markets; AEST/AEDT for the team; NSW/Victoria; E6 s7, E5, ticket v1 s9)

When	What	Owner
Tue Sep 29 AEST	Roles; trader + backup; AI-use log started; no-contact rule written down	team leader; log keeper
Wed Sep 30 AEST	Own-words lines for the votes; exam/travel calendar	all
Thu Oct 1 AEST	Gate (section 1): votes, Session Rules, listings, box test, labels, drafts copy-checked, two-reader check	trader; second reader; team leader
Fri Oct 2 ET (U.S. open 9:30 a.m. = 11:30 p.m. AEST Fri)	Target fills , one session. Screenshot every position (starts the tested window). Jobs report (SNIP date): check only, never trade	trader; data owner
Fri Oct 2 AEST	Repo-visibility decision (E6 D11)	team leader
Sun Oct 4	Daylight saving starts: U.S. open = 12:30 a.m. AEDT next day. AI-use log backfilled	log keeper
Wed Oct 7 AEDT	Submit the roster (due Sat Oct 10, 08:00 AEDT)	team leader
Fri Oct 9 ET	Hard stop for fills (ASM team rule)	trader
Wed Oct 14 ET	CPI release (SNIP date): check only	-
Oct 19-23	Duration check (trade only if the hedge drifts more than 0.25 years from its start, another day from any buy); choose the three notes	rates analyst

When	What	Owner
Tue Oct 20 close = Wed Oct 21, 07:00 AEDT	End of the tested window: screenshot every position	data owner
Oct 21-22 AEDT	Reflections written by students, without AI	note writers
Thu Oct 22 AEDT	Consistency table, word counts, number check; TN submitted	log keeper; team leader
Fri Oct 23, 5:00 p.m. ET = Sat Oct 24, 08:00 AEDT	TN deadline, no extensions (VRF)	-
Oct 27-28	FOMC meeting (VP via ticket v1), after the TN deadline	-
Tue Nov 3 AEDT	IPS words frozen; no "tidy-up" trades after this	all
Fri Nov 6 ET	Trading ends, portfolio freezes (Sat Nov 7 AEDT)	-

Brisbane is 1 hour behind Sydney from Oct 4; Perth 3 hours (E6 s7).

5. Which three executed trades become the notes (a menu, not a quota: TN guide L11-13, L29-30, VRF)

Slot	Trade	Shows (only if true)	Why this trade (INT)	The reflection's one extra job (E2 F9)
A	TLH (IEF bought with it as one decision)	Supported and tested	The operations job: the ten payments only her portfolio may fund	Tested: the three pre-registered numbers (section 7), then what the rule said to do (hold); small moves and poor tracking reported as they are
B	The building minimum (2032 note, IBTM or stand-in)	Refined , only if the decision log shows, dated before this order, the change from "a floor bought in 2031" to "the minimum bought when the 2028 money arrives", with the reason in students' words; otherwise supported	The most Laura-specific holding: a figure partners will rely on, sized by her own second deposit (E4 L6)	Refined: what changed and the research that changed it (at equal stock risk, about the same middle, better bad cases; at most one number). Supported: why a dated holding (an amount on a date, not only a moving price)
C	VT	Supported (carries out the plan as refined)	The growth job: money nobody else relies on	The true refinement story: growth-first (about 75% stocks) was tested and dropped because it could miss a payment (about 3 in 100 modelled futures; about 4 in 10 if the 2028 deposit never came; F-401, F-404, VRF, MODEL); now all money nobody relies on is in stocks, the "why so little stock" answer by goal. One number at most

- Never create a trade to fill a category. Three different jobs, three different role words (TN guide L11 "three investment decisions", VRF).
- Fallback (iii): A = TLH; B = a listed dated rung, else IEF only if the "one decision" acceptance is logged; C = SGOV/BIL leftover. Fallback ALT: A = TLH; B = VT; C = VGSH (ticket v1 s4).
- Final choice made Oct 19-20 from executed trades only (ticket v1 s4).

6. Per-note checklists (elements in noun phrases, never wording)

6.1 WInS note: common rules (E6 s5.6; Guide p.3 L84-86, VRF)

- [] **C1** one Guide role word + the team's own label (same label in every note, the log and the IPS).
- [] **C2** Laura's dated need, **with the scaling marker**.
- [] **C3** one dated, checkable fact, source named in words (no link).
- [] **C4** the rule or risk.
- [] **≤300 characters** until the box test says otherwise; if the box visibly allows more, extra detail up to about 600.
- [] **Trim order** if over: keep C2 (with marker), C3 and the role word; move C4 and the label to the reflection (AY1 D12 C1; T2 finding 6).
- [] No ticker, side or quantity needed (the trade record shows them).
- [] Paste, confirm nothing is cut **before** submitting; never finish a cut sentence with an added note (it carries a later time stamp); copy the saved note character for character into the decision log straight after.

6.2 Note elements and banned words by note

Note	C2 need (with marker)	C3 one fact (pick one)	C4 rule / risk	Banned in this note
A operations	The ten \$50,000 payments 2033-2042; her operations money after both deposits (about two-thirds of her total money, never "of the payments", E3-10)	The pair's rate sensitivity about 10 years, set to the payments' (issuer durations, dated; team's calculation); or TLH's own effective duration (iShares, dated)	Never sold after a rate rise or trimmed after a fall; re-mixed only if sensitivity drifts more than a quarter-year	match, matched, "mature in her payment years" (74.7% of TLH matures after 2042, VRF via ticket v1), stable, safe, reduce volatility, guaranteed, locked, operating reserve, "bought in January 2027", "fits in \$300k"
B building minimum	The least she will put toward the building in 2033, sized to repay her second deposit; "after her 2028 deposit"	The note matures 15 Nov 2032, 47 days before 1 Jan 2033 (Treasury, dated); or IBTM holds Treasuries maturing Apr-Sep 2032 and ends around December 2032 (iShares, dated). Stand-in: moves like a late-2032 Treasury, no date	Held to its end; never sold because rates moved	"repays \$X" for IBTM; guaranteed; risk-free; "locked 5%"; any 2031 range or top; promised; pledged; any date claim for a stand-in
C stock fund	The building's stretch and her flexibility in 2033, from money nobody else relies on; "after her 2028 deposit"	All of the plan's stock money, about 9% of her money after both deposits (team's calculation); or VT's stock count and share outside the U.S. (Vanguard, dated)	Held until 2033, never sold for market reasons; a fall shrinks only the stretch above the minimum, never the payments or the minimum	a return forecast, overvalued, CAPE, "halves AI exposure", "known/locked 5.2%", "Laura likes risk", the statistics degree, promised

Banned in every note and reflection (E6 s2, s5.1, s5.7, s5.9, s6.1 rank 6; ticket v1 s8): Laura quotes or the case pull quote as hers; "Laura's portfolio" for the WInS holdings; reserve size, facility range, co-sponsor range or any 2031 figure; "promised"/"pledged" for anything above the minimum; "guaranteed", "risk-free", "safe", "100%", "certain" without its qualifier; identity, heritage or Taiwan as a reason; "falling" or book-title puns; the statistics degree (E6 D8: zero uses); "our portfolio manager approves"; WInS profit or rank; probabilities (team judgement). Vocabulary to keep consistent: "certainty" for the payments, "confident" for the range, "credibility" for her standing, "moves like / stands in for" for WInS funds.

6.3 What each ≤100-word reflection must cover (students, after the trades, without AI; TN guide L46-52, VRF; E6 s5.7)

- [] The three official answers: **why** the decision; **how it aligned** with the strategy; **how it served** her goals, funding needs or risk.
- [] Which of supported / tested / refined it shows, true to what happened (section 5).
- [] Exactly one extra job (section 5).
- [] ≤95 words by two counters (higher count wins). At most one number, except the tested reflection (exactly three); every number dated and graded in words.
- [] One case-stated trait that passes the swap test (would it be true of any client?): A = fixed payments only her portfolio may fund (case L91-92); B = a figure partners will rely on (L114-116); C = "thoughtful risks" within "an appropriate balance" (L69-74) (VRF).
- [] Reflection A also covers the scaling in plain words: WInS shows her plan after both deposits, scaled; the funds stand in for the bonds she buys; in January 2027 almost all of her real first deposit buys the payments.
- [] Not in any reflection: Laura quotes; the pull quote as hers; her career story or regret line as a reason; "loss tolerance", "break-even", "house money", "composure"; the statistics degree; reserve size, facility range or 2031 figure; WInS profit or rank.
- [] Same rule names as the IPS will use (decision-log column).

7. Pre-registered "tested" data capture (E6 s5.5, s5.8; ticket v1 s0)

Item	Rule
Pre-register	Before the first fill, log: window = fill-date close to the Tue Oct 20 close; the three measures below; the rule's expected action (hold)
Measure 1	10-year Treasury rate move in bp (treasury.gov daily par curve)
Measure 2	The hedge's (IEF + TLH) % change, from WInS screenshots

Item	Rule
Measure 3	The payments' % change: value the ten payments on each date's curve with <code>A2_curve_recheck.py</code> or <code>D9_numbers.py</code> [3]; never <code>official_curve_pv.py</code> (fixed to 2026-09-25; AY2 D7 C5)
Screenshots	Every WInS position on the fill date and at the Oct 20 close (Wed Oct 21, 07:00 AEDT)
Distributions	Check IEF and TLH ex-dividend dates in the window; a ~0.4% distribution is large next to a ~1% signal
Labels	All three numbers: the team's own calculation from the WInS screen and the treasury.gov curve; percentages, not dollars
Odds	A 10bp-or-larger move: about 52% for an Oct 2 fill, about 40% for Oct 9 (T2 [4], ASM). Small moves are reported as they are; "tested" is optional

8. Student-written notes and outside readers (E6 s5.4e; ticket v1 s0d; PM-04)

Step	Check
1	A named student drafts offline from section 6 elements, with no research file or AI tool open
2	Second student restates the note's role in one sentence; mismatch = redraft
3	Five-word copy check: every five-word run of the draft searched in <code>research/</code> ; any hit rewritten. Example: <code>set -- \$DRAFT; while [\$# -ge 5]; do grep -rlif "\$1 \$2 \$3 \$4 \$5" research/ && echo "HIT: \$1 \$2 \$3 \$4 \$5"; shift; done</code> (strip punctuation first)
4	Draft pasted into a copy of the decision log before the order
5	Two readers outside the team read all three drafts and say what Laura's real money holds in January 2027. Either says "stocks" or "a building bond": fix the drafts. Two failures after redrafting: switch to (iii) (E6 D3)
6	Readers are also asked whether the WInS stocks contradict the plan (E6 s2 tests)
7	Reflections (Oct 21-22): same steps 1-3; no AI "polish"

9. Decision log and AI-use log (first names only; no surnames, emails or birthdays)

Decision log (start before the gate; E6 R7, R10; M021):

Column	Content
Date and time	AEST/AEDT and ET
Decision / trade	Includes every "decided not to trade" moment
Options considered	e.g. REC vs ALT; (ii)R vs (iii)
Vote	First names for/against
Reason	Students' own words
Rule name	Same name the IPS will use
Evidence and grade	Source and status label
Numbers	Prices, shares, \$ at fill
Saved WInS note	Character-for-character copy
Refined flag	For note B: the dated REC decision entry, written before the minimum order

AI-use log (start Tue Sep 29, backfill by Sun Oct 4; PM-08; Wharton AI policy: record in the Final Report's Works Cited, VP via brief):

Column	Content
Date	When used
Student	First name
Tool / file	e.g. Claude sessions, this pack, E6 spec, ticket v1, D9 draft checker
Purpose	Brainstorm, research, check
What was used	Ideas, numbers, labels proposed
What was not	Confirms no AI wording in notes or reflections

10. Submission checklist (TN guide L42-53, VRF; SMAApply)

- [] Exactly three notes, each from a trade **executed** in WInS ("Yes, we will verify this").
- [] Each note quoted **exactly as it appears in WInS** (copied from the WInS screen, not from a draft). Ask Wharton whether an "added" note counts (ticket v1 open item 4).
- [] Each reflection ≤ 100 words (planned ≤ 95 by two counters) and answers the three questions.
- [] No reserve size, facility contribution, projections or co-sponsor text (TN guide L20-22, VRF).
- [] Notes and reflections consistent with the IPS central idea (split of jobs) and labels (PM-06 consistency table).
- [] Submitted via the SMAApply (SurveyMonkey Apply) form on **Thu Oct 22 AEDT**; deadline Fri Oct 23, 5:00 p.m. ET = Sat Oct 24, 08:00 AEDT; no extensions (VRF).
- [] Screenshot of the submission confirmation logged.

11. Top TN risks and preventions (E5 top 10; T2; INT)

Risk	Prevention
PM-01 strategy not owned; voice reads AI-made	Own-words lines before each vote; explain-back; students draft from a blank page
PM-04 AI wording in permanent notes	Offline drafting from elements; five-word copy check; no AI polish
PM-06 notes contradict the IPS (book date, scaling, labels)	Split-of-jobs idea; scaling marker in notes; two-reader check; consistency table Oct 22
PM-08 AI use not logged	Dated AI-use log from Sep 29, backfilled
PM-05 process breakdown	Roles Sep 29; internal deadlines two days early
PM-09 generic notes (fail the swap test; the hedge note looks like Wharton's example)	One case-stated trait per reflection; Laura-specific building-minimum note
Note cut at an unseen limit	Zero-risk box test; 300-character drafts; trim order
Fewer than three executed trades (iii) or an order rejected at the open	(ii)R default; $\geq \$1,000$ float; recompute shares from the last close; (iii) T-bill order only after fills
Position-limit breach or rejection	Branch chosen before ordering; one point under the cap; never add near the cap
Day-trading or advisor-trade breach	Buys only on the first day; no same-day fixes ("run with it", VP); only the trader places orders
Overclaim words ("guaranteed", "match", "promised")	Banned-word lists in section 6.2
Late or wrong-time-zone submission	Submit Thu Oct 22 AEDT; deadline written in both zones

What this teaches

1. A trade note is permanent evidence: the reasoning is written before the trade, so the thinking has to be finished first.
2. Separating jobs (money others rely on vs money nobody relies on) makes three trades read as one plan, not three bets.
3. Deciding branches and tests in advance turns surprises (a cap, a small rate move) into honest evidence instead of scrambles.

IPS specification: what the 50-word pitch and the 500-word IPS must contain (no drafted text)

Summary (5 lines)

1. Central idea = a **split of jobs**: money other people will rely on (ten operating payments, the building minimum) is bought when her money arrives and held; money nobody relies on is fully in world stocks (E6 s2, INT).
2. The IPS = 10 content blocks on the four case dates (2027, 2028, 2031, 2033), ≤ 470 words, pitch ≤ 48 words, by two word counters (E6 s3, INT); every guide requirement is mapped below with its line (VRF).
3. Ten decision rules (R1-R10) must be fixed by vote and logged by Mon Oct 26 AEDT, because the strategy freezes at the deadline and the Final Report may only apply them (IPS guide L73-74, VRF).
4. Format breaches are fatal ("will not be considered for semifinal selection", L82-83, VRF): Letter, Times New Roman 12, Word "Double", 1-inch margins, exactly 3 PDF pages, ≤ 5 MB, no charts/links/footnotes/citations/quotes.
5. Due Fri Nov 6, 5:00 p.m. ET = Sat Nov 7, 09:00 AEDT (SMAApply L10, VRF; conversion DER, E5 [1]); team targets: words frozen Tue Nov 3, PDF check Wed Nov 4, submit Thu Nov 5 (ASM team rule, E6 s3.3).

Serves: IPS (Trading Notes only through the consistency checks in section (i)).

W2 writer, insight_v1 run, 2026-09-28. **Authority:** research/insight_v1/phase_E/E6_final_spec.md (cited "E6 s"). This file re-states its decisions for the IPS; it decides nothing new. **Nothing here is text to submit.** The six students write every pitch and IPS sentence in their own words; Wharton's AI policy requires any AI use to be credited in the Final Report's Works Cited (R-W46, VP via E6). No Laura quotes appear in the IPS (brief s16, binding). Everything below assumes the team adopts lock-early and **REC** (E6 decision D1-D2, both still open votes); if the team votes ALT, sections (c) R2/R4 and (k) change as E6 s5.10 and s7 say.

Status labels (brief s3): **VP** = VERIFIED-PRIMARY (read on the primary page); **VRF** = VERIFIED-REPO-FILE (official or primary file in the repo, with line); **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION (a chosen input or team rule); **MODEL** = output of an assumption-based simulation, never a forecast; **DER** = derived by arithmetic or a script from labelled inputs; **INT** = E6's or this writer's judgement, never a fact about Laura; **UNVERIFIED** = not yet checked. Line numbers: "L" alone = the IPS guide competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt; "case L" = Laura_Gao_2026_Client_Profile.txt.

Terms (defined once; for the team's understanding, not IPS vocabulary):

- **Treasury:** a bond issued by the U.S. government. **Zero-coupon Treasury (STRIPS):** pays one fixed dollar amount on one date and nothing before. **Treasury bills:** Treasuries maturing within a year, used here as a cash place.
- **Ladder / rung:** ten zero-coupon Treasuries, each maturing on 15 Nov before one \$50,000 payment (2033-2042); a rung is one of them (E6 terms).
- **Building minimum:** one zero-coupon Treasury bought in January 2028 that repays the second deposit's full dollar amount shortly before 1 Jan 2033 (E6 terms, INT design).
- **Stock fund:** one broad world stock fund (an **ETF**, a fund traded like a share) holding all money the ladder and the minimum do not need.
- **Kept money:** whatever the 2033 rule does not give to the building; Laura's flexibility.
- **Default:** the U.S. government failing to pay a Treasury in full and on time. **Nominal:** in plain dollars, not adjusted for rising prices (inflation).
- **Rate sensitivity (duration):** roughly how much a bond fund's price moves when interest rates move; "about 10 years" means about 10% per 1 percentage point.
- **bp:** 0.01 percentage point. **p5 / p50 / p95:** the value 5% / 50% / 95% of simulated paths fall below (p50 = median).
- **Co-sponsor:** a partner Laura will ask to help fund the facility from 2031 (case L109-116, VRF).
- **Derivative / forward:** a contract whose value depends on another price, e.g. an agreement to swap currencies later; banned in WInS (VP via E6 s8c).

(a) The central idea and the 50-word pitch

a1. Central idea (content, not wording) (E6 s2, INT)

- **A split of jobs.** Money others will rely on (the ten operating payments; the building minimum she will announce) is bought when her money arrives and held to its date, never taking stock-market risk. Money nobody relies on is invested fully in world stocks.
- **Consequences the idea must let a reader see:** once bought, the payments depend neither on markets nor on her next contract (her second deposit is career income: "using earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures", case L43-45, VRF); she tells partners only what she already owns, plus a stretch whose odds anyone can check.
- **Why not a time-worded ban** ("no stock risk until all ten payments are bought"): the WInS book buys world stocks on day one and the notes are read before the IPS, so a ban reads as a contradiction; a split of jobs is true of both Laura's plan and the WInS book (E2 C1/F4; AY2 rank 1; INT). The "payments first" ordering lives inside the 2027 and 2028 rules, not in the pitch.

a2. What the pitch must contain (official requirement: "Maximum 50 words" L33; "Quickly summarize your team's investment strategy. Clearly communicate the central idea behind your approach and how it is designed to support your client's financial goals and future funding needs." L35-36, VRF)

#	Element (the team writes the words)	Must carry	About (INT)
1	Purpose first	What the plan secures for the residency, named by what it secures (its ten years of operations), before any instrument or prohibition (E3-5; E2 F3; D13c checklist)	12-15 words
2	The Laura-specific rule	Each of her two deposits buys a promise when it arrives: the first, the operating years; the second, the building's minimum from her own earnings. Once bought, neither depends on markets or on her next contract	15-18 words
3	What the rest does + credibility	Only money nobody relies on is in world stocks, for a larger gift and her flexibility; she tells partners only what she owns	12-15 words

a3. Hard limits and banned content (E6 s2; D9 M062; D13 rules; E3 4c; all INT/ASM team rules unless marked)

- **Length:** ≤48 words, counted by two counters (e.g. Word and `D9_draft_checker.py --kind pitch`), the higher count wins (E5 PM-02). Official cap 50 (L33, VRF).
- **Numbers:** none; at most one if essential. **Tense:** future or conditional; nothing is bought in 2026.
- **Banned words and content:**
 - flat claims "bought in January 2027" or "fits inside \$300,000" (false in about 1 rate path in 3, MODEL 30-34%, E6 R1);
 - "guaranteed", "risk-free", "safe", "100%", "95%", a probability, or "certain" without its qualifier;
 - "promise"/"promised" for the stretch above the minimum;
 - jargon: barbell, LDI, duration, STRIPS, ladder, funded ratio, percentile, glide path, liability matching;
 - fund names, tickers, a return target;
 - any quotation, the case pull quote, "Laura said";
 - identity, heritage or Taiwan as a reason; book-title, "falling" or "balance" metaphors (the case's own "appropriate balance" is fine);
 - the statistics degree; "innovative".

a4. Tests a pitch draft must pass (E6 s2; D12 M234; INT)

- [] Six unpaid readers outside the team restate all three elements after one read (paraphrase test; unpaid because paid help is banned, R-W21/R-W28, VP via E5).
- [] No reader thinks anything is already bought.
- [] Shown the three WInS notes first, no reader thinks the WInS stocks contradict the pitch (E2 s5).
- [] Swap test: the pitch fails if it still reads true with another client's name in it (D6 M125, INT).
- [] Run `D9_draft_checker.py --kind pitch` (flags jargon/overclaim words; a flag is a prompt to look, not an error, ASM).

(b) IPS content map: guide requirement -> block -> word budget -> evidence

Whole-IPS rules (E6 s3, INT unless marked): at most 12 ideas in 10 blocks; one action and one "because" per case date; ≤470 IPS words (hard cap 500, L39, VRF); case facts only where a rule needs them; rule parameters in words ("half", "the full amount of her second deposit"); at most **one** rounded, dated computed figure (the forced-caution fact in B5); no percentiles, probabilities, dollar ranges, tables or lists; every forecast graded ("we assume", "history suggests"); every price dated ("at September 2026 prices"); at most about 15% of words on what could go wrong (ASM ceiling, E3-11); no headings inside the IPS ("You do not need to address these points separately", L66, VRF).

b1. The ten blocks (content only)

Block	Requirement it fills (quoted, VRF)	Content the team's sentences must carry (E6 s3.1)	Words (INT)	Evidence (file; status)
B1 Purpose + central idea	"What is the central idea behind your investment strategy?" (L16); "What your team is trying to accomplish for the client" (L55); "Your client's profile and financial goals should remain at the center" (L27-28)	What the plan secures (ten operating years, then the building minimum, then growth); the split of jobs; her second deposit is her own career income, so once bought the payments no longer depend on markets or her next contract	40	Case L43-45, L88-92 (VRF); E6 s2 (INT)
B2 Certainty	"desired degree of funding certainty" (L10); case: "define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty, and identify the assumptions" (case L97-99)	(a) bought, not forecast: each payment matched by a Treasury maturing shortly before its date, held to maturity; (b) evaluated by price: on the purchase date the money covers the market price of all ten, no model percentage; (c) residual risk: certain in U.S. dollars barring a U.S. Treasury default; (d) gap (i) January 2027 price; (e) gap (ii) fixed dollars vs Taiwan costs and exchange rate, named not solved	50	Case L90-92, L146-148 (VRF); brief s16 (binding); section (e) below
B3 Jan 2027: buy the payments	"How will the portfolio's asset allocation and composition change as future funding needs approach and payments are made?" (L22-23); "preparing for the operating commitment" (L48)	Buy all ten on arrival, no waiting for better yields (waiting is a bet on rates); if short, longest-dated first, because the waiting piece is smallest and least rate-sensitive and less stays unbought if the next money never came; counterpoint stated: the waiting piece is the first operating year; leftover waits in Treasury bills	35	Case L91-92 (VRF); rule R1; D1/AX1b (MODEL)
B4 Jan 2028: finish, minimum, stocks	"planned changes in the portfolio as future funding needs approach" (L46-47); "determining a responsible facility contribution" (L49)	Second deposit first completes any unbought payment; then buys a Treasury repaying the deposit's full dollar amount just before 2033 (the building minimum; the case's "meaningful personal contribution", case L111); the rest into one broad world stock fund left alone to 2033; one labelled stress clause (smaller or later deposit follows the same order; joint tail stated once)	45	Case L43-45, L101-104, L110-112 (VRF); rule R2; E4 [7] (MODEL)
B5 Risk and return, goal by goal	"risk tolerance" (L3); "How does your strategy balance growth, risk, liquidity, funding reliability, and financial flexibility?" (L20, also L59); "approach risk and return" (L46)	Forced caution: at September 2026 prices the payments cost about 98% of her first deposit (the one allowed number), so 2027 holds no stocks by arithmetic; chosen: the minimum takes no market risk because partners rely on it; stocks are for a larger gift and flexibility; the most markets can take = the stock fund; portfolio fact: her income already rides on publishing and speaking, so no themed or single-country bets; returns as a range across two published forecasters, not a target	45	Case L68-74 (VRF); E3 [1]: 97.4-98.1% (DER); section (d)

Block	Requirement it fills (quoted, VRF)	Content the team's sentences must carry (E6 s3.1)	Words (INT)	Evidence (file; status)
B6 Diversification + liquidity	"portfolio construction and diversification" (L46); "liquidity needs" (L10); SMAApply criterion "uses appropriate diversification" (SMAApply page L49, VRF)	Diversified by funding purpose and by date: ten dated payment bonds, one dated building bond, one fund of about 10,000 companies worldwide; U.S. Treasuries for the promises are deliberate (dollar-denominated, dated); liquidity: no withdrawals before 2033, each payment from its own maturing bond	25	Case L63-65 (VRF); VT 10,088 stocks, 37.7% non-U.S. at 8/31 (VP via ticket v1 s12; PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK; alternates VTI + VXUS, then ITOT + IXUS)
B7 Jan 2031: co-sponsor range (method only)	"How will your strategy protect the operating commitment while preserving flexibility for a responsible facility contribution?" (L25-26, L64-65); the final range is NOT expected (L68-69)	Nothing traded; she speaks only once payments and minimum are owned; bottom = the minimum owned; top = minimum plus half the stock fund's value that day; confidence both ways (below bottom only on a Treasury default; top reached if world stocks are no lower two years later, graded "history suggests"); words by tier: owned / expected if stocks hold their value, never "promised"; U.S. dollars binding	40	Case L109-120 (VRF); rule R4; numbers go to the Final Report (E6 s10)
B8 Jan 2033: reserve, gift, flexibility	"managing the operating reserve's asset composition over time, determining a responsible facility contribution, and preserving appropriate financial flexibility" (L48-50)	Reserve = the ladder, bought from 2027, named the operating reserve on 1 Jan 2033 before the first payment; held to maturity; composition changes only as bonds mature (case allows "if at all", case L96); gift = minimum + half the stock fund, never above the announced top; everything else kept in U.S. dollar Treasury bills, hers "as the project develops" (case L102-103); no New Taiwan dollar conversion or hedge before the facility decision; Taiwan building costs expected to rise, so a fixed-dollar gift buys less	45	Case L94-107 (VRF); rules R3, R5
B9 Governance + costs	"What principles will guide your investment decisions?" (L18); "disciplined decisions during changing market conditions" (L4-5); "How your strategy will guide portfolio construction and investment decisions as the client's needs change" (L61-62); "may not revise its investment strategy after the submission deadline" (L73-74)	Written rules run the portfolio; Laura makes the two calls the case gives her, within the rules (2031 announcement, never above what the rule allows; the 2033 amount and use of kept money); rules change only if her circumstances change, never because markets moved; stock fund never sold to protect a gain or wait for a recovery; no rebalancing between the three parts; bought Treasuries carry no ongoing charge; any cost paid from the stock fund	35	Case L101, L110 (VRF); rules R6-R8; CFA 2010 elements (VP via D8)
B10 The WInS sentence	"Your portfolio and Final Report should reflect the strategy established in your IPS." (L75)	WInS shows her plan on the day both deposits are in, scaled to \$300,000; the Treasury funds stand in for her bonds (move like them, do not mature on her dates); in January 2027 almost all of her real first deposit buys the payments	25	D10 1.5; ticket v1 s1; T2 16a (INT); rule R10
		Content words	about 385	E6 s3.1 (INT)
		Connecting words allowance	about 70	INT
		Target total (plan ≤470; hard cap 500)	about 455	ASM page-fit margin, section (h)

Paragraph plan (≤7 paragraphs, D10 2.3, ASM): (1) B1+B2; (2) B3; (3) B4; (4) B5+B6; (5) B7; (6) B8; (7) B9+B10.

b2. Coverage tick-list (every "should" has a home; tick each against a sentence number in the draft)

- [] Goals, horizons, liquidity, certainty, flexibility (L9-10) -> B1, four dates, B6, B2, B8.
- [] Central idea (L16); principles (L18) -> B1; B5, B9.

- [] Balance of growth, risk, liquidity, reliability, flexibility (L20, L59) -> B5, B6, B2, B8.
- [] Allocation changes as needs approach and payments are made (L22-23, L46-47) -> B3, B4, B7, B8.
- [] Protect the operating commitment while keeping flexibility for a responsible facility contribution (L25-26, L64-65) -> B7, B8.
- [] Risk tolerance (L3), risk and return (L46) -> B5. Construction and diversification (L46) -> B6.
- [] Preparing for the commitment; reserve composition over time; facility contribution; flexibility (L48-50) -> B3-B4; B8.
- [] Framework, not individual investments or detailed calculations (L51-52) -> number policy; no tickers.
- [] What / why / how as needs change (L53-62) -> B1, B9. Official record; portfolio reflects IPS (L73-75) -> B9, B10.
- [] Case: define and evaluate certainty, name assumptions (case L97-99) -> B2.
- [] Case: assumptions on performance, cash-flow timing, outside funding, inflation on projections and facility costs, flexibility (case L146-148) -> B5; B3-B4; B1-B2; B2+B8; B8.
- [] Case: favorable and unfavorable outcomes; range protects the payments (case L118-120) -> B7 (method only).
- [] Excluded on purpose (L68-69, VRF): final reserve calculation, projections, the final range, the co-sponsor draft.

(c) Decision rules to fix before Nov 6 (trigger / action / record) (E6 s4; all parameters fixed now, because a parameter chosen after Nov 6 is a redesign: L73-74 and Guide p.5 L156-159, VRF)

Vote each by **Mon Oct 26 AEDT** with first names, date and reason in the decision log (E5 s4, ASM team rule). The IPS states each rule in words only; the numbers in the right column are for the team's understanding and the Final Report.

Rule	Trigger	Action	Record	Evidence (status)
R1 Buy the payments (rates-fall rule)	First deposit arrives, 1 Jan 2027	Buy all ten rungs at that day's prices, all at once (no staging, no yield trigger). If short, buy longest-dated first; the unbought part is always the earliest payment and is the first use of the next money. Leftover waits in Treasury bills. Headline stays conditional: "if prices allow, otherwise when the second deposit arrives" (never "by 1 Jan 2028 at the latest")	Purchase-day ladder cost vs cash; which payments were bought; gap or leftover	Real Nov-15 ladder \$294,387 at the 2026-09-25 curve, headroom \$5,613 = 19bp (VRF inputs, ASM method). About 1 in 3 rate paths cost more than \$300k (MODEL 30-34%; history: ≥ 19 bp falls in 32-36% of 68-day windows since 1962/1990, VP data, DER). Staging: no reliable gain, 27-35% chance of needing the deposit (MODEL). Longest-first: +5% top-up price risk vs +15%, \$31.4k vs \$47.7k face unbought at -100bp if no deposit (MODEL, D1/AX1b)
R2 The 2028 deposit	Deposit arrives (1 Jan 2028 per case; each instalment if late)	In order: (1) complete any unbought payment; (2) buy a zero-coupon Treasury maturing before 1 Jan 2033 repaying the deposit's dollar amount (the building minimum), or all that is left if less; (3) everything else, including the 2027 leftover, into one broad world stock fund. Smaller or later: same order, minimum smaller or later. No minimum and no stock fund until all ten payments are bought. Joint tail (team's own stress case): rates fell before Jan 2027 and the deposit is smaller than the unbought part -> all of it to the payments, no minimum, no stock fund, unfunded part of the first payment stated	Deposit amount and date; top-up cost; minimum's face and price; stock fund's starting value	Case "will contribute" (case L43, VRF). Growth money \$158k; minimum costs \$118k at 5y 4.98% (ASM that it holds); stock fund \$40k = 8.7% of all money (MODEL). -50bp first: fund about \$20-23k; -100bp: about \$1-7k; -150bp: no fund, minimum about \$127-137k (E6_final_checks.py [8], MODEL; corrects E4 [7]'s \$31k/\$15k/\$147k). Joint tail with no deposit: \$11,957 (-50bp) or \$31,413 (-100bp) of the 2033 payment unfunded (MODEL, D1/D3)
R3 Operating reserve	1 Jan 2033, before the first payment or any facility contribution	Ladder named the operating reserve; pays the first \$50,000 at once. Held to maturity; each rung matures about 47 days before its payment and waits in bills; nothing sold early; no rebalancing between reserve, minimum and stock fund; the reserve empties itself by 1 Jan 2042	Reserve market value, cost and face on 1 Jan 2033	Case L94-97 (VRF). About \$395k market value on today's forwards, cost about \$292-294k, face \$500k (VRF inputs, ASM)

Rule	Trigger	Action	Record	Evidence (status)
R4 2031 co-sponsor range: METHOD only (no figures in the IPS)	1 Jan 2031, when she begins approaching co-sponsors (case L109)	Condition: all ten payments and the minimum owned, otherwise no range, only "aspiration". Nothing traded. Bottom = the minimum owned. Top = minimum + half the stock fund's market value on 1 Jan 2031, set once, not reset. Confidence both sides: below the bottom only on a U.S. Treasury default (and only if no fee is ever charged to it, R8); the top is reached if the stock fund is no lower on 1 Jan 2033 than on 1 Jan 2031 (history and model named in the Final Report). Words: owned / expected if stocks hold their value / kept; never "promised" or "pledged" for the stretch. U.S. dollars binding; any New Taiwan dollar figure dated and illustrative	Stock fund value on 1 Jan 2031; the stated range	Case L110-120 (VRF). Illustration only, NOT for the IPS: median bottom \$150k, top \$175k; top reached in 73% of model paths (67% on a Vanguard-type input) (MODEL); U.S. stocks' two-year total return ≥ 0 in 84% of 97 overlapping periods 1928-2025 (VP data, DER, E4 [5])
R5 2033 facility contribution + flexibility	1 Jan 2033, after R3	(1) Gift = minimum + half the stock fund's value, never above the announced top. (2) Kept money = the rest (other half plus any growth above the top), moved the same day to U.S. dollar Treasury bills, hers "as the project develops"; she may choose to add to the gift (her decision, not counted by co-sponsors). (3) No New Taiwan dollar conversion and no currency hedge before the facility decision (a forward is a derivative; banned in WInS and, on the safe reading of "Investments permitted (for BOTH contributions)", in the plan, R-AN15)	Gift, kept money, date of the facility decision	Case L101-107 (VRF; no contingency fund required, L106-107). Gift p5/p50/p95 \$165k/\$174k/\$188k; kept \$16k/\$32k/\$66k (MODEL). "Half" sized against Taiwan building-cost drift 3.54%/yr since 2021 (F-508, VP inputs; ASM sanity check, not a contingency fund)
R6 Stock fund conduct	Every day 2028-2032	One broad world stock fund; never rebalanced against the Treasuries; never sold before 1 Jan 2033, to protect a gain or to "wait for a recovery". A fall shrinks only the stretch above the minimum	Yearly check that holdings match R2-R6	Case L69-74 (VRF). Bottom-decile 2031-32: gift \$169k at the median, at or above the announced bottom in 100% of paths (MODEL, E6 [3])
R7 Governance + reviews	The four case dates, plus a yearly check	Who decides: the students' written rules; the advisor is administrator only, never "our portfolio manager approves" (R-W19, VP). Laura makes the two calls the case gives her, within the rules (case L101, L110). A rule changes only if Laura's circumstances change (e.g. a different deposit amount or date), never a market move; worded so it cannot read as permission to revise after Nov 6	Dated decision log: what fired, when, the numbers	L73-74 (VRF); Guide p.5 L156-159 (VRF); CFA 2010 elements 2a-2b (VP via D8/D10)
Rebalancing (inside R3/R6)	n/a	None between the three parts. Composition changes only as bonds mature. This is written down as a deliberate no-rebalance policy	Yearly check	CFA 2010 element 4c (VP via D8): a no-rebalance policy is documented; E6 conflict 35
R8 Costs	Any fee or trading cost	Ladder and minimum held to maturity with no ongoing charge; every cost paid from the stock fund, never from the payments or the minimum; the Final Report states the fee assumption, base and rate	The fee assumption	Case L43 "with an asset management firm" (VRF); case silent on fees, so a team assumption (ASM). 1% on the stock fund only costs about \$3k of the median total; the same rate on all assets paid from the fund about \$31k (MODEL, E6 [6])

Rule	Trigger	Action	Record	Evidence (status)
R9 Certainty definition + assumptions	Fixed in the IPS	B2's five parts; assumptions: returns a range across two published forecasters (graded); flows at the start of each year (case L59); no outside money for the payments (case L91-92); payments fixed in dollars (case L90); Taiwan building costs expected to rise (case L146-147); U.S. dollars binding; flexibility kept by R5	The IPS itself	Case L97-99, L146-148 (VRF)
R10 WInS conduct to Nov 6	Each trading day	Book (ii)R (E6 s5). The operations hedge is never sold after a rate rise or trimmed after a fall; re-mixed IEF-against-TLH only if its rate sensitivity drifts more than a quarter-year. Minimum and stock fund held. No day trades; no tidy-up trades after the words freeze (Tue Nov 3)	Every trade and every "decided not to trade" moment	L75 (VRF); User Guide "Day Trading: This is not permitted" (VP via A4). All securities PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK; alternates in <code>securities_and_allocation.md</code> s1-s2

(d) Risk tolerance, goal by goal (E6 B5, R4-R6; E3; content for B5, not wording)

Risk tolerance here = how much loss each pot of money can bear, given what it is for. The case gives the frame: "willing to take thoughtful risks" but wants "an appropriate balance between pursuing growth and protecting the capital required for her goals" (case L69-74, VRF).

Goal (case line, VRF)	Tolerance (INT)	Why (evidence)	What the IPS must show
Ten \$50k operating payments, 2033-2042, "high degree of certainty", no outside money (case L88-92)	None for market risk	Fixed nominal amounts others rely on; payments cost about 98% of the first deposit at September 2026 prices (DER, E3 [1]), so there is nothing spare to risk in 2027	Caution here is forced by arithmetic , not chosen timidity
Building minimum (her announced bottom)	None for market risk	Partners will rely on it; overpromising "could damage her credibility" (case L114-116)	Chosen caution, with the reason named (reliance)
Facility stretch and kept money	Full stock-market risk	Nobody relies on it; living costs are outside the portfolio and there are no withdrawals before 2033 (case L61-65); a five-year horizon (the stock fund is bought in January 2028 and held to January 2033)	What stocks are for (a larger gift, more flexibility) and the most markets can take (the stock fund, roughly what her deposits would have earned safely; never the payments or the minimum)
Her income (outside the portfolio)	Already exposed to publishing and speaking (case L43-45)	Adding themed or single-country bets would stack risk	A portfolio fact, not a diagnosis of her: no themed, AI or Taiwan tilts (D8 M081; brief s4 tokenism rule)

- **The honest cost (E6 s0, MODEL):** about 9% of her money is in stocks 2028-2032; the p95 is \$16k lower than the 2031-lock alternative; in exchange the p5 total is better (\$182k vs \$168k). The IPS names the trade-off in principle only; no percentiles (number policy).
- **Her regret line** (B1b-Q11, section f) is why the plan must answer "why so little stock" by goal, not by temperament; it is **not** quoted in the IPS (brief s16).
- **Return assumptions:** a range across two published forecasters (e.g. JPM 2026 LTCMA AC World 7.00% compound, VRF `JPM_LTCMA_2026_US_matrix_USD.pdf` p.2; the Vanguard-type input, 5.08% compound, an ASM hybrid built on VP inputs, E6 [4]), graded "we assume", never a target.
- **Banned in B5 (E6 s6.2; E3):** CFA "ability/willingness" labels, "loss tolerance", "break-even", "house money", "composure", "Laura likes risk", the statistics degree.

(e) Stating certainty and its named gaps honestly, without jargon (B2; brief s16 binding)

What a strong B2 must contain (elements, not wording):

1. **The definition:** "high certainty" means each payment is **bought**, not forecast: a U.S. Treasury maturing shortly before each payment date, held to maturity (case L97, VRF; E6 R9).

2. **How it was evaluated:** by **price**, not by a model: on the purchase date the money covers the market price of all ten. No percentage (INT; MODEL percentages are not "certainty").
3. **The residual risk:** certain in U.S. dollars, barring a U.S. Treasury default (INT).
4. **Gap (i), one clause:** if rates fall before January 2027 the ten can cost more than the first deposit (about 1 in 3 modelled paths, MODEL; the ladder cost more than \$300k on 173-176 of 185 trading days in 2026, VRF/DER), and the earliest payment then waits for the second deposit (career income, case L43-45, VRF). The IPS mentions the deposit only as the completer (E6 conflict 22). Background for the team only, never in the IPS or the notes: her own word for self-employed income (B8b-Q18, section f).
5. **Gap (ii), one clause:** each payment is a fixed U.S. dollar amount (case L90, VRF), so rising prices, Taiwan's costs and the exchange rate mean it buys less over time. Named, not solved. Illustration for the team only: \$50k is worth \$42,063 (2033) and \$33,681 (2042) in 2026 dollars at 2.5% inflation (ASM, F-114).
6. **The joint tail, once (B4):** what could still stop a payment = a rate fall before January combined with a missing deposit, or a U.S. default (D12 M234 test, E6 D10).

Plain-word rules: say "bought", "owned", "held to its date", "barring a U.S. government default"; avoid "guaranteed", "risk-free", "100%", "matched", "immunised", "funded ratio". Use "certainty" only for the payments, "confident" for the range, "credibility" for her standing (E6 s6.1 rank 6, INT). Keep all what-could-go-wrong content to about 15% of the words (ASM ceiling).

(f) The client understanding the IPS must show

f1. The client picture (case facts, VRF)

- \$300,000 at the start of 2027 "with an asset management firm"; +\$150,000 at the start of 2028 from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (case L43-45).
- Living costs outside the portfolio; no other additions or withdrawals before 2033 (case L61-65). Flows at the start of each year (case L59).
- "Thoughtful risks" with "an appropriate balance" between growth and "protecting the capital required for her goals" (case L68-74).
- A collaborative, community-oriented creative residency in Taiwan from 2033 (case L76-80).
- Ten fixed \$50k operating payments 2033-2042, not inflation-adjusted, portfolio-funded "with a high degree of certainty", no outside money (case L88-93).
- Operating reserve set aside at the start of 2033, composition "if at all" changing (case L94-97); facility contribution decided after that, with flexibility "as the project develops"; no contingency fund required (case L101-107).
- In 2031 she needs a credible **range** with stated confidence that protects the payments; overpromising damages credibility (case L109-120).

f2. What the case designers test (INT reading of the case's own "must" list, case L134-148, VRF)

1. Payments supported "with a high degree of certainty" (L136) -> **does the team define and evaluate certainty?** (B2)
2. "a responsible facility contribution" (L138) -> **a rule, not a wish** (R5).
3. "how investment uncertainty could affect both" (L140-141) -> **both goals under good and bad markets** (B5, B7).
4. Communicates to co-sponsors "clearly and credibly" (L143) -> **no overpromising; confidence both ways** (R4).
5. "Preserves appropriate financial flexibility" (L145) -> **kept money exists even when stocks fall** (why $s = \frac{1}{2}$, not 1; E6 s6.2).
6. Assumptions on performance, timing, outside funding, inflation on projections **and facility costs**, flexibility (L146-148) -> R9.
7. Hidden tests (INT): reconcile "bought 2027" with "set aside 2033" (E5 PM-24); the second deposit is career income (does the plan protect against a smaller one?); fixed U.S. dollars vs Taiwan costs; WInS gains not added to projections (case L132-133).

f3. Laura's values from her verified words (BACKGROUND FOR THE TEAM ONLY; never quoted in the IPS, brief s16)

All rows are D13a **VERIFIED-PRIMARY** (verbatim on the primary page, accessed 2026-09-27; phase_D/D13a_laura_quotes_verified.md). Readings are INT (D13c). None is a statement about investing: there is **no** verified Laura investment philosophy.

Id	Value it suggests (INT)	Source (URL), year	Context caveat
D13a-N01	Accountable to early backers -> "announce only what is bought"	Daily Pennsylvanian, https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a , 2016	Student-era ; about entrepreneurship, not money management
B1b-Q11	Regret-driven leaps -> the plan must answer "why so little stock"	Overachiever, https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid , 2021	About leaving a job for her art career
B8b-Q18 (= B8a-Q07)	She called self-employment income "unstable" -> why the 2028 deposit is not assumed and payments are bought first	Overachiever (same URL), 2021	Same career answer; use once, never speculate about her finances
B8b-Q38	Access first ("diverse voices ... in the room") -> operations first	Local News Matters, https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/ , 2026	About festival grants, not the residency
B8b-Q37	Growth after community support -> shape of the 2031 message	Local News Matters (same URL), 2026	About the comics festival
B8b-Q10	Validate, then build	Signal founder's note, https://signaltemp.github.io/ , c. 2022 (date ASM)	About a student-advice product
B8b-Q21	Meaning over metrics -> report what numbers secure	Overachiever (same URL), 2021	About social-media likes
B8b-Q13	Authenticity over clever concepts -> cut what is only for show	The Nerd Daily, https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/ , 2025	About novel drafts; never "No more gimmicks." alone (B8b-Q14)
B8b-Q12	Scraps what is wrong -> honest about the dropped growth-first plan	The Nerd Daily (same URL), 2025	About drafts of a novel
B8b-Q31	Deadlines by rule	Geeks OUT, https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/ , 2022	About posting art
B8b-Q35	Purpose statement first -> the IPS opens with purpose	Pulse Spikes, https://pulsespikes.org/story/laura-gao , 2021	About art pieces
B8a-Q19	Data inside a story	Pulse Spikes (same URL), 2021	About a comic
B9b-Q09	Labels unfinished results and grades them -> grade numbers, name gaps	https://lauragao.com/rewriting-herstory , undated	Her website project
B8a-Q13	Plans for the downside	Overachiever (same URL), 2021	About reactions to her comic
B2b-Q08	Taught financial literacy to high-schoolers -> plain, honest about costs	Poets&Quants, https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/ , 2018	Student-era resume line
B3a-Q04, B8b-Q39	Access on need ("Pro-bono visits")	Speaking guide, https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf , 2024	About her speaking fees

Never: "risk-adverse" (B8a-Q03, VP) as investment risk (it is about choosing a business major); the case pull quote as her words (VRF as the case's text only); a "floor-first leap" trait; any EXCLUDE-PRIVACY or PARAPHRASE-UNVERIFIED line (D13a; brief s16).

(g) What NOT to put in the IPS (E6 s3, s6.2; D13 rules; L68-69, L123)

- Final reserve calculation, projections, the final 2031 dollar range, the co-sponsor draft (L68-69, VRF: "not expected").
- Tables, bullet lists, charts, images, links, URLs, footnotes, citations, quotation marks around anyone's words (L123, VRF).
- Laura quotes, the pull quote, her career or regret story as a reason; identity, heritage or Taiwan as a reason; book-title puns; private details (brief s4, s16).
- Tickers, fund names, CUSIPs; percentiles, probabilities, stage-by-stage stock shares, a fee number, a New Taiwan dollar figure, a return target.

- House names or frameworks: barbell, LDI, "Rule 1-4 + G", CFA labels, "letter to Future You", "two clocks".
- "Our portfolio manager approves" or anything implying the advisor decides (R-W19, VP).
- WInS profit, loss or rank; WInS gains added to projections (case L132-133, VRF).
- "Only the facility shrinks" (false in the joint tail); "fully funded by 1 Jan 2028 at the latest"; "fits inside \$300,000"; "guaranteed"/"risk-free"/"safe"; "promised" for the stretch; the statistics degree; "innovative".
- The rejected alternatives and the growth-first history (Final Report and TN reflection C territory, E6 s5.5, s10).

(h) Format and compliance checklist ("Submissions that do not meet the requirements below will not be considered for semifinal selection", L82-83, VRF)

Title page (page 1, "1 page maximum", L84-85, VRF)

- [] "Official team name" (L88), "the same as what you submitted with your official team roster" (L90). Copy from the roster confirmation screenshot, character for character (PM-18, ASM process). Team name UNVERIFIED until the roster is submitted (due Oct 9 5 p.m. ET; team target Wed Oct 7 AEDT).
- [] "Finalized team member names", same as the roster (L92-94), "Format: First Name, Last Initial" (L96). Wharton's sample separates names with " | " (L130, VRF). (Names are not stored in this repo: privacy rule.)
- [] "WInS username" = "the team's username for logging into the WInS platform" (L98-100), **not** the team name (SMAApply FAQ, VP via A4); copy from the WInS login screen.

Pages 2-3 ("2-page maximum", L103-104, VRF)

- [] Headings as in the sample: "Investment Strategy Elevator Pitch" and "Investment Policy Statement" (L135, L139, VRF).
- [] Pitch ≤ 50 (plan ≤ 48) and IPS ≤ 500 (plan ≤ 470) words (L107-109, VRF), by two counters, higher wins.
- [] "Font: Times New Roman" (L112), "Size: 12-point" (L114) everywhere, headings included.
- [] "Spacing: Double-spaced" (L116): Word "Double"; paragraph spacing 0 before/after; no blank lines; ≤ 7 paragraphs.
- [] "Margins: 1 inch on all sides" (L118); U.S. Letter (E6 D9; paper size is team discretion, L101, VRF).
- [] No "Graphics, charts, images, attachments, external links, footnotes, and formal citations" (L123).

Page-fit evidence (why ≤ 470 words, not 500)

- Word "Double" for TNR 12 = 27.6 pt line (ASM: font line height 1.15 em); Letter with 1-inch margins holds 23 lines a page, 46 on pages 2-3 (DER, D9_communication.md L34-35).
- A full 50 + 500 words needs 44-46 lines -> **0-2 spare lines on Letter**; about 525 words max (DER, D9 table L387). A4 leaves 2-4 spare; Wharton's own sample is Letter at a 24 pt pitch (VRF measured with pypdf, D9 L380) and leaves 8-10.
- Audit re-run: Letter, 7 paragraphs, 0 pt spacing, 27.6 pt: median 100.0% of the two pages, so 500 words "does not reliably fit"; Word's default 8 pt after-paragraph spacing adds overflow (DER, D10 s2.3 and audit C3).
- Long words cut capacity (13.4 words/line average; 9.0 for 7+ letter words, DER), so run `.venv/bin/python research/insight_v1/scripts/D9_ips_page_fit.py --draft <file>` on the real text (first paragraph = pitch) and then trust only the exported PDF.
- "Exactly 24 pt" line spacing (Wharton's sample) is a documented last resort only, never the plan (E6 D9).

PDF check (two students, Wed Nov 4 AEDT; E6 s3.3, ASM team rule)

- [] Export to PDF and open the exported file itself: **exactly 3 pages**; text ends on page 3 with ≥ 2 spare lines.
- [] Font list shows Times New Roman only; "Maximum file size: 5 MB" (L122).
- [] Search the PDF for "http", "www", "[", "(" citations and quotation marks.
- [] "Submit as a PDF through SurveyMonkey Apply" (L120). Open the IPS form in SMAApply early and record its fields and who can submit (PM-23; by Mon Oct 12 AEDT, ASM).

Deadline: Fri Nov 6, 5:00 p.m. ET, no extensions (SMAApply L10, VRF) = **Sat Nov 7, 09:00 AEDT** Sydney (U.S. clocks change Nov 1; Brisbane 08:00, Perth 06:00) (DER, E5 [1]). Trading ends and the portfolio locks at the same deadline (CLAUDE.md, VRF via SMAApply). Team targets (ASM): words frozen Tue Nov 3; last rule-driven trade by Fri Oct 30 ET; PDF check Wed Nov 4; submit Thu Nov 5.

(i) Consistency checks between the Trading Notes and the IPS

Why: "maintains consistency with the team's IPS" (SMAApply criterion L49, VRF); the TN analysis "should be consistent with the strategic approach your team is developing and will later articulate in its IPS" (TN guide L30-31, VRF via E5); notes cannot be edited, only added to (VP via A4). Run on Thu Oct 22 and Tue Nov 3 AEDT (E5 PM-06, ASM).

- [] **Central idea:** the notes and the IPS both express the split of jobs; no time-worded ban anywhere.
- [] **Scaling:** every operations and stock note carries "after both deposits" / "after her 2028 deposit" inside the note; B10 states the scaling and the stand-in funds (E6 s5.1 conditions).
- [] **Labels:** the team's own labels for the three parts are identical in notes, log and IPS (E6 D5); one label per holding.
- [] **Rule names:** each note's rule matches the IPS rule (a decision-log column: note -> IPS rule -> label, D9 M212).
- [] **Vocabulary:** "certainty" / "confident" / "credibility" used the same way; "moves like / stands in for" for WInS funds; never "matched" or "operating reserve" for a WInS fund; never "promised" for the stretch.
- [] **Numbers:** any number in a reflection (e.g. about 9% stocks after both deposits; about 98% of the first deposit) matches the IPS's forced-caution fact and date.
- [] **Refined story:** if note B shows "refined", the log shows, dated before the order, the change from a 2031 floor to the 2028 minimum; the IPS describes only the current rule.
- [] **Frozen book reflects IPS (L75):** on Nov 4 compare WInS holdings with B10; no tidy-up trades after Tue Nov 3 (R10).
- [] **Two-reader check:** outside readers who read the notes and then the IPS describe one strategy, and say Laura's real January 2027 money holds bonds, not stocks (E3-10).

(j) Top IPS risks and preventions (E5, ranked; ratings INT/ASM)

Rank	Risk (E5 id)	Prevention	Owner (role) / by
1	Not the students' own; reads AI-made (PM-01, severe)	Own-words notes before every vote; blank-page drafting from this element list; explain-back of the four dates without notes	Team leader; IPS draft Mon Oct 26 AEDT
2	Format breach (PM-02, fatal)	Section (h); $\leq 470/\leq 48$ words by two counters; PDF check by two students	Format checker; every draft from Oct 26, final Wed Nov 4
3	Over-complex, jargon-dense (PM-03)	10 blocks, ≤ 12 ideas; four-date spine; 6-reader paraphrase test	IPS lead writer; idea list Sat Oct 24; tests Oct 26-30, Nov 2-3
4	Notes contradict the IPS (PM-06)	Section (i)	Log keeper + IPS lead; Oct 22, Nov 3
5	Reads timid, or as a study of her (PM-07)	Purpose first; forced vs chosen caution; what stocks are for; one stress clause for the \$150k	Client lead; Oct 24, Oct 30
6	Generic thesis (PM-09)	Swap test on each paragraph; lead with the three Laura-specific decisions	Client lead; Oct 26-Nov 3
7	Required element missing (PM-11)	Tick-list b2 with the sentence number that covers each item	IPS lead
8	Rules left open, so the Final Report looks like a redesign (PM-12)	Section (c) voted and logged	Team leader; Mon Oct 26
9	Overclaiming words (PM-10)	Banned lists (a3, e, g); <code>D9_draft_checker.py --kind ips</code>	Second reader; every draft
10	Title-page mismatch (PM-18, fatal) or missed deadline (PM-13, fatal)	Copy from roster and WInS login; submit Thu Nov 5 AEDT; two people with SMAApply access	Admin liaison; Nov 4-5

Rank	Risk (E5 id)	Prevention	Owner (role) / by
11	Frozen book does not reflect the IPS (PM-22)	R10; Nov 4 holdings check	Trader + IPS lead
12	AI use unlogged (PM-08)	Dated AI-use log, reconciled with repo history, credited later in Works Cited	Log keeper; weekly

(k) Open team decisions (default from E6 s7, INT; the team decides)

#	Decision	Default	Test for choosing	By
D1	Adopt lock-early	Adopt	Every student writes, with no AI file open, why the payments are bought first; vote logged with reasons	Gate, Thu Oct 1
D2	REC (minimum bought 2028) or ALT (2031 lock)	REC	(1) Values: which matters more for Laura, a minimum no market can lower from 2028 with better bad cases, or more stock 2028-30 with a worse bad case; (2) explain-back of the four dates in two minutes; if only one design passes for all six, choose it	Gate, before any building or stock order
D3	WlnS book	(ii)R	Two-reader check passes, and the scaling fits in about 25 IPS words; two failures -> (iii)	Gate
D5	The team's labels for the three parts	Team's own words	Survive the explain-back; identical in notes, log and IPS	Before the first note
D6	Share of the stock fund added to the gift (s)	One half	Median kept money $\approx$ three years of Taiwan building-cost drift on the gift (\$32k vs about 11% of \$174k, ASM check); plainest split to explain	Section-4 vote, Mon Oct 26
D7	Cost clause in the IPS	Keep (one clause)	If the PDF has <2 spare lines, compress inside governance; cut only if the fee base will never touch the promises	Oct 26; Nov 4
D8	Statistics degree	Zero uses	Use once only if it changes a decision	Oct 24
D9	Paper and line spacing	U.S. Letter, Word "Double"	Exported PDF ends on page 3 with ≥ 2 spare lines	Nov 4
D10	Joint tail in the IPS	One clause	A reader asked "what could still stop a payment?" answers: a rate fall before January with a missing deposit, or a U.S. default	Oct 26
D12	Roles	Assign now	Every student owns at least one IPS block (and one note or reflection)	Tue Sep 29

(D4 WlnS instrument and D11 repository visibility are E6 decisions that do not change IPS content; see E6 s7.)

What this teaches

- A policy statement is judged on whether a stranger can see the rules that will run the money, so the pitch leads with purpose and the IPS is organised around four dated decisions rather than around products.
- Honesty about certainty is a strength: saying how certainty was defined, how it was checked, and the two gaps that remain earns more trust than "guaranteed".
- A rule has to be fixed before the outcome is known. Once the strategy freezes, any parameter chosen afterwards looks like hindsight.
- The layout counts as much as the words: two pages at 27.6 pt leaves 0-2 spare lines, so a correct IPS that runs onto a fourth PDF page is never read.

Securities and allocation: final WInS holdings and Laura's instrument map (insight_v1, writer W1)

Summary (5 lines)

1. WInS book (ii)R (Laura's plan on 2 Jan 2028, scaled to \$300,000): IEF 23.5% + TLH 42.5% (operations hedge), building minimum 24.3% (2032 Treasury note, else IBTM, else VGIT/SPTI), VT 8.7%, cash about 1% (E6 s5.2; INT).
2. Fallbacks decided in advance: an undated stand-in for the minimum, book (iii), or the ALT book; position-limit branches before any order (E6 s5.3, s5.10).
3. Laura's real plan: a ten-rung STRIPS ladder in 2027; in 2028, completion, then a zero-coupon Treasury repaying the \$150,000, then one world stock fund held to 2033 (E6 s1, s4).
4. Nothing is traded in 2031; in 2033 the ladder becomes the operating reserve and runs down to 2042 (E6 R3-R5).
5. Every ticker is pending the WInS listing and position-limit check, and each has a same-type alternate (section 6 has the issuer facts).

Serves: Trading Notes (TN) (the WInS holdings the notes quote) **and IPS** (the instruments behind blocks B3, B4, B6, B8 and B10; the IPS itself names no ticker). Authority: `research/insight_v1/phase_E/E6_final_spec.md`; fund facts copied from `wins_now/securities_and_allocation_v1.md` s12 ("ticket v1"). Supersedes ticket v1 s1 and s11 for the book and the long-term map (E6 s5.1, s9 #1, #5, #7). PROVISIONAL until the team votes. AI-generated research; no deliverable text.

Labels: VP = verified on the primary page (agent and date named); VRF = official repo file; SNIP = snippet; ASM = assumption; MODEL = model output, not a forecast; DER = W1's arithmetic from labelled inputs; INT = judgement; UNVERIFIED = not checked. **Terms:** *ETF* = fund traded like a share. *Treasury* = U.S. government bond. *STRIPS / zero-coupon* = a Treasury paying one amount on one date. *Rung* = one ladder bond. *Duration* = about the % a bond fund falls if rates rise 1 percentage point. *Position limit* = the most WInS allows in one security. *Expense ratio* = the fund's yearly cost as a % of the money in it. *NAV* = net asset value per share. *bp* = 0.01 percentage point. *Accrued interest* = interest built up since the last coupon, paid by the buyer.

Every ticker below is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK (no separate approved list this season; "Any ETF available on WInS", VP via ticket v1).

1. Final WInS holdings: recommended book (ii)R (E6 s5.1-5.2)

Illustration at 2026-09-25 closes; recompute shares from the last close (T2 finding 1). \$ and shares DER.

Holding	Job / role word	Weight	\$ at \$300k	~Shares	Alternates	2025-26 list (history only, VRF)
IEF	Operations hedge: future funding	23.5%	\$70,500	~783	VGIT, then SPTI (re-solve: VGIT 16.6 / TLH 49.4)	yes
TLH	Operations hedge: future funding	42.5%	\$127,500	~1,365	IEF 40.9 + TLT 25.1 if TLH is missing	no (TLT yes)
4.125% Treasury note 15-Nov-2032 (CUSIP 91282CFV8), else IBTM	Building minimum: risk management or future funding (one word, recorded)	24.3%	\$72,900	note ≈ \$75k face (≈ \$72,570 at model 95.27 clean + 1.49 accrued, ASM, D1); IBTM ~3,336	VGIT (~1,281), then SPTI (~2,669), "moves like, not dated"	IBTM no; VGIT no
VT	Stock fund: growth	8.7%	\$26,100	~163	VTI + VXUS 62/38 (~42 / ~114 sh); then ITOT + IXUS	VT, VTI, VXUS yes; ITOT, IXUS not found (W1 grep, 2026-09-28)
Cash	Liquidity	~1%	~\$2,990 after 4 × \$25	-	at least \$1,000	-

- Why these weights: the plan on 2 Jan 2028 is ladder 65.9%, minimum 25.3%, stock fund 8.7% of about \$464k (MODEL, E6 [7]); the float comes out of the minimum (DER).
- Hedge mix 35.6/64.4 gives "about 10 years", like the payments (IEF 6.86y, TLH 11.58y on 9/25, VP via ticket v1; liability 9.90y, VRF).
- Wording constraints for the WInS holdings: never "Laura's portfolio" (Asset Manager Code F.2, VP via D10); the funds "stand in for" / "move like" her bonds.

2. Fallback books (E6 s5.10)

Trigger	Holdings at \$300k (illustration, DER)
No dated 2032 instrument listed	As section 1 with VGIT (else SPTI) as the minimum's stand-in; never claims a date
Two-reader check fails twice → (iii)	IEF 34.9% \$104,700 / TLH 63.1% \$189,300, re-priced to the trade-date ladder cost; then SGOV (else BIL, SHV) with the leftover only if $\geq \$1,000$ (~\$4,500); otherwise a listed dated payment rung; otherwise revert to (ii)R
Team votes ALT	IEF 23.5 / TLH 42.5 / VT 17.0 (\$51,000) / VGSH 16.0 (\$48,000) / cash 1; growth-money bond choice recorded before the VGSH order

3. Position-limit branches (E6 s5.3; ticket v1 s3)

Limit shown	Book (ii)R
None, or $\geq 43\%$	Section 1
25-42%	IEF 23.5 / TLH at cap – 1 / SPTL rest of hedge; minimum one point under the cap if the cap is under 25.3%. At 25%: IEF 23.5 / TLH 24 / SPTL 18.5 / IBTM 24 / VT 8.7 / cash ~1.3% (5 ETF trades, \$125)
Under 25%	IEF, TLH, SPTL at cap – 1; VGIT then VGLT take the rest; minimum: IBTM at cap – 1, rest in VGIT (VGIT's two parts count as one holding). 5 or more hedge funds needed (cap $\approx 17\%$ or less): stop and ask Wharton
Separate, higher bond limit	The 2032 note may carry the whole minimum

Never add within one point of a cap; a 20-24% stock fall (or 10-12% plus a 50bp rally) can lift TLH over a 25% cap (T2 [5], ASM).

4. The dated Treasury / IBTM item and its fallback (E6 s5.2, D4; ticket v1 s7)

Choice	Instrument	Facts	Rules
1	4.125% Treasury note, 15 Nov 2032, CUSIP 91282CFV8	Exists and is strippable (Treasury MSPD Table V, 2026-08-31, VP per S1 via D1); matures 47 days before 1 Jan 2033 (DER); model price 95.27 clean + ~1.49 accrued per 100 on 9/25 (ASM, D1); WInS shows its own price; bond prices update once a day (ticket v1 s7)	Find it in the Bonds drop-down by maturity; \$10 commission; buyer pays accrued interest. Listing is UNVERIFIED
2	IBTM (iShares iBonds Dec 2032 Term Treasury ETF)	NAV \$21.83 (9/25); expense 0.07%; effective duration 5.05y; 15 Treasuries maturing Apr-Sep 2032; about \$555m net assets; about 216k shares/day. Funds "will terminate on or about October or December 15 of the year in each Fund's name" and "do not seek to return any predetermined amount" (both re-read by E6 on ishares.com, 2026-09-28, VP)	Describe only as "a fund that ends in December 2032"; never "repays \$X". Listing is UNVERIFIED
3	VGIT, then SPTI (undated stand-in, about 4.8-4.9y)	See section 6	"Moves like a Treasury maturing in late 2032"; never a date. The real plan is unchanged

5. Laura's long-term instrument map (the real plan, not WInS; E6 s1, s4)

Date	Instrument	Size (MODEL/ASM at the 2026-09-25 curve)	Rule
1 Jan 2027	Ten Treasury STRIPS, each maturing 15 Nov 2032 ... 15 Nov 2041, before one \$50,000 payment (1 Jan 2033-2042). First rung: principal STRIP 912821KC8 (VP per S1 via D1)	Model cost \$294,387 for Nov-15 dates, headroom \$5,613 (19bp) (VRF inputs, ASM method; a model price, not a STRIPS quote). About 1 in 3 rate paths cost more than \$300k (MODEL 30-34%)	R1: buy all at once on arrival; if short, longest-dated first; leftover waits in Treasury bills

Date	Instrument	Size (MODEL/ASM at the 2026-09-25 curve)	Rule
1 Jan 2028	(1) Any unbought rung; (2) the building minimum: a zero-coupon Treasury maturing before 1 Jan 2033 repaying \$150,000 face (the 15-Nov-2032 STRIP is the natural candidate, INT); (3) one broad world stock fund (VT-type, about 10,000 companies; alternates VTI + VXUS, ITOT + IXUS)	Minimum costs about \$118k at the 5-year 4.98% (ASM); stock fund about \$40k = 8.7% of all money (MODEL). Rates -50bp first: fund ~\$20-23k; -100bp: ~\$1-7k; -150bp: no fund, minimum ~\$127-137k (E6_final_checks.py [8], MODEL; corrects E4 [7])	R2: same order if the deposit is smaller or late. Joint tail: if rates fell and the deposit is smaller than the unbought part, all of it completes payments and the unfunded part of the first payment is stated (\$11,957 at -50bp or \$31,413 at -100bp with no deposit, MODEL)
2028-2032	Stock fund held	-	R6: never rebalanced against the Treasuries, never sold before 1 Jan 2033
1 Jan 2031	Nothing traded	Range method only: bottom = minimum owned; top = minimum + half the stock fund's value that day (no figures here)	R4
1 Jan 2033	Ladder named the operating reserve; pays the first \$50,000 at once (2032 rung matured 15 Nov, waited in bills)	About \$395k market value on today's forwards (ASM), cost ~\$292-294k, face \$500k	R3: held to maturity; each rung matures about 47 days before its payment and waits in Treasury bills; nothing sold early
1 Jan 2033	Gift = minimum + half the stock fund, never above the announced top; kept money to U.S. dollar Treasury bills	-	R5: no New Taiwan dollar conversion or currency hedge before the facility decision (a forward is a derivative)
2033-2042	Reserve run-down: one rung pays each 1 Jan	Empties by 1 Jan 2042	R3

Permitted-instrument reading for the real plan: Treasuries (STRIPS and one zero for the minimum), one broad stock ETF and Treasury bills; no derivatives (R-AN15 safe reading, E6 s8c).

6. Fund facts (VP: read by T1 on the issuer pages, 2026-09-28 about 00:35 UTC, per ticket v1 s12; not re-opened by W1 except where stated)

Ticker	Issuer source	Expense	Duration (as of)	Close (as of)	Other
IEF	ishares.com/us/products/239456	0.15%	6.86y (9/25)	\$90.00 (9/25)	16 holdings; \$41.57bn; 30-day volume 8.83m
TLH	ishares.com/us/products/239453	0.15%	11.58y (9/25)	\$93.38 (9/25)	61 holdings maturing 2037-2046, 74.7% after 2042 (S1); \$10.53bn; 1.97m/day
TLT	ishares.com/us/products/239454	0.15%	14.84y (9/25)	\$79.32 (9/25)	47 holdings maturing 2044-2056
SPTL	ssga.com (SPDR Portfolio Long Term Treasury ETF)	0.03% gross	13.65y OAD (9/24)	\$24.27 (9/24)	110 holdings; \$10.95bn
SPTI	ssga.com (SPDR Portfolio Intermediate Term Treasury ETF)	0.03%	4.78y OAD (9/24)	\$27.31 (9/24)	-
VGIT / VGLT	investor.vanguard.com API	0.03%	4.9y / 13.5y (8/31)	\$56.90 / \$50.98 (9/25)	102 / 101 bonds
IBTM	ishares.com/us/products/328944	0.07%	5.05y (9/25)	\$21.85 close; NAV \$21.83 (9/25, E6 re-read 2026-09-28)	15 holdings; ends Dec 2032
VT	investor.vanguard.com API	0.06% (S2/S3)	n/a	\$160.03 (9/25)	10,088 stocks, 37.7% non-U.S. (8/31); top ten 21.7% (6/30)
VTI / VXUS	investor.vanguard.com (VXUS via S3)	0.03% / 0.05% (S2)	n/a	\$379.77 / \$86.35 (9/25)	-
VGSH / SHY	Vanguard API / ishares 239452	0.03% / 0.15%	1.9y (8/31) / 1.79y (9/25)	\$57.59 / \$81.21 (9/25)	ALT book only
SGOV / BIL / SHV	ishares 314116 / ssga.com / (S1)	0.09% / 0.1353% / 0.15%	~0.1y / 0.10y / 0.28y	\$100.66 (9/25) / \$91.58 (9/24)	(iii) only; BIL yield to maturity 3.94% (9/24)

ITOT, IXUS: no facts in ticket v1 (UNVERIFIED; read the issuer page before use).

What this teaches

1. A holding is chosen for its job and its date, not its name: the same Treasury fund can be "hedge" or "stand-in" only if the note says which.
2. Every pick needs a pre-agreed substitute, because a security you cannot buy on the day is a plan you do not have.
3. WInS shows one dated snapshot of a longer plan; saying the date out loud keeps the notes honest.

Why us: why would Laura choose our firm's strategy over the other firms?

1. Our edge is not "buy the payments first". That is the textbook answer, and many teams will give it (M011, INT). The edge is the Laura-specific rule: each deposit buys one promise when it arrives.
2. Her first deposit buys the ten operating payments. Her second deposit, which is her own career income, buys the building's minimum in 2028 (case L43-45, L91-92, VRF).
3. On one set of model paths, this design (REC) has about the same middle as the 2031-lock design (ALT) and better bad cases: total p5/p50/p95 \$182k/\$207k/\$250k against \$168k/\$210k/\$266k (MODEL, E6 [2], re-run 2026-09-28).
4. It loses on median gift, stock exposure and p95. The simulated client (E3) would choose the firm on substance but could be won by a rival with the same plan in plainer words. So the edge only counts if the words carry it.
5. Limits: every "Laura" here is a simulation grounded only in the case and her verified public words. A blind contest (section 0 below) ranked our strategy first with all 3 simulated evaluators; E2 scored the older E1 design, not this spec.

Serves: TN (Oct 23) and IPS (Nov 6). This file holds reasons and evidence only. None of it is text to submit, and the team writes every word.

Labels: VP = VERIFIED-PRIMARY; VRF = VERIFIED-REPO-FILE; SNIP = SNIPPET-UNVERIFIED; ASM = ASSUMPTION; MODEL = model output from assumptions, not a forecast; INT = judgement; SIM = simulated-client reaction (a kind of INT, never a fact about Laura). Case lines refer to `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`. E6 [n] is a section of `scripts/E6_final_checks.py`. W3 re-ran it on 2026-09-28 and every figure below that cites it matched. Every ticker named here is **PENDING WINS AVAILABILITY + POSITION-LIMIT CHECK**; primaries and alternates are in `securities_and_allocation.md` s1-s2. **R1-R8 in this file are reasons, not the spec's decision rules R1-R10** (those live in `ips_spec.md` s(c)).

0. Blind "Laura's Choice" contest (added 2026-09-28 by the main loop)

Serves: IPS (strategy choice) and TN (what the notes must show). Files: `phase_E/tournament/firm_A-F.md`, `fairness_log.md`, `eval_L1-L3.md`; numbers for the rival firms: `scripts/E7_tournament_field_numbers.py` (MODEL). Six anonymised one-pagers (300-380 words, same seven sections, loaded words and tells removed by a fairness auditor): ours (Firm D, the E6 REC design) and five rival approaches (A: typical strong team with a Monte Carlo 95% and staged reserve; B: growth-first 75% equity; C: themed/values tilts; E: generic 60/40 to 40/60; F: complex optimiser). Three evaluators (simulations of Laura, grounded ONLY in the case, D13a VERIFIED-PRIMARY lines and the D13c voice map) ranked all six, each reading them in a different order.

Evaluator	Ranking (1st -> 6th)	Score for ours (/10)	Next best
L1	D, A, E, F, C, B	7.5	A 6.0
L2	D, A, E, F, C, B	8.0	A 6.5
L3	D, A, E, B, F, C	8.0	A 6.0
Borda total (6 points for 1st ... 1 for 6th): D 18 (maximum) , A 15, E 12, F 8, B 5, C 5. Ours was first 3/3.			

Why they chose ours (their reasons, with the evidence they cited):

- The 2031 bottom is money she already owns, so she cannot overpromise (case p.3 credibility warning, L114-115).
- It is the only plan whose payments do not depend on markets until 2033: the market-based plans miss a payment in 0.3-3.2% of model paths, 7-14% if the 2028 deposit is \$75k, and 41-46% if it never arrives (MODEL, E7/strategy_mc) - and that deposit comes from income she herself called "unstable" (B8b-Q18, VERIFIED-PRIMARY, Overachiever 2021).

- It names its own costs and gaps (the ~1-in-3 January price risk, the joint tail, lower far upside), which matches how she talks about unfinished work and costs (B8b-Q18, B9b-Q09) and her line about not letting down "people who were your earliest supporters" (D13a-N01, student-era, Daily Pennsylvanian 2016).
- Medians are nearly equal across plans (ours \$207k vs A \$206k, E \$213k; MODEL), so the market plans mainly buy far upside at the price of payment risk.

What all three said ours must still fix (actions for the IPS; the first is the most important):

1. **"Appropriate balance" is not yet shown** (case L69-74; her "take the jump" line B1b-Q11): stocks average 7.5% of all her money 2028-32. State where the leap is (the residency itself; the growth money takes the market risk), give the price of certainty in one line (median about equal to the typical plan, give-up only at the far upside: p95 \$250k), and show one tested alternative with a smaller 2028 floor and more in stocks, so the choice is visibly a choice. -> team decision D-open (see open_questions.md).
2. **Call the \$150k bond the bought floor of the 2031 range, not a "minimum facility contribution"** fixed in 2028 (case p.3: no predetermined contribution; flexibility as the project develops) (L2).
3. **State the in-range confidence directly** (case L117-118): under the cap rule the 2033 contribution always lands between the owned bottom and the announced top, barring a U.S. default; keep ~73% (model) / 84% (history) only as the chance of reaching the top (L1, L3). Say how meaningful a ~\$150k-\$175k range is (case L111-112).
4. Already in the spec, but must be visible in the IPS: the joint-tail rule (payments first, the floor shrinks, stated in dollars; R1-R2) and both certainty gaps (nominal US\$ vs Taiwan costs; the inflation assumption; what a fixed \$50k buys by 2042; case L90, L146-147).

0. Honest limits (read first)

- **E3 is a simulation.** Its "client" was built only from the case (VRF) and from D13a VERIFIED-PRIMARY lines used in their context. It is not Laura, and nothing from it may be attributed to her (E3 "How to read").
- **E3 reviewed E1's design, not the final spec.** E1 was the 2031 lock (now ALT). The final REC design came from E4 and E6. So the client-seat verdict covers:
 - the shared core (payments first, certainty by price, named gaps);
 - wording findings that E6 adopted.

It does not cover REC's 2028 minimum, its cap or its lower stock share (INT).

- **E2 scored E1 as specified:** $7/7/7/5/6 = 32/50$, and about $38/50$ after its ten fixes (INT). E6 adopted most of those fixes, but no reader has scored the E6 spec.
- **A lean blind contest was run after this file was first written (see section 0).** Three independent simulated-Laura evaluators, each reading six anonymised one-page strategies in a different order, ranked ours first. It is still a simulation (3 evaluators, archetype rivals modelled on one engine; see `phase_E/tournament/fairness_log.md`), not Laura's view.
- **The "rival" became our recommendation.** E4's strongest rival design was adopted as REC. The rivals left to beat are archetypes (growth-first, all-Treasury, the 2031 lock, a same-plan team with better words), not real teams.
- **The model has thin tails** (AX1 C12, VRF audit). That is why history windows are shown beside it. The results also depend on the return input: JPM AC World 7.00% compound (VRF). On a Vanguard-like 5.08% input, REC is \$179k/\$202k/\$242k (E6 [4], ASM hybrid).

1. The reasons, with evidence, the rival each beats, and where it is close or loses

#	Reason (content, not wording)	Evidence (status)	Rival it beats	Where it is close or loses
R1	Once bought, the ten payments depend neither on markets nor on her next contract	Fixed \$50k payments funded only by the portfolio, with "high degree of certainty" (case L88-92, VRF). The 2028 deposit comes from advances, speaking and licensing (L43-45, VRF). Research only, not for the TN or IPS: she called self-employment "unstable income" (B8b-Q18, VP, 2021); letting down "earliest supporters" is worse than letting herself down (D13a-N01, VP, student-era 2016). Growth-first misses a payment in 3.2% of paths, 13.7% if the deposit is \$75k, and 40.8% if it is \$0; lock-early misses none in that model, which buys at the 2026-09-25 curve (MODEL, strategy_mc.py, VRF script). Lock-early is not miss-proof: if rates fall before January 2027 and the deposit never comes, part of the 2033 payment is unfunded (\$11,957 at -50bp, \$31,413 at -100bp; joint tail, E6 R2, MODEL)	Growth-first (about 75% stocks, reserve bought later; the team's original plan)	Losses on upside. Growth-first's 2033 surplus p50 is \$226k against \$207k and its p95 \$540k against \$273k (MODEL, strategy_mc.py; those lock-early figures are the council's 60%-equity design, not REC, whose p50/p95 on the E6 basis are \$207k/\$250k, E6 [2]). Close: every lock-early team has this reason, so it is not a differentiator alone (M011, INT). About 1 path in 3 costs more than \$300k in January 2027, so the "bought" date can slip to 2028 (MODEL 30-34%)
R2	Certainty is defined by a price she can check, and the plan names its own two gaps	Case L97-99 asks the team to define and evaluate certainty (VRF). E2 lists "defines certainty by a price, not a model" and "names its own two gaps" among three things most teams will not do (INT). Gap 1: the real Nov-15 ladder costs \$294,387, so headroom is \$5,613 (19bp) on the 2026-09-25 curve (VRF inputs, ASM method). Gap 2: a fixed \$50k is worth \$42,063 in 2033 and \$33,681 in 2042, in 2026 dollars at 2.5% inflation (ASM, F-114)	Rivals that state certainty as a model probability ("95%") or say "guaranteed"	Close: a strong rival can copy it once seen. Risk: too much caveat reads as a disclaimer. The spec caps it at about 15% of IPS words (E3-11, ASM)
R3	The building's minimum is fixed in January 2028, sized to her own second deposit, three years before she speaks to co-sponsors	\$150k deposit (L43-45, VRF); a "meaningful personal contribution" signals viability (L110-112, VRF); overpromising damages credibility (L114-116, VRF). REC bottom: \$150k fixed. ALT bottom: p5/p50/p95 \$134k/\$167k/\$211k, \$134k after a bad 2028-30, and \$93k in the 1929-start window (MODEL; VP data, derived; E6 [2], [3], [5])	ALT, the 2031 lock (the council plan in CLAUDE.md, and E1)	Losses on the typical case: ALT's bottom is above \$150k in 78% of paths (median \$167k), and its median gift is \$186-188k against \$174k (MODEL, E4 [2], E6 [2]). The WlnS stand-in (2032 note or IBTM) is UNVERIFIED as listed
R4	Same middle, better bad cases, in the model and in a century of history	Total money for the facility plus flexibility: REC \$182k/\$207k/\$250k, ALT \$168k/\$210k/\$266k. On the same paths ALT is ahead in 53% of cases, by a median \$1k, and behind by \$23k at p5 (MODEL, E6 [2]). Every 6-year window from 1928, rescaled to JPM: worst \$169k against \$120k, p10 \$185k against \$174k, median \$214k for both (VP data, derived, E6 [5]). No REC path ends below the \$157,736 put in; 1.73% of ALT paths do (MODEL)	ALT; any "balanced sleeve locked later" design	Losses at the top: p95 is \$16k lower and the stock share averages 7.5% against 9.7% over 2027-32 (MODEL). This is a different bet on when risk is taken, not a free improvement (E4 s0, INT)
R5	In 2031 she tells partners only what she owns, plus a stretch whose odds anyone can check	Case L109-120 asks for a range and a confidence that protect the payments (VRF). The top is the minimum plus half the stock fund's value that day, capped. It is reached in 73% of model paths (MODEL). U.S. stocks' two-year return was zero or positive in 84% of 97 periods from 1928 to 2025 (VP data, derived, E4 [5]). 86% of the median gift is already owned when she speaks (MODEL, E6 [2]). Before E6's wording fixes, E3 still found the 2031 method passes "never overpromise" (SIM, E3 finding 7)	Rivals with an uncapped model top: an "80%" top landed within the range 62-93% of the time across models and history (AX1 C12, D3 v2, VRF audit)	Losses on visible upside: the cap means strong markets raise her kept money, not the announced gift. D5 argued an uncapped top answers "favorable outcomes" (L118) more honestly (E4 X3, INT). Answer: the excess stays hers, and she may add it in 2033
R6	Risk sits only where she alone bears it, and all of that money is fully in stocks	The payments cost 97.4% (exact-date) to 98.1% (Nov-15 STRIPS) of the first deposit (MODEL on the 2026-09-25 curve, VRF). So 2027 holds no stocks by arithmetic (E3 finding 2, SIM). D13c V1/V2 read her two verified risk ideas (take the jump, B1b-Q11, VP 2021; do not fail early supporters, D13a-N01, VP 2016) as: her own risk is fine, risk on people who trusted her is not (INT)	A timid all-Treasury firm (no "pursuing growth", L72-73, VRF); a "balanced everywhere" firm	Close to all-Treasury: at 50% stocks, the growth money beats an all-Treasury benchmark by only about \$1k at the median and ends below it in about 48% of paths (MODEL, D3 M034). Looks timid: about 9% stocks overall from 2028. D13c N1 warns that a "take the jump" founder may read caution as unlike her (INT)

#	Reason (content, not wording)	Evidence (status)	Rival it beats	Where it is close or loses
R7	Simpler to read: four dated decisions, a case number and "half"	E2 counted about 55 ideas in E1 (INT). The spec has 10 blocks, ≤ 470 words, and one number (E6 s3). The growth-mix rule, band, reset, lock share and percentile are all removed (E6 8b). Last year's lesson: "over-complex for the insight" (CLAUDE.md, VRF)	Our own E1 version; rivals with dense frameworks	Close: a disciplined rival can be as simple. Simplicity only shows if the draft stays under the cap (PM-03, E5)
R8	The notes and the IPS say one thing	The central idea is a split of jobs, true of both the WinS book and the real plan. A ban worded in time ("no stocks until...") is broken by a day-one VT buy (E2 C1/F4; AY2 rank 1; VRF audit)	Any team whose notes contradict its later IPS (E5 PM-06)	Close: only works if every note carries its scaling marker and passes the two-reader check (E6 5.4e)

Net verdict (INT).

- **Against growth-first:** we win on the case's own test ("high degree of certainty") and lose only on upside.
- **Against the all-Treasury firm:** we win on "pursuing growth" and co-sponsor upside, but only narrowly on money.
- **Against the 2031 lock:** we trade about \$12-14k of median gift and some stock exposure for a bottom that is fixed and owned early, and a better worst case.
- **Against a same-plan rival with warmer, plainer words:** we have no structural edge. E3's main finding (SIM) is that this is where Laura would be lost.

2. What the simulated client (E3) and the reader (E2) add

- **E3 (SIM, anchored to case lines):**
 - would very likely choose the firm on substance;
 - hesitates because E1 is "written from the trading desk outward".

E3's fixes are all in the spec: never "promised" for the stretch (E3-1); forced versus chosen caution (E3-2); what stocks are for (E3-3); the cost clause protects the minimum too (E3-4); the pitch puts purpose first (E3-5); no behavioural diagnosis of her (E3-6); the longest-first counterpoint is named (E3-7).

- **E2 (INT):** substance is "of Top-50 quality", presentation is "not yet". Most of the +6 points from the fixes come from presentation and consistency, not strategy. Articulation was the weakest score (5/10), because the one true refinement story (growth-first dropped after testing) was unused. E6 now puts it in note C.
- **D13c (INT on VP quotes):** the plan feels like hers if it:
 - makes the leap certain for the people who will rely on it;
 - takes risk only where she alone bears the cost;
 - says its costs and gaps out loud;
 - sounds like a teacher, not a trading desk.

None of her words may appear in the TN or IPS (brief s16).

3. What the Trading Notes and IPS must make visible for these reasons to land

Trading Notes (Oct 23):

- [] Three notes, three jobs:
 - operations (TLH with IEF);
 - the dated building minimum (2032 note or IBTM);
 - the stock fund (VT).

Each note carries a Guide role word, one dated checkable fact and its rule (E6 5.6). This makes R3, R6 and R8 visible.

- [] Every note carries its scaling marker ("after both deposits" or "after her 2028 deposit"). Two outside readers still say Laura's real January 2027 money holds only bonds (R8).
- [] Note B names a date, not a price: "held to its end". Never "guaranteed" or "promised" (R3, R5).

- [] Reflection A is the tested one: three pre-registered numbers and "the rule said hold" (R2).
- [] Reflection C tells the true refinement: growth-first was tested and dropped because it could miss a payment, and all money nobody relies on is now in stocks. This covers R1, R6 and Articulation.
- [] No reserve size, 2031 figure, Laura quote or statistics degree anywhere.

IPS (Nov 6):

- [] The pitch puts purpose first and states the Laura-specific rule: each deposit buys one promise, and her second buys the building's minimum from her own earnings. No numbers, and no claim that anything is already bought (R1, R3).
- [] B2 defines certainty by price, names the residual risk and gives the two gaps one clause each (R2).
- [] B5 separates forced caution (the one allowed number: about 98%) from chosen caution. It says what stocks are for and the most markets can take (R6).
- [] B4 and B7 give the minimum rule, the 2031 method, confidence on both sides, and the words "owned / expected if stocks hold their value / kept" (R3, R5).
- [] B8 and B9 cover the cap, where kept money goes, "rules change only if her circumstances change", and no rebalancing between the parts (R4, R5).
- [] B10 is the WInS scaling sentence (R8).
- [] 10 blocks, ≤ 470 words by two counters, and six unpaid readers can restate the three pitch elements (R7).

What this teaches

- "Why us" is a comparison, so it needs named rivals, a common yardstick and an honest list of where we lose. A reason every team shares (buy the payments first) is necessary but does not win.
- A simulated client and a model can sharpen a plan, but they are not the client or the future. Label them, and let the written page show the reasoning.

Strategy changes: what the final spec (E6) changes against the current strategy

1. The core stays: the first deposit buys all ten payments before any money takes stock-market risk (brief s8 items 1-8).
2. The facility half changes. In January 2028 the second deposit buys a building minimum equal to its own dollar amount. Everything left goes into one world stock fund held to 2033, and nothing is traded in 2031. This replaces the ~60%-equity sleeve and the 80% lock in 2031.
3. For the TN: the WInS book changes from ~65/35 IEF/TLT/equity to (ii)R (operations hedge, dated building minimum, stock fund). The central idea becomes a split of jobs, and there is a gate before any order.
4. For the IPS: 10 blocks on four dated decisions; certainty defined by price with two named gaps; a capped 2031 range with the chance of reaching the top stated; "fits inside \$300k" and "bought in January 2027" dropped as headlines.
5. Every change is a proposal. The team has not voted (CLAUDE.md, VRF), and the gate votes come first (E6 5.4a).

Serves: TN (Oct 23) first, then IPS (Nov 6). Research only; the team decides and writes every word.

Labels (as in why_us.md): VP, VRF, SNIP, ASM, MODEL, INT. "Current" means brief s7 plus CLAUDE.md "Strategy status". Figures cited E6 [n] were re-run by W3 on 2026-09-28 and matched. Every ticker named here is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK**; primaries and alternates are in securities_and_allocation.md s1-s2.

1. Changes ranked: deadline order (TN first), then impact within each deadline

Tier 1: WInS now and Trading Notes (Oct 23)

#	Change (current -> new)	Reason	Numbers	Side effects	Rejected, and why
T1	Gate before any order (new). Three votes: lock-early; REC or ALT; book (ii)R or (iii). Each vote comes after every student writes 3-5 own-words lines. Also: roles, a decision log with rule-name and evidence-grade columns, and a back-filled AI-use log	"The team has not read the memos yet" (CLAUDE.md, VRF). Ownership and voice is the top semifinal risk (PM-01, INT). AI policy (R-W46/R-W51, VP)	Target fills Fri Oct 2 ET; hard stop Fri Oct 9 ET (E6 s0)	Hours this week	Trading before the votes: permanent notes would contradict a later choice (PM-06)
T2	Central idea: a timing ban -> a split of jobs. Money others rely on is bought and held to its date; money nobody relies on is fully in world stocks	A ban ("no stock risk until...") is broken on day one by the recommended book's VT buy (E2 C1/F4; AY2 rank 1, VRF audit)	None	15-20 characters of scaling marker per note	The ban wording (E1 P1): contradicted by the notes, which are read before the IPS
T3	WInS book: ~65% IEF/TLT + ~35% equity -> (ii)R: IEF 23.5 / TLH 42.5 / building minimum 24.3 (the 15-Nov-2032 note, else IBTM, else VGIT) / VT 8.7 / cash ~1. This is Laura's plan on 2 Jan 2028, scaled to \$300k	Three jobs give three distinct notes. It shows "funding purposes" diversification (R-W71, VP). The old 65/35 was hedge versus growth, not bonds versus stocks (F-409, VRF). The IEF/TLH mix gives about 10 years of rate sensitivity, like the payments (IEF 6.86y, TLH 11.58y issuer durations 9/25, VP; liability 9.90y, VRF)	Plan shares on 2 Jan 2028: 65.9 / 25.3 / 8.7% of \$464k (MODEL, E6 [7]). 4 trades, \$100 (VP commissions)	VT shows 8.7%, not 17%. The IBTM and 2032-note listings are UNVERIFIED. Position-limit branches decided in advance (E6 5.3)	(iii) literal 2027 book: two decisions plus a filler bills note that earns ~\$18 against a \$25 fee (T2 finding 4); kept as fallback. The 0.9% "middle path" VT sliver: a weak note (D10). A payment rung: a second future-funding note (AX1b 7). VGSH third note: ALT only

#	Change (current -> new)	Reason	Numbers	Side effects	Rejected, and why
T4	Notes written by students from elements: ≤300 characters until a zero-risk box test; a five-word copy check against research/ ; two outside readers; the team's own labels	AI wording in permanent, un-editable notes (PM-04, INT; "cannot be edited", VP)	The WInS limit is UNVERIFIED. A sister product caps at 300 (VP); Wharton's example is 413 characters (VRF)	None	Pasting spec wording (breaks the AI policy)
T5	Three-note plan: A = TLH, tested (three pre-registered numbers, fill to Oct 20 close); B = building minimum, refined only if the log shows the 2031->2028 change dated before the order, otherwise supported; C = VT, supported, carrying the growth-first story	E2 scored Articulation 5/10 and flagged the one true refinement as unused (E2 F8-F9, INT)	P(10-year moves ≥10bp in the window): ~52% for an Oct 2 fill, ~40% for an Oct 9 fill (T2 [4], ASM). Growth-first miss: 3.2%, 40.8% with no deposit (MODEL, VRF script)	Small moves must be reported as they are	A planned "discipline" rebalance trade (a band fires in ~0-2% of cases before Oct 23, AY2). Calling the SPTL cap split a refinement (it is pre-decided, AY2 D7 C1)
T6	One vocabulary: certainty (payments), confident (range), credibility (her standing), "moves like / stands in for". Never "match", "promised" or "guaranteed"	Case L110 "expects", L116 "potential contribution" (VRF); D9 M062; E3-1	74.7% of TLH matures after 2042 (VRF), so "match" is false in WInS	None	"Promised-at-risk" labels (E1): read as overpromising (E3 finding 1, SIM)
T7	Data capture: screenshots on the fill date and at the Oct 20 close; value the payments with A2_curve_recheck.py or D9_numbers.py	The tested reflection needs pre-registered numbers. official_curve_pv.py is fixed to 9/25 (AY2 D7 C5)	Ex-dividend of ~0.4% against a signal of ~1% (ASM)	Ten minutes	None

Tier 2: IPS (Nov 6; freezes the strategy)

#	Change (current -> new)	Reason	Numbers	Side effects	Rejected, and why
I1	Format stance (new): U.S. Letter, Word "Double", ≤470 IPS / ≤48 pitch words by two counters, PDF checks, title page copied from the roster	Non-compliance = "will not be considered" (IPS guide L82-83, VRF)	0-2 spare lines at 27.6 pt (D9/AX2 C3)	Fewer words than E1 budgeted	"Exactly 24 pt" as a plan: last resort only
I2	Architecture: ~60%-equity sleeve + 80% of it locked in a 2-year Treasury in 2031 -> building minimum in Jan 2028 + 100% world stock fund to 2033; nothing traded in 2031	Same middle, better bad cases, fewer rules (E6 s0; conflict 1). E1's rejection rested on a withdrawn figure (AX1 C6, VRF audit)	Total p5/p50/p95 \$182k/\$207k/\$250k against \$168k/\$210k/\$266k. Bottom fixed at \$150k against \$134k/\$167k/\$211k. History worst \$169k against \$120k, same median \$214k. Paths below money put in: 0% against 1.73% (MODEL; VP data, derived; E6 [2], [5])	Stock share ~9% each year 2028-32 (average 7.5% against 9.7%). p95 -\$16k. Median gift \$174k against \$186-188k. Kept money \$32k against \$21-24k	ALT (a bet on top of REC: p5 -\$23k, p50 +\$1k, p95 +\$32k; bottom \$93k in a 1929 window); kept as the documented alternative. Growth share ~60% / 50-60 debate: moot. Minimum as a share (2/3, 0.8): a free parameter. Ratchet: a second rule for \$2-6k

#	Change (current -> new)	Reason	Numbers	Side effects	Rejected, and why
I3	2031 range: "floor + upside with a stated model probability" -> bottom = owned minimum; top = minimum + half the fund's value that day, capped; the chance of reaching the top stated	Model-free and checkable. An uncapped model top landed within the range 62-93% of the time (AX1 C12, VRF audit)	Median \$150k-\$175k (width 1.17x). P(top) 73% model, 67% on the Vanguard-like input. U.S. two-year returns ≥ 0 in 84% of periods since 1928 (VP data, derived)	Strong markets raise kept money, not the gift	Uncapped p90 top (model-dependent); "give 90%, capped" (promises less than is bought, AX1 C3); $s = 1$ (nothing kept in falls); $s = \frac{3}{4}$ (kept p5 only \$8k)
I4	Certainty: "fully funded at market prices, cash-flow/duration matched" -> defined by price + residual risk + two named gaps; "fits inside \$300k" and "bought in January 2027" dropped as headlines	Brief s16 (binding); case L90, L97-99 (VRF)	Real Nov-15 ladder \$294,387, headroom \$5,613 (19bp; VRF inputs, ASM method). This supersedes CLAUDE.md's \$7.7k/27bp, which uses exact-date pricing. ~1 in 3 rate paths cost >\$300k (MODEL 30-34%; history 32-36% of 68-day windows, VP data). \$50k is worth \$42,063 (2033) and \$33,681 (2042) at 2.5% (ASM)	About 50 words	Model-probability certainty; TIPS (they track U.S. prices, not Taiwan costs)
I5	Risk by goal: forced versus chosen caution; what stocks are for; the most markets can take; statistics degree used zero times; no CFA labels or trap words	E3 findings 2, 3, 6 (SIM); D13c N1 (INT)	Payments = 97.4-98.1% of the first deposit (MODEL)	None	CFA three-factor wording in the IPS (E1 P6): "analysed, not understood"
I6	IPS structure: "only two rules" / "Rule 1-4 + G" -> 10 blocks on four dated decisions, ≤ 12 ideas, one number	E2 F1-F2 (INT); PM-03; last year's lesson (VRF)	~55 ideas in E1 (E2 count, INT)	Some true points move to the Final Report	E1's ~55-idea IPS
I7	Pitch: purpose first, three elements, no numbers, no date claims	E2 F3; E3-5; the date claim fails in ~1 path in 3 (MODEL)	≤ 48 words	Loses the vivid date	A ban-first pitch
I8	2027 rule stated with its "because": buy on arrival; longest-first; name the first-year counterpoint; leftover in bills	D1 with AX1b; E3-7	Longest-first: +5% top-up price rise against +15% for nearest-first; \$31.4k against \$47.7k of face unbought if the deposit never came, at -100bp (MODEL)	One clause	Staging or a yield trigger: no reliable gain, 27-35% chance of needing the deposit (MODEL). Nearest-first. Rolling long rungs short: +\$20k per -100bp (F-108)
I9	2028 rule: "only the facility shrinks" -> completion first, one labelled stress clause, joint tail named once	AY2 D8 C1; the case says "will" (R-AN35)	Joint tail with no deposit: \$11,957 (-50bp) or \$31,413 (-100bp) of the 2033 payment unfunded (MODEL, D1/D3). Rates-fall effect on REC (E6 [8], as in the spec's R2 row): stock fund about \$20-23k (-50bp), about \$1-7k (-100bp); at -150bp no fund and a minimum of about \$127-137k (the range depends on whether the 5-year rate also falls; MODEL). This corrects E4 [7]'s \$31k / \$15k / \$147k	None	Doubting the \$150k in three places (E2 C6)
I10	2033: "leftover mainly as project flexibility" -> reserve = the ladder, named on 1 Jan 2033; gift = minimum + half the fund, $\leq$ top; kept money in U.S.-dollar bills; no NT\$ conversion or hedge	Case L94-107 (VRF); R-AN15 (forwards are derivatives)	Reserve ~\$395k market value, \$500k face (ASM). Kept money median ~15% of post-reserve money (MODEL)	None	NT\$ hedge (~5.9% cost; the Taiwan 2-year yield is SNIP); kept money invested

#	Change (current -> new)	Reason	Numbers	Side effects	Rejected, and why
I11	Governance and costs: "two-clock sleeve management" -> no rebalancing between the parts; rules change only for her circumstances; all costs paid from the stock fund	CFA 2010 2a-2b, 4c (VP via D8); E3-4	1% fee on the stock fund only: -\$3k at the median; on all assets: -\$31k (MODEL, E6 [6])	About 15 words	Fee number in the IPS; "our portfolio manager approves" (R-W19)
I12	Diversification by purpose and date, plus liquidity (new clause)	IPS L46 (VRF); E2 F5	~10,000 stocks in VT (VP)	~25 words	EWT, AI or thematic tilts: tokenism and concentration
I13	Paraphrase test: 6 unpaid readers, two rounds, plus an explain-back	D12 M234 (digital.gov method, VP)	~2 student-hours per round	None	None

Kept unchanged from the current strategy: buy all ten payments first with Treasuries held to maturity; longest-first; the residual risk is a U.S. default; flexibility after 2033; the pre-mortem discipline (E5).

2. The 10 insights behind the changes (question -> answer -> evidence (status) -> deliverable)

- When designs tie in the middle, what chooses?** -> The better bad case, with fewer rules. -> REC p5 \$182k against \$168k, median \$207k against \$210k (MODEL, E6 [2]); history worst \$169k against \$120k (VP data, derived); M237, M238. -> IPS (B4, B7); TN note B.
- What job is Laura hiring a firm for?** -> Making promises that no market or future contract can take back: the payments first, then a building minimum from her own earnings. -> Case L43-45, L91-92, L114-116 (VRF); M143, M228 (INT). -> IPS pitch and B1.
- Is "high certainty" true, and where is it not?** -> It is bought, not forecast, but with two gaps. The January 2027 price can exceed \$300k (~1 in 3 paths). The payments are nominal U.S. dollars against Taiwan costs. -> \$294,387 ladder, 19bp headroom (VRF inputs, ASM); 30-34% (MODEL); \$33,681 real in 2042 (ASM); M004, M053, M084. -> IPS B2-B4; the pitch avoids date claims.
- Why so little stock for a founder who says "you should always take the jump"?** -> The caution is forced by arithmetic, not chosen for her temperament. All money nobody relies on is 100% stocks. -> 97.4-98.1% of the first deposit (MODEL); B1b-Q11 (VP, research only); D13c N1 (INT). -> IPS B5; TN reflection C.
- How is the 2031 range set without overpromising?** -> By a rule: bottom = what is owned, top = a market value on the day, capped. Excess stays hers, and the chance of reaching the top is reported. -> P(top) 73% (MODEL); 84% of U.S. two-year periods (VP data, derived); case L110-120 (VRF); M039, M040, M107, M191. -> IPS B7.
- What makes this more than the textbook lock-early?** -> The Laura-specific rule: each deposit buys one promise when it arrives. The second deposit's own amount sets the building minimum. -> Case L43-45, L110-112 (VRF); M011 (INT); E4 L6. -> Pitch element 2; TN note B.
- What should the WInS book show?** -> Her plan on the first day all three jobs exist, scaled, with the scaling said inside every note and in the IPS. -> E6 5.1; D10 1.5; IPS L75 (VRF); M005, M247. -> TN (all notes); IPS B10.
- Which words survive a permanent, un-editable note?** -> One vocabulary: never "match", "promised" or "guaranteed"; "moves like / stands in for". -> 74.7% of TLH matures after 2042 (VRF); case L110, L116 (VRF); M062. -> TN notes; IPS.
- How do the three notes show supported, tested and refined honestly?** -> A is tested on a pre-registered window with three numbers. B is refined only if logged. C tells the true growth-first story. -> Growth-first 3.2% / 40.8% miss (MODEL); E2 F8-F9 (INT); TN guide L12, L29-30 (VRF); M002. -> TN reflections.
- How dense can a 500-word IPS be?** -> About 12 ideas in 10 blocks, one number, and a paraphrase test. -> E2 counted ~55 ideas in E1 (INT); "over-complex for the insight" (CLAUDE.md, VRF); M036, M235, M234. -> IPS.

What this teaches

- A good plan is usually improved by removing things, not adding them. Here the payments-first core survived every audit. The parts built around it (a mixed sleeve, a later lock, a model percentile) were replaced by one dated purchase and one fund.
- Rank changes by what is due first and by what a reader can see. A perfect 2033 rule is worth less than notes that do not contradict the IPS.

Open questions: decisions, WInS data, unverified sources, parked questions, later items

1. There are 12 team decisions still open (E6 s7). The three gate votes come before any order: lock-early, REC or ALT, and book (ii)R. Each has a default and a test for choosing.
2. Nine things can only be seen inside WInS. The most important are the position limit, the IBTM and 2032-note listings, the note-box limit, and any activity minimum.
3. Several inputs are still UNVERIFIED or SNIP. One spec inconsistency was flagged and has since been fixed: R2's rates-fall figures now follow E6 [8].
4. 200 Phase C questions were parked, most as "already answered". They are listed below, grouped, one line each. This is a log, not a Q&A bank.
5. Final Report items wait until after Nov 9, one line each.

Serves: TN (Oct 23) and IPS (Nov 6). The "Later" list serves neither and is only parked for the Final Report.

Labels: VP, VRF, SNIP, ASM, MODEL, INT (as in `why_us.md`); UNVERIFIED = not yet checked on any source. Every ticker named here is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK**; primaries and alternates are in `securities_and_allocation.md` s1-s2.

1. Team decisions still open (E6 s7; defaults are INT)

#	Decision	Spec default	Test for choosing	By (AEST/AEDT)	Serves
D1	Adopt lock-early	Adopt	Every student writes why the payments come first, with no AI file open; the vote is logged with the reasons	Gate, Thu Oct 1	TN, IPS
D2	REC (minimum in 2028) or ALT (2031 lock)	REC	Values question plus a two-minute explain-back of the four dates. Choose the design all six can explain. Numbers: E6 [2], [5]	Gate, before any building or stock order	TN, IPS
D3	WInS book	(ii)R	Two-reader check, and the scaling said in about 25 IPS words. Two failures mean (iii)	Gate	TN
D4	Instrument for the minimum	2032 note, then IBTM, then VGIT	The first one listed	Gate	TN
D5	Labels for the three parts	The team's own words	They survive the explain-back and are identical in notes, log and IPS	Before the first note	TN, IPS
D6	Share of the stock fund added to the gift (s)	One half	Kept money at the median is about three years of Taiwan building-cost drift (ASM)	Mon Oct 26	IPS
D7	Cost clause	Keep one clause	Compress it if the PDF has fewer than 2 spare lines	Oct 26; Nov 4	IPS
D8	Statistics degree	Zero uses	Use it only if it changes a decision	Oct 24	TN, IPS
D9	Paper and spacing	Letter, Word "Double"	The PDF ends on page 3 with at least 2 spare lines	Nov 4	IPS
D10	Joint tail in the IPS	One clause	A reader can name what could still stop a payment	Oct 26	IPS
D11	Repository visibility until Dec 4	Team leader decides and logs it	PM-30 trade-off against the earlier acceptance (CLAUDE.md)	Fri Oct 2	TN, IPS (copy risk)
D12	Roles	Assign now	Each student owns at least one note or reflection and one IPS block	Tue Sep 29	TN, IPS

Also still open for the team (from E4 s8, INT):

- How much stock exposure for 2028-32: the default (45 percent-years) or the two-thirds alternate (59), which lowers the bottom to \$134k (MODEL).

2. Data to get from inside WInS (before the first order; serves TN)

- [] **Session Rules screenshot:**

- single-security position limit, and any bond or security-type limit;
- day-trading tooltip, trades allowed, commissions.

The limit value is UNVERIFIED; it is shown only in the portfolio summary (VP). It picks the branch in E6 5.3; below about 17%, stop and ask Wharton.

- [] **Logged-in Trading Details page:**
 - any activity minimum (undefined publicly, R-W23);
 - whether a first trade is due by Oct 10 (UNVERIFIED).
- [] **Listings:** search IEF, TLH, VT, IBTM, SPTL, VGIT, SPTI, VTI, VXUS, SGOV, BIL.
- [] **Bonds drop-down:** every U.S. Treasury maturing Jul-Dec 2032, with coupon, WInS price and accrued interest. Search by maturity. The 15-Nov-2032 note (CUSIP 91282CFV8) listing is UNVERIFIED.
- [] **Note-box character limit:** a zero-risk test typing about 450 characters, without confirming (UNVERIFIED; 300 in a sister product, VP; Wharton's example is 413, VRF).
- [] **Daily volumes of Vanguard funds,** checked against the 2x-volume rule.
- [] **WInS username** for the IPS title page, copied character for character.
- [] **Screenshots of every position** on the fill date and at the Oct 20 close.
- [] **Ex-dividend dates** for IEF and TLH in the test window.

3. Blocked or unverified sources and inputs

- **Network access varies.** CLAUDE.md lists treasury.gov, FRED, iShares, JPM, Wharton and SMaply as blocked by default. The insight_v1 run read several of them when allowed (for example, E6 re-read IBTM on ishares.com on 2026-09-28, VP). Check brief s2 before relying on any fresh fetch.
- **WInS instrument availability:** IBTM (not on the 2025-26 list, VRF), TLH (not on it), and any 2032 Treasury. All UNVERIFIED.
- **IBTM** "does not seek to return any predetermined amount" (VP). A note may never say "repays \$X" for it.
- **The January 2028 five-year rate** is held at 4.98% (ASM). At 4.0%, the stock fund is about \$35k, not \$40k (E4 [4], ASM).
- **Return inputs:** JPM AC World 7.00% compound (VRF). Vanguard-like 5.08% (ASM hybrid; the Vanguard VCMM page was cited via D-files, not re-opened by W3).
- **The model has thin tails** (AX1 C12). History windows are used beside it.
- **Taiwan 2-year yield** (used for the NT\$ hedge cost): SNIP.
- **NSW school holidays** for this team's calendar: UNVERIFIED (E5).
- **Semifinal basis:** treat all three deliverables as scored (R-AN41; SMaply). The exact weighting is not stated.
- **Laura quotes:**
 - 6 PARAPHRASE-UNVERIFIED lines;
 - 8 new verbatim lines in D13c s7d, not yet ruled on by D13a;
 - none is allowed in the TN or IPS anyway.
- **Spec inconsistency (for the main loop):** E6 s4 R2 prints E4 [7]'s rates-fall figures (stock fund \$31k at -50bp, \$15k at -100bp, minimum ~\$147k at -150bp). E6 [8], re-run 2026-09-28, corrects these to \$23k, \$7k, and \$0 with a minimum of \$137k (5-year unchanged). The corrected figures should win. **FIXED 2026-09-28 by the main loop** in E6_final_spec.md, ips_spec.md and securities_and_allocation.md, and by the phase G verifier in strategy_changes.md (range shown for 5-year rate unchanged vs also lower: fund \$20-23k at -50bp, \$1-7k at -100bp; at -150bp no fund and a minimum of \$127-137k).
- **CLAUDE.md headroom** (\$7.7k, 27bp) uses exact-date pricing. The real Nov-15 STRIPS ladder gives \$5,613 (19bp) (E6 R1; VRF inputs, ASM method).

4. Parked questions (Phase C; 200, grouped by domain, one line each; text truncated from `phase_C/parked.json`)

The high-priority parked items whose answers reached the spec (brief s16):

- M007 bad-year rule: E6 R6 (fund never sold) and R7.
- M008 2031 message order: E6 R4 and B7 (owned / expected / kept).

- M015 fundraising leads with the pre-funded ten years: the E6 s10 later list.
- M021 dated decision log: E6 5.4, R7.
- M033 whole-portfolio stock share: E6 B5 (forced versus chosen, ~9%).
- M035 diversification for a fixed-income reader: E6 B6.
- M040 range as a rule: E6 R4.
- M149 funded-status wording: E6 B2 (certainty by price).

Tags give the skeptics' kill reasons (already_answered, no_so_what, trivial, wrong_premise, unanswerable, too_complex). Items are sorted by priority within each group.

D1 Rates and the ladder (19)

- M048 [already_answered]: Should the IPS and first Treasury trade note say why locking is right in this rate world (real 10y...
- M056 [already_answered]: With the U.S. rated below AAA by all three agencies, what honest wording should replace 'certain...
- M084 [already_answered]: What is the real value of the ten fixed \$50k payments at the market's own inflation price and under...
- M244 [already_answered]: Does answering the reserve question with 'it is the 2027 ladder and it changes by itself' satisfy the...
- M029 [no_so_what/already_answered]: Which dated events or reversals before Laura's January 2027 purchase (an AI-equity bust, a fall in...
- M042 [trivial/already_answered]: Which market date should the IPS and Final Report numbers be priced on (25 Sep, the 6 Nov freeze, or...
- M098 [already_answered]: Which of our Treasury-matching and bond statements would an Aberdeen fixed-income professional mark...
- M114 [no_so_what/already_answered]: What would it cost to make the ten payments inflation-indexed, and should the Final Report show that...
- M222 [already_answered/trivial]: Does Laura accept that her facility absorbs every rate move before January 2027 one-for-one (each...
- M159 [already_answered]: What does insisting on Treasuries rather than high-grade corporate bonds cost Laura, and does the...
- M179 [already_answered]: Should co-sponsor material describe the operating commitment as US\$500,000 nominal or at its 2033...
- M240 [no_so_what/already_answered/wrong_premise]: In WInS, should the growth money's short-Treasury slot be a defined-maturity fund ending December...
- M115 [already_answered]: The case fixes every payment date, yet asks teams to state assumptions about 'the timing of cash...
- M163 [no_so_what/already_answered/wrong_premise]: Should the Final Report use Wharton's own Penn Wharton Budget Model estimate (25% chance of reaching...
- M164 [trivial/already_answered/no_so_what]: Does the projected debt-limit X-date of about February 2028, with Congress needing to act by January...
- M165 [already_answered]: Do the reserve's Nov-15 maturities (paying ~47 days before each Jan-1 payment; 2033-01-01 is a...
- M166 [trivial/already_answered/no_so_what]: If an agency downgrades the U.S. again, what will the plan do, and should the IPS pre-commit to no...
- M168 [trivial/already_answered]: Given the 2026 evidence (Fed independence upheld in Trump v. Cook, a 12-0 hike, 5y5y inflation...
- M170 [wrong_premise/already_answered]: Does the 20-year 'hump' in the Treasury curve (20y 5.54% above 30y 5.49%) make the 10-20 year sector...

D2 Equity, AI and return inputs (6)

- M010 [already_answered]: Should the team reject any 'AI hedge' tilt, describe Laura's own AI exposure as two-sided (lost...
- M047 [already_answered/trivial]: Should the human-capital 'underweight publishing, media and education' tilt (e.g. HarperCollins's...
- M173 [already_answered/trivial]: When the 2027 J.P. Morgan LTCMA appears (likely mid/late October 2026, between the Trading Notes and...

- M119 [already_answered/trivial]: Does 'objectives that reflect ... her passion for innovation' justify any innovation, tech or AI...
- M174 [already_answered/wrong_premise]: Is JPM's highest equity return (EM 7.8%) worth an emerging-market tilt when MSCI EM is now 26.85%...
- M175 [no_so_what/unanswerable]: Will AI mega-IPOs and hyperscaler equity raises push AI concentration in broad funds higher during...

D3 Quant model and design (20)

- M040 [already_answered/trivial]: Should the recommended co-sponsor 'dollar range' be a figure computed today (a 5-7-year forecast), a...
- M099 [already_answered]: Since readers may expect a probability for how certainty was evaluated (last season's strong teams...
- M061 [already_answered]: Should the Final Report evaluate implementation by how closely the WInS bond book tracked the present...
- M138 [already_answered]: How big is the 2028 deposit risk compared with market risk for the facility range, and should the...
- M085 [trivial/already_answered]: Which return should the sleeve's bonds earn in the model when JPM's LTCMA (4.00%, set at a 4.16% 10y)...
- M113 [already_answered]: Since the facility money is only what is left after the reserve, is it a residual claim? And should...
- M123 [already_answered]: With a statistics-degree client and an academic director who holds an M.A. in Statistics and runs the...
- M169 [already_answered/wrong_premise]: Should the post-2033 flexibility money (not the nominal reserve) be named as the buffer that lets...
- M232 [trivial]: What rounding rule should the deliverables use so that no number looks more precise than the model...
- M245 [already_answered]: Should the Final Report show that the reserve design many rivals will propose, an equity 'bucket' for...
- M248 [already_answered]: If rivals show \$300k+ facility contributions from 8-10% return assumptions, should our Final Report...
- M009 [unanswerable/already_answered]: What correlation between the 2028 \$150k deposit and 2027 equity returns should the model assume (her...
- M139 [already_answered/trivial]: Because the 2028 deposit is known by 2031, should the 2031 co-sponsor range exclude deposit risk and...
- M233 [no_so_what/trivial]: Should the Final Report attribute the WInS result to its causes (rate moves on the Treasury part,...
- M104 [already_answered/no_so_what/wrong_premise]: How late should the 'late 2028 deposit' stress case be (for example about 12 months), given that both...
- M086 [no_so_what]: Which stock-bond correlation should the model use (JPM's ~0, the 2016-2020 regime where bonds hedged...
- M151 [already_answered/no_so_what]: US public pensions discount promises at their assumed return (median 7.0%); at 7% Laura's payments...
- M162 [already_answered/trivial]: After the 2031 floor is locked, should the rest of the sleeve still hold equities, given that...
- M200 [already_answered]: Do the readers reward quantitative sophistication (algorithms, optimisation, ML) or restraint, and...
- M239 [already_answered]: Does any lock timing, for example buying the whole ladder with both deposits in January 2028, give...

D4 Taiwan, currency and building costs (14)

- M058 [already_answered]: What facility-cost inflation assumption should the Final Report use (Taiwan construction costs...
- M069 [already_answered]: Where exactly must inflation appear in the deliverables, and should every co-sponsor dollar figure be...
- M022 [wrong_premise]: Is a U.S.-China/Taiwan crisis (e.g. a blockade between 2031 and 2033) the single correlated scenario...
- M172 [already_answered]: Should real erosion of the fixed \$50k be measured with U.S. CPI, Taiwan CPI, or Taiwan wages (the...

- M176 [trivial]: Given the ~310bp U.S.-Taiwan short-rate gap, the partial natural hedge and the jump risk, should any...
- M215 [already_answered/wrong_premise]: How should the FR and co-sponsor draft describe Taiwan geopolitical risk, given that Laura's founding...
- M043 [already_answered]: Since the case never says why the residency is in Taiwan, should deliverables refuse to supply a...
- M088 [already_answered/unanswerable]: What share of the S&P 500 and of a world fund depends on TSMC's Taiwan fabs (direct weight plus U.S....
- M171 [too_complex/already_answered/unanswerable]: With a 19.6%-of-GDP current-account surplus, insurers' reduced FX hedging and central-bank smoothing,...
- M089 [already_answered/no_so_what]: Should the IPS carry a one-line rule for the locked payments if the residency is delayed or cannot...
- M090 [too_complex/no_so_what]: Do TWD and equities tend to fall together (TWD weakens in crises and jumps in good times), making FX...
- M130 [already_answered/unanswerable]: The 2024-25 case wrote '\$10,000 USD' for a Nigerian project, while the 2026-27 case names no currency...
- M177 [trivial/already_answered/no_so_what]: Should the sleeve's non-U.S. equities stay currency-unhedged, since USD/TWD moves with the broad...
- M182 [unanswerable]: Do Taiwan public subsidies require a self-funded share (自籌款), so that Laura's fixed US\$50k makes the...

D5 Co-sponsors, range and facility (26)

- M008 [already_answered]: Should Laura's 2031 message be ordered as an unconditional floor already bought, then a conditional...
- M015 [already_answered]: Since capital funders test whether operations are sustainable after the building is finished, should...
- M041 [too_complex/no_so_what]: Should the plan pre-schedule a late-2032 checkpoint (when annual public funders set 2033 budgets) at...
- M065 [already_answered]: With no facility cost given, what makes Laura's contribution 'meaningful' to co-sponsors: its dollar...
- M096 [already_answered/wrong_premise]: Should the 2031 pledge be tied only to money already bought, treating the bottom as the only promise...
- M014 [trivial/already_answered]: What should the Final Report say, without solving it, about the residency after 2042 (year eleven,...
- M027 [too_complex/already_answered/wrong_premise]: Does the case allow the facility contribution to be committed in 2033 but drawn in stages against...
- M146 [already_answered/trivial/wrong_premise]: Should the co-sponsor draft carry one line on what the range is NOT (not a pledge of the top figure,...
- M187 [already_answered]: Should the 2031 floor be bought before Laura's very first co-sponsor conversation, so the first...
- M192 [already_answered]: Since capital funders expect 'project-cost estimates are firm' but the case leaves facility cost...
- M193 [already_answered]: Should the fundraising part state a two-way 'no substitution' rule: co-sponsor money never replaces...
- M220 [already_answered/too_complex/unanswerable]: What does 'financial flexibility as the project develops' buy for someone who builds in stages...
- M024 [unanswerable/already_answered/no_so_what]: Who exactly will read her 2031 fundraising material (Taiwan's annual, project-based public culture...
- M126 [already_answered/no_so_what]: All six clients pay something forward, and Laura's residency serves 'artists, writers, designers,...
- M180 [already_answered/trivial]: How much of Laura's portfolio should co-sponsor material disclose to prove that the promise is...
- M183 [already_answered/no_so_what]: If the residency is delayed or cannot open, what happens to Laura's facility money, and should...
- M092 [too_complex/already_answered]: Should part of Laura's above-floor amount be offered as a 1:1 match or challenge for co-sponsor gifts...

- M093 [already_answered/no_so_what]: Since the seed-money effect is measured against a known campaign goal and the facility cost is...
- M105 [no_so_what/wrong_premise]: Given Laura's free-access record (a free festival, mini grants, pro-bono school visits), should the...
- M120 [already_answered/trivial]: Will co-sponsors read 'operating reserve' in its nonprofit sense (a few months of expenses held by...
- M135 [already_answered/no_so_what]: Should one clause of the fundraising draft cite Laura's record of building platforms and communities...
- M153 [already_answered/too_complex/wrong_premise]: Since Yale sets each year's spending from its endowment value 'from two years ago' for budget...
- M155 [too_complex/wrong_premise]: Should post-2033 'financial flexibility' be written as a draw rule on the leftover money (overruns...
- M178 [already_answered]: Will co-sponsors write Laura's stated low end into their own grant conditions, so that missing it...
- M186 [already_answered/trivial/no_so_what]: Should Laura's US\$50k payments be described as unrestricted general operating support, since flexible...
- M242 [already_answered/wrong_premise]: Is part of 'pursuing growth' the growth of what the residency can raise, since a bought, credible...

D6 Behaviour and client psychology (18)

- M007 [already_answered]: Which pre-committed, purpose-tied 'bad-year rule' should the IPS carry for Laura's emotions in a...
- M033 [already_answered]: Will a first-screen reader see the whole-portfolio equity share (~1.5% in 2027, ~20% in 2028-30, ~4%...
- M046 [no_so_what/unanswerable]: Should the IPS turn the case's 'understands that investing involves uncertainty and periods of market...
- M075 [no_so_what/unanswerable]: Does Laura-specific evidence (her lumpy creative income and working design studio as human capital;...
- M236 [already_answered]: Does the plan show Laura, in one table, how each goal fails softly (payments, the 2031 announcement,...
- M158 [already_answered]: Of the four standard 'get back on track' levers, which can Laura use, and should the IPS pre-commit...
- M017 [already_answered/wrong_premise]: Given the case's firm 'will contribute \$150,000' but no public next-book deal, a 2025 writing break...
- M045 [already_answered]: Should we rule out every identity- or interest-themed security (the EWT/Taiwan tilt, publishing or...
- M050 [already_answered]: What does the unattributed pull quote signal (self-reliance; private conviction turned into public...
- M054 [already_answered]: Should the Final Report leave Laura's 'continued business income' uncouneted as a flexibility source...
- M072 [no_so_what/already_answered]: Is Laura's 2028 income more exposed to political and budget cycles (corporate DEI/ERG event budgets,...
- M221 [already_answered/no_so_what]: Does Laura tolerate losses only she bears (she helped launch an Anti-Resume Project to normalize...
- M073 [already_answered/no_so_what]: How should the Final Report describe policy risk to her school and library channel (possibly her...
- M133 [already_answered]: The case says Laura 'has no shortage of ideas' but ties flexibility to the residency 'as the project...
- M141 [no_so_what/trivial]: What does 'licensing' cover for an author whose publisher bought only world English rights, and does...
- M160 [trivial]: Should the FR cite 'behaviour gap' or 'adviser's alpha' figures to justify pre-commitment rules,...
- M219 [already_answered]: Do domain-specific risk research and entrepreneurial-finance theory support reading 'Although'
- M224 [trivial]: Laura 'has no shortage of ideas' and the case bars withdrawals before 2033: should the IPS present...

D7 Wharton intent and judging (23)

- M021 [already_answered/trivial]: Should the team keep a dated decision log from today (disagreements, roles, mistakes, reasons for not...
- M149 [already_answered]: Should the IPS describe the portfolio as following a funded-status glide path (de-risk as the funded...
- M134 [already_answered]: Past clients praised teams that made them feel 'seen' (Jordan) and went beyond 'just ... the numbers'...
- M108 [already_answered]: Which rejected alternatives (growth-first, rolling short, equity-heavy, all-Treasury) should the...
- M206 [already_answered]: Which Final Report content must be collected during trading because it cannot be recreated after the...
- M016 [already_answered/unanswerable]: Who reads our written deliverables first (Wharton student evaluators or Aberdeen asset managers), how...
- M032 [wrong_premise/already_answered]: Should the WInS hedge be placed in dated steps under a rule written in advance (core Treasury trades...
- M068 [already_answered/wrong_premise]: Since Wharton replaced 'compelling and clever pitch' and 'win him/her over as a client' with...
- M124 [already_answered]: The committed outflow grew from 50% of the client's money (Jordan) to 111% (Laura), and this is the...
- M210 [trivial]: Should the team pre-build FR components before Nov 9, assuming a page cap near last season's 15 pages?
- M211 [unanswerable/already_answered]: Pre-mortem: if we miss the Top 50 with a correct strategy, which single cause is most likely, and...
- M067 [unanswerable/already_answered]: What will the median and top-decile 2026-27 report look like (how many teams will reuse last season's...
- M087 [no_so_what/already_answered/trivial]: How do we evidence discipline-by-inaction (not selling or trading the Treasury hedge as yields rise...
- M101 [already_answered]: Would a candid 'anti-resume' in the Articulation section (last season's rule violation and the...
- M128 [already_answered]: Top-50 lists show many client-themed team names, and a 2024 semifinal judge praised the ability to...
- M038 [already_answered]: Since this year's case names no values or ESG preference and every value word describes her work...
- M214 [trivial/already_answered]: How should the team describe Laura's quantitative background, given that the case/resume say...
- M116 [no_so_what/already_answered]: What does the way Wharton's cases have changed across seasons reveal about what this year's designers...
- M129 [trivial/no_so_what/unanswerable]: How did the 2021-22 finalists fund Jordan's ten scholarship payments (cash, dividend stocks, bonds),...
- M137 [no_so_what/trivial]: Should the team assign an owner to watch Wharton Global Youth and Wharton Magazine for a new Laura...
- M199 [already_answered/trivial]: Is the 2026-27 deliverable structure (three-note Trading Notes Analysis, 500-word IPS, strategy...
- M203 [no_so_what/already_answered]: What are the realistic odds of the Top 50, and what does the pattern of past semifinal schools say...
- M207 [unanswerable]: Can we find any readable public account of what 2025-26 finalists (FigCapital, HHA Investments,...

D8 Professional practice, fees, governance (17)

- M035 [already_answered]: How do we show 'appropriate diversification' and understanding of the investments to a fixed-income...
- M052 [already_answered]: Given Laura's CFPB, trading-firm (Belvedere Trading) and financial-literacy-teaching background, what...
- M223 [already_answered/trivial]: Should the Final Report include a short table 'what the profile does not tell us -> what we assumed...
- M225 [already_answered]: Does 'the capital required for her goals' grow in 2031, when the announced range's low end becomes a...

- M064 [already_answered]: Which time-horizon stages should the IPS name (2027 buy, 2028 top-up, 2031 announce, 2033 gift), each...
- M025 [already_answered]: Do practitioner yardsticks (the UK regulator's ~90% hedge for a mature 10-year-duration scheme, a nil...
- M037 [already_answered/trivial]: What liquidity need does the case actually give Laura (none before 2033), and what should the IPS's...
- M078 [trivial/already_answered]: Should the ten-payment reserve be held from 2027 in a separate, labelled sub-account with a written...
- M079 [already_answered]: Where does our 'bought' funding certainty sit next to published standards (Solvency II 99.5%, UK...
- M083 [already_answered]: Is the right professional analogue for Laura's ~152%-funded surplus a pension 'surplus glidepath'...
- M148 [wrong_premise/already_answered]: What surplus buffer above 100% funded do pension practitioners require before calling a plan...
- M150 [already_answered]: Can the plan borrow municipal finance's distinction between economic defeasance (Treasuries bought to...
- M080 [no_so_what/already_answered]: Does Australia's Retirement Income Covenant (income, stability, flexible access) mirror the IPS...
- M082 [trivial/already_answered]: Which named frameworks (cash-flow matching/LDI, funded-status de-risking, goals-based buckets,...
- M112 [trivial/already_answered]: Should the Final Report give Laura one recommendation, or a small menu of options for her to choose from?
- M154 [no_so_what/already_answered/wrong_premise]: Does the case's 'responsibly contribute' track UPMIFA's legal prudence factors for endowment spending...
- M157 [no_so_what/already_answered/trivial]: Would a pension CIO consider buying the promise from an insurer (a deferred 10-year period-certain...

D9 Communication and wording (38)

- M077 [already_answered]: Which of Laura's own verified ideas can the strategy echo honestly, and should every Laura-specific...
- M142 [already_answered]: Should the IPS pre-commit, in one sentence, what happens if the 2028 contribution is late or smaller?
- M057 [already_answered]: Should the plan's headline progress measure be what it secures ('all ten years of operating payments...
- M103 [already_answered]: Which elements of our strategy or presentation would Laura call 'gimmicks' (Taiwan equity tilt,...
- M184 [trivial/already_answered]: Should one Trading Note reflection tie an executed Treasury purchase to co-sponsor credibility...
- M188 [too_complex/already_answered]: Which confidence level should the 2031 statement carry, and should it always use 'dual format' (a...
- M201 [already_answered]: What is our cut list: which concepts stay in the Final Report's main text, which go to an appendix,...
- M227 [already_answered/trivial]: Should the FR's client section open with one table (goal -> required or desired return -> risk she...
- M049 [already_answered/trivial]: Should the team adopt the case's new vocabulary (certainty, co-sponsor, flexibility) and ban old...
- M070 [trivial]: Should the 500-word IPS be organised one-to-one on the IPS guide's five client items and the case's...
- M097 [already_answered]: Should the co-sponsor wording be written to survive translation into Mandarin (every probability...
- M100 [already_answered]: Would book-title puns and identity metaphors ('strong foundations for messy roots', 'lessons for...
- M117 [already_answered/trivial]: Should the fundraising draft be written as a 2031 document with clearly labelled illustrative figures...
- M127 [trivial/already_answered]: The case opens with Laura's belief that 'stories have the power to change how people see the world',...

- M136 [already_answered/trivial]: Since 'Laura Gao believes stories have the power to change how people see the world' follows...
- M161 [already_answered/trivial]: Should the fundraising draft report uncertainty as dollar outcomes in named scenarios (practice:...
- M189 [already_answered/trivial]: What exactly does our confidence percentage refer to (paths of a stated model, or past two-year...
- M190 [already_answered]: How should the two ends of the range be labelled so readers do not treat a result just outside a hard...
- M195 [already_answered]: Which two or three drivers of change should the 2031 statement name in advance, given evidence that...
- M198 [trivial/already_answered]: Should every sentence of the fundraising part pass a 'floor test' (still true if 2033 lands exactly...
- M205 [trivial]: Does the Final Report open with the strategy clearly stated up front, so a reader who reads only page...
- M229 [already_answered]: Should the IPS state one primary goal plus explicit guardrails that are never breached (all ten...
- M246 [already_answered]: Rivals who buy the reserve in 2033 at cautious rates will show reserves of about \$405-440k; will a...
- M076 [already_answered/trivial]: Should the Final Report use a few clean analytic charts that each carry a conclusion, meeting the...
- M131 [trivial/already_answered]: Because Laura taught financial literacy to high school students and the case calls her an educator,...
- M167 [trivial/already_answered]: Since no U.S.-dollar asset is safer than a Treasury and every rival plan shares that floor, should...
- M218 [already_answered/trivial]: Can the IPS central idea be framed as 'both, not either-or' (a locked promise AND growth), echoing...
- M106 [trivial]: How large should the Final Report's trade-offs and assumptions disclosure be: one compact, separate...
- M194 [trivial/already_answered]: How can the fundraising part sound confident without overpromising, given evidence that people prefer...
- M213 [already_answered]: Should the IPS open with a one-sentence purpose statement and treat each security as a replaceable...
- M020 [trivial/already_answered]: What voice and pronoun conventions should the deliverables use (her site lists she/her and they/them,...
- M132 [trivial/already_answered]: Since the 2023-24 prompt 'Explain what makes your investment strategy unique compared to other teams'...
- M144 [no_so_what]: Is there one principle drawn from her books' plots (not titles) that maps onto a real mechanism in...
- M147 [wrong_premise/trivial]: What readability level should the IPS and FR target, given that her books are rated 'Ages 14-up' and...
- M152 [trivial/no_so_what]: Is the lottery annuity (funded by U.S. Treasury securities, 'the higher the interest rates, the...
- M156 [too_complex/already_answered/wrong_premise]: Should the FR report risk as 'expected negative years out of 20' (Australia's Standard Risk Measure)...
- M204 [trivial/unanswerable]: Is there any sign that non-U.S. teams are disadvantaged in the written round, and does anything in...
- M217 [already_answered]: Must the team refuse to use a personal-advice line of Laura's as a finance metaphor? (Text withheld here: the line is D13a EXCLUDE-PRIVACY, B8b-Q34; answer: yes, never use it.)

D10 Compliance and WInS rules (19)

- M185 [already_answered]: Which single co-sponsor rule belongs in the 500-word IPS, given that the IPS excludes the final range...
- M216 [already_answered]: May the FR quote Laura's own public phrase 'self-employed, with unstable income' to justify the...
- M023 [already_answered]: How specific should our AI-use record be (what AI did, what students decided and redid by hand, each...

- M110 [trivial/already_answered]: When one strategic decision needs several trades (because of a position limit or a two-fund duration...
- M196 [trivial/already_answered]: Should the fundraising part attribute the range and confidence to Laura's investment team, with an...
- M202 [already_answered]: Can the semifinal readers see our WInS holdings and transaction history, and if so must every open...
- M051 [already_answered]: Is the case pull quote 'The only person who needs to believe in something is yourself.' traceable to...
- M074 [already_answered]: Which exact words may the deliverables use about Laura's books and honours ('bestselling', finalist...
- M118 [already_answered]: Does this year's Trading Notes Analysis still expect one stock, one ETF and one team choice, as last...
- M122 [already_answered/no_so_what]: Given that the FAQ still carries recycled text ('10 weeks of trading', 'long-term portion of your...
- M140 [already_answered]: Which claims in the team's two client documents must be removed before anything reaches a deliverable...
- M181 [trivial/already_answered]: Because the case bans withdrawals before 2033, which words accurately describe Laura's 2031 floor as...
- M208 [already_answered/unanswerable]: What is the unpublished 'required trading activity' rule, and could a low-turnover, few-holding...
- M145 [already_answered]: How much real 2025-26 news about her career (no announced next book, a 2025 writing break, her...
- M066 [already_answered]: What do the newly found 2026-27 Investment Competition Guide and Competition Infographic add to or...
- M102 [already_answered/trivial]: How should the team verify external facts about Laura when secondary sources conflict (dates, bios, a...
- M111 [no_so_what/already_answered]: Does the WInS rule 'no more than twice a security's current daily trading volume' use today's running...
- M209 [trivial/unanswerable]: Which word counter will Wharton use for the 50/500/100-word limits, and what internal caps protect us...
- M230 [no_so_what/already_answered]: Should the IPS include a roughly 15-word 'non-goals' clause (not beating the market, not timing...

5. Later (after Nov 9; Final Report only; one line each, not worked here)

- The 2031 range in dollars at weak, median and strong 2031 states, with the likely figure beside it.
 - The chance of reaching the top from U.S. history (two-year periods since 1928) and from the model, both named.
 - Named historical windows beside the model's p5 (E4 [5]; E6 [5]).
 - The all-Treasury benchmark and the rejected alternatives (growth-first, 2031 lock, uncapped percentile top).
 - The fee assumption, its base, and net-of-fee figures (E6 R8, [6]).
 - A New Taiwan dollar reference at a dated rate, and the building-power assumption line (D4).
 - The reserve's market value on 1 Jan 2033 with cost and face, and the rung-by-rung ladder (S3).
 - The fundraising excerpt order: operations funded; minimum owned since 2028; stretch and its odds; what is not promised; U.S. dollars binding.
 - The decision-log traceability table (M212) and the paraphrase-test log.
 - At most one or two dated, verified Laura lines, in context (D13a/D13c).
 - The residency after 2042 (M014), named but not solved.
 - Works Cited entry for all AI use (R-W46).
-

Added after the blind Laura's Choice contest (2026-09-28)

- **D-open (balance), decide before the IPS:** all three simulated evaluators ranked our plan first but said it reads as timid (7.5% average stock share 2028-32). Test one alternative with a smaller 2028 floor

(for example \$100-120k) and more in the world stock fund against REC on `scripts/E6_final_checks.py`'s engine, then choose and state the price of certainty in one line. Default: keep REC and state the price (median about equal to the typical plan; p95 \$250k).

- **Wording:** call the \$150k bond "the bought floor of the 2031 range", never a "minimum facility contribution".
- **In-range confidence:** state that the 2033 contribution lands inside the announced range by the cap rule (barring a U.S. default); report ~73% (model) / 84% (history) only as the chance of reaching the top.

What this teaches

- Keep open questions in three different piles: choices only the team can make, facts only the platform can show, and sources nobody has checked yet. Mixing them hides which ones block the next trade.
- Parking a question is a decision too, so log why. Most "important" questions were already answered somewhere.

Part 2 - The final strategy and the blind contest

The chief strategist's specification that Part 1 is built from, and the blind 'Laura's Choice' contest.

1. E6 Final strategy specification (chief strategist)
2. Contest: Firm A one-pager
3. Contest: Firm B one-pager
4. Contest: Firm C one-pager
5. Contest: Firm D one-pager
6. Contest: Firm E one-pager
7. Contest: Firm F one-pager
8. Contest: fairness audit log
9. Contest: evaluator L1 (simulated Laura)
10. Contest: evaluator L2 (simulated Laura)
11. Contest: evaluator L3 (simulated Laura)

E6 Chief Strategist: final strategy specification for the team vote (Trading Notes and IPS)

Agent E6 (Chief Strategist), insight_v1 run, Phase E. Written 2026-09-28. AI-generated research for Team Caplet: decisions to vote on, rules, numbers and checklists. **None of it is text to submit.** The six students decide every point and write every WInS note, reflection, pitch sentence and IPS sentence in their own words. Any AI use goes in the Final Report's Works Cited (Wharton AI policy, R-W46). Laura appears only through the case and her public professional record. Nothing here suggests contacting her (R-W16: contact means disqualification).

Scope (brief s17). This file serves WInS trading now (as the source of the notes), the Trading Notes Analysis (TN, due Oct 23) and the IPS (due Nov 6). The IPS freezes the strategy (IPS guide L73-75, VRF), and the Final Report must apply it without redesign (Guide p.5 L156-159, VRF). So section 4 fixes every decision **rule** the Final Report will apply. There is no Final Report content: no dollar range, projections, charts or fundraising text. **Relayed messages:** one, "Try again" (a request to re-run this task). It changes no scope; this file is a fresh run.

Status labels (brief s3): **VP** = VERIFIED-PRIMARY; **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **MODEL** = an assumption-based model output, never a forecast; **INT** = E6's judgement, never a fact about Laura. Ids: R- = case register, F- = fact register, SH-/BS- = stakeholder map, M- = Phase C question, P = E1 proposal, F1-F10 = E2 fix, E3-n = E3 change, PM- = E5 cause (all under research/insight_v1/). **Every security named is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK** (brief s14).

Scripts. New: .venv/bin/python research/insight_v1/scripts/E6_final_checks.py (about 2 seconds, no network; its docstring lists every input and label). It re-uses the E4 and D6 engines (seed 20260927, 200,000 paths, the verified lognormal fit), reproduces E1 P7 row 1 and E4's default row exactly, and puts both finalist designs on one basis (sections [1]-[7], cited below as E6 [n]). Re-run on 2026-09-28: E1_split_and_slices.py, E3_client_view_numbers.py, E4_rival_numbers.py; every figure E1-E4 quote reproduces. IBTM facts were re-read on the iShares page on 2026-09-28 with fetch_text.py --grep (both quoted sentences found verbatim).

Evidence read in full (audit corrections override the text above them): the brief; CLAUDE.md; the v4 prompt; the case, IPS guide, TN guide, Competition Guide, Infographic and SMaply page; phase_E/E1-E5; phase_D/ D1-D12, D3_model_v2_results, D3_quant, D13c, the D13a summary and rules, all five audit files and every appended correction list (AX1, AX1b, AX2, AY1, AY2); phase_D/trading_now_brief.md, wins_now/securities_and_allocation_v1.md, wins_now/T2_red_team.md (it exists: saved 01:09 UTC, after E1-E3 checked), S4's summary; phase_A/ case register (summary, R-C, R-I, R-AN, X), fact register, stakeholder map (summary, T-1 to T-8, BS-01 to BS-20, section 7), guardrails (summary); phase_C/survivors.json (all 48) and the parked items M007, M008, M015, M021, M033, M035, M040, M149; the council chair memo and round-2 ruling (history only).

Terms (labels for this file only; the team chooses its own words for anything it writes, T2 finding 8):

- **Operations ladder:** ten zero-coupon U.S. Treasuries (STRIPS), each maturing on 15 November before one \$50,000 payment (2033-2042). A **rung** is one of them.
- **Building minimum:** one zero-coupon Treasury bought in January 2028 that repays the full dollar amount of the second deposit shortly before 1 January 2033. It is the least Laura will put toward the building.
- **Stock fund:** one broad world stock fund holding everything the ladder and the minimum do not need.
- **Kept money:** whatever the 2033 rule does not give to the building. It is Laura's flexibility.
- **REC / ALT:** the recommended design (this file) and the alternative (E1's 2031 lock).
- **p5 / p50 / p95:** in a simulation, the value 5% / 50% / 95% of paths fall below (p50 = median). **bp:** 0.01 percentage point.

0. Summary (read this first)

The call (INT). Keep the heart of "lock early": the first deposit buys all ten payments before any money takes stock-market risk (brief s8 items 1-8 stand). Change the facility half:

- **January 2028:** the second deposit first finishes any unbought payment. It then buys a Treasury that repays the deposit's full dollar amount just before 2033: the building minimum. Everything left goes into one world stock fund, which is left alone until 2033.
- **2031:** nothing is traded. Laura announces the minimum she already owns, up to the minimum plus half of what the stock fund is worth that day.
- **2033:** the ladder becomes the operating reserve. The building gets the minimum plus half of the fund, never more than the announced top. The rest stays hers.

Why this beats E1's 2031 lock (MODEL on JPM AC World 7.00%, bonds 5.0%, no fee; E6 [2]-[5]):

1. **Same middle, better bad cases.** Total facility-plus-flexibility money p5/p50/p95 is \$182k/\$207k/\$250k, against \$168k/\$210k/\$266k for the 2031 lock. On the same paths the 2031 lock is ahead in only 53% of cases, by a median \$1k, and it loses up to \$23k at p5. Compared at equal stock exposure (E4's two-thirds variant): p5 +\$8k, median -\$2k, p95 -\$1k. Across every six-year window of U.S. history since 1928, rescaled to JPM: worst \$169k against \$120k, and the same median (\$214k). The plan already follows this rule elsewhere (E1 P5, D2): when the middle ties, take the better bad case.
2. **The bottom co-sponsors hear is fixed three years before she speaks.** It is set by a number the case gives: her second deposit, \$150,000 (case L43-45, VRF). Under the 2031 lock it is 80% of whatever 2028-30 markets leave: p5 \$134k, falling to \$93k in a 1929-type window.
3. **Simpler and model-free.** This removes the growth-mix rule (the 50/60 debate, band and yearly reset), the top percentile q, and the question of short versus dated bonds. The 2031 top is a market value on the day, and its odds can be checked against history, not only against a model.

What it costs (stated plainly):

- **Less stock.** About 9% of her money is in stocks from 2028 to 2032, against 17% in 2028-30 then 3.5% under the 2031 lock. Six-year averages: 7.5% against 9.7%. The p95 is \$16k lower.
- **A smaller median gift but more kept money.** The median gift is \$174k (against about \$186-188k) because half the stock fund is kept: kept money median \$32k (against about \$21-24k). Because of the cap, strong markets add to her kept money, not to the announced gift.
- **WInS listing.** The WInS stand-in for the minimum (the 15-Nov-2032 note or IBTM) is UNVERIFIED as listed. The fallback is in section 5.10.

This week (tier 1, before any order; target fills by Fri Oct 2 ET, hard stop Fri Oct 9 ET): the three gate votes (lock-early; REC or ALT; WInS book (ii)R), each preceded by every student's own-words note. Then the Session Rules screenshot, the IBTM search and the Bonds drop-down check, the zero-risk note-box test, roles, and the decision and AI-use logs. The note drafts are written by students from element lists, copy-checked, and read by two outside readers (section 5.4).

The biggest risks to the semifinal are not the strategy (E2 verdict; E5 top 10, INT). They are:

- ownership and voice (PM-01);
- a format breach (PM-02, fatal);
- density (PM-03: E1 specified about 55 ideas; this spec caps the IPS at 10 blocks and ≤ 470 words);
- AI wording in permanent notes (PM-04);
- notes that contradict the IPS (PM-06).

Section 6 ranks every change by deadline and impact.

1. The strategy in one plain paragraph

(This paragraph is for the team to understand and explain. It is not wording for any deliverable. Labels: case facts VRF; dollar sizes MODEL/ASM at the 2026-09-25 curve; fund size VP via ticket v1.)

Laura adds \$300,000 at the start of 2027 and \$150,000 at the start of 2028. She has promised her residency \$50,000 at the start of each year from 2033 to 2042. The first \$300,000 buys ten U.S. government bonds, each timed to pay just before one of those payments. From that day, nothing that happens to stock markets, or to her book and speaking income, can take the payments away. If bond prices rise before January 2027 and the money falls a little short, the latest payments are bought first, and the

second deposit finishes the job before it does anything else. When the \$150,000 arrives, the plan buys one more government bond that gives back the whole \$150,000 just before the residency opens. That is the least she will put toward the building, and markets cannot shrink it. Whatever is left, about \$40,000, goes into one fund that owns about 10,000 companies worldwide, and nobody touches it until 2033. In 2031, when she starts asking partners to join, she can say exactly what she already holds for the building, and how much more to expect if stocks simply keep their value. In 2033 the ten bonds become her operating reserve and pay the residency year by year. The building gets its minimum plus half of the stock fund. The rest stays with her for whatever the project needs next.

2. The central idea and what the 50-word pitch must contain (IPS; also governs every TN note)

Central idea (content, not wording): a split of jobs.

- Money other people will rely on is bought when her money arrives and held to its date. That means the ten operating payments, and the building minimum she will announce. It never takes stock-market risk.
- Money nobody relies on is invested fully in world stocks.
- Consequences:
 - once bought, the payments depend neither on markets nor on her next contract (her second deposit is career income, case L43-45, VRF);
 - she tells partners only what she already owns, plus a stretch whose odds anyone can check.

Why a split of jobs, and not a time-worded ban (E2 F4, C1; E4 L5; AY2 rank 1; VRF): a ban such as "no stock risk until all ten payments are bought" is visibly broken by the WInS book, which buys world stocks on day one, and the notes are read before the IPS exists. A split of jobs is true of Laura's real plan and of the WInS book. The ordering ("payments first") lives inside the 2027 and 2028 rules, not in the pitch.

Pitch specification (IPS guide L31-36, R-I16/R-I17, VRF: "Maximum 50 words", "Clearly communicate the central idea ... and how it is designed to support your client's financial goals and future funding needs").

#	Element (content; the team writes the words)	Must contain	About (INT)
1	Purpose first (E3-5; E2 F3; D13c checklist "purpose comes first")	What the plan secures for the residency, named by what it secures (its ten years of operations), before any instrument or prohibition	12-15 words
2	The Laura-specific rule (M143; E4 1.2)	Each of her two deposits buys a promise when it arrives: the first, the operating years; the second, the building's minimum from her own earnings. Once bought, neither depends on markets or on her next contract	15-18 words
3	What the rest does, plus credibility (M008; E1 P1 item 3)	Only money nobody relies on is in world stocks, for a larger gift and her flexibility. She tells partners only what she owns	12-15 words

- **Hard limits:** at most 48 words by two word counters, using the higher count (E5 PM-02). No numbers; at most one if it is essential. Tense stays future or conditional: nothing is bought in 2026 (E3-5).
- **Must not contain** (E1 P1/P17; D9 M062; D13 rules; E3 4c):
 - "bought in January 2027" or "fits inside \$300,000" as a flat claim;
 - "guaranteed", "risk-free", "safe", "100%", "95%", or "certain" without its qualifier;
 - "promise" or "promised" for the stretch above the minimum;
 - jargon: barbell, LDI, duration, STRIPS, ladder, funded ratio, percentile, glide path, liability matching;
 - fund names or tickers, or a return target;
 - any quotation, the case pull quote, "Laura said";
 - identity, heritage or Taiwan as a reason;
 - book-title, "falling" or "balance" metaphors (the case's own "appropriate balance" is fine);
 - the statistics degree; "innovative"; a probability.
- **Tests:**
 - six unpaid readers can restate all three elements after one read (D12 M234 protocol);
 - no reader thinks anything is already bought;
 - no reader thinks the WInS stocks contradict the pitch (ask after showing the three notes, E2 s5).

3. IPS content map (IPS, Nov 6): what fills each required element, and the word budget

Rules for the whole IPS (IPS guide L51-52, L68-70, L123; D9 M036; E2 F1-F2; E5 PM-02, PM-03; VRF / INT):

- **Structure:** at most 12 ideas in 10 blocks, built on the four case dates (2027, 2028, 2031, 2033). Each date gets one action and one "because" tied to Laura or the case.
- **Length:** at most 470 IPS words and at most 48 pitch words, by two counters.
- **Numbers:** case facts only where a rule needs them; rule parameters in words ("half", "the full amount of her second deposit"); at most one rounded, dated computed figure. The recommended one is the forced-caution fact (block B5). No percentiles, probabilities, dollar ranges, tables or lists.
- **Words:** no framework or house names, no tickers, no Laura quotes. Every forecast carries a grade word ("we assume", "history suggests"); every price carries "at [month] 2026 prices" (D12 M231).
- **Proportion:** at most about 15% of the words describe what could go wrong (E3-11, ASM ceiling). Each gap gets one clause.

3.1 The ten blocks (content elements only; the team writes every sentence)

Block	Answers (IPS guide lines; case lines; all VRF)	Content the team's sentences must carry	Words (INT)
B1 Purpose and central idea	IPS L16 (central idea), L18 (principles), L27-28 (client at the centre), L53-57 (what and why), L45-46 (tailored)	What the plan secures for the residency (ten operating years, then the building's minimum, then growth). The split of jobs (section 2). The Laura-specific benefit: her second deposit is her own career income, so once bought the payments no longer depend on markets or on her next contract, and her own earnings are kept whole for the building	40
B2 Certainty: definition, evaluation, residual risk, two gaps	IPS L9-10 ("desired degree of funding certainty"); case L91, L97-99 (define, evaluate, assumptions); L90 (not inflation-adjusted); L146-148; R-AN10 (no currency named)	(a) Bought, not forecast: each payment is matched by a Treasury maturing shortly before its date and held to maturity. (b) Evaluated by price: on the purchase date the money covers the market price of all ten, with no model percentage. (c) Residual risk: certain in U.S. dollars, barring a U.S. Treasury default. (d) Gap (i): if rates fall before January 2027, the price can exceed the first deposit, and the earliest payment then waits for the second deposit. (e) Gap (ii): each payment is a fixed U.S. dollar amount, so rising prices and Taiwan's costs and exchange rate mean it buys less over time. This gap is named, not solved	50
B3 January 2027: buy the payments	IPS L22-23 (planned changes), L48-50 (preparing for the operating commitment); case L91-92	Buy all ten on arrival, with no waiting for better yields, because waiting is a bet on rates. If money is short, buy the longest-dated payments first. Because: the piece left waiting is the smallest and least rate-sensitive, and less would stay unbought if the next money never came. Counterpoint, stated: the waiting piece is the residency's first operating year (E3-7; E2 F7c). Any leftover waits in Treasury bills	35
B4 January 2028: finish, then the building's minimum, then stocks	IPS L22-23, L46-47, L48-50; case L43-45, L101-104, L110-112	The second deposit first completes any unbought payment. It then buys a Treasury repaying the deposit's full dollar amount just before 2033 (the building minimum; the case's "meaningful personal contribution"). Everything else goes into one broad world stock fund, left alone until 2033. One labelled stress clause, as the team's own scenario (the case says she "will", R-AN35): a smaller or later deposit follows the same order, and the minimum is smaller or later. If rates fell first and the deposit is smaller than the unbought part, all of it completes the payments and the unfunded part of the first payment is stated (P4; AY2 rank 2)	45

Block	Answers (IPS guide lines; case lines; all VRF)	Content the team's sentences must carry	Words (INT)
B5 Risk and return, goal by goal	IPS L3 (risk tolerance), L20 and L59 (balance), L46 (risk and return); case L61-74; R-AN16, R-AN17	Forced caution: at September 2026 prices the payments cost about 98% of her first deposit (the one allowed number), so 2027 holds no stocks by arithmetic (E3-2). Chosen: the minimum takes no market risk because partners will rely on it; money nobody relies on takes full stock-market risk. What stocks are for: a larger gift and more flexibility if markets are good (E3-3). The most markets can take: the stock fund, which is roughly what her deposits would have earned safely. Never the payments or the minimum (E4 L10). Portfolio fact, not a diagnosis: her income already rides on publishing and speaking, so the plan adds no themed or single-country bets (D8 M081; E3 finding 6). Returns are a range across two published forecasters, not a target (D8 M109)	45
B6 Diversification and liquidity	IPS L46 (construction and diversification), L9-10 (liquidity); SMAApply criterion "appropriate diversification"; R-AN45 ("funding purposes", VP)	Diversified by funding purpose and by date: ten dated payment bonds, one dated building bond, and one fund of about 10,000 companies worldwide. Holding U.S. Treasuries for the promises is deliberate, because the bills are in U.S. dollars and dated. Liquidity: no withdrawals before 2033 (case L63-65), and each year's payment comes from its own maturing bond (E2 F5)	25
B7 January 2031: the co-sponsor range (method only)	IPS L25-26, L64-65; case L109-120	Nothing is traded, and she speaks only once the payments and the minimum are owned. Range: bottom = the minimum she owns; top = the minimum plus half of the stock fund's value that day. Confidence both ways: below the bottom only if the U.S. Treasury defaults; the top is reached if world stocks are no lower two years later (graded "history suggests"; the numbers go in the Final Report). Words by tier: owned / expected if stocks hold their value (never "promised", E3-1). U.S. dollars are binding	40
B8 January 2033: reserve, gift, flexibility	IPS L48-50 (reserve composition over time; responsible facility contribution; flexibility); case L94-107	The reserve is the ladder, bought from 2027 and named the operating reserve on 1 Jan 2033, before the first payment (P9). It is held to maturity, and its composition changes only as bonds mature. The case allows "if at all" (L96). The gift is the minimum plus half the stock fund, never above the announced top. Everything else is kept, moved to U.S. dollar Treasury bills, and hers to use "as the project develops" (L102-103). There is no conversion to New Taiwan dollars or currency hedge before the facility decision. Taiwan building costs are expected to rise, so a fixed dollar gift buys less (L146-147)	45
B9 Governance and costs	IPS L5, L61-62, L73-74; Guide p.5 L156-159; case L101, L110	The team's written rules run the portfolio. Laura makes the two calls the case gives her, within the rules: what she announces in 2031 (never more than the rule allows) and the 2033 amount and use of kept money. Rules change only if her circumstances change, never because markets moved. The stock fund is never sold to protect a gain or to wait for a recovery. There is no rebalancing between the three parts (CFA 2010 element 4c, VP via D8). The bought Treasuries carry no ongoing charge, and any cost is paid from the stock fund (E3-4)	35
B10 The WInS sentence	IPS L75 ("Your portfolio ... should reflect the strategy"); D10 1.5	WInS shows her plan on the day both deposits are in, scaled to \$300,000. The Treasury funds stand in for her bonds: they move like them but do not mature on her dates. In January 2027 almost all of her real first deposit buys the payments (ticket v1 s1; T2 16a)	25
		Content words, before connecting words	about 385
		Connecting words (INT allowance)	about 70
		Target total (hard cap 500; plan ≤470)	about 455

Paragraph plan (at most 7 paragraphs, D10 2.3):

1. B1 and B2;
2. B3;
3. B4;
4. B5 and B6;
5. B7;
6. B8;
7. B9 and B10.

The IPS guide says "You do not need to address these points separately" (L66, VRF), so no headings inside the IPS.

3.2 Coverage check (every IPS-guide and case "should" item has a home; tick against the draft)

Requirement (VRF)	Block
Goals, time horizons, liquidity needs, desired degree of funding certainty, need for flexibility (IPS L9-10)	B1 (goals; horizons via the four dates), B6 (liquidity), B2 (certainty), B8 (flexibility)
Central idea (L16); principles (L18)	B1; B5, B9
Balance of growth, risk, liquidity, funding reliability, flexibility (L20, L59)	B5, B6, B2, B8
How allocation and composition change as needs approach and payments are made (L22-23; L46-47 "planned changes")	B3, B4, B7, B8 (the four dates; rungs mature)
Protect the operating commitment while preserving flexibility for a responsible facility contribution (L25-26, L64-65)	B7, B8
Risk tolerance (L3); risk and return (L46)	B5
Portfolio construction and diversification (L46)	B6
Preparing for the commitment; reserve composition over time; responsible facility contribution; flexibility (L48-50)	B3-B4; B8; B8; B8
Framework, not individual investments or detailed calculations (L51-52)	Number policy above; no tickers
What, why, how guided as needs change (L53-62)	B1, B9
Official record; no revision after the deadline; portfolio reflects the IPS (L73-75)	B9, B10
Define high degree of funding certainty; how evaluated; assumptions (case L97-99)	B2
Assumptions on investment performance, timing of cash flows, outside funding, inflation's effect on projections and facility costs, flexibility (case L146-148)	B5 (graded returns); B3-B4 (timing); B1-B2 (no outside money for the payments); B2 and B8 (inflation, building costs, currency); B8 (flexibility)
Favorable and unfavorable outcomes; the range protects the payments (case L118-120)	B7 (method only; the numbers are in the Final Report)
Not expected in the IPS: final reserve calculation, projections, final range, co-sponsor draft (L68-69)	Excluded

3.3 Format checks (non-compliance = "will not be considered for semifinal selection", IPS guide L82-83, VRF)

- [] **Title page (page 1 only):**
 - the official team name, exactly as on the submitted roster;
 - members as "First Name, Last Initial", exactly as on the roster;
 - the **WInS username**, not the team name (IPS guide L84-100, VRF);
 - these are copied from the roster screenshot and the WInS login, character for character (PM-18).
- [] **Pages 2-3:** the headings "Investment Strategy Elevator Pitch" and "Investment Policy Statement", as in the sample. Pitch ≤48 words and IPS ≤470 words by two counters; the higher count wins.
- [] **Type and layout:**
 - Times New Roman 12 everywhere, headings included;
 - double-spaced using Word's "Double", on U.S. Letter with 1-inch margins;
 - paragraph spacing 0, no blank lines, at most 7 paragraphs.
- [] **No banned items:** no tables, bullet lists, charts, images, links or URLs, footnotes, citations, or quotation marks around anyone's words (L123).
- [] **PDF checks:**
 - export the PDF and open the exported file itself;
 - exactly 3 pages; the text ends on page 3 with at least 2 spare lines;
 - the font list shows Times New Roman; file ≤5 MB;
 - search it for "http", "www" and brackets.

- [] **Line spacing:** "Exactly 24 pt" (Wharton's sample pitch) is a documented last resort only, never the plan (D9; AY2; E5 C4).
- [] **Dates** (AEDT; E5 [1]): words frozen Tue Nov 3; two-student PDF check Wed Nov 4; submit Thu Nov 5. The deadline is Sat Nov 7 09:00 AEDT (Fri Nov 6, 5 p.m. ET).

4. Decision rules fixed before Nov 6 (IPS; the Final Report later shows each one "fired as written")

Every parameter is fixed here, because a rule whose parameter is chosen after Nov 6 is a redesign (D8 M006; IPS guide L73-74, VRF). "Record" is the decision-log entry that makes the later check possible.

Rule	Trigger	Action	Record	Anchor (VRF unless marked)
R1 Buying the payments (the rates-fall rule)	The first deposit arrives, 1 Jan 2027	(1) Buy the ten-rung ladder at that day's prices, all at once: no staging, no yield trigger. (2) If the money is short, buy the longest-dated payments first. The unbought part is always the earliest payment, and it is the first use of the next money. (3) If money is left over, it waits in Treasury bills until 2028. (4) Headline, conditional: bought in January 2027 if prices allow, otherwise when the second deposit arrives. Never "by 1 January 2028 at the latest" (AX1b item 3)	Purchase-day ladder cost against cash; which payments were bought; the gap or leftover	Case L91-92, L43-45. Numbers: the real Nov-15 ladder costs \$294,387 on the 2026-09-25 curve, headroom \$5,613 (19bp) (VRF inputs, ASM method). About 1 in 3 rate paths cost more than \$300k (MODEL 30-34%; history: ≥ 19 bp falls in 32-36% of 68-day windows since 1962/1990, VP data, derived). Staging gives no reliable gain (+0.2 to -0.5 points) and a 27-35% chance of needing the deposit (MODEL). Longest-first carries a +5% top-up price rise against +15% for nearest-first, and leaves \$31.4k of face unbought against \$47.7k if the deposit never came, at -100bp (MODEL; D1, AX1b)
R2 The second deposit	The deposit arrives (1 Jan 2028 per the case; each instalment if late)	In this order: (1) complete any unbought payment; (2) buy a zero-coupon Treasury maturing before 1 Jan 2033 that repays the deposit's dollar amount (the building minimum), or all that is left if less; (3) put everything else, including the 2027 leftover, into one broad world stock fund. Smaller or late: same order; the minimum is smaller or later. No building minimum and no stock fund exist until all ten payments are bought. Joint tail (the team's own stress case): if rates fell before January 2027 and the deposit is smaller than the unbought part, all of it goes to the payments, there is no minimum and no stock fund, and the unfunded part of the first payment is stated	Deposit amount and date; top-up cost; the minimum's face and price; the stock fund's starting value	Case L43-45 ("will contribute"), L110-112, L101-104. MODEL: growth money \$158k; the minimum costs \$118k at the 5-year 4.98% (ASM that it holds); stock fund \$40k = 8.7% of all money. A -50bp fall first leaves the fund at about \$20-23k; -100bp at about \$1-7k; -150bp leaves no fund and a minimum of about \$127-137k (E6 [8], which corrects E4 [7]'s \$31k / \$15k / \$147k; the range depends on whether the 5-year rate also falls; MODEL). Joint tail with no deposit: \$11,957 (-50bp) or \$31,413 (-100bp) of the 2033 payment unfunded (D1/D3). A deposit of about \$43k or more covers a 150bp fall, about \$51k the worst 98-day fall since 1990 if rates do not fall further (AX1 V4). \$75k deposit: minimum \$75k, total \$94k/\$109k/\$135k (E4 [4])
R3 The operating reserve	1 Jan 2033, before the first payment or any facility contribution	The ladder is named the operating reserve and pays the first \$50,000 at once (the 2032 rung matured on 15 Nov and waited in bills). It is held to maturity. Each rung matures about 47 days before its payment and waits in Treasury bills. Nothing is sold early, and there is no rebalancing between the reserve, the minimum and the stock fund. The reserve empties itself by 1 Jan 2042	The reserve's market value, cost and face on 1 Jan 2033 (the Final Report reports market value first)	Case L94-97 ("if at all", L96); IPS L48-50; P9. About \$395k market value on today's forwards (VRF inputs, ASM), cost about \$292-294k, face \$500k

Rule	Trigger	Action	Record	Anchor (VRF unless marked)
R4 The 2031 co-sponsor range (method; no figures in the IPS)	1 Jan 2031, when she begins approaching co-sponsors (case L109)	Condition: all ten payments and the minimum are owned; otherwise no range is stated (only "aspiration"). Nothing is traded. Bottom = the minimum she owns. Top = the minimum plus half of the stock fund's market value on 1 Jan 2031, set once and not reset later. Confidence, both sides: below the bottom only on a U.S. Treasury default (and only if no fee is ever charged to it, R8); the top is reached if the stock fund is no lower on 1 Jan 2033 than on 1 Jan 2031, with the history and the model named in the Final Report. Words: owned / expected if stocks hold their value / kept; never "promised" or "pledged" for the stretch (E3-1; BS-05). U.S. dollars are binding; any New Taiwan dollar figure is dated and illustrative	The stock fund's value on 1 Jan 2031; the stated range	Case L110-120; R-AN9; AX1 C2-C3 (a capped range is reported with the chance of reaching the top, never "100% within"). MODEL at the median: bottom \$150k, top \$175k (width 1.17x); 86% of the median gift already owned when she speaks; the top is reached in 73% of model paths (67% on the Vanguard-type input). U.S. stocks' two-year total return was zero or positive in 84% of 97 overlapping periods 1928-2025 (89% since 1950) (VP data, derived; E4 [5])
R5 The 2033 facility contribution and flexibility	1 Jan 2033, after R3	(1) Gift = the minimum plus half of the stock fund's value, never more than the announced top. (2) Kept money = everything else (the other half, plus any growth above the top). It moves the same day to U.S. dollar Treasury bills and is Laura's to use "as the project develops"; she may choose to add to the gift, which is her decision and not counted by co-sponsors (AX1 C15). (3) No conversion to New Taiwan dollars and no currency hedge before the facility decision: a forward contract is a derivative, banned in WInS and, on the safe reading of "Investments permitted (for BOTH contributions)", in the plan (R-AN15); early conversion would also be a currency bet	Gift, kept money and the date of the facility decision	Case L101-107 (she decides; flexibility; no contingency fund required), L118-119. MODEL: gift \$165k/\$174k/\$188k; kept \$16k/\$32k/\$66k, about 15% of the post-reserve money at the median, the size of three or four years of Taiwan building-cost drift (F-508 3.54%/yr since 2021; ASM sanity check, not a contingency fund, D5)
R6 The stock fund's conduct	Every day, 2028-2032	One broad world stock fund. It is never rebalanced against the Treasuries and never sold before 1 Jan 2033, whether to protect a gain or to "wait for a recovery". A fall shrinks only the stretch above the minimum, never the payments or the minimum	Yearly check that holdings match R2-R6	Case L69-74, L61-65. MODEL: the fund falls in about a quarter of years; in a bottom-decile 2031-32 the gift is \$169k at the median and stays at or above the announced bottom in 100% of paths (E6 [3])
R7 Governance	The four case dates, plus a yearly check	Who decides: the students' written rules. The advisor is administrator only; never write "our portfolio manager approves" (R-W19, VP). Laura makes the two calls the case gives her, within the rules (L101, L110). What may change a rule: a change in Laura's circumstances (for example, a different deposit amount or date), never a market move. Worded so that it cannot read as permission to revise after Nov 6 (AY2 D8 C12)	Dated decision log: what fired, when, the numbers	IPS L73-74; Guide p.5 L156-159; CFA 2010 elements 2a-2b (VP via D8/D10)

Rule	Trigger	Action	Record	Anchor (VRF unless marked)
R8 Costs	Any fee or trading cost	The bought Treasuries (the ladder and the minimum) are held to maturity with no ongoing charge. Every cost is paid from the stock fund, never from the payments or the minimum. The Final Report states the fee assumption, its base and its rate, and shows the effect	The fee assumption	Case L43 ("with an asset management firm"); Rules page to the Asset Manager Code F.4.d (VP via D8; "should provide"). MODEL (E6 [6]): a 1% fee charged on the stock fund only costs about \$3k of the median total; the same rate charged on all assets but paid from the fund costs about \$31k. It could still be paid from the fund in every model path (thin-tailed model). The fee base is the assumption that matters
R9 The certainty definition and the stated assumptions	Fixed in the IPS	Block B2's five parts. Assumptions: returns as a range across two published forecasters (graded); flows at the start of each year (case L59); no outside money for the payments (L91-92); payments fixed in dollars (L90), Taiwan building costs expected to rise (L146-147), U.S. dollars binding; flexibility kept by R5	None (the IPS is the record)	Case L97-99, L146-148
R10 WInS conduct (consistency with the IPS)	Each trading day to Nov 6	The book is (ii)R (section 5). The operations hedge is never sold after a rate rise or trimmed after a fall. It is re-mixed IEF-against-TLH only if its rate sensitivity drifts more than a quarter-year from its start (ticket v1 s8). The minimum and the stock fund are held. No day trades; no "tidy-up" trades after the words freeze (Tue Nov 3)	Every trade, and every "decided not to trade" moment	IPS L75; User Guide "Day Trading: This is not permitted" (VP via A4)

5. The WInS book and the three Trading Notes plan (tier 1: WInS now and TN, Oct 23)

5.1 Which book: (ii)R = Laura's plan on 2 January 2028, scaled to \$300,000

Decision (INT; team vote at the gate): (ii)R.

• **For:**

- It is the first day all three jobs exist, so the notes show three distinct decisions with three jobs: operations, the building minimum, growth (TN guide L4-6, L11, VRF).
- It shows "funding purposes" diversification in the holdings (R-W71, VP).
- It survives one trading session as written (T2 one-session test), and it copes with a position cap better than (iii) (ticket v1 s3).
- Its building-minimum note is the most Laura-specific holding in any book (E4 L6).

- **Against (iii):** with no listed dated rung it gives two decisions plus a T-bill leftover note that earns about \$18 against a \$25 commission, which reads as filler (T2 finding 4; E2 F4, ASM).

• **Conditions:**

1. the central idea is worded as a split of jobs (section 2);
2. each operations note and stock note carries its scaling marker inside the note ("after both deposits" / "after her 2028 deposit") (ticket C2);
3. block B10 is in the IPS;
4. no note calls the WInS holdings "Laura's portfolio" (Asset Manager Code F.2, VP via D10);
5. the two-reader check passes (5.4).

5.2 Orders (illustration at the 2026-09-25 closes; recompute every share count from the last close, T2 finding 1)

Order	Holding	Weight	~Shares	Job / Guide role word	Primary, then alternates (all PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK)	2025-26 list (history only, VRF)
1	IEF (7-10y Treasuries)	23.5%	~783	Operations: future funding	IEF; VGIT, then SPTI (ticket v1 s12)	IEF yes
1	TLH (10-20y Treasuries)	42.5%	~1,365	Operations: future funding (IEF and TLH are one duration-weighted decision)	TLH; IEF + TLT at duration weights (40.9/25.1) if TLH is missing	TLH no, TLT yes
2	Building minimum: the 4.125% Treasury note of 15-Nov-2032 (CUSIP 91282CFV8), or IBTM (iShares iBonds Dec 2032 Term Treasury ETF)	24.3%	note ≈ \$75k face; IBTM ~3,336	Building minimum: risk management (it takes the minimum out of market risk) or future funding; one word, recorded	1) the 2032 note (\$10 trade); 2) IBTM (\$25); 3) if neither is listed, an undated ~5-year stand-in (VGIT, then SPTI), noted as "moves like, not dated"	IBTM no; VGIT no
3	VT (world stocks)	8.7%	~163	Stock fund: growth	VT; VTI + VXUS (62/38); ITOT + IXUS	VT, VTI, VXUS yes
	Cash float	~1% (~\$2,990 after 4 × \$25)		Liquidity	Keep at least \$1,000	

- **Weights:** ladder 65.9%, minimum 25.3% and stock fund 8.7% of the plan's \$464k on 2 Jan 2028 (MODEL arithmetic, E6 [7]). The hedge mix IEF 35.6% / TLH 64.4% of the hedge gives "about 10 years" of rate sensitivity, like the payments' (issuer durations IEF 6.86y, TLH 11.58y on 9/25, VP; 9.90y liability, VRF).
- **Order of entry:** IEF and TLH first, then the minimum, then VT, all in one session. The time stamps then show promise-first (D7). Orders placed while the market is closed fill at the next open (R-W62, VP), so size them from the last close.
- **Instrument facts (VP):**
 - IBTM: NAV \$21.83 on 9/25; expense 0.07%; effective duration 5.05y; 15 Treasuries maturing Apr-Sep 2032; about \$555m of net assets; about 216k shares a day. Both sentences below were re-read on ishares.com on 2026-09-28: the funds "will terminate on or about October or December 15 of the year in each Fund's name", and they "do not seek to return any predetermined amount".
 - VT: 10,088 stocks, 37.7% outside the U.S. (8/31); expense 0.06% (ticket v1 s12).
 - The 2032 note's model price is 95.27 clean plus about 1.49 accrued per 100 on 9/25 (ASM, D1). WInS shows its own price.

5.3 Position-limit branches (read both Session Rules lines before any order; ticket v1 s3)

Single-security limit shown	Book (ii)R
None, or 43% or more	As in 5.2
25-42%	IEF 23.5 / TLH at the cap minus 1 / SPTL the rest of the hedge. The minimum is held one point under the cap if the cap is under 25.3%. At a 25% cap: IEF 23.5 / TLH 24 / SPTL 18.5 / IBTM 24 / VT 8.7 / cash about 1.3% (5 ETF trades, \$125) (E6 [7])
Under 25%	Hedge as ticket v1 s3 (IEF, TLH, SPTL at the cap minus 1; VGIT then VGLT take the rest). Minimum: IBTM at the cap minus 1, the rest in VGIT. If 5 or more hedge funds would be needed (a cap of about 17% or less), stop and ask Wharton (Contact Us) before trading
A separate, higher bond limit is shown	The 2032 note may carry the whole minimum

Never add to a holding within one point of the cap. A 20-24% stock fall, or a 10-12% fall with a 50bp rally, can lift TLH above a 25% cap (T2 [5], ASM that WInS re-checks the limit).

5.4 Gate: tick every box before the first order (owners are roles; E5 section 5 calendar)

- [] **(a) Votes**, each recorded with first names, date and reason, **after** every student has written 3-5 lines in their own words with no AI file open (R-W51; PM-01):

- lock-early;
- REC or ALT (section 7, D2);
- book (ii)R or (iii).
- [] **(b) Session Rules screenshot:** single-security limit, any bond or type limit, the day-trading tooltip, trades allowed, commissions. Also read the logged-in Trading Details page. If it shows an activity minimum or a cap on a whole security type, stop and ask Wharton; never add trades to meet a count (T2 finding 13; PM-15).
- [] **(c) Listings:** search IEF, TLH, VT, IBTM (and SPTL, VGIT). Check the Bonds drop-down for every U.S. Treasury maturing Jul-Dec 2032, recording coupon, price and accrued interest; search by maturity, not by a built-up symbol (AX1b item 10).
- [] **(d) Note-box test at zero risk:** type about 450 characters on the order-review screen, see whether they are cut, clear them, and do **not** press Confirm. Until a limit is seen or ruled out, notes are drafted to 300 characters (UNVERIFIED for WInS; a sister product caps at 300, VP; Wharton's own example is 413 characters, VRF).
- [] **(e) Notes** are drafted offline by students from the 5.6 elements, with no research file open (PM-04).
 - A second student restates each note's role in one sentence.
 - The second student then searches every five-word run of the draft in research/; any hit is rewritten.
 - Two readers outside the team read all three drafts and say what Laura's real money holds in January 2027. If either says "stocks" or "a building bond", fix the drafts (E3-10).
- [] **(f) Roles:**
 - one trader plus a backup (only they place orders);
 - note writers; a second reader; a log keeper; a data owner;
 - every student drafts at least one note or reflection and one IPS block (PM-05; R-W38, VP).
- [] **(g) Cash:** at least \$1,000 left after orders at the next open's prices. No sells on the day anything is bought.
- [] **(h) Labels:** the team names its own labels for the three parts in a meeting, and logs that the proposals came from an AI source (E2 F10; T2 finding 8).

5.5 The three notes: which trades, and supported / tested / refined (a menu, not a quota: TN guide L12, L29-30, VRF)

Slot	Trade	Shows	The reflection's one extra job (E2 F9)
A	TLH (operations; IEF was bought with it as one decision)	Supported and tested	Tested: over the pre-registered window (fill-date close to the Oct 20 close), exactly three numbers, each labelled as the team's own calculation from the WInS screen and the treasury.gov curve (T2 finding 7). They are: the 10-year rate move in bp; the hedge's % change; the payments' % change. Then what the rule said to do (hold). A 10bp-or-larger move has about a 52% chance for an Oct 2 fill and about 40% for an Oct 9 fill (T2 [4], ASM). Report small moves and poor tracking as they are (AY2 D7 C2)
B	The building minimum (2032 note, IBTM or the stand-in)	Refined , only if the decision log shows, dated before this order, the change from "a floor bought in 2031" (the council plan in CLAUDE.md) to "the minimum bought when the 2028 money arrives", with the reason in the students' words. Otherwise supported	Refined: what changed, and the research that changed it (tested at equal stock risk: about the same middle, better bad cases; at most one number). Otherwise: why a dated holding (an amount on a date, not only a price that moves)
C	VT (stock fund)	Supported (it carries out the plan as refined)	The true refinement story (E2 F8). The first plan (growth-first, about 75% stocks) was tested and dropped because it could miss a payment (about 3 in 100 modelled futures; about 4 in 10 if the 2028 deposit never came; F-401, F-404, VRF, MODEL). Now stock risk sits only where nobody relies on the money, and all of that money is in stocks: the "why so little stock" answer, stated by goal (ticket s6; E1 P6). One number at most

5.6 What each WInS note must contain (elements in noun phrases, not wording; ≤300 characters until the box test)

The Guide wants a note to capture "the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role" (Guide p.3 L84-86, VRF). The trade record already shows the ticker, side and quantity.

Every note has four cores.

- **C1:** one Guide role word, plus the team's own label.
- **C2:** Laura's dated need, including the scaling marker.
- **C3:** one dated, checkable fact, with its source named in words.
- **C4:** the rule or risk. If space runs short, C4 and the label move to the reflection (AY1 D12 C1).

Note	C2 need (with marker)	C3 one fact (pick one)	C4 rule / risk	Never say
A operations	The ten \$50,000 payments 2033-2042; her operations money after both deposits (about two-thirds of her total money; never of the payments, E3-10)	The pair's rate sensitivity is about 10 years, set to the payments' (issuer durations, dated; the team's calculation); or TLH's own effective duration (iShares, dated)	Never sold after a rate rise or trimmed after a fall; re-mixed only if the sensitivity drifts more than a quarter-year	match, matched, "mature in her payment years" (74.7% of TLH matures after 2042, VRF), stable, safe, reduce volatility, guaranteed, locked, operating reserve, "bought in January 2027", "fits in \$300k"
B building minimum	The least she will put toward the building in 2033, sized to repay her second deposit; "after her 2028 deposit"	The note: matures 15 Nov 2032, 47 days before 1 Jan 2033 (Treasury, dated). Or IBTM: holds Treasuries maturing Apr-Sep 2032 and ends around December 2032 (iShares, dated)	Held to its end; never sold because rates moved	"repays \$X" for IBTM; guaranteed; risk-free; "locked 5%"; any 2031 range or top; promised; pledged
C stock fund	The building's stretch and her flexibility in 2033, from money nobody else relies on; "after her 2028 deposit"	All of the plan's stock money, about 9% of her money after both deposits (the team's calculation); or VT's stock count and share outside the U.S. (Vanguard, dated)	Held until 2033, never sold for market reasons. A fall shrinks only the stretch above the minimum, never the payments or the minimum	a return forecast, overvalued, CAPE, "halves AI exposure", "known/locked 5.2%", "Laura likes risk", the statistics degree, promised

5.7 What each ≤100-word reflection must contain (written by students after the trades, without AI; TN guide L46-52, VRF)

- [] The three official answers: **why**; **how it aligned** with the strategy; **how it served** her goals, funding needs or risk. For A, the alignment answer includes the scaling in plain words: WInS shows her plan after both deposits, scaled; the funds stand in for the bonds she buys.
- [] Which of supported / tested / refined it shows, true to what happened (5.5).
- [] Exactly one extra job (5.5).
- [] ≤95 words by two counters. At most one number, except the tested reflection, which carries exactly three. Every number is dated and graded in words.
- [] One case-stated trait that passes the swap test (D6 M125):
 - A: fixed payments only her portfolio may fund (L91-92);
 - B: a figure partners will rely on (L114-116);
 - C: "thoughtful risks" within "an appropriate balance" (L69-74).
- [] **Not in any reflection:**
 - Laura quotes; the pull quote as hers; her career story or regret line as a reason;
 - behavioural-trap vocabulary ("loss tolerance", "break-even", "house money", "composure");
 - the statistics degree;
 - reserve size, a facility range or a 2031 figure;
 - WInS profit or rank.
- [] The same rule names the IPS will use (a decision-log column, D9 M212).

5.8 Data capture (the only thing that must be done on the day)

- Screenshot every WInS position on the fill date and at the Oct 20 close (Wed Oct 21, 07:00 AEDT).
- Curves can be pulled later from treasury.gov. Value the payments with `A2_curve_recheck.py` or `D9_numbers.py` [3], never `official_curve_pv.py`, which is fixed to 2026-09-25 (AY2 D7 C5).

- Check the IEF and TLH ex-dividend dates in the window; a distribution of about 0.4% is large next to a signal of about 1%.

5.9 Never (WInS)

- Never place an order before the gate, and never let the advisor place a trade ("Students should place trades on WInS, NOT the advisor", VP).
- Never buy and sell the same security on one U.S. trading day, or "fix" an error the same day ("run with it", VP).
- Never lengthen the hedge to bet on rates (EDV, ZROZ, GOVZ, TLT alone).
- Never buy leveraged, inverse, crypto-linked, thematic or single-country funds, ETNs, or stocks under \$5. No Taiwan or AI tilt.
- Never put a reserve size, facility range or 2031 figure in a note.
- Never trade for rank, and never tidy up after Tue Nov 3.

5.10 Fallbacks (decided in advance)

- **No dated 2032 instrument is listed.**
 - Use the undated stand-in (VGIT, then SPTI, about 4.8-4.9 years).
 - Note B says it moves like a Treasury maturing in late 2032; it never claims a date.
 - The real plan is unchanged: the minimum is a real Treasury bought in January 2028.
- **The two-reader check fails twice after redrafting.** Choose (iii), Laura's literal January-2027 book.
 - Orders: IEF 34.9 / TLH 63.1, re-priced on the trade date, plus the leftover in SGOV or BIL only if it is at least \$1,000 at trade-date prices. Otherwise a listed dated payment rung; otherwise revert to (ii)R (ticket v1 s2; T2 finding 1).
 - One reflection answers "why so little stock" with the forced-versus-chosen content.
- **The team votes ALT.** Use ticket v1 (ii) at 50/50: IEF 23.5 / TLH 42.5 / VT 17.0 / VGSH 16.0 / cash 1. Note C becomes the VGSH note, never "the floor" or "already owned". Record the growth-money bond choice (short or dated, AX1 C18) before the VGSH order.

6. Changes against the current strategy (brief s7 and CLAUDE.md), ranked; and what was rejected

6.1 Accepted changes, ranked by deadline (TN first), then by impact on reaching the semifinals (impact grades INT)

Rank	Change	Serves / by	Impact	Evidence and numbers	Side effects
1	Gate votes this week, each after own-words notes; roles; decision log (rule-name and evidence-grade columns); AI-use log backfilled	WInS/TN; Tue Sep 29-Thu Oct 1 AEST	High	CLAUDE.md "The team has not read the memos yet" (VRF); PM-01, PM-05, PM-08 (E5, INT); R-W46, R-W51 (VP)	Hours this week, during NSW school holidays if they apply (UNVERIFIED for the team, E5)
2	Central idea worded as a split of jobs; scaling markers inside notes; the two-reader check	TN and IPS; before the first note	High	E2 C1 and F4; AY2 rank 1; E3-10 (VRF/INT)	15-20 characters per note
3	WInS book (ii)R: operations hedge, dated building minimum, stock fund (replaces the "~65% Treasuries / ~35% equity" mirror, which was hedge/growth, not bonds/stocks, F-409)	WInS/TN; by Fri Oct 2 ET	High	Section 5.1; E4 L6; T2 one-session test	VT shows 8.7%, not 17%; IBTM or the note is unconfirmed
4	Notes drafted by students from elements; ≤300 characters; zero-risk box test; five-word copy check; the team's own labels	TN; every order	High (PM-04 is severe)	T2 findings 6 and 8; AY1 D12 C1-C2; R-W26/R-W49 (VP)	None

Rank	Change	Serves / by	Impact	Evidence and numbers	Side effects
5	Three-note plan: A tested, B refined only if logged, C carries the growth-first story; one extra job per reflection	TN; reflections Oct 21-22 AEDT	Medium-high (Articulation)	E2 F8-F9 (INT); D7 with AY2	None
6	One vocabulary: "certainty" for the payments, "confident" for the range, "credibility" for her standing; "moves like / stands in for"; never "promised" for the stretch	TN and IPS	Medium	D9 M062 with AY2 C2; E3-1	None
7	Data capture on the fill date and at the Oct 20 close	TN	Medium	AY2 D7 C5	Ten minutes
8	Format stance: Letter at Word "Double"; $\leq 470/\leq 48$ words by two counters; PDF checks; title page copied from the roster	IPS; every draft from Oct 26	Fatal if missed	IPS guide L82-123 (VRF); D9/AX2 C3 (0-2 spare lines at 27.6 pt); E5 PM-02, PM-18	Fewer words than E1 budgeted
9	At most 12 ideas in 10 blocks on four dated decisions (replaces "Rule 1-4 + G" and E1's ~55 ideas)	IPS; idea list Sat Oct 24	High	E2 F1-F2; PM-03; last year's lesson (CLAUDE.md, VRF)	Some true points move to the Final Report
10	Pitch: purpose first, three elements, no numbers, no date claims	IPS	High	E2 F3; E3-5; D7 M011 (the date claim fails in about 1 path in 3, MODEL)	Loses the vivid date
11	Architecture change: the building minimum (= the second deposit's dollar amount) is bought in January 2028; the rest is one world stock fund held to 2033; nothing is traded in 2031. This replaces "growth sleeve ~60% equity + 80% of the sleeve locked in a 2-year Treasury in 2031"	IPS (and WInS now)	High (strategy and credibility)	E6 [2]: total p5/p50/p95 \$182k/\$207k/\$250k vs \$168k/\$210k/\$266k; the bottom is fixed at \$150k vs \$134k/\$167k/\$211k. On the same paths ALT minus REC: p5 -\$23k, p50 +\$1k, p95 +\$32k. History (rescaled): worst \$169k vs \$120k; p10 \$185k vs \$174k; median equal (\$214k). No path ends below the money put in (\$157,736), against 1.7% (MODEL). AX1 C6 withdrew the old "\$101k vs \$165k" rejection	Stock share about 9% in every year 2028-32 (average 7.5% vs 9.7%); p95 -\$16k; median gift \$174k vs \$186-188k; kept money \$32k vs \$21-24k; the gift is capped at the top
12	2031 range: owned minimum up to minimum + half the fund's value that day, capped; the chance of reaching the top is stated (history and model named in the Final Report). Replaces "upside with a stated model probability" and E1's uncapped p90 top	IPS	High (Creativity and Presentation; Client Knowledge)	E6 [2]; E4 [5]; AX1 C2-C3; D5 finding 5 met by reporting P(top reached), never "100% within"	Strong markets raise kept money, not the gift; she may add in 2033
13	Certainty definition with two gaps, one clause each (the January 2027 price; nominal dollars against Taiwan costs and currency)	IPS	Medium-high	Brief s16 (binding); E1 P2; AY2 D9 C2-C3; F-114: \$50k is worth \$42,063 (2033) and \$33,681 (2042) in 2026 dollars at 2.5% (ASM)	About 50 words
14	Risk by goal: forced vs chosen, what stocks are for, the most markets can take; no CFA labels, trap words or diagnosis; the statistics degree used zero times	IPS; TN reflections	Medium-high (PM-07)	E3 findings 2, 3 and 6; E3-9; D3 AX1 C19; payments cost 97.4-98.1% of the first deposit (E3 [1])	None
15	2027 rule: buy on arrival, longest-first with its reason and the first-year counterpoint; no staging; leftover in bills	IPS	Medium	D1 with AX1b; E3-7; E2 F7c	One clause

Rank	Change	Serves / by	Impact	Evidence and numbers	Side effects
16	2028 deposit rule: completion first; one labelled stress clause; joint tail named once (replaces "only the facility shrinks")	IPS	Medium	AY2 D8 C1; P4; E2 C6 (the \$150k doubted once, R-AN35)	None
17	Diversification (purpose and date) and liquidity clause	IPS	Medium	IPS L46 (VRF); E2 F5; M035	About 25 words
18	2033 order of use: reserve composition over time; gift $\leq$ top; kept money in U.S. dollar bills; U.S. dollars binding; no New Taiwan dollar hedge; building-cost drift named	IPS	Medium	P8-P9; AX2 D4 C1, C3; IPS L48-50 (VRF)	None
19	Governance: what stays hers; never because markets moved; reviews on the four dates; no rebalancing between parts	IPS	Medium	P10; E3-8; CFA 2010 2a-2b, 4c (VP via D8/D10)	None
20	Cost principle kept as one clause (extended to protect the minimum)	IPS	Low-medium	E3-4; E6 [6]; BS-01	About 15 words
21	Paraphrase test (6 unpaid readers, two rounds) and the explain-back check	IPS	Medium-high	D12 M234 (digital.gov method, VP); E2 F10; PM-01, PM-03	About 2 student-hours per round

6.2 Rejected or not chosen (one line each, with the reason)

Proposal	Why not (evidence)
ALT: E1's 2031 lock (50/50 growth money reset yearly, then 80% bought in 2031)	On the same basis it is REC plus a bet: p5 -\$23k, median +\$1k, p95 +\$32k. Its bottom depends on 2028-30 markets (p5 \$134k; \$93k in a 1929 window). It needs a growth-mix rule, a band and a percentile. Its best points (more stock in 2028-30, a higher median bottom) are real; it stays as the documented alternative (section 7, D2)
Growth sleeve ~60% equity (CLAUDE.md call 1)	Superseded by the architecture. The shares D6's rule picks (55-70%) depend on the fee and the forecaster (AY1 C1); the median ties
50/50 vs 60/40 debate (E1 P5, D2, D6)	Moot under REC. For ALT, 50/50, owned as a choice of spread, with no fee-dependent claim (T2 finding 2)
E1's slices 80/10/10 or 70/15/15; top percentile q of 80/85/90	Moot: REC has no lock share and no model percentile
Uncapped gift with a model top (D5, E1 P7)	The top would rest on a model that has too few very bad years (AX1 C12): "80%" tops landed within the range 62-93% of the time across models and history. A capped market-value top needs no model
"Give 90%, capped" (D3 M039)	It promises less than is bought (AX1 C3)
A minimum sized as a share (two-thirds or 0.8) instead of the deposit amount	A free parameter. The deposit amount is case-anchored (E4 C2). Two-thirds is kept only as the equal-risk sensitivity (bottom \$134k)
$s = \frac{3}{4}$ or $s = 1$	$s = 1$ leaves nothing kept whenever stocks fall, failing case test 5 (L145; E4 2.1). $s = \frac{3}{4}$ matches E1's median gift but its kept money at p5 is only \$8k (E4 [2], MODEL)
Ratchet (lock in thirds, 2029-31)	A second rule for \$2-6k (AX1 C11)
Staged purchase or a yield trigger in 2027	No reliable gain; a 27-35% chance of needing the 2028 deposit (D1 [7], AX1b item 11)
Nearest-dated payments first	More top-up price risk (+15% vs +5%) and more face unbought if the deposit never comes (AX1b item 23)
Rolling the 2037-42 payments short ("tiering"); partial locks of the payments	+\$20k per -100bp (F-108); partial locks fail in 13-26% of paths without the deposit (D3 M243)
TIPS for the payments or the minimum	The promise is fixed in nominal dollars; TIPS track U.S. prices, not Taiwan's costs or the exchange rate (brief s8 item 10; E4 1.4)
Converting to, or hedging into, New Taiwan dollars in 2031	A forward is a derivative (banned in WInS; the safe reading for the plan, R-AN15); about 5.9% cost at today's rate gap (ASM; the Taiwan 2-year yield is SNIP)

Proposal	Why not (evidence)
EWT, AI, thematic or single-country tilts	Tokenism and concentration (D8 M081; D13c R3; S2)
The "middle path" WInS book (a 0.9% VT sliver)	A weak note, and it reverses the T-bill rule (D10)
(iii) as the default book	Two decisions plus filler without a rung (T2 finding 4; E2 F4). It stays the fallback (5.10)
A planned "discipline/rebalance" trade; the SPTL split as a "refinement"	A band fires before Oct 23 in about 0-2% of cases even in a crisis (AY2); a pre-planned split is not a refinement (AY2 D7 C1)
A payment rung under (ii)	A second future-funding note (AX1b item 7)
A funded-ratio "below 100%" trigger	It can never fire once the payments are bought (D10 finding 6)
A fee number, a New Taiwan dollar figure, stage-by-stage stock shares, a "letter to Future You", CFA labels, "our portfolio manager approves", the statistics degree, any Laura quote	Words that do not change a decision, or that break the D13, IPS or advisor rules (E1 P17; E2 F1; E3-6, E3-9; R-W19)

7. Team decisions still open: recommended default and the test for choosing

#	Decision	Default (INT)	Test for choosing	By (AEST/AEDT unless ET)
D1	Adopt lock-early (gate a)	Adopt	Every student writes, with no AI file open, why the payments are bought first. The vote is logged with those reasons	Gate, Thu Oct 1
D2	REC (minimum bought in 2028) or ALT (2031 lock)	REC	(1) Values: each student states which matters more for Laura. Either a building minimum no market can lower from 2028, with better bad cases for about the same middle outcome; or more stock in 2028-30 and a higher announced bottom in typical markets, at the price of a worse bad case. (2) Explain-back: every student explains the chosen design's four dates in two minutes without notes. If only one design passes for all six, choose it. The numbers the team sees are E6 [2] and [5]	Gate, before any building or stock order
D3	WInS book	(ii)R	The two-reader check (5.4e), and whether the team can state the scaling in its own words in about 25 IPS words (E1 P13 decision test). Two failures lead to (iii)	Gate
D4	WInS instrument for the minimum	The 2032 note, then IBTM, then VGIT	Whichever is listed, in that order (5.2)	Gate
D5	The team's labels for the three parts	The team's own words (no default wording is given)	The labels survive the explain-back and appear identically in the notes, the log and the IPS	Before the first note
D6	Share of the stock fund added to the gift (s)	One half	The kept money at the median is about three years of Taiwan building-cost drift on the gift ($s = \frac{1}{2}$: \$32k against about 11% of \$174k; ASM check, not a contingency fund). It is the plainest split to explain	Section-4 vote, Mon Oct 26
D7	Cost clause in the IPS	Keep (one clause)	If the PDF has fewer than 2 spare lines, compress it inside the governance sentence; cut it only if the fee base in the Final Report will never touch the promises (R8)	Oct 26; Nov 4
D8	Statistics degree	Zero uses in the TN and IPS	Use it once only if it changes a decision, never as a compliment (E3-9; AX1 C19)	Idea list, Oct 24
D9	Paper and line spacing	U.S. Letter, Word "Double"	The exported PDF ends on page 3 with at least 2 spare lines. "Exactly 24 pt" is a documented last resort	Nov 4
D10	Joint tail in the IPS	One clause	A reader answering "what could still stop a payment?" gives: a rate fall before January combined with a missing deposit, or a U.S. default (D12 M234)	Oct 26
D11	Repository visibility until Dec 4	The team leader decides; revisit and log	The PM-30 trade-off (copying, AI-record accuracy) against the team's earlier acceptance (CLAUDE.md)	Fri Oct 2

#	Decision	Default (INT)	Test for choosing	By (AEST/AEDT unless ET)
D12	Roles	Assign now	Every student owns at least one note or reflection and one IPS block	Tue Sep 29

Calendar anchors (E5 [1], derived with zoneinfo):

- Roster due Sat Oct 10 08:00 AEDT; submit it Wed Oct 7.
- TN due Sat Oct 24 08:00 AEDT; submit it Thu Oct 22.
- IPS due Sat Nov 7 09:00 AEDT; submit it Thu Nov 5. U.S. clocks change on Nov 1.
- Brisbane is 1 hour earlier; Perth 3 hours earlier.

8. Three final checks

(a) Every decision traces to Laura or the case: PASS

The two judgement calls are flagged (ASM/INT, each still anchored to the case).

Decision	Case line or Laura fact (VRF unless marked)
Buy all ten payments first, with Treasuries held to maturity	L88-92 (fixed \$50k; "high degree of certainty"; no outside money)
Buy on arrival; longest-first; completion from the second deposit	L91; L43-45 ("will contribute" at the start of 2028); L59 (flows at the start of the year)
Building minimum = the second deposit's dollar amount, bought when it arrives	L43-45 (her career earnings); L110-112 ("meaningful personal contribution" signals viability); L114-116 (overpromising damages credibility); L69-74 ("protecting the capital"). <i>Judgement call 1 (INT): reading "protecting the capital" as returning her contribution whole for the building</i>
All money nobody relies on in one world stock fund	L69-74 ("thoughtful risks", "pursuing growth"); L61-65 (living costs outside; no withdrawals); L85-86 vs L91-92 (others may help the facility, never the payments)
No tilts; broad funds	R-AN34 (no values or theme in the case); brief s4 tokenism rule
2031 range method and two-sided confidence	L109-120
Cap, and kept money for flexibility	L101-103 (she decides; "as the project develops"); L118-119. <i>Judgement call 2 (ASM): half, sized against Taiwan building-cost drift (F-508, VP inputs)</i>
Reserve = the ladder, named in 2033, changing only as bonds mature	L94-97 ("if at all")
Certainty by price, with two gaps; U.S. dollars binding; no New Taiwan dollar hedge	L97-99; L90; L146-148; R-AN10; R-AN15; WInS derivatives ban (VP)
Governance: she decides her two calls; rules change only for her circumstances	L101, L110; IPS L73-74; Guide p.5
Cost principle	L43 ("with an asset management firm"); R-W29 (VP). The case is silent on fees, so this is labelled a team assumption
WInS book (ii)R and the scaling sentence	L129 (WInS = implementation); IPS L75; R-W71 (VP)
No Laura quotes, no identity reasons, no degree hook	Brief s16 (D13 rules); D13c section 5

(b) Nothing is more complex than it needs to be: PASS, with one watched item

- **Money rules:** four dated decisions plus governance. **Parameters:** the minimum's size (set by a case number) and "half". The cap is a yes/no that replaces a model percentile.
- **Removed from E1:** the growth-mix rule, band and yearly reset; the lock share; q; the uncapped/cap debate; the VGSH-versus-dated question; six E1-C conflicts. The IPS drops from about 55 specified ideas to 10 blocks (E2 3.2, INT).
- **Complexity that stays, and why it earns its place:**
 - the rates-fall branch binds in about 1 path in 3 (MODEL);
 - the joint-tail clause is required for a frozen rule not to overclaim (AY2 rank 2);
 - the WInS scaling sentence is required for consistency (D10 1.5);

- the cap branches are forced by a rule the team cannot see yet.
- **Watched item:** book (ii)R costs 15-20 characters of scaling marker per note. That cost is earned only if the two-reader check passes; otherwise the fallback is (iii) (D3 test).

(c) Nothing requires trading outside the WInS rules: PASS, conditional on the gate

Rule (VP: SMapply Trading Details and FAQ; 2026-27 User Guide)	How the plan complies
Permitted: cash, stocks $\geq \$5$, "Any ETF available on WInS", government/Treasury bonds available on WInS	Only ETFs (IEF, TLH, SPTL, IBTM, VT; alternates VGIT, SPTI, VTI, VXUS), one possible Treasury note, and cash. Each is pending the listing check (gate c)
Banned: margin, short selling, stock-secured debt, crypto, derivatives, "anything else"	None used. No leveraged, inverse, options-based or crypto-linked fund (R-AN49), no ETN. Cash float $\geq \$1,000$ at next-open prices (T2 finding 1)
"Day Trading: This is not permitted."	First session is buys only. No same-day sells. Any later re-mix happens on another day
No trade above 2× a security's daily volume	The largest orders are tiny against volume (IBTM ~3,336 shares vs ~216k a day; TLH ~1,365 vs ~1.97m; VP volumes via ticket v1/E4). Vanguard volumes are read in WInS
Up to 200 trades; commissions \$25 per ETF and \$10 per bond	4-6 trades, \$50-150
Position limit (Session Rules)	Branches decided in advance (5.3); stop and ask below about 17%
Activity minimum (undefined publicly, R-W23)	If one exists: ask Wharton, and meet it only with plan-consistent, logged trades; never staged ones (PM-15)
The real plan (not WInS), under the safe reading of "Investments permitted (for BOTH contributions)" (R-AN15)	Treasuries (STRIPS and one zero for the minimum), one broad stock ETF and Treasury bills. No forwards or other derivatives

9. Conflict register: every conflict, which side wins, and why

#	Conflict (sources)	Winner	Why (evidence)
1	Floor timing, 2028 or 2031 (E4 C1; AX1 C6; E2 R2 and section 5; T2 finding 5; ticket gate b)	2028 (REC)	The same median with a better bad case and better history, on the same basis (E6 [2], [5]). This is the plan's own tie-break rule. It also removes three rules. E1's rejection rested on a withdrawn figure (AX1 C6)
2	Size of the bottom: the deposit amount or a share (E4 C2)	Deposit amount	Set by a case number, with no free parameter. The IPS can name it as a rule output
3	Capped market-value top or uncapped model top (E1 C3/C5; E4 C3; D3/D5; AY1 C11; AX1 C2-C3)	Capped market-value top, reporting P(top reached)	Model-free and checkable against history. D5's "100% within is empty" is met by reporting the chance of reaching the top. Money above the top stays hers (AX1 C15)
4	Growth split 50/50 or 60/40 (E1 C2; E3 K6; E5 C6; T1; T2 finding 2; D2 vs D6)	Moot under REC (50/50 for ALT)	The stock fund is 100% stocks. The split is set by the jobs, not a probability rule
5	Short or dated bonds for the growth money (AX1 C18; E2 R1; E4 L4)	Dated (the minimum matures before 2033)	The money has a known use-date; a dated instrument removes reinvestment risk. VGSH is dropped
6	WInS book (ii) or (iii) (E1 C1; E3 K3; E5 C1; D10 vs T1/D12; T2 finding 3)	(ii)R , with the two-reader check as the tie-breaker	Three decisions and jobs; funding-purpose diversification; survives one session; copes with caps. (iii)'s literal truth is kept as the fallback
7	The third note: VGSH or a dated rung (E1 C6; D1 vs D12; AX1b item 7)	The dated building minimum	A distinct job, Laura-specific and dated (E4 L6)
8	VGSH role word (E1 C7; AY1 C10)	Moot	VGSH is not in the REC book
9	Kept money after 2033: bills or invested (E1 C8)	U.S. dollar Treasury bills	Flexibility is for surprises; no case line supports growth for this money (D8 M081)
10	Where the longest-first counterpoint lives (E1 C9; E3 K8)	In the IPS rule's "because" (E3)	A client would ask which operating year is exposed (E3 Q2); it costs one clause
11	Note length (E1 C10)	Draft to 300 characters, and test at zero risk	The evidence is mixed (AY1 D12 C2)

#	Conflict (sources)	Winner	Why (evidence)
12	Fee clause (E1 C11/P12; AY2 C13; E3-4/K2; E5 C2)	Keep one clause (E3)	It protects both bought amounts (E3 condition 2), and the Final Report's fee base is decisive (E6 [6]). Under REC the growth reason no longer rests on a fee
13	Stock shares by stage in the IPS (E1 C12; E2 3.4)	Leave them out; use one forced-caution price fact	Four shares confuse a first reader (E2); under REC the share is about 9% in every year after 2027 anyway
14	Label of the 2031 middle slice (E3 K1)	Owned / expected / kept; never "promised"	Case L110, L116 ("expects", "potential contribution"); D9 M062
15	Statistics degree once or never (E1 P17; E3 K4; E5 C3; AY2 C6)	Never	It risks being a persuasion lever (AX1 C19); the certainty definition shows the respect anyway
16	Pitch order (E3 K5)	Purpose first	E3-5; E2 F3; the D13c communication checklist
17	Central idea as a timing ban or a split of jobs (E1 P1 vs E2 F4, E4 L5)	Split of jobs	True in the plan and in the WInS book (AY2 rank 1)
18	IPS structure: "Rule 1-4 + G" or four dated decisions (E1 P11 vs E2 F2)	Four dated decisions	The order a reader expects; the rule names live in the log
19	Density: E1's elements or about 12 ideas (E1 P16 vs E2 F1; PM-03)	10 blocks, ≤470 words	The page-fit ceiling (D9; AX2 C3); last year's lesson
20	Risk tolerance in CFA three-factor terms or in plain goal-by-goal terms (E1 P6 vs E3)	Plain, goal by goal; the CFA framework stays internal	The IPS bans citations; E3's "analysed, not understood" risk
21	Bad-year pre-commitments framed as behavioural traps or as the firm's discipline (E1 P10 vs E3-6)	The firm's discipline and her public word	Case L68-69: she already understands volatility
22	The \$150k doubted in three places or one (E1 P2-P4 vs P17; E2 C6)	One labelled stress clause	Case "will" (R-AN35). Gap (i) mentions the deposit only as the completer
23	"Fully funded by 1 Jan 2028 at the latest" (D1)	"When the second deposit arrives"	AX1b item 3
24	"Only the facility shrinks" (D8 rule 1; chair memo)	Joint tail stated	AY2 D8 C1; D1 finding 4
25	Guide p.5 as a reason against changes before Nov 6 (D1, D2)	It does not apply before Nov 6	AX2 D2 C3; AX1b item 4; Guide L89-90
26	The SPTL cap split as a "refinement" (D7)	Not a refinement	It is decided before the first order (AY2 D7 C1)
27	"The hedge is never traded" vs the ticket's re-mix band (D7)	Never sold or trimmed; only re-mixed	AY2 D7 C7
28	P(ladder > \$300k): 1 in 4 or 1 in 3	About 1 in 3 for the real Nov-15 ladder	AX1b item 1; AX1 C8/V5; 176 of 185 days in 2026
29	Hedge target 9.90 or 10.16 years	"About 10 years"	S4 finding 8: issuer durations sit 0.2-0.3y below full-revaluation durations
30	The all-Treasury comparison called "riskless" (D3)	"All-Treasury benchmark" (Final Report)	AX1 C5: not riskless seen from today
31	"Needs met at today's yields" (D8 headline)	Corrected: the payments need what the bought ladder earns; the facility has no required size	AY2 D8 C2
32	VGSH "already owned" / "the amount she can promise" (D9, ticket v0)	Moot	VGSH is not in the REC book; the wording ban stays for ALT (AY2 D9 C1)
33	Council's nearest-first vs longest-first (brief s10)	Longest-first	D1 [3]; AX1b item 23
34	"Fits inside \$300k" / "bought in January 2027" as headlines (brief s7)	Dropped	D7 M011; D1; true in only about 2 of 3 rate paths
35	Stock fund rebalanced or not (D8 rule 2 vs E4)	Not rebalanced; written down	Its risk is the risk stated (100% stocks in money nobody relies on); CFA 4c: a no-rebalance policy is documented

#	Conflict (sources)	Winner	Why (evidence)
36	Semifinal basis: IPS + Final Report or all three (E5 C7; X-8)	Treat all three as scored; weight hours to the IPS after Oct 23	R-AN41; the SMaply page
37	Paper and spacing (E5 C4-C5; AY2)	Letter, Word "Double"	Fitting under the strictest reading removes the question
38	Public repository (E5 C8)	Team leader's call; revisit and log	Outside strategy; PM-30
39	The rival's "stock fund 100% after she speaks" vs E1's lower post-2031 stock	Accepted	Bad 2031-32: the gift is \$169k at the median and stays at or above the bottom in 100% of paths (E6 [3]); the cost is the stretch only
40	E1's lean on (ii) at 17% VT vs E3/E5's timidity worry	Answered by framing, not by more stock	Forced vs chosen; "all money nobody relies on is fully in stocks"; the most markets can take is stated. The team may still choose ALT for more stock (D2)

10. Later (after Nov 9), one line each (Final Report; not worked here)

- The 2031 range in dollars at weak, median and strong 2031 states; the likely figure beside it, not the midpoint.
- The chance of reaching the top: U.S. history (two-year periods since 1928) and the model, both named.
- Named historical windows beside the model's p5 (E4 [5]; E6 [5]).
- The all-Treasury benchmark and the rejected alternatives (growth-first, the 2031 lock, the uncapped percentile top).
- The fee assumption, its base and net-of-fee figures (R8; E6 [6]).
- A New Taiwan dollar reference at a dated rate; the building-power assumption line (D4).
- The reserve's market value on 1 Jan 2033, with cost and face; the rung-by-rung ladder (S3).
- The fundraising excerpt order: operations funded for ten years; the minimum owned since 2028; the stretch and its odds; what is not promised; U.S. dollars binding (M008, M015, BS-17).
- The decision-log traceability table and the paraphrase-test log.
- At most one or two dated, verified Laura lines in context (D13a/D13c).

Sources (accessed 2026-09-27/28)

Official, in the repo (VRF):

- `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`: L16, L43-45, L59-74, L83-120, L123-133, L136-148, L161-169.
- `2026_WGY_Investment_Policy-FINAL.txt`: L3-28, L31-41, L45-78, L82-125.
- `2026_WGY_Trading_Notes_Analysis-FINAL.txt`: L4-31, L36-53.
- `2026_WGY_Investment_Competition_Guide.txt`: L70-94, L154-159, L181-186.
- `2026_WGY_Competition_Infographic.txt`.
- `SMaply_Deliverables_Page-2026-09-27.md`: L8-17, L49-57.

Market data (VRF): `competition/official_market_data/daily-treasury-rates-2026-09.csv` (row 09/25/2026); `JPM_LTCMA_2026_US_matrix_USD.pdf` p.2.

Primary web (VP):

- iShares IBTM, <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf>. Re-read by E6 on 2026-09-28: NAV \$21.83 (9/25), expense 0.07%, and both quoted sentences found verbatim.
- Carried by the cited files, not re-opened by E6:
 - SMaply Trading Details and FAQ, <https://wghsinvcomp.smaply.us/res/p/trading/> and [/res/p/faqs/](https://wghsinvcomp.smaply.us/res/p/faqs/);
 - Wharton Rules & Roles and the AI policy;
 - the 2026-27 WInS User Guide;
 - issuer pages for IEF, TLH, TLT, SPTL, VT, VGIT, VGSH, SGOV, BIL (ticket v1 s12);
 - Damodaran histretSP (via the D3 snapshot); FRED DGS10 and DEXTAUS;
 - the Vanguard VCMM page;

- CFA Institute, *Elements of an IPS* (2010) and *Asset Manager Code*;
- digital.gov paraphrase testing;
- the NSW term and HSC pages (via E5).

Research files (lower authority; audit corrections applied):

- research/insight_v1/phase_E/E1_change_proposals.md, E2_judge.md, E3_laura.md, E4_rival.md, E5_premortem.md.
- phase_D/ D1-D12, D3_quant and D3_model_v2_results (AX1), D13a, D13c, audit_*.md, trading_now_brief.md.
- wins_now/securities_and_allocation_v1.md, T2_red_team.md, S4_red_team.md.
- phase_A/case_register.md, fact_register.md, stakeholder_map.md, wins_week1_guardrails.md.
- phase_C/survivors.json, parked.json.
- research/council_2026-09-27/01_chair_memo.md, round2_referee_ruling.md (history); CLAUDE.md.

Scripts: research/insight_v1/scripts/E6_final_checks.py (new). Re-run: E1_split_and_slices.py, E3_client_view_numbers.py, E4_rival_numbers.py. Figures quoted from D1, D3, D4, D8, D9, T1 and T2 are as reproduced by their auditors.

What this teaches

1. **Keep the insight; change what earns nothing.** The winning idea was right from the council onward: buy the payments first. What changed is the part added around it. A balanced growth sleeve that is locked later turned out to be a coin flip on top of a simpler design, with the same middle and a worse bad case. When two designs tie in the middle, the one with the better bad case and fewer rules wins. That rule was already in the plan; this file only applied it consistently.
2. **Compare designs on one basis before choosing.** The early lock looked worse until it was compared at the same stock risk, on the same simulated paths and on a century of real history. Most arguments between experts dissolve once everyone uses the same yardstick.
3. **A number the client already knows is the most credible number.** Her second deposit is a figure she, her partners and the case all share. Building the bottom of the range on it needs no forecast. Only the top carries a chance, and that chance can be checked against history.
4. **Put all the risk where it belongs, and none where it doesn't.** "Half stocks, half bonds everywhere" dilutes twice. "Nothing at risk that others rely on; everything else fully invested" is easier to explain, and it answers "thoughtful risk" in a way a reader can see.
5. **The scored thing is the written page.** The strategy survived every audit. What decides the semifinal is whether six students own it, write it in their own words, keep the notes and the IPS saying one thing, and fit it on two pages. Most of this file is about those joints, because that is where plans usually fail.

Firm A

1. How the ten payments are protected

The portfolio is split into a Treasury bucket (intermediate U.S. Treasuries) and a stock bucket (broad U.S. and international stock funds). A glide path lowers the stock share each year: 65%, 60%, 50%, 40%, 30% and 20% of the portfolio in 2027-2032. From 2030 the Treasury bucket buys Treasuries maturing in 2033-2042 in stages, so the ten-year reserve (about \$395k at today's forward rates) is complete on 1 January 2033.

2. What "high certainty" means and how it was checked

High certainty means at least a 95% probability that the portfolio covers the reserve on 1 January 2033. Checked by a 200,000-path Monte Carlo on JPM's 2026 long-term assumptions (U.S. stocks 6.7% compound, Treasuries 4.0%) and a one-point uncertainty in 2033 interest rates. Model estimate: funded in 99.3% of paths; 91.4% if the 2028 deposit is \$75k.

3. How growth is pursued and with what money

All money not yet needed by the Treasury bucket sits in the stock bucket, rebalanced once a year to the glide path. The average stock share over 2027-2032 is 44%. Model estimate: median money left after the reserve is \$206k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

The range is the 10th to 90th percentile of the model's 2033 facility contribution, re-run with 2031 values, stated with 80% confidence. Model estimate from today: \$75k to \$302k, median \$175k.

5. Flexibility after 2033

The facility receives 85% of the money above the reserve. The other 15% is kept in Treasury bills for project changes.

6. What the competition portfolio (WInS) holds

The 2027 allocation: VTI 45%, VXUS 20%, IEF 30%, SHY 4%, cash 1%. Five ETF trades at \$25 each. Pending WInS availability check.

7. The main risks the firm accepts

- Payments depend on markets until the reserve is bought: model estimate 0.7% miss, 8.6% if the 2028 deposit is \$75k, 44% if it does not arrive.
- A one-point rate fall before the staged purchases adds up to about \$16k to the reserve's cost (model estimate).
- The 2031 range is wide: its bottom is less than half its median.

Firm B

1. How the ten payments are protected

The whole portfolio grows until 2033, then buys the reserve. Stocks are about 75% of the portfolio in 2027-2030, 60% in 2031 and 40% in 2032; the rest is intermediate U.S. Treasuries. On 1 January 2033 the firm buys a Treasury ladder covering the ten \$50,000 payments (about \$395k at today's forward rates).

2. What "high certainty" means and how it was checked

High certainty means at least a 95% probability that the portfolio covers the reserve on 1 January 2033. Checked by a 200,000-path Monte Carlo on JPM's 2026 long-term assumptions (U.S. stocks 6.7% compound, Treasuries 4.0%) and a one-point uncertainty in 2033 interest rates. Model estimate: funded in 96.8% of paths; 86.3% if the 2028 deposit is \$75k.

3. How growth is pursued and with what money

All of Laura's money pursues growth until 2031. The case describes her as willing to take risks, and a plan running to 2042 gives stocks time to earn their higher expected return (JPM: 6.7% against 4.0% for Treasuries). The average stock share over 2027-2032 is 67%. Model estimate: median money left after the reserve is \$226k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

The range is the 25th to 75th percentile of the model's 2033 facility contribution, re-run with 2031 values, stated with 50% confidence. Model estimate from today: \$118k to \$304k, median \$203k.

5. Flexibility after 2033

The facility receives 90% of the money above the reserve. The other 10% is kept in Treasury bills for project changes.

6. What the competition portfolio (WInS) holds

The 2027 allocation: VTI 50%, VEA 15%, VWO 10%, IEF 24%, cash 1%. Four ETF trades at \$25 each. Pending WInS availability check.

7. The main risks the firm accepts

- Payments depend on markets until 2033: model estimate 3.2% miss, 13.7% if the 2028 deposit is \$75k, 40.8% if it does not arrive.
- A stock fall in 2031-2032 lands just before the reserve is bought.
- A one-point rate fall by 2033 adds about \$16k to the reserve's cost (model estimate).

Firm C

1. How the ten payments are protected

Stocks are 65% of the portfolio in 2027-2029, 55% in 2030, 40% in 2031 and 25% in 2032; the rest is intermediate U.S. Treasuries. From 2030 the firm buys Treasuries maturing in 2033-2042 in stages, so the ten-year reserve (about \$395k at today's forward rates) is complete on 1 January 2033.

2. What "high certainty" means and how it was checked

High certainty means at least a 95% probability that the portfolio covers the reserve on 1 January 2033. Checked by a 200,000-path Monte Carlo on JPM's 2026 long-term assumptions (U.S. stocks 6.7% compound, Treasuries 4.0%) and a one-point uncertainty in 2033 interest rates. Model estimate: funded in 98.5% of paths; 89.4% if the 2028 deposit is \$75k.

3. How growth is pursued and with what money

Of the stock money, 70% is a broad world fund and 30% is split across four themes linked to Laura's public work: social media and the creator economy, Taiwan, clean energy, and an ESG-screened U.S. fund. The average stock share over 2027-2032 is 53%. Model estimate: median money left after the reserve is \$214k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

The range is the 10th to 90th percentile of the model's 2033 facility contribution, re-run with 2031 values, stated with 80% confidence. Model estimate from today: \$65k to \$335k, median \$182k.

5. Flexibility after 2033

The facility receives 85% of the money above the reserve. The other 15% is kept in Treasury bills for project changes.

6. What the competition portfolio (WInS) holds

The 2027 allocation: VT 45%, SOCL 5%, EWT 5%, ICLN 5%, ESGU 5%, IEF 34%, cash 1%. Six ETF trades at \$25 each. Pending WInS availability check.

7. The main risks the firm accepts

- Payments depend on markets until the reserve is bought: model estimate 1.5% miss, 10.6% if the 2028 deposit is \$75k, 42.6% if it does not arrive.
- Theme funds are narrower and more volatile than the model assumes.
- A one-point rate fall before the staged purchases adds up to about \$16k to the reserve's cost (model estimate).

Firm D

1. How the ten payments are protected

In January 2027 the first \$300,000 buys ten zero-coupon U.S. Treasuries, one maturing on 15 November before each \$50,000 payment (2033-2042); on the 2026-09-25 curve they cost \$294,387. If money falls short, the latest payments are bought first and the 2028 deposit completes the rest.

2. What "high certainty" means and how it was checked

High certainty means each payment is bought at market prices with a bond maturing before it: certain in nominal U.S. dollars unless the U.S. Treasury defaults. Checked on the official Treasury curve. Model estimate: about 1 in 3 rate paths push the cost above \$300,000 by January.

3. How growth is pursued and with what money

In January 2028 the \$150,000 deposit buys a zero-coupon Treasury repaying \$150,000 before 2033 (the minimum facility contribution; cost about \$118k). The remaining ~\$40k goes into one world stock fund. The average stock share over 2027-2032 is 7.5%. Model estimate: median money left after the reserve is \$207k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

Bottom: the minimum she owns (\$150k). Top: the minimum plus half the stock fund's value on 1 January 2031 (median \$175k, model estimate). The bottom holds barring a U.S. Treasury default; the top is reached if the fund is no lower in 2033 than in 2031 (model estimate 73%).

5. Flexibility after 2033

The facility receives the minimum plus half the stock fund, capped at the top; the rest is kept in Treasury bills for project changes. Model estimate (median): \$174k to the facility, \$32k kept.

6. What the competition portfolio (WInS) holds

The January 2028 plan scaled to \$300,000: IEF 23.5%, TLH 42.5%, the 15-Nov-2032 Treasury note or IBTM 24.3%, VT 8.7%, cash 1%. Four trades (\$25 per ETF, \$10 per Treasury note). Pending WInS availability check.

7. The main risks the firm accepts

- Less upside: model estimate 95th percentile of money left after the reserve \$250k.
- A rate fall before January 2027 plus no 2028 deposit leaves part of the 2033 payment unfunded (model estimate \$12k after a half-point fall, \$31k after a one-point fall).
- No 2028 deposit means no minimum contribution and no stock fund.

Firm E

1. How the ten payments are protected

A portfolio of 60% stocks and 40% bonds is rebalanced once a year. It moves to 55/45 in 2029, 50/50 in 2030 and 40/60 in 2031-2032. On 1 January 2033 the firm buys a Treasury ladder covering the ten \$50,000 payments (about \$395k at today's forward rates).

2. What "high certainty" means and how it was checked

High certainty means at least a 95% probability that the portfolio covers the reserve on 1 January 2033. Checked by a 200,000-path Monte Carlo on JPM's 2026 long-term assumptions (U.S. stocks 6.7% compound, Treasuries 4.0%) and a one-point uncertainty in 2033 interest rates. Model estimate: funded in 98.9% of paths; 90.4% if the 2028 deposit is \$75k.

3. How growth is pursued and with what money

Growth comes from the whole portfolio's stock share, split 60% U.S. and 40% international; bonds are a broad U.S. bond fund. The average stock share over 2027-2032 is 51%. Model estimate: median money left after the reserve is \$213k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

The range runs from the 10th percentile to the median of the model's 2033 facility contribution, re-run with 2031 values. Model estimate from today: \$66k to \$170k. The model gives a 90% chance of reaching at least the bottom and a 50% chance of exceeding the top.

5. Flexibility after 2033

The facility receives 80% of the money above the reserve. The other 20% is kept in Treasury bills for project changes.

6. What the competition portfolio (WInS) holds

The 2027 allocation: VTI 36%, VXUS 24%, BND 39%, cash 1%. Three ETF trades at \$25 each. Pending WInS availability check.

7. The main risks the firm accepts

- Payments depend on markets until 2033: model estimate 1.1% miss, 9.6% if the 2028 deposit is \$75k, 42.5% if it does not arrive.
- The bond fund holds corporate and mortgage bonds and is not matched to the payment dates.
- A one-point rate fall by 2033 adds about \$16k to the reserve's cost (model estimate).

Firm F

1. How the ten payments are protected

An optimiser (a formula that picks weights from expected returns, volatilities and correlations) resets twelve fund weights each quarter, subject to a loss limit: the worst 5% of outcomes above the reserve must average at least \$40k. Bond funds are weighted so each adds equal risk. Stocks fall from 55% of the portfolio in 2027 to 20% in 2032. The ten-year reserve (about \$395k at today's forward rates) is bought in stages in 2031-2032.

2. What "high certainty" means and how it was checked

High certainty means at least a 97.5% probability that the portfolio covers the reserve on 1 January 2033. Checked by a 200,000-path Monte Carlo on JPM's 2026 long-term assumptions (U.S. stocks 6.7% compound, Treasuries 4.0%) and a one-point uncertainty in 2033 interest rates. Model estimate: funded in 99.7% of paths; 93.1% if the 2028 deposit is \$75k.

3. How growth is pursued and with what money

Stock funds carry momentum, quality, minimum-volatility and value tilts. The average stock share over 2027-2032 is 38%. Model estimate: median money left after the reserve is \$201k; worst-5% average \$42k.

4. What Laura can tell co-sponsors in 2031 (range and confidence)

The range is the 5th to 95th percentile of the model's 2033 facility contribution, re-run with 2031 values, stated with 90% confidence. Model estimate from today: \$59k to \$317k, median \$171k.

5. Flexibility after 2033

The facility receives 85% of the money above the reserve. The other 15% is kept and managed by the same optimiser.

6. What the competition portfolio (WInS) holds

The 2027 allocation: VTI 20%, VEA 10%, VWO 5%, MTUM 5%, QUAL 5%, USMV 5%, VLUE 5%, IEF 15%, TLT 10%, SHY 8%, TIP 6%, GLD 5%, cash 1%. Twelve ETF trades at \$25 each. Pending WInS availability check.

7. The main risks the firm accepts

- Payments depend on markets until the reserve is bought: model estimate 0.3% miss, 6.9% if the 2028 deposit is \$75k, 45.8% if it does not arrive.
- Optimiser weights are sensitive to small errors in expected returns.
- A one-point rate fall before the staged purchases adds up to about \$16k to the reserve's cost (model estimate).

Fairness audit of the six firm summaries (firm_A.md ... firm_F.md)

Serves: IPS (a fair comparison before the team picks the strategy the IPS freezes on Nov 6). Relayed user message during this task: "just cut off the unnecessary bits." Applied: every file was trimmed, and filler shared by several firms was removed evenly. No scope change. Originals backed up outside the repo (scratchpad tournament_orig/).

Summary

- All six now have the same 7 section headings in the same order and exactly 3 risk bullets each.
- Word counts after the edits (Python `len(text.split())`, headings included): A 368, B 359, C 365, D 378, E 354, F 375. All are within 300-380. Before the edits: A 368, B 372, C 373, D 378 (428 after the first rewrite), E 379, F 376.
- Numbers checked against brief sections 6 and 14, re-running `research/insight_v1/scripts/E7_tournament_field_numbers.py` (A, B, C, E, F) and against `phase_E/E6_final_spec.md` / `ips_spec.md` R1-R5 (D). A, B, C and E matched exactly. One unsupported number was found and corrected (F's 97.5%, see F below). Every model output now carries a "model estimate" label.

Edits common to several firms

1. A, B, C, E, F: removed "Reserve bonds are held to maturity." from section 5. It was filler that said the same thing in every file.
2. A, B, C, E, F: added "(model estimate)" to the "\$16k per one-point rate fall" risk line. Rechecked: flat 5.26% -> 4.26% annuity-due of ten \$50k adds \$16,089, the same flat-curve method as `strategy_mc.py`.
3. A, C, F (which buy the reserve in stages): "A one-point rate fall by 2033 adds about \$16k" became "A one-point rate fall before the staged purchases adds up to about \$16k". The original overstated their rate risk, because buying in stages locks in part of the cost earlier. B and E buy in 2033, so their wording is unchanged.

Firm-by-firm edits

A: "safe bucket" / "growth bucket" -> "Treasury bucket" / "stock bucket" ("safe" is a loaded word). **B:** removed the quoted case line "has been willing to take thoughtful risks". It was the only direct quote in any file, which made B stand out. It is replaced by a neutral paraphrase ("The case describes her as willing to take risks"), and the reason now gives JPM's numbers (6.7% vs 4.0%), so the argument stays at full strength. Removed the "95th percentile \$540k" (section 3) and the "5th percentile \$20k" risk bullet. No other firm gives tail percentiles of the money left over, and they gave B one number that makes it look better and one that makes it look worse. The "money left over" figures are now equally specific: every firm gives only the median. **C:** reordered section 1 so the glide path comes first, as in A, B, E and F, and added "the rest is intermediate U.S. Treasuries". Added "a one-point uncertainty in 2033 interest rates" to the check description: it is true of the engine (`strategy_mc.py` `rate_sd = 0.010`), and A, B and E already said it. Removed the duplicated modelling caveat (it is kept in the risk bullet) and the phrase "They carry a values narrative alongside the core" (a sales line, not a mechanism). **D:** "The payments are bought, not forecast" (the council's own slogan) -> a plain definition in the same "High certainty means ..." form as the other firms. "building minimum" (a term the project invented) -> "minimum facility contribution". "gift" -> "contribution" / "to the facility", the terms the other firms use. Removed "U.S. stocks' two-year return was non-negative in 84% of periods since 1928": no other firm cites history, so it was extra evidence only D had. Added "Model estimate: median money left after the reserve is \$207k" to section 3, so D has the same number as the others (E6 total p50 \$207k). Section 5: 5th/50th/95th percentile figures -> median only (\$174k / \$32k), as for the other firms. Section 6: added the trade count and commissions (\$25 per ETF, \$10 per Treasury note), as every other firm has. Dropped "(duration-matched)" and "about 1%" (cash is exactly 1%: $23.5 + 42.5 + 24.3 + 8.7 = 99$). Risks: dropped "A fixed \$50,000 buys less each year in Taiwan". It applies equally to all six plans, and only D disclosed it. The vague ladder-cost bullet was replaced with the model's joint-tail amounts (\$12k at -50bp, \$31k at -100bp, from E6 R2) so that it has numbers like the other firms' miss rates. Kept: \$294,387 (the real 15-November ladder, E6 R1, VRF inputs; it is higher than the

brief's \$292,264 because the bonds mature about 6 weeks early), "about 1 in 3" (MODEL 30-34%), \$118k, ~\$40k, 7.5%, \$175k, 73%, \$250k. **E:** removed "balanced" and "conservative" (loaded words), leaving a neutral description of the range. Removed the fourth risk bullet ("A low range may lead co-sponsors...") so that E has 3 bullets like the rest. Section 4 already states the 50% chance of exceeding the top. **F:** CORRECTED A NUMBER. "funded in 97.5% of paths" (regime-switching model) had no source: no script or file computes a regime-switching result. The engine used for all five model firms gives 99.7% funded (0.3% miss) and 93.1% at a \$75k deposit (E7 output). F now reports those figures and the same method sentence as A, B, C and E. The regime-model claim and its risk bullet ("Results depend on the regime model's parameters") were removed. The base miss rate (0.3%) was added to F's first risk bullet, which A, B, C and E already had. Jargon defined or removed: "mean-variance optimiser" -> "an optimiser (a formula that picks weights from expected returns, volatilities and correlations)"; "CVaR limit" -> "loss limit"; "dynamic risk parity" -> "bond funds are weighted so each adds equal risk". Section 4: dropped the 25th and 75th percentiles so that F gives low / high / median like the other firms. Section 6: added "The 2027 allocation:" to match.

Checks that passed with no edit

Glide-path averages (A 44.2%, B 66.7%, C 52.5%, E 50.8%, F 38.3%); WInS weights sum to 100% and the stock shares match each 2027 glide weight; trade counts match the number of funds; miss rates, funded rates and medians match the E7 run (seed 20260927, 200k paths); ranges match the E7 gift percentiles (A \$75k/\$302k/\$175k; B \$118k/\$304k/\$203k; C \$65k/\$335k/\$182k; E \$66k/\$170k; F \$59k/\$317k/\$171k); \$395k reserve = brief s14 \$394,930; JPM 6.7%/4.0% = brief s6.

Known modelling differences NOT fixed (these are content, not presentation; flagged for whoever reads the results)

- The engine prices the 2033 reserve at a flat 5.26% (~\$401k), not the ~\$395k implied by forward rates (brief s11). This slightly overstates the miss rates of A, B, C, E and F. D buys its bonds directly and is not affected.
- The engine models A, C, E and F as plain stock/Treasury glide paths. C's themes, F's tilts and optimiser, E's bond fund (treated as Treasuries), and the staged purchases of A, C and F are not modelled. C and F state this in their risk bullets. E's and A's model numbers ignore it.
- D's numbers come from a different script (E6_final_checks.py). The same seed and JPM inputs are used, but the ladder is bought in 2027 rather than simulated to 2033.
- Stocks in every model are U.S. large cap only (JPM 6.7%/16.47%), including the international funds.

What this teaches

A fair comparison checks presentation as well as numbers. A quote, a slogan, an extra statistic or a single adjective can make one option look stronger before anyone reads the numbers. Keeping the structure, the level of numerical detail and the labels the same lets a reader judge the ideas themselves.

Eval L1: which firm would Laura choose? (simulation)

Evaluator L1, 2026-09-28. **This is a SIMULATION**, not Laura's view. It judges the six firm files only against: the case (competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt, cited as "case L"), her VERIFIED-PRIMARY words in phase_D/D13a_laura_quotes_verified.md (cited by id), and the readings in phase_D/D13c_voice_map.md (cited as D13c V/N/R). Every "she would" below is an inference. There is no verified statement by her about investing, so nothing here claims to be her investment philosophy (D13c V8). Serves: the IPS strategy choice (tier 2) and, through the WInS book, the Trading Notes. Quotes are evaluator evidence only: none of them goes into the Trading Notes or the IPS (brief s.16). Relayed user message: "just cut off the unnecessary bits". Applied: this file holds only the ranking, the reasons and the sign-off conditions.

Ranking

rank	firm	score /10	one-line reason
1	D	7.5	Buys the ten payments and the 2031 floor instead of forecasting them; same median as A, but far too little growth
2	A	6.0	Plain glide path, lowest miss of the simple plans, but payments still ride on markets and on the 2028 deposit
3	E	5.5	Simplest to understand; payments unmatched until 2033 and a range that is inside only ~40% of the time
4	F	4.0	Best-protected market plan, but a twelve-fund quarterly optimiser she could not check, for the lowest median
5	C	3.0	Uses her public work as a reason for theme funds, one of which rides on the same economy as her own income
6	B	2.5	Most payment risk and a 50%-confidence range: fails both of the case's hard lines

The deciding test (simulation): the case gives two hard lines. The payments must be funded "with a high degree of certainty" without outside money (case L91), and she must not promise co-sponsors "more than she can ultimately contribute" (case L114). Her own words rank letting down "people who were your earliest supporters and were the first to believe in you" above letting herself down (D13a-N01, student-era, 2016). The residents and the 2031 co-sponsors are those people. Only D keeps its promises to them without a forecast. The medians are nearly equal: D \$207k, A \$206k, F \$201k, E \$213k, C \$214k, B \$226k (each firm's own model estimate). So the market plans mostly buy upside at the far end, and they pay for it with a chance of missing payments that other people rely on.

Firm by firm

1. Firm D (7.5)

Earns her confidence:

- The payments are bought in January 2027, and "certain" is defined by what can be checked, with its one exception named (U.S. default). Case L91; D13c V9 (define certainty by what is observable, not by a model percentage).
- It names its own weak spots: a 1-in-3 chance the ladder costs more than \$300k, and a \$12k-\$31k joint-tail gap. This matches how she labels her own results "[unfinished]" (B9b-Q09) and names the cost before a leap (B8b-Q18).
- The 2031 bottom (\$150k) is owned, not forecast, so it cannot be walked back. Case L114-116; D13a-N01.
- A \$150k floor is a "meaningful personal contribution" (case L111). Growth money comes on top, and more is offered if markets allow. This is her festival pattern: more "as the festival grows" (B8b-Q39; D13c E3).
- Flexibility is kept, not everything goes to the building (case L102-103), and its WInS book uses four trades.

Loses her confidence:

- 7.5% average stocks over 2027-2032 for a client "willing to take thoughtful risks" who wants "an appropriate balance between pursuing growth and protecting the capital" (case L70-72). She tells herself "you should always take the jump because you're always going to regret not doing it" (B1b-Q11). The file never says where the leap is (D13c N1).
- Locking \$118k of the 2028 deposit into the floor five years early is the main cause of the low stock share, and the file does not weigh it against a smaller floor or a later purchase.
- The payments still depend partly on the 2028 deposit if rates fall first. That money comes from "publishing advances, speaking engagements, licensing" (case L44), income she herself called "unstable" (B8b-Q18).
- The confidence statement answers the wrong question. The case asks how confident the team is that the contribution "will fall within that range" (case L117-118). D gives only the chance of reaching the top (73%).
- Nothing on inflation or facility costs (case L147). "Certain in nominal dollars" is not the same as the same residency in 2042 (D13c N4 gap ii).

2. Firm A (6.0)

Earns her confidence:

- Plain and checkable: two buckets, one glide path, a yearly rebalance, five trades. D13c V12: she taught financial literacy to high-school students (B2b-Q08).
- The lowest miss of the simple market plans (0.7%). Staged Treasury buying from 2030 shows it has thought about timing.
- It admits its range is wide: "its bottom is less than half its median". This is the honest-framing habit in D13c V13 and B9b-Q09.

Loses her confidence:

- 44% chance of missing payments if the 2028 deposit does not arrive. That deposit is income she called "unstable" (B8b-Q18; case L44). The residents carry that risk (D13a-N01).
- "High certainty" is a model's 95% (D13c V9, N4). Its median (\$206k) equals D's, so A's certainty gap buys only upside.
- The 2031 bottom (\$75k) is a model percentile, not money she owns. The range therefore does not "protect the portfolio's ability to fund the ten-year operating commitment" (case L119) in a way she could show a co-sponsor.

3. Firm E (5.5)

Earns her confidence:

- The most familiar design (60/40, one yearly rebalance, three trades). It hides nothing (D13c R6).
- An underpromising range (bottom reached 90%, per the model). It errs on the side the case prefers (case L114).

Loses her confidence:

- The bond fund holds corporate and mortgage bonds that are "not matched to the payment dates" (E's own risk list). The whole reserve is then bought on a single day, 1 January 2033. Case L91.
- Its range misses the case's question. With 90% above the bottom and 50% above the top, the chance of falling within the range is only about 40% (my arithmetic from E's numbers). Case L117-118.
- 42.5% miss if the 2028 deposit does not arrive (B8b-Q18).

4. Firm F (4.0)

Earns her confidence:

- Best payment protection among the market plans (0.3% miss). A stated loss limit on the bad cases.

Loses her confidence:

- A twelve-fund optimiser with factor tilts, gold and quarterly resets. She could not check it, and F admits it is "sensitive to small errors". D13c V12 (plain rules) and V7 (cut what exists only for show).

- The most complexity for the lowest median (\$201k). The post-2033 flexibility money stays with "the same optimiser" rather than being held for the project (case L102-103).
- Its \$59k-\$317k range at 90% confidence is too wide to tell a co-sponsor much (case L116, "credible range").
- 45.8% miss if the 2028 deposit does not arrive (B8b-Q18).

5. Firm C (3.0)

Earns her confidence:

- The broad core fund is sound, and its miss rate (1.5%) sits mid-pack.

Loses her confidence:

- 30% of its stocks are in "themes linked to Laura's public work" (creator economy, Taiwan, clean energy, ESG). This uses her work and identity as a reason for a holding (D13c R3, strong; 5.1 swap test). The case names no values screen.
- SIMULATION inference: a creator-economy fund tends to move with the same economy as her 2028 income (case L44). It would add risk just where the plan is weakest.
- Hype-shaped themes (D13c V11; B8a-Q37, "how long will this novelty factor last?", student-era). C itself admits they are "narrower and more volatile than the model assumes".
- 42.6% miss without the 2028 deposit. The range bottom (\$65k) is a percentile.

6. Firm B (2.5)

Earns her confidence:

- It takes her appetite for risk seriously and uses the fewest funds.

Loses her confidence:

- It quotes half of the case line ("willing to take risks") and drops the other half, "appropriate balance ... protecting the capital" (case L70-72). D13c V1 warns against "Laura loves risk".
- 3.2% miss, 13.7% with a \$75k deposit, 40.8% with none: the most risk on money others rely on (case L91; D13a-N01).
- A 50%-confidence range is the case's overpromising risk (case L114). Half the time she would be outside what she told co-sponsors.
- 90% of the surplus goes to the building. Case L102-103 warns that committing everything limits flexibility.

What Firm D must change before I would sign (simulation)

1. **Answer the case's confidence question.** Under D's own cap rule, the 2033 contribution always lands between the owned bottom and the top unless the U.S. Treasury defaults (my reading of D section 5). Say that as the confidence the contribution falls within the range. Keep the 73% as the chance of reaching the top. Case L117-118.
2. **Say where the leap is, and what certainty costs.** The residency is the leap. The payments and the floor are bought for the people relying on them, and the growth money takes the risk. Put the price in one line: D's median is about the same as A's (\$207k v \$206k), and the give-up sits at the far upside (D's p95 \$250k). Case L70-72; B1b-Q11; D13c N1.
3. **Show you tested a bigger growth share.** Either keep more of the 2028 money in stocks (a smaller floor bought in 2028, or the floor bought in 2031, when she first speaks to co-sponsors), or show with one number why locking \$118k in 2028 is worth it. At 7.5% stocks, "appropriate balance" (case L72) is not yet shown.
4. **Close the 2028-deposit gap with a rule.** If rates fall before January 2027 and the deposit is late or smaller, state what gives way first. The payments come first and the minimum contribution shrinks, in dollars. That income is "unstable" in her own word (B8b-Q18; case L44).
5. **Name what nominal certainty does not cover.** A fixed \$50k buys less each year, and building costs are in Taiwan. One plain sentence, as the case asks about inflation and facility costs (case L147; D13c N4).

6. **Explain the WInS book in the Trading Notes.** It mirrors the 2028 plan (8.7% VT), not the 2027 book. One note must say why so little is in stocks. Every security remains PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK.

What this teaches

- A client with two risk ideas does not need one "risk level". She needs each idea to govern different money: her own leap can take risk, but money other people rely on should be bought, not forecast.
- Compare medians before you compare stories. Five of six plans end up near the same middle outcome. They differ mainly in who carries the bad cases.
- A "range with confidence" is a promise. The number that matters is the chance of landing inside it, not the chance of reaching the top.

Evaluator L2: Laura Gao simulation choosing a firm

Serves: IPS (which strategy to freeze by Nov 6). SIMULATION: every "she would" below is an inference from the case and her D13a VERIFIED-PRIMARY words, not a view she has stated. She has no verified statement about investing (D13c V8). Relayed user message noted: "just cut off the unnecessary bits". Applied: this file keeps only the ranking and the reasons.

Sources: the case, `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (cited as p.N); D13a quote ids; D13c value ids (V1-V13, R1-R12). Firm numbers are the firms' own model estimates, not checked here.

Ranking

rank	firm	score /10	one-line verdict
1	D	8.0	Buys the promise first and never announces more than it owns; too timid and leans on the 2028 deposit
2	A	6.5	A sensible glide path with dated Treasury purchases; payments still depend on markets until 2033, and its 2031 floor is not bought
3	E	6.0	Simple and honest about underpromising; its bond fund does not match the payment dates
4	F	4.5	The best certainty among the market-based plans, but it runs on an optimiser she cannot check
5	C	4.0	A sound base spoiled by theme funds "linked to Laura's public work"
6	B	3.0	Fails the case's core test: the payments ride on markets and on income she called "unstable"

Per firm

1. Firm D (8.0)

- Earns confidence:
 - All ten payments are bought with Treasuries in January 2027, and the firm states what "certain" means: bought at market prices, nominal dollars, barring a U.S. default. Case p.3: "must be funded ... with a high degree of certainty". Her quote D13a-N01 says letting down "people who were your earliest supporters" is worse than letting herself down.
 - The 2031 bottom (\$150k) is already owned, so she can never promise more than she has. Case p.3: "If Laura promises more than she can ultimately contribute, she could damage her credibility". D13c R11.
 - Money is kept back for "project changes" (case p.3 "financial flexibility as the project develops"; D13c V10).
 - It names its own costs: less upside, a 1-in-3 chance the ladder costs more than \$300k, and what happens with no 2028 deposit (D13c V1 "names the cost first"; quote B8b-Q18; D13c R5 "honest numbers").
- Loses confidence:
 - Stocks average 7.5% of the portfolio over 2027-2032. Against her own rule, "you should always take the jump because you're always going to regret not doing it" (B1b-Q11), and the case's "willing to take thoughtful risks" with an "appropriate balance between pursuing growth and protecting the capital" (p.2), this looks one-sided.
 - When the ladder costs more than \$300k, the latest payments wait for the 2028 deposit. That deposit comes from income she called "unstable" (B8b-Q18).
 - It earmarks \$150k in 2028 as the "minimum facility contribution", five years before she has to decide (p.3: "no predetermined facility contribution").
 - Inflation is not mentioned, although the case asks for its "effect ... on portfolio projections and facility costs" (p.4).

2. Firm A (6.5)

- Earns confidence:
 - A plain glide path (stocks fall from 65% to 20% of the portfolio), which reads as the case's "appropriate balance" (p.2).
 - Treasuries maturing on the payment dates are bought in stages from 2030.

- A 99.3% funding estimate, with the model named.
- Broad, low-cost funds and five trades (D13c V11, V12).
- Loses confidence:
 - The payments depend on markets until 2033, and the model misses 44% of the time if the 2028 deposit never comes. The case says the payments must not rely on outside money (p.3), and that deposit comes from income she called "unstable" (B8b-Q18).
 - The firm admits its 2031 range is wide and that its bottom is less than half the median. At 80% confidence, 1 year in 10 ends below the promised bottom. Case p.3 on overpromising; D13a-N01.

3. Firm E (6.0)

- Earns confidence:
 - The simplest plan: 60/40, three funds, three trades (D13c V12).
 - The 2031 range runs from the 10th percentile to the median, so she would underpromise. The firm states both probabilities: 90% to reach the bottom, 50% to exceed the top (D13c R11; B9b-Q09, where she grades her own results).
 - It keeps 20% for flexibility, the largest share of the six firms (p.3).
- Loses confidence:
 - The bond fund holds corporate and mortgage bonds and is "not matched to the payment dates". The firm says so itself, but it is still the wrong tool for ten fixed payments (p.3).
 - The whole reserve is bought on one day, 1 January 2033, at whatever rates hold then.
 - With no 2028 deposit the payments miss 42.5% of the time (B8b-Q18).
 - With the top set at the median, about half of all outcomes land above the range, so the range says little about where her contribution will actually fall (p.3 asks for a range and a confidence that the amount falls within it).

4. Firm F (4.5)

- Earns confidence:
 - The highest funding estimate among the market-based plans (99.7%).
 - An explicit limit on the worst 5% of outcomes, which is planning for the downside (B8a-Q13: "I expected a way worse reaction").
- Loses confidence:
 - Twelve funds, factor tilts, gold, and weights reset every quarter by an optimiser the firm itself calls "sensitive to small errors". A statistics graduate who labels her own results "[unfinished]" (B9b-Q09; case p.1 degree) would not sign weights she cannot check. D13c V7 (cut what exists only for show) and V12 (plain words).
 - Its 90% range, \$59k-\$317k, is too wide to tell co-sponsors anything (p.3 "clearly and credibly").
 - The money kept for flexibility is also run by the optimiser.

5. Firm C (4.0)

- Earns confidence:
 - A glide path and staged Treasury purchases, as in Firm A.
 - A 98.5% funding estimate.
- Loses confidence:
 - 30% of the stock money goes into themes "linked to Laura's public work": the creator economy, Taiwan, clean energy and an ESG screen. The case names no values screen and no link from her background to her portfolio. D13c R3 (her identity or interests used as a reason for a holding, strong evidence), V7 (B8b-Q13 "authentic, personal connections") and R12 (hype; B8a-Q37 "how long will this novelty factor last?").
 - The Taiwan tilt adds risk tied to the very place the residency is built.
 - The firm admits the theme funds are "more volatile than the model assumes".

6. Firm B (3.0)

- Earns confidence:
 - It takes her appetite for leaps seriously (B1b-Q11).
 - It names its miss rates openly.
- Loses confidence:

- The payments miss 3.2% of the time, 13.7% if the 2028 deposit halves and 40.8% if it never comes. That is not "a high degree of certainty" without outside money (p.3), and it rests on "unstable income" (B8b-Q18).
- It reads the case selectively. It quotes "willing to take risks" and drops "protecting the capital required for her goals" (p.2).
- Its 2031 range has only 50% confidence, and 90% of the surplus goes to the facility. That invites the overpromising the case warns against (p.3) and leaves little flexibility.
- A stock fall in 2031-2032 lands just before the reserve is bought.

What Firm D must change before she would sign (SIMULATION)

1. **Keep the payments independent of the 2028 deposit.** The ten payments should not wait on the 2028 deposit (B8b-Q18; p.3 "may not rely on ... outside funding"). The plan should say plainly which payments go unbought if the ladder costs more than \$300k, and what the fallback is if the deposit arrives smaller or not at all.
2. **Show she is not being made timid.** State the price of certainty in dollars next to one growth-heavier alternative. Consider giving the post-floor growth money a real stock share rather than 7.5% of the whole portfolio (B1b-Q11; p.2 "appropriate balance"; D13c V1).
3. **Stop fixing the facility gift in 2028.** Call the \$150k zero-coupon bond the bought floor of the 2031 range, and leave the actual facility decision to 2033 (p.3 "no predetermined facility contribution"; flexibility "as the project develops").
4. **Name both certainty gaps.** Payments are certain in U.S. dollars, but the costs are in Taiwan and a fixed \$50k buys less each year. Also state the inflation assumption for projections and facility costs (p.4).
5. **Use plain words.** Define "zero-coupon". Explain why the WInS portfolio copies the January 2028 plan rather than the 2027 one (D13c V12; p.4 says WInS is the implementation of the strategy).

What this teaches

A client does not pick the plan with the most return or the cleverest model. She picks the one that keeps the promise other people will rely on, that she can follow and check herself, and that names its own costs. The runner-up plans were not bad. Each lost her confidence on one of those three points: payments that depend on markets (A, E, B), a model she cannot check (F), or a portfolio that flatters her (C).

Eval L3: simulated Laura Gao picks a firm

SIMULATION. This is an AI playing Laura, not Laura. It uses only the case (competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt, cited as "case L"), D13a VERIFIED-PRIMARY quotes (cited by id) and the D13c voice map. Every "she would..." is an inference. We hold no verified statement by her about investing (D13c V8). Serves: IPS, because the 2031 range and certainty rules are frozen on Nov 6. Relayed user message: "just cut off the unnecessary bits". Applied by keeping this file short.

Ranking

rank	firm	score /10	one-line verdict (simulation)
1	D	8.0	The payments and the 2031 floor are bought before anything is promised, and the costs are stated. Growth is thin.
2	A	6.0	Plain and honest glide path. Payments stay exposed to markets until 2033, and the range is wide.
3	E	5.5	Simple, and it under-promises at the top of the range. The bonds do not match the payments, and payments rely on the 2028 deposit.
4	B	4.0	Takes her risk-taking at face value but puts the risk on the people who will rely on her. Its range is a coin flip.
5	F	3.5	A black box with a range too wide to announce.
6	C	3.0	Themed funds "linked to Laura's public work" turn her work into decoration.

The same gap appears in all six firms. None states an assumption about inflation or about facility costs in Taiwan, although the case asks for both (case L146-147). None says what a fixed \$50k (case L90) buys by 2042.

Firm by firm

1. Firm D (8.0)

Earns confidence:

- **Payments bought in January 2027**, each by a Treasury zero maturing before its payment date. Certainty is defined as "bought", not forecast. Case L91 asks for "a high degree of certainty", and case L97-99 asks teams to define it and say how they checked it. D13a-N01: letting down "people who were your earliest supporters" is worse than letting yourself down.
- **The 2031 floor (\$150k) is already owned when she speaks to co-sponsors.** The contribution rule (section 5) caps the gift at the top of the range. Case L114-115: "If Laura promises more than she can ultimately contribute, she could damage her credibility".
- **It names its own gaps:**
 - about 1 in 3 rate paths push the ladder above \$300k;
 - the unfunded part of the 2033 payment is \$12k after a half-point rate fall and \$31k after a one-point fall, if the 2028 deposit is also missing;
 - the price of certainty is a p95 surplus of \$250k.

This matches B8b-Q18 (she names the cost, "unstable income", before she leaps) and B9b-Q09 (she labels her own results "[unfinished]").

- **\$32k is kept in T-bills after 2033.** Case L102-103: "committing all remaining assets could limit her financial flexibility".

Loses confidence:

- **The average stock share is 7.5%**, with only about \$40k in one stock fund. Case L69-74 says she "has been willing to take thoughtful risks" and wants "an appropriate balance". B1b-Q11: "you should always take the jump". On paper this reads as timid unless the firm says where the leap is.
- **The whole 2031 floor depends on the 2028 deposit**, which comes from "publishing advances, speaking engagements, licensing" (case L44-45) and from income she herself called "unstable" (B8b-Q18). A missing deposit means no floor and no stock fund (D's own risk 3).

- **Spurious precision:** "\$294,387" and "7.5%" are stated to the dollar and the tenth of a point. D13c R5 warns against numbers more precise than the model behind them.

2. Firm A (6.0)

Earns confidence:

- **Easy to follow:** five funds, a published glide path, and Treasuries bought in stages from 2030 (D13c V12: she taught financial literacy to high-school students, B2b-Q08).
- **Candid about its range:** "its bottom is less than half its median".
- **Growth is real:** the average stock share is 44%, which answers "thoughtful risks" (case L69-74).

Loses confidence:

- **Payments depend on markets until 2033:** a 0.7% miss rate, 8.6% if the 2028 deposit is \$75k and 44% with no deposit. Case L91 requires funding "with a high degree of certainty", and the deposit is "unstable" income (B8b-Q18). D13a-N01 applies: the downside lands on residents, not on her.
- **The range bottom has a 10% model chance of being missed.** Case L114-115 (overpromising). The bottom is a model percentile, not money she owns.
- **"High certainty" rests on a model probability.** She would ask which model (B9b-Q09, D13c R5).

3. Firm E (5.5)

Earns confidence:

- **Its range tops out at the median,** so a result above the top is as likely as not. That under-promises (case L114-115).
- **Three ETFs,** which makes it the simplest firm.
- **It admits that its bond fund "is not matched to the payment dates."**

Loses confidence:

- **The in-range confidence is only about 40%** (90% reach the bottom, 50% exceed the top). Case L117-118 asks how confident the team is that her contribution "will fall within that range".
- **The miss rate is 1.1%, rising to 9.6% at a \$75k deposit** (B8b-Q18, D13a-N01, as for A).
- **The broad bond fund holds corporate and mortgage bonds,** which adds credit risk to money meant for a nominal promise.

4. Firm B (4.0)

Earns confidence:

- **Honest numbers,** including a 40.8% miss rate with no deposit.
- **Plain language and four trades.**
- **It takes her risk-taking seriously** (B1b-Q11).

Loses confidence:

- **It reads the case as "willing to take risks"** and drops "thoughtful" and "appropriate balance ... protecting the capital required for her goals" (case L69-74).
- **The miss rate is 3.2% (13.7% at \$75k),** the weakest payment protection of the six. It fails case L91 in spirit and puts the jump on the people who rely on her (D13a-N01).
- **The 2031 range has 50% confidence,** so half the time her contribution lands outside the range she announced (case L114-118).
- **A 2031-32 stock fall lands just before the reserve is bought.** Case L119-120 says the range must "protect the portfolio's ability to fund the ten-year operating commitment".

5. Firm F (3.5)

Earns confidence:

- **The best payment protection among the market-risk firms:** a 0.3% miss rate.
- **A defined loss limit.**

Loses confidence:

- **An optimiser with 12 funds, factor tilts and gold, reset every quarter**, which the firm itself calls "sensitive to small errors". D13c R4 and R5 warn against this. B8b-Q13 says she scrapped drafts that "lost the heart" of her work. Our extension is that complexity is not a virtue.
- **The range runs from \$59k to \$317k**. A top more than five times the bottom is not "a credible range" (case L115-116) that co-sponsors can plan around.
- **Nothing in the case calls for gold or factor tilts**. Hype risk (D13c R12, B8a-Q37).

6. Firm C (3.0)

Earns confidence:

- **70% of its stock money is a plain world fund.**
- **It admits that its theme funds are more volatile than its model assumes.**

Loses confidence:

- **30% of the stock money goes to funds "linked to Laura's public work"**: creator economy, Taiwan, clean energy and ESG. The case names no values screen. Clean energy does not appear in her public work. D13c R3: identity or heritage as a reason for a holding is a strong on-sight rejection.
- **A creator-economy fund doubles her career risk**. The 2028 deposit comes from publishing, speaking and licensing (case L44-45), so a creator-economy fund can fall just when that income weakens.
- **The range runs from \$65k to \$335k, and the miss rate is 1.5% (10.6% at \$75k)**, the same weaknesses as A, with more model error.

What Firm D must change before simulated Laura signs

1. **Say where the leap is, and price the growth share**. The residency is the leap. Show one alternative with a smaller 2028 floor and more in stocks, give its p5 and p50, and explain why D chose its version. Without that, the 7.5% stock share reads as timid (case L69-74; B1b-Q11).
2. **Add a rule for a smaller or late 2028 deposit**. Set the order: complete the payments first, buy the floor next, then fund the stock fund. The 2031 bottom is whatever the deposit actually bought. Announcing in 2031, after the deposit, keeps that honest (B8b-Q18; case L119-120).
3. **Give the in-range confidence directly** (case L117-118). Under D's cap rule the 2033 contribution always falls inside the range if the 2028 deposit arrives; the top is reached in about 73% of model paths. Also say how meaningful \$150k-\$175k is next to the cost of a facility (case L111-112), without estimating that cost (case L164-165).
4. **State inflation and Taiwan cost assumptions**, and what a fixed \$50k buys by 2042 (case L90, L146-147).
5. **Round the numbers and grade each one** as a market price or a model estimate. Drop "\$294,387"-style precision (D13c R5; B9b-Q09).

What this teaches

A client may admire risk-taking in her own career and still refuse risk that falls on people who trust her. The best plan here does not rank first for taking the least risk. It ranks first because it makes a promise only after the money for that promise is bought.

Part 3 - The research record, phase by phase

Everything the run produced, in the order it was made. Later files and audit corrections override earlier ones.

1. Run manifest (README)
2. Phase A - Source access check
3. Phase A - Case register (A1)
4. Phase A - Fact register (A2)
5. Phase A - Stakeholder map (A3)
6. Phase A - WInS week-1 guardrails (A4)
7. WInS - S1 Treasury and cash sleeve
8. WInS - S2 Growth sleeve
9. WInS - S3 Ladder and reserve detail
10. WInS - S4 Red team of ticket v0
11. WInS - Ticket v0 (week-1 allocation)
12. WInS - Trading-now brief
13. WInS - Ticket v1 (both book options)
14. WInS - T2 Red team of ticket v1
15. Phase B - B1a Case anomalies
16. Phase B - B1b Case anomalies
17. Phase B - B2a Why Laura / designer intent
18. Phase B - B2b Why Laura / designer intent
19. Phase B - B3a Laura's world and income
20. Phase B - B3b Laura's world and income
21. Phase B - B4a Practitioner benchmark
22. Phase B - B4b Practitioner benchmark
23. Phase B - B5a 2026 macro and markets
24. Phase B - B5b 2026 macro and markets
25. Phase B - B6a Co-sponsors and philanthropy
26. Phase B - B6b Co-sponsors and philanthropy
27. Phase B - B7a Competition and judges
28. Phase B - B7b Competition and judges
29. Phase B - B8a Voice of Laura
30. Phase B - B8b Voice of Laura
31. Phase B - B9a Client synthesis
32. Phase B - B9b Client synthesis

33. Phase B - B10a Contrarian
34. Phase B - B10b Contrarian
35. Phase C - Filter log
36. Phase C - Librarian notes
37. Phase D - D1 Rates and fixed income
38. Phase D - D2 Equity and AI concentration
39. Phase D - D3 Quant answers
40. Phase D - D3 Model v2 results
41. Phase D - D4 Taiwan, FX and geopolitics
42. Phase D - D5 Co-sponsors
43. Phase D - D6 Client psychology
44. Phase D - D7 Wharton intent
45. Phase D - D8 Professional practice
46. Phase D - D9 Communication
47. Phase D - D10 Compliance
48. Phase D - D12 Client-side overflow
49. Phase D - Audit: rates and quant (AX1)
50. Phase D - Audit: D1 rates (AX1b)
51. Phase D - Audit: markets and rules (AX2)
52. Phase D - Audit: client and co-sponsors (AY1)
53. Phase D - Audit: judges, practice, communication (AY2)
54. Phase D - D13a Laura's quotes verified
55. Phase D - D13b Other quotes verified
56. Phase D - D13c Voice-of-Laura map
57. Phase E - E1 Change proposals (architect)
58. Phase E - E2 Red team: semifinal reader
59. Phase E - E3 Red team: Laura (simulation)
60. Phase E - E4 Red team: rival strategist
61. Phase E - E5 Pre-mortem
62. Final verification of the deliverables

insight_v1: "Leave nothing unconsidered" research run (AI-generated brainstorming)

What this is. The output of the locked ultracode prompt v4 (`research/ultracode_prompt_v4_deep_strategy.md`), run on 2026-09-27 by Claude Code (an AI assistant) as a multi-agent research program for Team Caplet. It stress-tests the provisional "lock early" strategy for the 2026-27 client Laura Gao and turns what it finds into ranked, evidence-backed proposals **for the team to approve or reject**.

AI-use statement (Wharton policy). Everything in this folder is AI-generated research and brainstorming. None of it is submission-ready text, and none of its wording may be pasted into the Trading Notes, IPS or Final Report. If the team uses an idea from here, the Final Report should credit AI assistance (Claude Code) as Wharton's AI policy requires. The six students make every decision and write every deliverable.

How the run works

Phase	Folder	What happens
Setup	<code>_context</code> /	<code>brief.md</code> : the single context file every agent reads first
A Foundations	<code>phase_A/</code>	case register (every official requirement quoted), fact register, stakeholder map, WInS week-1 guardrails
B Questions	<code>phase_B/</code>	8 lenses x 2 agents generate anchored questions (plus Laura's own words)
C Filter	<code>phase_C/</code>	dedupe, then so-what / anchor / adversarial (2 of 3 skeptics) / simplicity tests; counts logged
D Research	<code>phase_D/</code>	specialists answer surviving questions with sources and Python; auditors re-check
E Integration	<code>phase_E/</code>	strategy architect, red team (judge, Laura, rival), pre-mortem, chief strategist
F Completeness	<code>phase_F/</code>	critic loop until two rounds add nothing new
Outputs	this folder	the 12 files listed in v4 Part 7
Scripts	<code>scripts/</code>	every Python model used; run with <code>.venv/bin/python</code> from the repo root

Status labels used everywhere

- **VERIFIED-PRIMARY**: read on the primary source (URL + access date given).
- **VERIFIED-REPO-FILE**: from an official or primary file in this repo (path given).
- **SNIPPET-UNVERIFIED**: from a search snippet or secondary source; not confirmed on the primary page.
- **ASSUMPTION**: a modelling or judgement input we chose, with the reason.
- **PARAPHRASE-UNVERIFIED**: words attributed to a person that could not be confirmed verbatim.

Source access

See `phase_A/source_access.md`. Network access was widened by the team on 2026-09-27, so primary sites were reachable.

What this teaches

A research program is only as good as its labels: a number without a source and status is an opinion. Separating "what the case says", "what we verified", "what we assumed" and "what we think it means" is the habit judges (and Laura, a statistics graduate) will look for.

Source access check

Checked 2026-09-27 11:35 UTC from the run's cloud container (curl through the egress proxy; WebFetch uses the same policy). Earlier the same day almost every primary site was blocked; the team widened network access before the run.

Host	Result
home.treasury.gov	200 reachable
fred.stlouisfed.org	200 reachable
www.ishares.com	200 reachable
am.jpmorgan.com	200 reachable
globalyouth.wharton.upenn.edu	200 reachable
magazine.wharton.upenn.edu	200 reachable
lauragao.com	200 reachable
en.wikipedia.org	200 reachable
poetsandquantsforundergrads.com	200 reachable
wghsinvcomp.smapply.us	200 reachable (public pages only; logged-in pages not accessible)
www.federalreserve.gov	200 reachable
nuvoices.com, legacy.diversebooks.org, www.bookweb.org, thenerddaily.com	reachable
www.apra.gov.au	200 reachable
www.futurefund.gov.au	403 refused by the site (also with a browser user agent)

Official competition files in the repo: both PDFs (client case, IPS guide) match the SHA-256 values in `competition/official/2026_27/manifest.yaml`; text extractions are complete.

Python: `.venv` (Python 3.12) with numpy, scipy, pandas, matplotlib, pypdf. Both verified scripts re-run and reproduce their recorded outputs (\$292,264; lock-early surplus p5/p50/p95 \$159k/\$207k/\$273k; growth-first 3.2% miss).

What this teaches

Always record where a fact came from and whether you could open the original. "I couldn't reach the source" is a finding, not a failure; guessing is the failure.

A1 Case Register: every requirement, number, definition, permission, prohibition and word choice

Agent: A1 (Case Lawyer), insight_v1 run, written 2026-09-27. Authority order: case > guides > criteria page > public web pages (brief section 1). This file makes no strategy decisions and contains no text meant for submission.

0. Summary (read this first)

1. **This year's WInS trading rules are public, and the team has not yet recorded them.** Two SMAApply resource pages open without logging in: "Trading Details" (<https://wghsinvcomp.smapply.us/res/p/trading/>) and "FAQs" (<https://wghsinvcomp.smapply.us/res/p/faqs/>). Both were read 2026-09-27 at about 11:40 UTC and are VERIFIED-PRIMARY. They state:
 - **\$300,000** starting virtual cash (R-W56, R-W85). This equals Laura's 2027 deposit. The \$150,000 "will not be added to WInS" (R-W59).
 - Trading runs Sept 28 to Nov 6, 2026 (R-W58).
 - **Up to 200 trades** (R-W65), and no single trade larger than twice a security's daily volume (R-W66).
 - Allowed: stocks priced at \$5 or more, **any ETF available on WInS**, and **Treasury bonds** (U.S., U.K., Germany, France, Italy, the Netherlands) (R-W67 to R-W69, R-W89).
 - **Not allowed:** margin, short selling, stock-secured debt, crypto, derivatives (R-W90).
 - **No required sector allocation or minimum number of sectors** (R-W70). This contradicts the third-party claim "sector minimum = team size" recorded in the brief and CLAUDE.md.
 - Commission: \$25 per stock trade, \$10 per Treasury-bond trade (R-W92).
 - Bond prices update once a day (R-W91).

I found no fixed "approved ETF list"; the rule is "any ETF available on WInS". The team must still check that each instrument is actually available in WInS, and confirm with Wharton anything ambiguous (R-AN15, R-AN49). Confirm against this year's WInS list and rules before trading.
2. **The repo's official PDFs match the live official files byte for byte.** I downloaded the three Box links shown on the public SMAApply pages today. Their SHA-256 hashes equal the repo files and `manifest.yaml` (VERIFIED-PRIMARY). The repo `.txt` extractions are faithful word for word. The only differences are whitespace, bullet control characters, one soft hyphen and two letter-spacing artefacts in pypdf output (section 1.2). The two photos in the case contain a little text that the text file does not have (R-C22).
3. **Official sources disagree on what decides the semifinals.** The Rules page says the Top 50 are picked "based on the strength of their Investment Policy Statement (IPS) and Final Reports" (R-W24). The SMAApply page says the Trading Notes Analysis, IPS and Final Report "are evaluated" (R-S9). The case says "Evaluators will consider the three deliverables together" (R-C91). See X-8.
4. **Source-citation duties are split across deliverables.** The IPS forbids "formal citations" (R-I55). The Code of Conduct requires teams to "properly cite any sources used" (R-W30). The AI policy page says AI use "must be recorded in your Works Cited pages" (R-W46). The only compatible reading (INTERPRETATION): the sources, and a record of how AI was used, go in the Final Report.
5. **Contacting the client means disqualification** (R-W16, R-W55). Wharton also says "While the client is real, the financial scenario is developed specifically for the competition" (R-W4). Laura's real finances and real plans are not case facts.
6. **Notable case-design facts** (details in section 7):
 - Contributions total \$450,000, less than the \$500,000 of nominal payments (R-AN13).
 - The co-sponsor confidence statement is about the contribution falling **within** a range, which has two ends (R-AN9).
 - The case never names a currency (R-AN10).
 - The first payment falls on the same date the reserve is set aside (R-AN2).
 - An SMAApply FAQ lists permitted investments "for BOTH contributions" (R-AN15).

7. **Field size, 2025-26** (Wharton news page, VERIFIED-PRIMARY): "more than 6,300 registered teams"; "some 2,300 teams from 79 countries submitted final reports" (R-W97 to R-W99). The main page counter gives 2,339 teams (R-W10). Last year's WInS cash was \$500,000; this year it is \$300,000.

Entry count: 324 register entries (C 97, I 61, T 36, S 30, W 100), plus 60 anomalies and 24 cross-document items.

Legend

- Status labels:
 - VERIFIED-REPO-FILE: quoted from the official file in `competition/official/2026_27/`.
 - VERIFIED-PRIMARY: quoted from the public Wharton or SMaply page, read by me on 2026-09-27 (URL given).
 - INTERPRETATION: my reading; never a fact.
 - DERIVED: arithmetic on quoted numbers.
- Location: page (p.) and line (L) in the repo `.txt` file. For the SMaply `.md` file, the md line number. For web pages, the page section.
- Types: requirement, number, definition, permission, prohibition, format, evaluation, framing, word-choice.
- Quotes keep the source's curly apostrophes (') where the source uses them.

1. Method and source checks

1.1 Files and hashes

- Official PDFs, repo SHA-256:
 - Client_Profile `bfb653cb...e1fead7`
 - IPS guide `d4aeaca7...06df6a`
 - Trading Notes guide `72bbc4a9...d05d52`

These match `manifest.yaml`. VERIFIED-REPO-FILE.

- Live downloads, 2026-09-27, <https://upenn.box.com/shared/static/<id>.pdf>:
 - Client case id `i4t4b2n4ycjkev7p78mhyjsvypombkxo`, linked from <https://wghsinvcomp.smaply.us/res/p/client/>
 - IPS id `0iib496jhswp30c12zhn6b2fret8dl30`
 - Trading Notes id `3pxn798a1lnlifkikj4g56srmg2ijuo6`

The last two are linked from <https://wghsinvcomp.smaply.us/res/p/deliverables/>. All three hashes are **identical** to the repo files. VERIFIED-PRIMARY.

- PDF metadata (pypdf):
 - Case: created 2026-09-10 15:19 (-04:00), Adobe InDesign 21.4, 4 pages.
 - IPS guide: created 2026-09-09 17:01 (-04:00), 6 pages.
 - Trading Notes guide: created 2026-09-10 15:21 (-04:00), 2 pages.

All are US Letter (612x792 pt) with no hyperlink annotations. VERIFIED-REPO-FILE.

- The SMaply Deliverables page is **publicly readable** at <https://wghsinvcomp.smaply.us/res/p/deliverables/> (read 2026-09-27). Its text matches the repo transcription `SMaply_Deliverables_Page_2026-09-27.md`. The only differences: the live page uses curly apostrophes, and its "Click Here" links point to the Box URLs above. VERIFIED-PRIMARY. The repo file's header calls it a "registered-team portal" page, but it opens without logging in.

1.2 Text-extraction check (case, IPS guide, Trading Notes guide)

I ran a word-level diff between the repo `.txt` files and a fresh pypdf extraction (`/home/user/caplet-investment/.venv/bin/python, pypdf`). No word is added, dropped or changed in meaning. VERIFIED-REPO-FILE. The differences:

File	Repo .txt	Fresh pypdf	Meaning changed?
Case p.1 L4	"teacher/advisor"	"teacher/ advisor" (line-wrap space)	No

File	Repo .txt	Fresh pypdf	Meaning changed?
Case p.1 L18	"per" (soft hyphen U+00AD) + "sonal"	"per-" + "sonal"	No (the word is "personal")
Case p.4, IPS p.1-2 bullets	Some bullets start with a BEL control character (U+0007) before the word	Plain bullet	No; a search for "\x07" hits these
IPS p.4 L128	"ONLY"	"ONL Y" (letter-spacing artefact)	No
Trading Notes p.1 L3, L32	"Team's", "ANALYSIS"	"T eam's", "ANAL YSIS" (kerning artefacts)	No
All	Blank lines between bullets	Bullets on consecutive lines	No

The images carry text that is not in the text files (checked by viewing the two embedded JPEGs):

- Case p.1 has a portrait photo with no text.
- Case p.2 has a photo of Laura at a book display. It shows the *Messy Roots* cover with the subtitle "A Graphic Memoir of a Wuhanese American" and a conference badge marked "SPEAKER".
- The pull quote on p.1 (R-C21) sits in the "MEET LAURA GAO" sidebar. The text layer gives it no attribution line.

2. Case: `Laura_Gao_2026_Client_Profile.txt` (R-C)

All entries VERIFIED-REPO-FILE unless marked.

Page 1: framing and biography

- **R-C1** · p.1-4 L2/35/82/122 · format · "Investment Competition Guide" — Every page of the case has this header, the same header as the Trading Notes guide. The IPS guide uses a different one.
- **R-C2** · p.1 L3 · framing · "You are a team of young analysts working at an asset management company." — The role-play frame: we are professionals at a firm.
- **R-C3** · p.1 L4-5 · framing · "Your portfolio manager (your team's teacher/advisor who makes the final investment decisions for your firm's portfolio)" — In the fiction, the advisor decides. The real rules forbid this (R-W19, R-W87; see R-AN4).
- **R-C4** · p.1 L5 · word-choice · "recently met with a potential client, Laura Gao" — "Potential": she has not hired the firm yet.
- **R-C5** · p.1 L5-6 · definition · "a bestselling author, illustrator, entrepreneur, and educator, who is planning the next chapter of her career" — Her roles as the case lists them. "Next chapter" is a book metaphor.
- **R-C6** · p.1 L6-7 · framing · "Laura has built a successful creative business by turning ideas into opportunities" — Her business is creative and built on ideas.
- **R-C7** · p.1 L7-8 · framing · "she is now seeking thoughtful financial planning to help her achieve her long-term goals." — "Thoughtful" is the first of three uses in the case (R-C24, R-C37).
- **R-C8** · p.1 L9-10 · framing · "Your team hopes to develop the investment strategy that Laura ultimately chooses as she works toward achieving her future vision." — She will compare strategies and choose one. Ours must win her choice.
- **R-C9** · p.1 L12 · framing · "Laura Gao believes stories have the power to change how people see the world." — The first thing the case says about her is a belief, not a fact about money.
- **R-C10** · p.1 L13-14 · definition · "Born in Wuhan, China, and raised in Texas, Laura is a bestselling graphic novelist, illustrator, entrepreneur, and educator whose work explores identity, belonging, and the power of storytelling." — Official bio and themes of her public work. The brief allows the themes; never use them to pick investments.
- **R-C11** · p.1 L14-15 · framing · "While studying at the Wharton School, she combined her passion for creativity with an entrepreneurial mindset and launched a small business." — She was an entrepreneur while still a student.
- **R-C12** · p.1 L16-17 · number/definition · "She graduated in 2018 with a degree in Statistics & Information Decisions Management before beginning her career as a product manager in the technology industry." — 2018; a quantitative degree; a tech product manager.

- **R-C13** · p.1 L18-19 · number · "In 2020, Laura published *The Wuhan I Know*, a comic inspired by her personal experiences during the COVID-19 pandemic." — 2020, her first public work named in the case. The line is broken by a soft hyphen.
- **R-C14** · p.1 L19-20 · framing · "Originally intended as a response to misinformation and anti-Asian racism" — The comic was made to correct misinformation.
- **R-C15** · p.1 L20-22 · framing · "the comic resonated with readers worldwide and ultimately inspired her bestselling graphic memoir, *Messy Roots*." — A small project grew into a larger one.
- **R-C16** · p.1 L22-23 · framing · "Today, her books are read in classrooms around the world and have sparked conversations about identity, belonging, and community." — Education reach. "Community" appears again at R-C40.
- **R-C17** · p.1 L24-25 · framing · "Although Laura's career began in business and technology, she has always approached creativity with an entrepreneurial mindset." — Business and creativity together.
- **R-C18** · p.1 L25-26 · framing · "she has combined artistic passion with strategic thinking and a willingness to pursue unconventional opportunities." — Willing to take unconventional paths.
- **R-C19** · p.1 L27 · framing · "That philosophy continues to guide Laura as she considers what comes next." — Her approach to career choices guides what comes next, but R-C37/R-C38 add a limit.
- **R-C20** · p.1 L28-30 · framing · "MEET LAURA GAO / Storyteller, Entrepreneur, and Creative Visionary" — Sidebar headline. "Storyteller" comes first.
- **R-C21** · p.1 L31-33 · framing · "The only person who needs to believe in something is yourself." — Pull quote in the sidebar, with no attribution line in the text layer (see R-AN30).
- **R-C22** · p.1-2 images · framing · Image text (not in the .txt): "A Graphic Memoir of a Wuhanese American" (the *Messy Roots* cover subtitle) and a "SPEAKER" badge — Shows the public roles of author and speaker. VERIFIED-REPO-FILE (viewed the embedded JPEGs).

Page 2: goals and money

- **R-C23** · p.2 L36-37 · framing · "Looking Ahead" / "Laura has no shortage of ideas for the future." — Many ideas; the portfolio serves the chosen ones.
- **R-C24** · p.2 L37-39 · framing · "she recognizes that achieving ambitious goals requires more than creativity alone. Thoughtful financial planning and long-term investing will play an important role in turning those ideas into reality." — She sees finance as the thing that turns her ideas into reality.
- **R-C25** · p.2 L39-41 · framing · "Working with your portfolio manager, Laura has identified several long-term financial objectives that reflect both her entrepreneurial mindset and her passion for innovation." — The goals are already set with the portfolio manager (PM). Teams do not choose her goals.
- **R-C26** · p.2 L43 · number · "At the beginning of 2027, Laura plans to invest \$300,000 with an asset management firm." — \$300,000 on 1 Jan 2027. "An" firm, not "your" firm.
- **R-C27** · p.2 L43-45 · number · "She will contribute an additional \$150,000 at the beginning of 2028, using earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures." — \$150,000 on 1 Jan 2028. The case says "will", not "may". All the named income sources come from her own work.
- **R-C28** · p.2 L46-47 · definition · "For purposes of the competition, teams should use the following timeline. Year 0 is 2026, the present, before any money has been invested." — Year 0 = 2026 = now.
- **R-C29** · p.2 L49-57 · number · "2026 is Year 0." "2027 is Year 1." "2028 is Year 2." "2031 is Year 5." "2033 is Year 7." — Only the event years are listed.
- **R-C30** · p.2 L58 · definition · "Any year not listed follows the same pattern: subtract 2026 from the calendar year to get its year number." — So 2042 = Year 16 (DERIVED).
- **R-C31** · p.2 L59 · definition · "All contributions and withdrawals occur at the beginning of the applicable year." — Every cash flow happens on 1 January.
- **R-C32** · p.2 L59-60 · word-choice · "The ten operating payments are therefore spaced exactly one year apart." — Mentions the payments before the case has introduced them (R-AN12).
- **R-C33** · p.2 L61-63 · definition · "Laura's living expenses and short-term financial needs will be covered by income and financial resources outside the portfolio." — No living-cost withdrawals. She has other resources.

- **R-C34** · p.2 L63-65 · prohibition · "Apart from the two contributions described above, she will neither add to nor withdraw from the portfolio before 2033." — No other money moves in or out before 2033.
- **R-C35** · p.2 L66-67 · requirement · "The portfolio should therefore be managed to support the goals described below." — The portfolio exists for the goals that follow.
- **R-C36** · p.2 L68-69 · framing · "Laura understands that investing involves uncertainty and periods of market volatility." — She accepts that markets move.
- **R-C37** · p.2 L69-71 · word-choice · "Although she has been willing to take thoughtful risks throughout her entrepreneurial career" — "Although" contrasts her career risk-taking with what she wants now (R-AN16).
- **R-C38** · p.2 L71-74 · requirement · "she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals." — The team recommends the balance. What is protected is the capital "required for her goals".
- **R-C39** · p.2 L76 · number · "In 2033, Laura plans to establish a collaborative creative residency in Taiwan." — 2033; Taiwan; "plans".
- **R-C40** · p.2 L76-80 · definition · "She envisions a small, community-oriented space where artists, writers, designers, entrepreneurs, and educators can temporarily live, work, teach, and collaborate." — Small and community-oriented; stays are temporary.

Page 3: the commitment, the reserve, the facility, co-sponsors

- **R-C41** · p.3 L83 · definition · "Establishing the residency will require funding for the facility as well as reliable support for its early operations." — Two needs: the facility, and operations. The ten years are called "early operations".
- **R-C42** · p.3 L84 · requirement · "The investment portfolio must fund the operating commitment described below" — The portfolio must fund it.
- **R-C43** · p.3 L84-85 · permission · "and may also cover part of the facility's cost." — The facility is optional and partial ("part").
- **R-C44** · p.3 L85-86 · permission · "Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources." — Outside money is allowed for the rest of the facility and for extra operating support. The list includes "continued business income".
- **R-C45** · p.3 L88-89 · number · "Laura will make ten annual payments of \$50,000 toward the residency's operating expenses, one at the beginning of each year from 2033 through 2042." — $10 \times \$50,000$, on 1 Jan 2033 to 1 Jan 2042.
- **R-C46** · p.3 L90 · definition · "For purposes of the competition, each payment is a fixed \$50,000 and is not adjusted for inflation." — Payments are fixed in nominal dollars (not raised for inflation).
- **R-C47** · p.3 L91 · requirement · "All ten payments must be funded by the investment portfolio with a high degree of certainty." — The core requirement. "High degree of certainty" is not defined (R-AN1).
- **R-C48** · p.3 L91-92 · prohibition · "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement." — No outside money for the ten payments.
- **R-C49** · p.3 L92-93 · definition · "Funding the residency beyond the final payment in 2042 is outside the scope of the competition." — Nothing after 2042.
- **R-C50** · p.3 L94-95 · requirement · "At the beginning of 2033, before making the first operating payment or contributing to the facility, Laura will set aside a portion of the portfolio to fund the ten payments." — Order on 1 Jan 2033: reserve first, then the payment and the facility (R-AN2).
- **R-C51** · p.3 L95 · definition · "This set-aside is called the operating reserve." — The official term.
- **R-C52** · p.3 L95-96 · requirement · "Teams must recommend its size and initial asset composition" — Two deliverable items: how big, and what it holds at first.
- **R-C53** · p.3 L96-97 · requirement · "and explain how its composition should change, if at all, as the annual payments approach and are made." — "If at all": a reserve that never changes is allowed (R-AN27).
- **R-C54** · p.3 L97-98 · requirement · "They should also define what they consider a high degree of funding certainty" — The team writes the definition. Note "funding certainty" here versus "certainty" at L91.

- **R-C55** · p.3 L98 · requirement · "explain how they evaluated that level of certainty" — The method must be shown.
- **R-C56** · p.3 L98-99 · requirement · "and identify the assumptions supporting their recommendation." — Assumptions must be listed.
- **R-C57** · p.3 L101-102 · requirement · "After establishing the operating reserve in 2033, Laura must decide how much of the remaining portfolio she can responsibly contribute toward the cost of establishing the residency facility." — The facility money comes only from what is left after the reserve. Laura decides.
- **R-C58** · p.3 L102-103 · framing · "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops." — Flexibility is about the project's future needs.
- **R-C59** · p.3 L104 · definition · "There is no predetermined facility contribution." — No target amount is given.
- **R-C60** · p.3 L104-106 · requirement · "Teams must recommend and justify the extent to which Laura can responsibly contribute, based on their investment strategy, projected portfolio outcomes, assumptions, and understanding of her goals." — The recommendation must rest on four bases. "Extent" could be a share or a rule, not only a dollar amount (R-AN20).
- **R-C61** · p.3 L106-107 · permission · "Teams are not expected to determine the size or investment composition of a separate contingency fund or endowment." — Not required. It is not forbidden either.
- **R-C62** · p.3 L109 · number · "Laura plans to begin approaching potential co-sponsors in 2031, two years before the residency is established." — 2031 = Year 5 = two years before 2033.
- **R-C63** · p.3 L110 · requirement · "At that time, she will need to describe how much she expects to contribute toward the facility in 2033." — A 2031 statement about 2033.
- **R-C64** · p.3 L110-112 · framing · "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute." — Three reasons a contribution helps. "May" is hedged.
- **R-C65** · p.3 L112-113 · framing · "In 2031, however, the value of her portfolio in 2033 remains uncertain." — The case states the uncertainty outright.
- **R-C66** · p.3 L114-115 · framing · "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors." — Overpromising is the named risk.
- **R-C67** · p.3 L115-116 · requirement · "Rather than promising one exact amount, she wants to communicate a credible range for her potential contribution." — A range, not a single number.
- **R-C68** · p.3 L117 · requirement · "Teams must recommend the dollar range Laura should communicate" — Expressed in dollars.
- **R-C69** · p.3 L117-118 · requirement · "and state how confident they are that her 2033 contribution will fall within that range." — A confidence level for landing "within" the range, which has two ends (R-AN9).
- **R-C70** · p.3 L118-119 · requirement · "They should explain how favorable and unfavorable market outcomes could affect the amount she can provide." — Good and bad scenarios.
- **R-C71** · p.3 L119-120 · requirement · "The proposed range must also protect the portfolio's ability to fund the ten-year operating commitment." — The promised range cannot put the payments at risk.

Page 4: strategy rules, deliverables, scope

- **R-C72** · p.4 L123-124 · requirement · "Each team must also draft part of Laura's fundraising materials describing this potential contribution for prospective co-sponsors." — Students write "part" of her fundraising material (a Final Report item; students write it themselves, per R-W26).
- **R-C73** · p.4 L126 · framing · "There is no single correct strategy for achieving Laura's goals." — Many valid answers.
- **R-C74** · p.4 L126-128 · permission · "Teams may reach different conclusions about risk, asset allocation, liquidity, return expectations, funding confidence, the facility contribution, and the financial flexibility Laura should preserve." — Seven dimensions where answers may differ.
- **R-C75** · p.4 L129 · definition · "The WInS portfolio represents each team's implementation of its investment strategy during the competition." — WInS = putting the strategy into practice.

- **R-C76** · p.4 L130 · definition · "It does not determine Laura's actual portfolio value at the beginning of 2027." — WInS results never become her starting value.
- **R-C77** · p.4 L130-132 · requirement · "For long-term projections, teams should begin with Laura's \$300,000 investment in 2027, add the \$150,000 contribution in 2028, and use reasonable return assumptions consistent with their strategy." — The projection recipe.
- **R-C78** · p.4 L132-133 · prohibition · "Gains or losses generated during the WInS trading period should not be added to or subtracted from Laura's portfolio projections." — Keep WInS profit and loss (P&L) out of projections.
- **R-C79** · p.4 L134 · requirement · "Each team must develop and justify an investment strategy that:" — Five tests follow.
- **R-C80** · p.4 L136 · requirement · "Supports the ten-year operating commitment with a high degree of certainty." — Test 1.
- **R-C81** · p.4 L138 · requirement · "Determines a responsible facility contribution for the residency." — Test 2.
- **R-C82** · p.4 L140-141 · requirement · "Addresses how investment uncertainty could affect both the operating commitment and the facility contribution." — Test 3: uncertainty for both goals.
- **R-C83** · p.4 L143 · requirement · "Communicates Laura's potential facility contribution to co-sponsors clearly and credibly." — Test 4. Almost the same words as the Creativity criterion (R-S28).
- **R-C84** · p.4 L145 · requirement · "Preserves appropriate financial flexibility when determining the facility contribution." — Test 5.
- **R-C85** · p.4 L146-148 · requirement · "Teams should identify and explain their assumptions about investment performance, the timing of cash flows, outside funding, the effect of inflation on portfolio projections and facility costs, and the financial flexibility Laura should preserve." — Five assumption areas, including inflation's effect on facility costs (R-AN8).
- **R-C86** · p.4 L148-149 · requirement · "They should also consider how favorable and unfavorable investment outcomes would affect their recommendations." — Scenario analysis.
- **R-C87** · p.4 L150 · definition · "Teams will demonstrate their work through three connected deliverables:" — "Connected".
- **R-C88** · p.4 L152-153 · definition · "Trading Notes Analysis. This shows how selected investment decisions reflected, tested, or refined the team's developing strategy." — The strategy may be "developing" at this stage.
- **R-C89** · p.4 L155-156 · definition · "Investment Policy Statement. This formally establishes the strategy and the team's decision-making framework." — "Formally establishes" and a "decision-making framework".
- **R-C90** · p.4 L158-160 · definition · "Final Report. This evaluates how that strategy was implemented and presents the team's analysis and recommendations for Laura's operating commitment, facility contribution, financial flexibility, and communication with co-sponsors." — The Final Report carries the numbers.
- **R-C91** · p.4 L161 · evaluation · "Evaluators will consider the three deliverables together." — The deliverables are judged as one package (compare R-W24).
- **R-C92** · p.4 L161-162 · requirement · "Teams should use each deliverable for its intended purpose while maintaining a clear and consistent investment strategy across all three." — One strategy across all three.
- **R-C93** · p.4 L162-163 · number · "Detailed Final Report requirements will be released at the beginning of Week 7." — Week 7 = 9 Nov (R-S19).
- **R-C94** · p.4 L164 · definition · "The total cost of the facility has not been determined." — Facility cost is unknown by design.
- **R-C95** · p.4 L164-166 · permission · "Teams are not expected to estimate that cost, prepare a construction budget, determine the project's total funding gap, or develop a detailed business plan for the residency." — Four things that are out of scope.

- **R-C96** · p.4 L166-167 · requirement · "Their primary focus should remain on the investment strategy and how much the portfolio can responsibly provide toward Laura's goals." — Stay on investing.
- **R-C97** · p.4 L168-169 · permission · "For purposes of the competition, teams do not need to account for personal income taxes, capital gains taxes, or the legal and regulatory requirements of establishing a residency in Taiwan." — Taxes and Taiwan law are out of scope. Currency is not mentioned (R-AN10).

Derived arithmetic from case numbers (DERIVED; no assumptions)

- $10 \times \$50,000 = \$500,000$ in nominal payments. Contributions total $\$300,000 + \$150,000 = \$450,000$. The gap is \$50,000 before any facility contribution.
- Payment dates: Years 7 to 16 (2033 to 2042). The last payment is 9 years after the first.
- From deposit to first payment: 6 years for the \$300,000 (Year 1 to Year 7) and 5 years for the \$150,000 (Year 2 to Year 7).
- $2031 \text{ to } 2033 = 2 \text{ years}$ (Year 5 to Year 7).
- Competition "Year 0" (2026) is when WInS trading happens.

3. IPS guide: 2026_WGY_Investment_Policy-FINAL.txt (R-I)

All entries VERIFIED-REPO-FILE.

Page 1: purpose and prompts

- **R-I1** · p.1-6 header · format · "Investment Policy Statement (IPS)" — The page header, which differs from the case and Trading Notes guide.
- **R-I2** · p.1 L3-4 · definition · "An Investment Policy Statement (IPS) outlines an investor's financial goals, risk tolerance, and guidelines for managing a portfolio." — The official definition includes "risk tolerance" and "guidelines".
- **R-I3** · p.1 L4-5 · definition · "It helps investors stay focused on long-term objectives and make disciplined decisions during changing market conditions." — An IPS is a tool for behaving well in bad markets.
- **R-I4** · p.1 L6-7 · framing · "Your investment strategy is the cornerstone of the Wharton Global High School Investment Competition, and your IPS is where you clearly articulate that strategy." — "Cornerstone". Repeated on the SMAApply page (R-S16).
- **R-I5** · p.1 L8-9 · requirement · "Your strategy should explain how you will invest your client's money to support their financial goals and future cash flow needs while managing risk and uncertainty." — Goals, cash flows, risk.
- **R-I6** · p.1 L9-10 · requirement · "It should reflect your client's goals, time horizons, liquidity needs, desired degree of funding certainty, and need for financial flexibility." — Five items to reflect. "Desired degree of funding certainty" belongs in the IPS itself.
- **R-I7** · p.1 L12 · word-choice · "There is no single "correct" investment strategy for your client." — "Correct" is in quotation marks.
- **R-I8** · p.1 L12-13 · requirement · "Your team should use research and analysis to develop an approach that reflects your client's financial goals, funding needs, and changing time horizons." — "Changing" time horizons.
- **R-I9** · p.1 L16 · requirement · "What is the central idea behind your investment strategy?" — Prompt 1: one central idea.
- **R-I10** · p.1 L18 · requirement · "What principles will guide your investment decisions?" — Prompt 2: principles.
- **R-I11** · p.1 L20 · requirement · "How does your strategy balance growth, risk, liquidity, funding reliability, and financial flexibility?" — Prompt 3: five things to balance.
- **R-I12** · p.1 L22-23 · requirement · "How will the portfolio's asset allocation and composition change as future funding needs approach and payments are made?" — Prompt 4 assumes the allocation changes over time (a glide path; compare the case's "if at all", R-C53).

- **R-I13** · p.1 L25-26 · requirement · "How will your strategy protect the operating commitment while preserving flexibility for a responsible facility contribution?" — Prompt 5: protection versus flexibility.
- **R-I14** · p.1 L27-28 · requirement · "Always remember who your strategy is designed to serve. Your client's profile and financial goals should remain at the center of your investment decisions." — Client first.
- **R-I15** · p.1 L30 · format · "Your IPS submission has two parts." — Pitch + IPS (the title page is extra).
- **R-I16** · p.1 L31-33 · format · "Investment Strategy Elevator Pitch" / "Maximum 50 words" — Hard limit: 50 words.
- **R-I17** · p.1 L35-36 · requirement · "Quickly summarize your team's investment strategy. Clearly communicate the central idea behind your approach and how it is designed to support your client's financial goals and future funding needs." — The pitch must carry the central idea plus how it serves her goals.
- **R-I18** · p.1 L37-39 · format · "Investment Policy Statement" / "Maximum 500 words" — Hard limit: 500 words.
- **R-I19** · p.1 L41 · requirement · "Clearly explain your team's investment strategy and the reasoning behind it." — Strategy plus reasons.

Page 2: content expectations and the freeze

- **R-I20** · p.2 L45-46 · requirement · "Your IPS should demonstrate how your strategy is tailored to your client's financial goals, time horizons, and needs." — Tailoring.
- **R-I21** · p.2 L46-47 · requirement · "Explain how your team will approach risk and return, liquidity, portfolio construction and diversification, and planned changes in the portfolio as future funding needs approach." — Four topics, including "planned changes".
- **R-I22** · p.2 L48-50 · requirement · "Your IPS should also establish your strategic approach to preparing for the operating commitment, managing the operating reserve's asset composition over time, determining a responsible facility contribution, and preserving appropriate financial flexibility." — The approach to the reserve and the facility belongs in the IPS. The numbers do not (R-I31).
- **R-I23** · p.2 L51-52 · requirement · "Focus on the strategy and decision-making framework that guide your portfolio rather than describing individual investments or presenting detailed financial calculations." — Rules and framework, not tickers or maths.
- **R-I24** · p.2 L53-55 · requirement · "Your IPS should make it clear:" / "What your team is trying to accomplish for the client" — Clarity item 1.
- **R-I25** · p.2 L57 · requirement · "Why your strategy is appropriate for the client and their financial goals" — Item 2.
- **R-I26** · p.2 L59 · requirement · "How your strategy balances growth, liquidity, funding reliability, risk, and financial flexibility" — Item 3. Same five words as R-I11, in a different order.
- **R-I27** · p.2 L61-62 · requirement · "How your strategy will guide portfolio construction and investment decisions as the client's needs change" — Item 4: rules that guide future decisions.
- **R-I28** · p.2 L64-65 · requirement · "How your approach will protect the required operating payments while supporting a responsible facility contribution" — Item 5. "Required" payments.
- **R-I29** · p.2 L66 · permission · "You do not need to address these points separately." — No checklist structure is needed.
- **R-I30** · p.2 L66-67 · evaluation · "Your goal is to present a clear, cohesive, and well-supported investment strategy." — Three quality words.
- **R-I31** · p.2 L68-69 · permission · "The IPS is not expected to include final operating-reserve calculations, detailed portfolio projections, a final facility-contribution range, or the draft communication to co-sponsors." — Four items that stay out of the IPS.
- **R-I32** · p.2 L69-70 · requirement · "These elements will be developed and supported in the Final Report using the strategy established in the IPS." — The Final Report numbers must follow from the IPS strategy.
- **R-I33** · p.2 L72 · evaluation · "Your IPS is the foundation of your Final Report and will be evaluated." — The IPS is scored.

- **R-I34** · p.2 L73 · definition · "Once submitted, the IPS becomes the official record of your team's investment strategy." — The IPS is the official record.
- **R-I35** · p.2 L73-74 · prohibition · "Your team may not revise its investment strategy after the submission deadline." — The strategy freezes on 6 Nov.
- **R-I36** · p.2 L75 · requirement · "Your portfolio and Final Report should reflect the strategy established in your IPS." — The WInS portfolio and the Final Report must match the IPS.
- **R-I37** · p.2 L75-78 · definition · "In the Final Report, your team will evaluate how effectively it implemented that strategy and present its analysis and recommendations for the operating commitment, facility contribution, financial flexibility, and communication with co-sponsors under varying investment outcomes." — The Final Report is self-evaluation plus recommendations under different scenarios.

Page 3: format rules (the disqualification rules)

- **R-I38** · p.3 L82 · requirement · "Please follow these directions carefully." — Emphatic.
- **R-I39** · p.3 L82-83 · prohibition · "Submissions that do not meet the requirements below will not be considered for semifinal selection." — Breaking a format rule rules the team out of the semifinals.
- **R-I40** · p.3 L84-85 · format · "Page 1: Title Page" / "1 page maximum" — One page.
- **R-I41** · p.3 L86-90 · format · "Official team name" / "This should be the same as what you submitted with your official team roster." — Must match the roster.
- **R-I42** · p.3 L92-94 · format · "Finalized team member names" / "These should be the same as what was submitted with your official roster." — Must match the roster.
- **R-I43** · p.3 L96 · format · "Format: First Name, Last Initial" — e.g. "Jordan A." (R-I59).
- **R-I44** · p.3 L98-100 · format · "WInS username" / "This is the team's username for logging into the WInS platform." — The username, not the team name (R-W86).
- **R-I45** · p.3 L101 · permission · "Formatting choices not specified below are left to the team's discretion." — Appears after the title-page list, and it is unclear whether it covers the title page or the whole file (R-AN37).
- **R-I46** · p.3 L101-102 · requirement · "Keep the presentation simple and focus on communicating the investment strategy clearly." — Simplicity is asked for directly.
- **R-I47** · p.3 L103-104 · format · "Pages 2 and 3: Investment Strategy" / "2-page maximum" — The pitch and the IPS together fit in 2 pages.
- **R-I48** · p.3 L105-109 · format · "Investment Strategy Elevator Pitch, maximum 50 words" / "Investment Policy Statement, maximum 500 words" — Limits restated.
- **R-I49** · p.3 L112 · format · "Font: Times New Roman" — Required font.
- **R-I50** · p.3 L114 · format · "Size: 12-point" — Required size.
- **R-I51** · p.3 L116 · format · "Spacing: Double-spaced" — Required spacing.
- **R-I52** · p.3 L118 · format · "Margins: 1 inch on all sides" — Required margins.
- **R-I53** · p.3 L120 · format · "Submit as a PDF through SurveyMonkey Apply." — PDF only.
- **R-I54** · p.3 L122 · number · "Maximum file size: 5 MB." — 5 MB cap.
- **R-I55** · p.3 L123 · prohibition · "Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted." — Seven banned elements.
- **R-I56** · p.3 L124 · permission · "Teams should use research and analysis to develop their strategy, but sources do not need to be cited in the IPS." — Research is expected but not cited in the IPS.
- **R-I57** · p.3 L125 · requirement · "Relevant sources and supporting evidence must be included in the Final Report." — Citations move to the Final Report.

Pages 4-6: sample layout

- **R-I58** · p.4 L128 · format · "ILLUSTRATIVE FORMAT EXAMPLE | PLACEHOLDER CONTENT ONLY" — The sample is for layout only.
- **R-I59** · p.4 L129-131 · format · "SAMPLE TEAM NAME" / "Jordan A. | Taylor B. | Casey C. | Morgan D." / "WInS Username: sample_team-1234567" — Sample title page: names separated by pipes, with the label "WInS Username:".

- **R-I60** · p.5 L135/L139 · format · Section headings "Investment Strategy Elevator Pitch" and "Investment Policy Statement" — The sample puts the pitch and the IPS on the same page under two headings.
- **R-I61** · p.5-6 L136-174 · number · Placeholder lengths: pitch 41 words; IPS placeholder about 394 words running onto p.6; 13-15 words per line. DERIVED (word count of the placeholder). This gives a rough guide to how many words fit per line (R-AN39).

4. Trading Notes guide: 2026_WGY_Trading_Notes_Analysis-FINAL.txt (R-T)

All entries VERIFIED-REPO-FILE.

- **R-T1** · p.1-2 header · format · "Investment Competition Guide" — Same header as the case.
- **R-T2** · p.1 L3 · framing · "What Was Your Team's Decision-Making Process?" — The deliverable's title question is about process.
- **R-T3** · p.1 L4-6 · definition · "The purpose of this deliverable is to demonstrate how your team's investment decisions reflected the strategy it developed for the client, including its approach to growth, risk, liquidity, funding reliability, financial flexibility, and future cash-flow needs." — Purpose, plus six topics.
- **R-T4** · p.1 L7 · framing · "Successful investing is not simply about selecting individual stocks." — A warning against stock picking for its own sake.
- **R-T5** · p.1 L7-8 · framing · "It requires making disciplined decisions that consistently support an overall investment strategy." — Discipline and consistency.
- **R-T6** · p.1 L8-9 · requirement · "Throughout the competition, your Trading Notes should document the research, analysis, and reasoning behind your investment decisions." — Every note should record research and reasoning.
- **R-T7** · p.1 L9-10 · evaluation · "Collectively, they provide evidence of how your team has put its strategy into practice." — The notes are evidence.
- **R-T8** · p.1 L11 · requirement · "For this analysis, select three investment decisions that best illustrate your team's strategic thinking." — Three decisions, chosen by the team.
- **R-T9** · p.1 L11-13 · requirement · "Explain the reasoning behind each decision, how it aligned with, tested, or refined your overall investment strategy, and how it supported the client's goals, funding needs, or risk considerations." — Three things to explain.
- **R-T10** · p.1 L14-15 · requirement · "Collectively, the decisions should demonstrate how your team used individual investments as part of a cohesive portfolio strategy." — The three together should show one portfolio logic.
- **R-T11** · p.1 L15-16 · evaluation · "The focus should be on the quality of your reasoning and the intended role of each decision, not simply whether an investment gained or lost value during the competition." — Reasoning over profit and loss.
- **R-T12** · p.1 L17 · number · "Your team will formally present its investment strategy in the Investment Policy Statement (IPS) in Week 6." — IPS = Week 6 (6 Nov).
- **R-T13** · p.1 L18-19 · permission/requirement · "Although your strategy may continue to evolve as you conduct additional research and evaluate new information, you should already be using a clear strategic approach to guide your investment decisions." — Change is allowed before the IPS, but a clear approach is expected already.
- **R-T14** · p.1 L20-22 · permission · "At this stage, your team is not expected to calculate the final size of the operating reserve, complete long-term portfolio projections, determine Laura's facility contribution, or draft her communication to co-sponsors." — Four items not expected.
- **R-T15** · p.1 L22 · definition · "These recommendations will be developed in the Final Report." — Where they go.
- **R-T16** · p.1 L22-23 · requirement · "The Trading Notes Analysis should focus on your investment decisions and their relationship with your developing strategy." — Focus.
- **R-T17** · p.1 L24-26 · permission · "Important" / "The Trading Notes you select do not need to correspond to investments that remain in your portfolio at the end of the competition." — The note trades may be sold later.

- **R-T18** · p.1 L26-27 · permission · "A decision can still demonstrate thoughtful analysis and strategic thinking even if your team later sells the investment or changes its approach." — Changing course is not penalised.
- **R-T19** · p.1 L28 · evaluation · "What matters is that each Trading Note provides evidence of your team's decision-making process." — The note is evidence of process.
- **R-T20** · p.1 L29-30 · requirement · "Your reflections should clearly explain how each decision supported, tested, or refined your overall investment strategy." — Supported / tested / refined.
- **R-T21** · p.1 L30-31 · requirement · "The analysis should be consistent with the strategic approach your team is developing and will later articulate in its IPS." — The notes must be consistent with the future IPS.
- **R-T22** · p.2 L35 · format · "Trading Note Example" — An official sample.
- **R-T23** · p.2 L36-37 · framing · "We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment." — Wharton's own example: a Treasury bond fund for the payments.
- **R-T24** · p.2 L37-38 · framing · "The position provides income and greater stability than equities, although interest-rate changes may affect its value." — The example names a risk of its own trade.
- **R-T25** · p.2 L38-39 · framing · "This trade supports our plan to balance continued growth with reliable future cash flows as the residency funding date approaches." — The example describes shifting towards safer assets as the date nears.
- **R-T26** · p.2 L36-39 · number · The example note is 59 words, about 415 characters. DERIVED. Gives a length scale for a WInS note. No WInS character limit is stated.
- **R-T27** · p.2 L42 · requirement · "Select three (3) Trading Notes from trades your team executed in WInS." — Exactly three, from executed trades.
- **R-T28** · p.2 L44 · requirement · "Include each Trading Note exactly as it appears in WInS." — Copy it word for word, with no edits.
- **R-T29** · p.2 L46 · format · "For EACH Trading Note, write a reflection of 100 words or fewer explaining:" — 100 words or fewer per reflection.
- **R-T30** · p.2 L48 · requirement · "Why your team made the investment decision." — Reflection item 1.
- **R-T31** · p.2 L50 · requirement · "How the decision aligned with your overall investment strategy." — Item 2.
- **R-T32** · p.2 L52 · requirement · "How the decision supported the client's goals, funding needs, or risk considerations." — Item 3.
- **R-T33** · p.2 L53 · requirement · "The Trading Notes must correspond to trades executed in WInS." — Must match real WInS trades.
- **R-T34** · p.2 L53 · word-choice · "Yes, we will verify this." — Unusually emphatic (R-AN42).
- **R-T35** · p.2 L54 · format · "Your team will submit their trading notes via the SurveyMonkey Apply form." — A form, not a file upload.
- **R-T36** · whole guide · format (omission) · The guide sets no font, file type, page limit or citation rule, and says nothing on whether the note text counts toward the 100 words. VERIFIED-REPO-FILE (absence). The form may add rules. Check the SMAApply form before 23 Oct.

5. SMAApply Deliverables page: `SMAApply_Deliverables_Page_2026-09-27.md` (R-S)

VERIFIED-REPO-FILE. Also checked live at <https://wghsinvc.com.smaply.us/res/p/deliverables/> on 2026-09-27 (VERIFIED-PRIMARY, identical text).

- **R-S1** · L3 · framing · "registered-team portal; text pasted by Team Leader Ray" — The repo note. The page is in fact public (section 1.1).
- **R-S2** · L6 · word-choice · "The 2026-2027 has four REQUIRED Deliverables:" — The noun after "2026-2027" is missing. "REQUIRED" is in capitals.
- **R-S3** · L8 · number · "Team Roster (Due: October 9; no later than 5:00 p.m. ET)" — 9 Oct, 5 pm Eastern Time.

- **R-S4** · L9 · number · "Trading Notes Analysis (Due: October 23; no later than 5:00 p.m. ET)" — 23 Oct.
- **R-S5** · L10 · number · "Investment Policy Statement (Due: November 6; no later than 5:00 p.m. ET)" — 6 Nov.
- **R-S6** · L11 · requirement · "IMPORTANT: Trading ends and your portfolio is locked." — The portfolio locks on the IPS due date.
- **R-S7** · L12 · number · "Final Report & Official School Documentation (Due: December 4; no later than 5:00 p.m. ET)" — 4 Dec.
- **R-S8** · L14 · requirement · "Team must submit all the deliverables, and meet their necessary requirements, in order to receive a digital credential." — Needed for the digital credential.
- **R-S9** · L15 · evaluation · "Your Trading Notes Analysis, Investment Policy Statement and Final Report are evaluated." — Three deliverables are scored. The roster is not.
- **R-S10** · L16 · requirement · "All deliverables must be submitted via your team's SurveyMonkey Apply account." — The submission channel.
- **R-S11** · L17 · prohibition · "We will not grant any deadline extensions." — No extensions.
- **R-S12** · L22 · prohibition · "Once your official team roster is submitted, teams may not change their members." — Members are locked after 9 Oct.
- **R-S13** · L23-32 · requirement · Roster fields: "Full Name - Every Student", "Email - Every Student (NOTE: A unique email must be provided for each student.)", "Birthday (Month, Day, Year) - Every Student", "Anticipated Graduation Year - Every Student", "Official Team Name", "School Country", "School State (U.S. school only)", "School City", "Official School Website (URL)" — Personal data. It must never go in the repo (brief privacy rule).
- **R-S14** · L35 · definition · "The purpose of this deliverable is to demonstrate how your team's investment decisions reflected the strategy it developed for the client." — Same as the guide, shortened.
- **R-S15** · L36 · format · Trading Notes instructions = Box link [3pxn798a...](#) — The same file as the repo PDF (hash match).
- **R-S16** · L39 · framing · "Your investment strategy is the cornerstone of the Wharton Global High School Investment Competition, and your Investment Policy Statement is where you clearly articulate that strategy." — Same as R-I4.
- **R-S17** · L40 · format · IPS instructions = Box link [0iib496j...](#) — Same file (hash match).
- **R-S18** · L43 · definition · "In your Final Report, your team will evaluate how it implemented its investment strategy and present its analysis and recommendations for achieving Laura's goals." — The Final Report's purpose.
- **R-S19** · L44 · number · "Detailed Final Report instructions and requirements will be shared on November 9." — 9 Nov = Week 7.
- **R-S20** · L45 · requirement · "We require official documentation from your school to ensure your team is in compliance with the Competition Rules." — School documentation is required.
- **R-S21** · L45 · requirement · "Do not wait until the last minute to request documentation." — Ask the school early.
- **R-S22** · L45 · format · "This will be submitted along with your Final Report as a separate file." — A separate file.
- **R-S23** · L46 · word-choice · "Sample Documentation: Coming soon" — Not yet published.
- **R-S24** · L49 · evaluation · "Investment Strategy: Presents a clear and creative investment thesis; demonstrates disciplined planning across Laura's changing time horizons and cash-flow needs; uses appropriate diversification; and maintains consistency with the team's IPS." — Criterion 1 has four parts. "Creative" and "appropriate diversification" are both named.
- **R-S25** · L51 · evaluation · "Client Knowledge and Objectives: Tailors the strategy to Laura's circumstances, priorities, risk considerations, and residency goals; demonstrates a thoughtful understanding of the client; and presents recommendations that can earn her confidence." — Criterion 2. "Earn her confidence" matches the case's "ultimately chooses" (R-C8).
- **R-S26** · L53 · evaluation · "Portfolio Analysis: Demonstrates understanding and effective use of investment concepts and tools; integrates quantitative and qualitative analysis; and uses reasonable

assumptions and projections to evaluate funding reliability, the facility contribution, and financial flexibility under varying market outcomes." — Criterion 3: numbers and words, and scenarios.

- **R-S27** · L55 · evaluation · "Articulation of Competition Experience: Demonstrates teamwork, communication, and learning; clearly explains the team's research and decision-making process; and reflects meaningfully on the team's growth and response to challenges." — Criterion 4: process and growth.
- **R-S28** · L57 · evaluation · "Creativity and Presentation: Presents a compelling, well-organized narrative in an authentic team voice; uses data effectively to support conclusions; demonstrates original thinking and meaningful reflection; and communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors." — Criterion 5. The co-sponsor message is scored here.
- **R-S29** · L48-57 · evaluation (omission) · No weights, no scoring scale, and no note on which criterion applies to which deliverable. VERIFIED-REPO-FILE (absence).
- **R-S30** · L6-17 · word-choice · The page lists four "REQUIRED" deliverables (including the roster and school documentation). The case speaks of "three connected deliverables" (R-C87). Both hold: three are evaluated, four are required.

6. Public Wharton and SMaply web pages (R-W)

All entries VERIFIED-PRIMARY, read 2026-09-27 between 11:38 and 11:45 UTC with curl through the session proxy. The text was extracted from the HTML. Quotes are exact apart from spacing around bold tags.

Page keys:

- [MAIN] <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/> (also reached from /investment-competition/ and /investment-competition/rules/ by redirect)
- [RULES] <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/>
- [FAQ] <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/faq/>
- [REG] <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/register-now/>
- [AI] <https://globalyouth.wharton.upenn.edu/ai-policy/>
- [NEWS] <https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/> (listed on MAIN as January 27, 2026)
- [ARCH] <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/archive/>
- [S-HOME] <https://wghsinvcomp.smaply.us/>
- [S-CLIENT] <https://wghsinvcomp.smaply.us/res/p/client/>
- [S-TRADE] <https://wghsinvcomp.smaply.us/res/p/trading/> (page title "Trading Details")
- [S-WINS] <https://wghsinvcomp.smaply.us/res/p/wins/>
- [S-FAQ] <https://wghsinvcomp.smaply.us/res/p/faqs/>
- [S-DELIV] <https://wghsinvcomp.smaply.us/res/p/deliverables/>

These slugs returned 404: trading-rules, case-study, resources, approved-securities, key-dates.

Main page [MAIN]

- **R-W1** · [MAIN] top · framing · "Registration for the 2026-2027 Competition is closed." — Registration is closed.
- **R-W2** · [MAIN] intro · definition · "a free, experiential investment challenge for high school students (9th to 12th grade) and teachers. Students work in teams of four to six, guided by a teacher from their school as their advisor, and have access to an online stock market simulator." — Basic format.
- **R-W3** · [MAIN] Important Dates · number · "September 28 Competition begins; first day of trading" ... "October 9 Deliverable: Official team roster" ... "October 23 Deliverable: Trading Notes Analysis" ... "November 6 Deliverable: Investment Policy Statement (IPS)" ... "December 4 Deliverable: Final report and school documentation" ... "TBD Semifinalists (top 50 teams) announced" ... "TBD Virtual Semifinals" ... "April 29 & 30 Learning Day & Global Finale" — Same dates as SMaply. Semifinal dates are TBD (to be decided).

- **R-W4** · [MAIN] About · definition · "Each year, teams are introduced to a new case study featuring a Wharton School alum. While the client is real, the financial scenario is developed specifically for the competition." — The scenario is invented. Do not treat Laura's real-life plans or finances as case facts.
- **R-W5** · [MAIN] About · requirement · "Using the client's background, financial goals, and their own research, teams develop a long-term investment strategy designed to help the client achieve their objectives." — Background, goals, and outside research are all expected.
- **R-W6** · [MAIN] About · definition · "Teams also use the Wharton Investment Simulator (WInS) to implement their investment strategy by building and managing a portfolio using virtual cash to buy and sell approved securities." — WInS = implementation.
- **R-W7** · [MAIN] About · evaluation · "Unlike other investment competitions, success is not determined by portfolio performance. Teams are evaluated on the quality of their investment strategy, how well it aligns with their client's objectives, the strength of their research and analysis, and their ability to clearly communicate and defend their recommendations." — Four judging themes. "Defend" points to the semifinal Q&A.
- **R-W8** · [MAIN] About · evaluation · "Following the submission of final reports, a panel of judges selects 50 semifinalist teams. Those teams are invited to present their investment strategies during the virtual Semifinals. The top 10 teams advance to the Global Finale" — Top 50 are chosen after 4 Dec. Top 10 go to the finale.
- **R-W9** · [MAIN] · framing · "Academic Director: Professor Michael Roberts" (a finance professor; his research listed there includes "the effect of interest rates on bank lending") — Context on who designs the competition. The page does not say he judges.
- **R-W10** · [MAIN] counters · number · "* 2025-2026 Participation Completion Numbers": Teams 2,339; Countries 79; Students 12,145; Advisors 1,625 — Read from the `data-counter-value` attributes in the HTML, because the page's display shows "0" until JavaScript runs. Last year about 2,300 teams completed.

Rules & Roles [RULES]

- **R-W11** · [RULES] Student Eligibility · requirement · "Each team must designate one student team leader who is at least 16 years old at the start of the Competition. No exceptions will be granted." — Team leader must be 16 or older.
- **R-W12** · [RULES] Student Eligibility · requirement · "Teams must be composed of current students from the same high school." — Same school.
- **R-W13** · [RULES] Teams · requirement · "Each team must consist of four to six students, including the student team leader, throughout the Competition." — 4 to 6 members at all times.
- **R-W14** · [RULES] Teams · definition · "Each team will share one WInS account." — One shared account.
- **R-W15** · [RULES] Teams · requirement · "All team members are responsible for: Developing and implementing the team's investment strategy; conducting research and analysis; collaborating with teammates and placing trades through the team's shared WInS account; and remaining actively engaged throughout the Competition and meeting all deadlines." — Students own the strategy and the trades.
- **R-W16** · [RULES] Teams · prohibition · "Teams may not contact the Competition client. Teams that violate this rule will be disqualified." — Never contact Laura.
- **R-W17** · [RULES] Teams · prohibition · "After the official team roster is submitted, teams may not add members." ... "Teams that fall below four members or exceed six members at any time during the Competition will be disqualified." — Roster lock; size rule.
- **R-W18** · [RULES] Advisors · requirement · "Each team must have one Advisor, who must be a teacher or educator at the high school attended by the team members." — Advisor must be from the school.
- **R-W19** · [RULES] Advisors · prohibition · "The Advisor's role is to provide guidance, encouragement, and educational support throughout the Competition. Advisors may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development." — Directly contradicts the case's fictional portfolio manager (R-C3; R-AN4).
- **R-W20** · [RULES] Advisors · definition · "Monitoring team engagement and portfolio activity." / "Serving as a sounding board without completing work on behalf of students." — What advisors do.

- **R-W21** · [RULES] Advisors · prohibition/permission · "Teams may seek guidance from a parent, industry professional, or other adult serving as a secondary advisor, provided the team also has a primary teacher Advisor. However, the use of paid advisors, education consultants, or other agents is prohibited. Similarly, no participant or team may enroll in any non-Wharton extracurricular course (online or in person) that claims to teach the Investment Competition." — Unpaid secondary advisors are allowed. Paid help and outside "competition courses" are not.
- **R-W22** · [RULES] Deliverables & Requirements · definition · "Trading Notes Analysis: Document your team's decision-making process." / "Investment Policy Statement (IPS): Define your client's investment objectives, risk tolerance, and overall investment strategy." / "Comprehensive Final Report: Present and justify your team's investment strategy, portfolio recommendations, and analysis." — The IPS description names "risk tolerance".
- **R-W23** · [RULES] Trading Requirements · requirement · "Teams must meet the required trading activity and portfolio management guidelines throughout the competition." — The public pages give no minimum amount of trading (R-AN36; X-18).
- **R-W24** · [RULES] Semifinals & Global Finale · evaluation · "The top 50 teams will be selected as semifinalists based on the strength of their Investment Policy Statement (IPS) and Final Reports." — Trading Notes are not named (X-8).
- **R-W25** · [RULES] Semifinals · requirement · "Semifinalists must also submit documentation on official school letterhead confirming that: The team has permission to participate in the Competition. All students and the advisor are affiliated with the school." — Letterhead confirmation.
- **R-W26** · [RULES] Use of Generative AI · permission/prohibition · "Teams may use generative AI tools (such as ChatGPT) for brainstorming and idea generation." ... "AI-generated work may not be submitted as your own." ... "Any AI-generated material included in your report must be properly cited, just like any other reference source." — Brainstorming with AI is allowed. AI text may not be submitted as the team's own, and any AI use must be cited.
- **R-W27** · [RULES] AI Policy · prohibition · "Beware that use may also stifle your own independent thinking and creativity. You may not submit any work generated by an AI program as your own." — Restated.
- **R-W28** · [RULES] Ethics · prohibition · "We trust that you will not lie about your investment decisions and outcomes, fabricate analyses, invent teamwork and experiential stories, compensate advisors, education consultants, or other agents, enroll in non-Wharton extracurricular courses that claim to "teach" the competition, or plagiarize existing strategies." — The reflections and the Articulation section must be true.
- **R-W29** · [RULES] Ethics · requirement · "All teams should review the CFA Institute's Asset Manager Code and operate by these standards. We expect you to read these rules and apply them to all you do on behalf of your potential client throughout the competition." — Names a professional ethics code. "Potential client" again.
- **R-W30** · [RULES] Code of Conduct · requirement · "Each team is required to properly cite any sources used and acknowledge ownership of all images and other media that it submits as part of a deliverable or a presentation that it (or a team member) does not own or did not solely develop." — A general citation duty (X-9).
- **R-W31** · [RULES] Code of Conduct · permission · "each team (and/or the students on such team) retains ownership of their own work." — Teams own their work.
- **R-W32** · [RULES] Recording Policy · prohibition · "Video recording, audio recording, photography, screenshots, or posting presentations or participants online during the Semifinals or Global Finale is strictly prohibited." — Applies at the semifinals and finale.
- **R-W33** · [RULES] Prizes & Recognition · number · "All students on teams that submit the required competition deliverables and meet the minimum competition requirements will receive a Digital Credential (participation badge) in February 2027." — The credential needs the "minimum competition requirements", which are not defined publicly.
- **R-W34** · [RULES] Participation & Travel · definition · "The Semifinals will be conducted virtually." / finalists "may participate either in person or virtually" — Format.

FAQ [FAQ]

- **R-W35** · [FAQ] Basics · number · "Registered teams will be able to practice on the Wharton Investment Simulator (WInS) from September 15 – 25, 2026 (4:00 p.m. ET). All practice portfolios will be removed at that time." — Practice period is over.
- **R-W36** · [FAQ] Basics · number · "The competition will officially begin on September 28, 2026 and end on December 4, 2026." — 28 Sep to 4 Dec.
- **R-W37** · [FAQ] Eligibility · framing · "so they can achieve our ultimate goal with this game, which is learning the nuances of investing." — Learning is the stated goal. "Game".
- **R-W38** · [FAQ] Team size · requirement · "All team members must play a contributing role." — Everyone contributes.
- **R-W39** · [FAQ] Time · number · "The competition is 10 weeks in length" — Week 1 starts 28 Sep. Week 6 ends 6 Nov, Week 7 starts 9 Nov and Week 10 ends 4 Dec. This matches R-T12 and R-C93 (DERIVED).
- **R-W40** · [FAQ] Judging · evaluation · "No. Teams will not receive individual feedback." — No feedback.
- **R-W41** · [FAQ] Judging · evaluation · "We are looking for teams that score exceptionally well on all five components of the criteria, and that embrace all aspects of the competition with passion and creativity, which is then reflected in various ways in their final reports." — All five criteria matter. The Final Report is the showcase.
- **R-W42** · [FAQ] Judging · evaluation · "your team's standings on WInS have little to do with the final outcome. Winners are selected on the strength and articulation of their overall investment strategies and competition experiences, not on the percentage growth of their portfolios" — Returns do not decide the winner.
- **R-W43** · [FAQ] Judging · evaluation · "There is no single winning investment strategy. Strong teams understand the client, work effectively together, conduct thoughtful research and analysis, and develop a clear and creative strategy designed around the client's goals. Their investment decisions reflect that strategy, and their deliverables connect into a cohesive, well-supported submission. Just as importantly, successful teams communicate their reasoning, assumptions, tradeoffs, and competition experience clearly and authentically." — Wharton's own description of a winning team. "Tradeoffs" and "authentically" are named.
- **R-W44** · [FAQ] Semifinals · requirement · "Your team will be expected to create and deliver a short presentation (via video conference for the Semifinals and live or via video conference for the Global Finale) about your team's strategy, analysis and experience over the past 10 weeks to a live panel of judges. All team members must participate." — Every member presents.

Register page [REG]

- **R-W45** · [REG] · definition · "Accounts are created per team, not per student. Each team shares one Wharton Investment Simulator (WInS) account." — Consistent with R-W14.

AI policy page [AI]

- **R-W46** · [AI] Investment Competition > AI Policy · requirement · "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages." — Any AI use must be recorded in a Works Cited list. This is stricter than the Rules page version. It implies the Final Report has a Works Cited section.
- **R-W47** · [AI] · requirement · "All material referenced or generated by AI should be cited like any other reference material (with due consideration for the quality of the reference, which may be poor)." — Cite AI, and treat it as a weak source.
- **R-W48** · [AI] Investment Competition > Ethics · prohibition · "...enroll in non-Wharton extracurricular courses that claim to "teach" the competition, pass AI generated writing or work off as your own, or plagiarize existing strategies." — This version of the Ethics text adds the AI clause, which the Rules page version lacks.
- **R-W49** · [AI] Statement of AI Use · requirement · "When students express themselves in the Wharton Global Youth Program, they must use their voice and words." / "AI-generated work should be cited like any other reference material, including how and where students used AI-generated information." — Students write in their own words.
- **R-W50** · [AI] Pre-baccalaureate section · prohibition (scope: the pre-bacc program) · "Don't use AI for personal reflection or opinion-based tasks." — Written for another program. Still a strong signal for the

Trading Notes reflections and the Articulation criterion (INTERPRETATION).

- **R-W51** · [AI] Checklist Questions · requirement · "Did I come to my current conclusion before or after using generative AI tools?" / "Did I double check the numbers and information are most up-to-date?" — A self-check list.

2025-26 season news and archive [NEWS], [ARCH]

- **R-W52** · [ARCH] · framing · Clients 2021-22 to 2025-26: "Nichole Jordan, WG'08"; "Peter Wang Hjemdahl, W'19"; "Hilary Ash, W'13"; "Ladi Ayoola, WG'22"; "Connor Barwin, WG'23" — Laura (a 2018 Wharton graduate per the case, so "W'18" in this notation, DERIVED) is the next alum client. The archive does not list her yet.
- **R-W97** · [NEWS] · number · "the competition welcomed more than 6,300 registered teams from around the world, an increase of 1,300+ from the previous year." — 2025-26 registered teams.
- **R-W98** · [NEWS] · number · "students were crafting compelling investment strategies and building investment portfolios with \$500,000 in virtual cash, up from \$100,000." — Last year's WInS cash was \$500,000. This year's is \$300,000 (R-W56).
- **R-W99** · [NEWS] · number · "On December 12, some 2,300 teams from 79 countries submitted final reports, an increase of nearly 30% over last year's competition." — The real field is teams that finish: about 2,300.

SMApplly public resource pages

- **R-W53** · [S-HOME] · word-choice · "An investment in knowledge pays the best interest." — Unattributed tagline on the portal.
- **R-W54** · [S-CLIENT] · definition · "Get to the know the client. The Client Case Study is the starting point for the competition." / "It introduces the client, her background, financial circumstances, and long-term goals. Teams should use this information to understand who they are investing for and to develop a thoughtful investment strategy tailored to her needs. The case study provides the foundation for the team's portfolio decisions, Investment Policy Statement, Trading Notes Analysis, and Final Report." — Includes the typo "to the know". The case is the foundation of all deliverables.
- **R-W55** · [S-CLIENT] · prohibition · "IMPORTANT: Teams are not permitted to contact Laura Gao." — Restated for this client by name.
- **R-W56** · [S-TRADE] Account and Trading Period · number · "Your team will begin the official trading period with \$300,000 in virtual cash." — **Starting cash is \$300,000.**
- **R-W57** · [S-TRADE] · definition · "Each team shares one WInS account. All team members will use the same username and password." — Shared login.
- **R-W58** · [S-TRADE] · number · "Official trading begins on September 28, 2026, and ends on November 6, 2026. When trading ends, your portfolio will be frozen, and your team may no longer trade or change its holdings." — Trading window and freeze.
- **R-W59** · [S-TRADE] Important · definition · "The \$300,000 is your team's WInS simulator balance. The additional \$150,000 contribution described in the Client Case Study will not be added to WInS." — WInS never receives the \$150,000.
- **R-W60** · [S-TRADE] Important · prohibition · "WInS gains and losses should not be added to or subtracted from the client's long-term portfolio projections. Those projections should use the cash flows provided in the Client Case Study and reasonable return assumptions consistent with your strategy." — Consistent with R-C77/R-C78.
- **R-W61** · [S-TRADE] How Trades Are Executed · permission · "You may enter buy and sell orders at any time during the official trading period. The timing of execution depends on the security, its market, and whether that market is open." — Orders are accepted at any hour.
- **R-W62** · [S-TRADE] U.S. Securities · definition · "Orders placed while the market is open are executed using real-time prices." / "Orders placed while the market is closed are filled at the security's opening price when the market reopens." / "Prices displayed for open positions may be delayed by 10 to 15 minutes." — How fills work. Relevant for a team in Australia trading overnight.
- **R-W63** · [S-TRADE] International Securities · definition · "International equity orders may experience a longer delay and are processed at the end of the applicable trading day." — Non-U.S. orders fill at the end of the day.

- **R-W64** · [S-TRADE] · definition · "WInS automatically completes foreign-currency conversions. You do not need to exchange currency yourself, but exchange-rate movements may affect the value of your holdings." — Currency (foreign-exchange, FX) risk exists in WInS.
- **R-W65** · [S-TRADE] Trading Limits · number · "Your team may make up to 200 trades during the competition." — **200-trade cap confirmed.**
- **R-W66** · [S-TRADE] Trading Limits · number · "Your team may trade an amount equal to no more than twice a security's current daily trading volume." — Size cap per trade, tied to trading volume.
- **R-W67** · [S-TRADE] Approved Securities · permission · "Stocks: Any stock available on WInS priced at \$5 or more, or the equivalent of \$5 in its local currency." — Stocks priced at \$5 or more.
- **R-W68** · [S-TRADE] Approved Securities · permission · "Exchange-Traded Funds: Any ETF available on WInS." — **Any ETF available on WInS.** There is no separate list.
- **R-W69** · [S-TRADE] Approved Securities · permission · "Treasury Bonds: Bonds available on WInS. The list includes Treasury bonds from the United States, United Kingdom, Germany, France, Italy, and the Netherlands." — **Individual government bonds can be bought.**
- **R-W70** · [S-TRADE] Diversification · permission · "There is no required sector allocation or minimum number of sectors. However, diversification remains an important investment principle." — **No sector minimum.** This replaces the unverified "sector minimum = team size" claim.
- **R-W71** · [S-TRADE] Diversification · framing · "Your team may consider diversification across asset classes, sectors, industries, market capitalizations, geographic regions, risk characteristics, and funding purposes. The goal is to build an intentional mix of investments appropriate for your strategy, not simply to own a large number of securities." — "Funding purposes" counts as a kind of diversification (R-AN45).
- **R-W72** · [S-TRADE] Trading Outside the U.S. · permission · "You are not expected to stay awake during the night to trade. Orders may be entered while the U.S. market is closed and will be filled when the market reopens." — Written with non-U.S. teams in mind.
- **R-W73** · [S-TRADE] · evaluation · "The competition does not require frequent or same-day trading. Make thoughtful, well-researched decisions and record the reasoning behind them in your Trading Notes." — Fewer, better trades.
- **R-W74** · [S-WINS] · definition · "WInS is a tool for implementing your investment strategy." — WInS = a tool.
- **R-W75** · [S-WINS] · evaluation · "You are not being evaluated based on: Your portfolio ranking / How many trades you make / Whether you outperform other teams / Whether your portfolio makes money during the competition" — Four things that are not judged.
- **R-W76** · [S-WINS] · evaluation · "A long-term investment strategy cannot be judged solely by what happens in the market over a few weeks. Instead, your portfolio provides evidence of the decisions your team made and how you put your strategy into practice." — The portfolio is evidence.
- **R-W77** · [S-WINS] · definition · "All account activity will be cleared after the practice session. Practice trades, portfolio activity, gains and losses, and account balances will be reset before official trading begins on September 28." — Practice trades cannot be used as notes.
- **R-W78** · [S-WINS] · number · "Official trading will end on November 6, when your Investment Policy Statement is due. At that time, your WInS portfolio will be frozen" — Freeze date.
- **R-W79** · [S-WINS] · framing · "Test ideas and evaluate how different investments may contribute to growth, liquidity, funding reliability, financial flexibility, or risk management." — The same vocabulary again (X-16).
- **R-W80** · [S-WINS] · requirement · "Your portfolio should reflect a cohesive strategy designed around the client's goals. Each investment decision should have a clear purpose within that strategy." — Every holding needs a purpose.
- **R-W81** · [S-WINS] · evaluation · "Gains and losses generated in WInS do not determine how your team is evaluated." — Restated.
- **R-W82** · [S-WINS] Remember Your Trading Notes · requirement · "Complete a Trading Note when your team makes an investment decision. Your Trading Notes should record the reasoning, research, and

intended strategic role of each decision at the time it is made." — Write every note at the time of the trade.

- **R-W83** · [S-WINS] · evaluation · "The goal is not simply to achieve the highest return, but to demonstrate what your team has learned and how effectively you applied that learning throughout the competition." — Learning is judged.
- **R-W84** · [S-WINS] · definition · "Access WInS: <https://edu.stocktrak.com/wharton/>" — The platform is run by Stock-Trak (R-W85).
- **R-W85** · [S-FAQ] · number · "Your team will be responsible for managing a portfolio of \$300,000 in virtual cash." / "Wharton Global Youth Program has contracted with Stock-Trak®" — \$300,000 confirmed a second time.
- **R-W86** · [S-FAQ] · definition · "Your team username is NOT the same as your team name." — The title page needs both (R-I41, R-I44).
- **R-W87** · [S-FAQ] Who can place trades · prohibition · "Students should place trades on WInS, NOT the advisor. A key component of the Investment Competition is teamwork and communication. It should be a team decision. We strongly encourage teams to identify different roles for each team member." — Advisor does not trade. Assigning roles is encouraged.
- **R-W88** · [S-FAQ] Placing trades · framing · "you should not be doing a lot of buying and selling of the stocks designated for the long-term portion of your portfolio ... you are not trying to prove the worthiness of your long-term strategy through how your portfolio performs over 10 weeks of trading." — Low turnover. "Long-term portion" implies other portions exist.
- **R-W89** · [S-FAQ] · permission · "DOES THE COMPETITION ALLOW ALL SECURITIES? No." / "Investments permitted (for BOTH contributions): Cash / Any stock from any exchange available on WInS priced at \$5 (or the equivalent of \$5 in the local currency) or higher. / Any Exchange-Traded Funds (ETFs) available on WInS. / Any Government/Treasury Bonds from any exchange available on WInS." — The permitted list, with the phrase "for BOTH contributions" (R-AN15).
- **R-W90** · [S-FAQ] · prohibition · "The following are NOT permitted: Margin trading / Short selling / Stock-secured debt / Crypto / Derivatives (options, futures, swaps, etc.) / Anything other than the approved investments listed above." — **Prohibited list.**
- **R-W91** · [S-FAQ] · definition · "The bond prices are updated once daily at U.S. market open, and pay out every six months." ... "When the competition ends, any bonds held will have their value factored into the teams final portfolio value." — Bond prices are stale within the day. Coupons are paid twice a year.
- **R-W92** · [S-FAQ] · number · "Each stock transaction is charged a flat \$25 commission and treasury bonds are charged \$10. Commission is charged only on a trade that clears." — Trading costs.
- **R-W93** · [S-FAQ] · definition · "WInS provides real-world corporate actions such as dividends, stock splits, bond coupons, and so on." — Income is credited.
- **R-W94** · [S-FAQ] · framing · "You can only cancel open orders." ... "If you make an error, we encourage you to run with it and not scramble to "fix" it. Trading errors are part of the learning process." — Errors are learning material (and a possible Articulation story, if true).
- **R-W95** · [S-FAQ] Judging · evaluation · "Gains and losses generated in WInS are not used to select winning teams and are not applied to the client's long-term portfolio projections. ... Teams are evaluated on how thoughtfully they develop, implement, evaluate, and communicate a cohesive investment strategy designed around the client's goals." — Four verbs: develop, implement, evaluate, communicate. These match the three deliverables.
- **R-W96** · [S-DELIV] · framing · The public page text is identical to the repo transcription (section 1.1). The Box downloads' SHA-256 values match the repo PDFs. — The repo copy is current as of 2026-09-27.
- **R-W100** · [S-FAQ] typo · word-choice · "DOES THE COMPETITION ALLOW MARGING TRADING OR SHORT SELLING? No." / "We will not provide your with information" — Editing slips on the FAQ (see R-AN47).

7. Notable word choices and anomalies (R-AN)

Every "possible purpose" below is INTERPRETATION: a neutral guess at why the designers wrote it this way. None is a fact. Quotes are VERIFIED-REPO-FILE or VERIFIED-PRIMARY as referenced.

- **R-AN1 "High degree of certainty" is never defined.**

- Case p.3 L91: "All ten payments must be funded by the investment portfolio with a high degree of certainty." L97-98: "define what they consider a high degree of funding certainty".
- Two wordings are used ("certainty" and "funding certainty"). No percentage or method is given.
- Possible purpose: tests whether teams can turn a client's plain-language wish into a definition that can be tested, and explain how they checked it. This fits a client with a statistics degree (R-C12).
- **R-AN2 The reserve comes before everything, on the same day as the first payment.**
 - Case p.3 L94-95: "At the beginning of 2033, before making the first operating payment or contributing to the facility, Laura will set aside a portion of the portfolio to fund the ten payments."
 - DERIVED: the first payment is also due "at the beginning of 2033" (L88-89). The set-aside must therefore cover ten payments, one of them paid immediately.
 - Possible purpose: sets a priority order (the promise first, the facility second) and checks whether teams notice that the first payment is due at once.
- **R-AN3 "Not adjusted for inflation" is stated outright.**
 - Case p.3 L90: "each payment is a fixed \$50,000 and is not adjusted for inflation."
 - Possible purpose: removes a common source of confusion and makes the commitment a fixed-dollar liability that bonds can match. The loss of real value (what \$50,000 can buy) is shifted elsewhere: "any operating support beyond Laura's commitment, may come from co-sponsors..." (L85-86). Inflation still enters through the facility (R-AN8).
- **R-AN4 The fiction says the advisor decides; the rules say the opposite.**
 - Case p.1 L4: "your team's teacher/advisor who makes the final investment decisions for your firm's portfolio".
 - [RULES]: "Advisors may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development."
 - [S-FAQ]: "Students should place trades on WInS, NOT the advisor."
 - Possible purpose: role-play of a firm's hierarchy (analysts recommend; a portfolio manager signs off), without changing who does the work.
 - Consequence (INTERPRETATION): the IPS "decision-making framework" (R-C89) must be the students' own. Any mention of the advisor must match the real rules.
- **R-AN5 A competitive pitch to a client who has not yet committed.**
 - Case p.1 L9-10: "the investment strategy that Laura ultimately chooses". L5: "a potential client". p.2 L43: "with an asset management firm". [RULES] Ethics: "your potential client".
 - Possible purpose: the task is persuasion as well as analysis. The judge acts as the client choosing among firms. The criterion "recommendations that can earn her confidence" (R-S25) points the same way.
- **R-AN6 Outside money may fund the facility but not her commitment.**
 - Case p.3 L85-86: "Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors...". L91-92: "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement."
 - Possible purpose: makes the facility the flexible amount and the payments the fixed one. Her personal promise has to rest on her own money, while shared costs can be shared.
- **R-AN7 2031 is exactly two years before 2033 and gets its own timeline line.**
 - Case p.2 L55: "2031 is Year 5." p.3 L109: "in 2031, two years before the residency is established."
 - Possible purpose: creates a forecasting problem with a fixed, short horizon. A two-year gap is short enough for part of the promise to be secured with a matching two-year instrument. Two-year U.S. Treasury notes are a standard maturity; that is general market knowledge, not something the case states.
- **R-AN8 Inflation's effect on facility costs must be explained, but the facility cost need not be estimated.**
 - Case p.4 L146-147: "the effect of inflation on portfolio projections and facility costs". L164-165: "The total cost of the facility has not been determined. Teams are not expected to estimate that cost".
 - Possible purpose: asks for an assumption about how rising prices shrink what a fixed-dollar contribution buys (and the real value of projections), without a construction budget. Relevant to how the 2031 range is explained to co-sponsors.
- **R-AN9 The confidence statement is two-sided.**

- Case p.3 L117-118: "state how confident they are that her 2033 contribution will fall within that range."
- A contribution above the top of the range is literally "outside" it.
- Possible purpose: a range that is wide enough to be safe becomes useless to co-sponsors, and one narrow enough to be useful becomes risky. The wording forces the team to size both ends. The credibility sentence (L114) shows the low end matters most.
- **R-AN10 The case never names a currency.**
 - "\$300,000", "\$150,000", "\$50,000", "dollar range" (L117). No "U.S. dollar" or "USD"; no mention of the New Taiwan dollar or exchange rates. Yet the residency is "in Taiwan" (L76).
 - [S-TRADE]: "WInS automatically completes foreign-currency conversions ... exchange-rate movements may affect the value of your holdings."
 - Possible purpose: a simplification (the U.S. dollar is implied for a U.S. competition) that leaves room for teams to notice currency risk in a Taiwan project. Whether to raise it is a team judgement. Taxes and Taiwan law are explicitly excluded, but currency is not mentioned at all.
- **R-AN11 The timeline lists only the event years.**
 - Case p.2 L49-57: 2026, 2027, 2028, 2031, 2033. 2029, 2030 and 2032 are missing, and so is 2042 (Year 16, the last payment).
 - Possible purpose: draws attention to the decision dates. The pattern rule (L58) covers the rest.
- **R-AN12 The payments are mentioned before they are introduced.**
 - Case p.2 L59-60: "The ten operating payments are therefore spaced exactly one year apart." This appears a page before "Operating Commitment" (p.3 L87).
 - Possible purpose: the timing convention exists to make the maths exact: yearly discounting, with no mid-year question.
- **R-AN13 The money put in is less than the money promised.**
 - DERIVED: $\$300,000 + \$150,000 = \$450,000$, while $10 \times \$50,000 = \$500,000$ in nominal payments.
 - The portfolio must grow before any facility money exists. At the 2026-09-25 Treasury curve, the payments' value on 1 Jan 2027 is \$292,264 (VERIFIED-REPO-FILE: [research/verified_2026-09-27/official_curve_pv.py](#), per the brief), so the present-day cost is below \$300,000 even though the nominal total is above \$450,000.
 - Possible purpose: makes time and interest rates central, and asks whether the promise can be secured early.
- **R-AN14 WInS starting cash equals Laura's first deposit exactly.**
 - [S-TRADE]: "\$300,000 in virtual cash" and "The additional \$150,000 contribution ... will not be added to WInS."
 - [NEWS]: last year "\$500,000 in virtual cash, up from \$100,000."
 - Possible purpose: this year's simulator is sized to be Laura's Year-1 portfolio. That bears on the open question in the brief of whether WInS should mirror the 2027 book or the post-2028 target. Neither document settles it. (Seed question 3's "\$500k = WInS cash" link does not hold this season.)
- **R-AN15 "Investments permitted (for BOTH contributions)".**
 - [S-FAQ]. Possible purpose, reading 1: the permitted-investment list also limits the long-term strategy for both Laura's \$300,000 and her \$150,000. Instruments outside it, such as derivatives, would then be off-limits even in the plan.
 - Reading 2: the wording is left over from an earlier year's case that had two contributions.
 - Treat reading 1 as the safe assumption. Confirm with Wharton via the Contact Us form if the plan relies on anything not on the list.
- **R-AN16 "Although ... thoughtful risks" and "the capital required for her goals".**
 - Case p.2 L69-74.
 - Possible purpose: signals that her career risk-taking does not simply carry over to the portfolio. What must be protected is the capital her goals require, not every dollar. "Thoughtful" appears three times in the case (L7, L38, L70). The risk wording is qualitative on purpose.
- **R-AN17 The case never states a risk tolerance, return target, benchmark or asset mix.**
 - Absent from the case: "risk tolerance", "benchmark", "rebalanc(e)", "return target". Word check, VERIFIED-REPO-FILE.

- Yet the IPS definition includes "risk tolerance" (R-I2), and so does the Rules page IPS description (R-W22).
- Possible purpose: teams must infer and justify her risk tolerance from the case, including the difference between how much risk she is willing to take and how much she can afford to take.
- **R-AN18 "Responsibly" and "responsible" are key words that are never defined.**
 - Case: "responsibly contribute" (L102, L105), "responsibly provide" (L167), "a responsible facility contribution" (L138). IPS guide: "responsible facility contribution" (3 times).
 - Possible purpose: the team must say what makes a contribution responsible: it protects the reserve and preserves flexibility (R-C84).
- **R-AN19 Flexibility "as the project develops", yet no contingency fund is required.**
 - Case p.3 L102-103 and L106-107.
 - Possible purpose: flexibility belongs to the residency project, not to her personal spending (living costs are outside the portfolio, L61-63). Teams must express it without sizing a separate fund, for example as a rule or a share (INTERPRETATION).
- **R-AN20 "The extent to which" and "no predetermined facility contribution".**
 - Case p.3 L104-105.
 - Possible purpose: permits a rule-based answer (a share of the surplus, a floor plus a share) and even a small amount. The dollar requirement applies to the 2031 range (L117), not to the 2033 contribution itself.
- **R-AN21 "Draft part of Laura's fundraising materials".**
 - Case p.4 L123. Only "part"; the audience is prospective co-sponsors. The IPS guide (R-I31) and the Trading Notes guide (R-T14) both exclude it, so it belongs in the Final Report. Criterion 5 scores it (R-S28).
 - Possible purpose: tests whether teams can explain uncertainty to people who are not investors. Under the AI policy it must be the students' own writing (R-W26, R-W49).
- **R-AN22 Year 0 is "before any money has been invested", yet WInS trading happens in Year 0.**
 - Case p.2 L47. p.4 L130: "It does not determine Laura's actual portfolio value at the beginning of 2027."
 - Possible purpose: separates the practice run (WInS, in 2026) from the plan (money from 2027).
- **R-AN23 The portfolio "may also cover part of the facility's cost".**
 - Case p.3 L84-85. "Part" signals no expectation that her portfolio pays for the whole facility.
 - Possible purpose: sets the expectation of a partial, credible contribution.
- **R-AN24 "Continued business income" is listed as a possible outside source.**
 - Case p.3 L86 lists it among sources for the facility and extra support. For the commitment, teams "may not rely on ... other outside funding" (L91-92).
 - Possible purpose: her future earnings may help the project but cannot back the promise. Whether it counts as "outside funding" is the team's reading; the stricter reading is safer.
- **R-AN25 The ten years are called "early operations".**
 - Case p.3 L83: "reliable support for its early operations". L92-93: after 2042 is out of scope.
 - Possible purpose: frames the commitment as start-up support. It explains why there is no need to fund the residency forever.
- **R-AN26 "A portion of the portfolio" and "the remaining portfolio".**
 - Case p.3 L95, L101.
 - Possible purpose: the reserve is one part of the portfolio. The remainder is for the facility and flexibility, which is the two-part structure the case expects.
- **R-AN27 "If at all": a reserve that never changes is allowed.**
 - Case p.3 L96. Contrast IPS p.1 L22: "How will the portfolio's asset allocation and composition change as future funding needs approach and payments are made?"
 - Possible purpose: the case allows a reserve that pays itself out as bonds mature, with no trading. The IPS prompt still expects the portfolio as a whole to change over time (X-2).
- **R-AN28 "Laura will set aside" in 2033, but nothing forbids earlier funding.**

- Case p.3 L94-95 defines the reserve at 2033. It does not say how the money is invested before then.
- Possible purpose: tests whether teams see that the set-aside is a label at a date, while the investing decision starts in 2027. The team must decide how to describe this reading; the case gives no ruling (open item in the brief).
- **R-AN29 The case's "must" versus "should" pattern.**
 - "Must" appears 9 times in the case. It covers:
 - the portfolio funding the commitment;
 - the size and composition of the reserve;
 - the facility recommendation;
 - the range and the confidence statement;
 - the range protecting the payments;
 - the fundraising draft;
 - the strategy tests.
 - "Should" covers the certainty definition (L97 "They should also define"), the assumptions and the scenarios.
 - Possible purpose: signals what is a hard requirement. In practice every "should" item is also scored under Portfolio Analysis.
- **R-AN30 The pull quote versus the co-sponsor task.**
 - Case p.1 L31-33: "The only person who needs to believe in something is yourself." Versus p.3 L114-115: overpromising could "damage her credibility and lose the confidence or participation of co-sponsors."
 - Possible purpose: her self-belief meets a 2031 task where others must believe her. The case may be testing whether teams can turn conviction into evidence. INTERPRETATION only. The quote has no attribution line in the text layer; B8/D13 should check it against a primary source before anyone presents it as her words.
- **R-AN31 Her roles are listed differently in each place, and all her income sources come from her own work.**
 - "bestselling author, illustrator, entrepreneur, and educator" (L5-6); "bestselling graphic novelist..." (L13); "author, illustrator, educator, and public speaker" (L25); "Storyteller, Entrepreneur, and Creative Visionary" (L29-30). The 2028 sources: "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (L44-45).
 - Possible purpose: shows a varied creative career whose income depends on her personal brand. The case states the \$150,000 as certain ("will"). Stress-testing it is a team choice, which the Portfolio Analysis criterion allows ("varying market outcomes").
- **R-AN32 A statistics degree.**
 - Case p.1 L16: "Statistics & Information Decisions Management".
 - Possible purpose: a client who can judge a probability statement. Loose phrases such as "95% safe" with no stated method would be exposed (INTERPRETATION).
- **R-AN33 Her first named work answered misinformation.**
 - Case p.1 L19-20: "Originally intended as a response to misinformation".
 - Possible purpose: fits a case whose central communication task is credibility and not overpromising. Use only as a link to her stated public work, never as decoration (INTERPRETATION).
- **R-AN34 What is conspicuously NOT asked or stated** (VERIFIED-REPO-FILE word checks across the four official documents unless noted):
 - No values-based or impact preference: 0 hits each for "ESG", "values", "impact", "sustainab" (compare last season's impact-led case, per the brief).
 - No benchmark, rebalancing rule, fee rule, return target, risk-tolerance number or confidence level (for example 95%).
 - No currency and no exchange-rate mention (R-AN10). No estate or legacy. No retirement.
 - No cash needs before 2033.
 - No instruction for what happens if the residency cannot open or moves.
 - No requirement to use WInS holdings in projections; this is forbidden, in fact (R-C78).
 - No facility cost and no contingency fund (both explicitly not expected).
 - No word on what happens to leftover money after 2042.

- The case does not mention technology or AI risk to her income.
- The case does not say whether the \$150,000 could be smaller.
- The public web pages give no minimum trading activity.

Possible purpose: keeps the case focused on one liability, one uncertain surplus and one communication problem. Anything a team adds from this list must earn its place (brief design principle).

- **R-AN35 "She will contribute an additional \$150,000".**
 - Case p.2 L43-45. The case uses "will", not "expects to".
 - Possible purpose: removes deposit risk from the base case. Teams who stress-test it go beyond the case and must label that as their own scenario.
- **R-AN36 Trading freezes in Week 6, but the competition runs 10 weeks.**
 - [S-TRADE]/[S-WINS]: trading ends 6 Nov. [FAQ]: the competition ends 4 Dec, "10 weeks".
 - Possible purpose: the Final Report is written about a frozen portfolio and must judge its implementation ("evaluates how that strategy was implemented", R-C90). The Rules page's "required trading activity" (R-W23) is not quantified anywhere public.
- **R-AN37 Scope of "Formatting choices not specified below".**
 - IPS p.3 L101: the sentence sits at the end of the title-page block but says "below".
 - Possible purpose: probably means that anything not listed (headings, alignment, page numbers) is the team's choice. Safe reading: follow the sample layout (R-I59, R-I60) and add nothing banned (R-I55).
- **R-AN38 No citations in the IPS versus citing everything.**
 - IPS p.3 L123-125: "formal citations are not permitted" and "sources do not need to be cited in the IPS". [RULES] Code of Conduct: "required to properly cite any sources used". [AI]: "must be recorded in your Works Cited pages".
 - Possible purpose: keeps the IPS a clean, persuasive 500 words and moves the evidence to the Final Report (IPS L125).
- **R-AN39 Two pages double-spaced for 550 words is tight but workable.**
 - IPS p.3 L103-118. The official placeholder averages 13-15 words per line (R-I61).
 - ASSUMPTION (estimate from the sample, not a measurement of a real document): 9 inches of usable height at double spacing gives about 23 lines per page, so about 46 lines. 550 words plus two headings needs roughly 42-44 lines.
 - Possible purpose: the page limit and the word limit together punish padding. Test-print the real PDF.
- **R-AN40 The Trading Notes come before the IPS.**
 - Case p.4 L150-160 order: Trading Notes → IPS → Final Report. TN guide L17-21: the strategy "may continue to evolve" but notes must be "consistent with the strategic approach your team is developing and will later articulate in its IPS."
 - Possible purpose: rewards teams that have a strategy before trading. The notes show how it was "reflected, tested, or refined".
- **R-AN41 Official sources disagree on what selects the semifinalists.**
 - [RULES]: "based on the strength of their Investment Policy Statement (IPS) and Final Reports". [S-DELIV]: "Your Trading Notes Analysis, Investment Policy Statement and Final Report are evaluated." Case L161: "Evaluators will consider the three deliverables together."
 - Possible purpose: none; probably different pages were updated at different times. Safe reading: treat all three as scored.
- **R-AN42 "Yes, we will verify this."**
 - TN p.2 L53. An unusually informal, emphatic sentence in a formal guide.
 - Possible purpose: a warning, likely prompted by teams inventing or editing notes. It pairs with "exactly as it appears in WInS" (L44).
- **R-AN43 Notes are locked when written.**
 - TN L44: "Include each Trading Note exactly as it appears in WInS." [S-WINS]: "record the reasoning, research, and intended strategic role of each decision at the time it is made."
 - Possible purpose: the note shows thinking at the time of the trade. The reflection (100 words or fewer) is the only place to add hindsight. Every note written from now on is a potential deliverable.

- **R-AN44 Wharton's own example note is a bond fund for the payments.**
 - TN p.2 L36-39: "an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment".
 - Possible purpose: shows the expected quality of a note: a purpose, a link to Laura, and a named risk ("although interest-rate changes may affect its value"). It also hints that preparing for the liability is a legitimate trading theme. Many teams will copy it, so matching it does not make a team stand out (INTERPRETATION).
- **R-AN45 "Funding purposes" counts as a kind of diversification.**
 - [S-TRADE]: "diversification across asset classes, sectors, industries, market capitalizations, geographic regions, risk characteristics, and funding purposes."
 - Possible purpose: lets a portfolio split by goal (for example, money for the payments versus money for growth) count as diversification. Relevant to criterion 1, "uses appropriate diversification" (R-S24).
- **R-AN46 "The long-term portion of your portfolio".**
 - [S-FAQ]. Implies the designers expect portfolios to have more than one portion.
 - Possible purpose: may be older FAQ wording, but it fits a split portfolio.
- **R-AN47 Editing slips on the web pages.**
 - "The 2026-2027 has four REQUIRED Deliverables" (R-S2); "Get to the know the client" (R-W54); "MARGING TRADING" and "provide your with information" (R-W100); "the teams final portfolio value" (R-W91).
 - Possible purpose: none. These pages are lightly edited, so avoid reading hidden meaning into single words there. The PDFs are the carefully edited documents.
- **R-AN48 The bond lists differ slightly.**
 - [S-TRADE]: Treasury bonds from six named countries. [S-FAQ]: "Any Government/Treasury Bonds from any exchange available on WInS."
 - Possible purpose: none. In practice, what WInS lists is what is allowed. Check in WInS before trading.
- **R-AN49 "Any ETF" versus "no derivatives".**
 - [S-TRADE] "Any ETF available on WInS" and [S-FAQ] "Derivatives (options, futures, swaps, etc.)" are "NOT permitted". Leveraged and inverse ETFs use derivatives internally.
 - Possible purpose: unclear. The safe course is to avoid leveraged, inverse or options-based ETFs unless Wharton confirms them. Confirm against this year's WInS list and rules before trading.
- **R-AN50 The \$5 stock rule has no stated timing.**
 - [S-TRADE]: "priced at \$5 or more". It does not say whether this applies only at purchase.
 - Possible purpose: blocks penny stocks. Likely irrelevant to a fund-and-bond strategy.
- **R-AN51 A 200-trade cap with flat commissions.**
 - [S-TRADE] and [S-FAQ]: \$25 per stock trade, \$10 per bond trade.
 - Possible purpose: discourages frequent trading. With \$300,000 the cost is small (DERIVED: 20 trades cost at most \$500). The trade count matters more than the cost.
- **R-AN52 Bond prices update once a day and coupons are paid every six months.**
 - [S-FAQ].
 - Possible purpose: explains why WInS bond values look stale within the day. Individual bonds behave like real Treasuries, coupons included, which matters for matching the payment dates.
- **R-AN53 The goals are already fixed.**
 - Case p.2 L39-41: "Working with your portfolio manager, Laura has identified several long-term financial objectives".
 - Possible purpose: teams build the strategy; they do not renegotiate the goals.
- **R-AN54 Three deliverables are evaluated; four are required; school documentation is extra.**
 - [S-DELIV] and case L150.
 - Possible purpose: administrative items (roster, school documentation) are pass/fail gates, not scored.
- **R-AN55 Week language versus dates.**
 - Case L163 "beginning of Week 7". TN L17 "Week 6". SMapply gives dates.
 - Possible purpose: the guides may be reused across years. The dates line up this year (R-W39).

- **R-AN56 "While the client is real, the financial scenario is developed specifically for the competition."**
 - [MAIN].
 - Possible purpose: tells teams that public facts about the person can inform understanding, but the money story is fiction. Do not import her real finances or real plans.
- **R-AN57 A no-contact rule, with disqualification.**
 - [RULES] and [S-CLIENT].
 - Possible purpose: fairness, and the client's privacy. Consistent with the brief's privacy rule.
- **R-AN58 Who receives the payments is left vague.**
 - Case p.3 L88: "payments of \$50,000 toward the residency's operating expenses". It does not say whether an entity or a bank account receives them.
 - Possible purpose: keeps legal structure out of scope (compare L168-169).
- **R-AN59 Nothing after 2042.**
 - Case L92-93. It says nothing about money left over after the last payment.
 - Possible purpose: bounds the problem. Leftover money is covered only by "flexibility" (R-C84).
- **R-AN60 Last season's field was much smaller than the headline registration number.**
 - [NEWS]: "more than 6,300 registered teams" versus "some 2,300 teams ... submitted final reports". [MAIN] counter: 2,339 teams.
 - Possible purpose: none. For the team, the realistic rival field is the teams that finish (about 2,300 last year), not everyone who registers. The North Star's "6,000 others" fits the registered count, not the finishing count.

8. Cross-document consistency (X)

Each item: what the IPS guide, Trading Notes guide or web pages say, and how it constrains or pulls against the case.

- **X-1 Horizons.** IPS L10 "time horizons" and L13 "changing time horizons"; criterion 1 "Laura's changing time horizons" (R-S24). The case has five dated points: 2027, 2028, 2031, 2033 and 2033-2042. Consistent. The team should name them explicitly.
- **X-2 Changing allocation versus "if at all".** IPS L22-23 (prompt 4) and L47 ("planned changes in the portfolio as future funding needs approach") versus case L96 ("if at all"). Mild tension. The case allows a static reserve; the IPS expects the overall portfolio's plan to change over time.
- **X-3 Where the numbers go.**
 - The IPS guide (L68-70) and the Trading Notes guide (L20-22) both exclude the final reserve calculations, projections, the final range and the co-sponsor draft. The case requires all of these, so they are Final Report items.
 - The IPS must still carry the approach to the reserve (L48-50) and the "desired degree of funding certainty" (L10). So the *definition* of certainty plausibly belongs in the IPS while the *calculation* does not (INTERPRETATION).
- **X-4 Individual investments.**
 - The IPS says describe the framework "rather than describing individual investments" (L51-52).
 - The Trading Notes are about individual decisions, but "as part of a cohesive portfolio strategy" (TN L14-15).
 - A clean division: the notes carry the specifics and the IPS carries the rules.
- **X-5 The freeze chain.** TN L18-19 (strategy may evolve) → IPS L73-74 ("may not revise its investment strategy after the submission deadline") → Final Report "evaluates how that strategy was implemented" (case L158). Criterion 1 requires "consistency with the team's IPS". Any strategy change must happen before 6 Nov and be explained in the Final Report.
- **X-6 The WInS freeze date is the IPS date.** [S-DELIV], [S-TRADE], [S-WINS]: the portfolio is frozen on 6 Nov. IPS L75: "Your portfolio and Final Report should reflect the strategy established in your IPS." The WInS holdings on 6 Nov will be read against the IPS.
- **X-7 WInS \$300,000 versus a two-deposit plan.**

- Case L129-133: WInS = implementation; its results stay out of projections. [S-TRADE]: the \$150,000 "will not be added to WInS".
- No document says whether WInS should show the Year-1 portfolio or the long-run target. This is a team choice to state and defend (open item in the brief).
- **X-8 Semifinal selection basis.** [RULES] says the IPS and Final Reports. [S-DELIV] says all three evaluated. Case L161 says the three together. The FAQ says "all five components of the criteria". Treat all three as scored.
- **X-9 Citations.** IPS L123-125 (no formal citations in the IPS; sources in the Final Report). [RULES] Code of Conduct (cite all sources). [RULES] and [AI] (cite AI; record AI use in "Works Cited pages"). Consistent only if the Final Report has a Works Cited section that includes how AI was used. The IPS has no citations but must not contain AI-written text (R-W26).
- **X-10 Advisor.** Case framing (the advisor decides) versus [RULES] and [S-FAQ] (the advisor may not decide or trade). The real rules govern conduct. The case framing governs only the story.
- **X-11 Risk tolerance.** IPS definition (L3) and the [RULES] IPS description ("risk tolerance") versus the case ("thoughtful risks", "appropriate balance"; no number). The IPS must state a risk tolerance that the team derives from the case.
- **X-12 Diversification.**
 - Criterion 1: "uses appropriate diversification". IPS L46-47: "portfolio construction and diversification". [S-TRADE]: no sector minimum; diversification by "funding purposes" counts.
 - Consistent: diversification is expected but defined broadly.
 - This removes the unverified "sector minimum" constraint from the brief (section 6).
- **X-13 Instruments.** The TN example uses a Treasury bond ETF. [S-TRADE] and [S-FAQ] also allow individual government bonds. Derivatives are banned. Both bond routes are open in WInS in principle. Confirm what is actually listed in WInS before trading.
- **X-14 Communicating with co-sponsors.** Case L143: "Communicates Laura's potential facility contribution to co-sponsors clearly and credibly." Criterion 5: "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors." Nearly the same words. The criterion adds "investment uncertainty".
- **X-15 Reasonable assumptions.** Case L132 "reasonable return assumptions consistent with their strategy"; criterion 3 "reasonable assumptions and projections"; [S-TRADE] and [S-WINS] repeat it. Consistent. Returns must be defensible, and consistent with the chosen strategy.
- **X-16 A shared vocabulary.** The same five words recur across documents: growth, risk, liquidity, funding reliability, financial flexibility (IPS L20 and L59; TN L5; [S-WINS]). The case uses "financial flexibility" and "funding confidence" (L127). These are the graders' own terms (INTERPRETATION).
- **X-17 Voice.** Criterion 5 "an authentic team voice"; [AI] "they must use their voice and words"; [RULES] Ethics "invent teamwork and experiential stories" is banned; [FAQ] "clearly and authentically". Consistent. The reflections and the Articulation section must be the students' own and true.
- **X-18 Trading activity.**
 - [RULES]: "must meet the required trading activity". [S-WINS]: not judged on "How many trades you make". [S-FAQ]: avoid "a lot of buying and selling". [S-TRADE]: cap of 200 trades.
 - The public pages set no minimum number. Some minimum may exist in the logged-in portal or in WInS weekly emails. Check.
- **X-19 Where the judging weight falls.** Case L161 says the deliverables are judged together. [FAQ] says criteria are "reflected in various ways in their final reports". The Final Report is the main showcase, but the IPS is the anchor.
- **X-20 Dates.** Case "beginning of Week 7" = [S-DELIV] "November 9". TN "Week 6" = IPS due 6 Nov. [FAQ] 10 weeks, 28 Sep to 4 Dec. Consistent.
- **X-21 Names.** IPS title page "First Name, Last Initial" versus the roster, which requires full names, emails and birthdays (R-S13). The repo keeps first names only (brief privacy rule).
- **X-22 Year 0 versus WInS.** Case Year 0 = 2026, "before any money has been invested", while WInS trades real market prices in 2026. Consistent with R-C76. WInS is a demonstration of the strategy, not Laura's money.

- **X-23 An IPS as a rulebook.** IPS L3-5 (guidelines; disciplined decisions in changing markets) and L61-62 ("guide portfolio construction and investment decisions as the client's needs change"), together with the case's "favorable and unfavorable" scenarios (L118, L148). The IPS should contain the decision rules the Final Report later tests (INTERPRETATION).
- **X-24 Corrections to the brief and CLAUDE.md** (VERIFIED-PRIMARY, pages cited above):
 - "Approved list, starting cash and WInS rules are UNKNOWN": now largely known. Starting cash \$300,000. Allowed: stocks at \$5 or more, any ETF on WInS, government bonds on WInS, cash. Banned: margin, shorting, stock-secured debt, crypto, derivatives. Cap of 200 trades. Volume cap of twice the daily volume. Commissions \$25 per stock trade and \$10 per bond trade.
 - "sector minimum = team size": contradicted ("no required sector allocation or minimum number of sectors").
 - "200-trade cap": confirmed.
 - "first trade by Oct 10": not found on any public page; still unverified.
 - "2025-26 starting cash \$500k": confirmed by [NEWS]. This season is \$300,000.
 - The historical 2025-26 ETF list is not needed this year, since any ETF on WInS is allowed.
 - Still to check inside WInS: which specific ETFs and bonds are actually listed, and whether any minimum trading activity rule exists.

9. Sources (all accessed 2026-09-27)

- Repo official files (VERIFIED-REPO-FILE):
 - `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` and `.pdf`
 - `competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt` and `.pdf`
 - `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` and `.pdf`
 - `competition/official/2026_27/SMAApply_Deliverables_Page_2026-09-27.md`
 - `competition/official/2026_27/manifest.yaml`
 - Box official PDFs (VERIFIED-PRIMARY, hash-matched):
 - <https://upenn.box.com/shared/static/i4t4b2n4ycjkev7p78mhyjsvypombkxo.pdf>
 - <https://upenn.box.com/shared/static/0iib496jhswp30c12zhn6b2fret8dl30.pdf>
 - <https://upenn.box.com/shared/static/3pxn798a1lnlifkikj4g56srmg2ijuo6.pdf>
 - Wharton pages (VERIFIED-PRIMARY), keys as in section 6:
 - <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/>
 - `.../rules-roles/`
 - `.../faq/`
 - `.../register-now/`
 - `.../archive/`
 - <https://globalyouth.wharton.upenn.edu/ai-policy/>
 - <https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/>
 - SMAApply public pages (VERIFIED-PRIMARY):
 - <https://wghsinvcomp.smapply.us/>
 - <https://wghsinvcomp.smapply.us/res/p/client/>
 - <https://wghsinvcomp.smapply.us/res/p/trading/>
 - <https://wghsinvcomp.smapply.us/res/p/wins/>
 - <https://wghsinvcomp.smapply.us/res/p/faqs/>
 - <https://wghsinvcomp.smapply.us/res/p/deliverables/>
 - Not reached: <https://edu.stocktrak.com/wharton/> (WInS itself; needs the team login, not attempted). The logged-in SMAApply pages (weekly updates, forms) are not public.
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What this teaches

1. Read a contract (and a case is a kind of contract) three ways:
 - Pull out the **hard rules** ("must", "may not").

- Pull out the **undefined words** ("high degree of certainty", "responsibly", "credible"). The team has to define these itself, and that is where teams win or lose marks.
 - Note what is **left out**. Currency, a risk-tolerance number and a confidence level are all missing, so any choice there must be explained.
2. Always check whether the official source says more than you were told. Here, the "unknown" trading rules were sitting on public pages. Knowing them (no sector minimum, individual Treasury bonds allowed, 200 trades, no derivatives) changes what the team can safely do.
 3. When official sources disagree (which deliverables pick the semifinalists; citations or none), do not pick the convenient one. Adopt the reading that keeps you compliant with all of them, and write down why.

Fact register (A2, Phase A): the run's single source of truth for numbers

AI-generated research (Claude Code, 2026-09-27) for Team Caplet. Not submission text: it is a list of checked numbers with their sources. The team decides what to use and writes every deliverable in its own words.

Summary (read this first)

- Both verified scripts reproduce exactly.** I re-ran them on 2026-09-27 and got the same results: \$292,264; lock-early 2033 surplus p5/p50/p95 \$159k/\$207k/\$273k; growth-first miss 3.2%. Every number in brief section 6 matches (F-101 to F-106, F-401 to F-408).
- NEW, high impact: this season's WInS rules are partly public.** The 2026-27 SMapply FAQs page can be opened without logging in. It says the team manages **\$300,000 of virtual cash**, the same as Laura's 2027 deposit. It also lists the permitted investments: cash, stocks priced at \$5 or more, "any ETFs available on WInS" and "any Government/Treasury Bonds ... available on WInS". Margin, shorting, crypto and derivatives are banned. Commissions are \$25 per stock trade and \$10 per bond trade (F-605 to F-608). This replaces "starting cash unknown" in the brief. The logged-in "Trading Details" page may add more rules, so check it and the WInS account itself before trading.
- "The ladder fits inside \$300k" is a September 2026 fact, not a normal one.** I priced the same ten payments on every 2026 Treasury curve. The ladder cost more than \$300k on 173 of 185 trading days: \$316.5k on Jan 2 and \$325.6k at the February low in rates. It fell below \$300k only in September (F-111). Using an ASSUMPTION of a zero-drift random walk at 2026 volatility, the chance it costs more than \$300k on 2027-01-01 is about 24% (F-112). That is within the council's 23-32%.
- Most brief section 6 "UNVERIFIED" items are now confirmed on primary sources:**
 - The FOMC raised rates 25bp to 3.75-4.00%, 12-0, on 2026-09-16. It was the first rise since July 2023 (F-005).
 - Taiwan CPI for August 2026 was +2.04% (DGBAS). The 2.4% figure is wrong (F-501).
 - The central bank of Taiwan held its discount rate at 2% on 2026-09-17 (F-506).
 - The latest USD/TWD on FRED is 31.82 (2026-09-18). Volatility over the last 10 years is 4.69% (F-510, F-512).
 - The current IEF/TLT durations are 6.86/14.88 years, which gives a duration match of 62.2/37.8 (F-203 to F-205).
 - iShares Treasury iBonds exist for Dec 2026-2036 but not for Dec 2037-2043 (F-206).
- Taiwan construction costs.** From the official DGBAS platform I derived +6.53% y/y (the snippet says 6.54%). That jump happened in 2026 alone: the index barely moved in 2024-25 and has averaged ~3.5% a year since 2021 (F-508). So the one-month "3.2x CPI" multiple should not be used as a trend (S-34).
- I resolved all 10 cross-file inconsistencies in brief section 10 and superseded 38 older numbers** (section H). The largest corrections are:
 - "Curve B" was the official curve of 2026-03-20, not a mystery. It was six months stale.
 - The coupon-reinvestment "dispute" was two correct sums for two different ladders.
 - The "65/35" WInS mirror is ladder/sleeve, not bonds/equity. Total equity after 2028 is about 17-24% (F-409).

How to read this file

- ID scheme:** F-0xx rates; F-1xx liability valuation; F-2xx instruments; F-3xx capital-market assumptions; F-4xx simulation outputs; F-5xx Taiwan/FX/inflation; F-6xx competition facts; S-xx superseded numbers.
- Status labels** follow brief section 3. **Derived numbers** are arithmetic by an A2 script on labelled inputs. They carry the weakest input's label plus "(derived, script)". "ASSUMPTION method" means the arithmetic depends on a modelling choice, for example how the curve is bootstrapped.
- Feeds** lists the later numbers that change if this one changes.
- Access date** for every web source is 2026-09-27 (between 11:38 and 12:15 UTC), from this run's container.
- Trading rule:** any item that mentions an instrument carries "PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading" (brief section 4, updated 2026-09-27). This register records facts about instruments; it does not pick securities.
- Note for the brief:** brief section 4 still says this year's starting cash and WInS rules are "UNKNOWN". F-605 to F-608 show that the public SMapply FAQs page already states the cash amount and the permitted-investment categories. The logged-in details are still unknown.

Plain-English terms used below

- **Par yield:** the coupon rate at which a new Treasury would sell for exactly its face value.
- **Basis point (bp):** 0.01%.
- **Discount factor / bootstrap:** turning par yields into the value today of \$1 paid on a future date.
- **Forward value:** what today's curve locks in for a later date.
- **DV01:** dollars gained or lost for a 1bp parallel rate move.
- **Duration:** the % price change for a 1% rate move.
- **Effective duration:** the fund manager's duration measure.
- **Yield to maturity (YTM):** the fund's average yield if its bonds are held to maturity.
- **30-day SEC yield:** a standardised recent income yield, after fees.
- **Compound (geometric) return:** the growth rate actually achieved over many years.
- **Arithmetic return:** the simple average of yearly returns. It is higher than the compound return when returns jump around.
- **Volatility:** the standard deviation of returns, per year.
- **Correlation:** how closely two assets move together, from -1 to +1.
- **Breakeven inflation:** nominal yield minus TIPS real yield, i.e. the inflation the market is "pricing".

Step 1: reproduction check (2026-09-27, .venv/bin/python, repo root)

Script	Brief section 6 value	Re-run output	Match
research/verified_2026-09-27/official_curve_pv.py	\$292,264; -25bp \$299,596; -50bp \$307,135; -100bp \$322,864; +50bp \$278,198; +100bp \$264,890; DV01 \$289/bp; duration 9.90y; headroom \$7,736 = 27bp; forward zeros 5.08/5.24/5.48%; 64.8/35.2	identical, digit for digit	Yes
research/verified_2026-09-27/strategy_mc.py (200k paths, seed 20260927)	L never misses; surplus p5/p50/p95 \$159k/\$207k/\$273k; floor \$127k/\$165k/\$217k; G 3.2% (13.7% at \$75k, 40.8% at \$0); G surplus \$20k/\$226k/\$540k; sleeve equity 50/60/70% median \$205k/\$207k/\$209k, p5 \$164k/\$159k/\$154k; equities 5.0%: L median \$200k, G miss 5.5%	identical	Yes
"Same payments valued at 2033-01-01 on forwards: ~\$394.9k" (not printed by either script)	~\$394.9k	\$394,930 (A2_curve_recheck.py)	Yes

New scripts written by A2 (all in `research/insight_v1/scripts/`, each with a docstring listing inputs, labels and the run command):

- `A2_curve_recheck.py`: curve and ladder numbers, 2026 history, IEF/TLT weights, council arithmetic checks.
- `A2_jpm_extract.py`: JPM LTCMA rows and correlations, read from the PDF.
- `A2_twd_vol.py`: USD/TWD level and volatility, live from FRED.
- `A2_taiwan_cci.py`: the Taiwan construction cost index, rebuilt from the DGBAS platform.

A. Rates and Treasuries

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-001	U.S. Treasury par yield curve	1M 4.04, 1.5M 4.14, 2M 4.20, 3M 4.24, 4M 4.32, 6M 4.33, 1Y 4.50, 2Y 4.81, 3Y 4.94, 5Y 4.98, 7Y 5.06, 10Y 5.17, 20Y 5.54, 30Y 5.49	% (semiannual bond-equivalent)	2026-09-25	competition/official_market_data/daily-treasury-rates-2026-09.csv row 09/25/2026; the live CSV at https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&format=csv has identical September rows	VERIFIED-REPO-FILE and VERIFIED-PRIMARY	F-011-F-014, all of B, F-205, F-401-F-409	The verified script ignores the 1.5M and 4M columns (no effect).
F-002	Is there a newer curve?	No. The latest date is 2026-09-25 (9/26-27 is a weekend)	-	checked 2026-09-27 11:40 UTC	same live CSV	VERIFIED-PRIMARY	-	Re-price on purchase day (January 2027).

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-003	Path of yields in 2026	10Y: 4.19 (Jan 2), low 3.97 (Feb 27), 4.44 (Jun 30), 4.79 (Sep 1), high 5.18 (Sep 24), 5.17 (Sep 25). 2Y: 3.47 (Jan 2) to 4.81. 30Y: 4.86 to 5.49	%	2026-01-02 to 09-25	live 2026 CSV (F-001)	VERIFIED-PRIMARY	F-111, F-112	The 10Y rose 38bp in September (4.79 to 5.17) and about 120bp from the February low.
F-004	How high is the 10Y historically?	5.18% on 2026-09-24 is the highest daily close since 2007-07-06 (5.19%); the 2008-2025 maximum was 4.98% (2023-10-19)	%	2026-09-24	FRED DGS10 (H.15), https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10	VERIFIED-PRIMARY	landscape	Snippets said "since June 2007"; the last higher close was 6 July 2007. A team note's "highest since late 2023" is wrong (S-13).
F-005	FOMC decision	Raised 1/4 point to "3-3/4 to 4 percent"; vote "12 - 0"; effective 2026-09-17	%	2026-09-16	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm	VERIFIED-PRIMARY	F-515	First increase since 2023-07-27 (to 5.25-5.50). The range was 3.50-3.75 from 2025-12-11 until this move (https://www.federalreserve.gov/monetarypolicy/openmarket.htm).
F-006	Implementation rates	Interest on reserve balances 3.90%; overnight reverse repo 3.75%; standing repo 4.0%; primary credit 4.0%	%	effective 2026-09-17	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a1.htm	VERIFIED-PRIMARY	F-515	-
F-007	Fed projections (Summary of Economic Projections), medians	Fed funds end-2026 4.1, 2027 4.1, 2028 3.9, 2029 3.6, longer run 3.2 (2026 central tendency 4.1-4.4); PCE inflation 2026 3.7, 2027 2.3, 2028 2.1, longer run 2.0; core PCE 2026 3.4; GDP 2026 2.3; unemployment 2026 4.1	%	2026-09-16	https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm	VERIFIED-PRIMARY	F-114	The 2026 median of 4.1 is the midpoint of 4.00-4.25, i.e. one more 25bp hike. These are projections, not promises.
F-008	TIPS real par yields	5Y 2.64, 7Y 2.73, 10Y 2.83, 20Y 3.08, 30Y 3.22 (10Y 2.85 on 9/24; 2.61 on 9/17)	%	2026-09-25	https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_real_yield_curve&field_tdr_date_value=2026&page&format=csv	VERIFIED-PRIMARY	F-009	Resolves the council's 2.85% vs 2.653% conflict (S-22).
F-009	Breakeven inflation (nominal minus real)	5Y 2.34, 7Y 2.33, 10Y 2.34, 20Y 2.46, 30Y 2.27	%	2026-09-25	F-001 minus F-008	VERIFIED-PRIMARY (derived)	F-114	The 7Y ($\approx$ to 2033) is 2.33%. This is market pricing, including risk premia; it is not a forecast.
F-010	10Y on the JPM LTCMA data date	4.16%	%	2025-09-30	FRED DGS10	VERIFIED-PRIMARY	F-317	Yields are about 100bp higher now than when JPM set its bond assumptions.

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-011	Spot zero rates for payment dates, from 9/25	Jan-2027 4.33%; Jan-2033 5.05%; Jan-2037 5.22%; Jan-2042 5.46% (annual compounding: 4.38/5.11/5.29/5.54)	% semiannual	2026-09-25	A2_curve_recheck.py method on F-001	VERIFIED-REPO-FILE inputs + ASSUMPTION method	S-10	Replaces the chair memo's 5.04-5.38%.
F-012	Forward zero rates from 2027-01-01	to 2033 5.08%; to 2037 5.24%; to 2042 5.48%	% semiannual	2026-09-25	official_curve_pv.py	VERIFIED-REPO-FILE inputs + ASSUMPTION method	strategy_mc (5.08% forward factor)	Reproduced.
F-013	2-year growth factor for the 2031 floor at today's 2Y	1.0985 (annual convention, used by strategy_mc) vs 1.0997 (semiannual bond-equivalent, Treasury's convention)	x	2026-09-25	A2_curve_recheck.py	ASSUMPTION (2031 2Y = today's 4.81%)	F-402	The 0.12% difference is trivial (\$166.1k vs \$166.3k at S=\$189k, a=0.8). Say which convention is used.
F-014	Realised volatility in 2026	10Y yield: 71bp a year; ladder cost (value at 2027-01-01): 7.24% a year	bp / %	2026-01-02 to 09-25	A2_curve_recheck.py on live 2026 CSV	VERIFIED-PRIMARY inputs (derived)	F-112	Council assumed 36-57bp over 0.27y, which is 69-110bp a year.

B. Liability valuation (the ten \$50,000 payments, 1 Jan 2033-2042)

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-101	Cost at 2027-01-01 (value locked by today's curve)	\$292,264	USD	curve 2026-09-25	official_curve_pv.py; A2_curve_recheck.py [1]	VERIFIED-REPO-FILE inputs + ASSUMPTION method	F-104, F-117, F-401-F-409	Reproduced. The single number to use (brief section 8.4).
F-102	Parallel-shift costs	-100bp \$322,864; -50bp \$307,135; -25bp \$299,596; +25bp \$285,133; +50bp \$278,198; +100bp \$264,890	USD	2026-09-25	same	same	rule-if-rates-fall analysis	-
F-103	DV01 / duration of the 2027 value	\$289/bp (289.2); 9.90 years	USD/bp; years	2026-09-25	same	same	F-205	-
F-104	Headroom under the \$300k deposit	\$7,736 = 27bp (26.7)	USD; bp	2026-09-25	same	same	F-112	-
F-105	Value today (spot) and its duration	\$288,924; 10.16 years	USD; years	2026-09-25	A2_curve_recheck.py [1]	same	F-205	For hedging today (WInS), the spot duration (10.16y) is the right target. The 9.90y figure is the duration of the January-2027 value.
F-106	Forward value of the remaining payments	at 2028-01-01 \$306,077; at 2031-01-01 \$356,384; at 2033-01-01 \$394,930	USD	2026-09-25	A2_curve_recheck.py [2]	same	F-409; "which number is the reserve"	Replaces the council's \$357-358k / \$395.8k-\$397k (S-09).
F-107	Cost of each rung at 2027-01-01	2033 \$37,002; 2034 \$35,095; 2035 \$33,262; 2036 \$31,497; 2037 \$29,792; 2038 \$28,154; 2039 \$26,579; 2040 \$25,065; 2041 \$23,607; 2042 \$22,210. 2033-36 total \$136,856; 2037-42 total \$155,408	USD	2026-09-25	same	same	long-rungs-first rule	The council said 2037-42 was "about \$154k" (S-09 family).

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-108	Extra cost of the 2037-42 rungs if rates fall before they are bought	-100bp +\$20,221; -200bp +\$43,232	USD	2026-09-25	same	same	tiering rejection (brief section 8.2)	Reproduces the council's \$20k/\$43k.
F-109	Flat-rate equivalents	A flat 5.359% a year reproduces \$292,264; a flat 5.26% gives \$295,054 (the old "\$295k")	%; USD	2026-09-25	A2_curve_recheck.py [3]	ASSUMPTION (flat curve)	F-110	5.26% is not the verified curve's equivalent; 5.36% is.
F-110	Reserve needed on 2033-01-01 if bought then (annuity-due, flat rate)	7% \$375,762; 5.36% \$399,778; 5.26% \$401,312; 5% \$405,391; 4.5% \$413,440; 4% \$421,767; 3% \$439,305	USD	-	same	ASSUMPTION (rates are illustrations)	growth-first comparison	strategy_mc prices G's 2033 reserve at 5.26% ± 1pp. That is about \$6.4k above the forward value (F-106), so it slightly overstates G's shortfall (brief section 11).
F-111	The same ladder priced on every 2026 curve date	Jan 2 \$316,546; max \$325,563 (Feb 27, 10Y 3.97%); Jun 30 \$312,974; Sep 1 \$302,391; min \$292,051 (Sep 24). Above \$300k on 173 of 185 trading days; at or below \$300k only on the 12 days since 2026-09-10	USD	2026-01-02 to 09-25	A2_curve_recheck.py [6] on live 2026 CSV	VERIFIED-PRIMARY inputs + ASSUMPTION method	rule-if-rates-fall	"Fits inside \$300k" rests on the September 2026 sell-off.
F-112	Chance the ladder costs more than \$300k on 2027-01-01	~24%; a one-standard-deviation move by then is about ±\$11.0k	%	2026-09-25	A2_curve_recheck.py [6]	ASSUMPTION (zero-drift lognormal at 2026 realised volatility of 7.24%/yr over 0.27y)	rule-if-rates-fall	A model number, not a forecast. The council gave 23-32% (38% at \$295k).
F-113	Coupon-reinvestment arithmetic (if coupon Treasuries are used instead of zero-coupon bonds)	Budget ladder (face = cost \$295,054): coupons \$15,520/yr in 2028-32; reinvesting 200bp lower loses \$5,211 by 2033. A fully matched-from-2033 coupon ladder needs \$381,258 of face, which pays \$20,054/yr; the loss is \$6,734	USD	-	A2_curve_recheck.py [4]	ASSUMPTION (flat 5.26% annual coupons, reinvested at 3.26%)	brief section 8.11 (zero-coupon preferred)	Both council sums are correct. The budget-ladder one (\$15.5k/\$5.2k) is the one that fits Laura's \$300k (S-16).
F-114	What a fixed \$50k is worth in today's (2026) money	2.0% inflation: 2033 \$43,528, 2042 \$36,422. 2.5% (JPM): \$42,063 / \$33,681. 3.7% (Fed's 2026 PCE median, used only as a stress): \$38,772 / \$27,958	USD (2026 dollars)	-	A2_curve_recheck.py [7]	ASSUMPTION (inflation rates)	real-erosion question (brief section 9)	U.S. dollars only; Taiwan costs are in F-5xx.
F-115	Discounting example	\$50k due in 6 years at 5.26% = \$36,761	USD	-	same [3]	ASSUMPTION (flat rate)	teaching examples	The referee's "\$36.9k" should read \$36.8k (S-17).
F-116	Team note's "\$626k by 2033"	implies a flat 5.99% a year	%	-	same [3]	ASSUMPTION check	-	Reproduces the council.
F-117	Funded ratio (assets ÷ Treasury price of the promise)	\$300,000 / \$292,264 = 1.026	ratio	2026-09-25	F-101	VERIFIED-REPO-FILE inputs + ASSUMPTION method	certainty definition	Any fall in rates before purchase lowers it (F-111).

C. Instruments (verified data only; PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading)

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-201	IEF fact sheet (7-10 year Treasury fund)	Effective duration 6.95 yrs; expense ratio 0.15%; 30-day SEC yield 4.29%; weighted average maturity 8.45 yrs; convexity 0.58; 13 holdings; 3y standard deviation 6.54%; equity beta 0.23; net assets \$47,093.37M; 1y NAV return 2.61%; 2022 return -15.23%. No yield to maturity is printed	various	2026-06-30	competition/official_market_data/IEF_fact_sheet_2026-06-30.pdf p.1 (pypdf)	VERIFIED-REPO-FILE	F-205	-
F-202	TLT fact sheet (20+ year Treasury fund)	Effective duration 15.31 yrs; expense ratio 0.15%; SEC yield 4.89%; weighted average maturity 26.03 yrs; convexity 3.27; 46 holdings; 3y standard deviation 13.74%; beta 0.58; net assets \$41,099.93M; 2022 return -31.41%. No yield to maturity printed	various	2026-06-30	.../TLT_fact_sheet_2026-06-30.pdf p.1	VERIFIED-REPO-FILE	F-205	-
F-203	IEF today	Effective duration 6.86 yrs ; YTM 5.17%; SEC yield 4.84%; 12m trailing yield 4.13%; weighted average maturity 8.44; convexity 0.57; weighted average coupon 4.24; YTD NAV total return -4.20% (all as of Sep 24). NAV \$89.94 on Sep 25 (52-week range 89.68-97.96); net assets \$41,570,344,536; expense 0.15%	various	2026-09-24/25	https://www.ishares.com/us/products/239456/ishares-710-year-treasury-bond-etf	VERIFIED-PRIMARY	F-205	Duration fell from 6.95 as yields rose.
F-204	TLT today	Effective duration 14.88 yrs ; YTM 5.54%; SEC yield 5.41%; trailing yield 4.90%; weighted average maturity 26.04; convexity 3.13; weighted average coupon 3.34; YTD -6.15% (as of Sep 24). NAV \$79.28 on Sep 25, at its 52-week low (range 79.28-92.05); net assets \$45,753,218,426; expense 0.15%	various	2026-09-24/25	https://www.ishares.com/us/products/239454/ishares-20-year-treasury-bond-etf	VERIFIED-PRIMARY	F-205	Mark-to-market losses matter for WInS only, not for a held-to-maturity ladder.
F-205	Duration-matching mix (a duration match only, NOT a cash-flow match)	June fact sheets, 9.90y target: 64.8% IEF / 35.2% TLT . Current durations, 9.90y: 62.2 / 37.8 . Spot 10.16y target: 61.6/38.4 (June) or 58.9/41.1 (current)	%	2026-09-24	A2_curve_check.py [5]	VERIFIED-PRIMARY inputs (derived)	WInS mirror	The mix drifts as durations and time move, so re-check before each hedge trade. PENDING APPROVAL CHECK: confirm both funds on this year's WInS approved list/rules before trading.
F-206	Which defined-maturity Treasury funds exist (iShares "iBonds Dec-20XX Term Treasury" family)	Dec 2026 through Dec 2036; then Dec 2044-2046 and Dec 2054-2056. None for Dec 2037-2043	-	2026-09-27	iShares product screener JSON (ishares.com, product-screener-v3.1)	VERIFIED-PRIMARY	brief section 8.12	Dec 2032-2036 funds pay out before the Jan 2033-Jan 2037 payments (5 payments). Jan 2038-2042 (5 payments) have no matching iShares fund. Other issuers were not checked, and availability inside WInS is unknown: PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading.

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-207	Example: the Dec 2034 fund in that family	Expense 0.07%; effective duration 6.44 yrs; YTM 5.15% (Sep 24)	various	2026-09-24	https://www.ishares.com/us/products/337745/ishares-ibonds-dec-2034-term-treasury-etf	VERIFIED-PRIMARY	-	Shows the fund type's cost level (the council's 0.07% snippet is now confirmed for this vintage).
F-208	Other instrument claims in groundwork/instruments.md (BulletShares Treasury 2027-31 launch; zero-coupon and T-bill fund expense ratios; VGIT/VGLT durations)	as stated there	-	-	council snippets	SNIPPET-UNVERIFIED (not re-checked by A2)	-	Low priority. Re-check on issuer pages before any of these is recommended (brief section 4, rule d).

D. Capital-market assumptions (J.P. Morgan 2026 LTCMA, U.S. dollar matrix)

Source for F-301 to F-314: [competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf](#), page 2 ("Source: J.P. Morgan Asset Management; data as of September 30, 2025"; "All returns are nominal"). Read with pypdf by [A2_jpm_extract.py](#). Status: **VERIFIED-REPO-FILE**.

The column order (compound 2026, arithmetic 2026, volatility, compound 2025) comes from the header positions. It is confirmed by the check $\text{arithmetic} \approx \text{compound} + \text{vol}^2/2$, which works only in this order. The horizon is "10- to 15-year" (<https://am.jpmorgan.com/us/en/asset-management/institutional/insights/portfolio-insights/ltcma/>, VERIFIED-PRIMARY). Laura's growth-sleeve horizon (2027-2033) is shorter than that.

ID	Asset class	Compound 2026	Arithmetic 2026	Volatility	Compound 2025 (prior edition)	Feeds / notes
F-301	U.S. Large Cap	6.70	7.94	16.47	6.70	strategy_mc equity input (brief section 8.14: compound)
F-302	AC World Equity	7.00	8.28	16.78	7.10	global alternative for the sleeve
F-303	EAFE Equity	7.50	8.90	17.63	8.10	-
F-304	Emerging Markets Equity	7.80	9.74	20.93	7.20	-
F-305	U.S. Small Cap / U.S. Mid Cap	6.90 / 7.00	8.89 / 8.55	21.10 / 18.56	6.90 / 7.00	-
F-306	AC Asia ex-Japan Equity	7.90	9.83	20.84	7.20	Closest JPM row to any Taiwan tilt; JPM has no Taiwan-only row
F-307	U.S. Intermediate Treasuries	4.00	4.06	3.48	3.80	strategy_mc bond input
F-308	U.S. Long Treasuries	4.90	5.69	13.02	4.30	-
F-309	TIPS	4.30	4.47	5.88	4.10	-
F-310	U.S. Aggregate Bonds	4.80	4.91	4.76	4.60	-
F-311	U.S. Cash	3.10	3.10	0.67	3.10	-
F-312	U.S. Inflation	2.50	2.52	1.77	2.40	F-114
F-313	Others (compound / arithmetic / vol)	Short-duration govt/credit 4.00/4.01/1.63; IG corporate 5.20/5.46/7.39; high yield 6.10/6.46/8.74; world govt hedged 4.30/4.38/4.02; U.S. minimum-volatility factor 7.00/7.79/13.16; U.S. REITs 8.80/10.15/17.40; commodities 4.60/6.15/18.32; gold 5.50/6.78/16.68				reference only

ID	Fact	Value	Status	Notes
F-314	JPM correlations	U.S. large cap vs intermediate Treasuries -0.01 ; vs long Treasuries 0.02; vs cash 0.01; vs inflation -0.01; vs TIPS 0.34; vs aggregate bonds 0.30. AC World vs intermediate 0.00, vs long 0.01. EAFE 0.02/0.02. EM 0.00/0.02. Small cap -0.08/-0.05. Intermediate vs long Treasuries 0.84. Large cap vs AC World 0.97, vs EAFE 0.87, vs EM 0.73. Inflation vs intermediate Treasuries -0.27, vs long -0.22	VERIFIED-REPO-FILE	Near-zero stock-bond correlation is JPM's long-run view. The council's +0.4 stock-rate stress remains an ASSUMPTION.
F-315	Is a newer JPM edition out?	No. The LTCMA page still shows the 2026 edition on 2026-09-27; the 2026 edition was released 2025-10-20 (search result)	VERIFIED-PRIMARY (page) / SNIPPET-UNVERIFIED (release date)	A 2027 edition may appear before the IPS (Nov 6) or Final Report (Dec 4). Re-check in October ; it will reflect yields about 100bp higher (F-010).
F-316	strategy_mc matches JPM	Equity log mean 0.0649 and log sd 0.1520 give an annual-return sd of about 16.5% (JPM 16.47%). Bond log sd 0.0340 gives about 3.5% (JPM 3.48%). Correlation -0.01	VERIFIED-REPO-FILE inputs (derived)	Inputs are faithful to JPM, but i.i.d. lognormal with no fat tails (brief section 11).
F-317	JPM bond assumptions vs today's yields	JPM intermediate Treasuries 4.00% compound (set when the 10Y was 4.16%) vs IEF YTM 5.17% today (F-203)	VERIFIED (derived comparison)	strategy_mc's sleeve bonds are probably conservative by about 1pp a year (ASSUMPTION interpretation).
F-318	Chance U.S. large cap loses money	~33% over 1 year; ~27% over 2 years (2031 to 2033)	ASSUMPTION (JPM compound/arithmetic, lognormal)	Replaces the chair memo's 36%/32% (16% vol, 5.5%). Model property, not a forecast.

E. Simulation outputs (`strategy_mc.py`; all ASSUMPTION-based model outputs, reproduced exactly 2026-09-27)

These are properties of the model, not of the world. The inputs are F-101, F-001 (2Y 4.81%), F-301, F-307 and F-314. The assumptions are listed in the script docstring and in brief section 11.

ID	Case	Lock-early (L)	Growth-first (G)
F-401	Base	Never misses (by construction). 2033 surplus p5/25/50/75/95 \$159k/\$186k/\$207k/\$232k/\$273k	Miss 3.2%. Surplus \$20k/\$131k/\$226k/\$338k/\$540k
F-402	2031 floor (80% of sleeve in a 2Y at 4.81%)	p5/50/95 \$127k/\$165k/\$217k	-
F-403	2028 deposit \$75k	Surplus p5/50/95 \$84k/\$109k/\$144k; floor \$67k/\$87k/\$114k	Miss 13.7%; surplus p5/50/95 -\$51k/\$125k/\$396k
F-404	2028 deposit \$0	Surplus \$8k/\$11k/\$15k; floor \$6k/\$9k/\$12k	Miss 40.8%; surplus -\$123k/\$25k/\$253k
F-405	Equities 5.0% compound	\$154k/\$200k/\$264k	Miss 5.5%; -\$4k/\$189k/\$483k
F-406	2033 rate shock sd 1.5pp	unchanged	Miss 3.5%; \$17k/\$226k/\$541k
F-407	Sleeve equity 50 / 60 / 70%	p5/50/95 \$164k/\$205k/\$260k; \$159k/\$207k/\$273k; \$154k/\$209k/\$287k	-

ID	Fact	Value	Status	Notes
F-408	Growth-first's 2033 reserve price in the model	Flat 5.26% ± 1pp gives \$401.3k at the centre, vs \$394.9k on the forward curve (F-106)	ASSUMPTION	Overstates G's shortfall slightly (brief section 11).
F-409	Total equity share after the 2028 deposit (ladder ≈ 66% of assets, sleeve ≈ 34%)	Sleeve equity 50/60/70% gives 17.0 / 20.4 / 23.8% of the whole portfolio	ASSUMPTION (sleeve grows 5% in 2027; ladder at forward value \$306,077)	The "65/35" WInS mirror is ladder/sleeve , not bonds/equity. Read as 65% Treasuries / 35% equity, it holds about 15pp more equity than the plan (resolves brief section 9, WInS mirror inconsistency).
F-410	Council model outputs A2 could not re-run (council scripts were in an earlier session's scratchpad, now gone)	No-lock shortfall 3.3-9.3%. Partial locks 50/80/90/95% fail 35/26/14/3.4% with no 2028 deposit. Median facility \$194-212k. G-L median gap -\$23k to +\$19k; p5 gap \$125-150k in L's favour. Twist gap \$3.3-4.4k	ASSUMPTION-based, NOT reproducible	Lower authority than F-401 to F-407. Quote only with "council model, unverified".

F. Taiwan, FX and inflation

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-501	Taiwan CPI, August 2026	+2.04% y/y ; -0.01% m/m (+0.13% seasonally adjusted); Jan-Aug average +1.84%	%	released 2026-09-08	https://eng.stat.gov.tw/News_Content.aspx?n=2317&s=236681 and https://www.stat.gov.tw/News_Content.aspx?n=3703&s=236675 (DGBAS)	VERIFIED-PRIMARY	F-509	The CEIC snippet's 2.4% is wrong (S-19). The snippet's "+0.13% m/m" was the seasonally adjusted figure.
F-502	Taiwan CPI, earlier months	Jul +2.54%, Jun +2.60%, May +2.20% y/y	%	2026	DGBAS release headlines (www.stat.gov.tw news list)	VERIFIED-PRIMARY	-	-
F-503	Taiwan producer and trade prices, Aug 2026	PPI +16.75% y/y; import prices (USD) +20.45%; export prices (USD) +23.05%	%	2026-09-08	F-501 pages	VERIFIED-PRIMARY	landscape	Strong upstream price pressure (AI boom, Middle East energy shock per DGBAS/CBC text).
F-504	DGBAS forecasts	CPI 2026 2.07% , 2027 1.90% ; real GDP 2026 11.05%, 2027 6.04%	%	2026-08-14	https://eng.stat.gov.tw/News_Content.aspx?n=2317&s=236587	VERIFIED-PRIMARY	F-509	Confirms the Focus Taiwan snippet.
F-505	Central bank (CBC) forecasts	CPI 2026 2.03%, core 2.16%; 2027 CPI 1.83%, core 1.89%; GDP 2026 11.48%, 2027 5.82%; Jan-Aug core CPI average 2.11%	%	2026-09-17	https://www.cbc.gov.tw/en/cp-448-192883-c5fa1-2.html	VERIFIED-PRIMARY	F-509	-
F-506	CBC policy rates	Held: discount rate 2% , secured-loan refinancing 2.375%, temporary accommodations 4.25%	%	2026-09-17	same	VERIFIED-PRIMARY	F-515	The discount rate has been 2% since 2024-03-22 (CBC home page), so this is the 10th quarterly hold in a row (derived count).
F-507	Taiwan market rates	Overnight interbank call-loan rate 0.812% ; M2 growth 6.79%	%	2026-09-24 / 09-23	https://www.cbc.gov.tw/en/mp-2.html	VERIFIED-PRIMARY	F-515	Taiwan market rates are far below the 2% discount rate.
F-508	Taiwan construction cost index (2021 = 100)	Derived total 119.50 (Aug 2026): +6.53% y/y (Jul +6.53%). Wage class 123.75, +8.12% y/y (published sub-index). Total excluding wages +6.02%. Machinery rental +2.64%. Index was flat in 2024-25 (110.3 in Jan 2024 to 112.9 in Dec 2025) and jumped in Jan-May 2026. Average since the 2021 base: 3.54%/yr ; Jan 2024-Aug 2026: 3.16%/yr	index; %	2026-08	DGBAS CCI platform https://www.stat.gov.tw/CCI/CCI_Site/CCIPrice/PartialPriceClassesList.aspx via A2_taiwan_cci.py	VERIFIED-PRIMARY (wage index) / derived total (fit error ≤0.023 points)	F-509, facility-cost framing	The headline 6.54% (udn snippet) is corroborated within 0.01pp but was not read on a primary page, because the ws.dgbas.gov.tw files were blocked.
F-509	Construction costs vs CPI	One month (Aug 2026): 6.54 / 2.04 = 3.2x . As a trend: CCI about 3.5%/yr since 2021 vs Taiwan CPI 2026-27 forecasts of about 1.8-2.1%	ratio	2026	F-501, F-504, F-505, F-508	derived; long-run CPI history SNIPPET-UNVERIFIED	co-sponsor range	Do not project 6.5%/yr or 3.2x forward; both are a one-year spike (S-34).

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-510	USD/TWD (FRED DEXTAUS)	31.82 TWD per USD	TWD/USD	2026-09-18 (updated Sep 21; next release Sep 28)	https://fred.stlouisfed.org/series/DEXTAUS (Fed H.10 noon buying rate)	VERIFIED-PRIMARY	F-512-F-514	Matches the council's mirror file.
F-511	NT\$/US\$ closing rate (CBC)	31.780	TWD/USD	2026-09-24	https://www.cbc.gov.tw/en/mp-2.html	VERIFIED-PRIMARY	-	The chair memo's "31.73 on 9/26" was a Saturday and is unverifiable (S-21).
F-512	USD/TWD volatility (daily log changes x $\sqrt{252}$)	1y 4.03%; 3y 5.64%; 5y 5.25%; 10y 4.69% ; full history since 1983 4.79%. 10y from month-end values 5.58% (monthly averages give 4.44%, which understates). 10y range 27.52-33.21; 10y change +0.14%/yr; last 12 months +5.61% (TWD weaker)	% a year	to 2026-09-18	A2_twd_vol.py (live FRED)	VERIFIED-PRIMARY inputs (derived)	F-513, co-sponsor range	Replaces the unsourced "5-7%" (S-20).
F-513	Size of real multi-year moves	Since 2006: 2-year sd 6.8% (p5/p95 -12.1%/+10.6%); 6.25-year sd 6.4% (-12.7%/+6.7%). Since 1983: 2-year sd 9.4%, 6.25-year sd 16.0% (includes the 1980s appreciation and 1997). $\sqrt{\text{time}}$ from 10y daily: 2y 6.6%, 6.25y 11.7%	%	to 2026-09-18	A2_twd_vol.py	VERIFIED-PRIMARY inputs (derived)	2031 range, TWD quote	The period chosen changes the answer a lot. State which one you use.
F-514	FX illustration	\$50,000 = NT\$1,591,000 at 31.82. If TWD per USD moves -5%/-10%, the same NT\$ cost \$52,632/\$55,556; +5%/+10% gives \$47,619/\$45,455	USD	2026-09-18	A2_twd_vol.py	ASSUMPTION (moves are illustrations)	-	-
F-515	U.S.-Taiwan short-rate gap	Policy: 3.75-4.00 minus 2.00 = 175-200bp (different instruments). Market: interest on reserve balances 3.90 minus Taiwan call rate 0.812 $\approx$ 310bp	bp	Sep 2026	F-005, F-006, F-506, F-507	VERIFIED-PRIMARY inputs (derived)	cost of converting the floor to TWD	The council's "~5.3% over two years" assumed a 2% Taiwan 2-year rate. With market rates the cost is probably ~6-8% over two years (ASSUMPTION). The Taiwan 2-year government yield is UNVERIFIED (S-35).
F-516	Taiwan 10-year yield	~1.70%	%	2026-07-14	snippet (council)	SNIPPET-UNVERIFIED	-	Not found on a primary page by A2.

ID	Fact	Value	Unit	As of	Source	Status	Feeds	Notes
F-517	Context stated by the official sources	DGBAS: a Middle East conflict since late February 2026 pushed energy costs up. CBC: the ECB hiked twice since June, the BoJ resumed hikes and the Fed "pivoted toward a policy rate increase"	text	Aug-Sep 2026	F-504, F-505 pages	VERIFIED-PRIMARY	landscape (Phase D)	Background only.

G. Competition facts

ID	Fact	Value	As of	Source	Status	Feeds	Notes
F-601	Client cash flows	\$300,000 at the beginning of 2027; \$150,000 at the beginning of 2028; ten fixed \$50,000 payments at the beginning of each year 2033-2042, "not adjusted for inflation"; Year = calendar year - 2026; all flows at the beginning of the year	case	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt lines 43-60, 88-92	VERIFIED-REPO-FILE	everything	Total paid out: \$500,000 nominal (derived).
F-602	Deadlines (5:00 p.m. ET)	Roster Oct 9; Trading Notes Oct 23; IPS Nov 6 ("Trading ends and your portfolio is locked"); Final Report instructions Nov 9; Final Report + school documentation Dec 4	2026	SMApplY_Deliverables_Page_2026-09-27.md; https://globalyouth.wharton.upenn.edu/competitions/investment-competition/; https://wghsinvcomp.smapplY.us/prog/2026-2027_wharton_global_high_school_investment_competition/	VERIFIED-REPO-FILE + VERIFIED-PRIMARY	plan timing	First trading day Sep 28, 2026; practice Sep 15-25; competition ends Dec 4; "10 weeks"; Global Finale April 29-30, 2027.
F-603	IPS format	Title page (1 page max); pitch ≤50 words; IPS ≤500 words; pages 2-3, 2-page max; Times New Roman 12; double-spaced; 1-inch margins; PDF ≤5 MB; no graphics, charts, images, attachments, external links, footnotes or formal citations	-	2026_WGY_Investment_Policy-FINAL.txt p.3	VERIFIED-REPO-FILE	-	Full register: A1.
F-604	Trading Notes	3 notes, "exactly as it appears in WInS"; reflection ≤100 words each; "Yes, we will verify this"	-	2026_WGY_Trading_Notes_Analysis-FINAL.txt p.2	VERIFIED-REPO-FILE	-	-
F-605	WInS virtual cash, 2026-27	"Your team will be responsible for managing a portfolio of \$300,000 in virtual cash."	2026-09-27 11:49 UTC	https://wghsinvcomp.smapplY.us/res/p/faqs/ (public; the page refers to the Oct 9, 2026 roster, so it is this season's)	VERIFIED-PRIMARY	WInS mirror, seed Q3	Same as Laura's 2027 deposit. Last season's was \$500,000 (F-609). Confirm in the team's WInS account.

ID	Fact	Value	As of	Source	Status	Feeds	Notes
F-606	Permitted investments (the page says "for BOTH contributions")	Permitted: cash; "Any stock from any exchange available on WInS priced at \$5 (or the equivalent of \$5 in the local currency) or higher"; "Any Exchange-Traded Funds (ETFs) available on WInS"; "Any Government/Treasury Bonds from any exchange available on WInS". NOT permitted: margin trading, short selling, stock-secured debt, crypto, derivatives (options, futures, swaps, etc.), "Anything other than the approved investments listed above"	same	same	VERIFIED-PRIMARY	compliance, instrument types	"For BOTH contributions" is not explained. It may mean WInS will also simulate the \$150k contribution; check the logged-in Trading Details page. Whether particular Treasury issues, zero-coupon Treasuries or defined-maturity funds are <i>available on WInS</i> is still unknown: PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading.
F-607	WInS mechanics	\$25 flat commission per stock transaction; \$10 per Treasury bond trade (only on trades that clear); bond prices updated once daily at U.S. market open; bonds pay every six months; non-U.S. equities clear at the end of the next trading day; dividends and coupons credited	same	same	VERIFIED-PRIMARY	trade plan	The FAQ says "you should not be doing a lot of buying and selling" of long-term holdings.
F-608	Required trading activity	The rules page says: "Teams must meet the required trading activity and portfolio management guidelines throughout the competition"; details are on the logged-in SMAApply "Trading Details" page	same	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/ ; SMAApply program page lists "Trading Details" under Pages	VERIFIED-PRIMARY (existence)	compliance	The specific requirements are NOT public. "Sector minimum = team size", "200-trade cap" and "first trade by Oct 10" remain SNIPPET-UNVERIFIED third-party claims.
F-609	2025-26 season size	More than 6,300 registered teams at the start of trading (2025-09-29; +1,300 on the prior year); \$500,000 virtual cash, the first change since 2012 (up from \$100,000); about 2,300 teams from 79 countries submitted final reports (Dec 12, 2025; +~30%); "More than 5,400 high school teams ... began" (Mar 20, 2026). Site counters "2025-2026 Participation Completion Numbers": 2,339 teams, 79 countries, 12,145 students, 1,625 advisors	2026-01-27 / 03-20 / live	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/ ; https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/ ; competition page	VERIFIED-PRIMARY	North Star framing	"Nearly 6,000 teams competed" was not found on the official pages I read (SNIPPET-UNVERIFIED; S-25). Top 50 ≈ 2.1% of 2,339 completers (derived).
F-610	2025-26 results	1st FigCapital (Stuyvesant HS); 2nd HHA Investments (Colégio Marista de Brasília); 3rd Riverhawk Traders (Farmington HS); 11 finalist teams; judges from Aberdeen, Achilles, King Street, Zen Capital Partners, WRDS	2026-04-30	https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	VERIFIED-PRIMARY	winners.md	Official pages disagree on the 2026 finale dates: "April 25-26" (Apr 30 article) vs "April 24 and 25" (Jan 27 article).
F-611	Judging	"success is not determined by portfolio performance"; the Top 50 are chosen "based on the strength of their Investment Policy Statement (IPS) and Final Reports" (rules page), while SMAApply says Trading Notes, IPS and Final Report "are evaluated"; "all five components of the criteria"; no individual feedback	2026-09-27	competition page; rules page; SMAApply FAQs	VERIFIED-PRIMARY	-	Treat all three as judged (case: "Evaluators will consider the three deliverables together").

ID	Fact	Value	As of	Source	Status	Feeds	Notes
F-612	Rules relevant to the plan	Teams of 4-6; team leader ≥ 16 ; "Teams may not contact the Competition client" (disqualification); advisors "may not make decisions on behalf of students or actively participate in ... trading decisions, or strategy development"; AI for brainstorming only; AI material must be cited	2026-09-27	rules page	VERIFIED-PRIMARY	Articulation section	Consistent with brief section 4.

H. Superseded numbers (old value -> correct value -> why)

ID	Old value (where)	Correct value	Why
S-01	Ladder cost "about \$295k ($\pm$ \$3k)" (chair memo 2-3)	\$292,264 (F-101)	One official curve, one method (brief section 8.4).
S-02	\$292,323 "Curve A" (referee 1; curve.md)	\$292,264	Curve A mixed estimated 1Y/7Y/20Y (4.49/5.11/5.51) with snippets.
S-03	\$297,610 (flat 5.17%); "~\$297k" (team notes)	retire	Treats a par yield as a zero rate.
S-04	\$314,465 "Curve B"	retire	Curve B is exactly the official curve of 2026-03-20 (1M 3.73 ... 10Y 4.39, 30Y 4.96): six months stale.
S-05	\$333,328 (flat 4%); team's "\$333k"; "risk-free floor ~\$140k"	illustration only	Flat 4% is not today's market.
S-06	DV01 "~\$297/bp", "\$290-297" (chair)	\$289/bp	Official curve.
S-07	Headroom "15-20bp" (chair); "\$7,677" (referee)	\$7,736 = 27bp	Official curve.
S-08	-100bp "~\$326k" (chair); \$322,931 (curve.md)	\$322,864	Official curve.
S-09	Reserve value at 1/1/2031 "\$357-358k", at 1/1/2033 "\$396-397k" (chair); "\$395.8k" (referee); 2037-42 rungs "\$154k"	\$356,384; \$394,930 ; \$155,408 (F-106, F-107)	Official curve.
S-10	Zero rates 2033-42 "5.04-5.38%" (chair)	5.05-5.46% semiannual (F-011)	Official curve; semiannual bootstrap.
S-11	Implied forward zeros "5.11/5.24/5.46" (curve.md)	5.08/5.24/5.48% (F-012)	Official curve.
S-12	20Y "5.45"/"5.51"; 5Y "4.99"; 30Y "5.48"; 1Y "4.49"; 7Y "5.11" (chair, curve.md)	20Y 5.54; 5Y 4.98; 30Y 5.49; 1Y 4.50; 7Y 5.06 (F-001)	Official CSV (5.45 was the 20Y on 9/23).
S-13	"10-year ~5.1% (highest since late 2023)" (team notes)	5.17% on 9/25; 5.18% on 9/24 was the highest close since 2007-07-06 (F-004)	The 2023 peak was 4.98%.
S-14	IEF duration "near 7.5 yr" (blog, instruments.md)	6.95 (June fact sheet); 6.86 (Sep 24)	Primary sources.
S-15	IEF/TLT "64.7/35.3" (referee), "69.3/30.7" (7.5y case); blueprint "~42% IEF + ~23% TLT" of 65%	64.8/35.2 on June durations; 62.2/37.8 on current durations ($\approx$ 40.4% + 24.6% of a 65% hedge)	Rounding of 9.90; durations have moved (F-205).
S-16	Coupon arithmetic "\$20k/yr, \$6.7k loss" (L defence) vs "\$15.5k, \$5.2k" (referee)	Both are correct sums: \$20.1k/\$6.7k for a ladder with \$381k of face, \$15.5k/\$5.2k for a \$295k ladder. The \$15.5k/\$5.2k framing fits Laura's budget (F-113)	Different ladders, not an error.
S-17	"\$50k in 6 years at 5.26% $\approx$ \$36.9k" (referee section 4)	\$36,761 $\approx$ \$36.8k	Arithmetic.
S-18	"If JPM 6.7% is arithmetic the median is 5.52%"; referee MC rows "6.7% arithmetic"	JPM 6.70% is compound (7.94% arithmetic) (F-301)	Primary PDF.
S-19	Taiwan CPI Aug "2.4%" (CEIC snippet, chair memo); "+0.13% m/m"	2.04% y/y; -0.01% m/m (+0.13% seasonally adjusted) (F-501)	DGBAS release.
S-20	TWD volatility "5-7%" (chair, unsourced); "monthly 4.4%" (taiwan.md)	4.0-5.6% over 1-10y windows (10y 4.69%); 10y month-end 5.58% (F-512)	The 4.4% used monthly averages, which smooth the series.
S-21	USD/TWD "31.73 (9/26)" (chair)	31.780 (CBC close 9/24); 31.82 (FRED 9/18)	9/26 was a Saturday.

ID	Old value (where)	Correct value	Why
S-22	10Y TIPS "2.85% / breakeven 2.28%" vs "2.653%" (chair)	Real 10Y 2.83% (9/25), 2.85% (9/24); breakeven 2.34% (9/25) (F-008, F-009)	2.653% was the 9/17 auction yield (snippet), a different measure and date.
S-23	Fed hike "verified in search results" (curve.md, taiwan.md)	VERIFIED-PRIMARY on federalreserve.gov (F-005)	Now read on the primary page.
S-24	"Final Report (due Nov 9)" (chair section 9)	Instructions Nov 9; report due Dec 4 (F-602)	SMAApply and Wharton pages.
S-25	2025-26 "nearly 6,000 teams competed"	>6,300 registered; >5,400 began; ~2,300 (counter: 2,339) completed final reports (F-609)	Official pages. "Why Laura would choose us over 6,000" is rhetoric; the field judged on final reports is about 2,300.
S-26	WinS starting cash "\$500k" (last season) / "unknown" (brief sections 4, 6)	\$300,000 (F-605)	2026-27 SMAApply FAQs (public).
S-27	"Put a number on high certainty: $\geq 95\%$ Monte Carlo + floor" (winners.md lesson 2)	The council's market-consistent definition (brief section 8.6)	A model probability moves with its inputs: G misses 3.2% or 5.5% depending on the equity assumption (F-401, F-405).
S-28	Liability PV \$313.8k, Macaulay 10.4y, modified 9.95y at a flat 4.5% (instruments.md)	Spot PV \$288,924; spot duration 10.16y (F-105)	Official curve.
S-29	Median 2033 facility sleeve "\$194-212k (four models)"	\$207k median (p5 \$159k, p95 \$273k) in the one reproducible model (F-401)	Reproducible script, JPM inputs.
S-30	G shortfall "3.3-9.3%", "3-8%", "4.3%"; 17% at \$75k; 46-55% at \$0	3.2%; 13.7%; 40.8% (F-401, F-403, F-404)	JPM compound inputs. Still model-dependent.
S-31	Rung order "nearest maturity first" (chair memo section 1.3, round 1)	Longest-dated first (round 2; brief section 7)	A design decision, not a number. The current strategy text uses longest-first.
S-32	"iBonds Dec 2033-2035 exist; later vintages unverified" (chair)	Dec 2026-2036 exist; none for Dec 2037-2043 (F-206)	iShares screener.
S-33	"Taiwan inflation ~1.8-2.2% (unverified)" (team notes)	2026: 2.07% (DGBAS) / 2.03% (CBC); 2027: 1.90% / 1.83% (F-504, F-505)	Primary forecasts.
S-34	"Construction costs ~3.2x CPI"; "x0.881 over two years if +6.54% persists" (referee)	3.2x is a one-month ratio. The CCI trend is ~3.5%/yr since 2021 (~0.93 over two years at that rate) (F-508, F-509)	The 2026 jump followed two flat years.
S-35	"Converting the floor to TWD costs ~5.3% over two years"; "2-year forward ~5% fewer TWD per USD"	Probably larger (~6-8%): Taiwan market short rates (0.81%) are well below the 2% discount rate used as a proxy (F-515)	Taiwan 2-year yield still UNVERIFIED.
S-36	Twist gap "\$3.3-4.4k" (referee)	unchanged, ASSUMPTION	Not recomputed; it rests on modelling IEF as an 8.5-9.5y bond and TLT as a 30y bond.
S-37	"One sd of two-year FX is $\pm 6.6\%$ " (referee)	Consistent: 6.6% by $\sqrt{\cdot}$ -time; 6.8% realised since 2006 (F-513)	Confirmed.
S-38	TWD 10-year range 27.52-33.21, +0.14%/yr, vol 10y/5y/3y/1y 4.7/5.2/5.6/4.0% (taiwan.md)	Confirmed from live FRED (F-512)	The mirror file matched the primary series.

Brief section 10 inconsistencies: resolution map. All resolved; the primary or verified value wins:

- Ladder \$295k vs \$292,264: resolved by S-01.
- Taiwan CPI 2.4% vs 2.04%: resolved by S-19.
- Rung order: resolved by S-31 (longest-first).
- 20Y 5.45/5.51 vs 5.54: resolved by S-12.
- TWD volatility 5-7% vs 4.7%: resolved by S-20.
- Coupon arithmetic: resolved by S-16.
- Fed hike: resolved by S-23.
- Chair memo's Nov 9 date: resolved by S-24.
- IEF 6.95 vs ~7.5: resolved by S-14 (now 6.86).
- winners.md " $\geq 95\%$ MC": resolved by S-27.
- 6,300+ vs nearly 6,000: resolved by S-25.

I. Still unverified or blocked (say so; never guess)

Item	What was tried	Status
DGBAS attachment files (CPI tables, CCI headline file) on ws.dgbas.gov.tw	curl: TLS "unable to get local issuer certificate", even with the proxy CA bundle; WebFetch: EGRESS_BLOCKED	Blocked. Headline CPI was read on eng.stat.gov.tw instead; the CCI total was derived (F-508).

Item	What was tried	Status
Exact CCI headline "+6.54%"	derived 6.53%	SNIPPET-UNVERIFIED (corroborated).
Taiwan CPI history 2022-2025; Taiwan 2-year and 10-year government yields	not found on reachable primary pages	SNIPPET-UNVERIFIED / UNVERIFIED.
SMAApply "Trading Details" page (logged-in): trade minimums, sector rules, first-trade deadline, what "for BOTH contributions" means	public pages only	UNVERIFIED. The team must read it.
Which specific securities (Treasury issues, zero-coupon Treasuries, defined-maturity funds, IEF/TLT) are <i>available on WInS</i>	not accessible	UNVERIFIED. Confirm against this year's WInS list/rules before trading.
List & Lucking-Reiley (2002) "~6x more giving from 10% to 67% seed"	SSRN 403; journal site 403	SNIPPET-UNVERIFIED (brief section 8.17: a mechanism only).
Council model outputs (F-410)	scripts not preserved	Not reproducible.
2027 JPM LTCMA	not yet published (F-315)	Re-check in October.
Festival attendance "2,000+" and other figures in the team's client documents	not checked by A2 (Laura facts belong to other agents)	SNIPPET-UNVERIFIED.

Sources (all accessed 2026-09-27)

Repo files

- `competition/official_market_data/` (Treasury CSV; IEF and TLT fact sheets p.1; JPM LTCMA p.2).
- `competition/official/2026_27/` (case, IPS guide, Trading Notes guide, SMAApply deliverables page).
- `research/verified_2026-09-27/*.py`.
- Council files as cited.

Web (primary)

- home.treasury.gov: nominal and real par-yield CSVs for 2026.
- fred.stlouisfed.org: DEXTAUS, DGS10.
- federalreserve.gov: [monetary20260916a.htm](#), [monetary20260916a1.htm](#), [fomcprojtabl20260916.htm](#), [openmarket.htm](#).
- ishares.com: IEF and TLT product pages, Dec 2034 iBonds page, product screener.
- am.jpmorgan.com: LTCMA landing page.
- eng.stat.gov.tw and www.stat.gov.tw: DGBAS releases, CCI platform.
- cbc.gov.tw: [en/mp-2.html](#), [en/cp-448-192883-c5fa1-2.html](#).
- globalyouth.wharton.upenn.edu: competition page, FAQ, rules-roles, three news articles.
- wghsinvcomp.smaply.us: program page, [/res/p/faqs/](#), [/res/p/deliverables/](#).

What this teaches

A number is only as strong as the path back to where it came from. In one afternoon this register found several things:

- A "mystery" yield curve was simply six months old.
- A heated argument about coupon reinvestment was two correct sums about two different ladders.
- The most important price in the plan, a ladder "fitting inside \$300k", has been true on only 12 trading days this year.
- The competition had already published how much virtual cash each team gets.

Checking the original source is not bureaucracy. It changes what you should do. The habit to keep is to write the source and its status next to every number when you first use it, and to re-check the few numbers your plan depends on (the curve, the rules, the durations) on the day you act.

A3 Stakeholder map and blind spots

Agent: A3 (Blind-spot Mapper), insight_v1 run, written 2026-09-27. This is AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It holds **no submission-ready prose**. Where it says what a sentence "must contain", that is a checklist. The six students write every word. It names **no securities**.

Scope: semifinals only (brief section 4). Finale judges get one line (SH-17) and are then parked.

0. Summary (read this first)

1. **Twenty-six stakeholders, one test.** Every stakeholder below reaches the deliverables through one of three readers: **Laura** (who chooses a firm), the **co-sponsors** (who read the fundraising draft) and the **semifinal readers** (who score all three deliverables together). The case turns each of them into a scored requirement:
 - "recommendations that can earn her confidence" (criterion 2, R-S25);
 - "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors" (criterion 5, R-S28);
 - "Evaluators will consider the three deliverables together" (case p.4 L161, R-C91).

All three are VERIFIED-REPO-FILE.

2. **The rules name a professional ethics code, and no research file uses it (BS-02).** The Rules page tells teams to "review the CFA Institute's Asset Manager Code and operate by these standards" (R-W29, VERIFIED-PRIMARY via A1). I read the Code itself (VERIFIED-PRIMARY, CFA Institute PDF, 2nd ed., ©2009, 2010). Its provisions line up almost one for one with the IPS guide:
 - know the client (B.6.a);
 - act only within the stated mandate (B.5.a);
 - make disclosures that are "truthful, accurate, complete, and understandable" (F.2);
 - disclose fees (F.4.d);
 - keep records (D.4).

This hands the fictional "firm" a ready-made governance backbone, and it costs almost no words.

3. **Nobody has modelled the firm's own fee (BS-01).** The case has Laura invest "with an asset management firm" (p.2 L43), and the Code requires fee disclosure (F.4.d). No file in `research/` mentions a management fee (grep, 2026-09-27), and the verified model runs with no fees (brief section 11).

Rough arithmetic (ASSUMPTION: a yearly fee on all assets, paid from the growth sleeve; ladder +5.1% a year, sleeve +5.9% a year): a 0.5% fee cuts the median 2033 surplus by about **\$17k** and a 1.0% fee by about **\$34k** (roughly 8% and 16%). The ten payments are unaffected, but only as long as the fee is paid from the sleeve.

4. **Credibility is one of Laura's income assets, not only a fundraising asset (BS-03).** The whole 2028 deposit comes from reputation-driven work: "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (p.2 L44-45, VERIFIED-REPO-FILE). If she misses a range she announced in public, the harm reaches her earning power as well as the co-sponsors. This is a second, Laura-specific reason the 2031 floor must be bought, not forecast. Existing files discuss credibility only with co-sponsors.
5. **Funders are reluctant to pay running costs, and Laura's promise is exactly that money (BS-17).** Gregory & Howard (SSIR, Fall 2009, VERIFIED-PRIMARY) describe funders' "unrealistic expectations about how much it costs to run a nonprofit". The case's split fits this: co-sponsors may fund the building, but Laura alone funds ten years of operations (R-AN6). For a co-sponsor, "running costs for the first ten years are already set aside" answers the question a building funder worries about most. No research file makes this link. It belongs in the Final Report fundraising draft.
6. **"Dollar range" is ambiguous to a Taiwanese reader (BS-04).** The case never names a currency (R-AN10), and Taiwan's currency is also a "dollar" (the New Taiwan dollar, NT\$). Existing work covers exchange-rate *risk*, not this *labelling* risk. The fundraising draft should write "US\$" every time, and give the NT\$ figure as a dated reference.

7. Pledge wording matters to co-sponsors (BS-05).

- The case says Laura will "describe how much she *expects* to contribute" and calls it her "potential contribution" (p.3 L110, L116; VERIFIED-REPO-FILE).
- Fundraisers routinely discount pledges. One search snippet says 10-30% are never collected (SNIPPET-UNVERIFIED).
- Whether a charitable pledge is legally binding varies by U.S. state (California Bar memo, 2010, VERIFIED-PRIMARY). That is not the team's question to answer.

The team's question is which verb goes with which tier: "set aside" for the bought floor, "expects" or "aims" for the stretch.

8. **Operators and participants feel the case's two simplifications most:** fixed nominal payments (R-C46) and nothing after 2042 (R-C49). Both are known open items (brief section 9). This map adds only who carries the cost: the staff's real budget shrinks each year, and by the case's own list the gap falls to "co-sponsors, grants, collaborators, program fees, continued business income" (p.3 L85-86).

9. **Rival teams may reach the same architecture (BS-10).** The council's claim that "most strong teams will write safe bucket + growth bucket" is untested (ASSUMPTION). Many teams will use AI tools, which readily suggest liability matching. Caplet's edge should therefore rest on Laura-specific reasoning and real process evidence, not on "lock early" being rare.

10. **One administrative gate has no owner: the school documentation (BS-11).** Its official wording: "Do not wait until the last minute to request documentation" (R-S21, VERIFIED-REPO-FILE). The semifinal letterhead confirmation (R-W25) has the same problem. Neither appears as an action in any research file.

Entry count: 26 stakeholder entries (SH-01 to SH-26), 20 blind spots (BS-01 to BS-20), 8 cross-stakeholder tensions (T-1 to T-8) and 14 candidate questions for Phase B (Q-A3-01 to Q-A3-14).

1. How to read this file

Status labels (brief section 3). I abbreviate them inside dense blocks:

- **VRF** = VERIFIED-REPO-FILE
- **VP** = VERIFIED-PRIMARY
- **SNIP** = SNIPPET-UNVERIFIED
- **ASM** = ASSUMPTION (a modelling input or a judgement; "ASM/judgement" marks interpretation)
- **PARA** = PARAPHRASE-UNVERIFIED

References.

- R-C / R-I / R-T / R-S / R-W / R-AN / X- are entries in A1's `phase_A/case_register.md`. Each carries its quote, its location and its status.
- F-xxx are entries in A2's `phase_A/fact_register.md`.
- W-n are web sources (section 8).
- Case locations are page (p.) and line (L) in `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`.

Deliverable tags.

- **TN** = Trading Notes Analysis, Oct 23 (tier 1)
- **IPS** = Investment Policy Statement, Nov 6 (tier 2)
- **FR** = Final Report, Dec 4 (tier 3)
- **ADMIN** = a pass/fail gate that is not scored

Terms used below.

- **Co-sponsor:** an organisation or person who funds part of the residency alongside Laura.
- **Lead gift:** the largest early gift in a fundraising campaign.
- **Pledge:** a promise to give in the future.
- **Mark-to-market:** valuing a holding at today's market price, not at what it will pay at maturity.
- **Human capital:** the value of a person's future earnings.
- **Funded ratio:** assets divided by the Treasury price of the promised payments.
- **AUM fee:** a yearly fee charged as a percentage of "assets under management".

2. Overview table

"Reaches us via": L = Laura, C = co-sponsors, J = semifinal readers, T = the team itself.

ID	Stakeholder	Reaches us via	Most important thing to them	Deliverable most affected
SH-01	Laura as investor (choosing a firm)	L	The promise is safe, and her risk appetite is used where it cannot hurt it	IPS, FR
SH-02	Laura as founder of the residency	L, C	A facility contribution she can stand behind, with flexibility left	FR
SH-03	Laura as author, educator and public voice	L	Nothing she announces publicly turns out untrue	FR
SH-04	Laura as statistics graduate	L, J	Every probability names its model, and "certainty" is testable	FR, IPS
SH-05	Private and family foundations	C	A floor that is real, and a plan for shortfalls and overruns	FR
SH-06	Public arts bodies (US and Taiwan)	C	Eligibility, local-currency budgets, public accountability	FR
SH-07	Taiwanese partners (host institutions)	C	NT\$ figures; their share of cost and risk	FR
SH-08	Corporate sponsors	C	Timing, visibility, reputational safety	FR
SH-09	Individual donors	C	A visible personal commitment (seed-money signal)	FR
SH-10	Wharton/Penn alumni networks	C, J	Rigor they recognise, and an alumna's credibility	FR
SH-11	Residency participants	(C)	The programme exists and continues	FR
SH-12	Future staff and operators	(C)	A predictable operating budget; who covers erosion; 2043	FR
SH-13	Taiwan host community	(C)	A good neighbour; local benefit	FR (one line at most)
SH-14	Portfolio manager (case fiction)	J	A defensible, rule-based framework the firm can sign	IPS
SH-15	Real teacher advisor (admin only)	T	Rule compliance; documentation; engagement	ADMIN, FR
SH-16	Semifinal readers	J	Five criteria, consistency across deliverables, an authentic voice	TN, IPS, FR
SH-17	Finale judges	(parked)	Out of scope for this run	none
SH-18	Rival teams	J (comparison)	Their median submission sets the bar	all
SH-19	Wharton Global Youth (organiser)	J	Learning, ethics, AI honesty, respect for a real alumna	all
SH-20	The six students	T	Learning, fairness of workload, no repeat of last year's rule break	all
SH-21	Laura's publishers and literary agent	(L)	Source of the 2028 deposit	FR
SH-22	Speaking hosts and licensees	(L)	Also a source of the 2028 deposit	FR
SH-23	Readers of her books, and classrooms	(L)	Her public voice; her books' classroom and library channel	FR (income risk)
SH-24	The team's school	T	Permission, letterhead documentation, reputation	ADMIN
SH-25	The "firm" and its ethics code	J	CFA Asset Manager Code (named in the Rules)	IPS, FR
SH-26	Laura after 2033 (running the residency)	L	Flexibility when her earnings may be lower	FR

3. Stakeholder entries

Each entry lists:

- **Interests:** what they care about.
- **Fears:** what they worry about.
- **Needs to see:** what would earn their trust.
- **Loses trust if:** what would lose it.
- **Anchor:** the official requirement that connects them (quote, location and status).
- **Implication:** a decision, number or sentence, tied to a deliverable.

Laura (four roles)

SH-01 Laura as investor (the client choosing among firms)

- **Interests:** she chooses one strategy out of many. The case says "the investment strategy that Laura ultimately chooses" (p.1 L9-10, R-C8, VRF). She wants "an appropriate balance between pursuing growth and protecting the capital required for her goals" (p.2 L71-74, R-C38, VRF).
- **Fears** (ASM/judgement, drawn from the case wording): a plan so cautious that it ignores "thoughtful risks" (p.2 L70, VRF). Or a plan that risks the promise. Or statements that show her balance falling while the payments are in fact safe (see BS-09).
- **Needs to see:**
 - Risk taken only with money the promise does not need.
 - One clear principle.
 - Her own goals in her own order: the reserve first, then the facility, then flexibility (p.3 L94-103, R-C50, R-C57, VRF).
- **Loses trust if:**
 - The plan is generic, e.g. an unexplained 60/40.
 - It is a return forecast dressed up as certainty.
 - The fees are hidden (BS-01).
- **Anchor:** criterion 2, "presents recommendations that can earn her confidence" (SMAApply L51, R-S25, VRF). IPS guide: "Always remember who your strategy is designed to serve" (IPS p.1 L27, R-I14, VRF).
- **Implication:**
 - IPS: the central idea must be stated as *her* trade-off, growth versus the capital her goals require, not as a finance label.
 - FR: report "payments covered" (the funded ratio) next to the market value (BS-09).
 - Decision still open: the sleeve's equity weight, 50/60/70% (brief section 9).

SH-02 Laura as founder of the residency

- **Interests:** a "small, community-oriented space" in Taiwan (p.2 L76-80, R-C40, VRF). She wants the largest contribution she can *responsibly* make (p.3 L101-106, R-C57, R-C60, VRF), and she wants to keep flexibility "as the project develops" (p.3 L102-103, R-C58, VRF).
- **Public record of building institutions (VP, lauragao.com/hey, read 2026-09-27):** she "co-organizes San Francisco's LGBTQ+ comics festival, Pride in Panels". This is cited only as evidence that she has built community institutions before. It is never a reason for any investment choice.
- **Fears:**
 - Over-committing in 2033 and having nothing left when costs overrun.
 - A co-sponsor walking away after a missed range (p.3 L114-115, R-C66, VRF).
- **Needs to see:**
 - A contribution rule that leaves a stated buffer.
 - What happens in good and bad markets (p.3 L118-119, R-C70, VRF).
 - What happens if the project is delayed or changes (BS-07).
- **Loses trust if:** the team commits "all remaining assets". The case warns against exactly this: "committing all remaining assets could limit her financial flexibility" (p.3 L102-103, VRF).

- **Anchor:** case test 5, "Preserves appropriate financial flexibility when determining the facility contribution" (p.4 L145, R-C84, VRF).
- **Implication (FR):**
 - Decide the share of the post-reserve surplus that is *not* contributed, and say why. The council's view is to keep it "mainly as project flexibility" (CLAUDE.md interview item 3; ASM).
 - A strong sentence names the buffer's purpose: project overruns and changes, not her living costs, which sit outside the portfolio (p.2 L61-63, VRF).

SH-03 Laura as author, educator and public voice

- **Interests:** her public credibility. The case says her books "are read in classrooms around the world" (p.1 L22-23, R-C16, VRF). Her first named work was "a response to misinformation" (p.1 L19-20, R-C14, VRF).
- **Public record (VP, lauragao.com/hey, 2026-09-27):**
 - She lists "Speaking engagements".
 - She is described as shaping "the next generation of creators as a Professor of Comics at California College of the Arts".
 - The case lists her as "educator" (p.1 L6, VRF).
 - Teaching income is not one of the case's 2028 sources (brief section 8.19).
- **Fears** (ASM/judgement): a promise she has made in public breaks; the plan contains a claim she could not repeat in front of a class.
- **Needs to see:** plain language, honest ranges, and no hype.
- **Loses trust if:**
 - "Guaranteed" or "100%" appears without its qualifiers.
 - Her identity is used as decoration (brief section 4, tokenism rule).
- **Anchor:** criterion 5, "an authentic team voice" and "clearly and credibly" (SMAApply L57, R-S28, VRF).
- **Implication:**
 - FR: every uncertainty statement must be one she could say out loud without a footnote.
 - BS-03: her credibility is part of her income.

SH-04 Laura as statistics graduate

- **Interests:** a degree in "Statistics & Information Decisions Management" (p.1 L16, R-C12, VRF). Her site has a "Visual Data Stories" section (VP, lauragao.com, 2026-09-27).
- **Fears** (ASM/judgement): a "95%" with no model behind it; confusion between the median and the mean; a confidence statement that is really one-sided.
- **Needs to see:**
 - "Certainty" defined in a way she can check herself: market-priced full funding, cash flows matched, residual risks named (brief section 8.6).
 - Every probability with its model and at most two significant figures.
 - The two-sided "within that range" wording handled properly (p.3 L117-118, R-AN9, VRF).
- **Loses trust if:** false precision; one model presented as the truth; a range whose confidence counts only the low end.
- **Anchors:**
 - "define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty" (p.3 L97-98, R-C54, R-C55, VRF).
 - Criterion 3, "uses reasonable assumptions and projections" (SMAApply L53, R-S26, VRF).
- **Implication:**
 - FR: state the confidence for *both* ends of the 2031 range: P(below floor) and P(above top).
 - IPS: one plain sentence defining certainty, with no numbers needed (R-I31).

Prospective co-sponsors (kept generic; the case names none)

Common anchors for all co-sponsor entries, all VRF:

- "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute" (p.3

L110-112, R-C64).

- "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors" (p.3 L114-115, R-C66).
- "Each team must also draft part of Laura's fundraising materials" (p.4 L123-124, R-C72).

SH-05 Private and family foundations

- **Interests:** due diligence. One foundation-practice bulletin (PKF O'Connor Davies, July 2019, VP, W-6) says a funder should check "whether the sources of potential revenue are reliable", and should "have a clear understanding of how the grantee would respond to any unexpected revenue shortfalls or cost overruns".
- **Fears:**
 - A founder's pledge that later shrinks.
 - Being asked for running costs indefinitely. Funders' "unrealistic expectations about how much it costs to run a nonprofit" are the first step of the "starvation cycle" (Gregory & Howard, SSIR, Fall 2009, VP, W-3).
- **Needs to see:**
 - A floor already set aside, not a forecast.
 - An explicit answer to "what if markets are bad", which is also a case requirement (R-C70).
 - What happens to her contribution if building costs overrun. Brief section 9 lists a "scope clause" as open.
- **Losses trust if:** the range is so wide that it is useless, or so narrow that it is risky (R-AN9); or the "certainty" turns out to cover only nominal USD and nobody said so.
- **Implication (FR fundraising draft):** a strong passage contains:
 1. what is already set aside (the floor) and in what form;
 2. the stretch figure with its stated probability and model;
 3. that ten years of operating payments are funded separately, from her portfolio alone (R-C48);
 4. what she will do if outcomes are unfavourable (a smaller contribution, never a raid on the operating reserve).

SH-06 Public arts bodies (U.S. arts councils; Taiwan's national and local culture agencies)

- **Interests:** public accountability, eligibility rules, local-currency budgets.
- **Evidence:**
 - Taiwan's Ministry of Culture funds artist villages and international residency exchange, and runs an "Arts Residency Network Taiwan" platform (SNIP, W-12; the site artres.moc.gov.tw returned HTTP 403 to curl on 2026-09-27).
 - Most U.S. state arts councils have grants for individual artists (Artist Communities Alliance page, VP, W-4).
- **Fears** (ASM/judgement): figures in a foreign currency; a private founder's commitment that depends on markets.
- **Needs to see:** figures in their own currency with a date and an exchange rate; a clear statement of which costs Laura's money covers.
- **Losses trust if:** "dollar" is left ambiguous (BS-04).
- **Anchor:** "Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources" (p.3 L85-86, R-C44, VRF).
- **Implication (FR):** quote US\$ as the binding figure, with an NT\$ reference at a stated date and rate. A2 has the numbers: USD/TWD 31.82 on 2026-09-18 and 10-year volatility 4.69% (F-510, F-512, VP).

SH-07 Taiwanese partners (host institutions, universities, cultural organisations)

- **Interests:** local costs are paid in NT\$, and construction costs are rising much faster than consumer prices. Taiwan's construction cost index was +6.53% y/y in August 2026 against CPI of +2.04% (F-508, F-501, VP).
- **Fears** (ASM/judgement): absorbing currency and building-cost risk that the founder's USD promise passes on to them.

- **Needs to see:**

- The contribution in NT\$ under strong-TWD and weak-TWD cases.
- Who carries cost overruns.
- That a USD promise is "certain in nominal USD", not in NT\$.

- **Losses trust if:** the materials call a USD figure "certain" without that qualifier.

- **Anchor:** "the effect of inflation on portfolio projections and facility costs" (p.4 L146-147, R-C85, R-AN8, VRF).
- **Implication (FR):** a sentence on cost inflation must use a building-cost assumption, not CPI (brief section 8.15). F-508 warns that the 6.5% figure is a 2026 jump; the 2021-2026 average is about 3.5% a year (VP).

SH-08 Corporate sponsors

- **Interests:** brand visibility, fit with their budget calendar, reputational safety.
- **Fears** (ASM/judgement): public association with a project that stalls.
- **Needs to see:** a firm commitment date (2033) and a founder commitment that is not contingent on markets.
- **Losses trust if:** the figures keep moving between 2031 and 2033 with no explanation.
- **Anchor:** R-C66 (credibility).
- **Implication (FR):** low priority. The floor-plus-stretch structure already serves them. No extra content is needed. Say so to keep complexity down.

SH-09 Individual donors (including readers and community supporters)

- **Interests:** a visible personal commitment. In List & Lucking-Reiley (2002), higher announced seed money raised small-donor giving. The brief treats this as a mechanism, not a prediction for institutions (brief section 8.17; council citation, SNIP-level for the details).
- **Fears:** giving to a project that never opens.
- **Needs to see:** a simple statement of what Laura herself puts in and when.
- **Losses trust if:** the numbers are too technical to repeat.
- **Implication (FR):** the fundraising draft should work for a non-specialist in one reading (criterion 5). Put technical detail in an appendix.

SH-10 Wharton/Penn alumni networks (as potential co-sponsors or connectors)

- **Interests:** Laura is a 2018 Wharton graduate (p.1 L16, VRF). Alumni readers will recognise, and test, finance rigour.
- **Fears** (ASM/judgement): a sloppy method attached to a Wharton name.
- **Needs to see:** a method they would accept: market pricing of the promise, stated models, sensitivity to rates.
- **Losses trust if:** they find the jargon without the substance.
- **Implication:** the same as SH-04. No separate content. (Recorded so that Phase F does not flag it as missed.)

The residency's people and place

SH-11 Residency participants (artists, writers, designers, entrepreneurs, educators)

- **Interests:** the case lists them and says they will "temporarily live, work, teach, and collaborate" (p.2 L78-80, R-C40, VRF).
 - The Artist Communities Alliance (VP, W-4) says: "Most artists ask: 'Where can I go that's free?' The short answer is, nothing is free." It also says residencies often subsidise fees through work-exchange, "teaching a workshop" among the examples.
 - Program fees are one of the case's permitted outside sources (p.3 L86, VRF).
 - **Fears** (ASM/judgement): a programme that shrinks or closes mid-way; fees that rise because the operating budget erodes.
- **Needs to see:** that the programme's base funding does not depend on annual fundraising for its first ten years.
- **Anchor:** "reliable support for its early operations" (p.3 L83, R-C41, VRF).

- **Implication (FR):** participants never read our deliverables. Their interest reaches us through co-sponsors (SH-05). One clause in the fundraising draft ("operations funded for ten years, independent of co-sponsors") serves them. Do not write a programme or business plan; the case says so explicitly (p.4 L164-166, R-C95, VRF).

SH-12 Future staff and operators, who depend on the \$50k a year

- **Interests:** a predictable budget. The ten payments are fixed at \$50,000 and arrive "at the beginning of each year" (p.3 L88-90, R-C45, R-C46, VRF). A lump sum on 1 January is good for cash planning.
 - An arts-funding commentator (Emil J Kang, Substack, 13 July 2026, VP, opinion piece, W-5): "None of this runs for free. A residency covers room and board, studio space, the staff who hold it together".
 - One funder at a 2007 Grantmakers in the Arts roundtable (VP, W-7): "When we look at proposals for staff support, we ask organizations if they have sustainable payroll and benefit packages, governed by the board." This is one speaker, and old.
- **Fears:**
 - **Real erosion.** At an assumed 2% a year, the fixed \$50k is worth about \$43.5k in 2026 money by 2033 and about \$36.4k by 2042 (council arithmetic; ASM inflation rate). Taiwan wages in construction rose +8.12% y/y (F-508, VP), although that is not a residency wage index.
 - **Currency.** Payments in USD, wages in NT\$.
 - **The 2043 cliff.** Nothing is funded after 2042 (p.3 L92-93, R-C49, VRF).
 - **Needs to see:** that someone has named who covers the erosion and what happens in 2043.
 - **Losses trust if:** the plan calls the operating budget "secure" without saying "in nominal USD".
 - **Anchor:** "each payment is a fixed \$50,000 and is not adjusted for inflation" (p.3 L90, R-C46, VRF). The "operating support beyond Laura's commitment" may come from other sources (p.3 L85-86, VRF).
- **Implication (FR):** one sentence that must contain:
 1. the payments are fixed in nominal USD, as the case specifies;
 2. their real value falls at an assumed rate;
 3. by the case's own terms the top-up belongs to co-sponsors, program fees and other sources;
 4. funding after 2042 is outside the plan (R-C49).

This is a known unanswered item (brief section 9, "real erosion... must be said aloud"). This map adds the operators as the people who bear it.

SH-13 Taiwan host community (neighbours, local creative scene, local government)

- **Interests** (ASM/judgement): local benefit, access to programmes, good-neighbour behaviour. The case calls the space "community-oriented" (p.2 L77, VRF). An Artist Communities Alliance research project on residencies as "a platform for engaging communities" is SNIP only; ACA says its reports are members-only (W-4 search result).
- **Fears / needs:** out of scope for an investment strategy. Taiwan's legal and regulatory requirements are explicitly excluded (p.4 L168-169, R-C97, VRF).
- **Implication:** none for TN or IPS. At most one clause in FR, if it helps the fundraising draft. Listed so that its omission is a deliberate choice (design principle: complexity must earn its place).

Competition stakeholders

SH-14 The portfolio manager (case fiction: "your team's teacher/advisor who makes the final investment decisions")

- **Interests:** the case sets up a firm with analysts and a PM who signs off (p.1 L3-5, R-C2, R-C3, VRF). A PM wants a policy that any member of the firm could apply the same way, and that the firm can defend.
- **Fears** (ASM/judgement): analysts proposing trades that break the mandate; recommendations with no stated basis.
- **Needs to see:** a written decision framework with rules: the lock rule, the rule if rates fall, the 2031 floor rule and the rebalancing bands.
- **Anchor:**
 - "This formally establishes the strategy and the team's decision-making framework" (p.4 L155-156, R-C89, VRF).

- Asset Manager Code B.5.a, "Take only investment actions that are consistent with the stated objectives and constraints" (VP, W-1).
- **Implication (IPS):** write the framework as pre-set rules, which also reads well to judges. Do *not* write that "our PM decided". The real rules forbid advisor decisions (SH-15, R-AN4).

SH-15 The real teacher advisor (administrative role only)

- **Interests:** the role is "to provide guidance, encouragement, and educational support". Advisors "may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development" (Rules page, R-W19, VP via A1). "Students should place trades on WInS, NOT the advisor" (R-W87, VP). The advisor also monitors "team engagement and portfolio activity" (R-W20).
- **Fears** (ASM/judgement): a rule breach that disqualifies the team; missing school documentation.
- **Needs to see:** the team making and logging its own decisions, and the paperwork done early (SH-24).
- **Implication:**
 - FR, Articulation: one true sentence on the advisor's administrative role (CLAUDE.md "Team"; R-W28 bans invented teamwork stories).
 - ADMIN: the advisor is the natural route to the school letterhead (BS-11).

SH-16 Semifinal readers (who pick the Top 50 from the written deliverables)

- **Interests:** the five criteria (SMAApply L49-57, R-S24 to R-S28, VRF). They check consistency: "maintaining a clear and consistent investment strategy across all three" (p.4 L161-162, R-C92, VRF). They check the trade record: "Yes, we will verify this" (TN p.2 L53, R-T34, VRF). Returns do not count: "your team's standings on WInS have little to do with the final outcome" (R-W42, VP).
- **Fears** (ASM/judgement):
 - AI-written text. "AI-generated work may not be submitted as your own" (R-W26, VP).
 - Jargon.
 - Mismatches between the notes, the IPS and the WInS portfolio.
 - Format breaches in the IPS, which are disqualifying (IPS p.3 L82-83, R-I39, VRF).
- **Needs to see:**
 - Laura-specific tailoring.
 - Reasoning over results: "The focus should be on the quality of your reasoning and the intended role of each decision" (TN p.1 L15-16, R-T11, VRF).
 - An authentic voice.
 - One strategy told the same way three times.
- **Loses trust if:**
 - A trading note's wording differs from WInS.
 - A reflection justifies a trade by its gain.
 - The IPS mirror (~65/35) contradicts the plan's ~17-24% total equity (F-409, derived).
- **Implication:**
 - TN: each reflection names which of Laura's needs the trade serves.
 - IPS: at most one rounded number.
 - FR: sources and a record of AI use in a Works Cited list (R-W46, VP).

SH-17 Finale judges. Out of scope for this run (brief section 4: semifinals only). No finale content is produced. The only carry-over: nothing in the written deliverables should be impossible to defend aloud (R-W7, "communicate and defend", VP). Parked.

SH-18 Rival teams

- **Scale:**
 - 2025-26 had "more than 6,300 registered teams". "Some 2,300 teams from 79 countries submitted final reports" (R-W97, R-W99, VP).
 - The realistic field is teams that finish, about 2,300 last year (R-AN60).
- **What they will likely submit** (ASM/judgement): "safe bucket + growth bucket" with a Monte Carlo "95%" (council judge panel view; untested). With AI tools common, a liability-matched Treasury ladder may also be common (BS-10).
- **What they fear:** the same things we do. Their median quality sets the bar.

- **Implication (all deliverables):** distinctiveness should come from:
 1. reasons specific to Laura (lumpy reputation-driven income, BS-03; the statistics degree, SH-04; the co-sponsor audience, SH-05 to SH-07);
 2. true process evidence (decision log, the trades that tested the plan);
 3. clarity.

It should not come from the architecture alone. The ethics rule bans plagiarising "existing strategies" (R-W28, VP).

SH-19 Wharton Global Youth (the organiser) and what it is teaching

- **Interests (VP, main competition page, read with curl 2026-09-27):**
 - Students "learn about strategy-building, teamwork, communication, risk, diversification, company and industry analysis, and many other aspects of investing."
 - The page promises "finance skills that will last a lifetime".
 - "Unlike other investment competitions, success is not determined by portfolio performance."
 - The FAQ says the "ultimate goal with this game... is learning the nuances of investing" (R-W37, VP), and the WInS page says "demonstrate what your team has learned" (R-W83, VP).
 - It points teams to a professional code (R-W29, VP; see SH-25).
 - It protects a real alumna: "While the client is real, the financial scenario is developed specifically for the competition" (R-W4, VP), and contacting her is forbidden (R-W16, R-W55, VP).
- **What this year's case seems designed to teach** (ASM/judgement; A1's R-AN entries give the evidence):
 1. separate a fixed promise from an uncertain surplus;
 2. turn a vague wish ("high degree of certainty") into a testable definition;
 3. communicate uncertainty honestly to people who are not investors.
- **Fears** (ASM/judgement): AI plagiarism, privacy breaches involving a real alumna, rule breaches.
- **Implication (FR, Articulation):** show what the team learned (e.g. why "lock early" beat "growth first", brief section 8.1). Credit AI use. Respect the privacy rule.

SH-20 The six students (learning goals)

- **Interests:**
 - Ray's stated goal is to learn markets and client analysis (CLAUDE.md, VRF).
 - The team's goal is the Top 50.
 - All members must contribute: "All team members must play a contributing role" (R-W38, VP).
 - Last year's lessons: over-complex packaging, and a trade outside the rules (v4 Part 2, VRF).
- **Fears:** repeating last year's rule break; an AI-written feel; uneven workload. Roles are still unassigned (CLAUDE.md open item).
- **Needs:** a strategy each of the six can explain in plain words; a decision log kept at trade time (R-W82, VP).
- **Implication:**
 - TN (tier 1): assign a trader and a note-writer before the next trade.
 - FR: the Articulation evidence must be real. Ethics bans invented "teamwork and experiential stories" (R-W28, VP).

Stakeholders the prompt missed

SH-21 Laura's publishers and literary agent (source of much of the 2028 deposit)

- **Evidence:**
 - Her site links a Publishers Weekly item titled "My book deal with HarperCollins" (VP, lauragao.com/hey, 2026-09-27). It also says she is represented by a literary agency (VP; named on the page, not needed here).
 - How advances are paid (Authors Guild, 20 July 2021, VP, W-8): "paying them out in three, and even up to six, installments". "Royalties are not paid to the author until the publisher has fully recouped the full amount of the advance." "The sales numbers of one book can affect the ability to sell the next project."

- **Interests** (ASM/judgement): the author's next books, delivered on schedule. A residency project competes for her time.
- **Why this matters:** the case says she "will contribute an additional \$150,000 at the beginning of 2028, using earnings from publishing advances..." (p.2 L43-45, R-C27, R-AN35, VRF). Instalment timing makes *late* as plausible as *smaller* (BS-13).
- **Implication (FR):** stress tests should label a "late 2028 deposit" case as well as a "smaller" one. Under the lock-early rule, lateness only delays the growth sleeve. The team's own scenario must be labelled as beyond the case (R-AN35).

SH-22 Speaking hosts, schools that book visits, and licensees

- **Interests:** the case lists "speaking engagements, licensing" as deposit sources (p.2 L44-45, VRF). School and library budgets and event demand drive them (ASM/judgement).
- **Implication (FR):** these sit in the same "human capital" bucket as SH-21. They matter only through the 2028 deposit and the wrong-way-risk question (brief section 9). No separate content.

SH-23 Readers of her books, and the classroom and library channel

- **Evidence:**
 - The case: "her books are read in classrooms around the world" (p.1 L22, VRF).
 - ALA (release April 2026, VP, W-9): "ALA's Office for Intellectual Freedom (OIF) tracked 4,235 unique titles challenged in 2025, the second highest ever documented by ALA." Also: "Of the unique titles challenged in 2025, 1,671 (40%) represent the lived experiences of LGBTQIA+ people and people of color."
 - **I have not verified whether any of Laura's titles were challenged. Do not claim they were.**
- **Why it matters:** the school and library channel for books on identity themes faces a documented challenge climate. That is a sector risk to the publishing income behind the 2028 deposit. It is **not** a reason to pick or avoid any investment (tokenism rule).
- **Implication (FR):** at most one clause in the human-capital discussion, framed as a risk to an income channel. Owner: B3/D6.

SH-24 The team's school

- **Interests:** its name is on the submission. It must provide "official documentation from your school" (SMAApply L45, R-S20, VRF). Semifinalists must also supply letterhead confirming "The team has permission to participate" and that all students and the advisor "are affiliated with the school" (R-W25, VP).
- **Fears:** a last-minute request; an administrative miss.
- **Implication (ADMIN):** request the documentation in October ("Do not wait until the last minute", R-S21, VRF). "Sample Documentation: Coming soon" (R-S23, VRF), so check SMAApply again for the sample. See BS-11.

SH-25 The "firm" (case fiction) and its ethics code

- **Evidence (Asset Manager Code, CFA Institute, 2nd ed., ©2009, 2010; PDF downloaded and read 2026-09-27, VP, W-1):**
 - Principle 5: "Communicate with clients in a timely and accurate manner."
 - B.4: "Have a reasonable and adequate basis for investment decisions."
 - B.6.a: "Evaluate and understand the client's investment objectives, tolerance for risk, time horizon, liquidity needs, financial constraints, any unique circumstances..."
 - D.4: "Maintain records for an appropriate period of time in an easily accessible format."
 - E.1: "Managers must not misrepresent the performance of individual portfolios or of their firm."
 - F.2: "Ensure that disclosures are truthful, accurate, complete, and understandable and are presented in a format that communicates the information effectively."
 - F.4.d: "Management fees and other investment costs charged to investors..."
 - **Anchor:** the Rules page, "All teams should review the CFA Institute's Asset Manager Code and operate by these standards" (R-W29, VP via A1).
- **Implication:** see BS-01 and BS-02.

SH-26 Laura after 2033, while she runs the residency

- **Interests (ASM/judgement):** once she runs a residency, her own earnings may change. The case lists "continued business income" among possible *outside* sources for the project (p.3 L86, R-AN24, VRF), and says nothing about her earnings after 2033.
- **Why it matters:** this is a Laura-specific reason for the flexibility the case asks for (R-C58, R-C84), besides building-cost overruns. It is an interpretation, not a case fact.
- **Implication (FR):** a flexibility sentence may give two purposes: project overruns and changes, and the founder's own reduced capacity to top up. Living costs stay outside the portfolio (p.2 L61-63, VRF). Owner: B3/D6.

4. Cross-stakeholder tensions (where pleasing one reader can cost another)

ID	Tension	Anchors (VRF unless marked)	How the current strategy resolves it	What is still open
T-1	Laura's "thoughtful risks" (SH-01) vs co-sponsors' need for a floor (SH-05)	p.2 L70; p.3 L114-115	Risk is taken only in the surplus sleeve; the floor is bought in 2031 (brief section 7)	Sleeve equity weight; the share of the sleeve locked as the floor (brief section 9)
T-2	The statistician's wish for method (SH-04) vs the IPS ban on calculations and charts (SH-16)	R-C55; IPS p.2 L51-52 and p.3 L123	Principle in the IPS, method in the FR	Which single number, if any, goes in the IPS
T-3	Operators' real budget (SH-12) vs fixed nominal payments (case)	R-C46; R-C44	Treasuries match the nominal promise (brief section 8.10)	The sentence on who bears erosion (brief section 9)
T-4	Taiwanese partners' NT\$ (SH-07) vs a USD promise	R-AN10; R-C85	Disclose; quote in two currencies (brief section 8.15)	Whether to convert part of the 2031 floor to NT\$ (brief section 9)
T-5	Founder's wish for a big gift (SH-02) vs flexibility (SH-26)	R-C57; R-C58	Leftover money kept mainly as flexibility (CLAUDE.md)	How much to hold back, as a rule or a share
T-6	The organiser's learning and AI honesty (SH-19) vs the team's wish to win (SH-20)	R-W26; R-W49	AI output is research only; students write	Works Cited AI record (R-W46) not yet drafted
T-7	Being distinctive against rivals (SH-18) vs the simplicity principle	R-S24 "creative"; v4 Part 2	Distinctiveness through client insight, not complexity	BS-10
T-8	The fictional PM decides (SH-14) vs the real rules, where students decide (SH-15)	R-C3; R-W19	Write the IPS as the team's rules; the advisor is admin only	Wording in the Articulation section

5. Blind spots: considerations that no current document in `research/` addresses

Method.

- On 2026-09-27 I searched every `.md` file under `research/` (excluding `_context/` and the case register's own quotes) for each topic's key words with `grep`. Examples: "management fee|advisory fee", "Asset Manager Code|CFA Institute", "enforceab|legally binding", "alumni", "staff|salar|payroll", "key person|successor", "custod|counterparty", "host community", "grant cycle|lead time".
- "Not addressed" means no strategy or research file treats the point. A topic that appears only as an unanswered seed question or an A1 anomaly entry counts as *not addressed*, and is marked "(seed only)".

Each blind spot gives: what it is; why it matters (status); the implication and deliverable; and the phase or agent best placed to answer it.

BS-01 The firm's own fee and fund costs are missing from every projection.

- The case says Laura will invest "with an asset management firm" (p.2 L43, VRF). The Code requires disclosure of "Management fees and other investment costs" (F.4.d, VP). The verified model has "no fees" (brief section 11).
- Rough effect, from my arithmetic:
 - Inputs (ASM): the fee is charged yearly on all assets and paid from the sleeve. The ladder grows at 5.1% a year from \$292,264. The sleeve starts at \$7,736, adds \$150k in 2028 and grows at 5.9% a

year.

- With no fee, the 2033 surplus is ~\$211k. At 0.25%, 0.5% and 1.0% it is ~\$202k, ~\$193k and ~\$177k.
- A fee therefore lowers the facility contribution, not the payments, because the ladder is untouched.
- **Implication:** FR projections must state a fee assumption (zero, fund expenses only, or an explicit rate) and its effect on the 2031 range. IPS: "low-cost" as a principle is consistent with Laura's likely preference (team review, ASM). Owner: D3 (model fees properly) and D8 (what fee is realistic for a portfolio this size; needs a source).

BS-02 The CFA Asset Manager Code is named in the rules but used nowhere.

- R-W29 (VP) tells teams to "operate by these standards". The Code's text (VP, W-1) maps onto the deliverables:
 - B.6.a "Evaluate and understand the client's investment objectives, tolerance for risk, time horizon, liquidity needs..." matches the IPS guide's list (IPS p.1 L9-10, VRF).
 - B.5.a "consistent with the stated objectives and constraints" matches the IPS freeze ("may not revise its investment strategy", R-I35).
 - F.2 matches criterion 5's clarity.
 - E.1 (do not misrepresent performance) matches the TN focus on reasoning over gains.
 - D.4 (records) matches the decision log.
- **Implication:** IPS, one clause: the framework follows a professional code's duties (know the client, act within the mandate, disclose plainly). FR: cite the Code in the Articulation or governance part. Do not over-quote it. This is a low-complexity credibility gain. Owner: D10 (compliance) and D8.

BS-03 Credibility is part of Laura's human capital.

- Her 2028 income sources are reputation-driven (p.2 L44-45, VRF). A missed public range damages co-sponsor confidence (R-C66, VRF) and, by inference, future advances, speaking and licensing (ASM/judgement). Existing files treat credibility only as a fundraising issue.
- **Implication:** FR, a second, Laura-specific reason the 2031 floor must be bought: breaking a public promise costs her twice. Also strengthens the wrong-way-risk discussion (brief section 9). Owner: B3/D6.

BS-04 "Dollar range" is ambiguous in Taiwan.

- The case uses "\$" and "dollar range" and never names a currency (R-AN10, VRF). NT\$ is the "New Taiwan dollar" (general knowledge; ASM that readers could confuse the two). Existing files cover exchange-rate *risk* (brief section 8.15, F-510), not the ambiguity of the label.
- **Implication:** FR fundraising draft: "US\$" on every figure; the NT\$ reference with a date and rate; one line saying which currency is binding. Owner: D5/D9.

BS-05 The words of the promise: "expects" versus "pledges".

- The case: "describe how much she expects to contribute" and "her potential contribution" (p.3 L110, L116, VRF).
- Fundraisers discount pledges. The "10-30% never collected" figure comes from vendor blogs (SNIP, W-11; primary pages refused with HTTP 403).
- Enforceability of charitable pledges differs by jurisdiction. Restatement (Second) of Contracts §90(2) makes charitable subscriptions binding without consideration in some places, while "charitable pledges are not enforceable in California unless the pledgor receives consideration" (California Bar Business Law Section memo, 2010, VP, W-10; law may have changed since; legal questions are not the team's to settle).
- **Implication:** FR draft: pick the verbs deliberately. The bought floor can be stated as set aside. The stretch is an expectation with a probability. No legal claims. Owner: D5/D9.

BS-06 Co-sponsors will read Laura's figure as a share of a total that nobody has estimated.

- Fundraising norms put a lead gift at roughly 15-30% of a campaign goal, and expect leaders to give first (SNIP, W-13; AFP and NonProfit PRO pages refused or redirected).
- But "Teams are not expected to estimate that cost... or determine the project's total funding gap" (p.4 L164-166, R-C95, VRF).
- **Implication:** FR: state that the contribution is sized from her portfolio, not as a share of a budget. Do not compute a campaign goal (that would breach the scope and add complexity). Owner: D5.

BS-07 What the plan does if the residency is delayed, moved or cancelled.

- The case gives no instruction (R-AN34, VRF). Seed question 24 asks about Taiwan Strait risk (seed only). No file answers it.
- Relevant fact: Treasury holdings can be sold at market price. The ladder's value moves about \$289 per 0.01% of rates (F-101 area, brief section 6), so a change of plan is possible at a market price, with no penalty beyond that.
- **Implication:** FR flexibility section, one sentence: the reserve sits in highly liquid government bonds, so a change of plan costs only the market move. Do not add scenarios beyond one line (scope). Owner: D1/D4.

BS-08 Governance continuity: can anyone apply the rules?

- The case fixes dates (2031, 2033). Nobody has specified who carries out the 2031 floor purchase and the 2033 designation, or with what records.
- Code D.4 (records) and D.3 ("accurate and complete" client information, with independent third-party review) apply (VP).
- **Implication:** IPS: write each rule so that any PM could apply it (trigger, action, record). This is also SH-14's need. Keep it generic. No personal speculation about Laura (privacy rule). Owner: D8.

BS-09 What Laura sees on her statements between 2027 and 2032.

- The locked ladder is marked to market. With a duration of 9.90 years (brief section 6), a +100bp rise lowers its value by roughly 9-10% even though every payment is still covered. At 2027-01-01 that is \$292,264 falling to \$264,890 (brief section 6, VRF script output). The council noted that this is "hard to explain to co-sponsors" (council 03_proposals_and_crossexam.md L259). No file sets a reporting rule.
- **Implication:**
 - FR: set a reporting metric ("payments covered: yes/no; funded ratio"), with the market value shown second.
 - TN: if a WInS Treasury fund falls, the reflection explains why that does not matter for a held-to-maturity promise, and notes that WInS funds are a duration proxy, not a held ladder (brief section 8.13).

Owner: D6.

BS-10 Rivals using the same AI tools may reach the same architecture.

- The distinctiveness claims in the council files are untested (ASM). No file considers that liability-driven investing (LDI) may be a common AI-suggested answer this season.
- **Implication (all deliverables):** measure distinctiveness by Laura-specific reasoning (BS-03, SH-04, SH-05 to SH-07) and by genuine process evidence. Owner: B7/E4.

BS-11 School documentation and the semifinal letterhead have no owner or date.

- Requirements: R-S20, R-S21, R-S23 (VRF) and R-W25 (VP). CLAUDE.md lists the Dec 4 due date only.
- **Implication (ADMIN):** assign an owner (with the advisor) and request the documentation in October. Owner: the team and D10.

BS-12 Taiwanese co-sponsors may need materials in their own language and currency. (Seed only: judge-panel translation remarks concern jargon, not language.)

- No file asks whether the fundraising draft's audience includes Mandarin-reading institutions.
- **Implication:** FR: at most, note that figures are given in US\$ with an NT\$ reference. Translation is out of scope and should not be attempted (complexity). Owner: D5.

BS-13 The 2028 deposit can be late, not only smaller.

- Advances are paid in instalments tied to milestones (Authors Guild, VP, W-8). The case fixes "beginning of 2028" (R-C27, VRF). Existing stress tests vary the size (brief section 6), not the timing.
- **Implication:** FR: one scenario label, "late (mid-2028)", alongside "smaller" and "missing". Its effect under lock-early is small: sleeve growth only. Owner: D3.

BS-14 The classroom and library channel risk to publishing income (SH-23).

- The ALA 2025 data are VP (W-9). The team's own documents raised book challenges as a risk (unverified), but no research file quantifies them.

- **Implication:** FR: one clause in the human-capital discussion; no investment consequence (tokenism rule). Owner: B3.

BS-15 Which co-sponsor types exist in Taiwan.

- Evidence is SNIP only: the Ministry of Culture funds artist villages and residency exchange (W-12, site refused curl with HTTP 403).
- **Implication:** FR: keep the co-sponsor audience generic ("foundations, public cultural agencies, local partners"). Name no organisation. Owner: D5 (only if it earns its place).

BS-16 Funders ask how a grantee responds to shortfalls and overruns (PKF 2019, VP, W-6).

- This matches the case's "favorable and unfavorable" requirement (R-C70) and the open "scope clause" (brief section 9).
- Partly known; what is new is the external evidence that this is a standard funder question.
- **Implication:** FR: the fundraising draft must answer "what if costs overrun" in one line: her contribution does not rise beyond the stated range; the scope adjusts. Owner: D5.

BS-17 Operating money is the money funders least like to give (SSIR 2009, VP, W-3; Kang 2026, VP opinion, W-5: "Funders can give unrestricted operating money to the residencies carrying the earliest risk").

- This explains the case's split (R-AN6) and turns Laura's ten-year operating commitment into a fundraising asset.
- **Implication:** FR fundraising draft must contain the fact that the first ten years of operating payments are funded from her portfolio alone, so co-sponsor money can go to the building. IPS: supports "promise first" as the central idea. Owner: D5.

BS-18 Operational risk in "certain barring U.S. default" is unspecified.

- The certainty definition names "operational error" as residual (council chair memo section 4). No file says which operational steps could fail (buying wrong maturities, missed reinvestment, custody) or how they are controlled.
- **Implication:** FR: one line naming the control (records and independent checks, per Code D.3 and D.4). This makes the certainty definition complete for a statistician (SH-04). Owner: D1/D10.

BS-19 Laura's capacity to top up after 2033 (SH-26). Not addressed. Interpretation only.

- **Implication:** FR flexibility rationale. Owner: B3/D6.

BS-20 The organiser protects a real alumna.

- The case scenario is fiction, but the person is real (R-W4, R-AN56). No file sets a rule for how the team *writes about* her identity and life in the FR. The privacy rule exists for research (brief section 4) but not as a writing checklist.
- **Implication:** FR: add a pre-submission check. Public professional record only; no speculation about her real finances; identity themes cited only as her public work and never as an investment reason. Owner: D10/D9.

Known open items this map touches but does not re-open (brief section 9): real erosion of the \$50k (SH-12); the joint tail of falling rates and a missing 2028 deposit (SH-21); wrong-way risk (SH-21, SH-22); the currency of the 2031 floor (SH-07); the scope clause (BS-16); post-2033 flexibility size (SH-02, SH-26).

6. Candidate questions for Phase B (each anchored; A3 does not answer them)

ID	Question	Anchor	Would change
Q-A3-01	What fee assumption should projections carry, and how much does it move the 2031 range?	Code F.4.d (VP); p.2 L43	FR numbers
Q-A3-02	Which provisions of the Asset Manager Code does our IPS framework already meet, and which one clause would show it?	R-W29	IPS sentence
Q-A3-03	Does a missed public range hurt Laura's income, and should that be part of the reason the floor is bought?	p.2 L44-45; R-C66	FR sentence
Q-A3-04	How should the fundraising draft label currency so that no reader confuses US\$ and NT\$?	R-AN10; p.3 L117	FR draft

ID	Question	Anchor	Would change
Q-A3-05	Which verb goes with the bought floor and which with the stretch?	p.3 L110, L116	FR draft
Q-A3-06	How do we stop co-sponsors reading the contribution as a share of an unestimated total?	R-C95	FR draft
Q-A3-07	What does the plan do if the residency is delayed by one to two years?	R-AN34	FR flexibility sentence
Q-A3-08	What metric do we report to Laura each year so that bond price falls do not look like broken promises?	R-S25; R-C38	FR; TN reflection
Q-A3-09	Is lock-early actually rare among AI-assisted rivals, and if not, what is our real edge?	R-S24 "creative"; R-W28	all
Q-A3-10	Who requests the school documentation and letterhead, and by when?	R-S21; R-W25	ADMIN
Q-A3-11	Should stress tests include a late 2028 deposit as well as a smaller one?	R-C27; R-AN35	FR scenarios
Q-A3-12	What single line answers a funder's "what if costs overrun"?	R-C70; PKF (VP)	FR draft
Q-A3-13	Does "operations funded for ten years" belong at the top of the fundraising draft?	R-AN6; R-C64	FR draft
Q-A3-14	Which operational controls complete the certainty definition?	R-C54 to R-C56	FR

7. Implications ranked by deadline

• Tier 1 (TN, Oct 23; WInS now):

- Assign a trader and a note-writer (SH-20).
- Each reflection names Laura's need and does not justify by gain (SH-16, BS-09).
- Keep the WInS mirror consistent with the plan (SH-16, F-409).
- Request the school documentation (BS-11, ADMIN).

• Tier 2 (IPS, Nov 6):

- The framework as pre-set rules any PM could apply (SH-14, BS-08).
- One clause on professional duties (BS-02).
- "Promise first" as the central idea, with the operating-money rationale (BS-17).
- A plain certainty definition (SH-04).
- "Low cost" as a principle (BS-01).

• Tier 3 (FR, Dec 4):

- A fee assumption in projections (BS-01).
- Two-sided confidence for the range (SH-04).
- The fundraising-draft checklist: floor set aside; stretch with its model; operations funded for ten years; US\$ labelled with an NT\$ reference; the overrun answer; no share-of-total (SH-05 to SH-07, BS-04 to BS-06, BS-16, BS-17).
- The erosion sentence (SH-12).
- The human-capital clause (BS-03, BS-13, BS-14).
- The flexibility rationale (SH-02, SH-26, BS-07).
- The reporting metric (BS-09).
- The privacy check (BS-20).
- Works Cited with the AI record (SH-16, R-W46).

8. Sources

All web sources were accessed 2026-09-27 from this run's container.

Official and repo files (VRF):

- `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`
- `2026_WGY_Investment_Policy-FINAL.txt`
- `2026_WGY_Trading_Notes_Analysis-FINAL.txt`

- SMaply_Deliverables_Page_2026-09-27.md
- research/insight_v1/phase_A/case_register.md (A1; its R-W entries are VP)
- research/insight_v1/phase_A/fact_register.md (A2)
- research/council_2026-09-27/*.md (prior AI brainstorming, lower authority)
- research/team_materials/review_why_laura_and_public_profile.md
- CLAUDE.md

The team's two client documents (outside the repo) were read as leads only. Nothing was copied from them. No personal details from them are used here.

Web sources:

ID	Source	Status
W-1	CFA Institute, <i>Asset Manager Code</i> , 2nd ed., ©2009, 2010. PDF https://rpc.cfainstitute.org/sites/default/files/-/media/documents/code/amc/asset-manager-code-and-guidance-2nd-ed.pdf (downloaded with curl, text extracted) and web page https://rpc.cfainstitute.org/codes-and-standards/asset-manager-code	VP
W-2	Laura Gao, "Hey" page, https://lauragao.com/hey/ (public professional facts only)	VP
W-3	Ann Goggins Gregory & Don Howard, "The Nonprofit Starvation Cycle", Stanford Social Innovation Review, Fall 2009, https://ssir.org/articles/entry/the_nonprofit_starvation_cycle	VP (read with curl)
W-4	Artist Communities Alliance, "Fees, stipends, and funding for residencies", https://artistcommunities.org/fees-stipends-and-funding-residencies (no date shown)	VP
W-5	Emil J Kang, "Artist Residencies Are Arts Infrastructure", Substack, 13 July 2026, https://emilkang.substack.com/p/artist-residencies-are-arts-infrastructure (opinion)	VP
W-6	PKF O'Connor Davies, "Best Practices: Due Diligence on Grantees", Private Foundations Bulletin, July 2019, https://www.pkfod.com/wp-content/uploads/2019/07/Best-Practices-Due-Diligence-on-Grantees-v4.pdf	VP
W-7	Grantmakers in the Arts, "Do Foundation Sustainability Strategies Work?", October 2007, https://www.giarts.org/article/do-foundation-sustainability-strategies-work (roundtable notes; old)	VP
W-8	The Authors Guild, "Everything You Need to Know About Book Advances", 20 July 2021, https://authorsguild.org/blog/everything-you-need-to-know-about-book-advances/	VP
W-9	American Library Association, "American Library Association releases 2025 Most Challenged Books List...", April 2026, https://www.ala.org/news/2026/04/american-library-association-releases-2025-most-challenged-books-list-national-library	VP
W-10	State Bar of California, Business Law Section, "Charitable Pledges" legislative proposal memo (file BLS-2010-03), https://www.calbar.ca.gov/sites/default/files/portals/0/documents/legislation/BLS-2010-03-charitable_pledges.pdf	VP for the memo's text; current law not checked
W-11	Pledge non-collection "10 to 30 percent" (vendor blogs, e.g. https://www.bonterratech.com/blog/unfulfilled-pledges , which returned 403)	SNIP
W-12	Taiwan Ministry of Culture residency support, https://artres.moc.gov.tw/en/about/index (HTTP 403 with curl)	SNIP
W-13	Capital-campaign lead-gift and board-giving norms: https://afpglobal.org/4-persistent-capital-campaign-myths-dispelled-data (403), https://www.nonprofitpro.com/post/find-right-board-giving-goal-capital-campaign/ (307 redirect, not followed), https://www.donorsearch.net/resources/gift-range-chart-guide/	SNIP
W-14	Wharton Global Youth, competition main page, https://globalyouth.wharton.upenn.edu/competitions/investment-competition/	VP (read with curl)

Access notes:

- WebFetch was refused ("EGRESS_BLOCKED") for every host I tried, including hosts the brief lists as reachable (lauragao.com, globalyouth.wharton.upenn.edu).
- Plain curl through the proxy worked for most hosts. It returned 403 for afpglobal.org, bonterratech.com, littlegreenlight.com, help.candid.org and artres.moc.gov.tw.
- I report these rather than guess what they say.

What this teaches

A client never stands alone. Laura's money serves a building, a staff, visiting artists, co-funders in another country, and her own reputation, and each of these reads the plan with a different fear. Mapping them does not mean pleasing everyone. It shows which promise matters to whom: operators need the nominal \$50k,

co-sponsors need a floor that exists, and Laura needs her word to hold. It also shows where a single sentence, such as "US\$", "set aside" or "funded for ten years", does more work than another model. The blind spots here are mostly small and cheap to fix, and that is the point. Judges and statisticians tend to find weak spots in the details nobody owned: fees, labels, paperwork and wording.

A4 WInS week-1 guardrails (early run of D10, Compliance Officer)

Agent: A4, insight_v1 run. Written 2026-09-27 (the day before trading opens). Official sources win over everything else here. This file is compliance research for the team. It has **no submission-ready text**: no trading notes, no IPS wording, no reflections. It names **instrument types only** (no tickers). Every trade idea must still be checked against this year's WInS rules and what WInS actually offers.

Status labels (brief section 3): **VERIFIED-PRIMARY** (I read it myself on the primary page on 2026-09-27; URL given), **VERIFIED-REPO-FILE** (official file in this repo; path and line given), **SNIPPET-UNVERIFIED** (search snippet or secondary website), **ASSUMPTION**, **INTERPRETATION** (my reading, not a fact). Season tags: **[2026-27]** = this season; **[PRIOR]** = an earlier season, for background only.

0. Summary: the ten things to know before the first trade

1. **This year's trading rules are public and I read them myself.** SMAApply "Trading Details" and "FAQs" pages (VERIFIED-PRIMARY [2026-27], read 2026-09-27 11:54 UTC) state: **\$300,000** virtual cash; trading **Sept 28 to Nov 6, 2026; up to 200 trades**; each trade no larger than **twice the security's daily volume**; allowed instruments are **stocks priced at \$5 or more, "Any ETF available on WInS", and Treasury bonds** (U.S., U.K., Germany, France, Italy, Netherlands); **not allowed**: margin, short selling, stock-secured debt, crypto, derivatives, and "Anything other than the approved investments listed above"; **no sector minimum**. These pages confirm and extend what A1 recorded in `phase_A/case_register.md` (R-W56 to R-W95).
2. **There is no separate "approved ETF list" this year.** The rule is "Any ETF available on WInS". Last season was different: the 2025-26 WInS User Guide said "You are only permitted to buy ETFs from the approved lists" (VERIFIED-PRIMARY [PRIOR]). The file `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` is therefore **history, not a rule**. INTERPRETATION: last year's rule break may have come from exactly this kind of list rule. This year the trap is different: WInS may let you place an order that the written rules still forbid. Two examples: stocks priced between \$3 and \$5 (the generic Stock-Trak default minimum is \$3; Wharton's is \$5), and crypto funds.
3. **New finding: this season's WInS User Guide exists.** A "2026-2027-WInS-Userguide.pdf" was uploaded to the Wharton WInS portal on 2026-09-15 (PDF creation date 2026-09-15). I rendered its pages to check which text is actually visible. The visible text says: "Day Trading: This is not permitted." and "Position Limit: This is how much of your portfolio you can invest in one single stock." (VERIFIED-PRIMARY [2026-27]). It also says to check **Portfolio Summary > Session Rules** before trading.
4. **Position limit: open, and important for the current strategy.** This year's number is **not published anywhere public**. Stock-Trak's generic Wharton-portal FAQ says "The default position limit is 25%" (VERIFIED-PRIMARY, generic platform page, not season-specific). If a 25% limit per security applies, the current WInS mirror (about 65% Treasuries split roughly 64.8/35.2 between an intermediate and a long Treasury fund, so about 42% in one fund) **would be blocked**. It would need three or more Treasury instruments. **Read Session Rules on day 1 before the first order.**
5. **Day trading is not permitted** [2026-27 user guide]. The guide does not define it. ASSUMPTION (safe reading): never buy and sell the same security on the same U.S. trading day. This matters for rebalancing and for fixing mistakes. The FAQ also says: "If you make an error, we encourage you to run with it" (VERIFIED-PRIMARY).
6. **Trading notes are written at trade time and quoted word for word later.** Official: "Include each Trading Note exactly as it appears in WInS." / "The Trading Notes must correspond to trades executed in WInS. Yes, we will verify this." (VERIFIED-REPO-FILE, Trading Notes guide p.2 L44, L53). SMAApply WInS page: notes "should record the reasoning, research, and intended strategic role of each decision at the time it is made" (VERIFIED-PRIMARY). **No character limit is published anywhere I could reach** (it is not in the 2026-27 guide or on the public pages). Stock-Trak's own 2017 description says students "cannot edit or delete their trade notes" (VERIFIED-PRIMARY, vendor blog, not season-specific). Assume the note is permanent.

7. **Only executed trades count.** An order placed after the U.S. close on Nov 6 would fill at the next open. By then the portfolio is frozen. INTERPRETATION: that order never executes, and a note on an unfilled order cannot be one of the three notes.
8. **The AI policy lets you brainstorm, requires AI use to be recorded, and bans AI-written deliverables.** The Wharton AI policy page adds a line the Rules page lacks: "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages." (VERIFIED-PRIMARY). See section 4.
9. **Disqualification and exclusion risks are listed with exact sources in section 5.** The biggest are: contacting Laura; IPS format errors ("will not be considered for semifinal selection"); missed deadlines (no extensions); team size below 4 or above 6; paid help; plagiarism and AI misuse; notes that do not match executed trades.
10. **The third-party "rules" in CLAUDE.md are outdated.** "Sector minimum = team size" and "first trade by a deadline" trace to the **2023-24** season (Lumiere page quoting "October 13, 2023"; SNIPPET-UNVERIFIED [PRIOR]). This year's primary page says the opposite: "There is no required sector allocation or minimum number of sectors." The "200 trades" claim is now confirmed by the primary page.

(a) What the official documents in the repo say

All quotes are VERIFIED-REPO-FILE [2026-27] from `competition/official/2026_27/`. Line numbers refer to the `.txt` extractions. A1 checked their PDFs against the live Box files by SHA-256 (see `case_register.md` section 0.2).

a1. Trading and the WinS portfolio

#	Exact quote	Location	What it means (INTERPRETATION)
a1.1	"The WinS portfolio represents each team's implementation of its investment strategy during the competition."	Case p.4 L129-130	WinS shows the strategy in action. Every trade must fit the strategy.
a1.2	"It does not determine Laura's actual portfolio value at the beginning of 2027."	Case p.4 L130	WinS results never become her starting value.
a1.3	"Gains or losses generated during the WinS trading period should not be added to or subtracted from Laura's portfolio projections."	Case p.4 L132-133	Keep WinS profit and loss (P&L) out of every projection.
a1.4	"Successful investing is not simply about selecting individual stocks. It requires making disciplined decisions that consistently support an overall investment strategy."	Trading Notes (TN) guide p.1 L7-8	Stock-picking stories are not the aim.
a1.5	"Throughout the competition, your Trading Notes should document the research, analysis, and reasoning behind your investment decisions."	TN p.1 L8-9	Every trade gets a note, not just the three you pick later.
a1.6	"you should already be using a clear strategic approach to guide your investment decisions."	TN p.1 L18-19	Trades from day 1 should already follow a stated approach.
a1.7	"The focus should be on the quality of your reasoning and the intended role of each decision, not simply whether an investment gained or lost value during the competition."	TN p.1 L15-16	Say what role each trade plays. Returns do not matter for the notes.

a2. Trading notes and verification against executed trades

#	Exact quote	Location
a2.1	"Select three (3) Trading Notes from trades your team executed in WinS."	TN p.2 L42
a2.2	"Include each Trading Note exactly as it appears in WinS."	TN p.2 L44
a2.3	"For EACH Trading Note, write a reflection of 100 words or fewer explaining: – Why your team made the investment decision. – How the decision aligned with your overall investment strategy. – How the decision supported the client's goals, funding needs, or risk considerations."	TN p.2 L46-52
a2.4	"The Trading Notes must correspond to trades executed in WinS. Yes, we will verify this."	TN p.2 L53
a2.5	"The Trading Notes you select do not need to correspond to investments that remain in your portfolio at the end of the competition."	TN p.1 L25-26
a2.6	"The analysis should be consistent with the strategic approach your team is developing and will later articulate in its IPS."	TN p.1 L30-31

#	Exact quote	Location
a2.7	"At this stage, your team is not expected to calculate the final size of the operating reserve, complete long-term portfolio projections, determine Laura's facility contribution, or draft her communication to co-sponsors."	TN p.1 L20-22
a2.8	Example note: "We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment. The position provides income and greater stability than equities, although interest-rate changes may affect its value. This trade supports our plan to balance continued growth with reliable future cash flows as the residency funding date approaches."	TN p.2 L36-39
a2.9	"Your team will submit their trading notes via the SurveyMonkey Apply form."	TN p.2 L54

DERIVED (my own count): the example note has 59 words and 413 characters. This is the only length signal in the official files. No character limit is stated in any repo file.

What "verify" implies (INTERPRETATION): Wharton can see the team's WInS account, so it can compare each quoted note with the trade it is attached to. A note quoted with any change, or taken from a practice trade or an unfilled order, would fail that check.

a3. The WInS username

#	Exact quote	Location
a3.1	"WInS username" / "- This is the team's username for logging into the WInS platform."	IPS guide p.3 L98-100
a3.2	Sample title page: "WInS Username: sample_team-1234567"	IPS guide p.4 L131

Public source, same topic (VERIFIED-PRIMARY [2026-27], SMaply FAQs, <https://wghsinvcomp.smaply.us/res/p/faqs/,read> 2026-09-27): "No. Your team's username is only used to access WInS. Your team username is NOT the same as your team name." The 2026-27 WInS User Guide p.4 also says "you will not be able to change your team's username."

a4. The portfolio lock

#	Exact quote	Location
a4.1	"Investment Policy Statement (Due: November 6; no later than 5:00 p.m. ET) * IMPORTANT: Trading ends and your portfolio is locked."	SMaply Deliverables transcription L10-11
a4.2	"Once submitted, the IPS becomes the official record of your team's investment strategy. Your team may not revise its investment strategy after the submission deadline."	IPS guide p.2 L73-74
a4.3	"Your portfolio and Final Report should reflect the strategy established in your IPS."	IPS guide p.2 L75
a4.4	"Your team will formally present its investment strategy in the Investment Policy Statement (IPS) in Week 6."	TN p.1 L17

a5. IPS format rules (breaking any = "will not be considered for semifinal selection")

#	Exact quote	Location
a5.1	"Please follow these directions carefully. Submissions that do not meet the requirements below will not be considered for semifinal selection."	IPS p.3 L82-83
a5.2	"Page 1: Title Page / 1 page maximum"	IPS p.3 L84-85
a5.3	"Official team name - This should be the same as what you submitted with your official team roster."	IPS p.3 L88-90
a5.4	"Finalized team member names - These should be the same as what was submitted with your official roster. • Format: First Name, Last Initial"	IPS p.3 L92-96
a5.5	"WInS username"	IPS p.3 L98
a5.6	"Pages 2 and 3: Investment Strategy / 2-page maximum"	IPS p.3 L103-104
a5.7	"Investment Strategy Elevator Pitch, maximum 50 words" / "Investment Policy Statement, maximum 500 words"	IPS p.3 L107-109
a5.8	"Font: Times New Roman" / "Size: 12-point" / "Spacing: Double-spaced" / "Margins: 1 inch on all sides"	IPS p.3 L112-118
a5.9	"Submit as a PDF through SurveyMonkey Apply." / "Maximum file size: 5 MB."	IPS p.3 L120-122
a5.10	"Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted."	IPS p.3 L123
a5.11	"sources do not need to be cited in the IPS. Relevant sources and supporting evidence must be included in the Final Report."	IPS p.3 L124-125

#	Exact quote	Location
a5.12	"Formatting choices not specified below are left to the team's discretion."	IPS p.3 L101
a5.13	Sample: "Jordan A. Taylor B. Casey C. Morgan D."	IPS p.4 L130

Note: 500 words, double-spaced, 12-point, on at most 2 pages together with a 50-word pitch is tight.

ASSUMPTION: at roughly 250-300 double-spaced words per page, 550 words with headings may not fit on two pages. The team must test the layout early.

a6. Final Report format

#	Exact quote	Location
a6.1	"Instructions: Detailed Final Report instructions and requirements will be shared on November 9."	SMAApply L44
a6.2	"Detailed Final Report requirements will be released at the beginning of Week 7."	Case p.4 L162-163
a6.3	"We require official documentation from your school ... This will be submitted along with your Final Report as a separate file." / "Sample Documentation: Coming soon"	SMAApply L45-46
a6.4	"Evaluators will consider the three deliverables together."	Case p.4 L161

The Final Report's format rules are not yet published. Nothing about its length, font or charts can be known before Nov 9.

a7. Deadlines and submission (all deliverables)

"All deliverables must be submitted via your team's SurveyMonkey Apply account." / "We will not grant any deadline extensions." / "Team must submit all the deliverables, and meet their necessary requirements, in order to receive a digital credential." (SMAApply L14-17). "Once your official team roster is submitted, teams may not change their members." (SMAApply L22).

(b) What the public web says (searched 2026-09-27, 11:50-12:00 UTC)

b1. VERIFIED-PRIMARY [2026-27]: read on the page myself

Pages were fetched with curl through the session proxy and the text was taken from the HTML. Access date for all: 2026-09-27.

SMAApply "Trading Details", <https://wghsinvcomp.smapply.us/res/p/trading/> (public, no login needed)

- Starting cash: "Your team will begin the official trading period with \$300,000 in virtual cash."
- Account: "Each team shares one WInS account. All team members will use the same username and password."
- Dates: "Official trading begins on September 28, 2026, and ends on November 6, 2026. When trading ends, your portfolio will be frozen, and your team may no longer trade or change its holdings."
- "The \$300,000 is your team's WInS simulator balance. The additional \$150,000 contribution described in the Client Case Study will not be added to WInS."
- "WInS gains and losses should not be added to or subtracted from the client's long-term portfolio projections."
- Execution: "You may enter buy and sell orders at any time during the official trading period." / "Orders placed while the market is closed are filled at the security's opening price when the market reopens." / "Prices displayed for open positions may be delayed by 10 to 15 minutes." / "International equity orders may experience a longer delay and are processed at the end of the applicable trading day." / "WInS automatically completes foreign-currency conversions ... exchange-rate movements may affect the value of your holdings."
- Trade limits: "Your team may make up to 200 trades during the competition." / "Your team may trade an amount equal to no more than twice a security's current daily trading volume."
- Approved securities: "Stocks: Any stock available on WInS priced at \$5 or more, or the equivalent of \$5 in its local currency." / "Exchange-Traded Funds: Any ETF available on WInS." / "Treasury Bonds: Bonds available on WInS. The list includes Treasury bonds from the United States, United Kingdom, Germany, France, Italy, and the Netherlands."

- Sectors: "There is no required sector allocation or minimum number of sectors. However, diversification remains an important investment principle." / "The goal is to build an intentional mix of investments appropriate for your strategy, not simply to own a large number of securities."
- "The competition does not require frequent or same-day trading. Make thoughtful, well-researched decisions and record the reasoning behind them in your Trading Notes."

SMApply "FAQs", <https://wghsinvcomp.smapply.us/res/p/faqs/>

- "Your team will be responsible for managing a portfolio of \$300,000 in virtual cash."
- "DOES THE COMPETITION ALLOW ALL SECURITIES? No. Investments permitted (for BOTH contributions): Cash / Any stock from any exchange available on WInS priced at \$5 (or the equivalent of \$5 in the local currency) or higher. / Any Exchange-Traded Funds (ETFs) available on WInS. / Any Government/Treasury Bonds from any exchange available on WInS. The following are NOT permitted: Margin trading / Short selling / Stock-secured debt / Crypto / Derivatives (options, futures, swaps, etc.) / Anything other than the approved investments listed above."
- "Each stock transaction is charged a flat \$25 commission and treasury bonds are charged \$10. Commission is charged only on a trade that clears."
- "The bond prices are updated once daily at U.S. market open, and pay out every six months."
- "You can only cancel open orders." / "If you make an error, we encourage you to run with it and not scramble to "fix" it. Trading errors are part of the learning process."
- "Students should place trades on WInS, NOT the advisor. ... It should be a team decision. We strongly encourage teams to identify different roles for each team member."
- "you should not be doing a lot of buying and selling of the stocks designated for the long-term portion of your portfolio"
- "When your team logs in to the shared WInS account, they'll find tutorial videos, a user guide, and FAQs under "Getting Started""
- "Additionally, teams must meet the required trading activity and portfolio management guidelines throughout the competition." (No minimum amount of activity is defined on any public page.)

SMApply "Wharton Investment Simulator (WInS)", <https://wghsinvcomp.smapply.us/res/p/wins/>

- "You are not being evaluated based on: Your portfolio ranking / How many trades you make / Whether you outperform other teams / Whether your portfolio makes money during the competition"
- "All account activity will be cleared after the practice session. Practice trades, portfolio activity, gains and losses, and account balances will be reset before official trading begins on September 28."
- "Official trading will end on November 6, when your Investment Policy Statement is due. At that time, your WInS portfolio will be frozen"
- "Your portfolio should reflect a cohesive strategy designed around the client's goals. Each investment decision should have a clear purpose within that strategy."
- "Complete a Trading Note when your team makes an investment decision. Your Trading Notes should record the reasoning, research, and intended strategic role of each decision at the time it is made."
- "Access WInS: <https://edu.stocktrak.com/wharton/>"

2026-2027 WInS User Guide (new finding), <https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf> (found through the portal's public media listing; upload date 2026-09-15; PDF creation date 2026-09-15; 18 pages; SHA-256 19d797dfa6269aeec312797e50cf0d98e7547d5b2bfb97f14b76f8d64229a84e). Caution: the file keeps hidden text from older versions, including the 2024-25 "approved lists" and "70%" rule. I rendered pages 6, 7, 10, 11 and 17 to images and quote **only the text that is visible**.

- p.5: "how many of your allotted 200 trades your team has used."
- p.6: "Before you start trading, you might want to check your specific trading parameters set by Wharton Global Youth Program ... go to 'Portfolio Simulation' and click 'Portfolio Summary'. Then, click 'Session Rules'." / "Wharton Global Youth Program sets the currency, starting cash, interest rates, commission charges, trading dates, and a few more rules: Position Limit: This is how much of your portfolio you can invest in one single stock. Day Trading: This is not permitted. Short Selling: This is not permitted. Allow trading on margin: This is not permitted." (The screenshot on this page is dated 9/17/2024. It shows

"Total Trades Allowed 200", "Day Trading Allowed No", and a bond "Position Limit (Single Position) 100%". It shows no equity position limit. It is [PRIOR] evidence.)

- p.7: "Remember: The Investment Competition primarily involves the buying and selling of stocks, exchange-traded funds (ETFs) and a limited number of treasury/government bonds."
- p.10: "Enter a Trade Note (these are important!) to discuss how the purchase fits into your overall strategy." The screenshot (dated 2023) shows the banner "You are required to write a TRADE NOTE explaining each trade." and a "View Example" link. No character counter is visible.
- p.11: "Once you are on the trading page, select the symbol of the treasury bond you would like to trade get a quote. ... When you buy a bond in between coupon dates you will have to pay accrued interest to the seller."
- p.4: "you will not be able to change your team's username. Your team's username is only used to access WInS. Your team username is NOT the same as your team name."

Wharton Global Youth public pages (globalyouth.wharton.upenn.edu):

- Main page, </competitions/investment-competition/>: "September 28 Competition begins; first day of trading" ... "November 6 Deliverable: Investment Policy Statement (IPS) must be submitted as instructed (5:00 p.m. ET)"; "a free, experiential investment challenge for high school students (9th to 12th grade)"; "Students work in teams of four to six, guided by a teacher from their school as their advisor".
- Rules & Roles, </competitions/investment-competition/rules-roles/> (eligibility and advisor role):
 - "Teams must be composed of current students from the same high school."
 - "Each team must designate one student team leader who is at least 16 years old at the start of the Competition. No exceptions will be granted. Team leaders may not be changed after the Competition begins on September 28, 2026 unless the Advisor submits a written request due to extenuating circumstances and receives approval".
 - "The Advisor's role is to provide guidance, encouragement, and educational support throughout the Competition. Advisors may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development."
 - Advisors are responsible for: "Monitoring team engagement and portfolio activity." / "Serving as a sounding board without completing work on behalf of students."
 - "Teams may seek guidance from a parent, industry professional, or other adult serving as a secondary advisor, provided the team also has a primary teacher Advisor. However, the use of paid advisors, education consultants, or other agents is prohibited."
 - "Teams must meet the required trading activity and portfolio management guidelines throughout the competition."
 - "To remain eligible for the Semifinals and Global Finale, teams must meet all competition requirements and submit the competition deliverables by the published deadlines."
 - "The top 50 teams will be selected as semifinalists based on the strength of their Investment Policy Statement (IPS) and Final Reports." (The SMAApply page says the Trading Notes Analysis is also evaluated. A1 logged this conflict as X-8.)
- FAQ, </competitions/investment-competition/faq/>: "Any student on the team may serve as the team leader, provided they are at least 16 years old on the first day of the Competition (September 28, 2026)." / "Copies of the weekly emails will also be posted on the Wharton Investment Simulator (WInS) for all team members to view."
- AI policy: see section (d).

Stock-Trak pages (the WInS vendor; platform facts, **not season rules**):

- Wharton WInS portal FAQ, <https://edu.stocktrak.com/wharton/portfolio-faq/> (page last modified 2023-08-10): "Trading Notes are comment and explanations that must be added to each trade." / "The default position limit is 25% which means you cannot put more than 25% of your portfolio value in a single security. Please note that only professors have the ability to change the position limit for their classes." / "Minimum Stock Price: The minimum stock price must be \$3.00 or higher" / "we only allow users to trade up to half of the volume of any security on the market." Use: this shows what the platform does by default. Wharton's written rules (a \$5 minimum) come first, and Session Rules shows this year's actual settings.
- "New Feature – Trade Notes", <https://www.stocktrak.com/new-feature-trade-notes/> (Stock-Trak blog, July 10, 2017): "students cannot edit or delete their trade notes ... they can add more notes to the same

trade (each note has its own timestamp for a record of when each note is added)." This conflicts with the 2024-25 and 2025-26 WInS guides ("Trade Notes will let you add/edit trade notes." [PRIOR]). The 2026-27 guide dropped that line. Treat every note as **permanent and time-stamped**.

b2. [PRIOR] facts from earlier seasons (history only; not this year's rules)

Fact	Source	Status
2025-26 virtual cash: "\$500,000 in virtual cash, up from \$100,000"	Wharton news, https://globallyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/ (Jan 27, 2026)	VERIFIED-PRIMARY [PRIOR]
2025-26: "Remember: You are only permitted to buy ETFs from the approved lists." / "Teams may use stocks (no approved list) and ETFs and treasury bonds from the approved lists"	2025 WInS User Guide, https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2025/09/2025-WInS-User-Guide.pdf , p.7 and p.11	VERIFIED-PRIMARY [PRIOR]
2024-25: "Teams must still have at least 70% of their total stock allocation from the approved list."	2024 WInS User Guide p.7	VERIFIED-PRIMARY [PRIOR]
2025-26 approved ETF list (100 ETFs)	competition/historical/2025_26/	VERIFIED-REPO-FILE [PRIOR]; not a 2026-27 rule
2024-25 Session Rules screenshot: trading 09/30/2024-12/06/2024 4:00 PM; 200 trades; no day trading; bond position limit 100%	2026-27 guide p.6 image	VERIFIED-PRIMARY image [PRIOR]

b3. SNIPPET-UNVERIFIED third-party claims (do not act on these)

Claim	Where it came from	Verdict
"Each team must invest in a number of sectors equal to the number of team members"	WebSearch summary (2026-09-27) citing third-party guides. Lumiere Education page (read directly; published 2023-08-05) gives it as a 2023-24 criterion: "the portfolio is invested in at least as many sectors as there are team members."	[PRIOR] and contradicted for 2026-27 by the primary "There is no required sector allocation or minimum number of sectors."
"First trade by" a deadline (CLAUDE.md: "by Oct 10")	Lumiere: "teams must fully execute at least one trade on WInS by close of the U.S. markets (4:00 p.m. ET) on October 13, 2023"	[PRIOR] . No 2026-27 primary page states a first-trade deadline. ASSUMPTION: the team trades in week 1 anyway, so it is safe either way.
"Each team starts with \$500,000" for 2026-27	WebSearch summary citing Aralia. I read the Aralia page directly and the figure is not on it .	Wrong : the primary figure is \$300,000.
"A minimum of four and a maximum of seven members"	WebSearch summary	Wrong for 2026-27 : primary says "four to six".
"trading period begins on September 30 at 9:30 AM"	WebSearch summary	[PRIOR] (2024 season). This year: Sept 28.
"200-trade cap"	Third-party	Now VERIFIED-PRIMARY [2026-27] .

b4. What I could not find

- **The 2026-27 position limit, the exact end time on Nov 6, and the trading-note character limit** are on no public page. They are probably on the logged-in Session Rules page and trade screen (see section f).
- No separate 2026-27 "approved list" exists on any public page. The public rule is "any ETF available on WInS".
- Blocked or unreachable: app.raena.ai (connection failed); www.aralia.com through WebFetch (egress blocked, but curl reached it). The logged-in SMAppl and WInS pages cannot be reached from here.

(c) "Before every trade" checklist (week 1 onward)

Print this. One person reads each line aloud and a second person ticks it. Instrument types only. **Every item means: confirm against this year's WInS rules and Session Rules.**

A. Is the instrument allowed?

1. [] The instrument type is one of: a stock priced at **\$5 or more** (local-currency equivalent), an **ETF available on WInS**, or a **government/Treasury bond available on WInS**, or cash. (*SMAppl Trading*)

Details and FAQs.) Confirm against this year's WInS list and rules.

2. [] It is **not** margin, a short sale, stock-secured debt, crypto, or a derivative (options, futures, swaps), and it is not "anything other than" the listed types. Do not trade it just because WInS lets you place the order. Types to avoid unless Wharton confirms in writing (INTERPRETATION of "ETF" and "Crypto"): exchange-traded notes (ETNs, which are not ETFs), closed-end funds, mutual funds, crypto-linked ETFs, and leveraged or inverse ETFs (they use swaps and futures). Confirm against this year's WInS rules.
3. [] It is not one of last year's list rules applied by mistake, **and** it is not assumed to be allowed because it was on last year's list. The 2025-26 list is history only.
4. [] For an individual Treasury bond: it appears in the WInS **Bonds** drop-down; you have noted the face value, price, last coupon date and **accrued interest** (the interest built up since the last coupon, which the buyer pays). Bond prices update only once a day at the U.S. open. Commission is \$10. Confirm against this year's WInS rules.

B. Do the rules allow this size and timing?

5. [] Session Rules checked this week (screenshot saved in the decision log): position limit per security **and** per security type, trades allowed, day trading, dates and times. After this trade, **no single position exceeds the position limit**. Confirm against this year's Session Rules.
6. [] **Not a day trade**: we have not bought or sold this same security earlier today (U.S. trading day). ASSUMPTION: this is the safe reading of "Day Trading: This is not permitted." Confirm against this year's WInS rules.
7. [] Order size is **no more than twice the security's daily volume**. ASSUMPTION: for a thinly traded fund, keep well below this.
8. [] Trade count: we are within 200 total and are keeping a reserve (ASSUMPTION: at least 50 left for Oct 23 to Nov 6).
9. [] Timing: from Australia, an order placed during the day fills at the **next U.S. open** (for example, 9:30 a.m. ET Monday 28 Sept = 11:30 p.m. Sydney time). No order depends on filling after the **U.S. close on Friday Nov 6** (08:00 Sat 7 Nov Sydney time, ASSUMPTION about the end time). For a trade meant to be one of the three notes, make sure it has **filled** (status "Filled" in Order History) well before Oct 23.
10. [] Order type chosen on purpose (market or limit). Remember the FAQ: "You can only cancel open orders" and "run with" errors. No quick buy-then-sell fixes.

C. Is the note right, before the order is sent?

11. [] **The note was drafted and agreed before the order was placed**, in the team's own words (no AI-written text; see section d). It says the instrument **type**, its **role** in the strategy, and how it serves Laura's goals, funding needs or risk. (*TN guide; SMAApply WInS page: "at the time it is made".*)
12. [] It uses the **same strategy words the IPS will use**. These are the official vocabulary: growth, risk, liquidity, funding reliability, financial flexibility, future cash-flow needs, operating commitment, operating reserve, facility contribution. (*TN p.1 L4-6; IPS p.1 L20-26.*) The team keeps a one-page glossary of its own words and uses it for every note.
13. [] **Nothing in the note contradicts the plan**: no hot-theme, momentum or single-stock story that the IPS will not support. No claim about WInS profit being added to Laura's projections. No reserve size, facility amount or co-sponsor range (not expected yet; TN p.1 L20-22). No "our advisor decided" (advisors may not make decisions).
14. [] No personal details about Laura. Nothing that looks like contacting her. Only case facts and public professional record.
15. [] Length: short and complete. The example is 59 words and 413 characters. **Assume the note cannot be edited after it is saved** until the team has confirmed otherwise in WInS.
16. [] Spell-checked in a separate document first, then pasted. What is in WInS is what will be quoted "exactly".

D. Record it (decision log)

17. [] After the fill: copy the **note text exactly as WInS shows it** (character for character) into the team decision log, plus a **screenshot** of Order History and Transaction History showing date/time, symbol, quantity, price, status "Filled", and the note. Export Transaction History to Excel weekly.
18. [] The log records: who proposed, who checked (two names), which checklist items were checked, the rule source, and whether this trade is a **candidate for the three notes** (does it show strategy being

applied, tested or refined?).

19. [] The log records **who placed the order**: a student, never the advisor ("Students should place trades on WinS, NOT the advisor").
20. [] If anything above is unclear: **do not trade**. Log the question and ask Wharton through the SMaply "Contact Us" route (the team leader is the contact; FAQ: "Please use the Contact Us form.").

(d) AI-use policy: exact wording and what it means for this research

d1. Exact wording (VERIFIED-PRIMARY [2026-27], read 2026-09-27)

Rules & Roles, "Use of Generative AI" (<https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/>):

"Teams may use generative AI tools (such as ChatGPT) for brainstorming and idea generation. However: AI-generated content may be inaccurate, incomplete, or misleading and should be used with caution. / AI-generated work may not be submitted as your own. / Any AI-generated material included in your report must be properly cited, just like any other reference source. / Plagiarism, misuse of AI, or any other form of academic dishonesty will be handled in accordance with the Competition Ethics and Code of Conduct policies."

Rules & Roles, "AI Policy" (same page):

"You may not submit any work generated by an AI program as your own. If you include material generated by an AI program, it should be cited like any other reference material (with due consideration for the quality of the reference, which may be poor). Any plagiarism or other form of cheating will be dealt with severely under relevant Investment Competition policies."

AI Policy page, Investment Competition section (<https://globalyouth.wharton.upenn.edu/ai-policy/>):

"All material referenced or generated by AI should be cited like any other reference material (with due consideration for the quality of the reference, which may be poor). If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages."

and its Ethics text adds: "...enroll in non-Wharton extracurricular courses that claim to "teach" the competition, pass AI generated writing or work off as your own, or plagiarize existing strategies."

Statement of AI Use (same page): "When students express themselves in the Wharton Global Youth Program, they must use their voice and words. Using somebody else's work without crediting the source – including generative AI — is plagiarism. ... AI-generated work should be cited like any other reference material, including how and where students used AI-generated information."

Checklist Questions (same page): "Did I come to my current conclusion before or after using generative AI tools?" / "Did I verify the information I gathered from generative AI tools, whether it is the research it mentions, or the source it cites?" / "Did I double check the numbers and information are most up-to-date?" / "Did I understand the key learnings and how to arrive at the answers without the assistance of generative AI tools?"

Written for Wharton's pre-baccalaureate courses, not the competition, but relevant (INTERPRETATION): "Don't use AI for personal reflection or opinion-based tasks."

Ethics (Rules & Roles): "Plagiarism and any kind of academic cheating are grounds for dismissal from the competition."

d2. What this means for how the team uses this research (INTERPRETATION plus ASSUMPTION)

1. **Everything in research/insight_v1/ is AI brainstorming.** It is allowed as idea generation and must be **recorded**. Keep a running "AI use log": date, tool (Claude Code), what it was asked, what the team kept, what the team checked itself. This becomes the Works Cited / AI-use entry in the Final Report. The IPS cannot carry citations, so the record lives in the Final Report ("Relevant sources and supporting evidence must be included in the Final Report", IPS p.3 L124-125).

2. **No AI text in any deliverable.** Trading notes, reflections, the pitch, the IPS and the Final Report are written by the students in their own words. Do not paste or lightly reword AI sentences. The **trading notes are the highest risk**, because they are typed into WinS now, time-stamped, and later quoted "exactly".
3. **Reflections and the Articulation section must be the team's own experience.** They cannot be invented: "We trust that you will not ... fabricate analyses, invent teamwork and experiential stories". AI must not write them.
4. **Verify before relying.** Every number the team uses from this research should be re-checked against its primary source (URL/file given here) or recomputed by a student. The policy's checklist asks exactly this.
5. **Decide before or after AI?** The team should write down its own view first (for example, the strategy choice), then use AI output to test it. The log should show that order.
6. **Charts and images:** if the Final Report uses a chart that AI helped make, cite it. The Code of Conduct requires teams to "acknowledge ownership of all images and other media ... it (or a team member) does not own or did not solely develop."
7. Wharton warns that "use may also stifle your own independent thinking and creativity". The judging criteria ask for "an authentic team voice". The safest and strongest path is the same: AI for questions and checks, students for words and decisions.

(e) Disqualification and penalty risks across all deliverables

"Stated consequence" quotes the source exactly. Where no consequence is written down, I say so.

#	Risk	Exact rule	Source	Stated consequence
e1	Contacting Laura	"Teams may not contact the Competition client. Teams that violate this rule will be disqualified." / "IMPORTANT: Teams are not permitted to contact Laura Gao."	Rules & Roles; SMAApply Client page	Disqualified
e2	Team size outside 4-6	"Teams that fall below four members or exceed six members at any time during the Competition will be disqualified."	Rules & Roles	Disqualified
e3	Removing a member without approval	"Teams that remove a member without prior approval will be disqualified." Roster locked after submission: "teams may not change their members."	Rules & Roles; SMAApply L22	Disqualified
e4	Paid help or outside competition courses	"Teams suspected ... of using paid advisors, education consultants, extracurricular coursework, or other agents will be disqualified."	Rules & Roles	Disqualified
e5	Changing advisor without approval	"Teams that change Advisors without prior approval will be disqualified."	Rules & Roles	Disqualified
e6	Advisor making decisions or trading	"Advisors may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development." / "Students should place trades on WinS, NOT the advisor."	Rules & Roles; SMAApply FAQs	None stated; a rule breach
e7	Plagiarism or AI text submitted as own work	"Plagiarism and any kind of academic cheating are grounds for dismissal from the competition." / "will be dealt with severely"	Rules & Roles (Ethics; AI Policy)	Dismissal possible
e8	AI use not recorded	"how you use it must be recorded in your Works Cited pages"	AI policy page	Treated as misuse of AI (e7)
e9	Missing a deadline	"no later than 5:00 p.m. ET" / "We will not grant any deadline extensions." / "To remain eligible for the Semifinals ... submit the competition deliverables by the published deadlines."	SMAApply L8-17; Rules & Roles	Loses semifinal eligibility; no credential

#	Risk	Exact rule	Source	Stated consequence
e10	IPS: any format rule broken (1-page title page; 2-page strategy; 50/500-word limits; TNR 12; double spacing; 1-inch margins; PDF of 5 MB or less; no graphics, charts, images, attachments, external links, footnotes or formal citations)	"Submissions that do not meet the requirements below will not be considered for semifinal selection."	IPS guide p.3 L82-125	Not considered for semifinals
e11	IPS title page: team name different from roster; names not "First Name, Last Initial" or different from roster; team name given instead of WInS username	IPS p.3 L88-100; FAQ "Your team username is NOT the same as your team name."	IPS guide; SMAApply FAQs	Same as e10
e12	Trading notes changed, paraphrased, or not from an executed WInS trade	"Include each Trading Note exactly as it appears in WInS." / "The Trading Notes must correspond to trades executed in WInS. Yes, we will verify this."	TN p.2 L44, L53	None stated; an integrity failure that could fall under e7 (INTERPRETATION)
e13	Reflection over 100 words; not exactly three notes	"Select three (3) Trading Notes" / "a reflection of 100 words or fewer"	TN p.2 L42, L46	None stated; a requirement failure
e14	Trading outside the permitted instruments (last year's problem)	"The following are NOT permitted: ... Anything other than the approved investments listed above." / "Teams must meet the required trading activity and portfolio management guidelines throughout the competition."	SMAApply FAQs; Rules & Roles	None stated; risk to "meet all competition requirements" (eligibility)
e15	Day trading or exceeding the position limit	"Day Trading: This is not permitted."	2026-27 WInS guide p.6	Usually blocked by the platform; a rule breach if not
e16	Changing the strategy after Nov 6; portfolio or Final Report that does not follow the IPS	"Your team may not revise its investment strategy after the submission deadline." / "Your portfolio and Final Report should reflect the strategy established in your IPS." / criterion: "maintains consistency with the team's IPS"	IPS p.2 L73-75; SMAApply criteria	Scored down (Investment Strategy criterion); the rule itself forbids revision
e17	Trades after the lock	"your portfolio will be frozen, and your team may no longer trade or change its holdings"	SMAApply Trading Details	Impossible after the freeze; unfilled orders are not trades
e18	WInS P&L put into projections	"should not be added to or subtracted from Laura's portfolio projections"	Case p.4 L132-133	Scored down (a case error)
e19	Missing school documentation	"Final Report & Official School Documentation (Due: December 4)" / "Semifinalists must also submit documentation on official school letterhead"	SMAApply; Rules & Roles	Not eligible (e9); sample "Coming soon"
e20	Team leader under 16 or changed after Sept 28	"at least 16 years old at the start of the Competition. No exceptions will be granted."	Rules & Roles	Change needs written approval
e21	Inappropriate team name	"reserves the right to require a team to change an inappropriate team name or disqualify the team"	Rules & Roles	Disqualification possible
e22	Lying about decisions or inventing teamwork stories	"We trust that you will not lie about your investment decisions and outcomes, fabricate analyses, invent teamwork and experiential stories"	Rules & Roles (Ethics)	Ethics breach (e7)
e23	Sources not cited in the Final Report	"Each team is required to properly cite any sources used" / "Relevant sources and supporting evidence must be included in the Final Report."	Rules (Code of Conduct); IPS p.3 L124-125	"will not be accepted" applies to materials that break laws or IP rules
e24	Final Report format	Unknown until Nov 9	SMAApply L44	Check on Nov 9

Deadline clock for Australia (DERIVED with Python `zoneinfo`; ASSUMPTION: the school is in Sydney, AEDT from Oct 4):

Deadline (5:00 p.m. ET)	UTC	Sydney	Brisbane	Perth
Roster, Fri Oct 9	21:00 Fri	08:00 Sat Oct 10	07:00 Sat	05:00 Sat
Trading Notes, Fri Oct 23	21:00 Fri	08:00 Sat Oct 24	07:00 Sat	05:00 Sat
IPS + portfolio lock, Fri Nov 6	22:00 Fri (U.S. market close 21:00)	09:00 Sat Nov 7 (close 08:00)	08:00 Sat	06:00 Sat
Final Report, Fri Dec 4	22:00 Fri	09:00 Sat Dec 5	08:00 Sat	06:00 Sat
Working rule (ASSUMPTION): submit by Friday evening Australian time, a full night early.				

(f) Open compliance questions only the team can answer

When logged in to WinS on day 1, before the first order:

1. **Session Rules page** (Portfolio Simulation > Portfolio Summary > Session Rules): record, with a screenshot, the starting cash (expect \$300,000), trading begin and end date/time, allowed security types, total trades allowed (expect 200), day trading, margin, short selling, commissions, and **position limits (single position and all positions of one security type)** for equities/ETFs and for bonds. If the single-position limit is below about 45%, the planned two-fund Treasury mix breaks. The team then needs a strategy decision (more Treasury instruments of the same type), not a workaround.
2. **Dashboard**: does it show \$300,000 cash and 200 trades remaining? Were practice trades cleared?
3. **Trade note box**: is a note required? Is there a character limit or counter? Can a saved note be edited, or only added to? If several notes exist on one trade, which one would count as "the" note? (Ask Wharton if unclear.)
4. **Which instrument types WinS actually lists**: the U.S. Treasury bonds in the Bonds drop-down (which maturities? Are there bonds maturing 2033-2042?). Search the trade screen for Treasury ETFs of different durations, defined-maturity Treasury ETFs, zero-coupon/STRIPS ETFs, and broad equity ETFs. Record what exists. Do not assume.
5. **Getting Started / weekly emails** inside WinS and SMaply: any rule not on the public pages (for example, a minimum-activity rule behind "required trading activity and portfolio management guidelines")?
6. **Exact WinS username** as it appears at login (case, hyphens, digits): needed on the IPS title page.
7. **Official team name** and the "First Name, Last Initial" forms exactly as the roster (due Oct 9) will list them.
8. **Who holds the login and who places trades** (a student "trader" role), and the two-person check.
9. **Which Australian time zone** applies (it changes every deadline hour; see the table in e).
10. **"Investments permitted (for BOTH contributions)"** in the SMaply FAQ: does this list also limit what the team may recommend for Laura's long-term portfolio (the \$300,000 and \$150,000)? INTERPRETATION: probably a leftover phrase, but it matters if the plan recommends instruments WinS does not list (for example, STRIPS). Consider asking Wharton through Contact Us.
11. **Ambiguous ETF types** (leveraged/inverse, crypto-linked, ETNs): does the team want any at all? If yes, ask Wharton first. If no, record "not used" in the decision log.
12. **How "day trading" is defined** in Session Rules (the tooltip "?" next to it).
13. Whether the \$25 commission applies to ETFs as "stock transactions" (budget for 200 trades: at most \$5,000, about 1.7% of \$300,000; DERIVED).
14. **AI use log**: who keeps it, and where (it feeds the Works Cited pages).

Sources (all accessed 2026-09-27, 11:50-12:00 UTC unless noted)

- Repo, VERIFIED-REPO-FILE: [competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt](#), [2026_WGY_Investment_Policy-FINAL.txt](#), [2026_WGY_Trading_Notes_Analysis-FINAL.txt](#), [SMaply_Deliverables_Page_2026-09-27.md](#); historical [competition/historical/2025_26/README.md](#).
- SMaply public: <https://wghsinvcomp.smaply.us/res/p/trading/> ; <https://wghsinvcomp.smaply.us/res/p/faqs/> ; <https://wghsinvcomp.smaply.us/res/p/wins/> ; <https://wghsinvcomp.smaply.us/res/p/deliverables/>

; <https://wghsinvcomp.smapply.us/res/p/client/>

- Wharton Global Youth: <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/> ; .../rules-roles/ ; .../faq/ ; <https://globalyouth.wharton.upenn.edu/ai-policy/> ; news (Jan 27, 2026) URL in b2.
- Stock-Trak (WInS vendor): <https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf> ; .../2025/09/2025-WInS-User-Guide.pdf ; .../2024/09/2024-WInS-User-Guide.pdf ; <https://edu.stocktrak.com/wharton/wp-json/wp/v2/media> (media listing used to find the 2026-27 guide) ; <https://edu.stocktrak.com/wharton/portfolio-faq/> ; <https://www.stocktrak.com/new-feature-trade-notes/> (2017).
- Third-party (SNIPPET-UNVERIFIED): <https://www.lumiere-education.com/post/wharton-global-youth-program-s-investment-competition-a-comprehensive-guide> (published 2023-08-05, modified 2025-09-23); <https://www.aralia.com/helpful-information/guide-to-the-wharton-global-high-school-investment-competition/> ; <https://internshala.com/competitions/wharton-global-high-school-investment-competition-2026-27/> ; WebSearch summaries.

What this teaches

Compliance is a habit you practise, not a document you read once. Three lessons carry over to real investing. First, **the rulebook beats the platform**: a system that lets you click "buy" does not make a trade allowed. Real fund managers work under an investment mandate (a written list of what they may hold), and "the broker let me" is no defence. Second, **the record is made at the moment of the decision**. A time-stamped note cannot be improved with hindsight. That is why professionals write the reason before they trade, and why the notes will show honest reasoning only if they are honest from day 1. Third, **check which season a rule comes from**. Half the "rules" circulating online were true in 2023 and false now. Always ask where a rule came from, and whether it applies this year.

S1: Treasury and cash sleeve (WInS hedge, cash proxy, Laura's real ladder)

Agent S1, insight_v1 run, 2026-09-27. **PROVISIONAL week-1 research.** The team has not approved the strategy, and Phases B-E may change it. These are research notes and numbers. They are not text to submit. Every WInS ticker below is **PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading.** Script: `research/insight_v1/wins_now/S1_hedge_weights.py`. Run it from the repo root with `.venv/bin/python research/insight_v1/wins_now/S1_hedge_weights.py`. It reproduces the \$292,264 liability value, the 9.90y duration and the \$289/bp DV01. Data snapshots used by the script, all VERIFIED-PRIMARY and downloaded 2026-09-27:

- `S1_fund_holdings_snapshot.csv`: every bond held by 15 funds.
- `S1_mspd_table5_2026-08-31_fixed_2032plus.csv`: Treasury's table of strippable securities.

Summary (short version)

1. **The better WInS hedge is IEF + TLH, not IEF + TLT.** Weights from today's issuer durations: IEF 35.7% / TLH 64.3%. IEF holds notes maturing May 2033 to Aug 2036. TLH holds bonds maturing Feb 2037 to Aug 2046. Together their bonds sit closer to Laura's ten payment dates (2033-2042) than IEF + TLT's do; 74.7% of TLH's bond weight matures after the last payment, so this is a closer fit, not a year-by-year match [corrected by S3 after the S4 red team, 2026-09-27]. IEF + TLT is a "barbell": 7-10y notes plus 20-30y bonds, with nothing in between. Under a 50bp twist of the curve, IEF+TLT misses the liability by up to **\$3,619**. IEF+TLH misses by up to **\$1,352**. Both mixes stay within about \$1.3k of the liability under ± 100 bp parallel moves (VERIFIED-PRIMARY inputs, model output).
2. **The IEF/TLT weights in the brief are out of date.** Durations on the 2026-09-24 issuer pages are IEF 6.86y and TLT 14.88y. The 2026-06-30 fact sheets said 6.95y and 15.31y. The duration match therefore moves from 64.8/35.2 to **62.1/37.9**. Recompute the weights every time you trade.
3. **A third fund does not pay for itself.** The best 3-fund mix (IBTP 34.8 / TLH 62.0 / TLT 3.1) has a worst twist error of \$1,620, which is worse than the best 2-fund mixes. No long-only 3-fund mix removes the twist error. The liability is a "bullet": all its cash flows fall between 6 and 16 years. Constant-maturity funds cannot reproduce that exactly. Only a real ladder can.
4. **Cash/floor proxy:** BIL. It was on the 2025-26 list, costs 0.1353% and holds \$47.9bn. Alternates are SGOV (0.09%, \$111.8bn, not on the 2025-26 list) and SHV (on the list). If the team wants a "2031 floor" illustration trade, use VGSH (0.03%, on the 2025-26 list) or SHY.
5. **Laura's real ladder:** use ten Treasury STRIPS, \$50,000 face each, maturing Nov 15 of the year before each Jan 1 payment. Principal STRIPS exist for 7 of those dates. For Nov 15 2036, 2037 and 2038, use interest ("coupon") STRIPS. On the 2026-09-25 curve the ladder costs **\$294,387 at 2027-01-01**. That is \$2,124 more than the \$292,264 exact-date figure, because each \$50k arrives about 47 days early. **Headroom under the \$300k deposit is about \$5.6k (~19bp), not \$7.7k (27bp).** Most of the \$2.1k comes back if the early proceeds sit in T-bills for six weeks.
6. The iShares **iBonds Dec 2037-2043 Treasury funds do not exist**. iShares' full product list jumps from IBTR (Dec 2036) to IBGA (Dec 2044) (VERIFIED-PRIMARY). So iBonds can fund only the 2033-2037 payments.

Terms used (plain English)

- **Duration:** roughly the % a bond's price falls if interest rates rise by 1 percentage point. Duration 9.90 means about -9.9% for +1%. **Effective duration** is the issuer's version, which allows for how cash flows change when rates move.
- **DV01:** dollars gained or lost per 0.01% (1 basis point, "bp") move in rates. The liability's DV01 is \$289/bp.
- **Key-rate duration (KRD):** duration split by maturity. It shows how much value changes if only the 2y, or only the 10y, or only the 30y yield moves. Two portfolios can have the same total duration but very different KRDs.

- **Parallel shift:** every yield moves by the same amount.
- **Twist:** short and long yields move in opposite directions. In a **steepener**, long yields rise relative to short ones. A **flattener** is the reverse.
- **Tracking error (\$)** here means: the change in the hedge's value minus the change in the liability's value, for a hedge sized at \$292,264. A positive number means the hedge did better than the liability.
- **STRIPS:** Treasury bonds split into separate zero-coupon pieces. Each piece pays one amount on one date. Principal STRIPS come from the final repayment. Interest STRIPS come from a coupon payment.
- **Defined-maturity fund** (iBonds, BulletShares): holds bonds maturing in one year. It pays out and closes around Dec 15 of that year.
- **SEC yield:** a standard 30-day yield measure after fees. **YTM** (yield to maturity) is the average return if the holdings are held to maturity, before fees.

1. Candidate funds (issuer primary pages, accessed 2026-09-27)

Liquidity is shown as the 30-day average share volume times price, about \$/day. "2025-26 list" is historical only ([competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt](#)). Line numbers are the ticker lines in that file.

1a. Intermediate and long Treasury funds (constant maturity: the fund keeps rolling, so it never matures)

Fund	What it holds (index)	Eff. duration (as of)	Expense	YTM / 30-day SEC	Net assets; liquidity	2025-26 list	Status
IEF iShares 7-10Y	16 Treasury notes, maturities May-2033 to Aug-2036 [corrected by S3 after the S4 red team, 2026-09-27] (ICE US Treasury 7-10Y)	6.86y (9/24)	0.15%	5.17% / 4.84%	\$41.6bn; ~8.8m sh/day (~\$0.8bn); spread 0.01%	Yes (line 1053, #88)	VERIFIED-PRIMARY
TLT iShares 20+Y	47 bonds, maturities May-2044 to Aug-2056 [corrected by S3 after the S4 red team, 2026-09-27] (ICE US Treasury 20+Y)	14.88y (9/24)	0.15%	5.54% / 5.41%	\$45.8bn; ~35.9m sh/day (~\$2.8bn); 0.01%	Yes (line 1041, #87)	VERIFIED-PRIMARY
TLH iShares 10-20Y	61 bonds, maturities Feb-2037 to Aug-2046 [corrected by S3 after the S4 red team, 2026-09-27] (ICE US Treasury 10-20Y)	11.59y (9/24)	0.15%	5.49% / 5.29%	\$10.5bn; ~2.0m sh/day (~\$0.18bn); 0.01%	No	VERIFIED-PRIMARY
GOVT iShares US Treasury	218 notes/bonds, Aug-2027 to May-2056 (ICE US Treasury Core)	5.45y (9/24)	0.05%	5.09% / 4.83%	\$41.6bn; ~10.9m sh/day (~\$0.24bn); 0.05%	Yes (line 1089, #91)	VERIFIED-PRIMARY
IEI iShares 3-7Y	83 notes (ICE US Treasury 3-7Y)	4.20y (9/24)	0.15%	5.06% / 4.65%	\$17.2bn; ~2.0m sh/day	No	VERIFIED-PRIMARY
VGIT Vanguard Interm. Treasury	102 bonds (Bloomberg US Treasury 3-10Y)	4.9y (8/31)	0.03% (as of 2025-12-19)	YTM 4.5% (8/31) / SEC 4.78% (9/24)	\$48.3bn total fund (8/31)	No	VERIFIED-PRIMARY
VGLT Vanguard Long Treasury	101 bonds (Bloomberg US Long Treasury)	13.5y (8/31)	0.03%	5.2% (8/31) / 5.41% (9/24)	\$15.1bn total (8/31)	No	VERIFIED-PRIMARY

Fund	What it holds (index)	Eff. duration (as of)	Expense	YTM / 30-day SEC	Net assets; liquidity	2025-26 list	Status
SPTI SPDR Portfolio Interm. Treasury	110 holdings (Bloomberg 3-10Y US Treasury)	4.78y option-adjusted (9/24)	0.03%	5.08% / 4.83%	\$10.3bn; 138k sh prior day; spread 0.04%	No	VERIFIED-PRIMARY
SPTL SPDR Portfolio Long Treasury	110 holdings (Bloomberg Long US Treasury)	13.65y (9/24)	0.03%	5.52% / 5.48%	\$10.9bn; 1.09m sh prior day; 0.04%	No	VERIFIED-PRIMARY
SCHR Schwab Interm. Treasury	3-10y Treasuries	not found	0.03%	not found	\$12.3-13.5bn (conflicting snippets)	No	SNIPPET-UNVERIFIED (schwabassetmanagement.com blocked by egress proxy)
SCHQ Schwab Long Treasury	Treasuries >10y	"often exceeding 15 years" (snippet, vague)	0.03%	not found	not found	No	SNIPPET-UNVERIFIED (blocked)

1b. Zero-coupon / STRIPS funds (all far longer than the liability)

Fund	Holds	Duration	Expense	YTM / SEC	Assets; liquidity	2025-26 list	Status
EDV Vanguard Extended Duration	80 Treasury STRIPS, 20-30y (Bloomberg US Treasury STRIPS 20-30Y Equal Par)	24.0y (8/31)	0.05%	5.4% (8/31) / 5.64% (9/24)	\$4.2bn total (8/31)	No	VERIFIED-PRIMARY
GOVZ iShares 25+Y STRIPS	20 principal STRIPS, Aug-2051 to May-2056	26.45y (9/24)	0.15% gross, 0.10% net (fee waiver)	5.54% / 5.64%	\$311m; ~585k sh/day (~\$19m); spread 0.03%; premium 0.36%	No	VERIFIED-PRIMARY
ZROZ PIMCO 25+Y Zero Coupon	STRIPS with 25+ years left	27.92y (snippet)	0.15%	not found	not found	No	SNIPPET-UNVERIFIED (pimco.com 403/404)

1c. Defined-maturity Treasury funds (iShares iBonds; fee 0.07%; data as of 9/24-9/25)

Fund	Holdings (maturities)	Eff. duration	YTM / SEC	Assets; ~volume	Funds the payment on	2025-26 list
IBTM Dec 2032	15 notes, Apr-Sep 2032	5.05y	5.09% / 4.79%	\$555m; 216k sh	Jan 1 2033	No
IBTO Dec 2033	11 notes, Apr-Nov 2033	5.71y	5.12% / 4.83%	\$452m; 226k sh	Jan 1 2034	No
IBTP Dec 2034	4 notes, Aug-Nov 2034	6.44y	5.15% / 4.90%	\$317m; 111k sh	Jan 1 2035	No
IBTQ Dec 2035	4 notes, Aug-Nov 2035	7.08y	5.18% / 4.93%	\$255m; 100k sh	Jan 1 2036	No
IBTR Dec 2036	4 notes/bonds, Feb-Aug 2036	7.59y	5.19% / 4.93%	\$34m (launched 2026-03-25); 69k sh	Jan 1 2037	No
(Dec 2037-2043)	none exist	-	-	-	2038-2042 uncovered	-
IBGA Dec 2044	8 bonds, Aug-Nov 2044	11.74y	5.54% / 5.43%	\$74m	too late	No

All VERIFIED-PRIMARY:

- iShares product pages and holdings CSVs.
- The iShares screener JSON, which lists iBonds Treasury funds IBTG-IBTR (2026-2036) and IBGA/IBGB/IBGC/IBGK/IBGL/IBGM (2044/45/46/54/55/56) and nothing for 2037-2043.

The pages state that "in the final year" holdings "will mature and the proceeds will be held in cash equivalents until the liquidation of the fund", and that funds "terminate on or about October or December

15 of the year in each Fund's name" (IBTR page). The iBonds are thin: IBTR has \$34m in assets.

Invesco BulletShares Treasury (BSGR 2027, BSTS 2028, BSGT 2029, BSTU 2030, BSTV 2031) are reported as launched 2026-06-10 at 0.07%. That is **SNIPPET-UNVERIFIED**: invesco.com returned HTTP 406 to curl and was blocked for WebFetch. They mature 2027-2031, so they cannot fund the 2033+ payments. At most they could hold the 2031 floor.

1d. T-bill and floating-rate funds (cash)

Fund	Holds	Duration	Expense	YTM / SEC	Assets; liquidity	2025-26 list	Status
BIL SPDR 1-3 Month T-Bill	33 T-bills (Bloomberg 1-3M T-Bill)	0.10y	0.1353%	3.94% / 3.59% (9/24)	\$47.9bn; spread 0.01%	Yes (line 1101, #92)	VERIFIED-PRIMARY
SGOV iShares 0-3 Month	T-bills maturing Oct-Nov 2026 etc. (ICE 0-3M US Treasury)	0.11y	0.09%	3.97% / 3.67%	\$111.8bn; ~22.5m sh/day (~\$2.3bn); 0.01%	No	VERIFIED-PRIMARY
SHV iShares 0-1 Year	T-bills and notes to May-2027	0.28y	0.15%	4.14% / 3.71%	\$21.4bn; ~2.9m sh/day	Yes (line 1077, #90)	VERIFIED-PRIMARY
TFLO iShares Treasury Floating Rate	8 Treasury floating-rate notes, Oct-2026 to Jul-2028	0.01y	0.15%	4.28% / 3.77%	\$6.9bn; ~1.9m sh/day	No	VERIFIED-PRIMARY
USFR WisdomTree Floating Rate Treasury	Treasury floating-rate notes (Bloomberg US Treasury FRN)	~0	0.15%	not found	\$18.76bn (snippet)	Yes (line 1124, #94)	SNIPPET-UNVERIFIED (wisdomtree.com blocked)
SHY iShares 1-3Y (floor proxy)	91 notes (ICE US Treasury 1-3Y)	1.79y	0.15%	4.86% / 4.43%	\$26.2bn; ~4.9m sh/day	Yes (line 1029, #86)	VERIFIED-PRIMARY
VGSH Vanguard Short Treasury (floor proxy)	92 bonds (Bloomberg US Treasury 1-3Y)	1.9y (8/31)	0.03%	4.3% (8/31) / 4.55% (9/24)	\$39.3bn total (8/31)	Yes (line 1113, #93)	VERIFIED-PRIMARY

2. Duration-matched mixes and how well they track the liability

Method (script):

- The same official 2026-09-25 par curve and bootstrap as `research/verified_2026-09-27/official_curve_pv.py`.
- The liability is valued at 2027-01-01 (\$292,264; model duration 9.90; DV01 \$289).
- Each fund is valued from its **actual holdings' cash flows** on that curve, so its key-rate profile is real, not assumed.
- Weights solve (weight × issuer duration) = 9.90 using the issuer's current effective duration, as the task asks. Scenario definitions (ASSUMPTION, chosen to be simple and symmetric):
- Parallel ±100bp.
- 2s30s twist: 2y -25bp and 30y +25bp for a steepener, straight line in between (pivot about 16y).
- 2s10s twist: 2y -25bp and 10y +25bp, flat beyond 10y. Other ASSUMPTIONS:
- Shocks are instantaneous, with no fees or bid/ask.
- SPTI/SPTL use VGIT/VGLT holdings as proxies (same Bloomberg index families).

What the liability itself does:

Scenario	+100bp	-100bp	Steepener 2s30s	Flattener 2s30s	Steepener 2s10s	Flattener 2s10s
Liability change	-\$27,373	+\$30,600	+\$1,693	-\$1,666	-\$7,522	+\$7,630

Key-rate durations (model; they sum to about the total duration). The liability sits on the 7y, 10y and 20y points and has **zero** 30y exposure:

	7y	10y	20y	30y	model / issuer dur.
Liability	1.58	6.74	2.53	0.00	9.90

	7y	10y	20y	30y	model / issuer dur.
IEF	3.42	3.58	0	0	6.90 / 6.86
TLH	-0.25	3.59	8.75	0	11.87 / 11.59
TLT	-0.18	-0.93	5.94	10.55	15.22 / 14.88
IBTR	1.01	6.73	0	0	7.64 / 7.59
VGLT (=SPTL proxy)	-0.21	1.16	7.26	5.65	13.67 / 13.5

2-fund mixes. Tracking error is in \$ for a \$292,264 hedge; a positive number means the hedge beat the liability.

Mix (weights)	Fee	+100bp	-100bp	Worst 50bp twist	Twist detail (2s30s steep/flat; 2s10s steep/flat)	Both on 2025-26 list?
IEF 35.7 / TLH 64.3	0.150%	-244	+991	1,352	-1,352 / +1,333; +612 / -555	IEF yes, TLH no
IBTR 42.2 / TLH 57.8 (best found)	0.116%	-244	+919	1,176	-1,176 / +1,160; +197 / -149	no
IEF 62.1 / TLT 37.9 (today's durations)	0.150%	+165	+1,274	3,619	-3,584 / +3,619; +1,266 / -1,161	both yes
IEF 64.8 / TLT 35.2 (brief, June durations)	0.150%	+726	+499	3,291	-3,258 / +3,291; +1,479 / -1,388	both yes
GOVT 52.8 / TLT 47.2	0.097%	+423	+1,904	5,929	-5,895 / +5,929; +1,386 / -1,183	both yes
VGIT 41.9 / VGLT 58.1	0.030%	+500	+1,033	3,849	-3,835 / +3,849; +1,606 / -1,469	no
SPTI 42.3 / SPTL 57.7	0.030%	+595	+910	3,816	similar to VGIT/VGLT	no

3-fund mixes (weights ≥ 0 summing to 1, duration 9.90, chosen to minimise key-rate gaps):

Mix	Worst twist	$\pm 100\text{bp}$ worst
IBTP 34.8 / TLH 62.0 / TLT 3.1 (best 3-fund)	\$1,620	\$1,050
IEF 40.4 / TLH 53.0 / TLT 6.7	\$1,744	\$1,041
IBTR 51.0 / TLH 38.3 / TLT 10.7	\$1,767	\$984
iBonds IBTM-IBTR 6.5% each + TLH 67.6% (6 funds, reference)	\$1,471	\$1,037

Asking a 3-fund mix to match value, duration **and** the 2s30s twist exactly gives a negative weight for every combination (script output). No long-only fund mix can do this.

What the numbers say:

- **Parallel moves are handled by any duration match.** The $\pm 100\text{bp}$ error is at most about \$1.3k (0.4% of the liability) for IEF/TLH and IEF/TLT.
 - Most of it is convexity: the barbell gains a little more when rates fall.
 - Some of it is the gap between the issuer's durations and the par-curve model's durations (model TLT 15.22 vs issuer 14.88).
 - **Twists separate the choices.** The liability has no 30y exposure. TLT's weight sits on the 20y and 30y points, while the liability's biggest single exposure is the 10y point. So the IEF/TLT barbell is short the 10y and long the 30y by about 4 duration-years each. This matches the brief's "twist gap ~\$3-4k", now recomputed with real holdings: \$3.3-3.6k.
- TLH puts the long leg on the 10y and 20y points, where the liability is. That cuts the worst twist error by about 63% **with the same number of funds and the same fee**. This is complexity-free precision.
- Scale check: a 50bp twist in the 6 weeks of WInS trading would be unusual. Realistic 10-20bp twists mean errors of a few hundred dollars for IEF/TLH, against about \$1k for IEF/TLT. The case for TLH is the logic (its bonds sit closer to the payment dates than TLT's), not the dollars. The earlier wording "the bonds mature when Laura pays" was wrong for most of TLH [corrected by S3 after the S4 red team, 2026-09-27].
- **Weights drift.** As rates rose, TLT's duration fell from 15.31 to 14.88 and IEF's from 6.95 to 6.86. That moved the match by 2.7 points of weight. Rule for the team: recompute weights from the issuer page on every trade date.
 - IEF weight = $(D_{\text{TLH}} - 9.90) / (D_{\text{TLH}} - D_{\text{IEF}})$; TLH weight = $1 - \text{IEF weight}$ [corrected by S3 after the S4 red team, 2026-09-27].

- Note that the liability's own duration also shortens by about 0.25 a quarter as time passes (ASSUMPTION: rough roll-down; not modelled here).

If the WlnS "mirror" is about 65% Treasuries (brief section 7; not approved), IEF/TLH comes to about **23.2% IEF + 41.8% TLH** of the total WlnS portfolio. The IEF/TLT fallback comes to about 40.4% IEF + 24.6% TLT.

3. Laura's real (non-WlnS) ladder: what exactly to buy in January 2027

The source is Treasury's Monthly Statement of the Public Debt, Table V ("holdings of Treasury securities in stripped form"), record date 2026-08-31, from api.fiscaldata.treasury.gov (VERIFIED-PRIMARY, accessed 2026-09-27). It lists every strippable note and bond with its principal-STRIPS CUSIP and the amount already stripped. TreasuryDirect confirms three points (VERIFIED-PRIMARY):

- STRIPS are bought, held and sold "only through a financial institution, a broker, or dealer".
- Par must be in "multiples of \$100".
- Each stripped piece "is a zero-coupon security that matures separately" with its own CUSIP.

Costs are for \$50,000 face, valued at 2027-01-01 on the 2026-09-25 curve. ASSUMPTION: STRIPS are priced on the par-derived zero curve. Real STRIPS quotes can differ by a few bp, and a 10bp difference on a 10-year zero is about 1% of that rung's cost.

Payment	Recommended rung (maturity)	Instrument (principal-STRIPS CUSIP; underlying; \$ already stripped 8/31)	Cost at 2027-01-01	Alternates (cost)
Jan 1 2033	Nov 15 2032	Principal STRIP 912821KC8 of the 4.125% note 91282CFV8 (\$26m)	\$37,253	Dec 31 2032 principal 912821TN5 (\$1.6m stripped, thin) \$37,007; Aug 15 2032 principal 912821JN6 (\$169m) \$37,747; IBTM ~\$37,218
Jan 1 2034	Nov 15 2033	Principal STRIP 912821NP6 of the 4.5% note 91282CJJ1 (\$17m)	\$35,336	Aug 15 2033 912821LW3 (\$27m) \$35,812; May 15 2033 912821LG8 (\$225m) \$36,293; IBTO ~\$35,346
Jan 1 2035	Nov 15 2034	Principal STRIP 912821QX6 of the 4.25% note 91282CLW9 (\$209m)	\$33,495	Aug 15 2034 912821QH1 \$33,952; IBTP ~\$33,550
Jan 1 2036	Nov 15 2035	Principal STRIP 912821TF2 of the 4% note 91282CPJ4 (\$246m)	\$31,720	Aug 15 2035 912821SR7 (\$1,183m, deepest) \$32,162; IBTQ ~\$31,826
Jan 1 2037	Nov 15 2036	Interest STRIP due 11/15/2036 (no Nov-2036 principal on 8/31; see note)	\$30,007	May 15 2036 principal 912821UK9 (\$40m) \$30,861; Aug 15 2036 912821UZ6 (\$1m) \$30,432; IBTR ~\$30,227
Jan 1 2038	Nov 15 2037	Interest STRIP due 11/15/2037 (no principal STRIPS Jun-Dec 2037)	\$28,362	May 15 2037 principal 912803DA8 (5% bond, \$504m) \$29,183
Jan 1 2039	Nov 15 2038	Interest STRIP due 11/15/2038 (no principal STRIPS Jun-Dec 2038)	\$26,779	May 15 2038 principal 912803DD2 (4.5% bond, \$559m) \$27,569
Jan 1 2040	Nov 15 2039	Principal STRIP 912803DJ9 of the 4.375% bond 912810QD3 (\$1,893m)	\$25,257	Aug 15 2039 912803DH3 \$25,635
Jan 1 2041	Nov 15 2040	Principal STRIP 912803DP5 of the 4.25% bond 912810QL5 (\$969m)	\$23,791	912803FU2 (1.375% bond, same date, \$118m) \$23,791
Jan 1 2042	Nov 15 2041	Principal STRIP 912803DU4 of the 3.125% bond 912810QT8 (\$520m)	\$22,387	912803GD9 (2% bond, same date) \$22,387
Total			\$294,387	exact-date value \$292,264

Notes on the ladder:

- **Why Nov 15:** Treasury notes and bonds mature on the 15th (Feb/May/Aug/Nov) or at month-end. Nov 15 is the last common maturity before each Jan 1.
- **Interest STRIPS for Nov 15 2036-2038 exist.** \$272.1bn of fixed-rate notes and bonds that pay coupons every May 15 and Nov 15 until 2042-2056 is already held in stripped form (Table V, 38 securities; VERIFIED-PRIMARY). Stripping a bond creates an interest piece for every remaining coupon date. Table V does not list interest-STRIPS CUSIPs or amounts by date, so the exact CUSIPs and how

deep the market is are **not verified**. In market practice, interest pieces with the same payment date share one CUSIP ("fungible"); that is SNIPPET-UNVERIFIED general knowledge. A broker quote would confirm.

- **Nov 15 2036 principal:** the 10-year note sold at the November 2026 refunding would mature Nov 15 2036. The pattern in Table V (10-year notes dated 2035-11-15, 2036-02-15, 2036-05-15, 2036-08-15) supports this, but it is an ASSUMPTION until the Treasury auction announcement. If it happens, its principal STRIP (same \$30,007 cost) can replace the interest STRIP for Jan 2037.
- **The ladder costs \$2,124 more than the exact-date \$292,264**, because each rung pays about 47 days early (Nov 15 vs Jan 1). The rest of the gap: the early cash earns T-bill interest for about 47 days, roughly \$250 per rung at ~4% (ASSUMPTION about future bill rates). If the team uses May-dated principal STRIPS for 2038 and 2039 instead of interest STRIPS, add about \$1.6k.
- **Updated headroom:** $\$300,000 - \$294,387 = \$5,613$, which is about 19bp of rates (DV01 \$289). The verified model's 27bp assumes exact-date zeros.
- **Transaction cost:** broker mark-ups on small STRIPS lots are unknown (UNVERIFIED). Ten \$50k-face lots could plausibly cost 0.1-0.5% each (ASSUMPTION, not sourced). Get a quote before relying on the headroom.
- **Alternatives:**
 - iBonds IBTM-IBTR for 2033-2037. Costs are similar (\$37.2k-\$30.2k per rung, using issuer YTM less the 0.07% fee; ASSUMPTION: YTM is earned).
 - Advantage: a fund is easier to hold.
 - Disadvantages: monthly coupon distributions must be reinvested; the fund pays out around Dec 15, so the cash sits for two weeks; IBTR has only \$34m in assets.
 - Nothing exists for 2038-2042.
 - Coupon notes and bonds maturing each Nov 15, sized so that principal plus coupons cover each payment.
 - This is possible, but the coupons received in 2027-2032 arrive before any payment and must be reinvested at unknown rates. The brief estimates about \$5k of coupon reinvestment risk.
 - STRIPS avoid this, which is why they are the primary pick (consistent with brief section 8 item 11).

4. Recommendations

Every WlnS pick below is **PENDING APPROVAL CHECK: confirm on this year's WlnS approved list/rules before trading**.

(a) WlnS hedge sleeve (the "promise" sleeve; mirrors the ten fixed payments)

- **Primary: IEF 35.7% + TLH 64.3% of the hedge sleeve.** Recompute from issuer durations on the trade date.
 - Key facts: IEF 0.15%, 6.86y, 16 notes due 2033-2036, \$41.6bn; TLH 0.15%, 11.59y, 61 bonds due 2037-2046, \$10.5bn. Both 9/24-9/25, ishares.com, accessed 2026-09-27.
 - Why it fits Laura: the funds own bonds that mature in the years she pays (2033-2042). This is the closest a 2-fund ETF mix gets to her actual ladder, and it has the smallest twist error of the liquid choices (\$1,352 vs \$3,619).
 - Tie to the case: "a high degree of certainty" for fixed \$50k payments.
 - Trading Notes angle: Wharton's own example note buys an intermediate-term Treasury ETF "to reduce portfolio volatility and begin preparing for Laura's future operating commitment" (Trading Notes guide p.2).
 - 2025-26 list: IEF yes; TLH **no**.
- **Alternate 1: IEF 62.1% + TLT 37.9%.** Both were on the 2025-26 list, so this is the most likely to be approved.
 - Worst twist error \$3.6k; TLT is very liquid (\$45.8bn).
 - Use this if TLH is not approved.
- **Alternate 2: SPTI 42.3% + SPTL 57.7%** (or VGIT 41.9% + VGLT 58.1%). Cheapest at 0.03%, but the twist error is about \$3.8k and neither pair was on the 2025-26 list.
- Not recommended:

- STRIPS funds (EDV, GOVZ, ZROZ): durations of 24-28y put 12-29% of the sleeve (in the duration-matched pairs) on the 30y point, where the liability has nothing.
- GOVT/TLT: worst twist \$5.9k.
- A third fund: it adds no accuracy.

(b) WlnS cash / floor proxy

- **Primary: BIL.** 0.1353%, 0.10y duration, 33 T-bills, \$47.9bn, spread 0.01% (ssga.com, 9/24).
 - Pick it over SGOV because it was on the 2025-26 list (line 1101). Over six weeks, the 0.045% fee gap costs about \$5 per \$100k.
 - Last year's lesson was that the team traded outside the rules, so the higher chance of approval wins.
- **Alternates:** SGOV (0.09%, \$111.8bn, 0.11y; not on the 2025-26 list); SHV (0.15%, 0.28y; on the 2025-26 list).
 - TFLO and USFR are floating-rate options (USFR was on the 2025-26 list; its facts are SNIPPET-UNVERIFIED).
- **Optional "2031 floor" illustration** (the floor is bought in a 2-year Treasury): VGSH (0.03%, 1.9y, on the 2025-26 list) or SHY (0.15%, 1.79y, on the list).
 - Use only if the team wants a discipline/floor trade for the Trading Notes (blueprint trade 3).
 - Keep it small: the real floor decision is in 2031.

(c) Laura's real ladder (January 2027; not a WlnS trade, but still subject to the team's approval of the strategy)

- **Primary:** ten Treasury STRIPS, \$50,000 face each, maturing Nov 15 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040 and 2041.
 - Use principal STRIPS where they exist (2032-2035 and 2039-2041); use interest STRIPS for 2036-2038, or the new Nov-15 2036 note's principal once it is issued.
 - Cost is about **\$294.4k** at the 9/25 curve.
 - Buy the longest rungs first if rates fall before January (brief section 7).
 - **Alternate 1:** iBonds IBTM-IBTR for the 2033-2037 payments plus STRIPS for 2038-2042.
 - **Alternate 2:** May- or Aug-dated principal STRIPS where Nov-dated ones are missing or thin: 2037-05-15 (\$504m stripped) and 2038-05-15 (\$559m). This adds about \$1.6k.

5. Open issues for Phases B-E

- **This year's WlnS list.** Is TLH on it? Can WlnS trade iBonds, STRIPS funds or individual Treasuries at all? (UNKNOWN.)
- **TLH not approved.** IEF/TLT is the fallback. Tell the team the twist gap is about \$3.6k per 50bp and explain why.
- **Interest-STRIPS CUSIPs and depth** for 11/15/2036-2038, and broker mark-ups on \$50k lots: get a real quote. (UNVERIFIED.)
- **Nov 2026 refunding:** confirm the new 10-year note matures 2036-11-15. (ASSUMPTION.)
- **Snippet-only facts:** Schwab, WisdomTree, PIMCO and Invesco pages were blocked (HTTP 403/406, egress proxy). SCHR, SCHQ, USFR, ZROZ and BSxx remain SNIPPET-UNVERIFIED.
- **Vanguard data age:** Vanguard durations are from 2026-08-31, a month older than the iShares/SSGA figures. Rates rose in September, so they are slightly overstated.
- **Liability duration drift:** the liability's duration shortens over time, and the 2027-01-01 anchor gets closer. Recompute the 9.90 target before each rebalance.

Sources (all accessed 2026-09-27)

- iShares product pages (key facts, portfolio characteristics) and "latest-holdings.csv" for each fund:
 - IEF <https://www.ishares.com/us/products/239456/ishares-710-year-treasury-bond-etf>
 - TLT <https://www.ishares.com/us/products/239454/ishares-20-year-treasury-bond-etf>
 - TLH <https://www.ishares.com/us/products/239453/ishares-1020-year-treasury-bond-etf>
 - GOVT <https://www.ishares.com/us/products/239468/ishares-us-treasury-bond-etf>
 - IEI <https://www.ishares.com/us/products/239455/ishares-37-year-treasury-bond-etf>

- GOVZ <https://www.ishares.com/us/products/315911/ishares-25+-year-treasury-strips-bond-etf>
- IBTM <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf>
- IBTO <https://www.ishares.com/us/products/332291/ishares-ibonds-dec-2033-term-treasury-etf>
- IBTP <https://www.ishares.com/us/products/337745/ishares-ibonds-dec-2034-term-treasury-etf>
- IBTQ <https://www.ishares.com/us/products/342105/ishares-ibonds-dec-2035-term-treasury-etf>
- IBTR <https://www.ishares.com/us/products/350028/ishares-ibonds-dec-2036-term-treasury-etf>
- IBGA <https://www.ishares.com/us/products/337747/ishares-ibonds-dec-2044-term-treasury-etf>
- SGOV <https://www.ishares.com/us/products/314116/ishares-0-3-month-treasury-bond-etf>
- SHV <https://www.ishares.com/us/products/239466/ishares-short-treasury-bond-etf>
- TFLO <https://www.ishares.com/us/products/260652/ishares-treasury-floating-rate-bond-etf>
- SHY <https://www.ishares.com/us/products/239452/ishares-13-year-treasury-bond-etf>
- iShares product screener (full US fund list, used for "no Dec 2037-2043 iBonds Treasury"): <https://www.ishares.com/us/product-screener/product-screener-v3.1.jsn?dcrPath=/templatedata/config/product-screener-v3/data/en/us-ishares/ishares-product-screener-backend-config&siteEntryPassthrough=true>
- Vanguard (the data behind the investor.vanguard.com fund profile pages):
 - Endpoints: <https://investor.vanguard.com/vmf/api/{VGIT,VGLT,EDV,VGSH}/{profile,characteristic,price,expense}> and [.../portfolio-holding/bond.json](https://investor.vanguard.com/vmf/api/{VGIT,VGLT,EDV,VGSH}/portfolio-holding/bond.json)
 - Profile page: <https://investor.vanguard.com/investment-products/etfs/profile/vgit>
- SSGA:
 - SPTI <https://www.ssga.com/us/en/intermediary/etfs/spdr-portfolio-intermediate-term-treasury-etf-spti>
 - SPTL <https://www.ssga.com/us/en/intermediary/etfs/spdr-portfolio-long-term-treasury-etf-sptl>
 - BIL <https://www.ssga.com/us/en/intermediary/etfs/spdr-bloomberg-1-3-month-t-bill-etf-bil>
- U.S. Treasury:
 - Monthly Statement of the Public Debt, Table V (record date 2026-08-31): https://api.fiscaldata.treasury.gov/services/api/fiscal_service/v1/debt/mspd/mspd_table_5?filter=record_date:eq:2026-08-31
 - TreasuryDirect STRIPS page: <https://www.treasurydirect.gov/marketable-securities/strips/>
- Snippets only (WebSearch, 2026-09-27; primary pages blocked):
 - SCHR: <https://www.schwabassetmanagement.com/products/schr>, <https://cbonds.com/etf/1447/>
 - SCHQ: <https://cbonds.com/etf/2767/>
 - ZROZ: <https://www.morningstar.com/funds/arcx/zroz/quote>, <https://www.aaii.com/etf/ticker/ZROZ>
 - USFR: <https://www.wisdomtree.com/us/products/fixed-income/usfr>
 - BulletShares Treasury: <https://etfdb.com/innovative-etfs-content-hub/invesco-grows-bulletshares-suite-treasury-etfs/>, <https://www.aaii.com/etf/ticker/BSTV>
- Repo files:
 - `competition/official_market_data/daily-treasury-rates_2026-09.csv` (row 09/25/2026)
 - `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` (lines 1029-1124)
 - `research/verified_2026-09-27/official_curve_pv.py`
 - `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` (example note, p.2)

What this teaches

Two portfolios with the same duration can behave differently. Duration only promises protection when every interest rate moves by the same amount. Laura's payments fall between 2033 and 2042. IEF + TLH owns bonds much closer to those years than IEF + TLT does (though most of TLH matures after 2042 [corrected by S3 after the S4 red team, 2026-09-27]), so it follows her payments even when short and long rates move apart. IEF + TLT gets the same total duration by pairing short bonds with 30-year bonds she does not need, and the gap shows up when the curve twists. Adding a third fund did not help. The simplest fix was choosing the right second fund. For her real money, the best "hedge" is not a fund at all: ten zero-coupon Treasuries that each pay \$50,000 just before a payment date. Real instruments carry real frictions (they mature on Nov 15, not Jan 1, and some dates exist only as interest STRIPS), and those frictions cost about \$2k of the headroom. Checking what actually exists changes the numbers.

S2: Growth sleeve fund research (WInS week 1, PROVISIONAL)

Agent S2 (growth-sleeve researcher), insight_v1 run, written 2026-09-27. This is AI-generated research for Team Caplet. It is not submission text; the team decides and writes everything. Phases B-E may change it, and the team has not yet approved the strategy.

Script: `research/insight_v1/scripts/S2_growth_sleeve.py`, with data in `research/insight_v1/scripts/data/S2/`. Run it from the repo root with `.venv/bin/python research/insight_v1/scripts/S2_growth_sleeve.py`.

Terms used below:

- **Expense ratio:** the fund's yearly fee as a share of the money invested.
- **Top-10 concentration:** the share of the fund held in its ten biggest stocks.
- **Look-through:** what you end up owning, stock by stock, once all your funds are added together.
- **Active share:** how far a portfolio's stock weights differ from an index. 0% means identical and 100% means nothing in common.

0. Summary

1. **The WInS rules page removes the sector problem.** I read Wharton's public SMapply "Trading Details" page myself on 2026-09-27. It says: "There is no required sector allocation or minimum number of sectors." It also says: "Exchange-Traded Funds: Any ETF available on WInS." Starting cash is \$300,000, and the team gets up to 200 trades (VERIFIED-PRIMARY, <https://wghsinvcamp.smapply.us/res/p/trading/>). The FAQ adds a flat \$25 commission on each stock or ETF trade (VERIFIED-PRIMARY, <https://wghsinvcamp.smapply.us/res/p/faqs/>). This agrees with A1's case register (`phase_A/case_register.md`, R-W65 to R-W70, R-W92). So the "sector minimum = team size" claim is **contradicted**. Section 4 keeps a sector-fund fallback only as a contingency.
2. **Recommended pick for both the WInS growth sleeve and Laura's long-term sleeve: VT (Vanguard Total World Stock ETF).** PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading. VT costs 0.06%, holds 10,048 stocks, and its top-10 concentration is 21.7% (issuer fact sheet 2026-06-30). The S&P 500's top-10 concentration is 38.8% (issuer holdings 2026-09-24). VT was on the 2025-26 list.
 - **Alternate 1:** VTI + VXUS, about 62/38.
 - **Alternate 2:** ITOT + IXUS, the iShares twins at the same weights.
3. **The real case for global over U.S.-only is concentration, not return.** J.P. Morgan's 2026 LTCMA (repo PDF) gives AC World 7.00% compound against U.S. large cap 6.70%. On a ~\$95k equity sleeve over 2028-2033, that gap is worth only about **\$1.9k**. Expected volatility is almost the same (a VT-like mix: 16.42%; U.S. large cap: 16.47%). What changes a lot is how much rides on one theme:
 - A 12-stock "AI-linked mega-cap" basket is **40.7% of the S&P 500** today.
 - That basket is **~22% of VT**, counting the top 10 only.
4. **Going global does not remove AI exposure. It moves some of it to Asia.** VXUS's four largest holdings are TSMC, Samsung, SK hynix and ASML, together 10.8%. All four make AI chips or chip equipment. Emerging-market funds hold **13-16% TSMC**. The team should say this plainly rather than claim that "global = diversified away from AI".
3. **Taiwan: keep market weight and add no tilt.** VT holds Taiwan at 3.5% and TSMC at 1.7%.
 - On a ~\$95k equity sleeve, that is about **\$3.3k of Taiwan stocks**. A 2033 facility contribution has a median of about \$207k, so this is about 1.6% of it. That is far too small to hedge Taiwan-dollar facility costs.
 - Taiwan's market is about 75% technology (EWT fact sheet). A tilt toward Taiwan would therefore be an AI bet, and choosing it because the residency is in Taiwan would be tokenism.
 - Drop EWT (0.59% fee, 76 holdings, 23% 3-year standard deviation).
4. **A human-capital tilt (underweighting publishing and media) does not earn its complexity.**

- Book publishing is almost absent from the index. News Corp, which owns HarperCollins (publisher of *Messy Roots*), is **0.021% of the S&P 500**, about \$20 on the sleeve.
- Alphabet and Meta make up about 80% of "Communication Services". Underweighting that sector would really be an anti-AI bet with the wrong label.
- Timing also rules it out. Laura's income matters to the portfolio mainly through the 2028 deposit, and that arrives before the sleeve holds much stock.

5. The equity-weight inconsistency needs a team decision.

- The plan implies roughly 17-24% total equity after 2028 (50-70% of the sleeve).
- The WInS mirror in the blueprint uses ~35% equity.
- My suggestion: mirror the plan, about **21% VT / 79% Treasuries** in WInS. The Trading Notes then match the IPS numbers. Section 6 sets out the options.

1. Candidate table (issuer primary data, accessed 2026-09-27)

"2025-26 list" is the historical list only (competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt, VERIFIED-REPO-FILE; it holds 100 ETFs). This year's rule is "any ETF available on WInS" (VERIFIED-PRIMARY, SMapply). The approval check for this year is therefore "is it searchable and tradable in WInS?".

Top-10 figures from issuer fact sheets are as of 2026-06-30 unless marked otherwise. The "AI/semis in top 10" column adds up the top-10 names that belong to the AI-linked set: the Magnificent 7 plus AVGO, MU, AMD and ORCL, and outside the U.S. TSMC, Samsung, SK hynix, ASML, Tokyo Electron, MediaTek, Delta and Hon Hai. That set is my own definition (ASSUMPTION), so the figure is a **lower bound**.

Ticker	Role	Index	Expense ratio	Holdings	Top-10 %	Largest holding	AI/semis in top 10	Taiwan % / TSMC %	2025-26 list
VTI	U.S. total	CRSP US Total Market (being renamed Morningstar, see note a)	0.03%	3,531	33.4	NVDA 6.4	32.0	0 / 0	Yes (0.03)
ITOT	U.S. total	S&P Total Market Index	0.03%	2,460 (9/24)	32.11	NVDA 6.63	32.11	0 / 0	No
SCHB	U.S. total	(Dow Jones U.S. Broad Stock Market, snippet)	0.03% SNIPPET-UNVERIFIED	~2,400 SNIPPET-UNVERIFIED	n/a	n/a	n/a	0 / 0	No
VOO	U.S. large	S&P 500	0.03%	506	37.9	NVDA 7.5	36.4	0 / 0	Yes (0.03)
IVV	U.S. large	S&P 500	0.03%	504 (9/24)	36.37	NVDA 7.51	36.37	0 / 0	No
SPYM (formerly SPLG, note b)	U.S. large	S&P 500	0.02% gross	506	36.67 (6/30); 38.82 (9/24)	NVDA 8.19 (9/24)	40.73 (9/24, full holdings, 12 names)	0 / 0	No (neither ticker)
VT	Global all-country	FTSE Global All Cap	0.06%	10,048	21.7	NVDA 4.0	21.7	3.5 / 1.7	Yes (0.06)
ACWI	Global all-country	MSCI ACWI	0.32%	2,196 (9/24)	23.19	NVDA 4.54	23.19	3.30 / 1.83	No
VEA	Developed ex-U.S.	FTSE Developed All Cap ex US (includes Canada and Korea)	0.03%	3,868	15.0	Samsung 3.5	8.9	0 / 0	Yes (0.05 then)
IEFA	Developed ex-U.S.	MSCI EAFE IMI (no Canada, no Korea)	0.07%	2,627 (9/24)	12.28	ASML 3.03	3.87	0 / 0	No
SCHF	Developed ex-U.S.	FTSE Developed ex US (snippet)	0.03% SNIPPET-UNVERIFIED	n/a	13.7 SNIPPET-UNVERIFIED	n/a	n/a	n/a	No

Ticker	Role	Index	Expense ratio	Holdings	Top-10 %	Largest holding	AI/semis in top 10	Taiwan % / TSMC %	2025-26 list
VXUS	Total ex-U.S.	FTSE Global All Cap ex US	0.05%	8,755	15.0	TSMC 4.3	10.8	9.1 / 4.3	Yes (0.07 then)
IXUS	Total ex-U.S.	MSCI ACWI ex USA IMI	0.07%	4,585 (9/24)	14.95	TSMC 4.36	10.72	9.06 / 4.36	No
VWO	Emerging	FTSE Emerging Markets All Cap China A Inclusion (no Korea)	0.06%	6,332	27.7	TSMC 16.3	19.8	34.3 / 16.3	Yes (0.08 then)
IEMG	Emerging	MSCI EM IMI (includes Korea)	0.09%	2,861 (9/24)	35.27	TSMC 13.22	31.5	27.41 / 13.22	No
RSP	U.S. equal weight	S&P 500 Equal Weight	0.20% (2025-26 list; issuer page refused)	~500 (by design)	~2 (by design)	~0.2 each	~2.4 (12 x ~0.2)	0 / 0	Yes (0.2)
EWT (for reference)	Single country	MSCI Taiwan 25/50	0.59%	76 (9/24)	50.51	TSMC 20.71	most of it (IT 75.11%)	~100 / 20.71	Yes (0.58)

Notes and status:

- **Vanguard (VTI, VOO, VT, VEA, VXUS, VWO): VERIFIED-PRIMARY.**

- Source: fact sheets as of 2026-06-30 at <https://fund-docs.vanguard.com/F0970.pdf> (VTI), F0968.pdf (VOO), F3141.pdf (VT), F0936.pdf (VEA), F3369.pdf (VXUS) and F0964.pdf (VWO).
- I also checked the current expense ratios for VTI (0.03), VT (0.06), VXUS (0.05) and VWO (0.06) on the product pages <https://investor.vanguard.com/investment-products/etfs/profile/{ticker}>.
- "Holdings" means the fund's "Number of stocks".
- Vanguard's sector labels follow ICB, a different classification from the S&P/MSCI "GICS". That is why VTI shows "Technology 41.0%".
- (a) The VTI product page is now titled "VTI-Vanguard **Morningstar** Total Stock Market ETF". The fact sheet says the CRSP benchmarks are being rebranded as Morningstar after "Morningstar's recent acquisition of CRSP", with the change "expected to occur in July 2026". The ticker is unchanged. If WinS shows the new name, it is the same fund.

- **iShares (ITOT, IVV, ACWI, IEFA, IXUS, IEMG, EWT): VERIFIED-PRIMARY.**

- Sources: fact sheets as of 2026-06-30 (e.g. <https://www.ishares.com/us/literature/fact-sheet/itot-ishares-core-s-p-total-u-s-stock-market-etf-fund-fact-sheet-en-us.pdf>) and product pages <https://www.ishares.com/us/products/{239724, 239726, 239600, 244049, 244048, 244050, 239686}>.
- The product pages give the expense ratio and "Number of Holdings" as of 2026-09-24, and the "30 Day Median Bid/Ask Spread" of **0.01% for all seven** as of 2026-09-25. That is a liquidity check.
- The CSV holdings download returned an HTML page instead of data, so iShares top-10 figures come from the fact sheets.

- **SPYM: VERIFIED-PRIMARY.**

- Sources: the fact sheet <https://www.ssga.com/us/en/intermediary/library-content/products/factsheets/etfs/us/factsheet-us-en-spy.pdf> (gross expense ratio 0.02%, CUSIP 78464A854) and the daily holdings file (holdings as of 24-Sep-2026) at <https://www.ssga.com/us/en/intermediary/library-content/products/fund-data/etfs/us/holdings-daily-us-en-spy.xlsx>.
- (b) The old SPLG holdings file stops at "As of 29-Oct-2025" (VERIFIED-PRIMARY, same URL pattern with "splg").
- Search snippets say the ticker changed from SPLG to SPYM on 2025-10-31 (SNIPPET-UNVERIFIED, MIAx alert <https://www.miaxglobal.com/alert/2025/10/30/>...). **In WinS, search "SPYM", not "SPLG".**

- **Schwab (SCHB, SCHF): the issuer site refused access.** <https://www.schwabassetmanagement.com/products/schb> returned HTTP 403 "Access Denied". The figures shown are SNIPPET-UNVERIFIED

(WebSearch 2026-09-27, e.g. aaii.com, etfdb.com). Neither fund is recommended until someone verifies them.

- **Invesco (RSP): the issuer site refused access.** invesco.com returned HTTP 406 for the product page and the fact sheet. The 0.20% comes from the 2025-26 list (VERIFIED-REPO-FILE, historical). "\$89B assets" and "neared \$100 billion ... July 2026" are SNIPPET-UNVERIFIED.
- **Sizing check:** WInS limits each trade to twice a security's daily volume (SMAApply, VERIFIED-PRIMARY). ITOT alone traded 1.67M shares (about \$280M) on 2026-09-25 (VERIFIED-PRIMARY). A ~\$60-110k order is tiny next to that, so the limit cannot bind for any core fund here. VT's own daily volume is not verified (ASSUMPTION that it is ample; VT ETF net assets were \$77.6B on 2026-06-30).

2. U.S.-only or global? (JPM 2026 LTCMA, [competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf](#), page 2)

2.1 JPM inputs (VERIFIED-REPO-FILE; data as of 2025-09-30, USD, nominal)

Asset class	Compound	Arithmetic	Volatility	(Arithmetic - cash 3.10) / volatility
U.S. large cap	6.70%	7.94%	16.47%	0.294
AC World equity	7.00%	8.28%	16.78%	0.309
EAFE equity	7.50%	8.90%	17.63%	0.329
Emerging markets equity	7.80%	9.74%	20.93%	0.317
U.S. mid cap / small cap	7.00% / 6.90%	8.55% / 8.89%	18.56% / 21.10%	0.295 / 0.274
U.S. quality factor / min-vol factor	6.60% / 7.00%	7.64% / 7.79%	15.11% / 13.16%	0.300 / 0.356

- **Correlations** (a correlation of 1 means two assets move together perfectly): U.S. large cap with EAFE 0.87, U.S. with EM 0.73, EAFE with EM 0.86, U.S. with AC World 0.97.
- The last column is a simple reward-per-risk ratio (Sharpe-style). It is my calculation from JPM's numbers.

2.2 Mixes, computed by the S2 script

A portfolio's compound return is roughly its arithmetic return minus half its variance (ASSUMPTION: the standard approximation).

Mix	Arithmetic	Volatility	≈ Compound
100% U.S. large cap	7.94%	16.47%	6.58% (JPM states 6.70%)
62/28/10 U.S./EAFE/EM (VT-like; VT is 61.9% U.S.)	8.39%	16.42%	7.04% (matches JPM AC World 7.00%)
70/30 U.S./EAFE	8.23%	16.34%	6.89%

2.3 What this means

1. **Return:** global is expected to beat U.S.-only by about 0.3 points a year. Take \$94,800 of equity (60% of a ~\$158k sleeve; ASSUMPTION from brief section 7) held for 5 years (2028 to 2033):
 - at 6.70% it grows to \$131.1k;
 - at 7.00% it grows to \$133.0k;
 - **the difference is +\$1.9k.**

This fits the earlier finding in brief section 8, item 5: more equity or different equity mainly changes the range, not the median. **Return is not a reason to choose.**

2. **Volatility:** by JPM's numbers, going global barely reduces volatility (16.42% vs 16.47%). Currency risk offsets the diversification. **Do not claim "global lowers risk" in the volatility sense.**
3. **Concentration:** here the difference is large, and the numbers come from issuer holdings.

	S&P 500 (SPYM, 9/24)	VTI/ITOT (6/30)	VT/ACWI (6/30)
Top-10 concentration	38.8%	32-33%	21.7-23.2%
Largest single holding	NVDA 8.19%	~6.5%	NVDA 4.0-4.5%
AI-linked basket (12 names)	40.7%		

One theme sits behind 40% of a U.S. large-cap fund. That is the honest risk argument for Laura.

4. **Valuation context:** the fact sheets show price-to-earnings (P/E) ratios of 29.2x for ITOT, 30.2x for IVV, 18.6x for IEFA and 19.9x for IEMG (VERIFIED-PRIMARY, 2026-06-30). A P/E ratio is the share price divided by a year of earnings; a higher number means investors pay more for each dollar of profit. My reading (ASSUMPTION): JPM expects higher returns outside the U.S. mainly because those markets start from lower prices relative to earnings. That is a market-consistent reason. It is not a forecast that AI will fail.
5. **Consistency flag:** `strategy_mc.py` models the sleeve as U.S. large cap only (brief section 11). If the team picks VT, the projection input should become AC World (7.00% compound, 16.78% volatility). That adds roughly +\$1.9k to the median. The change is small, but the IPS, the model and the WInS holding should all use one assumption.

2.4 Global funds still carry AI exposure, now in Asia

VXUS's top four holdings are TSMC 4.3, Samsung 2.6, SK hynix 2.2 and ASML 1.7. IEMG's top three are TSMC 13.22, Samsung 7.15 and SK hynix 6.70. All are AI-hardware suppliers (VERIFIED-PRIMARY, fact sheets). Adding ex-U.S. stocks lowers single-stock concentration but still leaves the portfolio tied to the same AI build-out cycle.

What a strong sentence must contain:

- the verified top-10 numbers (38.8% vs 21.7%);
- the fact that JPM expects the world to earn slightly more than the U.S.;
- an admission that AI chip exposure remains through TSMC and Korea.

2.5 Taiwan exposure and the residency (no tokenism)

Facts (VERIFIED-PRIMARY, issuer fact sheets as of 2026-06-30):

Fund	Taiwan %	TSMC %	Other
VT	3.5	1.7	
ACWI	3.30	1.83	
VXUS	9.1	4.3	
IXUS	9.06	4.36	
VWO	34.3	16.3	
IEMG	27.41	13.22	
EWT	~100	20.71	information technology 75.11%

Size in Laura's plan (ASSUMPTION inputs):

- A ~\$94.8k equity sleeve in VT holds about \$3.3k of Taiwan stocks and about \$1.6k of TSMC.
- The median 2033 surplus is \$207k (brief section 6). The Taiwan holding is about 1.6% of it.

Why no Taiwan tilt:

- **Currency.** The residency is in Taiwan, but the case never names a currency (A1, R-AN10). A rise in the Taiwan dollar would push up facility costs measured in USD, and would also lift Taiwan stocks measured in USD. At market weight this offset is worth only a few thousand dollars against a facility bill of hundreds of thousands. It is a fact to disclose, not a hedge to build.
- **What a tilt really buys.** Adding EWT would mostly be adding TSMC and the AI supply chain: 75% information technology, a 23.22% 3-year standard deviation and a 0.59% fee. That is more AI concentration, not a hedge.
- **Shared shocks.** A Taiwan-specific shock (geopolitics, an earthquake) could hit the residency project and the holding at the same time. The link could run either way, and it is too small to model usefully.
- **Tokenism.** Choosing Taiwan stocks because the residency is in Taiwan is the kind of decoration the brief forbids.

What the team can honestly say: "the world index already owns Taiwan at its market weight; we chose not to add more".

3. Should the sleeve hold RSP (equal weight) or a factor fund?

- **RSP.** It cuts the AI basket from ~40.7% to ~2.4% (about 0.2% × 12 names). That is a large active bet in its own right. It leans toward smaller and value stocks, costs 0.20% (versus 0.03-0.06%), trades back to

equal weights every quarter, and has no JPM return assumption to anchor a projection. VT already cuts top-10 concentration from 38.8% to 21.7% while staying market-weighted at 0.06%. **Verdict: RSP does not earn a place.** Keep it only as an optional answer if the team explicitly wants to act on an AI-bubble view, and then call it a view.

- **Quality factor (QUAL).** JPM gives the U.S. quality factor 6.60% compound, below U.S. large cap's 6.70%, so it does not earn a place.
- **Minimum volatility (USMV).** JPM gives the min-vol factor a higher reward-per-risk (0.356). The sleeve's job is growth, though, and the brief's "risk only in the surplus" design already controls risk. **Not recommended; listed so the team knows it was considered.**

4. Sector-rule fallback (contingency only; the rule appears NOT to exist)

Status. SMAApply "Trading Details" says: "There is no required sector allocation or minimum number of sectors." (VERIFIED-PRIMARY, 2026-09-27). Use this section only if Wharton later says otherwise, or if WInS were to reject broad ETFs. A broad fund such as VT already holds all 11 GICS sectors, the standard 11-way split of the stock market.

Fallback set. These are the 11 Select Sector SPDRs, weighted like today's S&P 500.

- I mapped each of the 504 SPYM holdings (2026-09-24) to the sector fund that holds it; 0.00% was left unmapped.
- All 11 funds have a 0.08% gross expense ratio (issuer fact sheets as of 2026-06-30, VERIFIED-PRIMARY).
- 10 of the 11 were on the 2025-26 list. XLRE was not; VNQ was.

Fund	Sector	Weight	Fund	Sector	Weight
XLK	Information Technology	39%	XLI	Industrials	8%
XLF	Financials	12%	XLP	Consumer Staples	4%
XLC	Communication Services	10%	XLE	Energy	3%
XLV	Health Care	9%	XLU	Utilities	2%
XLY	Consumer Discretionary	9%	XLB	Materials	2%
			XLRE (or VNQ)	Real Estate	2%

Alternates. Vanguard sector ETFs, 0.09% (VGT, VFH, VDC, VDE, VIS, VOX, VCR, VHT; VNQ 0.13%) from Vanguard fact sheets as of 2026-06-30 (VERIFIED-PRIMARY). VPU and VAW were not checked.

How the story changes (script output):

1. **It is not the market.** The sector funds cap their largest stocks: XLK holds NVDA 15.3%, AAPL 13.8% and MSFT 10.4%, and XLC holds META 23.2%. So the 11-fund mix owns only 29.7% of the AI basket, against 40.7% in the index.
 - NVDA falls from 8.19% to 6.00%.
 - Alphabet (both share classes) falls from 5.46% to 2.05%.
 - The mix has an active share of **13.5%** against the S&P 500.

The team would be making a partly hidden anti-mega-cap bet and would have to disclose it.

2. **It is U.S.-only.** No Select Sector fund holds TSMC or Europe, so the global argument in section 2 is lost unless the team adds VXUS as a 12th fund.
3. **It costs more to run.** WInS charges a \$25 commission per trade (VERIFIED-PRIMARY). Eleven buys cost \$275 against \$25-50 for one or two funds. Every rebalance also uses up to 11 of the 200 trades.
4. **Complexity is visible to judges.** Eleven tickers make the equity sleeve look like sector picking, which works against the plan's message that "risk lives only in the surplus".

5. Optional human-capital hedge (underweight publishing/media): not recommended

The idea. Laura's 2028 deposit comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (case p.2, lines 43-45, VERIFIED-REPO-FILE). If her income depends on the

publishing and media economy, perhaps the portfolio should own less of it.

Facts:

- *Messy Roots* is published by HarperCollins. Source: the harpercollins.ca product page for ISBN 9780063067769, VERIFIED-PRIMARY 2026-09-27. Wikipedia (secondary) says the same: publisher HarperCollins, March 8, 2022.
- HarperCollins appears among News Corp's businesses on <https://newscorp.com/> (VERIFIED-PRIMARY 2026-09-27).

Size (script, SPYM holdings 2026-09-24):

- News Corp (NWSA + NWS) is **0.021%** of the S&P 500. On a \$94.8k equity sleeve that is **about \$20**.
- Communication Services is 10.08% of the index. Alphabet and Meta alone are 8.04% of the index, **80% of the sector**.
- Halving Communication Services would move about \$4.8k. About 80% of that would come out of Alphabet and Meta, not publishing.

Why it fails:

1. **Wrong target.** The listed market barely contains trade publishing. A "media underweight" is really a bet against AI platforms.
2. **Wrong timing.** Her income reaches the portfolio mainly through the 2028 deposit. In 2027 the sleeve is only the ~\$7.7k left after the Treasury purchase (brief section 6). At 60% equity that is ~\$4.6k of stock, and halving Communication Services would move about \$230. By the time the sleeve is large, the deposit has already arrived.
3. **The strategy already hedges the human-capital risk that matters.** Under lock-early, a missing 2028 deposit hits the facility contribution, not the ten payments (brief section 8, item 8). That is the strong human-capital argument, and it needs no tilt.

Verdict: no tilt. A strong sentence would name:

- the 0.021% figure;
- the 80% Alphabet/Meta share;
- the fact that the lock-early structure is her real hedge against career risk.

This also serves the "willingness vs ability" point from the team's client insight.

6. Recommendations

Every ticker below carries this label: **PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading.** This year the check means confirming that the ticker is available and tradable in WInS (SMAApply: "Any ETF available on WInS").

6.1 WInS growth sleeve

Primary: VT. PENDING APPROVAL CHECK.

- Global all-cap, FTSE Global All Cap index, 0.06%, 10,048 stocks, top-10 concentration 21.7%, ETF net assets \$77.6B (fact sheet 2026-06-30). It was on the 2025-26 list.
- It is U.S.-listed, so it fills at real-time U.S. prices. WInS delays non-U.S. equity orders until the end of the day (SMAApply, VERIFIED-PRIMARY).
- **Trace:**
 - The case says she wants "an appropriate balance between pursuing growth and protecting the capital required for her goals" (case p.2). The Treasuries protect the capital; VT is the growth part.
 - One cheap, transparent fund means one sentence can explain the growth sleeve.
 - VT matches the JPM AC World assumption used for projections, so WInS, the IPS and the model tell the same story.

Alternate 1: VTI + VXUS at about 62/38. PENDING APPROVAL CHECK.

- Blended fee ~0.038%. Both were on the 2025-26 list.
- Same market exposure as VT. Two funds also give the team a visible U.S./international rebalance, which could become Trading Note 3 ("discipline").

Alternate 2: ITOT + IXUS at about 62/38. PENDING APPROVAL CHECK.

- iShares, 0.03% and 0.07%, bid/ask spread 0.01% (9/25). Neither was on the 2025-26 list.
- Use it if Vanguard tickers are missing from WInS.

Not recommended:

- **ACWI:** same exposure at 0.32%, about 5x VT's fee.
- **VOO/IVV/SPYM alone:** 38.8% top-10 concentration.
- **RSP, EWT, the sector set:** see sections 3-4.

6.2 Laura's long-term growth sleeve (2028-2033, then post-2033 flexibility)

Primary: VT. Label as above. The reasons are the same, and it is a single holding she can follow herself. A statistics-trained client can check one index against one JPM line.

Alternate 1: VTI + VXUS. It saves about 2 basis points a year, about \$21 a year on \$95k (a basis point is 0.01%). It also allows a U.S./international rebalancing band if the team wants one.

Alternate 2: ITOT + IXUS.

The equity share inside the sleeve (50/60/70%) is the brief's open question. S2 takes no position beyond the brief: that choice changes the range, not the median.

6.3 Resolve the equity-weight inconsistency (team decision; do not hide it)

Figures computed from the brief's inputs (ASSUMPTION):

- At 2028-01-01 the Treasury ladder is worth about \$305k: \$292.3k grown at about the 1-year rate of 4.50%.
- The sleeve is about \$158k: \$7.7k plus the \$150k deposit.
- At 60% sleeve equity, stocks are about \$95k of about \$463k, **≈ 20.5% of the total**. At 50% and 70% sleeve equity the total is **≈ 17%** and **≈ 24%**.

WInS option	Equity (VT)	Treasuries	For	Against
A: literal January 2027 book	~2-3%	~97%	Exactly what her \$300k would hold in 2027	Almost no growth trade to write a note about; looks like "all bonds"
B: mirror the post-2028 plan (suggested)	~21% (~\$63k)	~79% (~\$237k)	Same percentages as the IPS and the model, so the three deliverables agree	Needs one sentence explaining that WInS's \$300k stands in for the post-2028 mix
C: blueprint's 35%	~35% (~\$105k)	~65%	More visible growth activity	Implies an equity share the plan never holds; judges read the notes and the IPS together

My suggestion is B. The team decides.

What a strong Trading Note on the VT buy must contain (elements only; the team writes it):

- the action and share of the portfolio;
- "growth sleeve = money the ten payments don't need";
- why world rather than U.S.-only (top-10 concentration 21.7% vs 38.8%);
- the risk accepted (stock-market falls reduce the facility range, not the payments).

Final Report chart idea (charts are not allowed in the IPS): a bar chart of top-10 concentration for the S&P 500, VTI, VT and VXUS, with the AI-linked share highlighted. The data is the table in section 1.

7. Open issues passed on

1. Check in WInS that VT, VTI, VXUS, ITOT and IXUS are available, and that SPYM appears under its new ticker.
2. The team chooses the WInS equity mirror: option A, B or C in section 6.3.
3. Switch the projection's equity assumption from U.S. large cap to AC World if VT is chosen (strategy_mc.py, brief section 11).
4. Verify SCHB, SCHF and RSP on issuer pages. The sites refused this session (Schwab HTTP 403, Invesco HTTP 406).

5. The iShares holdings CSV endpoint returned HTML, so the AI-basket shares for non-S&P funds are top-10 lower bounds only.

Sources (all accessed 2026-09-27)

- **Wharton SMaply:**
 - <https://wghsinvcomp.smaply.us/res/p/trading/> (sector rule, "any ETF", \$300,000, 200 trades, volume cap)
 - <https://wghsinvcomp.smaply.us/res/p/faqs/> (\$25 commission)
- **Vanguard:**
 - fact sheets <https://fund-docs.vanguard.com/F0970.pdf>, F0968.pdf, F3141.pdf, F0936.pdf, F3369.pdf, F0964.pdf
 - sector fact sheets F0958, F0957, F0955, F0951, F0953, F0959, F0954, F0956, F0986
 - product pages <https://investor.vanguard.com/investment-products/etfs/profile/vti> (and vt, vxus, vwo, voo, vea)
- **iShares:**
 - fact sheets <https://www.ishares.com/us/literature/fact-sheet/{itot-ishares-core-s-p-total-u-s-stock-market-etf, ivv-ishares-core-s-p-500-etf, acwi-ishares-msci-acwi-etf, iefa-ishares-core-msci-eafe-etf, ixus-ishares-core-msci-total-international-stock-etf, iemg-ishares-core-msci-emerging-markets-etf, ewt-ishares-msci-taiwan-etf}-fund-fact-sheet-en-us.pdf>
 - product pages <https://www.ishares.com/us/products/239724/> (ITOT), 239726 (IVV), 239600 (ACWI), 244049 (IEFA), 244048 (IXUS), 244050 (IEMG), 239686 (EWT)
- **State Street:**
 - fact sheets <https://www.ssga.com/us/en/intermediary/library-content/products/factsheets/etfs/us/factsheet-us-en-{spym,spy,xlk,xlf,xlv,xly,xlc,xli,xlp,xle,xlu,xlb,xlre}.pdf>
 - daily holdings <https://www.ssga.com/us/en/intermediary/library-content/products/fund-data/etfs/us/holdings-daily-us-en-{spym,splg,xlk,...}.xlsx> (saved as CSV in `research/insight_v1/scripts/data/S2/`)
- **Refused:** <https://www.schwabassetmanagement.com/products/schb> (HTTP 403); [invesco.com](https://www.invesco.com) RSP pages (HTTP 406).
- **Repo files:**
 - `competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf`, page 2
 - `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt`
 - `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`, lines 43-45 and 68-74
- **Publisher facts:** <https://www.harpercollins.ca/9780063067769/messy-roots-a-graphic-memoir-of-a-wuhanese-american/> ; <https://newscorp.com/> ; https://en.wikipedia.org/wiki/Messy_Roots (secondary)
- **Snippets (UNVERIFIED):** WebSearch results for SCHB, SCHF, RSP and SPLG→SPYM (aaii.com, etfdb.com, miaxglobal.com, etf.com).

What this teaches

Picking the growth fund is mostly about matching the fund to a forecast you actually use. It is less about hunting for extra return. The U.S. vs world choice is worth about \$2k to Laura's median outcome. Its larger effect is on how much of her surplus rides on one theme: 40% of the S&P 500 against about 22% of the world index. The best check on any "clever" tilt (Taiwan, equal weight, sectors, a media underweight) is to size it in dollars and see what it really buys. Here each one turned out to be tiny, mislabeled, or an AI bet under another name.

Final-Report research: the January 2027 STRIPS ladder and the reserve run-down

Agent S3, insight_v1 run, 2026-09-27. **PROVISIONAL.** The team has not approved the strategy, and Phases B-E may change it. This is AI-generated research (tables, numbers, caveats), not text to submit. The team decides and writes every word.

Why this file exists. These two tables were sections E.2 and E.5 of securities_and_allocation_v0.md. The S4 red team (finding 6) said they are Final-Report material (due Dec 4), not week-1 trading material. The Trading Notes guide says teams are "not expected to calculate the final size of the operating reserve" at the Trading Notes stage (competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt line 20, VERIFIED-REPO-FILE). Nothing here belongs in a WInS note.

None of these instruments is traded in WInS. They are Laura's real (non-WInS) portfolio. If the team ever uses one in WInS, it is **PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading** (this season: WInS availability + Session Rules position limit).

Numbers come from research/insight_v1/wins_now/S1_hedge_weights.py (ladder), research/insight_v1/scripts/S3_allocation_numbers.py section 6 (run-down) and research/insight_v1/scripts/S4_red_team_checks.py section 5 (rate-fall costs). Run each from the repo root with .venv/bin/python <path>.

Terms

- **STRIPS:** zero-coupon Treasuries. Each pays one amount on one date and nothing before.
- **Rung:** one STRIPS in the ladder; it pays one of Laura's ten \$50,000 payments.
- **Principal / interest STRIP:** a STRIP cut from a bond's final repayment, or from one of its coupons.
- **Forward value:** what something is worth on a future date if today's curve predicts future rates correctly (ASSUMPTION wherever used).

1. The January 2027 ladder, rung by rung (was E.2)

Source: S1, from Treasury MSPD Table V (record date 2026-08-31, api.fiscaldata.treasury.gov, VERIFIED-PRIMARY per S1, accessed 2026-09-27). Costs are valued at 2027-01-01 on the 2026-09-25 official curve (model output; ASSUMPTION: STRIPS price on the par-derived zero curve; real quotes differ by a few bp). STRIPS are bought "only through a financial institution, a broker, or dealer" in \$100 multiples (treasurydirect.gov, VERIFIED-PRIMARY per S1).

Pays the payment on	Rung matures	Instrument	Cost at 2027-01-01	Alternate
Jan 1 2033	Nov 15 2032	Principal STRIP 912821KC8	\$37,253	IBTM (iBonds Dec 2032) ~\$37,218
Jan 1 2034	Nov 15 2033	Principal STRIP 912821NP6	\$35,336	IBTO ~\$35,346
Jan 1 2035	Nov 15 2034	Principal STRIP 912821QX6	\$33,495	IBTP ~\$33,550
Jan 1 2036	Nov 15 2035	Principal STRIP 912821TF2	\$31,720	IBTQ ~\$31,826
Jan 1 2037	Nov 15 2036	Interest STRIP due 11/15/2036 (CUSIP not in Table V: broker quote)	\$30,007	IBTR ~\$30,227 (\$34m fund); May 15 2036 principal 912821UK9 \$30,861
Jan 1 2038	Nov 15 2037	Interest STRIP due 11/15/2037	\$28,362	May 15 2037 principal 912803DA8 \$29,183
Jan 1 2039	Nov 15 2038	Interest STRIP due 11/15/2038	\$26,779	May 15 2038 principal 912803DD2 \$27,569
Jan 1 2040	Nov 15 2039	Principal STRIP 912803DJ9	\$25,257	Aug 15 2039 912803DH3 \$25,635
Jan 1 2041	Nov 15 2040	Principal STRIP 912803DP5	\$23,791	912803FU2 (same date)
Jan 1 2042	Nov 15 2041	Principal STRIP 912803DU4	\$22,387	912803GD9 (same date)

Pays the payment on	Rung matures	Instrument	Cost at 2027-01-01	Alternate
Total			\$294,387	Exact-date value \$292,264; headroom under \$300k ~\$5.6k (~19bp)

Caveat (S4 finding 9, script section 5; ASSUMPTION: zero-drift lognormal cost, 7.24% a year volatility, the brief's section 14 method):

- The real Nov-15 ladder costs more than \$300,000 on 2027-01-01 in about **3 of 10** rate paths (30.7%), against 24.3% for exact-date zeros.
- Cost if rates fall before the purchase: -25bp \$301,676; -50bp \$309,169; -100bp \$324,793.
- In those paths the rule "buy the longest rungs first and the rest from the 2028 deposit" means certainty for the **first** rungs depends on the 2028 deposit arriving. With a missing or late deposit, that is the brief's unanswered "joint tail" (section 9). Passed to Phases B-E.
- STRIPS dealer mark-ups are unknown (UNVERIFIED) and would add to the cost.

2. The operating reserve after 2033, as payments are made (was E.5)

Values on today's forward curve (ASSUMPTIONS: forwards are realised; rungs treated as exact-date zeros). The Nov-15 rungs pay about 47 days early and each earns about \$258 of T-bill interest while it waits (ASSUMPTION: 4% bill rate).

Jan 1	Payments left (face)	Holdings just before the payment	Market value	Value of STRIPS still held after paying
2033	10 (\$500k)	\$50k T-bills + 9 STRIPS	\$394,930	\$344,930
2034	9 (\$450k)	\$50k T-bills + 8 STRIPS	\$363,676	\$313,676
2035	8 (\$400k)	\$50k T-bills + 7 STRIPS	\$330,957	\$280,957
2036	7 (\$350k)	\$50k T-bills + 6 STRIPS	\$296,706	\$246,706
2037	6 (\$300k)	\$50k T-bills + 5 STRIPS	\$260,825	\$210,825
2038	5 (\$250k)	\$50k T-bills + 4 STRIPS	\$223,086	\$173,086
2039	4 (\$200k)	\$50k T-bills + 3 STRIPS	\$183,342	\$133,342
2040	3 (\$150k)	\$50k T-bills + 2 STRIPS	\$141,396	\$91,396
2041	2 (\$100k)	\$50k T-bills + 1 STRIP	\$97,042	\$47,042
2042	1 (\$50k)	\$50k T-bills	\$50,000	\$0

How the composition changes (answers the case's "how composition changes as payments approach"):

- Treasuries only, getting shorter and more cash-like every year.
- Nothing is sold before maturity, so rate moves change the market value but never the payment.
- The small interest earned while each rung waits goes to project flexibility.
- Brief section 8 item 9: report the reserve at its 2033-01-01 market value (\$394.9k) and reconcile "funded 2027, set aside 2033" in one sentence.

Final Report chart idea: a stacked bar per year 2033-2042 (T-bills waiting vs STRIPS still held), from the table above. The IPS bans charts; the Final Report does not.

Sources

- `research/insight_v1/wins_now/S1_treasury_sleeve.md` and `S1_mspd_table5_2026-08-31_fixed_2032plus.csv` (MSPD Table V, api.fiscaldata.treasury.gov, accessed 2026-09-27 by S1).
- `competition/official_market_data/daily-treasury-rates_2026-09.csv` row 09/25/2026 (VERIFIED-REPO-FILE).
- `research/insight_v1/wins_now/S4_red_team.md` finding 9; `research/insight_v1/scripts/S4_red_team_checks.py`.

What this teaches

A plan that is "certain" at today's prices can still depend on something uncertain before it is bought. The ladder fits inside \$300,000 today, but only just, and in about three rate paths out of ten it will not. Saying

exactly which payments then lean on the 2028 deposit is more honest, and more useful to Laura, than a single "fully funded" label.

Part 3 - The research record, phase by phase · research/insight_v1/wins_now/S4_red_team.md

S4 red team: attacking `securities_and_allocation_v0.md` (S3's WInS week-1 plan)

Agent S4 (adversarial red team), insight_v1 run, written 2026-09-27. **PROVISIONAL.** The team has not approved the strategy, and Phases B-E may change it. This is AI-generated research: findings, evidence and fixes. It is not text to submit, and none of the fixes below is a drafted Trading Note. The team decides and writes every word.

Every ticker named here is **PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading** (this season that means: listed in WInS, and within the Session Rules position limit).

Files attacked: `wins_now/securities_and_allocation_v0.md` (S3), `wins_now/S1_treasury_sleeve.md`, `wins_now/S2_growth_sleeve.md`, `wins_now/S1_hedge_weights.py`, `scripts/S3_allocation_numbers.py`. New script: `research/insight_v1/scripts/S4_red_team_checks.py` (run from the repo root with `.venv/bin/python research/insight_v1/scripts/S4_red_team_checks.py`). It reproduces every new number below.

Summary

The core of S3's plan survives the attack:

- every fund fact on the primary picks matches the issuer page;
- the arithmetic reproduces;
- the 2025-26 list look-ups are right;
- the insight that "65/35" was really hedge/sleeve, not bonds/stocks, is correct.

It is **not ready to trade as written**. Two findings are blocking:

1. **No approval gate.** The plan places orders at the first U.S. open (Mon Sep 28 ET) on a strategy the team has not approved. Trading notes are permanent and Wharton checks them.
2. **A false maturity story.** The Note-1 story "IEF and TLH hold bonds that mature when Laura pays" is wrong for three-quarters of TLH. **74.7% of TLH's bond weight matures in 2042-2046, after the last payment.**

Five findings are important:

- the headline allocation is blocked under the only documented position limit (25%), and S3's preferred fallback is its most complex one;
- VT is bought on Day 1 while VGSH waits for the same decision;
- Note 3 would write an open parameter (the "80%" floor) into a permanent note;
- option (ii) is defensible but oversold;
- the file is far too long for a Day-1 trade ticket.

The minor findings show one thing: debates about precision (9.90 vs 10.16 vs 8.92 years; \$1,352 vs \$1,544) are smaller than the measurement noise in the duration numbers themselves. **"About 10 years" is the honest precision.**

Verdict: keep the IEF/TLH + VT + VGSH design and all its fund facts. Before any order, apply the fixes to findings 1-4 and cut the file to a one-page ticket (finding 6).

Terms used (plain English)

- **Duration:** roughly the percentage a bond's price falls if interest rates rise by 1 percentage point.
- **Spot vs forward duration:**
 - The ten payments are worth \$288,924 **today**. Their duration today is 10.16 years (the spot duration).
 - Valued as of 2027-01-01 (the forward value) they are worth \$292,264, with a duration of 9.90 years.
- **Tracking error:** how many dollars the hedge's change in value misses the payments' change by in a rate scenario.

- **Twist:** short-term and long-term rates move in opposite directions.
- **Position limit:** the most WInS lets you hold in one security, as a share of the portfolio (Session Rules).
- **Gate:** a condition that must be met before an action is allowed.
- **STRIPS:** zero-coupon Treasuries that pay one amount on one date.

1. What I re-checked, and what held

1a. Fund facts on the issuer's primary pages (S4 re-opened them with curl on 2026-09-27, ~12:28-12:35 UTC)

Fund	S3's claim	Issuer page today	Status
IEF	0.15%; 6.86y (9/24); 16 notes; \$41.57bn; 8,830,020 sh/day; SEC yield 4.84%	0.15%; effective duration 6.86 yrs (Sep 24); 16 holdings; net assets \$41,570,344,536 (Sep 25); 30-day average volume 8,830,020; 30-day SEC yield 4.84%; YTM 5.17%; closing price \$90.00	Match. VERIFIED-PRIMARY, https://www.ishares.com/us/products/239456/
TLH	0.15%; 11.59y; 61 bonds; \$10.53bn; 1,969,419 sh/day; 5.29%	0.15%; 11.59 yrs (Sep 24); 61 holdings; \$10,528,708,140; 1,969,419; 5.29%; 3-year standard deviation 11.35%; closing price \$93.38	Match. VERIFIED-PRIMARY, https://www.ishares.com/us/products/239453/
TLT (alt.)	0.15%; 14.88y; 47 bonds	14.88 yrs (Sep 24); 47 holdings	Match. VERIFIED-PRIMARY, /239454/
SHY (alt.)	0.15%; 1.79y; 91 notes	1.79 yrs; 91 holdings	Match. VERIFIED-PRIMARY, /239452/
VT	0.06% (as of 2026-02-27); 10,088 stocks (8/31); 37.7% non-U.S.; \$81.9bn; \$160.03	expenseRatio 0.0600 as of 2026-02-27; numberOfStocks 10088 (2026-08-31); foreignHolding 37.7; share class \$81.9 billion; market price 160.03 (9/25)	Match. VERIFIED-PRIMARY, https://investor.vanguard.com/vmf/api/VT/{expense,characteristic,price}
VT top-10	21.7% (fact sheet 6/30)	"Top ten as % of total net assets 21.7%"; Taiwan 3.5%; TSMC 1.7%; U.S. 61.9% (as of June 30, 2026)	Match. VERIFIED-PRIMARY, https://fund-docs.vanguard.com/F3141.pdf
VGSH	0.03% (2025-12-19); 1.9y (8/31); 92 bonds; \$34.8bn; SEC yield 4.55%	0.0300 as of 2025-12-19; averageDuration 1.9 years (8/31); numberofBonds 92; \$34.8 billion; yieldPct 4.55 (9/24); \$57.59	Match. VERIFIED-PRIMARY, https://investor.vanguard.com/vmf/api/VGSH/{...}
BIL (optional)	0.1353%; 0.10y OAD; 33 T-bills; \$47.92bn; renamed "State Street SPDR..."	0.1353% gross; fund OAD 0.10 years (index 0.09); 33 holdings; \$47,922.72M (Sep 24); page title "State Street® SPDR® Bloomberg 1-3 Month T-Bill ETF"	Match. VERIFIED-PRIMARY, ssga.com (redirected URL)

Holdings maturity dates could not be re-downloaded. The iShares holdings endpoint returned an HTML page to my curl today. For maturities I used S1's snapshot, `S1_fund_holdings_snapshot.csv` (iShares holdings as of 2026-09-24, VERIFIED-PRIMARY per S1). On that data **S3 and S1 misquote the ranges** (finding 2).

1b. The 2025-26 list (historical only, VERIFIED-REPO-FILE)

I searched `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` by ticker and by name. Every line number S3 cites is correct:

- IEF #88 (line 1053); TLT #87 (1041); VT #23 (273, 0.06); VTI #22 (261); VXUS #51 (603, 0.07);
- VGSH #93 (1113, merged with its name, 0.04); SHY #86 (1029); BIL #92 (1101); SHV #90 (1077); VNQ #21 (249).

TLH, SGOV, SPTI, SPTL, VGIT, VGLT, ITOT and IXUS are absent. No "10-20" Treasury fund appears anywhere on the list.

1c. WInS rules (SMApply Trading Details, re-read by S4 2026-09-27, VERIFIED-PRIMARY)

S3's quotes are exact:

- \$300,000 virtual cash;
- "The additional \$150,000 ... will not be added to WInS";
- 200 trades;
- "no more than twice a security's current daily trading volume";
- "Any ETF available on WInS";
- "There is no required sector allocation or minimum number of sectors";

- "not simply to own a large number of securities".

1d. The arithmetic

I re-ran `S1_hedge_weights.py` and `S3_allocation_numbers.py` with `.venv/bin/python`. Both reproduce:

- \$292,264, 9.90y and DV01 \$289/bp;
- IEF/TLH 35.7/64.3 and IEF/TLT 62.1/37.9;
- twist errors \$1,352 / \$3,619;
- ladder cost \$294,387;
- 2028 plan split 65.9 / 20.4 / 13.6;
- all dollar amounts, share counts, band triggers (-18.5% / +23.8%) and reserve values.

I also checked the hand formulas: $0.357 \times 6.86 + 0.643 \times 11.59 = 9.90$, and the portfolio duration of 6.77y.

What failed is a method choice and a wording claim (findings 2 and 8), not the arithmetic.

2. Findings

Finding 1 (BLOCKING): there is no gate between "strategy not approved" and "Day-1 orders"

Evidence.

- S3's calendar puts IEF, TLH and VT at "Day 1 (orders fill at the Mon Sep 28 ET open)". That is 11:30 p.m. AEST on Sep 28, about a day and a half from now.
- CLAUDE.md says the strategy is "NOT YET DECIDED BY THE TEAM. The team has not read the memos yet." (VERIFIED-REPO-FILE).
- Trade notes are permanent. The Trading Notes guide says "Include each Trading Note exactly as it appears in WInS ... Yes, we will verify this" (VERIFIED-REPO-FILE, p.2). Stock-Trak says students "cannot edit or delete their trade notes" (via A4).
- No rule requires a Day-1 trade. SMaply says "The competition does not require frequent or same-day trading. Make thoughtful, well-researched decisions" (VERIFIED-PRIMARY, re-read today). "First trade by Oct 10" is still unverified.
- WInS gains and losses are not judged (R-W75). Waiting a few days therefore costs nothing that counts.
- A hasty note is permanent. It is also the exact failure the team had last year: rules and complexity got ahead of understanding (CLAUDE.md, Team).
- S3's Step 0 only asks the team to choose among mirror options (i)/(ii)/(iii). That assumes lock-early is already approved.

Exact fix. Replace Step 0 with a gate of five boxes, all ticked before the first order:

- (a) the team has read a short version of the plan and recorded a vote to adopt "lock early" (or not), with names and date;
- (b) the team has chosen (ii) or (iii) (finding 5);
- (c) a Session Rules screenshot is saved (position limit for single positions and by security type; day trading);
- (d) each WInS note has been drafted **by a student, in the student's words**, from the checklists, and read by a second student;
- (e) the fallback in finding 3 has been picked in advance.

Target the first orders for **any U.S. session in week 1 once the gate is passed** (for example, fills by Fri Oct 2 ET), not "Day 1". Change the "Why this order" cell for step 1 so it does not imply urgency.

Finding 2 (BLOCKING before any hedge note is written): "the bonds mature when Laura pays" is false for most of TLH

Evidence (S1 holdings snapshot 2026-09-24, VERIFIED-PRIMARY per S1; computed by `S4_red_team_checks.py` section 3):

- **TLH:** 61 bonds maturing 2037-02-15 to **2046-08-15**. By weight: 2037 0.9%, 2038 1.2%, 2039 3.0%, 2040 8.6%, 2041 11.6%, then 2042 13.3%, 2043 20.1%, 2044 17.0%, 2045 14.7%, 2046 9.3%. So **74.7% matures after the last payment (2042-01-01)**.
- **IEF:** 16 notes maturing **2033-05-15 to 2036-08-15**.
- S3's section A.2 and S1's table say "Aug-2033 to May-2036" and "Feb-2037 to May-2046". Both ranges are misquoted against S1's own data.
- In the combined hedge (IEF 35.7% / TLH 64.3%), about **52%** of the bonds mature in 2033-2041 and about **48%** mature after 2042. The first four payments are 46.8% of the payments' value, but IEF is only 35.7% of the hedge.
- The duration match still works because the late TLH bonds make up for the shortfall in 2037-2041 bonds. It works "on average", not "year by year".
- The wording that overclaims:
 - S3 A.2: TLH is "Operating-commitment hedge, part 2: owns bonds maturing Feb-2037 to May-2046, covering the later payments";
 - S1 summary: "Together they own bonds that mature in the same years as Laura's ten payments";
 - S1 "What this teaches": "owns bonds maturing in exactly those years".
- Also, IEF and TLH are constant-maturity funds. They never mature and keep rolling into newer bonds. A reader who opens TLH's holdings page (the note is "verified") can check all of this.

What stays true (the honest case for TLH over TLT):

- Laura's payments are 6-16 years away.
- TLH's bonds are 10-20 years away. TLT's are 18-30 years away (TLT: 2044-05-15 to 2056-08-15).
- TLH sits closer to the payments. Its twist error is about 60% smaller than IEF/TLT's in both measurement methods: \$1,544 vs \$3,812 forward-consistent; \$1,352 vs \$3,619 S1 method.

Exact fix.

1. S3 A.2 and the F.1 tables: correct the ranges to IEF "May-2033 to Aug-2036" and TLH "Feb-2037 to Aug-2046". Drop "covering the later payments" and replace it with "10-20-year bonds, closer to the payment dates than TLT".
2. S1 summary item 1 and "What this teaches": replace "in the same years"/"exactly" with "closer to"/"spanning".
3. Note-1 checklist (S3 B.2). Replace "Why this fund: its bonds mature 2037-2046 ..." with these elements:
 - the fund's bonds are *closer to* Laura's payment dates than a 20+ year fund's;
 - the pair is blended to move like her payments when rates change (about 10 years);
 - the funds are a stand-in: they never mature, and the real plan buys a ladder in January 2027.

Must not claim that the funds' bonds mature in the payment years, or that the funds "match" the payments.

Finding 3 (IMPORTANT): the headline allocation is blocked under the only documented position limit, and the preferred fallback is the most complex

Evidence.

- Stock-Trak's generic FAQ says "The default position limit is 25% which means you cannot put more than 25% of your portfolio value in a single security" (VERIFIED-PRIMARY via A4, `phase_A/wins_week1_guardrails.md` line 282).
- This season's value is unknown. The 2024-25 screenshot showed only a bond limit of 100% [PRIOR].
- TLH at 42.4% breaks a 25% limit, and so does IEF at 41.0% in the TLT fallback.
- S3's section 0 calls the capped mix "the best one": IEF 22.6% / TLH 24.0% / **the 3.125% Treasury due 2041-11-15** 19.4%. That mix needs three things that are uncertain or awkward:
 - (a) bonds must have their own higher limit (UNVERIFIED);
 - (b) the bond must be in the WlnS drop-down (UNVERIFIED);
 - (c) a second order screen, with prices that "are updated once daily".
 - The ETF-only version has four Treasury funds, one of which (SPTI 4%, about \$12k) exists only to fix the duration.

- So on Day 1, at 11:30 p.m. AEST, the team would walk a three-level decision tree.

Exact fix. Pre-choose one rule that works for any cap from 25% to 43%:

- IEF stays at its formula weight (23.6%, under a 24% working cap);
- TLH is held one point under the cap;
- SPTL takes the rest of the 66% (VGLT as its alternate; both PENDING APPROVAL CHECK).

Forward-consistent results (script section 2; the same S1 scenarios; ASSUMPTION: instantaneous shocks):

Cap	Mix (share of total portfolio)	Hedge duration (issuer)	Worst 50bp twist	Worst $\pm 100\text{bp}$
none / $\geq 43\%$	IEF 23.6 / TLH 42.4	9.90y	\$1,544	\$451
40%	IEF 23.6 / TLH 39 / SPTL 3.4	10.00y	\$1,775	\$472
35%	IEF 23.6 / TLH 34 / SPTL 8.4	10.16y	\$2,117	\$967
25%	IEF 23.6 / TLH 24 / SPTL 18.4	10.47y	\$2,801	\$1,956
25% (S3 row 2, for comparison)	IEF 24 / TLH 24 / SPTL 14 / SPTI 4	9.89y	\$2,269	\$765

- Every error is below 1% of a \$292k hedge. Because WInS P&L is not judged (R-W75), one extra ticker beats two extra tickers.
- Every duration is still "about 10 years", so the note wording does not change.
- Delete D.1b row 1 (the individual bond) and the IBGA row from the week-1 plan. Keep the bond idea only as a Final-Report illustration, if at all.

Finding 4 (IMPORTANT): VT is bought on Day 1 but VGSH waits for "the same decision", and Note 3 locks an open parameter into a permanent note

Evidence.

- VGSH is held in cash until week 2 because "the 2031 rule and the sleeve split (50/60/70) are open team decisions" (S3 B.1).
- But VT's size depends on the **same** split: 17% / 20.5% / 24% of the portfolio (S3 A.1). Buying VT at 60% on Day 1 presumes the split that VGSH supposedly waits for.
- S3 itself says "Do not create a decision just to get a note". The staggered timing does exactly that.
- The Note-3 checklist suggests naming the rule "about 80% of the sleeve locked two years before 2033".
 - The 2031 floor share is listed as **open/disputed** (brief section 9).
 - The Trading Notes guide says teams are "not expected to ... determine Laura's facility contribution" at this stage (VERIFIED-REPO-FILE, p.1).
 - A permanent WInS note that names 80% forces the IPS (Nov 6) to match it.
- The "2031 floor proxy" label overclaims:
 - in 2026 the VGSH position is 40% of the sleeve;
 - the floor in 2031 is about 80% of the sleeve, and a different instrument (a 2-year note).

Exact fix.

1. Buy VT and VGSH together, once the gate (finding 1) is passed, at the provisional 60/40 split. Record the split as provisional (CLAUDE.md shows the team accepted 60% provisionally).
2. Relabel VGSH everywhere as "the short-Treasury part of the growth money". Say its intended future use (it can become a guaranteed amount before Laura talks to co-sponsors) without a percentage.
3. Delete "about 80% ..." from the Note-3 checklist.
4. The third note can be the VGSH buy itself (a distinct job: flexibility and credibility), or a genuine later decision:
 - a duration refresh;
 - a change of the split after the team's research;
 - the (ii)/(iii) decision, if it is made after a first trade.

Log any "we did not trade" decisions for the Final Report.

Finding 5 (IMPORTANT): option (ii) is defensible but oversold; its rationale mixes dates and overstates what the IPS needs

Evidence.

- The case says: "The WInS portfolio represents each team's implementation of its investment strategy during the competition. It does not determine Laura's actual portfolio value at the beginning of 2027" (case p.4, VERIFIED-REPO-FILE). SMAApply adds that the \$150,000 "will not be added to WInS" (VERIFIED-PRIMARY).
- Read literally, this supports (iii), the ~97% ladder book, at least as much as (ii).
- S3 says (iii) "describes 1 of the 6 pre-2033 years". But WInS runs in Sep-Nov 2026, the eve of the January 2027 purchase. (iii) describes exactly that moment.
- (ii) takes its **size** from 2028-01-01 (hedge 65.9%) but its **duration** from 2027-01-01 (9.90y). The brief (section 14) says to hedge the WInS book to the spot duration (10.16y). So three dates are in play.
- S3 A.1 reason 4 says the WInS weights, the IPS percentages and the model inputs "must be one set of numbers". The IPS guide asks teams to "Focus on the strategy and decision-making framework ... rather than describing individual investments or presenting detailed financial calculations" (VERIFIED-REPO-FILE, p.2). The consistency needed is about rules and rough shares, not a table of weights.

Exact fix.

- Keep (ii) as S3's recommendation if the team wants to show growth and flexibility, but present (ii) and (iii) as equals. The team picks with a stated test, for example: "Which reading of 'implementation during the competition' would we defend in the Final Report?"
- If (ii), the one-sentence explanation (the team's words) must contain:
 - WInS holds the target mix Laura's portfolio reaches after both deposits, scaled to \$300k;
 - the Treasury part is set to the payments' interest-rate sensitivity today, about 10 years;
 - in January 2027 almost all of her real \$300k buys the ladder.

The IPS must say the last point too, so a reader is not surprised that WInS holds about 20% stocks.

- Delete "must be one set of numbers" from A.1. Replace it with "must follow the same rules and roughly the same shares".

Finding 6 (IMPORTANT): unearned complexity. The file is a research dossier, not a trade ticket

Evidence.

- S3's file is 538 lines (55 KB). It contains:
 - three mirror options × three cash levels, although cash is known to be \$300k;
 - three capped hedges plus a rejected IBGA row;
 - a 12-fund sector fallback for a rule that does not exist;
 - a CUSIP-level STRIPS ladder;
 - a ten-row reserve run-down;
 - twist errors quoted to the dollar.
- SMAApply: "The goal is to build an intentional mix of investments appropriate for your strategy, not simply to own a large number of securities" (VERIFIED-PRIMARY).
- Last year's lesson: "strategy packaging was over-complex for the insight" (CLAUDE.md).

Exact fix.

1. Cut:

- the \$100k and \$500k columns (A.2, A.4, B);
- A.4 entirely;
- D.1b rows 1 and 3 and the IBGA row (use finding 3's one-line rule);
- D.2 (replace with one line: "no sector rule exists; a broad fund already holds all 11 sectors");
- D.3 (keep one checklist line: record which U.S. Treasuries are in the drop-down);
- C's illustrative duration drifts;
- every twist figure except one comparison sentence.

2. **Move** E.2 (CUSIP ladder) and E.5 (reserve run-down) to a Final-Report research file. They are Dec-4 material. The Trading Notes guide says the reserve size is "not expected" now.
3. **Put a one-page ticket at the top** with:
 - the gate (finding 1);
 - Session Rules → mix A (no cap) or the one-line capped rule;
 - four orders (IEF, TLH, VT, VGSH);
 - three note checklists;
 - the three "never" rules: never sell the hedge after a rate rise, never trim it after a fall, never day-trade.

Finding 7 (IMPORTANT): a semifinal reader could see this as a generic "Treasury ETF + world index" portfolio unless the notes carry the Laura-specific logic

Evidence.

- Wharton's own example note buys "an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment" (Trading Notes guide p.2, VERIFIED-REPO-FILE).
- Thousands of teams read the same example. A Treasury-ETF note is therefore the most common note in the field. VT is a default global fund.
- WInS under (ii) is about 79% Treasuries. The case calls Laura someone who "has been willing to take thoughtful risks" (case p.2). A skim can read 79% Treasuries as timid.
- What is genuinely distinctive and correct here:
 - (a) the hedge is sized to a specific promise (ten fixed \$50,000 payments, valued at about \$292k), not to "lower volatility";
 - (b) the second fund is chosen by distance to the payment dates (finding 2's honest version);
 - (c) a rule is committed in advance: no selling the hedge after rate rises, because the payments got cheaper too;
 - (d) risk is measured on the surplus: 60% of the money the promise does not need is in world stocks;
 - (e) the official phrase "funding purposes" (R-W71) as a kind of diversification.

Exact fix. Each hedge note must contain (a) and (c), and must avoid "reduce volatility"/"stable" (S3 already warns about this). The growth note must frame equity as a share of the free money (d), not of the total. One note or reflection should use (e). Laura's statistics degree ("Statistics & Information Decisions Management", case p.1 line 16, VERIFIED-REPO-FILE) supports "one index she can check against one published assumption". That is a real Client Knowledge link. Use it once, not as decoration.

Finding 8 (MINOR): the 9.90-year anchor is right in practice but justified wrongly; S1's tracking method mixes spot and forward

Evidence (script sections 1-2; model outputs on VERIFIED inputs):

- A hedge bought **today** must match the payments' **spot** duration, 10.16y. Its forward value moves 0.27y less, just as the payments' forward value does. S3's reason ("the sensitivity of that forward cost is 9.90") is therefore not the right test. This agrees with brief section 14.
- But the issuer's effective durations run below full-revaluation durations: TLH 11.59 issuer vs 11.87 model; IEF 6.86 vs 6.90. So the issuer-weighted "9.90" mix is **10.09y** in model terms, close to 10.16.
- Measured forward-consistently (hedge forward value vs liability forward value):
 - the S3 mix misses by at most **\$451 per ±100bp**;
 - weighting to 10.16 on issuer numbers misses by \$1,035;
 - weighting to 8.92 (the 2028 view) misses by \$3,260.
- S1's `tracking()` compares fund **spot** returns with the liability's **forward** change. Recomputed consistently:
 - IEF/TLH worst twist is \$1,544 (not \$1,352);
 - IEF/TLT is \$3,812 (not \$3,619);
 - ±100bp is \$451 / \$860 (not \$991 / \$1,274).
- The ranking TLH > TLT > SPTI/SPTL is unchanged.

Exact fix.

- Keep the issuer-weighted 9.90 target. Replace S3 A.3's reason with: "issuer durations sit about 0.2-0.3 years below full-revaluation durations, so the 9.90 issuer mix is about 10.1 years, close to the payments' spot duration of 10.16; it tracks within about \$500 per 100bp".
- Delete the 8.92 discussion, which tracks worst.
- The main loop should add one line to brief section 14 so the two files agree.
- Optionally switch S1's `tracking()` to the forward-consistent version in `S4_red_team_checks.py`.
- In every deliverable say "about 10 years".

Finding 9 (MINOR for WInS; IMPORTANT for the Final Report): the real ladder fits inside \$300k less often than the brief says

Evidence (script section 5; ASSUMPTION: the brief's zero-drift lognormal with 7.24% volatility):

- S1's Nov-15 STRIPS ladder costs \$294,387, so the headroom is about 19bp, not 27bp.
- $P(\text{cost} > \$300\text{k on } 2027-01-01)$ is **30.7%**, against 24.3% for exact-date zeros.
- Cost at -25 / -50 / -100bp: \$301,676 / \$309,169 / \$324,793.
- In those paths, S3 E.1's rule ("buy the longest rungs first and the rest from the 2028 deposit") means the nearest payments depend on the 2028 deposit. That is the brief's unanswered "joint tail" (section 9). STRIPS dealer mark-ups are unknown and would add to the cost (S1, UNVERIFIED).

Exact fix. Add one line to S3 E.1: "in about 3 of 10 rate paths the ladder needs part of the 2028 deposit; certainty for the first rungs then depends on that deposit arriving". Pass the joint tail to Phases B-E. None of this belongs in a week-1 note.

Finding 10 (MINOR): the model and WInS hold different assets

Evidence.

- `research/verified_2026-09-27/strategy_mc.py` models the sleeve's bonds as JPM U.S. intermediate Treasuries (4.00% compound, 3.48% volatility; line 27) and its equity as U.S. large cap.
- WInS holds VGSH (1.9y) and VT (global). S2 and S3 both flag this (S3 open issue 4).

Exact fix. Before the Final Report, the main loop changes two model inputs:

- equity: JPM AC World 7.00% / 16.78%;
- sleeve bonds: an explicit short-Treasury ASSUMPTION (S3 suggests about 3.5-3.9%).

Keep VGSH in WInS: short Treasuries are what the 2031 floor will be built from. The effect on the median is under about \$2k (S2, S3).

Finding 11 (MINOR): the volume check uses the wrong denominator

Evidence.

- The rule is "no more than twice a security's **current daily** trading volume" (SMAApply, VERIFIED-PRIMARY, re-read today).
- S3 compares orders with 30-day **average** volume (e.g. "0.035% of TLH's volume cap").
- An overnight order that fills at the open may be tested against the volume traded so far that day.
- At these sizes (largest 1,362 TLH shares against about 2m a day) it should still pass, but the stated check is not the rule.

Exact fix. Reword F.2 item 5: read today's volume in WInS. If an order is rejected for volume, split it across days. Never switch to a thinner fund to get it through.

Finding 12 (MINOR): the calendar misses the U.S. clock change

Evidence.

- U.S. daylight saving ends Sunday Nov 1, 2026 (the first Sunday of November).
- From Mon Nov 2 the U.S. open is **1:30 a.m. AEDT**. The Nov 6 close (4:00 p.m. EST) is **8:00 a.m. AEDT on Sat Nov 7**.
- S3's calendar stops at the Oct 4 Australian change (ASSUMPTION: the team is in an AEDT state; its state is not recorded).

Exact fix. Add the line to B.1. It matters only if the team trades in the last week, which S3 already advises against.

Finding 13 (MINOR): stale or internally inconsistent statements in S3

Evidence and fixes.

- S3 section 1 says "CLAUDE.md and brief sections 6 and 9 still carry the old third-party claim" about a sector minimum. S3 open issue 5 asks the main loop to add the \$300k cash. CLAUDE.md and brief section 14 already say both (VERIFIED-REPO-FILE). **Fix:** delete both statements.
- S3 E.1 puts the 2027 leftover (~\$5.6k) in T-bills. But option (iii) holds VT 1.5%, and the script grows the leftover at the 60/40 sleeve rate. **Fix:** pick one rule. The simplest is T-bills in 2027 and VT from 2028; then (iii) holds VT about 0%.
- The label is still brief section 4's wording. **Fix:** use brief section 14's "PENDING WinS AVAILABILITY + POSITION-LIMIT CHECK".
- S1 section 2 labels "(D_TLH – 9.90)/(D_TLH – D_IEF)" as the "IEF/TLH weight". It is the **IEF** weight. **Fix:** relabel it.
- S1 gives TLT's range as "Aug-2044 to May-2056". The snapshot says 2044-05-15 to 2056-08-15. **Fix:** correct it.

Finding 14 (MINOR): the hedge rebalancing band is narrower than the uncertainty in the duration number

Evidence. The band is $9.90 \pm 0.25y$ (S3 C). The issuer-vs-model gap for TLH alone is 0.28y (finding 8).

Exact fix. Keep the band, since it will rarely trigger and that is the point. Never quote the hedge's duration to two decimals in a deliverable, and never trade to fix a gap smaller than 0.25y.

3. What survived the attack (keep)

- **All primary-pick fund facts** (section 1a) and the 2025-26 list look-ups (1b).
- **The hedge/sleeve decomposition:** 65.9% hedge / 20.4% equity / 13.6% sleeve bonds at 2028 (reproduced). This is the real fix for the brief's 35%-vs-20% inconsistency.
- **TLH over TLT** as the second hedge fund. It still wins in a consistent recomputation, as long as it is justified by "closer to the payments", not "matures when she pays".
- **VT as the single growth fund:** 0.06%, top-10 21.7% against 38.8% for the S&P 500 (S2). No Taiwan tilt, no sector funds, no leverage.
- **The "never" rules:** never sell the hedge after a rate rise, never trim it after a fall, never move money between the sleeves, never day-trade, never trade for ranking.
- **The Session Rules check before the first order**, and copying each note into the decision log.
- **Cash over BIL** for short waits (the \$50 round trip needs about 14 days of BIL yield).

4. Which findings each deliverable depends on

Deliverable	Findings that must be fixed first
WinS week 1 (now)	1, 2, 3, 4 (blocking/important), then 6
Trading Notes (Oct 23)	2, 4, 7
IPS (Nov 6)	5, 8 ("about 10 years"), 10
Final Report (Dec 4)	9, 10; the moved E-sections from finding 6

Sources (all accessed 2026-09-27 by S4 unless marked)

- iShares product pages (VERIFIED-PRIMARY, curl ~12:28 UTC): <https://www.ishares.com/us/products/239456/> (IEF), [/239453/](https://www.ishares.com/us/products/239453/) (TLH), [/239454/](https://www.ishares.com/us/products/239454/) (TLT), [/239452/](https://www.ishares.com/us/products/239452/) (SHY). The holdings endpoint `.../1467271812596.ajax?fileType=csv` returned an HTML page to curl today, so maturities come from S1's snapshot.

- Vanguard (VERIFIED-PRIMARY): [https://investor.vanguard.com/vmf/api/VT/{profile,expense,characteristic,price},the same for VGSH; fact sheet https://fund-docs.vanguard.com/F3141.pdf \(as of June 30, 2026\).](https://investor.vanguard.com/vmf/api/VT/{profile,expense,characteristic,price},the same for VGSH; fact sheet https://fund-docs.vanguard.com/F3141.pdf (as of June 30, 2026).)
- State Street (VERIFIED-PRIMARY): [https://www.ssga.com/us/en/intermediary/etfs/state-street-spdr-bloomberg-1-3-month-t-bill-etf-bil \(redirected from the old spdr... URL\).](https://www.ssga.com/us/en/intermediary/etfs/state-street-spdr-bloomberg-1-3-month-t-bill-etf-bil (redirected from the old spdr... URL).)
- SMAApply Trading Details (VERIFIED-PRIMARY): <https://wghsinvcomp.smapply.us/res/p/trading/>
- Repo files (VERIFIED-REPO-FILE):
 - `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (lines 16, 68-74, 88-103, 114, 129-133, 161);
 - `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` (pp.1-2);
 - `competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt` (p.2);
 - `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` (lines as cited);
 - `research/verified_2026-09-27/strategy_mc.py` (line 27);
 - `research/insight_v1/phase_A/wins_week1_guardrails.md` (lines 33-42, 234-239, 282-283);
 - `research/insight_v1/phase_A/case_register.md` (R-W66, R-W71, R-W75);
 - `research/insight_v1/wins_now/S1_fund_holdings_snapshot.csv`;
 - `CLAUDE.md`.
- Scripts re-run: `research/insight_v1/wins_now/S1_hedge_weights.py`, `research/insight_v1/scripts/S3_allocation_numbers.py`. New: `research/insight_v1/scripts/S4_red_team_checks.py`.

What this teaches

1. **Check a story against the holdings.** "The bonds mature when she pays" sounded right and was repeated in three places. Adding up the holdings by year took one line of code and showed that three-quarters of TLH matures after Laura's last payment. The fund was still the right choice, but for a different reason: its bonds are closer to her dates than the alternative's.
2. **Precision has a floor.** The team debated 9.90 vs 10.16 vs 8.92 years. The gap between two honest ways of measuring the same fund's duration (0.28 years for TLH) is as big as that debate. "About 10 years" is the true precision. Anything finer is decoration.
3. **Order of operations is part of compliance.** Last year the team traded outside the rules. This year the rules are clearer, but the same failure can come back as permanent notes written before the team understands its own strategy. The cheapest fix is a gate: approve, check Session Rules, draft the note, then trade.
4. **A red team should say what survived.** Most of this plan is right. The work is to remove the few claims that would not survive a careful reader, and to cut everything a high-school team cannot explain in its own words.

Securities and allocation v1: WInS week-1 ticket, and Laura's long-term instrument map

Agent S3 (Allocation Architect), insight_v1 run. First written 2026-09-27; **revised the same day after the S4 red team** (research/insight_v1/wins_now/S4_red_team.md). This is a **PROVISIONAL week-1 recommendation**. The team has not approved the strategy, and Phases B-E may change it. Everything here is AI-generated research: tables, numbers, reasons and checklists. None of it is text to submit. The team decides and writes every word, including every WInS note.

Every WInS security in this file is PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading. This season there is no separate approved list; "Any ETF available on WInS" is allowed (SMAApply Trading Details, VERIFIED-PRIMARY, read 2026-09-27). So the check is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK**: is the ticker listed in the team's WInS account, and does the order fit the Session Rules position limit?

Scripts (run from the repo root):

- .venv/bin/python research/insight_v1/scripts/S3_allocation_numbers.py (weights, dollars, share counts, bands; section 10 is the capped-hedge rule added after the red team);
- .venv/bin/python research/insight_v1/scripts/S4_red_team_checks.py (maturity profiles, forward-consistent tracking).

Moved out of this file: the CUSIP-level STRIPS ladder and the 2033-2042 reserve run-down now live in research/insight_v1/wins_now/S3_final_report_ladder_and_reserve.md (Final Report material, not week-1).

THE TICKET (one page)

Gate: tick all five before the first order

- [] **(a) Team vote on "lock early"**. The team has read a short version of the plan and recorded a vote to adopt it (or not), with first names and date, in the team's decision log. If the vote is "no", stop: this ticket does not apply.
- [] **(b) Option chosen: (ii) or (iii)** (section 2), using the stated test, and recorded.
- [] **(c) Session Rules screenshot saved** (Portfolio Summary > Session Rules): position limit (single position, and by security type if shown), day trading, trades allowed, commissions.
- [] **(d) Each WInS note drafted by a student, in that student's own words**, from the checklists below, and read by a second student before submitting. Notes cannot be edited after submission (Stock-Trak, via A4).
- [] **(e) Position-limit branch picked in advance** (the rule below), so nobody improvises at 11:30 p.m.

When: aim for fills in **week 1 once the gate is passed**, for example by **Fri Oct 2 ET**. There is no reason to rush for Mon Sep 28. SMAApply says "The competition does not require frequent or same-day trading" (R-W73, VERIFIED-PRIMARY), and WInS gains and losses are not judged (R-W75). The "first trade by Oct 10" claim is still UNVERIFIED: check the logged-in Trading Details page.

Session Rules branch (decided in advance)

Single-security limit shown in Session Rules	Hedge orders (share of the \$300,000)
44% or more, or no limit	IEF 23.6% / TLH 42.4%
Under 44% (any cap from 25% up)	IEF 23.6% / TLH one point under the cap / SPTL the rest of the 66% (alternate VGLT). At a 25% cap: IEF 23.6 / TLH 24.0 / SPTL 18.4

The hedge stays 66% of the portfolio and "about 10 years" in every branch.

Orders, option (ii), at \$300,000 (share counts at 2026-09-25 closes; recompute on the trade date)

#	Order	Weight	Dollars	~Shares	Job
1	BUY IEF	23.6%	\$70,744	~786	The promise (hedge), part 1

#	Order	Weight	Dollars	~Shares	Job
2	BUY TLH (capped: TLH ~771 sh + SPTL ~2,276 sh)	42.4%	\$127,256	~1,362	The promise (hedge), part 2
3	BUY VT	20.5%	\$61,500	~384	Growth: 60% of the money the promise does not need
4	BUY VGSH	12.5%	\$37,500	~651	Short-Treasury part of the growth money
	Cash float	1%	\$3,000		Commissions; no margin allowed

All four go in **one session**, hedge orders entered first. Overnight orders all fill at the next U.S. open. Cost: 4-5 trades, \$100-125. **Option (iii)** instead: IEF 35.0% / TLH 63.0% / cash 2.0%, with no VT or VGSH (section 2).

Three note checklists (elements only; the team writes the words)

1. **Hedge note (TLH, or SPTL if capped).** Sized to a specific promise: ten fixed \$50,000 payments, 2033-2042, which cost about \$292k to lock in for January 2027 at today's Treasury prices. TLH's bonds are *closer* to her payment dates than a 20+ year fund's. The pair moves like her payments when rates change (about 10 years). The funds stand in for the ladder she buys in January 2027. The rule: never sell the hedge after a rate rise. **Never** say the funds "match" the payments, that their bonds mature in her payment years, or "stable"/"reduce volatility".
2. **Growth note (VT).** Equity is 60% of the money the promise does not need, not a share of the total. It serves the facility contribution and flexibility. One global index: a concentration argument, not a return forecast. The risk: a fall shrinks the facility range, never the payments.
3. **Short-Treasury note (VGSH).** A different funding purpose: money that can later become an amount Laura can promise co-sponsors before 2033, because it is already owned. Split recorded as provisional (60/40). **No floor percentage in the note.**

Three "never" rules

1. Never sell the hedge after a rate rise (the payments got cheaper too).
2. Never trim the hedge after a rate fall (it grew because the payments cost more).
3. Never day-trade: never buy and sell the same security on the same U.S. trading day ("Day Trading: This is not permitted", 2026-27 WlnS User Guide p.6, via A4).

Terms used (plain English)

- **Hedge (the "promise" money):** the Treasury funds that stand in for the ten fixed \$50,000 payments due 2033-2042. Their value moves with the cost of those payments.
- **Growth money (the "sleeve"):** everything the payments do not need. It pays for the facility contribution and for flexibility.
- **Duration:** roughly the % a bond fund's price falls if interest rates rise by 1 percentage point. The payments' duration is about 10 years.
- **Basis point (bp):** 0.01 percentage point.
- **Position limit:** the most WlnS lets you hold in one security, as a share of the portfolio (Session Rules).
- **Twist:** short-term and long-term rates move in opposite directions.
- **Stand-in:** in WlnS, funds that behave like Laura's real ladder without being it. The funds never mature; they keep rolling into newer bonds.
- **Rebalancing band:** a range around a target. The team trades only when a weight leaves it.

1. The securities (primary, alternates, facts)

Key facts are from issuer primary pages, **accessed 2026-09-27** (as-of dates in brackets). "Re-read" means S3 opened the page again during this revision (12:40-12:50 UTC); other facts were read by S3/S4 earlier the same day. 2025-26 list = `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` (historical only, VERIFIED-REPO-FILE; searched by ticker and by name, since some lines merge them).

Role	Primary (PENDING APPROVAL CHECK)	Key facts (issuer page)	Alternates, same type (PENDING APPROVAL CHECK)	2025-26 list
Hedge, part 1	IEF iShares 7-10 Year Treasury Bond ETF	0.15%; effective duration 6.86y [9/24]; 16 notes maturing 2033-05-15 to 2036-08-15 (S1 holdings snapshot 9/24); net assets \$41.57bn; 30-day avg volume 8,830,020; close \$90.00 [9/25]. ishares.com/us/products/239456 , re-read, VERIFIED-PRIMARY	VGIT (0.03%, 4.9y [8/31], 102 bonds; Vanguard API, re-read). SPTI (0.03%, 4.78y; ssga.com , S1)	IEF yes (#88, line 1053). VGIT, SPTI no
Hedge, part 2	TLH iShares 10-20 Year Treasury Bond ETF	0.15%; 11.59y [9/24]; 61 bonds maturing 2037-02-15 to 2046-08-15, 74.7% of bond weight after 2042-01-01 (S4 script on S1 snapshot); \$10.53bn; 1,969,419/day; close \$93.38 [9/25]. ishares.com/us/products/239453 , re-read, VERIFIED-PRIMARY	TLT (0.15%, 14.88y, 47 bonds maturing 2044-05-15 to 2056-08-15; \$45.75bn; re-read). SPTL/VGLT (below)	TLH no ; TLT yes (#87, line 1041)
Hedge, part 3 only if capped	SPTL State Street SPDR Portfolio Long Term Treasury ETF	0.03% gross [8/31]; option-adjusted duration 13.65y [9/24]; 110 holdings; Bloomberg Long U.S. Treasury Index (10+ years); AUM \$10,949.50M [9/24]; prior-day exchange volume 1,088,651; close \$24.27. ssga.com (spdr-portfolio-long-term-treasury-etf-sptl) , re-read 12:42 UTC, VERIFIED-PRIMARY	VGLT Vanguard Long-Term Treasury ETF (0.03%; 13.5y [8/31]; 101 bonds; \$15.1bn; price ~\$50.94 [9/25]; Vanguard API, re-read)	SPTL, VGLT no
Growth	VT Vanguard Total World Stock ETF	0.06% [as of 2026-02-27]; 10,088 stocks, 37.7% non-U.S. [8/31]; FTSE Global All Cap Index; fund total net assets \$101.7bn [8/31]; market price \$160.03 [9/25]; top-10 21.7% [fact sheet 6/30, S2/S4]. investor.vanguard.com/vmf/api/VT , re-read. Vanguard does not publish volume (check in WlnS)	VTI 62% + VXUS 38% of the equity (two funds); ITOT + IXUS in the same split (S2, iShares pages)	VT yes (#23, line 273); VTI yes (#22); VXUS yes (#51); ITOT, IXUS no
Short Treasuries in the growth money	VGSH Vanguard Short-Term Treasury ETF	0.03% [as of 2025-12-19]; 1.9y [8/31]; 92 bonds; Bloomberg U.S. 1-3 Year Treasury Index; fund total net assets \$39.3bn [8/31]; SEC yield 4.55% [9/24]; price \$57.59 [9/25]. Vanguard API, re-read. Volume not published	SHY (0.15%, 1.79y, 91 notes, \$26.18bn; ishares.com/239452 , re-read). A WlnS-listed ~2-year U.S. Treasury note (\$10 commission; prices update once a day)	VGSH yes (#93, line 1113, merged with name); SHY yes (#86, line 1029)
Cash float	Cash	none	BIL (0.1353%, 0.10y; ssga.com ; page now titled "State Street SPDR Bloomberg 1-3 Month T-Bill ETF"), SGOV (0.09%)	BIL yes (#92); SGOV no

Why these and not others (each traced to Laura or the case):

- **Low-cost, broad, transparent, liquid** (brief section 4). No leverage, inverse, thematic or single-country funds: nothing in the case justifies them. A Taiwan tilt would be tokenism (S2).
- **TLH over TLT**: Laura's payments are 6.3-15.3 years away. TLH's bonds are 10.4-19.9 years away; TLT's are 17.6-29.9 years away. So TLH sits **closer** to her dates. Measured consistently, its worst 50bp-twist error is about 60% smaller than IEF/TLT's (\$1.5k against \$3.8k on a \$292k hedge; S4 method). It is a closer fit, **not** a year-by-year match: three-quarters of TLH matures after her last payment.
- **VT as one fund**: 10,088 stocks and a 21.7% top-10 share (S2). Laura holds a statistics degree ("Statistics & Information Decisions Management", case line 16, VERIFIED-REPO-FILE). One index she can check against one published assumption fits that background. Use this once, in one place, not as decoration.
- **VGSH**: short Treasuries are what any future promised amount would be built from; they keep part of the growth money safe and available.

2. What the \$300,000 represents: (ii) and (iii) as equals

Plan numbers at 2028-01-01 (script section 3; ASSUMPTION: today's forwards and JPM returns are realised): hedge 65.9%, equity 20.4%, short/other sleeve bonds 13.6% at a 60/40 sleeve. So the old "65/35" was hedge/sleeve, not bonds/stocks. Option (i) (34% equity) is dropped: no version of the plan holds it.

	(ii) The mix after both deposits, scaled to \$300k	(iii) The literal January 2027 book
Hedge (about 10 years)	66%	98% (ladder \$294,387 of \$300,000)
VT / VGSH / cash	20.5% / 12.5% / 1%	0% / 0% / 2% (the 2027 leftover waits in T-bills; VT starts in 2028)
Case support	"implementation of its investment strategy during the competition" (case line 129): shows the whole strategy, including growth and flexibility	"It does not determine Laura's actual portfolio value at the beginning of 2027" (case line 129-130) and the \$150,000 "will not be added to WInS" (SMApply, VERIFIED-PRIMARY): WInS is the \$300k on the eve of the January 2027 purchase
Notes it gives	Three different funding purposes (promise, growth, short Treasuries); SMApply lists "funding purposes" as a kind of diversification (R-W71)	Hedge notes only; the growth argument has to come from the IPS
Under a 25% position limit	5 positions (IEF, TLH, SPTL, VT, VGSH)	5 Treasury funds: IEF 24 / TLH 24 / SPTL 24 / VGIT 17.6 / VGLT 8.4 (script section 10; twist error \$2.8k, about 1% of the hedge)
Main weakness	WInS shows about 20% stocks while Laura's real 2027 money is almost all ladder; must be explained	A reader may see about 98% Treasuries as timid for a client who "has been willing to take thoughtful risks" (case line 70)

The test the team applies: "Which reading of 'implementation during the competition' would we defend in the Final Report, in our own words?" Pick one and record why. S3's lean is (ii), because it shows growth and flexibility as well as the promise, but (iii) is fully defensible.

If (ii), the team's one explanatory sentence (IPS or Final Report) must contain three things:

1. WInS holds the mix Laura's portfolio reaches after both deposits, scaled to \$300,000;
2. the Treasury part is set to the payments' sensitivity to rates today, about 10 years;
3. in January 2027 almost all of her real \$300,000 buys the ladder.

Consistency across the three deliverables means **the same rules and rough shares**, not one table of weights. The IPS guide asks teams to "Focus on the strategy and decision-making framework ... rather than describing individual investments or presenting detailed financial calculations" (2026_WGY_Investment_Policy-FINAL.txt line 51, VERIFIED-REPO-FILE).

3. The hedge: weights, duration, behaviour, capped rule

- **Weights:** IEF weight = $(D_{TLH} - 9.90) / (D_{TLH} - D_{IEF})$ with the issuer "effective duration" on the trade date. Today: IEF 35.7% / TLH 64.3% of the hedge (23.6% / 42.4% of the total).
- **Why a 9.90 target, and why that is right (reason corrected after the red team):** a hedge bought today should match the payments' **spot** duration, 10.16 years (brief section 14). Issuer effective durations sit about 0.2-0.3 years below full-revaluation model durations (TLH 11.59 issuer vs 11.87 model). So the mix weighted to 9.90 on issuer numbers is about 10.1 years in model terms, close to 10.16, and tracks the payments within about \$450 per 100bp (forward-consistent, S4 script). Weighting to 10.16 on issuer numbers would overshoot (\$1,035 per 100bp). **One line for brief section 14:** "use spot duration (10.16) as the target; on issuer durations, the 9.90 weights hit it."
- **Precision:** the gap between two honest duration measures (0.28 years for TLH) is as large as the debate about the target. **Say "about 10 years" in every deliverable; never quote the hedge's duration to two decimals.**
- **What it does in WInS:** the \$198k hedge moves about \$20k per 1 percentage point of rates (first-order). The value of Laura's payments moves by about the same percentage in the same direction. Judged against the payments the hedge is close to risk-free; judged in dollars alone it is not.
- **Capped rule (only if Session Rules show a limit under 44%).** IEF stays at 23.6%; TLH one point under the cap; SPTL takes the rest. Script section 10, forward-consistent tracking:

Cap	Mix (share of total)	Duration (issuer)	Worst 50bp twist error
none	IEF 23.6 / TLH 42.4	about 10y	\$1,544
35%	IEF 23.6 / TLH 34.0 / SPTL 8.4	about 10y	\$2,120
25%	IEF 23.6 / TLH 24.0 / SPTL 18.4	about 10.5y	\$2,803

Every error is under 1% of the hedge. One extra fund beats the old four-fund or individual-bond variants, which are dropped from the week-1 plan. If SPTL's share would be under 2% of the portfolio (a cap of 43-

44%), put that sliver in IEF instead (duration moves under 0.1 year). ASSUMPTION: the limit is checked when an order is placed; if WInS applies it continuously, the 1-point buffer on TLH lasts until rates fall about 1 percentage point (script arithmetic).

- **If TLH is not in WInS:** IEF 41.0% / TLT 25.0% (both on the 2025-26 list; twist error \$3.8k). If TLH is missing **and** a 25% cap applies: IEF 24 / SPTL 24 / VGIT 9.5 / VGLT 8.5 (about 10 years; twist error \$3.3k; script section 10). That is the only four-fund case, and it arises only if two checks fail together.

4. Sequence and calendar

Times: U.S. open 9:30 a.m. ET. For the team (ASSUMPTION: the team's Australian state is not recorded):

- until Sat Oct 3: 11:30 p.m. AEST the same date;
- from Sun Oct 4 (NSW/Victoria daylight saving; Queensland does not change): 12:30 a.m. AEDT the next date;
- **from Mon Nov 2 (U.S. daylight saving ends Sun Nov 1, 2026): 1:30 a.m. AEDT** (12:30 a.m. AEST in Queensland). The Nov 6 close (4:00 p.m. EST) is **8:00 a.m. AEDT on Sat Nov 7**. Orders entered in the Australian daytime fill at the next U.S. opening price (SMAApply FAQ, VERIFIED-PRIMARY).

Step	When	Action
0	Before any order	The five-box gate; the WInS checks in section 7
1	Week 1 once the gate is passed (target fills by Fri Oct 2 ET)	Orders 1-4 in one session, hedge first. Split recorded as provisional 60/40
2	Weekly (same weekday)	Check bands (section 5); trade only if one is breached
3	Oct 19-23	Duration refresh: recompute IEF/TLH weights; trade only if the hedge is outside 9.65-10.15 on issuer numbers
4	Oct 23	Trading Notes due: choose the best 3 executed trades
5	Nov 6	IPS due; WInS freezes. No "tidy-up" trades in the last days

Where the third note comes from: the VGSH buy itself (a distinct funding purpose). A genuine later decision can replace it: a duration refresh, a change of the split after the team's research, or a band trade. Do not create a decision just to get a note. Log "we decided not to trade" moments for the Final Report.

5. Rebalancing rule for WInS

What	Target	Band	How to fix
Hedge duration	about 10y (9.90 on issuer numbers)	9.65-10.15y; never trade to fix a smaller gap	IEF against TLH (or SPTL) only
Hedge vs growth money	set once at purchase	No band. The hedge's share rises when rates fall because the payments cost more; that is correct	Never move money between them
Inside the growth money	VT 60% / VGSH 40% (provisional)	VT at 55-65%	VT against VGSH only. Triggers only if stocks move about -18.5% or +23.8% relative to bonds
Cash float	about 1%	at least \$1,000, at most about 2%	Excess distributions go to VGSH

The full "never" list (the ticket's three plus): never sell the hedge to buy stocks; never lengthen duration to bet on rates (TLT alone, EDV/GOVZ/ZROZ, foreign government bonds whose currency does not match a USD promise); no thematic, leveraged, inverse or single-country funds; no trading for ranking or in the last days; never write or edit a note after the fact; the advisor never places trades (R-W87); cash never negative (no margin); never push a position above the Session Rules limit; never re-weight to the June fact-sheet durations.

6. Substitutes if a pick is not in WInS

If this fails	First substitute	Second	2025-26 list
TLH	IEF 41.0 / TLT 25.0 (capped at 25%: IEF 24 / SPTL 24 / VGIT 9.5 / VGLT 8.5)	SPTI + SPTL, or VGIT + VGLT, at duration weights	TLT yes; others no
IEF	VGIT (re-weight to about 10y with TLH)	SPTI	no / no
SPTL (capped branch)	VGLT	TLT (duration longer; re-run section 10)	VGLT no; TLT yes

If this fails	First substitute	Second	2025-26 list
VT	VTI 62% + VXUS 38% of the equity	ITOT + IXUS	VTI, VXUS yes
VGSH	SHY	a WInS-listed ~2-year U.S. Treasury note	SHY yes

No sector rule exists this season ("There is no required sector allocation or minimum number of sectors", SMAApply, VERIFIED-PRIMARY); a broad fund already holds all 11 sectors, so no sector-fund fallback is needed.

7. Checks in WInS before the first order

- [] 1. Official account, not practice (practice trades were reset, R-W77). A student places every trade (R-W87).
- [] 2. Session Rules screenshot (gate box c). Pick the branch.
- [] 3. Re-read SMAApply Trading Details and FAQ that morning; note any change to cash, trades, volume or commissions.
- [] 4. Search each ticker: it must show as an **ETF**, with the issuer's full name (BIL is now "State Street SPDR..."; SPTL is "State Street SPDR Portfolio Long Term Treasury ETF"). All picks are U.S.-listed.
- [] 5. **Volume:** read **today's** volume in WInS for each ticker; the rule is "no more than twice a security's current daily trading volume" (SMAApply, VERIFIED-PRIMARY). An overnight order fills at the open, when the day's volume so far is smallest. If an order is rejected, split it across days. Never switch to a thinner fund to get it through.
- [] 6. Commission shown (\$25 per ETF trade, ASSUMPTION that ETFs count as "stock"); cash stays positive after all orders.
- [] 7. Note field: check for a character limit (none published). Write the note before pressing submit; copy it word for word into the decision log straight after.
- [] 8. Record which U.S. Treasuries appear in the bond drop-down (for the Final Report only).
- [] 9. Check whether WInS pays interest on cash (decides cash vs BIL for money that waits more than about 3 weeks).
- [] 10. Recompute the IEF/TLH weights from the issuer pages on the trade date; record numbers and time.
- [] 11. Anything ambiguous: ask Wharton through the official channel. Never contact Laura (R-W16).

8. Laura's long-term instrument map (not WInS)

Date (start of year)	Money	Instruments	Rule
Jan 2027	\$300,000	Ten Treasury STRIPS, \$50k face each, maturing each Nov 15 2032-2041, ~\$294,387 (detail in the Final-Report file). The ~\$5.6k left over waits in T-bills (SGOV/BIL)	If rates fall first and the ladder costs more than \$300k, buy the longest rungs first and the rest from the 2028 deposit
Jan 2028	+\$150,000	Growth money (~\$158k): 60% VT / 40% VGSH (provisional)	Rebalance once a year, band 55-65
Jan 2031	Growth money	Part of it moves into a 2-year Treasury note maturing 2032-12-31 (the promised amount for co-sponsors). The share is an open team parameter (brief section 9; the model uses 80%). It never appears in a WInS note	The promised amount is bought, not forecast
Jan 2033	Ladder (~\$394.9k market value on forwards) + note + rest	The ladder is the operating reserve; the note matures into cash for the facility decision	Payments come from maturing rungs only
2033-2042	Reserve runs down	Each Nov-15 rung waits about 47 days in T-bills, then pays Jan 1	Nothing sold early

- **Caveat (red-team finding 9):** in about 3 of 10 rate paths the real ladder needs part of the 2028 deposit; certainty for the first rungs then depends on that deposit arriving (ASSUMPTION-based: zero drift, 7.24% volatility). The joint tail (rates fall **and** the deposit is missing) is passed to Phases B-E. Not a week-1 note topic.
- **2031 note maturity:** Treasury's auction data show the December 2025 two-year note matures 2027-12-31 (CUSIP 91282CPS4; api.fiscaldata.treasury.gov, VERIFIED-PRIMARY 2026-09-27). That the pattern

holds in December 2030 is an ASSUMPTION. Alternates: the Nov 15 2032 principal STRIP (912821KC8); iBonds IBTM (Dec 2032).

- **Model inputs to align before the Final Report (finding 10):** `strategy_mc.py` should use JPM AC World (7.00% / 16.78%) for equity and an explicit short-Treasury ASSUMPTION (about 3.5-3.9%) for sleeve bonds, so the model holds what WInS holds. Effect on the median is under about \$2k (S2, S3).

Changes after red team

#	Severity	Finding (short)	What S3 did
1	Blocking	No gate before Day-1 orders	Fixed. Five-box gate at the top of the ticket; target "week 1 once the gate is passed (e.g. fills by Fri Oct 2 ET)", not Day 1
2	Blocking	"Bonds mature when Laura pays" false; ranges misquoted	Fixed. Ranges corrected (IEF May-2033 to Aug-2036; TLH Feb-2037 to Aug-2046; TLT May-2044 to Aug-2056); 74.7% after-2042 figure stated; Note-1 checklist rewritten to "closer to / moves like / stand-in", with a "never claim" line. Same wording fixed in S1's summary, fund table, scale check and "What this teaches" (marked in S1)
3	Important	42.4% TLH blocked by a 25% limit; fallback too complex	Fixed. One pre-chosen rule for any cap (IEF 23.6 / TLH cap-1 / SPTL rest; VGLT alternate), computed in script section 10. Individual 2041 bond, four-fund mix, two-issuer mix and IBGA rows removed. Added a 2% sliver rule and the continuous-limit ASSUMPTION
4	Important	VT on Day 1, VGSH waits; "80%" in Note 3; "floor proxy" overclaims	Fixed. VT and VGSH bought together at a provisional 60/40; VGSH relabelled "short-Treasury part of the growth money"; no percentage in Note 3; the third note is the VGSH buy or a genuine later decision
5	Important	Option (ii) oversold	Fixed. (ii) and (iii) presented as equals with a stated test; three-element sentence for (ii); "one set of numbers" replaced by "same rules and rough shares"; option (i) dropped (no plan version holds 34% equity)
6	Important	538-line dossier, not a ticket	Fixed. One-page ticket at the top; \$100k/\$500k columns, A.4, D.1b rows, D.2 sector table, D.3, drift illustrations and all but one twist comparison cut; E.2 and E.5 moved to <code>S3_final_report_ladder_and_reserve.md</code>
7	Important	Could read as a generic Treasury + world-index portfolio	Fixed. Checklists require promise-sizing (~\$292k) and the no-sell rule in the hedge note, ban "stable"/"reduce volatility", frame equity as 60% of the free money; "funding purposes" used once (section 2); statistics degree used once (section 1)
8	Minor	9.90 anchor justified wrongly; mixed spot/forward tracking	Fixed. New reason (issuer vs model durations); 8.92 discussion deleted; one reconciling line for brief section 14; "about 10 years" rule; figures now forward-consistent (\$1,544 / \$451). S1's own <code>tracking()</code> left unchanged (optional; its ranking is the same)
9	Minor	Ladder fits under \$300k less often (30.7%)	Fixed. One caveat line in section 8 and in the Final-Report file; joint tail passed to Phases B-E
10	Minor	Model holds different assets	Passed on. Section 8 states the two model changes for the main loop; S3 does not edit <code>strategy_mc.py</code> (verified script, outside S3's path)
11	Minor	Volume check used 30-day averages	Fixed. Check 5 now reads today's volume in WInS; split across days if rejected; never a thinner fund
12	Minor	Calendar missed the U.S. clock change	Fixed. Nov 2 open 1:30 a.m. AEDT; Nov 6 close 8:00 a.m. AEDT Sat Nov 7
13	Minor	Stale statements, 2027 leftover, label, S1 formula and TLT range	Fixed. Stale CLAUDE.md/brief statements and open issue deleted; one leftover rule (T-bills in 2027, so (iii) holds VT 0%); label now "PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK" alongside the brief section 4 wording; S1 formula relabelled "IEF weight" and TLT range corrected
14	Minor	±0.25y band narrower than duration uncertainty	Fixed. Band kept; "never trade to fix a smaller gap" and "never quote two decimals in a deliverable" added

Nothing was rejected.

Open issues for Phases B-E and the team

1. Team vote on lock-early, then (ii) vs (iii) (gate boxes a and b).
2. Session Rules position limit and the day-trading definition; WInS availability of TLH, SPTL, VT, VGSH; note-field limit; interest on cash.
3. Sleeve equity 50/60/70% and the 2031 share: both open; 60/40 is provisional.
4. Joint tail: rates fall before January 2027 **and** the 2028 deposit is missing or late.
5. Model alignment (AC World equity; short-Treasury sleeve bonds) before the Final Report.
6. "First trade by Oct 10": still UNVERIFIED.

Sources (all accessed 2026-09-27)

- **SMApply (VERIFIED-PRIMARY, curl):** <https://wghsinvcomp.smapply.us/res/p/trading/> ; <https://wghsinvcomp.smapply.us/res/p/faqs/> (via `phase_A/case_register.md` R-W56-R-W94).
- **iShares (VERIFIED-PRIMARY, re-read 12:45 UTC):** <https://www.ishares.com/us/products/239456/ishares-710-year-treasury-bond-etf> (IEF), [/239453/ishares-1020-year-treasury-bond-etf](https://www.ishares.com/us/products/239453/ishares-1020-year-treasury-bond-etf) (TLH), [/239454/ishares-20-year-treasury-bond-etf](https://www.ishares.com/us/products/239454/ishares-20-year-treasury-bond-etf) (TLT), [/239452/ishares-13-year-treasury-bond-etf](https://www.ishares.com/us/products/239452/ishares-13-year-treasury-bond-etf) (SHY). Holdings maturities: `wins_now/S1_fund_holdings_snapshot.csv` (iShares holdings as of 2026-09-24, per S1).
- **State Street (VERIFIED-PRIMARY, re-read 12:42 UTC):** <https://www.ssga.com/us/en/intermediary/etfs/spdr-portfolio-long-term-treasury-etf-sptl> (SPTL); <https://www.ssga.com/us/en/intermediary/etfs/state-street-spdr-bloomberg-1-3-month-t-bill-etf-bil> (BIL, S3/S4 earlier).
- **Vanguard (VERIFIED-PRIMARY, re-read):** <https://investor.vanguard.com/vmf/api/{VT,VGSH,VGLT,VGIT}/{profile,expense,characteristic,price}>; VT fact sheet <https://fund-docs.vanguard.com/F3141.pdf> (June 30, 2026; via S2/S4).
- **U.S. Treasury (VERIFIED-PRIMARY):** <https://www.treasurydirect.gov/marketable-securities/treasury-notes/> ; https://api.fiscaldata.treasury.gov/services/api/fiscal_service/v1/accounting/od/auctions_query (2-year notes).
- **Repo files (VERIFIED-REPO-FILE):** `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (lines 16, 70-74, 114, 129-133); `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (lines 14, 20, 30, 36, 44, 53); `2026_WGY_Investment_Policy-FINAL.txt` (line 51); `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt` (lines as cited); `research/insight_v1/phase_A/wins_week1_guardrails.md` (A4: user guide, Session Rules, Stock-Trak 25% default and note permanence); `research/insight_v1/wins_now/S1_treasury_sleeve.md`, `S2_growth_sleeve.md`, `S4_red_team.md`.

What this teaches

1. **Check the story against the holdings.** "The bonds mature when she pays" sounded right and was wrong for three-quarters of TLH. The fund was still the right choice, for a humbler reason: its bonds sit closer to her dates than the alternative's. A note that a reader can check must say only what the holdings show.
2. **Decide the rule before the surprise.** A position limit could appear at 11:30 p.m. on the first trading night. Choosing one simple fallback in advance ("TLH one point under the cap, SPTL takes the rest") turns a scramble into a look-up, and costs under 1% of the hedge in tracking.
3. **Approval comes before orders.** Trade notes are permanent. The cheapest protection is a gate: the team agrees, checks the rules, and writes its own notes, and only then trades. Waiting a few days costs nothing that is judged.
4. **Honest precision is "about 10 years".** When two fair ways of measuring the same fund differ by 0.28 years, quoting 9.90 versus 10.16 is decoration. Simple, checkable words carry more weight with a reader than extra decimals.

Trading-now brief: read this before the first WInS order

Every ticker named in this brief is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK (listed in the team's WInS account, and inside the Session Rules position limit; alternates are in the ticket, section 12).

Agent T1 (Trading-Now editor), insight_v1 run, written 2026-09-28 about 01:00 UTC; revised the same day after the T2 red team (research/insight_v1/wins_now/T2_red_team.md; see "Changes after T2 red team" at the end). U.S. trading for the season opens at 9:30 a.m. ET today (13:30 UTC; 11:30 p.m. AEST). **Serves: Trading Notes (TN, due Oct 23) and, where noted, the IPS (Nov 6). No Final Report work (brief s17).** AI-generated research: decisions to take, evidence and checklists. It is not text to submit; the six students decide and write every note, reflection and IPS sentence (Wharton AI policy). The orders themselves are in research/insight_v1/wins_now/securities_and_allocation_v1.md (the ticket). No relayed user messages arrived during this task. T1 did not commit anything.

Labels: **VP** verified on the primary page; **VRF** verified in a repo file; **SNIP** snippet or secondary, unverified; **ASM** assumption or model output (not a forecast); **INT** our interpretation (a judgement, not a fact). A file name means "see that file", with the label it carries there. Where a specialist file ends with "Audit corrections", the corrections are applied here.

1. Decisions to take before the first order (all team votes; record first names, date and reason in the log)

#	Decision	Recommendation (T1)	Must be decided by
1	Adopt "lock early" (gate box a)	Adopt. Growth-first misses a payment in 3.2% of modelled paths, 13.7% if the 2028 deposit is \$75k and 40.8% if it is \$0 (strategy_mc.py , VRF/ASM). Lock-early never misses by construction once the ladder is bought. A full lock is the only design whose certainty rests on market prices on the purchase date, not a forecast (and, in about 1 in 3 modelled rate paths, partly on the 2028 deposit, because the ladder may cost more than \$300k in January 2027; AX1b item 1, ASM); partial locks fail in 13-26% of paths if the deposit is missing (D3 M243, ASM)	Before any order
2	Which book WInS shows: (ii) the mix after both deposits, scaled to \$300k, or (iii) the literal January-2027 book (gate box b)	(ii), medium-low confidence; a close call after T2 added the official hints for (iii). Both sides are in section 3. Choose (iii) instead if the team cannot state the (ii) scaling in its own words in about 30 IPS words, or if it reads Wharton's "\$150,000 will not be added to WInS" as naming the date WInS shows	Before any order
3	If (ii): growth split 50/50 or 60/40 (world stocks / short Treasuries)	50/50, low confidence (downgraded after T2: D6's rule does not separate the two without a fee assumption), decided now. It is a choice of spread, not of median. Both evidence sets are in section 4. If the team cannot agree, buy the recorded provisional 60/40, say "provisional" in the growth note, and name a decision date	Before the VT order
3b	If (ii): floor timing (January 2028 or January 2031) and the growth money's bonds (VGSH or a dated Dec-2032 Treasury fund) (D3 AX1 C6, C18, VRF; T2 finding 5)	Record the team's choice before the VGSH order. If undecided, the VGSH note uses only the "facility contribution and flexibility" need and never refers to a later promised amount. Both questions also go to the IPS list (section 6)	Before the VGSH order
4	Position-limit branch, including a cap under 25% (gate box e)	Use the pre-set branch table in the ticket (section 3 there). Also pre-decide: under a cap below 25%, (iii) needs 5-6 Treasury funds, so switch to (ii), unless a separate, higher bond limit lets one listed Treasury carry the excess	Before any order
5	Note format and words (gate boxes c and d)	Test the note-box length at zero risk before any real note (ticket gate box c). Until a limit is seen or ruled out, draft every note to 300 characters or fewer, using the ticket's trim order. The team chooses its own label words (D9's are proposals) and logs the AI source; use the banned-word list (section 2). A second student restates each note's role and checks that no clause repeats ticket wording before it is submitted	Before any order

#	Decision	Recommendation (T1)	Must be decided by
6	Optional dated Treasury "rung" (one bond near a payment date)	Under (iii): buy it if the Bonds drop-down lists a suitable bond. Under (ii): optional; skip unless the team wants it (D1 M044, corrected by AX1b)	At the gate session
7	Data owner for the "tested" evidence	One named student screenshots WInS position values on the fill date and at the Oct 20 close (D7 M002 as corrected by AY2)	Before any order

Timing (ASM team rule): there is no reason to trade at today's open. SMaply says "The competition does not require frequent or same-day trading" (VP, via `phase_A/wins_week1_guardrails.md`), and WInS gains are not judged (Guide p.3, VRF). Target fills by **Fri Oct 2 ET** (11:30 p.m. AEST that day), with a hard stop of **Fri Oct 9**. The "first trade by Oct 10" claim is a 2023-24 rule (SNIP); "required trading activity" is undefined on public pages (VP). Read the logged-in Trading Details page before the first order. Earlier fills also give the "tested" window more days: about a 52% chance of a 10bp-or-larger rate move by the Oct 20 close for an Oct 2 fill, about 40% at the Oct 9 hard stop (T2 script [4], ASM driftless model). Orders entered while the U.S. market is closed fill at the next open (SMaply Trading Details, R-W62, VP), so recompute every share count from the last close before entering orders.

2. Findings that change what you do in WInS or write in a note this week

1. **Notes are permanent and may be cut at 300 characters.** Notes cannot be edited, only added to (Stock-Trak, VP vendor page, not season-specific). A sister Stock-Trak product caps notes at "300 characters" (VP for that product; **UNVERIFIED for WInS**, D12; evidence mixed, D12 AY1 C2). Wharton's own example is 413 characters (VRF count), which is counter-evidence. *Decision:* test the box length **at zero risk before any real note** (ask Stock-Trak Live Chat or Wharton, or type a long test text on the order-review screen and do not press Confirm; ticket gate box c). Until a limit is seen or ruled out, draft to 300 characters, and use the ticket's trim order (section 5): keep the need (with the (ii) scaling marker), the dated fact and the Guide role word; move the risk/rule and the team label to the reflection (D12 AY1 C1). Never finish a cut sentence with an added note (it carries a later timestamp, and the note is quoted "exactly as it appears in WInS", TN guide L44, VRF). *Deadline:* the first order.
 - **Conflict resolved (D9 vs D12).** D9 asks for five elements in about 90 words; D12 fits four in 300 characters.
 - Resolution: a four-element core that must fit in 300 characters (role, Laura's dated need, one dated research fact, the risk or rule), with a trim order if it does not fit. The ticket (section 5) lists the elements.
 - The trade record already shows the ticker, side and quantity, so the note need not repeat them.
 - Add D9's extra detail only if the box visibly allows it. 2. **Under option (iii) you need a named third trade, or you may have only two executed trades on Oct 23 (AX2 blocking, D10 C1).** The TN guide needs three notes "from trades your team executed in WInS" ("Yes, we will verify this", L42/L53, VRF).
 - With no cap, (iii) is two orders (IEF, TLH).
 - The plan's own rule supplies the third: the 2027 leftover "waits in T-bills". So buy SGOV or BIL as part of the same gate plan, so that it reads as implementation, not a note-making trade.
 - **The leftover is real only at trade-date prices (T2 finding 1, blocking for (iii); fixed in ticket section 2).** On the trade date, re-price the ladder (A2 script plus about \$2.1k for the Nov-15 dates; hand check about \$294,400 plus \$292 per 1bp fall since Sep 25, ASM). Hedge = that cost. T-bill leg = \$300,000 minus the cost, minus the \$1,000 float, minus commissions. **If the leg is under \$1,000** (the curve about 12bp or more below Sep 25: about an 11% chance by Oct 2 and 19% by Oct 9, ASM), (iii) has no genuine third trade unless a dated rung is listed, and the vote reverts to (ii). Never buy a token sliver.
 - **Order of entry under (iii):** IEF and TLH first; the T-bill order only after both show "Filled", sized to the cash actually left. The \$1,400 float is used up by a 5bp overnight fall (about 1 day in 8, ASM), so entering all three orders at once risks the T-bill order being rejected.
 - The SGOV/BIL note gives the leftover as a dollar amount at the trade-date price, never "about 2%".
 - Honest cost: it earns about \$18 in the WInS window against a \$25 commission (ASM, T1 script [5]). It is justified by the plan (about \$220 over the real one-year wait), never by WInS profit.

- **Deadline:** the gate vote. 3. **Decide the book with both sides visible (section 3).** No official rule picks a date (D10 M005, VRF). The frozen Nov-6 portfolio must "reflect the strategy established in your IPS" (IPS guide L75, VRF). So the IPS names the date WInS shows under either choice. The "middle path" is dropped: a ~1% VT sliver adds a weak note and reverses the T-bill rule (D10).

2. Words that must never appear in a note (audit corrections AX2 and AY2).

- "known" or "locked" for the ~5.2% the curve implies for 2028-33. It is market-implied, not lockable, and forwards are derivatives (D2 C1).
- "halves AI exposure": top-10 concentration falls about 44%, and VT holds Asian chip makers outside its top ten (D2 C5).
- "half = the promise", "the 2031 floor", "already owned" or "the amount she can promise co-sponsors" for VGSH (D2 C6; D9 AY2 C1).
- "match"/"matched", "mature in her payment years", "stable", "safe", "guaranteed", "risk-free", "operating reserve" for WInS funds (S4 finding 2; D9 M062). Three-quarters of TLH's bonds mature after 2042 (VRF, S1 snapshot).
- "reduce volatility" for the **promise money** (S4 finding 7; D9 M012). For VGSH, the idea of lower swings in the growth money is allowed only with its number beside it, presented as the ordinary effect of holding Treasuries in the growth money, not a VGSH property (D12 AY1 C10; T2 finding 14). Wharton's own example note uses the phrase (TN guide L36, VRF).
- "the rule picks 50% under both forecasts" unless the fee assumption is stated (D6 AY1 C1; see section 4).
- "bought in January 2027" or "fits inside \$300,000" stated flatly: the real ladder would cost more than \$300k in **about 1 in 3** modelled rate paths ("1 in 4" is for the idealised exact-date ladder; AX1b item 1, ASM).
- **No Laura quotes** in any note or reflection (D13 rule). No "floor-first leap" trait. No statistics-degree hook in the notes; spend it at most once, in the IPS (AY2 C6).

3. **One label per job (D9 M001/M062 with AY2 fixes).** The words below are D9's **proposals** ("INT: proposal, the team may rename", D9 M001, VRF), not fixed text. The team chooses its own label words once before trade 1, logs that the proposals came from an AI source, and never switches between notes.

Label	Guide role word
Promise money (IEF, TLH, SPTL)	future funding (risk management only "measured against the payments")
Growth money (VT)	growth
Short-Treasury part of the growth money (VGSH)	risk management, then liquidity
The 2027 leftover (SGOV/BIL)	liquidity

6. "Supported, tested, or refined" is a menu, not a quota (D7, VRF: TN guide L12, L29-30; case L152).

- *Tested* lives in a reflection. The window is pre-registered: fill date to the Oct 20 close. Report the rate move (bp), the hedge's % change and the payments' % change: exactly three numbers, dated and labelled as the team's own calculation (ticket section 6). A move of 10bp or more has about a 52% chance for an Oct 2 fill and about 40% for an Oct 9 fill (T2 script [4], ASM). Never use the "83% at some close" figure: that is cherry-picking (AY2 C2).
- *Refined* is honest only as a pre-announced split decision under (ii), or an order rejected by a limit nobody could see beforehand. The pre-planned SPTL split is **not** a refinement (AY2 C1).
- Drop the blueprint's planned "discipline/rebalance trade". No sensible band fires before Oct 23 (about 0-2% even in a crisis, ASM). Never stage a trade.

7. Data capture (AY2 C5). Only the WInS screenshots must be taken on the day: position values on the fill date and at the Oct 20 close. Treasury curves can be pulled later from the treasury.gov yearly CSV.

- Value the payments with `D3_funded_status_log.py`, `A2_curve_recheck.py` or `D9_numbers.py` [3], **not** `official_curve_pv.py` (fixed to 2026-09-25).
- IEF and TLH pay monthly distributions (about 0.4%), so compare total returns or note the ex-dates. 8. **Position limit unknown; read both lines in Session Rules (A4, VP).** The 2026-27 User Guide defines the limit per "single stock"; a 2024 screenshot showed a separate bond limit of 100% [PRIOR]. The Stock-Trak default is 25% (VP, generic).
- Lower caps hurt (iii) far more than (ii). At 25%, (iii) needs 5 Treasury funds and (ii) needs 3. At 20%, (iii) needs 6 and (ii) needs 4 plus a VTI/VXUS split if VT is at 20.5%. At 15%, (iii) is not feasible with

the listed funds (T1 script [3]).

- Every capped mix stays at about 10-10.6 years (issuer numbers), with a worst twist error of 0.5-1.1% of the hedge (ASM, S4's forward-consistent method). A "twist" is short and long rates moving in opposite directions.

3. Book (ii) vs (iii): both sides and T1's call (the team chooses, gate box b)

	(ii) the mix after both deposits, scaled	(iii) the literal January-2027 book
Rules	Complies under no cap, 35% and 25% caps (D10; AX2 C2: Vanguard volumes and caps below 25% are unchecked)	Same
Trades before Oct 23	4 natural orders; survives a single trading session as written (T2 one-session test)	2, plus the named T-bill leftover (AX2 blocking fix), which exists only if the leftover is at least \$1,000 at trade-date prices (about 89% of modelled rate paths for an Oct 2 fill, 81% for Oct 9, ASM) and the T-bill order is entered after the hedge fills (T2 finding 1); a rung only if listed. Without a listed rung: three trades but two decisions (IEF and TLH are one duration-weighted decision; TN guide L11 asks for "three investment decisions"; T2 finding 4)
Official hints	The Investment Strategy criterion "uses appropriate diversification" (SMAApply L49, VRF), and SMAApply's own list of diversification kinds includes "funding purposes" (R-W71, VP): only (ii) shows three funding purposes in the WInS holdings (promise, growth, short Treasuries)	WInS cash equals Laura's first deposit, and "The additional \$150,000 contribution described in the Client Case Study will not be added to WInS" (R-W56, R-W59, VP). The case: WInS "does not determine Laura's actual portfolio value at the beginning of 2027" (case L130, VRF). Read literally, these support (iii) at least as much as (ii) (S4 finding 5, INT); WInS runs on the eve of the January 2027 purchase
What the holdings show on their own	"Her whole plan, in miniature": three jobs side by side (D10 1.4)	"Promise first": the plan's central priority is visible in the holdings themselves, before any growth money exists (D10 1.4)
Note roles	Three distinct roles: future funding, growth, risk management/liquidity (D12). This fits the TN guide's first sentence, "approach to growth, risk, liquidity, funding reliability, financial flexibility, and future cash-flow needs" (VRF)	Future funding plus liquidity only; every note is a promise or cash note (D12; D10 C1)
Truth at a date	Holds +20.5 points of equity at 60/40 (+17.0 at 50/50) versus Laura's ~0% in January 2027 (D10, AX2 C7)	True at a stated date without scaling (D10)
Words needed	Three-element IPS sentence of about 25-40 words: the post-2028 mix; scaled to \$300k; in January 2027 almost all of her real \$300k buys the ladder. Plus the stand-in element: WInS holds stand-in funds that move like her payments, not the ladder itself (D10 1.5, VRF; T2 finding 16a). Plus a short scaling marker folded into each hedge and growth note (AY2 C1; D9 M012). Otherwise the pitch rule "no stock risk until the payments are bought" looks contradicted	About 12-20 IPS words for the date sentence, including the stand-in element (D10 1.5). Plus a T-bill leftover clause (about 8-10 words) , because the SGOV note relies on the rule "any leftover waits in T-bills until 2028" (T2 finding 16b). The hedge note sizes to the payments' value on the trade-date curve in one clause (D9)
Reader risk	A reader may think the team holds about 20% stocks although the payments cost 97% of the first deposit	A reader may see about 98% Treasuries as timid for a client who "has been willing to take thoughtful risks" (case L69-71, VRF). One reflection must answer "why so little equity?" (D13c N1)
Leaning	S3 ticket: (ii). D12: a TN reason for (ii)	D10: (iii) at medium-low confidence, "not decision-ready" until C1 was fixed (AX2). S4 finding 5: present the two as equals

T1 call, re-checked after the two added rows: still (ii), medium-low confidence, and closer than before (INT).

- The TN is scored (SMAApply; case), but it may not count for semifinal selection: the Rules & Roles page says the Top 50 are chosen on the IPS and Final Reports (conflict X-8, [phase_A/case_register.md](#)). If it does count, (ii) gives three distinct roles and three distinct decisions; (iii) without a listed rung gives two decisions.
- (ii) is the only book that shows diversification by funding purpose in the holdings (R-W71), which the Investment Strategy criterion names.

- (ii) survives one trading session as written; (iii) needs a trade-date re-price, a two-step order entry and a fallback to (ii) in about 1 case in 9 to 1 in 5 (T2 finding 1, ASM).
- The extra IPS words for (ii) (about 15-20 more than (iii), before the (iii) T-bill clause) fit inside AY2's word budget, but the IPS page barely fits (AX2 C3), so (iii)'s economy may count for more than the TN if X-8 is read strictly.
- A cap below 25% would make (iii) clumsy.
- **Where (iii) wins:** literal truth at a stated date, Wharton's own hint (WInS holds exactly her first deposit, and the \$150k is never added), "promise first" visible in the holdings, and IPS economy. If the team gives those more weight, or finds the scaling hard to explain in its own words, (iii) is fully defensible, with the section 2 item 2 rules.

4. Growth split if (ii): 50/50 vs 60/40 (both evidence sets; the team chooses)

- **For 50/50 (D2, as corrected by AX2; D8/AY2; D3 M034):**
 - The median 2033 facility money barely moves: -\$1.5k to +\$0.4k.
 - The bad case (p5) improves by \$6-7.5k, and the 2031 floor's p5 by \$5-6k. The good case (p95) gives up \$11-14k.
 - The stock premium over the Treasury rate the curve implies for her dates is small and uncertain: about -1.2 to +2.4 points a year across sources (AX2 C2).
 - On one input set, stocks at 50-70% add about +\$11-14k in the middle but cost about \$41-56k at p5 versus Treasuries only (ASM).
 - The 50/50 case rests on D2's evidence alone (the four bullets above). D6's rule (hold the most stock that still leaves about a 19-in-20 chance that the growth money ends 2033 with at least every dollar Laura put in) does **not** pick 50% without a fee assumption. After its audit (D6 AY1 C1, VRF) it gives about 55-70% depending on inputs: 60-65% under JPM, 55% under Vanguard's midpoint, 65-70% with fat tails; 50% only if a yearly fee of 0.5% or more is assumed (E1 P5 found the same; the case is silent on fees). It is a reason for a choice in the 50-70% band and does not separate 50/50 from 60/40 (ASM model outputs).
- **For 60/40 (D6; D13c; D3 M034 with AX1; CLAUDE.md):**
 - It is the team's recorded provisional call.
 - D6's rule (as corrected, D6 AY1 C1) gives 60-65% under JPM with no fee; 60% is inside its 55-70% band.
 - Medians understate what equity is expected to add (D3 AX1 C4, VRF; MODEL): under JPM the **mean** gain of 60% growth-money equity over an all-Treasury benchmark is about \$6.5-9.4k, against a median of about \$3k; under Vanguard's midpoint it is about zero. (This compares 60% with 0%; the 50-vs-60 mean difference was not computed.)
 - The whole portfolio already looks cautious (17-20.5% stocks after 2028). A growth share set near the minimum "to look safe" should be avoided. The reason for holding stock is the case's "appropriate balance between pursuing growth and protecting the capital", plus the stated rule; Laura's career "take the jump" words are a tension the plan must answer (why so little equity), never a justification for a higher share (D6 AY1 C3; brief s16).
 - D3: 40-60% are all defensible.
- **T1 call: 50/50, low confidence (downgraded after T2).** The evidence does not separate the two; it is a choice of spread. The reason left for 50/50 is D2's: the median is flat across both published houses while the bad case (p5) and the 2031 floor's p5 improve, and the growth money feeds a promise made in 2031. It is also simpler to explain (stocks are exactly half of the money the promise does not need). 60/40 is equally defensible on the mean gain under JPM and as the team's recorded call. Never write that "the rule picks 50% under both forecasts" unless the fee assumption is stated.
 - The IPS reason must rest on what does not change (the 2031 promise; a flat median across houses), not on this year's valuations (AX2 C7).
 - The Guide's "should not be rewritten simply because markets move" applies after the IPS, so a research-based change before Nov 6 is allowed (AX2 C3). Never change it because prices moved.
 - Orders (ticket): 50/50 means VT 17.0% (~318 sh) and VGSH 16.0% (~833 sh); 60/40 means VT 20.5% (~384 sh) and VGSH 12.5% (~651 sh). Band: 45-55 or 55-65, measured as VT ÷ (VT + VGSH + cash float), starting at about 50% or 60% (ASM, 9/25 closes; recompute from the last close).

5. What NOT to do this week

- Do not place any order before all five gate boxes are ticked. Do not trade because the season "opened".
- Do not let the advisor place trades or appear as a decider in a note ("Students should place trades on WInS, NOT the advisor", VP; advisors "may not make decisions", VP).
- Do not buy anything outside cash, \$5+ stocks, ETFs and government bonds listed in WInS. No leveraged, inverse, crypto-linked or thematic funds, no ETNs, and no Taiwan or AI tilt (A4; D8 M081; D13c R3).
- Do not day-trade, and do not "fix" an error by selling the same day ("run with it", SMAApply FAQ, VP).
- Do not sell the promise money after a rate rise, trim it after a fall, or move money between promise money and growth money.
- Do not put a reserve size, facility amount, co-sponsor range or 2031 floor share in any note. They are not expected now (TN guide L20-22, VRF), and permanent notes would bind the IPS.
- Do not paste AI text into WInS, and do not copy ticket phrases into a note. Running the D9 draft checker counts as AI use: log it (AY2 D9 C12).
- Under (iii), do not buy a token T-bill sliver to get a third note: if the trade-date leftover is under \$1,000, the vote reverts to (ii) unless a dated rung is listed (section 2 item 2).

6. Hand-off to the IPS spec: rules to fix before Nov 6 (owner: ips_spec; serves IPS, and TN consistency)

Items raised by T2 and by the audit sections T1 had not applied. Specifications only; the team writes the words.

1. **Floor timing:** is the facility floor bought in January 2028 or January 2031? (D3 AX1 C6, VRF: the barbell is not dominated; "no barbell" survives only as a judgement.) Under (ii), settle it before the VGSH order, or keep the VGSH note away from any later promised amount (decision 3b).
2. **Growth-money bonds:** short Treasuries (VGSH-type) or a dated Dec-2032 Treasury fund? (D3 AX1 C18, VRF; any fund named is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK.)
3. **Date sentence, either option:** it must contain that WInS holds stand-in funds that move like her payments, not the ladder itself (D10 1.5, VRF; T2 finding 16a).
4. **(iii) only:** a T-bill leftover clause, about 8-10 words ("any leftover waits in T-bills until 2028" as content, D1 M004 rule 7), added to AY2's IPS word budget (`audit_judges_practice_comms.md` s5; T2 finding 16b).
5. **Growth split reason:** state D6's rule only with its fee assumption, or rest the reason on D2's flat median and better bad case (D6 AY1 C1; E1 P5). IPS: "about 50%" or "about 60%", never a decimal.
6. **Whole-portfolio stock share by stage** (about 0% in 2027, 17-20% in 2028-30, a few percent after the 2031 floor), so a first reader sees where the risk is (D6 AY1 C4 item 8; parked M033; E1 P6).
7. **Pitch/IPS item "the payments are bought in January 2027"** only with its condition: if prices have risen, the rest is completed from the 2028 deposit before any growth money (D6 AY1 C4; D8 rule 1). And the second certainty gap named in one clause: "certain" is in nominal US\$, and a fixed \$50,000 buys less each year in Taiwan (D12 AY1 C4; brief s16).
8. **2031 range method wording:** the bottom is "the amount already bought", "bought, not forecast", "in US\$, barring a U.S. Treasury default", never "guaranteed" (D5 AY1 C1, C3; D3 AX1 C3, C14); one top percentile across files (D5 AY1 C10). Method only; no dollar figures (brief s17).

Evidence files (all under `research/insight_v1/`)

`wins_now/securities_and_allocation_v0.md`, `wins_now/S4_red_team.md`, `phase_A/wins_week1_guardrails.md`, `phase_D/D1_rates.md` (+AX1b), `D2_equity_ai.md` (+AX2), `D3_quant.md` (+AX1), `D5_cosponsors.md` (+AY1), `D6_behavioural.md` (+AY1), `D7_wharton_intent.md` (+AY2), `D8_practice.md` (+AY2), `D9_communication.md` (+AY2), `D10_compliance.md` (+AX2), `D12_overflow_client.md` (+AY1), `D13c_voice_map.md`, `audit_markets_rules.md`, `audit_judges_practice_comms.md`, `phase_C/survivors.json` (tier-1 items M001, M002, M004, M005, M012, M028, M034, M044, M062, M212, M238, M247). Official files: `competition/official/2026_27/` (TN guide, Guide, case, IPS guide). New script: `scripts/T1_ticket_v1_numbers.py` (fund facts re-read 2026-09-28 on `ishares.com`, `ssga.com`, `investor.vanguard.com`). Notes on gaps (corrected after T2 finding 2): the first

version said D5, D6, D12 and D3 had no appended audit section and that `phase_D/audit_rates_D1.md` was not in the repo. That was wrong for all but D3: file times show the AY1 sections on D5, D6 and D12 and `audit_rates_D1.md` saved at 00:34-00:37 UTC, 11-14 minutes before T1's files (00:48). Only D3's AX1 section (about 01:00) came later. The AX1b section appended to D1 (24 items) is the digest of `audit_rates_D1.md`, so D1's corrections were applied. After T2, T1 re-read D5 AY1, D6 AY1, D12 AY1 and D3 AX1; what changed is listed in "Changes after T2 red team" below. Added evidence files: `audit_client_cosponsors.md`, `audit_rates_D1.md`, `audit_rates_quant.md`, `phase_E/E1_change_proposals.md` (P5, P6; written after T1, not yet audited), `wins_now/T2_red_team.md`, `scripts/T2_red_team_checks.py`.

Changes after T2 red team (2026-09-28)

T2 found 1 blocking, 7 important and 12 minor issues in this brief and the ticket. All blocking and important ones are fixed; the minor ones are fixed where they touch this file. The ticket has its own list.

T2 #	Severity	What changed in this brief
1	blocking (iii)	Section 2 item 2: the (iii) leftover is re-priced on the trade date; T-bill leg formula; "under \$1,000 = no genuine third trade unless a rung is listed, revert to (ii)"; IEF and TLH first, T-bill only after both fill; dollar leftover in the note. Timing: recompute share counts from the last close. Section 5: no token sliver
2	important	Section 4: D6's rule restated as 55-70% (50% only with a 0.5%+ fee); the 50/50 case rests on D2 alone; D3 AX1 C4 (mean vs median) added to the 60/40 side; T1's 50/50 lean downgraded to low confidence; "Notes on gaps" corrected
3	important	Section 3: rows "Official hints" and "What the holdings show on their own"; the TN-first reason re-worded around X-8; lean re-checked (still (ii), medium-low, closer)
4	important	Section 3, (iii) "Trades before Oct 23": three trades but two decisions without a rung
5	important	Decision 3b (floor timing, growth-money bonds before the VGSH order); section 6 items 1-2
6	important	Decision 5 and section 2 item 1: zero-risk length test before any real note; trim order
7	important	Section 2 item 6: a tested reflection carries exactly three numbers (details in ticket section 6)
8	important	Decision 5 and section 2 item 5: labels are D9's proposals; the team picks its own words, logs the AI source; second-student check against ticket wording
10	minor	Availability label line under the title
12	minor	Section 4: band denominator
14	minor	Section 2 item 4: "reduce volatility" banned for the promise money only
15	minor	Decision 1: "market prices on the purchase date ... partly on the 2028 deposit in about 1 in 3 paths"
16a-b	minor	Section 3 "Words needed" (stand-in element, (iii) T-bill clause); section 6 items 3-4
18	minor	Timing paragraph and section 2 item 6: 52% (Oct 2 fill) vs 40% (Oct 9)
20	minor	"Notes on gaps" corrected; fill-at-next-open cited to Trading Details (R-W62)
9, 11, 13, 16c, 17, 19	minor	Ticket only

What else changed after re-reading the audits T1 had missed (beyond T2's list):

- D6 AY1 C3: Laura's "take the jump" words removed as a reason for a higher stock share (section 4); they are a tension to answer.
- D6 AY1 C4, D12 AY1 C4, D5 AY1 C1/C3/C10, D3 AX1 C3/C14: IPS items handed off in section 6 (items 6-8).
- D6 AY1 C2: already consistent (under (ii) the promise note gives rate sensitivity and the scaling, not ~\$292k; the dollar cost is (iii)-only or reflection material).
- D12 AY1 C3: already addressed by the named T-bill trade, now with T2's re-price rule.
- D3 AX1 C8 ("about 1 in 3"; the ladder cost more than \$300k on almost every day of 2026): already used; unchanged.

What this teaches

A decision made at the moment of action is judged by what you knew and wrote then, so read the corrections, not just the conclusions: one stale sentence about a rule tilted a vote here, and the audit that fixed it was already in the repo. This week that means three things. Settle the few choices that make notes permanent before you trade: which book, which split, which words. Plan your evidence in advance, and let

the market supply the test instead of inventing a trade. Say plainly what your holdings are not: stand-ins, not the ladder itself.

WInS ticket v1: orders for options (ii) and (iii), position-limit branches and the trading-notes plan

Agent T1 (Trading-Now editor), insight_v1 run, written 2026-09-28 (before the first U.S. open of the season, 13:30 UTC); revised the same day after the T2 red team (see "Changes after T2 red team" near the end). This file **supersedes** wins_now/securities_and_allocation_v0.md for trading. That file's internal heading already said "v1" after the S4 red team; this is the next revision.

Serves: WInS trading as the source of the Trading Notes (TN, due Oct 23), plus the one IPS consistency point that depends on it (brief s17). No Final Report work.

Status: PROVISIONAL. The team has not voted on the strategy. This is AI-generated research: orders, numbers, checklists and "must contain / must not say" lists. **It contains no note, reflection or IPS text.** The students decide and write every word.

Every security here is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK. There is no separate approved list this season; "Any ETF available on WInS" is allowed (SMAApply Trading Details, VP). No relayed user messages arrived during this task. Read the decisions first in research/insight_v1/phase_D/trading_now_brief.md.

Numbers: .venv/bin/python research/insight_v1/scripts/T1_ticket_v1_numbers.py (run from the repo root). Section numbers in brackets below point to its output.

- Prices are closes of **2026-09-25** (Fri), re-read on the issuers' pages on 2026-09-28 (VP). SPTL, SPTI and BIL show "as of Sep 24".
- Durations are issuer "effective duration": IEF 6.86y and TLH 11.58y as of Sep 25 (TLH was 11.59y on Sep 24).
- Why the target is 9.90 and not the 10.16-year spot duration (S4 finding 8, VRF; added after T2 finding 18): issuer durations sit about 0.2-0.3 years below full-revaluation durations, so a mix set to 9.90 on issuer numbers is about 10.1 years in full revaluation, close to the payments' 10.16-year spot duration (ASM model). In notes: "about 10 years".
- Share counts are rounded down. **Recompute every share count from the last close before entering orders** (T2 finding 1): the counts below use the Sep-25 closes and are illustrations only.
- The ladder figure "\$294,387" is a **model price** on the Sep-25 par curve for Nov-15 STRIPS dates, not a STRIPS market quote (D3 AX1 C25, VRF). Say "the payments' value on the [date] Treasury curve (our calculation)".

Labels: VP = verified on the primary page. VRF = verified in a repo file. SNIP = snippet or secondary, unverified. ASM = assumption or model output, not a forecast. INT = our interpretation.

Terms:

- *Promise money / hedge*: the Treasury funds that stand in for the ten fixed \$50,000 payments (2033-2042).
- *Growth money*: everything the payments do not need.
- *Duration*: roughly the % a fund falls if rates rise 1 percentage point. About 10 years here.
- *Position limit / cap*: the most WInS allows in one security.
- *Twist*: short and long rates moving in opposite directions.
- *Rung*: one Treasury bond that matures near one payment date.

THE TICKET

0. Gate: tick all five before the first order (unchanged in spirit from v0; items added)

- [] **(a) Lock-early vote** recorded: first names, date, reason. If the vote is "no", stop; this ticket does not apply.
- [] **(b) Book chosen:** (ii) or (iii). If (ii), also the growth split (50/50 or 60/40, or "provisional 60/40, decision by [date]"). T1 recommends (ii) at medium-low confidence and 50/50 at low confidence (after

T2 the two splits are close to a coin-flip on evidence; the reasoning is in the brief). Also record:

- **If (ii):** before the VGSH order, the team's choice on two points the D3 audit reopened (AX1 C6, C18, VRF): "is the facility floor bought in January 2028 or January 2031?" and "growth-money bonds in short Treasuries (VGSH) or a dated Dec-2032 Treasury fund?". If either is undecided, the VGSH note uses only the "facility contribution and flexibility" need (section 5) and never refers to a later promised amount (T2 finding 5).
- **If (iii) and no dated rung is listed:** the team accepts, in the vote, that two of its three notes (IEF and TLH) describe one duration-weighted decision, although the TN guide asks for "three investment decisions" (L11, VRF). Record that acceptance in the decision log (T2 finding 4). If the team will not accept it, choose (ii).
- [] **(c) Session Rules screenshot** (Portfolio Summary > Session Rules). Record:
 - the single-security limit **and** any separate bond or security-type limit;
 - the day-trading definition (the "?" tooltip);
 - trades allowed and commissions.

Also read the logged-in Trading Details page for any "required trading activity" and first-trade date.

- ****If the logged-in pages show an activity minimum (a number of trades) or a cap on a whole security type (all ETFs, all bonds): stop and ask Wharton through Contact Us before any order. Never add trades to meet a count****
 (T2 finding 13; "required trading activity" is undefined on public pages, Rules & Roles, VP).
 - ****Note-box length, checked at zero risk before any real note**** (D12 AY1 C2, VRF; T2 finding 6): ask Stock-Trak
 Live Chat or Wharton (Contact Us), ****or**** type a long test text (about 450 characters of filler) on the order-review screen, see whether it is cut or counted, then clear it and do ****not**** press Confirm. Record the result. The 300-character plan stays until a limit is seen or ruled out (UNVERIFIED for WInS; evidence mixed: a sister product caps at 300, Wharton's own example is 413 characters).

- [] **(d) Each WInS note drafted by a student in the student's own words** from section 5, to **300 characters or fewer** (or the length found in box (c)). A second student restates its role in one sentence (D12), checks that **no clause repeats this ticket's wording** (T2 finding 8), and the draft is pasted into a copy of the decision log first. Label words ("promise money", "growth money" ...) are D9's proposals: the team picks its own and logs that the proposals came from an AI source.
- [] **(e) Branch picked in advance** from section 3, including the cap-under-25% rule and the cap-to-option switch.

When: aim for fills by **Fri Oct 2 ET** (11:30 p.m. AEST that day); hard stop Fri Oct 9 (ASM team rule). Orders placed while the market is closed fill at the security's opening price when the market reopens (SMAApply **Trading Details**, R-W62, VP), so an order sized at the last close can cost more at the open. Recompute every share count from the last close before entering orders.

- **Option (ii):** all orders go in **one session**, promise money first. The \$3,000 cash float survives about a 12bp overnight rate fall plus a 1% stock rise (T2 script [3], DER/ASM; about a 0.4% chance of failing with counts from the last close, 8-9% if the Sep-25 counts are used).
- **Option (iii):** enter IEF and TLH first. Enter the T-bill order **only after both show "Filled"**, sized to the cash actually left (section 2). Same U.S. session if someone is awake, otherwise the next one; it is the plan's own rule, not a note-making trade. Reason: the (iii) float (about \$1,400) is used up by a 5bp overnight fall (about 1 day in 8; T2 script [3], ASM), and the order entered last would be the one rejected (margin is banned, VP).

Screenshot every position on the fill date: this starts the "tested" window. A fill by Oct 2 gives about a 52% chance of a 10bp-or-larger rate move by the Oct 20 close; a fill at the Oct 9 hard stop about 40% (T2 script [4], ASM driftless model). Either is fine: "tested" is a menu option, not a quota.

1. Option (ii): the mix Laura's portfolio reaches after both deposits, scaled to \$300,000

The hedge is 66% of the book (the plan's 65.9% promise share after 2028, ASM). VT equals the plan's whole-portfolio stock share (20.4% at 60/40, 17.0% at 50/50). The 1% cash float comes out of the short-

Treasury part.

#	Order	Split 60/40: weight, \$, ~shares	Split 50/50: weight, \$, ~shares	Role word (Guide) and team label
1	BUY IEF	23.5%, \$70,500, ~783	same	future funding: promise money, part 1
2	BUY TLH	42.5%, \$127,500, ~1,365	same	future funding: promise money, part 2
3	BUY VT	20.5%, \$61,500, ~384	17.0%, \$51,000, ~318	growth: growth money
4	BUY VGSH	12.5%, \$37,500, ~651	16.0%, \$48,000, ~833	risk management, then liquidity: short-Treasury part of the growth money
	Cash	1.0%, \$3,000 (about \$2,900 after 4 x \$25)	same	liquidity: cash float

- Hedge: IEF share = $(11.58 - 9.90) / (11.58 - 6.86) = 35.6\%$ of the hedge [1]. It is "about 10 years" on issuer numbers, and its worst 50bp twist error is about \$1.5k, 0.5% of a \$292k hedge [2] (ASM).
- Band inside the growth money, measured as **VT ÷ (VT + VGSH + cash float)** (the cash float comes out of the short-Treasury part, so it counts as growth money; D3 AX1 C17 and T2 finding 12): VT at **45-55%** (50/50) or **55-65%** (60/40). With whole shares it starts at about 50% (49.9%) or 60% (60.3%) (T2 script [1], DER). Check weekly and trade VT against VGSH only. It is very unlikely to fire before Oct 23 (about 0-2% even in a crisis; D7 as corrected by AY2, ASM).

2. Option (iii): the literal January-2027 book, with its named third trade

The hedge is set on the trade date, not fixed at 98% (T2 finding 1, blocking; fixed here). On the day of the orders, re-price the ladder: run `.venv/bin/python research/insight_v1/scripts/A2_curve_recheck.py` (it downloads the live treasury.gov curve and prints "[1] Ten payments valued at 2027-01-01 on the [latest] curve"; exact-date method) and add about \$2,100 for the real Nov-15 STRIPS dates (the gap was \$2,123 on the Sep-25 curve, ASM). Hand check: about \$294,400 plus about \$292 for every 1bp the curve has fallen since Sep 25, using the 10-year par yield as the gauge (minus the same for a rise; parallel-shift ASM from S4 finding 9's model prices). Then:

- **Hedge (IEF + TLH)** = that ladder cost, split with the section 10 formula.
- **T-bill leg** = \$300,000 minus ladder cost minus \$1,000 float minus 3 × \$25 commissions.
- **Pre-set rule: if the T-bill leg is under \$1,000** (the curve about 12bp or more below Sep 25), (iii) has **no genuine third trade** unless a dated rung is listed (section 7). Then the gate vote reverts to (ii). **Never buy a token sliver.** (T2 script [2], ASM driftless 4.45bp/day: about an 11% chance by an Oct 2 fill and 19% by Oct 9; no leftover at all about 6% / 13%; the ladder tops \$300,000 after about a 19bp fall.)

Illustration at the Sep-25 closes only (model ladder cost \$294,387; recompute everything on the trade date):

#	Order	Weight	\$	~Shares	Role word and team label
1	BUY IEF	34.9%	\$104,700	~1,163	future funding: promise money, part 1
2	BUY TLH	63.1%	\$189,300	~2,027	future funding: promise money, part 2
3	BUY SGOV (alternate BIL, then SHV), only after 1 and 2 show Filled	~1.5%	~\$4,500	~45 SGOV / ~49 BIL	liquidity: the 2027 leftover
	Cash	~0.5%	~\$1,400 after 3 x \$25 (keep at least \$1,000)		liquidity: cash float

- The T-bill ETF is bought as part of the same gate session plan (the next U.S. session at the latest), sized to the cash actually left after the hedge fills, minus the \$1,000 float and its \$25 commission. If that is under \$1,000, the pre-set rule above applies. It is part of the book, not a trade made later to get a note.
- The SGOV/BIL note states the leftover as a **dollar amount at the trade-date price** (our calculation), never "about 2%".
- Honest cost: about \$18 of interest in the WInS window against a \$25 commission. In the real plan the leftover waits a year, about \$220 (ASM; BIL yield 3.94% on Sep 24, VP) [5]. The reason is the plan's rule, never WInS profit.

- **If no T-bill ETF is listed**, the third trade must be the dated rung (section 7). If neither exists, (iii) cannot promise three executed trades by Oct 23, so **choose (ii)**.

3. Position-limit branch (read both limit lines; decided before the first order)

Each capped fund is held **one point under the cap** (ASM: the limit is checked when the order is placed). Figures are shares of the \$300,000 [3]. The twist error is the worst 50bp twist, measured forward-consistently (S4 method, ASM). The (iii) column uses the Sep-25 hedge of 98% as an illustration; on the trade date the hedge is the re-priced ladder cost (section 2), so re-solve the capped weights with the same fill-order rule.

Single-security limit shown	Option (ii): hedge 66%	Option (iii): hedge 98%
None, or 65% or more	IEF 23.5 / TLH 42.5 (about 9.9y; \$1.5k)	IEF 34.9 / TLH 63.1 (about 9.9y; \$1.5k)
44-64%	same as above	IEF 34.9 / TLH at the cap minus 1 / SPTL the rest (at 50%: TLH 49 / SPTL 14.1; about 10.2y)
25-43%	IEF 23.5 / TLH at the cap minus 1 / SPTL the rest (v0 rule). At 35%: TLH 34 / SPTL 8.5 (10.2y; \$2.1k). At 25%: TLH 24 / SPTL 18.5 (10.5y; \$2.8k). If SPTL would be under 2% (a 43% cap), put the sliver in IEF	36-43%: IEF 34.9 / TLH at the cap minus 1 / SPTL the rest. 25-35%: IEF and TLH at the cap minus 1, SPTL the rest up to the cap minus 1, then VGIT and VGLT solved to about 10 years. At 35%: IEF 34 / TLH 34 / SPTL 30 (10.6y; \$3.0k). At 25%: IEF 24 / TLH 24 / SPTL 24 / VGIT 17.6 / VGLT 8.4 (9.9y; \$2.8k)
Under 25% (new rule)	IEF, TLH and SPTL at the cap minus 1; VGIT, then VGLT, take the rest so the hedge stays about 10 years. At 20%: IEF 19 / TLH 19 / SPTL 19 / VGIT 9 (9.9y; \$2.6k). At 60/40 VT (20.5%) exceeds 19%, so split the equity into VTI 12.7% (~100 sh) and VXUS 7.8% (~270 sh). At 50/50 VT (17%) fits down to an 18% cap. If 5 or more hedge funds would be needed (a cap of about 17% or less), stop and ask Wharton (Contact Us) before trading.	Switch to (ii) and re-vote gate box (b). (iii) needs 6 funds at 20% and is not feasible at 15% with the listed Treasury ETFs. Exception: Session Rules show a separate, higher bond limit and the Bonds drop-down lists a Jul-Dec 2037-2041 Treasury. Then keep the ETFs at the cap minus 1 and carry the rest of the hedge in that bond, re-solved to about 10 years on the trade date

- SPTL's alternate is VGLT. Every branch is described in notes as "about 10 years"; never quote two decimals.
- **Continuous limit (ASM, D10):** a holding placed at 24% under a 25% cap crosses 25% only after rates fall about 101bp (TLH, ii) or 118bp (SPTL, iii), or rise about 128bp (IEF, iii). **A stock fall does it too** (T2 script [5], DER; ASM that WInS re-checks the limit as prices move): under (ii) at a 25% cap, TLH at 24% crosses 25% after a 20% stock fall (60/40) or 24% (50/50) with rates flat, or after only a 10-12% stock fall combined with a 50bp rate fall (a typical flight to safety). Never add to a position near the cap.
- **If an order is rejected by a limit nobody could see beforehand**, the response is a genuine rule-driven trade and may be a "refined" note (AY2 C1). A branch chosen in advance is **not** a refinement.
- **If IEF is not listed:** VGIT, then SPTL (section 12), re-solved with the section 10 formula (TLH rises to about 49-50% under (ii) and 73-74% under (iii), so re-check the cap branch).
- **If TLH is not listed:** (ii) IEF 40.9 / TLT 25.1; (iii) IEF 60.7 / TLT 37.3. These are recomputed at the Sep-25 durations with the ticket's formula; TLT 14.84y. If a cap also applies, re-solve with the same fill-order rule before trading.

4. Which trades are the candidates for the three notes (choose the final three on Oct 19-20)

"Supported, tested, **or** refined" is a menu, not a quota (TN guide L12 and L29-30; Guide p.3; case L152; all VRF). A trade is never created to fill a category.

Slot	Option (ii)	Option (iii)	Supported / tested / refined
A	TLH (or SPTL if capped): the promise money	TLH: the promise money	Supported at the trade. Its reflection can show tested : the rate move from the fill to the Oct 20 close, against the hedge's % change and the payments' % change

Slot	Option (ii)	Option (iii)	Supported / tested / refined
B	VT: growth money	The dated rung, if listed (section 7); otherwise IEF, which is the same decision as TLH (a duration-weighted pair bought together), so only if the team accepted that at gate box (b) and logged it (TN guide L11 asks for "three investment decisions"; T2 finding 4)	Supported. The VT reflection can show tested: the growth money moved, the payments did not
C	VGSH (a distinct role), or the split-decision trade if a pre-announced decision changes the split (refined), or the rung	SGOV/BIL: the 2027 leftover	Supported, or refined (ii only)

Prefer three different role words, at least one honest "tested" reflection, and "refined" only if it really happened. **Under either option**, one reflection answers **why so little equity** (D13c N1(e), VRF; T2 finding 16c). Under (iii) the content is: in January 2027 almost all of her real \$300,000 buys the payments, and her room for stock risk starts with the 2028 money (D13c N1; D10). Under (ii) (17-20.5% stocks) it is: risk stated goal by goal (promise money: no stock risk; growth money: the chosen share) (E1 P6).

5. What every WInS note must contain (elements only; ≤300 characters planned)

The four elements the Guide asks for (p.3 L84-86, VRF) are the reasoning, the alignment with strategy, the supporting research or analysis, and the expected role. D9 M001 and D12 M247 map them to a **core that fits in 300 characters**. The trade record already shows the side, ticker and quantity, so name the instrument type in a few words at most. Mapping: C1 and C2 carry the reasoning and the alignment with the strategy, C1 names the role, C3 is the research, and C4 names the risk the team accepts.

Every entry below is a content element described in noun phrases, not wording. Any phrase that still reads like a sentence is illustrative only: do not copy it into WInS (notes are permanent and quoted exactly; Wharton AI policy: students "must use their voice and words", VP; T2 finding 8).

- [] **C1 Role:** one Guide role word, plus the team's own label for the job (tables above; D9's labels are proposals, the team picks its own words). Never switch labels between notes.
- [] **C2 Laura's dated need:** the ten \$50,000 payments 2033-2042, or the facility contribution and flexibility in 2033, or the small 2027 leftover waiting for the 2028 deposit. **Under (ii), fold the scaling into this element:** a marker that the promise money is her mix "after both deposits", and the growth money is money "after her 2028 deposit" (AY2 C1; D9 M012).
- [] **C3 One research fact, dated, with its source named in words** (no link). Prefer a price or an issuer figure over a forecast (D12 grades).
- [] **C4 The main risk accepted, or the pre-set rule that governs it.**
- [] **Trim order if a draft is over the limit** (D12 AY1 C1, VRF; T2 finding 6; the (ii) promise note's required element phrases alone use about 258 characters, T2 script [7]): keep C2 (with the (ii) scaling marker), C3 and the Guide role word; move C4 and the team label to the reflection. Keep "promise money first" as the order of entry (the time stamps are evidence of it, D7); never reorder trades to test the box.
- [] If the box visibly allows more than 300 characters (gate box (c) test): add D9's detail (a second fact or clause), up to about 600 characters. Never use a second, added note to finish a sentence. Paste, check nothing is cut, then submit.

Note	C2 need	C3 research fact (pick one)	C4 risk or rule (element, not wording)	Must never say
Promise money (TLH / IEF / SPTL)	The ten payments; (ii): the promise's share of her portfolio after both deposits (about two-thirds)	(iii): the payments' value on the [trade-date] Treasury curve (our calculation; re-price with A2_curve_recheck.py, see section 2), a model price, not a STRIPS quote (D3 AX1 C25). Either option: the pair's rate sensitivity (about 10 years, issuer durations, dated) set to the payments'. Why TLH (reflection if no room): its maturity range sits closer to her dates than a 20+ year fund's, with the caveat that about three-quarters still mature after 2042, so never imply they mature in her years	The rule after a rate rise (no sale, and why: the payments' value fell too) and the only change allowed (an IEF/TLH re-mix if the hedge's duration moves more than 0.25 years)	match, matched, mature in her payment years, locked, stable, reduce volatility, safe, guaranteed, operating reserve, "bought in January 2027" or "fits in \$300k" as flat facts
Growth money (VT; ii only)	Facility contribution and flexibility in 2033, from the growth money after her 2028 deposit	First option (T2 finding 17): the rate the [date] Treasury curve implies for her 2028-2033 dates (about 5.2% a year on the Sep-25 curve; D2 with AX2 C1), which the stock share has to beat, worded as "implied by the [date] curve". Or: the split and how it was set (if provisional: that fact and the decision date). Or: VT's stock count (10,088, Aug 31 2026) and top-ten share (21.7%, Jun 30 2026), both Vanguard (VP), which any team can cite	The risk element: what a stock fall changes (the facility range) and what it does not (the payments). The no-tilt reason (content only): her concentrated exposure is her career, which already drives the 2028 deposit (D8 AY2 item 14)	a return forecast, "overvalued", CAPE, "halves AI exposure", "known/locked 5.2%", "Laura likes risk", the statistics degree, "the rule picks 50% under both forecasts" (true only with a 0.5%+ fee, D6 AY1 C1)
Short-Treasury part (VGSH; ii only)	Facility contribution and flexibility. Only if gate box (b) recorded "floor bought in 2031 from the growth money": also its short Treasuries as part of what a later bought amount can come from (never "the floor"; D6 AY1 C7)	Duration 1.9 years (Aug 31, VP): about -1.9% per +1 point. The growth money's fall in a 20% stock fall: about 10% (50/50) or 12% (60/40) instead of 20% (D12 arithmetic, ASM), presented as the ordinary effect of holding Treasuries in the growth money, not a VGSH property (D12 AY1 C10)	The rebalancing rule: traded only against VT, within the band	"the 2031 floor", "already owned", "the amount she can promise co-sponsors", any floor %, "half = the promise". The idea of lower swings in the growth money is allowed here only with the number beside it (T2 finding 14)
2027 leftover (SGOV/BIL; iii only)	The small amount left after the payments are bought, waiting for the 2028 deposit (case L61-65: no other flows)	The leftover as a dollar amount at the trade-date price (our calculation, section 2), never "about 2%"; the fund's maturity (about 0.1-year T-bills, issuer, dated)	The rule: not invested for growth until 2028	"earns", "cash buffer for living costs" (living costs are outside the portfolio), "reserve"
Dated rung (section 7)	The payment it stands for (e.g. 1 Jan 2033)	Its maturity date and face (Nov 15, 47 days before 1 Jan 2033; \$50,000 = one payment under (iii), scaled under (ii))	The hold-to-maturity rule (what it fixes: the amount repaid) and where the other payments' rate sensitivity sits (the ETFs, about 10 years)	"the whole ladder is in WInS", "risk-free" in WInS value terms; for IBTM only "a fund that ends in December 2032"
Split-decision trade (refined; ii only; only if it happens)	Same as the growth money	Which research changed the split, and that the decision date was announced in the first growth note	The reason: the 2031 promise and a flat median across two published forecasts (D2); never a price move	"because stocks rose/fell", "overvalued"

6. What every ≤100-word reflection must cover (written by students after the trades; not in WInS)

- [] The three official questions (TN guide L46-52, VRF): **why** the team made the decision; **how** it aligned with the strategy; **how** it supported Laura's goals, funding needs or risk considerations.
- [] Which of **supported** / **tested** / **refined** it shows, true to what happened.

- [] **At most one number, except a "tested" reflection, which carries exactly three** (T2 finding 7): the rate move in bp, the hedge's % change and the payments' % change. Every number is dated, with its kind said in words ("at [date] Treasury prices" for a price), and the tested three are labelled as the team's own calculation from the WInS screen and the treasury.gov curve (D3 AX1 C7, VRF).
- [] **If tested:** the pre-registered window (fill date to the Oct 20 close); the three numbers above (percentages, not dollars, under (ii)). About 20-25 words inside the three answers. Report small moves or poor tracking as they are (AY2 C2, C15).
- [] **Under (ii):** the scaling in plain words, at least in the promise-money reflection. The content: WInS shows her mix after both deposits, scaled to \$300k; WInS holds stand-in funds, not the ladder; in January 2027 almost all of her real \$300k buys the payments.
- [] **Under (ii), in the growth reflection:** risk stated goal by goal (promise money: no stock risk; growth money: the chosen share), which is the (ii) answer to "why so little equity" (D13c N1(e); E1 P6; T2 finding 16c).
- [] **Under (iii):** the "why so little equity" answer, once (section 4).
- [] One case-stated trait, and it must pass the swap test (would it still be true for any client?). For the promise: fixed payments only her portfolio may fund (case L91-92). For growth: "thoughtful risks" within "an appropriate balance" (L69-74). For VGSH or the leftover: flexibility "as the project develops" (L102-103) or the two-deposit timing (D6 M125).
- [] Not in any reflection: Laura quotes; the "floor-first" trait; reserve size, facility amount, co-sponsor range or 2031 floor share (TN guide L20-22); probabilities (a team judgement, D1); WInS profit or rank.
- [] The same rule names the IPS will use (decision-log column, D9 M212). The words are the students'; any AI help, including the D9 draft checker, is logged.

7. Optional dated rung (D1 M044, with the AX1b corrections)

Use it only if the Bonds drop-down lists a suitable U.S. Treasury. Search by maturity, not by a built-up symbol. \$10 commission. Accrued interest is about \$768-824 on \$50,000 face; the buyer pays it. Bond prices update once a day. **Price the rung in the drop-down first, then size IEF and TLH in the same session** from the hand formula (T2 finding 11; "fund it from IEF" was wrong for rows 2-3, where the re-solve lowers TLH more than IEF): $\text{IEF share of hedge} = ((1 - r) \times D_TLH + r \times D_rung - 9.90) / (D_TLH - D_IEF)$, where r is the rung's share of the hedge (AX1b item 16); TLH takes the rest. **A rung found after IEF/TLH have filled** is bought with a separate re-mix on a later U.S. day; never sell a fund on its fill day (day-trading rule, User Guide p.6, VP).

Order of choice: stop at the first "yes".

Order of choice	(iii): face \$50,000	(ii): face ~\$34,000	Notes
1. A Jul-Dec 2032 Treasury (4.125% 15-Nov-2032)	~\$48.4k (16.1%); IEF 13.6 / TLH 68.3	~\$32.9k (11.0%); IEF 9.0 / TLH 46.0	Pays for the first payment. It raises TLH , so skip it under any cap below about 69% (iii) or 47% (ii)
2. A Jul-Dec 2033-2041 Treasury (likeliest 15-Nov-2039 or 15-Nov-2040)	~\$45-46k (15.1-15.5%); IEF 28-30 / TLH 53-55	2039: ~\$31.6k (10.5%); IEF 18.8 / TLH 36.6	About 10-year duration; lowers TLH . Under a cap below 44%, allowed only if the re-solve needs no new fund
3. IBTM (iBonds Dec 2032) ETF	~\$37k (12.3%); IEF 17.9 / TLH 67.8	~\$25k (8.4%); IEF 11.9 / TLH 45.7	A fund, not a bond. It returns no fixed amount (iShares, VP). It raises TLH like row 1 (T2 script [6], DER; IBTM 5.05y VP), so skip it under any cap below about 69% (iii) or 47% (ii)
4. None listed	no rung	no rung	Record "no suitable dated Treasury listed". Do not force a Feb/May bond

Figures are D1's at the Sep-25 curve (ASM model prices). Under (ii) the rung repeats the "future funding" role, so it is optional; under (iii) it is the natural distinct note (AX1b item 7). The reason to hold it is implementation: the promise is dated, so the book holds one dated bond where WInS lists one (AX1b item 8).

8. Never

- Never place an order before the gate. Never let the advisor place a trade. Never day-trade (User Guide p.6, VP): no buy and sell of the same security on the same U.S. trading day, and no same-day "fix" of an error.
- Never sell the promise money after a rate rise or trim it after a fall. Re-mix IEF against TLH only if the hedge's duration moves **more than 0.25 years from its starting value** (this band works in every cap

branch). Never move money between promise money and growth money.

- Never lengthen the hedge to bet on rates (EDV, ZROZ, GOVZ, TLT alone). No leveraged, inverse, crypto-linked, thematic or single-country funds, and no ETNs. No Taiwan or AI tilt.
- Never put the reserve size, facility amount, co-sponsor range or 2031 floor share in a note. No Laura quotes. No words from the banned lists (section 5 and brief section 2 item 4).
- Never trade for rank, and never "tidy up" in the last days before Nov 6. Never push a position above the limit.

9. Calendar (U.S. Eastern Time; Australian times assume NSW/Victoria)

Date	What
Now	Gate; logged-in checks (section 10); votes (a) and (b); notes drafted
By Fri Oct 2	Fills. U.S. open 9:30 a.m. = 11:30 p.m. AEST until Sat Oct 3, then 12:30 a.m. AEDT the next day from Sun Oct 4. Screenshot the positions
Oct 2 jobs report, Oct 14 CPI (SNIP dates)	Check only, never trade
By Fri Oct 16	Only if the split was provisional: decide, then trade once or log "decided not to trade"
Oct 20 close	End of the "tested" window: screenshot the positions
Oct 19-23	Duration check (trade only on a >0.25y move); choose the three notes; write the reflections; team read-through
Fri Oct 23, 5:00 p.m.	TN due (8:00 a.m. Sat Oct 24 AEDT). Submit on Friday evening Australian time
Oct 27-28	FOMC meeting (VP), after the TN deadline
From Mon Nov 2	U.S. open is 1:30 a.m. AEDT
Fri Nov 6	Portfolio freezes with the IPS (close = 8:00 a.m. Sat Nov 7 AEDT)

10. WInS checks before the first order (from v0 s7, A4 and D1; tick in the session)

1. Official account (practice trades were reset). A student places every trade.
2. Session Rules screenshot (gate box c). Pick the branch in section 3.
3. Re-read SMAApply Trading Details and the FAQ that morning, and the logged-in Trading Details / Getting Started pages (required activity; first-trade date).
4. Search each ticker: it must show as an **ETF** with the issuer's full name. SPTL and BIL now show "State Street SPDR ...".
5. Bonds drop-down: list every U.S. Treasury maturing 2032-2042 (maturity, coupon, accrued interest). Check whether a volume rule or an order-type limit appears for bonds. "No volume limit on bonds" and "market orders only" come from a generic Stock-Trak page (VP), UNVERIFIED for WInS.
6. Today's volume in WInS for each ETF. The rule is "no more than twice a security's current daily trading volume" (VP). Every planned order is below 0.2% of that cap on published volumes (D10, AX2 C5); Vanguard funds publish no volume. If an order is rejected, split it across days; never switch to a thinner fund.
7. The commission is shown (\$25 per ETF trade, ASM that ETFs count as "stock"; \$10 per bond). Cash stays positive.
8. Note box: the zero-risk length test is done at gate box (c), before any real note. On each real order, paste the draft and confirm nothing is cut **before** submitting. Straight after, copy the saved note character for character into the decision log.
9. Does WInS pay interest on cash? Record it. It affects only how the T-bill leg is described, not the plan's rule.
10. Re-read the IEF and TLH effective durations on ishares.com that day. Recompute: $\text{IEF share of hedge} = (\text{D_TLH} - 9.90) / (\text{D_TLH} - \text{D_IEF})$.
11. Anything unclear: ask Wharton through Contact Us. Never contact Laura (disqualification, VP).

11. What WlnS stands in for (so every note is true; the rule wording belongs to the IPS spec)

- **Jan 2027:** \$300k buys the ten payments as a Treasury STRIPS ladder (about \$294k on the Sep-25 curve, a model price, not a STRIPS quote; ASM, D3 AX1 C25). Then:
 - If it costs more, buy the longest-dated payments first. The 2028 deposit completes the ladder before any growth money (D1, D8 rule 1).
 - The joint tail: if rates fall **and** the deposit is smaller than the gap, part of the first payment stays unfunded. This must be named, not hidden (AY2 on D8).
 - Any leftover waits in T-bills.
 - **Jan 2028:** the \$150k plus the leftover is the growth money, at 50/50 or 60/40, reset yearly within its band.
 - **Jan 2031:** a share of the growth money (an open parameter; the model uses 80%, D5 suggests 70%) is bought as a 2-year Treasury. This never appears in a WlnS note.
- **Jan 2033:** the ladder is the operating reserve. The WlnS Treasury funds are never called that.

12. Securities (facts re-read 2026-09-28; all PENDING WlnS AVAILABILITY + POSITION-LIMIT CHECK)

Ticker	Role	Expense	Duration (as of)	Close (as of)	Other (issuer page)	2025-26 list (history only)
IEF	promise, part 1	0.15%	6.86y (9/25)	\$90.00 (9/25)	16 holdings; \$41.57bn; 30-day volume 8.83m	yes
VGIT, then SPTI	IEF alternates (restored from v0 s6; T2 finding 10)	0.03% / 0.03%	4.9y (8/31) / 4.78y fund OAD (9/24)	\$56.90 (9/25) / \$27.31 (9/24)	102 bonds / -; re-solve with the section 10 formula using the substitute's duration: no cap, VGIT 25.1% of the hedge (ii: VGIT 16.6 / TLH 49.4; iii at 98%: 24.6 / 73.4), SPTI 24.7% (ii: 16.3 / 49.7; iii: 24.2 / 73.8) (DER); TLH rises, so re-check the cap branch	no / no
TLH	promise, part 2	0.15%	11.58y (9/25)	\$93.38 (9/25)	61 holdings, maturing 2037-2046, 74.7% after 2042 (S1 snapshot); \$10.53bn; 1.97m/day	no
SPTL	promise, if capped	0.03% gross	13.65y fund OAD (9/24)	\$24.27 (9/24)	110 holdings; \$10.95bn; 1.09m prior day	no
VGIT / VGIT	cap alternates	0.03%	13.5y / 4.9y (8/31)	\$50.98 / \$56.90 (9/25)	101 / 102 bonds	no / no
TLT	only if TLH is missing	0.15%	14.84y (9/25)	\$79.32 (9/25)	47 holdings, maturing 2044-2056	yes
VT	growth	0.06%	n/a	\$160.03 (9/25)	10,088 stocks, 37.7% non-U.S. (8/31); top ten 21.7% (6/30)	yes
VTI + VXUS	VT alternate or low-cap split (62/38)	0.03% / 0.05% (per S2)	n/a	\$379.77 (9/25) / \$86.35 (9/25, via S3)		yes / yes
VGSH	short-Treasury part	0.03%	1.9y (8/31)	\$57.59 (9/25)	92 bonds; SEC yield 4.55% (9/24)	yes
SHY	VGSH alternate	0.15%	1.79y (9/25)	\$81.21 (9/25)	91 holdings	yes
SGOV	2027 leftover (iii)	0.09%	0.10y (9/25)	\$100.66 (9/25)	\$111.8bn; 30-day volume 22.5m	no
BIL / SHV	leftover alternates	0.1353% / 0.15%	0.10y (9/24) / 0.28y (via S1)	\$91.58 (9/24) / -	BIL yield to maturity 3.94% (9/24)	yes / yes
IBTM	rung fallback 3	0.07%	5.05y (9/25)	\$21.85 (9/25)	15 holdings; ends Dec 2032	no

Sources, VP, read 2026-09-28 about 00:35 UTC with curl:

- ishares.com/us/products/239456 (IEF), 239453 (TLH), 239454 (TLT), 239452 (SHY), 314116 (SGOV), 328944 (IBTM);
- ssga.com/spdr-portfolio-long-term-treasury-etf-sptl, [spdr-portfolio-intermediate-term-treasury-etf-spti](https://ssga.com/spdr-portfolio-intermediate-term-treasury-etf-spti), [state-street-spdr-bloomberg-1-3-month-t-bill-etf-bil](https://ssga.com/state-street-spdr-bloomberg-1-3-month-t-bill-etf-bil);
- investor.vanguard.com/vmf/api/{VT,VGSH,VGLT,VGIT,VTI}/{price,characteristic}.

Not re-read today: VXUS (the API returned an error; S3's 9/25 read is used); VT top-ten share and expense ratios (S2 and S3, VP there); SHV (S1).

13. Changes from v0

#	Change	Why (source)
1	Option (iii) gets a named third trade : the ~2% 2027 leftover into SGOV or BIL, bought in the same session. It falls back to the rung, then to switching to (ii)	AX2 blocking / D10 C1
2	Option (ii) shows both splits : 60/40 (VT 20.5 / VGSH 12.5) and 50/50 (VT 17.0 / VGSH 16.0), with bands 55-65 or 45-55	D2 (+AX2 C3, C4, C7), D6, D3 M034
3	IEF/TLH re-solved at the Sep-25 durations: 23.5 / 42.5 (ii) and 34.9 / 63.1 (iii), was 23.6 / 42.4	Issuer pages, VP
4	Cap-under-25% rule added, plus the cap-to-option switch ((iii) under 25% means switch to (ii)), the VT/VTI+VXUS split, a stop at 5+ hedge funds, and the individual-bond exception	AX2 C2; T1 script [3]-[4]
5	Duration band now ±0.25y from the starting mix . The fixed 9.65-10.15 band clashed with the 25%-cap mix at 10.5y	T1 (inconsistency in v0 s3 vs s5)
6	Notes planned for 300 characters , split into a note core and a reflection	D12 (UNVERIFIED for WlnS) vs D9
7	VGSH relabelled: role word "risk management", then "liquidity". No "already owned", no "amount she can promise", no floor %	D12; AY2 on D9 (C1); AX2 on D2 (C6)
8	Banned words extended: known or locked 5.2%; halves AI exposure; half = the promise; match; stable or reduce volatility; flat "bought in January 2027" or "fits in \$300k"	AX2 C1, C5, C6; S4 finding 2; D9 M062; D7 M011; AX1b
9	Notes plan by supported / tested / refined. "Refined" only via a pre-announced split decision or an unforeseen limit, not SPTL. "Tested" on a pre-registered window with data capture	D7 + AY2 C1, C2, C5
10	Planned "discipline/rebalance" trade dropped	D7 M002
11	Optional dated rung , conditional and corrected: choice order, IBTM wording, cap rule, role clash under (ii)	D1 M044 + AX1b items 5-10, 14, 16
12	Statistics degree removed from the VT note (one place across TN and IPS; the IPS owner decides)	AY2 C6; D6 M125
13	Under (ii), the scaling is folded into each hedge or growth note's "need" element; under (iii), one reflection answers "why so little equity"	AY2 C1; D9 M012; D13c N1
14	Ladder-above-\$300k odds stated as "about 1 in 3" for the real Nov-15 ladder ("1 in 4" is the idealised ladder) and never put in a note	AX1b item 1; D3
15	The Guide p.5 line ("should not be rewritten simply because markets move") is no longer cited as forbidding changes before Nov 6	AX2 C3; AX1b item 4
16	Gate box (c) now covers both limit lines, the day-trading tooltip and the logged-in activity rules. Box (d) adds the 300-character draft and the second student's restatement	A4; D12
17	The long-term map is cut to four lines. The model-input note (sleeve bonds "3.5-3.9%") is withdrawn in favour of D2's market-consistent ~4.8% ASM; that is Final Report work, for later	AX2 C8; brief s17
18	TLH-missing weights recomputed (IEF 40.9 / TLT 25.1 (ii); IEF 60.7 / TLT 37.3 (iii))	T1, issuer durations 9/25

Changes after T2 red team (2026-09-28; source: [wins_now/T2_red_team.md](#), script [scripts/T2_red_team_checks.py](#))

T2 found 1 blocking, 7 important and 12 minor issues. All blocking and important ones are fixed here or in the brief; the minor ones that touch this file are fixed too. The file keeps its path and the name "v1".

T2 #	Severity	What changed in this ticket
1	blocking (iii)	Section 2: the (iii) hedge is set to the ladder cost re-priced on the trade date (A2 script + about \$2.1k for Nov-15 dates, or the \$292/bp hand check); T-bill leg = \$300,000 minus cost minus \$1,000 float minus commissions; pre-set rule "leg under \$1,000 = no genuine third trade unless a rung is listed, revert to (ii); never a token sliver". "When": IEF and TLH first, T-bill order only after both show Filled; all share counts recomputed from the last close. SGOV note C3: dollar leftover at the trade-date price
2, 3	important	Brief only (vote tables). Gate box (b) now says 50/50 is a low-confidence lean
4	important	Gate box (b) and section 4 slot B: under (iii) with no listed rung, IEF + TLH are one decision; the team accepts that explicitly at the vote and logs it, or chooses (ii)
5	important	Gate box (b): under (ii), record floor timing (Jan 2028 or Jan 2031) and the growth-money bond choice (VGSH or a dated Dec-2032 fund) before the VGSH order; otherwise the VGSH note uses only the facility/flexibility need. Both questions handed to ips_spec (brief section 6)
6	important	Gate box (c): zero-risk note-length test before any real note. Section 5: trim order (keep C2 with the scaling marker, C3 and the role word; move C4 and the label to the reflection)
7	important	Section 6: "at most one number, except a tested reflection, which carries exactly three", dated and labelled as the team's own calculation
8	important	Section 5 recast as noun-phrase elements with an "illustrative, do not copy" warning; gate box (d): the second student checks no clause repeats ticket wording; label words are the team's own
9	minor	Section 3: the stock-fall path to the continuous limit
10	minor	Section 12 and section 3: IEF alternates VGIT, then SPTI, with re-solved weights
11	minor	Section 7: price the rung first, size IEF/TLH from the formula; IBTM row gets its split and cap rule; a rung found later needs a re-mix on a later day
12	minor	Section 1: band denominator $VT \div (VT + VGSH + \text{cash float})$, starting at about 50% or 60%
13	minor	Gate box (c): stop-and-ask branch for an activity minimum or a security-type cap
14	minor	Section 5: "reduce volatility" banned for the promise money only; for VGSH the idea is allowed only with its number, as the ordinary effect of holding Treasuries
16c	minor	Sections 4 and 6: the "why so little equity" answer under either option; under (ii), risk stated goal by goal in the growth reflection
17	minor	Section 5: the curve-implied ~5.2% for her 2028-2033 dates is the growth note's first research option
18	minor	Header: S4 finding 8's line reconciling 9.90 (issuer) with the 10.16-year spot duration. "When": 52% vs 40% tested-window odds
19	minor	Header, sections 5 and 11: "\$294,387" labelled a model price on the curve (our calculation)
20	minor	"When": the fill-at-next-open citation is SMapply Trading Details (R-W62), not the FAQ
15, 16a-b	minor	Brief only

Open items (for the team, today)

1. Votes (a) and (b), and the split if (ii); under (ii) also floor timing and the growth-money bond choice; under (iii) without a rung, the "two notes, one decision" acceptance. 2. The Session Rules numbers (both lines), the zero-risk note-box test, the bond list and interest on cash. 3. Who places trades and who captures the data. 4. Ask Wharton whether an "added" trade note counts. 5. The "first trade by Oct 10" claim stays UNVERIFIED until the logged-in page is read. 6. Under (iii): re-price the ladder on the trade date and apply the \$1,000 T-bill-leg rule.

What this teaches

1. **A snapshot needs a date.** WInS can show Laura's plan on one day only: January 2027 or after 2028. Both are honest if the date is said out loud, and each has a price in words or in note variety.
2. **Check that the deliverable is possible, not only allowed.** A book can obey every trading rule and still leave the team one executed trade short on Oct 23. The fix came from the plan itself, where the leftover already waits in T-bills, not from inventing a trade.
3. **A limit changes the best design.** A 25% or 20% cap costs the literal book three or four extra funds, but the scaled book only one or two. Deciding the branch before the session turns a midnight scramble into a look-up.
4. **A fix needs its own failure test.** The T-bill trade solved "only two trades" on Friday's prices, but a 12bp rate fall or a thin cash float at the open can make it vanish. Writing the rule down before trading

(re-price, then buy only if the leftover is real) turns a fragile trade into an honest one.

5. **Short boxes force clear thinking.** If a note may hold only 300 characters, each word must carry the role, the need, one fact or the rule. The fuller story belongs in the reflection, written later but never rewritten in the note.

T2 red team: attacking the trading-now brief and WInS ticket v1

Agent T2 (adversarial red team), insight_v1 run, written 2026-09-28 (about 01:10-02:00 UTC, before the season's first U.S. open at 13:30 UTC). Files attacked: `research/insight_v1/phase_D/trading_now_brief.md` ("the brief") and `research/insight_v1/wins_now/securities_and_allocation_v1.md` ("the ticket"), both by T1.

Serves: WInS trading as the source of the Trading Notes (TN, Oct 23) and the IPS consistency points that depend on it (IPS, Nov 6). No Final Report work (brief s17). **Status:** PROVISIONAL, AI-generated research (findings, evidence, fixes). It contains no note, reflection or IPS text; every fix below is a specification change to T1's files. The six students decide and write every word. No relayed user messages arrived during this task. Nothing was committed.

Every ticker named here is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK (brief s14): listed in the team's WInS account, and inside the Session Rules position limit. Alternates are the ticket's (section 12) unless stated.

Labels: VP = verified on the primary page; VRF = verified in a repo file (path given); SNIP = snippet or secondary; ASM = assumption or model output, not a forecast; INT = interpretation (a judgement); DER = derived by the T2 script from labelled inputs.

Script: `.venv/bin/python research/insight_v1/scripts/T2_red_team_checks.py` (run from the repo root; sections [1]-[7] are cited below). I also re-ran `T1_ticket_v1_numbers.py`: every figure in the ticket reproduces.

Terms: *hedge / promise money* = the Treasury funds standing in for the ten \$50,000 payments. *Growth money* = what the payments do not need. *Float* = uninvested cash kept for commissions. *Gap* = the price jump between the last close and the next open. *Position limit / cap* = the most WInS allows in one security. *bp* = 0.01 percentage point.

Summary

Verdict: the ticket's arithmetic, rule compliance and day-trading discipline hold (section "What survived"). The plan survives a single trading session under option (ii) as written. It does **not** yet survive one under option (iii): its "named third trade" can disappear or fail at the fill. Two decision tables the team will vote on are also tilted by stale evidence: T1 did not apply audit corrections that were already in the repo.

20 findings: **1 blocking** (option (iii) only), **7 important**, **12 minor**.

#	Severity	Serves	One line
1	blocking (option iii only)	WInS-now, TN	The (iii) T-bill "third trade" vanishes if rates fall ~12-16bp before the fill, and can fail for lack of cash if all orders go in before the open
2	important	WInS-now, TN, IPS	The 50/50 case quotes D6's rule as "picks 50% under both houses"; after its audit (AY1 C1) it picks 55-65% with no fee
3	important	WInS-now, TN, IPS	The (ii)-vs-(iii) table leaves out the strongest official hint for (iii), one criterion for (ii), and overweights the TN for semifinal selection
4	important	TN	Under (iii) with no listed rung, the three notes show only two investment decisions (TN guide asks for three)
5	important	WInS-now, TN, IPS	Option (ii)'s VGSH order and note lock in two choices the D3 audit (AX1 C6, C18) has reopened
6	important	WInS-now, TN	The 300-character plan: mixed evidence, no zero-risk check, no trim order, and the (ii) promise note's required phrases use ~258 characters
7	important	TN	The reflection checklist says "at most one number" but the "tested" reflection needs three
8	important	TN, IPS	Paste-ready clauses and AI-coined labels sit in the ticket; notes are permanent and quoted exactly
9-20	minor	as marked	Continuous-limit omission, missing IEF alternate and labels, rung mechanics, band denominator, unbranched Session Rules outcomes, over-broad ban, one overclaim, IPS-consistency gaps, a generic growth fact, timing/label gaps, stale notes and citations

What survived the attack (keep)

- **Arithmetic** (DER, script [1]; VP prices of 2026-09-25): every order reproduces with whole shares. (ii) 60/40: IEF 783 / TLH 1,365 / VT 384 / VGSH 651 shares, cash left \$3,024 after \$100 of commissions. (ii) 50/50: VT 318 / VGSH 833, cash \$3,104. (iii): IEF 1,163 / TLH 2,027 / SGOV 45, cash \$1,444 after \$75. Cap branches, share counts, the TLH-missing weights (IEF 40.9 / TLT 25.1; IEF 60.7 / TLT 37.3), the VTI/VXUS split and the \$18 vs \$25 T-bill economics all reproduce (T1 script re-run).
- **Rule compliance** (VP rules via brief s14): every planned order is cash, an ETF or a U.S. Treasury; none is leveraged, inverse, crypto-linked or an ETN; every price is above \$5; the largest order is under 0.2% of the 2x-volume cap on published volumes (AX2 C5); commissions fit; no margin is planned (but see finding 1 on gaps).
- **No day trading planned:** every first-session order is a buy. The later trades the ticket allows (a split decision, a duration re-mix, a limit response) involve different securities or later days, and the ticket bans same-day round trips and same-day "fixes" (User Guide p.6 "Day Trading: This is not permitted", VP via A4).
- **The gate** (five boxes), the cap-branch table, the calendar and time-zone conversions (Oct 4 AEDT start; Nov 1 U.S. change; Oct 23 5 p.m. ET = 8 a.m. Sat AEDT), "menu, not quota" for supported/tested/refined, no planned rebalance trade, no Laura quotes in notes, the statistics degree out of the VT note, and the banned-word lists all match the audited specialist files (D7/AY2, D9/AY2, D2/AX2, S4).
- **Scope:** no Final Report narrative, chart specs or Works Cited plans. Section 11 of the ticket keeps only rule content the IPS needs (brief s17 allows that).

Findings

1. BLOCKING (option iii only): the named third trade is fragile in two independent ways

Evidence.

- The ticket fixes the (iii) hedge at 98% (\$294,000) because the Nov-15 STRIPS ladder cost \$294,387 on the **Sep-25** curve (ticket L77, ASM on a VRF curve). The T-bill leg is whatever is left: about \$4,538 after a \$1,000 float and 3 commissions.
- Re-priced for a parallel move before the trade date (DER, script [2]; ladder prices from S4 finding 9, ASM): the T-bill leg falls under \$1,000 after a **12.2bp** fall, reaches zero after **15.7bp**, and the ladder exceeds \$300,000 after **19.3bp**. With 2026's realised 4.45bp/day (D7, VP data; driftless normal ASM), the chance of a fall that shrinks the leg under \$1,000 is about **11% by an Oct 2 fill and 19% by the Oct 9 hard stop**; the chance that no leftover exists at all is about 6% / 13%.
- In that case the literal January-2027 book has **no leftover**, so a SGOV/BIL note saying the leftover "waits for the 2028 deposit" would state something false at that date's prices, in a permanent note. If the team instead keeps 98% / \$4.5k as printed, the book is no longer the literal January-2027 book the IPS will name.
- **Cash at the fill** (DER, script [3]): orders entered while the U.S. market is closed fill at the next open (Trading Details R-W62, VP; R-W72: teams "are not expected to stay awake"). Sized at the last close, the (iii) orders cost about \$291 more per 1bp fall at the open. The \$1,444 float is used up by a **5bp** overnight fall (about 13% of days; about 29% if the share counts are left at the Sep-25 closes and filled on Oct 1). The T-bill order is entered last, so it is the order most likely to be rejected, or to push cash negative (margin is banned, VP). Either way (iii) is back to the two-trade problem that AX2 called blocking (D10 C1).
- Option (ii) is not exposed: its float (\$3,024-3,104) survives about a 12bp fall plus a 1% stock rise, if the share counts are recomputed from the last close (about a 0.4% chance of failing; 8-9% if the Sep-25 counts are used).

Exact fix (ticket section 2 and brief s2 item 2; before any (iii) order):

1. Replace "The hedge is 98% of the book" with: under (iii), on the trade date re-price the ladder with `D9_numbers.py` [3] or `A2_curve_recheck.py`; hedge = that cost; T-bill leg = \$300,000 minus that cost minus the \$1,000 float minus commissions.

2. Add a pre-set rule: if that T-bill leg is under \$1,000 (rates about 12bp or more below Sep 25), (iii) has no genuine third trade unless a dated rung is listed (section 7); then the gate vote reverts to (ii). Never buy a token sliver.
3. Add to "When": under (iii), enter IEF and TLH first; enter the T-bill order only after both show "Filled", sized to the cash actually left (same U.S. session if someone is awake, otherwise the next one; it is still the plan's rule, not a note-making trade). Under both options, recompute share counts from the last close before entering orders.
4. SGOV/BIL note, element C3: the leftover as a dollar amount at the trade-date price, never "about 2%".

2. IMPORTANT: the 50/50 case misstates D6's rule after its audit, and T1 missed three audit sections

Evidence.

- Brief L133-135: D6's rule "picks 60% under JPM and 50% under Vanguard's midpoint. Applied under both published forecasts, it picks 50%".
- AY1 C1 on D6 (VRF, `D6_behavioural.md` audit section; `audit_client_cosponsors.md` L54-56): on a 5-point grid the same rule picks **60% (JPM, 4% bonds), 65% (JPM, 5% bonds), 55% (Vanguard midpoint), 65-70% with fat tails**, 55-60% with a 0.5% fee and 50-55% with a 1.0% fee. "The rule gives a checkable reason for a choice in the 50-70% band; it does not confirm 60%." The 50% reading is a 10-point-grid artefact.
- E1 (`phase_E/E1_change_proposals.md` P5, written after T1, not yet audited) reaches 50% under both houses **only with a fee of up to 0.5%**; with no fee both central cases give 55%. The case is silent on fees and the fee clause is "the first thing to cut" (D8 AY2 item 13, VRF).
- The 60/40 side also omits D3 AX1 C4 (VRF, D3 audit section): medians understate what equity is expected to add. Under JPM the *mean* gain of 60% sleeve equity over an all-Treasury benchmark is \$6.5-9.4k against a median of about \$3k; under Vanguard's midpoint it is about zero. It does not decide 50 vs 60, but it belongs on the scale.
- Cause: brief L172-173 says "D5, D6, D12 and D3 had no appended audit section" and `audit_rates_D1.md` "was not in the repo". File times (VRF, `ls --time-style=full-iso`, 2026-09-28) show the AY1 sections on D5, D6 and D12 and `audit_rates_D1.md` saved at 00:34-00:37 UTC, 11-14 minutes **before** T1's files (00:48). Only D3's AX1 section (00:57) came later.

Exact fix (brief section 4; before the VT note is drafted, because a note may record "how the split was set"):

- Replace L133-135 with: D6's rule (AY1 C1) gives about 55-70% depending on inputs: 60-65% under JPM, 55% under Vanguard's midpoint, 50% only if a 0.5%+ yearly fee is assumed. It is a reason for a choice in the 50-70% band and does not separate 50/50 from 60/40.
- Keep the 50/50 case on D2's evidence alone (flat median across houses; p5 +\$6-7.5k; p95 -\$11-14k; 2031 floor p5 +\$5-6k). Add AX1 C4 (mean vs median gain) to the 60/40 side.
- Update L172-173, re-read the AY1 sections on D5, D6 and D12 and the AX1 section on D3, and list what changed.
- Never let a note or the IPS say the rule "picks 50% under both forecasts" unless the fee assumption is stated.

3. IMPORTANT: the (ii)-vs-(iii) table is not balanced

Evidence.

- Missing for (iii): Wharton set WInS cash equal to Laura's first deposit, and "The additional \$150,000 contribution ... will not be added to WInS" (R-W56, R-W59, VP). S4 finding 5 (VRF): read literally, this and the case's "It does not determine Laura's actual portfolio value at the beginning of 2027" support (iii) "at least as much as (ii)"; WInS runs on the eve of the January 2027 purchase. D10 1.4 (VRF): (iii) shows the plan's central priority in the holdings themselves.
- Missing for (ii): the Investment Strategy criterion "uses appropriate diversification" (SMAApply L49, VRF) and SMAApply's own list of diversification kinds, which includes "funding purposes" (R-W71, VP). Only (ii) shows three.

- T1's first reason for (ii) (brief L117) is that the TN is "the first judged deliverable". The Rules & Roles page says the Top 50 are chosen on "the strength of their Investment Policy Statement (IPS) and Final Reports" (VP; conflict X-8 in `phase_A/case_register.md`). The TN's weight in semifinal selection is uncertain, so (iii)'s IPS economy (12-20 words vs 25-40; the IPS barely fits, AX2 C3) may count for more than the brief implies (INT).

Exact fix (brief section 3 table): add two rows, "Official hints" and "What the holdings show on their own", with the points above on each side. Re-word L117 to "The TN is scored (SMAApply, case) but may not count for semifinal selection (Rules & Roles, X-8); if it does, (ii) gives three roles". Keep T1's lean if it still holds after the rows are added.

4. IMPORTANT (option iii): three notes, but only two investment decisions

Evidence. The TN guide asks the team to "select three investment decisions that best illustrate your team's strategic thinking" (L11, VRF) and to show "a cohesive portfolio strategy" (L14-15). Under (iii) with no listed rung, the ticket's slots are TLH, IEF ("weaker, same role", ticket L121) and SGOV. IEF and TLH are one decision (a duration-weighted pair bought together); a reader sees two notes with the same need, fact type and rule. D12 (VRF) already said (iii) notes can differ only by supported/tested/refined, not by role.

Exact fix. Brief section 3, column (iii), row "Trades before Oct 23": add "without a listed rung: three trades but two decisions (TN guide L11)". Ticket section 4 slot B under (iii): "the dated rung if listed; otherwise the team must accept, at the gate vote, that two notes describe one decision". Record that acceptance in the decision log.

5. IMPORTANT (option ii): the VGSH order and note pre-commit two choices the D3 audit reopened

Evidence. D3's AX1 section (VRF, saved after T1): C6 "The barbell is not dominated" and adds a team choice for the IPS: "Is the facility floor bought in January 2028 or January 2031?"; C18 logs "VGSH or a dated Dec-2032 Treasury fund for the growth money's bonds?" as an open question. Under (ii), the ticket buys VGSH now and its note element C2 is "the kind of asset a later promised amount can be built from" (ticket L148), which assumes a floor bought later from short Treasuries. Notes are permanent and must be consistent with the IPS (TN guide L30-31, VRF).

Exact fix. Gate box (b), if (ii): the team also records "floor bought in 2031, growth-money bonds in short Treasuries" (or its alternative) before the VGSH order. If it has not decided, the VGSH note uses only the "facility contribution and flexibility" need, with no reference to a later promised amount. Add both questions to the IPS "rules to fix before Nov 6" list (owner: ips_spec).

6. IMPORTANT: the 300-character plan needs a zero-risk check and a trim order

Evidence.

- AY1 C2 on D12 (VRF): the 300 cap comes from a sister product only; Wharton's own 413-character example is counter-evidence; "Verify at zero risk before the first real note" (ask Stock-Trak Live Chat or Wharton, or type a long test text on the order-review screen and do not press Confirm). The ticket's check is made while typing the first real note (brief L40; ticket check 8), and the first note is the promise note, the longest.
- The ticket's own required element phrases for a (ii) promise note (role word, label, need, scaling marker, dated research fact with source, rule) use about **258 characters before any connecting words** (DER, script [7]; phrases taken from ticket section 5, not a drafted note). That leaves about 40 characters for grammar under a 300 cap.
- AY1 C1 on D12 (VRF) sets a priority: need, alignment, research, role word; "the risk accepted moves to the reflection if space is short". The ticket keeps C4 (risk/rule) in the core and gives no trim order.

Exact fix. Gate box (c): add AY1 C2's zero-risk test before any order. Ticket section 5: add a trim order for any note over the limit: keep C2 (need, with the (ii) scaling marker) and C3 (dated fact) and the Guide role word; move C4 and the team label to the reflection. Keep "promise money first" as the order of entry (D7 uses the time stamps as evidence of promise-first); do not reorder trades to test the box.

7. IMPORTANT: the reflection checklist contradicts itself

Evidence. Ticket L157: "At most one number". Ticket L158-160: a tested reflection carries the rate move in bp, the hedge's % change and the payments' % change, which is three numbers. D7 AY2 C15 (VRF) fits those three inside the official questions in about 20-25 words.

Exact fix. Replace L157 with: at most one number, except a "tested" reflection, which carries exactly three (bp move; hedge % change; payments' % change), each dated, and labelled as the team's own calculation from the WInS screen and the treasury.gov curve (D3 AX1 C7, VRF).

8. IMPORTANT: paste-ready clauses and AI-coined labels in a ticket for permanent notes

Evidence.

- The ticket says it "contains no note, reflection or IPS text" (L11), but section 5 contains complete clauses a student could type into WInS unchanged: "A fall shrinks the facility range, never the payments." (L147); "Never sold after a rate rise (the payments got cheaper too); only re-mixed to keep about 10 years" (L146); "Held to maturity, so later prices do not change what it repays" (L150).
- D9 AY2 C4 and D7 AY2 C13 (VRF) classed the same pattern as important: a template the team fills in "becomes AI-written text in a submitted deliverable" (Wharton AI policy: students "must use their voice and words", VP via A4). Notes are quoted "exactly as it appears in WInS" (TN guide L44, VRF) and cannot be edited.
- The fixed labels ("promise money", "growth money", "short-Treasury part of the growth money", "the 2027 leftover") are D9's proposals ("INT: proposal, the team may rename", D9 M001, VRF). The brief presents them as fixed (L75).

Exact fix. Ticket section 5: recast each quoted clause as an element in noun phrases (for example "risk element: what a stock fall changes (the facility range) and what it does not (the payments)"), and mark every remaining phrase "illustrative, do not copy". Brief item 5: "the team chooses its own label words (D9's are proposals) and logs the AI source". Gate box (d): the second student checks that no clause in the draft repeats ticket wording.

9. MINOR: the continuous-limit check ignores a stock fall

Evidence (DER, script [5]; ASM that WInS re-checks the limit as prices move): under (ii) at a 25% cap, TLH placed at 24% crosses 25% after a **20%** stock fall (60/40) or **24%** (50/50) with rates unchanged, or after only a **10-12%** stock fall combined with a 50bp rally (a typical flight to safety). Ticket L106 names only a 101bp rate fall. **Exact fix.** Add the stock-fall path to L106-107. The rule "never add to a position near the cap" is unchanged.

10. MINOR: IEF has no alternate in v1, and the brief carries no availability label

Evidence. v1 supersedes v0 for trading (ticket L4) but dropped v0's substitute table: TLH, SPTL, VT, VGSH and SGOV have alternates, IEF has none (v0 section 6 had VGIT, then SPTI). The brief names tickers throughout and never carries the "PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK" label (0 occurrences, DER). **Exact fix.** Ticket section 12: add "IEF alternate: VGIT (4.9y), then SPTI (4.78y); re-solve with the section 10 formula using the substitute's duration". Brief: one label line under the title.

11. MINOR: dated-rung mechanics

Evidence. "Fund it from IEF" (ticket L176) is wrong for rows 2-3: re-solved with the ticket's own formula, a 2039/2040 bond lowers TLH more than IEF. The IBTM row (L185) gives no IEF/TLH split or cap rule: it leaves TLH at about **67.8%** under (iii) and **45.7%** under (ii) (DER, script [6]; IBTM 5.05y VP), so it breaks the same caps as row 1. If a rung is added after IEF/TLH have filled, "funding" it means selling them, which must never happen on their fill day (day-trading rule, VP). **Exact fix.** Replace "Fund it from IEF" with "size IEF and TLH after pricing the rung in the drop-down, in the same session, from the formula". Copy row 1's cap rule to row 3. Add: "a rung found later is bought with a separate re-mix on a later day; never sell a fund on its fill day".

12. MINOR: the growth band's denominator is undefined

Evidence (DER, script [1]): at 60/40 VT is 62.1% of VT+VGSH but 60.3% including the cash float; at 50/50, 51.5% vs 49.9%. The 55-65 band starts 2.9 points from its top on the first reading. D3 AX1 C17 (VRF) found the same convention gap. **Exact fix.** Ticket L73: "VT ÷ (VT + VGSH + cash float); starts at about 60% or 50%".

13. MINOR: two Session Rules outcomes have no branch

Evidence. "Teams must meet the required trading activity and portfolio management guidelines" (Rules & Roles, VP) is undefined publicly; a Stock-Trak setting may cap a whole security type (A4 f.1 asks to record

it). The plan's low-trade design has no branch for either. **Exact fix.** Gate box (c): "If the logged-in pages show an activity minimum or a cap on a whole security type (all ETFs, all bonds), stop and ask Wharton (Contact Us) before any order; never add trades to meet a count".

14. MINOR: "reduce volatility" is banned for the one fund it describes

Evidence. The ban covers all WInS funds (brief L68; ticket L146). For VGSH, the ticket's own C3 number (the growth money falls about 10-12% instead of 20% in a 20% stock fall) is exactly lower volatility, and Wharton's example note uses the phrase (TN guide L36, VRF). The ban is right for the promise money (S4 finding 7, D9 M012). **Exact fix.** Scope the ban to the promise money; for VGSH allow "lowers the growth money's swings" only with the number beside it (AY1 C10: it is the ordinary effect of holding Treasuries, not a VGSH property).

15. MINOR: one overclaim in decision 1

Evidence. Brief L21: "the only design whose certainty rests on today's prices". The ladder is bought at January 2027 prices, and in about 1 in 3 rate paths part of it waits for the 2028 deposit (AX1b item 1, ASM; brief s16). **Exact fix.** "whose certainty rests on market prices on the purchase date, not a forecast (and, in about 1 in 3 rate paths, partly on the 2028 deposit)".

16. MINOR (IPS consistency): three gaps between permanent notes and the IPS spec

Evidence and exact fixes.

- (a) D10 1.5 (VRF) requires the IPS date sentence to say WInS holds **stand-in funds, not the ladder**, under either option. Brief section 3 "Words needed" omits it. Add it to both columns.
- (b) Under (iii), the SGOV note depends on the rule "any leftover waits in T-bills until 2028" (D1 M004 rule 7). AY2's IPS word budget (`audit_judges_practice_comms.md` s5, VRF) has no line for it. Add "(iii) only: T-bill leftover clause, about 8-10 words" to the budget handed to ips_spec.
- (c) D13c N1(e) (VRF) says "either way" one reflection answers "why so little equity?". The ticket asks for it under (iii) only (L125, L163). Under (ii), at 17-20.5% stocks, add to the growth reflection checklist: risk stated goal by goal (promise money: no stock risk; growth money: the chosen share), as E1 P6 also proposes.

17. MINOR: the growth note's research element is the most generic one available

Evidence. Ticket L147 offers VT's stock count and top-ten share: any team can cite them. The Laura-specific, dated, checkable market fact is the rate the Treasury curve implies for her 2028-2033 growth-money dates, about 5.2% a year (D2 E2 with AX2 C1: "market-implied, not locked"; VRF curve), which is what the stock share must beat. **Exact fix.** Add it as the first C3 option for the growth note, with the words "implied by the [date] curve" and the existing ban on "known/locked".

18. MINOR: timing and labelling details for the "tested" evidence

Evidence. The chance of a 10bp or larger move from the fill to the Oct 20 close is about 52% for an Oct 2 fill but about **40%** at the Oct 9 hard stop (DER, script [4]; ASM). The brief says earlier fills give "more days" but not how much the test weakens. Brief s14 still says "use spot duration (10.16y)" while the ticket targets 9.90 on issuer numbers; S4 finding 8 (VRF) reconciled them but its one line was never added. **Exact fix.** Brief timing paragraph: add the 52% vs 40% figures. Ticket header: add S4's line "9.90 on issuer durations is about 10.1 in full revaluation, close to the 10.16 spot duration".

19. MINOR: model prices shown as prices

Evidence. "\$294,387" and the (iii) note's "\$292-294k at [trade-date] Treasury prices" are par-curve model prices, not STRIPS quotes (D3 AX1 C25, VRF). **Exact fix.** Ticket section 5 promise row: "the payments' value on the [date] Treasury curve (our calculation)".

20. MINOR: stale notes and one citation

Evidence and exact fixes. Brief L172-173 (see finding 2): correct it. Ticket L57 cites "SMAApply FAQ" for fills at the next open; the source is SMAApply Trading Details (R-W62, VP). Correct the citation.

The one-session test (does the plan survive if the team trades only once this week?)

	Option (ii)	Option (iii)
Orders in one session	4 buys (5-7 under a cap); all fill at one open	3 buys (4 with a rung)
Three executed trades by Oct 23?	Yes; three distinct roles	Only if the T-bill leg exists at trade-date prices and the last order fits the cash (finding 1: roughly 1 in 9 to 1 in 5 that the honest leg is under \$1,000; about 1 in 8 that a 5bp overnight fall leaves too little cash if all orders go in before the open; ASM)
Three distinct decisions?	Yes	Only with a listed rung (finding 4)
Day-trade risk	None (buys only)	None (buys only)
"Tested" window	~52% chance of a 10bp+ move (Oct 2 fill); ~40% at the Oct 9 hard stop	same
Needs a second session later?	Only if the split was provisional (decision by Oct 16) or an order was rejected	Only for the fix in finding 1 step 3

Reading (INT): with findings 1 and 6 fixed, both options survive one session. As written, only (ii) does.

Sources (accessed 2026-09-28)

- T1 files attacked (VRF): `research/insight_v1/phase_D/trading_now_brief.md`, `research/insight_v1/wins_now/securities_and_allocation_v1.md`; script `scripts/T1_ticket_v1_numbers.py` (re-run).
- Inputs read (VRF, with their own labels): `wins_now/securities_and_allocation_v0.md`, `wins_now/S4_red_team.md`, `phase_A/wins_week1_guardrails.md`, `phase_A/case_register.md` (R-W56, R-W59, R-W62, R-W71, R-W72, X-8); `phase_D/D1_rates.md` (+AX1b), `D2_equity_ai.md` (+AX2), `D3_quant.md` (+AX1), `D5_cosponsors.md` (+AY1), `D6_behavioural.md` (+AY1), `D7_wharton_intent.md` (+AY2), `D8_practice.md` (+AY2), `D9_communication.md` (+AY2), `D10_compliance.md` (+AX2), `D12_overflow_client.md` (+AY1), `D13c_voice_map.md`, `audit_markets_rules.md`, `audit_client_cosponsors.md`, `audit_judges_practice_comms.md`, `audit_rates_D1.md`; `phase_C/survivors.json` (M001, M002, M004, M005, M012, M028, M034, M044, M062, M212, M238, M247); `phase_E/E1_change_proposals.md` (P5, P13; written after T1, not yet audited).
- Official (VRF): `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L11-16, L30-31, L36-53), `2026_WGY_Investment_Competition_Guide.txt` (p.3, p.5), `2026_WGY_Investment_Policy-FINAL.txt` (L73-75), `Laura_Gao_2026_Client_Profile.txt` (L61-74, L91-92, L102-103, L129-130), `SMAApply_Deliverables_Page_2026-09-27.md`.
- Primary web rules (VP, via Phase A and the audits, not re-fetched by T2): SMAApply Trading Details and FAQ; Wharton Rules & Roles; 2026-27 WInS User Guide.
- New script: `research/insight_v1/scripts/T2_red_team_checks.py`.

What this teaches

1. **A fix can have its own failure mode.** The T-bill trade solved "only two trades" on Friday's prices. Re-pricing it for a small rate move showed it can vanish, so the fix needs its own rule written down before trading.
2. **Timing is part of the arithmetic.** An order typed in Sydney fills hours later at a new price. A thin cash float turns an ordinary overnight move into a rejected order.
3. **Read the corrections, not just the conclusions.** One stale sentence ("the rule picks 50% under both forecasts") was enough to tilt a vote. The audit that fixed it was already in the repo, fourteen minutes old.
4. **A fair comparison lists the other side's best argument.** For the WInS book, that argument sits in Wharton's own words: the simulator holds exactly her first deposit, and the second one is never added.

B1a Case Anomalies: a line-by-line interrogation of the four official documents

Agent: B1a (lens "Case Anomalies", starting angle: read the case, IPS guide, Trading Notes guide and SMApply criteria page line by line, in order, and ask of every sentence "why is it worded this way?"). insight_v1 run, Phase B, written 2026-09-27. AI-generated research (Claude Code) for brainstorming by Team Caplet. **Nothing here is text to submit.** Where a hypothesis says what a sentence "should contain", that is a checklist for the students, who write every word themselves. Securities named here are examples for specialists to check, and every one is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK: confirm on this year's WInS approved list/rules before trading.**

Map used: research/insight_v1/phase_A/case_register.md (R-ids), fact_register.md (F-ids), stakeholder_map.md (BS/Q-A3 ids), wins_week1_guardrails.md (G ids). I did not re-ask anything in brief section 8. Where I touch an item from brief section 9 (open), I say what evidence is new.

0. Summary (read this first)

29 questions, ordered by deadline: 7 for WInS trading now and the Trading Notes (Oct 23), 9 for the IPS (Nov 6), 13 for the Final Report (Dec 4). The six I think matter most, and that I expect almost every other team to miss:

1. **Laura's 2033 facility contribution is a decision she makes, not a market outcome (Q17).** The case asks how confident the team is that "her 2033 contribution will fall within that range" (case p.3 L117-118). Most teams will answer with a forecast interval for portfolio value. But she chooses the amount in 2033. If the plan pairs a bought floor with a rule that caps the gift at the top of the range (anything above the cap stays with her as flexibility), then the contribution lands in the range whenever the portfolio can afford the bottom. For a bought floor, that is close to certain. This removes the "two-sided range" trap (R-AN9) and links the range rule to the flexibility rule (Q22).
2. **The recommended "dollar range" is computed in 2026, but Laura says it in 2031 (Q18).** A fixed number worked out today will be out of date by 2031. The case's own timeline supports giving her a rule to apply in 2031, with today's projection shown as an illustration.
3. **Wharton's own example trading note explains the bond trade in the language of lowering portfolio volatility (Q2).** Our hedge holds longer-duration Treasuries, and their prices swing more than the example's intermediate fund. So our notes must explain in a sentence that price volatility is not risk to Laura's promise: the hedge moves with the value of the ten payments. A team that copies the example's logic undercuts its own strategy.
4. **Once the IPS is submitted, the strategy cannot be revised, so every "what if" rule has to be written in advance (Q8).** "Your team may not revise its investment strategy after the submission deadline" (IPS p.2 L73-74). The Final Report must still give recommendations "under varying investment outcomes" (IPS p.2 L75-78). Every contingent rule (rates fall before January 2027; a smaller or late 2028 deposit; the 2031 floor; the 2033 gift) has to be in the IPS, or the Final Report will look like a revision.
5. **The WInS rules point to treating WInS as a rehearsal of Laura's 2027 deposit (Q1; new evidence on an open item).** WInS cash was cut from \$500,000 to exactly \$300,000 and "The additional \$150,000 ... will not be added to WInS" (R-W56, R-W59). The case says WInS is implementation "during the competition", which is Year 0. That evidence favours the literal 2027 book, which is almost all ladder. The current draft instead mirrors the post-2028 target (wins_now/securities_and_allocation_v0.md). The team has to choose and state the choice.
6. **Past cases show how this case was designed (Q28, new primary evidence).** The 2021-22 case already had ten annual payments (a \$5,000 scholarship "for at least 10 years", with "Should she keep \$50,000 in cash?"). The 2024-25 case gave teams facility cost estimates; this year's case removes them on purpose. The "portfolio manager ... who makes the final investment decisions" line is left over from older cases, where it referred to the firm's own \$100,000,000 portfolio, not the client's (Q9). VERIFIED-PRIMARY, Wharton archive pages read 2026-09-27.

1. Method

I read each document in authority order (case, IPS guide, Trading Notes guide, SMapply page) one sentence at a time. For each sentence I asked five things: what does it allow; what does it forbid; what does it leave undefined; what number or word is odd; and what would a careless team get wrong because of it. Then I compared the wording with three earlier Wharton cases (2021-22, 2022-23, 2024-25), which I read on Wharton's public archive pages. The aim was to see which sentences are new this year, and so were probably written on purpose. I checked a few instrument facts on primary pages (TreasuryDirect, iShares) where a sentence depends on them.

Terms used below:

- **Liability-driven investing (LDI):** investing so that assets move with the value of a promised future payment.
- **Surplus:** assets minus the market value of the promise.
- **Duration:** how much a bond's price changes when interest rates move by 1%.
- **Floor:** an amount already secured by a bought bond.
- **Residual claim:** whatever is left after a prior claim has been paid.
- **Stage gate:** a checkpoint at which a project moves on only if set conditions are met.

2. The questions, with reasoning

Status labels follow brief section 3. "Hypothesis" is always my guess, never a finding.

Tier 1: WInS trading now and the Trading Notes (Oct 23)

Q1. Should the WInS portfolio be the literal 2027 book (almost all ladder), not the post-2028 target mix? Seeds 3 and 4; new evidence on the brief section 9 open item.

- Anchors: R-W56, R-W59, R-C75, R-C76, R-AN14, R-AN22, X-7, F-101, F-605.
- Official words: "The WInS portfolio represents each team's implementation of its investment strategy during the competition. It does not determine Laura's actual portfolio value at the beginning of 2027." (case p.4 L129-130); "The \$300,000 is your team's WInS simulator balance. The additional \$150,000 contribution described in the Client Case Study will not be added to WInS." (S-TRADE).
- Reasoning:
 - "During the competition" is Year 0 (2026).
 - The second sentence only makes sense if the designers expected some teams to treat WInS as the period just before Laura's 2027 deposit.
 - Last year's cash was \$500,000 (R-W98). This year's was cut to exactly the Year-1 deposit.
 - Seed 3's premise ("500k = WInS cash") no longer holds; see R-AN14. The new link is 300k = the 2027 deposit.
 - Under lock-early, the 2027 book is about 292k of ladder and about 8k of surplus (F-101, F-104). Its equity share is tiny.
 - The current draft instead recommends the post-2028 mirror: hedge about 66%, VT about 20% (VT is Vanguard Total World Stock ETF, a global equity fund), and a floor proxy.
- Hypothesis (guess):
 - The case text favours the 2027 book.
 - The criterion "uses appropriate diversification" (R-S24) and the value of showing the growth sleeve in notes pull the other way.
 - A defensible middle path: hold the 2027 book, and put a small, clearly labelled "surplus sleeve" (about 3% of 300k) in equity. Every note says which bucket the trade belongs to.
 - Either way, the IPS must say in one sentence which book WInS represents.
- Would change: a **decision** (the WInS weights now; the equity weight differs by about 15-20 points) and a sentence in the IPS and Final Report.

Q2. How do our notes explain that long-duration Treasuries, whose prices swing more than the example's intermediate fund, reduce risk to Laura's promise?

- Anchors: R-T23, R-T24, R-T25, R-AN44, F-203, F-204, F-205.

- Official words: "We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment. ... This trade supports our plan to balance continued growth with reliable future cash flows as the residency funding date approaches." (TN p.2 L36-39).
- Reasoning:
 - The example uses two ideas:
 - "Reduce portfolio volatility": the asset-only view.
 - "As the residency funding date approaches": a glide path, the gradual shift to safer assets over time.
 - Our strategy holds long Treasuries now; the draft has TLH/TLT near 40% of WInS. TLT's 3-year standard deviation is 13.74% (F-202), about twice IEF's 6.54% (F-201).
 - By the example's logic, that trade raises volatility. By LDI logic, it lowers surplus risk.
 - If our note copies the example's words, a judge could see a contradiction. If it explains the difference, it shows that we understand the promise, not just the assets.
- Hypothesis (guess): each hedge note needs one plain clause saying that the bond's value rises and falls with the value of the ten payments, so the gap between them stays small. Avoid "reduce volatility" as the stated purpose.
- Would change: a **sentence** in every hedge-trade WInS note written from now on (notes cannot be edited later) and in the Oct 23 reflections.

Q3. Should every WInS note carry one dated, checkable data point, a link to Laura, and one named risk, going beyond Wharton's example, which contains no number?

- Anchors: R-T6, R-T11, R-T28, R-W82, R-AN43, G (guardrails section 0 item 6).
- Official words: "your Trading Notes should document the research, analysis, and reasoning behind your investment decisions" (TN p.1 L8-9); "record the reasoning, research, and intended strategic role of each decision at the time it is made" (S-WINS).
- Reasoning:
 - The guide asks for "research", but the example (59 words, R-T26) contains no data.
 - Notes are fixed at trade time and quoted "exactly as it appears in WInS" (TN p.2 L44).
 - A dated fact, for example "10-year Treasury 5.17% on 25 Sep" (F-001), proves the reasoning came before the outcome. Hindsight cannot fake that.
 - A statistics graduate client (R-C12) would value this.
- Hypothesis (guess): yes. Spec per note: action; which bucket; one dated number with its source named in words; the Laura goal it serves; one named risk. Aim for 60-90 words. No character limit is published (G, section a2).
- Would change: a **sentence** spec for every WInS note from day 1 (WInS-now); it feeds the TN.

Q4. When one strategic decision takes several trades (a position limit forces a split, or two funds make up one duration match), which trade carries the note, and should each note stand alone when quoted?

- Anchors: R-T8, R-T27, R-T33, G (guardrails section 0 item 4), R-W66.
- Official words: "select three investment decisions that best illustrate your team's strategic thinking" (TN p.1 L11) versus "Select three (3) Trading Notes from trades your team executed in WInS." (TN p.2 L42).
- Reasoning:
 - The guide uses "decision" and "trade" as if they were the same thing.
 - With a possible 25% per-security position limit (UNVERIFIED for this season, G), the hedge alone needs 3 or more trades.
 - If the "real" decision's reasoning sits on a companion trade's note, the quoted note may look thin.
- Hypothesis (guess):
 - Write a full note on every trade.
 - On multi-trade decisions, each note names its companion trades ("one of three trades that together match...").
 - Choose the lead trade (the largest, or the first executed) as the TN candidate.
- Would change: a **decision** on how notes are written now and which trade is picked for the TN.

Q5. Should the three TN picks between them cover the six named dimensions, and should at least one be a decision that "tested" or "refined" the strategy rather than one that only "reflected" it? Extends seed 13.

- Anchors: R-T3, R-T9, R-T17, R-T18, R-T20, R-C88, R-AN40.
- Official words: "including its approach to growth, risk, liquidity, funding reliability, financial flexibility, and future cash-flow needs" (TN p.1 L5-6); "reflected, tested, or refined the team's developing strategy" (case p.4 L152-153); "A decision can still demonstrate thoughtful analysis and strategic thinking even if your team later sells the investment or changes its approach." (TN p.1 L26-27).
- Reasoning:
 - Seed 13 asked why the notes come before the IPS. The answer the text gives: the designers want to see a strategy being developed.
 - The words "tested" and "refined", and the explicit protection for later sells, invite at least one note that shows the strategy being adjusted, for example:
 - a rebalance after a rate move;
 - a trade split forced by a rule the team discovered;
 - a sell.
 - Three "we bought X for the plan" notes read as a static list.
- Hypothesis (guess): pick one note each for the promise (funding reliability and future cash-flow needs), for growth and risk, and for a real refinement (liquidity or flexibility). This needs a genuine refinement trade before Oct 23. Never create one only for show: fabrication is banned (R-W28).
- Would change: a **decision** (which three trades; whether a planned rebalance or refinement happens before Oct 23).

Q6. Does the WInS rule "no more than twice a security's current daily trading volume" use today's running volume, so that an overnight order from Australia, filled at the open in a thin Treasury fund or an individual bond, could be rejected or only partly filled?

- Anchors: R-W62, R-W66, R-W72, R-W91, F-206.
- Official words: "Your team may trade an amount equal to no more than twice a security's current daily trading volume." (S-TRADE); "Orders placed while the market is closed are filled at the security's opening price when the market reopens." (S-TRADE).
- Reasoning:
 - "Current" is undefined. At the open, today's volume is close to zero.
 - The team is in Australia, so its orders mostly arrive overnight (U.S. time).
 - Defined-maturity funds are small:
 - iShares iBonds Dec 2036 (IBTR): 30-day average volume 68,915 shares at \$23.63, about \$1.6m a day; net assets \$34.2m (VERIFIED-PRIMARY, ishares.com, 2026-09-27).
 - Dec 2034 (IBTP): 111,113 shares at \$24.22, about \$2.7m a day (VERIFIED-PRIMARY).
 - On a daily-average basis, a \$30-40k order is far inside 2x. On an intraday basis at the open, it might not be.
 - Individual Treasury bonds in WInS price once a day (R-W91). How the rule applies to them is unknown.
- Hypothesis (guess): Stock-Trak probably uses the previous day's or an average volume. Test with one small order in a thin fund before a large one, and read Session Rules.
- Would change: a **decision** (order type, timing and sizing in WInS now; whether thin iBonds-type funds are used).

Q7. How do we show "appropriate diversification" when about two-thirds or more of the book is Treasuries?

- Anchors: R-S24, R-W70, R-W71, R-AN45, X-12, F-409.
- Official words: "uses appropriate diversification" (criterion 1); "diversification across asset classes, sectors, industries, market capitalizations, geographic regions, risk characteristics, and funding purposes. The goal is to build an intentional mix ... not simply to own a large number of securities." (S-TRADE).
- Reasoning:

- A grader skimming holdings could read "concentrated in Treasuries".
- Wharton's own page lists "funding purposes" as a kind of diversification. The hedge and the growth sleeve are exactly that.
- Inside the growth sleeve, geographic breadth (for example a global fund) covers the rest.
- Hypothesis (guess): state in the IPS and Final Report that diversification is by purpose first (promise versus growth), then by geography and asset within the growth sleeve. Use Wharton's own term "funding purposes".
- Would change: a **sentence** in the IPS (diversification line); the growth-sleeve **decision** (global versus U.S. only).

Tier 2: the IPS (Nov 6)

Q8. Which contingent rules must be written into the IPS now, because the strategy cannot be revised after Nov 6 and the Final Report must cover "varying investment outcomes"? Extends seed 4.

- Anchors: R-I34, R-I35, R-I36, R-I37, R-I27, X-5, X-23, F-111, F-112, F-403.
- Official words: "Once submitted, the IPS becomes the official record of your team's investment strategy. Your team may not revise its investment strategy after the submission deadline." (IPS p.2 L73-74); "present its analysis and recommendations ... under varying investment outcomes" (IPS p.2 L77-78); "How your strategy will guide portfolio construction and investment decisions as the client's needs change" (IPS p.2 L61-62).
- Reasoning:
 - The money arrives in January 2027, after the freeze. The ladder cost more than \$300k on 173 of 185 trading days in 2026 (F-111), and $P(\text{cost} > \$300\text{k on } 2027-01-01)$ is about 24% (F-112, ASSUMPTION).
 - A rule for that case written only in the Final Report is a revision. Written in the IPS, it is the strategy being applied.
 - The same holds for:
 - a smaller or late 2028 deposit (F-403);
 - the 2031 floor share;
 - the 2033 gift rule;
 - rebalancing bands.
- Hypothesis (guess): the IPS needs about 4-5 one-line "if X, then Y" rules; about 120-150 of the 500 words. Priority: January 2027 purchase rule; 2028 shortfall rule; 2031 floor rule; 2033 contribution rule; band rule.
- Would change: a **decision** (the IPS structure and word budget) and **numbers** (rule thresholds).

Q9. Who decides what: which decisions are automatic rules, which does the firm make, and which does Laura make? Extends seed 2 with new primary evidence.

- Anchors: R-C3, R-C25, R-C57, R-AN4, R-AN53, R-W19, R-W87, X-10, R-C89.
- Official words: "Your portfolio manager (your team's teacher/advisor who makes the final investment decisions for your firm's portfolio)" (case p.1 L4-5); "Laura must decide how much of the remaining portfolio she can responsibly contribute" (case p.3 L101-102); "This formally establishes the strategy and the team's decision-making framework." (case p.4 L155-156).
- New evidence (VERIFIED-PRIMARY, read 2026-09-27):
 - The 2021-22 and 2022-23 cases say: "The members of your [analyst] team hope to one day become portfolio managers who make the final investment decisions for WGAM's portfolio".
 - In those cases the firm (Wharton Global Asset Management, WGAM) "currently manages a \$100,000,000 portfolio".
 - So the portfolio manager line is inherited boilerplate. "Your firm's portfolio" originally meant the firm's own \$100m book, not the client's strategy.
 - This year the firm name and the \$100m book were dropped, which left the phrase orphaned.
- Hypothesis (guess):
 - The IPS "decision-making framework" should allocate decision rights. Laura approves the policy and makes the two big choices: the 2031 announcement and the 2033 gift. Pre-set rules handle

rebalancing and the hedge.

- The team (as the firm) recommends, and reports yearly.
- Say nothing that implies the real advisor decides.
- Would change: a **sentence** (the governance clause in the IPS).

Q10. In an IPS that bans charts, which numbers and rules must appear in words, and are small tables or informal source mentions allowed? Extends seed 14.

- Anchors: R-I23, R-I31, R-I45, R-I55, R-I56, R-AN37, R-AN38, F-603.
- Official words: "Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted." (IPS p.3 L123); "Focus on the strategy and decision-making framework ... rather than describing individual investments or presenting detailed financial calculations." (IPS p.2 L51-52); "The IPS is not expected to include ... a final facility-contribution range" (IPS p.2 L68-69).
- Reasoning:
 - Tables are not on the banned list, but a reader may treat a table as a "graphic". That is an exclusion risk, and exclusion means "will not be considered for semifinal selection".
 - "Final" in "a final facility-contribution range" implies that a provisional rule is fine.
 - Professional IPSs normally carry target weights and ranges. Those are not "detailed calculations".
- Hypothesis (guess):
 - No tables.
 - Allow about 6-10 numbers in words: target bucket weights and bands; the certainty test (for example "funded ratio of at least 100% at market prices"); the floor share; the key dates.
 - Name data sources informally in the text (for example "at September 2026 Treasury yields"); do not cite them.
 - Count words with the same tool the team will use for the final PDF, and leave a safety margin.
- Would change: a **decision** (IPS layout) and **numbers** (which figures appear).

Q11. Should the deliverables tie each of the case's three wordings (certainty, confidence, reliability) to one object, and use them only that way?

- Anchors: R-C47, R-C54, R-C69, R-C74, R-I6, R-I11, X-16, R-AN1.
- Official words:
 - "a high degree of certainty" and "a high degree of funding certainty" (case p.3 L91, L97-98): the payments.
 - "state how confident they are that her 2033 contribution will fall within that range" (p.3 L117-118): the facility range.
 - "Teams may reach different conclusions about ... funding confidence" (p.4 L127).
 - "funding reliability" (IPS L20, L59; TN L5; S-WINS).
- Reasoning:
 - The case uses "certainty" for the payments and "confident" for the facility range.
 - "Funding confidence" appears in the list of things teams may disagree on. So the confidence level applies to the facility, while the payments need high certainty.
 - Mixing the words (for example "95% confidence for the payments") signals a careless reading to a statistics graduate.
- Hypothesis (guess): certainty goes with the payments (bought: market-consistent, not a model percentage); confidence goes with the facility range (a stated probability); reliability is the portfolio property the IPS balances.
- Would change: **sentences** throughout the IPS and Final Report (the glossary line in the Final Report).

Q12. The case says Laura has "financial resources outside the portfolio". Does that raise her capacity to take risk with the surplus, and should the IPS say this portfolio is a dedicated goal portfolio, not her whole wealth?

- Anchors: R-C33, R-C34, R-C37, R-C38, R-AN16, R-AN17, F-407, F-409.
- Official words: "Laura's living expenses and short-term financial needs will be covered by income and financial resources outside the portfolio." (case p.2 L61-63).
- Reasoning:

- "Financial resources", not only "income", means she has other assets.
- Risk capacity (the ability to bear loss) for the surplus is therefore higher than for a whole-wealth portfolio.
- The promise is ring-fenced by the ladder. This bears on the open sleeve-equity question (50/60/70%; F-407: median \$205k/\$207k/\$209k, p5 \$164k/\$159k/\$154k).
- Goals-based wealth management treats each goal as its own sub-portfolio.
- Hypothesis (guess): yes. This is case evidence for the upper end of the sleeve band (60-70%) and for calling the portfolio "goal-dedicated" in the IPS. It does not touch the payments.
- Would change: a **number** (sleeve equity weight) and a **sentence** (risk-tolerance line: willingness from the "thoughtful risks" line, capacity from outside resources).

Q13. Should the Final Report give Laura one recommendation or a small menu of options?

- Anchors: R-C8, R-C38, R-S25, R-AN5.
- Official words: "she wants her investment team to recommend an appropriate balance" (case p.2 L71-74) versus "the investment strategy that Laura ultimately chooses" (p.1 L9).
- Reasoning:
 - She chooses among firms; within a firm, she asked for a recommendation.
 - A menu can look like a failure to decide. A single number with no alternatives can look rigid.
 - Professional practice (D8) often shows one recommendation plus the cost of the alternatives.
- Hypothesis (guess): one recommendation, with a single line on what she would give up or gain by choosing one step more cautious or one step bolder (for example sleeve equity 50% versus 70%, using F-407 numbers). That respects her choice without abdicating.
- Would change: a **decision** (the structure of the Final Report's recommendation section).

Q14. What liquidity need does the case actually give Laura, and what should the IPS's "liquidity" sentence therefore say?

- Anchors: R-I11, R-I21, R-C31, R-C33, R-C34, R-T3.
- Official words: "How does your strategy balance growth, risk, liquidity, funding reliability, and financial flexibility?" (IPS p.1 L20); "she will neither add to nor withdraw from the portfolio before 2033." (case p.2 L63-65).
- Reasoning:
 - "Liquidity" appears in every guide, yet the case gives no cash need before 2033 (living costs are outside the portfolio).
 - Liquidity needs are therefore dated: Jan 1 each year from 2033 to 2042, plus the facility gift in 2033.
 - That lets the plan hold bonds to maturity and avoid a cash drag, but it must be said explicitly. Otherwise a reader may think liquidity was ignored.
- Hypothesis (guess): one sentence stating the dated liquidity needs, and that each is met by a maturing bond, not by selling assets at uncertain prices.
- Would change: a **sentence** in the IPS.

Q15. Would a former tech product manager recognise the plan as hers if it were framed as stage gates with decision dates (2027 buy, 2028 top-up, 2031 announce, 2033 gift) and conditions for each?

- Anchors: R-C12, R-C17, R-C18, R-C29, R-AN11.
- Official words: "before beginning her career as a product manager in the technology industry" (case p.1 L16-17); the timeline lists only 2026, 2027, 2028, 2031 and 2033 (p.2 L49-57).
- Reasoning:
 - The case lists only the years when a decision falls (R-AN11).
 - Product managers work with roadmaps, milestones and go/no-go gates.
 - Framing the IPS as dated gates with rules is both professional (X-23) and in her own working language. It is not identity decoration, because the frame comes from her stated career.
- Hypothesis (guess): yes, as the organising frame for the IPS and the Final Report timeline chart. D6 should check her public professional writing for how she talks about product and project planning. Do not overclaim.

- Would change: a **decision** (the organising frame of the IPS and Final Report).

Q16. The case names no values or ESG preference this year, but every recent case had a social goal. Should our deliverables say that Laura's impact comes through the payout (the residency), not through screening the portfolio? Extends seed 12 with new evidence.

- Anchors: R-AN34, R-C39, R-C40, R-W52.
- Official words: 0 hits for "ESG", "values", "impact", "sustainab" in the four documents (R-AN34, VERIFIED-REPO-FILE).
- New evidence:
 - 2021-22: a scholarship for women of colour in engineering (VERIFIED-PRIMARY).
 - 2022-23: a start-up fund and a wellness centre (VERIFIED-PRIMARY).
 - 2024-25: housing and a youth centre in Nigeria (VERIFIED-PRIMARY).
 - 2025-26: "think beyond financial returns and consider how their strategies could support long-term community impact" (VERIFIED-PRIMARY, finalists page).
 - This year the impact is in the goal. The investment instructions are silent on it.
- Hypothesis (guess):
 - Do not add ESG screens.
 - One line in the Final Report: the promise is secured with the plainest instruments, so that every dollar of impact reaches the residency.
 - Tokenism rule: never pick investments by identity.
- Would change: a **decision** (no screens) and a **sentence** (the Final Report rationale).

Tier 3: the Final Report (Dec 4)

Q17. Because the 2033 contribution is Laura's decision rather than a market outcome, should the confidence statement be the probability that the portfolio can afford the bottom of the range, combined with a rule that caps the gift at the top? Extends seed 9.

- Anchors: R-C57, R-C67, R-C68, R-C69, R-C71, R-AN9, R-AN20, F-402.
- Official words: "state how confident they are that her 2033 contribution will fall within that range" (case p.3 L117-118); "Laura must decide how much of the remaining portfolio she can responsibly contribute" (p.3 L101-102).
- Reasoning:
 - "Her 2033 contribution" is chosen by her in 2033.
 - Rule: contribute the floor, plus a set share of anything above it, up to the top of the range. Anything beyond the top stays as flexibility. Under this rule, the contribution falls within the range whenever the residual is at least the floor.
 - If the floor is bought in 2031, that confidence is close to certain. The upper end then becomes a stated aspiration with its own probability, not a risk of "missing" the range.
 - Most teams will present a portfolio-value forecast interval and report something like "90% confident", which mixes up a decision with a forecast.
- Hypothesis (guess):
 - Report two numbers: $P(\text{contribution} \geq \text{floor})$, close to 100% once bought; and $P(\text{reaching the top of the range})$, from the model.
 - Say that the range cannot be exceeded by design.
 - D3 should compute both from `strategy_mc` under the rule.
- Would change: **numbers** (confidence levels) and a **sentence** (the range statement in the Final Report and the co-sponsor draft).

Q18. Should the recommended "dollar range" be a fixed number computed today, or a rule Laura applies in 2031, with today's projection shown as an illustration?

- Anchors: R-C62, R-C63, R-C65, R-C68, R-C77, F-402.
- Official words: "Laura plans to begin approaching potential co-sponsors in 2031"; "In 2031, however, the value of her portfolio in 2033 remains uncertain."; "Teams must recommend the dollar range Laura should communicate" (case p.3 L109-117).
- Reasoning:

- The Final Report is written in 2026. The 2031 portfolio value will be known in 2031.
- A fixed 2026 range (for example "\$150-250k") will almost certainly be wrong in 2031, one way or the other. F-402: the floor's p5/p50/p95 are \$127k/\$165k/\$217k.
- The case asks for "the dollar range", so a rule alone is not enough.
- Hypothesis (guess):
 - Give the rule. For example: announce the bought floor as the bottom; the top is the floor plus a share of the remaining sleeve.
 - Also give the 2026-vantage illustration: the median case range, and how the range shifts in a bad or good 2027-2031, using p5/p95 from F-402.
 - This answers "favorable and unfavorable market outcomes" (L118-119) directly.
- Would change: a **decision** (the form of the recommendation) and **numbers** (the illustrative range).

Q19. The case says Laura will "begin" approaching co-sponsors in 2031. Should the plan schedule range updates that narrow the range between 2031 and 2033, and when would the rest of the floor be bought? Extends seed 8.

- Anchors: R-C62, R-AN7, F-013, F-402.
- Official words: "Laura plans to begin approaching potential co-sponsors in 2031, two years before the residency is established." (case p.3 L109).
- Reasoning:
 - "Begin" implies that fundraising runs from 2031 to 2033, with more conversations along the way.
 - Seed 8's point (a 2-year Treasury matches the gap) is settled in brief section 8.7.
 - New angle: a range that narrows at pre-announced dates (for example in 2032) matches how capital campaigns update donors. It lets more of the sleeve be locked as the date nears, and it keeps credibility because updates only move upward from a bought floor.
- Hypothesis (guess):
 - One pre-announced update in 2032. The floor never falls.
 - The top of the range can narrow.
 - D5 should check fundraising practice for pledge updates.
- Would change: a **decision** (the 2031-2033 rule in the IPS) and a **sentence** (the co-sponsor draft).

Q20. With the facility cost unknown, what makes Laura's contribution "meaningful" to co-sponsors: its dollar size, its share of her capacity, or the certainty that it will be delivered? Extends seed 6.

- Anchors: R-C64, R-C94, R-C95, R-AN6, BS-17, Q-A3-06, Q-A3-13.
- Official words: "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute." (case p.3 L110-112).
- Reasoning:
 - "Meaningful" is undefined. The cost is deliberately unknown (R-C94).
 - The three signals the case lists are viability, commitment and willingness. They depend on credibility and on "skin in the game" (a personal stake) relative to what she has, not on share of cost.
 - Seed 6 (why co-sponsors may fund the building but not operations) is partly answered by BS-17: funders avoid running costs.
 - New link: Laura's secured ten years of operations is itself a "meaningful" contribution, because a building with no operating money is a stranded asset. So the operating promise counts in the co-sponsor message, not only the facility gift.
- Hypothesis (guess): the co-sponsor draft should lead with total secured support: \$500,000 of operating payments (nominal), plus the facility floor. It should state her gift as a share of her investable portfolio, never as a share of the (unknown) cost.
- Would change: a **sentence** (the headline of the fundraising draft) and a **decision** (whether to express the gift as a share of her capacity).

Q21. Since the facility money is only what is left after the reserve, is it a residual claim? And should the Final Report show co-sponsors that matching the reserve removes the leverage that would otherwise make the gift swing more than the whole portfolio? Extends seed 5.

- Anchors: R-C50, R-C57, R-AN2, R-AN26, F-401, F-106.
- Official words: "At the beginning of 2033, before making the first operating payment or contributing to the facility, Laura will set aside a portion of the portfolio to fund the ten payments." (case p.3 L94-95); "how much of the remaining portfolio" (L101).
- Reasoning:
 - Seed 5 (promise first, then the facility) is confirmed by the word order and by "remaining".
 - New angle: if the reserve is bought only in 2033 at an unknown price (about \$395k on today's forwards, F-106), the residual is total assets minus a moving claim. A 10% move in total assets moves the residual by roughly 3 times as much in percentage terms.
 - Brief section 8 already gives the growth-first surplus: p5/p50/p95 \$20k/\$226k/\$540k, versus lock-early \$159k/\$207k/\$273k.
 - The co-sponsor-facing explanation of why the range is narrow is new: "the promise is already paid for, so her gift depends only on the growth money".
- Hypothesis (guess): one Final Report chart (the waterfall on 1 Jan 2033: reserve, then first payment, then gift, then flexibility) and one sentence for co-sponsors. Data: F-106 and F-401.
- Would change: a **sentence** and a Final Report **chart** specification.

Q22. How should "financial flexibility" be expressed when teams need not size a contingency fund? Is the flexibility simply the upside above the top of the range? Extends seed 11.

- Anchors: R-C58, R-C61, R-C84, R-AN19, R-AN20, R-AN59.
- Official words: "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops." (case p.3 L102-103); "Teams are not expected to determine the size or investment composition of a separate contingency fund or endowment." (L106-107).
- Reasoning:
 - "Not expected" is not the same as "not allowed". But a fund sized with no cost estimate would be arbitrary.
 - "The extent to which" (L104-105) permits a share or a rule as the answer.
 - With the Q17 cap rule, flexibility falls out automatically: whatever exceeds the capped gift stays with Laura for "as the project develops" (overruns, a slower start, needs after 2042).
 - One rule then answers two case requirements: a responsible contribution, and preserved flexibility.
- Hypothesis (guess): state flexibility as a rule (for example, "at least X% of the post-reserve surplus is never pledged"). Report its dollar value at p5/p50/p95. D3 should pick X from F-401 so that p5 still leaves a buffer.
- Would change: a **number** (X) and a **decision** (flexibility defined by rule, not by a separate fund).

Q23. What would it cost to make the ten payments inflation-indexed, and should the Final Report show that the phrase "not adjusted for inflation" is what makes the promise affordable? Extends seed 10.

- Anchors: R-C46, R-AN3, R-AN8, F-008, F-009, F-101, F-109, F-114.
- Official words: "For purposes of the competition, each payment is a fixed \$50,000 and is not adjusted for inflation." (case p.3 L90).
- Reasoning:
 - Brief section 8.10 settles that TIPS are not the core hedge. That is not what this question asks.
 - The qualifier "For purposes of the competition" appears three times in the case (timeline, payments, taxes). It marks simplifications that the designers know are unrealistic.
 - Quick arithmetic (ASSUMPTION: flat 5.359% nominal, which reproduces \$292k per F-109, and 2.35% inflation, close to the 2.33-2.34% breakevens in F-009):
 - fixed payments cost **\$292,273** at 2027-01-01;
 - payments indexed from 2033 (first payment \$50k, then rising with inflation) cost **\$321,971**, about **+\$30k**, more than the \$300k deposit;
 - payments indexed in 2027 dollars cost **\$370,121** (+\$78k);
 - the real value of the 2042 payment in 2033 dollars is \$40,568.
 - So the fixed nominal design is what makes the promise fit inside the first deposit. The cost of inflation passes to "any operating support beyond Laura's commitment" (L85-86).

- Hypothesis (guess):
 - One Final Report sentence and one number pair: real value falls from \$50k to about \$40.6k (2033 dollars) by 2042.
 - Indexing would have cost about \$30k more than she has in 2027.
 - Say it plainly to co-sponsors (brief section 9 says this "must be said aloud").
 - D1 should re-price on the actual TIPS real curve (F-008) instead of the flat-rate shortcut.
- Would change: a **number** and a **sentence** (Final Report assumptions; co-sponsor draft).

Q24. Funding after 2042 is out of scope, but a co-sponsor reading a ten-year pledge will ask about year eleven. What should the Final Report say without solving it? Extends seed 7.

- Anchors: R-C49, R-C41, R-AN25, R-AN59.
- Official words: "Funding the residency beyond the final payment in 2042 is outside the scope of the competition." (case p.3 L92-93); "reliable support for its early operations" (L83).
- Reasoning:
 - The case calls the ten years "early operations", which assumes other money comes later.
 - A matched reserve runs to exactly zero in 2042 by design.
 - Silence could read as a cliff. Solving it breaks scope.
- Hypothesis (guess): one sentence marking the boundary: the commitment is start-up support, and it ends on schedule. Point to the unpledged flexibility (Q22) as her optional capacity, not a promise. No numbers after 2042.
- Would change: a **sentence** (the co-sponsor draft; the Final Report scope note).

Q25. The case fixes every payment date, yet asks teams to state assumptions about "the timing of cash flows". Which timing assumptions does it want?

- Anchors: R-C31, R-C32, R-C85, R-AN12, F-107, F-113.
- Official words: "All contributions and withdrawals occur at the beginning of the applicable year." (case p.2 L59); "Teams should identify and explain their assumptions about investment performance, the timing of cash flows, outside funding..." (p.4 L146-147).
- Reasoning:
 - Real Treasuries do not mature on 1 January.
 - TreasuryDirect (VERIFIED-PRIMARY, 2026-09-27): 2-, 5- and 7-year notes are issued on the last day of the month. 10-year notes and 30-year bonds are issued on the 15th of Feb/May/Aug/Nov. Maturity equals issue date plus term, so matching instruments pay out on Nov 15 or Dec 31. S1 already costs the Nov-15 gap at +\$2,124 (about 47 days of bill reinvestment).
 - iBonds funds "terminate on or about October or December 15" of their year (VERIFIED-PRIMARY, ishares.com).
 - The other timing items the case leaves open:
 - whether the 2028 deposit arrives on time (Q-A3-11 flags that it could be late);
 - whether the facility gift is one payment in 2033 or instalments during construction.
- Hypothesis (guess): list four timing assumptions in the Final Report:
 - maturities about 6 weeks before each payment, with cash held in bills;
 - the 2028 deposit on 1 Jan, plus a late-deposit stress;
 - the facility gift paid in one lump in 2033;
 - the 2031 floor note maturing 31 Dec 2032.
- Would change: a **sentence** (the assumptions list) and a **number** (the ladder cost, \$294.4k versus \$292.3k).

Q26. Which market date should the Final Report's numbers be priced on: 25 Sep, 6 Nov (the freeze), or the latest curve before 4 Dec?

- Anchors: R-I36, R-AN36, X-6, F-001, F-002, F-111.
- Official words: "Your portfolio and Final Report should reflect the strategy established in your IPS." (IPS p.2 L75); "Trading ends and your portfolio is locked." (SMAApply).
- Reasoning:

- The 10-year Treasury yield moved about 120bp in 2026 (F-003). The ladder cost ranged from \$292k to \$326k (F-111).
- If the Final Report's numbers come from a different date than the IPS and the WInS freeze, the three deliverables may disagree. For example, the IPS says "fits within \$300k" but the late-November curve says it doesn't. That breaks "a clear and consistent investment strategy across all three" (case p.4 L162).
- Hypothesis (guess): price the base case on 6 Nov (the freeze date, matching the frozen WInS book). Add one line showing the latest curve before submission as a sensitivity. The IPS should avoid a hard dollar claim that one rate move could falsify.
- Would change: **numbers** (every Final Report figure) and a **sentence** (the IPS "fits inside" wording).

Q27. The case never says why the residency is in Taiwan. Should our deliverables refuse to supply a reason, and guard against confusing Taiwan with her Wuhan birthplace?

- Anchors: R-C10, R-C39, R-AN10, R-AN56.
- Official words: "Born in Wuhan, China, and raised in Texas" (case p.1 L13); "In 2033, Laura plans to establish a collaborative creative residency in Taiwan." (p.2 L76).
- Reasoning:
 - The case gives a place but no reason.
 - A tempting sentence (for example "returning to her roots") would:
 - invent a motive;
 - risk tokenism;
 - conflate China and Taiwan, which is factually wrong and politically sensitive.
 - Wharton says the financial scenario is invented (R-W4).
 - Public interviews mention time she spent working in Taiwan (SNIPPET-UNVERIFIED). Even if verified, it bears only on currency and cost context, not on motive.
- Hypothesis (guess): use no motive sentence. Treat Taiwan as a location with currency (Q-A3-04) and cost facts only (F-508, F-510).
- Would change: a **sentence** (deleted or never written: the client-knowledge section and the co-sponsor draft).

Q28. What does the way cases have changed across seasons reveal about what this year's designers want rewarded?

- Anchors: R-W52, R-AN34, R-C94, R-C95, R-C61, F-609.
- Evidence (VERIFIED-PRIMARY, Wharton archive pages, 2026-09-27, unless noted):
 - 2021-22: "provide a small scholarship of \$5,000 per year ... for at least 10 years. ... Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends?"
 - 2022-23: "use the interest from that investment to sustain the fund".
 - 2024-25: "it will cost about \$10,000 USD per housing unit; however, material costs and market dynamics are always changing" and "Operating the youth center for one year should cost at least a quarter of your original \$100,000".
 - 2025-26: grow \$500,000 to at least \$1.5m by 2036 (SNIPPET-UNVERIFIED, Scribd summary; page blocked).
 - 2026-27: a promise with "a high degree of certainty", no cost estimate, and no contingency fund or endowment expected.
- Reasoning:
 - Four years ago, ten payments were a side goal. This year, the promise is the core.
 - The designers removed:
 - the cost estimates teams used to anchor on;
 - endowment-style "live off interest" framing (explicitly not expected);
 - last year's growth target.
- Hypothesis (guess): the designers want to reward:
 - a precise definition of certainty;
 - honest communication of uncertainty;
 - restraint (a responsible, not maximal, gift).

They do not want to reward return-chasing or construction budgeting. D7 should find how 2021-22's winners funded the scholarship, and whether judges then preferred bonds or dividend stocks.

- Would change: a **decision** (where Final Report pages are spent: less projection theatre, more on the definition and communication).

Q29. Should the fundraising draft be written as a 2031 document, using clearly labelled illustrative figures from today's projection, or as a template with blanks?

- Anchors: R-C72, R-AN21, R-S28, X-14.
- Official words: "Each team must also draft part of Laura's fundraising materials describing this potential contribution for prospective co-sponsors." (case p.4 L123-124); criterion 5: "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors".
- Reasoning:
 - The draft is used in 2031 but written in 2026.
 - A template with blanks is safe but lifeless. A dated 2031 document with invented "actual" numbers would be fabrication, unless it is labelled as an illustration.
 - "Part" means the team chooses which part: probably the paragraph on her financial commitment.
- Hypothesis (guess): write it as a 2031 draft with every figure marked as an illustration from the median case, plus one line on how the figures will be updated. Students write it themselves (R-W26). Confirm against the Nov 9 instructions.
- Would change: a **decision** (the draft's form) and **sentences** (the labelling line).

3. Parked (asked, but failing the "would it change anything?" test or already covered)

- Why the case says "plans to invest" for 2027 and "will contribute" for 2028. There is no strategic consequence beyond R-AN35.
- Headers ("Investment Competition Guide" versus "IPS"), PDF creation dates, typos on the web pages (R-AN47). These are editing artefacts.
- The "Wuhanese American" subtitle in the case image (R-C22). It shows her public work and has no investment consequence.
- The wording of the Nov-15 maturity gap. Solved by S1 ([wins_now/S1_treasury_sleeve.md](#)); kept only as part of Q25.
- Word counts, and whether the 100-word reflection includes the quoted note. Covered by R-T36 and the guardrails; add to the day-1 checklist.

4. Evidence found in this pass (with status)

#	Fact	Source	Status
E1	2021-22 case: \$5,000 scholarship "for at least 10 years"; "Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends?"; firm WGAM "currently manages a \$100,000,000 portfolio"; "hope to one day become portfolio managers who make the final investment decisions for WGAM's portfolio"	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/ (accessed 2026-09-27)	VERIFIED-PRIMARY
E2	2022-23 case: same WGAM boilerplate; "use the interest from that investment to sustain the fund"; "compete against other student teams ... trying to land the same client"; "How will you create a compelling and clever pitch to convince Peter to choose your strategy?"	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2022-2023/ (2026-09-27)	VERIFIED-PRIMARY
E3	2024-25 case: "\$10,000 USD per housing unit; however, material costs and market dynamics are always changing"; youth centre operations "at least a quarter of your original \$100,000 investment"	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/ (2026-09-27)	VERIFIED-PRIMARY

#	Fact	Source	Status
E4	2025-26 case: grow \$500,000 into at least \$1.5m by 2036	Scribd summary via search (https://www.scribd.com/document/965855349/Case-Study-Wharton-Global-Youth-Program); page would not load	SNIPPET-UNVERIFIED
E5	2025-26 finalists page: "think beyond financial returns and consider how their strategies could support long-term community impact"; 49 semifinalists; semifinal judges include Stock-Trak's CEO, a UBS wealth-management director and a family-office COO	https://globallyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/ (2026-09-27)	VERIFIED-PRIMARY
E6	Treasury issue pattern: 2-, 5-, 7-year notes issued on the last day of the month; 3-year on the 15th; 10-year notes and 30-year bonds on the 15th of Feb/May/Aug/Nov	https://www.treasurydirect.gov/auctions/general-auction-timing/ (2026-09-27)	VERIFIED-PRIMARY (maturity = issue + term is an ASSUMPTION)
E7	IBTR (iBonds Dec 2036): net assets \$34,216,407; close \$23.63; daily volume 85,109; 30-day average 68,915; 4 holdings; funds "terminate on or about October or December 15 of the year in each Fund's name"	https://www.ishares.com/us/products/350028/ishares-ibonds-dec-2036-term-treasury-etf (2026-09-27; data as of 2026-09-25)	VERIFIED-PRIMARY
E8	IBTP (iBonds Dec 2034): net assets \$316,979,783; close \$24.22; daily volume 51,024; 30-day average 111,113; expense 0.07%	https://www.ishares.com/us/products/337745/ishares-ibonds-dec-2034-term-treasury-etf (2026-09-27)	VERIFIED-PRIMARY
E9	Cost of indexing the payments: fixed \$292,273; indexed from 2033 \$321,971; indexed in 2027 dollars \$370,121; 2042 payment worth \$40,568 in 2033 dollars	inline Python in this pass: flat 5.359% nominal (F-109), 2.35% inflation (about F-009)	ASSUMPTION (flat rates)
E10	"For purposes of the competition" appears 3 times in the case (L46, L90, L168)	case .txt	VERIFIED-REPO-FILE

Reproduce E9 (repo root): `.venv/bin/python -c "r=0.05359;pi=0.0235;print(round(sum(50000/(1+r)**(6+k) for k in range(10))),round(sum(50000*(1+pi)**k/(1+r)**(6+k) for k in range(10))),round(sum(50000*(1+pi)**(6+k)/(1+r)**(6+k) for k in range(10))))"`

5. Leads for the specialists

- **D1 (rates):**
 - Re-price Q23 on the real TIPS curve (F-008, Treasury real-yield CSV) instead of the flat shortcut.
 - For Q26, compute the ladder cost on the 6 Nov curve as soon as it is published (same Treasury CSV, F-001).
 - For Q2, give the correlation between a duration-matched hedge and the liability value versus a cash benchmark, as one number a note can quote.
- **D3 (quant):**
 - Q17/Q18/Q22: add a contribution rule to `strategy_mc.py`: `gift = min(top, floor + s × (residual – floor))`; the remainder is flexibility.
 - Report $P(\text{gift} \geq \text{floor})$, $P(\text{gift} = \text{top})$, and flexibility at p5/p50/p95 for $s = 0.5/0.7/1.0$.
 - Q21: show the residual's percentage volatility under growth-first versus lock-early.
- **D4 (Taiwan/FX):**
 - Q27: confine Taiwan to currency and cost facts (F-508, F-510, F-514).
 - Check her public professional pages only for any stated reason for Taiwan, and record none unless it is a professional statement she made publicly.
- **D5 (philanthropy):**
 - Q19: find how capital campaigns update donors on pledges between announcement and payment.
 - Q20: find what "lead gift" or "meaningful founder commitment" means in capacity terms. Sources to try: CCS Fundraising and the Giving USA public summaries (not yet checked).
- **D6 (client psychology):**
 - Q15: find her own public words on product management, planning and milestones (lauragao.com, reachable).
 - Q12: goals-based wealth management literature (Chhabra's "wealth allocation framework"; Das, Markowitz et al. 2010 on mental accounting) for "dedicated goal portfolio" language.

- **D7 (Wharton intent):**
 - Q28: the 2021-22 winners' approach to the ten scholarship payments (archive page; news items from 2022).
 - Retry the 2025-26 case on an accessible mirror.
 - Note that the 2023-24 and 2025-26 archive URLs return 404 in the `previous-winners/case-study-for-YYYY-YYYY/` pattern.
- **D8 (professional practice):**
 - Q10/Q13: CFA Institute IPS elements (a sample IPS with target weights and ranges).
 - Private-bank practice of one recommendation plus alternatives.
- **D9 (plain English):**
 - Q11: a three-word glossary (certainty, confidence, reliability).
 - Q2: a one-clause explanation of surplus risk that a 16-year-old can say aloud.
- **D10 (compliance):**
 - Q6: read the Session Rules tooltip for the volume rule; test one small overnight order in a thin fund.
 - Q4: note-writing protocol for split trades.
 - Q10: the tables question. If in doubt, no tables.

6. Sources (all accessed 2026-09-27)

- Official repo files (VERIFIED-REPO-FILE): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`, `2026_WGY_Investment_Policy-FINAL.txt`, `2026_WGY_Trading_Notes_Analysis-FINAL.txt`, `SMAApply_Deliverables_Page_2026-09-27.md`.
 - Phase A files: `research/insight_v1/phase_A/case_register.md`, `fact_register.md`, `stakeholder_map.md`, `wins_week1_guardrails.md`. Also `research/insight_v1/wins_now/securities_and_allocation_v0.md` and `S1_treasury_sleeve.md`.
 - Wharton archive pages (VERIFIED-PRIMARY):
 - <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/>
 - <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2022-2023/>
 - <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/>
 - <https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/>
 - <https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/>
 - TreasuryDirect (VERIFIED-PRIMARY): <https://www.treasurydirect.gov/auctions/general-auction-timing/>
 - iShares (VERIFIED-PRIMARY): <https://www.ishares.com/us/products/350028/ishares-ibonds-dec-2036-term-treasury-etf> ; <https://www.ishares.com/us/products/337745/ishares-ibonds-dec-2034-term-treasury-etf>
 - Blocked or not loaded: Scribd 2025-26 case (the page script would not load, so only the snippet is used).
 - Search-only leads (SNIPPET-UNVERIFIED): interview snippets about Laura's time in Taiwan (for example themaverickshow.com podcast 190). Not used as fact. No personal details recorded.
-

What this teaches

- **Read an official document like a contract, then like a designer.** The contract reading finds the rules ("must", "may not"). The designer reading asks why this sentence is here this year. Comparing with older cases showed which lines are inherited boilerplate (the portfolio manager "who makes the final investment decisions") and which were deliberately added or removed (no cost estimate, no endowment, "a high degree of certainty").
- **Watch for words that hide a decision.** "Her 2033 contribution" sounds like a forecast, but it is a choice Laura makes. When the thing being forecast is partly under the client's control, a rule can turn a risky forecast into a near-certain promise.

- **Timing words matter.** "During the competition", "begin approaching in 2031", "may not revise after the submission deadline": each one tells you when a decision happens, and so where in the three deliverables it has to be written.
- **An official example is a sample, not a template.** Wharton's example note is well built, but it rests on an asset-only view of risk. Copying it would contradict a strategy built around Laura's promise.

B1b Case Anomalies: what is MISSING or CHANGED versus the last two seasons

Agent: B1b (lens "Case Anomalies", starting angle "missing or changed"), insight_v1 Phase B, written 2026-09-27. AI-generated research (Claude Code) for Team Caplet, for brainstorming only. No submission-ready prose: where a question says what a sentence "must contain", that is a checklist, and the six students write every word. Securities are not named here (not needed for this lens). Scope: semifinal written deliverables only (brief section 4).

Status labels as in brief section 3. INTERPRETATION = my reading, never a fact. Every "hypothesis" is a GUESS.

0. Summary (read this first)

1. **Two official 2026-27 documents are missing from the repo.** The public SMaply page "Investment Competition Guide" (<https://wghsinvcamp.smaply.us/res/p/guide/>, read 2026-09-27) links two Box PDFs that are not in `competition/official/2026_27/` or `manifest.yaml`, and that no Phase A file cites:
 - a 5-page "**2026-2027 Investment Competition Guide**" (<https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf>, SHA-256 3638e84692ed0d826660a65e89cad0ce7eea8c1907b6a8f00ce02c0d51fb7add);
 - a 1-page "**Competition Phases Infographic**" (<https://upenn.box.com/shared/static/l2p0l26svbbptmcizhmswyq5f7rbyp44.pdf>, SHA-256 2827f92a989d6d72fbc253d80d975eb06bd00c86f631e83850ae17eec544ed10).

Both are VERIFIED-PRIMARY (downloaded and read 2026-09-27). They add rules and wording that bear directly on the Trading Notes and the IPS (section 2). **Action for the main loop:** add both to `competition/official/2026_27/` and to the manifest (I may write only this file, so I have not).

2. **This season inverts last season's case.** In 2025-26 (Connor Barwin, MTWB) the headline goal was a **growth target**: grow \$500,000 to at least \$1.5 million by 2036. That is about 11.6% a year (DERIVED from a SNIPPET-UNVERIFIED copy), plus small annual grants from 2029. The case also stated a values rule: "Every dollar invested reflects MTWB's values". This season the headline is a **fixed promise that must be certain**. There is no return target and no values language, and the capital project is the leftover amount. Last season's typical report habits (stock picks, "9% annualized" targets, ESG/thematic funds) are the wrong template this year. That is the clearest sign of what most of the ~2,300 finishing teams will get wrong (Q8, Q7).
3. **Wharton also changed the machinery, and every change points the same way.** Compared with last season:
 - the approved ETF list is gone, and "Any Government/Treasury Bonds" are now allowed;
 - WInS cash is cut from \$500k to exactly Laura's \$300k deposit;
 - trading now freezes on the IPS date, four weeks before the Final Report;
 - Trading Notes are a separate early deliverable, and last season's one-stock/one-ETF/team-choice pattern is gone;
 - "clever pitch" is replaced by "authentic team voice" and "earn her confidence";
 - "WInS can only accurately assess ... short-term investments" is replaced by "The WInS portfolio represents each team's implementation".

INTERPRETATION: the designers built a liability-matching case and then opened the tools and deadlines needed to show one.

4. **Top five questions** (ranked by effect on a deadline):
 - Q2, trade-note role tags now;
 - Q3/Q4, how the three notes should be chosen;
 - Q12, what the IPS must pre-commit, because the new Guide says "you may not redesign your strategy after observing the results";
 - Q5, whether WInS should show the Year-1 book;
 - Q20, whether the 2031 range is a number computed today or a rule applied in 2031.

5. **What is missing this year**, confirmed by word checks and comparison. No return target, no drawdown limit, no risk-tolerance number, no values/ESG preference, no living-cost figure, no facility cost, no contingency sizing, no inflation adjustment on the payments, no firm size or name, no "Your Investment Challenge" section, no stock/ETF language anywhere in the case, and no currency.

6. **What is new this year:**

- co-sponsors;
- a 2031 range with a confidence statement;
- an "operating reserve" whose meaning clashes with the nonprofit term (Q18);
- a Year-0 timing convention;
- a residency in Taiwan;
- a fundraising draft;
- the IPS as a frozen "official record";
- Trading Notes due before the IPS.

1. Comparison table: 2024-25 vs 2025-26 vs 2026-27

Sources: 2024-25 case page (VERIFIED-PRIMARY, live Wharton page). 2025-26 case: the Wharton page is retired; the text comes from a Scribd copy of the Wharton page (SNIPPET-UNVERIFIED: a secondary copy, with some lines truncated in the copy). 2025-26 welcome email: a Scribd copy (SNIPPET-UNVERIFIED). 2026-27: repo official files (VERIFIED-REPO-FILE) and the new Guide (VERIFIED-PRIMARY).

Element	2024-25 (Ladi Ayoola)	2025-26 (Connor Barwin)	2026-27 (Laura Gao)	Change and likely reason (INTERPRETATION)
Money in	"\$100,000 investment a baseline"	\$500,000	\$300,000 (2027) + \$150,000 (2028)	First two-deposit case; timing matters
Headline goal	"generate a return" for 3 build phases (2030, 2040)	"at least \$1.5 million" by 2036 ($\approx 11.6\%/yr$, DERIVED)	10 × \$50,000 fixed, "high degree of certainty"	Flipped from a return target to a certain liability
Costs given	"\$10,000 USD per housing unit; however, material costs and market dynamics are always changing"; centre "at least \$50,000"; a year of operations "at least a quarter of your original \$100,000"	Project list, no costs (from the copy)	"total cost of the facility has not been determined"; not to be estimated	Cost estimating removed; inflation <i>discussion</i> kept (R-C85)
Payouts	Operating year 2040	From "the end of year three (2029)", $\sim$ \$10,000 and "can rise above"	Start of each year 2033-42, "spaced exactly one year apart", Year 0 = 2026	Last year's timing was ambiguous (end of which year?); this year's is exact
Values	Impact quote in the client's voice	"Every dollar invested reflects MTWB's values — community-first, locally rooted, and sustainability-minded."	None (0 hits for ESG/values/impact/sustainab, R-AN34)	Values language removed on purpose (Q7)
Firm	"WGAM" named	"The firm manages a diversified portfolio currently worth \$100,000,000."	"an asset management company", unnamed and unsized	Firm context removed (Q10)
Asset language	n/a	"analyze industries and companies to identify appropriate stocks and Exchange-traded Funds (ETFs)"; "long-term and short-term profitability"	Case never says stock, ETF or bond (0 hits); TN guide: "not simply about selecting individual stocks"	Stock picking de-emphasised (Q4, Q8)
Persuasion	n/a	"compelling and clever pitch to convince Connor"; "win Connor's business"	"authentic team voice"; "recommendations that can earn her confidence"	Tone moved from pitch to trust (Q9)
WInS role	n/a	"your 10 weeks on WInS can only accurately assess the performance of short-term investments"	"The WInS portfolio represents each team's implementation of its investment strategy"	WInS moved from test bed to implementation (Q5)

Element	2024-25 (Ladi Ayoola)	2025-26 (Connor Barwin)	2026-27 (Laura Gao)	Change and likely reason (INTERPRETATION)
WInS cash	\$100,000	\$500,000	\$300,000 (= deposit 1)	Sized to Laura's Year-1 book
Instruments	Approved list (70% of stocks from it, per a 2024 guide: [PRIOR])	"ETFs and treasury bonds from the approved lists" (2025 WInS guide, VERIFIED-PRIMARY [PRIOR])	"Any ETF"; "Any Government/Treasury Bonds from any exchange available on WInS"	Tools opened for liability matching (Q6)
Deliverables	Final report (trading ended Dec 6, 2024)	Roster Oct 17; "Midterm Report" Oct 31 ("will be evaluated"); final report Dec 12; "Teams will have one portfolio"	TN Analysis Oct 23; IPS Nov 6 (trading freezes); Final Report Dec 4	Strategy frozen before the report; the notes come first (Q12, Q2)
Disclaimer	n/a	"investment scenarios ... have been embellished"	Web: "the financial scenario is developed specifically for the competition" (R-AN56)	Same idea
Portfolio-manager sentence	n/a	Identical wording: "Your portfolio manager (your team's teacher/advisor who makes the final investment decisions for your firm's portfolio)"	Same	Boilerplate, not a new signal (Q10)

Evidence that the redesign is deliberate: "The Wharton Global Youth team is doing some reimagining of our signature competition" (Wharton news, April 2025, VERIFIED-PRIMARY). Typical 2025-26 reports (SNIPPET-UNVERIFIED Scribd copies, lower authority, used only to gauge the field):

- "Profit Gurus" had a "Trading Notes Analysis" section with "ETF Selected: SPY / Stock Selected: AAPL / Team Choice: GOOGL" and cash of \$422k of \$495k.
- Another team built an "Enhanced ESG" portfolio "targeting 9% annualized returns" with clean-energy and infrastructure thematic ETFs.

2. The two official documents not in the repo (VERIFIED-PRIMARY, read 2026-09-27)

What they add beyond the case and the two guides (exact words):

- The Guide, p.2, gives the list of things to consider: "Your team will need to consider the client's goals, time horizons, required cash flows, funding commitments, liquidity needs, desired degree of funding certainty, risk tolerance, and communication with potential co-sponsors." "Risk tolerance" is named here; the case never names it (R-AN17).
- The Guide, p.2, endorses a portfolio split by purpose: "how its components will support different needs".
- The Guide, p.3, sets what a trade note should capture: "When your team executes a trade in WInS, the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding."
- The Guide, p.3, gives the timing: "The Trading Notes Analysis is due at the end of Week 4 (October 23), two weeks before your IPS is due." It also says: "Do not treat Trading Notes as an afterthought."
- The Guide, p.3, says the step "UNDERSTAND" includes studying "sources of uncertainty".
- The Guide, p.5, on the freeze: "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." And: "You may identify decisions you would make differently, but you may not redesign your strategy after observing the results."
- The Guide, p.5, lists what is scored: "The competition recognizes thoughtful strategy, research and analysis, client understanding, disciplined decisions, management of risk and uncertainty, communication, and creativity." And: "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions."

- The Infographic: "Strategy guides every decision." "Strong submissions connect the client's goals, research, portfolio decisions, and final recommendations through one clear and cohesive investment strategy." "There is no single correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions."
- The Guide, p.5: "September 15: Competition Materials Released" (the WInS practice period also began then).

3. The questions (ranked; the same content as the structured output)

Each entry gives: anchor, then hypothesis (a GUESS), then what would change, then the deliverable and domain, then the North Star link.

Q1. What do the two unrecorded official documents (the 5-page Guide and the Infographic) change in our Phase A conclusions?

- Anchor: R-new. Guide p.5: "you may not redesign your strategy after observing the results"; Guide p.3: "expected role in growth, liquidity, risk management, or future funding".
- Hypothesis (GUESS): they change little in the facts, but three things matter.
 - They make pre-commitment in the IPS explicit (Q12).
 - They give the four role words for trade notes (Q2).
 - They name "risk tolerance" and "tradeoffs", so the IPS should state both.
- Would change: a **decision**. Add both to the official set and re-run A1's cross-document check (X-items) against them. Deliverables: WInS-now, TN, IPS, FR. Domain: D10. Seed: none.
- North Star: a team that reads every official page designs to the judges' own checklist.

Q2. Should every WInS trade note from now on name its role in the Guide's four words and link it to one dated cash flow of Laura's?

- Anchor: R-T11, R-W82, R-AN43 (notes cannot be edited). Guide p.3 quoted above.
- Hypothesis (GUESS): yes. Notes are locked when written. Any three can become the Oct 23 notes. The Guide defines the content: alignment, research, and role (growth / liquidity / risk management / future funding).
 - A checklist for each note: role word; which cash flow it serves (the 2033-42 payments / the 2031 range / the 2033 facility); the research fact; the named risk.
 - Wharton's own sample note (R-AN44) has all four elements.
 - Would change: a **decision**, the note template used from the next trade. Deliverables: WInS-now, TN. Domain: D10. Seed: 13.
- North Star: every trade visibly traces to one of Laura's dated needs, which a stock-picking team cannot show.

Q3. Should the three Trading Notes deliberately show the three verbs "supported / tested / refined", so that at least one real trade before Oct 23 tests or refines the strategy?

- Anchor: R-C88 ("reflected, tested, or refined"), R-T9, R-T20. Infographic: "supported, tested, or refined the developing strategy".
- Hypothesis (GUESS): three "buy" notes for the initial allocation show only "supported".
 - A "tested" or "refined" note shows thinking. Examples: a duration-hedge rebalance after a rate move; a correction after learning a WInS rule.
 - It must be a real rule-triggered decision, not a trade made for the notes. The FAQ warns against "a lot of buying and selling". So write the rebalance rule into the notes now, and let the market trigger it.
- Would change: a **decision**, which trades to plan before Oct 23 and a rebalancing rule. Deliverables: WInS-now, TN. Domain: D7. Seed: 13.
- North Star: shows a firm that updates on evidence, which a statistics-trained client values.

Q4. Does this year's Trading Notes Analysis still expect one stock, one ETF and one team choice, as last season's reports show, or may all three notes be bond/ETF trades?

- Anchor: R-T4 ("not simply about selecting individual stocks"), R-T8, R-T28. Last season's format (SNIPPET-UNVERIFIED, Scribd): "ETF Selected ... Stock Selected ... Team Choice".
- Hypothesis (GUESS): there is no quota this year. The official TN guide says "select three investment decisions that best illustrate your team's strategic thinking", with no asset-type rule.
 - A lock-early portfolio may hold few or no single stocks, and that is allowed.
 - Confirm on the logged-in SMAApply form (it may show separate fields).
 - Would change: a **decision**, whether we need a stock position at all for the notes. Deliverables: WInS-now, TN. Domain: D10. Seed: 13.
- North Star: freed from a stock-pick ritual, every note can serve Laura's promise.

Q5. Given the WInS cash cut from \$500k to exactly Laura's \$300k deposit, and "WInS ... short-term" replaced by "represents each team's implementation", should the WInS book show her Year-1 (2027) portfolio rather than the post-2028 target?

- Anchor: R-AN14, R-C75, R-W85, X-7. The item is open in brief section 9. **New evidence:** the change in wording from the 2025-26 case, and the Guide p.3: "your portfolio provides evidence of the decisions your team made".
- Hypothesis (GUESS): the designers sized WInS to the first deposit.
 - A Year-1 mirror is the most literal "implementation": about 97% in liability-matching bonds, the surplus in growth.
 - It is also the easiest to defend in the IPS.
 - The post-2028 mirror (~65/35) is defensible only if it is stated as "the target mix after the second deposit".
 - The team must pick one, and the Final Report must say which, in one sentence.
- Would change: a **decision** (the WInS weights now) and a **sentence** in the IPS. Deliverables: WInS-now, TN, IPS, FR. Domain: D7. Seed: 3.
- North Star: the portfolio judges see is Laura's own first year, not a generic model.

Q6. Did Wharton open "Any Government/Treasury Bonds from any exchange" and drop the approved list so that liability-matching instruments can be used, and which 2032-2042 maturities does the WInS bond list actually offer?

- Anchor: R-W67-R-W69, R-AN48, F-606. [PRIOR] 2025 WInS guide: "ETFs and treasury bonds from the approved lists" (VERIFIED-PRIMARY). The 2025-26 list had no defined-maturity or zero-coupon funds (historical README).
- Hypothesis (GUESS): the opening was deliberate. The WInS bond drop-down may still list only some maturities. Check inside WInS before relying on a cash-flow match.
- Would change: a **decision**, individual bonds versus a fund mix in WInS. Deliverables: WInS-now, TN. Domain: D1. Seed: none.
- North Star: using the tools Wharton just opened shows we understood why they opened them.

Q7. Why did Wharton remove all values language, and should our deliverables say explicitly that no values screen is applied, or stay silent?

- Anchor: R-AN34. 2025-26: "Every dollar invested reflects MTWB's values" (SNIPPET-UNVERIFIED). 2024-25: "When I'm working or doing anything, I want to feel like I'm making an impact" (VERIFIED-PRIMARY). 2026-27: none.
- Hypothesis (GUESS): removed on purpose, to keep the case about certainty.
 - Teams that carry over ESG or thematic funds impose an unstated preference on the client.
 - Recommendation: say nothing in the IPS (words are scarce).

- In the Final Report, one clause under client knowledge: we imposed no screen the case does not name, and gave the reason (cost and tracking of a fixed liability).
- Never cite her identity as a reason for a screen.
- Would change: a **sentence** (FR client-knowledge section) and a **decision** (no thematic sleeve). Deliverables: IPS, FR. Domain: D6. Seed: 12.
- North Star: it respects Laura's actual stated wishes over a template.

Q8. How many teams will reuse last season's growth-target template, and what one visible feature of our deliverables proves we did not?

- Anchor: R-AN17, R-C73. 2025-26 goal "\$1.5 million by 2036" (SNIPPET-UNVERIFIED); $\approx 11.6\%/yr$ (DERIVED).
- Hypothesis (GUESS): most teams will. Last year's sample reports lead with return targets and stock names.
 - Our visible difference: the first line of the pitch states the promise as already bought, with its price and date, and no return target.
 - A strong pitch must contain: who it serves; the promise; how it is secured; what the surplus is for.
- Would change: a **sentence** (the elevator pitch specification). Deliverables: IPS. Domain: D7. Seed: 1.
- North Star: the direct answer to "why us": we solved her case, not last year's.

Q9. Since "compelling and clever pitch" and "win Connor's business" became "authentic team voice" and "earn her confidence", should our deliverables drop pitch devices in favour of adviser-style disclosure?

- Anchor: R-S25, R-S28, R-C8. 2025-26 wording (SNIPPET-UNVERIFIED).
- Hypothesis (GUESS): yes. Replace metaphors and branded framework names with three plain parts: what we know, what we assume, what could go wrong.
 - This fits a client whose first public work answered misinformation (R-AN33).
 - Judge Hahn's "hiding behind fancy terms" remark (SNIPPET-UNVERIFIED, winners.md) agrees.
 - Would change: a **sentence**, the style checklist for the IPS and FR. Deliverables: IPS, FR. Domain: D9. Seed: none.
 - North Star: Laura chooses the firm she can trust, not the firm with the best slogan.

Q10. Since the "portfolio manager ... makes the final investment decisions" sentence is copied word for word from 2025-26, and the firm is now nameless and sizeless, should the IPS spend any words on governance beyond rules that any reviewer can apply?

- Anchor: R-C2, R-C3, R-AN4, X-10, BS-01, BS-02.
- Hypothesis (GUESS): the sentence is boilerplate.
 - The IPS should not describe an org chart or the advisor.
 - Its "decision-making framework" should be pre-committed rules (Q12).
 - Firm terms (fee, reporting) belong in the FR as ASSUMPTIONS, following CFA Asset Manager Code F.4.d.
- Would change: a **decision**, the IPS word budget. Deliverables: IPS, FR. Domain: D10. Seed: 2.
- North Star: rules Laura can check herself beat an authority she must trust.

Q11. Why does every official document list "liquidity" when the case removes every liquidity need before 2033, and what should "liquidity" mean in our IPS?

- Anchor: R-I6, R-I11, R-I21, R-C33, R-C34, R-C74; Guide p.2 "liquidity needs".
- Hypothesis (GUESS): here "liquidity" means two things:
 - (a) each Jan 1 payment 2033-42 is paid on time from maturing bonds;
 - (b) the growth sleeve can be turned into cash for the 2033 facility without a forced sale in a bad year.

There is no cash buffer before 2033, since living costs are outside the portfolio. A generic "keep 5% cash for liquidity" shows the team did not read the case. The WInS cash weight should be small and explained.

- Would change: a **sentence** in the IPS and a **number** (the WInS cash weight). Deliverables: WInS-now, IPS. Domain: D10. Seed: none.
- North Star: shows we read her case line by line.

Q12. Which rules must the IPS pre-commit so the Final Report can "evaluate" them without "redesign[ing] your strategy after observing the results"?

- Anchor: R-I34, R-I35, X-5, X-23; Guide p.5 (quoted in section 2).
- Hypothesis (GUESS): the IPS needs one-line rules for:
 - (1) rates moving before the January 2027 purchase;
 - (2) a smaller or late 2028 deposit;
 - (3) how the 2031 range is set;
 - (4) the 2033 order of use: reserve including payment 1, then facility, then flexibility (seed 5);
 - (5) sleeve rebalancing bands;
 - (6) what happens to money left over after 2033.

The Final Report may then show "the rule fired as written". Anything not pre-committed and later changed looks like a redesign.

- Would change: a **decision**, the IPS content checklist. Deliverables: IPS, FR. Domain: D10. Seed: 5.
- North Star: a firm that commits in advance protects Laura from its own hindsight.

Q13. The case gives no 2027 prices ("Year 0 ... before any money has been invested"), so what price date should projections use, and how is that stated?

- Anchor: R-C28, R-C77, R-AN22, F-101, F-111, F-112.
- Hypothesis (GUESS): use the latest official curve (2026-09-25: \$292,264) as a labelled ASSUMPTION. Show the $\pm 50\text{bp}$ band (\$278k-\$307k) and $P(\text{cost} > \$300\text{k}) \approx 24\%$ (ASSUMPTION model). Point to rule (1) in Q12.
- Would change: a **number** and a **sentence** in the FR assumptions. Deliverables: FR, IPS. Domain: D1. Seed: 4.
- North Star: honest about the one thing we cannot know yet.

Q14. Does "objectives that reflect ... her passion for innovation" justify any innovation, tech or AI tilt, or does "innovation" describe the residency?

- Anchor: R-C25 (case p.2 L39-41; the only use of "innovation").
- Hypothesis (GUESS): it describes her goals, not asset choices. Many teams will buy AI or tech names on this word. Any tech weight should come only from broad-market weights (section 9's AI-exposure item).
- Would change: a **decision**, the sleeve composition (no thematic tilt). Deliverables: WInS-now, IPS. Domain: D2. Seed: 20.
- North Star: we invest for her goals, not for her adjectives.

Q15. Why may co-sponsors fund the facility but not her commitment, and should the fundraising draft lead with the fully pre-funded ten years of operations, putting the facility range second?

- Anchor: R-C44, R-C48, R-AN6, BS-17.
- Hypothesis (GUESS): funders least like to pay for operations. Her covering them removes their biggest worry. So the pre-funded operations are the credibility anchor, and the facility range is the invitation.
- Would change: a **sentence**, the order of the fundraising draft. Deliverables: FR. Domain: D5. Seed: 6.
- North Star: turns a case rule into Laura's strongest fundraising fact.

Q16. Will co-sponsors read "operating reserve" in its nonprofit sense (months of expenses held by the organisation), and should the fundraising draft use a different label?

- Anchor: R-C51, R-new.

- Evidence: the nonprofit benchmark ratio is "above 25% (equivalent to three months) is better" (nonprofitaccountingbasics.org, VERIFIED-PRIMARY).
- Hypothesis (GUESS): yes, there is a risk of misreading.
 - The case's reserve is ten years of her personal commitment, about 40 times the sector norm.
 - Keep "operating reserve" in the IPS and FR, because it is the case term.
 - In co-sponsor text, describe it plainly: ten payments already funded.
 - Note that the residency may still need its own short-term reserve, which is outside her commitment.
- Would change: a **sentence** (the vocabulary of the fundraising draft). Deliverables: FR. Domain: D5. Seed: none.
- North Star: we speak the co-sponsors' language on Laura's behalf.

Q17. Why does the case name an "endowment" next to the contingency fund, and how should money left over after 2033 be described so co-sponsors do not assume it is a perpetual fund for years after 2042?

- Anchor: R-C61, R-C49, R-AN59.
- Hypothesis (GUESS): the designers expected teams to propose endowments. Describe the leftover as uncommitted flexibility governed by a rule, with no promise beyond 2042.
- Would change: a **sentence** in the FR and the fundraising draft. Deliverables: FR. Domain: D5. Seed: 7.
- North Star: it prevents an accidental overpromise, which is exactly what the case warns about.

Q18. Does "recommend the dollar range Laura should communicate" in 2031 ask for a range computed today (a 5-7-year forecast) or a rule Laura applies in 2031 (a 2-year forecast), and how do we give both without looking evasive?

- Anchor: R-C62, R-C65, R-C68, R-AN7.
- Hypothesis (GUESS): the case wants numbers ("dollar range"), but the true decision happens in 2031.
 - Give the 2031 rule: floor = what is already secured in a 2-year Treasury; top = a stated percentile.
 - Also give today's projected range under that rule, e.g. median floor $\approx$ \$165k (F-402, ASSUMPTION model), with the spread across scenarios.
 - The two differ a lot in width.
- Would change: a **number**, the headline range in the FR. Deliverables: FR. Domain: D3. Seed: 8, 9.
- North Star: gives Laura a method she can reuse, not a guess that ages.

Q19. Because Laura decides her 2033 contribution, can "will fall within that range" be made partly a policy outcome, so that the stated confidence depends only on the low end?

- Anchor: R-C69, R-AN9, R-C58, R-C84.
- Hypothesis (GUESS): yes, with a rule: contribute at least the floor, and at most the top of the range. Anything above the top stays as flexibility.
 - "Above the range" then cannot happen by design.
 - Confidence = $P(\text{available} \geq \text{floor})$. With a bought floor, that is near-certain (barring U.S. default).
 - Disclose that the top is a cap Laura chooses, not a forecast.
 - Would change: a **number** (the confidence %) and a **sentence** (range definition). Deliverables: FR. Domain: D3. Seed: 9.
- North Star: a confidence statement that a statistician would accept.

Q20. Given that the payments are nominal, where exactly must inflation appear, and should every co-sponsor dollar figure be tagged "2033 dollars" or "today's dollars"?

- Anchor: R-C46, R-C85, R-AN3, R-AN8, F-114, F-508.
- Evidence: 2024-25 "material costs and market dynamics are always changing" (VERIFIED-PRIMARY).
- Hypothesis (GUESS): inflation belongs in three places:
 - (1) the real value of the projections;
 - (2) facility costs: Taiwan construction costs $\approx$ 3.5%/yr since 2021, not the 6.5% spike;

- (3) the real erosion of the fixed \$50k: \$42.1k in 2033 and \$33.7k in 2042 at 2.5% (in 2026 dollars). Who absorbs that loss is open in section 9.
- Would change: a **sentence** and a **number** (a two-column range in the FR). Deliverables: FR. Domain: D4. Seed: 10.
- North Star: shows Laura what her money will actually buy in Taiwan.

Q21. How do we express "financial flexibility" as a rule or ratio, instead of a contingency fund the case says we need not size?

- Anchor: R-C58, R-C61, R-C84, R-AN19.
- Hypothesis (GUESS): use a share of the post-reserve surplus kept uncommitted until named project milestones, stated in the IPS as a rule and sized in the FR.
- Would change: a **number** (the share) and a **decision**. Deliverables: IPS, FR. Domain: D8. Seed: 11.
- North Star: flexibility "as the project develops" expressed in Laura's own words.

Q22. With charts, citations and "detailed financial calculations" excluded, which few numbers must the 500-word IPS still carry to be testable, and what counts as "detailed"?

- Anchor: R-I55, R-C92, X-3.
- Hypothesis (GUESS): 3-6 numbers are enough, e.g. \$50,000 × 10 from 2033, the funded-ratio test, the sleeve equity band, rebalancing bands and the flexibility share. Leave out tables and projections.
- Would change: a **sentence**, the IPS numbers list. Deliverables: IPS. Domain: D9. Seed: 14.
- North Star: words alone must prove the plan is checkable.

Q23. Which three tradeoffs should the IPS name, since the Guide, the Infographic and the FAQ all say that strong teams explain "reasoning, assumptions, and tradeoffs"?

- Anchor: R-W43, R-AN17, X-11; Infographic quote.
- Hypothesis (GUESS):
 - (a) certainty of the promise vs the size of the facility;
 - (b) sleeve growth vs a higher 2031 floor;
 - (c) locking now vs waiting for rates.

Each tradeoff must say what Laura gives up.

- Would change: a **sentence**, the IPS content checklist. Deliverables: IPS. Domain: D9. Seed: none.
- North Star: shows Laura the choices were hers to see.

Q24. What makes a personal contribution "meaningful" to co-sponsors when no facility cost exists: its dollar size, its share of Laura's own post-reserve assets, or how certain it is?

- Anchor: R-C64, BS-06.
- Hypothesis (GUESS): funders read a lead gift by commitment relative to capacity, and by how firm it is. So present the floor as a share of what remains after the ten-year commitment is secured.
- Would change: a **sentence** (the framing of the fundraising draft). Deliverables: FR. Domain: D5. Seed: none.
- North Star: a contribution that is meaningful because it is certain, which matches her credibility.

Q25. Given that the FAQ still says "10 weeks of trading" and "long-term portion of your portfolio" (recycled text), does "Investments permitted (for BOTH contributions)" really limit Laura's long-term plan to instruments listed in WInS?

- Anchor: R-AN15, R-AN46, F-606.
- **New evidence:**
 - Trading this year is about 6 weeks, so the FAQ text is recycled.

- The 2025-26 welcome email says "Teams will have one portfolio" (SNIPPET-UNVERIFIED), which implies that earlier seasons had more than one.
- Hypothesis (GUESS): "BOTH" is probably a leftover. The safe course is still to name instrument types available in WInS, and to mark any STRIPS recommendation for the real plan as not tradable in WInS.
- Would change: a **decision** (whether the long-term plan names STRIPS). Deliverables: FR, IPS. Domain: D10. Seed: none.
- North Star: stay compliant without crippling the real-world plan.

Q26. Last season's written reports were screened by "Wharton School student evaluators" plus "a professional review panel of asset managers from Aberdeen Investments": how much fixed-income precision and plain English should our written deliverables target?

- Anchor: F-611, R-W24 (the Rules page on Top 50 selection). The news quote is VERIFIED-PRIMARY.
- Hypothesis (GUESS): two readers.
 - A student screener: needs plain English, one glossary line per term.
 - A bond professional: will check durations, cash-flow match and Treasury maturity dates. So precision matters and errors are costly.
 - Confirm whether the same set-up holds for 2026-27 (not stated).
- Would change: a **decision** about the depth and the glossary in the FR. Deliverables: FR, IPS. Domain: D7. Seed: none.
- North Star: written for the people who actually decide the Top 50.

Q27. Which "sources of uncertainty" should our deliverables name, and does the case's list of where the 2028 money comes from invite a deposit stress test despite "will"?

- Anchor: R-C27, R-AN31, R-AN35, BS-13; Guide p.3 "sources of uncertainty".
- Hypothesis (GUESS): name five:
 - markets;
 - rates before 2027;
 - the size and timing of the 2028 deposit;
 - inflation and facility costs;
 - the currency.

Stress the deposit only as a labelled team scenario.

- Would change: a **decision** (the FR scenario set) and a **sentence** in the IPS. Deliverables: IPS, FR. Domain: D6. Seed: 16.
- North Star: shows we understand her income comes from her own brand.

Q28. Why is the residency in Taiwan when the case gives Laura no link to Taiwan, and may we cite her public professional statement that she worked as a comic artist from Taipei?

- Anchor: R-C39, R-AN10.
- Evidence: the 2021 Overachiever Magazine interview (VERIFIED-PRIMARY, 2021-03-15): "I'm currently in Taiwan, Taipei. My current day job is as a comic artist."
- Hypothesis (GUESS): Taiwan is a real professional link, not tokenism.
 - Use it at most as one client-knowledge clause.
 - The interview also contains personal-life material; do not use it (brief privacy rule).
- Would change: a **sentence** in the FR client section. Deliverables: FR. Domain: D6. Seed: 19.
- North Star: shows we researched her beyond the case, without using her identity as decoration.

4. Parked (logged, not pursued)

- The case's "Investment Competition Guide" page header (R-C1) also names the new Guide. Probably shared branding; no deliverable effect.
- 2025-26 had a first-trade deadline (Oct 10, 2025, SNIPPET-UNVERIFIED welcome email). No 2026-27 primary page states one (guardrails b3). Trading in week 1 makes it moot.
- The field-size wording differs: 2025-26 "more than 5,400 high school teams ... began" (finalist article, VERIFIED-PRIMARY) vs "more than 6,300 registered" vs ~2,300 finished. The effect on deliverables is nil; R-AN60 covers it.
- The "Year numbers" convention itself (R-AN12) is answered. The only residual is a dated cash-flow table as the FR's first exhibit (FR chart spec for D9).

5. Leads for the specialists

- **D10 (compliance):**
 - Ingest the Guide and Infographic (section 0.1 URLs and hashes).
 - Check the logged-in TN form for asset-type fields (Q4) and the WInS bond drop-down for 2032-42 maturities (Q6).
 - The FAQ's recycled phrases ("10 weeks of trading", "long-term portion", "for BOTH contributions") support reading 2 of R-AN15.
- **D7 (Wharton intent):**
 - 2024-25 case page (live, VERIFIED-PRIMARY): <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/>
 - The 2025-26 case text survives only as a Scribd copy: <https://www.scribd.com/document/965855349/Case-Study-Wharton-Global-Youth-Program> (curl works; the text is in the HTML `<p>` blocks, some lines truncated).
 - 2025-26 welcome email copy: <https://www.scribd.com/document/933222558/25-26-Welcome-Email>
 - Sample 2025-26 team reports (field-habit evidence only): <https://www.scribd.com/document/980515751/Full-Report-Connor-MTWB> and <https://www.scribd.com/document/977481046/Updated-Investment-Strategy-for-Connor-Barwin>
 - "Reimagining" quote: the BAM 2025 article.
- **D5 (philanthropy):**
 - Nonprofit operating-reserve norm (3 months = 25%): <https://www.nonprofitaccountingbasics.org/internal-reporting/operating-reserve-ratio> (VERIFIED-PRIMARY). Also the NORI definition (SNIPPET-UNVERIFIED via search; the candid.org page is blocked).
 - Needed: lead-gift and "leadership gift" practice for how a founder's pledge is stated as a range.
- **D6 (client):**
 - The Overachiever interview (2021-03-15, VERIFIED-PRIMARY) has professional risk-attitude lines: "you should always take the jump because you're always going to regret not doing it"; and, on leaving tech, "It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""
 - Use only these professional lines; the page also has private material.
 - The case pull quote is still unattributed. It was not found in this interview or by search (SNIPPET search, 2026-09-27).
- **D3 (quant):** Q18/Q19 need the 2031 range computed two ways: from 2027 (today's view) and as a 2031 rule (conditional two-year view). They also need a confidence definition under a contribution cap.
- **D1 (rates):** the price-date assumption (Q13) and a list of the WInS bonds available.
- **D4 (Taiwan):** use the construction-cost trend of ~3.5%/yr (F-508), not the one-year 6.5% spike, for the two-currency, two-dollar-date tags (Q20).
- **D9 (communication):** the tone shift (Q9); the tradeoff sentences (Q23); the IPS numbers list (Q22); the reserve-label clash (Q16).

6. Sources (all accessed 2026-09-27)

Source	URL / path	Status
2026-27 case, IPS guide, TN guide, SMAApply page	competition/official/2026_27/	VERIFIED-REPO-FILE
2026-27 Investment Competition Guide (5 pp)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf via https://wghsinvcomp.smaply.us/res/p/guide/	VERIFIED-PRIMARY
2026-27 Competition Phases Infographic	https://upenn.box.com/shared/static/l2p0l26svbbptmci zhmswyq5f7rbyp44.pdf	VERIFIED-PRIMARY
SMAApply FAQ ("10 weeks of trading", "BOTH contributions")	https://wghsinvcomp.smaply.us/res/p/faqs/	VERIFIED-PRIMARY
2024-25 case (Ladi Ayoola)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/	VERIFIED-PRIMARY
Archive of clients and winners	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/archive/	VERIFIED-PRIMARY
2025-26 case text (Connor Barwin)	https://www.scribd.com/document/965855349/Case-Study-Wharton-Global-Youth-Program	SNIPPET-UNVERIFIED (secondary copy; the Wharton original is retired)
2025-26 welcome email	https://www.scribd.com/document/933222558/25-26-Welcome-Email	SNIPPET-UNVERIFIED
2025-26 sample team reports	Scribd 980515751, 977481046	SNIPPET-UNVERIFIED (field evidence only)
2025-26 season article (evaluators; 6,300; 2,300)	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/ (Jan 27, 2026)	VERIFIED-PRIMARY
2026 finalists article ("5,400 ... began")	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/	VERIFIED-PRIMARY
2026 champions article	https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	VERIFIED-PRIMARY
2025 BAM article ("reimagining")	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/	VERIFIED-PRIMARY
2025 WInS User Guide [PRIOR]	https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2025/09/2025-WInS-User-Guide.pdf	VERIFIED-PRIMARY [PRIOR]
Nonprofit operating-reserve ratio	https://www.nonprofitaccountingbasics.org/internal-reporting/operating-reserve-ratio	VERIFIED-PRIMARY
Overachiever interview with Laura Gao (2021-03-15)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	VERIFIED-PRIMARY (professional lines only)
Blocked or failed	Wayback CDX (connection reset); the 2025-26 Wharton case page (retired, 404 on the guessed slug); lauragao.com/about (404)	—

What this teaches

- Comparing a document with its earlier versions is one of the fastest ways to find what the authors care about. What they **removed** (return targets, values language, stock-picking instructions, cost figures) tells you which old habits they now want you to drop. What they **added** (certainty, co-sponsors, a frozen strategy) tells you what will be scored.
- Always check every link on an official page. Two official documents were sitting one click away, unread.
- A word can mean different things to different readers ("operating reserve" to a nonprofit, "liquidity" in a case with no cash needs). Good advisers translate for each audience rather than copying the source's wording.

B2a Why Laura / Case-Designer Intent: questions from six seasons of client history

Agent B2a, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: "Why Laura / Case-Designer Intent". Starting angle: the client history 2021-2027 (Jordan, Hjemdahl, Ash, Ayoola, Barwin, Gao). I read each past case and each season's winner and finalist news on globallyouth.wharton.upenn.edu, built one table of case shapes, and asked which trend Laura's case continues or breaks and what that implies the semifinal readers will reward.

This file is AI-generated research for Team Caplet, for brainstorming only. It holds questions, evidence and checklists. It contains no text meant for submission. The six students decide and write every deliverable in their own words, and any AI use must be recorded in the Final Report's Works Cited (R-W46). Laura is covered only through the case and her public professional record. Past clients are covered only through Wharton's own case pages and news items; I left out their personal and family details on purpose.

Status labels (brief section 3): VERIFIED-PRIMARY (I read it on the primary page, URL + 2026-09-27), VERIFIED-REPO-FILE, SNIPPET-UNVERIFIED, ASSUMPTION, PARAPHRASE-UNVERIFIED. INTERPRETATION marks my reading, never a fact. A "hypothesis" in a question is a guess.

0. Summary (read this first)

- Laura's promise problem has been asked before, in a zero-rate world.** The 2021-22 case (Nichole Jordan) asked teams to fund a \$5,000 scholarship "for at least 10 years" and wrote: "Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends? Another option?" (VERIFIED-PRIMARY). On the day that season began, the 10-year Treasury yielded 1.48% (FRED, VERIFIED-PRIMARY). Buying the ten payments in advance would then have cost about 95% of their face value. Today the same kind of promise costs about 58% (\$292,264 for \$500,000, brief section 6). INTERPRETATION: the designers have brought back an old question at a time when interest rates change the answer. At 2021 rates, Laura's ten payments would cost about \$429,000, almost all of her \$450,000 (ASSUMPTION: flat 1.48%). (Q3)
- The \$300,000 first deposit is almost exactly the price of the promise.** At a flat rate, \$300,000 buys the ten payments at 5.09% (DERIVED). Through most of 2026 it did not quite reach: the same ladder cost more than \$300k on 173 of 185 trading days (F-111). The case PDF was created on 2026-09-10 (case_register 1.1), the first day the ladder cost fell under \$300k. INTERPRETATION: the designers probably meant the first deposit to cover roughly the promise, and the second deposit to fund the facility and flexibility. That argues for (a) a lock rule written to depend on rates, not "it fits inside \$300k", and (b) a two-deposits, two-jobs story (Q4, Q5).
- The committed outflow grew from half the client's money to more than all of it.** Jordan's payments were 50% of her \$100,000. Laura's \$500,000 of payments is 111% of her \$450,000 (DERIVED). This is the first case with no return target and the first to use the word "certainty" (0 hits before, 6 this year; my word count, below). "Stock" drops from 2-4 hits per case to 0. "Co-sponsor" has 9 hits, "flexib" 5, "credib" 3, "uncertain" 3, where every past case had 0. INTERPRETATION: the scoring weight has moved from picking investments to reliability, flexibility and credibility (Q6, Q7).
- This is the first case with no personal-life hooks at all.** Every past case described hobbies (Peloton, yoga, scuba, Muay Thai, modelling; volleyball for Ash, per Wharton news). Teams turned those hobbies into stock picks (for Ash, "Nike ... caters to Ms. Ash's love of sports", Teen Ink 2023-24 report, SNIPPET-UNVERIFIED). Laura's case has 0 hits for "enjoy", "love", "free time", "family", "lives" and "partner" (VERIFIED-REPO-FILE word check). Only her professional mindset and her goals remain. INTERPRETATION: "tailoring" this year has to run through her cash flows, constraints and way of thinking, not through her tastes (Q2).
- Clients and judges have said the same things for five years (all VERIFIED-PRIMARY).**
 - Clients reward feeling understood. Jordan (2022): she "had never felt more "seen" in her life". Ayoola (2025): "not just about the numbers, but about understanding and connecting with people".
 - Judges reward restraint and clear reasoning: "Cut slides, cut words, cut minutes" (2022); "a lot of depth and accuracy" (2024); "clear thoughtfulness and reasoning behind all the decisions" (2025).
 - Judges have also named two gaps that most teams leave open: "consider behavioral finance and their clients' emotions during a market downturn" and "take a look at this person holistically" (2024, the second from an Aberdeen fixed-income manager).

Our plan does not yet cover these two gaps explicitly (Q1, Q10, Q11).

- Two facts correct the team's "Why Wharton chose Laura" document.**
 - Laura is not the youngest recent client. The 2022-23 case states "Peter, who is 25 years old" (VERIFIED-PRIMARY).
 - "Unstable income is new" needs one correction. Her living costs are outside the portfolio, so the instability matters only through the 2028 deposit. What is genuinely new is that this is the first case in six seasons with a scheduled second contribution, and that contribution depends on creative income (Q11).
- 19 questions follow (section 3),** ranked by deadline tier within groups. The five I rate highest for "99% of teams will miss": Q4 (the \$300k calibration and a rate-conditional lock rule), Q3 (why locking is right *now*: the 2021 comparison), Q2 (no hobby hooks, so no identity-themed securities), Q10 (the two gaps judges keep naming), Q5 (two deposits, two jobs).

1. Method

- Read Wharton's archive page and every linked news item for the finals and semifinals 2022-2026. Read the three past case pages that are still live (2021-22, 2022-23, 2024-25). Read the 2025-26 case from a Scribd copy of the retired Wharton page. For 2023-24 I found no case page, so the case shape comes from a team report on Teen Ink. URLs are in section 6.
- Word-counted the same key words in each case text (Python regex, lower case). The 2025-26 count uses the Scribd copy, which is partly truncated. The 2024-25 page may end before any boilerplate. So the counts are indicative, not exact.
- Took season-start 10-year yields from FRED DGS10. Priced the 2021-22 scholarship at the 2021-09-27 FRED curve (ASSUMPTION: par yields used as zero rates, payments at mid-year). Solved for the flat rate at which \$300,000 buys Laura's ten payments. All inline Python is in section 5.
- Checked 19 key quotes word for word with `fetch_text.py --grep` (all "FOUND VERBATIM: YES").
- Checked Phase A (R, F, SH ids) and the four Phase B files already written (B1a, B1b, B7a, B7b) so I would not re-ask their questions. Where I overlap, I say what is new.

2. Client history table (2021-2027)

Season	Client (Wharton)	Money in	What the case asked	Shape of the liability	WlnS cash	Words that framed success	Winners	Finished reports
2021-22	Nichole Jordan, WG'08 (SVP at Via, TransitTech)	\$100,000	\$5,000/yr scholarship from 2022 "for at least 10 years"; build generational wealth and fund education for young relatives	10+ fixed annual payments = 50% of assets, starting at once; "Should she keep \$50,000 in cash? ... dividends? Another option?"	\$100,000	"long-term and a small amount of mid-term profitability"; "WlnS can only accurately assess ... short-term"	Sailing to Success (AZ); Sky Investments (Bergen County Academies); M&R (Marvin Ridge)	~1,300
2022-23	Peter Wang Hjemdahl, W'19 (rePurpose Global)	\$100,000	\$20,000 now to grow ≥\$10,000 in 5 years, then yearly \$5-10k gifts to start-ups "use the interest"; wellness centre in Miami within 15 years	Return target plus a perpetual-style gift paid from income	\$100,000	"compelling and clever pitch to convince Peter"; "trying to land the same client"	DMV's Finest (Thomas Jefferson HS); Amity 7 Chakras; NovyPorg_1 (Prague)	~1,400
2023-24	Hilary Ash, W'13 (LA28 Olympic and Paralympic Games)	not found (case page retired)	Per a team report (SNIPPET-UNVERIFIED): renovate a property in South America within 5 years; launch a sports consulting firm within 15 years	Two dated goals, return-driven	not found	Volleyball, "a passion highlighted in Hilary Ash's client profile", inspired team creativity (Wharton news)	Spark Investments (Bergen County Academies); Wreckers (Staples HS); DMV's Finest	>1,600
2024-25	Ladi Ayoola, W'22 (Visa; co-founder of Plugtent)	"\$100,000 investment a baseline"	Phase 1: ≥5 housing units by 2030 at "about \$10,000 USD per housing unit"; Phase 2 (2040): storefront + youth centre ≥\$50,000, plus one year of operations ≥ a quarter of \$100k; Phase 3 later	Dated cost targets; first mention of operating costs	\$100,000	"generate a return"; "impact" (2 hits)	BAM Investing (Deerfield); Finance from France (Chicago); FA Quakers	~1,800
2025-26	Connor Barwin, WG'23 (Make the World Better foundation)	\$500,000	Grow to ≥\$1.5m by 2036 for a capital project; community contributions from "the end of year three (2029)" of ~\$10k that "can rise"	Growth target + small rising grants	\$500,000	"Every dollar invested reflects MTWB's values"; "win Connor's business"; Wharton: "think beyond financial returns"	FigCapital (Stuyvesant); HHA (Brasília); Riverhawk (Farmington)	~2,300
2026-27	Laura Gao, W'18 (author, illustrator, entrepreneur, educator)	\$300,000 (2027) + \$150,000 (2028)	Ten fixed \$50,000 payments 2033-2042 "with a high degree of certainty"; operating reserve set aside in 2033; a "responsible" facility contribution; a 2031 dollar range + confidence for co-sponsors; a fundraising draft	Fixed nominal liability = 111% of contributions; deferred 6 years; no return target	\$300,000 (= first deposit)	"earn her confidence"; "authentic team voice"; "certainty" x6; "co-sponsor" x9	TBD	TBD

Sources: 2021-22, 2022-23 and 2024-25 case pages, VERIFIED-PRIMARY. 2023-24 goals from the Teen Ink team report, SNIPPET-UNVERIFIED as to the case wording. 2025-26 from the Scribd copy, SNIPPET-UNVERIFIED. Winners and report counts from the archive and finale/semifinal articles, VERIFIED-PRIMARY. 2026-27 from the repo case, VERIFIED-REPO-FILE.

Word counts across the case texts (my count; indicative only, see Method):

Case	return	profit	risk	certainty*	thoughtful	impact	values	stock	inflation	uncertain	credib	flexib	co-sponsor	pitch/convince/"win"
2021-22	0	2	0	0	0	0	0	4	0	0	0	0	0	1
2022-23	1	1	1	0	0	1	0	2	0	0	0	0	0	4
2024-25	2	0	0	0	0	2	0	0	0	0	0	0	0	0

Case	return	profit	risk	certainty*	thoughtful	impact	values	stock	inflation	uncertain	credib	flexib	co-sponsor	pitch/convince/"win"
2025-26 (copy)	1	2	1	0	0	3	2	3	0	0	0	0	0	8
2026-27	2	0	2	6	3	0	0	0	2	3	3	5	9	0

*In 2021-22 and 2022-23 the regex hit "certain factors", which is not the concept, so those cells are set to 0. "Volatil" appears only in 2026-27 ("Laura understands that investing involves uncertainty and periods of market volatility", VERIFIED-REPO-FILE p.2).

Season-start 10-year Treasury yield (FRED DGS10, VERIFIED-PRIMARY; 2026 from the repo Treasury CSV, F-001):

Season start	2021-09-27	2022-09-26	2023-09-25	2024-09-23	2025-09-29	2026-09-25
10y yield	1.48%	3.88%	4.55%	3.75%	4.15%	5.17%

What the table shows (INTERPRETATION):

- **Trends that continue.**
 - Every client is a Wharton alum paying something forward to others. Jordan: "opening doors or creating opportunities for others like me". Hjemdahl: "pay forward the gratitude". Ayoola: "it truly takes a village". Barwin: community spaces. Laura: a residency for "artists, writers, designers, entrepreneurs, and educators", the same roles the case gives her.
 - Three cases in a row fund a physical, community-facing place: Ayoola, Barwin, Gao.
 - Horizons stay long (10-16 years).
 - The "portfolio manager ... makes the final investment decisions" line is boilerplate from 2021 onward.
- **Trends that break.**
 - The return target disappears.
 - Certainty, flexibility, credibility and a third-party audience (co-sponsors) appear.
 - Hobbies and values language vanish.
 - The liability grows from 50% to 111% of the client's money.
 - Cash cannot fund the promise, because \$450k is less than \$500k.
 - The timing is exact ("Year 0 is 2026"). Earlier cases were loose: "starting in 2022", "the end of year three".
 - WInS cash equals the first deposit.
 - The instruments open up to any ETF and any government bond.
- **What that implies readers will reward.** An answer to a *promise* problem that is reliable, plainly explained and honest about uncertainty to outsiders. That fits every judge remark since 2022 (restraint, accuracy, reasoning, the client's emotions, the whole person). A stock-screening showcase does not fit it. Past finales celebrated "stock-filtering algorithms", "endless Environmental, Social & Governance considerations" (2023) and "customized algorithms, AI innovations, portfolio optimizations" (2025).

3. The questions

Each gives: the question; its anchor; the hypothesis (a GUESS) with evidence; what it would change and where; the deliverables; the domain; the seed; the North Star link.

Tier 1: WInS now and Trading Notes (Oct 23)

Q4. Was the \$300,000 first deposit sized to be roughly the price of the ten payments at 2026 rates, so the IPS lock rule should say "buy as much of the ladder as the 2027 deposit affords and complete it from the 2028 deposit" rather than rely on today's "it fits inside \$300k"?

- Anchor: F-111 ("Above \$300k on 173 of 185 trading days"), R-AN13, R-AN14, R-C26/R-C27. Case PDF created 2026-09-10 (case_register 1.1).
- Hypothesis (GUESS): yes. At a flat rate, \$300k buys the promise at 5.09% (DERIVED). At 4.0-4.5%, the rates most people would call "reasonable" in 2025-26, the promise costs \$317-333k. At those rates the first deposit falls just short and the two deposits together cover it easily. The designers probably wrote the case when the ladder cost \$302-325k (Jan-Sep 2026, F-111). Our "fits inside \$300k with \$7.7k to spare" depends on a September 2026 rate peak. The 10y is at its highest close since 2007 (F-004).
- Would change (decision + sentence):
 - The IPS rule wording must depend on rates and name the 2028 deposit as the planned completion source.
 - The WInS promise book should be sized as "what \$300k buys today" (about 97%), with the gap stated.
 - One Trading Note reflection could say the rule was tested by rates.
- Deliverables: WInS-now, TN, IPS. Domain: D1. Seed: none (extends brief section 9, "Rule if rates fall").
- North Star: Laura sees a plan that works in the world the case was written for, not one that needed lucky rates.

Q5. Should our narrative spine be "the first deposit secures the promise; the second deposit and growth fund the facility and flexibility", the one idea a reader remembers, as Wharton's WInS cash (\$300k = deposit 1) seems to invite?

- Anchor: R-AN14; R-W59 ("The additional \$150,000 ... will not be added to WInS"); R-C8 ("the investment strategy that Laura ultimately chooses"). Judge, 2022 finale (VERIFIED-PRIMARY): "Cut slides, cut words, cut minutes".

- Hypothesis (GUESS): yes, as long as it is stated with Q4's rate condition. It gives the 50-word pitch a structure that is easy to remember. It explains the WInS book as the Year-1 promise book, which settles the open question (Year-1 book vs post-2028 target, brief section 9) in favour of Year 1, with a small growth sleeve. It maps onto the case's order: reserve first, facility second (R-C50).
- New versus B1b Q5: the case history shows WInS cash was never before tied to a client deposit (\$100k × 13 seasons, then \$500k). Q4's calibration shows the tie is also economic.
- Would change: decision (WInS allocation mirrors the Year-1 book) + the sentence spec for the pitch.
- Deliverables: WInS-now, TN, IPS. Domain: D9. Seed: 1.
- North Star: the simplest true story of her money; each deposit has a job she can repeat to a co-sponsor.

Q2. Laura's case is the first in six seasons with no hobbies, family or city. Did Wharton remove the "hobby hooks" that past teams turned into stock picks, and should we therefore rule out every identity- or interest-themed security (Taiwan tilt, publishing, "creator economy", Asian-American themes)?

- Anchor: R-AN34 (no values or impact language), R-C10 (her themes are her public work), brief section 4 (tokenism). Word check (VERIFIED-REPO-FILE): 0 hits for enjoy, love, free time, family, lives, partner, San Francisco.
- Evidence:
 - The 2021-22 case listed Peloton, yoga, wine tasting. The 2022-23 case listed scuba, Muay Thai, yoga. Both VERIFIED-PRIMARY.
 - 2023-24: "volleyball, a passion highlighted in Hilary Ash's client profile that inspired great creativity" (VERIFIED-PRIMARY). A team bought Nike because it "caters to Ms. Ash's love of sports" (Teen Ink, SNIPPET-UNVERIFIED).
 - The Top 50 lists are full of client-themed team names: "Serve and Set Finances" (2024); "End Zone Equity", "Lumen Field Investments" (2026) (VERIFIED-PRIMARY).
- Hypothesis (GUESS): yes, the removal is deliberate. Tailoring must run through cash flows, constraints and her professional mindset. The EWT/Taiwan tilt (open, brief section 9) should be dropped unless a cash-flow reason exists; I see none, because the promise is in USD.
- Would change: decision (drop themed positions from WInS now) + a sentence in the IPS saying how we tailored.
- Deliverables: WInS-now, TN, IPS. Domain: D7. Seed: 12.
- North Star: she is a person with a plan, not a theme; a firm that tailors to her goals shows it listened.

Q18. The case is the first to say the client "understands that investing involves uncertainty and periods of market volatility". Should our risk-tolerance statement, and one Trading Note, use that line to set a stated sleeve drawdown she accepts, while the payments carry no market risk?

- Anchor: case p.2 L68-74 (R-C37/R-C38 area; VERIFIED-REPO-FILE): "Laura understands that investing involves uncertainty and periods of market volatility. Although she has been willing to take thoughtful risks ...". "Volatil" has 0 hits in the four earlier case texts (my check).
- Hypothesis (GUESS): yes. Willingness is stated in the case, and ability is set by the promise. A number such as "the growth sleeve could fall about a third in a bad year; the ten payments would not change" turns an adjective into something she can hold us to. Wharton's example trading note (R-AN44) speaks of reducing volatility; our note should say why volatility in the promise hedge is not risk to her.
- Would change: a number (the stated sleeve drawdown, for D3 to size) + a sentence in the IPS risk section + one TN reflection.
- Deliverables: TN, IPS, FR. Domain: D6. Seed: 18.
- North Star: it answers her own stated understanding of risk, not a generic risk questionnaire.

Q11. Laura is the first client in six seasons with a scheduled second contribution, and it depends on reputation-driven creative income. Is a "human-capital" underweight (publishing, media, consumer names) large enough to matter in a broad-index sleeve, or should we only state the link and do nothing (complexity must earn its place)?

- Anchor: R-C27 ("publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"), R-AN31, SH-03/BS-03 (credibility is income). Earlier clients: "both with steady jobs and incomes" (2024-25); retirement already covered (2021-22, 2022-23) (VERIFIED-PRIMARY). 2024 judge: "take a look at this person holistically" (VERIFIED-PRIMARY).
- Hypothesis (GUESS): publishing and media are well under 2% of a U.S. broad index (to verify). An underweight would add trades and explanation for almost no protection. The real link to her income is the equity market as a whole (a 2027 crash also hits advances and speaking), and that is the wrong-way deposit risk already open in brief section 9. So: state it, do not trade it.
- New versus brief section 9 ("human-capital underweight" open): the history shows this is the first case where the client's future income feeds the portfolio. That is why readers will look for the "holistic" point.
- Would change: decision (keep or drop the underweight idea) + a number (sector weight).
- Deliverables: WInS-now, IPS, FR. Domain: D2. Seed: none.
- North Star: shows we see her whole financial life without adding complexity she has to pay for.

Tier 2: IPS (Nov 26)

Q3. Wharton asked a ten-payment question before, in 2021 at a 1.48% 10-year yield, when pre-funding cost about 95% of face. Should our IPS and FR say why locking the promise is right for Laura now (it costs about 58% of face at 2026 rates), so readers see a choice made for this rate world and not a reflex for safety?

- Anchor: R-AN13 (\$450k in vs \$500k out). 2021-22 case (VERIFIED-PRIMARY): "wants to be able to offer the scholarship for at least 10 years. Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends? Another option?"

- Evidence (ASSUMPTION arithmetic, section 5):
 - Jordan's ten \$5,000 payments at the 2021-09-27 curve were worth about \$47.3k (0.946 of face).
 - Laura's ten payments at the 2026-09-25 curve are worth \$292,264 (0.585 of face).
 - At a flat 1.48%, Laura's promise would cost about \$429k.
- Hypothesis (GUESS): yes. In 2021, "keep cash" and "buy bonds" were almost the same thing. In 2026, locking earns about \$208k of interest for her, so the case works only in a high-rate world. The designers made cash impossible (\$450k < \$500k) and interest rates the main variable. Saying so in one clause shows judgement, not caution.
- Would change: an IPS sentence (why now) and an FR chart (cost of the promise against the yield, marking 2021 and 2026).
- Deliverables: IPS, FR. Domain: D1. Seed: none.
- North Star: Laura sees that we understood why her promise is affordable today, not only that it is safe.

Q10. Judges and clients have named the same two gaps since 2024: the client's emotions in a downturn, and the whole person. Which pre-commitment rule in the IPS answers the first, and which sentence answers the second?

- Anchor: criterion "Client Knowledge and Objectives" (R-S25, "recommendations that can earn her confidence"). 2024 finale (VERIFIED-PRIMARY):
 - "urging them (as investment advisors) to consider behavioral finance and their clients' emotions during a market downturn";
 - "When analyzing someone's investments, take a look at this person holistically". Guide p.5 (via B7a/B1b, VERIFIED-PRIMARY): "you may not redesign your strategy after observing the results".
- Hypothesis (GUESS):
 - (a) A one-line rule: "a fall in the growth sleeve never changes the promise book; we rebalance the sleeve inside bands and do not sell into a fall". The IPS must pre-commit it, because it cannot be added later.
 - (b) One sentence that joins her income (the 2028 deposit), her credibility and the portfolio.

Neither is in the current strategy text (brief section 7).

- Would change: an IPS rule and sentence; an FR scenario ("what Laura sees in a 2029 bear market").
- Deliverables: IPS, FR. Domain: D6. Seed: none (connects to seed 27, which belongs to another lens).
- North Star: she chooses the firm that has already planned how she will feel on a bad day.

Q8. Laura has a statistics degree, and the competition's academic director has an M.A. in Statistics and runs the Wharton Financial Analytics Initiative. What would a statistically trained reader reject on sight in our "high degree of certainty" definition and our 2031 confidence statement?

- Anchor: R-C12, R-C54/R-C55, R-AN32, R-AN9. Main page (VERIFIED-PRIMARY): "his M.A. in Statistics and Ph.D. in Economics from the University of California at Berkeley"; "founded and heads the Wharton Financial Analytics Initiative".
- Also: Laura describes her degree as "a degree in business analytics" (Wharton Magazine, 6 Apr 2022, VERIFIED-PRIMARY). The case chose the word "Statistics".
- Hypothesis (GUESS): the reader would reject:
 - a bare "95%" with no model named;
 - calling a model probability a "confidence" without saying what is random;
 - a two-ended range whose top end was never tested;
 - independent lognormal returns presented as reality;
 - no sample size or seed.

The fix is a four-line "model card" in the FR (model, inputs with dates, what is not modelled, what the number means), plus the market-priced definition of certainty already chosen.

- New versus brief section 8 item 6: the reader profile (academic director) is new evidence that raises the bar on method disclosure, not on the definition.
- Would change: an FR sentence/box; the IPS certainty sentence must name its test.
- Deliverables: IPS, FR. Domain: D3. Seed: 18.
- North Star: a statistics graduate trusts a firm whose numbers say what they are and are not.

Q6. The committed outflow has grown from 50% of the client's money (Jordan) to 111% (Laura), and this is the first case with no return target. Should the Final Report spend most of its analysis on funding reliability, the facility rule and the co-sponsor range, and much less on security selection?

- Anchor: R-C79-R-C84 (the five tests: none is about returns), R-S26 ("evaluate funding reliability, the facility contribution, and financial flexibility under varying market outcomes"), R-AN34. History table (section 2).
- Hypothesis (GUESS): yes. A rough split: about two-thirds on the promise, the facility and the co-sponsors, and about one-third on implementation and security rationale. Past finales celebrated stock screens (2023, 2025), so most repeat-school teams will over-weight security selection.
- Would change: decision (page allocation in the FR) + the IPS order of topics.
- Deliverables: IPS, FR. Domain: D7. Seed: none.
- North Star: her problem is a promise; the firm that spends its pages on her problem is the one she picks.

Q7. The case's vocabulary moved: "certainty" (0 → 6 hits), "co-sponsor" (0 → 9), "flexib" (0 → 5), "credib" and "uncertain" (0 → 3), while "stock" (2-4 → 0) and "pitch/convince/win" (up to 8 → 0) vanished. Should we adopt the

new words as our own terms and ban the old ones, and use the word table in the FR's Articulation section as evidence of how we read the case?

- Anchor: R-S27 ("clearly explains the team's research and decision-making process"), R-S28 ("authentic team voice"). Word table in section 2 (my count).
- Hypothesis (GUESS): yes to the vocabulary. Words like "beat the market", "alpha", "pitch" or "win her business" read like last season's template. The word table is an honest process artefact for the Articulation criterion, and it takes one sentence or one small chart.
- New versus B1b Q9 (tone): this adds counts and makes them a reusable checklist.
- Would change: a vocabulary checklist for all three deliverables; a possible FR chart.
- Deliverables: TN, IPS, FR. Domain: D9. Seed: 12.
- North Star: we speak the language of her case (certainty, flexibility, credibility), which is the language she chose.

Q14. The 2026-27 Guide opens by defining the whole competition as helping a client "achieve specific financial goals and meet future cash-flow needs", which the public main page does not say. Is Wharton re-centring the competition on cash-flow planning, and should our first exhibit (and every Trading Note) name the dated cash flow it serves?

- Anchor: R-new. Guide p.1 (VERIFIED-PRIMARY): "designed to help a client achieve specific financial goals and meet future cash-flow needs". Main page R-W5: "designed to help the client achieve their objectives". Criterion 1 (R-S24): "disciplined planning across Laura's changing time horizons and cash-flow needs".
- Hypothesis (GUESS): yes. The Guide is written generically ("a client") and so is likely to stay for future seasons. A one-row-per-year table (2027 to 2042: deposits, payments, reserve date, range date) is the cheapest way to score on "disciplined planning".
- New versus B1b Q2: the Guide's opening sentence is new evidence that this is the competition's frame, not only this case's.
- Would change: FR exhibit 1; a TN habit (each note names its date); IPS wording "cash-flow needs".
- Deliverables: TN, IPS, FR. Domain: D7. Seed: none.
- North Star: Laura's plan is a calendar of promises; showing the calendar shows we understood her.

Q1. Past clients said what won them over: feeling "seen" (Jordan, 2022) and plans "not just about the numbers" (Ayoola, 2025). With no personal hooks in Laura's case, which case-stated facts about how she thinks should each recommendation visibly mirror, so she would feel understood rather than analysed?

- Anchor: R-S25; R-C6, R-C12, R-C18, R-C24 ("Thoughtful financial planning ... turning those ideas into reality"), R-C37/R-C38; seed 32. Quotes VERIFIED-PRIMARY: "she had never felt more "seen" in her life"; "not just about the numbers, but about understanding and connecting with people".
- Hypothesis (GUESS), five mirrors, all from the case:
 - (1) Ideas become reality through planning: the plan makes the residency fundable.
 - (2) Statistics: certainty is defined and tested.
 - (3) "Thoughtful risks": risk only where the promise is safe.
 - (4) A product manager: dated milestones (2027, 2028, 2031, 2033) with decision rules.
 - (5) A public voice whose credibility is an asset: nothing overpromised.

Every "tailored" claim should cite one of these. Any claim that cites none is decoration.

- Would change: a checklist for the IPS client paragraph and for each TN reflection's "how it serves Laura" line.
- Deliverables: TN, IPS, FR. Domain: D6. Seed: 32.
- North Star: this is the direct answer: she recognises her own way of thinking in our plan.

Tier 3: Final Report (Dec 4)

Q9. The pull quote "The only person who needs to believe in something is yourself." is unattributed, while every past case quoted the client in the first person. Last year's client framed success as "confidence ... and humility to know that you don't have all the answers". Is the case built on turning private conviction into public credibility, and how should the fundraising draft use the quote, if at all?

- Anchor: R-C21, R-AN30, R-C66. Past quotes (VERIFIED-PRIMARY):
 - Jordan: "What drives me is opening doors or creating opportunities for others like me."
 - Barwin (2026): "approach every problem with confidence that you can figure it out, and humility to know that you don't have all the answers".

My search found no source for the pull quote. It is not on the Wharton Magazine, Overachiever, Mochi or Wikipedia pages (fetch_text --grep: NO); Goodreads timed out. PARAPHRASE-UNVERIFIED as her words.

- Hypothesis (GUESS): yes to the theme. Her self-belief is what starts the project, and co-sponsors need evidence. The draft should not quote the line as hers unless it is verified. Instead it can show conviction as money already committed ("ten years of operations are already funded"), and humility as a range with a stated floor.
- Would change: a decision (quote or not); the structure of the FR fundraising draft.
- Deliverables: FR. Domain: D6. Seed: 15.
- North Star: we turn what she believes into what others can check, which is exactly what she needs in 2031.

Q12. Laura's case-stated career includes product manager. Would framing the 2031 range as "committed (already bought) plus stretch (with a stated probability)" feel like her own professional language, and more credible to co-

sponsors, than "p10-p90"?

- Anchor: R-C12 ("product manager in the technology industry"), R-C67-R-C71, R-AN9.
- Hypothesis (GUESS): yes for the co-sponsor audience. Committed-versus-stretch is plain English, and it matches the floor-plus-upside design (brief section 8 item 7). It is professional, not identity, so it passes the tokenism rule. Percentile language belongs in the FR's method box (Q8).
- Would change: a sentence and structure spec for the FR fundraising draft.
- Deliverables: FR. Domain: D9. Seed: 32.
- North Star: she reads her range in words she already uses to promise things.

Q13. All six clients pay something forward, and Laura's residency is for "artists, writers, designers, entrepreneurs, and educators", her own roles. Are the residents, not only Laura, the people who rely on the ten payments, and does that raise the bar for "high degree of certainty" and for how we describe it to co-sponsors?

- Anchor: R-C40 (the residency), R-C41 ("reliable support for its early operations"), R-C47. Rules page R-W29 (CFA Asset Manager Code). History (VERIFIED-PRIMARY): "others like me" (Jordan), "pay forward" (Hjemdahl), "it truly takes a village" (Ayoola).
- Hypothesis (GUESS): yes. Residents will plan their lives around a residency place, so a broken operating promise harms third parties, not just Laura. That is a Laura-specific reason (beyond the case's "must") for market-priced certainty. It is also the strongest co-sponsor line: funders fear running costs (stakeholder map BS-17), and here running costs are already secured.
- Would change: a sentence in the IPS objective ("why certainty"); the FR fundraising lead.
- Deliverables: IPS, FR. Domain: D5. Seed: 15.
- North Star: we protect the people her residency serves, which is why she is building it.

Q19. The case's first line about Laura is a belief, "stories have the power to change how people see the world", and her first named work answered misinformation. Should the co-sponsor draft be built as a short story backed by numbers (problem, promise, proof, range) rather than a table-first term sheet?

- Anchor: R-C9, R-C14, R-AN33, R-S28 ("compelling, well-organized narrative"; "clearly and credibly to prospective co-sponsors").
- Hypothesis (GUESS): yes, with care. Use a four-beat story, each beat carrying one verified number. The tokenism risk is low because this is her case-stated belief about communication, not her identity. The failure mode is puns on her book titles, which the field will overuse (B7b).
- Would change: the structure of the FR fundraising draft; one Creativity-criterion decision.
- Deliverables: FR. Domain: D9. Seed: 15.
- North Star: she will recognise a firm that tells the truth as a story, the way she does.

Q17. Semifinal lists show many client-themed team names and "creative" themes, and a 2024 semifinal judge said the client-hobby creativity "resonated". Under this year's "authentic team voice" criterion, which creative element is authentic for us, and which is decoration?

- Anchor: R-S28. 2024 semifinal judge (VERIFIED-PRIMARY): "The one thing that stood out to me was the ability to connect the client's objectives to an investment strategy." Top 50 names, 2024 and 2026 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): creativity that carries analysis is authentic. Examples: the 2021-vs-2026 cost-of-the-promise chart (Q3), or one line on how Australia's Future Fund or superannuation handle long promises, if it earns its place (v4 Part 2 lesson 4). Book-title themes are decoration. Note the judge's own words: "connect the client's objectives", not hobbies.
- Would change: decision on the FR's creative device.
- Deliverables: FR. Domain: D7. Seed: none.
- North Star: our creativity is about her problem, so it reads as ours and as hers.

Q16. How did the 2021-22 finalists fund Jordan's ten scholarship payments (cash, dividend stocks, bonds)? Does that show readers accept an answer suited to its rate regime, so a Treasury match in 2026 will not look "too simple"?

- Anchor: 2021-22 case (VERIFIED-PRIMARY, quoted in Q3). 2022 finale (VERIFIED-PRIMARY): teams showed "client-focused talk of dividends and growth stocks, ESG and risk management".
- Hypothesis (GUESS): the finalists mostly used dividend stocks and growth stocks. Nobody locked Treasuries, because at 0-1.5% there was little to lock. If so, our FR can include a short "alternatives we rejected" line: cash (cannot reach \$500k from \$450k); dividend stocks (dividends were cut in 2008 and 2020, a fact to verify); growth-first (brief section 8).
- Would change: an FR "alternatives considered" sentence; confidence in the simple core.
- Deliverables: IPS, FR. Domain: D7. Seed: none.
- North Star: shows Laura we tested the obvious answers and chose the one that fits her year.

Q15. The 2024-25 case, for a Nigerian project, wrote "\$10,000 USD per housing unit". The 2026-27 case, for a Taiwan project, never names a currency. Does that make currency a deliberate simplification (one disclosure line) rather than a hidden test (a modelled TWD scenario)?

- Anchor: R-AN10, BS-04. 2024-25 case (VERIFIED-PRIMARY): "Ladi believes it will cost about \$10,000 USD per housing unit; however, material costs and market dynamics are always changing."
- Hypothesis (GUESS): simplification. Last year the designers named USD because they gave costs. This year they give no costs ("not expected to estimate that cost"). So the currency's job is only in the co-sponsor labelling (write US\$) and one sentence

on NT\$ purchasing power. Brief section 8 item 15 already says "disclose, quote in two currencies". What is new is the prior-case contrast, which argues for keeping it to about one sentence.

- Would change: the number of FR words on TWD (a cap); no TWD positions in WInS.
- Deliverables: FR. Domain: D4. Seed: none.
- North Star: we handle the Taiwan detail with care but keep her plan simple.

4. Evidence found in this pass (with status)

#	Fact	Source	Status
E1	2021-22 case: "Starting in 2022, Nichole would like to provide a small scholarship of \$5,000 per year ... for at least 10 years. Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends? Another option?"; "\$100,000"; "\$100,000 in virtual cash"; lists hobbies	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/	VERIFIED-PRIMARY
E2	2022-23 case: "Peter, who is 25 years old"; \$20,000 fund, "at least a \$10,000 return", "use the interest"; wellness centre "in the next 15 years"; "compelling and clever pitch"	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2022-2023/	VERIFIED-PRIMARY
E3	2023-24 goals: South American property within 5 years; sports consulting firm within 15 years; a Nike pick justified by her love of sports; a mid-term report existed	https://www.teenink.com/opinion/all/article/1215809/ (team report)	SNIPPET-UNVERIFIED (secondary, team text)
E4	2024-25 case: "\$10,000 USD per housing unit"; Phase 2 in 2040 "at least \$50,000"; one year of operations "at least a quarter of your original \$100,000"	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/	VERIFIED-PRIMARY
E5	2025-26 case: grow \$500,000 to at least \$1.5 million by 2036; community contributions from "the end of year three (2029)"; "win Connor's business"	https://www.scribd.com/document/965855349/Case-Study-Wharton-Global-Youth-Program	SNIPPET-UNVERIFIED (copy of the retired Wharton page)
E6	2022 finale: Jordan "had never felt more "seen" in her life"; judge: "Cut slides, cut words, cut minutes"; teams' "dividends and growth stocks, ESG and risk management"; ~1,300 final reports	https://globalyouth.wharton.upenn.edu/news/top-teams-sail-to-success-in-the-2022-wharton-investment-competition-global-finale/ (28 Apr 2022)	VERIFIED-PRIMARY
E7	2023 finale: "stock-filtering algorithms ... endless Environmental, Social & Governance considerations"; "we will build on the success of the Investment Competition with changes and new competitive experiences"	https://globalyouth.wharton.upenn.edu/news/the-2023-investment-competition-global-finale-ends-in-sweet-victory-for-dmvs-finest/ (26 Apr 2023)	VERIFIED-PRIMARY
E8	2024 finale judges: "a lot of depth and accuracy"; "consider behavioral finance and their clients' emotions during a market downturn"; "take a look at this person holistically" (Zoe McCormick, abrdn North American fixed income); "doing some reimagining of our signature competition"	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024-investment-competition-global-finale/ (23 Apr 2024)	VERIFIED-PRIMARY
E9	2024 semifinals: "volleyball, a passion highlighted in Hilary Ash's client profile"; judge: "the ability to connect the client's objectives to an investment strategy"	https://globalyouth.wharton.upenn.edu/news/wharton-global-youth-hosts-five-virtual-events-to-announce-the-2024-investment-competition-finalists/ (25 Mar 2024)	VERIFIED-PRIMARY
E10	2025 finale: Ayoola: "not just about the numbers, but about understanding and connecting with people"; judge (Aberdeen credit): "clear thoughtfulness and reasoning behind all the decisions"; teams used "customized algorithms, AI innovations, portfolio optimizations"	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/ (1 May 2025)	VERIFIED-PRIMARY
E11	2026 finalists: Barwin's "confidence ... and humility" quote; his case "encouraged students to think beyond financial returns"	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/ (20 Mar 2026)	VERIFIED-PRIMARY
E12	Academic Director: Prof. Michael Roberts; "M.A. in Statistics"; heads the Wharton Financial Analytics Initiative; research includes "the effect of interest rates on bank lending"	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/	VERIFIED-PRIMARY
E13	Guide p.1: "achieve specific financial goals and meet future cash-flow needs"	https://upenn.box.com/shared/static/b9kh53xila9cxyznk1xv5yegl3snlc1.pdf	VERIFIED-PRIMARY
E14	Season-start 10y: 1.48 / 3.88 / 4.55 / 3.75 / 4.15 (2021-2025); 2021-09-27 curve 1y 0.09, 2y 0.31, 3y 0.56, 5y 0.98, 7y 1.30, 10y 1.48	https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10 (and DGS1/2/3/5/7)	VERIFIED-PRIMARY
E15	Jordan payments ≈ \$47.3k at the 2021 curve (0.946 of face); Laura \$292,264 (0.585); flat break-even for \$300k = 5.087%; promise cost at flat 1.48% ≈ \$428.9k, at 4% \$333.3k, at 4.5% \$317.5k, at 5% \$302.5k	inline Python, section 5	ASSUMPTION (par-as-zero, flat rates)
E16	Laura: "After I graduated from Penn in 2018 with a degree in business analytics, I worked in tech as a product manager."	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/ (6 Apr 2022)	VERIFIED-PRIMARY (professional line only; the page also has personal material, which I did not record)
E17	Pull quote not found on Wharton Magazine, Overachiever, Mochi or Wikipedia pages	fetch_text --grep "believe": NO on each; Goodreads timed out	Search result: unattributed
E18	Case word checks: "volatil" 1 (2026-27) vs 0 in four past cases; hobbies/family words 0	repo case .txt + past case texts	VERIFIED-REPO-FILE / my count

5. Reproduce the numbers (repo root)

```
.venv/bin/python -c "
import numpy as np; from scipy.optimize import brentq
m=[1,2,3,5,7,10]; y=[0.09,0.31,0.56,0.98,1.30,1.48] # FRED 2021-09-27
pv=sum(5000/(1+np.interp(t,m,y)/100)**t for t in np.arange(10)+0.5); print(round(pv), round(pv/50000,3))
f=lambda r: sum(50000/(1+r)**(6+k) for k in range(10))
print(round(brentq(lambda r: f(r)-300000,0.001,0.2)*100,3), [round(f(r)) for r in (0.0148,0.04,0.045,0.05)])&quot;
```

Output: 47297 0.946 / 5.087 [428906, 333328, 317478, 302509].

6. Sources (all accessed 2026-09-27)

Source	URL / path	Status
Wharton archive (clients and winners 2021-2026)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/archive/	VERIFIED-PRIMARY
Case 2021-22, 2022-23, 2024-25	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/ (and -2022-2023/, -2024-2025/)	VERIFIED-PRIMARY
Case 2023-24, 2025-26 on the same URL pattern	same pattern	404 (retired)
Case 2025-26 copy	https://www.scribd.com/document/965855349/Case-Study-Wharton-Global-Youth-Program	SNIPPET-UNVERIFIED
2023-24 team report	https://www.teenink.com/opinion/all/article/1215809/Profiteering-Peninsula-Panthers-2023-2024-Wharton-Global-High-School-Investment-Competition-Final-Report	SNIPPET-UNVERIFIED
Finale articles 2022-2026	the five /news/ URLs in E6-E11, plus https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	VERIFIED-PRIMARY
Semifinal/finalist articles 2022-2026	/news/announcing-the-top-10-teams-advancing-to-our-2022-investment-competition-grand-finale/; /news/celebrating-the-top-10-teams-in-the-2022-2023-wharton-global-high-school-investment-competition/; /news/wharton-global-youth-hosts-five-virtual-events-to-announce-the-2024-investment-competition-finalists/; /news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/; the four Top-50 announcement pages (2022-2026)	VERIFIED-PRIMARY
Competition main page (academic director)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/	VERIFIED-PRIMARY
2026-27 Competition Guide	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf	VERIFIED-PRIMARY
FRED DGS1-DGS10	https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10	VERIFIED-PRIMARY
Wharton Magazine interview (2022)	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	VERIFIED-PRIMARY (professional lines only)
Global Youth "case study" page	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/case-study/ ("More information coming soon.")	VERIFIED-PRIMARY
Global Youth RSS feed (no Laura article yet; latest item 30 Apr 2026)	https://globalyouth.wharton.upenn.edu/feed/	VERIFIED-PRIMARY
Blocked or failed	poetsandquantsforundergrads.com (Cloudflare challenge page today, although Phase A listed it as reachable); Goodreads quotes (timeout); Wayback (not tried; blocked per B1a)	—
Repo	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt; Phase A registers; B1a, B1b, B7a, B7b	VERIFIED-REPO-FILE / prior agents

7. Leads for the specialists

- **D1 (rates):**
 - Q4: price the ladder on every 2025 curve too (Treasury CSV). This shows whether \$300k was ever enough while the case was likely being drafted (2025-26).
 - Q3: build the FR chart "cost of Laura's promise vs flat yield 1%-6%", marking 1.48% (2021) and 5.17% (2026).
 - Write the rate-conditional lock rule and its fallback (long rungs first, per brief section 7).
- **D2 (equity):** Q11. Weight of publishing, media and entertainment names in IVV/VTI (iShares/Vanguard holdings files, reachable), to decide whether a human-capital underweight is material.
- **D3 (quant):**
 - Q8: draft a four-line "model card" spec (model, inputs with dates, what is not modelled, how to read the number).
 - Q18: size a plain sleeve-drawdown number for the IPS: the historical worst 12-month fall for a 60/40 sleeve, or the p5 one-year sleeve return from strategy_mc.py.
- **D4 (Taiwan/FX):** Q15. Keep TWD to one disclosure sentence plus a US\$ label. The 2024-25 "USD" contrast is the evidence.
- **D5 (co-sponsors):** Q13. Find evidence that funders value pre-funded operations (BS-17), and wording on who relies on a residency's operating budget (residents, partner institutions).
- **D6 (client):**
 - Q1: turn the five case-stated mirrors into a checklist.
 - Q10: pre-commitment language used by private banks.

- Q9: keep the pull quote unattributed unless a primary source is found. The other B2 agent should keep searching her public talks and pages.
 - **D7 (Wharton intent):**
 - Q16: find any 2021-22 semifinalist report (Scribd, Teen Ink, school news) showing how the scholarship was funded.
 - Q17: check 2025-26 Top-50 report styles for themes.
 - Recover the 2023-24 Ash case text (Scribd search "Hilary Ash case study") to fill the table's missing row.
 - **D9 (plain English):**
 - Q7: turn the word table into a ban list and a use list.
 - Q12: committed/stretch wording.
 - Q19: a four-beat story template (problem, promise, proof, range) with one number per beat.
 - **D10 (compliance):** Q2. If the Taiwan tilt is dropped, make sure any existing WInS position in it is sold before the notes are chosen, with its own note explaining why (the notes cannot be edited later, R-AN43).
 - **Correction for the team's "Why Wharton chose Laura" document:**
 - "Youngest client in recent years" is wrong; the 2022-23 client was stated as 25 (E2).
 - "Three clients in a row fund community infrastructure" holds only for 2024-25, 2025-26 and 2026-27. The 2023-24 goals were a property renovation and her own firm (E3, SNIPPET-UNVERIFIED).
-

What this teaches

- **Compare the case with its own history.** Six seasons side by side show what the designers kept, what they removed and what they added. What they removed (return targets, hobbies, values words, stock language) tells you which old habits no longer score. What they added (certainty, co-sponsors, flexibility) tells you what now does.
- **The same question can have a different right answer in a different year.** A ten-payment promise cost about 95% of its face value in 2021 and about 58% in 2026. Interest rates, not cleverness, changed the answer. Good advisers say *why now*.
- **Listen to what clients and judges repeat.** Across five finales they asked for the same things: feel understood, cut the clutter, reason clearly, plan for the client's emotions in a downturn, see the whole person. A pattern that repeats across years is stronger evidence than any single quote.
- **Tailoring is not decoration.** When a case removes personal details, the way to tailor is through the client's goals, cash flows and way of thinking. Buying something because the client likes it was never the point.

B2b Why Laura / Case-Designer Intent: questions from Wharton's own framing

Agent B2b, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: "Why Laura / Case-Designer Intent". Starting angle: Wharton's OFFICIAL framing. That means the exact case wording, the 2026-27 Competition Guide, Wharton Global Youth and Wharton Magazine pieces, and Wharton's own pages for earlier seasons, which are still in the site's source code. The aim is to work out what the designers want students to learn and what a semifinal reader will look for first.

This file is AI-generated research for Team Caplet, for brainstorming. It contains questions, evidence and checklists. It contains no text meant for submission. The six students write every word of every deliverable, and the Final Report's Works Cited must record how AI was used (R-W46). Laura is covered here only through the case and her PUBLIC PROFESSIONAL record. I left out personal details that some sources contain (family, hometown, school, private life, a personal loss). Nothing here suggests contacting her.

Status labels follow brief section 3. Every "hypothesis" is a GUESS. INTERPRETATION means my reading, never a fact.

0. Summary (read this first)

I found **new primary evidence** that no earlier file has. It changes how the team should read "the investment strategy that Laura ultimately chooses".

1. **Wharton's own pages still carry last season's case, the old judging page and the old deliverables page.** They sit inside the WordPress page source, in hidden page sections, and the site's public REST interface serves them (VERIFIED-PRIMARY, read 2026-09-27). This upgrades several SNIPPET-UNVERIFIED items in B1b/B7b to primary, and it lets us compare wording line by line:
 - 2025-26 case: "Your team hopes to **shape the winning** investment strategy that Connor and the Make the World Better foundation ultimately choose". The case also had a section asking "How will you create a **compelling and clever pitch** to convince Connor to choose your strategy?"
 - 2026-27 case: "Your team hopes to **develop** the investment strategy that Laura ultimately chooses". The pitch section is gone.
 - Old judging criterion (the 2023-24 and 2022-23 versions are on the page): "strategy is tailored uniquely to the client's financial goals; **would win him/her over as a client.**" The 2026-27 criterion reads: "presents recommendations that can **earn her confidence**" (R-S25).
 - The main competition page used to say teams meet the client's goals "as they try to win his or her business". It now says teams "develop a long-term investment strategy designed to help the client achieve their objectives" (R-W5).
 - INTERPRETATION: Wharton has moved deliberately from *persuasion* to *trust*. The North Star question ("why would Laura choose us?") should be answered as "because this is the plan she can trust", not "because our pitch is the cleverest" (Q1, Q2).
2. **The pull quote follows a house pattern.** Recent cases each print a client quote that states the season's value theme:
 - 2024-25 (Ayoola): "I want to feel like I'm making an impact". That season's case was impact-led.
 - 2025-26 (Barwin): "Wharton made me think more about the sustainability of a nonprofit, and the importance of building strong partnerships to keep the work going."
 - 2026-27 (Gao): "The only person who needs to believe in something is yourself."

INTERPRETATION: this year's theme is self-reliance. That fits the case's hardest rule: Laura alone must fund the ten payments, and no co-sponsor money may count (R-C48). Co-sponsors are invited to believe in the facility only. Barwin's quote even reads like a brief for this year's design: *sustainability* (ten funded operating years) plus *partnerships* (co-sponsors) (Q3, Q4).

3. **Laura's public professional record is more financial than the case says** (Poets&Quants 2018 profile, read via the publisher's own API, VERIFIED-PRIMARY). While at Wharton she:
 - was a "Trading Intern at Belvedere Trading";
 - was a "Policy Analyst Intern at Consumer Financial Protection Bureau";
 - was a "Business Data Analyst Intern at Amazon";

- was a mentor for MoneyThink, where she "taught ... financial literacy to high school students in West Philadelphia".

Her stated "biggest lesson" from business school: "If what you want doesn't exist, create it yourself." A client who has worked at a trading firm and a consumer-protection regulator, and who has taught high schoolers about money, will spot sales talk, hidden costs and unlabelled probabilities at once (Q6, Q7).

- Laura has designed a case competition herself.** Wharton Magazine (30 Oct 2017, VERIFIED-PRIMARY): the Amazon HQ2 to Philly Case Competition "was the brainchild of ... Laura Gao W18", and "Gao asked city officials what problem areas they needed help with, and these problem areas became the basis for the case competition's four subtopics." She builds a case around the client's own problem list. That suggests our Final Report should answer the case's five required tests one for one (Q8).
- Her one documented career "thoughtful risk" was taken after a contract was secured.** Inverse (7 Mar 2022, journalist's account, VERIFIED-PRIMARY as article text): "Gao had misgivings about leaving a steady job in tech for life as an artist ... So when she got the book deal with HarperCollins, she gave Twitter her notice." INTERPRETATION: secure the floor first, then take the leap. That is the same shape as "lock the promise, then take risk with the surplus". Use it with care, as one line of evidence and never as decoration (Q5).
- This is the first case in the archive (2017-18 to 2026-27) with explicit risk-attitude and certainty language.** Word check of Wharton's case pages: no earlier case uses "certain", "volatility" or "uncertainty". The only earlier risk line is 2020-21's "help to balance the riskiness of his portfolio". The 2026-27 Guide adds "management of risk and uncertainty" to "What Are We Looking For?". What the designers are testing is how teams handle and explain uncertainty (Q9).
- The case numbers were probably set when the promise cost MORE than \$300,000.** The same ten-payment ladder cost over \$300k on 173 of 185 trading days of 2026, and it has cost \$300k or less only since 2026-09-10 (F-111). The case PDF was created at 15:19 on that same day, 2026-09-10 (case_register section 1.1). At any flat rate from 3.5% to 5.0%, the 2027 cost is \$302k-\$350k (ASSUMPTION arithmetic, section 3). INTERPRETATION: the designers probably expected the promise to use up nearly all of the money, and part of the 2028 deposit too. Being able to lock the whole promise in January 2027 is a gift of September 2026 yields, not the case's design. This is new evidence on an open item in brief section 9 (Q10).

Top five questions for the next two deadlines (full list in section 2):

- Q1: persuasion versus trust. This sets the tone of every note, the pitch and the IPS.
- Q3: the self-reliance reading of the pull quote. This could supply the IPS's central idea.
- Q5: Laura's "floor first, then leap" pattern. This supports the IPS risk-tolerance sentence.
- Q10: the designers' calibration. This shapes the IPS rule for the case where rates fall before January 2027.
- Q12: the IPS word budget for client background ("Don't use precious space repeating known facts").

1. Evidence found (what Wharton itself says, by source)

1.1 Wharton's pages for earlier seasons (VERIFIED-PRIMARY; historical, not this year's rules)

Read 2026-09-27 through <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=...> (the site's own public REST interface). The text sits in page sections switched off on the live page (disable_element="yes"), so a browser does not show it.

- **2025-26 case (Connor Barwin)**, in the source of /competitions/investment-competition/case-study/ (modified 2026-05-07):
 - "Your team hopes to shape the winning investment strategy that Connor and the Make the World Better foundation ultimately choose to make a lasting community impact from an initial \$500,000 investment."
 - "Connor Barwin (WG'23) believes in the power of community." Compare 2026-27: "Laura Gao believes stories have the power to change how people see the world." The "[client] believes ..." opening is house style.
 - Pull quote: "Wharton made me think more about the sustainability of a nonprofit, and the importance of building strong partnerships to keep the work going."

- "Every dollar invested reflects MTWB's values — community-first, locally rooted, and sustainability-minded."
- "As you strive to win Connor's business, how will your team stand out against competing teams? How will you create a compelling and clever pitch to convince Connor to choose your strategy?"
- "Remember, your 10 weeks on WInS can only accurately assess the performance of short-term investments."
- "The investment scenarios included in this case study have been embellished for the purpose of the competition."

This upgrades B1b's Scribd copy (SNIPPET-UNVERIFIED) to VERIFIED-PRIMARY.

- **2024-25 case (Ladi Ayoola)**, </investment-competition/previous-winners/case-study-for-2024-2025/>:
 - pull quote: "When I'm working or doing anything, I want to feel like I'm making an impact, either directly or indirectly, in people's lives."
 - "He and Mosun, both with steady jobs and incomes". Laura's case says nothing like this; her 2028 money comes from "publishing advances, speaking engagements, licensing".
- **2022-23 case (Peter Wang Hjemdahl)**:
 - pull quotes: "Waste is a global issue that requires a global solution." and "Without Wharton, I would have never been able to launch and scale my social enterprise."
 - "Peter has heard great things about the creativity and investing knowledge of WGAM's analyst teams".
- **2021-22 case (Nichole Jordan)**: a scholarship "for at least 10 years"; "Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends?" (already in B1a).
- **2020-21 case (Florian Hagenbuch)**: "Given his cosmopolitan upbringing, Florian expects some level of geographic diversity in his investments. It would be appropriate to consider a blend of growth and value stocks to help balance the riskiness of his portfolio."
 - This is the only earlier case that links a client's background to an investment choice, and it states the link outright. Laura's case states no such link (see Q11).
 - **Word check across all archived case texts** (2017-18 to 2025-26, extracted to my scratchpad): 0 hits for "certain", "volatil" and "uncertain". "risk" appears only in 2020-21 ("riskiness"). VERIFIED-PRIMARY (the absence). The 2023-24 case text is not on the site; its archive URL returns 404 (B1a).
- **Old "Judging and Evaluation" page**, </investment-competition/judging-and-evaluation/> (modified 2024-06-03; holds the 2023-24 version and an older Peter/2022-23 version):
 - "Client knowledge and objectives: strategy is tailored uniquely to the client's financial goals; would win him/her over as a client."
 - "Creativity and presentation: compelling narrative and visually pleasing presentation of data; many creative elements; clearly presented with an authentic team voice."
 - "Excessive investing jargon doesn't necessarily dazzle the judges. Your report should reflect your team's voice."
 - "Don't use precious space repeating known facts about your client and defining concepts. However, it is okay to demonstrate that you understand your client beyond the case study."

This upgrades B7b's third-party transcription to VERIFIED-PRIMARY.

- **Old "Deliverables" page, 2023-24 (Hilary Ash)**, </investment-competition/deliverables/>:
 - midterm prompt: "What do you think Hilary's financial goals are? Don't just repeat the case study, show us that you have thought about how her personal circumstances will determine her financial needs."
 - "Final Report Audience: Your report should be directed to Wharton Global Asset Management's (WGAM) portfolio manager (your team's teacher/advisor)."
 - summary page: "Explain what makes your investment strategy unique compared to other teams in a maximum of 75 words."
 - Works Cited in APA style. For AI: "In your citation, please include 1-2 sentences on how you used that AI tool to achieve your goal."
 - "Our judges are instructed not to open any links within your report."

1.2 This season's official framing (VERIFIED-PRIMARY or VERIFIED-REPO-FILE)

- **Case (R-C8):** "Your team hopes to develop the investment strategy that Laura ultimately chooses as she works toward achieving her future vision."
- **Case (R-C6):** "turning ideas into opportunities". R-C24: "achieving ambitious goals requires more than creativity alone. Thoughtful financial planning and long-term investing will play an important role in turning those ideas into reality."
- **Case (R-C36 to R-C38):** "Laura understands that investing involves uncertainty and periods of market volatility. Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals."
- **2026-27 Competition Guide** (Box PDF b9kh53xi..., read 2026-09-27; first found by B1b/B7a):
 - "The competition recognizes thoughtful strategy, research and analysis, client understanding, disciplined decisions, management of risk and uncertainty, communication, and creativity."
 - "Your work should demonstrate an understanding of the client, informed investment decisions, and clear communication of how your approach supports the client's goals while addressing risk and uncertainty."
 - "The case study is your starting point. It is up to your team to determine what additional research and analysis you need to develop and support your strategy."
 - The Guide's final stage: "explaining why your strategy is appropriate for the client". The words "choose", "win" and "pitch" do not appear anywhere in the Guide (word check).
- **Main competition page** (modified 2026-09-11): "develop a long-term investment strategy designed to help the client achieve their objectives". The old version of the page (</investment-competition/>, modified 2024-07-11) said: "as they try to win his or her business".
- **No Wharton article about Laura exists yet this season.** The Global Youth site search (REST API, "Gao") returns no post about her. Wharton Magazine has two items: the 2022 book interview and the 2017 Amazon pitch piece; its 2020 list of alumni COVID efforts only mentions her in passing. For comparison, Wharton published "6 Questions for Hilary Ash" on 2024-01-02, after that season's final reports were due. VERIFIED-PRIMARY.

1.3 Past clients' own reactions (VERIFIED-PRIMARY; finale context, used only for what they say clients value)

- Nichole Jordan, 2022 finale article: she "concluded that she had never felt more "seen" in her life as students worked to meet and exceed her specific financial goals." (Wharton's paraphrase with one quoted word.)
- Ladi Ayoola, 2025 finale article: "Thank you for making the investment plans and proposals not just about the numbers, but about understanding and connecting with people." (already in B7a)
- Hilary Ash, 2024: "Congratulations to all of you because what you have done is incredibly impressive". This says little about what she valued.
- Judges, 2024 (Balchunas, McCormick) and 2025 (Jain): already in B7a/B7b; not repeated here.

1.4 Laura's public professional record from Wharton-linked sources (VERIFIED-PRIMARY)

- **Wharton Magazine, 30 Oct 2017, "Wharton's Pitch to Amazon"** (by Sara Albert):
 - the Amazon HQ2 to Philly Case Competition "was the brainchild of former Amazon intern Laura Gao W18";
 - "Gao asked city officials what problem areas they needed help with, and these problem areas became the basis for the case competition's four subtopics".
- **Poets&Quants for Undergrads, 2018 Best & Brightest profile** (post dated 2018-03-30; the HTML page returned a Cloudflare 403, so it was read through the publisher's WordPress API, post 11618). Professional items only:
 - internships: "Business Data Analyst Intern at Amazon", "Product Management Intern at SmartyPal", "Trading Intern at Belvedere Trading in Chicago", "Policy Analyst Intern at Consumer Financial Protection Bureau in Washington, D.C.";
 - "Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia.";

- founder of The Sign.al, "an organization that encourages the exploration of unconventional career paths";
- "What is the biggest lesson you gained from studying business? If what you want doesn't exist, create it yourself.";
- her advice on studying business: "this lack of structure and certainty can be off-putting for students who are easily influenced by those around them or are risk-adverse" (sic);
- the nomination by Wharton's Director of Student Life: "For carving her own path while at Wharton and for helping her peers follow their own dreams and passions".
- The profile gives her major as "Economics with a concentration in Business Analytics". The case says "a degree in Statistics & Information Decisions Management", and Wharton Magazine 2022 says "a degree in business analytics". The case wins (see Q11).
- **Wharton Magazine, 6 Apr 2022, "A Wharton Grad From Wuhan's Exploration of Identity"** (Braden Kelner). Professional quotes only:
 - "By letting your voice shine above the doubts, you'll realize the right people and places will naturally gravitate toward you";
 - on her work habit while a product manager: "I'd push myself to doodle a short 10-minute comic of a memorable moment from that day. Soon enough, I had a comics diary that I could not only decompress into, but also take pride in how far I'd come when I reread it."
 - **Inverse, 7 Mar 2022** (not Wharton; journalist's account; verified verbatim with `fetch_text.py --grep`): "Still, Gao had misgivings about leaving a steady job in tech for life as an artist. ... So when she got the book deal with HarperCollins, she gave Twitter her notice." I omit the sentence between these two, which describes a private event.
- **Pull quote search:** "The only person who needs to believe in something is yourself." Two web searches and greps of about ten interview pages found no primary source. It stays PARAPHRASE-UNVERIFIED as Laura's words (R-AN30). D13 or B8 should keep looking.

2. The questions

Ordered by deadline: WInS now and the Trading Notes (Oct 23), then the IPS (Nov 6), then the Final Report (Dec 4). Each entry gives the question, the anchor, a hypothesis (a GUESS), what would change, the deliverables, the domain, the seed and the North Star link.

Tier 1: WInS now, the Trading Notes and the tone of everything (Oct 23)

Q1. Since Wharton replaced "shape the winning investment strategy", "compelling and clever pitch" and "would win him/her over as a client" with "develop the investment strategy that Laura ultimately chooses" and "recommendations that can earn her confidence", should every deliverable aim to earn trust rather than to persuade?

- Anchor: R-C8, R-S25, R-AN5. New: 2025-26 case and old judging page (VERIFIED-PRIMARY, section 1.1).
- Hypothesis (GUESS): yes. "Chooses" survives, but the persuasion scaffolding around it was removed. Signs of trust that a reader can check:
 - claims are stated with their assumptions;
 - uncertainty is stated honestly;
 - each note names the risk of its own trade (Wharton's example note does this, R-T24);
 - there are no superlatives and no claims about beating other teams.
 - Would change (sentence): the tone rules for TN reflections, the pitch and the IPS. Ban "best", "optimal", "guaranteed" and "unlike other teams". Every claim carries its basis. The North Star is restated as "the plan she can trust".
- Deliverables: TN, IPS, FR. Domain: D7. Seed: 1.
- North Star: Laura does not choose the most persuasive firm. She chooses the one whose claims hold up.

Q2. What will a semifinal reader look for FIRST, given that Wharton's Guide never says "choose" or "win" and asks teams to explain "why your strategy is appropriate for the client"?

- Anchor: R-new (Guide p.5, "explaining why your strategy is appropriate for the client"; "What Are We Looking For?"), R-I17, R-T9.

- Hypothesis (GUESS): a reader first checks whether the team has found Laura's binding constraint. Here that is a fixed, nominal promise that no one else may fund. The reader then checks whether each decision follows from it. In the TN that means the first line of each reflection names which of her needs the trade serves. In the IPS it means the pitch's first clause names the promise.
- Would change (decision): the order of content in each TN reflection and in the 50-word pitch, with the constraint first and the instrument second.
- Deliverables: TN, IPS. Domain: D7. Seed: 1.
- North Star: a firm that names her real constraint in its first sentence shows it listened.

Q3. Wharton's recent pull quotes state each season's value theme ("impact" in 2024-25; "sustainability ... strong partnerships" in 2025-26). Does "The only person who needs to believe in something is yourself" signal self-reliance, meaning that the part only Laura must believe in (the ten payments) is funded by her alone and the part others join (the facility) is where co-sponsors come in?

- Anchor: R-C21, R-AN30, R-C48 ("Teams may not rely on co-sponsors..."), R-C44. New: the 2024-25 and 2025-26 pull quotes (VERIFIED-PRIMARY).
- Hypothesis (GUESS): yes. It turns the tension in R-AN30 into a design:
 - private conviction = the promise she funds herself, locked;
 - public proof = a range others can rely on.
 - In the TN this reads as "we secure what only Laura can promise".
- Would change (sentence): the IPS central idea (prompt 1, R-I9) and the order of the fundraising draft. Check the quote's attribution first (Q4).
- Deliverables: TN, IPS, FR. Domain: D7. Seed: 15.
- North Star: a plan built on her own stated principle, that the promise rests on her alone, is one she would see as hers.

Q4. Until the pull quote is traced to a primary source, how may our deliverables use it, given that Wharton's past cases printed pull quotes as the client's own words?

- Anchor: R-C21, R-AN30. New: the pull-quote pattern of 2022-23, 2024-25 and 2025-26 (VERIFIED-PRIMARY); my search found no primary source (section 1.4).
- Hypothesis (GUESS): it is probably hers, given the house pattern. Still, safe wording attributes it to the case ("the case profile's quote"), not to Laura, until D13 finds the source.
- Would change (sentence): the attribution wording wherever the quote appears. It avoids a credibility error in a case whose central risk is overpromising.
- Deliverables: TN, IPS, FR. Domain: D7. Seed: 15.
- North Star: getting her words exactly right is the smallest proof that we will get her money right.

Q5. Laura's one documented career "thoughtful risk" was taken after a book contract was secured: she left a "steady job in tech" when "she got the book deal". Is that the evidence for how she understands "thoughtful" risk (floor first, then leap), and may the IPS use it to justify "lock the promise, then take risk with the surplus"?

- Anchor: R-C37, R-AN16, R-C18. New: Inverse 2022 (VERIFIED-PRIMARY as article text; a journalist's paraphrase, not a quote from her).
- Hypothesis (GUESS): yes, as INTERPRETATION stated once and labelled. It shows the balance she asks for (R-C38) matches a choice she has already made in her own career. Do not over-read one anecdote. Do not use any private detail.
- Would change (sentence): the IPS risk-tolerance sentence, which should say her willingness is high and her capacity where the promise is concerned is near zero. It also gives a Laura-specific reason for the trades that build the Treasury core in the TN.
- Deliverables: TN, IPS, FR. Domain: D6. Seed: 33 (also 17).
- North Star: our structure is the one she has already used in her own career.

Q6. Laura interned at a trading firm (Belvedere Trading) and at a consumer-protection regulator (the CFPB). What would such a client reject on sight in a TN reflection or an IPS: an

unlabelled probability, a cost left out, or a product-sales tone?

- Anchor: R-C12, R-AN32, SH-04, BS-01, BS-02 (CFA Asset Manager Code). New: Poets&Quants 2018 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): all three.
 - Every probability must name its model, or be avoided in favour of "bought, not forecast" (brief 8.6).
 - Costs must appear: WInS commissions now (\$25 per stock/ETF trade, \$10 per bond trade, F-607); a fee assumption in the Final Report (BS-01).
 - The tone must be that of an adviser disclosing, not a seller.
 - This extends seed 18 with new evidence. The client is not only a statistics graduate but someone with trading and regulatory experience.
- Would change (decision): add a one-line cost and fee disclosure to the FR projections, with a Laura-specific reason, and apply the CFA Code's "truthful, accurate, complete, and understandable" disclosure standard (F.2, via A3). Keep "%" claims out of the TN unless their basis is shown.
- Deliverables: TN, IPS, FR. Domain: D8. Seed: 18.
- North Star: a client who has seen finance from the trading desk and the regulator trusts the firm that discloses first.

Tier 2: the IPS (Nov 6; the strategy freezes)

Q7. Laura taught financial literacy to high school students (MoneyThink), her books are "read in classrooms", and the case calls her an "educator". Should every key concept in our deliverables (the certainty definition, the 2031 range) come with a one-line plain-English version Laura could repeat to a co-sponsor?

- Anchor: R-C5, R-C16, R-S28 ("communicates ... clearly and credibly to prospective co-sponsors"), R-W37 ("learning the nuances of investing"). New: Poets&Quants 2018.
- Hypothesis (GUESS): yes. Wharton picked a client-educator in a program for high schoolers. The criterion asks for clarity to non-investors. A concept Laura could teach is one she can defend in 2031.
- Would change (decision): a D9 check that each technical term in the IPS has a plain version, and that the FR carries a "how Laura would say it" line for the certainty definition and the range. The students write these lines.
- Deliverables: IPS, FR. Domain: D9. Seed: 32.
- North Star: she chooses the plan she can explain to others herself.

Q8. Laura built a case competition around "what problem areas" the client "needed help with". Should our IPS and Final Report be organised one to one around the case's own problem list (the five tests, R-C80 to R-C84, and the IPS guide's five clarity items, R-I24 to R-I28)?

- Anchor: R-C79 to R-C84, R-I24 to R-I28, R-I29 ("You do not need to address these points separately"). New: Wharton Magazine 2017 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): for the Final Report, yes: one section or one table row per test, in the case's order. For the IPS, no separate headings are needed (R-I29), but each test should be answered by an identifiable sentence.
- Would change (decision): the FR table of contents, and the IPS paragraph checklist for Nov 6.
- Deliverables: IPS, FR. Domain: D9. Seed: null.
- North Star: a client who has designed a case reads for "did they answer my problems", and ours visibly does.

Q9. This is the first archived case (2017-2026) with explicit certainty, volatility and uncertainty language, and the 2026-27 Guide adds "management of risk and uncertainty". Is Wharton testing the gap between Laura's willingness to take risk and her capacity to take it, so that the IPS must state both explicitly?

- Anchor: R-C36 to R-C38, R-AN17, X-11, R-I2 ("risk tolerance"). New: archive word check and Guide p.5 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): yes. The IPS should give her risk tolerance in two parts:
 - willingness: high, as her career shows;
 - capacity: zero for the promise, moderate for the surplus.

It should then show that the equity weight follows from this, not from a generic age or horizon rule. This is partly open in brief section 9 (sleeve equity weight). What is new is evidence that this distinction is the season's designed test.

- Would change (sentence): the IPS risk sentence; the FR explanation of the sleeve weight.
- Deliverables: IPS, FR. Domain: D7. Seed: 18.
- North Star: separating what she is willing to risk from what she cannot afford to lose is the core of being understood.

Q10. The ladder cost more than \$300,000 on 173 of 185 trading days in 2026 and fell to \$300k or less only on 2026-09-10, the day the case PDF was created. At any flat 3.5%-5.0% rate it costs \$302k-\$350k. Did the designers mean the promise to use up nearly all of the money, and should our IPS treat the rule for falling rates (top up from the 2028 deposit) as the main path rather than a contingency?

- Anchor: R-AN13, F-111, F-112, F-110, case_register section 1.1 (PDF metadata). Brief section 9 already lists "Rule if rates fall" as open. What is NEW is the design-time calibration evidence.
- Hypothesis (GUESS): yes. At design time a full lock with the first deposit alone was probably not affordable. The designers probably expected the facility to be thin and to depend on growth. A plan whose surplus comes mainly from September 2026 yields must say so. If rates return to their 2026 average before January, it must still work with a written top-up rule. There is a ~24% chance the lock costs more than \$300k (F-112, ASSUMPTION).
- Would change (decision and sentence):
 - the IPS rule: if the ladder costs more than the first deposit, fund the longest payments first and fill the rest from the 2028 deposit, stated as the base rule;
 - an FR sentence: today's surplus partly reflects unusually high yields;
 - D1/D3 should re-run the surplus at the 2026 average curve.
- Deliverables: IPS, FR. Domain: D7 (numbers: D1, D3). Seed: null (extends seed 4).
- North Star: a plan that also works in the rate world the designers probably had in mind is robust, not lucky.

Q11. The 2020-21 case spelled out the link between the client's background and an investment choice ("Given his cosmopolitan upbringing ... geographic diversity"). Laura's case states no such link, even though the residency is in Taiwan. Should our deliverables refuse to turn her biography into investment choices (no Taiwan, China or "creative-economy" tilt justified by who she is), and use the case's own wording for her degree and roles?

- Anchor: R-C10, R-C12, R-AN34, R-W4 ("the financial scenario is developed specifically for the competition"). New: 2020-21 case text (VERIFIED-PRIMARY); the degree wording differs between Poets&Quants, Wharton Magazine and the case.
- Hypothesis (GUESS): yes. When Wharton wants a background-to-portfolio link, it writes one. The absence this year is a signal against tokenism. The degree is cited as the case gives it ("Statistics & Information Decisions Management"). Any Taiwan exposure must be justified by a cash-flow reason (NT\$ costs, D4), not by biography.
- Would change (decision): drop or re-justify any Taiwan/EWT tilt (brief section 9 open item). Fix the degree wording in all drafts.
- Deliverables: IPS, FR, WInS-now. Domain: D7. Seed: 19.
- North Star: respecting her story means not using it as a stock-picking signal.

Q12. Wharton's old guidance said "Don't use precious space repeating known facts about your client", and the 2023-24 midterm said "Don't just repeat the case study, show us that you have thought about how her personal circumstances will determine her financial needs". In a 500-word IPS, how many words go to Laura's background, and which beyond-the-case facts earn a place?

- Anchor: R-I18, R-I20, R-S25 ("demonstrates a thoughtful understanding of the client"). New: old judging page and 2023-24 deliverables page (VERIFIED-PRIMARY).
- Hypothesis (GUESS): at most one or two sentences, and only facts that change a decision:
 - her income is reputation-driven and uneven, so the 2028 deposit is uncertain (BS-03, BS-13);

- her quantitative training, so certainty is defined as bought, not modelled;
- "floor first" (Q5).

No biography retelling.

- Would change (number): a word budget of about 30-50 of the 500 words for client-specific reasoning, placed inside the rules rather than in a separate biography paragraph.
- Deliverables: IPS, FR. Domain: D9. Seed: 32.
- North Star: showing that we understand how her life shapes her needs beats reciting her life.

Q13. The 2023-24 summary page asked teams to "Explain what makes your investment strategy unique compared to other teams". That prompt has been dropped. Should the 50-word pitch contain any comparison with other approaches, or only the central idea and how it serves her goals (R-I17)?

- Anchor: R-I16, R-I17. New: 2023-24 deliverables page (VERIFIED-PRIMARY).
- Hypothesis (GUESS): no comparison. The pitch spends its 50 words on the promise, the rule and the balance. Any distinctiveness should be visible, not claimed.
- Would change (decision): a pitch checklist: the constraint, the central rule, the growth-versus-protection balance, and the link to her residency. Nothing about rivals.
- Deliverables: IPS. Domain: D9. Seed: 1.
- North Star: she chooses the plan that talks about her, not about other teams.

Q14. In 2023-24 the Final Report's audience was "WGAM's portfolio manager (your team's teacher/advisor)". This year the audience is unstated until Nov 9, but the case frames Laura as the chooser and the co-sponsors as the readers of the fundraising draft. Whose voice and pronouns should the IPS use, given that it freezes before the FR instructions appear?

- Anchor: R-C3, R-C8, R-C72, R-I34 ("official record"), R-AN4. New: 2023-24 deliverables page (VERIFIED-PRIMARY).
- Hypothesis (GUESS): write the IPS as the firm's policy for Laura ("the portfolio will ..."; "Laura" in the third person), with decisions made by the students' framework. The real advisor makes no decisions. This fits all likely FR audiences. D10 should re-check on Nov 9.
- Would change (sentence): IPS voice and pronouns; the FR address line.
- Deliverables: IPS, FR. Domain: D9. Seed: null (extends seed 2).
- North Star: a policy written for her, in a consistent voice, reads as a firm she could hire.

Q15. The case describes Laura as someone "turning ideas into opportunities" who "has no shortage of ideas", and it ties flexibility to the residency "as the project develops". Should our flexibility rule reserve the leftover money for the residency project, the thing she has chosen, rather than for her next idea?

- Anchor: R-C6, R-C23, R-C24, R-C58, R-C84, R-AN19.
- Hypothesis (GUESS): yes. The case links flexibility to "the project", and a client full of ideas benefits from a rule that protects the chosen project from new ones. Laura, not the firm, can override the rule, and the override is written down. This differs from B1a Q22, which asks how to *express* flexibility. This question asks what flexibility is *for*.
- Would change (decision): the purpose clause in the IPS flexibility rule; how the FR describes money left after 2033.
- Deliverables: IPS, FR. Domain: D6. Seed: null.
- North Star: a firm that guards her chosen project against her many other good ideas is the one an idea-rich founder needs.

Q16. Every value word in the 2026-27 case describes Laura's WORK (stories, identity, belonging, community, misinformation). None describes her money, whereas 2025-26 said "Every dollar invested reflects MTWB's values". Should her values show up in how we communicate (accuracy, no overpromising, plain words) rather than in security selection?

- Anchor: R-C9, R-C10, R-C14, R-AN33, R-AN34. New: the 2025-26 values sentence (VERIFIED-PRIMARY). B1a Q16 and B1b Q7 ask whether to *mention* the missing screen. This question asks where the values *go* instead.

- Hypothesis (GUESS): yes. Her first named work was "a response to misinformation" (R-C14). The value that carries into finance is accuracy, and it lands on the co-sponsor range and the certainty definition, not on an ESG fund.
- Would change (sentence): one FR sentence tying the honesty standard of the range to her own public work, labelled as INTERPRETATION. No screen in the IPS.
- Deliverables: IPS, FR. Domain: D7. Seed: 12.
- North Star: honouring her values means not overpromising, which is exactly what her work fights.

Tier 3: the Final Report (Dec 4)

Q17. Barwin's 2025-26 case quote ("sustainability of a nonprofit ... strong partnerships to keep the work going") reads like the brief for this year's design: ten funded operating years plus co-sponsors. Should the fundraising draft present Laura's operating commitment as the thing that makes the building sustainable, and invite co-sponsors as partners in the facility?

- Anchor: R-C41 ("reliable support for its early operations"), R-C44, R-AN6, BS-17. New: the 2025-26 pull quote (VERIFIED-PRIMARY). B1b Q15 asks whether to lead with operations. What is new is evidence that this is the designers' through-line from last season.
- Hypothesis (GUESS): yes. Co-sponsors will read funded operating years as proof that the facility will not become an unfunded building. That lowers the co-sponsor's risk before any facility dollar is named.
- Would change (sentence): the order and framing of the FR fundraising draft (operations first, then the facility range, then the invitation to partner). The students write it.
- Deliverables: FR. Domain: D5. Seed: null (extends seed 6).
- North Star: we turn her private promise into the public proof co-sponsors need.

Q18. Past clients praised teams that made them feel "seen" (Jordan) and that went beyond "just ... the numbers" (Ayoola). But the semifinal readers are Wharton student evaluators and Aberdeen asset managers, not Laura. What written evidence shows those readers that the plan "can earn her confidence"?

- Anchor: R-S25, SH-16, R-W41. New: 2022 and 2025 finale articles (VERIFIED-PRIMARY).
- Hypothesis (GUESS): traceability. Each major decision cites the case line or public professional fact that drove it, in a small "decision → why it is hers" table in the FR. Flattery, identity references and biography do not count.
- Would change (decision): add a traceability table (FR only; the IPS bans graphics and tables are risky there) and check every row against the case.
- Deliverables: FR. Domain: D7. Seed: 32.
- North Star: a reader who is not Laura can see, row by row, why each choice is hers.

Q19. Laura's professional record shows her building platforms and communities: she founded The Sign.al and organised a city-backed case competition. Should one line of the fundraising draft use this track record as evidence that the residency will be run well, given the case says a contribution signals "commitment" and "viability"?

- Anchor: R-C64 ("signal that the residency is financially viable, demonstrate Laura's commitment"), R-C72, SH-05. New: Poets&Quants 2018 and Wharton Magazine 2017 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): perhaps one clause at most, and only if the team's draft needs it. The case asks teams to draft the part "describing this potential contribution", so money comes first. A track-record line risks padding and must not overstate (these were student projects).
- Would change (sentence): at most one clause in the fundraising draft; otherwise leave it out.
- Deliverables: FR. Domain: D5. Seed: null.
- North Star: co-sponsors back founders with a record, and her record is building places where people explore their paths.

Q20. Laura kept a daily "comics diary" as a product manager so she could "take pride in how far I'd come when I reread it" (Wharton Magazine 2022). Does that give a Laura-specific reason for our dated decision log and the FR Articulation section, or is it decoration?

- Anchor: R-S27 (Articulation), R-T6, R-W82 (notes "at the time it is made"). New: Wharton Magazine 2022 (VERIFIED-PRIMARY).
- Hypothesis (GUESS): the parallel is real but easy to overdo. Keep the dated decision log because the rules reward it. Mention her habit only if a student honestly finds it meaningful. Otherwise leave it out.
- Would change (decision): keep a dated team decision diary from now on (WInS-now). The FR mention is optional.
- Deliverables: WInS-now, FR. Domain: D6. Seed: 17.
- North Star: a record of how our thinking changed is something she values in her own work.

Q21. The case opens "Laura Gao believes stories have the power to change how people see the world", which is house style ("Connor Barwin ... believes in the power of community"), and her sidebar lists "Storyteller" first. Does criterion 5's "compelling ... narrative" call for a literary device, or for the plan's own story told on a timeline from 2027 to 2042?

- Anchor: R-C9, R-C20, R-S28. New: the 2025-26 opener (VERIFIED-PRIMARY) shows the "believes" line is a template.
- Hypothesis (GUESS): the plan's story: what happens in 2027, 2028, 2031, 2033 and 2042, and under good and bad markets, shown as one timeline chart in the FR. Book metaphors ("messy roots") are what rivals will use (B7b). Do not over-read the opener.
- Would change (decision): FR structure built around one timeline visual; no book-title metaphors in headings.
- Deliverables: FR. Domain: D9. Seed: null.
- North Star: the story she cares about is her residency's future, told clearly.

Q22. Should the team watch Wharton Global Youth and Wharton Magazine for a new Laura interview or a Wharton piece on the residency theme before Dec 4, and set a rule now for using anything that appears after the Nov 6 freeze?

- Anchor: R-I35 ("may not revise its investment strategy after the submission deadline"), R-new (Guide p.5, "you may not redesign your strategy after observing the results"). New: no Laura article exists yet (REST search, 2026-09-27); the Ash interview appeared on 2024-01-02.
- Hypothesis (GUESS): an interview is unlikely before Dec 4, going by past timing. If one appears, it may inform FR sentences about communication but cannot change the frozen strategy.
- Would change (decision): a monthly check owner and a one-line usage rule.
- Deliverables: FR. Domain: D7. Seed: null.
- North Star: using her newest public words, correctly and within the rules, keeps the plan current and hers.

3. Working arithmetic behind Q10 (ASSUMPTION: flat annual rates; not a curve valuation)

Ten \$50,000 payments at the start of 2033-2042, valued at the start of 2033 and discounted six years to 2027:

Flat rate	Value at 2033	Value at 2027	Shortfall under \$300k, carried to 2028
3.50%	\$430,384	\$350,118	\$51,872
4.00% (JPM intermediate Treasuries, F-307)	\$421,767	\$333,328	\$34,661
4.50%	\$413,440	\$317,478	\$18,265
5.00%	\$405,391	\$302,509	\$2,635
5.36% (flat equivalent of 25 Sep curve, F-109)	\$399,762	\$292,245	\$0

Consistent with F-110 (value at 2033) and F-111 (curve-based 2026 history: Jan 2 \$316.5k, Jun 30 \$313.0k, Sep 1 \$302.4k, below \$300k only from Sep 10). Computed 2026-09-27 with the repo Python.

4. Leads for the specialists

- **D7 (Wharton intent historian):**

- Use the REST interface https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=<slug>&_fields=content to read hidden page sections. Slugs used: `case-study` (holds the full 2025-26 case), `judging-and-evaluation` (old criteria), `deliverables` (2023-24 and 2022-23 deliverables, including "Final Report Audience"), `case-study-for-2024-2025`, `case-study-for-2022-2023`, `case-study-for-2021-2022`, `previous-case-study-20XX-20XX`.
- The 2023-24 case text is still missing. Try other slugs, or the Reviewr/SMApply archives.
- Add the 2025-26 case and the old judging page to the repo's historical folder, with the retrieval method (main loop's decision).
- **D13 / B8 (quotes):**
 - The pull quote is still unsourced. Next places to try: her newsletter or Substack, speaking-bureau pages, podcast transcripts (NüVoices #72, The Maverick Show #190, First Principles), and her 2024 book's interviews.
 - Wharton Magazine 2022 and Poets&Quants 2018 hold verified professional quotes (section 1.4).
 - The Poets&Quants HTML page is Cloudflare-blocked (403). Use <https://poetsandquantsforundergrads.com/wp-json/wp/v2/posts/11618>.
- **D6 (client psychology):** Inverse 2022 ("floor first, then leap") and the Poets&Quants "risk-adverse ... certainty" advice line. Both are professional-record items. Keep the private sentences out.
- **D8 (practice benchmark):** the CFA Asset Manager Code's disclosure standard (A3, BS-02), plus Laura's CFPB and trading-firm internships (Poets&Quants), give a Laura-specific reason for fee and cost disclosure.
- **D1 / D3:** re-run the lock-early surplus at the 2026-average curve (and at the Jun 30 curve, \$313.0k) to show the plan with a partial lock plus a 2028 top-up (Q10). F-111 has the daily curve history.
- **D5 (co-sponsors):** the Barwin quote as the designers' through-line (Q17). Track-record evidence (Q19): The Sign.al and the Amazon HQ2 case competition (Wharton Magazine 2017).
- **D9 (communication):**
 - old judging lines "Excessive investing jargon doesn't necessarily dazzle the judges" and "Don't use precious space repeating known facts about your client";
 - the 2023-24 prompt "Don't just repeat the case study";
 - the IPS word budget (Q12) and the pitch checklist (Q13).
- **D10 (compliance):** the 2023-24 AI citation rule ("include 1-2 sentences on how you used that AI tool") and "judges are instructed not to open any links". These are the likely shape of this year's Final Report rules (check on Nov 9). Use the case's degree wording.

5. Sources (all accessed 2026-09-27)

Source	URL	Date of piece	Status
2025-26 case (hidden section of current case-study page)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study (page https://globalyouth.wharton.upenn.edu/competitions/investment-competition/case-study/)	page modified 2026-05-07	VERIFIED-PRIMARY
Old Judging and Evaluation page	https://globalyouth.wharton.upenn.edu/investment-competition/judging-and-evaluation/ (via REST, slug <code>judging-and-evaluation</code>)	modified 2024-06-03	VERIFIED-PRIMARY (historical)
Old Deliverables page (2023-24, 2022-23)	https://globalyouth.wharton.upenn.edu/investment-competition/deliverables/ (via REST)	modified 2024-05-30	VERIFIED-PRIMARY (historical)
2024-25 case	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/	2025-06-26	VERIFIED-PRIMARY
2022-23 case	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2022-2023/	2023-05-18	VERIFIED-PRIMARY
2021-22, 2020-21, 2019-20, 2018-19, 2017-18 cases	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/ (slugs in section 4)	2021-2022	VERIFIED-PRIMARY
Main competition page (current and 2024 versions)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/ ; https://globalyouth.wharton.upenn.edu/investment-competition/	2026-09-11 / 2024-07-11	VERIFIED-PRIMARY

Source	URL	Date of piece	Status
2026-27 Competition Guide	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf (linked from https://wghsinvcomp.smapply.us/res/p/guide/)	created 2026-09-09	VERIFIED-PRIMARY
SMAApply client page	https://wghsinvcomp.smapply.us/res/p/client/	live	VERIFIED-PRIMARY
2022 finale (Jordan "seen")	https://globalyouth.wharton.upenn.edu/news/top-teams-sail-to-success-in-the-2022-wharton-investment-competition-global-finale/	2022-04-28	VERIFIED-PRIMARY
2023 finale	https://globalyouth.wharton.upenn.edu/news/the-2023-investment-competition-global-finale-ends-in-sweet-victory-for-dmvs-finest/	2023-04-26	VERIFIED-PRIMARY
2024 finale (Ash)	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024-investment-competition-global-finale/	2024-04-23	VERIFIED-PRIMARY
2025 finale (Ayoola)	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/	2025-05-01	VERIFIED-PRIMARY
2026 finale (Barwin)	https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	2026-04-30	VERIFIED-PRIMARY
6 Questions for Hilary Ash	https://globalyouth.wharton.upenn.edu/articles/sports-and-media/6-questions-for-hilary-ash-of-the-los-angeles-2028-olympic-and-paralympic-games/	2024-01-02	VERIFIED-PRIMARY
Global Youth post search ("Gao"; recent posts)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/posts?search=Gao	-	VERIFIED-PRIMARY (absence)
Wharton Magazine, "Wharton's Pitch to Amazon"	https://magazine.wharton.upenn.edu/digital/whartons-pitch-to-amazon/	2017-10-30	VERIFIED-PRIMARY (quote checked with --grep)
Wharton Magazine, "A Wharton Grad From Wuhan's Exploration of Identity"	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	2022-04-06	VERIFIED-PRIMARY
Poets&Quants for Undergrads, 2018 Best & Brightest: Laura Gao	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/ (HTML 403; read via https://poetsandquantsforundergrads.com/wp-json/wp/v2/posts/11618)	2018-03-30	VERIFIED-PRIMARY (publisher API)
Inverse, Messy Roots feature	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel	2022-03-07	VERIFIED-PRIMARY (article text; journalist's paraphrase)
Pull quote source search	web searches + about 10 interview pages	-	not found; PARAPHRASE-UNVERIFIED as Laura's words
Wayback Machine	web.archive.org	-	connection reset through proxy; not used

Repo files used: `research/insight_v1/_context/brief.md`; `research/ultracode_prompt_v4_deep_strategy.md`
Parts 1-3, 5-6; `research/insight_v1/phase_A/case_register.md`, `fact_register.md`, `stakeholder_map.md`, `wins_week1_guardrails.md`; `research/insight_v1/phase_B/B1a_*.md`, `B1b_*.md`, `B7a_*.md`, `B7b_*.md` (to avoid duplicates); `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (VERIFIED-REPO-FILE).

What this teaches

- 1. Compare this year's words with last year's.** One word can carry the whole season's intent: "shape the *winning* strategy" became "develop the strategy", and "win him/her over" became "earn her confidence". You only see it by putting the two versions side by side. The old versions were still on Wharton's own server, and the site's public data feed showed them.
- 2. Separate house style from signal.** "[Client] believes ..." opens every recent case, so it is template, not a clue. A risk sentence that appears for the first time in nine seasons is a clue.

3. **Date your evidence.** The case numbers were fixed when rates were lower. A strategy that only works at today's rates is lucky, not robust, and a careful client (or judge) will notice.
4. **Use a real person's record with restraint.** Her public professional facts (trading and regulator internships, teaching high schoolers about money, designing a case competition) explain how she will read our work. Her private life does not belong in an investment plan.

B3a Laura's World & Human Capital: questions from her professional record and the risks to the 2028 deposit

Agent B3a, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: "Laura's World & Human Capital". Starting angle: her PUBLIC PROFESSIONAL record (checked on primary pages), and each income source the case names for the 2028 deposit ("publishing advances, speaking engagements, licensing, and other entrepreneurial ventures", R-C27), mapped to industry evidence on how volatile it is and whether it falls when markets fall (wrong-way risk).

This file is AI-generated research for Team Caplet, for brainstorming only. It holds questions, evidence and checklists, not text for submission. The six students decide and write every deliverable in their own words, and any AI use must be recorded in the Final Report's Works Cited (R-W46). Laura is covered only through the case and her public professional record. The team's two client documents contain personal-life details (a birth name, family members, a partner, a home city). I did not copy any of them here. Wharton says "While the client is real, the financial scenario is developed specifically for the competition" (R-AN56), so nothing here estimates her real income or wealth; industry data is used only as base rates for stress tests.

Status labels (brief section 3): VERIFIED-PRIMARY (VP: I read it on the primary page, 2026-09-27), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SU), ASSUMPTION (ASM), PARAPHRASE-UNVERIFIED. INTERPRETATION marks my reading, never a fact. Every "hypothesis" is a GUESS. Web searches ran out (the session limit of 200 was reached) partway through, so some leads are left for Phase D.

Plain-English terms used below

- **Human capital:** the value of a person's future earnings. Laura's books, talks and licences are her human capital.
- **Wrong-way risk:** two bad things tending to happen together. Here: her 2028 deposit shrinking in the same bad market that shrinks the portfolio.
- **Correlation (rho, ρ):** a number from -1 to +1 saying how closely two things move together. 0 means unrelated.
- **Advance:** money a publisher pays an author before a book earns royalties, usually in instalments.
- **Idiosyncratic risk:** risk specific to one person or book (a deal comes late) and not to the whole market.
- **Growth sleeve:** the part of the portfolio left after the Treasury ladder is bought; the only part that takes market risk.

0. Summary (read this first)

1. **For the facility, Laura's 2028 income matters about as much as five years of markets (NEW number).** From the verified simulation (F-401, F-403, F-404): every \$1 of the 2028 deposit adds about **\$1.31** to the median 2033 surplus (207 to 109 to 11 thousand dollars as the deposit goes 150 to 75 to 0 thousand). A 50% smaller deposit moves the median by **\$98k**. The whole market spread (p5 to p95) at a full deposit is **\$114k**. Under lock-early the ten payments do not depend on the deposit at all. INTERPRETATION: "her savings buy the promise; her earnings fund the building" is a true description of the plan, and a Laura-specific one (Q1).
2. **By 2031 the deposit is known.** The 2031 co-sponsor range is set three years after the deposit arrives. So the deposit's uncertainty belongs in today's projections (IPS/Final Report), not in the 2031 range (Q2).
3. **The evidence says her deposit is only weakly tied to the stock market, and mostly through timing.**
 - U.S. book sales fell only **1.8% in 2009** (AAP via Publishers Weekly, VP).
 - Comics and graphic novel sales rose from **\$1.1 billion (2019) to \$2.01 billion (2022)**, straight through the 2020 crash, and reached **\$2.2 billion in 2025** (ICv2, VP).
 - A small publisher said "the effect of the success (or failure) of just a couple of books will far outweigh macroeconomic effects" (quoted by T. McIlroy, 2019, VP on that page).

- The sharp exception: in November 2008 Houghton Mifflin Harcourt "temporarily stopped acquiring manuscripts" (SU). A crash can freeze NEW deals. It does not cancel money already contracted.
- Her own career: the only market crash in it (2020) is the year her comic went viral and HarperCollins bought two books "at auction" (Publishers Weekly, July 2020, VP) (Q3, Q12).

4. The recent shocks to her channels were political, not market-driven.

- Her speaking guide sells to "Companies and DEI Organizations" (it lists Asana, Adobe and a Google LGBTQ+ group) as well as schools and libraries (VP).
- In 2025, U.S. corporate diversity programmes were cut back (SU).
- The federal library agency (IMLS) had its grants terminated; they were reinstated after a November 2025 court ruling (ALA, VP).
- Book challenges hit 4,235 titles in 2025 (ALA, VP via A3).
- These shocks are real, but they are not tied to the S&P 500. That makes the deposit risk mostly *diversifying*, and is one more reason the lock must not depend on the deposit (Q7, Q8).

5. Generative AI cuts both ways for her.

- 26% of illustrators surveyed had lost work to generative AI, and 37% said their income had fallen in value (Society of Authors, 2024, VP).
- Her own publisher offered \$5,000 per nonfiction backlist title for an AI-training licence, split 50/50 with the author, opt-in (Authors Guild, VP).
- Her studio has designed for an "AI-powered online learning software" start-up (her resume, VP).
- So "AI threatens her income" is half the story. Her personal stance is not on public record; do not assume one (Q9).

6. The team docs' "underweight publishing/media" idea probably changes nothing, or the wrong thing. Publishers are a tiny index weight. The big "media" names are AI platforms, the only part of the index that might offset her AI risk. D2 to check (Q10).

7. Profile corrections for the deliverables (primary-checked):

- "Bestselling" is the case's word. Her resume says Messy Roots "Debuted on #9 on Indies Bestsellers List".
- Her awards are mostly **finalist** honours ("Finalist for Harvey Award, Goodreads Choice Awards, and California Booksellers Award"). Say "finalist", not "winner", unless naming an award her own guide lists as won.
- CCA is not "acquired by Vanderbilt". Vanderbilt is to "acquire the campus of CCA after the wind-down of CCA's operations in 2027".
- Wikipedia dates "The Wuhan I Know" to 2021, which is wrong against the case (2020) (Q13).

8. The joint tail in brief section 9 (rates fall AND the deposit is missing) looks small, because only "smaller" hurts the ladder, not "late". At -100bp the ladder gap is about \$22.9k (F-101 family). A payment is left unfunded only if the deposit is cut by more than about 85% (to below about \$23k). A deposit that arrives up to a year late still lands four years before the first unfunded 2033 rung. D3 to confirm (Q5).

20 questions follow (section 2), ranked roughly by deadline and impact.

1. Evidence base: what I verified

1.1 Her public professional record (checked against the team docs)

#	Claim (team docs or case)	What the primary source says	Source	Status
E1	Wharton, Statistics & Information Decisions	Resume: "Bachelor of Science in Statistics and Information Decisions, Concentration in Product Design", summa cum laude	https://lauragao.com/s/Laura-Gao-Resume.pdf	VP
E2	Tech PM 2018-2020	"Twitter Product Manager 2018 – 2020", Abuse & Safety and Growth ML	resume	VP
E3	Data analyst at Amazon	"Amazon.com Data Analyst 2017"	resume	VP

#	Claim (team docs or case)	What the primary source says	Source	Status
E4	Trading and CFPB internships (team docs)	Poets&Quants 2018 profile lists "Trading Intern at Belvedere Trading in Chicago" and "Policy Analyst Intern at Consumer Financial Protection Bureau in Washington, D.C."	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/ (raw HTML; the text helper fails on this page)	VP (raw HTML)
E5	"LG Studios" freelance design	"LG Studios Freelance Product Designer, Author & Illustrator 2020 – Present"; clients include "Retainit, AI-powered online learning software" and a 3D-printed mask start-up	resume	VP
E6	HarperCollins two-book deal (2020)	"Donna Bray at HarperCollins/Balzer + Bray has acquired, at auction, world English rights to debut YA graphic memoir The Wuhan I Know, plus a second graphic work, by Laura Gao." First book "planned for summer 2021"	Publishers Weekly Rights Report, week of July 20, 2020	VP
E7	"Bestselling"	Resume: "Debuted on #9 on Indies Bestsellers List". Speaking guide: "the bestselling graphic memoir, Messy Roots"	resume; speaking guide PDF	VP
E8	Awards	Resume: "Finalist for Harvey Award, Goodreads Choice Awards, and California Booksellers Award". Bio page: "A Harvey and Goodreads Choice Award honoree". The speaking guide also lists "GOLDEN POPPY AWARD WINNER", "CYBIL AWARD WINNER", "NYPL BEST BOOKS OF 2022", and state reader-choice nominations (Texas Maverick List finalist, Green Mountain, Black-Eyed Susan)	resume; https://lauragao.com/hey/ ; speaking guide	VP
E9	Professor at CCA; festival co-organiser	Bio: "shapes the next generation of creators as a Professor of Comics at California College of the Arts and co-organizes San Francisco's LGBTQ+ comics festival, Pride in Panels"	https://lauragao.com/hey/	VP
E10	"CCA acquired by Vanderbilt, completing 2027" (team docs)	Vanderbilt: "Vanderbilt and the California College of the Arts have agreed for Vanderbilt to acquire the campus of CCA after the wind-down of CCA's operations in 2027"; CCA "intends to continue instruction and operations through the 2026-27 academic year". The Art Newspaper: CCA "will close permanently in 2027", with a "\$20m deficit"	https://news.vanderbilt.edu/2026/01/13/vanderbilt-university-to-establish-full-time-academic-campus-in-san-francisco/ ; https://www.theartnewspaper.com/2026/01/13/california-college-arts-closing-vanderbilt-university-takeover	VP. The team docs' wording is inexact: the college closes; Vanderbilt buys the campus
E11	Speaking market segments	Guide headings: "Schools, Libraries and Bookstores"; "Festivals and Trade Conferences"; "Companies and DEI Organizations" (previous talks: "APAPI Author Talk with Asana, Doodle Therapy with Adobe, Drawing Your Life Workshop with Google LGBTQ+ Group"). "Rates will differ based on travel, length of event, and books purchased. Pro-bono visits can be offered on need-basis." One popular topic: "Taking the Leap to Pursue Art Full-Time from a Traditional Tech Job"	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	VP
E12	Messy Roots collaboration	Wikipedia: "written and illustrated by Laura Gao, with coloring by Weiwei Xu". The guide asks: "What's the process of collaboration with two artists like?"	https://en.wikipedia.org/wiki/Messy_Roots ; guide	VP (Wikipedia is secondary)
E13	China market	Wikipedia: "According to Gao, the book will not likely be for sale in Mainland China due to the LGBTQ themes." (cites NPR 2022)	https://en.wikipedia.org/wiki/Messy_Roots	VP on Wikipedia; SU for the underlying NPR words
E14	Date of "The Wuhan I Know"	Case: 2020 (R-C13). Wikipedia's Messy Roots page says "In 2021, while working for Twitter...", which conflicts with the case and with the July 2020 PW deal	Wikipedia; PW	Case wins. Do not cite Wikipedia for dates
E15	Book challenge of Messy Roots	A YouTube title reads "Interview: Laura Gao on Messy Roots book ban and anti-LGBTQ backlash"	search snippet only	SU. Do not claim her book was banned

#	Claim (team docs or case)	What the primary source says	Source	Status
E16	Festival attendance figures ("2,000+", "125+ exhibitors")	not re-checked by me	team docs	SU (the brief already flags these)

1.2 Industry evidence for each named income source

Source (case words)	What drives it	Evidence	Status	Tie to markets (INTERPRETATION)
"publishing advances"	New contracts; paid in instalments on signing, delivery and publication	Authors Guild (2021, via A3, W-8): advances paid "in three, and even up to six, installments"; "The sales numbers of one book can affect the ability to sell the next project"	VP (A3)	Money already contracted is not cut by a crash. NEW deals can freeze: HMH "temporarily stopped acquiring manuscripts" in Nov 2008 (SU)
(book sales behind royalties and future advances)	Consumer and library demand	AAP: "Book sales fell 1.8% in 2009"; adult hardcover had declined 13% in 2008 (PW, VP). FRED bookstore sales (MRTSSM451211USN): 2008 -2.1%, 2009 -6.0%, 2020 -31.0%, 2021 +29.7% (my calculation from the live FRED CSV; the store channel also has a long structural decline). ICv2: comics and graphic novels \$1.1bn (2019), \$2.01bn (2022), \$1.87bn (2023, -7%), \$2.2bn (2025, a new high)	VP	Low beta (sensitivity to the stock market); the graphic-novel category grew through the 2020 crash
(author income level)	Very skewed	Authors Guild 2023 survey: median full-time author book income \$10,000 (2022); graphic novelists' median book income \$15,000, combined author income \$25,000, second only to romance	VP	Says nothing about Laura (fiction scenario); shows the case's \$150k in one year is far in the right tail of author incomes
"speaking engagements"	Event budgets: schools and libraries (public budgets), festivals, corporations	2009: "cancelled contracts and clients whose budgets suddenly included no money for speakers", but publisher speaker bureaus were "surprisingly recession-resilient", with fees of "\$5,000 and \$20,000" split with authors (Publishing Perspectives 2009, via snippet; the page did not load)	SU	Corporate segment: high beta plus 2025 political (DEI) risk. School and library segment: public budgets that lag recessions (ASM)
(library channel)	Federal and state funding; challenges	IMLS grants terminated in 2025, then reinstated "as a direct result of a November 21 federal court decision" (ALA, Dec 2025, VP); DOJ-ALA settlement April 2026 (SU); 4,235 titles challenged in 2025 (ALA, VP via A3)	VP/SU	Political, not market-driven
"licensing"	Translation rights, AI-training licences, film/TV options, art/print licences	HarperCollins sold only "world English rights" (PW, VP), so translation rights sit with the author's side (INTERPRETATION of standard practice). AI: "\$5,000 fee per title, which will be split 50-50 between an author who chooses to participate and HarperCollins" (Authors Guild, VP; applies to nonfiction backlist per snippets)	VP/SU	AI licences are a new, market-linked income; film/TV options are cyclical (not verified here)
"other entrepreneurial ventures"	Freelance product design and illustration (LG Studios), prints, workshops	Resume (VP). Society of Authors 2024: "A quarter of illustrators (26%)... have already lost work due to generative AI"; 37% said income decreased in value (EWC summary, VP)	VP	Freelance tech design follows start-up funding (high beta, ASM) and is the most AI-exposed

1.3 Numbers I derived from verified files (not new simulations)

Item	Value	Derivation	Status
Median 2033 surplus per \$1 of 2028 deposit	≈ \$1.31	$(207 - 109)/75 = (109 - 11)/75 = 1.307$, using F-401, F-403, F-404	VRF inputs, derived
Median shift if the deposit is 50% smaller	-\$98k	207 - 109	derived
Market spread p5 to p95 at a full deposit	\$114k	273 - 159 (F-401)	derived

Item	Value	Derivation	Status
Ladder gap if rates fall 100bp before Jan 2027	≈ \$22.9k	\$322,864 - \$300,000 (F-101 family, brief section 6)	derived
Ladder gap if rates fall 50bp	≈ \$7.1k	\$307,135 - \$300,000	derived

2. Questions (20), with reasoning

Ranking: tier 1 (WInS now, Trading Notes) first, then IPS, then Final Report.

Q1. How big is the 2028 deposit risk compared with market risk for the facility range, and should the Final Report show the facility range as a table by deposit size?

- Anchors: R-C27, R-C77, R-C70, F-401, F-403, F-404.
- Evidence: section 1.3. Every \$1 of deposit ≈ \$1.31 of 2033 median surplus; a 50% cut moves the median by \$98k, close to the whole p5-p95 market spread (\$114k).
- Hypothesis (guess): yes. Her own earnings are as big a driver of the facility as markets. A three-row table (deposit \$150k / \$75k / \$0, each with p5/p50/p95) is simpler and more honest than folding both risks into one probability.
- Changes: a number and a chart (FR: facility range by deposit size); a sentence (IPS: "the promise does not depend on the 2028 contribution").
- Deliverables: IPS, FR. Domain: D3. Seed: 16.
- North star: it shows we understood where her plan's risk really sits: in her own creative career, not only in the stock market.

Q2. Because the deposit is known by 2031, should the 2031 co-sponsor range ignore deposit risk and show only 2031-33 market risk on the portfolio she actually has then?

- Anchors: R-C62, R-C65, R-C69, R-C71.
- Evidence: the case timeline (2028 deposit, 2031 range; VRF).
- Hypothesis (guess): yes. The FR should give the 2031 rule ("floor bought + share of the sleeve"), not a 2026 estimate of the 2031 numbers. Deposit uncertainty only widens today's projection.
- Changes: a decision (how the FR frames the 2031 range) and a sentence.
- Deliverables: FR. Domain: D3. Seed: 16.
- North star: a statistics graduate will notice whether we mix risks that will already be settled by 2031.

Q3. What correlation between the 2028 deposit and 2027 equity returns should D3 assume, source by source?

- Anchors: R-C27, R-AN31, R-AN35, brief section 11 ("2028 deposit independent of markets").
- Evidence: section 1.2. Contracted advances are not cut by a crash; new deals can freeze (HMH 2008, SU). Book sales fell only 1.8% in 2009 (VP). Graphic novels grew through 2020 (VP). Corporate speaking and freelance design are the most cyclical parts (SU/ASM).
- Hypothesis (guess): overall ρ about +0.2 to +0.3, with idiosyncratic timing risk dominating. Test $\rho = 0, 0.3, 0.5$ and show the facility median barely moves (under lock-early the payments never move).
- Changes: a number (ρ in the model) and a sentence in the FR risk section.
- Deliverables: FR. Domain: D3 (with D6). Seed: 16.
- North star: replaces a generic "wrong-way risk" worry with evidence from her own industries.

Q4. Against industry base rates, how should the team treat the case's firm "will contribute \$150,000" without speculating about her real finances?

- Anchors: R-C27, R-AN35, R-AN56.
- Evidence: Authors Guild 2023: graphic novelists' median combined author income \$25,000 (VP). \$150k saved from one year's earnings, after living costs covered elsewhere, is far in the right tail. The case chose a "bestselling" client for whom that is plausible.
- Hypothesis (guess): keep "will" as the base case. Use industry skew only to justify a labelled stress case. Never write anything about her actual advances or income.
- Changes: a sentence (FR assumptions list) and a compliance decision (no real-finance inference).

- Deliverables: FR. Domain: D10 (with D6). Seed: 16.
- North star: respects both the case and her privacy, which is how a professional firm behaves.

Q5. If rates fall before January 2027 and the ladder costs more than \$300k, how likely is it that a late or smaller deposit leaves a payment unfunded (the brief section 9 joint tail)?

- Anchors: F-111, F-112, F-107, R-C27, R-C31.
- Evidence: gap $\approx$ \$7.1k at -50bp and $\approx$ \$22.9k at -100bp (section 1.3). $P(\text{cost} > \$300\text{k}) \approx 24\%$ (F-112, ASM).
- Hypothesis (guess): very small. "Late" does not hurt, because a deposit up to a year late is still four years before the unfunded 2033 rung (only about a year of extra rate risk on a $\sim$ \$23k gap). Only a cut of more than $\sim 85\%$ leaves a hole. So $P(\text{joint tail}) \approx P(\text{rates} -100\text{bp}) \times P(\text{deposit} < \sim \$23\text{k})$, which is well under 1% on any reasonable inputs (ASM).
- Changes: a number (joint-tail probability in the FR) and a sentence (IPS contingency).
- Deliverables: IPS, FR. Domain: D3 (with D1). Seed: 16.
- North star: answers the one open hole in "high certainty" with arithmetic she can check.

Q6. WInS holds only the \$300k (the \$150k "will not be added to WInS", R-W59). Should the WInS book mirror the 2027 book (ladder $\approx 97\%$, small sleeve) rather than the post-2028 target, with the human-capital logic as the reason?

- Anchors: R-W56, R-W59, R-C75, F-409, F-605.
- Evidence: the WInS balance equals her savings deposit, not her earnings deposit (VP). F-409: "65/35" is ladder/sleeve, not bonds/equity.
- Hypothesis (guess): mostly the 2027 book, with a small equity sleeve that the Trading Notes explain: "her savings secure the promise; growth waits for her 2028 earnings". Risk: a near all-Treasury book gives fewer decisions to analyse (Portfolio Analysis criterion). The team decides.
- Changes: a decision (WInS target weights now) and the Trading Note rationale.
- Deliverables: WInS-now, TN, IPS. Domain: D7 (with D10). Seed: 16.
- North star: every WInS weight traces to where Laura's money actually comes from.

Q7. Were the 2025-26 shocks to her speaking market political (corporate DEI rollbacks, library funding) rather than market-driven, and does that make the deposit risk diversifying rather than wrong-way?

- Anchors: R-C27, R-AN31, SH-22.
- Evidence: her guide's "Companies and DEI Organizations" segment (VP). Corporate Pride and heritage-month cutbacks in 2025 (SU: Axios 403; Advocate snippet "39%" of executives plan to reduce Pride engagement). IMLS terminations and reinstatement (VP).
- Hypothesis (guess): yes. Recent risk to her channels came from policy, not the market, while U.S. equities rose (the 2025 index return is UNVERIFIED; D2 to check).
- Changes: a sentence (FR human-capital paragraph); an input to Q3's p.
- Deliverables: FR. Domain: D6. Seed: 16.
- North star: shows we know her actual markets (schools, libraries, companies), not a textbook "author".

Q8. Is the school and library channel the largest identifiable risk to her publishing income, and how can the FR say so in one clause without claiming her book was banned?

- Anchors: R-C16 ("read in classrooms around the world"), SH-23, BS-14.
- Evidence: state reader-choice nominations and a guide section for "Schools, Libraries and Bookstores" (VP). 4,235 titles challenged in 2025 (VP via A3). IMLS grant turmoil (VP). Messy Roots challenge: SU only (E15).
- Hypothesis (guess): yes, as a sector risk. One clause in the FR, framed as a channel risk; never a reason to pick or avoid any investment.
- Changes: a sentence (FR).
- Deliverables: FR. Domain: D6 (with D10). Seed: 19.
- North star: understanding her world without tokenism.

Q9. Should the FR describe her exposure to generative AI as two-sided (lost illustration work vs new AI-licensing income) and leave any AI tilt in the portfolio as her own choice?

- Anchors: R-C27 ("licensing", "other entrepreneurial ventures"), R-AN34 ("The case does not mention technology or AI risk to her income").
- Evidence: SoA 2024 26%/37% (VP). HarperCollins AI licence, \$5,000 per title split 50/50 (VP). Her studio designed for an AI-powered start-up (VP). No public statement by her on AI (B8b L-8).
- Hypothesis (guess): yes. A broad index already carries large AI weight. State the exposure in one sentence, offer no tilt, and never state her view.
- Changes: a sentence (FR) and a decision (no AI tilt or screen).
- Deliverables: FR, IPS (one clause at most). Domain: D2 (with D6). Seed: 20.
- North star: she is a data-literate creator; she would reject both "AI is doom" and "buy AI to hedge" without evidence.

Q10. Does "underweight publishing, media and education" (team docs) change anything in a broad-index portfolio, or does it mainly cut AI platforms?

- Anchors: R-C38 (balance growth vs protection), design principle (brief section 0), R-S26.
- Evidence: publishers are small or absent in large-cap indexes (ASM). The "communication services" sector is dominated by large internet platforms (ASM; D2 to check sector weights on an issuer fact sheet).
- Hypothesis (guess): the tilt is a no-op or counterproductive. Drop it and explain in one FR sentence: her human capital is a reason for the promise to be locked, not for sector bets.
- Changes: a decision (no sector tilt) and a sentence.
- Deliverables: WInS-now, FR. Domain: D2. Seed: 20.
- North star: complexity that does not earn its place is removed, and we say why.

Q11. Is the U.S.-China and Taiwan risk to Laura concentrated in the residency and the equity sleeve rather than her income, and does the Treasury core act as the hedge in that joint scenario?

- Anchors: R-C39, R-C85, R-AN34 ("No instruction for what happens if the residency cannot open or moves").
- Evidence: Messy Roots "will not likely be for sale in Mainland China" (Wikipedia, VP secondary). HarperCollins holds world English rights (VP). Her Taiwan media coverage is listed in her guide (VP).
- Hypothesis (guess): yes. A Taiwan crisis would hit the growth sleeve (large Taiwanese chip makers sit in global indexes; D4 to check weights) and the residency plan together, while U.S. Treasuries would likely hold or rise (ASM). Her direct China-market income is small.
- Changes: a sentence (FR risk register) and possibly the EWT/Taiwan-tilt decision (drop).
- Deliverables: FR, WInS-now. Domain: D4. Seed: 19.
- North star: belonging across borders is handled as a cash-flow and risk fact, not as decoration.

Q12. In the one market crash of her career (2020), her creative income stepped up. Should the FR avoid assuming her income automatically falls with markets?

- Anchors: R-C13, R-C15, R-C37.
- Evidence: PW July 2020: two-book deal "at auction" (VP). Comics market \$1.1bn (2019) → \$2.01bn (2022) (VP). The 2020 S&P fall itself is UNVERIFIED here (D2).
- Hypothesis (guess): yes, but with $n = 1$ and survivorship caveats. Evidence for a strong negative link is weak, so the plan's safety must not rely on either sign. Lock-early does not.
- Changes: a sentence (FR); the p range in Q3 includes 0.
- Deliverables: FR. Domain: D6. Seed: 16.
- North star: using her real record instead of stereotypes about "artists' income".

Q13. Which facts about her must the deliverables phrase precisely?

- Facts in question: "bestselling"; finalist vs winner; the CCA status; the comic's date; "professor" in 2027.
- Anchors: R-C5, R-C10, R-C13, R-S25; CFA Asset Manager Code F.2 ("truthful, accurate, complete", SH-25).

- Evidence: E7, E8, E10, E14.
- Hypothesis (guess):
 - Use the case's own words ("bestselling", "educator").
 - Say "finalist" for Harvey and Goodreads Choice.
 - Do not mention CCA at all: teaching income is not in the case (brief 8.19), and CCA's closure is about her employment, not the plan.
 - Never cite Wikipedia for dates.
- Changes: sentences (IPS title/pitch context, FR client section).
- Deliverables: IPS, FR. Domain: D10 (with D9). Seed: null.
- North star: one wrong fact about a real alumna costs more credibility than any insight gains.

Q14. Which claims in the team's two client documents must be removed before anything reaches a deliverable?

- Types of claim: personal, unverified, or contradicted by the case.
- Anchors: brief section 4 (privacy); R-AN56, R-AN57; brief section 8.19.
- Evidence: the docs include personal-life material (categories named in the header). They also include unverified figures (festival counts) and case-contradicting advice (dated facility liability, income-gap cash buffer, values screens).
- Hypothesis (guess): a one-page "safe facts" list (section 1.1, E1-E13) replaces the docs as the team's source. The docs stay out of the repo.
- Changes: a decision (the team's fact base).
- Deliverables: TN, IPS, FR. Domain: D10. Seed: null.
- North star: she would recognise a firm that knows her work and respects her privacy.

Q15. She interned on a trading desk and at the CFPB. Should the IPS carry a one-line statement on costs, commissions and conflicts, and should the WInS plan keep trades few?

- Anchors: R-C12, SH-25 (Asset Manager Code F.4.d, fees), BS-01 (no fee assumption anywhere), F-607 (\$25/\$10 commissions).
- Evidence: E4 (P&Q 2018, VP).
- Hypothesis (guess): yes. One IPS clause ("low-cost, liquid, transparent instruments; costs disclosed"). The FR adds a fee line to the projections. In WInS, fewer, larger trades.
- Changes: a sentence (IPS); a number (fee assumption in FR projections); a decision (trade count).
- Deliverables: WInS-now, IPS, FR. Domain: D8 (with D10). Seed: 34.
- North star: a client who knows how the finance industry earns money chooses the firm that shows its costs.

Q16. She herself gives a public talk titled "Taking the Leap to Pursue Art Full-Time from a Traditional Tech Job". Can the IPS describe her risk tolerance using that professional framing (contract first, then the leap)?

- Anchors: R-C37, R-AN16, R-AN17; B8b Q3 (floor-first career sequencing).
- NEW vs B8b: the framing is one of her four advertised speaking topics (E11, VP). It is her own public account of her career risk, not our reading of her life. The PW deal (July 2020) precedes her leaving her job (Wikipedia, VP secondary).
- Hypothesis (guess): yes, in the FR's client section, as a paraphrase of her public career facts. Keep it out of the IPS's 500 words unless one clause fits.
- Changes: a sentence (FR; possibly the IPS risk-tolerance clause).
- Deliverables: IPS, FR. Domain: D6 (with D9). Seed: 33.
- North star: her "thoughtful risks" described in terms she already uses in public.

Q17. Which of the four named income sources is most tied to markets, and should D3 weight "other entrepreneurial ventures" (freelance tech design) as the most cyclical?

- Anchors: R-C27, R-AN31.
- Evidence: her studio's clients are start-ups (resume, VP). Freelance design depends on start-up funding (ASM) and is the most AI-exposed (SoA, VP).

- Hypothesis (guess): rank by market sensitivity:
 1. freelance design/ventures (high);
 2. corporate speaking (medium-high);
 3. licensing (medium);
 4. contracted advances (low in the short run);
 5. school and library speaking (lagged).
- Changes: a number (per-source p in D3's model, if the model splits sources at all). Otherwise one sentence.
- Deliverables: FR. Domain: D3 (with D6). Seed: 16.
- North star: the stress test follows her actual business lines.

Q18. What does "licensing" in the case cover for an author who kept her translation rights, and does it add market-linked income (AI licences, film/TV options)?

- Anchors: R-C27.
- Evidence: PW "world English rights" only (VP); HarperCollins AI licence (VP); film/TV option markets UNVERIFIED.
- Hypothesis (guess): translation, AI-training and adaptation options; lumpy and partly market-linked. One FR clause at most. Do not speculate about any actual licence.
- Changes: a sentence (FR assumptions list).
- Deliverables: FR. Domain: D6. Seed: 20.
- North star: shows we read the case's word "licensing" as a real business, not filler.

Q19. Should the FR name her "continued business income" as a second source of project flexibility after 2033 (R-C44), or treat it as uncounted because her human capital is uncertain?

- Anchors: R-C44, R-AN24, R-C58, SH-26.
- Evidence: the case lists "continued business income" as a possible outside source for the facility (VRF). Her income is USD while residency costs are NT\$ (F-512 FX volatility).
- Hypothesis (guess): mention it as possible upside, never count it. Size the flexibility money without it. It carries both career risk and currency risk (USD income against NT\$ costs).
- Changes: a sentence (FR flexibility section) and a decision (flexibility sizing excludes her income).
- Deliverables: FR. Domain: D5 (with D4). Seed: 19.
- North star: the plan never leans on her future work to keep a promise, which is exactly the case's rule.

Q20. Should the IPS pre-commit, in one sentence, what happens if the 2028 contribution is late or smaller?

- Anchors: R-I (IPS "formally establishes the strategy and the team's decision-making framework", R-C89); R-C85 ("timing of cash flows"); the strategy freezes Nov 6.
- Evidence: Q1 and Q5 numbers; advances paid in instalments (VP via A3).
- Hypothesis (guess): yes. "The operating reserve is funded from the first contribution; the second contribution funds growth and the facility; if it is late or smaller, the facility range scales and the payments do not change." This is guidance on content, not wording: the students write the sentence.
- Changes: a sentence (IPS decision framework).
- Deliverables: IPS. Domain: D9 (with D10). Seed: 16.
- North star: she can see her own earnings' role in the plan in one line of the official record.

3. Parked or merged (did not pass my own so-what test)

- Speaking-fee levels and her pro-bono policy: bears on her real income (R-AN56). Merged into Q7/Q17 as channel facts only.
- Colorist or collaborator income split (E12): about her real finances. Parked.
- Book metaphors (Kirby's climbing, "falling"): B8b Q15/Q17 already own "gimmicks" and identity metaphors. Not re-asked.
- Credibility as human capital (BS-03): already answered by A3. Not re-asked.
- Whether her works are part of any AI copyright settlement: speculative about her finances. Parked.

4. Leads for the specialists

- **L-1 (D3):** use F-401/F-403/F-404 to build the deposit table (Q1). Add a one-factor deposit model: $\text{deposit} = \$150k \times (1 - \text{cut})$, where cut is idiosyncratic plus $p \times (2027 \text{ equity shock})$. Run $p \in \{0, 0.3, 0.5\}$. Add a "12 months late" case. The joint-tail arithmetic for Q5 uses the brief section 6 DV01 and ladder numbers.
- **L-2 (D6):** industry sources for the FR human-capital paragraph:
 - Authors Guild 2023 income survey: <https://authorsguild.org/news/key-takeaways-from-2023-author-income-survey/> (VP).
 - ICv2 2023: <https://icv2.com/print/article/57351> and ICv2 2025: <https://icv2.com/articles/news/view/61573/> (VP).
 - AAP 2009 (PW): <https://www.publishersweekly.com/pw/by-topic/industry-news/bookselling/article/42783-aap-book-sales-dipped-in-2009.html> (VP).
 - McIlroy recession review: <https://thefutureofpublishing.com/2019/04/the-impact-of-recessions-on-publishing-industries-part-iii-books/> (VP).
 - FRED MRTSSM451211USN bookstore sales (VP, live CSV).
- **L-3 (D6/D10):** re-verify the SU items before use:
 - HMH acquisition freeze 2008 (try the Horn Book page <https://www.hbook.com/story/houghton-mifflin-harcourt-update>).
 - 2009 speaking-bureau piece (Publishing Perspectives, 2009-08; did not load).
 - Axios on Google calendar heritage months (403).
 - The Advocate "39%" Pride figure.
 - DOJ-ALA IMLS settlement (April 2026).
 - The Messy Roots "book ban" video (do not use unless a primary news or PEN America entry names the title).
- **L-4 (D2):** check sector weights on an issuer fact sheet for a broad U.S. and a global index fund. Needed: the communication-services share and its largest holdings (Q10), and the Taiwan weight in global/EM funds (Q11). Also check the S&P 500 2020 drawdown and 2025 total return (Q7, Q12), from a primary index page.
- **L-5 (D4):** the Taiwan joint scenario (Q11): sleeve loss, residency delay and Treasury behaviour together.
- **L-6 (D9/D10):** a "safe facts" list for the team (section 1.1 E1-E13) with exact wording. Say "finalist" (resume), and use the case's "bestselling" and "educator".
- **L-7 (D8):** the Asset Manager Code fee disclosure (F.4.d) plus her trading/CFPB background (E4) supports Q15.
- **Blocked or failed:** axios.com (403); the Publishing Perspectives 2009 page (no text returned); Poets&Quants (the text helper fails; the raw HTML works). WebSearch budget exhausted for this session.

Sources (all accessed 2026-09-27)

Source	URL	Status
Laura Gao bio page	https://lauragao.com/hey/	VP
Publicity & Speaking Guide (PDF)	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	VP
Speaking events page	https://lauragao.com/speaking-events	VP
Resume (PDF; professional lines only used)	https://lauragao.com/s/Laura-Gao-Resume.pdf	VP
Kirby's Lessons page	https://lauragao.com/kirbys-lessons-for-falling-in-love	VP
Poets&Quants 2018 Best & Brightest	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	VP (raw HTML)
Publishers Weekly Rights Report, July 2020	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.html	VP (verbatim check)
Wikipedia: Messy Roots; Laura Gao	https://en.wikipedia.org/wiki/Messy_Roots ; https://en.wikipedia.org/wiki/Laura_Gao	read; secondary (one date error)

Source	URL	Status
Vanderbilt news, 2026-01-13	https://news.vanderbilt.edu/2026/01/13/vanderbilt-university-to-establish-full-time-academic-campus-in-san-francisco/	VP
The Art Newspaper, 2026-01-13	https://www.theartnewspaper.com/2026/01/13/california-college-arts-closing-vanderbilt-university-takeover	VP
Authors Guild 2023 income survey takeaways	https://authorsguild.org/news/key-takeaways-from-2023-author-income-survey/	VP
Authors Guild on HarperCollins AI licence	https://authorsguild.org/news/harpercollins-ai-licensing-deal/	VP
SoA AI survey 2024 (EWC summary)	https://europeanwriterscouncil.eu/soa-survey-uk-ai-2024/	VP (summary page of the SoA survey)
ICv2 2023 market	https://icv2.com/print/article/57351	VP
ICv2 2025 market	https://icv2.com/articles/news/view/61573/icv2-white-paper-comics-graphic-novel-sales-reached-new-high-2025	VP
PW / AAP 2009 book sales	https://www.publishersweekly.com/pw/by-topic/industry-news/bookselling/article/42783-aap-book-sales-dipped-in-2009.html	VP
McIlroy, recessions and books (2019)	https://thefutureofpublishing.com/2019/04/the-impact-of-recessions-on-publishing-industries-part-iii-books/	VP (secondary compilation)
FRED bookstore sales	https://fred.stlouisfed.org/graph/fredgraph.csv?id=MRTSSM451211USN	VP (live CSV; annual sums computed by me)
ALA on IMLS grant reinstatement, Dec 2025	https://www.ala.org/news/2025/12/ala-welcomes-reinstatement-all-federal-impls-grants-libraries	VP
HMH acquisition freeze, 2008	search results (Horn Book, WSWS, Wikipedia)	SU
Publishing Perspectives 2009 speakers	https://publishingperspectives.com/2009/08/by-turning-authors-into-speakers-publishers-profit-even-in-recession/	SU (page returned no text)
Axios / Advocate on 2025 corporate DEI and Pride cutbacks	https://www.axios.com/2025/02/12/google-calendar-holidays-heritage-months-remove-diversity ; https://www.advocate.com/news/companies-that-stopped-sponsoring-pride	SU (403 / figure not found in text)
Phase A files	research/insight_v1/phase_A/*.md	VRF

What this teaches

A client's income is an asset too, and it has its own risks. Before assuming "when markets crash, her income crashes", look at the evidence for each income stream.

- Books held up in 2009.
- Graphic novels grew through 2020.
- The recent shocks to her markets came from politics, not the stock market.

The biggest lesson is structural. If the promise is paid for with money she already has, uncertainty about money she has not earned yet can only change the size of the gift, never the safety of the promise. And when you describe a real person, precision is respect: "finalist" is not "winner", and "closing" is not "acquired".

B3b Laura's World & Human Capital (starting angle: her books, her public values, and the income behind the 2028 deposit)

Agent B3b, insight_v1 run, Phase B. Written 2026-09-27. This is AI-generated research for Team Caplet and is not text for submission. It generates questions. It does not answer them for the team, and it drafts no deliverable prose.

Privacy rule applied: this file uses only Laura Gao's public professional record: her books, publisher and review copy, her website's professional pages, her résumé's work history, and press about her work. Several sources I opened also contain family, relationship, name-history, contact and hobby details. I left all of them out on purpose. Nothing here suggests contacting her (R-W55).

Relation to other Phase B files: B8a and B8b (Voice of Laura) already collected her own words on risk, gimmicks, purpose statements, staged institution-building, late books and the unattributed pull quote. I do **not** re-ask those questions. Where one of my questions touches theirs, I say what new evidence I bring. My lens starts from the **books themselves** (what the publisher, reviewers and she say they are about) and from the **economics of the income** that is meant to fund the \$150,000 deposit in January 2028 (publishing contracts, advances, the classroom and library channel, speaking, licensing, generative AI, and US-China tension).

0. Summary (read this first)

- 1. The one market crash of her career coincided with her career's biggest boost.** "The Wuhan I Know" went viral in late March/April 2020 (NPR, 2020-04-04). The S&P 500 fell 33.9% from 19 Feb to 23 Mar 2020 (FRED, VERIFIED-PRIMARY). HarperCollins bought her first two books "at auction" in the week of 20 July 2020 (Publishers Weekly, VERIFIED-PRIMARY). North American comics and graphic-novel sales then rose from \$1.1bn (2019) to \$2.01bn (2022) (ICv2, VERIFIED-PRIMARY). So the simple story ("a crash also shrinks her advances") has one real counter-example. Her income is driven by events, publishing contracts and policy more than by the stock market. This bears directly on the open "wrong-way risk" item in brief section 9 (Q04).
- 2. The 2020 two-book contract appears to be used up, so any "publishing advance" in January 2028 must come from a contract that is not public yet.** Publishers Weekly (2020) records "world English rights to debut YA graphic memoir The Wuhan I Know, plus a second graphic work ... The second graphic work is about a queer girl's many loves". That description matches *Kirby's Lessons for Falling (in Love)* (HarperAlley, March 2025). Her résumé says she "wrote and illustrated two award-winning graphic novels" for HarperCollins. The Authors Guild's typical six-part advance schedule ends "12 months after publication", which for Kirby would be about March 2026 (ASSUMPTION: a generic schedule, not her contract). Implication: the deposit is **contract-cycle risk**, and the case's "will" (R-AN35) carries more hidden work than teams will notice (Q03). This adds new evidence to B8b Q13.
- 3. The deposit sits in the right tail of author incomes.** \$150,000 in one year is 6.0 times the 2022 median *combined* author income of full-time graphic novelists (\$25,000; Authors Guild 2023 survey, VERIFIED-PRIMARY). The team must not doubt the client (the case says "will"). But it should know that any part of the *promise* that rests on the deposit rests on right-tail career success. That happens in exactly one place: the rule for "rates fall before January 2027 and the ladder costs more than \$300k" (Q02).
- 4. Her human capital has a tech leg, which is labour flexibility in the Bodie-Merton-Samuelson sense.** Her résumé lists "LG Studios | Freelance Product Designer, Author & Illustrator 2020 - Present". It also lists earlier work as a Twitter product manager and an Amazon data analyst. A client who can switch back to paid product work has more capacity to take risk with money the promise does not need (BMS 1992, VERIFIED-PRIMARY abstract). This is a Laura-specific input to the open sleeve equity weight (50/60/70%) (Q05).
- 5. On AI she is not a simple "artist versus AI" story.** Her own studio's clients include "Retainit, AI-powered online learning software" (résumé, VERIFIED-PRIMARY). HarperCollins, her publisher, offers

authors an opt-in AI-training licence of \$5,000 per title, split 50-50 (Authors Guild, VERIFIED-PRIMARY). I found no public statement by her on generative AI. So any "AI hedge" story in the portfolio would be invented. AI exposure should be disclosed and left as her choice (Q09). A separate point teams will miss: **a client who is a professional illustrator should get a Final Report with no AI-generated images and no imitation of comic art** (Q10, Q11).

6. **Her classroom and library channel faces policy risk, and that risk is now documented on primary pages.** In 2025 Executive Order 14238 directed the elimination of the Institute of Museum and Library Services (IMLS); a court reversed this in November 2025 (ALA, VERIFIED-PRIMARY). In June 2026 the FCC advanced a proposal asking whether E-Rate "should be terminated or limited to only rural areas" (ALA, VERIFIED-PRIMARY). A Montana school board banned *Messy Roots* in January 2024 (Book Riot, VERIFIED-PRIMARY). Both books are rated "Ages 14-up" (Publishers Weekly). These are risks to her income, **not** reasons to tilt any investment (Q06, Q20).

7. **Factual traps that most teams will fall into.**

- (a) Linking her Wuhan birth to Taiwan ("returning to her roots"). The case never does this, and it is wrong (Q08).
- (b) Inflating her accolades. Her own résumé gives the exact claims: "Debuted on #9 on Indies Bestsellers List" and "Finalist for Harvey Award, Goodreads Choice Awards, and California Booksellers Award" (Q14).
- (c) Using 2026 news about her real career, such as her college "will close permanently in 2027" (The Art Newspaper), even though R-W4 says the scenario is fictional (Q15).

8. **What she would recognise as hers is a benefit, not a metaphor.** For a client who calls her own income "unstable" (Overachiever 2021, verified by B8a), the most personal thing our plan can offer is plain: **once the ladder is bought, her promise to the residency no longer depends on her next book deal, speaking season or licence** (Q01). Book-title puns and plot metaphors are optional at best (Q13).

Counts: 21 questions (section 3); 24 evidence items (section 2); 9 items parked as duplicates (section 4).

1. Method and source access

- Every web page was read with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL`. Quotes were checked with `--grep "phrase"` and returned `PHRASE FOUND VERBATIM: YES`. Access date for every source: **2026-09-27** (about 15:30-16:30 UTC).
- WebSearch was used three times and then hit the session's search budget (200 calls used by the run). After that I reached pages by known URLs, by links on pages already opened, and through the run's shared fetch cache (`/tmp/insight_v1_fetch_cache/`, pages other agents had already fetched).
- **Blocked or not found:**
 - `search.books.com.tw` and `readmoo.com` (Taiwan bookstores): HTTP 403. I could not check whether a Traditional Chinese (Taiwan) edition of her books exists.
 - Open Library lists English editions only for both books (a check, not proof).
 - Google Books API: no result.
 - No 2024-2026 ICv2 market-size page was reachable without search. The latest verified figure is for 2023.
 - Corporate DEI and heritage-month speaking cuts in 2025: no primary page read (SNIPPET-UNVERIFIED lead only).
 - PEN America's ban index: not reached (search budget).
 - No public statement by Laura about generative AI was found in the ~40 cached pages about her. Absence of evidence is not evidence.
- **Anomaly:** Wikipedia's *Messy Roots* article says she made the comic "In 2021 ... as a response to ... the 2021 Atlanta spa shootings". The case (R-C13: "In 2020") and NPR (2020-04-04) date the comic to 2020. The Wikipedia sentence is wrong on the year. Do not cite Wikipedia for dates.

2. Evidence found (E-ids used in the questions)

Status: VP = VERIFIED-PRIMARY (read on the page, phrase verified); VRF = VERIFIED-REPO-FILE; SU = SNIPPET-UNVERIFIED; ASM = ASSUMPTION; DER = derived arithmetic.

2a. The books (what they are, who they are for)

E	Fact (exact words where quoted)	Source	Status
E01	<i>Messy Roots: A Graphic Memoir of a Wuhanese American</i> , "HarperCollins/Balzer + Bray, \$22.99 (272p)"; "Ages 14-up"; review opens "In this fresh, frank, and tender debut"; closes "A multidimensional, thoroughly entertaining account of growing into queer Asian American identity." Reviewed 2022-01-06	publishersweekly.com/9780063067769	VP
E02	<i>Kirby's Lessons for Falling (in Love)</i> : "Laura Gao. HarperAlley, \$26.99 (304p) ... \$18.99 paper"; "Ages 14-up"; plot: a competitive climber whose "dynamic move at the Texas Youth Fall Invitational goes awry", who must do another activity while her arm heals so she can still reach a university climbing team; the review praises Gao for "delicately balancing the pros and cons of Kirby's close-knit Christian immigrant community". Reviewed 2024-12-12; published March 2025	publishersweekly.com/9780063067806	VP
E03	Her site's Kirby page: "Once dubbed the Queen of Balance as her school's top rock climber, Kirby Tan suffers an injury that sidelines her for the rest of the season." Kirkus line on the page: "A refreshingly raw and vulnerable exploration of grief and hope."	lauragao.com/kirbys-lessons-for-falling-in-love	VP (publisher and review copy, not her voice)
E04	<i>Messy Roots</i> page: "3 starred reviews · 12+ awards & honors"; SLJ: "lifts the story of Wuhan beyond COVID". Her Educators guide lists accolades including "GOLDEN POPPY AWARD WINNER", "CYBIL AWARD WINNER", "HARVEY AWARDS FINALIST", "GOODREADS CHOICE AWARDS FINALIST", "NYPL BEST BOOKS OF 2022"	lauragao.com/messyroots/ ; lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	VP
E05	Her résumé: "Debuted on #9 on Indies Bestsellers List, and featured on NPR and New York Times"; "Finalist for Harvey Award, Goodreads Choice Awards, and California Booksellers Award". Her /hey page: "A Harvey and Goodreads Choice Award honoree"	lauragao.com/s/Laura-Gao-Resume.pdf ; lauragao.com/hey	VP
E06	Her Educators guide (her note): the activities "exercise students' creativity in less daunting ways than traditional essay writing"	Educators guide PDF	VP
E07	Horn Book (as quoted by Wikipedia): "Gao personalizes her experiences with insight and humor"; "candid depiction of the universal search for one's place in the world"	en.wikipedia.org/wiki/Messy_Roots	Wikipedia page VP; Horn Book original SU
E08	"It's not published in Mandarin" (her answer, NPR 2022-04-24, on whether the book reached China; she adds she is unsure it ever will be, because of censorship rules on LGBTQ content: paraphrase of the same answer)	npr.org/sections/goatsandsoda/2022/04/24/1093992912/...	VP (first sentence); rest PARAPHRASE of a VP page

2b. The contract and income economics

E	Fact	Source	Status
E09	"Donna Bray at HarperCollins/Balzer + Bray has acquired, at auction, world English rights to debut YA graphic memoir <i>The Wuhan I Know</i> , plus a second graphic work, by Laura Gao." "The second graphic work is about a queer girl's many loves." (Rights Report, week of 20 July 2020)	publishersweekly.com/.../83911-rights-report-week-of-july-20-2020.html	VP
E10	Résumé: "HarperCollins, wrote and illustrated two award-winning graphic novels"	Résumé PDF	VP
E11	Authors Guild (2021-07-20): advances are paid "in three, and even up to six, installments"; a common six-part payout is "on signing the contract, three to six months after contract signing, on delivery of the manuscript, on acceptance of the manuscript, upon publication date, and one part 12 months after publication"	authorsguild.org/blog/everything-you-need-to-know-about-book-advances/	VP
E12	Authors Guild 2023 survey (2022 incomes, full-time authors): "Graphic novelists ranked second, earning a median book income of \$15,000 and a combined income of \$25,000 when including other author-related income." Also: "most authors are earning half of their writing-related income from sources other than their books."	authorsguild.org/news/key-takeaways-from-2023-author-income-survey/	VP
E13	$\$150,000 \div \$25,000 = 6.0\times$ the median combined income of a full-time graphic novelist	E12 + case R-C27	DER

E	Fact	Source	Status
E14	ICv2 (2024-07-15): U.S./Canada comics and graphic-novel sales "around \$1.87 billion in 2023, down 7% from \$2.01 billion in 2022"; 2019 "when sales were \$1.1 billion". The Covid surge "crested in 2022"	icv2.com/print/article/57351	VP
E15	S&P 500: 3,386.15 (2020-02-19) → 2,237.40 (2020-03-23) = -33.9% ; 3,257.30 on 2020-07-21 (the week of her deal), -3.8% from the peak	fred.stlouisfed.org SP500 CSV	VP (DER for the %)
E16	Résumé: "LG Studios	Freelance Product Designer, Author & Illustrator 2020 - Present"; "Founded professional studio for product design and illustration work"; "Clients include: Retainit, AI-powered online learning software". Earlier: Twitter product manager 2018-2020; Amazon data analyst 2017	Résumé PDF
E17	Her /projects page lists game, CRM and UI/UX work (e.g. "UI/UX work for a retail media adtech company"); her site sells prints and has a "Visual Data Stories" section	lauragao.com/projects ; lauragao.com	VP
E18	Bodie, Merton & Samuelson (1992): "The ability to vary labor supply ex post induces the individual to assume greater risks in his investment portfolio ex ante."	nber.org/papers/w3954	VP (abstract)
E19	CFA Institute Research Foundation, <i>Lifetime Financial Advice: Human Capital, Asset Allocation, and Insurance</i> (2007): individuals "must take into account human capital" when setting asset allocation	rpc.cfainstitute.org/research/foundation/2007/lifetime-financial-advice-human-capital-asset-allocation-and-insurance	VP (summary page)

2c. Channel and policy risks (to income, never to investment choice)

E	Fact	Source	Status
E20	ALA (Dec 2025): "On March 14, President Trump issued Executive Order 14238, which directed the elimination of the agency" (IMLS). Grants were reinstated after a 21 Nov 2025 court ruling. ALA news list, 2026-06-25: the FCC advanced a proposal that "asks whether the E-Rate program should be terminated or limited to only rural areas"	ala.org/news/2025/12/ala-welcomes-reinstatement-all-federal-imls-grants-libraries	VP
E21	Book Riot (2024-01-12): "Laurel Public Schools (MT) school board just banned six books", including <i>Messy Roots</i>	bookriot.com/book-banning-will-not-stop-at-schools/	VP (Book Riot's report)
E22	ALA: 4,235 unique titles challenged in 2025 (via stakeholder map SH-23)	A3 stakeholder map, W-9	VP (by A3)
E23	Authors Guild (2024-11-19): "The HarperCollins licensing deal provides for a \$5,000 fee per title, which will be split 50-50 between an author who chooses to participate and HarperCollins." It covers nonfiction titles and is opt-in	authorsguild.org/news/harpercollins-ai-licensing-deal/	VP
E24	The Art Newspaper (2026-01-13): California College of the Arts "will close permanently in 2027"; her /hey page says she is "a Professor of Comics at California College of the Arts"	theartnewspaper.com/2026/01/13/california-college-arts-closing-vanderbilt-university-takeover ; lauragao.com/hey	VP. For team awareness only. See Q15: do not use in deliverables.

Already verified by B8a/B8b and relied on here: "unstable income" (Overachiever 2021); "No more gimmicks." (Nerd Daily 2025); "I'm taking a short break from writing" (Nerd Daily 2025); *Messy Roots* planned for "summer 2021" but published 8 March 2022; past talks at corporate heritage and employee-group events (her speaking guide).

3. Questions (21), with reasoning

Format: question · anchors · anchor quote · hypothesis (a GUESS unless it cites evidence) · what would change · deliverables · domain · seed · north star. Ranked by deadline tier, then impact.

Tier 1 and 2 (WInS now, Trading Notes, IPS)

Q01. Should the IPS name, as the client benefit, that from the Treasury purchase onward her ten-year promise stops depending on her own future earnings (book deals, speaking, licensing)?

- Anchors: R-C27, R-C33, R-C38, R-S25, SH-01, BS-03.

- Anchor quote: "she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals" (case p.2).
- Hypothesis (guess):
 - Yes. It is the most Laura-specific benefit available, and it needs no metaphor.
 - Her income is event-driven and contract-driven (E09-E14). She has called it "unstable" (B8a #1).
 - Most teams will describe safety as "low volatility". Ours can describe it as **independence from her career's ups and downs**.
 - That independence is complete in January 2027 if the ladder costs $\leq$ \$300k. It is complete in January 2028 if a top-up is needed (see Q02).
- Would change: a sentence. It sets what the pitch and the IPS's opening must contain (the team writes it).
- Deliverables: IPS, FR, TN (the first Treasury note's "why"). Domain: D9. Seed: 33.
- North star: she chooses the firm whose plan makes her dream stop depending on her next good year.

Q02. If rates fall before January 2027 and the ladder costs more than \$300k, how much of the promise may rest on the 2028 deposit? Should the IPS rule say so in one line?

- Anchors: R-C27, R-C47 (portfolio-funded "with a high degree of certainty"), R-AN35, brief section 9 (joint tail), F-111.
- Anchor quote: "All ten payments must be funded by the investment portfolio with a high degree of certainty."
- New evidence: the deposit is right-tail income (E13: 6.0x the median) and depends on a contract that is not yet public (E09-E11).
- Hypothesis (guess, DER from brief section 6):
 - At -50bp the gap is \$7,135 (4.8% of the deposit). At -100bp it is \$22,864 (15.2%). P(cost > \$300k) is about 24% (Phase A, ASSUMPTION model).
 - Relying on a deposit the case says "will" arrive is allowed.
 - A statistics-trained client would still want the dependency named: "up to \$X of the promise waits for the 2028 deposit; if it is late, [rule]".
 - Candidate rules for D1/D3 to compare:
 - (a) long rungs first; the gap sits in the 2033 rung, the least rate-sensitive, and the deposit fills it;
 - (b) also hold the gap as a T-bill "pending" line;
 - (c) accept it and state it.
- Would change: a decision (the January 2027 rule) and a number (maximum dependence in \$ and %).
- Deliverables: IPS, FR. Domain: D1. Seed: 16.
- North star: we say exactly how much of her promise rests on her own career, and for how long. That is usually zero, and never unstated.

Q03. Is the HarperCollins two-book contract of 2020 fully delivered and paid? If so, should the team's labelled stress case treat the "publishing advances" part of the 2028 deposit as uncontracted, and size a "late" case and a "smaller" case on that basis?

- Anchors: R-C27, R-AN35, BS-13, SH-21.
- Anchor quote: "She will contribute an additional \$150,000 at the beginning of 2028, using earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures."
- New evidence beyond B8b Q13:
 - The 2020 deal was for "The Wuhan I Know, plus a second graphic work ... about a queer girl's many loves" (E09).
 - Her résumé counts "two" HarperCollins novels (E10). Kirby came out March 2025 (E02).
 - A typical last instalment is "12 months after publication" (E11), i.e. about March 2026 (ASM).
- Hypothesis (guess):
 - Yes, as a *labelled* scenario only. The case base (\$150k, Jan 2028) is unchanged.
 - A new deal signed in 2026-27 would pay only its signing instalment(s) by January 2028. Those are small shares of a six-part advance.
 - So "late" (12 months) and "smaller" (\$75k) are equally reasonable, and both are already in the model.

- This evidence justifies them in the FR without guessing at her private finances.
- Would change: a sentence (the FR's justification for the stress cases). Possibly a number (the weight given to the "late" case).
- Deliverables: FR. Domain: D6. Seed: 16.
- North star: our stress tests come from how her industry pays, not from generic shocks.

Q04. Given that her career's biggest boost came during the 2020 crash, what correlation between the 2028 deposit and 2027 equity returns should the model assume? Should the FR state it as an explicit assumption?

- Anchors: R-C85 ("timing of cash flows, outside funding"), R-C86, brief sections 9 and 11 ("2028 deposit independent of markets").
- Anchor quote: "They should also consider how favorable and unfavorable investment outcomes would affect their recommendations."
- New evidence:
 - E15 (-33.9% crash, Feb-Mar 2020) came at the same time as the viral comic and the July 2020 auction deal (E09).
 - The comics market grew +83% from 2019 to 2022 (E14, DER).
 - Books are low-price goods (\$18.99-\$26.99, E01/E02).
 - Advances are paid on contract milestones (E11), and school and library budgets are set by policy (E20).
- Hypothesis (guess):
 - The correlation is close to zero, and the sign is uncertain.
 - The real joint tail is a *policy or contract-cycle* shock (a missing deal, cuts to school and library budgets, fewer corporate heritage-month events), not a 2027 equity crash.
 - Keep zero as the base, with a +0.3 stress (ASM) to show the plan survives it. Under lock-early it does, by construction, once the ladder is bought.
 - This probably closes the open "wrong-way risk" item with one sentence and no extra model.
- Would change: a number (the correlation input in D3's model) and a sentence (FR assumptions list).
- Deliverables: FR, IPS (assumptions line). Domain: D6 (with D3). Seed: 16.
- North star: we tested the risk that is actually hers, not a textbook one.

Q05. Does her human capital argue for the low or the high end of the growth-sleeve equity weight (50/60/70%)?

- Anchors: brief section 9 (sleeve weight open), R-C37, R-C38, R-AN16, R-S25 ("risk considerations").
- Anchor quote: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career".
- Evidence: two forces pull in opposite directions.
 - Towards **more** equity: labour flexibility (E18). She has a working product-design studio and a PM/data-analyst record (E16, E17).
 - Towards **less** equity: lumpy, contract-driven creative income (E11-E14). Human capital belongs in asset allocation (E19).
- Hypothesis (guess):
 - The promise is locked and living costs sit outside the portfolio (R-C33). So the sleeve is "surplus" money, and labour flexibility dominates.
 - That supports 60% or a little higher, never lower for human-capital reasons.
 - The sleeve weight barely moves the median (brief 8.5), so this is a *reasoning* gain more than a number gain.
- Would change: a decision (the weight, or at least the reason the IPS gives for it).
- Deliverables: IPS, FR. Domain: D6 (D8 to benchmark). Seed: 16.
- North star: the risk level is set by who she is professionally (a flexible, multi-skilled founder), not by an age or a quiz.

Q06. Should the WInS portfolio carry any tilt meant to "hedge" her human capital (e.g. underweighting media and publishing stocks such as HarperCollins's parent, News Corp)? Or should that idea be dropped now?

- Anchors: brief section 9 ("human-capital underweight of publishing/media"), R-S24 ("appropriate diversification"), G-ids in the WInS guardrails (compliance first).
- Anchor quote: "uses appropriate diversification; and maintains consistency with the team's IPS" (SMApply criterion 1).
- Hypothesis (guess): drop it.
 - Her income drivers are contracts, policy (E20-E22), and events such as corporate heritage-month bookings (SU). None of these is well tracked by any listed stock.
 - Publishing is a tiny share of broad indices. HarperCollins is only one division of News Corp (general knowledge, ASM).
 - A tilt adds a trade, a commission (\$25, F-607) and an explanation, and it hedges almost nothing. That fails "complexity must earn its place".
 - The hedge that works is structural: the promise is locked (Q01).
- Would change: a decision (no human-capital sector tilt in WInS; one FR line explaining why).
- Deliverables: WInS-now, IPS, FR. Domain: D2. Seed: 16.
- North star: we considered her career risk and chose the simplest hedge that actually works.

Q07. Is a US-China/Taiwan crisis the one scenario that hits her promise, facility, residency and income at the same time? Should the FR name it as the single "correlated" scenario, while keeping the direction of the income effect honest?

- Anchors: R-C39, R-C58, R-C86, brief section 9, seed 24.
- Anchor quote: "In 2033, Laura plans to establish a collaborative creative residency in Taiwan."
- Evidence:
 - Her books are English-only: "It's not published in Mandarin" (E08). World English rights only (E09).
 - Her subject matter is China, Wuhan and the diaspora (E01).
 - The residency is in Taiwan (case).
 - Broad U.S. equity indices depend on Taiwan-made chips (D2 to verify).
 - Her 2020 surge came *during* a spike in anti-Asian sentiment (R-C14), so her income may **rise**, not fall, in a crisis.
- Hypothesis (guess):
 - Yes. One named scenario: equities fall, the TWD moves, residency operations are disrupted, and the facility range is cut.
 - The Treasury-locked promise is unaffected (it is in USD, paid by the U.S. Treasury).
 - Her income effect is "uncertain in sign" (E15 precedent).
 - One scenario beats five generic shocks.
- Would change: a sentence (FR scenario list and co-sponsor draft content spec). Possibly a decision (whether the facility range carries a "location contingency" clause).
- Deliverables: FR. Domain: D4. Seed: 19.
- North star: we found the one risk that links all her goals and showed the promise survives it.

Q08. How do we make sure no deliverable links her Wuhan birth to the Taiwan location (e.g. "returning to her roots"), which would be a factual error and politically loaded?

- Anchors: R-C10, R-C39, brief 8.19 (residency ≠ her home), R-S28 ("clearly and credibly").
- Anchor quote: "Born in Wuhan, China, and raised in Texas" (case p.1).
- Hypothesis (guess):
 - This is a high-frequency error among rival teams (ASM).
 - The case gives no reason for Taiwan. Her public professional link is only that she has lived and worked there (B8a #42, AI transcript).
 - Treat "China ≠ Taiwan; Wuhan ≠ Taiwan" as a checklist item, and never explain the location choice on her behalf.
- Would change: a sentence (checklist rule for all deliverables).
- Deliverables: TN, IPS, FR. Domain: D9 (D4 to confirm wording). Seed: 19.
- North star: the client who fought a one-sided story about her hometown will notice a careless one about Taiwan.

Q09. Should the team reject any "AI hedge" argument (holding AI-heavy stocks to offset AI's threat to her career), disclose the index's AI concentration plainly, and leave any AI tilt as her choice?

- Anchors: seed 20, R-S24, R-C18, B8a Q20 (disclosure).
- Anchor quote: "she has combined artistic passion with strategic thinking and a willingness to pursue unconventional opportunities" (case p.1).
- New evidence beyond B8a Q20:
 - She designs *for* an AI-powered learning company (E16).
 - Her publisher offers an opt-in AI-training licence, author's share \$2,500 per nonfiction title (E23). *Kirby* is fiction; whether *Messy Roots* qualifies is unknown and none of our business.
 - Her main income channels (in-person speaking, school visits, teaching, festivals, human-authored books) are among the least AI-substitutable (ASM).
 - No public statement by her on generative AI was found.
- Hypothesis (guess):
 - Yes, reject the hedge story. It would attribute a view to her that she has not stated, and it is weak economics.
 - Keep the broad index. State the concentration once. No thematic AI fund in WInS.
- Would change: a decision (no AI tilt either way; no "AI hedge" wording) and a sentence (FR disclosure).
- Deliverables: WInS-now, IPS, FR. Domain: D2. Seed: 20.
- North star: we did not put words in her mouth about AI. We showed the facts and left the choice with her.

Q10. Given that she is a professional illustrator and her industry body calls unlicensed AI training "flagrant theft", should the team commit to no AI-generated images in any deliverable, and to a specific AI-use log in the Works Cited?

- Anchors: R-W46, R-W47, R-S27, R-S28 ("authentic team voice"), E23.
- Anchor quote: "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages."
- Evidence: the Authors Guild page (E23, VP) says "AI companies' ongoing, flagrant theft of books and journalism". The comics community is openly hostile to generative AI art (Broken Frontier series title, SU).
- Hypothesis (guess):
 - Yes. Many teams will decorate a Final Report with an AI "residency in Taiwan" image or a cartoon of Laura. For this client that is the single most self-defeating visual choice.
 - The AI log should say what AI did (research help, fact-checking) and did not do (no text, no images).
- Would change: a decision (a no-AI-images rule for the FR, and the content spec of the AI log).
- Deliverables: FR (TN and IPS have no images). Domain: D10. Seed: 20.
- North star: we respected her craft in how we made the report, not only in what it says.

Q11. Should the Final Report avoid imitating comic style (panels, speech bubbles, drawn characters) and instead meet the standard of her own "Visual Data Stories": a few clean, honest charts?

- Anchors: R-S28 ("uses data effectively"), R-C12, SH-04, E17.
- Anchor quote: "Presents a compelling, well-organized narrative in an authentic team voice; uses data effectively to support conclusions" (SMAApply criterion 5).
- Hypothesis (guess):
 - Yes. Comic-style layouts imitate her professional craft (a tokenism and quality risk). Her data-journalism section and statistics degree set the relevant bar.
 - Two or three charts, each proving one claim (payments covered; the 2031 range with both tails; the flexibility buffer).
- Would change: a decision (the FR's visual style) and chart specs.
- Deliverables: FR. Domain: D9. Seed: 34.
- North star: we meet her as a data storyteller, not as a cartoon.

Tier 3 (Final Report and co-sponsor materials), plus team-wide writing rules

Q12. Should the team adopt a written "swap test" for every Laura-specific sentence? The test: keep it only if it depends on her professional record (statistics degree, event-driven income, institution-building, public credibility); cut it if it works only because of her ethnicity, sexuality or birthplace.

- Anchors: brief section 4 (tokenism), R-C10, R-S25 ("thoughtful understanding of the client").
- Anchor quote: "whose work explores identity, belonging, and the power of storytelling" (case p.1).
- Hypothesis (guess):
 - Yes. It is a one-line, checkable rule that turns "no tokenism" into practice.
 - Most personalisation that survives is about **how she earns, decides and communicates**, not who she is.
- Would change: a decision (a checklist item applied to TN, IPS and FR drafts).
- Deliverables: TN, IPS, FR. Domain: D9 (D6). Seed: 34.
- North star: personal without being tokenistic. That line decides whether she feels understood or used.

Q13. Is there one principle drawn from her books' *plots* (not their titles) that maps onto a real mechanism in our plan? If there is, where, if anywhere, may it appear?

- Anchors: seed 17, R-C9 ("stories have the power..."), B8a Q22 and B8b Q17 (titles and identity quotes as metaphors).
- Anchor quote: "Laura Gao believes stories have the power to change how people see the world."
- New evidence (plot level, E02/E03):
 - Kirby, dubbed the "Queen of Balance", falls, is sidelined, and keeps her goal by taking a different route while she heals.
 - *Messy Roots* "lifts the story of Wuhan beyond COVID" (E04): the full story rather than the headline.
- Hypothesis (guess):
 - "Keep the goal, change the route after a fall" matches our pre-committed contingency rules (rates fall before 2027; a late deposit; a bad sleeve year).
 - At most one mention, in the FR's Articulation section. Never in the IPS.
 - Probably better left out entirely. She scrapped drafts for being gimmicky (B8b). The plan should *be* goal-fixed/route-flexible, not *say* so.
- Would change: a decision (use or drop the metaphor).
- Deliverables: FR. Domain: D9. Seed: 17.
- North star: if she recognises her book in our plan, it should be because of how the plan behaves, not a pun.

Q14. Which exact words may the deliverables use about her books and honours, so that nothing is overstated?

- Anchors: R-C10, R-C15 ("bestselling"), R-W28 (no lying), CFA Code F.2 (SH-25), SH-03.
- Anchor quote: "a bestselling graphic novelist, illustrator, entrepreneur, and educator" (case p.1).
- Evidence: E04/E05.
 - Her own résumé says "#9 on Indies Bestsellers List" (not the New York Times list).
 - She was a *finalist* for the Harvey and Goodreads Choice awards. "Award-winning" is supported by the Golden Poppy and Cybils wins.
- Hypothesis (guess): use the case's words ("bestselling") and at most her own verified phrasings. Never write "New York Times bestselling" or "Harvey Award winner".
- Would change: a sentence (bio line in the FR and co-sponsor draft).
- Deliverables: FR. Domain: D9 (D10). Seed: 33.
- North star: accuracy about her is the first test of whether our numbers can be trusted.

Q15. How much real 2025-26 news about her career may the FR use, given that "the financial scenario is developed specifically for the competition"? Examples: no announced next book; a 2025 writing break; her college closing in 2027.

- Anchors: R-W4, R-W5 ("their own research"), R-AN35, brief privacy rule.
- Anchor quote: "While the client is real, the financial scenario is developed specifically for the competition."

- Hypothesis (guess):
 - Use such facts only to *motivate* labelled, generic stress cases ("publishing timelines slip; author income is lumpy"), citing industry sources (E11-E14).
 - Never state or imply anything about her current employment or finances. So E24 (her college's closure) should not appear in any deliverable.
 - Rewarded research is research into her public work and industry, not into her circumstances.
- Would change: a decision (the boundary rule) and sentences (how the stress cases are justified).
- Deliverables: FR. Domain: D10. Seed: null.
- North star: deep research that never becomes intrusive, which is what a real adviser would do.

Q16. Should the co-sponsor draft carry one line on what the range is NOT (not a pledge of the top figure, not an estimate of facility cost, not a promise in NT\$)? This applies her founding principle of pre-empting misreading.

- Anchors: R-C14 ("a response to misinformation"), R-C66, R-C72, R-S28, BS-04.
- Anchor quote: "If Laura promises more than she can ultimately contribute, she could damage her credibility".
- Hypothesis (guess):
 - Yes, one line. Her first public work was built to correct a misreading. The main credibility risk in 2031 is that a co-sponsor reads the top of the range as a pledge.
 - Pre-empting that protects both the fundraising and her reputation, and her reputation is her income (BS-03).
- Would change: a sentence (content spec for the fundraising draft).
- Deliverables: FR. Domain: D5 (D9). Seed: 33.
- North star: her credibility is protected in the exact way she protected her hometown's story.

Q17. What readability level should the IPS and FR target, given that her books are rated "Ages 14-up" and she designs teaching material to be "less daunting" than essays?

- Anchors: R-C16 ("read in classrooms around the world"), R-S28, E01, E02, E06.
- Anchor quote: "Today, her books are read in classrooms around the world".
- Hypothesis (guess):
 - Aim for about a 14-16-year-old reader (roughly U.S. grade 9-10) for narrative text, with each technical term defined once.
 - The team is her books' own readership. Writing so a classmate could follow is authentic, and it is her standard.
 - D9 to pick a measurable target, e.g. a Flesch-Kincaid grade ≤ 10 for non-technical paragraphs (ASM).
- Would change: a number (the readability target) and a checklist item.
- Deliverables: TN, IPS, FR. Domain: D9. Seed: 34.
- North star: she writes for teenagers. A plan a teenager can follow is a plan she can trust.

Q18. Should the FR avoid naming her "continued business income" as the backstop for facility overruns, and size the flexibility buffer without counting on it?

- Anchors: R-C44, R-AN24, R-C58, SH-26.
- Anchor quote: "Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources."
- Hypothesis (guess):
 - Yes. Her income is event-driven (E09-E15), and her time after 2033 goes to running the residency (SH-26, ASM).
 - Counting on her earnings would rebuild, for the facility, the dependence we removed from the promise.
 - The buffer is sized from portfolio money only. Her income is upside.
- Would change: a sentence, and possibly a number (the uncommitted share after the facility contribution).
- Deliverables: FR. Domain: D6 (D5). Seed: 16.
- North star: the same independence principle is applied to her dream as to her promise.

Q19. Is "licensing" in the case partly translation and foreign rights? Could a Taiwanese partner that funds comics translation or residencies be a natural co-sponsor type for the fundraising draft?

- Anchors: R-C27 ("licensing"), R-C44, R-C72, SH-07.
- Anchor quote: "using earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures".
- Evidence: the 2020 deal was for "world English rights" (E09), so other-language rights stay with the author. As of 2022, "It's not published in Mandarin" (E08). No Traditional Chinese edition was found; the Taiwan bookstores were blocked.
- Hypothesis (guess):
 - Possibly. Taiwan's cultural-content agencies fund translation and international comics (UNVERIFIED, D5 to check).
 - Low priority: this changes at most the list of co-sponsor *types* the draft addresses. Never name an organisation as committed.
- Would change: a sentence (co-sponsor audience in the draft).
- Deliverables: FR. Domain: D5. Seed: 19.
- North star: the fundraising draft speaks to partners who plausibly exist in her world.

Q20. How should the FR describe the policy risks to her classroom and library channel? Should it cite the verified 2025-26 events in one clause, or stay generic?

- Anchors: R-C16, R-C27, SH-23.
- Anchor quote: "her books are read in classrooms around the world and have sparked conversations about identity, belonging, and community."
- New evidence beyond B8a Q09: EO 14238 and its reversal; the June 2026 FCC E-Rate proposal; ALA's 4,235 challenged titles; the Laurel ban (E20-E22).
- Hypothesis (guess):
 - One generic clause ("school and library budgets and book challenges can affect an author's speaking and sales") in the human-capital paragraph.
 - Do not name her banned book in a document written for her. It adds nothing to the decision and could read as unkind.
 - Evidence goes in the team's notes, not the report.
- Would change: a sentence (length and specificity).
- Deliverables: FR. Domain: D6. Seed: 16.
- North star: we understand her industry's risks without dramatising her own experience.

Q21. Reviewers repeatedly praise her work as frank, candid, humorous and tender (E01, E07). Should the FR's "authentic team voice" allow one or two moments of warmth or humour, as long as the numbers stay exact?

- Anchors: R-S28 ("authentic team voice"), seed 34, B8b Q28 (voice).
- Anchor quote: "Presents a compelling, well-organized narrative in an authentic team voice".
- Hypothesis (guess):
 - Yes, sparingly, and never about her or her identity: e.g. honest humour about the team's own mistakes in the Articulation section.
 - Candour about uncertainty is the professional form of "frank".
- Would change: a decision (tone guidance for the FR).
- Deliverables: FR. Domain: D9. Seed: 34.
- North star: a report that sounds like people she would want to work with, not a finance brochure.

4. Parked or not re-asked (already covered elsewhere)

Item	Where covered	Why not re-asked
Is the pull quote hers?	B8a Q01, B8b Q1	Settled as UNATTRIBUTED; nothing new found
"Secured base, then leap" as the risk pattern	B8a Q02, B8b Q3	Q01 here uses it as a <i>benefit statement</i> , not as evidence

Item	Where covered	Why not re-asked
Late-deposit size (9-12 months)	B8a Q08, B8b Q12	Q03 adds contract-completion evidence only
Gimmicks / finance-bro language	B8a Q03, Q07; B8b Q15, Q16	Covered
Purpose statement / holding "jobs"	B8a Q04; B8b Q18	Covered
"Both, not either-or"	B8b Q21 (same quote, E06-adjacent)	Covered
Taiwan risk framing (full story)	B8a Q11	Q07 adds the "one correlated scenario" angle only
Rural vs city inflation index	B8a Q10	Not my lens
Pronouns	B8a Q21	Covered (she/her and they/them on her /hey page, VP)

5. Leads for the specialists

- D1 (rates):** Q02. Use the brief's shock table (\$299,596 / \$307,135 / \$322,864 at -25/-50/-100bp). Express the top-up as a share of the \$150k deposit (DER: 0% / 4.8% / 15.2%). Compare the rules (a)-(c).
- D3 (quant):** Q04. Add a deposit-equity correlation parameter (base 0, stress +0.3; ASM) and a "late 12 months" switch. FRED SP500 2020 (E15) is the one real-life data point from her career; do not fit a model to it.
- D6 (client psychology):**
 - Q03-Q05, Q18, Q20.
 - Primary sources: Authors Guild advances page (E11) and 2023 income survey (E12); BMS 1992 abstract (E18); CFA RF *Lifetime Financial Advice* (E19); her résumé's LG Studios line (E16).
 - Fuller survey data: the Guild's PDF (linked from E12, not opened).
- D2 (equity/AI):** Q06, Q09. Verify the share of Taiwan-made chips in the index's largest holdings for Q07 (not researched here). Check whether News Corp is in the broad index and its weight (tiny, ASM).
- D4 (Taiwan/geopolitics):** Q07, Q08. Also check whether a Traditional Chinese edition of *Messy Roots* exists. books.com.tw and readmoo.com gave 403 from this container; try a publisher catalogue or the National Central Library ISBN search.
- D5 (co-sponsors):** Q16, Q19. Check Taiwan's cultural-content agency (TAICCA) programmes for comics translation and residencies (UNVERIFIED; not opened).
- D9 (communication):** Q01, Q08, Q11-Q14, Q17, Q21. Exact accolade wording is in E04/E05. Book ages: "Ages 14-up" (E01/E02).
- D10 (compliance):** Q10, Q15. The AI policy wording is in `phase_A/wins_week1_guardrails.md` section (d).
- Still unverified (do not use as fact):**
 - 2025 corporate DEI and heritage-month event cuts (affecting her past talk types).
 - 2024-2026 comics market size (ICv2 publishes yearly; only 2023 was reached).
 - PEN America's ban index entries for her titles.
 - Whether U.S. tariffs affect China-printed full-colour graphic novels. This is a possible US-China channel into publishing costs, and so into advances: a lead only, no evidence read.
- Fetch-cache tip:** other agents' pages are in `/tmp/insight_v1_fetch_cache/`. Grep them for `og:url` or `canonical` to find what is already there before spending searches.

6. Sources (all accessed 2026-09-27)

Source	URL	Piece date	Status
Publishers Weekly review, <i>Messy Roots</i>	https://www.publishersweekly.com/9780063067769	2022-01-06	VP
Publishers Weekly review, <i>Kirby's Lessons for Falling (in Love)</i>	https://www.publishersweekly.com/9780063067806	2024-12-12	VP
Publishers Weekly Rights Report, week of 20 July 2020	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.html	2020-07	VP

Source	URL	Piece date	Status
Laura Gao, Kirby page	https://www.lauragao.com/kirbys-lessons-for-falling-in-love	current	VP
Laura Gao, Messy Roots page	https://lauragao.com/messyroots/	current	VP
Laura Gao, /hey, /projects, /speaking-events	https://lauragao.com/hey ; https://lauragao.com/projects ; https://lauragao.com/speaking-events	current	VP
Laura Gao, résumé PDF (professional lines only used)	https://lauragao.com/s/Laura-Gao-Resume.pdf	undated, current	VP
Messy Roots Educators & Book Clubs Guide	https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	c. 2024 (says Kirby "will be released in 2025")	VP
NPR, "The pandemic inspired a cartoonist..."	https://www.npr.org/sections/goatsandsoda/2022/04/24/1093992912/the-pandemic-inspired-a-cartoonist-to-explore-their-wuhanese-roots-and-queer-ide	2022-04-24	VP (professional line only)
Wikipedia, <i>Messy Roots</i> and <i>Laura Gao</i>	https://en.wikipedia.org/wiki/Messy_Roots ; https://en.wikipedia.org/wiki/Laura_Gao	current	VP for the page; secondary; one date error noted
Authors Guild, book advances	https://authorsguild.org/blog/everything-you-need-to-know-about-book-advances/	2021-07-20	VP
Authors Guild, 2023 income survey	https://authorsguild.org/news/key-takeaways-from-2023-author-income-survey/	2023-09-27 (upd. 2023-10-25)	VP
Authors Guild, HarperCollins AI licensing	https://authorsguild.org/news/harpercollins-ai-licensing-deal/	2024-11-19	VP
ICv2 market size 2023	https://icv2.com/print/article/57351	2024-07-15	VP
FRED SP500	https://fred.stlouisfed.org/graph/fredgraph.csv?id=SP500	data to 2020-07-24 used	VP
NBER w3954 (Bodie, Merton, Samuelson)	https://www.nber.org/papers/w3954	1992	VP (abstract)
CFA Institute RF, <i>Lifetime Financial Advice</i>	https://rpc.cfainstitute.org/research/foundation/2007/lifetime-financial-advice-human-capital-asset-allocation-and-insurance	2007	VP (summary)
ALA on IMLS reinstatement (and news list)	https://www.ala.org/news/2025/12/ala-welcomes-reinstatement-all-federal-impls-grants-libraries	2025-12	VP
Book Riot, Book Censorship News	https://bookriot.com/book-banning-will-not-stop-at-schools/	2024-01-12	VP
The Art Newspaper, CCA closing	https://www.theartnewspaper.com/2026/01/13/california-college-arts-closing-vanderbilt-university-takeover	2026-01-13	VP (team awareness only; Q15)
Broken Frontier, comics and generative AI (title only)	https://www.brokenfrontier.com/raging-against-the-machine-comics-and-generative-ai-art-part-1/	unknown	SU (search result title)
Blocked	search.books.com.tw, readmoo.com (403); Google Books API (no result)	-	not read

What this teaches

A person's job is an asset too. Economists call it **human capital**: the value of what you will earn in the future. When we plan Laura's money we have to ask how her *income* behaves, not just her portfolio. Hers comes in lumps, from contracts, book releases, speaking seasons and school budgets. It does not move in step with the stock market. In 2020 her career took off while stocks crashed. That changes the advice. The best protection for her residency promise is not a clever trade that "hedges" publishing stocks. It is buying the payments in Treasuries so that the promise stops depending on her next good year at all. The other lesson is about respect. Understanding a client means getting her facts exactly right (which list, which award, which city). It means using her professional record rather than her identity, and presenting the work in a way that honours her craft: no AI images for an illustrator, and clear charts for a statistics graduate.

B4a Top-Practitioner Benchmark: institutional practice (pensions, insurers, endowments, foundations, Australian funds)

Agent: B4a, insight_v1 run, Phase B (question generation). Written 2026-09-27. Starting angle: how institutions that owe fixed nominal amounts fund them: cash-flow matching, immunisation, funded-ratio glide paths, hedge ratios, surplus risk, sponsor backing, spending and pledge policies. The other B4 agent starts from private banks and goals-based wealth management; I have stayed on the institutional side.

This file is AI-generated research for Team Caplet, for brainstorming only. It holds **no submission-ready prose**. Where it says what a sentence "must contain", that is a checklist; the students write every word. No securities are recommended here.

Status labels follow brief section 3. Short forms in dense places: **VP** = VERIFIED-PRIMARY (I read it on the page on 2026-09-27; URL in section 4), **VRF** = VERIFIED-REPO-FILE, **SNIP** = SNIPPET-UNVERIFIED, **ASM** = ASSUMPTION, **INT** = INTERPRETATION (my reading, not a fact). R-, F-, BS-, SH- and X- ids point to the Phase A files.

0. Summary (read this first)

I wrote 25 questions. Six stand out because they rest on primary sources nobody in the repo has used, and because their answers would change something the team does in the next four weeks:

1. **Q3. The UK pension regulator's own yardstick says Laura's promise is already "mature".** The UK Defined Benefit Funding Code (November 2024, VP) says a scheme with no cash-balance benefits reaches "significant maturity" when its liability duration is **10 years**. By then the regulator expects a "low dependency" portfolio, hedged to "at least 90%". The duration of Laura's ten payments is **about 10 years today** (spot 10.16y, F-105; 9.90y at 2027-01-01, F-103). A regulator's threshold, reached on day one, is an outside argument for hedging at once instead of drifting down a time-based glide path. It needs no model.
2. **Q2. Laura has no "sponsor", and the case built it that way.** Pensions may take investment risk when an employer can make up shortfalls: the UK code says "more risk can be allowed for where the scheme has access to sufficient employer cash flows" (VP). Australia's Future Fund chases CPI + 4-5% because the Commonwealth stands behind the pensions it pre-funds (VP; its drawdowns start "at least 2032-33", the same year as Laura's first payment). The case bans outside money for the payments, and "continued business income" is on the outside list (R-C44, R-C48, R-AN24). So Laura's sponsor backing after 2028 is nil. This gives one plain reason to hedge that a judge would recognise, and it is honest about why a famous long-horizon fund does the opposite.
3. **Q5. Fees can quietly un-fund a "fully funded" ladder.** JPMorgan's 2021 pension paper warns that hibernation strategies "will burn through capital over time" (VP). For Laura the headroom is only \$7,736 (F-104). ASM arithmetic: a 0.15% fund expense charged on the reserve costs about **\$4.4k** in 2027 money (57% of the headroom). A 0.5% advisory fee on the reserve costs about **\$14.7k** (1.9 times the headroom). A3 (BS-01) modelled fees on the sleeve only. The practitioner answer is to hold the reserve as directly owned Treasuries, charge every fee to the sleeve, and say so in the IPS.
4. **Q20. Measure the WInS Treasury sleeve the way a pension CIO does: funded status and hedge effectiveness, not return.** A weekly log of the change in bond value against the change in the ten payments' present value (\$289 per basis point, F-103) turns a falling bond fund into evidence that the strategy was "tested". That is the guide's own word (R-T20). It must start with the first trade to be usable by Oct 23.
5. **Q6. Australia already has a written rule for "what if the promise is under-funded at purchase".** APRA's SPS 160 (VP) requires a board-set "shortfall limit" and a "restoration plan" that restores full funding "within ... three years". That is the missing rule-if-rates-fall-first (brief section 9; P(cost > \$300k) about 24%, F-112) in regulator form: trigger, deadline, source of top-up (the 2028 deposit). An Australian angle that earns its place.

6. **Q12. Yale sets each year's spending from its endowment value "from two years ago"** (VP, Yale Provost). The case makes Laura name her 2033 contribution two years early, in 2031 (R-C62, R-AN7). The world's best-known spending rule uses the same two-year lag, and uses it to make budgets predictable. That is a recognisable precedent for stating the 2031 range as a rule rather than a forecast.

Other new evidence:

- The case's list of assumptions (R-C85) nearly matches the legal "prudence" factors for endowment spending under UPMIFA (Q13).
- Municipal finance separates "economic" from "legal" defeasance, with an independent "Verification Agent" report (Q7). This is new evidence for brief section 8 item 9 ("funded 2027, set aside 2033").
- Morningstar's 2025 study finds that taxable-bond fund investors captured "only about half" of their funds' returns (Q19). That is a reason to hold the reserve as individual bonds, not funds.
- Australia's Standard Risk Measure reports risk as "years of negative annual returns over any 20-year period" (Q21). ASM: the 60/40 sleeve comes out at about 5.4 of 20, while a hold-to-maturity ladder shows negative years even though its payments are certain.

Seed questions extended: 22 (Q9, Q25), 26 (Q1, Q2, Q3), 27 (Q19), 28 (Q24).

1. Method

1. Read the brief, v4 Parts 1-3, 5 and 6, CLAUDE.md, the official case, the IPS and Trading Notes guides, and the Phase A files (case register, fact register, stakeholder map, WInS guardrails). I also read the council's judge-panel file and team notes, so as not to repeat them.
2. Checked brief section 8 (already answered). Where a question touches an answered item, I say what is new.
3. Searched for institutional frameworks with WebSearch, then read each page with `fetch_text.py` or `curl`. Anything I could not open is labelled SNIP.
4. Did small arithmetic in Python on verified inputs (F-101, F-104, F-107, F-110, F-301, F-307). No script was saved: the calculations are one-liners, reproduced in section 3. Every result is labelled ASM.
5. Blocked or refused sources: the FSC-ASFA Standard Risk Measure guidance PDF (HTTP 403); AustLII (403); the Future Fund site (403 per brief; not retried); Powerball's own FAQ gave no matching text (read via Marketplace instead); the S&P defeasance criteria were not found on a primary page.

2. The questions

Format for each: the question; anchors; the official words or verified fact it hangs on; hypothesis (a guess, with evidence); what would change; deliverables; specialist; seed; north star.

A. Is the promise "funded" by a practitioner's standard? (WInS-now, TN, IPS)

Q1. What surplus buffer above 100% funded do pension practitioners require before they call a plan "hibernated", and is Laura's 102.6% (a 27bp cushion) enough to call the promise "high certainty" before the 2028 deposit arrives?

- Anchors: F-117 (funded ratio 1.026), F-104 (headroom \$7,736 = 27bp), F-111 (the ladder cost more than \$300k on 173 of 185 trading days in 2026), R-C47.
- Hypothesis (guess):
 - NISA (2021, VP) argues plans "greater than 95% funded" should hibernate now, and that a "105% funded plan has essentially no potential contributions (as measured at a 95% confidence level)".
 - Those plans have sponsors who can add money; Laura has none after 2028 (Q2).
 - So a practitioner would call 102.6% "funded but thin", and would count the 2028 deposit as the buffer that turns "funded" into "high certainty".
 - The answer probably changes one IPS sentence: certainty is reached when the ladder is bought *and* a stated buffer exists (the 2028 deposit, then the sleeve). It does not change the architecture.
- Would change: **sentence** (IPS certainty definition; FR evaluation of certainty). Possibly a **number**: a stated buffer target, e.g. ladder cost at most X% of assets.
- Deliverables: IPS, FR. Specialist: D8. Seed: 26.

- North star: shows Laura that "fully funded" was tested against the people who do this for a living, not declared.

Q2. Because the case forbids outside money for the payments (including "continued business income"), is Laura's "sponsor backing" effectively nil after 2028? If so, does practitioner logic (the UK code's "employer covenant", and the contrast with Australia's Future Fund) say the promise must be hedged rather than invested for growth?

- Anchors: R-C48, R-C44, R-AN24.
- Official words: "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement." (case p.3, VRF)
- Hypothesis (guess): yes.
 - UK DB Funding Code, para 29(a) (VP): risk must be "dependent on their assessment of the employer covenant, where more risk can be allowed for where the scheme has access to sufficient employer cash flows and contingent assets to support this level of risk".
 - Future Fund (VP, Treasurer's release): "the benchmark return rate will remain at between four and five per cent above CPI per annum over the long term". It pre-funds public-servant pensions that the Commonwealth still guarantees. Its drawdowns start "at least 2032-33", the same year as Laura's first payment.
 - One fund can chase growth because a government stands behind it. Laura's promise has nobody behind it. That is the one-line reason for hedging.
 - The Australian team angle earns its place here: it is a real, dated, checkable contrast, not decoration.
- Would change: **sentence** (the IPS "why this strategy fits her"; an FR alternatives paragraph).
- Deliverables: IPS, FR. Specialist: D8. Seed: 26.
- North star: explains why her plan is not a copy of any famous fund. It follows from her one hard constraint.

Q3. The UK regulator treats a pension with a liability duration of 10 years as "significantly mature" and expects it to be in a low-dependency, at-least-90%-hedged portfolio. Laura's promise has a duration of about 10 years from day one. Should the team use this outside yardstick to justify full hedging in 2027, instead of the gradual time-based de-risking that Wharton's own Trading Note example suggests?

- Anchors: F-105 (spot duration 10.16y), F-103 (9.90y at 2027-01-01), R-T25, R-I12, X-2.
- Official words: "This trade supports our plan to balance continued growth with reliable future cash flows as the residency funding date approaches." (TN guide p.2, VRF)
- Hypothesis (guess): yes, as Final Report evidence, never as IPS jargon.
 - DB Funding Code para 22(a) (VP): "For schemes with no cash balance benefits, the duration is 10 years."
 - Para 20 (VP): "we expect schemes to target a minimum level of interest rate and inflation hedging of at least 90%".
 - SI 2024/462 reg 5 (VP) defines the low-dependency allocation as "highly resilient to short-term adverse changes in market conditions so that further employer contributions are not expected to be required".
 - Laura's case is the no-employer version, so the logic applies more strongly.
 - Caveat: the UK measures duration on its own funding basis. Laura's figure is on Treasury rates. The claim should be "about 10 years, at the regulator's threshold", not an exact match.
- Would change: **decision** (confirms hedge-now over glide). **Sentence** in the FR on why the plan departs from the Wharton example's gradual path.
- Deliverables: FR, IPS. Specialist: D8 (with D1 for the duration basis). Seed: 26.
- North star: a stranger with authority (a regulator) reaches the same answer as our model, which is what a statistics graduate would want to see.

Q4. Should the IPS say the portfolio follows a funded-status glide path (it de-risks as the funded ratio rises), not a calendar glide path, and then say plainly what still changes over time?

- What still changes: the reserve pays itself out; the sleeve moves toward the 2031 floor.

- Anchors: R-I12, R-I21, R-C53, R-AN27, X-2, R-T25.
- Official words: "How will the portfolio's asset allocation and composition change as future funding needs approach and payments are made?" (IPS p.1, VRF)
- Hypothesis (guess):
 - Institutional glide paths are usually triggered by funded status. NISA (VP) writes of "the near ubiquity of glidepaths among our clients"; the pension de-risking literature (Russell, BlackRock; SNIP) describes funded-status triggers.
 - Laura starts at 102.6% funded, so a funded-status path says "hedge fully now".
 - The Wharton example assumes a calendar path. The IPS must answer the prompt ("changes as funding needs approach") without contradicting lock-early.
 - Answer: name the trigger (funded status) and list three planned changes: 2028 top-up, 2031 floor, post-2033 run-off.
- Would change: **sentence** (IPS answer to prompt 4). **Decision:** which three changes the IPS lists.
- Deliverables: IPS, TN. Specialist: D7 (designer intent) with D8. Seed: 28.
- North star: answers the IPS prompt head-on while keeping the promise safe. Most teams will write a calendar glide path by habit.

Q5. What fees will Laura pay on the reserve, and do they un-fund a ladder that has only \$7.7k of headroom? Should the IPS say the reserve is held as directly owned Treasuries and that all fees come out of the growth sleeve?

- Anchors: BS-01 (A3), F-104, F-107.
- Rules page (via A1, R-W29, VP): teams operate by the CFA Asset Manager Code. That Code requires fee disclosure (F.4.d, per A3).
- What is new vs BS-01: A3 modelled a fee on the sleeve and said the payments are safe "only as long as the fee is paid from the sleeve". Here is the size of the risk if it is not.
- Hypothesis (guess, ASM arithmetic):
 - Method: present value of a yearly fee f on the ladder $\approx f \times \Sigma(\text{rung cost} \times \text{years held}) = f \times \$2,933,301$ (F-107 rung costs; value grows at the curve).
 - Results: 0.07% $\rightarrow$ \$2.1k; 0.15% (IEF/TLT expense) $\rightarrow$ \$4.4k; 0.5% $\rightarrow$ \$14.7k; 1.0% $\rightarrow$ \$29.3k. Against \$7,736 headroom.
 - JPMorgan (Oct 2021, VP): "Hibernation strategies will burn through capital over time". Same mechanism.
 - In WInS, funds are fine (a six-week hedge). In Laura's real plan, the reserve should be individual Treasuries or STRIPS, with no fund expense, and the firm's fee charged to the sleeve.
- Would change: **number** (the funded ratio net of fees; the FR fee assumption). **Sentence** (an IPS cost principle).
- Deliverables: IPS, FR. Specialist: D3 (numbers), D10 (Code disclosure). Seed: null.
- North star: an honest firm shows Laura what it will cost her, and a rival that hides fees overstates her facility money.

Q6. Should the IPS adopt APRA SPS 160's two-part rule as its "rule if rates fall before January 2027"? The parts: (a) a stated "shortfall limit" (how far below 100% funded the ladder may start); (b) a "restoration plan" that returns the promise to 100% funded by a fixed date (1 Jan 2028, from the 2028 deposit first).

- Anchors: F-111, F-112 (about 24% chance the ladder costs more than \$300k on 2027-01-01, ASM), F-102, R-C34.
- Official words: "Apart from the two contributions described above, she will neither add to nor withdraw from the portfolio before 2033." (case p.2, VRF)
- Hypothesis (guess): yes.
 - SPS 160 (VP) defines a shortfall limit as the dip from which the fund can "reasonably expect ... to be restored to a satisfactory financial position within one year".
 - It requires a restoration plan "within a time period that is reasonable in the circumstances of the fund but which must not exceed three years".
 - For Laura the restoration deadline is shorter and fixed: the 2028 deposit is the only new money.

- The rule fits in about 25 IPS words and closes an open item (brief section 9). The council's "waterfall" already has the substance; SPS 160 gives it a recognised name and a trigger.
- Would change: **decision** (the rule's parameters: which rungs first, the trigger level). **Sentence** in the IPS.
- Deliverables: IPS, WInS-now (if rates fall during trading, the WInS book shows the same stress), FR. Specialist: D8 with D1. Seed: null.
- North star: shows Laura the team has a pre-written plan for the one thing that could go wrong before she even invests. That is how trustees, not traders, behave.

Q7. Can the plan borrow municipal finance's distinction between "economic defeasance" (Treasuries bought to cover the payments) and "legal defeasance" (a formal escrow checked by an independent verification agent)? It would explain "funded in 2027, set aside in 2033". Should the Final Report include a one-table "sufficiency check" (each rung's maturity value against each \$50k payment), the way a Verification Agent report does?

- Anchors: R-AN28, R-C50, R-C55.
- Official words: "explain how they evaluated that level of certainty" (case p.3, VRF)
- New evidence for brief section 8 item 9: a named institutional precedent for "bought early, formally set aside later".
- Hypothesis (guess): yes.
 - NABL (VP): "The Verification Agent is responsible for independently confirming that the investments purchased for the Escrow Fund will be sufficient to fund the Debt Service payments".
 - NABL (VP): "Once Bonds are legally defeased, they are generally considered no longer outstanding".
 - Laura's 2027 purchase is the economic step; the 2033 "set aside" is the formal step. The table is simple, can be checked by hand, and answers "how they evaluated" with no model.
- Would change: **sentence** (FR reconciliation). **Decision** (add the sufficiency table to the FR).
- Deliverables: FR. Specialist: D8, D9. Seed: 22.
- North star: a statistics graduate trusts a check she can redo in five minutes more than a simulated percentile.

Q8. Should the reserve be ring-fenced from 2027: held in a separate, labelled account, with a written rule that it pays only the ten payments, and a named person who can override the rule (Laura) and under what condition? Institutions use trusts and escrow accounts for exactly this.

- Anchors: R-C50, R-C26 ("with an asset management firm"), BS-18 (operational risk), BS-08 (governance continuity).
- Official words: "Laura will set aside a portion of the portfolio to fund the ten payments. This set-aside is called the operating reserve." (case p.3, VRF)
- Hypothesis (guess):
 - Yes, as a one-line governance rule. The legal form (trust, escrow) is out of scope, and Taiwan law is excluded.
 - What matters is that the money cannot drift into the facility when the facility runs over budget. That is the most likely real-world failure.
 - NABL (VP): escrow for defeased bonds is an "irrevocable Pledge".
- Would change: **decision** (IPS governance clause). **Sentence** (the FR co-sponsor draft can say the operating money is held separately).
- Deliverables: IPS, FR. Specialist: D10, D8. Seed: 2.
- North star: protects Laura from her own generosity under pressure, and co-sponsors can see the separation.

Q9. Where should "high degree of funding certainty" sit next to published institutional standards, and should the Final Report show that comparison at all? Standards: EU insurers must survive one year with 99.5% confidence; UK low-dependency pensions must be at least 90% hedged; planners' "confidence zones" of 75-90% are for adjustable goals.

- Anchors: R-C54, R-AN1, F-117.
- What is new vs brief section 8 item 6: verified external benchmarks to place the model-free definition against.

- Hypothesis (guess):
 - Solvency II Directive (VP): capital to meet obligations with "a confidence level of 99,5 % over a one-year period".
 - TPR code (VP): hedging "at least 90%".
 - The planner confidence zone of 75-90% is SNIP.
 - A bought, cash-flow-matched ladder sits above all three in nominal terms, because it is not a one-year probability.
 - The honest framing: "these standards are probabilities; ours is a purchase". The comparison belongs in the FR, one line, to stop a judge asking "why no 95%?". It does not belong in the IPS.
- Would change: **sentence** (FR certainty section).
- Deliverables: FR. Specialist: D8, D9. Seed: 22.
- North star: speaks the language of a statistician: which standard, over which horizon, measured how.

Q10. US public pensions discount their promises at their assumed return (median 7.0%, NASRA April 2026). At 7%, Laura's ten payments would "cost" about \$250k instead of \$292k. Should the Final Report show this \$42k gap to explain why growth-first plans under-reserve?

- Anchors: F-101, F-110, R-C77.
- Official words: "use reasonable return assumptions consistent with their strategy" (case p.4, VRF)
- New vs brief section 8 item 1: that item decided lock-early. This names the professional mistake behind the alternative and prices it.
- Hypothesis (guess, ASM arithmetic):
 - PV at 2027-01-01 of the ten payments at a flat 7%: \$250,386. At JPM's 6.7% equity return: \$257,485. At the Treasury curve: \$292,264.
 - The understatement is \$41.9k (14.3%) at 7%.
 - NASRA (VP): "the investment return assumption for most plans also serves as the discount rate used to determine the present value of the plan's liabilities" and "A rate set too high will understate liabilities".
 - One FR sentence, possibly one chart bar, makes the case for pricing the promise at Treasury rates without jargon.
- Would change: **number** and **sentence** (FR portfolio analysis; a chart candidate: three bars for the "cost of the promise" at 7%, 6.7% and the Treasury curve).
- Deliverables: FR. Specialist: D3, D8. Seed: null.
- North star: shows Laura the trap many professionals fell into, and that her plan avoids it on purpose.

Q11. Is the lottery annuity the plainest public precedent for "bought, not forecast"? Powerball funds annuity jackpots with U.S. Treasury securities, and "the higher the interest rates, the higher the advertised Grand Prize". Would using it with co-sponsors and judges pass the "no gimmick" test?

- Anchors: R-C83, R-S28, R-AN21.
- Official words: "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors" (SMApply criterion 5, VRF)
- Hypothesis (guess):
 - Useful only as a teaching line in the FR (or in the team's own learning), not in the co-sponsor draft. A lottery image sits badly next to a residency's credibility.
 - Marketplace (2022-11-08, VP) reports the annuity is paid "over a span of 30 years and accrues interest from investments in U.S. Treasury bonds".
 - It quotes the Powerball FAQ: "The higher the interest rates, the higher the advertised Grand Prize." (FAQ wording SNIP: not found on powerball.com today.)
 - The muni escrow (Q7) is the more dignified precedent.
- Would change: **sentence** (whether any everyday analogy appears, and which one).
- Deliverables: FR. Specialist: D9. Seed: null.
- North star: plain English that a creative founder would repeat to a partner. Also a warning against a flashy analogy she would find cheap.

B. The facility, the 2031 range and flexibility: how endowments and foundations do it (FR, some IPS)

Q12. Yale sets each year's spending from the endowment's value "from two years ago", so budgets are known in advance. Should the 2031 range be framed as a Yale-style lagged rule: known in 2031, applied in 2033, part locked, part smoothed? And would naming the precedent help co-sponsors trust it?

- Anchors: R-C62, R-AN7, R-C67, R-C69.
- Official words: "Laura plans to begin approaching potential co-sponsors in 2031, two years before the residency is established." (case p.3, VRF)
- Hypothesis (guess):
 - Yale Provost page (VP): "we apply the targeted spending rate (5.25%) to the endowment's year-end value from two years ago. Then we take 20% of that amount and add it to 80% of the total amount spent in the most recent fiscal year."
 - The two-year lag is a deliberate predictability device. The 2031 floor bought in a 2-year Treasury is the same idea, taken further (fully locked).
 - Likely use: one FR line citing the precedent. Maybe a rule design where the stretch is a stated share of the unlocked sleeve.
- Would change: **decision** (range rule design). **Sentence** (FR, co-sponsor section).
- Deliverables: FR. Specialist: D5, D8. Seed: 8.
- North star: co-sponsors hear a method used by a famous endowment, not a hope.

Q13. Does the case's word "responsibly" track the legal prudence standard for endowment spending (UPMIFA)? Its seven spending factors include "general economic conditions", "effect of inflation or deflation", "expected total return" and "other resources", almost the case's assumption list. Should the team define "responsible facility contribution" against those factors?

- Anchors: R-AN18, R-C85, R-C60, R-C84.
- Official words: "Teams should identify and explain their assumptions about investment performance, the timing of cash flows, outside funding, the effect of inflation on portfolio projections and facility costs, and the financial flexibility Laura should preserve." (case p.4, VRF)
- Hypothesis (guess):
 - NACUBO's UPMIFA summary (VP) lists: "1) duration and preservation of the endowment fund; 2) the purposes of the institution and the endowment fund; 3) general economic conditions; 4) effect of inflation or deflation; 5) the expected total return from income and the appreciation of investments; 6) other resources of the institution; and, 7) the investment policy of the institution."
 - The overlap with R-C85 is close (INT: probably not deliberate, but a ready-made checklist).
 - A definition built from these factors is defensible and short: "responsible = protects the reserve, allows for inflation, relies on stated returns, keeps other resources".
- Would change: **sentence** (FR definition of "responsible"; maybe an IPS clause).
- Deliverables: FR, IPS. Specialist: D8, D5. Seed: 28.
- North star: turns an undefined word into a standard that professionals and courts already use.

Q14. Should Laura's 2031 message have two legally distinct parts: an unconditional floor (already bought) and a conditional stretch (released only if the portfolio reaches a stated level)? Under US nonprofit accounting (ASU 2018-08), the residency can book the first as a pledge now and the second only when its condition is met.

- Anchors: R-C69, R-AN9, BS-05 (A3: which verb for floor vs stretch), R-C66.
- Official words: "If Laura promises more than she can ultimately contribute, she could damage her credibility" (case p.3, VRF)
- Hypothesis (guess):
 - Yes. It gives the floor/stretch split a precise meaning that a co-sponsor's finance officer already uses.
 - ASU 2018-08 (SNIP, several CPA-firm summaries): a conditional contribution needs "a barrier" and "a right of return" or release, and is "recognized when the condition or conditions are substantially met".

- Taiwan accounting differs, and Taiwan law is out of scope, so use the idea, not the standard's name, in the co-sponsor draft.
- Would change: **decision** (range structure). **Sentence** (FR fundraising draft checklist).
- Deliverables: FR. Specialist: D5 (with D10 for scope). Seed: 9.
- North star: co-sponsors can count her floor as real money, which is exactly what the case says a meaningful contribution should do.

Q15. Does the case allow the facility contribution to be committed in 2033 but paid in stages against construction milestones, as foundations pay capital grants? If so, does staging raise the responsible amount (the undrawn money earns Treasury yield and stays available for overruns) or change the 2031 range?

- Anchors: R-C57, R-C58, R-C63, R-AN19.
- Official words: "After establishing the operating reserve in 2033, Laura must decide how much of the remaining portfolio she can responsibly contribute toward the cost of establishing the residency facility." (case p.3, VRF)
- Hypothesis (guess):
 - The case fixes the decision in 2033, not the payment date. Staging looks allowed but is not stated (INT).
 - Milestone disbursement is standard grant practice (SNIP; not researched here).
 - If allowed, it lets "flexibility" be a mechanism (unpaid tranches) instead of a sized fund, which the case says teams need not size (R-C61).
 - Risk: it complicates the one-number story; the simplicity test may cut it to one FR sentence.
- Would change: **decision** (contribution timing assumption). **Sentence** (FR flexibility).
- Deliverables: FR. Specialist: D10 (case reading), D5. Seed: 11.
- North star: matches how a real building is paid for, which an entrepreneur would recognise at once.

Q16. Should the growth sleeve lock part of the facility floor whenever it crosses pre-set levels between 2028 and 2031, the way pension plans de-risk on funded-status triggers, instead of locking only in 2031? How would that change the 2031 floor distribution (p5/p50) and the chance the 2033 contribution lands inside the stated range?

- Anchors: F-402 (2031 floor p5/p50/p95 \$127k/\$165k/\$217k), R-C67 to R-C71, R-C70.
- Official words: "They should explain how favorable and unfavorable market outcomes could affect the amount she can provide." (case p.3, VRF)
- Hypothesis (guess):
 - Ratcheting lifts the p5 floor and cuts the median slightly. It is "lock in good years", the practitioner answer to "what if 2029 is great and 2030 crashes?".
 - Not yet modelled. The council's ratchet was for the liability hedge, not the facility floor.
 - It may fail the simplicity test if the gain is under ~\$5-10k at p5.
- Would change: **number** (2031 floor p5/p50; confidence of landing within the range). **Decision** (keep one 2031 lock, or ratchet).
- Deliverables: FR (IPS sentence only if adopted). Specialist: D3. Seed: 9.
- North star: shows the plan protects Laura's good luck as well as her promise.

Q17. Should "financial flexibility" after 2033 be written as a draw rule on the leftover money, not as a dollar size? For example: released first for facility overruns, then capped at X% a year or smoothed, endowment-style (Yale's 4.0-6.5% band, UPMIFA's 7% presumption).

- Anchors: R-C58, R-C61, R-AN19, R-C84.
- Official words: "Teams are not expected to determine the size or investment composition of a separate contingency fund or endowment." (case p.3, VRF)
- Hypothesis (guess):
 - A rule satisfies "preserve flexibility" without sizing a fund.
 - NACUBO (VP): UPMIFA's optional "rebuttable presumption of imprudence if an institution expends an amount greater than seven percent of fair market value of a fund, calculated in an averaging formula over three years".

- Yale's 4.0%/6.5% band is SNIP (a search snippet of the Yale page; the Provost page I read gives only the 5.25% rate).
- Likely one FR sentence.
- Would change: **decision** (how flexibility is expressed). **Sentence** (FR).
- Deliverables: FR. Specialist: D5, D8. Seed: 7, 11.
- North star: gives Laura room to adapt "as the project develops" with a guard rail she can explain to partners.

C. Governance, behaviour and reporting (IPS, FR, TN, WInS-now)

Q18. Which governance elements from professional IPS templates must the 500-word IPS carry, and which can be cut? Candidate elements: who decides and who monitors; dated review points (2028 deposit, 2031, 2033); event triggers (funded ratio below 100%, deposit short or late, residency delayed); a rebalancing process. The UK requires pension investment statements to be reviewed "at least every three years" and "without delay after any significant change".

- Anchors: R-I3, R-C89, X-23, R-AN4, BS-08.
- Official words: "It helps investors stay focused on long-term objectives and make disciplined decisions during changing market conditions." (IPS p.1, VRF)
- Hypothesis (guess):
 - CFA Institute, Elements of an IPS for Individual Investors (2010, VP): "2b. Describe the process for reviewing and updating the IPS." and "4c. Define the process for rebalancing portfolios to target allocations."
 - UK Occupational Pension Schemes (Investment) Regulations 2005 reg 2 (VP): review "(a)at least every three years; and (b)without delay after any significant change in investment policy."
 - The IPS probably needs just three governance lines: event-based review triggers, one rebalancing rule, and "the team recommends; the portfolio manager signs off; students decide in WInS" (matching the real rules, R-W19).
- Would change: **decision** (which clauses get IPS words).
- Deliverables: IPS. Specialist: D10, D8. Seed: 2.
- North star: an IPS that still works when nobody from our firm is in the room is what a client hires.

Q19. What would a private banker pre-commit to for a 2029 bear market? And does holding the reserve as individual bonds (not bond funds) remove the biggest behaviour trap? Morningstar 2025 finds bond-fund investors captured only about half of their funds' returns.

- Anchors: R-I3, R-C36, R-C37.
- Official words: "Laura understands that investing involves uncertainty and periods of market volatility." (case p.2, VRF)
- Hypothesis (guess):
 - Morningstar Mind the Gap 2025 (VP): the average dollar earned "7.0% per year" vs funds' "8.2%", a "1.2 percentage point 'investor return gap'".
 - Same study: "taxable-bond and municipal-bond fund investors earned the smallest share of their funds' aggregate total returns, capturing only about half".
 - Same study: allocation-fund investors captured "nearly 97%".
 - Vanguard 2022 (VP): behavioural coaching worth "0 to > 200" bps.
 - The pre-commitments would be: report the promise's funded status first; never move money between the hedge and the sleeve; rebalance the sleeve inside a band; hold the reserve as individual Treasuries so a bond-price fall is never a "loss" she can act on.
 - Morningstar also recommends "slightly haircutting projections" for this drag. That is a candidate FR assumption.
- Would change: **decision** (IPS behaviour rule). **Number** (a possible haircut on sleeve return assumptions). **Sentence** (FR).
- Deliverables: IPS, FR. Specialist: D6. Seed: 27.
- North star: Laura hires a firm for the bad year, and this shows we planned her bad year in advance.

Q20. From the first WInS trade, should the team keep a weekly "hedge effectiveness" log? It would record the Treasury sleeve's value change against the change in the ten payments' present value

(spot curve, about \$289 per basis point). A reflection could then say: "rates rose X bp, our bonds fell \$Y, the promise's cost fell \$Z, funded status held".

- Anchors: R-T20, R-T9, F-103, F-105, R-W76, R-W82.
- Official words: "Your reflections should clearly explain how each decision supported, tested, or refined your overall investment strategy." (TN guide p.1, VRF)
- Hypothesis (guess):
 - Yes; this is the highest-value WInS-now item in this file.
 - Pension CIOs report funded status and hedge ratio, not return (NISA and JPM papers, VP, frame everything in funded status).
 - Treasury funds may lose money during WInS (the 10y is at its highest since 2007, F-004). Without the log, a note looks like a loss. With it, the same note is evidence the hedge worked.
 - Needs: the daily par curve (treasury.gov, reachable), the verified PV method (`official_curve_pv.py`), and WInS position values. D1/D3 can script it.
- Would change: **decision** (start the log now). **Sentence** (TN reflections; FR "how effectively it implemented that strategy", R-I37).
- Deliverables: WInS-now, TN, FR. Specialist: D1, D3. Seed: 30.
- North star: proves in six weeks, with real data, that the plan behaves as promised. Nobody can do that with a growth-first book.

Q21. Should the Final Report report risk as "expected negative years out of 20", as Australian super funds must (the Standard Risk Measure)? It would also say why this label and volatility both misstate the reserve's risk: a held-to-maturity ladder can show negative years while every payment stays certain.

- Anchors: R-S26, R-S28, R-C82.
- Official words: "uses reasonable assumptions and projections to evaluate funding reliability, the facility contribution, and financial flexibility under varying market outcomes" (criterion 3, VRF)
- Hypothesis (guess):
 - ART (VP): the SRM is "based on the number of expected years of negative annual returns over any 20-year period".
 - ASM arithmetic with JPM inputs (F-301, F-307, F-314; lognormal): 100% U.S. large cap $\approx$ 6.7 negative years in 20; the 60/40 sleeve $\approx$ 5.4; intermediate Treasuries $\approx$ 2.5.
 - SRM bands (1 = fewer than 0.5 years ... 7 = 6 or more) are SNIP; the guidance PDF refused access.
 - Useful as one plain-English line for Laura and co-sponsors. The contrast (a market-value measure vs payment certainty) is itself an insight about what "risk" means for her.
- Would change: **number** and **sentence** (FR uncertainty communication).
- Deliverables: FR. Specialist: D9, D3. Seed: 18.
- North star: reports uncertainty in words a non-investor understands, and shows why the usual risk number would mislead her.

Q22. Australian law (the Retirement Income Covenant, 2022) makes super trustees "achieve and balance" three objectives: maximise expected income, manage risks to its "sustainability and stability", and keep "flexible access". These are Laura's growth, funding certainty and flexibility. Should the IPS's central idea state a priority order among the three (promise first, then flexibility, then growth), not a "balance"? And would naming the covenant as the team's inspiration (in the Articulation section) earn its place?

- Anchors: R-I11, R-I26, R-C38, X-16.
- Official words: "How does your strategy balance growth, risk, liquidity, funding reliability, and financial flexibility?" (IPS p.1, VRF)
- Hypothesis (guess):
 - Treasury's explanatory memorandum (exposure draft 2021, VP): trustees must "assist beneficiaries to achieve and balance three objectives: maximizing their expected retirement income; managing expected risks to the sustainability and stability of their expected retirement income; and having flexible access to expected funds during retirement."
 - The enacted wording is not checked (AustLII 403); SNIP summaries match.

- A strict order is stronger than a balance, because the case itself orders things (the reserve comes "before" the payment and the facility, R-AN2).
- The covenant is a genuine, dated Australian parallel for the Final Report's story of how the team thought. Leave it out of the IPS.
- Would change: **sentence** (IPS central idea; FR Articulation).
- Deliverables: IPS, FR. Specialist: D8, D7. Seed: 28.
- North star: our home country's law on long promises supports Laura's order of priorities. That is an authentic team-voice detail no other team has.

Q23. Chhabra's framework says "risk allocation should precede asset allocation" across personal, market and aspirational risk. Is Laura's aspirational-risk bucket already full (her creative business, future books, the residency itself)? If so, should the portfolio hold only personal-risk assets (the reserve) and market-risk assets (a broad sleeve), and refuse aspirational-style bets (themes, single stocks, country tilts)?

- Anchors: R-C37, R-AN16, R-C27, R-AN31.
- Official words: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career" (case p.2, VRF)
- Hypothesis (guess):
 - Yes. This challenges the team notes' mapping of "flexibility = aspirational bucket".
 - Chhabra (2005, Journal of Wealth Management; SNIP abstract via search): personal, market and aspirational risk; "risk allocation should precede asset allocation".
 - Her entrepreneurial risk lives outside the portfolio, and her 2028 income depends on it. So the portfolio should not add more of the same kind of risk. This is the institutional mirror of the human-capital argument (B3 lens), stated in a framework a private-bank judge would recognise.
- Would change: **decision** (no thematic or Taiwan/AI tilts in WInS or the plan). **Sentence** (IPS principle).
- Deliverables: WInS-now, IPS, FR. Specialist: D6, D2. Seed: 17, 20.
- North star: "you already take the big bets in your life; our job is to make sure they never threaten your promise". That reads as her, not generic.

Q24. Which named frameworks map to which part of the plan, and which (if any) should be named in the IPS? Candidates: liability-driven cash-flow matching (reserve), funded-status de-risking (timing), goals-based buckets (reserve / sleeve), Yale-style lagged rules (2031 range), UPMIFA prudence (facility, flexibility).

- Anchors: R-S26 ("effective use of investment concepts and tools"), R-I23, R-S24.
- Official words: "Focus on the strategy and decision-making framework that guide your portfolio rather than describing individual investments or presenting detailed financial calculations." (IPS p.2, VRF)
- Hypothesis (guess):
 - The IPS names none of them and describes each in plain words. The council judge panel already warned against "LDI" and "defeasance" in the IPS.
 - The FR carries one mapping table (a chart-free visual) showing each rule's professional ancestor.
 - This shows "effective use of investment concepts" without jargon in the scored 500 words.
- Would change: **decision** (naming policy per deliverable). **Sentence**.
- Deliverables: IPS, FR. Specialist: D8, D9. Seed: 28.
- North star: judges see professional depth in the FR, while Laura reads an IPS in her own language.

Q25. Would a pension CIO at least consider buying the promise from an insurer (a deferred 10-year period-certain annuity, the "buyout" option)? And what one reason rejects it for Laura?

Candidate reasons: insurer credit risk above Treasury risk; permitted-investment wording "for BOTH contributions"; no liquidity if the residency changes; less transparency.

- Anchors: R-AN15, R-C74, R-C73.
- Official words: "Teams may reach different conclusions about risk, asset allocation, liquidity, return expectations, funding confidence, the facility contribution, and the financial flexibility Laura should preserve." (case p.4, VRF)
- Hypothesis (guess):

- Rejected. The SMaply FAQ lists permitted investments "for BOTH contributions" (R-AN15; reading 1 = the list binds the plan). An annuity is not on it.
- A Treasury ladder can be sold if the residency is delayed or changes (BS-07). An annuity usually cannot.
- Worth one FR line under "alternatives considered". It pre-empts the obvious "why not just buy an annuity?".
- Would change: **sentence** (FR alternatives).
- Deliverables: FR. Specialist: D8, D10. Seed: 26.
- North star: shows the team chose Treasuries over a real alternative for Laura-specific reasons: flexibility, and certainty that does not depend on an insurer.

3. Evidence and arithmetic

3.1 Primary-source quotes used (all read 2026-09-27)

#	Exact text	Source	Status
E1	"For schemes with no cash balance benefits, the duration is 10 years."	TPR, Defined benefit funding code of practice, Nov 2024, "Significant maturity" para 22(a)	VP
E2	"we expect schemes to target a minimum level of interest rate and inflation hedging of at least 90% on a low dependency funding basis."	same code, para 20	VP
E3	"dependent on their assessment of the employer covenant, where more risk can be allowed for where the scheme has access to sufficient employer cash flows and contingent assets to support this level of risk"	same code, journey planning para 29(a)	VP
E4	"Low dependency investment allocation means the assets of a scheme are invested in such a way that the value of the assets relative to the value of the scheme's liabilities is highly resilient to short-term adverse changes in market conditions so that further employer contributions are not expected to be required to make provision for the scheme's liabilities."	SI 2024/462 reg 5 (legislation.gov.uk)	VP
E5	"a 'shortfall limit' is the extent to which an RSE licensee considers that a fund can be in an unsatisfactory financial position with the RSE licensee still being able to reasonably expect that, because of corrections to temporary negative market fluctuations in the value of fund assets, the fund can be restored to a satisfactory financial position within one year."	APRA SPS 160 (APRA handbook)	VP
E6	"within a time period that is reasonable in the circumstances of the fund but which must not exceed three years from the valuation date"	APRA SPS 160	VP
E7	"the Government won't start any drawdowns from the Fund until at least 2032-33" / "the benchmark return rate will remain at between four and five per cent above CPI per annum over the long term"	Treasurer's media release "Future of the Future Fund" (date not shown in the captured text; search indicates late Nov 2024, SNIP)	VP (text)
E8	"confidence level of 99,5 % over a one-year period."	Directive 2009/138/EC (Solvency II), EUR-Lex HTML	VP
E9	"The Verification Agent is responsible for independently confirming that the investments purchased for the Escrow Fund will be sufficient to fund the Debt Service payments on the Refunded Bonds"	NABL, Bond Basics: Verification Agent	VP
E10	"Once Bonds are legally defeased, they are generally considered no longer outstanding from the Issuer's perspective."	NABL, Bond Basics: Defeasance	VP
E11	"To determine the year's spending, we apply the targeted spending rate (5.25%) to the endowment's year-end value from two years ago. Then we take 20% of that amount and add it to 80% of the total amount spent in the most recent fiscal year."	Yale Office of the Provost FAQ	VP
E12	"1) duration and preservation of the endowment fund; 2) the purposes of the institution and the endowment fund; 3) general economic conditions; 4) effect of inflation or deflation; 5) the expected total return from income and the appreciation of investments; 6) other resources of the institution; and, 7) the investment policy of the institution."	NACUBO, UPMIFA topic page	VP

#	Exact text	Source	Status
E13	"a rebuttable presumption of imprudence if an institution expends an amount greater than seven percent of fair market value of a fund, calculated in an averaging formula over three years."	NACUBO, UPMIFA topic page	VP
E14	"Because the investment return assumption for most plans also serves as the discount rate used to determine the present value of the plan's liabilities" ... "A rate set too high will understate liabilities" ... "the median return assumption declined from 8.0 percent to 7.0 percent. The median rate has remained at that level since FY 21"	NASRA Issue Brief, April 2026	VP
E15	"for plans that are greater than 95% funded[2], we see a compelling case to move to a hibernation portfolio – now."	NISA, "95% is the new 105%" (c. 2021)	VP
E16	"Hibernation strategies will burn through capital over time, squandering hard-won pension funding gains."	J.P. Morgan AM, "Rethinking the pension plan endgame", Oct 2021	VP
E17	"That 1.2 percentage point 'investor return gap'" / "taxable-bond and municipal-bond fund investors earned the smallest share of their funds' aggregate total returns, capturing only about half." / "slightly haircutting projections"	Morningstar, Mind the Gap 2025 (US)	VP
E18	"Behavioral coaching 0 to > 200" (bps)	Vanguard, "Putting a value on your value: Quantifying Vanguard Advisor's Alpha" (2022)	VP
E19	"2b. Describe the process for reviewing and updating the IPS." / "4c. Define the process for rebalancing portfolios to target allocations."	CFA Institute, Elements of an IPS for Individual Investors (2010)	VP
E20	"(a)at least every three years; and (b)without delay after any significant change in investment policy."	UK SI 2005/3378 reg 2	VP
E21	"The Standard Risk Measure assigns a Risk Label from Very low to Very high, and a corresponding Risk Band from 1 to 7 for each option, based on the number of expected years of negative annual returns over any 20-year period."	Australian Retirement Trust, SRM page	VP (industry page, not regulator)
E22	"assist beneficiaries to achieve and balance three objectives: maximizing their expected retirement income; managing expected risks to the sustainability and stability of their expected retirement income; and having flexible access to expected funds during retirement."	Treasury (Aus), Explanatory Memorandum, exposure draft, Sept 2021	VP (exposure draft; enacted wording SNIP)
E23	"The higher the interest rates, the higher the advertised Grand Prize. ...," says an FAQ on the website for the game." / "accrues interest from investments in U.S. Treasury bonds"	Marketplace, 2022-11-08	VP (news page); Powerball FAQ wording SNIP

3.2 Arithmetic (ASM; inputs verified)

- **Fee drag on the reserve (Q5).** Rung costs at 2027-01-01 (F-107) × years to payment (6..15): $\Sigma = \$2,933,301$ (implied Macaulay duration 10.04y). PV of a yearly fee $f \approx f \times \$2,933,301$ (continuous approximation; the fee is charged on the market value, which grows at the curve).
 - $f = 0.07\%$ (a defined-maturity Treasury fund, F-207): \$2,053
 - $f = 0.15\%$: \$4,400
 - $f = 0.5\%$: \$14,667
 - $f = 1.0\%$: \$29,333
 - Compare the headroom of \$7,736 (F-104).
 - **Discount-rate comparison (Q10).** PV at 2027-01-01 of $10 \times \$50k$ (2033-2042) at a flat 7%: \$250,386; at 6.7%: \$257,485; at 5.36% flat: \$292,245 ($\approx$ F-101 \$292,264, as F-109 says). The 7% understatement vs the curve is \$41,878 (14.3%).
- **SRM-style counts (Q21).** $P(\text{negative calendar-year return}) \times 20$, lognormal:
 - U.S. large cap (6.70% compound, log sd 0.152 per F-316): $P = 0.335 \rightarrow 6.7$ of 20.
 - Intermediate Treasuries (4.0%, 3.4%): $P = 0.124 \rightarrow 2.5$.
 - 60/40 sleeve (arithmetic 6.39%, sd 9.97%, correlation -0.01): $P = 0.269 \rightarrow 5.4$.
 - These are model properties, not forecasts.
- **Duration vs UK threshold (Q3).** Macaulay $\approx 10.04y$ at 2027 (above); modified 9.90y (F-103); spot 10.16y (F-105). The UK threshold is 10 years (E1), measured on its own funding basis, so the comparison is approximate.

4. Sources (all accessed 2026-09-27)

Source	URL	Status
TPR DB funding code of practice (Nov 2024)	https://www.thepensionsregulator.gov.uk/-/media/thepensionsregulator/files/import/pdf/db-funding-code-of-practice.ashx	VP
UK SI 2024/462 reg 5	https://www.legislation.gov.uk/uksi/2024/462/regulation/5/made	VP
UK SI 2005/3378 reg 2 (SIP review)	https://www.legislation.gov.uk/uksi/2005/3378/regulation/2	VP
APRA SPS 160 Defined Benefit Matters	https://handbook.apra.gov.au/standard/sps-160	VP
Retirement Income Covenant EM (exposure draft)	https://treasury.gov.au/sites/default/files/2021-09/c2021-209553-explan_memo.pdf	VP
Retirement Income Covenant enacted text (AustLII)	http://www6.austlii.edu.au/cgi-bin/viewdoc/au/legis/cth/num_act/ccivfaoma2022678/sch9.html	Refused (403)
Future Fund release (Treasurer)	https://ministers.treasury.gov.au/ministers/jim-chalmers-2022/media-releases/future-future-fund	VP
Future Fund (Wikipedia, background only)	https://en.wikipedia.org/wiki/Future_Fund	VP (secondary)
Future Fund Investment Mandate Direction 2024 explanatory statement	https://www.finance.gov.au/sites/default/files/2024-11/F2024L01477ES.pdf	Fetch returned almost no text; not verified
Solvency II Directive 2009/138/EC	https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32009L0138	VP
NABL Verification Agent / Defeasance	https://www.nabl.org/bond-basics/verification-agent/ ; https://www.nabl.org/bond-basics/defeasance/	VP
Yale Provost spending FAQ	https://provost.yale.edu/how-exactly-does-spending-policy-work	VP
NACUBO UPMIFA	https://www.nacubo.org/Topics/Endowment-Management/Uniform-Prudent-Management-of-Institutional-Funds-Act	VP
NASRA return assumptions brief (Apr 2026)	https://www.nasra.org/files/Issue%20Briefs/NASRAInvReturnAssumptBrief.pdf	VP
NISA "95% is the new 105%"	https://www.nisa.com/perspectives/95-is-the-new-105-why-plans-should-consider-accelerating-their-glidepath/	VP
JPM "Rethinking the pension plan endgame" (Oct 2021)	https://am.jpmorgan.com/content/dam/jpm-am-aem/global/en/institutional/insights/portfolio-insights/portfolio-strategy/Rethinking-the-pension-plan-endgame.pdf	VP
Morningstar Mind the Gap 2025	https://www.morningstar.com/content/cs-assets/v3/assets/blt9415ea4cc4157833/blt2c5c4d9171638c42/689b424311f3880edc4b4813/US_Mind_the_Gap_2025.pdf	VP
Vanguard Advisor's Alpha 2022	https://www.vanguardsouthamerica.com/content/dam/intl/americas/documents/latam/en/2022/08/mx-sa-2335954-putting-a-value-on-your-value-quantifying-vanguard-advisors-alpha.pdf	VP
CFA Institute IPS elements (individual)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf	VP
CFA Institute IPS elements (institutional)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-institutional-investors.pdf	SNIP (not opened)
ART Standard Risk Measure	https://www.australianretirementtrust.com.au/disclaimers-and-disclosures/standard-risk-measure	VP
FSC-ASFA SRM guidance 2011	https://www.superannuation.asn.au/wp-content/uploads/2023/09/FSC-ASFA_StandardRiskMeasures_July2011.pdf	Refused (403)
Marketplace on Powerball and Treasuries (2022-11-08)	https://www.marketplace.org/story/2022/11/08/why-high-interest-rates-are-partly-responsible-for-the-2-04-powerball-jackpot	VP
ASU 2018-08 summaries (CPA Journal, JMCO)	https://www.cpajournal.com/2019/05/10/fasbs-new-guidance-for-contributions-received-and-contributions-made/ ; https://www.jmco.com/articles/nonprofit/conditional-contributions-asu-2018-08/	SNIP
Chhabra, Beyond Markowitz (2005)	https://www.ssrn.com/abstract=925138	SNIP (SSRN blocked)
Planner "confidence zone" 75-90%	search results (T. Rowe Price, MoneyGuidePro manuals)	SNIP

Source	URL	Status
Pension glide-path guides (Russell, Cambridge Associates, BlackRock)	https://russellinvestments.com/-/media/files/us/insights/institutions/defined-benefit/pension-de-risking-glide-paths.pdf ; https://www.cambridgeassociates.com/insight/liability-hedging-handbook-a-guide-to-best-practices-for-us-pension-plans/	SNIP (not opened)

5. Leads for the specialists

- **D1 / D3 (rates, quant): Q20 hedge-effectiveness log.** Inputs: the daily par curve CSV (home.treasury.gov, reachable), the method in `research/verified_2026-09-27/official_curve_pv.py`, and WInS position values. Output: a weekly table of Δ (Treasury sleeve value), Δ (PV of the ten payments, spot, \$289/bp now) and the funded ratio. Start with the first trade date.
- **D3: Q5 fee drag.** Replace my continuous approximation with year-by-year fees on the forward value path (F-106 values). Report net-of-fee funded ratios for 0.07/0.15/0.5/1.0%.
- **D3: Q16.** Add funded-status triggers to the sleeve in `strategy_mc.py`: lock X% of the sleeve into Treasuries maturing Jan 2033 whenever the sleeve exceeds pre-set levels, 2028-2031. Compare the 2031 floor p5/p50 and P(2033 contribution within the range) with the single 2031 lock (F-402).
- **D8: Q2/Q3.** The TPR code's "Significant maturity" and "Low dependency investment allocation" modules are the cleanest outside benchmark. Paragraph numbers: 17-22 (LDIA; hedging "at least 90%") and 22(a) of the maturity module (10 years). The "up to 15% growth assets" figure is only in a search snippet (Gateley summary of the consultation); verify it in TPR's "Regulatory approach" document before use.
- **D8 / D1: Q6.** SPS 160 paragraphs on "Shortfall limit" and "Unsatisfactory financial position" (APRA handbook). SPG 160 (the practice guide PDF on apra.gov.au) gives worked shortfall-limit examples; not read.
- **D8: Q7.** NABL bond-basics pages (reachable). If a real escrow verification report is wanted as a template, the City of Austin 2021 escrow agreement came up in search (services.austintexas.gov); not opened.
- **D5: Q12/Q14/Q17.** Yale Provost FAQ (VP). The Yale slide deck "Introduction to the Endowment Spending Policy" (your.yale.edu, Nov 2024) holds the 4.0%/6.5% band; not opened. NACUBO UPMIFA page (VP). For ASU 2018-08, read the FASB standard or a Big-4 guide before quoting.
- **D6: Q19.** Morningstar Mind the Gap 2025 PDF (VP): the investment-style section has the bond-fund "about half" finding. Vanguard 2022 Advisor's Alpha PDF (VP): the behavioural coaching section cites Bennyhoff (2018).
- **D9: Q21.** ART page (VP) for the SRM definition. The band thresholds need the FSC-ASFA 2011 guidance (403 here); try an APRA or fund PDS copy.
- **D7: Q4/Q22.** Wharton's own TN example (R-T23 to R-T25) implies a calendar glide path. Check whether past Wharton cases or guides used "glide path" wording, to judge whether the designers expect it.
- **D10: Q15/Q25.** Case reading on staged facility payments (R-C57, R-C63). The "for BOTH contributions" FAQ line (R-AN15) decides whether an insurer annuity is even in scope.
- **Blocked:** the Future Fund site and AustLII refuse (403). The Treasurer's release and the Treasury EM are good substitutes.

What this teaches

- Institutions that owe fixed amounts ask two questions before choosing investments: "Is the promise already paid for at market prices?" and "Who pays if we fall short?" When the answer to the second is "nobody", as for Laura, professionals hedge, and they take risk only with the surplus.
- The same idea shows up under many names: pensions ("low dependency", "hibernation"), insurers ("99.5% over one year"), municipal finance ("defeasance" with a "verification agent"), endowments ("spending rules") and Australian super ("shortfall limit", "restoration plan", the three-way "balance"). Recognising one idea in five disguises is what makes a framework feel familiar to a judge.
- Small costs matter most when the margin is small: a fee that looks tiny (0.15%) can eat more than half of a thin safety buffer over ten years.

- How you measure risk changes what you see. A bond held to maturity can "lose money" on paper in a year while its payment stays exactly as promised.

B4b Top-Practitioner Benchmark: questions from private wealth practice

Agent: B4b, insight_v1 run, Phase B (question generation). Written 2026-09-27. Lens: "Top-Practitioner Benchmark". Starting angle: private wealth practice (goals-based wealth management, the wealth allocation framework, private-bank coaching, probability-of-goal reporting, IPS governance). The other B4 agent starts from institutions; I touch pensions only where private-wealth evidence pointed there.

This is AI-generated research (Claude Code) for brainstorming. It holds **no submission-ready prose**. Where a question says "a sentence must contain", that is a checklist item; the six students write every word. Any practitioner source named here must be cited in the Final Report's Works Cited, together with how AI was used (R-W46).

0. Summary (read this first)

- 1. Our architecture is what top practice now recommends, and there is a fresh 2026 source that says so.** Russell Investments (13 July 2026) describes a "surplus glidepath": "lock down the benefits promised, then grow surplus assets with a prudent process". It adds that "any surplus strategy should preserve benefit security first, with additional risk taken only on assets above a defined surplus threshold" (VERIFIED-PRIMARY). That is lock-early plus a growth sleeve in a pension professional's words. The team can cite a named 2026 practitioner rather than asserting "this is what professionals do".
- 2. The best-known critique of full hedging does not apply to Laura, and saying so is a rare, strong move.** J.P. Morgan Asset Management (Oct 2021) argues against pension "hibernation" (full hedging). Its reasons are credit downgrades of corporate bonds (about 50bp a year), rising longevity and "experience losses" (VERIFIED-PRIMARY). None of these exists for ten fixed nominal payments matched with U.S. Treasuries on known dates. Laura's promise is the textbook case where full hedging works as designed (Q07).
- 3. The team's own notes mislabel Chhabra's wealth allocation framework.** They put the "aspirational" bucket on the flexibility money. In Chhabra's framework the aspirational bucket holds concentrated, high-risk bets such as "owning a small business, or leveraging unique human capital" (SNIPPET-UNVERIFIED). Laura's own creative business is her aspirational bucket, and so is the residency itself. A judge who knows the framework would spot the misread. The corrected mapping is also a Laura-specific reason for the portfolio's moderate risk (Q04).
- 4. Brunel's goal-probability table gives a practitioner scale for "certainty" and for the 2031 confidence.** "Needs" sit at 90-95%, "wants" at 80-85%, "wishes" at 65-75% and "dreams" at 50-60% (Das et al. 2018, VERIFIED-PRIMARY). Our payments sit above even the "needs" level, because they are bought, not forecast. With a bought floor, the only way to land outside the co-sponsor range is to land *above* it. Our confidence figure therefore equals the percentile we choose for the top of the range, so a "wants"-level 80-85% gives a principled top end (Q09, Q15).
- 5. The CFA Institute's own IPS standard supplies the governance and rebalancing items the case never states.** The case names no rebalancing rule, review cycle or risk metric (R-AN17, R-AN34). The CFA document says "If the policy is not to rebalance, this policy should be documented in the IPS". It also warns against "different metrics to highlight or disguise certain risks" (VERIFIED-PRIMARY). Both fit in one clause each of the 500 words (Q01, Q12, Q18).
- 6. WInS now: a pre-written rebalancing band turns market noise into a "tested" trading note.** The Trading Notes guide asks how decisions "supported, tested, or refined" the strategy (R-T20). If the team writes a band rule before the first trade, any market move during the WInS window produces a rule-driven trade. That trade is concrete evidence of discipline, and few teams will have one (Q01).
- 7. Two behavioural facts need care with a statistics-trained client.** Vanguard values "behavioural coaching" at about 150bp a year (VERIFIED-PRIMARY). Morningstar measures a 1.2 percentage-point "investor return gap" (VERIFIED-PRIMARY). But a 2026 *Financial Analysts Journal* paper finds that poor timing costs "only 0.10% per year" on the same sample (VERIFIED-PRIMARY abstract). Quoting the big

numbers as fact would be the kind of unexamined statistic Laura would reject. The case for pre-commitment rules should rest on her own decision dates instead (Q19).

Question count: 25 (Q01-Q25). By deadline: WInS-now/TN 4, IPS 10, FR 11. By domain: D8 11, D3 3, D1 3, D6 3, D9 2, D10 2, D7 1.

1. How to read this file

- **Anchors:** R-ids from `phase_A/case_register.md`, F-ids from `phase_A/fact_register.md`, BS/SH ids from `phase_A/stakeholder_map.md`, G ids from `phase_A/wins_week1_guardrails.md`.
- **Status labels** (brief section 3): VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SNIP), ASSUMPTION (ASM), PARAPHRASE-UNVERIFIED (PARA). Every "hypothesis" below is **a guess**, labelled as such.
- **Seeds** (v4 Part 6) for this lens: 22, 26, 27, 28. Where a question extends another seed, that seed is named.
- **Not re-asked** (brief section 8): lock-early vs growth-first, tiering, surplus-based risk, curve pricing, "equity buys range", model-free certainty, bought floor, which number is the reserve, TIPS, STRIPS. Q07, Q08, Q09 and Q10 touch the lock decision only through **new external evidence**, and each says what is new.

Terms used below

- **Goals-based wealth management (GBWM):** splitting a client's money into separate sub-portfolios, one per goal, each with its own time horizon and required chance of success. "Risk" becomes the chance of missing a goal.
- **Wealth allocation framework (WAF):** Ashvin Chhabra's version (2005). It has three "buckets": personal/safety (protect from disaster), market (diversified market risk) and aspirational (concentrated bets for a big upside).
- **Liability-driven investing (LDI):** investing so that assets move like a known future obligation. **Hibernation:** an LDI end-state where a fully funded pension hedges its obligations completely.
- **Funded ratio:** assets divided by the market price of the promised payments (F-117: 1.026).
- **Surplus:** assets above the price of the promise.
- **Rebalancing band:** a pre-set range around a target weight. A trade happens only when the weight leaves the band.
- **Capacity for loss:** the UK regulator's term for how big a fall in value a client can absorb without harming their standard of living. **Risk tolerance / attitude to risk** is how much risk the client is *willing* to take.
- **AUM fee:** a yearly fee charged as a percentage of assets under management.
- **Trigger event:** a named event that forces a review of the plan.

2. The questions

Each entry gives: the question; anchors; the exact official words or fact; the hypothesis (a guess) and evidence; what would change; deliverables; domain; seed; and the North Star link.

A. WInS now and the Trading Notes (tier 1)

Q01. Should the team write a rebalancing band for the WInS book before the first trade, so that a market move produces a rule-driven trade whose note shows the strategy being "tested"?

- Anchors: R-T20, R-I3, R-AN17, R-W73, G (day-trading ban, guardrails section 0 item 5).
- Official words: "Your reflections should clearly explain how each decision supported, tested, or refined your overall investment strategy." (TN p.1 L29-30, VRF). "It helps investors stay focused on long-term objectives and make disciplined decisions during changing market conditions." (IPS p.1 L4-5, VRF).
- Hypothesis (guess):

- Yes. A simple band would do: equity share of the book outside target ± 5 percentage points, checked once a week, trades never on the same U.S. day as a buy of that fund. It costs nothing to write and creates a note-worthy trade whenever markets move.
- CFA Institute: "Outside boundaries of acceptable variations from targets ... should be documented in the IPS ... If the policy is not to rebalance, this policy should be documented in the IPS." (VP, CFA Institute, *Elements of an IPS for Individual Investors*, May 2010, section 4c).
- Vanguard's rebalancing research: annual or semiannual monitoring with thresholds of "5% or so" is enough for most portfolios (SNIP, search summary of *Best practices for portfolio rebalancing*; PDF not opened).
- The 10Y moved 38bp in September alone (F-003). A ± 5 -point band on a 35% equity sleeve can plausibly be hit in a six-week window, but that is not certain (ASM).
- Would change: **decision**. WInS-now: write the band into the team's decision log before the first order; the first band-triggered trade becomes a Trading Note candidate ("tested"). IPS: one clause on the rebalancing rule.
- Deliverables: WInS-now, TN, IPS. Domain: D8. Seed: 27.
- North Star: Laura sees a firm that already behaves by rules in week 1, not one that writes rules after the fact.

Q02. When the market falls during WInS or in a future bear market, which of the four standard "get back on track" levers is Laura allowed to use? Should the IPS pre-commit never to use the "add more equity to catch up" lever?

- Anchors: R-C34, R-C38, R-C70, R-I3.
- Official words: "Apart from the two contributions described above, she will neither add to nor withdraw from the portfolio before 2033." (case p.2 L63-65, VRF).
- Hypothesis (guess):
 - Morgan Stanley lists four levers: "stretching your time horizon, scaling back the goal, increasing your equity investment, or saving more" (VP, morganstanley.com, "Goals-Based Planning: Stay on Track", read 2026-09-27).
 - For Laura, "save more" is closed by the case (no other contributions), and "stretch the horizon" is closed because the dates are fixed (2031, 2033). Of the two levers left, "scale back the goal" means the facility contribution, which is the case's designed shock absorber (R-AN6, R-AN20). "Increase equity" would be doubling down.
 - Pre-committing against it in the IPS is a clear, Laura-specific rule. WInS-now: after a WInS loss, do not add equity outside the band (Q01).
- Would change: **sentence**. One IPS clause: the facility contribution is the only variable that absorbs bad markets; the risk level is never raised to recover losses. Possible TN reflection content if a down week occurs.
- Deliverables: WInS-now, TN, IPS. Domain: D6. Seed: 27.
- North Star: it tells Laura in advance exactly what gives way in a bad market (her facility number, never her promise), which is the answer a founder most needs before she speaks to co-sponsors.

Q03. Should every WInS trade note name the goal ("funding purpose") the trade serves, as goals-based practitioners label each sub-portfolio, so that the three chosen notes read as one goals-based portfolio?

- Anchors: R-W71, R-AN45, R-W80, R-T10.
- Official words: "Your team may consider diversification across asset classes, sectors, industries, market capitalizations, geographic regions, risk characteristics, and funding purposes." (SMAApply Trading Details, VP via A1). "Each investment decision should have a clear purpose within that strategy." (R-W80, VP).
- Hypothesis (guess):
 - Yes. Wharton's own list includes "funding purposes", the core idea of goals-based practice.
 - Brunel builds "module-built portfolios, each of which is driven by a client's expressed goals" (VP, Brunel, "Goals-Based Wealth Management in Practice", CFA Institute Conference Proceedings, March 2012).

- So a label on every note (promise / 2031 floor / growth) doubles as evidence of "appropriate diversification" (R-S24). Notes cannot be edited later (G, guardrails item 6), so the label vocabulary must be fixed before the first trade.
- What is new versus the council's "say which bucket": Wharton's own wording makes goal-labelled diversification an official, scored idea, not only a style choice.
- Would change: **decision**: a fixed three-label vocabulary for all notes from trade 1.
- Deliverables: WInS-now, TN. Domain: D8. Seed: 28.
- North Star: every trade visibly answers "which of Laura's goals does this serve?", which is what a client chooses a firm for.

Q04. Is the team's Chhabra mapping wrong? Should Laura's career (and the residency itself) be named as her "aspirational" bucket, so the portfolio holds only the "safety" and "market" buckets?

- Anchors: R-C37, R-AN16, R-C33, R-C27, R-C18.
- Official words: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals." (case p.2 L69-74, VRF).
- Hypothesis (guess):
 - Yes, the mapping is wrong. The team notes (`research/council_2026-09-27/team_notes_2026-09-27.md` L15-18) map "Aspirational risk bucket = flexibility reserve (stays invested for growth; dry powder)".
 - In Chhabra's framework, aspirational investments include "holding concentrated stock positions ... owning a small business, or leveraging unique human capital" (SNIP, search summary of Chhabra's framework; SSRN and the journal page were blocked or rate-limited).
 - Chhabra (quoted by CFA Institute, 5 Aug 2015): "For those things you must do, the emphasis is going to be on lowering the risk of your not achieving these goals ... The goals you aspire to may require a somewhat higher level of risk." (VP, CFA Institute Enterprising Investor blog). The same blog notes his emphasis on "human alpha: your earning potential and expertise in your work" (VP).
 - Read that way, the case's "Although" says her aspirational risk already lives in her career and her residency, so the portfolio is the safety and market buckets. That is a Laura-specific, framework-backed reason for a moderate sleeve and no thematic bets.
- Would change: **sentence**. IPS risk-tolerance clause: why a risk-taking founder gets a moderate portfolio. The FR framework section: correct the bucket names. The team's notes need correcting before any deliverable uses them.
- Deliverables: IPS, FR. Domain: D8. Seed: 28.
- North Star: it shows we understood *why* a bold entrepreneur wants a careful portfolio, in a framework she may already know from Wharton.

B. The IPS (tier 2)

Q05. Using the UK regulator's split between willingness and capacity, is Laura's capacity for loss on the growth sleeve actually *high*? If so, is ~60% sleeve equity too low, or does the credibility risk of 2031 cap it?

- Anchors: R-C33, R-C38, R-AN17, X-11, F-407, F-409, BS-03.
- Official words: "Laura's living expenses and short-term financial needs will be covered by income and financial resources outside the portfolio." (case p.2 L61-63, VRF).
- Hypothesis (guess):
 - FCA/FSA guidance FG11/05 (VP, [fca.org.uk](https://www.fca.org.uk/publications/consultations/consultation-on-the-fca-approach-to-the-adequacy-of-capital-for-firms) PDF): "By 'capacity for loss' we refer to the customer's ability to absorb falls in the value of their investment. If any loss of capital would have a materially detrimental effect on their standard of living, this should be taken into account".
 - Laura's living standard is outside the portfolio by case design. So on the FCA definition her capacity for loss on the *sleeve* is high. The team's client profile (not in repo) rates capacity "moderate to low", citing career risk. The two views conflict.
 - Guess at the resolution: capacity is high for money that has no promise attached. It is limited only by the 2031 credibility statement (BS-03). So the sleeve's equity weight is set by her willingness ("thoughtful" risk) and by the 2031 rule, not by capacity. This would support 60% and argue against going below 50%.

- Would change: **number** (sleeve equity 50/60/70%, an open item in brief section 9) and the IPS risk sentence (separate willingness from capacity explicitly).
- Deliverables: IPS, FR. Domain: D6. Seed: 28.
- North Star: separating "can afford" from "wants" is exactly what a client who hires a professional expects; most teams will write only "moderate risk tolerance".

Q06. Does the 2026 pension "surplus glidepath" benchmark support our ~20% total equity after 2028, or does it imply the surplus (funded ratio ~1.5) could carry more return-seeking assets?

- Anchors: F-409, F-117, F-106, R-C38, R-C58.
- Fact: total equity after 2028 is 17.0/20.4/23.8% for sleeve equity 50/60/70% (F-409, ASM-based).
- Hypothesis (guess):
 - Russell Investments, 2026-07-13 (VP): at 110% funded, if surplus has no value, "the optimal portfolio would be a 20/80 ... or 30/70". Once a "meaningful surplus cushion" exists, return-seeking exposure "begins to rise modestly". Any increase should come "only after the plan has reached a clearly defined surplus threshold, and it should be implemented incrementally".
 - Laura's surplus has full value to her (it is the facility money). DERIVED (ASM inputs as in F-409): after the 2028 deposit, assets $\approx$ \$306.1k ladder + $\approx$ \$158.1k sleeve $\approx$ \$464k against a \$306.1k promise, a funded ratio $\approx$ 1.52.
 - A pension at that level would, per Russell, run *at least* 20-30% return-seeking. So our ~20% is at the careful edge of practice, not beyond it.
 - Counter-weight: the sleeve's horizon is 5 years and it carries a public 2031 statement. A pension surplus has no announcement date. Net guess: 60% sleeve equity is defensible. The benchmark gives the team the language: "a defined surplus threshold, incremental risk".
- Would change: **number** (sleeve equity, possibly 60% $\rightarrow$ 65-70% if the team weighs the benchmark heavily) and an FR sentence that cites the practice.
- Deliverables: IPS, FR. Domain: D8 (with D3 to rerun F-407 if the weight changes). Seed: 26.
- North Star: "we did what the best pension professionals now do, adapted to your 2031 promise" is a credible reason to choose us over a generic 60/40 team.

Q07. The best-known professional critique of full hedging ("hibernation") lists three structural leaks. Do any of them apply to Laura's ten payments? If none do, should the FR say so?

- Anchors: R-C46, R-C47, F-101, F-117.
- Official words: "each payment is a fixed \$50,000 and is not adjusted for inflation." (case p.3 L90, VRF).
- Hypothesis (guess):
 - J.P. Morgan AM, "Rethinking the pension plan endgame", October 2021 (VP): hedging portfolios suffer from "Downgrades and defaults ... we estimate this headwind has cost approximately 50 basis points (bps) of return per year", "Longevity extension", and "Experience losses".
 - Laura's liability has no lives (no longevity), no benefit formula (no experience losses) and is matched with Treasuries (no corporate downgrades). So none of the three applies. Her case is the rare one where "a fixed income strategy structured to precisely match liabilities" does not leak.
 - New versus brief section 8.1: 8.1 compared lock-early with *our own* alternatives. This is the strongest *outside professional* objection, and it fails here for stated reasons.
- Would change: **sentence**. FR: one sentence pre-empting "professionals say full hedging is inefficient", with the three reasons it does not bite. IPS: none (too technical).
- Deliverables: FR. Domain: D8 (D1 to check the three reasons). Seed: 26.
- North Star: judges reward teams that answer the best counter-argument; this shows we know the professional debate, not just one side.

Q08. Brunel, a leading goals-based practitioner, raises the return assumption (takes more risk) for needs 6-15 years away. Why is it right for us not to do the same for the 2037-42 payments, and what sentence says why?

- Anchors: R-C46, R-C47, R-C48, F-107, F-108.
- Official words: "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement." (case p.3 L91-92, VRF).

- Hypothesis (guess):
 - Brunel (VP, CFA Conference Proceedings, March 2012): for lifestyle needs in years 1-5 "we build a portfolio composed principally of fixed-income securities". For years 6-15 "we will raise the return assumption to 6 percent because we can take a little more risk".
 - This is a named top practitioner doing roughly what brief section 8.2 rejected (keeping later needs in riskier assets). A judge who knows goals-based practice may ask.
 - Guess at the answer: Brunel's family can cut lifestyle spending and refill from other assets. Laura's payments are fixed, cannot draw on outside funding and are the certainty test. Brunel's own warning fits: "a random fluctuation in asset values can be turned into a permanent loss of capital" when a spending portfolio must be refilled after a fall (VP, same source).
 - New evidence: a practitioner benchmark, not a new calculation. The \$20k/\$43k re-pricing costs (F-108) stand.
- Would change: **sentence** (FR, where the lock is justified against goals-based practice).
- Deliverables: FR. Domain: D8. Seed: 26.
- North Star: it proves the plan is tailored (fixed, non-negotiable payments) rather than copied from a standard goals-based template.

Q09. Top goals-based practice treats "needs" as 90-95% goals. We fund the payments at "bought, barring U.S. default". How do we explain that choice as a *client* decision with a *price*, not as a model result?

- Anchors: R-C47, R-C54, R-C55, R-AN1, F-401, F-117.
- Official words: "They should also define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty" (case p.3 L97-98, VRF).
- Hypothesis (guess):
 - Brunel's goal-probability table (VP, Das, Ostrov, Radhakrishnan & Srivastav, *Journal of Investment Management* 2018, Table 1): Needs/Nightmares 90-95; Wants/Fears 80-85; Wishes/Worries 65-75; Dreams/Concerns 50-60.
 - Our certainty sits above the top of that scale. Its price is visible in the verified model: growth-first median surplus \$226k vs lock-early \$207k, i.e. about \$19k of median facility money to go from a ~96.8% model pass rate to "bought" (F-401). What it protects: p5 surplus \$159k vs \$20k.
 - New versus section 8.6: 8.6 defines certainty. This question places it on a practitioner scale and names its price in the client's own terms. That is what a goals-based adviser would show her.
- Would change: **sentence**. IPS: the certainty definition gains a phrase showing it is her "needs" goal. FR: one line with the ~\$19k median price and the p5 protection.
- Deliverables: IPS, FR. Domain: D8 (D3 for the numbers). Seed: 22, 26.
- North Star: a statistics graduate sees a certainty level chosen on purpose and priced, not a "95%" pulled from software.

Q10. What does insisting on U.S. Treasuries (rather than high-grade corporate bonds, which many liability hedgers use) cost Laura, and does the saving survive expected downgrade losses?

- Anchors: R-C47, F-101, F-102, F-313.
- Fact: ICE BofA U.S. Corporate index option-adjusted spread 0.79%; AA sub-index 0.58% (FRED BAMLC0A0CM, BAMLC0A2CAA, 2026-09-24, VP).
- Hypothesis (guess):
 - DERIVED (ASM: a flat spread added to the Treasury curve, interpolating F-102): a ladder yielding 58-79bp more would cost about \$16k-22k less than \$292,264.
 - J.P. Morgan's downgrade-and-default drag of "approximately 50 basis points (bps) of return per year" for high-grade hedge portfolios (VP, Oct 2021) would eat most of that. The expected saving is maybe \$2-8k (ASM), and it adds real default risk to the one goal that must not fail.
 - So "Treasuries only" is a deliberate choice costing little in expectation. The FR can state it with a number. WInS permits corporate bond ETFs but only government bonds as individual bonds (R-W68, R-W69).
- Would change: **sentence** (FR: "why Treasuries" with its price) and confirms a **decision** (no corporates in the reserve).

- Deliverables: FR. Domain: D1. Seed: 22.
- North Star: showing the price of each safety choice is how a professional earns trust; almost no team will quantify what "safe" costs.

Q11. After Moody's downgraded the U.S. to Aa1 in May 2025, all three major agencies rate the U.S. below AAA. How do top practitioners word "risk-free" today, and what honest sentence should replace "certain barring U.S. default"?

- Anchors: R-C47, R-C54, R-AN32, BS-18, brief section 7 wording.
- Fact: Moody's downgraded the United States to Aa1 from Aaa on 2025-05-16 (SNIP; moodys.com press release not opened). The S&P (2011) and Fitch (2023) downgrades are background knowledge, UNVERIFIED in this run.
- Hypothesis (guess):
 - Practitioners still use Treasuries as the liability-hedge and "risk-free" benchmark. Their wording shifts to "backed by the U.S. government, the highest-quality and most liquid asset available in U.S. dollars", with default named as a remote residual rather than denied.
 - Honest version for Laura: certain in nominal U.S. dollars, with the remaining risks named (U.S. default, the operational errors in BS-18, and loss of buying power to inflation). Each risk is one clause.
- Would change: **sentence** (the certainty definition in IPS and FR).
- Deliverables: IPS, FR. Domain: D1 (D10 for wording). Seed: 22.
- North Star: honest uncertainty language is the case's core skill (credibility with co-sponsors) and what a statistician trusts.

Q12. What governance section would a CFA-standard IPS contain here? Specifically: who decides, how often the IPS is reviewed, and which trigger events force a review (2028 deposit late or smaller; rates fall before purchase; residency delayed; the 2031 announcement)?

- Anchors: R-C3, R-C89, R-I27, R-AN4, R-W19, BS-08, BS-13.
- Official words: "Investment Policy Statement. This formally establishes the strategy and the team's decision-making framework." (case p.4 L155-156, VRF).
- Hypothesis (guess):
 - The CFA standard's section 2 covers "who is responsible for determining investment policy, executing investment policy, and monitoring the results", and "the process for reviewing and updating the IPS". Its example is to review "no less frequently than annually" (VP, CFA 2010).
 - One IPS sentence can carry four things: the firm recommends and the PM approves (case fiction, consistent with the real rules because the students write it); an annual review; three to four named triggers; and each rule written so any PM could apply it (BS-08).
 - The case's "decision-making framework" wording invites exactly this, and few teams will name triggers.
- Would change: **sentence** (IPS governance clause) and a **decision**: which triggers to name.
- Deliverables: IPS, FR. Domain: D10 (with D8). Seed: 2 (case design), 27.
- North Star: Laura is choosing a firm to trust for 16 years; a named review process is what she would be signing.

Q13. Which fee assumption would a real adviser disclose for a \$450k client, and must the IPS say that fees come out of the growth sleeve so the locked payments are never touched?

- Anchors: R-C26, BS-01, Q-A3-01, F-401.
- Official words: "At the beginning of 2027, Laura plans to invest \$300,000 with an asset management firm." (case p.2 L43, VRF).
- Hypothesis (guess):
 - The median human-adviser AUM fee is around 1.0% at \$500k-\$1M (SNIP: search summary citing Kitces Research and Cerulli). Private banks such as J.P. Morgan Private Bank and Goldman Sachs Private Wealth generally require about \$10 million (SNIP). So Laura is a wealth-management client, not a private-bank client. The realistic benchmark is a fee-only adviser or an asset manager's managed account.
 - A3 estimated that a 0.5% or 1.0% fee lowers the median 2033 surplus by about \$17k or \$34k (ASM).

- Practitioner logic suggests a split: a held-to-maturity Treasury ladder needs little management, so a full AUM fee on it would be hard to justify (ASM/judgement).
- The IPS must say fees are paid from the surplus. If they came from the ladder, the lock would break.
- Would change: **number** (fee line in FR projections and the 2031 range) and an IPS **sentence** ("costs are paid from the growth money; the reserve is never sold to pay fees").
- Deliverables: IPS, FR. Domain: D8 (D3 to model). Seed: null.
- North Star: the CFA Asset Manager Code requires fee disclosure (BS-02). A firm that shows its own cost is the one she can trust.

Q14. If rates fall before January 2027 so the ladder costs more than \$300k, what do professional hedgers actually do: hedge in full when affordable, buy in stages, or wait for yield triggers?

- Anchors: F-104, F-111, F-112, R-C26, brief section 9 (open rule).
- Fact: the ladder cost more than \$300k on 173 of 185 trading days of 2026, and fits only since 2026-09-10 (F-111). $P(\text{cost} > \$300\text{k on } 2027-01-01) \approx 24\%$ (F-112, ASM).
- Hypothesis (guess):
 - Pension practice uses funded-status and yield triggers to raise the hedge ratio step by step (SNIP: Russell and Milliman search results).
 - Russell 2026 (VP) stresses a "clearly defined" threshold, implemented "incrementally".
 - For Laura the professional default would be: hedge the longest rungs first up to the \$300k available, buy the rest from the 2028 deposit, and never wait for a better yield (waiting is a rate bet). This matches the current longest-first rule (S-31). The benchmark adds a named, practitioner-style rule format: trigger, action, record.
- Would change: **decision** (confirm or replace the "rates fall first" rule) and an IPS clause.
- Deliverables: IPS. Domain: D1 (with D8). Seed: 4 (case design), 26.
- North Star: a pre-written rule for the one market move that could break the plan before it starts shows foresight most teams lack.

Q15. Should the IPS state the risk tolerance the way the CFA standard's example does: a stated loss the client finds intolerable, plus an anchor to her known liabilities?

- Anchors: R-I2, R-AN17, X-11, R-W22.
- Official words: "An Investment Policy Statement (IPS) outlines an investor's financial goals, risk tolerance, and guidelines for managing a portfolio." (IPS p.1 L3-4, VRF).
- Hypothesis (guess):
 - CFA 2010 (VP): "Where possible, the IPS should account for known liabilities to lend some quantitative basis to the risk tolerance assessment." It adds that approaches may "define multiple levels of risk associated with avoiding financial catastrophe, maintaining a current standard of living, meeting a specific future financial goal, or developing significant further wealth". Its example: "an absolute loss in any 12-month period of more than 33 percent is intolerable".
 - For Laura: risk tolerance = zero for the payments (bought); for the sleeve, a stated tolerable fall. For example, the sleeve can fall about a third without touching the payments or the bought 2031 floor (ASM; D3 to size).
 - The CFA's "multiple levels of risk" line is a professional endorsement of goal-by-goal risk. It is quotable evidence that our structure is standard, not invented.
- Would change: **sentence** (IPS risk tolerance) and possibly a **number** (the sleeve loss we say is tolerable).
- Deliverables: IPS. Domain: D8 (D3 for the number). Seed: 28.
- North Star: the case never states her risk tolerance (R-AN17). Deriving it in a measurable form shows client knowledge.

Q16. From 2027, should the promise assets be held and reported as a separate account, as goals-based practice does, so that "set aside in 2033" is a label on money already segregated?

- Anchors: R-C50, R-AN28, R-AN26, brief section 8.9.
- Official words: "At the beginning of 2033, before making the first operating payment or contributing to the facility, Laura will set aside a portion of the portfolio to fund the ten payments." (case p.3 L94-95,

VRF).

- Hypothesis (guess):
 - Yes. Goals-based practice runs "module-built portfolios" per goal (Brunel 2012, VP). UBS's "Liquidity" strategy is likewise a separate pot "to fund expenditure and meet liabilities for the next two to five years" (SNIP; UBS pages returned HTTP 403).
 - Brief section 8.9 settled *which number* is the reserve. This asks about *governance*: a separate sub-account from 2027 makes the reconciliation sentence concrete and gives Laura one line to watch.
- Would change: **sentence** (IPS: the reserve is a separate sub-account from purchase). It also fixes how FR charts show two pots.
- Deliverables: IPS, FR. Domain: D8. Seed: 28.
- North Star: Laura can point to "the money for the residency's first ten years" as a thing that exists, which is also what co-sponsors want to see (BS-17).

Q17. Which named frameworks fit which part of the plan? Which would a semifinal reader recognise, and which should be left out (for example endowment spending rules)?

- Anchors: R-S24, R-S26, R-W43, R-I46, v4 Part 2 lesson 1.
- Official words: "Demonstrates understanding and effective use of investment concepts and tools" (Portfolio Analysis criterion, R-S26, VRF). "Keep the presentation simple" (IPS p.3 L101-102, VRF).
- Hypothesis (guess), a proposed map:
 - Goals-based wealth management (Brunel; Das et al.) fits the overall structure.
 - Cash-flow matching or LDI fits the reserve.
 - Chhabra's framework fits the risk budget (corrected per Q04).
 - A surplus glidepath (Russell 2026) fits the sleeve.
 - The CFA IPS elements fit governance.
 - Endowment spending rules do *not* fit. They spend a percentage of a fluctuating fund forever, while Laura owes fixed dollars for ten years. Leave them out, and at most say why in the FR.
 - IPS: no framework names (jargon, R-I46). FR: one table with at most four.
- Would change: **sentence** (FR framework table; IPS stays jargon-free) and a **decision** (which names to drop).
- Deliverables: IPS, FR. Domain: D7 (D8 support). Seed: 28.
- North Star: using the right tool for each goal, and saying why one famous tool does not fit, shows judgement rather than name-dropping.

C. The Final Report (tier 3)

Q18. Should the plan commit to one progress measure (for example "payments covered: yes; funded ratio") and use it the same way in every deliverable and every future report to Laura, as the CFA standard asks?

- Anchors: R-C92, R-S24, BS-09, Q-A3-08.
- Official words: "Teams should use each deliverable for its intended purpose while maintaining a clear and consistent investment strategy across all three." (case p.4 L161-162, VRF).
- Hypothesis (guess):
 - CFA 2010 (VP): "Consistent use of metrics to assess and evaluate the risk profile of investment portfolios is important for meaningful comparisons to be made over time. Inappropriate use of different metrics to highlight or disguise certain risks should be avoided."
 - Morgan Stanley's goals system reports "Progress to Goals" (VP, same article as Q02).
 - One goal-progress metric, stated in the IPS and reused in TN reflections and FR charts, answers BS-09: bond price falls do not look like broken promises. It also prevents "metric shopping".
- Would change: **sentence** (IPS: the one measure) and FR chart design (the same metric each year 2027-2033).
- Deliverables: TN, IPS, FR. Domain: D9 (D8 support). Seed: 27.
- North Star: a single honest gauge that Laura can read in five seconds is what separates a trusted adviser from a report generator.

Q19. Should the FR cite "behaviour gap" or "adviser's alpha" figures to justify pre-commitment rules, given that a 2026 peer-reviewed paper says the famous 1.2%-a-year gap overstates timing losses about twelvefold?

- Anchors: R-AN32, R-S26, R-W47.
- Fact:
 - Morningstar *Mind the Gap 2025* (VP): the average dollar earned "about 1.2 percentage points per year less than these funds' 8.2% aggregate annual total return".
 - Vanguard (VP, *Putting a value on your value*, June 2020 brief): "Behavioural coaching 150" bps.
 - Fulkerson, Jordan, Riley & Yan, *Financial Analysts Journal* 2026 (VP abstract): "poor timing by mutual fund investors costs them only 0.10% per year".
- Hypothesis (guess):
 - Do not lean on either big number. Laura is a statistics graduate (R-AN32), and the AI policy warns about reference quality (R-W47).
 - Justify rules by her own decision dates instead: 2031 (public statement) and 2033 (reserve and facility). Those are the moments when an emotional decision would be costly.
 - Vanguard's own line can support the idea without the number: the IPS "sows the seed for future behavioural coaching opportunities ... Having a clearly set out investment policy will allow you to defend against these common behavioural pitfalls" (VP).
- Would change: **sentence** (FR behavioural rationale). It removes a statistic that could backfire.
- Deliverables: FR. Domain: D6. Seed: 27.
- North Star: refusing a flattering but contested statistic is itself evidence of the rigour a statistics-trained client is looking for.

Q20. With the 2031 floor bought, the only way to land outside the co-sponsor range is to land above it. Should the top of the range therefore be set at a stated percentile (for example Brunel's "wants" level, 80-85%), making the confidence figure equal to that percentile?

- Anchors: R-C69, R-AN9, R-C66, F-402, brief section 8.7.
- Official words: "state how confident they are that her 2033 contribution will fall within that range." (case p.3 L117-118, VRF).
- Hypothesis (guess):
 - Yes. $P(\text{below floor}) \approx 0$ by purchase, so $P(\text{within}) = P(\text{at or below the top end})$. Choosing the top at the 80th-85th percentile of the 2033 outcome distribution (computed in 2031) gives an 80-85% confidence statement.
 - This is Brunel's "wants" band (VP, Das et al. 2018 Table 1), and it links directly to the case's two-sided wording.
 - Planning software treats 75-90% as the "Confidence Zone" for adjustable goals (SNIP: MoneyGuidePro description via search; primary page not opened).
 - A 95% top would make the range wide and less useful; a 50% top would make "outside" too likely. This turns an open parameter into a principled number.
- Would change: **number** (the 2031 range's top end and the stated confidence) in the FR.
- Deliverables: FR. Domain: D3 (D8 support). Seed: 9 (case design), 26.
- North Star: a range whose confidence comes from a stated, professional rule is the most credible thing Laura can say to co-sponsors.

Q21. How should the fundraising draft report uncertainty: dollar outcomes in named scenarios (practice now prefers "outcomes, not probabilities"), with one probability at most, and wording that pre-empts the "wrong side of maybe" reaction?

- Anchors: R-S28, R-C70, R-C72, X-14, BS-05.
- Official words: "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors." (criterion 5, R-S28, VRF).
- Hypothesis (guess):
 - Kitces (VP, kitces.com): "outcomes, not probabilities, are what matter to clients". The "wrong side of maybe fallacy" is "our tendency to interpret a projection as 'wrong' if the outcome is inconsistent with the most likely outcome forecasted" (VP).

- Das et al. (VP) frame goals with two reference points: a Target Wealth and a Loss Threshold. These map onto the range's two ends.
- Guess at the recipe:
- The floor is stated as money already set aside (a fact).
- The top end is stated as an expectation with its percentile.
- Two or three dollar scenarios (weak, typical, strong markets) are each tied to what the residency can do.
- "Never below" is used only for the bought part.
- Would change: **sentence** (the structure of the FR fundraising draft; the students write it).
- Deliverables: FR. Domain: D9 (D5 support). Seed: 9, 15.
- North Star: co-sponsors are not statisticians. A draft that gives them outcomes they can plan around protects Laura's credibility, which the case names as the key risk.

Q22. Once the range is communicated in 2031, what may change it before 2033? Should the IPS pre-commit that the floor never moves down and only the upside is revisited, like a guardrail rule?

- Anchors: R-C66, R-C67, R-C71, BS-03.
- Official words: "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors." (case p.3 L114-115, VRF).
- Hypothesis (guess):
 - Yes. With Kitces-style "guardrails", advisors give clients in advance "the portfolio value that would trigger spending changes and the magnitude that would be prescribed for such changes", instead of recomputing a probability each year (VP, <https://www.kitces.com/blog/retirement-income-guardrails-monte-carlo-client-communication/>).
 - For Laura: the floor is fixed because it is bought. The top end is revised only upward, or at one pre-announced date. The downside can never become a broken promise. This turns BS-03 (credibility as income) into a rule.
- Would change: **sentence** (IPS clause on the 2031 rule; FR communication plan).
- Deliverables: IPS, FR. Domain: D8 (D5 support). Seed: 27.
- North Star: a promise designed so it can only be beaten, never broken, is what a public figure would choose.

Q23. How much should Laura hold back after the facility contribution? Do practitioner benchmarks (Brunel's 10% for "opportunistic goals"; UBS's "Legacy" pot for assets beyond needs) give a defensible rule?

- Anchors: R-C58, R-C84, R-AN19, R-C61, brief section 9 (post-2033 flexibility).
- Official words: "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops." (case p.3 L102-103, VRF).
- Hypothesis (guess):
 - Brunel's worked example (VP, 2012): "The family also wants to reserve 10 percent of their total wealth (\$3.5 million) for what they call 'opportunistic goals.'"
 - UBS's Legacy strategy covers "assets in excess of what is needed" (SNIP).
 - One case study is weak evidence. Guess: state flexibility as a share of the post-reserve surplus (for example 10-25%) held back from the facility. Justify it by project risk (overruns, delay), not by a benchmark alone. The case says no contingency fund need be sized (R-C61), so a percentage rule stays within scope.
- Would change: **number** (share held back; the facility contribution) in the FR.
- Deliverables: FR. Domain: D8. Seed: 11 (case design).
- North Star: it shows we protect her ability to keep acting like an entrepreneur after 2033, which is who she is.

Q24. After the 2031 floor is locked, should the rest of the sleeve still hold equities, given that goals-based practitioners keep money needed within about 3-5 years mainly in fixed income?

- Anchors: R-C62, R-C63, R-C65, F-402, brief section 9 (2031 rule parameters).
- Fact: the model locks 80% of the sleeve in a 2-year Treasury in 2031 (F-402). The other 20% stays invested to 2033.

- Hypothesis (guess):
 - Brunel (VP, 2012): "To provide for immediate lifestyle needs (within the next three to five years), we build a portfolio composed principally of fixed-income securities." UBS's Liquidity strategy covers "the next two to five years" (SNIP).
 - The unlocked 20% is not a *need*. It is upside she has not promised. So keeping it in equities is consistent with practice as long as the range's top end is set by rule (Q20), not by hope.
 - New versus brief section 9: the benchmark says the *promised* part must be out of equities (already done), and lets the unpromised part stay in. So the 80% parameter should equal "whatever is promised as the floor", not an arbitrary share.
- Would change: **number** (2031 lock share, defined by the floor promised) and an IPS clause.
- Deliverables: IPS, FR. Domain: D3 (D8 support). Seed: 26.
- North Star: a rule that links "what she says in 2031" to "what is locked in 2031" makes her words and her money match, which is what credibility is.

Q25. Does an Australian practitioner benchmark earn a place? Australia's Retirement Income Covenant makes superannuation trustees "balance" three objectives that mirror the IPS guide's trio. The Future Fund shows why a long-dated promise *without* government backing must be matched.

- Anchors: R-I11, R-I26, X-16, R-S27, v4 Part 2 lesson 4.
- Official words: "How does your strategy balance growth, risk, liquidity, funding reliability, and financial flexibility?" (IPS p.1 L20, VRF).
- Hypothesis (guess):
 - APRA/ASIC 2023 report (VP): trustees must "achieve and balance the three objectives of: maximising expected retirement income; managing expected risks to the sustainability and stability of retirement income; and having flexible access to expected funds during retirement". That is growth, funding reliability and flexibility, the case's own words, written into Australian law since 2022.
 - The Future Fund (Wikipedia, verbatim check, VP of the article text; futurefund.gov.au refused with 403) was set up to provide for public-servant pension liabilities. Its mandate targets "at least the Consumer Price Index + 4 to 5 per cent per annum". It can take growth risk against a long-dated promise because the Commonwealth stands behind it; Laura has no taxing power behind her. That contrast explains lock-early in one sentence.
 - Guess: one sentence in the FR's Articulation section ("as Australians we noticed..."), true and short. No more.
- Would change: **sentence** (FR Articulation / authentic voice).
- Deliverables: FR. Domain: D8. Seed: null (v4 Part 2 lesson 4).
- North Star: an authentic team voice (R-S28) grounded in a real regulatory parallel is distinctive among ~2,300 finishers and is honest about who we are.

3. Where our plan matches or falls short of top practice (benchmark table)

Practice element (source, status)	Our plan today	Match?	Gap to close (question)
Goal-by-goal sub-portfolios (Brunel 2012, VP; Das et al. 2018, VP)	Reserve + growth sleeve	Match	Label every trade by goal (Q03); separate account (Q16)
Risk = chance of missing a goal (Das et al., VP)	Certainty market-consistent; sleeve measured on surplus	Match	Place on the practitioner scale and price it (Q09)
Risk allocation before asset allocation (Chhabra via princetoninfo.com, SNIP)	Implicit	Partial	Correct the bucket mapping (Q04)
Lock benefits, grow surplus (Russell 2026, VP)	Lock-early + sleeve	Match	Cite it; test the equity weight against it (Q06)
Answer the hibernation critique (JPM 2021, VP)	Not addressed	Gap	Q07
Willingness vs capacity assessed separately (FCA FG11/05, VP)	Team docs conflate them	Gap	Q05, Q15
Governance, review cycle, triggers (CFA 2010 section 2, VP)	None written	Gap	Q12

Practice element (source, status)	Our plan today	Match?	Gap to close (question)
Rebalancing policy documented (CFA 2010 section 4c, VP)	None written	Gap	Q01
One consistent risk/progress metric (CFA 2010 section 4b, VP)	Not fixed	Gap	Q18
Fee disclosure (CFA Asset Manager Code F.4.d, VP via A3)	No fee in model	Gap	Q13
Outcomes, not bare probabilities, to clients (Kitces, VP)	Probability-led in council drafts	Partial	Q21
Pre-committed response to bad markets (Morgan Stanley levers, VP; Vanguard, VP)	Implied	Partial	Q02, Q22
Near-term needs in fixed income (Brunel, VP)	Floor bought in 2031	Match	Tie the lock share to the promised floor (Q24)
Longer needs take more risk (Brunel, VP)	We do not	Deliberate difference	Explain why (Q08)

4. Leads for the specialists

Sources read in this run (access 2026-09-27, UTC ~12:00-12:30), with what each is good for:

- **CFA Institute, *Elements of an Investment Policy Statement for Individual Investors*, May 2010** (VP): <https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf>
 - Sections: 1 Scope and Purpose; 2 Governance (2a-2e); 3 Objectives (3c risk tolerance); 4 Risk Management (4a reporting, 4b metrics, 4c rebalancing).
 - Key lines: "Perhaps most importantly, the IPS serves as a policy guide that can offer an objective course of action to be followed during periods of market disruption when emotional or instinctive responses might otherwise motivate less prudent actions." / "Where possible, the IPS should account for known liabilities to lend some quantitative basis to the risk tolerance assessment." / "If the policy is not to rebalance, this policy should be documented in the IPS." For D8, D10.
- **Brunel, "Goals-Based Wealth Management in Practice", CFA Institute Conference Proceedings Quarterly, March 2012** (VP; a copy hosted on squarespace): https://static1.squarespace.com/static/59e8d89d914e6b37450c946a/t/5c643e15104c7b43f7b84187/1550073366884/CFA+Goals+Based+WM_PDOC.pdf
 - Useful content: 3-5-year needs in fixed income; 6-15 years at a higher return assumption; the "declining-balance portfolio"; "a random fluctuation in asset values can be turned into a permanent loss of capital"; the 10% for "opportunistic goals"; and the risk definition as "the probability of not achieving goals" (reporting Das, Markowitz, Scheid & Statman 2010). For D8, D3.
- **Das, Ostrov, Radhakrishnan & Srivastav, "A New Approach to Goals-Based Wealth Management", *Journal of Investment Management* 16(3), 2018** (VP, author's site): <https://srdas.github.io/Papers/GBWM.pdf>
 - Table 1, Brunel's goal-probability table (Needs 90-95, Wants 80-85, Wishes 65-75, Dreams 50-60). Also the two reference points, Target Wealth and Loss Threshold. For D3, D8, D9.
- **CFA Institute Enterprising Investor, "Ashvin B. Chhabra: Understanding Goals Is the Value Proposition", 2015-08-05** (VP): <https://blogs.cfainstitute.org/investor/2015/08/05/ashvin-chhabra-on-wealth-management-the-value-proposition-lies-in-understanding-client-goals/>
 - Includes the three questions "What do you need for immediate safety? / What do you need to be comfortable? / What do you aspire to?" and the "human alpha" point.
- **Princeton Info, on Chhabra's "Beyond Markowitz"** (read verbatim, secondary source; quotes the 2005 paper: "risk allocation must precede asset allocation"): <https://princetoninfo.com/business/beat-the-market-consider-your-risks-and-asset-allocation/>
 - The original paper (*Journal of Wealth Management* 7(4), Spring 2005, pp. 8-34) was blocked: SSRN 403 and pm-research 429. Treat the quotes as SNIPPET-UNVERIFIED until D8 reads the paper or the book *The Aspirational Investor* (2015).
- **Russell Investments, "Pension surplus investing: Rethinking the value of overfunding", 2026-07-13** (VP): <https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/>

pension-surplus-investing-value-overfunding.html

- Key lines: "lock down the benefits promised, then grow surplus assets with a prudent process"; the 20/80-30/70 optimum at 110% funded; incremental re-risking above a defined threshold. For D8, D3.
- **J.P. Morgan AM, "Rethinking the pension plan endgame: Hibernation, termination or stabilization?", October 2021** (VP): <https://am.jpmorgan.com/content/dam/jpm-am-aem/global/en/institutional/insights/portfolio-insights/portfolio-strategy/Rethinking-the-pension-plan-endgame.pdf>
- Structural impediments: downgrades (~50bp a year), longevity, experience losses. For D1, D8.
- **FRED corporate spreads** (VP): BAMLC0A0CM 0.79 and BAMLC0A2CAA 0.58 on 2026-09-24, <https://fred.stlouisfed.org/graph/fredgraph.csv?id=BAMLC0A0CM>. For D1 (Q10).
- **FSA/FCA FG11/05, Assessing suitability, 2011** (VP): <https://www.fca.org.uk/publication/finalised-guidance/fsa-fg11-05.pdf> (capacity-for-loss definition, footnote 3). For D6.
- **Morgan Stanley, "Goals-Based Planning: Stay on Track"** (VP; undated on the page as read): <https://www.morganstanley.com/articles/goals-based-financial-planning-stay-on-track> (four levers; "Progress to Goals Reporting"). The GPS sample-report PDF gave an empty reply (unread). For D6, D9.
- **Kitces.com** (VP): <https://www.kitces.com/blog/monte-carlo-guardrails-probability-of-adjustment-success-client-communication-dynamic-retirement-spending/> ("outcomes, not probabilities") and <https://www.kitces.com/blog/the-wrong-side-of-maybe-fallacy-and-the-interpretation-of-monte-carlo-analysis/> ("wrong side of maybe"). For D9.
- **Vanguard, Putting a value on your value: Quantifying Vanguard Adviser's Alpha, research brief June 2020** (VP): <https://www.ch.vanguard/content/dam/intl/europe/documents/en/putting-a-value-on-your-value-quantifying-vanguard-adviser-alpha-eu-en-pro.pdf> (behavioural coaching 150bp; the IPS "sows the seed" passage). For D6.
- **Morningstar, Mind the Gap 2025** (VP): https://www.morningstar.com/content/cs-assets/v3/assets/blt9415ea4cc4157833/blt2c5c4d9171638c42/689b424311f3880edc4b4813/US_Mind_the_Gap_2025.pdf
 - Contrast it with Fulkerson et al., *FAJ* 2026 (VP abstract): <https://rpc.cfainstitute.org/research/financial-analysts-journal/2026/bad-timing-does-not-cost-investors-funds-returns>
 - For D6.
 - **APRA/ASIC, Implementation of the retirement income covenant, July 2023** (VP): <https://www.apra.gov.au/sites/default/files/2023-07/Information%20report%20-%20Implementation%20of%20the%20retirement%20income%20covenant-Findings%20from%20the%20APRA%20and%20ASIC%20thematic%20review%20July%202023.pdf> (the three objectives). For D8 (Q25).
- **Wikipedia, "Future Fund"** (VP of article text only; futurefund.gov.au refused with 403): https://en.wikipedia.org/wiki/Future_Fund ("CPI + 4 to 5 per cent"). D8 should confirm on the Future Fund's own mandate page or legislation.gov.au if Q25 is used.

Blocked or unread (do not guess their content):

- UBS Wealth Way PDFs (HTTP 403); the "two to five years" Liquidity wording is SNIPPET-UNVERIFIED.
- Chhabra 2005 on SSRN (403) and on pm-research (429).
- The Morgan Stanley GPS sample report (empty reply).
- MoneyGuidePro "Confidence Zone" 75-90% (search snippets only).
- Vanguard *Best practices for portfolio rebalancing* (not opened; "5% or so" is SNIPPET-UNVERIFIED).
- Moody's 2025 downgrade press release (not opened; SNIP).
- Kitces/Cerulli "~1% median fee at \$500k-\$1M" and "private bank minimum \$10M" (SNIP).

Facts for researchers to reuse:

- DERIVED from F-102 (ASM: a flat spread over the Treasury curve): Treasury ladder at +58bp ≈ \$276k, at +79bp ≈ \$270.5k.
- Post-2028 funded ratio ≈ 1.52 (F-106 + F-409 inputs).
- Brunel's table (percentages) is in section 2, Q09.

Suggested owner hand-offs:

- D8: Q04, Q06, Q07, Q08, Q12-Q17, Q22, Q23, Q25.
- D3: Q20, Q24 (compute the 80th-85th percentile top end on the 2031 conditional distribution; size the tolerable sleeve loss in Q15).

- D1: Q10, Q11, Q14.
- D6: Q02, Q05, Q19.
- D9: Q18, Q21.
- D10: Q12 wording.
- D7: Q17.

5. Parked (logged, not pursued; failed the "would change something" filter or out of scope)

- Endowment smoothing rules (e.g., Yale-style) for the 2031 stretch. It adds complexity, and the floor is bought. Folded into Q17 as "does not fit".
 - Private-bank structured products or "principal-protected notes" for the floor. Derivative-based and banned in WInS (R-W90); simpler Treasuries already do the job.
 - The family-office "family governance/education" modules in Brunel. They do not fit a single client with a fixed project.
 - Finale presentation framing of the benchmark (out of scope: semifinals only).
-

What this teaches

Top professionals do not beat markets by guessing. They win client trust by doing four things that fit on one page:

- give every goal its own pot and its own required chance of success;
- lock what must happen and take risk only with what is left over;
- write down in advance what they will do when markets fall, and what changes (here: only the facility amount);
- report one honest measure the same way every time.

Our plan already does the first two. The gaps are mostly in writing the rules down: governance, rebalancing, fees and one progress measure. Those gaps are cheap to close and rare among competitors.

Two further lessons:

- Borrowing a famous framework only helps if you use it correctly. Chhabra's "aspirational" bucket is Laura's career and her residency, not her spare cash.
- A popular statistic (the "behaviour gap") can be much weaker than it looks. Checking the counter-evidence before quoting it is what a client with a statistics degree would expect from the firm she chooses.

B5a: 2026 Macro & Markets Landscape (starting angle: U.S. rates, the Fed, fiscal position and Treasury credit)

Agent B5a, insight_v1 run, Phase B (question generation), written 2026-09-27. AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It holds **no submission-ready prose**. Where it says what a sentence "must contain", that is a checklist; the six students write every word. Any instrument named here is **PENDING Wins AVAILABILITY + POSITION-LIMIT CHECK** (brief section 14) and is research only.

Status labels (brief section 3): VERIFIED-PRIMARY (I read it on the page myself, 2026-09-27), VERIFIED-REPO-FILE, SNIPPET-UNVERIFIED, ASSUMPTION, PARAPHRASE-UNVERIFIED. "Derived" = my own arithmetic on labelled inputs (script steps are in section 5 so anyone can re-run them).

0. Summary (read this first)

I started where the plan is most exposed: the price of the ten-payment Treasury ladder and the phrase "certain, barring U.S. default". Seven findings drive most of the 24 questions below. As far as I can tell, no file in `research/` touches any of them (grep for "debt limit", "term premium", "Penn Wharton", "X-date" and "real yield" on 2026-09-27 found nothing beyond snippet mentions of TIPS yields).

1. **Higher inflation expectations did not raise rates in 2026. Higher real yields and a higher term premium did.** From the 2026 low (Feb 27) to Sept 24 the 10-year yield rose 121bp:
 - 113bp came from the real (inflation-protected) yield: 1.72% to 2.85%.
 - Only 8bp came from breakeven inflation: 2.25% to 2.33%.
 - The Fed Board's Kim-Wright 10-year term premium rose 50bp to 0.96%.
 - Sources: FRED DFII10, T10YIE, THREEFYTP10 (VERIFIED-PRIMARY inputs, derived).
 - The 2.85% real 10-year yield is the highest since 2008-11-24. The term premium is the highest since 2011-02-11.
 - Plain English: the market is paying long-term lenders more, in real terms, than at any time in 18 years. Laura needs to be a long-term lender anyway, because her promise is long-dated. **The ladder became affordable because of a real-yield window, not an inflation scare.** That is a "why now" no other file states.
 - 2. **"Barring U.S. default" has three layers, and the ladder only fully covers one.**
 - (a) **Outright default.** All three rating agencies are one notch below AAA, with stable outlooks (VERIFIED-PRIMARY). The ladder cannot protect against this, but no U.S.-dollar asset can.
 - (b) **Technical delay.** In 1979 some Treasury bills were paid late, amid a debt-limit fight and a back-office failure (SNIPPET-UNVERIFIED). Debt-limit "X-dates" (the day the Treasury can no longer pay all its bills on time) recur.
 - (c) **"Implicit" default through inflation.** The Penn Wharton Budget Model (Wharton's own unit) describes default "of either explicit (Treasury) debt or implicit (pay-as-you-go) debt" becoming "a near certainty on a real (inflation-adjusted) basis" at its debt outer bound. It puts "a 25% chance of reaching it in 14 years" (VERIFIED-PRIMARY, June 4, 2026). Fourteen years from 2026 is about 2040, inside Laura's 2033-2042 payment window.
 - Honest answer: the nominal dollars are close to certain. What those dollars will buy is not.
 - 3. **The debt limit will likely bind again right at Laura's second deposit.** The National Taxpayers Union (Aug 20, 2026, VERIFIED-PRIMARY on ntu.org; secondary analysis) projects:
 - the limit is reached around May 2027;
 - the X-date falls in February 2028;
 - Congress "would likely need to increase the debt limit by January 1, 2028".

That is the date the \$150,000 arrives and the top-up purchase happens. It also reports that the Senate leader suggested handling it in the lame-duck session "after the November election". That session falls between the IPS lock (Nov 6) and Laura's January 2027 purchase.

4. **A dated calendar of events stands between today and the January 2027 purchase.** The ladder's cushion under \$300k is 27bp (F-104). Events that can move it:

- FOMC Oct 27-28 and Dec 8-9 (VERIFIED-PRIMARY);
- Treasury refunding announcement Nov 4, two days before the IPS lock (VERIFIED-PRIMARY);
- U.S. midterms Nov 3;
- the stopgap funding bill (continuing resolution, CR) runs out Dec 11 (VERIFIED-PRIMARY secondary page, cra.org);
- the Iran war / Strait of Hormuz oil shock: Brent crude \$114.89 on Sept 22, up from \$61.98 on Jan 2 (FRED, VERIFIED-PRIMARY).

The 2-year Treasury yield sits ~100bp above the fed funds midpoint, so markets already price more hikes. A rally of more than 27bp probably needs a shock: de-escalation that drops oil, a growth scare, or an equity crash.

5. Official forecasters are about 100bp below the market.

- CBO (Feb 2026) projected a 10-year yield averaging 4.1% in 2026 and 4.4% in 2031-35, with 2026 PCE inflation of 2.7% (SNIPPET-UNVERIFIED via CRFB; cbo.gov refused, HTTP 403).
- JPM's LTCMA bond numbers were set when the 10-year was 4.16% (F-010).
- Today: 10-year 5.17%; Fed 2026 PCE median 3.7%.
- Any "reasonable return assumption" (R-C77) that cites these sources as they are understates bond returns and misstates the equity premium.

6. Stocks and bonds fell together in 2026. The daily correlation between S&P 500 returns and changes in the 10-year yield was **-0.44 in 2026**, versus -0.06 in 2024 and +0.14 in 2025 (FRED SP500/DGS10, derived). So Treasury prices and stocks moved together: when yields rose, both fell. JPM's long-run assumption is ~0 (F-314). The growth sleeve's diversification is weaker in this regime than the model assumes.

7. Taiwan construction costs, measured in U.S. dollars, barely rose.

- In New Taiwan dollars the cost index rose 6.53% y/y (F-508).
- Over the same year the NT dollar weakened from 30.15 to 32.02 per USD (August averages, FRED).
- So the **USD cost of Taiwan construction rose about +0.3% y/y**, and about **+0.8% a year since 2021** (3.6%/yr in TWD, less 2.7%/yr TWD depreciation). Derived, VERIFIED-PRIMARY inputs.
- Laura's contribution is in dollars. So the case's "effect of inflation on ... facility costs" (R-C85) is, so far, mostly a currency story, and the currency can reverse.

Questions are ranked by deadline tier (WinS-now/TN first, then IPS, then FR) within value. The table in section 2 is the short form; section 3 gives the reasoning and evidence for each.

1. Terms used (plain English)

- **Real yield:** the yield on inflation-protected Treasuries (TIPS), i.e. the return above inflation.
- **Breakeven inflation:** nominal yield minus real yield; roughly the inflation rate the market is pricing (it also contains risk premia, so it is not a pure forecast).
- **Term premium:** the extra yield investors demand for lending long instead of rolling short loans. The Kim-Wright estimate on FRED (THREIFYTP10) comes from a Fed Board model; it is an estimate, not an observed price.
- **Debt limit / X-date:** the legal cap on U.S. federal borrowing; the X-date is when the Treasury can no longer pay every bill on time without new borrowing authority.
- **Technical default:** a late or missed payment caused by a legal or operational failure, not by inability to pay.
- **Implicit default:** paying debts in full in dollars that have lost much of their value to inflation.
- **Funded ratio:** assets divided by the market price of the promise ($\$300,000 / \$292,264 = 1.026$ on 9/25, F-117).
- **CAPE:** price divided by 10-year average inflation-adjusted earnings; a slow-moving valuation gauge.
- **Stock-bond correlation:** whether stock returns and bond returns move together (+) or opposite (-).

2. The questions (short form)

#	Question	Anchor	Would change	Deliverable	Domain	Seed
Q01	Should the first Treasury trade note name the real-yield window (real 10y 2.85%, highest since 2008) as the reason the ladder is affordable now?	F-004, F-008, F-009, F-111, R-T9	sentence (TN note #1, IPS "why")	WInS-now, TN, IPS	D1	21
Q02	Which dated events before Jan 2027 could erase the 27bp cushion, and should the IPS rule be a funded-ratio trigger rather than a forecast?	F-104, F-112, F-005, F-007	decision (IPS purchase rule)	IPS, FR	D1	21
Q03	Which of the three layers of "default" (outright, technical delay, inflation) does the ladder cover, and which must the certainty definition name?	R-C47, R-C54, R-C55	sentence (IPS certainty definition)	IPS, FR	D1	22
Q04	Should the team use Wharton's own Penn Wharton Budget Model (25% chance of the debt outer bound by ~2040) to state honestly how certain "certain" is?	R-C56, R-C85	sentence + number (FR assumptions)	FR	D1	22
Q05	Does the projected Feb 2028 X-date, right at the Jan 2028 deposit, require a clause in the 2028 top-up rule?	F-601, R-C77	decision (FR operating rule)	FR, IPS	D1	null
Q06	Do Nov-15 maturities (47 days early; 2033-01-01 is a Saturday) double as a buffer against a technical payment delay?	R-C45, R-C47	decision (maturity choice) + sentence	FR	D1	22
Q07	What will the plan do if the U.S. is downgraded again, and should the IPS pre-commit to "hold, do not sell"?	R-C89, R-I10	sentence (IPS decision rule)	IPS	D1	22
Q08	Since no U.S.-dollar asset is safer than a Treasury, should the IPS spend one clause on sovereign risk and use the words on risks the ladder does not cover?	R-I18, R-C54	sentence (IPS word budget)	IPS	D9	22
Q09	What is the real value of the ten payments at market breakeven vs a named stress, and who bears the erosion?	R-C46, R-C85, R-C44	number (FR)	FR	D1	23
Q10	Is a 1970s-style inflation regime a base case or a stress case for 2033-2042, given the 2026 evidence on Fed independence?	R-C85, F-007, F-009	sentence (FR assumption)	FR	D1	23
Q11	Should the post-2033 flexibility money (not the reserve) be the named inflation buffer for the operating budget, and partly in TIPS?	R-C58, R-C84, R-C44	decision (flexibility composition)	FR	D3	23
Q12	Which return source is "reasonable" when JPM and CBO sit ~100bp below today's yields: a published CMA or market yields?	R-C77, F-010, F-317	number (projection inputs, 2031 range)	IPS, FR	D3	null
Q13	At today's yields the equity premium over Treasuries is ~1.7pp and CAPE ~41: does 60% sleeve equity still earn its p5 cost?	R-C38, F-301, F-407	number (sleeve equity %)	IPS, FR	D3	null
Q14	Should sleeve risk and the 2031 range be stressed at a positive stock-bond return correlation, as realised in 2026?	F-314, R-S26	number (sleeve p5, 2031 range)	FR, WInS-now	D3	null
Q15	Should the team pre-commit, in a trade note, not to trade the hedge around the Oct 28 FOMC and Nov 4 refunding?	R-T9, R-W73, R-W76	decision (WInS conduct)	WInS-now, TN	D1	null
Q16	Should every Treasury trade note record that day's ladder cost/funded ratio as a dated test of the strategy?	R-T9, F-111, F-117	decision (note format)	WInS-now, TN	D9	null
Q17	Does the 20-year "hump" (20y 5.54% > 30y 5.49%) make the 10-20y sector the cheapest place to buy Laura's long rungs?	F-001, F-107	decision (instrument)	WInS-now, TN, FR	D1	null
Q18	Should the WInS book hold individual Treasuries (which lock a yield) instead of only constant-maturity ETFs, to show the ladder itself?	F-606, F-607, R-C75	decision (WInS instrument)	WInS-now, TN	D10	null
Q19	Is a 2027 U.S. recession the common cause linking falling yields and a smaller 2028 deposit, and how big is the joint tail?	R-C86, F-403, F-404	number (FR joint-tail row)	FR	D3	null
Q20	How does an AI-equity crash feed into Treasury yields and the Jan 2027 cushion, and which way does it cut for Laura?	R-C82, F-111	sentence + decision	FR, IPS	D2	null

#	Question	Anchor	Would change	Deliverable	Domain	Seed
Q21	Is the growth sleeve secretly doubling Laura's Taiwan bet through AI-chip supply-chain concentration in the index?	R-C39, R-S24, R-W71	decision (sleeve index choice)	IPS, WInS-now	D4	24
Q22	Measured in U.S. dollars, how fast have Taiwan construction costs risen, and does that change the facility-inflation sentence?	R-C85, F-508, F-512	sentence + number	FR	D4	25
Q23	Is Taiwan's 2026 construction-cost jump an AI-boom wage spike that will fade before 2033, or a new trend?	F-508, F-504, R-C94	number (facility inflation assumption)	FR	D4	25
Q24	If the residency cannot operate in Taiwan, is the USD Treasury reserve portable, and should the promise be tied to the residency rather than the place?	R-C39, R-C45, R-C48	sentence (FR, co-sponsor draft)	FR	D4	24

3. Questions with reasoning and evidence

Q01. Why is the ladder affordable now? (seed 21; tier 1)

- **Question:** Should the first Treasury trade note name the real-yield window as the reason the ladder became affordable, instead of "rates are high"? The real 10y is 2.85%, the highest since Nov 2008, and the term premium is the highest since 2011.
- **Anchors:** F-004 ("5.18% on 2026-09-24 is the highest daily close since 2007-07-06"), F-008, F-009, F-111, R-T9.
- **Evidence** (FRED, VERIFIED-PRIMARY inputs, derived; script in section 5):

Date	10y nominal	10y real (DFII10)	10y breakeven	Kim-Wright term premium
2026-01-02	4.19	1.94	2.25	0.58
2026-02-27 (low)	3.97	1.72	2.25	0.46
2026-06-30	4.44	2.20	2.24	0.70
2026-09-01	4.79	2.44	2.35	0.90
2026-09-24	5.18	2.85	2.33	0.96

Real 10y 2.85% last matched or beaten 2008-11-24. The term premium 0.96% last matched 2011-02-11.

- **Hypothesis (guess):** Yes. In plain terms: the market is paying lenders an unusually large reward above inflation, and Laura is a natural long-term lender because her promise is long-dated. So she earns the premium without taking on a risk she does not already have: she holds to maturity, so price swings do not matter to her.
 - This is a "why now" rooted in her liability, not a market forecast.
 - It also warns that the window may close (Q02).
 - **Would change:** the sentence in Trading Note #1 (why this trade, why now) and one IPS reason. A strong sentence must contain: the dated real yield or funded ratio; "held to maturity"; and a link to the ten payments. It must not forecast rates.
- **North star:** it shows Laura that we bought her promise when the market was paying us to, and that we are not guessing at rates.

Q02. What could erase the 27bp cushion before January 2027, and what form should the rule take? (seed 21, extends open item "rule if rates fall")

- **Question:** Which dated events could push long yields down by more than 27bp before Laura's cash arrives? And should the IPS write the rule as a funded-ratio trigger ("buy longest rungs first until funded; top up in 2028") rather than any rate view?
- **Anchors:** F-104 (headroom \$7,736 = 27bp), F-112 (~24% chance cost > \$300k, ASSUMPTION), F-005, F-007.
- **Evidence (new vs brief section 9):**
 - Event calendar (VERIFIED-PRIMARY unless noted):
 - FOMC Oct 27-28 and Dec 8-9 (federalreserve.gov calendar).
 - Treasury refunding announcement Nov 4 (treasury.gov, sb0590).

- Midterms Nov 3 (SNIPPET-UNVERIFIED as a date; widely known).
- CR expiry Dec 11 (cra.org, secondary; SNIPPET for congress.gov).
- A possible lame-duck debt-limit vote (NTU, quoting Sen. Thune; PARAPHRASE-UNVERIFIED for Thune's own words).
- Oil: Brent \$114.89 on 9/22, peak \$138.21 on 4/7 (FRED DCOILBRENTU, VERIFIED-PRIMARY).
- The 2y (4.87% on 9/24, FRED DGS2) is ~100bp above the fed funds midpoint of 3.875%. The market expects further tightening.
- SEP 2027 fed funds range: 3.1-4.4 (VERIFIED-PRIMARY). Some FOMC participants already see cuts.
- CBO's Feb 2026 path (10y at 4.1% in 2026) shows what a "return to normal" looks like. It is ~107bp below today.
- **Hypothesis (guess):**
 - The most plausible >27bp rally is energy de-escalation (Hormuz reopening), a growth scare, or an equity crash. Every one of these would also show up in the news before January.
 - Because no one can forecast these, the rule should be mechanical: buy on arrival; longest rungs first; the shortfall comes from the 2028 deposit; and state the funded ratio on the purchase date.
 - The IPS locks on Nov 6, before most of these events, so the rule must be written to survive them.
- **Would change:** the IPS purchase-rule sentence: a trigger stated as a ratio, not a date or forecast. Also the FR scenario table (a named "de-escalation rally" scenario at -50bp = \$307,135, F-102).
- **North star:** she gets a rule she can follow without trusting anyone's rate forecast.

Q03. What does "barring U.S. default" honestly cover? (seed 22)

- **Question:** "Default" has three layers: outright non-payment, a technical delay, and an inflation-driven "implicit" default. Which does the ladder cover, and which must the certainty definition name?
- **Anchors:** R-C47 "All ten payments must be funded by the investment portfolio with a high degree of certainty."; R-C54, R-C55.
- **Evidence:**
 - **Ratings:** Moody's downgraded the U.S. to Aa1 on May 16, 2025 (VERIFIED-PRIMARY, moodys.com). Moody's definition of a downgrade: the borrower is "relatively less likely to meet its debt obligations on time and in full" (VERIFIED-PRIMARY). S&P affirmed AA+ stable (Fortune, Jun 27, 2026, VERIFIED-PRIMARY on fortune.com, secondary report): "All three major ratings agencies have the US pegged one level below Triple-A with stable outlooks". S&P also warned the rating "could slip over the next two years if deficits increase" (paraphrase in the article).
 - **Technical delay:** late T-bill payments in April-May 1979 (Zivney & Marcus 1989; SNIPPET-UNVERIFIED, since taxpolicycenter.org refused with HTTP 403). The snippet reports that bill yields rose ~60bp afterwards.
 - **Market price of default:** U.S. 5y CDS ~36bp in week 37 of 2026 (MacroMicro snippet, SNIPPET-UNVERIFIED). This is a risk-neutral price. It is not a real-world probability.
 - **Implicit default:** see Q04.
- **Hypothesis (guess):**
 - Outright default is not covered, and cannot be covered by any USD plan.
 - Technical delay is largely covered by maturities that pay before the due date (Q06).
 - Implicit default is not covered, and falls on the residency's real budget (Q09).

Checklist for the definition sentence:

- "in nominal U.S. dollars";
- "held to maturity";
- that it rests on the U.S. Treasury paying on time;
- that inflation risk to what \$50k buys is disclosed, not hedged.

- **Would change:** the IPS certainty-definition sentence and the FR assumptions list (R-C56).
- **North star:** a statistics graduate will trust a definition that says exactly what it does not cover.

Q04. Should Wharton's own budget model set the honest limit of "certain"? (seed 22)

- **Question:** Should the FR use the Penn Wharton Budget Model's June 2026 estimate to answer "how certain is certain" in real terms? It gives a 25% chance of reaching the U.S. debt outer bound within 14 years (~2040), and 25% by 2039 in its lower-openness case.
- **Anchor:** R-C56 ("identify the assumptions supporting their recommendation"), R-C85.
- **Evidence** (VERIFIED-PRIMARY, <https://budgetmodel.wharton.upenn.edu/p/2026-06-02-when-does-federal-debt-reach-unsustainable-levels/>, Smetters & He, dated June 4, 2026 on the page; quotes checked verbatim with `--grep`):
 - "Under historical excess cost growth in healthcare, this outer limit is likely reached within 20 years; there is a 25% chance of reaching it in 14 years."
 - The bound is where "default of either explicit (Treasury) debt or implicit (pay-as-you-go) debt becomes a near certainty on a real (inflation-adjusted) basis."
 - Lower-openness sensitivity: "there is already a 25 percent chance of reaching the ceiling by 2039, just 13 years from now."
 - PWBM also says the calculation assumes capital is efficiently priced and that "a sudden devaluation (e.g., AI bubble burst)" would tighten the limit.
 - It is an outer bound, "not a forecast". Median closure years are 2045-2051.
 - **Hypothesis (guess):**
 - Use it once in the FR as the reason the plan says "nominal" out loud. It should not be a headline.
 - Citing Wharton's own model shows research beyond the case, which matters for criterion 2.
 - Risk: over-dramatising. The sentence must say "outer bound, not a forecast".
 - **Would change:** one FR assumption sentence and possibly one number (the 25%-by-~2040 tail as the named reason for an inflation stress case in Q09).
- **North star:** an honest limit, drawn from her own university's research, is more credible to Laura and to co-sponsors than "risk-free".

Q05. The 2028 X-date: does the top-up rule need a debt-limit clause?

- **Question:** The debt limit is projected to bind around May 2027, with an X-date around Feb 2028 and a need to act "by January 1, 2028". Does the plan's 2028 step (deposit arrives; buy any missing long runs) need a clause for a debt-limit standoff at that exact time?
- **Anchor:** F-601 (\$150,000 at the beginning of 2028), R-C77.
- **Evidence:**
 - NTU (Aug 20, 2026; VERIFIED-PRIMARY on ntu.org; think-tank analysis, not official): "The X-Date is likely to be reached in February 2028"; "Congress would likely need to increase the debt limit by January 1, 2028"; headroom "under \$1.4 trillion ... as of August 10, 2026"; the limit was raised to \$41.104 trillion on July 4, 2025.
 - Time (Sept 23, 2026) headline: "A Debt Ceiling Fight Looms in 2027" (SNIPPET-UNVERIFIED; the page returned HTTP 406).
 - 2028-01-01 is a Saturday (derived).
- **Hypothesis (guess):**
 - Mostly a non-issue for STRIPS maturing 2032-2041, since X-date risk sits in bills maturing near the X-date.
 - The 2028 cash, if parked in T-bills while waiting, should avoid bills maturing in the X-date window.
 - Long yields in past standoffs often fell (flight to quality), which would make the top-up dearer (ASSUMPTION; needs 2011/2023 data).
 - Probably one FR line, not an IPS rule.
- **Would change:** an FR operating rule (where 2028 cash waits; buy long runs promptly).
- **North star:** shows we checked the real 2027-28 calendar against her cash-flow dates.

Q06. Do early maturities buffer a technical delay? (seed 22)

- **Question:** The reserve's Nov-15 maturities pay ~47 days before each Jan-1 payment (S1 file). Do they also act as a buffer against a technical Treasury payment delay? Should that be the stated reason to prefer them over instruments that pay on or after Jan 1?
- **Anchor:** R-C45 ("one at the beginning of each year from 2033 through 2042"), R-C47.
- **Evidence:**

- S1 found the STRIPS ladder costs \$2,124 more because the cash arrives early (wins_now/S1_treasury_sleeve.md, VERIFIED-PRIMARY inputs).
- 2033-01-01 is a Saturday, so the first payment date is not a business day (derived).
- 1979 delays lasted days to weeks (SNIPPET-UNVERIFIED).
- **Hypothesis (guess):** Yes, and it is nearly free. The 47-day lead turns "early cash" from a \$2k cost into a control against both the non-business-day problem and a short technical delay. It fits the operational-risk gap flagged in BS-18.
- **Would change:** the maturity-choice decision (Nov 15 vs Dec funds vs Feb 15) and one FR line.
- **North star:** her operators get paid on time even if Washington is late.

Q07. What does the plan do after another downgrade?

- **Question:** If an agency cuts the U.S. again (for example to AA), what will the plan do, and should the IPS pre-commit to "no forced selling of held-to-maturity Treasuries on a rating change"?
- **Anchor:** R-C89 ("formally establishes the strategy and the team's decision-making framework"), R-I10.
- **Evidence:** S&P says the rating could slip "over the next two years if deficits increase" (Fortune, Jun 27, 2026). CBO's baseline deficit is 5.8% of GDP in 2026, rising to 6.7% by 2036 (SNIPPET-UNVERIFIED; cbo.gov 403; CRFB summary read, secondary).
- **Hypothesis (guess):**
 - A downgrade changes none of the ladder's cash flows. Selling would turn a headline into a real loss.
 - One clause of pre-commitment shows decision discipline, and it costs ~15 words.
 - Private-banker coaching logic (seed 27) applies here.
- **Would change:** an IPS decision-rule clause (or leave it to the FR if words are short).
- **North star:** Laura sees we will not panic on her behalf.

Q08. How many IPS words should sovereign risk get?

- **Question:** Every U.S.-dollar plan from every rival shares the Treasury as its floor of certainty. Should the IPS spend one clause on sovereign risk and use the saved words on risks the ladder does not cover (inflation, TWD, operations)?
- **Anchor:** R-I18 (500-word maximum), R-C54.
- **Evidence:** INTERPRETATION. Bank deposits above the insured limit are riskier than Treasuries. Deposit insurance itself relies on the U.S. government. Ratings: sovereign AA+/Aa1/AA+ (Q03).
- **Hypothesis (guess):** Yes: one clause in the IPS, one paragraph in the FR. Most teams will either ignore the risk or over-dramatise it.
- **Would change:** the IPS word budget.
- **North star:** clear priorities read as judgement, not jargon.

Q09. Who bears inflation's erosion of the fixed \$50k? (seed 23)

- **Question:** What is the real value of the ten payments at the market's own inflation price (breakeven), and at one named stress case? Who bears the gap: the residency, co-sponsors, or Laura's flexibility money?
- **Anchor:** R-C46 ("each payment is a fixed \$50,000 and is not adjusted for inflation"), R-C85, R-C44 ("operating support beyond Laura's commitment, may come from co-sponsors, grants ...").
- **Evidence** (derived, section 5; inflation rates are ASSUMPTIONS except the breakeven, which is VERIFIED-PRIMARY):

U.S. inflation a year	2033 payment in 2026 \$	2042 payment in 2026 \$	Ten payments in 2026 \$
2.34% (10y breakeven, 9/25)	42,526	34,534	384,066
3.7% (Fed 2026 PCE median, stress)	38,772	27,958	331,037
5.0%	35,534	22,906	288,104
7.0% (≈ worst post-war 16-year U.S. CPI rate, 7.12% to 1982)	31,137	16,937	234,005

The worst rolling 16-year CPI rate (7.12%/yr) and 10-year rate (8.84%/yr), both ending 1982, come from FRED CPIAUCSL (derived). Headline CPI is 3.71% y/y (Aug 2026) but core CPI only 2.76% (derived, FRED), so the energy shock drives the gap.

- **Hypothesis (guess):**
 - The residency bears it by design: the payment is fixed, and co-sponsors may fund operating support beyond Laura's commitment.
 - The plan's honest move is to state the real value at market breakeven, plus one stress, and say who covers the gap.
 - F-114 already has 2.0/2.5/3.7%. New here: the market-implied rate and a historically grounded worst case.
- **Would change:** the FR inflation-assumption numbers (R-C85) and one co-sponsor-draft fact.
- **North star:** she sees we told her what \$50k will really buy in 2042.

Q10. Is high inflation the base case or the stress case?

- **Question:** Does the 2026 evidence support treating 1970s-style inflation as a stress, not a base case, for 2033-2042?
 - Fed independence was upheld in *Trump v. Cook* (June 29, 2026).
 - The FOMC hiked 12-0 and wrote "The Committee will deliver price stability."
 - The 5y5y forward inflation rate is anchored at 2.34%.
 - Against that: Kevin Warsh was confirmed in the most partisan chair vote on record.
 - **Anchors:** R-C85, F-007, F-009.
 - **Evidence:**
 - FOMC statement 2026-09-16 (VERIFIED-PRIMARY; quote checked).
 - *Trump v. Cook*, decided June 29, 2026; "The Government's application is denied" (VERIFIED-PRIMARY, supremecourt.gov PDF). News reports a 5-4 vote and for-cause protections upheld (SNIPPET-UNVERIFIED).
 - Warsh confirmed 54-45 on May 13, 2026 (SNIPPET-UNVERIFIED, CNN/CNBC). The Fed press release of May 15, 2026 names Powell chair pro tempore "until Kevin M. Warsh is sworn in" (VERIFIED-PRIMARY).
 - 5y5y (T5Y1FR) 2.34% on 9/25; its 2026 range was 2.05-2.36 (FRED, VERIFIED-PRIMARY).
 - SEP longer-run PCE 2.0% (F-007).
 - **Hypothesis (guess):** Market and institutional evidence says stress, not base. PWBM (Q04) says the tail is real and lies inside the window. Recommended framing: base = market breakeven; stress = one named high-inflation path.
- **Would change:** the FR assumption sentence and the choice of stress rate in Q09.
- **North star:** balanced, sourced realism instead of doom or complacency.

Q11. Is the flexibility money the inflation buffer? (seed 23; NEW angle, not a re-ask of settled item 8.10)

- **Question:** Should the post-2033 flexibility money be named as the buffer that lets Laura top up the operating budget if inflation erodes it? And should part of it sit in TIPS? The reserve stays nominal: brief 8.10 is not reopened.
- **Anchors:** R-C58 ("committing all remaining assets could limit her financial flexibility as the project develops"), R-C84, R-C44.
- **Evidence:**
 - TIPS real par yields: 5Y 2.64%, 7Y 2.73%, 10Y 2.83% (F-008).
 - A top-up to the operating budget is voluntary, since only the \$50k is committed (R-C45).
 - Brief 8.10 says TIPS are not the core hedge because the liability is nominal. That still holds for the reserve.
- **Hypothesis (guess):**
 - Possibly yes, as a principle ("flexibility money is partly real-return protected") rather than a sized fund. The case says teams need not size a contingency fund (R-C61).
 - Complexity test: it must fit in one FR sentence, or drop it.

- **Would change:** the decision on the composition of flexibility money, plus one FR sentence.
- **North star:** it answers the question a co-sponsor will ask ("what if \$50k is not enough in 2040?") without touching the promise.

Q12. Which return assumptions are "reasonable" when forecasters lag the market?

- **Question:** JPM (set when the 10y was 4.16%) and CBO (10y 4.1% for 2026) both sit ~100bp below today's yields. Should projections take sleeve bond returns from today's market yields and cite JPM only for equities, or wait for the 2027 LTCMA?
- **Anchor:** R-C77 ("use reasonable return assumptions consistent with their strategy"), F-010, F-317.
- **Evidence:**
 - IEF YTM 5.17%, TLT 5.54% (F-203, F-204).
 - JPM intermediate Treasuries 4.00% (F-307).
 - CBO 10y 4.1% (2026) and 4.4% (2031-35) (SNIPPET-UNVERIFIED via CRFB).
 - The 2027 JPM LTCMA is not yet out (F-315). Last year's edition came out ~Oct 20, which is before the IPS.
- **Hypothesis (guess):**
 - Use market yields for bonds (a held bond's return is roughly its yield) and JPM for equities. Say so in one line.
 - Mixing sources needs one sentence of justification.
 - This likely raises the sleeve's median and the 2031 floor slightly (F-317: ~1pp/yr on the bond part).
- **Would change:** the projection inputs; the 2031 range and facility numbers (FR); one IPS phrase on assumptions.
- **North star:** assumptions tied to prices she can check, not to a stale forecast.

Q13. Does 60% sleeve equity still earn its p5 cost? (extends open item 9, with NEW evidence)

- **Question:** At today's yields, JPM's 6.70% equity return beats a ~5.0% 6-year Treasury by only ~1.7pp, and the S&P 500 CAPE is ~41. Does 60% sleeve equity still earn its downside cost, or is 40-50% the "thoughtful risk" answer?
- **Anchor:** R-C38 ("recommend an appropriate balance between pursuing growth and protecting the capital required for her goals"), F-301, F-407.
- **Evidence:**
 - CAPE 41.48 (multpl.com, read on page; secondary data site; mean 17.42).
 - Forward P/E 19.32 on 9/24 (StreetStats snippet, SNIPPET-UNVERIFIED). That is an earnings yield of ~5.2%, roughly equal to the 10y nominal.
 - S&P 500 7,743 on 9/25, +17.2% over 1y, record 7,799 on 8/13 (FRED SP500, VERIFIED-PRIMARY).
 - Brief 8.5: more sleeve equity buys range, not median. New here: at today's yields, the expected premium itself is thin.
 - Rough arithmetic (ASSUMPTION): moving 20% of a ~\$158k sleeve from equity to bonds for ~5 years at a 1.7pp premium costs ~\$2-3k of median wealth.
- **Hypothesis (guess):** 50% may be the better-balanced number, with a small median cost and a better p5. But complexity and consistency with WInS matter more than 10pp either way. Phase D should re-run `strategy_mc.py` with bond returns at market yields.
- **Would change:** the sleeve equity % (IPS) and the 2031 range (FR).
- **North star:** "thoughtful risk" means taking equity risk only where it is paid.

Q14. Should risk be stressed at a positive stock-bond correlation? (extends open item 9, with NEW evidence)

- **Question:** In 2026, stocks and Treasury prices fell together (daily $\text{corr}(\text{S\&P return}, \Delta 10\text{y}) = -0.44$). Should the sleeve's p5 and the 2031 range be stressed at a positive stock-bond return correlation instead of JPM's ~0?
- **Anchor:** F-314 (JPM large cap vs intermediate Treasuries -0.01), R-S26.

- **Evidence:** 2024 -0.06, 2025 +0.14, 2026 -0.44, since June 2026 -0.48 (FRED SP500 and DGS10, derived). A negative number here means stocks fell when yields rose, i.e. both fell together.
- **Hypothesis (guess):**
 - A +0.3 to +0.4 stress lowers the sleeve's p5 by a few thousand dollars. The ladder is unaffected, because it is held to maturity.
 - WInS: the equity and Treasury holdings will fall together on inflation news. Expect it, and do not "fix" it by trading.
- **Would change:** the FR sleeve p5 and 2031 range numbers; the WInS expectation-setting note.
- **North star:** we tested the plan against the market she actually lives in.

Q15. Should the team pre-commit not to trade around the big October-November events?

- **Question:** Should the team state in a trade note, before the fact, that the Treasury hedge will not be traded around the Oct 28 FOMC or the Nov 4 refunding? Then any mark-to-market loss becomes evidence that the strategy was "tested", not a regret.
- **Anchor:** R-T9 ("how it aligned with, tested, or refined your overall investment strategy"), R-W73 ("does not require frequent or same-day trading"), R-W76.
- **Evidence:** FOMC Oct 27-28 (no new projections; the next SEP is Dec 9) and refunding Nov 4 both fall inside the WInS window (VERIFIED-PRIMARY). TLT sits at its 52-week low (F-204).
- **Hypothesis (guess):** Yes. It costs nothing and gives a ready "tested" note. The day-trading ban also argues for holding (brief section 14).
- **Would change:** a WInS conduct decision and the choice of the three notes.
- **North star:** she sees discipline under pressure, the behaviour she is hiring for.

Q16. Should each Treasury note record the day's funded ratio?

- **Question:** Should every WInS Treasury trade note record that day's ladder cost and funded ratio? The three notes would then show a live, dated test of the strategy.
- **Anchor:** R-T9, F-111 (ladder > \$300k on 173 of 185 days), F-117 (1.026).
- **Evidence:** The ladder was above \$300k for most of 2026 and below it only since Sept 10 (F-111). A note written at trade time is permanent (brief section 14).
- **Hypothesis (guess):** Yes, as one number per note, from the treasury.gov curve run through `official_curve_pv.py`. It is the simplest quantitative thread linking the TN, IPS and FR.
- **Would change:** the format of the trade notes (WInS-now) and the TN reflections.
- **North star:** one number Laura could track herself, since she is a statistics graduate.

Q17. Is the 10-20 year sector the cheapest place for Laura's long rungs?

- **Question:** The 20y par yield (5.54%) is above the 30y (5.49%). Does that make the 10-20y sector the cheapest point on the curve for the 2037-2042 rungs? Does it add a price reason, not only a matching reason, to S1's IEF+TLH hedge?
- **Anchor:** F-001, F-107 (2037-42 rungs cost \$155,408).
- **Evidence:** 20Y 5.54 vs 30Y 5.49 on 9/25 (F-001, VERIFIED-REPO-FILE). S1 recommends IEF+TLH on matching grounds (PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK).
- **Hypothesis (guess):** Yes. The 20y sector has traded cheap (higher yield) since the 20y bond was re-introduced (SNIPPET-level knowledge; verify with the treasury.gov history). Laura's long rungs fall exactly there.
- **Would change:** the instrument decision (supports TLH-type exposure) and a TN reason.
- **North star:** her money buys the promise at the best available price.

Q18. Should WInS hold individual Treasuries, not only ETFs?

- **Question:** WInS permits "Any Government/Treasury Bonds ... available on WInS" at \$10 a trade. Should the WInS book buy one or more individual Treasuries maturing Nov 2032-Nov 2041? That would show a locked yield (a ladder) rather than only constant-maturity ETFs, whose yield never locks.

- **Anchors:** F-606 ("Any Government/Treasury Bonds from any exchange available on WInS"), F-607, R-C75.
- **Evidence:** permitted type VERIFIED-PRIMARY; which issues are listed is UNVERIFIED (logged-in check). An ETF's duration and yield drift (F-205), while an individual bond's yield to maturity is fixed at purchase.
- **Hypothesis (guess):** Yes, if WInS lists them: one or two individual bonds, as a "demonstration rung". The trade note is then literally the strategy. Check the position limit and the 2x-daily-volume rule for bonds.
- **Would change:** the WInS instrument decision; Trading Note choice.
- **North star:** the WInS book shows her the ladder itself, not a proxy.

Q19. Is a 2027 recession the common cause of the joint tail? (extends unanswered open item)

- **Question:** Is a 2027 U.S. recession the common cause that both lowers yields (dearer long rungs) and shrinks the 2028 deposit (advances, speaking)? If so, how large is the joint tail?
- **Anchors:** R-C86, F-403, F-404.
- **Evidence:** needs data. The 10y fell sharply in 2001, 2008 and 2020 (FRED DGS10, known pattern; not computed here). Publishing and speaking income in recessions: no source found yet.
- **Hypothesis (guess):**
 - Yes, the recession link is plausible.
 - Lock-early limits the damage: only rungs not yet bought are exposed. In January 2027, with a 27bp cushion, that is none unless rates fall first.
 - The joint tail is Jan 2027 rally + 2028 deposit cut. Up to ~\$23-33k is unfunded (brief section 9, unverified). Phase D should price it with a scenario, not a correlation guess.
- **Would change:** the FR joint-tail row, and whether the 2031 floor must hold back money for unfunded rungs.
- **North star:** we tested the one scenario where her income and her markets fail together.

Q20. What does an AI-equity crash do to Treasuries and to Laura?

- **Question:** How does an AI-equity crash feed into Treasury yields and the January 2027 cushion, and which way does it cut for Laura?
- **Anchors:** R-C82 ("how investment uncertainty could affect both the operating commitment and the facility contribution"), F-111.
- **Evidence:**
 - Top 10 $\approx$ 38-41% of the S&P 500 (SNIPPET-UNVERIFIED).
 - PWBM: "a sudden devaluation (e.g., AI bubble burst)" tightens U.S. debt capacity (VERIFIED-PRIMARY).
 - The 2026 correlation regime (Q14).
- **Hypothesis (guess):**
 - Before Jan 2027: a crash probably lowers yields (flight to quality), so the ladder gets dearer. Laura holds cash, not stocks, so her only exposure is the cushion.
 - After 2027: a crash hits only the sleeve, while the ladder's market value rises (a good hedge).
 - Long run: a crash worsens fiscal capacity (PWBM), which argues for "nominal" honesty.
- **Would change:** one FR scenario sentence; possibly the IPS "buy on arrival" rationale.
- **North star:** she sees how her promise behaves in the scenario everyone fears in 2026.

Q21. Is the sleeve doubling Laura's Taiwan bet? (seed 24; sleeve decision; open item "EWT keep or drop")

- **Question:** A cap-weighted U.S. index is heavy in AI firms that depend on Taiwan chip supply, and the residency is in Taiwan. Should the sleeve deliberately avoid adding Taiwan exposure (drop any Taiwan tilt; consider broader or global equity)?

- **Anchor:** R-C39 ("In 2033, Laura plans to establish a collaborative creative residency in Taiwan."); R-S24 ("uses appropriate diversification"); R-W71 (diversification by "funding purposes").
- **Evidence:**
 - Taiwan's defence ministry (Sept 2026) says PLA drills focus on blockade (SNIPPET-UNVERIFIED, Taiwan News / Seoul Economic Daily).
 - Taiwan real GDP forecast +11.05% in 2026 on the AI boom (F-504).
 - Index concentration (Q20).
- **Hypothesis (guess):**
 - Drop the Taiwan tilt. The project already carries Taiwan risk. The case gives no reason for identity-based tilts, and brief section 4 bans tokenism.
 - A global or broad index lowers AI concentration (D2 to size).
- **Would change:** the sleeve index decision (IPS; WInS-now).
- **North star:** her portfolio does not bet twice on the place her dream already depends on.

Q22. Measured in dollars, how fast have Taiwan construction costs risen? (seed 25)

- **Question:** Measured in U.S. dollars (Laura's currency), how fast have Taiwan construction costs risen? Does that change how the FR explains "the effect of inflation on ... facility costs"?
- **Anchor:** R-C85 ("the effect of inflation on portfolio projections and facility costs"), F-508, F-512.
- **Evidence** (derived; inputs VERIFIED-PRIMARY):
 - CCI Aug 2026 +6.53% y/y in TWD (F-508).
 - USD/TWD monthly averages: Aug 2025 30.154, Aug 2026 32.023, 2021 average 27.937 (FRED DEXTAUS).
 - Result: USD-measured CCI **+0.31% y/y**. Since the 2021 base: **3.57%/yr in TWD** less **2.72%/yr TWD depreciation**, which gives **+0.82%/yr in USD**.
 - ASSUMPTION: monthly-average FX is matched to the monthly index, and the 2021 annual average is the base.
- **Hypothesis (guess):**
 - The headline "costs up 6.5%" overstates the pressure on a USD contribution.
 - But the offset depends on the TWD staying weak. A TWD rebound, e.g. the Fed cutting while the CBC holds (F-515), would reverse it.
 - A strong FR sentence contains: both currencies, the offset, and the reversal risk.
 - This refines brief 8.15; it does not contradict it.
- **Would change:** the FR facility-inflation sentence and numbers; the co-sponsor range explanation.
- **North star:** co-sponsors in Taiwan and a USD donor both see their own number.

Q23. Will Taiwan's 2026 construction-cost jump fade before 2033? (seed 25)

- **Question:** Is the 2026 jump an AI-boom wage spike that will fade before 2033, or a new trend?
- **Anchor:** F-508 (wage class +8.12% y/y; the index was flat in 2024-25), F-504, R-C94 ("The total cost of the facility has not been determined.").
- **Evidence:** Taiwan real GDP 2026 +11.05% (F-504). The energy shock (F-517) and Hormuz (Q02) raise material costs. Taiwan imports most of its energy (general knowledge; not verified here).
- **Hypothesis (guess):** A mix: the wage pressure from fab construction may persist while AI capex lasts, and energy may fade. Use the since-2021 rate (~3.5%/yr in TWD) as the base, not the 2026 spike.
- **Would change:** the facility-inflation assumption number (FR).
- **North star:** realistic, sourced facility expectations protect her credibility.

Q24. What happens to the payments if the residency cannot operate in Taiwan? (seed 24)

- **Question:** If a blockade or quarantine stops the residency operating in Taiwan, is the USD Treasury reserve portable? Should the plan tie the promise to the residency's operating expenses rather than to the place?

- **Anchors:** R-C39; R-C45 ("toward the residency's operating expenses"); R-C48.
- **Evidence:** The case promises payments to the residency's operating expenses; it does not name a Taiwan bank or currency (R-AN10). Treasuries are held in USD and can be paid anywhere.
- **Hypothesis (guess):** The reserve is location-agnostic by construction, which is a strength worth one FR line. A relocation plan is out of scope (Taiwan legal issues are excluded, R-C97). Do not dramatise.
- **Would change:** one FR sentence; the co-sponsor draft (what happens in a disruption).
- **North star:** her promise survives even the worst geopolitical case, calmly stated.

Parked (logged; failed the "would it change something" filter)

- 2031 2y yield uncertainty for the floor growth factor ($\pm$ ~\$5k; F-013 already labels it an ASSUMPTION).
- Treasury auction-size changes "in early 2027" (Yahoo snippet): no decision depends on it.
- U.S. dollar broad index: -8.1% from its Jan 2025 high but flat over 1y (FRED DTWEXBGS). It matters only through TWD (Q22).
- Investment-grade credit spreads 0.79% (FRED BAMLC0A0CM): calm. The stress is in rates, not credit. No plan change.

4. Sources (all accessed 2026-09-27 from this container)

Primary (read on the page; VERIFIED-PRIMARY)

- FOMC statement 2026-09-16: <https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm> (quote "The Committee will deliver price stability." checked verbatim).
- SEP tables 2026-09-16 (June comparison rows): <https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm>
 - Fed funds medians 4.1/4.1/3.9/3.6/LR 3.2 (June: 3.8/3.6/3.4/LR 3.1).
 - 2027 range 3.1-4.4.
 - PCE 2026 range 2.9-3.8.
- FOMC calendar: <https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm> (Oct 27-28; Dec 8-9 with SEP).
- Fed press release 2026-05-15 (Powell chair pro tempore pending Warsh): <https://www.federalreserve.gov/newsevents/pressreleases/other20260515a.htm>
- Treasury quarterly refunding statement Aug 5, 2026: <https://home.treasury.gov/news/press-releases/sb0590> (sizes steady "for at least the next several quarters"; next announcement Nov 4, 2026).
- Moody's U.S. rating page: <https://www.moody's.com/web/en/us/about-us/usrating.html> (downgrade to Aa1, May 16, 2025; downgrade definition quote checked).
- Supreme Court, *Trump v. Cook*, No. 25A312, decided June 29, 2026: https://www.supremecourt.gov/opinions/25pdf/25a312_5468.pdf
- Penn Wharton Budget Model, "When Does Federal Debt Reach Unsustainable Levels? Spring 2026 - Onward", Smetters & He, June 4, 2026: [https://budgetmodel.wharton.upenn.edu/p/2026-06-02-when-does-federal-debt-reach-unsustainable-levels/\(three quotes checked verbatim\)](https://budgetmodel.wharton.upenn.edu/p/2026-06-02-when-does-federal-debt-reach-unsustainable-levels/(three%20quotes%20checked%20verbatim)).
- PWBM, "Supreme Court Tariff Ruling: IEEPA Revenue and Potential Refunds", Feb 20, 2026: [https://budgetmodel.wharton.upenn.edu/p/2026-02-20-supreme-court-tariff-ruling/\(6-3 ruling; up to \\$175bn refunds\)](https://budgetmodel.wharton.upenn.edu/p/2026-02-20-supreme-court-tariff-ruling/(6-3%20ruling%20up%20to%20$175bn%20refunds)).
- FRED CSVs (<https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES>): THREEFYTP10, DFII10, T10YIE, T5YIFR, DGS2, DGS10, DGS30, SP500, DTWEXBGS, DEXTAUS, CPIAUCSL, CPILFESL, PCEPILFE, DFEDTARU, DCOILBRETEU, BAMLC0A0CM.

Secondary but read on the page (the claim is the author's, not official):

- National Taxpayers Union, "Projecting the Debt Limit", Aug 20, 2026: <https://www.ntu.org/publications/detail/projecting-the-debt-limit> (quotes checked verbatim).
- Fortune, "S&P keeps U.S. sovereign rating at AA+ with stable outlook", Jun 27, 2026: <https://fortune.com/2026/06/27/sp-global-us-sovereign-credit-rating-aa-stable-outlook-debt-deficits/>

- CRFB summary of CBO Feb 2026 outlook: <https://www.crfb.org/papers/cbos-february-2026-budget-and-economic-outlook>
- CRA government-affairs blog on the FY27 CR (to Dec 11): <https://cra.org/govaffairs/blog/2026/09/fy2027-sept-update/>
- multpl.com Shiller PE (41.48; mean 17.42): <https://www.multpl.com/shiller-pe>

Blocked or refused (not read; nothing inferred from them)

- cbo.gov publication page and PDF (HTTP 403).
- bipartisanpolicy.org (403).
- congress.gov CRS (403).
- time.com (406).
- taxpolicycenter.org (403).
- streetstats.finance (no text returned).

SNIPPET-UNVERIFIED (search results only)

- Warsh confirmed 54-45 on May 13, 2026 (CNN, CNBC).
- *Trump v. Cook* 5-4 majority details.
- U.S. 5y CDS ~36bp (MacroMicro).
- Forward P/E 19.32 (StreetStats).
- S&P 500 top-10 $\approx$ 38-41% (various).
- Iran war since Feb 28, 2026, and Hormuz closure (Wikipedia, World Bank blog titles).
- PLA blockade focus (Taiwan News, Sept 3, 2026).
- 1979 T-bill delays (+~60bp; Zivney & Marcus 1989).
- Treasury auction sizes expected to change early 2027 (Yahoo).

5. How the derived numbers were made (re-runnable)

1. Download the FRED CSVs listed above with `curl -sSL "https://fred.stlouisfed.org/graph/fredgraph.csv?id=SERIES"`.
2. Yield decomposition (Q01): values of DGS10, DFII10, T10YIE and THREEFYTP10 on or before each date. The "last time" checks find the latest earlier date with a value at or above today's.
3. Correlation (Q14): daily `SP500.pct_change()` against daily `DGS10.diff()`, aligned on common dates, by calendar year.
4. Real value (Q09): $50000 / (1+i)^{*(\text{year}-2026)}$. Worst rolling U.S. CPI: $(\text{CPI}_t / \text{CPI}_{\{t-12k\}})^{*(1/k)} - 1$ for $k = 10$ and 16 years.
5. USD construction cost (Q22): `CCI/FX` with the monthly-average DEXTAUS. CCI Aug 2025 = 119.50/1.0653 (F-508). Base 2021 = 100 against the 2021 average FX. Period length 5.08 years (mid-2021 to Aug 2026).

These were run inline in the scratchpad. Phase D should save a script under `research/insight_v1/scripts/` before any number goes into a deliverable.

6. Leads for the specialists

D1 (rates and fixed income)

- Use the FRED decomposition (Q01) as the "why now" fact base. Add NY Fed ACM term premium as a second estimate (newyorkfed.org, not tried).
- Price the event calendar (Q02): the ladder cost at -25/-50bp is already in F-102.
- Q03-Q07 need one table: layer of default $\rightarrow$ covered? $\rightarrow$ control $\rightarrow$ words used.
- Check the 20y "hump" history on treasury.gov (Q17).
- Price the 2028 X-date: find 2011 and 2023 10y/30y moves around the X-dates (FRED DGS10/DGS30).
- CBO's current outlook: try the CBO data page via another route, or the CBO GitHub data (not tried). The Aug/Sept 2026 CBO update, if any, was not found.

D2 (equity and AI concentration)

- Get primary index weights: S&P DJI fact sheet or an iShares IVV holdings CSV, the same method as S2. Get the earnings yield from a primary source.
- PWBM's "AI bubble burst" line links AI to fiscal capacity (Q20).

D3 (quant)

- Re-run `strategy_mc.py` with sleeve bonds at market yields (~5.0%) and a stock-bond return correlation of +0.3/+0.4 (Q12-Q14).
- Build the joint-tail scenario (Q19).

D4 (Taiwan, FX, geopolitics)

- Q22 is the headline: repeat it with DGBAS monthly CCI levels via `A2_taiwan_cci.py`, not the y/y ratio.
- Test windows (1y, 3y, since 2021) and the reversal case (TWD at 28-29 as in 2021).
- Check Taiwan's energy import share on a primary page (Bureau of Energy).

D6 / D9

- Q07, Q08, Q15 and Q16 are pre-commitment and communication choices. Each should fit in ≤ 15 words of IPS or one TN line.

D10

- Q18 needs the logged-in WInS check: are individual U.S. Treasuries listed, and do bonds count toward the position limit?

D7

- Citing the Penn Wharton Budget Model (Q04) in a Wharton competition is unusual. Check whether past winners cited Wharton research.

What this teaches

A yield is not one number. It is a stack: what the market expects the Fed to do, plus inflation expectations, plus a term premium for lending long. In 2026 almost all of the rise came from the real part, not from inflation fears. That tells you something useful: Laura was being paid well to lend long at exactly the moment her promise required it.

"Risk-free" is also a stack. Treasuries are as safe as U.S. dollars get:

- in nominal terms, near certain;
- against a late payment, mostly protected if the money arrives early;
- against inflation, not protected at all.

The honest skill is to say which layer your plan covers and which it does not. Then back each claim with a primary source you read yourself, whether that is the Fed's own statement, the Treasury's own schedule or Wharton's own budget model.

B5b: 2026 macro and markets, seen from equities, AI concentration, Taiwan, China and FX

Agent B5b, insight_v1 run, Phase B (question generation), written 2026-09-27. This is AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It contains **no submission-ready prose**. Where it says what a sentence "must contain", that is a checklist; the students write every word. Any security named is for fact context only and carries "PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK". Record this file in the team's AI-use log (Works Cited).

Lens: "2026 Macro & Markets Landscape". My starting angle (distinct from B5a, who starts from rates and fiscal): what a broad fund really owns in 2026 (AI and Taiwan), valuations, Taiwan prices and wages, the Taiwan dollar, Strait risk, and what all of it means for Laura. Seeds extended: 21, 22, 23, 24, 25.

Status labels follow brief section 3. I read every "VERIFIED-PRIMARY" item myself on 2026-09-27. Where I read a reputable secondary page in full (news agency, data aggregator) but not the primary document, I say "SECONDARY-READ" and it counts as SNIPPET-UNVERIFIED for decision purposes.

0. Summary (read this first)

Twenty-three questions follow (section 2). The eight that I think most teams will miss:

1. **The equity "premium" over Laura's own Treasury ladder is thin in 2026.** The S&P 500's forward earnings yield ($1/19.1 = 5.24\%$) is about equal to the 10-year Treasury (5.17%). The CAPE earnings yield ($1/41.48 = 2.41\%$) is below the 10-year TIPS real yield (2.83%): a negative "excess CAPE yield" of about -0.4pp. JPM's 6.70% for U.S. large caps is only 1.2-1.6pp above the ladder's forward zero rates (5.08-5.48%). "Thoughtful risk" in 2026 means taking equity risk for a small expected reward, and the team should price that out loud (Q1, Q2).
2. **A "U.S.-only" index is not Taiwan-free.** About 20% of the S&P 500 by weight is in companies whose leading chips are made mainly by TSMC (NVDA 8.18, AAPL 7.37, AVGO 2.51, AMD 1.55, MRVL 0.34, QCOM 0.31; SPYM holdings 2026-09-24; the supplier link is an ASSUMPTION to verify). Add VT's direct 3.5% Taiwan weight. Laura's portfolio, her residency and much of the world index all depend on one island (Q3, Q4).
3. **The Taiwan dollar tends to move in Laura's favour in bad times and against her in good times.** USD/TWD rose (TWD weaker) by 7.2% in the 1995-96 Strait crisis, 6.4% in Aug-Sep 2022 and 14.5% in 2008-09. In calm boom times it can jump the other way: -6.7% in two trading days (1-5 May 2025) and -13.0% from 1 Apr to 3 Jul 2025, which raised the USD cost of a NT\$ budget by 15%. Monthly USD/TWD changes correlate -0.38 with S&P 500 returns (2016-2026; FRED, computed). So the facility contribution may be **less** uncertain in NT\$ than in US\$ (Q6, Q7).
4. **The TWD looks "held weak".** Taiwan's current-account surplus was 19.6% of GDP in 2025 and the U.S. Treasury keeps Taiwan on its monitoring list; life insurers have cut currency hedges and the central bank leaned against appreciation (SECONDARY-READ). The jump risk is skewed toward a stronger TWD (Q8).
5. **The wrong inflation index is the default.** The \$50k is spent in Taiwan, mostly on people. Taiwan's minimum wage rises 4.745% on 2027-01-01 (11th straight rise) and regular earnings grew 3.18% y/y in June 2026, versus Taiwan CPI ~2%. At 3.5% a year the 2042 payment is worth about \$28.8k in 2026 money; at Taiwan CPI 2.07%, about \$36.0k (ASSUMPTION rates). U.S. CPI (JPM 2.5%) is neither (Q9, Q18).
6. **The September 2026 cheap ladder is partly an AI-boom side effect.** Hyperscaler capex is heading past \$690bn in FY26, now partly debt-financed (debt = 32% of capex, LTM mid-2026; FactSet); the FOMC says "capital investment is robust"; the 10-year term premium hit 0.97% on 2026-09-16, the highest since 2011-02-11 (FRED). An AI bust before January 2027 is a growth shock: yields and stocks would likely fall together, making the ladder dearer exactly when the sleeve shrinks (Q10).
7. **The stock-bond relationship has flipped.** Daily correlation of S&P 500 returns with 10-year yield changes: +0.41 in 2016-2020 (bonds hedged stocks), -0.44 in 2026 (stocks fall when yields rise, so bonds and stocks fall together). JPM's long-run -0.01 sits in between. Wharton's own sample note

justifies a Treasury ETF "to reduce portfolio volatility" - the weakest reason in this regime; matching the liability is the strong one (Q11, Q19).

8. **An EM tilt is a double bet on the Strait.** Taiwan is now 26.85% of MSCI EM (TSMC alone more than 14%), China is also a large weight, and JPM's highest equity return (EM 7.8%) comes with that concentration. Russia 2022 shows what "uninvestable" means for an index: MSCI removed it "at a price that is effectively zero" (Q14).

1. Key evidence gathered (with status)

1.1 Valuation and concentration (U.S.)

Fact	Value	Source (accessed 2026-09-27)	Status
S&P 500 forward 12-month P/E	19.1 (5-yr avg 19.8, 10-yr avg 19.0; 20.4 on 30 Jun)	FactSet Earnings Insight, 11 Sep 2026, https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_091126.pdf	VERIFIED-PRIMARY
Trailing 12-month P/E	25.9 (5-yr avg 24.4)	same	VERIFIED-PRIMARY
CY2026 / CY2027 EPS growth	31.6% / 15.1%; Q2 2027 only +1.3%	same	VERIFIED-PRIMARY
Semiconductors' Q3 2026 EPS growth	125%; IT sector 63.0%, only 24.0% without semis	same	VERIFIED-PRIMARY
S&P 500 close	7,591.70 (10 Sep; FactSet); 7,743.41 (25 Sep; FRED SP500); 6,845.50 end-2025 → +13.1% YTD; 2026 low 6,343.72 on 30 Mar	FactSet; https://fred.stlouisfed.org/series/SP500	VERIFIED-PRIMARY (derived)
Shiller CAPE	41.48 (25 Sep 2026); 40.90 (1 Sep); max 44.19 (Dec 1999); mean 17.42	https://www.multpl.com/shiller-pe (aggregator of Shiller data)	SECONDARY-READ; Shiller's own file not read
S&P 500 top-10 weight; 12-name AI basket; VT top-10	38.8%; 40.7%; 21.7%	research/insight_v1/wins_now/S2_growth_sleeve.md (issuer files)	VERIFIED-PRIMARY via S2
TSMC-dependent S&P 500 weight	~20.3% (NVDA, AAPL, AVGO, AMD, MRVL, QCOM)	computed from research/insight_v1/scripts/data/S2/spym_holdings.csv	weights VERIFIED-PRIMARY via S2; "TSMC-dependent" list is an ASSUMPTION
Semis + equipment in S&P 500	~18.3% (19-name list)	same file	ASSUMPTION list

1.2 AI capex, rates and the regime

Fact	Value	Source	Status
FOMC 16 Sep 2026	"Productivity growth is strong, and capital investment is robust." / "Inflation remains elevated." Hike to 3.75-4.00%, 12-0	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm	VERIFIED-PRIMARY
Hyperscaler capex	>\$690bn FY26 (>80% y/y); debt 9% of capex FY24 → 32% LTM mid-2026; Alphabet \$84.75bn equity raise June 2026; "anticipated IPOs from Anthropic and OpenAI in 2026 and 2027"; S&P cut Oracle to BBB- on 9 Jul	FactSet Insight, 23 Jul 2026, https://insight.factset.com/hyperscalers-tap-external-financing-as-ai-capex-outruns-cash-flow	VERIFIED-PRIMARY (FactSet's own analysis)
10-yr term premium (Kim-Wright)	0.9719% on 2026-09-16; last higher reading 2011-02-11	https://fred.stlouisfed.org/series/THREIFYTP10	VERIFIED-PRIMARY
Stock-yield correlation	daily corr(S&P return, Δ10y): 2016-20 +0.41; 2021 +0.10; 2022 -0.18; 2023-25 +0.01; 2026 -0.44. Monthly: 2016-21 +0.31; 2022-26 -0.54	FRED SP500 + DGS10, computed in this session	VERIFIED-PRIMARY inputs (derived)
Broad dollar index	129.28 end-2024 → 119.75 end-2025 (-7.4%) → 119.51 (18 Sep 2026)	https://fred.stlouisfed.org/series/DTWEXBGS	VERIFIED-PRIMARY (derived)
Moody's U.S. rating	Downgraded to Aa1 from Aaa, 16 May 2025 (all three agencies now below top tier)	https://www.moody's.com/web/en/us/about-us/usrating.html	VERIFIED-PRIMARY. A search snippet dating it 16 May 2026 is wrong.

Fact	Value	Source	Status
Fed chair	Kevin Warsh chair since 22 May 2026 (confirmed 54-45)	search snippets (thehill.com, spectrumlocalnews.com)	SNIPPET-UNVERIFIED
CBO	FY2026 deficit \$1.9tn; net interest \$1.0tn = 3.3% of GDP, rising to 4.6% by 2036	snippets of https://www.cbo.gov/publication/61882	SNIPPET-UNVERIFIED
CNBC "10-year 5.23% on Friday"	conflicts with Treasury par 5.17% on 25 Sep (different measure/time)	cnbc.com snippet	SNIPPET-UNVERIFIED; use F-001

1.3 Taiwan, China, FX

Fact	Value	Source	Status
Taiwan index weights after Aug 2026 review	ACWI 3.08 → 3.13%; EM 26.6 → 26.85%; AC Asia ex Japan 29.75 → 30.03%; TAIFEX up "nearly 60 percent this year" (46,021 on 13 Aug)	Taipei Times 14 Aug 2026, https://www.taipetimes.com/News/biz/archives/2026/08/14/2003862448	SECONDARY-READ (MSCI statement not read)
TSMC within MSCI EM	>14% of EM; 58.33% of MSCI Taiwan	search snippets (Taipei Times May 2026)	SNIPPET-UNVERIFIED
China weight in MSCI EM	snippet says 41.03% (31 Aug 2026) alongside Taiwan 20.39% - contradicts the 26.85% Taiwan figure above	finance.yahoo.com snippet	SNIPPET-UNVERIFIED, likely a different index variant; do not use
Taiwan current account / FX report	CA surplus 19.6% of GDP (2025), 14.1% (2024); US goods+services surplus \$145bn; net FX purchases \$7.7bn (0.8% of GDP); intervention "largely aimed at easing the rapid appreciation"; on U.S. monitoring list (July 2026)	Focus Taiwan/CNA 24 Jul 2026, https://focustaiwan.tw/business/202607240006	SECONDARY-READ (Treasury report not read)
Insurer hedging	hedge ratio down to ~45% (from 70% pre-COVID)	CommonWealth Magazine / insurancejournal snippets	SNIPPET-UNVERIFIED
USD/TWD shocks	1-5 May 2025: 32.07 → 29.93 (-6.7%); 1 Apr → 3 Jul 2025: 33.21 → 28.88 (-13.0%); 1995-96 crisis: 25.70 → max 27.55 (+7.2%); 25 Jul → 26 Sep 2022: 29.91 → 31.83 (+6.4%); Aug 2008 → Mar 2009: +14.5%; April 2025 and Dec 2025 drills: ≤0.4%	https://fred.stlouisfed.org/series/DEXTAUS , computed	VERIFIED-PRIMARY (derived)
USD/TWD long-run drift	2006→2026 -0.20%/yr; 2016→2026 +0.17%/yr	same	VERIFIED-PRIMARY (derived)
FX co-movement (monthly log changes, 2016-2026)	USD/TWD vs broad dollar corr 0.63 (beta 0.65); vs USD/KRW 0.71; vs S&P 500 return -0.38 (2021-26: -0.47)	FRED DEXTAUS, DTWEXBGS, DEXKOUS, SP500; computed	VERIFIED-PRIMARY inputs (derived)
Taiwan minimum wage	NT\$29,500 → NT\$30,900 (+4.745%) from 2027-01-01, pending Cabinet approval; "11th consecutive annual" rise; committee: projected 2.07% CPI "fully reflected"	Focus Taiwan 24 Sep 2026, https://focustaiwan.tw/society/202609240022	SECONDARY-READ
Taiwan regular earnings	NT\$49,343 in June 2026, +3.18% y/y; real regular wages +1.27% Jan-May	Focus Taiwan / Taipei Times snippets	SNIPPET-UNVERIFIED
Strait scenarios	Bloomberg Economics: war ≈ \$10.6tn, ~9.6% of world GDP in year one (Feb 2026 update); one-year blockade: Taiwan GDP -12.2%, China -8.9%, U.S. -3.3%, world -5%	bloomberg.com (paywalled) and defenceconnect snippets	SNIPPET-UNVERIFIED
PLA drills	"Justice Mission 2025", 29-30 Dec 2025, rehearsed a maritime blockade (130 aircraft, 28 ships in 24h per snippet)	thediomat.com / defensenews.com snippets	SNIPPET-UNVERIFIED
Russia precedent	MSCI reclassified Russia "at a price that is effectively zero and as of the close of March 9, 2022"	https://www.msci.com/eqb/pressreleases/archive/PR_Russia_Classification.pdf	VERIFIED-PRIMARY (verbatim)
U.S.-Taiwan trade deal	signed 15 Jan 2026: reciprocal tariff ≤15%; ≥\$250bn Taiwanese direct investment + ≥\$250bn credit guarantees	commerce.gov / ustr.gov fact sheets (snippets)	SNIPPET-UNVERIFIED
Laura and generative AI	a search summary linked her to criticism of an AI art project; the linked page is about a different artist	brokenfrontier.com / wikipedia snippets	UNVERIFIED and likely a conflation: do not use

2. The questions (with reasoning)

Format: question → anchor → current guess (labelled) → what would change → deliverables → specialist (domain) → seed → North Star link. Ranked roughly by value and deadline (WInS/TN first).

Q1. Is the equity premium over Laura's own ladder still worth taking?

- **Question:** How many extra percentage points a year does the growth sleeve's equity really expect over simply buying more Treasuries in 2026, and does that change the 50/60/70% sleeve equity weight?
- **Anchor:** R-C37/R-C38 ("willing to take thoughtful risks", "appropriate balance between pursuing growth and protecting the capital required for her goals"); F-301, F-012, F-008.
- **Guess:** JPM 6.70% minus ladder forward zeros 5.08-5.48% = +1.2-1.6pp. Valuation-based views are lower: forward earnings yield 5.24% $\approx$ 10-year 5.17%; CAPE earnings yield 2.41% < TIPS 2.83%. My guess is +0.5 to +1.5pp a year. On ~\$95k of stock that is roughly \$0.5-1.4k a year of expected extra return for a one-year sd of about \$15.6k.
- **Would change:** number (sleeve equity %, and whether 50% should be the base case); a sentence in the IPS risk section ("the reward for risk is small this year, so we take a measured amount").
- **Deliverables:** IPS, FR. **Domain:** D2 (with D3). **Seed:** 21.
- **Why it matters:** "thoughtful" risk, for a statistics graduate, means the price of risk was checked.

Q2. Which valuation number goes in the report: CAPE 41 or forward P/E 19?

- **Question:** Why do CAPE (41.48, second-highest since 1871, only Dec 1999 higher) and the forward P/E (19.1, near its 10-year average) disagree so sharply in 2026, and which one should anchor the sleeve's return assumption?
- **Anchor:** R-S26 ("uses reasonable assumptions and projections"); F-301.
- **Guess:** Earnings have boomed (CY2026 EPS +31.6%; semis +125%), so the forward P/E looks normal and CAPE, which averages ten years of earnings, looks extreme. The real risk is an earnings peak in semiconductors (Q2 2027 growth +1.3%), not only a price bubble. Show both, and pick one for the model with a stated reason.
- **Would change:** sentence in FR portfolio analysis; possibly the base equity assumption (use F-405's 5.0% case as a named stress).
- **Deliverables:** FR. **Domain:** D2. **Seed:** 21.
- **Why it matters:** shows the team read the market itself instead of repeating a headline.

Q3. How much of a "broad" fund is one Taiwanese factory?

- **Question:** What share of the S&P 500 and of VT depends on TSMC's fabs, counting both the direct Taiwan weight and U.S. companies whose leading chips TSMC makes?
- **Anchor:** R-S24 ("uses appropriate diversification"); R-W71 ("geographic regions, risk characteristics").
- **Guess:** Direct: VT 3.5% Taiwan, 1.7% TSMC. Indirect: ~20% of the S&P 500 (NVDA, AAPL, AVGO, AMD, MRVL, QCOM; ASSUMPTION list). A "global = diversified" claim needs this caveat.
- **Would change:** a sentence in the FR diversification analysis; a possible FR chart (direct vs indirect Taiwan exposure by fund).
- **Deliverables:** TN, FR. **Domain:** D2. **Seed:** null.
- **Why it matters:** Laura is building in Taiwan; telling her plainly how much of her portfolio already rides on Taiwan is honest client knowledge, not tokenism.

Q4. In a Strait blockade, which of Laura's pieces fail together?

- **Question:** If a blockade happens between 2031 and 2033, what happens to the Treasury ladder, the growth sleeve, the TWD and the residency, and so what must the low end of the 2031 range be?
- **Anchor:** R-C65, R-C66, R-C71 ("must also protect the portfolio's ability to fund the ten-year operating commitment"); R-C86.
- **Guess:** The ladder (U.S. Treasuries held to maturity) survives; its market value could rise (flight to safety) or fall (chip-shortage inflation). The sleeve falls (Bloomberg Economics: world GDP -5% in a one-year blockade, SNIPPET). The TWD weakens (+7.2% USD/TWD in 1995-96). The project may be delayed. So the co-sponsor low end must be the bought floor, and the upside should be stated as conditional.

- **Would change:** decision (share of the sleeve locked as the 2031 floor); sentence in the FR co-sponsor draft.
- **Deliverables:** FR. **Domain:** D4. **Seed:** 24.
- **Why it matters:** co-sponsors in Taiwan know this risk; naming it calmly earns credibility.

Q5. What happens to the committed payments if the residency cannot operate in Taiwan?

- **Question:** Should the IPS contain a one-line "delay or relocation" rule for payments already locked in Treasuries?
- **Anchor:** R-AN34 ("No instruction for what happens if the residency cannot open or moves"); R-C39 ("plans"); BS-07.
- **New evidence vs BS-07:** the December 2025 blockade rehearsal (SNIPPET) and Bloomberg's 2026 update make it a live scenario, not a hypothetical.
- **Guess:** Yes, one sentence: the payments stay in the ladder until the residency operates (wherever); Treasuries are liquid, so the only cost of a change of plan is the market move. No new instruments.
- **Would change:** sentence in the IPS flexibility section.
- **Deliverables:** IPS, FR. **Domain:** D4. **Seed:** 24.
- **Why it matters:** she is committing ten years of money to one place; a firm that has thought about "what if" is one she can trust.

Q6. When does the Taiwan dollar hurt Laura?

- **Question:** Is the USD cost of NT\$ spending more likely to rise in good times (TWD jumps, as in May 2025) and fall in crises (TWD weakened in 1995-96, 2008-09, 2022)?
- **Anchor:** R-AN10 (no currency named); R-C85; F-512, F-513.
- **Guess:** Yes. Monthly USD/TWD vs S&P returns: -0.38 (2016-26). The FX risk bites hardest when markets and Taiwanese co-sponsors are doing well, which is when the gap is easiest to fill.
- **Would change:** sentence in the FR co-sponsor draft (the FX risk is partly self-hedging); input to the "convert part of the floor to TWD" decision (brief section 9).
- **Deliverables:** FR. **Domain:** D4. **Seed:** 25.
- **Why it matters:** turns a scary unknown into an understood, partly offsetting risk.

Q7. Is the facility contribution less uncertain in NT\$ than in US\$?

- **Question:** Because stock falls and TWD falls tend to coincide, is the 2031-2033 range narrower when measured in NT\$ than in US\$?
- **Anchor:** R-C67/R-C68 (credible range); R-C69; F-513.
- **Guess:** Probably slightly narrower. Illustration (ASSUMPTION: 2-year sleeve sd 10%, FX sd 5%, corr -0.38): NT\$ sd $\approx$ 9.3% vs 10% in US\$, and vs 11.2% if a model treats FX as independent. Worth a proper simulation.
- **Would change:** number (range width); decision on quoting a NT\$ range alongside the US\$ range.
- **Deliverables:** FR. **Domain:** D3. **Seed:** 25.
- **Why it matters:** a statistics-trained client will value a correlation-aware range over two separate guesses.

Q8. Is the TWD being held weak, and is the jump risk one-sided?

- **Question:** With a 19.6%-of-GDP current-account surplus, insurers' reduced hedging and central-bank smoothing, is the main 2027-2033 FX risk a sudden TWD appreciation (like -13% in USD/TWD in Apr-Jul 2025)?
- **Anchor:** R-C85; R-C70 ("favorable and unfavorable market outcomes"); F-512.
- **Guess:** Yes, skewed toward a stronger TWD. Daily-vol statistics (4-5.6%) understate jumps. Stress case: "+15% USD cost of the NT\$ budget" in one quarter.
- **Would change:** number (FX stress size in the scenario table); sentence in FR assumptions.

- **Deliverables:** FR. **Domain:** D4. **Seed:** 25.
- **Why it matters:** shows research on her project's country beyond the case.

Q9. Which inflation erodes the fixed \$50k?

- **Question:** Should real erosion be measured with U.S. CPI (JPM 2.5%), Taiwan CPI (~1.8-2.1%) or Taiwan wages (minimum wage +4.745% for 2027; regular earnings +3.18%), converted at USD/TWD?
- **Anchor:** R-C46 ("not adjusted for inflation"); R-C85 ("the effect of inflation on portfolio projections and facility costs"); R-AN3; F-114, F-504.
- **Guess:** Running costs are mostly people and rent, so a wage-weighted index (~3-3.5%/yr) fits best. The 2042 payment in 2026 money: ~\$28.8k at 3.5%, ~\$36.0k at Taiwan CPI 2.07%, ~\$33.7k at U.S. 2.5% (ASSUMPTION rates). The USD/TWD drift was about zero over 20 years, so Taiwan inflation passes through almost directly.
- **Would change:** number (real-value table in the FR); sentence telling co-sponsors that the operating support shrinks in real terms (brief section 9 says this is unanswered).
- **Deliverables:** FR. **Domain:** D4. **Seed:** 23.
- **Why it matters:** few teams will notice that the money is spent in another country's labour market.

Q10. Is the cheap ladder a side effect of the AI boom?

- **Question:** If an AI bust (a growth shock) arrived before January 2027, would yields fall and stocks fall together, pushing the ladder above \$300k just as the sleeve shrinks?
- **Anchor:** F-111, F-112 (ladder above \$300k on 173 of 185 days; ~24% chance); R-C26.
- **Guess:** Partly yes. AI capex (>\$690bn, more debt-funded) and a 0.97% term premium are part of today's high yields. The 2026 regime is inflation-type (yields up, stocks down), but a bust would be growth-type. The rule for "rates fall before January" should be written with this scenario in mind.
- **Would change:** decision (the pre-January rule: staged purchase vs long-rungs-first); WInS-now sentence on why duration is bought now.
- **Deliverables:** WInS-now, TN, IPS. **Domain:** D1 (with D2). **Seed:** 21.
- **Why it matters:** links the headline story of 2026 (AI) to her specific promise.

Q11. Which stock-bond correlation should the model use?

- **Question:** JPM's -0.01, the 2016-2020 regime (bonds hedge stocks) or the 2022-2026 regime (they fall together)?
- **Anchor:** F-314; brief sections 9 and 11 (stock-rate correlation open).
- **New evidence:** daily corr(stock return, $\Delta 10y$ yield) +0.41 (2016-20) vs -0.44 (2026); monthly -0.54 (2022-26).
- **Guess:** Lock-early's payments do not care. The sleeve's p5 and any "bonds cushion stocks" claim do. Run the sleeve at +0.3 stock-bond return correlation as a stress.
- **Would change:** number (sleeve p5, 2031 floor p5); sentence in FR assumptions.
- **Deliverables:** IPS, FR. **Domain:** D3. **Seed:** 21.
- **Why it matters:** assumptions that match the world she lives in.

Q12. What does "certain barring U.S. default" exclude in 2026?

- **Question:** After all three agencies moved below AAA (Moody's Aa1, 16 May 2025) and with the term premium at a 15-year high, is the realistic residual risk a **delayed** payment in a debt-limit standoff rather than a loss?
- **Anchor:** R-C47, R-C54, R-C55, R-AN1.
- **Guess:** Yes: a delay, not a loss, is the credible residual. Payments fall on 1 January (a federal holiday), so the bonds that fund them mature before that date, which gives a buffer of days to weeks. The larger honest risk is real value (Q9), not default.
- **Would change:** sentence in the IPS certainty definition.
- **Deliverables:** IPS, FR. **Domain:** D1. **Seed:** 22.
- **Why it matters:** an honest answer to "how certain is certain" is what a statistician respects.

Q13. Which JPM edition does the plan use?

- **Question:** When the 2027 LTCMA appears (2026 edition released 2025-10-20 per snippet, so probably mid/late October 2026, between the Trading Notes and the IPS), will it change the plan's numbers, and which edition must the IPS and Final Report use?
- **Anchor:** F-315, F-317; R-C92 (consistency across deliverables); R-S24 ("maintains consistency with the team's IPS").
- **Guess:** Bond assumptions up (yields ~100bp higher than the 2026 data date); U.S. equities probably down (S&P +13% YTD, CAPE higher). The equity premium shrinks (Q1). Decide now: freeze one edition and date it.
- **Would change:** numbers across FR; decision on the edition.
- **Deliverables:** IPS, FR. **Domain:** D2. **Seed:** null.
- **Why it matters:** consistency across the three deliverables is scored.

Q14. Should the sleeve hold any emerging-market tilt?

- **Question:** Is JPM's highest equity return (EM 7.8%) worth it when EM is 26.85% Taiwan (TSMC >14%) plus a large China weight, which makes EM a double bet on the Strait?
- **Anchor:** R-S24; F-304, F-306.
- **Guess:** No tilt; keep VT's market weight (~10% EM). Russia 2022 showed an index market can be removed "at a price that is effectively zero".
- **Would change:** decision (no EM tilt), sentence in FR.
- **Deliverables:** FR. **Domain:** D2 (with D4). **Seed:** 24.
- **Why it matters:** avoids a tilt that would load her portfolio onto the same risk as her project.

Q15. Will AI mega-IPOs push concentration higher during the sleeve years?

- **Question:** If Anthropic (2026) and OpenAI (2027) list and join indices, and hyperscalers keep issuing equity, will the AI share of broad funds rise further in 2027-2033, and does the IPS need a simple review trigger?
- **Anchor:** R-I21 ("planned changes"); R-S24.
- **Guess:** Likely higher. A review trigger (not an automatic trade) costs one clause. Possibly not worth the words. Disclosure: the team's research tool is made by one of those companies; the AI-use log should say so.
- **Would change:** sentence (IPS monitoring); or a decision not to add it.
- **Deliverables:** IPS. **Domain:** D2. **Seed:** null.
- **Why it matters:** "we will watch what you really own" is a service promise.

Q16. How do we tell Laura what a broad fund owns, without a values screen?

- **Question:** What one-sentence disclosure of AI (~40% of the S&P 500) and Taiwan exposure is truthful and complete without implying a values preference the case never states?
- **Anchor:** R-AN34 (no ESG/values words); R-S25 ("earn her confidence"); BS-02 (CFA Asset Manager Code F.2, "truthful, accurate, complete, and understandable").
- **Guess:** Disclose and size, don't exclude. The choice to change it is hers.
- **Would change:** sentence in a TN reflection and the FR.
- **Deliverables:** TN, FR. **Domain:** D9 (with D6). **Seed:** null.
- **Why it matters:** respects her judgement; that is a reason to pick a firm.

Q17. Do Laura's income, portfolio and project all ride the same cycle?

- **Question:** Are her income (publishing and creative work, which generative AI may disrupt), her equity (AI-heavy) and her project (Taiwan, the AI-chip island) exposed to one AI/Taiwan cycle, leaving the Treasury ladder as the only piece outside it?
- **Anchor:** R-C27 (2028 deposit sources); R-C38; BS-03.

- **Guess:** Partly yes. This is the strongest Laura-specific reason for Treasuries under the payments. Do not claim anything about her personal views on AI (none verified).
- **Would change:** sentence in the IPS thesis (why lock-early fits her).
- **Deliverables:** IPS, FR. **Domain:** D6 (with D4). **Seed:** null (touches 20, which is another lens).
- **Why it matters:** "we built the plan around your life, not a template."

Q18. If inflation returns, who bears it, and how do we show it?

- **Question:** With the Fed's 2026 PCE median at 3.7%, the Middle East energy shock and Taiwan PPI +16.75%, who bears real erosion of the fixed payments, and should the Final Report show co-sponsors a real-value line?
- **Anchor:** R-C46; R-C85; F-007, F-009, F-503.
- **New evidence:** Taiwan upstream prices (PPI +16.75%, import prices +20.45%) and wage decisions (Q9).
- **Guess:** The residency bears it; co-sponsors, fees and grants fill the gap (R-C44). Show the ten payments in nominal and real terms, using the market's 2.33% breakeven as neutral and 3.7% as a stress.
- **Would change:** sentence (FR co-sponsor draft, assumptions).
- **Deliverables:** FR. **Domain:** D4 (with D1). **Seed:** 23.

Q19. Does the "reduce volatility" reason for Treasuries still hold in 2026?

- **Question:** Wharton's sample note buys a Treasury ETF "to reduce portfolio volatility". In 2026 stocks rose 13% while IEF and TLT fell 4-6%, and the two tend to fall together. Should our Treasury notes lead with "matches her payments" instead?
- **Anchor:** R-T23 (sample note); R-T11 ("quality of your reasoning"); F-203, F-204.
- **Guess:** Yes. In an inflation regime Treasuries may not cut WInS volatility; their job is to lock the promise.
- **Would change:** sentence in the Trading Note written at trade time (WInS now) and its reflection (Oct 23).
- **Deliverables:** WInS-now, TN. **Domain:** D1 (with D9). **Seed:** null.
- **Why it matters:** a sharper reason than the one most teams will copy from the example.

Q20. Is converting part of the 2031 floor to NT\$ worth ~6% over two years?

- **Question:** Given a ~310bp short-rate gap (IORB 3.90% vs Taiwan call rate 0.81%), the partial natural hedge (Q6) and the jump risk (Q8), should any of the floor be held in NT\$?
- **Anchor:** F-515, S-35; brief section 9 (open).
- **Guess:** No conversion in the base plan; quote both currencies instead. Reconsider only if the facility budget is firmly in NT\$ by 2031.
- **Would change:** decision (2031 rule parameter); sentence in FR.
- **Deliverables:** FR. **Domain:** D4. **Seed:** 25.

Q21. Why are U.S. rates at 2007 highs, and which piece matters for January?

- **Question:** Of the 10-year's 5.17%, how much is expected policy rates, inflation compensation (breakeven 2.34%) and term premium (0.97%), and which piece, if it reversed before January 2027, would push the ladder above \$300k?
- **Anchor:** F-001, F-004, F-008, F-009; F-104 (27bp headroom).
- **Guess:** Real yield 2.83% + breakeven 2.34%. The term premium rose ~0.5pp in a year. A term-premium collapse (e.g. a flight to safety or an AI bust) is the fastest way to lose 27bp.
- **Would change:** decision (pre-January rule); sentence in a TN reflection.
- **Deliverables:** WInS-now, TN, IPS. **Domain:** D1. **Seed:** 21.

Q22. Hedged or unhedged foreign stocks?

- **Question:** Should the sleeve's non-U.S. equities stay currency-unhedged, since USD/TWD moves with the broad dollar (monthly corr 0.63; the broad dollar fell 7.4% in 2025), so unhedged foreign holdings partly offset NT\$ cost rises?

- **Anchor:** R-C85; R-AN10; F-302.
- **Guess:** Unhedged (simpler and a mild natural hedge). This is a second, Laura-specific reason for a global fund beyond concentration.
- **Would change:** decision (fund choice, unhedged); sentence in FR.
- **Deliverables:** FR. **Domain:** D4 (with D2). **Seed:** 25.

Q23. What would a 1999-2002 path do to the plan?

- **Question:** If the sleeve followed the path after the only higher CAPE reading (Dec 1999), would the 2031 floor and the 2033 facility contribution still hold?
- **Anchor:** R-C86 ("favorable and unfavorable investment outcomes"); F-401, F-402.
- **Guess:** The floor holds (bought); the upside collapses (S&P about -49% peak to trough in 2000-02, from memory, to verify). A named historical stress is easier for co-sponsors to grasp than a model percentile.
- **Would change:** a scenario row and FR chart; the co-sponsor sentence.
- **Deliverables:** FR. **Domain:** D3. **Seed:** null.

Parked (fail the "would it change something" filter)

- China's own macro (deflation, yuan) as a separate item: no path to a decision beyond Q4/Q14.
- Anything about Laura's personal background and China: privacy and tokenism risk; not a portfolio input.
- Taiwan construction cost index: already answered (brief section 8.15; F-508). Only reopen if the facility cost question changes.

3. Leads for the specialists

- **D2 (equity):**
 - FactSet Earnings Insight PDFs (weekly), e.g. the 11 Sep 2026 issue: forward P/E, CY26/27 growth, sector detail.
 - multpl.com CAPE table (secondary); try Shiller's own data file for a primary source.
 - `scripts/data/S2/spym_holdings.csv` for look-through; S2's issuer tables.
 - FactSet hyperscaler financing note (23 Jul 2026).
 - JPM 2027 LTCMA: check am.jpmorgan.com from mid-October.
- **D1 (rates):**
 - FRED THREEFYTP10 (term premium) and DFII10.
 - Treasury real par curve (F-008).
 - FOMC statement 16 Sep 2026 wording.
 - Moody's rating page.
 - For Q12: check Treasury maturity dates (15th of month / month-end) versus 1-January payments, and what WInS offers.
- **D3 (quant):**
 - FRED SP500 (from 2016 only), DEXTAUS (from 1983), DTWEXBGS, DEXKOUS, DGS10.
 - Correlations computed here: stock-yield by regime; USD/TWD vs S&P -0.38.
 - A simulation of the range in NT\$ vs US\$ (Q7).
 - The S&P 2000-02 path needs a primary source (FRED SP500 starts 2016).
- **D4 (Taiwan/FX):**
 - Taipei Times 14 Aug 2026 (MSCI weights).
 - Focus Taiwan 24 Jul 2026 (Treasury FX report); the report itself on home.treasury.gov is not yet read.
 - Focus Taiwan 24 Sep 2026 (minimum wage).
 - DGBAS earnings releases; CBC mp-2 page (call rate).
 - FRED DEXTAUS crisis windows above.
 - MSCI Russia press release (verbatim quote).
 - Bloomberg Economics scenario numbers are paywalled: cite as SNIPPET only.
- **D6/D9:**
 - CFA Asset Manager Code F.2 (via stakeholder map BS-02) for the disclosure sentence (Q16).

- Do **not** use the search claim linking Laura to an AI-art controversy: it looks like a conflation with another artist.
- **D7 (competition):**
 - Most AI-assisted rivals will cite the S&P's AI concentration. Few will price the equity premium against their own ladder (Q1), use Taiwan wages (Q9) or the TWD/stock correlation (Q6-Q7). That is where the edge is.

4. Sources (all accessed 2026-09-27)

- **Verified primary:**
 - https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_091126.pdf
 - <https://insight.factset.com/hyperscalers-tap-external-financing-as-ai-capex-outruns-cash-flow>
 - <https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm>
 - <https://fred.stlouisfed.org/series/THREEFYTP10>
 - FRED series SP500, DGS10, DEXTAUS, DTWEXBGS and DEXKOUS (fredgraph.csv)
 - <https://www.moodys.com/web/en/us/about-us/usrating.html>
 - https://www.msci.com/eqb/pressreleases/archive/PR_Russia_Classification.pdf
- **Secondary, read in full:**
 - <https://www.multpl.com/shiller-pe>
 - <https://www.taipeitimes.com/News/biz/archives/2026/08/14/2003862448>
 - <https://focustaiwan.tw/business/202607240006>
 - <https://focustaiwan.tw/society/202609240022>
- **Snippets only:**
 - Bloomberg Taiwan war/blockade estimates
 - thediplomat.com and defensenews.com (Justice Mission 2025)
 - CBO 2026-2036 outlook
 - Fed chair transition (thehill.com, spectrumlocalnews.com)
 - insurer hedging (CommonWealth, insurancejournal)
 - U.S.-Taiwan trade deal (commerce.gov, ustr.gov)
 - Taiwan earnings (Focus Taiwan)
 - China EM weight (yahoo; conflicting)
- **Repo:**
 - competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (L70-73, 90-91, 118, 146-148)
 - 2026_WGY_Trading_Notes_Analysis-FINAL.txt (L36)
 - SMAApply_Deliverables_Page_2026-09-27.md (L49)
 - phase_A/*, wins_now/S2_growth_sleeve.md

What this teaches

Macro facts matter only when they reach the client's numbers. In 2026 the same story (AI) explains three things at once:

- why U.S. stocks are concentrated;
- part of why U.S. bond yields are high (heavy borrowing for AI and a higher term premium);
- why Taiwan is booming while its currency is held down.

For Laura, those three facts meet in one place: her payments are safest in U.S. Treasuries precisely because everything else she has (her income, her stocks, her Taiwan project) is tied to the same cycle. A good question does not ask "what will markets do?". It asks "when this moves, which of her numbers moves with it, and in which direction?".

Part 3 - The research record, phase by phase · research/insight_v1/phase_B/B6a_cosponsors.md

B6a Co-sponsors and Philanthropy: the co-funder's due-diligence view

Agent: B6a, insight_v1 run, Phase B (question generation). Written 2026-09-27. Lens: "Co-sponsors & Philanthropy". Starting angle (the other B6 agent starts elsewhere): **the co-funder's side of the table**. What does a foundation, capital-campaign funder, public culture agency or accountant actually *do* with a founder's pledge: what counts, what is discounted, what is written into their own grant conditions, and on what calendar? Seed questions extended: 6, 8, 9, 15 (v4 Part 6).

This file is AI-generated research for Team Caplet, for brainstorming only. It holds **no submission-ready prose**. Where it says what a sentence or paragraph "must contain", that is a checklist; the six students write every word (Wharton AI policy, R-W26/R-W49). No securities are recommended here. Laura's public professional record only; nothing here suggests contacting her.

Status labels (brief section 3). Short forms: **VP** = VERIFIED-PRIMARY (read on the page by me on 2026-09-27; URL in section 5), **VRF** = VERIFIED-REPO-FILE, **SNIP** = SNIPPET-UNVERIFIED, **ASM** = ASSUMPTION, **INT** = INTERPRETATION (my reading, not a fact). R-, F-, SH-, BS- ids point to the Phase A files.

Terms used below (defined once):

- **Pledge / promise to give**: a promise to give money later. **Unconditional** = nothing has to happen first. **Conditional** = the money is owed only if a stated hurdle is cleared.
- **Lead gift / leadership gift / seed money**: the big early gifts a campaign secures before it asks the public.
- **Challenge gift**: in the economics papers used here, an *unconditional* lead gift announced to other donors.
- **Matching gift**: "I give \$1 for each \$1 others give, up to a cap" (a conditional promise).
- **Capital campaign**: fundraising for a building. **Quiet phase**: the period before the public announcement, when lead gifts are collected.
- **Gift chart**: a funder's table of how many gifts of each size a campaign expects.
- **Present value (PV)**: what money due in the future is worth today, after discounting at an interest rate.
- **Backstop**: a named reserve source that covers an expected-but-uncertain source of money if it fails.

0. Summary (read this first)

I wrote 20 questions (section 2). Six matter most, because they rest on primary sources no file in the repo has used (I checked with grep: none cites Kresge, FASB ASU 2018-08, PwC's NFP guide, Rondeau & List, Karlan & List, Huck & Rasul or Andreoni) and because their answers change something in the Final Report (FR) or the IPS:

1. **Professional funders count only the unconditional part of a pledge (Q1)**. Kresge's capital-campaign guide counts "Written pledges or cash" (VP). The U.S. nonprofit accounting standard treats a statement that "specified an intention to give" as *not* a promise to give (FASB ASU 2018-08, example 958-605-55-51, VP). In the seed-money papers the effect is produced by a "challenge gift", defined as "an unconditional commitment by a donor" (Rondeau & List 2008, VP). **INT**: only the bought floor of Laura's 2031 range is a lead gift in any funder's books. The rest of the range is information, not money. That sets how the two ends of the range should be labelled.
2. **Co-sponsors may write her low end into their own conditions (Q2)**. Grant agreements often say "we pay if the project raises X". A matching grant is "a common form of a conditional promise to give" (ASU 2018-08, 958-605-55-17, VP). Kresge wants other sources "committed, imminent or backstopped" (VP). **INT**: if a co-sponsor writes "conditional on the founder contributing at least US\$X", the bottom of Laura's range becomes someone else's contract trigger. A second, outside reason (beside R-C66) why the low end must be *bought*, not a model percentile.
3. **The experiments say: announce a firm lead gift; do not turn her upside into a match (Q3)**. Huck & Rasul (2011, VP abstract): charities "maximize donations given by simply announcing the presence of a lead gift". Linear matching "partially crowd[s] out" donors' own gifts. Rondeau & List (2008, VP): "challenge gifts positively influence contributions in the field, but matching gifts do not". Karlan & List (2007, VP): a match raises response, but 3:1 and 2:1 add nothing over 1:1. A match would also tie Laura's 2033 amount to other donors' behaviour, which no team can model with a stated confidence.
4. **Capital funders test operating sustainability after the building is done (Q5, seed 6)**. Kresge's first seven questions to any capital applicant include "Do you have a plan for sustaining support gained during the campaign after it concludes?" (VP). Andreoni (1998, VP abstract) models seed money as solving a *fixed-cost* problem. Laura's ten pre-funded operating years answer the building funder's worry before it is asked. This is new primary evidence for BS-17, and it explains the case's asymmetry (R-AN6) from the funder's side.
5. **With an 80% lock, the 2031 range is almost a point (Q7)**. ASM illustration (my calculation, section 3): at the median 2031 sleeve, locking 80% in a 2-year Treasury leaves 2033 capacity at about **US\$199k-218k (p5-p95)**, of which **US\$165k** is already bought. That is roughly $\pm 5\%$ around the middle. It is honest and very credible, but it comes close to the "one exact amount" the case steers away from (R-C67). The lock share (brief section 9) is therefore also a *communication* parameter, not only a risk parameter.
6. **The case itself stops the 2031 floor from being paid in early (Q10)**. "She will neither add to nor withdraw from the portfolio before 2033" (R-C34, VRF). So in 2031 Laura can only *promise*, backed by Treasuries she still owns. She cannot hand money over, escrow it or transfer it. Words like "set aside for the residency" could overstate this to a funder. The honest label is "a written commitment backed by Treasury holdings that mature in January 2033" (checklist item, not prose).

Also new: Taiwan's Ministry of Culture artist-village operating subsidy is **annual**. Applications run each November for the following year (VP, county government notice of the MOC call). Public culture money in Taiwan therefore cannot offer ten-year operating certainty. Laura's commitment complements public money; it does not compete with it (Q11). A fundraising-paragraph checklist is in Q18.

Not re-asked (already in brief section 8 or other Phase B files): the List & Lucking-Reiley mechanism (8.17; I add new evidence in Q15 and Q19); range computed today vs a 2031 rule (B1a Q18, B1b Q18); "above the range" as a policy cap (B1b Q19, B8b Q10); staged facility payments (B4a Q15; I add one new fact in section 4); currency labelling (BS-04); cost-overrun answer (BS-16); expects-vs-pledges verbs (BS-05, which Q1 and Q10 sharpen with new evidence).

1. What I read (repo)

Brief (all sections, section 14 supersedes); v4 prompt Parts 1-3, 5, 6; CLAUDE.md; Phase A `case_register.md` (R-C41 to R-C72, R-AN1 to R-AN34, R-I22, R-I31, R-T14, R-S24 to R-S28), `fact_register.md` (F-101 to F-117, F-401 to F-409, F-510 to F-514), `stakeholder_map.md` (SH-05 to SH-10, BS-03 to BS-17), `wins_week1_guardrails.md` (headings); Phase B files B1a, B1b, B4a, B8b (to avoid duplicates, by grep).

2. The questions

Format for each: question; anchors; hypothesis (a GUESS, with evidence); what would change; deliverables; domain; seed; north star.

Q1. Which part of Laura's 2031 range would a professional co-funder actually count as money: only the bought floor, stated as an unconditional commitment, or the whole range?

- Anchors: R-C67, R-C69, R-C66, R-AN9. Official words: "Rather than promising one exact amount, she wants to communicate a credible range for her potential contribution." (case p.3, VRF)
- Evidence:
 - Kresge counts as initial funds "Written pledges or cash from individuals, corporations, businesses and foundations" (VP).
 - ASU 2018-08, 958-605-55-51: a written statement that "specified an intention to give" meant the recipient "did not receive a promise to give and did not recognize a contribution received" (VP).
 - Rondeau & List: "A challenge gift is an unconditional commitment by a donor, or set of donors, to provide a given sum of money to the cause." (VP)

- Kresge's gift chart separates the "prospect pool" from "dollars expected" (VP). Funders count expected yield, not hopes.
- Hypothesis (GUESS): only the floor counts. The upper part is read like a "prospect", discounted or ignored.
 - So the range should be built as two labelled tiers: a committed floor (bought) and an expected upper part (with its probability and model).
 - The confidence statement then splits in two: P(contribution $\geq$ floor) is about 100% barring U.S. default, by construction; P(reaching the top) is a model number.
- Would change: **decision**. It sets how the FR range is structured and labelled, and which number is the "planning figure" a co-sponsor can budget on (the floor, not the median).
- Deliverables: FR. Domain: D5. Seed: 9.
- North star: Laura's range would read the way funders' own books read it, so no co-sponsor has to discount it.

Q2. Will co-sponsors write Laura's stated low end into their own grant conditions ("conditional on the founder contributing at least US\$X"), so that missing it releases them and her shortfall cascades?

- Anchors: R-C66, R-C71, R-C64.
- Evidence:
 - ASU 2018-08, 958-605-55-17: "A challenge (matching) grant is a common form of a conditional promise to give." (VP)
 - Indicators of a barrier include "a matching requirement" (VP).
 - Kresge: sources in the plan "should be committed, imminent or backstopped" (VP). It advises applicants to "backstop the expected government funding" (VP).
- Hypothesis (GUESS): likely for at least some institutional co-sponsors, especially capital funders.
 - If so, the low end is a *trigger* in other people's contracts, not just a reputational number.
 - This is a second, outside reason (beside "damage her credibility", R-C66) that the low end must equal the bought 2031 floor and never a model percentile such as p5.
 - It also argues against stating a low end *below* the floor "to be safe". That would under-trigger co-sponsor matches.
- Would change: **decision** (the low end = the bought floor; IPS rule wording "announce only what is already bought"). **Sentence** (FR: why the floor is bought).
- Deliverables: IPS, FR. Domain: D5 (practice), D10 (make sure no legal claim is made; Taiwan law is out of scope, R-C97). Seed: 9.
- North star: we protect Laura's partners' contracts, not only her reputation.

Q3. Should any of Laura's upside above the floor be offered to co-sponsors as a matching or challenge offer instead of a plain range?

- Anchors: R-C64, R-C69, R-C70. Official words: "make potential co-sponsors more willing to contribute" (VRF).
- Evidence:
 - Huck & Rasul 2011 (14,000 Bavarian State Opera patrons; VP abstract): linear matching "raise[s] the total donations received including the match value, but partially crowd[s] out the actual donations given excluding the match". Charities "maximize donations given by simply announcing the presence of a lead gift".
 - Rondeau & List 2008 (Sierra Club BC; VP): "challenge gifts positively influence contributions in the field, but matching gifts do not".
 - Karlan & List 2007 (50,000+ prior donors; VP): "the match offer increases both the revenue per solicitation and the probability that an individual donates", but "larger match ratios (i.e., \$3:\$1 and \$2:\$1) relative to smaller match ratios (\$1:\$1) had no additional impact".
- Hypothesis (GUESS): no match.
 - The evidence favours a firm, announced lead gift.
 - A match makes Laura's 2033 amount depend on how others behave. The team cannot give that a stated confidence (R-C69), and it adds complexity (design principle).
 - At most, the FR can say in one line that a match was considered and rejected, and why (Articulation criterion, R-S27).
- Would change: **decision** (FR structure of the offer to co-sponsors).
- Deliverables: FR. Domain: D5. Seed: null.
- North star: a choice backed by field experiments, which few of the ~2,300 final-report teams will have weighed.

Q4. Does the leadership-gift literature show that a founder's belief persuades others only when it is costly and unconditional, and if so is the case's pull quote the private starting point with the bought floor as its public proof?

- Anchors: R-C21 ("The only person who needs to believe in something is yourself.", VRF), R-AN30, R-C64.
- Evidence:
 - Rondeau & List (VP), footnote 3: "when a large donor believes in the project by giving a gift this informational value affects an agent's beliefs regarding the value of the public good". They cite signalling theory (Vesterlund 2003; I did not read Vesterlund: RePEc shows "No abstract is available").
 - Andreoni's alternative: seed money as "an elimination device rather than as a credibility device" (quoted in Rondeau & List, VP).
- Hypothesis (GUESS): yes.
 - In economics, belief becomes evidence for others only when it costs the believer something they cannot take back.
 - This links seed 15 (private conviction vs public proof) to a mechanism a judge can check, without tokenism.
 - Caveat: the pull quote has no attribution in the PDF text layer (brief section 14). Verify it before presenting it as her words.
- Would change: **sentence** (the FR narrative linking her voice to the floor; possibly the 50-word pitch's framing).
- Deliverables: FR, IPS (framing only). Domain: D6 (client psychology), D9. Seed: 15.
- North star: it turns her own words into the design rule, which is what "recognise as hers" means.

Q5. Do capital funders explicitly test whether operations are sustainable after the building is finished, which would make Laura's ten pre-funded operating years the thing that unlocks facility money?

- Anchors: R-AN6, R-C41, R-C44, R-C48; BS-17 (Phase A, SSIR evidence).
- Evidence (new primary evidence beyond BS-17):
 - Kresge's seven campaign questions include "Do you have a plan for sustaining support gained during the campaign after it concludes?" and "build a larger support base of private giving to address annual operating needs" (VP).
 - Andreoni 1998 (VP abstract) models charities with "fixed costs", where "sometimes only a small amount of seed money can grow into a substantial charity".
- Hypothesis (GUESS): yes. The case's split (outside money may fund the building; only Laura funds the ten operating years) mirrors the capital funder's two questions: can you build it, and can you run it?
 - The operating commitment should come *first* in the co-sponsor material, before the facility range.
 - B1b Q15 suggested the same ordering from the case text; this adds the funder's own checklist as evidence.
 - Would change: **sentence** (FR fundraising draft order; one Articulation line explaining the asymmetry).
 - Deliverables: FR. Domain: D5. Seed: 6.
 - North star: we answer the question a building funder asks before they ask it.

Q6. What funder calendar makes 2031 the right year, and does it imply a second checkpoint in late 2032, when annual public funders set 2033 budgets?

- Anchors: R-C62, R-AN7, R-AN11.

- Evidence:
 - Kresge: "Typically, we suggest at least 20 percent and sometimes as much as 50 percent of your private fundraising goal be raised before applying" (VP). It also advises: "Have a significant amount of your campaign pledges committed before the general public ever hears about the campaign." (VP)
 - Rondeau & List citing Lawson (2001): consultants "advise seeking at least 10% and up to 80% of the overall objective of a capital campaign through leadership gifts before embarking on the public phase" (VP).
 - Taiwan MOC artist-village operating subsidy: the ministry selects grantees "每年" (every year). The 2022 round took applications from 1 to 30 November 2021 (VP, Lienchiang County notice of the MOC call).
- Hypothesis (GUESS): 2031 is the quiet-phase year for lead gifts; public and annual funders commit only in the budget cycle just before opening (about November 2032 for 2033).
 - So a simple IPS rule may need two dates: 2031, announce the bought floor and the upper tier; late 2032, reconfirm, when the upper tier is nearly known.
 - This extends B1a Q19 (range updates) with a concrete funder calendar. It may fail the complexity test. If so, one FR sentence is enough.
 - Would change: **decision** (does the IPS framework include a 2032 reconfirmation step?).
 - Deliverables: IPS, FR. Domain: D5, D4 (Taiwan budget cycle). Seed: 8.
 - North star: the plan follows how her partners actually budget, not only how markets move.

Q7. What share of the 2031 sleeve should be locked as the floor, once we see that locking 80% makes the 2033 range almost a single number, while funders also reward "stretch" gifts?

- Anchors: R-C67 ("Rather than promising one exact amount", VRF), R-C58, R-C84, F-402, brief section 9 (lock share open).
- Evidence:
 - ASM illustration (section 3): at the median 2031 sleeve (about US\$188k), an 80% lock gives a bought floor of US\$165k. 2033 capacity is then p5/p50/p95 US\$199k/207k/218k.
 - Kresge asks whether leaders' gifts are "the largest gifts they have ever made to your organization" (VP). Funders read commitment relative to capacity.
 - The case warns that "committing all remaining assets could limit her financial flexibility" (R-C58, VRF).
- Hypothesis (GUESS): 80% may be too high for communication, even if it is right for risk.
 - A lock of 60-70% would leave a visible upper tier, so the range reads as a range, while the floor is still most of the stated amount.
 - D3 should tabulate floor, p50 and p95 of 2033 capacity for 50/60/70/80% locks, and P(reaching the top) for each.
- Would change: **number** (lock share; range ends).
- Deliverables: FR (numbers), IPS (rule). Domain: D3, D5. Seed: 9.
- North star: a range that is visibly a range, but whose bottom never moves, is what "credible" means to both a statistician and a funder.

Q8. Should the co-sponsor material describe the ten-year operating commitment as US\$500,000 (nominal) or at its value in 2033 (about US\$395k), given that professional recipients book multi-year pledges at present value?

- Anchors: R-C45, R-C46, F-106 (US\$394,930 on 2033-01-01, VRF/derived).
- Evidence:
 - PwC's NFP guide (VP): "The present value of the future cash flows ... is the valuation technique typically used when measuring the fair value of a contribution due more than one year from the date of the promise." Its example books a US\$100 promise due in five years at US\$50.
- Hypothesis (GUESS): state both.
 - Give the nominal total (the case's figure).
 - Give the amount already invested to meet it in 2033 (about US\$395k at forward rates).
 - A finance-literate co-sponsor who sees only "US\$500,000" may think the team does not know about discounting.
 - Reconciles with brief 8.9 (report the market value on 1/1/2033).
- Would change: **number** and **sentence** (which figure headlines the operating commitment in the FR draft).
- Deliverables: FR. Domain: D1 (valuation), D5. Seed: null.
- North star: we show her promise the way her partners' accountants will see it.

Q9. When a recipient judges whether a pledge will be collected, it looks at "the donor's creditworthiness". How much of Laura's portfolio should the co-sponsor material disclose to prove that the promise is Treasury-backed, without exposing her whole finances?

- Anchors: R-C47, R-C83, R-S28; SH-05.
- Evidence:
 - PwC (VP): collectability factors include "the donor's creditworthiness, the NFP's past collection experience, and the NFP's policies concerning the enforcement of promises to give".
 - PKF 2019 (VP): a foundation checks "whether the sources of potential revenue are reliable".
- Hypothesis (GUESS): disclose the *form* and *timing* (Treasures maturing each January 2033-2042; floor in a 2-year Treasury maturing January 2033), not the portfolio total or the growth-sleeve holdings.
 - Funders do due diligence on the grantee (the residency), not on the donor's wealth.
 - What makes a founder's pledge collectible is the named asset behind it.
 - Would change: **sentence** (which facts about the portfolio appear in the co-sponsor draft).
 - Deliverables: FR. Domain: D5, D10 (privacy/disclosure; CFA Asset Manager Code F.2 "truthful, accurate, complete, and understandable", from BS-02). Seed: null.
- North star: we give co-sponsors proof, and give Laura privacy.

Q10. Because the case bans withdrawals before 2033, Laura's 2031 floor can only be a promise backed by assets she still owns. Which words describe that accurately, so "set aside for the residency" does not overstate it?

- Anchors: R-C34 ("she will neither add to nor withdraw from the portfolio before 2033", VRF), R-C63, BS-05.
- Evidence:
 - ASU 2018-08 distinguishes a promise with "rights and obligations" from an "intention" (VP; example 958-605-55-51 quoted in Q1).
 - Kresge's "organizational funds" are "funds set aside by the organization and earmarked for the campaign" (VP). That is money the *recipient* holds, which Laura's floor is not.
- Hypothesis (GUESS): the accurate category is "a stated commitment, backed by named Treasury holdings in her own portfolio that mature in January 2033".
 - It is not a transfer, an escrow or a restricted fund, because R-C34 forbids moving money out before 2033.
 - BS-05 chose "set aside" for the floor. This question tests whether that verb implies a transfer to a funder.
 - Would change: **sentence** (verb and noun choices in the FR draft and in any IPS rule wording).
 - Deliverables: FR, IPS. Domain: D10 (case reading), D9 (plain words). Seed: 9.
 - North star: one precise verb protects her credibility better than any extra model.

Q11. Are Taiwan's public culture funders annual and project-based, so they can never give multi-year operating certainty? If so, which co-sponsor type is the range message really for?

- Anchors: R-C44 (list of outside sources, VRF), R-C62; SH-06, BS-15.

- Evidence:
 - MOC artist-village operating subsidy: yearly selection. Grantees subsidised for several years in a row must submit a "自我提升計畫" (self-improvement plan) (VP, Lienchiang County notice, 2021).
 - Subsidy-rate tiers up to 90% of a project's budget: SNIP (law.moc.gov.tw returned HTTP 403).
 - NCAF (National Culture and Arts Foundation) runs an "Arts & Business" partnership platform for private sponsorship (SNIP; its home page loaded but gave no detail).
- Hypothesis (GUESS): public bodies fund programmes year by year; private foundations, corporate sponsors and individuals fund buildings.
 - The facility range mainly addresses private and corporate capital funders.
 - The operating commitment is what lets annual public grants be *additional* (programmes, participants) instead of life support.
 - Keep the audience generic in the FR (BS-15); this only changes emphasis.
 - Would change: **decision** (FR audience and emphasis: who the range is written for).
 - Deliverables: FR. Domain: D4 (Taiwan), D5. Seed: null.
 - North star: we show we know how arts money in Taiwan actually flows, without naming or contacting anyone.

Q12. Do Taiwan public subsidies require the applicant to fund part of the budget itself (a "自籌款" self-funded share)? If so, is Laura's fixed US\$50k what makes the residency eligible for public operating money?

- Anchors: R-C44, R-C48 (no reliance on outside money for the commitment), R-AN3 (real erosion of the fixed US\$50k falls to other sources).
- Evidence:
 - SNIP only: a MOC cultural-heritage subsidy rule caps grants at 50-70% of total project cost, so recipients provide "相對足額之自籌款" (matching self-funding). This is from a different MOC programme, not the artist-village one.
 - The artist-village rules page returned HTTP 403.
- Hypothesis (GUESS): likely yes, in some form. Her payments would then *leverage* public money that can top up the real erosion of the fixed US\$50k (brief section 9, "must be said aloud").
 - Careful: this can appear only as "beyond her commitment". It must never support the payments (R-C48).
 - Would change: **sentence** (FR: who covers the real erosion of the US\$50k; the co-sponsor value of the operating commitment).
- Deliverables: FR. Domain: D4. Seed: 6.
- North star: her fixed promise becomes the anchor that others' money attaches to.

Q13. If the residency is delayed or cannot open, what happens to Laura's facility money, and should the co-sponsor material say so, given that a refund promise raised giving in the seed-money experiment?

- Anchors: R-AN34 ("No instruction for what happens if the residency cannot open or moves", VRF); BS-07; R-C58.
- Evidence:
 - List & Lucking-Reiley 2002 (VP abstract): "imposing a refund increased contributions by a more modest 20%" (the seed-money change produced "nearly a sixfold increase").
 - Bagnoli & Lipman's refund logic, as described there: donors give more when they know their money is safe if the goal is not met.
- Hypothesis (GUESS): one line. Her facility contribution is released only when the project proceeds. Until then it stays in Treasuries and forms part of her flexibility. The operating reserve is untouched in any case.
 - This is the donor-side mirror of a refund policy. It needs no new model.
 - Would change: **sentence** (FR flexibility and co-sponsor draft).
 - Deliverables: FR. Domain: D5, D4 (Taiwan Strait scenario, seed 24). Seed: null.
 - North star: co-sponsors and Laura both see that no dollar is spent on a project that does not happen.

Q14. Will co-sponsors see Laura's US\$50k a year as the residency's single biggest revenue source, and ask how that dependence falls before 2043? How can the FR answer in one line without writing a business plan?

- Anchors: R-C49 (after 2042 out of scope), R-C95 (no business plan expected), R-AN25 ("early operations").
- Evidence:
 - Kresge asks applicants "Will you be able to change the mix of revenues that support your organization?" (VP).
 - PKF 2019 (VP): the funder must understand "how the grantee would respond to any unexpected revenue shortfalls".
- Hypothesis (GUESS): the answer is the case's own framing. The ten years are start-up support ("early operations", R-AN25); outside sources are named in the case (R-C44); post-2042 is outside the plan (R-C49). One sentence can name the post-2033 flexibility money as optional capacity, not a promise.
 - Extends B1a Q24 (the 2043 question) with the funder-side reason it will be asked.
 - Would change: **sentence** (FR co-sponsor draft; scope note).
 - Deliverables: FR. Domain: D5. Seed: 7.
 - North star: we answer a funder's standard concentration worry in Laura's favour.

Q15. The seed-money effect in the literature is a share of a known campaign goal, and the facility cost is deliberately unknown. What can the FR honestly claim about why her contribution matters?

- Anchors: R-C94, R-C95 (facility cost not estimated), R-C64 ("may signal"), brief 8.17.
- New evidence beyond brief 8.17:
 - List & Lucking-Reiley's effect is "a change in seed money from 10% to 67% of the campaign goal" (VP abstract). It needs a goal.
 - Andreoni's model rests on a threshold/fixed cost (VP abstract).
 - Huck & Rasul show that the "mere existence of a substantial lead donor" matters (VP abstract). That needs no goal.
- Hypothesis (GUESS): the FR cannot claim any percentage effect.
 - It can claim what does not need a goal: her contribution is firm, early and first, and the operating years are already funded.
 - "Meaningful" (R-C64) should be defined by firmness and timing, not size. This extends B1a Q20 and B1b Q24 with the reason: the size-based evidence needs a denominator the case withholds.
- Would change: **sentence** (FR: the justification for the contribution; avoids an over-claim a judge could catch).
- Deliverables: FR. Domain: D7 (what judges accept), D5. Seed: 9.
- North star: no over-claim, which is itself the case's test (R-C66).

Q16. Should one Trading Note reflection tie a Treasury purchase to co-sponsor credibility ("locking the promise first is what lets her later range be honest"), even though the co-sponsor draft itself is not expected yet?

- Anchors: R-T14 ("At this stage, your team is not expected to calculate the final size of the...", VRF), R-T16, R-T11; R-C88 ("developing strategy").
- Evidence: the TN guide asks each reflection to say how the trade serves "Laura's goals, funding needs, or risk" (brief section 5, VRF). It also excludes the co-sponsor draft (R-T14). A reason is not a draft.
- Hypothesis (GUESS): yes, in at most one clause of one note. It shows Client Knowledge early (R-S25) and makes the three deliverables visibly "connected" (R-C87). Keep the numbers out (R-T14).
- Would change: **sentence** (TN reflection content for a Treasury trade the team places in WInS now).

- Deliverables: TN, WlnS-now (choose which executed Treasury trade carries that note). Domain: D9, D7. Seed: 13.
- North star: from the first deliverable, judges see a firm that trades for Laura's promises, not for returns.

Q17. Which single co-sponsor rule belongs in the 500-word IPS, given that the IPS excludes the final range but must set the approach to the facility contribution?

- Anchors: R-I22 ("determining a responsible facility contribution", VRF), R-I31 (no "final facility-contribution range" or "draft communication to co-sponsors", VRF), R-I23, R-I35 (strategy frozen after Nov 6).
- Evidence: Q1, Q2 and Q10 above; B7b Q10 asks which rules the IPS must pre-declare in general.
- Hypothesis (GUESS): one rule, no numbers. For example, its *content* would be: the public figure never exceeds what is already bought when announced; the upper tier carries a stated probability; the operating reserve is never a source.
 - Because the IPS freezes the strategy, a rule set now protects every FR range from looking invented later.
 - Would change: **decision** (IPS content; roughly 25-40 of the 500 words).
 - Deliverables: IPS. Domain: D10 (freeze and consistency), D9. Seed: 9.
 - North star: the frozen IPS already contains the promise-keeping rule the FR later applies.

Q18. What is the minimum checklist a funder-grade "founder contribution statement" contains, and which items can be dropped to keep the FR "part" short?

- Anchors: R-C72 ("draft part of Laura's fundraising materials", VRF), R-C83, R-S28 ("investment uncertainty clearly and credibly", VRF).
- Evidence: Q1 to Q15; SH-05 checklist (Phase A); Kresge fact-sheet and gift-chart logic (VP); PKF (VP).
- Hypothesis (GUESS), as a checklist and **not** prose. A strong passage contains:
 1. what is committed (the floor), with its form and maturity (Q1, Q9, Q10);
 2. what is expected on top, with its probability and the model named in plain words (Q1, Q7);
 3. the date money moves (2033) and the currency (US\$ binding; NT\$ reference with date and rate; BS-04);
 4. that the ten operating years are already funded, separately, from her portfolio alone (Q5, Q8);
 5. what happens under unfavourable markets (the contribution falls toward the floor; the reserve is never touched; R-C70, R-C71);
 6. what happens if the project is delayed (Q13);
 7. what she is not promising: facility cost, overruns, post-2042 (R-C95, R-C49; BS-16);
 8. when she will update the figures (Q6). Candidates to drop if space is tight: item 8, then item 6.
- Would change: **sentence** (FR structure). This is the lens's "what a fundraising paragraph must contain" output.
- Deliverables: FR. Domain: D9, D5. Seed: null.
- North star: co-sponsors get everything they need in one reading, which is criterion 5 in practice.

Q19. Should the FR cite institutional readiness rules (for example, Kresge's "20 to 50 percent raised before applying") rather than the small-donor seed-money experiments as its evidence that Laura's contribution helps?

- Anchors: brief 8.17 ("a mechanism, not a prediction for institutions"); R-C64; R-S26 ("uses reasonable assumptions").
- New evidence: institutions encode the same logic directly in their own rules. Kresge wants private lead gifts raised first and counts only written pledges (VP). This answers brief 8.17's objection (the experiments are about households) with an institutional source.
- Hypothesis (GUESS): yes. One institutional source is more relevant to co-sponsors than lab or mail experiments.
 - Caveat: the Kresge guide is dated June 2011. It says the challenge grant "is no longer the centerpiece of our work" (VP). Cite it as established practice, not current Kresge policy.
- Would change: **sentence** (FR citations; Works Cited).
- Deliverables: FR. Domain: D7, D5. Seed: null.
- North star: the right evidence for the right audience shows judgement, not just research.

Q20. Should Laura's US\$50k payments be described as unrestricted general operating support, since operating money that the residency can spend freely is what arts funders say is scarcest?

- Anchors: R-C45 ("toward the residency's operating expenses", VRF), R-C41 ("reliable support for its early operations"); BS-17 (Kang 2026, VP in Phase A: "Funders can give unrestricted operating money to the residencies carrying the earliest risk").
- Evidence:
 - PwC (VP): future-dated promises carry an "implied time restriction" unless the donor says otherwise. So how the promise is worded changes how the recipient can use it.
- Hypothesis (GUESS): yes, within the case's wording ("operating expenses"). Say that the payments are flexible within operations. Program choices stay with the residency. This does not change any number.
- Would change: **sentence** (FR co-sponsor draft; one word, "unrestricted" or not).
- Deliverables: FR. Domain: D5. Seed: 6.
- North star: her money is the kind funders value most, and we say so in their language.

3. Evidence and small calculation

Range width at the 2031 decision point (ASSUMPTION illustration, not a verified model output). Inputs:

- the median 2031 floor of US\$165k (F-402, VRF) implies a 2031 sleeve of about US\$187.8k, since the floor is 80% of the sleeve $\times 1.0985$ (F-013);
- the other 20% is invested 60% U.S. large cap (log mean 0.0649, sd 0.152; F-316) and 40% intermediate Treasuries (4%, sd 3.4%) for two years;
- i.i.d. lognormal returns, 200,000 draws, seed 20260927, no fees.

Result:

- the unlocked part in 2033 is p5/p50/p95 US\$34.1k/41.9k/52.8k;
- total 2033 capacity is p5/p25/p50/p75/p90/p95 US\$199.1k/203.4k/206.9k/210.9k/215.1k/217.8k;
- chance of reaching a top set at p75 / p90 / p95: 25% / 10% / 5%.

This holds only at the median 2031 sleeve. The level itself is uncertain from today (floor p5-p95 US\$127k-217k, F-402). Run inline with `.venv/bin/python -c` (not saved as a script; D3 should rebuild it properly in `strategy_mc.py` style if Q7 is taken up).

Numbers used from Phase A (VRF): F-106 (US\$394,930 value of payments on 2033-01-01); F-402 (floor p5/50/95 US\$127k/165k/217k); F-401 (surplus p5/50/95 US\$159k/207k/273k); F-013 (2-year growth factor 1.0985).

4. Leads for the specialists

D5 (philanthropy and co-sponsors):

- Kresge "A Guide to the Challenge Grant" (updated 30 June 2011), VP. Useful passages:
 - seven readiness indicators ("Typically, we suggest at least 20 percent and sometimes as much as 50 percent...");

- "What Gifts Count?" ("Written pledges or cash"; "All pledges must be paid within five years from your fundraising end date");
- "committed, imminent or backstopped";
- glossary: "Backstopping", "Challenge grant - Money pledged by Kresge that will be paid when grant conditions are met".

New fact for B4a Q15 (staged facility payments): capital funders accept pledges *paid* over up to five years, so a staged payment of Laura's facility contribution is normal campaign practice (VP).

- Rondeau & List (2008), NBER w13728, VP (full text read). Gives definitions of challenge vs matching gifts; the Lawson (2001) 10-80% leadership-gift advice; footnote 3 on belief as information.
- Huck & Rasul (2011) JPE 95(5-6):351-362, VP abstract (RePEc). Karlan & List (2007) AER 97(5), VP abstract (NBER w12338).
- List & Lucking-Reiley (2002) JPE 110(1), VP abstract (RePEc). Full-text copies failed: UCSD (TLS error), davidreiley.com (403), SSRN (403, brief section 2).
- Andreoni (1998) JPE 106(6), VP abstract (RePEc). Vesterlund (2003) JPubE: not read (no abstract on RePEc).
- Still to find (my search budget ran out): CASE campaign counting standards (how campaigns count revocable vs irrevocable commitments); evidence on grant agreements conditioned on founder or board giving (Q2); AFP data on pledge write-offs (Phase A: afpglobal.org returns 403).

D10 (compliance, case reading) and D1:

- FASB ASU 2018-08 (June 2018), VP:
 - 958-605-25-5C: "A probability assessment about whether the recipient is likely to meet the stipulation is not a factor";
 - 958-605-55-17: the matching-grant example;
 - 958-605-55-51: "intention to give".
- PwC Viewpoint, Not-for-profit entities guide, section 7.3 (present value of multi-year promises, collectability factors, implied time restriction), VP.
- Legal enforceability is not the team's question (BS-05), and Taiwan law is out of scope (R-C97). Use these sources only as "how a professional reader values a promise", never as legal claims.

D4 (Taiwan):

- Lienchiang County government notice of the MOC "補助藝術村營運扶植計畫作業要點" 2022 call (VP). Key facts: annual selection; online applications 1-30 November 2021; multi-year grantees must file a self-improvement plan.
- law.moc.gov.tw (the rules text) returned HTTP 403; artres.moc.gov.tw 403 (as in Phase A). Subsidy-rate tiers and the "自籌款" rule are SNIP only.
- IFACCA D'Art report 45 (2013, VP, dated) has survey shares on how arts agencies support residencies. Example: "64% of respondents own, or provide ongoing maintenance of, residencies" (agency survey). Use only as background.

D3 (quant): Q7 table (lock share 50-80% → floor, capacity p5/p50/p95, P(top)). Also, for Q8, the PV of the operating commitment at 2031-01-01 (US\$356,384, F-106), in case the draft is dated 2031.

D7 / D9: Q4 (belief-as-signal framing), Q15 (no percentage over-claim), Q16 (TN clause), Q18 (checklist).

5. Sources (all accessed 2026-09-27)

#	Source	URL
S1	Kresge Foundation, "A Guide to the Challenge Grant", updated June 30, 2011	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
S2	FASB, Accounting Standards Update No. 2018-08, Topic 958, June 2018	https://storage.fasb.org/ASU%202018-08.pdf
S3	PwC Viewpoint, NFP guide 7.3 "Unconditional promises to give cash"	https://viewpoint.pwc.com/dt/us/en/pwc/accounting_guides/not-for-profit-entities/Not-for-profit-entities/nfp07/7_3_unconditional.html
S4	Rondeau & List, "Matching and Challenge Gifts to Charity", NBER w13728 (2008); Experimental Economics 11(3)	https://www.nber.org/system/files/working_papers/w13728/w13728.pdf ; https://www.nber.org/papers/w13728
S5	Karlan & List, "Does Price Matter in Charitable Giving?", NBER w12338 (2006); AER 97(5) 2007	https://www.nber.org/papers/w12338
S6	Huck & Rasul, "Matched fundraising: Evidence from a natural field experiment", JPubE 95(5-6) 2011	https://ideas.repec.org/a/eee/pubeco/v95y2011i5p351-362.html
S7	List & Lucking-Reiley, "The effects of seed money and refunds on charitable giving", JPE 110(1) 2002	https://ideas.repec.org/p/feb/natura/00301.html

#	Source	URL
S8	Andreoni, "Toward a Theory of Charitable Fund-Raising", JPE 106(6) 1998	https://ideas.repec.org/a/ucp/jpolec/v106y1998i6p1186-1213.html
S9	PKF O'Connor Davies, "Best Practices: Due Diligence on Grantees", July 2019	https://www.pkfod.com/wp-content/uploads/2019/07/Best-Practices-Due-Diligence-on-Grantees-v4.pdf
S10	Lienchiang County Government (Matsu), notice of MOC artist-village operating subsidy call, 2021 (page title 文化部「補助藝術村營運扶植計畫作業要點」111年受理申請公告)	https://www.matsucc.gov.tw/%E6%96%87%E5%8C%96%E9%83%A8%E3%80%8C%E8%A3%9C%E5%8A%A9%E8%97%9D%E8%A1%93%E6%9D%91%E7%87%9F%E9%81%8B%E6%89%B6%E6%A4%8D%
S11	MOC regulations database, artist-village subsidy rules (FL049953)	https://law.moc.gov.tw/law/LawContent.aspx?id=FL049953
S12	IFACCA D'Art Report 45, "International Perspectives on Artist Residencies", May 2013	https://ifacca.org/media/filer_public/1e/19/1e199fee-6d12-4218-a6a2-f969126b5298/dart45residenciesmay2013.pdf
S13	NCAF home page	https://www.ncafroc.org.tw/engIndex.html
S14	TransArtists funding tips; artouch.com Taiwan sponsorship article; Arts Residency Network Taiwan	transartists.org , artouch.com , artres.moc.gov.tw

WebSearch budget for the session ran out part-way through (200/200). Remaining leads are listed in section 4.

What this teaches

A promise is worth what the listener can count, not what the speaker hopes. Funders, accountants and the economists who run fundraising experiments agree on one simple rule: only an unconditional commitment moves other people's money. Everything else is information. That is why Laura's 2031 range should have a bought bottom, a clearly labelled top, and exact words about which is which. Two further lessons:

- Know your listener's calendar and contracts. A co-sponsor may write your number into its own grant conditions and budget one year at a time.
- Evidence must fit the audience. A mail experiment with small donors does not prove what a foundation will do, but that foundation's own rulebook does.

B6b Co-sponsors and philanthropy: the communication side of the 2031 range

Agent: B6b (lens "Co-sponsors & Philanthropy", starting angle: communication). insight_v1 run, written 2026-09-27. AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It holds **no submission-ready prose**: no fundraising paragraph, no IPS or Final Report text. Where it says what a passage "must contain", that is a checklist. The six students write every word, and any use of this file must be recorded in the Works Cited (AI policy, R-W46). No securities are named here.

Status labels follow brief section 3: VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SNIP), ASSUMPTION (ASM), PARAPHRASE-UNVERIFIED, INTERPRETATION (INT, my reading, never a fact).

0. Summary (read this first)

1. **The case asks for a forecast statement to non-specialists, and there is a research field on exactly that.** Forecast-communication studies (weather, climate, intelligence, company earnings) give tested rules for how to state a range and a confidence level. None of the earlier research files use it (grep of research/, 2026-09-27: no hits for IPCC, Budescu, Teigen, Gigerenzer, van der Bles). Five findings matter most (all VP, abstracts read):
 - People read verbal probability words ("likely", "very likely") as **closer to 50%** than intended. Adding the number next to the word fixes most of this, **in 17 languages** (Budescu et al. 2014). That matters for Taiwanese readers (Q1, Q21).
 - A bare percentage is ambiguous unless you say **what it is a percentage of** (Gigerenzer et al. 2005) (Q2).
 - Readers treat a range as a **yes/no category**: a result just outside a narrow edge counts as "wrong", even when the range was only a 50% band (Teigen, Løhre & Hohle 2018, the "boundary effect") (Q6).
 - Stating uncertainty **as numbers** barely lowers trust in the source. Verbal hedges lower it more (van der Bles et al. 2020). After an "unlikely" outcome happens, communicators who used numbers keep more credibility (Jenkins, Harris & Lark 2019) (Q14, Q15).
 - Keeping a promise earns as much credit as exceeding it; breaking it is punished (Gneezy & Epley 2014, via a university press release). So **the bottom of the range is the only real promise**, and a high top earns nothing (Q7, Q23).
2. **NEW NUMBER: by 2031 about 97% of the uncertainty is gone.** Using the verified model's own assumptions (strategy_mc.py, lock-early, 60% equity sleeve, 80% floor), I re-ran the paths. Only **2.8%** of the 2026-view variance of the 2033 surplus is still unresolved on 1 Jan 2031 (ASM-based, derived; script in section 5). From a 2031 starting point the 90% band is only about **±4-5% of the median** (\$199k-\$217k at the median 2031 state), against **\$159k-\$273k** in the 2026 view. The \$114k-wide range everyone quotes is a 2026 artefact. The range Laura actually says in 2031 can be narrow and honest (Q3, Q18).
3. **NEW NUMBER: locking 100% of the sleeve in 2031 would cost only about \$1k of median 2033 value** under the same assumptions (\$206.1k fully locked vs about \$207k at 80% locked). If that holds when D3 checks it, the 2031 "range" comes mostly from Laura's own flexibility choice and the exchange rate, not from markets (Q4).
3. **NEW: in NT\$ terms, currency swamps market risk.** The USD band at the median 2031 state is about -4%/+5%. The 2-year USD/TWD move since 2006 has p5/p95 of -12.1%/+10.6% (F-513, VP-derived). Any NT\$ figure needs its own label or band (Q5).
4. **How a capital funder actually counts a founder's money.** The Kresge Foundation's *Guide to the Challenge Grant* (2011, VP) counts only "Written pledges or cash" as money raised. Other sources "should be committed, imminent or backstopped", and it wants "a significant amount" pledged "before the general public ever hears about the campaign". Read against the case, the **floor** is what a funder can count, **the date the floor is bought** matters, and the top of the range counts for nothing (Q9, Q10).
5. **Matching gifts.** In a 50,000-letter field experiment, offering a match raised both giving and response rates, but "larger match ratios ... had no additional impact" (Karlan & List 2007, VP). This opens a design

option: offer a small 1:1 match on the part above the floor only. It still has to pass the complexity test (Q12).

6. **WInS-now item.** Trade notes cannot be edited. The team should fix a short certainty vocabulary now (what is "locked", what is "expected", what is "possible"), so that no note written this week says "guaranteed" in a way the IPS and Final Report must later walk back (Q19).

Question count: 23 (section 2). Leads for specialists: section 4. Sources: section 6.

1. What earlier work already covers (so I do not re-ask it)

Already covered in Phase A or other Phase B files. I only add **new evidence** to these:

- Currency labelling "US\$" (BS-04); which verb goes with the floor and which with the stretch (BS-05); share of an unestimated total (BS-06); overrun answer (BS-16); operating money is what funders least like to give (BS-17).
- Cap the gift at the top of the range, so the confidence depends only on the low end (B1a Q17, B1b Q19).
- A rule in 2031 versus a dollar range fixed today (B1a Q18, B1b Q18). My Q3 adds the 97%-resolved number.
- Pre-announced updates between 2031 and 2033 (B1a Q19).
- Nonprofit sense of "operating reserve" (B1b Q16). Conditional versus unconditional pledge under ASU 2018-08 (B4a Q14). Yale's two-year lag (B4a Q12). "Committed plus stretch" in her product-manager language (B2a Q12).
- Brief section 8.7 (floor bought, not a model percentile) and 8.17 (seed-money effect is a mechanism only).

My lens is distinct: **how** the range and confidence are worded, labelled, attributed, dated and shown, and what a funder can actually count.

2. Questions (23), grouped by deadline tier

Format for each: question; anchors; official words; hypothesis (a GUESS, with evidence and status); what would change; deliverables; specialist; seed; North Star.

Tier 1: WInS now and the Trading Notes (Oct 23)

Q19. Before more WInS notes are saved (they cannot be edited), should the team fix a three-tier certainty vocabulary: words for bought/locked amounts, for model-based expectations, and for mere possibilities? The aim is that no trade note uses "guaranteed" or "safe" in a way the IPS and Final Report must later qualify.

- Anchors: R-T28 ("Include each Trading Note exactly as it appears in WInS."), R-T21, R-C92, A4 checklist item 12 (strategy glossary), brief section 7 ("residual risk U.S. default").
- Official words: "Teams should use each deliverable for its intended purpose while maintaining a clear and consistent investment strategy across all three." (case p.4, VRF)
- Hypothesis (GUESS):
 - Yes. A4's glossary covers strategy words (growth, liquidity, and so on). It does not cover **certainty words**. Those are the words co-sponsors, a statistician client and the judges will test.
 - The team's own certainty definition is "certain in nominal USD, barring U.S. Treasury default" (brief section 7). So "guaranteed" is already too strong for a note. "Locked in at today's prices" or "matched" is defensible.
 - Evidence: numeric and precise wording keeps credibility when surprises come (Jenkins et al. 2019, VP abstract). Readers treat range edges as right/wrong lines (Teigen et al. 2018, VP).
 - This costs nothing and prevents an inconsistency that cannot be fixed later.
- Would change: **sentence** (wording of every trade note from now on); **decision** (a one-page certainty glossary shared by TN, IPS and FR).
- Deliverables: WInS-now, TN, IPS, FR. Specialist: D9 (with D10). Seed: null.
- North Star: Laura, a statistics graduate, sees the same careful words in October that she sees in December.

Tier 2: the IPS (Nov 6), which fixes the rules

Q4. Locking 100% of the sleeve in 2031 would cost only about \$1k of median 2033 value under the verified model's assumptions. Should the 2031 lock share rise from 80% toward 100%, so that the co-sponsor range reflects Laura's flexibility choice and the exchange rate rather than market risk?

- Anchors: R-C65, R-C71, F-402, F-013, brief section 9 ("share of sleeve locked as floor").
- Official words: "In 2031, however, the value of her portfolio in 2033 remains uncertain." (case p.3, VRF)
- Hypothesis (GUESS):
 - My re-run (section 5, derived from the VERIFIED script with its ASSUMPTIONS): at the median 2031 state, 80% locked gives 2033 p5/p50/p95 of about \$199k/\$207k/\$217k. 100% locked gives **\$206.1k with no market spread**. The median cost is about \$1k.
 - The reason: over two years, the unlocked 20% (about \$37k) earns only a little more in expectation than a 4.81% two-year Treasury.
 - Caveat (ASM): the model fixes the 2031 two-year yield at today's 4.81%. If 2031 yields are much lower, locking gives up more. D3 should test 2031 yields of 3% and 4%.
 - If the result holds, the case's "remains uncertain" is answered by a design choice: the USD amount is almost known in 2031. The range then describes how much Laura *chooses* to keep as flexibility (R-C58), plus currency for NT\$ readers.
- Would change: **number** (floor share 80% vs 90-100%); **decision** (IPS 2031 rule).
- Deliverables: IPS, FR. Specialist: D3 (with D1). Seed: 8.
- North Star: Laura can tell co-sponsors a number that is already bought, which is the strongest possible credibility.

Q10. Should the 2031 floor be bought before Laura's first co-sponsor conversation (on the first trading days of 2031), so that the first number any funder hears is already "committed, imminent or backstopped"? And what document would prove it to a funder's due-diligence team?

- Anchors: R-C62 ("begin approaching potential co-sponsors in 2031"), R-C66, R-AN7, BS-08 (records).
- Official words: "Laura plans to begin approaching potential co-sponsors in 2031, two years before the residency is established." (case p.3, VRF)
- Hypothesis (GUESS):
 - Yes. Kresge's guide (VP) advises: "Have a significant amount of your campaign pledges committed before the general public ever hears about the campaign". It also says non-gift sources "should be committed, imminent or backstopped".
 - A floor bought in, say, March 2031, after conversations have begun, means her first stated number was a forecast.
 - Proof of the floor would be an account statement or holding record. Its content (not its form) belongs in the IPS rule as "trigger, action, record" (BS-08).
- Would change: **decision** (date of the 2031 purchase in the IPS rule); **sentence** (FR: "already set aside on [date]" as a required element).
- Deliverables: IPS, FR. Specialist: D5 (with D8). Seed: 8.
- North Star: she never has to walk back a number, because nothing is said before it is bought.

Tier 3: the Final Report (Dec 4), fundraising part and range

Q1. Which confidence level should the 2031 statement carry, and should it use "dual format": a calibrated word plus a number (the IPCC's "very likely" with its numeric band), never a word alone?

- Anchors: R-C69, R-AN32 (statistics degree), R-S28 (criterion 5: "clearly and credibly").
- Official words: "state how confident they are that her 2033 contribution will fall within that range." (case p.3, VRF)
- Hypothesis (GUESS):
 - A numeric level of about 90%, always shown with its number. 90% matches the IPCC's "very likely" band.
 - Evidence:

- Budescu et al. 2014 (VP abstract), 25 samples, 24 countries, 17 languages: "laypeople interpret IPCC statements as conveying probabilities closer to 50% than intended". Supplementing the words "with numerical ranges increases the correspondence".
- Budescu et al. 2009 (VP abstract): readers deviated from the guidelines "even when the respondents had access to these guidelines".
- IPCC AR5 guidance note (VP): "When there is sufficient information, it is preferable to specify the full probability distribution or a probability range (e.g., 90-95%) without using the terms in Table 1."
- The exact IPCC table values ("very likely" 90-100%, "likely" 66-100%) were garbled in the PDF text layer. The paragraph I read confirms "likely" = "≥66% ... to 100%". The other bands are SNIP until D9 reads the table.
- A bare "likely" would be read as about a coin flip, which undersells a bought floor.
- Would change: **number** (the stated confidence level); **sentence** (the dual-format rule in the draft checklist).
- Deliverables: FR. Specialist: D9. Seed: 9.
- North Star: Laura and a co-sponsor read the same probability from the same words.

Q2. What exactly does the confidence percentage refer to (paths of a stated model? past two-year periods?), and must the fundraising part name that reference class?

- Anchors: R-C55 ("explain how they evaluated that level of certainty"), R-C69, R-AN32.
- Official words: "They should also define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty" (case p.3, VRF)
- Hypothesis (GUESS):
 - Yes, in one clause, for example "in N of 10 simulated market paths under [named assumptions]". That is a frequency with its source. The team writes the wording.
 - Evidence: Gigerenzer et al. 2005 (VP abstract): "even numerical probabilities can be interpreted by members of the public in multiple, mutually contradictory ways ... experts need to specify the reference class".
 - Natural frequencies ("9 in 10") are easier for non-specialists. The percentage and method can sit in the FR method box for Laura, the statistician.
- Would change: **sentence** (FR method line and fundraising checklist element).
- Deliverables: FR. Specialist: D9 (D3 supplies the class). Seed: 9.
- North Star: a statistician client sees her own discipline, "what is the denominator?", applied to her promise.

Q3. How much of the 2033 uncertainty is still unresolved when Laura speaks in 2031? Should the Final Report show the 2031 range as a narrow band for each of three 2031 starting states (about ±5% of the median) instead of the 2026-view band (\$159k-\$273k)? (New evidence on B1a Q18 / B1b Q18.)

- Anchors: R-C62, R-C65, R-C68, R-C70, F-401, F-402.
- Official words: "They should explain how favorable and unfavorable market outcomes could affect the amount she can provide." (case p.3, VRF)
- Hypothesis (GUESS, derived numbers labelled):
 - My re-run of the verified model's paths (ASM inputs as in `strategy_mc.py`) finds that only **2.8%** of the 2026-view variance remains after 2030.
 - Conditional on the 2031 sleeve, 2033 p5-p95 is:

2031 state	2033 p5	2033 p95	90% width	Bought floor
Weak (p10 sleeve)	\$162k	\$177k	\$15k	\$134k
Median	\$199k	\$217k	\$18k	\$165k
Strong (p90 sleeve)	\$246k	\$269k	\$23k	\$204k

The 90% width is about 8.9% of the median in each state. - So the "favorable and unfavorable market outcomes" the case asks about are mostly 2027-2030 outcomes, which Laura already knows by 2031. - A three-row table is simple, and it answers both the "dollar range" and the "rule" readings of the case. - Would change: **number** (the range width in the FR); **decision** (present a three-state table, not one wide band).

- Deliverables: FR. Specialist: D3. Seed: 8.
- North Star: shows that two years was enough time to make her promise precise, which is what the timing of the case allows.

Q5. In NT\$ terms, does exchange-rate risk over 2031-2033 swamp the market uncertainty left in the USD range? Should any NT\$ figure therefore get its own wider band, or be labelled as an illustration at a stated date and rate?

- Anchors: R-AN10, F-510, F-512, F-513, F-514, BS-04, SH-06, SH-07.
- Official words: "Teams must recommend the dollar range Laura should communicate" (case p.3, VRF). No currency is named anywhere (R-AN10).
- Hypothesis (GUESS):
 - Yes. The USD band at the median 2031 state is about -3.9%/+4.8% (Q3). USD/TWD 2-year moves since 2006 have p5/p95 of -12.1%/+10.6% (F-513, VP-derived).
 - Combined as independent (ASM), the NT\$ band is roughly $\pm 12\%$, about 2.5 times the USD band.
 - This does not change the USD promise. It changes what a Taiwanese partner should be told: the binding figure is US\$. NT\$ is a dated illustration, and its swing is larger than the market swing.
- Would change: **sentence** (FR currency element); possibly **number** (an NT\$ band).
- Deliverables: FR. Specialist: D4. Seed: null.
- North Star: respects partners who budget in NT\$, without tokenism: a practical currency fact, not an identity gesture.

Q6. How should the ends of the range be labelled? Readers treat a result just outside a hard edge as a broken promise (the "boundary effect"), yet a soft word like "can" or "up to" makes the top sound likely.

- Anchors: R-C66, R-C67, R-AN9.
- Official words: "Rather than promising one exact amount, she wants to communicate a credible range for her potential contribution." (case p.3, VRF)
- Hypothesis (GUESS):
 - Teigen, Løhre & Hohle 2018 (VP abstract): outcomes "falling outside a narrow range were deemed to be incorrectly predicted, in proportion to the magnitude of deviation". But "When the upper limit of a range is described as a value that 'can' occur ..., outcomes both below and beyond this value were regarded as consistent with the forecast."
 - Teigen & Filkuková ("Can > will"): "can" statements pick extreme values that listeners still judge probable. SNIP: title and topic from a University of Oslo project listing only; D9 should read the paper.
 - Guess: the bottom is a hard edge (bought). The top is a soft edge described **with its probability of being reached**, not with "up to" or "can", which readers take as the likely amount.
- Would change: **sentence** (labels of the two ends in the draft).
- Deliverables: FR. Specialist: D9. Seed: 9.
- North Star: nothing she says can be read later as a broken promise.

Q7. Exceeding a promise earns no more credit than keeping it, while breaking it is punished. So should the draft treat the bottom as the only promise, and should it show a "best estimate" at all?

- Anchors: R-C66, R-C64, R-AN9, BS-03 (credibility is part of her income).
- Official words: "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors." (case p.3, VRF)
- Hypothesis (GUESS):
 - The ScienceDaily release of Gneezy & Epley 2014 (VP for the release; the paper itself SNIP, SSRN blocked) says: "exceeding a promise isn't viewed any more highly than keeping a promise". Also: "Promises can be hard to keep, and promise makers should spend their effort keeping them wisely."
 - So a high top adds no credibility. Its only use is to help co-sponsors plan.
 - Dieckmann, Peters & Gregory 2015 (VP abstract): "including a best estimate increases perceptions that the distribution is roughly normal". A central number helps understanding, but it may become the number people remember.

- Guess: show the floor and the range with its probability in the co-sponsor part. Keep the median in the FR analysis for Laura.
- Would change: **decision** (best estimate shown or not); **sentence**.
- Deliverables: FR. Specialist: D6. Seed: 15.
- North Star: the plan protects her from the most expensive mistake a founder can make in public.

Q8. Should the confidence statement give the chance of missing the range on each side separately (below the floor, above the top), following the IPCC's advice to give reciprocal statements?

- Anchors: R-AN9, R-C69, R-C66, R-C71.
- Official words: "state how confident they are that her 2033 contribution will fall within that range." (case p.3, VRF)
- Hypothesis (GUESS):
 - The IPCC AR5 note (VP), paragraph 4: "a 10% chance of dying is interpreted more negatively than a 90% chance of surviving ... Consider reciprocal statements".
 - At the median 2031 state, a range from the bought floor (\$165k) to the conditional p90 (\$215k) gives, in my re-run (ASM, derived): P(inside) 0.900, **P(below) 0.000**, P(above) 0.100.
 - A single "90%" hides that every miss is on the harmless side. Two separate numbers make that visible.
 - This complements B1a Q17 (cap at the top) with the communication form.
- Would change: **sentence and number** (two numbers instead of one).
- Deliverables: FR. Specialist: D9 (numbers from D3). Seed: 9.
- North Star: co-sponsors see that the only surprise possible is a good one.

Q9. Would a capital funder count any part of Laura's range as money already raised, or only a written floor? Should the fundraising part therefore lead with the floor, not the range or its midpoint?

- Anchors: R-C64, R-C67, BS-05, B4a Q14.
- Official words: "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute." (case p.3, VRF)
- Hypothesis (GUESS):
 - Kresge's counting list (VP) starts with "Written pledges or cash from individuals, corporations, businesses and foundations". Its glossary defines a challenge grant as "Money pledged by Kresge that will be paid when grant conditions are met."
 - A forecast range is not on the list. The part a funder can count is the floor, if it is stated as a written commitment.
 - This is the same logic as ASU 2018-08 (B4a Q14, SNIP) from a funder's practice rather than from accounting.
 - So the "meaningful" signal (R-C64) is carried by the size of the floor. That links this question back to Q4.
- Would change: **sentence** (the headline number of the draft); **decision** (floor size matters more than the midpoint).
- Deliverables: FR. Specialist: D5. Seed: 9.
- North Star: her number is one a funder can put in its own spreadsheet.

Q11. Kresge-style funders expect "project-cost estimates are firm" before a capital grant, yet the case says the facility cost is undetermined. What must the fundraising part say so her contribution reads as sized from her portfolio, not as a share of an unknown budget? And what happens to it if costs change? (New evidence on BS-06 and BS-16.)

- Anchors: R-C94, R-C95, R-C60, BS-06, BS-16.
- Official words: "The total cost of the facility has not been determined." (case p.4, VRF)
- Hypothesis (GUESS):
 - Kresge's readiness indicator 2 (VP): "Your project-cost estimates are firm and construction timetables have been established."

- In 2031 Laura will be early in that process. The draft should not pretend otherwise.
- Elements: sized from what her portfolio can responsibly give; independent of the final cost; does not rise if costs rise; the project scope adjusts. No cost estimate or budget (scope).
- Would change: **sentence** (two checklist elements).
- Deliverables: FR. Specialist: D5. Seed: 6.
- North Star: she sounds like a founder who knows what a funder will ask, without inventing a budget.

Q12. Should part of Laura's above-floor amount be offered as a 1:1 match for co-sponsor gifts? Field evidence shows a match raises giving, but larger ratios add nothing.

- Anchors: R-C64 ("make potential co-sponsors more willing to contribute"), R-C44 (outside money may fund the facility), R-C71.
- Official words: "Any remaining facility cost ... may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources." (case p.3, VRF)
- Hypothesis (GUESS):
 - Karlan & List 2007 (VP, NBER abstract): "the match offer increases both the revenue per solicitation and the probability that an individual donates ... larger match ratios (i.e., \$3:\$1 and \$2:\$1) relative to smaller match ratios (\$1:\$1) had no additional impact."
 - That result is for direct-mail donors, not institutions. Brief 8.17 already warns against over-applying such results.
 - Guess: at most one line offering a 1:1 match on the above-floor part, and only up to an amount already bought. It must never touch the operating reserve.
 - This may fail "complexity must earn its place". D5 should decide whether institutions respond to founder matches. If unclear, drop it.
- Would change: **decision** (match or no match).
- Deliverables: FR. Specialist: D5. Seed: null.
- North Star: turns her upside into a lever for others' giving, but only if that is simple and honest.

Q13. Should the fundraising part state a two-way "no substitution" rule: co-sponsor money never replaces Laura's ten operating payments, and her operating reserve is never drawn for the building?

- Anchors: R-C48, R-C44, R-AN6, R-C71, BS-17.
- Official words: "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement." and "The proposed range must also protect the portfolio's ability to fund the ten-year operating commitment." (case p.3, VRF)
- Hypothesis (GUESS):
 - Yes, as one element. The case's asymmetry is a firewall in both directions.
 - Funders of buildings fear being asked for running costs (BS-17, SSIR, VP via A3). Founders' personal promises can also be quietly "backfilled" by donor money. Stating both directions answers both fears.
 - "Supplement, not supplant" is a U.S. public-grant phrase for the same idea (ASM: general knowledge, not verified here; D10 to check before any use).
- Would change: **sentence** (one checklist element).
- Deliverables: FR. Specialist: D5 (with D10). Seed: 6.
- North Star: shows we understood why the case splits the money the way it does.

Q14. How can the fundraising part sound confident without overpromising? People prefer confident advisers but do not dislike advice given as ranges, and numeric uncertainty barely dents trust in the source.

- Anchors: R-AN30 (pull quote vs co-sponsor task), R-S28, R-C66.
- Official words: "communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors" (SMApply criterion 5, VRF)
- Hypothesis (GUESS):
 - Gaertig & Simmons 2018 (VP abstract): participants "did not dislike advisers who expressed uncertainty by providing ranges of outcomes, giving numerical probabilities", while advisers who were "not sure" were judged worse. The authors conclude: "Advisors benefit from expressing themselves with confidence, but not from communicating false certainty."

- van der Bles et al. 2020 (VP abstract): "only a small decrease in trust in numbers and trustworthiness of the source, and mostly for verbal uncertainty communication".
- Rule: be confident about the method and the floor, and give uncertainty as numbers. Ban vague hedges ("hopefully", "should be able to", "we are not sure").
- This resolves seed 15 in communication terms: private conviction becomes public proof through a number already bought. The pull quote itself is still unattributed (R-AN30); do not quote it as hers until verified.
- Would change: **sentence** (tone rules for the draft).
- Deliverables: FR. Specialist: D9 (with D6). Seed: 15.
- North Star: sounds like Laura at her best, sure of the plan and honest about the numbers.

**Q15. Which two or three drivers of change should the 2031 statement name in advance?
Explaining uncertainty up front protects trust when the numbers later change.**

- Anchors: R-C70, R-C86, R-C65.
- Official words: "They should explain how favorable and unfavorable market outcomes could affect the amount she can provide." (case p.3, VRF)
- Hypothesis (GUESS):
 - Dries et al. 2024 (VP abstract): "communicating uncertainty buffers against a loss of trust when evidence changes. Moreover, explaining uncertainty does not appear to harm trust."
 - Jenkins, Harris & Lark 2019 (VP abstract, RePEc): perceptions were "least affected by the 'erroneous' prediction when it was expressed numerically".
 - Candidate drivers, if Q4 keeps some market exposure: returns on the unlocked part; for NT\$ readers, the exchange rate; and Laura's own flexibility choice.
 - Name them once, with the direction each one moves the amount. Nothing more.
- Would change: **sentence** (the "what moves it" element).
- Deliverables: FR. Specialist: D9. Seed: null.
- North Star: if a number changes in 2032, co-sponsors were told why it might.

Q16. Should the fundraising part attribute the range and confidence to Laura's investment team, with a date, the main assumptions and the limits of the projection, the way regulated advisers must disclose hypothetical projections?

- Anchors: R-C72, R-C69 ("how confident *they* are": the team's confidence), R-W29 (CFA Asset Manager Code), SH-25.
- Official words: "Each team must also draft part of Laura's fundraising materials describing this potential contribution for prospective co-sponsors." (case p.4, VRF)
- Hypothesis (GUESS):
 - The case places the confidence with the team ("how confident they are"). The materials are Laura's. One attribution line ("assessment prepared by her investment team, as of [date]") resolves who speaks.
 - An independent-looking assessment is also more credible than a founder grading herself (INT).
 - Disclosure standards give a ready checklist:
 - The CFA Asset Manager Code requires disclosures that are "truthful, accurate, complete, and understandable" (VP via A3, stakeholder map).
 - The SEC Marketing Rule requires advisers showing hypothetical performance to disclose "criteria used and assumptions made" and "risks and limitations" (SNIP; sec.gov returned 403, eCFR 406).
 - The Marketing Rule does not apply to a fundraising letter (INT). Its elements are borrowed as a checklist, not as a legal claim.
- Would change: **sentence** (attribution and as-of line); **decision** (whose voice).
- Deliverables: FR. Specialist: D10 (with D8). Seed: null.
- North Star: a fictional firm acting like a real, regulated one, which the Rules page asks for (R-W29).

Q17. What minimum checklist must the "part of Laura's fundraising materials" meet to be credible to a funder's finance reader and a non-specialist at once, and what should be left out?

- Anchors: R-C72, R-AN21, R-S28, R-C83, R-C64-R-C71.
- Official words: "Communicates Laura's potential facility contribution to co-sponsors clearly and credibly." (case p.4, VRF)

- Hypothesis (GUESS), a checklist, not text:
 - **Contains:** 1. The floor: amount in US\$, "already set aside" on a stated date, and in what form (Q9, Q10). 2. The range above it, with the probability of reaching the top in dual format and its reference class (Q1, Q2, Q6). 3. The chance of missing on each side (Q8). 4. What moves the amount, named in advance (Q15). 5. That ten years of operating payments are funded separately from her portfolio alone, with the two-way firewall (Q13). 6. That it is sized from her portfolio, not from a facility budget, and does not rise with cost overruns (Q11). 7. Currency: US\$ binding; NT\$ as a dated illustration (Q5). 8. Who prepared the assessment and when, plus key assumptions and limits (Q16). 9. When it will next be updated (B1a Q19).
 - **Leave out:** facility cost or funding gap (scope); model jargon (percentiles, lognormal) in the co-sponsor part (keep it in the FR method box); any claim about legal enforceability (not the team's question, BS-05); any personal detail about Laura (privacy); drafted text from AI (AI policy).
 - Length test: a non-specialist should get elements 1-5 in one reading (SH-09).
- Would change: **sentence** (structure of the FR fundraising part).
- Deliverables: FR. Specialist: D5 (with D9). Seed: 9.
- North Star: a co-sponsor finishes reading knowing exactly what Laura will give, how sure it is, and what is already done.

Q18. How wide can the stated range be before non-specialists find it useless?

- Anchors: R-AN9, R-C67, R-C69.
- Official words: "she wants to communicate a credible range for her potential contribution." (case p.3, VRF)
- Hypothesis (GUESS):
 - Evidence:
 - Du, Budescu, Shelly & Omer 2011 (VP abstract): investors' preference for imprecision "peaks for low levels of imprecision and diminishes when the range gets wider". Investors "favor forecasts that are as precise as warranted by the information available, but not more precise".
 - Yaniv & Foster 1995 (SNIP, from a search summary): people often prefer an informative wrong interval to a correct wide one.
 - Derived ratios (top divided by bottom):
 - the 2026-view band \$159k-\$273k: about 1.72;
 - the median-2031 band from floor to p90 (\$165k-\$215k): about 1.30;
 - p5-p95 at the median 2031 state: about 1.09.
 - Guess: a ratio well under about 1.3 reads as informative; 1.7 reads as "we don't know". That threshold is ASM until D6 finds evidence.
- Would change: **number** (range width and how the FR presents it).
- Deliverables: FR. Specialist: D6 (with D3). Seed: 9.
- North Star: a range narrow enough to plan with is the difference between a pledge and a shrug.

Q20. Which single Final Report chart shows co-sponsors the floor, the range and the confidence most clearly: a fan chart from 2027, or a bar (bought floor) with a range whisker for three 2031 starting states?

- Anchors: R-S28 ("uses data effectively"), R-C70, brief section 13 (charts in the FR only).
- Official words: "uses data effectively to support conclusions" (SMAApply criterion 5, VRF)
- Hypothesis (GUESS):
 - Dieckmann et al. 2015 (VP abstract): "simple graphical representations can decrease the variance in distributional perceptions".
 - The Fed's projection fan charts show 70% bands "based on historical forecast errors" (VP, SEP 2026-09-16 notes). That is a model, but a fan from 2027 visually repeats the misleading 2026-view width (Q3).
 - Guess: the bar plus whisker at three 2031 states is for co-sponsors. A fan chart, if any, goes in the method section for Laura. Data: the three rows in Q3 plus the floor.
- Would change: **decision** (which chart; data needed).
- Deliverables: FR. Specialist: D9. Seed: null.
- North Star: one picture a co-sponsor could redraw from memory.

Q21. Numbers survive translation and verbal probability words do not. Should every probability in the fundraising part be numeric, so a Mandarin-reading Taiwanese partner reads the same confidence as an English reader, without the team translating anything?

- Anchors: R-AN10, BS-12, R-C69, SH-06, SH-07.
- Official words: "In 2033, Laura plans to establish a collaborative creative residency in Taiwan." (case p.2, VRF)
- Hypothesis (GUESS):
 - Budescu et al. 2014 (VP abstract): under the dual format, "interpretations of the terms in various languages are more similar".
 - Translation stays out of scope (BS-12). Numeric probabilities make the text translation-proof at zero cost.
 - This is a practical point, not identity decoration (tokenism rule).
- Would change: **sentence** (a rule in the checklist).
- Deliverables: FR. Specialist: D9 (with D4). Seed: null.
- North Star: the plan works for the people in Taiwan who will actually read it.

Q22. Should the 2031-2033 band be cross-checked against actual past two-year outcomes of a sleeve-like mix (as the Fed's projection bands use past forecast errors), not only the lognormal model, so that both a statistician and a funder's finance reader accept the confidence number?

- Anchors: R-C55, R-S26 ("reasonable assumptions"), R-AN32, brief section 11 (i.i.d. lognormal, no fat tails).
- Official words: "explain how they evaluated that level of certainty, and identify the assumptions supporting their recommendation." (case p.3, VRF)
- Hypothesis (GUESS):
 - The Fed notes (VP) say its band "is based on root mean squared errors of various private and government forecasts made over the previous 20 years", and warn it "may not reflect ... current assessments".
 - With only about 20% of the sleeve unlocked for two years (Q4), even a 2008-type fall should move the 2033 amount by a few percent. D3 computes it from a long history of monthly returns (for example, FRED or published index data).
 - A historical worst case beside the model p5 is cheap and persuasive.
- Would change: **number** (the stated confidence, and the bottom of the market band if not fully locked).
- Deliverables: FR. Specialist: D3. Seed: 9.
- North Star: Laura can check our "9 in 10" against history herself.

Q23. Should every sentence of the fundraising part pass a "floor test": still true if 2033 lands exactly at the floor? This would be a one-line pre-submission check against overpromising.

- Anchors: R-C66, R-C71, R-AN9.
- Official words: "If Laura promises more than she can ultimately contribute, she could damage her credibility" (case p.3, VRF)
- Hypothesis (GUESS):
 - Yes. It turns the Gneezy & Epley asymmetry (Q7) and the boundary effect (Q6) into a check a student can run in five minutes.
 - Also run a "top test": no sentence implies the top is expected.
- Would change: **decision** (review rule for the FR draft).
- Deliverables: FR. Specialist: D9. Seed: null.
- North Star: nothing in the letter she sends could ever become untrue.

Parked (fail the "would change" filter or out of scope; logged only)

- Legal enforceability of a founder's pledge in Taiwan or the U.S. (out of scope: Taiwan legal; BS-05).
- Translating the draft into Mandarin (out of scope and complexity; Q21 covers it at zero cost).
- Donor-recognition and naming rights (no case anchor).
- Finale-style questions on how judges would challenge the range (scope: semifinals only).

3. Evidence found (claims with status)

#	Claim	Status	Source
E1	Laypeople read IPCC probability words as closer to 50% than intended; numbers alongside words fix it across 17 languages	VP (abstract)	Budescu, Por, Broomell & Smithson, <i>Nature Climate Change</i> 4:508-512, 2014-04-20
E2	Respondents deviated from IPCC guidelines "even when the respondents had access to these guidelines"	VP (abstract via Europe PMC)	Budescu, Broomell & Por, <i>Psychological Science</i> , 2009
E3	"likely" = "≥66% ... to 100%"; prefer "a probability range (e.g., 90-95%)" when information allows; "Consider reciprocal statements"	VP (paragraphs 4 and 10; the Table 1 text layer is garbled)	IPCC AR5 Uncertainty Guidance Note (Mastrandrea et al. 2010)
E4	Reference class must be specified for single-event probabilities	VP (abstract via Europe PMC)	Gigerenzer et al., <i>Risk Analysis</i> , 2005
E5	Boundary effect; "can" label softens the upper edge	VP (abstract, open access)	Teigen, Løhre & Hohle, <i>Judgment and Decision Making</i> 13(4):309-321, July 2018
E6	"Can" predictions are extreme but believed probable	SNIP	Teigen & Filkuková (UiO project listing)
E7	Numeric uncertainty causes only a small drop in trust, mostly for verbal formats	VP (abstract)	van der Bles et al., <i>PNAS</i> 117:7672-7683, 2020
E8	Numbers protect credibility after "unlikely" outcomes	VP (abstract on RePEc)	Jenkins, Harris & Lark, <i>Journal of Risk Research</i> 22(5):537-554, 2019
E9	Uncertainty communicated up front buffers trust loss when evidence changes	VP (abstract)	Dries et al., <i>Public Understanding of Science</i> , 2024
E10	People do not dislike uncertain advice given as ranges or probabilities; "not sure" advisers are judged worse	VP (abstract)	Gaertig & Simmons, <i>Psychological Science</i> , 2018
E11	Preference for range precision peaks at low imprecision	VP (abstract on RePEc)	Du, Budescu, Shelly & Omer, <i>OBHDP</i> 114(2):179-189, 2011
E12	Informativeness is often preferred over accuracy (730-780 vs 700-1500 miles)	SNIP	Yaniv & Foster, <i>JEP: General</i> , 1995
E13	A best estimate shifts perceived distribution toward normal; simple graphics reduce variance of perceptions	VP (abstract)	Dieckmann, Peters & Gregory, <i>Risk Analysis</i> , 2015
E14	Exceeding a promise is not valued above keeping it	VP for the press release; paper SNIP	ScienceDaily / Chicago Booth release, 2014-07-18
E15	Match offers raise giving; 2:1 and 3:1 no better than 1:1	VP (abstract)	Karlan & List, <i>NBER w12338 / AER</i> 97(5), 2007
E16	Kresge counted "Written pledges or cash"; other sources "committed, imminent or backstopped"; pledges committed before the public hears; cost estimates "firm"; 20-50% of the private goal raised before applying	VP (PDF read)	Kresge Foundation, <i>A Guide to the Challenge Grant</i> , updated 2011-06-30
E17	Fed SEP bands: 70% intervals from 20 years of forecast errors, with a caveat on current conditions	VP	FOMC SEP accessible version, 2026-09-16
E18	SEC Marketing Rule: disclose "criteria used and assumptions made" and "risks and limitations" of hypothetical performance	SNIP (sec.gov 403; eCFR 406)	search summaries of Rule 206(4)-1
E19	2.8% of 2026-view variance unresolved in 2031; conditional 90% widths \$15k/\$18k/\$23k; full lock costs about \$1k median; P(below floor)=0, P(above p90)=0.10 at the median state	Derived from VRF script; inputs ASM	section 5 script (scratchpad), <code>strategy_mc.py</code>

4. Leads for the specialists

D5 (philanthropy & co-sponsors)

- Kresge *Guide to the Challenge Grant*: <https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf> (VP, 2026-09-27).
 - Readiness indicators 1-7, pages 5-6 of the PDF text.
 - "What Gifts Count?" is page 8.
 - Glossary: "Backstopping" and "Challenge grant".
 - Use these for Q9, Q10, Q11.

- Kresge centennial page on its first challenge grants (1929): <https://kresge.org/centennial/> (SNIP; not opened).
- Karlan & List 2007 abstract: <https://www.nber.org/papers/w12338> (VP) for Q12. Check whether any study covers institutional (not individual) responses to founder matches. If none, recommend dropping Q12's match.
- A3's sources W-3 (SSIR starvation cycle) and W-6 (PKF funder due-diligence bulletin) pair with Q13 and Q11.

D9 (communication & plain English)

- IPCC AR5 guidance note PDF: https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf. Paragraphs 4 and 10 are VP. Table 1 needs reading from the rendered page (the text layer is garbled).
- Budescu 2014: <https://www.nature.com/articles/nclimate2194> (abstract VP).
- Budescu's IPCC 2016 slides: https://www.ipcc.ch/site/assets/uploads/2016/02/Budescu_IPCC_Communication_Meeting_OSLO_February_2016.pdf (not opened).
- Teigen et al. 2018 (open access, CC-BY): <https://www.cambridge.org/core/journals/judgment-and-decision-making/article/boundary-effect-perceived-post-hoc-accuracy-of-prediction-intervals/AED4A7230EA737078997644325739C38> Experiment 5 is the "can" label result.
- Europe PMC REST API works through the proxy where PubMed blocks. For example: `curl "https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=TITLE%3A%22...%22&format=json&resultType=core"`. It returns full abstracts.
- Nature Climate Change 2025 "Negative verbal probabilities undermine communication of climate science": <https://www.nature.com/articles/s41558-025-02472-1> (title only, SNIP). Relevant to directional words ("not certain" vs "possible").
- Irwin 2023 *Risk Analysis* on verbal vs numeric formats for expert and non-expert readers: <https://onlinelibrary.wiley.com/doi/abs/10.1111/risa.14009> (title only, SNIP).

D3 (quant modelling)

- Re-run section 5 with the 2031 two-year yield at 3%, 4% and 5%. Report the median cost of a 100% lock (Q4).
- Add a historical two-year worst case for a 60/40 sleeve (Q22).
- Report the three-state table (Q3) and the side-split miss probabilities (Q8).
- Label every output as model-based with ASSUMPTION inputs.

D4 (Taiwan & FX)

- Combine F-513's 2-year USD/TWD distribution with the Q3 USD band to give an NT\$ band (Q5). State the independence ASSUMPTION or estimate the correlation of USD/TWD with U.S. equities.

D6 (behavioural)

- Gneezy & Epley paper (SSRN <https://www.ssrn.com/abstract=2233670> blocked; try the SAGE DOI 10.1177/1948550614533134) for Q7.
- Du et al. 2011 and Yaniv & Foster 1995 (PDF: <https://deanfoster.net/research/YanivFoster1995JEPG.pdf>, not opened) for a width threshold (Q18).

D10 (compliance)

- SEC small-business guide on adviser marketing (403 here): try another route to the rule text (for example federalregister.gov, final rule release IA-5653) to verify E18 before Q16 cites it.
- Confirm that A3's CFA Code F.2 quote is exact.

5. Script used (kept in the scratchpad; the numbers are reproducible)

b6b_vantage.py (scratchpad, not committed): it imports `research/verified_2026-09-27/strategy_mc.py` (VRF), regenerates the same 200,000 paths (seed 20260927, JPM LTCMA inputs, lognormal ASM, 2031 two-year yield 4.81% ASM, 60% equity sleeve, 80% floor) and reports:

Measure	Result
2026-view 2033 sleeve p5/p50/p95	\$159k / \$207k / \$273k (matches F-401)

Measure	Result
Variance still unresolved on 1 Jan 2031	2.8%
Weak 2031 (p10 sleeve \$153k): floor, p5-p95	\$134k; \$162k-\$177k
Median 2031 (sleeve \$188k): floor, p5-p95	\$165k; \$199k-\$217k
Strong 2031 (p90 sleeve \$232k): floor, p5-p95	\$204k; \$246k-\$269k
Floor share 50% / 80% / 100% at the median state	p5-p95 \$188k-\$234k / \$199k-\$217k / \$206.1k exactly
Range [floor, conditional p90] at the median state	P(in) 0.900, P(below) 0.000, P(above) 0.100

Main loop (for D3 to re-create): s = sleeve grown 2027-2030 with the 2028 deposit; g_2 = two-year sleeve growth for 2031-2032; $\text{total}_{2033} = f*s*1.0481^{**2} + (1-f)*s*g_2$; conditional bands use quantiles of g_2 at fixed s .

6. Sources (all accessed 2026-09-27)

Source	URL	Status
Official case, IPS guide, TN guide, SMAApply page	competition/official/2026_27/	VRF
Phase A registers (R-, F-, SH-/BS-, G ids)	research/insight_v1/phase_A/	repo
Budescu et al. 2014	https://www.nature.com/articles/nclimate2194	VP (abstract)
Budescu et al. 2009; Gigerenzer et al. 2005; van der Bles et al. 2020; Gaertig & Simmons 2018; Dries et al. 2024; Dieckmann et al. 2015	Europe PMC REST API, https://www.ebi.ac.uk/europepmc/webservices/rest/search	VP (abstracts)
IPCC AR5 Uncertainty Guidance Note	https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf	VP (paragraphs 4, 10)
Teigen, Løhre & Hohle 2018	cambridge.org URL in section 4	VP (abstract and intro)
Jenkins, Harris & Lark 2019	https://ideas.repec.org/a/taf/jrskr/v22y2019i5p537-554.html	VP (abstract)
Du et al. 2011	https://ideas.repec.org/a/eee/jobhdp/v114y2011i2p179-189.html	VP (abstract)
Gneezy & Epley 2014 (press release)	https://www.sciencedaily.com/releases/2014/07/140717094556.htm	VP (release); paper SNIP
Karlan & List 2007	https://www.nber.org/papers/w12338	VP (abstract)
Kresge <i>Guide to the Challenge Grant</i> (2011)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf	VP
FOMC SEP 2026-09-16	https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm	VP
Teigen & Filkuková "Can > will"	https://www.sv.uio.no/psi/forskning/prosjekter/subjektive_sannsynligheter/ (search listing)	SNIP
Yaniv & Foster 1995	https://philpapers.org/rec/YANGOJ (search summary)	SNIP
SEC Marketing Rule summaries	https://www.sec.gov/resources-small-businesses/small-business-compliance-guides/investment-adviser-marketing (403)	SNIP
Blocked or refused	sec.gov (403), eCFR (406), PubMed (browser check), SSRN (per brief)	reported, not guessed

Web search budget ran out during this task (200/200). Further lookups used direct fetches and the Europe PMC API.

What this teaches

A forecast is only as good as the way it is heard. Research on weather, climate and earnings forecasts shows three things:

- People read words like "likely" as a coin flip.
- They treat the edges of a range as a pass/fail line.
- They reward a kept promise as much as a beaten one.

So the credible way to state Laura's 2031 contribution is:

- a bottom that is already bought;
- a range narrow enough to plan with (and by 2031 it can be, because about 97% of the uncertainty has already played out);
- probabilities written as numbers with what they are "out of";
- each side of a possible miss stated separately.

Funders count written, committed money, not forecasts. That is why the floor, and the date it is bought, matter more than how high the top of the range is.

B7a Competition Meta & Judges: questions from the evidence on past semifinalists, winners and readers

Agent B7a (Phase B, lens "Competition Meta & Judges"), insight_v1 run, written 2026-09-27 (about 12:00-12:40 UTC). Starting angle: what past semifinalists and winners did in their WRITTEN work, and what the semifinal readers of written deliverables have said they reward. Semifinals only: nothing here is for the finale pitch or Q&A. Finale articles are used only where a judge comments on the quality of analysis or writing, and they are labelled as such.

This file is AI-generated research for Team Caplet. It contains questions, evidence and checklists. It contains no text meant for submission. The team decides and writes every deliverable in its own words, and must record AI use in the Final Report's Works Cited (R-W46).

0. Summary (read this first)

1. **NEW OFFICIAL DOCUMENTS the repo and Phase A do not have.** The public SMAApply page "Investment Competition Guide" (<https://wghsinvcomp.smapply.us/res/p/guide/>, read 2026-09-27, VERIFIED-PRIMARY) links two 2026-27 PDFs:
 - "2026-2027 Investment Competition Guide", 5 pages, <https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf> (downloaded with curl, text read with pypdf, key quotes checked verbatim with `fetch_text.py --grep`; full SHA-256 in section 3.1).
 - "2026_WGY_Competition Inforgraphic-FINAL.pdf", 1 page, Box shared link <https://upenn.box.com/s/l2p0l26svbbptmcizhmswyq5f7rbyp44> (downloaded via the Box shared-file download).

They add scoring signals that are in no other official file. Examples (all VERIFIED-PRIMARY, exact words):

```
- "Judges want to understand not only what your team decided to do, but also the reasoning,
assumptions, and
tradeoffs behind those decisions." (Guide p.5)
- "The competition recognizes thoughtful strategy, research and analysis, client understanding,
disciplined
decisions, management of risk and uncertainty, communication, and creativity." (Guide p.5)
- A WInS trade note "should capture the reasoning behind the decision, including its alignment with
your strategy,
the supporting research or analysis, and its expected role in growth, liquidity, risk management,
or future
funding." (Guide p.3)
- "Do not treat Trading Notes as an afterthought." (Guide p.3)
- "A long-term strategy may include planned adjustments as funding dates approach, but it should not
be rewritten
simply because markets move or hindsight reveals a different outcome. You may identify decisions
you would make
differently, but you may not redesign your strategy after observing the results." (Guide p.5)
- "Strong submissions connect the client's goals, research, portfolio decisions, and final
recommendations through
one clear and cohesive investment strategy." and "CORE PRINCIPLE Strategy guides every decision."
(Infographic)
```

These should join the official set (main loop's decision; I wrote nothing outside my file). Several questions below hang on them (anchor "R-new").

2. **Who reads the written work.** The 2025-26 Top 50 were picked "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia" (Wharton news, 27 Jan 2026, VERIFIED-PRIMARY). abrdn (now Aberdeen) has supplied semifinal-round evaluators "from across the investment spectrum" since 2012 (abrdn press release, 30 Jan 2023, VERIFIED-PRIMARY). In 2018, 14 Aberdeen professionals from "equity, fixed-income and alternative investing, as well as marketing and distribution" read 160 reports (about 11 each) (Wharton news, 4 Apr 2018, VERIFIED-PRIMARY). With about 2,300 reports now, a student first

screen is likely (INTERPRETATION). So there are two reader types with different eyes: Wharton students applying the rubric, then asset managers, some of them fixed-income specialists.

3. **The deliverable format looks new this season.** Wharton said in 2024 and 2025 it was "doing some reimagining of our signature competition" (VERIFIED-PRIMARY). The 2025-26 season had a "mid-term report" (Wharton news, 27 Jan 2026: "Teams that competed at least through the mid-term report submissions will receive a participation badge", VERIFIED-PRIMARY). A third-party tracker lists 2025-26 stages as "required trade; roster; midterm report; final report" (CompeteMap, SNIPPET-UNVERIFIED). INTERPRETATION: the 3-note Trading Notes Analysis + 500-word IPS + freeze structure is probably new in 2026-27, so past winners' written reports are weak templates, and the 2026-27 rubric and Guide are the best evidence of what is rewarded.
4. **What readers have said, verified verbatim.** Semifinal judge Melissa Ko Hahn, who read "written reports and the videos" in 2025: "one of the tendencies of people who are just starting out is to fall into this trap of hiding behind fancy terms and formulas... Oftentimes, the best solutions are made up of simple, elegant ideas." The 2023 champions (DMV's Finest), about drafting the final report: "we really needed to cut things down... less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once." The competition's academic director, Prof. Michael Roberts (2022 interview): "the jargon and acronyms are just obfuscating the elegant simplicity of finance". Older Wharton guidance: a strategy "buried somewhere in your experience, rather than clearly stated, defined and articulated up front" will "not be among the strongest teams".
5. **The only full past final report I could read is a useful anti-model.** A 2023-24 report published on Teen Ink (team not in that year's Top 50) is a sector-by-sector stock tour. It maps the client's hobbies to tickers and cites short-term gains as proof ("The team's decision was proven with great success", "our portfolio grew 4.2%, affirming our updated strategy"). This year's rubric scores against both habits (R-T11, R-W75).
6. **The odds and the field.** Top 50 share of final reports fell from 3.9% (2022-23) to 2.1% (2025-26: 49 listed of about 2,339). One slot per year goes automatically to a team from the previous winner's school (Stuyvesant this season, INTERPRETATION from the stated rule). Most semifinal schools are new each year: 7 of 49 schools in the 2026 list were also in the 2025 list; 11 of 50 in 2025 were in 2024 (DERIVED from the official lists). Only one Australian team appears in the five Top-50 lists 2022-2026 (Gryphon Group, Brisbane State High School, 2022).
7. **26 questions follow** (section 2), ranked by deadline tier within each group. The five I rate highest for "99% of teams will miss": Q2 (trade-note contents from the Guide), Q12 (grade implementation by tracking the liability, not by P&L; start collecting data now), Q15 (the hindsight rule means every "if rates move" rule must be in the IPS), Q4 (fixed-income readers will check the hedge), Q20 (whether readers see the WInS holdings).

1. Method

- Read the brief, v4 Parts 1-3, 5, 6, CLAUDE.md, Phase A (case_register R-ids, fact_register F-ids, stakeholder_map, wins_week1_guardrails), the council's winners.md and judge panel, and the Trading Notes blueprint.
 - Web: WebSearch to find URLs; every page quoted here was opened with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL` and key quotes were checked with `--grep` (all returned "PHRASE FOUND VERBATIM: YES"). Box PDFs were downloaded with curl and read with pypdf.
 - Blocked or unreadable: Scribd (all three documents return a JavaScript wall), Medium (HTTP 403), The Croton Chronicle (paywall stub). Their content is not used beyond search snippets, which are labelled.
 - Derived numbers (selection rates, repeat-school overlap) come from counting the official lists; method in section 3.
 - Privacy: no student surnames recorded; winners are named by team and school only, as Wharton publishes them.
-

2. The questions

Format per question: text; anchors; anchor quote; hypothesis (a GUESS, with evidence); what would change; deliverables; domain; seed; north star. Anchors "R-new" point to the 2026-27 Guide or Infographic

(section 0 item 1).

Group A: what the semifinal readers are told to look for (tier 1-2)

Q1. Should the 2026-27 Investment Competition Guide and Infographic be added to the official set, and which of their sentences change our checklists?

- Anchors: R-new (Guide, Infographic); R-S24 to R-S28; R-W43.
- Anchor quote: "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions." (Guide p.5)
- Hypothesis (guess): yes. The Guide is an official 2026-27 document linked from the public SMAApply page and uses the same vocabulary as the IPS and Trading Notes guides. It adds four things the other files lack: a note-content list (Q2), a hindsight rule (Q15), a seven-item "recognizes" list that names "disciplined decisions" and "management of risk and uncertainty" separately from the five criteria, and the word "tradeoffs" as something judges want shown.
- Would change: decision (the main loop adds the two PDFs, with hashes, to `competition/official/2026_27/` and the case register) and sentences (each deliverable checklist gains "state the tradeoff" and "state the assumption").
- Deliverables: WInS-now, TN, IPS, FR. Domain: D10. Seed: null.
- North star: Laura chooses the firm that shows her the tradeoffs it made for her; the Guide says judges score exactly that.

Q2. Does every WInS trade note from now on contain the four elements the Guide lists (strategy link, supporting research or analysis, and a named role in growth, liquidity, risk management or future funding), and does our Trading Notes blueprint match?

- Anchors: R-new (Guide p.3); R-T28; R-AN43; R-W82.
- Anchor quote: "the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding."
- Hypothesis (guess): the council blueprint (`research/blueprints/01_trading_notes_blueprint.md` section 3) covers purpose and the link to Laura but not "the supporting research or analysis" as an explicit element. Notes cannot be edited (A4 guardrails), and the three chosen notes are quoted verbatim, so a note that lacks the evidence element cannot be repaired later. Most teams will copy Wharton's 59-word example, which names a role and a risk but no research. A note that names one checked fact (e.g., the fund's duration from the issuer page, or the Treasury curve date) would stand out.
- Would change: sentence-level spec for every WInS note written from today (four-element checklist), and the TN blueprint section 3.
- Deliverables: WInS-now, TN. Domain: D10. Seed: 13.
- North star: a note that shows its evidence proves to Laura that our firm decides on research, not on mood.

Q3. Who reads our submission first (Wharton student evaluators or Aberdeen professionals), how long do they spend on it, and what do they see first?

- Anchors: R-W24; R-W8; F-609; R-AN60.
- Anchor quote: "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia" (Wharton news, 27 Jan 2026, VERIFIED-PRIMARY).
- Hypothesis (guess): a two-stage screen. Student evaluators apply the rubric to all ~2,300 submissions; Aberdeen professionals read a shortlist. Evidence: 14 Aberdeen readers handled 160 reports in 2018 (about 11 each, VERIFIED- PRIMARY); 2,300 reports at that load would need ~200 professionals (DERIVED, ASSUMPTION that the load per reader is similar). If so, the first reader may spend minutes, not an hour, and may read the 2-page IPS and the first page of the Final Report most closely.
- Would change: decision on Final Report structure (strategy, certainty definition and headline numbers on page 1; a student evaluator must be able to tick each criterion without hunting) and how much effort goes into the IPS pitch.
- Deliverables: IPS, FR. Domain: D7. Seed: 31.

- North star: if the first reader cannot see in two minutes why Laura would choose us, the second reader never will.

Q4. What would an Aberdeen fixed-income professional check first in our Treasury-matching story, and which of our current claims would they mark wrong or loose?

- Anchors: F-203 to F-206; brief section 8 items 11-13; R-T23, R-T24.
- Anchor quote: Aberdeen readers came "from different areas of the company, including equity, fixed-income and alternative investing, as well as marketing and distribution" (Wharton news, 4 Apr 2018, VERIFIED-PRIMARY). Recent Aberdeen judges were "U.S. Credit Investment Manager" (2026) and "senior investment manager, North American fixed income" (2024) (VERIFIED-PRIMARY, finale articles).
- Hypothesis (guess): they will look for three things: (a) the difference between a bond fund (constant duration, never matures) and a ladder held to maturity; our WInS book of IEF/TLH-type funds matches duration today but drifts, and saying "locked" about a fund would be marked loose; (b) reinvestment of coupons; (c) the precision of "certain": nominal USD, barring U.S. default, if held to maturity. Individual Treasuries are allowed in WInS (R-W69), so a professional may ask why a liability-matching team did not buy any.
- Would change: sentences in the IPS and FR ("duration-matched in WInS; cash-flow-matched in Laura's real plan"), and possibly a WInS decision to hold at least one individual Treasury (see Q24).
- Deliverables: WInS-now, IPS, FR. Domain: D1. Seed: 29.
- North star: the professional reader is a stand-in for the adviser Laura would ask to check our work; passing that check is what makes the plan trustworthy.

Q5. Is the 2026-27 deliverable structure (three-note Trading Notes Analysis, 500-word IPS, strategy freeze at the IPS) new this season, and if so how much weight should past winners' reports get as templates?

- Anchors: R-AN40; R-AN36; X-8; R-new.
- Anchor quote: "The Wharton Global Youth team is doing some reimagining of our signature competition" (Wharton news on the 2025 winners, VERIFIED-PRIMARY; the same sentence appears in the 2024 winners article).
- Hypothesis (guess): yes, new. 2025-26 had a mid-term report and a final report (VERIFIED-PRIMARY mention of "mid-term report submissions"; stage list SNIPPET-UNVERIFIED). A 2023-24 report shows the old structure: two trading notes with reflections, a "WInS Portfolio and Recommended Portfolio" section, and an advisor's comment (Teen Ink, VERIFIED-PRIMARY as a page, content is one team's report). The Rules page wording "based on the strength of their Investment Policy Statement (IPS) and Final Reports" (R-W24) may therefore be new text, not stale text.
- Would change: decision to weight the 2026-27 Guide and rubric above any past-winner pattern; and it removes a feared disadvantage: repeat schools have no practice with a 500-word frozen IPS either.
- Deliverables: TN, IPS, FR. Domain: D7. Seed: 29.
- North star: this year every team starts equal on the new formats; the team that reads the new rules most closely gains the most.

Q6. Should the IPS state Laura's risk tolerance explicitly, split into ability (low for the promise) and willingness (higher for the surplus), even though the case never states one?

- Anchors: R-I2; R-W22; X-11; R-AN17; R-new (Guide p.2).
- Anchor quote: "Your team will need to consider the client's goals, time horizons, required cash flows, funding commitments, liquidity needs, desired degree of funding certainty, risk tolerance, and communication with potential co-sponsors." (Guide p.2). The Rules page defines the IPS as "Define your client's investment objectives, risk tolerance, and overall investment strategy." (R-W22)
- Hypothesis (guess): yes. Three official texts use "risk tolerance" (IPS guide definition, Rules page, Guide), while the case gives only "thoughtful risks" and "an appropriate balance". Readers trained on standard IPS templates will look for a stated risk tolerance; a split statement shows client understanding and explains why the plan is cautious with \$292k and bolder with the surplus.
- Would change: one IPS sentence (content spec, not wording) and the FR client section.
- Deliverables: IPS, FR. Domain: D6. Seed: null.

- North star: Laura has said she takes thoughtful risks; naming where she can and cannot afford risk shows we heard both halves.

Group B: what the typical team will submit, and where we can differ (tier 1-2)

Q7. What will the median and the top-decile 2026-27 report look like, now that the case has an explicit liability and Wharton's own example note buys a Treasury ETF?

- Anchors: R-AN44; R-T23; SH-18; BS-10.
- Anchor quote: "We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment." (TN guide p.2, VERIFIED-REPO-FILE)
- Hypothesis (guess): the median team will write "safe bucket + growth bucket" with a Monte Carlo percentile and buy a Treasury ETF because the example does; many will still add a stock-picking tour (the pre-2026 habit in the Teen Ink report). The top decile will have liability matching too (AI tools suggest it readily). So "lock early" alone will not separate us; the separator is Laura-specific reasoning, honest uncertainty and clarity (supports SH-18).
- Would change: where the team invests its writing effort (Laura-fit and co-sponsor credibility over architecture).
- Deliverables: IPS, FR. Domain: D7. Seed: 29.
- North star: if 6,000 firms offer Laura the same bond ladder, she chooses the one that explains it in her terms.

Q8. When the right answer (match the payments with Treasuries) is textbook, what counts as "clear and creative" to the readers?

- Anchors: R-S24; R-S28; R-W43.
- Anchor quote: "Presents a clear and creative investment thesis" (R-S24) and "demonstrates original thinking and meaningful reflection" (R-S28).
- Hypothesis (guess): creativity is scored on the thesis and its fit to Laura, not on exotic instruments. Candidate sources of genuine originality: the 2031 co-sponsor rule (a bought floor plus a stated stretch), a certainty definition that needs no model, and reporting the plan as a funded ratio. Older Wharton guidance: "Your strategy should be unique to your team" and "Be creative as you develop your unique team strategy!" (developing-strategy page, older season, VERIFIED-PRIMARY). The 2018 readers' "memorable moments" were metaphors and honest reflections (Wharton news, 4 Apr 2018).
- Would change: the pitch's central idea (which one creative element leads), and which idea gets a Final Report chart.
- Deliverables: IPS, FR. Domain: D9. Seed: 29.
- North star: an idea Laura has not seen from other firms, which is also correct, is the reason to choose us.

Q9. Do readers reward mapping the client's interests to holdings (a Taiwan ETF, publishing or media stocks), or does that read as tokenism, and should the team drop the EWT/Taiwan tilt for that reason?

- Anchors: R-S25; brief section 4 (identity rule); brief section 9 (EWT open item); R-AN34.
- Anchor quote: "Tailors the strategy to Laura's circumstances, priorities, risk considerations, and residency goals" (R-S25).
- Hypothesis (guess): the rubric names circumstances and goals, not interests. The Teen Ink report bought Nike and Marriott because the client liked sports and travel; it did not reach the Top 50 (list checked). Past clients praised being understood ("not just about the numbers, but about understanding and connecting with people", Ladi Ayoola, 2025, VERIFIED-PRIMARY; finale context). INTERPRETATION: tailoring that changes the risk plan (lumpy book income, the co-sponsor audience) is rewarded; tailoring that only changes the ticker list is not, and for Laura it risks tokenism.
- Would change: decision on the EWT/Taiwan tilt (keep only if it earns its place on risk grounds, e.g., facility costs in TWD) and one FR sentence on why no interest-based picks.
- Deliverables: WInS-now, IPS, FR. Domain: D6. Seed: null.
- North star: Laura would recognise a plan built around her promise; she might resent one built around her ethnicity.

Q10. Do the readers reward quantitative sophistication (algorithms, optimisation, machine learning) or restraint, and how many quantitative tools should the Final Report show?

- Anchors: R-S26; R-I23; R-I46.
- Anchor quote: "Demonstrates understanding and effective use of investment concepts and tools; integrates quantitative and qualitative analysis" (R-S26).
- Hypothesis (guess): mixed signals, resolved by depth. For: finale articles praise "customized algorithms, AI innovations, portfolio optimizations" (2025) and "next level" algorithms (2024). Against: Hahn's "fancy terms and formulas" (a semifinal judge who read the written reports); DMV's Finest's "less is more"; Roberts' "elegant simplicity"; the IPS guide's "Keep the presentation simple". The 2024 finale judge Vikas Keswani said the differentiator was sometimes "a little bit of it, but a lot of depth and accuracy" (VERIFIED-PRIMARY, finale context). Guess: one deeply checked analysis (the ladder priced on the official curve, with a rate-sensitivity table) plus one honest simulation for the surplus beats five methods.
- Would change: decision on the FR analysis list (keep, appendix, drop), e.g., whether bootstrapped forwards, Vasicek-type models or four-model comparisons appear at all.
- Deliverables: FR. Domain: D3. Seed: 29.
- North star: Laura studied statistics; she will trust one analysis she can check over five she cannot.

Q11. What is our "cut list": which of our concepts stay in the Final Report's main text, which go to an appendix, and which are dropped, and what test decides?

- Anchors: R-I46; R-S28; brief section 8 (concepts already settled); Part 2 lesson 1.
- Anchor quote: "we laid everything out in a rough draft of our final report and everything clicked — we really needed to cut things down. It also led us to reflect: less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once." (DMV's Finest, 2023 champions; Wharton news, VERIFIED- PRIMARY)
- Hypothesis (guess): the test is "does this change a recommendation for Laura or a number a co-sponsor will hear?" Terms like DV01, bootstrapping, forward zeros and wrong-way risk fail the test in the main text; the idea behind each can survive in plain words. The council list of 12 concepts is probably too many for the main text.
- Would change: decision on FR structure; the IPS vocabulary list (words allowed and banned).
- Deliverables: IPS, FR. Domain: D9. Seed: 14.
- North star: last year's report lost the insight in the packaging; a plan Laura can repeat to a co-sponsor in one breath is the plan she will choose.

Group C: how to show a 16-year strategy through six weeks of trades (tier 1)

Q12. How should the Final Report evaluate implementation, if not by P&L: can we grade the WInS book by how closely its value tracked the value of Laura's ten payments, and must we start recording the data now?

- Anchors: R-I37; R-W76; R-W60; R-new (Guide p.4: "Evaluate implementation of the IPS strategy").
- Anchor quote: "In the Final Report, your team will evaluate how effectively it implemented that strategy" (R-I37).
- Hypothesis (guess): yes. The natural implementation test for a matching strategy is tracking error against the liability: each day, WInS bond-sleeve value versus the present value of the ten payments on that day's official curve. If the ratio stayed near constant while rates moved, the hedge worked, whatever the P&L. Few teams will think of this. It needs daily data from Sept 28 (WInS values, Treasury par curves), which cannot be rebuilt exactly later if the team does not save WInS values as it goes.
- Would change: WInS-now process decision (save a daily or weekly WInS value snapshot and the curve date); a FR chart specification (funded ratio of the WInS book over the six weeks; data: WInS values, treasury.gov daily par curves, the verified PV script).
- Deliverables: WInS-now, FR. Domain: D3. Seed: 30.
- North star: showing that the promise stayed funded while markets moved is the proof Laura needs, far better than a return number.

Q13. Which three trades best show strategic thinking, given that hundreds of teams will copy Wharton's Treasury-ETF example, and should at least one be a decision that "tested or refined" the strategy?

- Anchors: R-T8; R-T9; R-T20; R-AN40; R-AN44; R-new (Infographic).
- Anchor quote: "Show how three investment decisions supported, tested, or refined the developing strategy." (Infographic) and "Your research, analysis, Trading Notes, and experience implementing your strategy should all contribute to the IPS." (Guide p.2)
- Hypothesis (guess): a set of three that covers three roles (the promise, the surplus, a rule in action) and includes one genuine "refined" decision (for example a change forced by the position limit or by a rate move) reads as a living process. Three "supported" notes read as a shopping list. The refinement must be real: invented stories are banned (R-W28).
- Would change: decision on the trade plan to Oct 23; the selection rule for the three notes.
- Deliverables: WInS-now, TN. Domain: D7. Seed: 13.
- North star: Laura hires a process, not a portfolio; three notes that show the process working are the evidence.

Q14. Can one WInS trade before Oct 23 be a pre-set rule firing (a rebalancing band or a rate trigger), so the notes show time and discipline, not just a starting allocation?

- Anchors: R-I3; R-I12; R-new (Guide p.5 "planned adjustments"); R-W73.
- Anchor quote: "It helps investors stay focused on long-term objectives and make disciplined decisions during changing market conditions." (R-I3)
- Hypothesis (guess): yes, if a band is set now and the market happens to cross it; it must not be forced. A trade that says "our rule said X, so we did X" is the best six-week evidence of a 16-year plan. If no band is crossed, a written "no trade, because the rule did not trigger" record is still Articulation evidence for the FR.
- Would change: decision to set and record bands before the next trade; the note spec for a rule-triggered trade.
- Deliverables: WInS-now, TN, IPS. Domain: D3. Seed: 30.
- North star: pre-commitment is how an adviser protects a client from panic; showing it in real time is proof.

Q15. Because the Guide forbids redesigning the strategy after seeing results, must every contingency rule we might want in the Final Report (rates fall before January 2027; 2028 deposit smaller or late; 2031 floor rule; 2033 reserve rule) already be named in the IPS?

- Anchors: R-new (Guide p.5); R-I35; R-I32; brief section 9 (rule if rates fall).
- Anchor quote: "you may not redesign your strategy after observing the results." (Guide p.5) and "These elements will be developed and supported in the Final Report using the strategy established in the IPS." (R-I32)
- Hypothesis (guess): yes. The ladder cost more than \$300k on 173 of 185 trading days of 2026 (F-111), and the chance it costs more on 2027-01-01 is about 24% (F-112, ASSUMPTION-based). If rates fall between Nov 6 and Dec 4 and the FR then introduces a "buy long rungs first" rule that the IPS never mentioned, a reader may see a post-hoc redesign. Named in the IPS as a planned adjustment, the same rule is evidence of foresight.
- Would change: decision on the IPS rule list (which rules appear, in plain words, within 500 words).
- Deliverables: IPS, FR. Domain: D10. Seed: 31.
- North star: Laura needs to know what we will do before it happens; the rules say so too.

Q16. Which single statement reconciles the WInS book with Laura's 16-year plan, so a reader does not see two different strategies?

- Anchors: R-W59; F-605; R-AN14; X-7; brief section 9 (WInS mirror open item); R-I36.
- Anchor quote: "Your portfolio and Final Report should reflect the strategy established in your IPS." (R-I36)
- Hypothesis (guess): the new evidence is that WInS cash (\$300k) equals the 2027 deposit exactly (F-605), so a reader may expect WInS to be the 2027 portfolio. If WInS instead mirrors the post-2028 target, one sentence in the IPS or FR must say which date WInS represents and why. The allocation work

(wins_now v0) shows the post-2028 mirror and the plan's total equity can be the same portfolio once "65/35" is read as hedge/sleeve (F-409).

- Would change: decision on which snapshot WInS shows; one reconciling sentence (spec) in the IPS and FR.
- Deliverables: WInS-now, IPS, FR. Domain: D7. Seed: 30.
- North star: one strategy told the same way three times is what the rubric calls cohesive, and what makes Laura confident.

Q17. Should at least one WInS holding be an individual U.S. Treasury bond maturing near a payment date, to show cash-flow matching rather than only duration matching?

- Anchors: R-W69; R-W91; R-AN52; F-206; brief section 8 items 11-13.
- Anchor quote: "Treasury Bonds: Bonds available on WInS. The list includes Treasury bonds from the United States" (R-W69).
- Hypothesis (guess): yes, if WInS lists a suitable maturity: it is rare (most teams will use ETFs), it answers the fixed-income reader's first question (Q4), and it turns "we would buy a ladder in 2027" into a demonstrated action. Costs: \$10 commission, daily-updated bond prices, the position limit. It could also make a strong trading note.
- Would change: WInS trade plan decision (PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK).
- Deliverables: WInS-now, TN. Domain: D1. Seed: 30.
- North star: showing, not telling, that the promise can be bought today.

Group D: process evidence and honesty (tier 1-3)

Q18. What real process evidence should the team start capturing this week (dated decision log with disagreements, roles, a mistake we "ran with", why we did not trade), so the Articulation criterion has true material in December?

- Anchors: R-S27; R-W28; R-W94; R-W87; SH-20.
- Anchor quote: "Demonstrates teamwork, communication, and learning; clearly explains the team's research and decision-making process; and reflects meaningfully on the team's growth and response to challenges." (R-S27)
- Hypothesis (guess): readers value the journey; the 2018 Aberdeen lead reader "enjoyed reading about their educational journey over the course of the competition" (VERIFIED-PRIMARY). Invented stories are banned, so only logged, dated events can be used. Teams that start logging in November will have thin evidence.
- Would change: WInS-now process decision (who keeps the log, what fields, weekly).
- Deliverables: WInS-now, FR. Domain: D7. Seed: 31.
- North star: Laura is hiring people; the log is the evidence of how those people decide.

Q19. How detailed and honest should the Final Report's record of AI use be, and could a specific, verified disclosure become a trust signal rather than a weakness?

- Anchors: R-W46; R-W47; R-W49; R-W26; X-17.
- Anchor quote: "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages." (R-W46)
- Hypothesis (guess): many teams will under-disclose. A specific record (tool, date range, purpose: brainstorming and fact-finding; what the team checked itself; that all text is the students') fits "treat AI as a weak source" (R-W47) and supports "authentic team voice". No public evidence on how readers score disclosures (UNKNOWN).
- Would change: WInS-now decision to keep an AI-use log from today; FR Works Cited spec.
- Deliverables: WInS-now, FR. Domain: D10. Seed: 31.
- North star: Laura's first book answered misinformation (R-AN33); a firm that is honest about its own tools earns her trust.

Q20. Can the semifinal readers see our WInS holdings and transaction history, and if so, must every open position at the freeze have a stated purpose?

- Anchors: R-T34; R-W80; R-AN42; R-S6.

- Anchor quote: "Your portfolio should reflect a cohesive strategy designed around the client's goals. Each investment decision should have a clear purpose within that strategy." (R-W80) and "Yes, we will verify this." (R-T34)
- Hypothesis (guess): Wharton can see every team account (it verifies notes), so readers can probably check the frozen book. Any leftover test position, odd lot or unexplained cash balance on Nov 6 becomes an inconsistency.
- Would change: WInS-now decision (a pre-freeze clean-up check in the week of Nov 2; every position mapped to a role).
- Deliverables: WInS-now, IPS, FR. Domain: D10. Seed: 31.
- North star: a tidy book is the visible proof that the strategy runs the portfolio.

Group E: pre-mortem (why a correct strategy could still miss the Top 50) (tier 2-3)

Q21. What are the realistic odds, and what does the pattern of past semifinal schools say about which controllable features matter most?

- Anchors: F-609; R-AN60; R-W8; R-W24.
- Anchor quote: "On December 12, some 2,300 teams from 79 countries submitted final reports" (R-W99).
- Hypothesis (guess): about 49 open slots among ~2,300 finishers (~2.1%, DERIVED), with one automatic slot for a team from Stuyvesant (INTERPRETATION of the stated rule). Most semifinal schools are new each year (7 of 49 repeated from 2025; 11 of 50 from 2024; DERIVED), so pedigree is not decisive; controllable features (rubric coverage, clarity, consistency, format compliance) are. A correct strategy can still miss by failing a format rule (R-I39), burying the strategy, or being inconsistent across the three deliverables.
- Would change: decision on where the last two weeks before each deadline go (compliance and consistency review, not new analysis).
- Deliverables: TN, IPS, FR. Domain: D7. Seed: 31.
- North star: Laura never sees a plan that was screened out on format; compliance is the entry ticket.

Q22. Is there any sign that non-U.S. teams are disadvantaged in the written round, and does anything in our writing (spelling, date format, currency labels, Australian references) create avoidable friction for U.S. readers?

- Anchors: R-AN10; Part 2 lesson 4; R-W72.
- Anchor quote: "You are not expected to stay awake during the night to trade." (R-W72; written with non-U.S. teams in mind)
- Hypothesis (guess): no structural disadvantage: the 2025-26 list includes teams from Turkey, India, China, the Czech Republic, Italy, Thailand, Singapore, Germany and Brazil. But only one Australian team appears in five years of lists (2022). Low Australian participation is the likelier cause (UNKNOWN). Small frictions are cheap to remove: "US\$" on every dollar figure (the case never names a currency), unambiguous dates (4 December 2026), and one consistent spelling standard.
- Would change: a style-sheet decision for all three deliverables.
- Deliverables: TN, IPS, FR. Domain: D9. Seed: 31.
- North star: nothing in the writing should make a reader work harder to see why Laura would choose us.

Q23. Does the Final Report open with the strategy "clearly stated, defined and articulated up front", so a reader who reads only page 1 can state our strategy, our certainty definition and our facility range?

- Anchors: R-S28; R-I30; R-W41.
- Anchor quote: "if you provide us with a team strategy that is buried somewhere in your experience, rather than clearly stated, defined and articulated up front, you will not be among the strongest teams." (developing-strategy page, older season text, VERIFIED-PRIMARY)
- Hypothesis (guess): this is the most common failure among strong teams that write chronologically ("first we researched, then we traded"). A page-1 test with someone outside the team would catch it.
- Would change: decision on FR structure; a page-1 test before Dec 4.

- Deliverables: FR (also the IPS pitch). Domain: D9. Seed: 31.
- North star: Laura (and a co-sponsor) will read the first page; it must answer her question.

Q24. Which Final Report content must be collected during trading because it cannot be recreated after the Nov 6 freeze, given the instructions arrive only on Nov 9?

- Anchors: R-S19; R-C93; R-AN36; R-I57.
- Anchor quote: "Detailed Final Report instructions and requirements will be shared on November 9." (R-S19)
- Hypothesis (guess): past final reports had a fixed format (a search snippet says 7-11 pages, double-spaced, Times New Roman 12, with a portfolio breakdown graphic including "team member responsibilities"; SNIPPET-UNVERIFIED) and included trading notes, a WInS vs recommended portfolio and an advisor comment (Teen Ink 2023-24 report). Items that are lost if not captured now: WInS value snapshots (Q12), screenshots of Session Rules, the decision log (Q18), the AI-use log (Q19), sources with access dates (R-I57). Teacher advice: "Chunk the final report... have students begin working on the final report as early as you can" (Wharton Essential Educator blog, VERIFIED-PRIMARY).
- Would change: a WInS-now collection checklist and owners.
- Deliverables: WInS-now, FR. Domain: D7. Seed: 31.
- North star: the Final Report proves the process; the proof must be gathered while the process happens.

Q25. Can we find any public account of what FigCapital, HHA Investments or other 2025-26 semifinalists wrote for the Barwin case, to calibrate the most recent "strong team" before our IPS is frozen?

- Anchors: F-610; R-W52.
- Anchor quote: "Stuyvesant High School (New York City, NY) — FigCapital" (2026 champions article, VERIFIED-PRIMARY).
- Hypothesis (guess): unlikely to exist in readable form. FigCapital was a 2025 top-10 finalist ("Fig Capital", VERIFIED-PRIMARY) and won in 2026, so repeat experience helps. Scribd holds documents titled "Updated Investment Strategy For Connor Barwin" and "Wharton Competition - Final Report" (titles only; pages blocked; SNIPPET-UNVERIFIED; authorship and ranking unknown). If a verified semifinal report is found, it recalibrates Q7; if not, Q5's conclusion stands (use the 2026-27 rubric). Never copy: "plagiarize existing strategies" is banned (R-W28).
- Would change: the benchmark in Q7 (a number: how common liability matching was among 2025-26 finalists).
- Deliverables: IPS, FR. Domain: D7. Seed: 29.
- North star: knowing the strongest rival lets us make sure our difference is about Laura, not about fashion.

Q26. Should the Final Report explain how the team would keep Laura on plan in a bad market (a behavioural rule), and treat her whole financial picture (book income) as part of the risk plan?

- Anchors: R-I3; R-S25; BS-03 (human capital); SH-04.
- Anchor quote: "urging them (as investment advisors) to consider behavioral finance and their clients' emotions during a market downturn" and "When analyzing someone's investments, take a look at this person holistically" (2024 winners article; judges Balchunas and McCormick; finale context; VERIFIED-PRIMARY).
- Hypothesis (guess): yes, one rule each. McCormick worked in North American fixed income at abrdn, the firm that supplies semifinal readers, so her view probably reflects theirs (INTERPRETATION). For Laura, "holistically" means her income depends on her reputation and publishing cycles, so the 2028 deposit can be late or smaller; the portfolio should not add risk that moves with her career.
- Would change: an IPS principle (discipline in downturns) and an FR section spec (human-capital paragraph; behaviour rule).
- Deliverables: IPS, FR. Domain: D6. Seed: null.
- North star: Laura chooses an adviser who plans for the bad year before it comes and sees her as more than an account.

3. Evidence found (with status)

3.1 New official 2026-27 documents (VERIFIED-PRIMARY, read 2026-09-27)

- SMaply page "Investment Competition Guide": <https://wgshinvcomp.smaply.us/res/p/guide/> : "Start by reviewing the Investment Competition Guide with your entire team." It links:
 - Guide PDF <https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf>, 5 pages, 1,715 words extracted, SHA-256 3638e84692ed0d826660a65e89cad0ce7eea8c1907b6a8f00ce02c0d51fb7add.
 - Infographic <https://upenn.box.com/s/l2p0l26svbbptmcizhmswyq5f7rbyp44> (file "2026_WGY_Competition Infographic-FINAL.pdf", 1 page), SHA-256 2827f92a989d6d72fbc253d80d975eb06bd00c86f631e83850ae17eec544ed10.
 - Neither file is in `competition/official/2026_27/`, and neither appears in the Phase A registers (checked with grep).
 - Other Guide sentences worth registering (exact):
 - "The goal is to develop a cohesive strategy in which your investment decisions work together to support the client's objectives across a range of possible market outcomes." (p.2)
 - "WInS is a tool for implementing your investment strategy. It is not the competition scorecard" (p.3)
 - "The Trading Notes Analysis is due at the end of Week 4 (October 23), two weeks before your IPS is due." (p.3)
 - ARTICULATE step: "Clearly communicate the investment strategy your team developed, including how it addresses risk and uncertainty, as well as the framework that guides its investment decisions." (p.4)
 - "The competition deliverables are not separate assignments." (p.4)
 - Key dates: "September 15: Competition Materials Released" (p.5).
 - Infographic: "There is no single correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions."

3.2 Who reads the written deliverables

- 2025-26: "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia" (news, 27 Jan 2026). VERIFIED-PRIMARY (grep).
- 2024-25: "After careful consideration by our internal reviewers and a group of professional asset managers" (news, 2025 semifinalists). 2023-24: "our internal reviewers and a group of professional asset managers from abrdn in Philadelphia". 2022-23: "internal reviewers and a group of professional asset managers from abrdn". 2021-22: "internal reviewers and a group of professional asset managers from Aberdeen Standard Investments". All VERIFIED-PRIMARY.
- abrdn press release, 30 Jan 2023: abrdn "provides evaluators from across the investment spectrum for the competition's semifinal round, reviewing more than 1,400 investment strategy proposals"; Eli Lesser: "our investment competition mission to promote client-focused objectives, a long-term-investing mindset, thoughtful analysis, and strategic decision-making." VERIFIED-PRIMARY.
- 2018 Region 3: "assembled a group of 13 other Aberdeen professionals to read and evaluate the Region 3 reports"; 160 reports; readers from "equity, fixed-income and alternative investing, as well as marketing and distribution"; "we enjoyed reading about their educational journey over the course of the competition." VERIFIED-PRIMARY (news, 4 Apr 2018).
- Semifinal judge who read written reports (2025): Melissa Ko Hahn quotes in section 0 item 4. VERIFIED-PRIMARY.

3.3 Winners, semifinalists and the field (all from official Wharton news pages, VERIFIED-PRIMARY)

Season	Final reports	Semifinalists	Share (DERIVED)	Winner (1st)
2021-22	~1,300	50	3.8%	Sailing to Success (Arizona)
2022-23	~1,400	55	3.9%	DMV's Finest (Thomas Jefferson HS, VA)

Season	Final reports	Semifinalists	Share (DERIVED)	Winner (1st)
2023-24	>1,600	50	≤3.1%	Spark Investments (Bergen County Academies, NJ)
2024-25	>1,800	50	≤2.8%	BAM Investing (Deerfield Academy, MA)
2025-26	~2,300 (counter 2,339)	49 listed ("Top 50"; "49 semifinalist teams advanced")	2.1%	FigCapital (Stuyvesant HS, NY)

- Automatic slot: "one team from Deerfield Academy will automatically advance to next year's semifinals (as long as they submit all the deliverables)" (2025 winners article); same rule stated in 2023 and 2024. For 2026-27 the rule implies a Stuyvesant team (INTERPRETATION; not announced).
- Repeat schools (DERIVED by matching school names in consecutive official lists): 2026 vs 2025: 7 of 49 (Amity International, Deerfield, Lycée Français de Chicago, Nový PORG, Richard Montgomery, Stuyvesant, Western Canada); 2025 vs 2024: 11 of 50. Repeat teams can climb: DMV's Finest reached the 2022 final, won in 2023; Fig Capital was a 2025 finalist and won in 2026.
- Australia: Gryphon Group, Brisbane State High School (2022 list) is the only Australian team in the 2022-2026 lists (grep of all five lists). New Zealand: Kristin School in 2022 and 2023.
- Minor anomaly: the Jan 2026 "Top 50" list has 49 entries (count of list markers), matching the March article's "49 semifinalist teams advanced". No decision rests on it.

3.4 What past winners and readers said about written work and analysis quality (VERIFIED-PRIMARY)

- DMV's Finest (2023 champions) on the final report draft: quoted in Q11.
- Alex Lamon (teacher-advisor): "It is easier to develop a strategy first and buy stocks second... Remember, winning isn't about portfolio gains. It is about having a coherent strategy." (Tales from the 2023 Teams). Also, from his Essential Educator blog: "The final report is the most important deliverable students produce." and "Chunk the final report." The same blog is the likely origin of the false "sector minimum = team size" claim: "Students need to invest in as many industries as they have team members" (an older-season rule; this season has no sector minimum, R-W70).
- Developing-strategy page (older season; mentions an approved stock list and "He is introduced in the Case Study"): "Evaluators want to understand why you are doing what you are doing" and the "buried" warning (Q23).
- Prof. Michael Roberts (academic director; 2022 interview): "It rests on two intuitive principles: costs and benefits, and risk."; "to get 90% of it you really only need basic arithmetic and maybe a little bit of probability and statistics."
- Finale-context judge remarks (used only for what they say about analysis quality): Chirag Jain (Aberdeen, 2025): "What distinguished managers was clear thoughtfulness and reasoning behind all the decisions that they put together."; Vikas Keswani (2024): "a little bit of it, but a lot of depth and accuracy"; Balchunas and McCormick (2024): Q26.

3.5 A typical (non-semifinal) past report

- "Profiteering Peninsula Panthers 2023-2024 Wharton Global High School Investment Competition Final Report", Teen Ink, <https://www.teenink.com/opinion/all/article/1215809/> (page states "This is the final report submitted to Wharton Global High School Investment Competition 2023-2024."). The team is not in the 2024 Top 50 list (checked). Features: sector-by-sector stock descriptions; picks justified by the client's interests ("caters to Ms. Ash's love of sports"); results used as proof ("The team's decision was proven with great success"; "our portfolio grew 4.2%, affirming our updated strategy"); two trading notes with reflections; a WInS vs recommended portfolio section; an advisor comment; a works-cited list. VERIFIED-PRIMARY as a page; it is one team's work, so treat it as an example, not a statistic.

3.6 Snippet-only or blocked

- 2025-26 stages "required trade; roster; midterm report; final report" (CompeteMap, third party, "last checked 22 Jul 2026"). SNIPPET-UNVERIFIED as a description of official rules.
- 2025-26 "trade execution by October 10... midterm and final reports by December 12" (search summary of a Scribd "25-26 Welcome Email"). SNIPPET-UNVERIFIED.

- Past final report "7–11 page document (double-spaced, 12-point Times New Roman, 1-inch margins...)" with a "portfolio breakdown graphic (sectors, team member responsibilities, stock holdings, client-specific allocations)" (search summary, source not identified). SNIPPET-UNVERIFIED.
- "Nearly 6,000 teams competed" is on the Stuyvesant school blog (talos.stuy.edu, read 2026-09-27), a school source, not a Wharton page. It explains the origin of the North Star's "6,000".
- Blocked: Scribd documents 979966514, 832434822, 977481046 (JavaScript wall); Medium (403); Croton Chronicle (paywall).

4. Sources (all accessed 2026-09-27)

#	Source	URL	Date of piece	Status
1	SMApplly "Investment Competition Guide" page	https://wghsinvcamp.smapplly.us/res/p/guide/	2026-27	VERIFIED-PRIMARY
2	2026-27 Investment Competition Guide (PDF)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf	2026-27	VERIFIED-PRIMARY
3	2026-27 Competition Infographic (PDF)	https://upenn.box.com/s/l2p0l26svbbptmci zhmswyq5f7rbyp44	2026-27	VERIFIED-PRIMARY
4	Top 50 2026 article	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/	2026-01-27	VERIFIED-PRIMARY
5	11 teams advance 2026	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/	2026-03-20	VERIFIED-PRIMARY
6	2026 champions	https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	2026-04-30	VERIFIED-PRIMARY
7	2025 semifinalists	https://globalyouth.wharton.upenn.edu/news/announcing-the-semifinalists-in-the-2025-wharton-global-high-school-investment-competition/	2025 (Jan)	VERIFIED-PRIMARY
8	2025 top 10 (Hahn quote)	https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/	2025 (Mar)	VERIFIED-PRIMARY
9	2025 winners (BAM)	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/	2025 (Apr)	VERIFIED-PRIMARY
10	2024 semifinalists	https://globalyouth.wharton.upenn.edu/news/here-are-the-top-50-teams-advancing-to-wharton-global-youths-2024-investment-competition-semifinals/	2024	VERIFIED-PRIMARY
11	2024 winners (Spark)	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024-investment-competition-global-finale/	2024 (Apr)	VERIFIED-PRIMARY
12	2023 semifinalists	https://globalyouth.wharton.upenn.edu/news/congratulations-to-our-2023-investment-competition-semifinalists/	2023	VERIFIED-PRIMARY
13	2023 winners (DMV's Finest)	https://globalyouth.wharton.upenn.edu/news/the-2023-investment-competition-global-finale-ends-in-sweet-victory-for-dmvs-finest/	2023 (Apr)	VERIFIED-PRIMARY
14	Tales from the 2023 teams	https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition- Tales-from-the-2023-teams/	2023 (Jun)	VERIFIED-PRIMARY
15	2022 semifinalists	https://globalyouth.wharton.upenn.edu/news/introducing-the-2022-semifinalists-in-the-wharton-global-high-school-investment-competition/	2022	VERIFIED-PRIMARY
16	2022 finale	https://globalyouth.wharton.upenn.edu/news/top-teams-sail-to-success-in-the-2022-wharton-investment-competition-global-finale/	2022 (Apr)	VERIFIED-PRIMARY
17	Developing a Strategy (older-season page)	https://globalyouth.wharton.upenn.edu/developing-strategy/	undated	VERIFIED-PRIMARY
18	Essential Educator: 10 tips (Lamon)	https://globalyouth.wharton.upenn.edu/essential-educator-blog/the-essential-educator-10-tips-for-teaching-the-wharton-investment-competition/	undated (after 2019-20)	VERIFIED-PRIMARY

#	Source	URL	Date of piece	Status
19	Prof. Michael Roberts interview	https://globalyouth.wharton.upenn.edu/news/prof-michael-roberts-wants-global-youth-students-to-discover-the-elegant-simplicity-of-finance/	2022	VERIFIED-PRIMARY
20	Region 3 finalists (Aberdeen readers)	https://globalyouth.wharton.upenn.edu/news/kwhs-investment-competition-region-3-finalists/	2018-04-04	VERIFIED-PRIMARY
21	abrdn press release	https://www.aberdeinplc.com/en-gb/news/all-news/abrdn-enters-second-decade-supporting-wharton-investment-competition-for-students	2023-01-30	VERIFIED-PRIMARY
22	Teen Ink 2023-24 final report	https://www.teenink.com/opinion/all/article/1215809/Profiteering-Peninsula-Panthers-2023-2024-Wharton-Global-High-School-Investment-Competition-Final-Report	2024	VERIFIED-PRIMARY (page)
23	Stuyvesant blog	https://talos.stuy.edu/cms/pages/stuyvesant-blog/stuyvesant-high-school-wins-the-2026-wharton-global-high-school-investment-competition-26/	2026	VERIFIED-PRIMARY (school source)
24	Wharton story (2026 finale)	https://www.wharton.upenn.edu/story/wharton-global-high-school-investment-competition-hosts-future-business-leaders/	2026	VERIFIED-PRIMARY
25	Delbarton news (2025 finalist)	https://www.delbarton.org/about-us/news-detail/~board/homepage/post/delbarton-is-finalist-in-whartoninvestment-competition	2025	VERIFIED-PRIMARY (school source)
26	CompeteMap listing	https://competemap.com/competitions/cmwr8ls8y0001vnne2cnjon1n	checked 2026-07-22	SNIPPET-UNVERIFIED (third party)
27	Scribd documents 979966514, 832434822, 977481046, 942363888, 820640088	scribd.com	various	BLOCKED (titles from search only)
28	Medium "My experience with the Wharton..."	abhinav-penagalapati.medium.com	unknown	BLOCKED (403)

Repo files used: research/insight_v1/_context/brief.md; research/ultracode_prompt_v4_deep_strategy.md; CLAUDE.md; research/insight_v1/phase_A/{case_register,fact_register,stakeholder_map,wins_week1_guardrails}.md; research/council_2026-09-27/groundwork/winners.md; research/council_2026-09-27/02_judge_panel.md; research/blueprints/01_trading_notes_blueprint.md; research/insight_v1/wins_now/securities_and_allocation_v0.md.

Corrections to winners.md (council): the Hahn quote is now VERIFIED-PRIMARY and comes from the 2025 top-10 article (source 8), not the 2022 finale page; "Less is more" is verified for DMV's Finest (source 14) and, separately, for judge Andrea Vittorelli at the 2022 finale ("Cut slides, cut words, cut minutes", source 16, finale context). The Aquino quote was not found on any page I read (still UNVERIFIED).

5. Leads for the specialists

- **D10 (compliance) / main loop:** add sources 2 and 3 (hashes in section 3.1) to the official set and the case register; re-check every checklist against the Guide's note-content list (Q2) and hindsight rule (Q15). The Guide says the Evaluation Criteria are "located under the Pages tab => Deliverables" (same as R-S24 to R-S28).
- **D7 (competition history):** the official news archive has Top-50 lists for every season since 2021-22 (sources 4, 7, 10, 12, 15). The 2018 Region 3 page (source 20) quotes a dozen passages from final papers that the organisers found memorable: the only public sample of what readers liked in written reports. Try a school-news search for each 2025-26 finalist (Hooke Investments, Westminster; Perse Capital Group; Asgard Assets; Better Future Fund; Malo Management) for descriptions of their written strategy. The Wharton "Meet the Experts" videos include a past winner (2018 champion), per source 18.
- **D1 (fixed income):** expect Aberdeen fixed-income readers (sources 11, 20, 6). Prepare the fund-versus-ladder distinction and check which individual U.S. Treasuries appear in WInS (R-W69) and their maturities near 2033-2042 payment dates.
- **D3 (quant):** specify the implementation-tracking chart (Q12): daily WInS bond-sleeve value divided by the PV of the ten payments on the same day's official par curve (reuse `research/verified_2026-09-27/official_curve_pv.py`). The team must save WInS values from Sept 28 onward.

- **D9 (plain English):** use the verified quotes of Hahn, DMV's Finest and Roberts (section 3.4) as the editing standard; build the cut list (Q11) and page-1 test (Q23).
- **D6 (client psychology):** the rubric's "earn her confidence" (R-S25) plus the 2024 judges' "clients' emotions during a market downturn" and "holistically" (source 11) support a behaviour rule and a human-capital paragraph.
- **Everyone:** the Teen Ink report (source 22) is a clear picture of the habits this year's rubric penalises: returns as proof, hobbies as tickers, many shallow stock descriptions.

What this teaches

A competition is judged by people, and the best evidence of what they reward is what they have written down, not what people guess. Here, the readers (Wharton students first, then professional asset managers, some of them bond specialists) and the organisers keep repeating the same few ideas: state the strategy up front, show the reasoning, assumptions and tradeoffs, keep it simple, and let every decision follow from one idea about the client. Checking the official site closely also found two official 2026-27 documents nobody had filed yet, which is a reminder that the cheapest insight is often reading the rules one more time. Finally, the numbers teach humility: about 2 in every 100 finishing teams reach the Top 50, and most of them are new schools each year. The teams that make it are not the ones with the fanciest maths, but the ones whose written work a busy reader can understand, check and trust.

B7b Competition Meta & Judges: simulate the field, then find where we can be different and still right

Agent B7b, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: Competition Meta & Judges (semifinals only: the written Trading Notes Analysis, IPS and Final Report). Starting angle: first write down what a typical strong team will submit for the Laura Gao case, then ask where readers will see sameness, what distinctive but correct moves exist, and how each deliverable's mechanics reward or punish each approach.

This file is research for the team. It contains questions, evidence and specifications. It contains no text meant for submission. Every factual claim carries a status label (brief section 3). Anything about Laura is limited to the case and her public professional record.

0. Summary (read this first)

1. **Two official 2026-27 documents are missing from the repo and from Phase A.** The public SMAApply page "Investment Competition Guide" (<https://wghsinvcamp.smaply.us/res/p/guide/>) links two Box PDFs:
 - a 5-page "**2026-2027 Investment Competition Guide**" (created 2026-09-09);
 - a 1-page "**Competition Phases**" infographic (created 2026-09-14).

I downloaded and read both (VERIFIED-PRIMARY, 2026-09-27). They contain the clearest official statements yet of what readers want. Four examples:

- "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions."
- "Do not treat Trading Notes as an afterthought."
- The Trading Note should capture the decision's "expected role in growth, liquidity, risk management, or future funding."
- "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome. You may identify decisions you would make differently, but you may not redesign your strategy after observing the results."

The main loop should add both files to `competition/official/2026_27/` with their hashes (section 6). I did not write outside my assigned file.

2. **Who reads the written deliverables is now known for last season (VERIFIED-PRIMARY).** The 2026 Top 50 were picked "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia". The 2025 Top 50 used "internal reviewers and a group of professional asset managers from abrdn". A 2026 finale judge from Aberdeen held the title "U.S. Credit Investment Manager". So a bond professional may read our reports, and a Wharton student may read them first. Nobody has published this season's reader panel yet.
3. **The rubric changed since 2023-24, and the changes point away from decoration.** A third-party transcription of the 2023-24 criteria (SNIPPET-UNVERIFIED as to Wharton's wording) included "visually pleasing presentation of data; many creative elements" and "would win him/her over as a client". The 2026-27 criteria (VERIFIED-REPO-FILE) instead say "uses data effectively to support conclusions; demonstrates original thinking", "recommendations that can earn her confidence", "disciplined planning across Laura's changing time horizons and cash-flow needs", and "maintains consistency with the team's IPS". The sector-count rule was dropped.
4. **The deliverable format is new this season, so nobody has a template.** In 2025-26 the deliverables were a mid-term report (Oct 31) and a final report (Dec 12), plus a first trade by Oct 10 (SNIPPET-UNVERIFIED, Scribd search summary). A 2025-26 semifinalist described "the required 15-page report" (VERIFIED-PRIMARY news article; the rule itself is secondary). The Trading Notes Analysis, the 50-

word pitch and the 500-word IPS appear to be new for 2026-27. Repeat schools cannot copy last year's structure for these two deliverables.

5. **What a strong rival looked like last year (VERIFIED-PRIMARY, KXAN via Yahoo).** Vandegrift's 2025-26 semifinalist team used a four-part stock screen, "thousands of Monte Carlo simulations", stress tests against the dot-com crash, 2008 and COVID, and reported "a 70% chance of reaching the long-term goal". Expect this year's strong teams to bring the same toolkit to a case that asks for "a high degree of certainty". Many will print a Monte Carlo "95%".
6. **Field arithmetic (DERIVED from VERIFIED-PRIMARY pages).**
 - About 2,339 teams finished in 2025-26.
 - The Rules page lets the 2026 champion school (Stuyvesant) fast-track one team to this season's semifinals, so about 49 seats are open, roughly 2.1% of finishers.
 - About 9 of the 2026 Top 50 teams came from schools that were also in the 2025 Top 50, so repeat schools matter.
 - No Australian school appears in either year's Top 50 list.
7. **The field simulation (section 2) finds the sameness hot-spots.** Expect a "three buckets + glide path" structure, a Monte Carlo "95%", a p10-p90 range, clones of Wharton's own Treasury-ETF example note, puns on her book titles, a Taiwan-ETF "connection" trade, and WInS return measured against the S&P 500. Our lock-early architecture is correct but may not be rare, because AI tools at 5% yields may also suggest a Treasury ladder (BS-10). The durable edge is therefore likely to be proof: rules executed in WInS, a hedge that visibly tracked the promise, and an evaluation measured against the liability.
8. **27 questions follow (section 4),** ranked by deadline tier. The top five for WInS now:
 - Q8: log the value of the promise against the WInS hedge daily, from now on.
 - Q27: write each WInS note in Wharton's own role vocabulary.
 - Q6: plan one note each for "supported", "tested" and "refined".
 - Q7: a decision *not* to trade can never become a Trading Note.
 - Q25: check the unpublished "required trading activity" rule.

1. Method

- Read the brief, the v4 prompt Parts 1-3, 5 and 6, CLAUDE.md, all official files, and the Phase A files (case register, fact register, stakeholder map, WInS guardrails). Also read the council judge panel (02_judge_panel.md), the winners groundwork (groundwork/winners.md), the Trading Notes blueprint and the S1 securities file.
- Web: WebSearch to find URLs, then `research/insight_v1/scripts/fetch_text.py` or curl to read each page. Every quote marked VERIFIED-PRIMARY was read on the page itself, and the key ones were checked with `--grep`. Scribd, SlideShare, web.archive.org and KXAN refused or needed JavaScript. Those items are marked accordingly.
- Field simulation: my judgement (ASSUMPTION), built from:
 - last season's semifinalist evidence (KXAN, HW Chronicle);
 - finale descriptions ("sentiment, SWOT and Pestel analyses, customized algorithms, AI innovations, portfolio optimizations");
 - third-party prep guides (Aralia, Lumiere);
 - what a generic AI answer to this case would likely contain.

It has not been tested. Q1 proposes the test.

2. Simulating the field: what a typical strong team will submit (ASSUMPTION unless labelled)

2.1 The obvious strategy

- **Three buckets:** growth now (60-80% equity until about 2030), a glide path from 2030 to 2033, and a bond reserve bought in 2031-2033 or built gradually.
- **Reserve method:** present value of ten \$50k payments at a flat 4-5%, which gives about \$395-420k at 2033 (compare F-110). Often the timing is wrong, with no annuity-due and no check on when the first

payment falls.

- **Certainty:** "Monte Carlo, 10,000 simulations, 95% probability of funding all payments". Evidence: last season's semifinalist used Monte Carlo and quoted a 70% probability (VERIFIED-PRIMARY, KXAN via Yahoo).
- **2031 range:** the 10th to 90th, or 25th to 75th, percentile of simulated 2033 surplus. Confidence stated as the percentile width (80% or 50%).
- **Holdings:**
 - 15-40 individual stocks, often with SWOT or PESTEL write-ups, because third-party guides teach this: Aralia's "Investment Thesis → Evidence → Risks → Expected Outcome → Client Fit" (VERIFIED-PRIMARY Aralia page, a third-party source);
 - sector ETFs;
 - one or two bond ETFs, copying Wharton's example note;
 - a Taiwan ETF or TSMC as the "Laura connection";
 - AI and big-tech names;
 - possibly an ESG screen carried over from last year's impact-led case.
- **Tools:** efficient frontier, Sharpe ratio, beta, stress tests on 2008, COVID and 2022.

2.2 The obvious story

- "Balance her entrepreneurial risk-taking with the certainty her promise needs."
- Metaphors from her book titles ("messy roots", "deep roots", "next chapter"), from storytelling, and from Taiwan or belonging.
- In the fundraising draft: an emotional paragraph about community, plus a percentile range.

2.3 The obvious charts (Final Report only; the IPS bans charts)

Allocation pies, a Monte Carlo fan chart, a glide-path area chart, sector bars, an efficient frontier, and WInS performance against the S&P 500.

2.4 The obvious Trading Notes

- Note 1 is a near-copy of the official example: "intermediate-term Treasury ETF to reduce volatility and prepare for the operating commitment".
- Note 2 is a tech or AI stock.
- Note 3 is a Taiwan ETF.
- Many teams will have written short or empty notes on early trades, not knowing they must quote them "exactly as it appears in WInS" (R-T28).

2.5 Where readers will see sameness (hot-spots)

Hot-spot	Why it will be common	Is our current plan inside it?
"Three buckets + glide path"	Standard goals-based advice; natural AI answer	Partly (ladder + sleeve reads like two buckets)
Monte Carlo "95%"	Taught in prep guides; used last season	No: we define certainty by what is bought (brief 8.6). Risk: a checklist reader may want a number (Q5)
p10-p90 range	Default percentile output	No: floor bought + stretch (brief 8.7)
Clone of Wharton's Treasury-ETF example note	The official example (R-T23, R-AN44)	Yes, unless our note names payment years and the match (Q12)
Book-title puns and identity metaphors	Easy "Client Knowledge" signal	Risk if we add them (Q20)
Taiwan ETF as "connection"	Easy tie-in	Open (brief 9: EWT keep or drop)
WInS return vs S&P 500 in the FR	Default benchmark	Not yet decided (Q11)
Treasury ladder / lock-in at 5% yields	AI tools at today's rates may suggest it (BS-10)	Yes: our core. The edge must come from evidence and client fit, not the idea alone

2.6 Distinctive but correct moves (candidates, each tied to an official requirement)

1. Evaluate implementation against the promise, not the S&P 500. Case: "evaluates how that strategy was implemented" (R-C90); criterion: "funding reliability" (R-S26). (Q8, Q11)
2. Execute the plan's own rules inside WInS and quote them in notes. Infographic core principle: "Strategy guides every decision" (VERIFIED-PRIMARY). (Q6, Q9)
3. Pre-declare in the IPS the rules that will produce the FR numbers, so the FR is an application, not a redesign. Guide p.5. (Q10)
4. Use few holdings, each with a job, and show look-through diversification. Trading Details: "not simply to own a large number of securities" (R-W71). (Q13)
5. Keep the vocabulary identical across all three deliverables, in Wharton's own words (X-16; Guide p.3). (Q27)
6. Tell a true, well-evidenced learning story, including an honest AI-use record (R-W46, R-W28). (Q21, Q22)

3. How each deliverable's mechanics reward or punish approaches

Mechanic (official)	Rewards	Punishes	Status
3 notes quoted "exactly as it appears in WInS"; "Yes, we will verify this" (R-T28, R-T34)	Notes written well at trade time; teams that planned which trades would become notes	Teams that wrote weak notes in week 1; any edit; notes on unfilled or practice trades	VERIFIED-REPO-FILE
Notes must correspond to executed trades (R-T33)	Discipline shown by a rule-triggered trade	Discipline shown only by <i>not</i> trading (cannot be a note) (Q7)	VERIFIED-REPO-FILE; inference
Reflection ≤100 words, three required parts (R-T29 to R-T32)	One concrete need of Laura's per note; one trade-off	Generic "diversification"; justification by gain (R-T11)	VERIFIED-REPO-FILE
"supported, tested, or refined" (R-T20; Guide; infographic)	A set of notes that shows development	Three "supported" notes that look like a shopping list	VERIFIED-REPO-FILE + VERIFIED-PRIMARY
TN due end of Week 4, "two weeks before your IPS" (Guide p.3)	Teams with a strategy before trading	Teams whose strategy changes after Oct 23 without explanation	VERIFIED-PRIMARY
50-word pitch; 500-word IPS; TNR 12, double-spaced, 2 pages; no charts, citations, footnotes or links; breach = "will not be considered" (R-I16 to R-I55)	Plain words; if-then rules; one central idea	Numbers that need a chart to believe; jargon; any format slip	VERIFIED-REPO-FILE
"Focus on the strategy and decision-making framework... rather than describing individual investments or presenting detailed financial calculations" (R-I23)	A rulebook	A ticker list	VERIFIED-REPO-FILE
"may not revise its investment strategy after the submission deadline" (R-I35) + Guide p.5 "may not redesign your strategy after observing the results"	Rules pre-declared in the IPS that later produce the FR numbers	A FR whose range or reserve method was not foreshadowed in the IPS (Q10)	VERIFIED-REPO-FILE + VERIFIED-PRIMARY
WInS frozen Nov 6; FR written about a frozen portfolio (R-AN36)	A FR that evaluates the frozen book against the IPS and the liability	A FR that apologises for P&L	VERIFIED-PRIMARY
FR rules published Nov 9; due Dec 4; last year "15-page report" (KXAN)	Pre-built components	A four-week crunch during school exams	FR rules UNKNOWN; 15 pages = secondary
Readers: Wharton student evaluators + Aberdeen asset managers (2026)	Clear first page; bond terms used precisely	Loose bond language ("risk-free", "matched" for a duration mix) (Q4)	VERIFIED-PRIMARY (last season)

4. The questions (27), with reasoning

Ordering: tier 1 (WInS now + TN) first, then IPS, then FR. "Anchor" uses Phase A ids. "R-new" means the anchor is a newly found official text quoted in full.

Tier 1: WInS now and Trading Notes (Oct 23)

Q1. Sameness test: if a team pastes this case into a mainstream AI tool, how much of our architecture comes back, and which of our elements come back rarely or never?

- Anchor: R-S24 ("clear and creative investment thesis"); R-W28 (no plagiarising "existing strategies"); BS-10.
- Reasoning: the council's claim that "most strong teams will write safe bucket + growth bucket" was never tested (SH-18, BS-10). At a 5.17% 10-year yield (F-001), an AI answer may well say "lock in a Treasury ladder". If so, lock-early is not our differentiator. The differentiators then have to be evidence, rules and fit to Laura.
- Method: 5-10 fresh prompts with the case text, no steering. Tabulate each answer's certainty definition, range method, reserve timing and WInS mix. Record this AI use for Works Cited (R-W46).
- Hypothesis (guess): about half will propose buying bonds only in 2030-2033. A minority will propose locking early. Almost none will propose a bought 2031 floor or liability-relative evaluation.
- Would change: which one or two ideas lead the pitch and the IPS's first paragraph.

Q6. Should the three Trading Notes be chosen as one "supported", one "tested" and one "refined" decision, and does the team need to plan now for a "tested" or "refined" trade to happen before about Oct 20?

- Anchor: R-T20; R-C88; R-AN40; Guide p.3 (R-new): "reflect on how those decisions supported, tested, or refined your strategy".
- Reasoning: all three official texts repeat the same three verbs. A typical team will submit three "we bought X because..." notes, which show support only. A "tested" note needs something to happen, such as a band breach or a rate move that triggers a rule. So the trigger has to be written before the event.
- Hypothesis (guess): readers reward visible development. One note per verb is the cleanest way to show it.
- Would change: the TN selection plan (`research/blueprints/01_trading_notes_blueprint.md` section 2) and the week 2-3 trade calendar.

Q7. A decision not to trade cannot be a Trading Note. How do we evidence discipline-by-inaction inside the three notes, for example not selling Treasuries as yields rose?

- Anchor: R-T33 ("must correspond to trades executed in WInS"); R-W73; R-W88 (low turnover).
- Reasoning: the most professional behaviour for a liability hedge is to hold still. Holding still leaves no trade, so it leaves no quotable note. Options:
 - a small rule-driven rebalance whose note states the rule and what was deliberately not done;
 - keep a dated decision log and use it in the FR's Articulation section.
 - Hypothesis (guess): write the rebalancing rule into the first note now. Later rebalance notes can then quote it. Inaction goes in the FR log.
- Would change: the wording of the next WInS notes; the start of a decision log today.

Q8. Should the team start today a daily log of (a) the value of Laura's ten payments on that day's Treasury curve and (b) the WInS hedge's value, so the FR can show that the hedge tracked the promise over the six weeks?

- Anchor: R-C90; R-S26 ("funding reliability... under varying market outcomes"); F-001; F-105 (spot value \$288,924, duration 10.16y); R-W76 ("your portfolio provides evidence").
- Reasoning: rates moved 38bp in September alone (F-003). A bond-heavy WInS book may show a loss that a student reader misreads. A chart of hedge value against the liability value shows whether the gap stayed small. That converts profit and loss into evidence that the strategy worked. It is also a way to show a 16-year strategy through six weeks: the six weeks are literally the "rates move before January 2027" risk (brief section 9).
- Data risk: the Treasury curve can be rebuilt later. WInS position-level daily values may not be recoverable, so the log must start now.
- Hypothesis (guess): tracking gap within about \$1-4k on a roughly \$289k liability (S1 twist numbers). Readers who are bond professionals would find this persuasive.
- Would change: a WInS-now logging task; one FR chart; a number in the FR ("the promise's cost moved \$X; our hedge moved \$Y").

Q9. Which of the plan's future decision points can be rehearsed by a real WInS trade in six weeks, and which only in words?

- The decision points: January 2027 lock; the rule if rates fall; the 2028 top-up; the rebalancing band; the 2031 floor in a 2-year Treasury; the 2033 reserve.
- Anchor: seed 30; R-C75 ("implementation of its investment strategy"); R-I12; R-W69 (individual Treasuries allowed); infographic "Strategy guides every decision" (R-new).
- Reasoning: the \$150k deposit never enters WInS (R-W59), so the 2028 top-up cannot be rehearsed. A 2-year Treasury bought for the floor can be, and so can a band rebalance. Each rehearsal must earn its complexity; the design principle says one clear rehearsal beats five token ones.
- Hypothesis (guess): two rehearsals are worth doing (the lock and the floor purchase). The rest go in words in the IPS and FR.
- Would change: the WInS trade list; which trades become notes.

Q12. The official example note is a Treasury ETF bought for the operating commitment, and hundreds of teams will clone it. What must our Treasury note contain so that a reader who has seen many clones recognises ours as different?

- Anchor: R-T23; R-AN44; R-T28 (quoted verbatim, so the note text is permanent).
- Reasoning: note text written now is frozen. It is also the reader's first impression, because the TN is the first deliverable.
- Hypothesis (guess). Elements (not wording):
 - which payment years it protects;
 - how its sensitivity to rates compares with the promise's;
 - the rule it follows;
 - one honest limit (for example, "moves like the payments, but is not a date-by-date match").
- Would change: the text of the next Treasury trade's WInS note.

Q27. Should every WInS note name its role using Wharton's own words ("growth, liquidity, risk management, or future funding") and the same labels the IPS and FR will use, so the reader's cross-check is effortless?

- Anchor: X-16 (shared vocabulary); R-C92 ("clear and consistent investment strategy across all three"); Guide p.3 (R-new): the note should capture its "expected role in growth, liquidity, risk management, or future funding".
- Reasoning: readers compare the deliverables. Identical labels (for example one fixed name each for the promise holdings, the growth sleeve and the floor) make consistency visible at a glance. Labels that drift ("hedge", "safe bucket", "reserve", "ladder") look like three strategies.
- Hypothesis (guess): yes. Choose three labels now and never vary them.
- Would change: every note from today; a glossary line in the FR.

Q25. What is the "required trading activity" in the Rules, and could a low-turnover, few-holding portfolio fail an unpublished eligibility gate?

- Anchor: R-W23 / F-608 ("Teams must meet the required trading activity and portfolio management guidelines"); R-W33; X-18.
- Reasoning: pre-mortem item. A correct strategy is worthless if the team becomes ineligible. Earlier seasons had a first-trade deadline ([PRIOR], A4 b3). The logged-in Trading Details page may set a minimum.
- Hypothesis (guess): at most a first-trade-by date. Still unverified.
- Would change: a WInS-now check today; possibly the timing of the first trade.

Q28. Which dated market events fall between Sept 28 and Oct 20 and could fire a pre-written rule, so that a "tested" note can exist before the Oct 23 deadline?

- Examples: U.S. CPI release, Treasury auctions, a band breach. The next FOMC meeting is probably Oct 27-28, after the TN deadline but before the IPS (UNVERIFIED; D1 to check on federalreserve.gov).
- Anchor: R-T8; R-T13; Guide p.3.
- Reasoning: the calendar decides whether a test can happen in time. If nothing fires, the team needs a fallback "refined" note, for example switching IEF+TLT to IEF+TLH after checking durations (S1).

- Hypothesis (guess): a rate or equity band trigger is more likely to fire in three weeks than any macro event.
- Would change: the trigger thresholds written into the first notes.

Tier 2: IPS (Nov 6)

Q10. Which decision rules must the IPS pre-declare, so that the FR's reserve size, 2031 range and facility share read as "applications" of the IPS, not as a "redesign after observing the results"?

- Anchor: R-I32; R-I35; X-5; Guide p.5 (R-new): "it should not be rewritten simply because markets move or hindsight reveals a different outcome. You may identify decisions you would make differently, but you may not redesign your strategy after observing the results."
- Reasoning: the FR numbers will be computed after Nov 6. If the IPS never mentions a bought 2031 floor or a liability-first reserve rule, the FR's range looks new. The IPS has 500 words, so every pre-declared rule costs words.
- Hypothesis (guess): four rules are enough:
 - the promise is priced and bought first;
 - risk is taken only in the surplus, within a stated band;
 - the 2031 statement is a bought floor plus a stated stretch;
 - the facility contribution is a share of what is left after the reserve, with a flexibility margin.

Each needs roughly 30-50 words.

- Would change: the IPS word budget and outline.

Q18. With no charts or citations, what must the IPS words alone prove, and which (if any) numbers survive?

- Anchor: seed 14; R-I55; R-I23; R-AN38; R-I6 ("desired degree of funding certainty").
- Reasoning: the reader cannot check a source in the IPS. So any number must be believable on its face or carry its own method in a clause. If-then rules can be checked later against the WInS record and the FR. Prior guidance says "at most one rounded number" (SH-16). The new point is that the IPS is the anchor the FR is audited against (Q10).
- Hypothesis (guess): the IPS proves three things in words:
 - the priority order (promise before facility);
 - the certainty definition in plain English;
 - the rules for change as dates approach.

One number, the promise's cost against the first deposit, is worth its words.

- Would change: the IPS outline and number policy.

Q19. In 50 words, most pitches will say "balance growth and security". What is the one claim only our strategy can make truthfully, and does it need a clause specific to Laura to avoid being generic?

- Anchor: R-I16, R-I17 ("central idea... how it is designed to support your client's financial goals"); R-S24 ("creative investment thesis"); North Star.
- Reasoning: the pitch is the first thing a screening reader sees on page 2. Sameness is most visible there.
- Hypothesis (guess): the distinctive, true claim is about when certainty is achieved (bought at the start, not hoped for at the end). The Laura clause should tie to her stated wish for "an appropriate balance". Book-title imagery should not stand in for it (Q20).
- Would change: the pitch content specification (not its wording).

Q16. Will a reader see about 20% total equity after 2028 (F-409) as ignoring Laura's "thoughtful risks", and does the IPS need one explicit sentence saying where her risk appetite lives?

- Anchor: R-C37, R-C38, R-AN16; R-S25 ("risk considerations"); F-409; brief section 9 (equity weight open).
- Reasoning: this is a "correct strategy misses the Top 50" risk. A student evaluator skimming a WInS book that is about two-thirds Treasuries may score "creative thesis" and "diversification" low. The case's

"Although..." wording supports protection, but only if the IPS says so explicitly.

- Hypothesis (guess): one sentence is enough. It should say that all risk-taking sits in the money the promise does not need, and that the extra equity there "buys range, not median" (brief 8.5).
- Would change: one IPS sentence; possibly the sleeve equity weight if Phase D finds readers penalise low equity.

Q17. Which word counter will Wharton use for the 50/500/100-word limits, and what internal caps protect us from a counting dispute?

- Anchor: R-I16, R-I18, R-I39 ("will not be considered for semifinal selection"); R-T29; R-T36 (does the quoted note count toward 100?).
- Reasoning: numbers ("292,264"), year ranges ("2033-2042"), hyphenated words and headings count differently in different tools. A breach disqualifies. Nobody publishes the counting rule.
- Hypothesis (guess): Wharton will use a word-processor count. An internal cap of about 475 words (IPS), 46 words (pitch) and 95 words (reflection) is a cheap margin (ASSUMPTION).
- Would change: the internal caps and a pre-submission checklist line.

Q5. Last season's strong teams reported Monte Carlo probabilities, and readers rewarded them. Will this season's readers expect a probability when the case says "explain how they evaluated that level of certainty", and should a number sit beside our model-free definition?

- Anchor: R-C54, R-C55; R-AN1; R-AN32; brief 8.6 (model-free certainty).
- What is new against brief 8.6: evidence of reader behaviour, which is not a modelling argument. A 2025-26 semifinalist's headline was "a 70% chance" (VERIFIED-PRIMARY, KXAN via Yahoo). Readers may use a checklist in which "a stated confidence level" is a box.
- Hypothesis (guess): keep the model-free definition as the definition. Add one evaluated number that a checklist reader can tick, for example the share of payment dollars matched by bonds held to maturity, or zero misses across all simulated paths labelled "by construction". Do not add a 95% forecast.
- Would change: one IPS sentence ("desired degree of funding certainty", R-I6); the FR certainty section.

Tier 3: Final Report (Dec 4; instructions Nov 9)

Q2. The rubric moved from "visually pleasing presentation of data; many creative elements" (2023-24, third-party transcription) to "uses data effectively to support conclusions; demonstrates original thinking" (2026-27). Should our FR use a few analytic charts, each carrying a conclusion, instead of many decorative ones?

- Anchor: R-S28; R-S24; Lumiere 2023-24 transcription (SNIPPET-UNVERIFIED as Wharton's text).
- Reasoning: rubric deltas show designer intent more reliably than the rubric's absolute words.
- Hypothesis (guess): yes. About 4-6 charts, each with a one-line takeaway. Candidates: liability against hedge (Q8); surplus range; the bought 2031 floor against the stretch; the effect of the 2028-deposit risk.
- Would change: the FR chart plan and page budget.

Q3. Who reads first, and what is the screening surface?

- Anchor: F-611; R-W8; R-W24; 2026 news (R-new): "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia"; 2025 news (R-new): "internal reviewers and a group of professional asset managers from abrdn".
- Reasoning: about 2,300 reports go to a small panel. If Wharton students screen first, the pitch, the FR's first page and the section headings do most of the work. The reading order is not published.
- Hypothesis (guess):
 - two stages, students first, then professionals on a shortlist;
 - the FR's first page should answer the case's must-list in order: reserve size, composition, certainty, facility, range with confidence, fundraising draft.
- Would change: FR page 1 and heading structure.

Q4. If an Aberdeen fixed-income professional reads our work, which bond statements would lose credibility, and which would gain it?

- Loses: "risk-free"; "matched" used for a duration-only ETF mix; ignoring reinvestment; confusing par and zero yields.
- Gains: a real ladder of Treasury STRIPS; a stated tracking gap; nominal-only certainty with its caveats.
- Anchor: R-new (Aberdeen review panel); F-610 (finale judge "U.S. Credit Investment Manager", VERIFIED-PRIMARY); brief 8.11, 8.13.
- Reasoning: a specialist reader turns small wording errors into large credibility losses, and turns precise wording into a signal of real understanding (criterion "Portfolio Analysis").
- Hypothesis (guess): use precise words ("duration-matched, not cash-flow-matched"). Show the twist gap honestly.
- Would change: sentences in the TN, IPS and FR; a bond-terms checklist for D9.

Q11. What benchmark should the FR use to evaluate implementation: WInS return against the S&P 500 (what most teams will do), or funded status and hedge tracking against Laura's promise, plus an audit of whether we followed our own IPS rules?

- Anchor: R-C90 ("evaluates how that strategy was implemented"); R-I37; R-W75 (not judged on outperforming); Guide p.4 "Evaluate your implementation".
- Reasoning: benchmarking against equities makes a liability-first plan look like a loser for the wrong reason. A rule-compliance audit ("we set N rules; we followed N; here is the one we broke and why") is rare and fits the Articulation criterion.
- Hypothesis (guess): make the primary metric liability-relative. Give the S&P comparison one line with the reason it is not the test.
- Would change: FR section structure and charts.

Q13. How do we show "appropriate diversification" and "understanding of the investments" with a WInS book of 3-6 positions, when rivals hold 20-40?

- Anchor: R-S24 ("uses appropriate diversification"); R-W70, R-W71 ("not simply to own a large number of securities"; "funding purposes"); R-AN45; F-409.
- Reasoning: a skimming reader may equate the number of holdings with diversification. Look-through facts (companies, countries and bond maturities inside each fund) and the "funding purposes" wording, which is Wharton's own, answer that. In the FR, the rule can be quoted; in the IPS it can only be paraphrased.
- Hypothesis (guess): one FR table of look-through counts, with a column naming each position's job.
- Would change: an FR table; the growth-note reflection wording.

Q14. Should the team pre-build FR components before Nov 9, assuming a page cap near last year's 15 pages?

- Components: the liability chart, the surplus distribution, the 2031 floor table, the fundraising-draft elements, and the assumptions list including inflation's effect on facility costs.
- Anchor: R-C93; R-S19; KXAN (2025-26 "required 15-page report", VERIFIED-PRIMARY article, secondary on the rule); a search summary citing "7-11 pages, double-spaced" (SNIPPET-UNVERIFIED, older season).
- Reasoning: instructions arrive with only 25 days left. A page cap forces triage. Pre-built parts reduce the crunch, which is the pre-mortem's most likely execution failure.
- Hypothesis (guess): yes. Build parts so they can be cut to 10 or 15 pages.
- Would change: the team's October-November work plan.

Q15. Pre-mortem (seed 31): if we miss the Top 50 with a correct strategy, which single cause is most likely, and what dated countermeasure prevents it?

- Anchor: R-I39; R-C92; R-W43; R-AN60; F-609.
- Candidates (my ranking, ASSUMPTION):
 1. The report reads as "too conservative / only bonds" to a skimming reader (Q16, Q13).
 2. Jargon or "fancy terms and formulas". The judge's exact warning, VERIFIED-PRIMARY: "hiding behind fancy terms and formulas".
 3. FR crunch after Nov 9 (Q14).

4. Visible inconsistency between week-1 notes, the IPS and the frozen WlnS book (Q27, Q10).
 5. Missing one of the case's must-items (for example, inflation's effect on facility costs, R-C85).
 6. An IPS format breach (Q17).
 7. A voice that sounds like AI (R-W49).
 8. Client research about the wrong "Laura Gao" (Q23).
- Hypothesis (guess): #1 and #2 together are the largest risk, because they hit the screening reader.
 - Would change: the team's risk checklist and deadlines.

Q20. Book-title puns and identity metaphors will be common. Will readers, and Laura, read them as tokenism? What is the most our deliverables should use?

- Anchor: R-C10, R-C16; brief 4 (identity never as decoration); R-S28 ("authentic team voice"); seed 34.
- Reasoning: sameness plus risk. A reader who has seen 50 "messy roots" portfolios discounts the 51st. Laura's public work is about identity, which is exactly why decoration can offend.
- Hypothesis (guess): zero puns. At most one functional reference, tied to a decision rather than to decoration.
- Would change: a style rule for all three deliverables.

Q21. Should the FR's Articulation section tell the true story of a past-season rule violation and the compliance routine built in response?

- Anchor: R-S27 ("reflects meaningfully on the team's growth and response to challenges"); R-W28 (no invented stories); v4 Part 2 lessons.
- Reasoning: true, specific failure-and-response stories are rare and authentic. The risk is that a reader focuses on the violation. Only part of the team was involved last season, so the story must be told accurately.
- Hypothesis (guess): one short, factual mention helps if it is linked to a verifiable routine (the pre-trade checklist) and the team agrees to include it.
- Would change: an FR Articulation sentence (a team decision).

Q22. Could a detailed, honest AI-use record become an Articulation strength rather than a risk?

- The record would say what AI did (brainstorming, checking), what the students decided, and what they redid by hand.
- Anchor: R-W46 ("how you use it must be recorded in your Works Cited pages"); R-W49; R-W51; X-9.
- Reasoning: many teams will under-disclose. Readers who suspect AI voice may discount a whole report. A specific record builds trust in the student voice. That fits "Did I come to my current conclusion before or after using generative AI tools?" (R-W51).
- Hypothesis (guess): yes, if concise (for example a half-page Works Cited subsection) and backed by a decision log.
- Would change: the FR Works Cited design; a logging habit starting now.

Q23. Every external fact about Laura must trace to the case's Laura. Do all team sources match the case identifiers (Wharton 2018, Statistics & Information Decisions Management, author of *The Wuhan I Know* and *Messy Roots*)?

- Anchor: R-C10, R-C12, R-C13; R-W4 (the scenario is invented); brief 13.
- Reasoning: a web search for her name returns other people with the same name, including a different student profile linked to Penn (search results, 2026-09-27; not opened, not recorded, for privacy). One misattributed fact in "Client Knowledge" is the kind of error a reader never forgives.
- Hypothesis (guess): the team docs already contain unverified claims (brief section 1). A source-match check is cheap.
- Would change: a verification step before any external Laura fact enters a deliverable.

Q24. Which pricing date should the IPS and FR numbers use, since rates will have moved by the time they are read (Dec-Jan)?

- Candidates: the Sept 25 curve, the Nov 6 freeze date, or the latest curve before Dec 4.
- Anchor: F-001, F-002, F-111 (the ladder was above \$300k on 173 of 185 days in 2026); R-C77.

- Reasoning: "fits inside \$300k" is a September 2026 fact (F-111). If rates fall 30bp before Nov 6, an IPS claim built on Sept 25 is wrong on the day it is submitted. A dated number with its rule ("if the cost exceeds the deposit, then ...") survives either way.
- Hypothesis (guess): state the rule, not the number, in the IPS. Re-price in the FR on the freeze date and show the six-week path (Q8).
- Would change: the IPS number policy and the FR's valuation date.

Q26. Where will readers look for the three-deliverables-together check, and what single table would let a reader verify consistency in one minute?

- A candidate table: rule in the IPS → WInS trade that applied it → note quoted → FR evaluation.
- Anchor: R-C91, R-C92; R-S24 ("maintains consistency with the team's IPS"); infographic (R-new): "Strong submissions connect the client's goals, research, portfolio decisions, and final recommendations through one clear and cohesive investment strategy."
- Reasoning: readers are told to consider the deliverables together but get no tool for it. Giving them one is distinctive and low-complexity.
- Hypothesis (guess): one FR table of about 6 rows replaces pages of narrative on consistency.
- Would change: one FR table (and it disciplines the notes now).

5. Evidence found this pass (all accessed 2026-09-27)

5.1 New official 2026-27 documents (VERIFIED-PRIMARY; not in repo)

- **Investment Competition Guide 2026-2027**, 5 pages.
 - URL: <https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf>, linked from <https://wghsinvcomp.smapply.us/res/p/guide/>.
 - PDF created 2026-09-09 (InDesign 21.4). SHA-256 3638e846...51fb7add (full hash in section 6).
 - Key text (exact):
 - p.2: "Your team will need to consider the client's goals, time horizons, required cash flows, funding commitments, liquidity needs, desired degree of funding certainty, risk tolerance, and communication with potential co-sponsors."
 - p.2: "The goal is not simply to identify investments you think will perform well. The goal is to develop a cohesive strategy in which your investment decisions work together to support the client's objectives across a range of possible market outcomes."
 - p.2: "Your research, analysis, Trading Notes, and experience implementing your strategy should all contribute to the IPS."
 - p.3: "When your team executes a trade in WInS, the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding."
 - p.3: "The Trading Notes Analysis is due at the end of Week 4 (October 23), two weeks before your IPS is due."
 - p.3: "Do not treat Trading Notes as an afterthought. They provide a record of how your team put its strategy into practice."
 - p.4: "The competition deliverables are not separate assignments. Together, they show how your team developed, implemented, articulated, and evaluated its investment strategy."
 - p.5: "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome."
 - p.5: "You may identify decisions you would make differently, but you may not redesign your strategy after observing the results."
 - p.5: "The competition recognizes thoughtful strategy, research and analysis, client understanding, disciplined decisions, management of risk and uncertainty, communication, and creativity."
 - p.5: "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions."
- **Competition Phases infographic**, 1 page.
 - URL: <https://upenn.box.com/s/l2p0l26svbbptmcizhmswyq5f7rbyp44>. PDF created 2026-09-14.

- Key text: "CORE PRINCIPLE Strategy guides every decision."; "ONE STRATEGY CONNECTS THE COMPETITION"; "Strong submissions connect the client's goals, research, portfolio decisions, and final recommendations through one clear and cohesive investment strategy."; "There is no single correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions."

5.2 Readers and judges (VERIFIED-PRIMARY)

- 2026 Top 50 article (Jan 27, 2026):
 - "with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia".
 - "Teams that competed at least through the mid-term report submissions will receive a participation badge".
 - <https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/>
- 2025 Top 50 article (Jan 30, 2025): "After careful consideration by our internal reviewers and a group of professional asset managers from abrdn in Philadelphia". <https://globalyouth.wharton.upenn.edu/news/announcing-the-semifinalists-in-the-2025-wharton-global-high-school-investment-competition/>
- 2025 finalists article (published about March 2025; the text dates the semifinals to the week of March 17, 2025). Melissa Ko Hahn, semifinal judge:
 - "A lot of work went into putting your written reports and the videos together, and it clearly came across,"
 - "Overall, one of the tendencies of people who are just starting out is to fall into this trap of hiding behind fancy terms and formulas. I would encourage you to fight against that. Oftentimes, the best solutions are made up of simple, elegant ideas."

Checked with `--grep`. This upgrades the winners.md paraphrase to VERIFIED-PRIMARY. <https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/>

- 2025 champions article. Chirag Jain, then "U.S. credit research analyst at Aberdeen Standard Investments": "What distinguished managers was clear thoughtfulness and reasoning behind all the decisions that they put together." (Context: finale, not semifinal; used here only as a signal of the Aberdeen reviewers' values.) <https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/>
- 2026 champions article (Apr 30, 2026): finale judges included "Chirag Jain, U.S. Credit Investment Manager, Aberdeen Investments". <https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/>
- 2026 finalists article (Mar 20, 2026): "49 semifinalist teams advanced to the virtual semifinal rounds, where they presented their strategies to panels of Wharton alumni and industry experts." Note 49, not 50. <https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/>
- Rules page: "The Global Champion school will be permitted to fast-track one team to the semifinal round of the 2027-2028 competition." (This season's text. By the same practice the 2026 champion school fast-tracks one team in 2026-27; INTERPRETATION.) <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/>

5.3 What past winners and semifinalists did (written work)

- Vandegrift ("The Compounders", 2026 semifinalist). KXAN article mirrored on Yahoo, read on the page (VERIFIED-PRIMARY for the article; the team's claims are self-reported):
 - "built a four-part screening system that focused on U.S.-traded securities, high-growth potential, low volatility, and alignment with the non-profit's mission";
 - "ran their portfolio through thousands of Monte Carlo simulations and stress-tested it against historic downturns, including the dot-com crash, the 2008 recession and COVID-19 market conditions";
 - "Their model showed a 70% chance of reaching the long-term goal";
 - "All four worked together on the required 15-page report."
 - <https://www.yahoo.com/news/articles/female-vandergrift-team-advances-wharton-174003734.html> (the KXAN original returned 403)

- Harvard-Westlake Chronicle (VERIFIED-PRIMARY article):
- advisor Rob Levin on the 2025-26 case: "His goal was to invest money for ongoing community recreation programs and long-term facilities construction."
- a student: "I formulated the idea for our strategy and programmed our model."
- Last season's case also paired ongoing operations with a facility, so returning teams and AI tools carry templates for this structure.
- <https://hwchronicle.com/112806/news/students-compete-in-yearly-wharton-stock-competition/>
- 2023 "tales" article (VERIFIED-PRIMARY):
- Brian Z, DMV's Finest (2022-23 champions): "less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once."
- Alex Lamon: "Remember, winning isn't about portfolio gains. It is about having a coherent strategy."
- <https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-tales-from-the-2023-teams/>
- Wharton "developing strategy" page (undated legacy page, VERIFIED-PRIMARY): "if you provide us with a team strategy that is buried somewhere in your experience, rather than clearly stated, defined and articulated up front, you will not be among the strongest teams." <https://globalyouth.wharton.upenn.edu/developing-strategy/>
- 2025 finale: teams showed "sentiment, SWOT and Pestel analyses, customized algorithms, AI innovations, portfolio optimizations, performance and readiness indices" (VERIFIED-PRIMARY, BAM article). This is finale content, used only as a picture of the field's toolkit.
- Posted past reports (Scribd "Wharton Competition - Final Report", "Mid-Term Report", "Final Report template"; SlideShare "Wharton final report") could not be read: the sites need JavaScript. Status: NOT READ.

5.4 Rubric history

- Lumiere Education (third party, page published for 2023-24, read directly) quotes the then-current criteria, for example:
 - "Creativity and presentation: compelling narrative and visually pleasing presentation of data; many creative elements; clearly presented with an authentic team voice."
 - "Client knowledge and objectives: strategy is tailored uniquely to the client's financial goals; would win him/her over as a client."

Status: VERIFIED-PRIMARY as Lumiere's text; SNIPPET-UNVERIFIED as Wharton's wording (the Wayback Machine was unreachable). <https://www.lumiere-education.com/post/wharton-global-youth-program-s-investment-competition-a-comprehensive-guide>

- 2025-26 schedule (Scribd "25-26 Week One Email", search summary only): mid-term report due Oct 31, final report due Dec 12, first trade by Oct 10, roster Oct 17. SNIPPET-UNVERIFIED.

5.5 Field arithmetic (DERIVED)

- 2025-26: 2,339 finishers (F-609). With one fast-track seat, about 49 open seats, about 2.1%.
- School overlap: 7 schools appear in both the 2025 and the 2026 Top 50 lists (Amity International, Deerfield, Nový PORG, Lycée Français de Chicago, Richard Montgomery, Stuyvesant, Western Canada HS). They account for about 9 of the 2026 Top 50 teams. My parser matched 49 of the 50 2026 lines, so treat the count as approximate.
- 27 of the parsed 2026 lines are U.S. schools. There are 0 Australian schools in either list.
- The 2025 list includes one Taiwanese school (Kang Chiao International School).

6. Sources (all accessed 2026-09-27; status as marked)

Source	URL	Status
Investment Competition Guide 2026-27 (5 pp)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf ; SHA-256 3638e84692ed0d826660a65e89cad0ce7eea8c1907b6a8f00ce02c0d51fb7add	VERIFIED-PRIMARY
Competition Phases infographic (1 p)	https://upenn.box.com/s/l2p0l26svbbptmci zhmswyq5f7rbyp44 ; SHA-256 2827f92a989d6d72fbc253d80d975eb06bd00c86f631e83850ae17eec544ed10	VERIFIED-PRIMARY

Source	URL	Status
SMAppl Guide page	https://wghsinvcomp.smappl.us/res/p/guide/	VERIFIED-PRIMARY
2026 Top 50 article	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investment-competition-teams-revealed/	VERIFIED-PRIMARY
2025 Top 50 article	https://globalyouth.wharton.upenn.edu/news/announcing-the-semifinalists-in-the-2025-wharton-global-high-school-investment-competition/	VERIFIED-PRIMARY
2025 top-10 article (Hahn quote)	https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/	VERIFIED-PRIMARY
2025 champions article	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-global-high-school-investment-competition/	VERIFIED-PRIMARY
2026 finalists article	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition-global-finale/	VERIFIED-PRIMARY
2026 champions article	https://globalyouth.wharton.upenn.edu/news/2026-investment-competition-global-champions/	VERIFIED-PRIMARY
Rules & Roles	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/	VERIFIED-PRIMARY
Developing strategy page	https://globalyouth.wharton.upenn.edu/developing-strategy/	VERIFIED-PRIMARY (undated)
2023 tales article	https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-tales-from-the-2023-teams/	VERIFIED-PRIMARY
KXAN/Yahoo Vandegrift article	https://www.yahoo.com/news/articles/female-vandergrift-team-advances-wharton-174003734.html	VERIFIED-PRIMARY (article); team claims self-reported
Harvard-Westlake Chronicle	https://hwchronicle.com/112806/news/students-compete-in-yearly-wharton-stock-competition/	VERIFIED-PRIMARY (article)
Stuyvesant blog (FigCapital)	https://talos.stuy.edu/cms/pages/stuyvesant-blog/stuyvesant-high-school-wins-the-2026-wharton-global-high-school-investment-competition-26/	VERIFIED-PRIMARY; no strategy detail
Aralia 2026-27 guide (third party)	https://www.aralia.com/helpful-information/guide-to-the-wharton-global-high-school-investment-competition/	VERIFIED-PRIMARY as third-party text
Lumiere 2023-24 guide (third party)	https://www.lumiere-education.com/post/wharton-global-youth-program-s-investment-competition-a-comprehensive-guide	VERIFIED-PRIMARY as third-party text; Wharton wording SNIPPET-UNVERIFIED
Scribd 25-26 week-one email; mid-term report; final report template; SlideShare final report	scribd.com/document/934687491,/842933720,/820640088,/942363888; slideshare.net/slideshow/wharton-final-report/262037395	NOT READ (JavaScript wall); summaries SNIPPET-UNVERIFIED
Web search summary "7-11 page Final Report, double-spaced"	search result, 2026-09-27	SNIPPET-UNVERIFIED (older season)
Wayback Machine	web.archive.org	BLOCKED (connection reset)
KXAN original	kxan.com	REFUSED (HTTP 403)

Repo files used: the official case, IPS guide, Trading Notes guide and SMAppl page; the Phase A files (case_register.md, fact_register.md, stakeholder_map.md, wins_week1_guardrails.md); 02_judge_panel.md; groundwork/winners.md; 01_trading_notes_blueprint.md; wins_now/securities_and_allocation_v0.md; wins_now/S1_treasury_sleeve.md.

7. Leads for the specialists

- **D10 / main loop:** add the two new official PDFs (section 6 hashes) to `competition/official/2026_27/` and `manifest.yaml`, and register their text as new R-ids. Several of my anchors are "R-new" because of them.
- **D10:** the logged-in "Trading Details" and Session Rules pages are needed for Q25 ("required trading activity") and for the note character limit. Also check the SMAppl TN form for word-count enforcement and whether the quoted note counts toward the 100 words (Q17).
- **D7:**
 - The 2026 reader panel is not yet announced. Watch the SMAppl updates.

- Past reports on Scribd and SlideShare need a human with a browser to read (team members can open them). Priority: "Wharton Competition - Final Report" (document 942363888), which is likely 2025-26.
 - Compare 2024-25 and 2025-26 semifinalist news items (school papers) for any description of written reports.
 - **D1:** confirm the Oct 2026 FOMC dates and the CPI and Treasury auction calendar for Sept 28 to Oct 20 (Q28). Draft the precise bond-wording checklist for a credit-professional reader (Q4).
 - **D3:** build the daily "promise value vs WlnS hedge" log. The inputs are the Treasury curve CSV (daily) and WlnS position values (manual daily capture). Report the tracking gap (Q8, Q24).
 - **D9:**
 - the sameness test protocol (Q1), with the AI use recorded for Works Cited;
 - the fixed three-label vocabulary (Q27);
 - pitch and IPS content specifications (Q18, Q19);
 - a no-pun style rule (Q20).
 - **D6:** whether ~20% total equity reads as ignoring "thoughtful risks" (Q16). Any evidence on how readers see conservative plans.
 - **D5:** none from this lens beyond the Q3 note that readers include asset managers, not philanthropy specialists.
 - **Source-match check (Q23):** every external Laura fact must match the case identifiers. Do not record other people's details.
-

8. Parked (logged, not pursued)

- Team names that reference the client: several 2026 semifinalists and 2 of 11 finalists had names referencing the client's NFL career. Correlation only, and the roster name is set by Oct 9. Parked.
 - Finale presentation format, judge Q&A: out of scope (semifinals only).
 - Spelling (Australian vs U.S.): the case uses U.S. spelling; a minor style choice for D9, not a question.
-

What this teaches

- **Before trying to be different, write down what "normal" looks like.** Only then can you see where readers will be bored (a Monte Carlo "95%", book-title puns, a copy of the official example) and where they will stop and read.
- **The rules of a deliverable are part of the strategy.** A Trading Note is frozen when you type it. It must come from a trade that actually executed. A decision not to trade can never be one. The IPS cannot show a chart. The Final Report cannot "redesign" after results. Each rule changes what you should do today, not just what you write later.
- **Know your reader.** Last year the written reports were screened with help from Wharton students and read by professional asset managers, including bond specialists. Simple, precise words beat "fancy terms and formulas". The winning 2022-23 team's own lesson was "less is more".
- **Look for the missing documents.** Two official guides were sitting one click away on a public page, and nobody had read them. Checking every link on an official page is cheap, and it can change the whole plan.

B8a Voice of Laura (lens: her own words; starting angle: long-form conversations)

Agent B8a, insight_v1 run, Phase B. Written 2026-09-27. AI-generated research for Team Caplet, not submission text. It collects Laura Gao's own public, professional words. It maps them to her values, how she decides and how she handles risk, and it turns them into questions that could change a decision, a number or a sentence. The team writes every deliverable in its own words. Quotes are for the team's understanding. Before quoting any of them in a deliverable, check the Works Cited and AI rules (R-W30, R-W46).

Privacy rule applied: I read pages that also contain personal and family material. I left all of it out: family members, partner, birth name, health, relationships and coming-out stories. Only her public professional statements and facts about her work are recorded here.

0. Summary (read this first)

1. **The case pull quote may not be her words.** "The only person who needs to believe in something is yourself." (R-C21) did not appear in any of the 25 public sources I opened (interviews, podcast transcript, her website, her speaking guide, her resume). An exact-phrase web search found only generic self-help pages. **Status: UNATTRIBUTED.** Treat it as the case's words ("the case profile highlights..."), never as "Laura says..." (Q01).
2. **Her real risk pattern: she lined up a funded base first, then leapt.** She calls the move "such a drastic move from an outside perspective" and says "you should always take the jump". But the reporting shows that she had "made a tentative plan" before the comic went viral, and she "gave Twitter her notice" only "when she got the book deal with HarperCollins" (Inverse, 2022, VERIFIED-PRIMARY). She also named the cost plainly: giving up a "cushy, paying job with health insurance to be self-employed, with unstable income" (Overachiever, 2021). **Our lock-early design (buy the promise first, take risk only with the surplus) matches this pattern.** This is the strongest Laura-rooted argument for it, and it is also an argument against "she likes risk, so give her more equity" (Q02).
3. **She rejects gimmicks and over-engineering in her own work.** She wrote "I scrapped three completely different drafts" and "So, I scrapped everything and went back to my roots. No more gimmicks." (Nerd Daily, 2025). This is last year's lesson ("complexity must earn its place") in her own voice. Every branded framework, thematic tilt or book-title pun has to pass this test (Q03, Q22).
4. **She writes a purpose statement for her work and treats the medium as replaceable.** "I can't think of many art pieces that I didn't also write some kind of purpose statement for." "I've always said that my main purpose is to tell a story" (Pulse Spikes, 2021). The mapping to an IPS is direct: purpose first, instruments as swappable "media" (primary pick plus alternates) (Q04).
5. **Her 2028 income is exposed to politics and timing more than to the stock market.**
 - Her books have been published late twice. *Messy Roots* was "planned for summer 2021" (Publishers Weekly, July 2020) and went on sale 8 March 2022. *Kirby* was hoped for "2024" (Inverse, 2022) and came out in 2025.
 - In March 2025 she said she was "taking a short break from writing".
 - Her past speaking clients include corporate DEI/ERG events at Asana, Adobe and a Google LGBTQ+ group.
 - A Montana school board banned *Messy Roots* in January 2024 (Book Riot).
 - Implication: the 2028 deposit is more likely to be *late* or cut by a *political/budget* shock than to move with equities. That changes how we write about human-capital and wrong-way risk (Q08, Q09).
6. **She has built community institutions in stages, kept them free, and given out small grants.**
 - Pride in Panels: "way beyond our initial passion project"; "We definitely hope to offer more as the festival grows"; the event is free and gives mini-grants to 20 creators (Local News Matters, Feb 2026).
 - She finds community through co-living and co-working spaces (Maverick Show, 2022).

- This supports three things: the case's operations-first order, a flexibility buffer, and a range that can grow as support is proven (Q12, Q13, Q23).

7. Taiwan appears in her own words.

- "Taiwan is like one of my top places to live in and to travel to all time."
- Among her favourite places she names Yilan, noting "some art residencies at the top of the mountains there" (Maverick Show transcript, June 2022).
- The case scenario is fictional (R-W4), so never present this as her plan. It still tells us that a Taiwan residency is plausible for her, and a rural setting would change which cost index is relevant (Q10).

8. Her first public act was correcting a one-sided risk story about a place. "It's disheartening to know that all they know are the bad parts of it" (NPR, 2020). "focus on the full human story" (HuffPost, 2020). So a Taiwan section that reduces Taiwan to "Strait war risk" is exactly the framing she fought. Name the risk precisely and completely, once (Q11).

9. She has shown both sides of finance: the trading floor and consumer protection. She was a Trading Intern at Belvedere Trading and a Policy Analyst Intern at the CFPB. She also taught "financial literacy to high school students in West Philadelphia" (Poets&Quants 2018). A high-school team is writing for a former financial-literacy teacher of high-schoolers. That raises the bar on fee disclosure, plain words and honesty (Q06, Q07).

10. Her working style matches the Trading Notes design. "Always start small and post that work, no matter how crappy it looks!" "I give myself a deadline for when I must post the art, finished or not." (Honey Pop, 2022). She kept a daily "comics diary". She co-founded an "Anti-Resume Project" to "normalize failure". Implications: an early small first trade, notes written as an experiment log, and a candid Articulation section (Q15-Q17).

Counts: **25 questions** (section 5). **48 numbered quotes** in the quotes table (section 2), all VERIFIED-PRIMARY (#4, #5 and #18 are reporter text), plus a few reported facts. 68 phrases in total were confirmed verbatim (list in section 7).

1. Method and source access (what I opened, how quotes were verified)

- Access date for every page: **2026-09-27** (between about 14:00 and 15:30 UTC), from this run's container.
- Verification: each quote was checked with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL --grep "phrase"` and returned `PHRASE FOUND VERBATIM: YES`.
- **One exception (Poets&Quants page), handled as follows:**
 - The helper's Chrome user agent got HTTP 403 from poetsandquantsforundergrads.com. Plain curl (default user agent) returned 200, so I saved that response into the helper's cache.
 - The helper's HTML parser then printed almost no text. It drops the page body on this WordPress page.
 - I verified these quotes with a scratchpad variant, `ft2.py`. It uses the helper's own `fetch()` and `norm()` functions and only swaps the tag stripper for a regex. It also returned `PHRASE FOUND VERBATIM: YES`.
 - The fix for others: add a fallback extractor to `fetch_text.py` (see Leads).
- **The Maverick Show transcript is AI-generated.** The page says "We produce our transcripts using Outcast AI". So its wording is a machine transcription of speech, not text she edited. The quotes are verbatim to the transcript, not necessarily to her speech. They are labelled VERIFIED-PRIMARY (transcript) with this caveat.
- **Anomaly:** the show's "Read the Transcript" link for Ep 190 (Laura) serves the transcript of Ep 189 (another guest). The correct PDF is `.../uploads/2025/10/190_Laura_Gao_Transcript.pdf`. I found it by URL pattern, and it opens with her introduction.

Blocked or not useful:

Source	Result
legacy.diversebooks.org Q&A	Captcha page (HTTP 202, sgcaptcha); not read

Source	Result
ampersand-mag.com/laura-gao (2021 profile)	TLS error; not read
nuvoices.com/2022/03/17/...	404; the alternate URL /podcast/nuvoices-podcast-72-... opened, but it is show notes only, with no transcript. Topics are mostly personal, so nothing was used
Wainao (Taiwan-based Chinese-language outlet)	Video only, with no transcript
YouTube "Laura Gao on Messy Roots book ban"	Title seen in search only; video not watched (SNIPPET)
lauragao.com/s/Interactive-School-Visits-Guide-Laura-Gao.pdf	Returned almost no text

2. Quotes table (her own words; all VERIFIED-PRIMARY unless marked)

Columns: # · exact text (copied character-for-character, original punctuation) · source (URL, piece date) · context · value it shows · implication. Access date for all: 2026-09-27.

Sources (short names used below):

- **PQ** = Poets&Quants for Undergrads, "2018 Best & Brightest: Laura Gao, Wharton School", 2018-03-30, <https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/>
- **WM** = Wharton Magazine, "A Wharton Grad From Wuhan's Exploration of Identity", 2022-04-06, <https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/>
- **OA** = Overachiever Magazine, "Interview with Laura Gao", 2021-03-15, <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>
- **IN** = Input/Inverse, "How a viral tweet turned into a queer coming-of-age graphic novel", 2022-03-07, <https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel>
- **PS** = Pulse Spikes, "Laura Gao: A Comic Artist Tackling Misconceptions...", 2021-06-18, <https://pulsespikes.org/story/laura-gao>
- **MV** = The Maverick Show Ep 190 transcript (AI transcription), episode 2022-06-23, https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf
- **HP** = The Honey Pop, "Exclusive Interview: Laura Gao on Messy Roots...", 2022-06-02, <https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/>
- **ND** = The Nerd Daily, "Q&A: Laura Gao, Creator of 'Kirby's Lessons for Falling (In Love)'", 2025-03-08, <https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/>
- **BW** = BookWeb (ABA), "An Indies Introduce Q&A With Laura Gao", 2022-02-07, <https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512>
- **NPR** = NPR Goats and Soda, "'The Wuhan I Know': A Comic About The City Behind The Coronavirus Headlines", 2020-04-04, <https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines>
- **HUF** = HuffPost, "Artist Illustrates What Her Hometown Of Wuhan Is Really Like", 2020-04-30, https://www.huffpost.com/entry/chinese-american-illustrator-life-in-wuhan_n_5ea9c739c5b63115cec2be5a
- **LNM** = Local News Matters (Bay City News), "SF queer comics fest expands, stays free in second year", 2026-02-12, <https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/>
- **DP16** = The Daily Pennsylvanian, "Student Spotlight Q&A: Laptop decal designer and Wharton sophomore Laura Gao", 2016-01-28, <https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a>
- **SG** = her "Messy Roots Publicity & Speaking Guide" PDF (undated; it says Kirby "will be released in 2025", so it was written in 2024 or earlier), <https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf>
- **RES** = her resume PDF (undated, current), <https://lauragao.com/s/Laura-Gao-Resume.pdf>
- **HEY** = <https://lauragao.com/hey/> (current)

2a. Risk, leaving tech, jumping

#	Exact text	Src	Context	Value	Implication
1	"It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""	OA	On leaving Twitter for full-time comics	She is aware of the downside and says the cost out loud	She calls her own income "unstable". That supports the case's split: career risk sits outside, and the promise is protected inside the portfolio (R-C37/R-C38)
2	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	OA	Same answer	Regret-minimising courage	She respects bold moves. Our "bold" part is the growth sleeve and the residency itself, not gambling the payments
3	"I feel like I've been jumping almost my whole life."	OA	On switching careers	Serial pivots	Her identity is someone who jumps. The portfolio's job is to be the platform that makes jumping safe
4	"Even before her comic went viral on Twitter, she'd made a tentative plan to travel around the world, teach art part-time, and focus on her own creative endeavors."	IN (reporter, not her words)	Reporter's account	She planned before she leapt	Evidence for the pattern "plan, then leap"
5	"So when she got the book deal with HarperCollins, she gave Twitter her notice."	IN (reporter)	Reporter's account	She leapt once a contract was signed	This is the key: a secured base, then risk . It is the same logic as lock-early
6	"So, I can't say no to this opportunity"	MV (AI transcript)	After her agent approached her	Acts decisively when a real opportunity arrives	Supports a pre-set decision rule (e.g. buying the ladder when the money arrives) over waiting for a perfect moment
7	"I was a bit hesitant. I was like, I know this is kind of a sensitive topic. I don't know what I'll get from it"	NPR	On posting "The Wuhan I Know"	She takes risk under uncertainty, with purpose	"Thoughtful risk" = uncertain outcome, clear purpose, downside she can bear
8	"When I first posted the comic, I expected a way worse reaction"	OA	On online abuse	Plans for the worst case	A pre-mortem mindset. Show the bad-case numbers first
9	"This is who you're trying to help, and this is the positive effect you actually have."	OA	How she steadied herself after hostile comments (a saved note)	Pre-commitment to purpose	A drawdown rule anchored in purpose (the ten payments are already bought) is her own coping style
10	"However, this lack of structure and certainty can be off-putting for students who are easily influenced by those around them or are risk-adverse."	PQ	Advice on studying business	She names risk aversion without contempt	She understands that some people need structure. "Certainty" and "structure" are her words for the protected part

2b. Building things, creating what is missing

#	Exact text	Src	Context	Value	Implication
11	"If what you want doesn't exist, create it yourself."	PQ	"Biggest lesson" from studying business	Founder mindset	The residency is this principle again. A strategy built around <i>her</i> problem, not a template, fits it
12	"However, I realized transferring was nothing more than running away from the problem."	PQ	On nearly leaving Wharton	Stays and fixes rather than exits	Stay-the-course rule. Fix a problem (e.g. rates fall before Jan 2027) with a rule, not an exit
13	"Well, if people aren't gonna give it to us, I'll just make one."	OA	On her second book	Build what is missing	Same as #11
14	"but why wasn't there any single event where we could celebrate that community and bring everyone together?"	LNM	Origin of Pride in Panels (2026 article)	Community infrastructure builder	She has already founded a community institution with partners. Co-sponsors are her normal way of working
15	"All of it is way beyond our initial passion project."	LNM	Festival growth	She grows things in stages	Staged ambition: floor first, upside later (range design)
16	"We definitely hope to offer more as the festival grows."	LNM	On mini-grants to creators	Scale after support is proven	A contribution that can grow with results, not a single fixed bet

#	Exact text	Src	Context	Value	Implication
17	"Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on"	LNM	On grants	Access for all	Program fees may be low, so the operating money matters more (interpretation)
18	"remaining free for exhibitors and attendees"	LNM (reporter)	The festival stays free	Free access	Same. Evidence for a low-fee model (interpretation, not a case fact)
19	"Was a simple, 10-panel comic enough to accomplish such a feat?"	WM	When asked to write more	Starts small and doubts scale	Small things can grow. It also mirrors a small reserve decision growing into ten years of operations

2c. Craft, simplicity, purpose, evidence

#	Exact text	Src	Context	Value	Implication
20	"I scrapped three completely different drafts before landing on this one."	ND	Writing Kirby	She will discard work that is not right	Discard ideas that do not earn their place
21	"So, I scrapped everything and went back to my roots. No more gimmicks."	ND	Same	Anti-gimmick	The complexity test. Drop tilts, branded names and puns that do not add value
22	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	PS	On writing and art	Purpose first	An IPS is a purpose statement. Lead with what the money is for
23	"I've always said that my main purpose is to tell a story"	PS	Across her media	Purpose fixed, medium flexible	Instruments are the "medium": primary pick plus alternates, same purpose
24	"That [story] was something I could prove better with data"	PS	Her "Rewriting Herstory" data project	Evidence-backed storytelling	Final Report: one chart that proves one point. The IPS must prove it in words alone (R-I55)
25	"I've always been an artist, and I believe that comics are a great way to marry imagery with the power of words to express a story"	NPR	On comics	Words plus pictures	The Final Report charts should carry the story, not decorate it
26	"Always start small and post that work, no matter how crappy it looks!"	HP	Advice to young artists	Ship early, iterate	WInS: a small first trade early in the window. The notes show a developing strategy (R-C88)
27	"I give myself a deadline for when I must post the art, finished or not."	HP	Same	Self-imposed deadlines	Internal deadlines ahead of Oct 23 and Nov 6
28	"Ultimately my goal is to tell a story; I don't need to be perfect to be impactful."	HP	Same	Impact over perfection	Plain and correct beats polished and complex
29	"don't put too much pressure on getting everything perfect"	MV (AI transcript)	On her creative process	Same	Same
30	"Soon enough, I had a comics diary that I could not only decompress into, but also take pride in how far I'd come when I reread it."	WM	Nightly comic habit while at Twitter	A process record	Keep a dated decision diary; the Final Report shows growth from it
31	"My biggest advice is to be OK with the fact that your significant projects will never be finished."	DP16	As a 19-year-old founder	Iterate forever	Tension with "the IPS freezes the strategy" (R-I35): freeze the principles and rules, and let execution iterate inside them
32	"how long will this novelty factor last?"	DP16	On her own launch hype	Sceptical of hype	She would question an AI or thematic hype trade
33	"I wish I wrote about these nuances and tradeoffs made in an author's note at the end. But at the same time, I wish I didn't always have to explain myself."	BW	On translation choices in Messy Roots	Values disclosed trade-offs, and brevity	One compact "trade-offs and assumptions" note, not pages of caveats

2d. Money, freedom, flexibility, success

#	Exact text	Src	Context	Value	Implication
34	"Now I get to fit work around life instead of the other way around, which is such a game-changer"	IN	On full-time art	Autonomy	Flexibility is a core value, not just a case requirement (R-C58)
35	"knowing that I don't have to own anything or be tied to any material possessions"	MV (AI transcript)	Impact of travel	Travels light, stays flexible	She prefers optionality to ownership. It argues against committing "all remaining assets" to a building
36	"My advice is to avoid siloing yourself into an end-all-be-all career situation."	PQ	Advice on studying business	Keep options open	Same. It is diversification in life language
37	"I did 50 things, and maybe I didn't like fully excel at each, but I got to try all of them"	OA	How she wants to look back at 80	Breadth over one bet	Diversification is her own life philosophy. A broad index sleeve, not concentrated picks
38	"mainly because I'm so interested in them and I don't want to pigeon-hole myself into one thing."	PS	On many projects	Same	Same
39	"it's great having thousands of likes, but at the end of the day, it's just a number."	OA	On online feedback	Meaning over metrics	Express results in what they secure (years of operations funded), not only dollars
40	"each has juggled passion against practicality, social pressure against personal identity, and countless bouts of career pivoting"	PQ	About the alumni The Sign.al profiled	Passion vs practicality balance	The case's "balance between pursuing growth and protecting the capital" (R-C38) in her own words
41	"Every character is grappling with how to balance society's expectations with their own desires."	ND	Kirby's theme	Balance	Same. Balance as a stated rule, not a vibe

2e. Taiwan, community, how she lives and works (professional-relevant only)

#	Exact text	Src	Context	Value	Implication
42	"Taiwan is like one of my top places to live in and to travel to all time"	MV (AI transcript)	She lived mostly in Taipei while writing the second half of Messy Roots	Real connection to Taiwan	The residency location is plausible for her. The scenario is still fiction (R-W4); do not state her real plans
43	"Or there's also a little like some art residencies at the top of the mountains there."	MV (AI transcript)	Recommending Yilan, Taiwan	She knows Taiwanese art residencies	Rural vs city matters for which cost index (construction vs Taipei property) is relevant
44	"join a co living space even just for like a week or a month"	MV (AI transcript)	Her travel tip	Community through shared spaces	The residency model ("temporarily live, work, teach, and collaborate", R-C40) is how she herself finds community
45	"I visited and started making community there."	MV (AI transcript)	Co-living and co-working in Lisbon	Community	Same
46	"It's disheartening to know that all they know are the bad parts of it"	NPR	On headlines about Wuhan	Against one-sided risk stories	Taiwan risk must be framed completely, not as a flashpoint caricature
47	"focus on the full human story"	HUF	On how to discuss Wuhan	Complete picture	Same
48	"There wasn't much that was incorrect as much as there just wasn't any news about the overall picture of Wuhan outside of the coronavirus"	HUF	Same	She objects to <i>incomplete</i> framing more than errors	A section can be accurate and still fail her by being incomplete

2f. Facts about her professional record (her own documents; VERIFIED-PRIMARY)

- "Trading Intern at Belvedere Trading in Chicago". She was also a "Policy Analyst Intern at Consumer Financial Protection Bureau". She was a "Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia." (PQ)

- "Bachelor of Science in Statistics and Information Decisions, Concentration in Product Design" (RES). Her own words in WM: "a degree in business analytics". PQ 2018: "Major: Economics with a concentration in Business Analytics". These are consistent with a Wharton BS in Economics with concentrations (ASSUMPTION; see Q05).
- "Experimented and shipped 10 features that resulted in +4.5M DAU" (RES, Twitter PM). She is an experimenter (A/B testing, SQL, R, Tableau).
- Speaking guide (SG):
 - "Rates will differ based on travel, length of event, and books purchased." "Pro-bono visits can be offered on need-basis."
 - "Previous talks: APAPI Author Talk with Asana, Doodle Therapy with Adobe, Drawing Your Life Workshop with Google LGBTQ+ Group".
 - Listed topic question: "What motivated you to quit a secure job to become an artist-writer in the middle of a pandemic?"
- HEY: she uses "she/her and they/them pronouns" (the SG says "she/they"). She signs off with "Let's make something together."
- Her Kirby page (publisher copy on her site): "Once dubbed the Queen of Balance".
- "I'm taking a short break from writing to focus on teaching my Writing Comics course at California College of the Arts!" (ND, March 2025).
- DP 2019-03-17 (reporter): "her anti-resume includes receiving 33 job rejections". PS (reporter): "To destigmatize alternate career paths and normalize failure, Laura helped launch the Anti-Resume Project".

2g. Publishing timing evidence (VERIFIED-PRIMARY on the pages named)

Book	Planned date	Source	Actual date	Source
<i>Messy Roots</i>	"Publication for the first book is planned for summer 2021"	Publishers Weekly Rights Report, week of July 20, 2020	"On Sale: 3/8/2022"	BookWeb, 2022
<i>Kirby</i>	"She's hoping to finish in time for a 2024 release."	Inverse, 2022	Published in 2025	SG says "will be released in 2025"; search snippets give 4 March 2025 (SNIPPET-UNVERIFIED for the day)

- The first book deal was "acquired, at auction" (PW).
- Book Riot (2024-01-12): "Laurel Public Schools (MT) school board just banned six books", *Messy Roots* among them (VERIFIED-PRIMARY for Book Riot's report). This closes the stakeholder map's open item SH-23 ("I have not verified whether any of Laura's titles were challenged"): one ban is now reported.

2h. Unverified or unattributed

Text	Status	Note
"The only person who needs to believe in something is yourself."	UNATTRIBUTED	Case pull quote (R-C21). Not found in any source above. An exact-phrase WebSearch returned only generic motivational pages (SNIPPET). It could come from her books or a talk I could not read. Until found, it is the case's wording
"Laura Gao cares so passionately about making a difference..."	VERIFIED-PRIMARY	This is a Wharton staff member's recommendation on the PQ page, not her words. Do not attribute it to her
Belvedere Trading is an options market-making firm	ASSUMPTION (general knowledge)	Not checked on a primary page

3. What her words say about her (values, decisions, risk)

Values, ranked by how often they recur across sources (my judgement; the quote numbers are the evidence):

1. **Build what is missing, with others** (#11, #13, #14, #44). She founds things: a club, a startup project, a festival, and in the case a residency.
2. **Purpose first; the medium is replaceable** (#22, #23, #39).

3. **Authenticity, no gimmicks** (#20, #21, #28; PQ "the Laura Gao I knew loved building apps, not excel models").
4. **Optionality and freedom** (#34-#38). She keeps options open and does not like being tied down.
5. **Honesty about failure and trade-offs** (#33; the anti-resume; "33 job rejections").
6. **Complete, fair stories about places and people** (#46-#48).
7. **Access and community** (#17, #18; pro-bono "on need-basis").

How she decides (evidence-based pattern, INTERPRETATION):

- She plans quietly first (#4).
- She commits when a concrete, funded opportunity appears (#5, #6).
- She starts small, ships to a deadline and iterates (#19, #26, #27, #31).
- She will scrap work that is not right (#20, #21).
- She uses data as proof inside a story (#24).
- She is an experimenter (A/B tests at Twitter).

How she handles risk (INTERPRETATION):

- High willingness, but she names the cost first (#1).
- She prepares for the worst case (#8).
- She anchors herself to purpose when things go badly (#9).
- Her "jumps" were cushioned by a signed contract and a community (MV: "with enough trust in people around me").
- This is "thoughtful risk" (R-C37) in her own life: **risk on top of a secured base**, not risk to the base.

What she would find inauthentic (seed 34), from her own contrasts:

- The "ideal Whartonite" voice she rejected: "excel models", "corporate coffee chats", peers asking "why aren't you going for McKinsey or Goldman Sachs" (PS, reporter quoting her).
- Hype ("novelty factor", #32).
- Gimmicks (#21).
- One-sided risk stories (#46).
- Numbers without human meaning (#39).
- Using her identity or book titles as decoration. This is the brief's tokenism rule, and it also fits #21.

What would earn her trust (seed 32):

- A purpose statement she could have written.
- Risk taken only where it cannot hurt the promise.
- Trade-offs disclosed briefly.
- Evidence shown, not asserted.
- An honest failure log.
- Room left for the project to grow.

Which of her ideas our strategy already echoes (seed 33):

- Lock-early = "secured base, then leap" (#4, #5).
- The growth sleeve = "take the jump" with money the promise does not need (#2).
- The broad index sleeve = "50 things" (#37).
- Flexibility after 2033 = "fit work around life" and not being tied down (#34, #35).
- The 2031 floor-plus-upside range = staged growth (#15, #16).

4. Questions already answered elsewhere (not re-asked)

Brief section 8 items 1-19 are not re-asked. Q02 adds **new evidence** for item 1 (lock-early). The new evidence is her own decision pattern, not a model result. Q09 adds new evidence to the open wrong-way-risk item (brief section 9).

5. Questions (25), with reasoning

Format: question · anchor · hypothesis (a GUESS unless marked) · what changes · deliverables · domain · seed.

Q01. Is the case pull quote Laura's own words, and how may the team use it?

- Anchor: R-C21, R-AN30: "The only person who needs to believe in something is yourself." (no attribution line in the text layer).
- Hypothesis (guess): it is Wharton's framing, or a line from her books or a talk. It is not in 25 of her public sources (UNATTRIBUTED).
- Why it matters: if it is hers, the co-sponsor task reads as her own conviction being turned into public proof. If it is Wharton's, it is the case designer's theme.
- Her record in fact shows reliance on others: "with enough trust in people around me" (MV) and co-organised, partner-funded projects. So the quote may over-state her solo self-belief.
- Changes: every deliverable sentence that uses it says "the case profile highlights...", not "Laura says...". Remove any pitch that hangs on the quote.
- Deliverables: IPS, FR · D7 · seed 15.

Q02. Does Laura's own leap (a tentative plan first, notice given only after the book deal was signed) show that her "thoughtful risk" means risk on top of a secured base? And should the IPS describe her risk tolerance that way instead of "high risk appetite, so more equity"?

- Anchor: R-C37 ("willing to take thoughtful risks"), R-C38; quotes #1, #4, #5.
- Hypothesis (guess, strong): yes.
 - This is new evidence for brief section 8.1 (lock-early). It is Laura's revealed behaviour, not a simulation.
 - It also argues for the lower end of the open 50/60/70% sleeve-equity choice. Her risk appetite is spent on her career and the residency, and the median barely changes (F-4xx: \$205k/\$207k/\$209k).
- Changes (sentence): the IPS risk sentence frames willingness vs capacity with this pattern.
- Changes (decision): the sleeve equity weight. Also one TN reflection could name the "secure first, then grow" logic.
- Deliverables: IPS, TN, FR · D6 · seed 33.

Q03. Which parts of our plan would she call "gimmicks"?

- Candidates: EWT/Taiwan tilt, "two-clock sleeve" labels, 200k-path Monte Carlo in the main text, thematic ETFs.
- Anchor: R-S24 ("clear and creative investment thesis"); quote #21 "No more gimmicks."
- Hypothesis (guess): a Taiwan equity tilt and any branded framework names fail her test. Plain names (the promise portfolio and the growth portfolio) pass.
- Changes (decision): keep or drop EWT (brief section 9) and branded labels, applied now in WInS. Also the complexity budget of the FR.
- Deliverables: WInS-now, IPS, FR · D9 · seed 34.

Q04. Should the IPS open with a one-sentence purpose statement, and treat each security as a replaceable "medium" (primary pick plus alternates) serving a fixed purpose?

- Anchor: R-I9 ("What is the central idea behind your investment strategy?"), R-I14; quotes #22, #23.
- Hypothesis (guess): yes. It mirrors her craft habit and the IPS guide's focus on "strategy and decision-making framework ... rather than describing individual investments" (R-I23).
- Changes (sentence order): the IPS opening. TN reflections name the purpose each trade serves.
- Deliverables: IPS, TN · D9 · seed 32.

Q05. What is the right way to describe her quantitative background?

- The sources differ: case/resume "Statistics & Information Decisions", her 2022 words "a degree in business analytics", PQ 2018 "Economics with a concentration in Business Analytics".
- Anchor: R-C12.
- Hypothesis (guess): all are consistent with a Wharton BS in Economics with concentrations. Use the case wording.
- The deeper point: she is a data *practitioner* (SQL, R, Tableau, A/B tests, "+4.5M DAU"). She will value testable claims and experiment-style evidence more than theory.

- Changes (sentence): "as a statistics graduate, Laura..." framing in the FR. Avoid over-claiming what her degree means.
- Deliverables: FR · D7 · seed null.

Q06. What would a former financial-literacy teacher of high-schoolers (MoneyThink), consumer-protection intern (CFPB) and trading intern (Belvedere) check first in a student plan?

- Anchor: R-S25 ("presents recommendations that can earn her confidence"); R-W29 (CFA Asset Manager Code); BS-01 (no fee assumption); PQ facts.
- Hypothesis (guess): fees and costs, conflicts, and whether a 16-year-old could explain every sentence.
- Changes (decision): add an explicit fee assumption (about \$17k lower median surplus at 0.5%/yr, per BS-01) and a jargon ban list. Changes (number): the projections net of fees.
- Deliverables: IPS, FR · D8 · seed null.

Q07. Which common student-pitch phrases sound like the "ideal Whartonite" voice she rejected, and should the team ban them?

- Examples: "maximise returns", "alpha", "conviction picks", "beat the market", "aggressive growth".
- Anchor: R-S28 ("an authentic team voice"); PQ "loved building apps, not excel models"; PS on "McKinsey or Goldman Sachs".
- Hypothesis (guess): yes. A short banned-phrases checklist for TN reflections and the pitch costs nothing and removes the most common authenticity failure.
- Changes (sentence): style checklist for all three deliverables.
- Deliverables: TN, IPS, FR · D9 · seed 34.

Q08. Given two publication slips (about 9 and 12 months) and a 2025 "short break from writing", should the stress tests include a 2028 deposit that arrives late (for example in 2029)? And what does lock-early do in that case?

- Anchor: R-C27 ("She will contribute an additional \$150,000 at the beginning of 2028, using earnings from publishing advances..."); BS-13; section 2g.
- Hypothesis (guess): the case says "will", so it is a sensitivity, not the base case.
 - Under lock-early, lateness only delays and shrinks the growth sleeve.
 - Exception: if rates fall before January 2027 and the ladder is topped up from the 2028 deposit, a late deposit leaves the top-up unfunded for a year. This is the brief section 9 joint-tail item.
- Changes (number): add a "deposit 12 months late" scenario. Changes (sentence): "timing of cash flows" assumption in the IPS/FR.
- Deliverables: IPS, FR · D3 · seed 16.

Q09. Is Laura's 2028 income more exposed to political and budget cycles than to the stock market?

- Evidence: corporate DEI/ERG talks at Asana, Adobe and Google; school visits; a 2024 school-board ban of *Messy Roots*.
- Anchor: R-C27; brief section 9 (wrong-way risk); SG "Previous talks..."; Book Riot 2024-01-12.
- Hypothesis (guess): partly. Her channels are sensitive to DEI rollbacks (2025 corporate cuts: SNIPPET-UNVERIFIED) and book challenges, which correlate only loosely with equities.
 - So an equity "media/publishing underweight" hedges little.
 - The real hedge is what we already do: the promise is independent of her income after January 2027.
 - Changes (decision): drop the human-capital sector underweight as a trade (less complexity).
 - Changes (sentence): the FR's human-capital paragraph names political/budget risk to the deposit.
- Deliverables: FR, IPS · D6 · seed 16.

Q10. If the residency is rural (she recommends Yilan and its mountain art residencies), which inflation index should the FR use for facility costs?

- Anchor: R-C39 ("In 2033, Laura plans to establish a collaborative creative residency in Taiwan."); case p.4 "effect of inflation on ... facility costs"; F-508 (Taiwan construction cost index); quotes #42, #43.

- Hypothesis (guess): the national construction cost index (about 3.5%/yr since 2021, F-508) is the defensible choice, with a stated caveat.
 - The case does not give a location.
 - Her public words make a non-Taipei setting plausible, but R-W4 says the scenario is fictional. So the FR must not say where she will build.
 - Changes (sentence): the facility-cost inflation assumption and its caveat.
 - Deliverables: FR · D4 · seed 19.

Q11. How should the FR and the co-sponsor draft describe Taiwan geopolitical risk?

- The problem: her founding public work was a protest against reducing a place to its worst headline.
- Anchor: R-C72 (fundraising draft), R-S28 ("clearly and credibly"); quotes #46-#48.
- Hypothesis (guess): name the risks once and precisely: currency, construction costs, operational continuity. Say what the plan does about each, and keep the "full picture" (why Taiwan is a strong location).
 - A dramatic "invasion scenario" paragraph would read to her as the framing she fought.
 - Staying silent would fail "credibly".
 - Changes (sentence): the Taiwan risk section's framing and length.
 - Deliverables: FR · D4 · seed 19.

Q12. Her institutions grow in stages after support is proven (Pride in Panels). Does that support describing the 2031 range as a bought floor plus an upside that grows with results?

- Anchor: R-C68 ("Teams must recommend the dollar range Laura should communicate"), R-C71; quotes #15, #16.
- Hypothesis (guess): yes for the floor-plus-upside framing. But the case fixes the decision at "the beginning of 2033" (R-C57), so any staging beyond 2033 is an FR suggestion only.
- Changes (sentence): the co-sponsor draft uses "a committed floor now; more as results come in", which is her own festival logic. Possible decision: whether to recommend part-payment of the facility contribution.
- Deliverables: FR · D5 · seed null.

Q13. If the residency follows her free-access ethos (free festival; pro-bono visits "on need-basis"), program fees will be small. Does that make the operating commitment and the post-2033 flexibility buffer more important?

- Anchor: R-C44 ("Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors, grants, collaborators, program fees..."); BS-17; quotes #17, #18.
- Hypothesis (guess, interpretation): yes. Operating money is the hardest to raise (BS-17), and a low-fee model reduces one listed source. That is one more reason to keep a real flexibility share.
- Changes (sentence): the flexibility purpose sentence. Changes (number): possibly the share of surplus kept (see Q14).
- Deliverables: FR · D5 · seed null.

Q14. Her most consistent stated value is keeping options open. Should the recommended flexibility share of the post-reserve surplus be higher than a typical team's, and by how much?

- Evidence: "avoid siloing yourself", "don't have to own anything", "fit work around life".
- Anchor: R-C58 ("committing all remaining assets could limit her financial flexibility as the project develops"), R-C84; quotes #34-#36.
- Hypothesis (guess): keep roughly 25-40% of the post-reserve surplus uncommitted. The exact number should come from the contribution-rule simulation (B1a lead: $\text{gift} = \min(\text{top}, \text{floor} + s \times (\text{residual} - \text{floor}))$). Her values support a lower s (e.g. 0.6-0.7 rather than 1.0).
- Changes (number): the facility share parameter s and the kept-flexibility dollars.
- Deliverables: FR · D3 · seed 33.

Q15. Should the team place a small first trade early and treat the Trading Notes as a dated experiment log (hypothesis, test, refine)?

- Her method: "start small and post", deadlines "finished or not", A/B testing.

- Anchor: R-C88 ("reflected, tested, or refined the team's developing strategy"), R-T9, R-W82 ("at the time it is made"); quotes #26, #27.
- Hypothesis (guess): yes. The first ladder/duration trade early (after checking the position limit) gives a real note to refine later. Waiting for a perfect plan wastes the window (Sep 28-Nov 6).
- Changes (decision): timing and sizing of the first WInS trades. The note template gets a "what we are testing" line.
- Deliverables: WInS-now, TN · D10 · seed null.

Q16. Would a candid "anti-resume" in the Articulation section earn more of her respect than a smooth success story?

- Content: last year's rule breach, rejected ideas, WInS mistakes.
- Anchor: R-S27 ("reflects meaningfully on the team's growth and response to challenges"), R-W83; DP 2019 ("33 job rejections"); PS on the Anti-Resume Project.
- Hypothesis (guess): yes, provided it is specific and shows what changed. She co-built a project to "normalize failure".
- Changes (sentence/decision): the FR Articulation structure. Possibly one TN reflection on a trade that did not work as intended.
- Deliverables: FR, TN · D7 · seed null.

Q17. Should the team keep a dated decision diary from day 1 (trade notes plus a weekly log), as she kept a daily comics diary, so the FR can show growth from primary records?

- Why now: WInS notes cannot be edited.
- Anchor: R-T28 ("Include each Trading Note exactly as it appears in WInS."), R-W82; quote #30.
- Hypothesis (guess): yes. It is low cost and feeds Articulation (R-S27) with evidence rather than memory.
- Changes (decision): a process rule this week.
- Deliverables: WInS-now, TN, FR · D10 · seed null.

Q18. Should the FR express results in what they secure (for example "all ten years of operating payments already bought") next to dollars?

- Anchor: R-S28 ("communicates Laura's potential facility contribution and investment uncertainty clearly and credibly"); BS-09; quote #39 "it's just a number".
- Hypothesis (guess): yes, as long as it stays inside the case. No residency budget or resident counts, because teams "are not expected to ... develop a detailed business plan".
- Changes (sentence): headline metrics in the FR and the co-sponsor draft ("years of operations funded", funded ratio).
- Deliverables: FR · D9 · seed 32.

Q19. How big should the trade-offs and assumptions disclosure be?

- She wanted an author's note on trade-offs but also "wish I didn't always have to explain myself".
- Anchor: case p.4 ("Teams should identify and explain their assumptions..."), R-I31; quote #33.
- Hypothesis (guess): one compact, clearly labelled trade-offs box in the FR, and one sentence in the IPS. No pages of caveats.
- Changes (sentence/decision): FR structure and word budget.
- Deliverables: FR, IPS · D9 · seed null.

Q20. She doubted her own launch hype at 19. Should the team state the growth sleeve's AI and mega-cap concentration plainly and leave the choice to her, rather than tilt toward or away from AI?

- Anchor: R-S24 ("appropriate diversification"); quote #32 "how long will this novelty factor last?"; seed 20.
- Hypothesis (guess): yes. Broad index, a plain disclosure of the concentration share, and no thematic AI fund.
- Changes (decision): no AI thematic ETF in WInS. Changes (sentence): one disclosure line in the FR.
- Deliverables: WInS-now, IPS, FR · D2 · seed 20.

Q21. Which pronoun should the deliverables use?

- Facts: her site lists "she/her and they/them"; the case uses "she".
- Anchor: R-C10 (case biography uses "she"); HEY page.
- Hypothesis (guess): use "she", consistently. It matches the case and one of her stated pronouns. Never mix within a document.
- Changes (sentence): the style-guide line. It is a small thing that careless teams get wrong.
- Deliverables: TN, IPS, FR · D9 · seed null.

Q22. Would using her book titles as metaphors read as authentic or as decoration?

- Examples: "strong foundations for messy roots", "lessons for falling".
- Anchor: R-S28 ("authentic team voice"); brief section 4 (tokenism rule); quote #21.
- Hypothesis (guess): decoration.
 - The prior team review suggested such frames. Her own "No more gimmicks" and the tokenism rule argue against puns in the pitch.
 - Echo her *principles* in plain words instead (secure first, then leap; start small).
- Changes (decision): drop book-title puns from the elevator pitch and headings.
- Deliverables: IPS, FR · D9 · seed 17.

Q23. Does her own way of finding community (co-living and co-working spaces) support stating the case's reserve-first order as a people-first design?

- Anchor: R-C40 ("temporarily live, work, teach, and collaborate"), R-C50/R-C57 order; quotes #44, #45.
- Hypothesis (guess): yes. The operating money pays for the community; the building is the container. That makes the operations-first order read as hers, not only as a rule.
- Changes (sentence): the IPS protection principle.
- Deliverables: IPS, FR · D5 · seed 17.

Q24. Should the IPS include one pre-committed "bad-year rule" tied to purpose, instead of generic "stay the course" language?

- Her coping habit: she kept a supporter's note "saved for every time I felt bad".
- Example rule: "if equities fall X%, we do nothing to the promise portfolio; we rebalance the sleeve at the next review".
- Anchor: R-I3 ("make disciplined decisions"), R-I13; quotes #8, #9.
- Hypothesis (guess): yes. One rule, one number, one reminder that the ten payments are already bought. That is her own pre-commitment style.
- Changes (sentence): the IPS decision rule. Changes (number): the rebalancing band (brief section 9 open item).
- Deliverables: IPS, FR · D6 · seed 27.

Q25. How should the team handle conflicts between secondary sources about her?

- Examples:
 - Kirkus places the comic after a March 2021 event, although the comic was posted in March 2020.
 - Some bios say she moved at four, Inverse says three.
 - The Maverick transcript link serves another episode.
 - A Wharton staff quote on the PQ page is easy to mistake for hers.
- Anchor: R-S25 ("demonstrates a thoughtful understanding of the client"); R-W30 (cite sources).
- Hypothesis (guess): cite only her own words or primary pages, and flag conflicts rather than pick silently. Client Knowledge is scored, and errors about the client are the cheapest way to lose her confidence.
- Changes (sentence): the FR's client section cites primary sources only.
- Deliverables: FR · D10 · seed null.

Parked (fail the "would change something" filter now):

- Her favourite books and travel lists.
- Her identity themes as investment screens (the tokenism rule; the case names no screens, brief section 8.19).

- Her real-life residence (privacy).
- Finale Q&A (out of scope).

6. Leads for the specialists (sources and facts to use)

- **D6 (client psychology):**
 - OA 2021 (#1-#3, #8, #9, #13, #37, #39) and IN 2022 (#4, #5, #34) are the core risk-pattern sources.
 - The MV transcript PDF (AI transcription) covers the leap decision, Taiwan and community.
 - Use these for Q02 and Q24. Behavioural terms to research: "regret minimisation", "pre-commitment devices", "mental accounting / goals-based buckets".
 - **D3 (quant):**
 - Q08: add a "2028 deposit 12 months late" scenario and the joint tail "rates fall before Jan 2027 + late deposit".
 - Q14: run the contribution-rule $s = 0.5/0.7/1.0$ from B1a and report the kept-flexibility dollars.
 - Publication-slip evidence is in section 2g.
 - **D5 (philanthropy):**
 - The Pride in Panels model: free event, library partner (SFPL Hormel Center), mini-grants to 20 creators, biennial, and expanded in 2026 (LNM 2026-02-12, VERIFIED-PRIMARY; 48hills.org 2026-02, same facts).
 - Use it for Q12, Q13 and Q23. Also research how small artist residencies fund operations with low fees.
 - **D4 (Taiwan):**
 - MV transcript: she lived mostly in Taipei while writing, rode the northern and eastern Huandao, and recommends Yilan and its mountain art residencies.
 - Keep to cost and currency facts (F-508, F-510). Never state her real plans (R-W4).
 - Framing guidance for Q11: NPR 2020 and HuffPost 2020 quotes #46-#48.
 - **D2 (equity):** Q20 hype scepticism (DP16 #32). Q09 says the human-capital sector underweight hedges little. Check the broad index's top-10 weight for the disclosure line.
 - **D7 (Wharton intent):**
 - Q01: try to locate the pull quote in *Messy Roots* or *Kirby* text (library preview or Google Books search), or in a Wharton talk.
 - Wharton Magazine tag page <https://magazine.wharton.upenn.edu/tag/laura-gao/> lists two items (2022-04-06 interview; "Alumni Book Roundup: Spring 2022", 2022-05-03).
 - **D8 (practice):** Q06. She has trading-floor, consumer-protection (CFPB) and financial-literacy teaching experience (PQ). Pair with the CFA Asset Manager Code F.2/F.4.d disclosures (stakeholder map W-1).
 - **D9 (plain English):** Q07 banned-phrase list; Q21 pronoun; Q22 no book-title puns; Q19 one trade-offs box; Q04 purpose-first opening.
 - **D10 (compliance and process):**
 - Q15 early small first trade (after reading the Session Rules position limit); Q17 decision diary; Q25 source discipline.
 - Tooling: `fetch_text.py` returns 403 on poetsandquantsforundergrads.com with its Chrome user agent, and its HTMLParser drops the body text on that page. Suggested fix: retry with curl's default user agent, and fall back to regex tag stripping when the output is under ~20 lines.
 - **Not read (retry if access widens):**
 - Diverse Books Q&A (captcha).
 - Ampersand Magazine 2021 profile (TLS error).
 - YouTube interview on the book ban (video).
 - Wainao video (Taiwan outlet).
-

7. Sources (all accessed 2026-09-27; status)

Source	URL	Piece date	Status
Poets&Quants for Undergrads, 2018 Best & Brightest	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	VERIFIED-PRIMARY (via ft2 fallback; see section 1)
Wharton Magazine interview	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	2022-04-06	VERIFIED-PRIMARY
Overachiever Magazine interview	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	VERIFIED-PRIMARY
Input/Inverse feature	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel	2022-03-07	VERIFIED-PRIMARY
Pulse Spikes profile	https://pulsespikes.org/story/laura-gao	2021-06-18	VERIFIED-PRIMARY
Maverick Show Ep 190 show notes	https://www.themaverickshow.com/podcast/190-messy-roots-queer-identity-the-beauty-of-wuhan-and-writing-a-graphic-memoir-while-traveling-the-world-with-laura-gao/	2022-06-23	VERIFIED-PRIMARY
Maverick Show Ep 190 transcript (AI)	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23	VERIFIED-PRIMARY (machine transcript)
The Honey Pop interview	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	2022-06-02	VERIFIED-PRIMARY
The Nerd Daily Q&A	https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	VERIFIED-PRIMARY
BookWeb Indies Introduce Q&A	https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512	2022-02-07	VERIFIED-PRIMARY
NPR "The Wuhan I Know"	https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines	2020-04-04	VERIFIED-PRIMARY
HuffPost	https://www.huffpost.com/entry/chinese-american-illustrator-life-in-wuhan_n_5ea9c739c5b63115cec2be5a	2020-04-30	VERIFIED-PRIMARY
Kirkus interview	https://www.kirkusreviews.com/news-and-features/articles/laura-gao-messy-roots-interview/	2022-03-06 (per Wikipedia citation; SNIPPET for the date)	VERIFIED-PRIMARY (read; mostly personal, not quoted)
Local News Matters, Pride in Panels 2026	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	2026-02-12	VERIFIED-PRIMARY
48 Hills, Pride in Panels 2026	https://48hills.org/2026/02/pride-in-panels-fest-inks-in-queer-comics-brilliance/	2026-02	VERIFIED-PRIMARY (context only)
Daily Pennsylvanian Q&A (Draw Street Journal)	https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a	2016-01-28	VERIFIED-PRIMARY
Daily Pennsylvanian, anti-resumes	https://www.thedp.com/article/2019/03/penn-anti-resume-ocr-recruiting-failure-signal	2019-03-17	VERIFIED-PRIMARY
34th Street, The Signal	https://www.34st.com/article/2018/04/the-signal-penn-club-squirrels-without-morality-cards-against-humanity-pre-professional	2018-04	VERIFIED-PRIMARY (no Laura quote)
Publishers Weekly Rights Report	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.html	2020-07-21	VERIFIED-PRIMARY
Book Riot, book bans	https://bookriot.com/book-banning-will-not-stop-at-schools/	2024-01-12	VERIFIED-PRIMARY (Book Riot's report)
lauragao.com: hey, projects, data-journalism, speaking-events, messyroots, kirby, editorial	https://lauragao.com/hey/ etc.	current	VERIFIED-PRIMARY
Publicity & Speaking Guide PDF	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	~2024 (undated)	VERIFIED-PRIMARY
Resume PDF	https://lauragao.com/s/Laura-Gao-Resume.pdf	current (undated)	VERIFIED-PRIMARY
Wikipedia, Laura Gao	https://en.wikipedia.org/wiki/Laura_Gao	edited 2026-08-31	Read; secondary; used only for citation dates
NüVoices #72 page	https://nuvoices.com/podcast/nuvoices-podcast-72-messy-roots-a-conversation-with-laura-gao	2022-03-17	Show notes only

Source	URL	Piece date	Status
Corporate DEI cuts 2025 (HR Dive, Forbes, etc.)	search results only	2025	SNIPPET-UNVERIFIED
Pull-quote exact-phrase search	WebSearch	-	SNIPPET: no match to Laura
Diverse Books Q&A	https://legacy.diversebooks.org/qa-with-laura-gao-messy-roots-a-graphic-memoir-of-a-wuhanese-american/	-	BLOCKED (captcha)
Ampersand Magazine profile	https://ampersand-mag.com/laura-gao	2021-03-04 (per Wikipedia)	BLOCKED (TLS)

Verbatim checks: 68 phrases were checked; all returned PHRASE FOUND VERBATIM: YES. The list is in the scratchpad file B8a_quotes.tsv (not in the repo). Section 2 holds every phrase used.

What this teaches

- A client's own words are data, but only if you check them. Three lessons came out of this lens:
- The most famous line in the case could not be traced to her, so it must be quoted as the case's words.
 - A person's *story* ("she took a big risk") can hide their *behaviour* ("she took it after the contract was signed"). Behaviour is the better guide to what she will trust.
 - The strongest client insight here is not a clever metaphor. It is noticing that the plan already works the way she has lived: secure the base, then leap; start small; keep options open; and tell the full story, including the trade-offs.

B8b Voice of Laura: her own platforms, author Q&As and festival press

Agent B8b, insight_v1 run, written 2026-09-27. Lens: Voice of Laura. Starting angle: her own website (lauragao.com and the PDFs it hosts), publisher and author Q&As, organisations she founded or co-runs, and her words quoted in reputable press. Professional and public statements only. I left out family, partner, name history, contact details and other personal-life material, even where a source contains it. Nothing here is text for submission. The team writes its own deliverables.

0. Summary (read this first)

1. **The case's pull quote is not verified as Laura's words.** The line is "The only person who needs to believe in something is yourself." (R-C21). I looked for it on 14 primary pages where she speaks (her site, 5 PDFs and pages on it, 9 interviews and features). I also searched all 195 pages cached by this run and ran two web searches. The phrase was not found anywhere. The nearest verified idea is from Wharton Magazine (2022): "it's not for anyone else to judge". Until D13 finds a source, treat the line as Wharton's sidebar copy. Do not present it as "Laura said".
2. **Her biggest career risk was taken floor-first.** She named the "unstable income" of self-employment herself (2021). Input reported (2022) "So when she got the book deal with HarperCollins, she gave Twitter her notice." She had also planned to "teach art part-time". So her "thoughtful risks" (R-C37) were jumps made **after** a contract secured the downside. This mirrors lock-early (promise bought first, risk only with the surplus). It is the single strongest "she would recognise this as hers" link I found. That sentence is interpretation; the facts are VERIFIED-PRIMARY.
3. **"No more gimmicks."** (Nerd Daily, 2025.) She scrapped three drafts of her second book because she "lost the heart of my writing—authentic, personal connections". This is her own test for inauthenticity. It supports the design principle directly: themed branding, book puns and extra model layers are exactly what she discards.
4. **She writes a purpose statement for her work.** Pulse Spikes (2021): "I can't think of many art pieces that I didn't also write some kind of purpose statement for." An IPS is a purpose statement. A trading note that opens with the holding's job fits how she already thinks.
5. **Her books ran late, and she paused writing in 2025.** Messy Roots was announced for "summer 2021" and published 8 March 2022. Kirby was hoped for "a 2024 release" and came out by March 2025. In March 2025 she said "I'm taking a short break from writing". I found no public announcement of a new book deal as of 2026-09-27; absence of evidence is not evidence. The case says the \$150k "will" come from publishing advances and other sources in January 2028 (R-C27). This gives a Laura-specific size for the "late deposit" stress case (BS-13): about 9-12 months. The case base stays unchanged.
6. **She has already built a community institution with partners, with free access.** Pride in Panels is "free for exhibitors and attendees", run with the San Francisco Public Library, and gave "mini grants to 20 comic creators". Her stated rule: "Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on". Her speaking guide offers "Pro-bono visits". Implication for the case's list of outside sources (R-C44): the residency's "program fees" may be deliberately low. More of the extra operating support would then fall on co-sponsors, which argues for keeping flexibility.
7. **Her institutions start small and scale after proof.** On The Sign.al: "Only after we realized EVERY student deserved to hear their words of wisdom did we decide to build something". On the festival: "All of it is way beyond our initial passion project." Her advice: "Always start small". This suggests a staged facility contribution (a firm floor now, more "as the project develops") rather than one lump. D5 to test.
8. **What she would find inauthentic** (from her own words and choices): polish over substance ("I don't need to be perfect to be impactful"), gimmicks, numbers without the human behind them ("it's just a number"), pressure to conform to an "ideal Whartonite" (P&Q, snippet-level), and borrowing her identity-context advice for finance. One verified quote maps perfectly onto "buy the floor before the 2031 reveal", but it is coming-out advice. Using it as a finance metaphor is the tokenism trap. Echo her **career** sequencing instead.

28 questions follow (section 3), 40 quotes (section 2), and leads for the specialists (section 5).

1. Method

- Pages read with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL`. Every quote marked VERIFIED-PRIMARY returned `PHRASE FOUND VERBATIM: YES` with `--grep` on 2026-09-27 (all access dates below are 2026-09-27).
- **Exceptions:** Poets&Quants loads with curl (HTTP 200, 169 kB), but the helper's text parser returns almost nothing and its grep says NO. I read the quotes in the raw HTML by hand. By the run's rule they are SNIPPET-UNVERIFIED until D13 re-checks them (lead L-1).
- Two kinds of wording are marked in the table: journalists' sentences about her (not her voice) and publisher copy on her site (not necessarily her voice).
- **Blocked or empty:**
 - legacy.diversebooks.org (WNDB Q&A): an sgcaptcha redirect (HTTP 202), despite the brief listing it as reachable.
 - mochimag.com: captcha.
 - harpercollins.com and harpercollins.ca: HTTP 403.
 - ampersand-mag.com: TLS error.
 - medium.com: 403.
 - prideinpanels.org home and /about: 404.
 - nuvoices.com podcast post: 404.
 - thesign.al: empty page.
 - The Maverick Show and NüVoices are audio only, with no transcript found.
 - SLJ's comic is an image.

2. Quotes register (exact wording; status; context; value; implication)

Status key: VP = VERIFIED-PRIMARY (helper grep YES, 2026-09-27); SU = SNIPPET-UNVERIFIED; PU = PARAPHRASE-UNVERIFIED (attribution unverified). "Not her voice" marks a journalist's or publisher's words.

#	Exact text	Source (piece date)	Status	Context	Value	Implication
V1	"The only person who needs to believe in something is yourself."	Case PDF p.1 sidebar (R-C21); no web source found	PU (case text is VERIFIED-REPOFILE; attribution to Laura unverified)	Sidebar pull quote with no attribution line	Self-conviction	Do not write "Laura says..." (Q1).
V2	"living proof that doodling on your homework can actually "get you somewhere" (take that, Mr. Dutton)!"	lauragao.com/hey (live page, ©2026)	VP	Her own bio	Self-deprecating humour	Her register is warm and plain; a stiff tone reads as not-her (Q28).
V3	"Their work explores the hilariously messy intersections of queer immigrant stories"	lauragao.com/hey	VP	Own bio; uses they/their	Messiness plus humour	She uses she/her and they/them; the case uses "she" (Q28).
V4	"What motivated you to quit a secure job to become an artist-writer in the middle of a pandemic?"	lauragao.com Publicity & Speaking Guide PDF (undated; says Kirby "will be released in 2025", so before 2025)	VP	A talk topic she advertises: "Taking the Leap..."	She frames her story as leaving security	Her leap is her public brand; the portfolio should not add a second leap on the promise (Q3).
V5	"Pro-bono visits can be offered on need-basis."	Publicity & Speaking Guide PDF	VP	Speaking fees	Access over revenue	Speaking income is partly given away; residency fees may be low (Q8, Q13).
V6	"Tired of authors talking your students to sleep? Choose an interactive hands-on activity!"	Publicity & Speaking Guide PDF	VP	Pitch to schools	Humour; hates dull talks	A wall of text will not earn her confidence (Q19, Q28).

#	Exact text	Source (piece date)	Status	Context	Value	Implication
V7	"Why must graphic novels be in competition with prose novels? How can kids benefit from both?"	lauragao.com Educators & Book Clubs Guide PDF, "A note from the author" (c.2022)	VP	Her reply to educators who question teaching comics	Both/and thinking	The safe (non-identity) echo for "growth AND protection" (Q21).
V8	"Write a letter to Future You."	Educators Guide PDF, ch.10 exercise, among "activities I've created"	VP	Classroom exercise she designed	Future-self pre-commitment	Written rules for a future bad year may feel natural to her (Q27). Gimmick risk.
V9	"Our bold interactive projects aim to help students discover their passions and live deliberately."	lauragao.com/editorial	VP	Describes The Sign.al, which she co-founded	Deliberate choice	"Deliberate" is her word for a good life; the IPS is a deliberate rulebook (Q18).
V10	"Only after we realized EVERY student deserved to hear their words of wisdom did we decide to build something to cultivate it."	signaltemp.github.io (Sign.al 5-year page, linked from lauragao.com/editorial; undated, about 2022, ASSUMPTION from a Jan-2017 launch)	VP	Founder's note	Validate, then scale	Staged facility contribution (Q6).
V11	"I couldn't be more proud of the current team keeping the legacy strong."	Same page	VP	Founder's note	Institutions that outlast the founder	Frame flexibility money as helping the residency last (Q11).
V12	"Can Kirby pull off her greatest balancing act yet?"	lauragao.com Kirby page (publisher-style copy)	VP (not necessarily her voice)	Book blurb; heroine is "the Queen of Balance"	Balance	Do not build a pun on it (Q15); note only.
V13	"I scrapped three completely different drafts before landing on this one."	The Nerd Daily Q&A, 2025-03-08	VP	Writing her second book	Iterates and discards	Showing discarded strategies is her own habit (Q24).
V14	"But in the process, I lost the heart of my writing—authentic, personal connections."	Nerd Daily, 2025-03-08	VP	Why she scrapped the drafts	Authenticity	Her test for inauthentic work (Q15).
V15	"No more gimmicks."	Nerd Daily, 2025-03-08	VP	Same passage	Anti-gimmick	Strongest support for the design principle (Q15, Q16).
V16	"Every character is grappling with how to balance society's expectations with their own desires."	Nerd Daily, 2025-03-08	VP	Theme of Kirby	Balance under pressure	Echo only lightly (Q21).
V17	"I'm taking a short break from writing to focus on teaching my Writing Comics course at California College of the Arts!"	Nerd Daily, 2025-03-08	VP	Plans after Kirby	Income mix shifted toward teaching	The 2028 advance pipeline is unclear (Q13).
V18	"I wish I wrote about these nuances and tradeoffs made in an author's note at the end. But at the same time, I wish I didn't always have to explain myself."	BookWeb Indies Introduce Q&A, 2022-02-07	VP	What she would change in Messy Roots	Wants trade-offs disclosed, but not over-explained	A short trade-off note, kept out of the main text (Q20).
V19	"It's not an easy thing to say, 'let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.'"	Overachiever Magazine, 2021-03-15	VP	On leaving Twitter	Knows her income is unstable	Her own words support human-capital reasoning (Q3, Q13).

#	Exact text	Source (piece date)	Status	Context	Value	Implication
V20	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	Overachiever, 2021-03-15	VP	Same	Regret of omission	Regret framing for the growth sleeve (Q4, Q5).
V21	"I did 50 things, and maybe I didn't like fully excel at each, but I got to try all of them and have so much fun with them."	Overachiever, 2021-03-15	VP	How she wants to look back at 80	Optionality, many ventures	Keep part of the surplus uncommitted (Q7).
V22	"it's great having thousands of likes, but at the end of the day, it's just a number. You never really understand the human connection behind it"	Overachiever, 2021-03-15	VP	On online reactions	Humans over metrics	Report dollars as years of residency funded (Q19).
V23	"I'd just be like, 'Hey, this is why you made this. This is who you're trying to help, and this is the positive effect you actually have.'"	Overachiever, 2021-03-15	VP	How she handled online abuse: re-reading a reader's note	Purpose-anchoring in bad times	Bad-year reports lead with "payments still covered" (Q19, Q27).
V24	"I thought, 'Well, if people aren't gonna give it to us, I'll just make one.'"	Overachiever, 2021-03-15	VP	On her next book	Build what is missing	Matches R-C24/R-C40 (the residency). Context only.
V25	"I'm currently in Taiwan, Taipei."	Overachiever, 2021-03-15	VP	Where she was working as a comic artist	Worked in Taiwan	Public professional fact; never infer where she lives now (Q10 lead; B1b Q28 already asks).
V26	"Instead, I stand up to the beast by admitting I don't yet know the 'right' answer, but whatever it is, it's not for anyone else to judge."	Wharton Magazine, 2022-04-06	VP	Describing the climax of Messy Roots	Honest about uncertainty; self-judged	The nearest verified relative of V1. Admitting "we don't know the exact 2033 number" is on-brand (Q1).
V27	"Soon enough, I had a comics diary that I could not only decompress into, but also take pride in how far I'd come when I reread it."	Wharton Magazine, 2022-04-06	VP	Daily comic habit while a PM	A progress log	A funded-ratio log for Laura (BS-09) fits her habit (Q19).
V28	"Now I get to fit work around life instead of the other way around, which is such a game-changer"	Input Mag, 2022-03-07 (her quote in a feature)	VP	Life after going full-time	Autonomy	Flexibility = freedom to choose (Q7).
V29	"So when she got the book deal with HarperCollins, she gave Twitter her notice."	Input Mag, 2022-03-07	VP (not her voice: reporter)	Sequence of her career jump	Floor-first leap	Key evidence for Q3.
V30	"She's hoping to finish in time for a 2024 release."	Input Mag, 2022-03-07	VP (reporter)	Second book timing; it came out by March 2025 (V17 page)	Publishing timelines slip	Size of the "late deposit" case (Q12).

#	Exact text	Source (piece date)	Status	Context	Value	Implication
V31	"Publication for the first book is planned for summer 2021"	Publishers Weekly Rights Report, week of 2020-07-20	VP (trade press)	HarperCollins deal; the book was published 2022-03-08	Timelines slip	Q12.
V32	"The quicker you get over your perfectionism, the faster you'll finish projects, get feedback, improve, and overcome imposter syndrome or "artist stage fright"."	Geeks OUT, 2022-05-11 (same words in The HoneyPOP, 2022-06-02)	VP	Advice to artists	Ship, then iterate	Place first WInS trades early and refine (Q25).
V33	"I give myself a deadline for when I must post the art, finished or not."	Geeks OUT, 2022-05-11	VP	Same	Deadlines over perfection	A no-timing purchase rule (Q5).
V34	"Ultimately my goal is to tell a story; I don't need to be perfect to be impactful."	Geeks OUT, 2022-05-11	VP	Same	Substance over polish	Anti-over-engineering (Q15).
V35	"I lived nomadically last year, splitting my time between Taiwan and Europe"	Geeks OUT, 2022-05-11	VP	Work pattern 2021	Worked in Taiwan	Same handling as V25.
V36	"Always start small and post that work, no matter how crappy it looks!"	The HoneyPOP, 2022-06-02	VP	Advice to webcomic artists	Start small	Staged facility contribution (Q6).
V37	"Go at your own pace and always make sure to be in an environment in which you feel safe and supported by allies before you make any big reveals."	The HoneyPOP, 2022-06-02	VP	Coming-out advice to young LGBTQ+ readers	Safety before a public step	DO NOT use as a finance metaphor (Q17). Record only.
V38	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	Pulse Spikes, 2021-06-18	VP	Her creative process	Purpose-first	IPS and trading notes as purpose statements (Q18).
V39	"That [story] was something I could prove better with data,"	Pulse Spikes, 2021-06-18	VP	Her "Rewriting Herstory" data project	Evidence through data	Charts must prove, not decorate (Q22).
V40	"All of it is way beyond our initial passion project. We're thrilled that the community really supported us and wanted us to continue to grow the festival,"	Local News Matters (Bay City News), 2026-02-12	VP	Pride in Panels 2026	Grew after community backing	Upside surprises; start small (Q6, Q10).
V41	"Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on,"	Local News Matters, 2026-02-12	VP	On the festival's mini grants	Access first	Residency program fees likely low (Q8).
V42	"We definitely hope to offer more as the festival grows."	Local News Matters, 2026-02-12	VP	Mini grants	Grow as resources allow	"As the project develops" (R-C58) (Q6).

#	Exact text	Source (piece date)	Status	Context	Value	Implication
V43	"There's a vibrant community of queer comic creators here in San Francisco, but why wasn't there any single event where we could celebrate that community and bring everyone together?"	Local News Matters, 2026-02-12	VP	Why she co-founded the festival	Build what is missing	Context for the residency.
V44	"That's just pretty unfair. For folks like us, we are made with the essence of the combinations that we have grown up in."	Mission Chronicle (Mission High School student newspaper; workshop 2025-03-06)	VP (student transcription)	About being between two cultures	Rejects either/or	Identity context: do not use for finance (Q17, Q21).
P1	"If what you want doesn't exist, create it yourself."	Poets&Quants, 2018 Best & Brightest	SU (read in raw HTML via curl; helper parser failed)	Biggest lesson from business school	Builder	Well known; many teams will quote it (Q16).
P2	"However, this lack of structure and certainty can be off-putting for students who are easily influenced by those around them or are risk-adverse."	Poets&Quants, 2018	SU (same)	Advice on studying business	She names the trade-off of uncertainty and risk aversion herself	She thinks about who can live with uncertainty (Q3).
P3	"Frankly, my first shock after arriving at Wharton was learning that "bankers" were not the people who exchanged checks for cash or chained pens to desks."	Poets&Quants, 2018	SU (same)	What surprised her	Humour about finance insiders	Finance jargon is foreign to her world (Q16).

(Quotes V1-V44 and P1-P3 = 47 rows. Forty of them go to the structured output, dropping lower-value rows.)

Facts used that are not Laura's words (VP unless stated):

- F-a: 48 Hills, Feb 2026: "California College of the Arts had been bought by Tennessee-based Vanderbilt University". The same article says Vanderbilt "officially swallows the school in 2027". This upgrades the team docs' CCA claim from UNVERIFIED to press-verified. Teaching is not a case income source (brief section 8.19).
- F-b: 48 Hills, Feb 2026, co-organiser Justin Hall: "The folks at the library have been super-generous, and this year we have enough money to bring in two special guests." Also: "the library approached the organizers to make it a biennial event". So an institutional partner funded growth after proof.
- F-c: Local News Matters, 2026-02-12: the festival stays "free for exhibitors and attendees" and gives "mini grants to 20 comic creators". The 2024 edition drew "More than 2,000 people" (this also verifies the team docs' "2,000+").
- F-d: Laura's public resume PDF (lauragao.com): "Experimented and shipped 10 features"; "Created weekly reports for leadership of findings and data visualizations using Tableau". She knows A/B testing and regular reporting. The resume does not list the Belvedere Trading or CFPB internships. The 2018 P&Q page does (SU).
- F-e: Timeline of Messy Roots: "planned for summer 2021" (PW, July 2020) → "coming out in January 2022" (Overachiever, March 2021) → "Next February" (Pulse Spikes, June 2021) → published 8 March 2022 (Publicity guide PDF: "Publication Date: March 8th, 2022"). Kirby: "2024 release" hoped (Input, March 2022) → "out now" (Nerd Daily, 2025-03-08).
- F-f: The Publicity guide lists coverage by "WAINAO MEDIA TAIWAN". She has had Taiwan-market press.

3. Questions (28), with reasoning

Format: Q, anchors, hypothesis (a GUESS unless stated), what it would change, deliverables, domain, seed, north star.

Q1. Is the case's pull quote Laura's own words, and may the team attribute it to her?

- Anchors: R-C21, R-AN30.
- Hypothesis (guess): it is Wharton's sidebar copy, or it comes from an unpublished conversation. It is not found on any of the 14 primary pages or 195 cached pages. The nearest verified idea is V26 ("it's not for anyone else to judge").
- Changes: a sentence. "The case highlights..." instead of "Laura says..." in the IPS pitch and FR. Also whether any theme is built on it.
- Deliverables: IPS, FR. Domain: D10 (with D13). Seed: 15.
- North star: a team that misattributes a quote to a statistics-trained, misinformation-fighting author loses her trust at once.

Q2. (Seed 15, new evidence.) When has Laura needed others to believe her, and what did she give them?

- Anchors: R-AN30, R-C64, R-C66.
- Hypothesis (guess): evidence first. The viral comic and the reader's note came before the book deal. The first festival's 2,000+ attendance came before the library's offer to expand. The Sign.al's interviews came before the club. So her pattern is "private conviction → small public proof → partners join". A bought 2031 floor is the finance version of "small public proof".
- Changes: a sentence. The FR fundraising logic becomes "proof, not promise".
- Deliverables: FR. Domain: D5. Seed: 15.
- North star: it shows our 2031 method is her own method.

Q3. Should the IPS define her risk tolerance as "high willingness, taken only after the downside is secured"?

- Anchors: R-C37, R-C38, R-AN16, R-AN17.
- Hypothesis (guess): yes. Evidence:
 - She left Twitter only once the HarperCollins offer existed (V29).
 - She had a fallback plan to "teach art part-time" (Input, VP).
 - She names "unstable income" herself (V19).
 - In 2018 she wrote about people who are "risk-adverse" facing uncertainty (P2, SU).
- Changes: the IPS risk-tolerance sentence (a decision on how risk tolerance is stated). It also justifies taking risk only in the surplus.
- Deliverables: IPS, FR. Domain: D6. Seed: 33.
- North star: this is the most "hers" reason for lock-early. It maps onto the one big risk she actually took.

Q4. Does her regret framing (V20: "you're always going to regret not doing it") argue for the upper end of the sleeve equity range, 60-70% rather than 50%?

- Anchors: brief section 9 (sleeve weight open); R-C38.
- Hypothesis (guess): it argues against the minimum. The verified sims show that 50/60/70% moves the median only \$205k/\$207k/\$209k, but moves p5 \$164k/\$159k/\$154k. Regret of omission is about missing upside (p95), which rises with equity. So a 60% base with the reason stated in her terms may be right. D6 must weigh the regret of a smaller facility against the embarrassment of a missed range.
- Changes: a number (the sleeve equity weight).
- Deliverables: IPS, FR, WInS-now. Domain: D6. Seed: null.
- North star: a weight chosen for her psychology, not a textbook 60/40.

Q5. Which rule for rates moving before January 2027 would she recognise as hers?

- Anchors: brief section 9 ("Rule if rates fall before January 2027"); R-C26.
- Hypothesis (guess): "buy on the deposit date, no timing", given V33 ("deadline... finished or not") and V20. A staged or wait-for-better-rates rule looks like perfectionism to her. The top-up then comes from the 2028 deposit (long rungs first).
- Changes: a decision (the IPS purchase rule).
- Deliverables: IPS, FR. Domain: D6 (with D1 for the cost numbers). Seed: 4.

- North star: a rule she can repeat in one breath.

Q6. Should the facility contribution be a staged rule rather than one lump sum on 1 Jan 2033?

- Anchors: R-C57, R-C58, R-C60, R-AN20, R-AN23.
- Hypothesis (guess): yes. "Always start small" (V36). Validate-then-build (V10). Growth as partners join (V40, V42; F-b). "Extent" allows a rule (R-AN20). The rule would be a floor contribution in 2033, with a stated share of what remains released as milestones are met.
- Changes: a decision on the facility-contribution structure (FR), and a sentence in the fundraising draft.
- Deliverables: FR. Domain: D5. Seed: null.
- North star: it matches how every institution she built actually grew.

Q7. What share of the post-reserve surplus should stay uncommitted?

- Anchors: R-C58, R-C84, R-C23 ("no shortage of ideas").
- Hypothesis (guess): a meaningful share, perhaps 20-40% (a GUESS for D5/D6 to test). The case warns against "committing all remaining assets". She says she wants to have "done 50 things" (V21) and to "fit work around life" (V28).
- Changes: a number (FR).
- Deliverables: FR. Domain: D5. Seed: 11.
- North star: flexibility defined by her own stated way of living, not by an arbitrary buffer.

Q8. Will the residency charge low or no program fees, given her free-access record?

- Anchors: R-C44 (the case lists "program fees" as a possible outside source).
- Evidence: the festival is free (F-c); her mini-grant priority (V41); pro-bono visits (V5).
- Hypothesis (guess): fees are likely modest. Extra operating support beyond her \$50k would then rely more on co-sponsors and grants. That strengthens keeping flexibility and pitching operating certainty to co-sponsors.
- Changes: a sentence in the FR assumptions ("we assume fees are not a material source"). Possibly the flexibility share (Q7).
- Deliverables: FR. Domain: D5. Seed: 6.
- North star: few teams will notice that her values shape the case's funding mix.

Q9. Should the fundraising draft be written for institutional partners (libraries, universities, cultural foundations) rather than wealthy individuals?

- Anchors: R-C72, R-C64, R-C83.
- Evidence: her one known co-funded institution runs on a public library partner that "approached the organizers" and funded guests (F-b).
- Hypothesis (guess): institutional readers want multi-year operating certainty and governance. That is exactly what her ten fully funded payments offer.
- Changes: a sentence (the audience and emphasis of the FR draft).
- Deliverables: FR. Domain: D5. Seed: null.
- North star: it shows we know who her partners actually are.

Q10. What do we tell co-sponsors she will do if 2033 lands above the top of the range?

- Anchors: R-C69, R-C70, R-AN9.
- Evidence: her projects have beaten expectations (V40; "wowed everyone", VP).
- Hypothesis (guess): state a rule. Any excess above the top goes to the facility in stages, or is kept as flexibility. Over-delivery then helps credibility instead of looking like a bad estimate.
- Changes: a sentence or decision in the fundraising draft.
- Deliverables: FR. Domain: D5. Seed: 9.
- North star: we handle both tails, as a statistician would expect.

Q11. Should the flexibility money be framed as helping the residency outlast her ten years?

- Anchors: R-C49, R-AN25, R-C58.
- Evidence: she values legacy (V11).

- Hypothesis (guess): yes, as one clause. Funding beyond 2042 is out of scope, but co-sponsors will ask about 2043.
- Changes: a sentence (FR).
- Deliverables: FR. Domain: D5. Seed: 7.
- North star: it speaks to what she is proud of, institutions that last.

Q12. How late should the "late 2028 deposit" stress case be?

- Anchors: R-C27, R-AN35, R-C85 ("timing of cash flows"); BS-13.
- Evidence: her two books arrived about 8-9 and about 12 months after the first announced timing (V30, V31, F-e).
- Hypothesis (guess): 9-12 months, i.e. the deposit arrives in early 2029. Under lock-early the payments stay safe; the facility range narrows by the lost sleeve growth.
- Changes: a number (the scenario delay) in D3's model and an FR sentence. The case base stays as written and the scenario is labelled as our own.
- Deliverables: FR. Domain: D3. Seed: 16.
- North star: stress tests drawn from her own record, not generic shocks.

Q13. With no public next-book announcement found (as of 2026-09-27), a writing break announced in March 2025, and her college employer changing hands in 2027 (F-a), what confidence label should the FR put on the 2028 \$150k?

- Anchors: R-C27, R-C44, R-AN31, R-AN35.
- Hypothesis (guess): keep the case's "will" as the base. Add a labelled scenario: the payments stay safe and only the facility range moves. Say so in one sentence using her own description of "unstable income" (V19), as a fact she stated, not a judgement. Also check for any 2026 deal announcement (D6/B3 lead).
- Changes: a sentence (FR). Possibly the probability weight in D3's wrong-way model.
- Deliverables: FR. Domain: D6. Seed: 16.
- North star: it respects her words and the case at the same time.

Q14. May the FR quote her own words about unstable income without it reading as intrusive?

- Anchors: R-S25 ("thoughtful understanding of the client"); brief privacy rule.
- Hypothesis (guess): yes, but only her public professional statement (V19), and only once. Never speculate about her finances.
- Changes: a sentence (FR).
- Deliverables: FR. Domain: D10. Seed: null.
- North star: depth without intrusion.

Q15. Which parts of our strategy or presentation would she call "gimmicks" (V15)?

- Anchors: R-S28 ("authentic team voice", "original thinking"); brief design principle.
- Candidates: themed names like "Messy Roots Portfolio", book puns ("balancing act", "falling safely"), identity-themed funds, a Monte Carlo layer where "bought" already answers the question, multi-factor jargon.
- Hypothesis (guess): cut all book-title branding. Keep one plain idea.
- Changes: decisions (a cut list). This also covers the team docs' "Strong foundations for messy roots" theme.
- Deliverables: TN, IPS, FR. Domain: D9. Seed: 34.
- North star: most teams will theme on her book titles; she explicitly rejected gimmicks.

Q16. Which words and moves would read as "finance-bro" to her?

- Anchors: R-S28; R-S25 ("earn her confidence").
- Evidence: V15; P3 (SU); she nearly left Wharton over the "ideal Whartonite" mould (P&Q, SU); her Anti-Resume and "Anti-Culture Culture Club" (V9).
- Hypothesis (guess): banned list: "alpha", "beat the market", "crush", "guaranteed" without qualifiers, bragging about P&L, unexplained acronyms. P1 ("create it yourself") will be quoted by many teams, so it is not distinctive.

- Changes: a style-guide decision.
- Deliverables: TN, IPS, FR. Domain: D9. Seed: 34.
- North star: sounding like her world, not Wall Street's.

Q17. Must the team avoid using her identity-context advice (V37, V44) as finance metaphors, even when they fit?

- Anchors: brief section 4 (tokenism rule); R-C10.
- Hypothesis: yes. V37 maps exactly onto "secure the floor before the 2031 reveal", but it is coming-out advice to young people. Using it for money is the inauthenticity she would notice first. Use her career sequencing (V29) and teaching note (V7) instead.
- Changes: a decision (a do-not-use list).
- Deliverables: IPS, FR. Domain: D9. Seed: 34.
- North star: the difference between understanding her and decorating with her.

Q18. Should every WInS holding and every Trading Note open with a one-line purpose ("this holding exists to...")?

- Anchors: R-T9, R-T10, R-I3, R-S24.
- Evidence: she writes a purpose statement for almost every art piece (V38).
- Hypothesis (guess): yes. It also enforces "cohesive portfolio" (R-T10) and is cheap to do.
- Changes: a decision on the TN structure and on the WInS trade-note habit now. WInS notes cannot be edited, only added to (G-ids), so the habit must start with the first trade.
- Deliverables: WInS-now, TN, IPS. Domain: D9. Seed: 32.
- North star: our documents are built the way she builds hers.

Q19. Should the FR's headline metric be "years of residency operations secured" (and "payments covered"), with dollars second?

- Anchors: R-C83, R-S28 ("uses data effectively"), BS-09.
- Evidence: V22 ("it's just a number"); V23 and V27 (progress logs, purpose anchors).
- Hypothesis (guess): yes. "10 of 10 years secured" plus the facility range.
- Changes: a sentence or number format (FR; a possible chart).
- Deliverables: FR. Domain: D9. Seed: 32.
- North star: numbers she can feel.

Q20. Should the FR carry a short "trade-offs we made" note, kept apart from the main argument?

- Anchors: R-C85, R-C86, R-S26.
- Evidence: V18 (she wished she had written an author's note on trade-offs, but dislikes always explaining herself).
- Hypothesis (guess): yes: 3-5 lines, e.g. what growth-first would have given at the median and what it risked.
- Changes: a decision (FR structure).
- Deliverables: FR. Domain: D9. Seed: null.
- North star: honest and uncluttered, both of which she asked for.

Q21. Can the central idea be stated as "both, not either-or" (a promise locked AND growth), echoing her teaching note (V7) rather than identity quotes?

- Anchors: R-C38, R-I17.
- Hypothesis (guess): yes. The case itself frames a "balance between pursuing growth and protecting". Our structure lets both run at once. V7 is her own non-identity phrasing.
- Changes: a sentence (the content spec of the pitch's central idea; the team writes it).
- Deliverables: IPS. Domain: D9. Seed: 17.
- North star: the thesis in her own logic.

Q22. Which single FR chart would a data-story maker find most honest?

- Anchors: R-S28, R-S26.
- Evidence: V39; F-d (weekly Tableau reporting).

- Hypothesis (guess): a 2027-2033 fan chart of the surplus, with the bought 2031 floor as a flat line, the model named on the chart, and the median and both ends labelled.
- Changes: a decision (the FR chart spec and its data needs).
- Deliverables: FR. Domain: D9 (D3 builds it). Seed: 18.
- North star: data that proves rather than decorates.

Q23. Can a Trading Note be written as an experiment?

- The shape: hypothesis → test → result, e.g. "does our IEF/TLT mix track the value of the ten payments within X% over the trading weeks?"
- Anchors: R-T9 ("tested, or refined"), R-C88.
- Evidence: F-d ("Experimented and shipped 10 features").
- Hypothesis (guess): yes. D3 can compute the tracking gap daily from WInS prices against the payments' present value (PV).
- Changes: a number (the tracking error to report) and TN framing.
- Deliverables: WInS-now, TN. Domain: D3. Seed: 30.
- North star: shows the "developing strategy" as a real test, in the language of her PM years.

Q24. How much of the FR's Articulation section should show the strategy we scrapped (growth-first) and why?

- Anchors: R-S27, R-T18.
- Evidence: V13 (she narrates scrapping three drafts).
- Hypothesis (guess): one clear paragraph plus one number (the shortfall odds).
- Changes: a decision (FR).
- Deliverables: FR. Domain: D7. Seed: 30.
- North star: authentic process, which the criteria reward and she practises.

Q25. Do her ship-then-iterate habits (V32, V33) argue for placing the core Treasury trades in WInS in week 1 and refining later, so the Trading Notes show a real sequence?

- Anchors: R-T13, R-T18, R-C88.
- Hypothesis (guess): yes. Place the core promise trades early, then document a refinement (e.g. a duration re-match as IEF/TLT durations drift).
- Changes: a decision (WInS trade timing now).
- Deliverables: WInS-now, TN. Domain: D7. Seed: 13.
- North star: the notes tell a true story of iteration.

Q26. How should the team credit AI use and teammates' roles, given that she credits collaborators prominently?

- Anchors: R-W46 (AI use must be recorded in Works Cited), R-S27.
- Evidence: "Gao and Xu's debut" in her own guide; the festival is co-organised.
- Hypothesis (guess): a plain "who did what, and how we used AI" box in the FR.
- Changes: a decision (FR compliance and articulation).
- Deliverables: FR. Domain: D10. Seed: null.
- North star: transparency she already practises.

Q27. Should the IPS pre-commit, in writing, what the team will do in a market crash?

- Anchors: R-I3 ("make disciplined decisions during changing market conditions").
- Evidence: V8 (a future-self letter exercise she created); V23 (purpose-anchoring).
- Hypothesis (guess): yes, as rules (e.g. "the promise is untouched; the sleeve rebalances within bands"). Do NOT make it an actual "letter to future Laura" (gimmick risk, Q15).
- Changes: a decision (IPS rules).
- Deliverables: IPS, FR. Domain: D6. Seed: 27.
- North star: behavioural coaching in a form she already uses with students, without the costume.

Q28. What voice should the team write in?

- Anchors: R-S28 ("authentic team voice"), R-C10 (the case uses "she").

- Evidence: her public voice is warm, plain and funny (V2, V6). She uses she/her and they/them (V3; NPR).
- Hypothesis (guess): plain and warm, with no forced jokes. Use "she" as the case does (one of her stated pronouns).
- Changes: a decision (the style guide).
- Deliverables: TN, IPS, FR. Domain: D9. Seed: null.
- North star: readable by her and by a co-sponsor.

4. Mapping: values → decisions → communication

Her value (evidence)	How she decides and handles risk	Strategy echo	Communication rule
Floor-first leaps (V19, V29, P2)	Jumps big, but after the downside is covered	Lock the promise, risk only the surplus	Say it as her sequence, not as finance jargon
Start small, scale on proof (V10, V36, V40, V42)	Pilots first	Staged facility contribution; bought 2031 floor	"Proof, not promise" to co-sponsors
Purpose statements (V38, V9)	Every piece has a stated why	Every holding has a job	One-line purpose per note
Anti-gimmick, anti-perfection (V14, V15, V32-V34)	Scraps what loses heart; ships on deadline	Simplest correct plan; no timing	Cut puns, themes, extra models
Humans over metrics (V22, V23, V27)	Anchors on who is helped	Metric = years secured	Lead with "payments covered" in bad years
Access over revenue (V5, V41, F-c)	Gives work away where needed	Assume low program fees; keep flexibility	Operating certainty is the pitch
Optionality (V21, V28)	Many ventures	Keep part of the surplus uncommitted	Name the purpose of the buffer
Honest uncertainty (V26, V18)	Admits "I don't yet know"	Ranges with two-sided confidence	A trade-offs note, short

5. Leads for the specialists

- **L-1 (D13):** the Poets&Quants page loads (HTTP 200) but `fetch_text.py` extracts almost no text. Re-verify P1-P3 by pointing the helper at a saved copy, or fix the parser. Raw-HTML reading confirmed the wording on 2026-09-27.
- **L-2 (D13/D10):** search for the pull quote V1 in any Wharton Global Youth video or transcript. The case PDF image layer has no attribution (R-C22). Try the SMAApply client page's video, if any.
- **L-3 (D3):** publication-timing evidence F-e (announced vs actual). Use it for the late-deposit case (Q12). Use the Authors Guild advance-installment source in the stakeholder map (W-8).
- **L-4 (D6/B3):** check for any 2026 book-deal announcement (Publishers Weekly Rights Reports; her site shows none as of 2026-09-27). Also F-a (CCA→Vanderbilt, 2027).
- **L-5 (D5):** Pride in Panels partnership facts (F-b, F-c: library partner, biennial, free, mini grants). This is a live model of how her co-funded institutions grow. prideinpanels.org returns 404; try prideinpanels.net/details/ or the SFPL event pages.
- **L-6 (D9):** her Publicity & Speaking Guide and Educators Guide PDFs (lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf; [/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf](http://s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf)) show how she presents herself to institutions. They are good tone models for the fundraising draft. Do not copy their text.
- **L-7 (B1b Q28 owner):** Taiwan work facts are VP in three places (V25, V35, HoneyPOP "including living in Taiwan and Portugal"). Use only as "she has worked in Taiwan". Never state or infer where she lives now.
- **L-8 (D2/B3):** no public statement by her on generative AI was found (one search). The team docs are right not to assume one.
- **Blocked sources to retry:** WNDB Q&A (captcha), Mochi (captcha), HarperCollins author and product pages (403), Ampersand (TLS), NüVoices (404), The Sign.al Medium (403).

Sources (all accessed 2026-09-27)

Source	URL	Status
lauragao.com home, /hey, /messyroots, /kirbys-lessons-for-falling-in-love, /speaking-events, /projects, /data-journalism, /editorial	https://lauragao.com/ (and the paths)	VP
Publicity & Speaking Guide PDF	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	VP
Educators & Book Clubs Guide PDF	https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	VP
Resume PDF (professional lines only)	https://lauragao.com/s/Laura-Gao-Resume.pdf	VP
Signal 5-year page	https://signaltemp.github.io/	VP
The Nerd Daily Q&A (2025-03-08)	https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	VP
BookWeb Indies Introduce Q&A (2022-02-07)	https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512	VP
Overachiever Magazine (2021-03-15)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	VP
Wharton Magazine (2022-04-06)	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	VP
Input Mag (2022-03-07)	https://www.inputmag.com/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel	VP
Publishers Weekly Rights Report (July 2020)	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.html	VP
Geeks OUT (2022-05-11)	https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/	VP
The HoneyPOP (2022-06-02)	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	VP
Pulse Spikes (2021-06-18)	https://pulsespikes.org/story/laura-gao	VP
Local News Matters (2026-02-12)	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	VP
48 Hills (Feb 2026)	https://48hills.org/2026/02/pride-in-panels-fest-inks-in-queer-comics-brilliance/	VP
NPR via WHRO mirror (2022-04-24)	https://www.whro.org/national-news/2022-04-24/the-pandemic-inspired-a-cartoonist-to-explore-their-wuhanese-roots-and-queer-identity	VP (read; only pronoun fact used)
Kirkus interview (2022-03)	https://www.kirkusreviews.com/news-and-features/articles/laura-gao-messy-roots-interview/	read; mostly personal, not used
Mission Chronicle (workshop 2025-03-06)	https://missionchronicle.org/1956/features/laura-gao-teaches-their-messy-roots/	VP (student paper)
Poets&Quants 2018	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	SU (helper parse fail; raw HTML read)
Wikipedia, Laura Gao	https://en.wikipedia.org/wiki/Laura_Gao	read; secondary
Maverick Show ep.190 page	https://www.themaverickshow.com/podcast/190-messy-roots-queer-identity-the-beauty-of-wuhan-and-writing-a-graphic-memoir-while-traveling-the-world-with-laura-gao/	read; host's description only

What this teaches

A client's own words are data. Read them the way you would read a fact sheet: exactly, with a date, and with a check of who is actually speaking (her, a reporter, or a publisher). Three lessons from this lens:

1. **Behaviour beats adjectives.** The case says she takes "thoughtful risks". Her record shows what that means: she jumped only after a contract secured the downside. Behaviour tells you how to design a plan; adjectives do not.
2. **Verify before you quote.** The most quotable line in the case could not be traced to her, and a misattributed quote would hurt us with exactly this client.
3. **Understanding someone is not borrowing their identity.** The best-fitting quote for our 2031 idea is personal advice about coming out. Using it would be tokenism, so we leave it out and use her career

story instead.

B9a Client Synthesis: Laura on the professional IPS client grid

Agent B9a, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: "Client Synthesis (deep client understanding)". Starting angle: the professional IPS client framework (the CFA Institute approach of objectives and constraints). The objectives are return and risk, with risk split into willingness and ability. The constraints are time horizon, liquidity, legal, taxes and unique circumstances. I filled each cell for Laura, first from the case and then from her verified public professional record. For every cell I asked what is known, what is inferred and what a question to her portfolio manager (PM) would resolve. The other B9 agent starts somewhere else.

This file is AI-generated research for Team Caplet, for brainstorming only. It holds questions, evidence, numbers and checklists. It has **no submission-ready prose**. The six students decide and write every deliverable in their own words, and any AI use must be recorded in the Final Report's Works Cited (R-W46). Laura appears here only through the case and her public professional record. I left out every personal-life detail, even where a source contains it. Nothing here suggests contacting her; that would mean disqualification (R-W16).

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (I read it on the primary page on 2026-09-27; URL in section 6); **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **PARA** = PARAPHRASE-UNVERIFIED; **INT** = INTERPRETATION (my reading, never a fact); **DER** = derived arithmetic (script in Appendix A; its inputs carry their own labels). Every "hypothesis" below is a guess.

Terms used (defined once):

- **Funded ratio**: money available divided by today's market price of the promised payments.
- **Required return**: the smallest yearly return that still meets a goal.
- **Hurdle**: the return an alternative would have given without risk. A risky choice has to beat it to be worth taking.
- **Certainty equivalent**: the sure amount a person would accept in place of a risky one.
- **Forward rate**: the interest rate that today's Treasury curve locks in for a future period.
- **Affordable loss**: an entrepreneur's rule of deciding first how much you can afford to lose, instead of starting from how much you hope to gain.
- **Sleeve**: the part of the portfolio outside the Treasury ladder; it holds the risk.

0. Summary (read this first)

I wrote **21 questions** (section 4). The eight below matter most, because each brings a number or a primary source that no Phase A/B file has used (I checked all 12 Phase B files on 2026-09-27):

1. **Seed 26 answered with a number: Laura is 103% funded in 2027 and about 152% funded once the 2028 deposit is counted** (Q1). Priced on the official curve, the \$150k due on 2028-01-01 is worth \$143,230 at 2027-01-01. So $(\$300,000 + \$143,230) / \$292,264 = 151.7\%$ (DER; F-117 gives only the 102.6% first-deposit figure). A pension at 103% would lock everything. A pension at 150% would take risk with the surplus. **Laura's ability to take risk switches on only when the 2028 deposit arrives**. For WInS this is direct evidence: WInS holds exactly her \$300k Year-1 money (R-AN14), and at that moment her risk ability is almost nil (surplus \$7.7k, 2.6%).
2. **The promise needs almost no return** (Q2). With both deposits, a flat **1.05% a year** funds all ten payments (DER). With the first deposit alone the figure is 5.09%, which is today's yield (DER; matches B2a). The CFA IPS standard states return as a *requirement* tied to obligations ("a required real growth rate of 4 percent to satisfy her future obligations", VP). So Laura's return objective should be written goal by goal: the payments need the Treasury rate, already locked; only the facility has a return *desire*.
3. **"Thoughtful risk" has an exact price this year** (Q3, Q4). Locking the entire surplus in Treasuries at today's forward rates would give about **\$204k in 2033** with almost no market risk (DER). The 60%-equity sleeve has a median of **\$207k-\$210k** and a p5 of **\$159k-\$162k**. It ends **below the locked \$204k in 42-46% of paths** (DER, both on the verified simulation). Equity therefore adds about \$3-6k at the median and costs about \$42-45k at the 5th percentile. The question for Laura is her *certainty*

equivalent: would she swap a sure ~\$204k for that spread? That is the real test of "thoughtful risks" (R-C37), and none of the files puts it to her that way.

4. **A consistency flaw that a statistics graduate would catch** (Q4). The verified simulation prices the ladder at the curve's 5.08% forward rate but lets the sleeve's bonds earn JPM's 4.00%. Its all-bond sleeve therefore ends at a median **\$195k, below the \$204k the same money locks in today** (DER). Every sleeve and range number in the Final Report inherits this bias. F-317 flagged "about 1pp conservative"; this file puts a dollar figure on the consequence.
5. **Her own entrepreneur's rule translates "thoughtful risks"** (Q6). Effectuation, the best-known research framework on how expert entrepreneurs decide, lists "Affordable Loss: 'Don't risk more than you can lose.' Instead of focusing on potential gains, decide what you're willing to lose." (VP). Sizing the sleeve's risk from a stated dollar loss Laura can afford, not from expected return, uses entrepreneur logic that she would recognise. It also matches her documented floor-first leap (B8a/B8b). Its "Crazy Quilt" principle ("Find people who'll commit to your idea", VP) describes her 2031 co-sponsor task (Q7).
6. **Research separates career risk from investing risk** (Q5). The DOSPERT scale (a standard test of risk attitude that asks about each domain separately) measures risk-taking by domain, "financial decisions (separately for investing versus gambling), health/safety, recreational, ethical, and social" (VP). Entrepreneurial-finance theory says "Non-diversified entrepreneurs demand both systematic and idiosyncratic risk premium" (Chen, Miao & Wang, NBER w14848, VP). Both support reading the case's "Although ... thoughtful risks" as: her career risk does not carry over to this portfolio. If anything it argues for less market risk.
7. **"Flexibility as the project develops" means different things for someone who ships in stages** (Q8, Q9). The money could be an overrun buffer (cash, 1-3 years), an option to expand after proof (some growth, 3-5 years) or an option to pivot (liquid). Each use calls for a different asset mix. Her record shows staged scaling ("Always start small", a festival that grew "way beyond our initial passion project", "shipped 10 features"; B8a, VP). That points to "option to expand", which would let the held-back money take some growth risk after 2033. No file has set the held-back money's horizon or mix from its *use*.
8. **The facility absorbs every rate move before January 2027, one for one, at about \$9.9k of 2033 facility per 25bp** (Q12; DER). This is the one place the "two goals competing for the same money" become real, and Laura has never been asked whether she accepts it.

What the team still does not understand about her (these are fact-find gaps; each becomes a stated assumption, Q15):

- how she ranks a bigger facility against more held-back money (Q14);
- what the held-back money is *for* (Q8);
- whether her tolerance differs for losses only she bears and shortfalls others see (Q9);
- the size of her "financial resources outside the portfolio" (already asked in B1a Q12; not re-asked here);
- whether her outside income shrinks if she runs the residency from 2033 (Q20).

1. The grid: Laura on the IPS objectives-and-constraints framework

Columns: **Known** (case, VRF) · **Inferred** (INT or DER, labelled) · **What a question to her PM would resolve** (her PM is the case's fictional portfolio manager; in practice the team states an assumption, Q15).

1.1 Return objective

- **Known**: there is no return target, benchmark or risk number (R-AN17, R-AN34). Projections use "reasonable return assumptions consistent with their strategy" (R-C77). Money in: \$300k in 2027 and \$150k in 2028 (R-C26, R-C27). Money out: ten fixed nominal \$50k payments, 2033-2042 (R-C45, R-C46).
- **Inferred**:
 - Required return for the payments alone, with both deposits: **1.05% a year** (DER).
 - First deposit alone: **5.09%** (DER), which is about today's yields (F-012).
 - The facility has a *desire*, not a requirement, because "There is no predetermined facility contribution" (R-C59).
 - Return needed for a given 2033 surplus (flat yearly return on the \$7,736 of 2027 surplus plus the \$150k): \$200k needs 4.81%, \$225k needs 7.29%, \$250k needs 9.55%, \$300k needs 13.57% (DER).

- So a surplus above about \$225k needs more than JPM's 6.70% compound on U.S. large caps (F-301) on every dollar.
- **PM question:** "Is there a facility amount below which Laura would rather not make a contribution at all?" (That would give the facility a minimum, which would make it a requirement.)

1.2 Risk: willingness (how much risk she wants to take)

- **Known:**
 - "Laura understands that investing involves uncertainty and periods of market volatility." (R-C36)
 - "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance..." (R-C37, R-C38)
 - She hands the choice of balance to the team.
- **Inferred:**
 - The career risk she took was floor-first (B8a quotes #4 and #5, B8b summary 2; VP there).
 - She names her own income as "unstable" (B8a #1, VP).
 - Domain-specific research says risk appetite in one domain does not predict it in another (DOSPERT, VP) (Q5).
 - The effectuation "affordable loss" rule is the entrepreneur's own risk grammar (Q6).
- **PM question:** "What 2033 facility outcome would Laura call a failure?" (This is the affordable-loss floor, and it sizes the equity.)

1.3 Risk: ability (how much risk she can afford to take)

- **Known:**
 - Living costs are outside the portfolio (R-C33).
 - There are no withdrawals before 2033 (R-C34).
 - Outside money may not back the payments (R-C48).
 - The 2031 range "must also protect" the payments (R-C71).
- **Inferred (DER), ability by stage:**

Stage	Funded ratio / state	Ability to bear sleeve loss
2027 (only \$300k exists)	102.6% funded; sleeve \$7.7k	Almost nil
2028-2030 (after the deposit)	~152% funded on today's curve; sleeve ~\$155k and growing	High: no promise rests on the sleeve yet
2031-2033 (range announced)	The announced low end becomes a public commitment	Low for the floor, moderate for the rest
2033+ (reserve set aside)	Held-back money only	Depends on what the money is for (Q8, Q9)

- **PM question:** "If the 2028 deposit is late or smaller, does Laura want the plan to wait for it or to act without it?" (B8b Q12 already sizes the lateness; the question here is her preference.)

1.4 Time horizon (multi-stage)

- **Known:**
 - The official dated points are 2026 (Year 0), 2027, 2028, 2031, 2033 and 2033-2042 (R-C29, R-C30).
 - The criterion wording is "disciplined planning across Laura's changing time horizons" (R-S24).
 - The IPS guide says "changing time horizons" (R-I8).
- **Inferred:** there are five horizons, plus one the case only hints at.
 - 2026: WInS, 6 weeks.
 - 2027-2028: build, with deposit risk.
 - 2028-2031: growth.
 - 2031-2033: promise-keeping. The range is public.
 - 2033-2042: payout.
 - Hinted: a facility-development window after 2033 ("as the project develops", R-C58).
 - Two paradoxes:
 - The longest-horizon money (the reserve, up to 2042) is the safest.
 - The facility money's real decision point is 2031, only **4 years** after the first deposit and **3** after the second.

- **PM question:** "When would Laura expect to need the held-back money: during construction (2033-2035) or later?"

1.5 Liquidity

- **Known:** there is no liquidity need before 2033 (R-C33, R-C34). On 1 Jan 2033 the reserve is set aside and the first \$50k is paid on the same day (R-AN2), and the facility contribution follows it (R-C57).
- **Inferred:** the only liquidity date that matters is 2033-01-01. The ladder's first rung and the 2031 floor (a 2-year Treasury) both mature then. Nothing new here; B1a Q14 and B2b Q11 cover it.
- **PM question:** none needed.

1.6 Legal and regulatory

- **Known:**
 - Taxes and Taiwan legal requirements are out of scope (R-C97).
 - The Rules page tells teams to operate by the CFA Institute Asset Manager Code (R-W29).
 - The FAQ says "Investments permitted (for BOTH contributions)" (R-AN15).
 - Who receives the payments is left vague (R-AN58).
- **Inferred:** in the IPS this cell needs one clause and no more (Q16).
- **PM question (parked):** whether the payments go to an entity she controls. Out of scope (R-C95, R-C97).

1.7 Taxes

- **Known:** out of scope (R-C97). One clause in the IPS saying so is enough.

1.8 Unique circumstances

- **Known:**
 - Her roles: author, illustrator, entrepreneur and educator (R-C5).
 - The 2028 money comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (R-C27).
 - "Laura has no shortage of ideas for the future." (R-C23)
 - She is "planning the next chapter of her career" (R-C5).
 - A Statistics & Information Decisions Management degree (R-C12).
- **Inferred:**
 - Her income depends on reputation, so credibility is part of her human capital (A3 BS-03).
 - She is an experimenter who ships in stages: "Experimented and shipped 10 features" (her resume; B8a 2f, VP).
 - Many ideas compete for the same money after 2033 (Q17).
 - Her outside income may change if she runs the residency (Q20).
- **PM questions:** "Can the held-back money fund her other ideas, or only the residency?" and "Will she run the residency herself?"

2. Evidence found in this run (new to the repo)

#	Evidence	Status	Where it bites
E1	2028 deposit PV at 2027-01-01 = \$143,230; funded ratio 102.6% → 151.7%	DER (official curve, VRF)	Q1
E2	Break-even flat return: 1.05% (both deposits), 5.09% (first only)	DER	Q2
E3	Surplus locked at forwards: \$203,998 in 2033 ($7,736 \times 1.3513 + 150,000 \times 1.2903$). Rate risk on the 2028 part: \$183k-\$201k for ± 100 bp on the 2028 five-year rate	DER; ASM (the 2028 rate is a forward, not locked today)	Q3
E4	Verified MC, lock-early sleeve: equity 0/40/60/80% → median \$195k/\$204k/\$207k/\$211k; $P(\text{surplus} < \$204\text{k})$ 78%/51%/46%/44%	DER (strategy_mc.py, verified)	Q3, Q4
E5	Same with sleeve bonds at 5.08% (curve forward): median \$202k/\$208k/\$210k/\$212k; p5 \$182k/\$173k/\$162k/\$149k; $P(< \$204\text{k})$ 57%/43%/43%/43%	DER; ASM (bond vol kept at 3.48%)	Q3, Q4

#	Evidence	Status	Where it bites
E6	P(U.S. large cap compound return < 5.08% locked) over 3/4/6/10 years: 43%/42%/40%/38%	DER; ASM (JPM lognormal, F-301)	Q10, Q19
E7	Pre-2027 rate move → 2033 facility: -25bp ≈ -\$9.9k; -50bp ≈ -\$20.1k; -100bp ≈ -\$41.4k	DER from F-102 × forward factor 1.3513 (ASM: factor held fixed)	Q12
E8	\$204k in 2033 in 2026 cost terms: \$178k at 2.0% CPI; \$160k at 3.54%/yr (Taiwan construction average since 2021, F-508); \$131k at 6.53%/yr (the 2026 spike)	DER; ASM (rates)	Q13
E9	A -30% equity year just before 2031 with 60% sleeve equity cuts the sleeve, and so the bought 80% floor, by ~18%	DER (bond return 0 assumed)	Q10
E10	CFA Institute, <i>Elements of an IPS for Individual Investors</i> (2010): "a required real growth rate of 4 percent"; "defining acceptable paths of risk may also be important"; "'volatility' as a descriptive measure of risk may be irrelevant beyond an absolute level of loss that would completely derail an investment portfolio"; "establishing a minimum time horizon for achievement of performance objectives clarifies when action may be needed to resolve underperformance issues"	VP (all four phrases confirmed verbatim with the helper)	Q2, Q6, Q11
E11	Effectuation 101: "Affordable Loss / "Don't risk more than you can lose." / Instead of focusing on potential gains, decide what you're willing to lose. Start small and build gradually."; "Crazy Quilt / "Build partnerships early." / Find people who'll commit to your idea."; "Lemonade / "Surprises are opportunities." / Turn unexpected events into new opportunities."	VP (verbatim)	Q6, Q7, Q8
E12	DOSPRT: "assesses general and domain-specific dimensions of risk preference: financial decisions (separately for investing versus gambling), health/safety, recreational, ethical, and social decisions." Citation: Weber, Blais & Betz (2002); Blais & Weber (2006)	VP	Q5
E13	Chen, Miao & Wang, NBER w14848 (April 2009): "Entrepreneurs face significant non-diversifiable business risks." ... "Non-diversified entrepreneurs demand both systematic and idiosyncratic risk premium."	VP	Q5
E14	Investor.gov (SEC): "Risk tolerance is your ability and willingness to lose some or all of your original investment in exchange for potentially greater returns." and "Those with a shorter time horizon may prefer to take on less risky or less volatile investments."	VP (second phrase verbatim)	Q19, Q10
E15	Hombert et al., NBER w20717: a French reform "provided generous downside insurance for unemployed individuals starting a business"	VP (abstract via the NBER API)	lead for Q6 (downside insurance changes who takes entrepreneurial risk)

Could not verify (reported, not guessed):

- **Heaton & Lucas (2000)**, "Portfolio Choice and Asset Prices: The Importance of Entrepreneurial Risk" (Journal of Finance). It is often cited for the finding that households with volatile business income hold less stock. I could not open it: Wiley returned 403, and NBER search did not return it. Status: **UNVERIFIED LEAD**; do not cite it until D6 reads it.
- A reference-class base rate for cost overruns on small cultural buildings (Flyvbjerg's work). Wikipedia says only "Cost overruns are common in infrastructure, building, and technology projects" (VP, general). No building-specific number was found. **Lead for D5.**
- The web-search budget for this session ran out before this agent started searching (tool message, 2026-09-27). All sources above were reached by direct URL with the fetch helper.

3. How the questions were filtered

- **Kept** only questions whose answer changes a decision, a number or a sentence in the TN, IPS or FR, or in WInS now.
- **Checked against brief section 8 and against the 12 Phase B files.** Where a question overlaps, I name the other file and say what is new. Not re-asked:
 - willingness vs capacity in general (B4b Q05, Q15; B1a Q12; B8b Q3; B7a Q6);
 - floor-first leap (B8a Q02);
 - late deposit (B8b Q12);
 - purpose statement (B8a Q04);

- staged facility (B8b Q6, B4a Q15);
- flexibility share (B8b Q7, B4b Q23).
- **Parked** (logged, not asked): the legal recipient of the payments (out of scope); CCA/teaching income (the brief says teaching income is not part of the case; the team docs' CCA claim is unverified); her age (privacy; not needed).

4. Questions

Tier 1: WInS now and Trading Notes (Oct 23)

Q1. (Seed 26) Laura is 103% funded in 2027 but about 152% funded once the 2028 deposit arrives. Should the WInS book, which holds exactly her Year-1 \$300k, therefore carry almost no risk sleeve, because her ability to take risk only switches on in 2028?

- Anchors: R-C26, R-C27, R-AN14, R-AN35, F-117, F-101, F-409. Quote: "At the beginning of 2027, Laura plans to invest \$300,000 with an asset management firm." (case p.2 L43, VRF)
- Evidence: E1.
 - $(\$300,000 + \$143,230) / \$292,264 = 151.7\%$ with the deposit; 102.6% without (DER).
 - WInS cash is "\$300,000", and the \$150,000 "will not be added to WInS" (R-W56, R-W59, VP via A1).
- Hypothesis (guess): yes, in principle. In WInS the "true" Year-1 book is about 97% ladder and 3% sleeve.
 - A 65/35 mirror shows her post-2028 state, which does not exist yet in WInS.
 - The team may still hold a small equity sleeve as a *rehearsal* of the 2028 book. But the note must then say so, and it must say that in Year 1 her ability is nil.
 - This is client-lens evidence for the open "WInS mirror" item (brief section 9), which B1b Q5 and B1a Q1 ask from the rules side.
- Would change: **decision** (WInS equity weight now: ~3% vs ~21% vs ~35%). **Sentence** (the TN reflection on why the equity sleeve is sized as it is).
- Deliverables: WInS-now, TN, IPS. Domain: D3. Seed: 26.
- North Star: a firm that sizes risk to *her* balance sheet on each date, not to a generic mix, is the firm that understood the 2028 deposit is the switch.

Q2. Laura's payments need only about 1.05% a year across both deposits (5.09% on the first alone). Should the IPS state a return requirement per goal (payments: today's Treasury rate, already locked; facility: a stated hurdle) instead of one portfolio return target?

- Anchors: R-I2, R-AN17, R-C38, R-C59. Quote: "An Investment Policy Statement (IPS) outlines an investor's financial goals, risk tolerance, and guidelines for managing a portfolio." (IPS p.1 L3-4, VRF)
- Evidence: E2, E10. The CFA example states the requirement tied to obligations: "a required real growth rate of 4 percent to satisfy her future obligations" (VP).
- Hypothesis (guess): yes.
 - One clause in the IPS: the payments need no growth beyond the Treasury rate the plan locks, and the facility has a return *desire* measured against a hurdle (Q11).
 - This separates "required" (safe) from "desired" (risky) in the language professionals use. It also answers "why so little equity?" in five words: the promise does not need it.
- Would change: **sentence** (IPS return objective); possibly the **number** stated as the growth hurdle.
- Deliverables: IPS, FR. Domain: D8. Seed: null.
- North Star: a client who studied statistics sees that the firm knows which of her goals *need* return and which merely *want* it.

Q3. Locking the whole surplus in Treasuries at today's forward rates would give about \$204k in 2033. The 60%-equity sleeve gives a median of about \$207-210k and a p5 of about \$159-162k, and it ends below \$204k in 42-46% of paths. Would Laura, whose certainty equivalent nobody has estimated, choose the risky version, and what equity weight follows?

- Anchors: R-C37, R-C38, R-C74, F-407, F-012, F-402. Quote: "she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals." (case p.2 L71-74, VRF)

- Evidence: E3, E4, E5 (DER on the verified script). The "locked" figure carries 2028 rate risk: the \$150k is only invested in 2028, so its five-year rate is a forward, not locked today ($\pm 100\text{bp}$ → \$183k-\$201k for that part, ASM).
- Hypothesis (guess): she would take *some* risk, for three reasons: the upside (p95 \$273-277k) is where a "meaningful" contribution (R-C64) grows, "thoughtful risks" is her self-description, and a fully locked surplus leaves nothing for "Lemonade". But the price is poor this year: +\$3-6k at the median for -\$42-45k at p5.
 - This supports 40-60% sleeve equity and argues against 70%.
 - B5b Q1 asks the same in percentage points of premium. What is new here: the client-choice framing (a sure \$204k vs a spread), and the 42-46% chance of ending below the sure amount.
 - Would change: **number** (sleeve equity 40/50/60%, open item in brief section 9). **Sentence** (IPS risk line: why any equity at all).
- Deliverables: IPS, FR (a chart: a "sure \$204k" line over the sleeve distribution). Domain: D6 (elicitation), D3 (numbers). Seed: null.
- North Star: showing Laura the sure alternative and letting the risk justify itself against it is what an adviser who respects her does. Most teams will show only the risky version.

Q4. (Seed 18) Should the verified simulation's sleeve bonds earn the curve's forward rate (about 5.08%) instead of JPM's 4.00%? As it stands, its all-bond sleeve (median \$195k) ends below simply locking the same money today (\$204k), which a statistics graduate would spot.

- Anchors: F-316, F-317, F-012, R-C77, R-AN32. Quote: "use reasonable return assumptions consistent with their strategy" (case p.4 L131-132, VRF).
- Evidence: E4 vs E5.
 - JPM's 4.00% was set when the 10-year Treasury was 4.16% (F-010, F-317).
 - The ladder in the same model grows at 5.08% (strategy_mc.py line with `FWD_FACTOR_27_33`, VRF).
- Hypothesis (guess): yes, for internal consistency. Use the curve for bonds bought now and JPM only for equity and for bonds bought later, and state it as an assumption. Effect: sleeve medians rise by about \$2-7k, and the 2031 floor and range numbers move with them. Brief section 11 lists other model gaps but not this one.
- Would change: **number** (every sleeve, floor and range figure in the FR).
- Deliverables: FR (IPS only if a number is quoted). Domain: D3. Seed: 18.
- North Star: a client with a statistics degree reads the assumptions first. An inconsistency she finds herself costs more trust than any chart earns.

Tier 2: IPS (Nov 6)

Q5. Do domain-specific risk research and entrepreneurial-finance theory support reading "Although she has been willing to take thoughtful risks..." as "her career risk does not carry over to this portfolio"? And should the IPS risk sentence say so without any model?

- Anchors: R-C37, R-AN16, R-AN31. Quote: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance" (case p.2 L69-72, VRF).
- Evidence:
 - E12 (DOSPERT measures investing risk separately from other domains, VP).
 - E13 (non-diversified entrepreneurs demand extra risk premium, VP).
 - Her own words that her income is "unstable" (B8a #1, VP).
 - UNVERIFIED LEAD: Heaton & Lucas (2000), household evidence that business-income risk lowers stock holdings.
- Hypothesis (guess): yes. The professional reading is that her human capital already carries concentrated, undiversified risk (books, speaking, licensing), so the financial portfolio should not add correlated risk. A3 BS-03 and the team review say the same thing in words. This adds two citable sources: a test that measures risk attitude by domain, and a theory result.
- Would change: **sentence** (IPS willingness/ability line); FR citation.
- Deliverables: IPS, FR. Domain: D6. Seed: 16 (the human-capital side of wrong-way risk).

- North Star: it explains why a bold entrepreneur can rationally want a careful portfolio, in a way she would recognise rather than resent.

Q6. Should "thoughtful risks" be translated into the entrepreneur's own "affordable loss" rule? That means stating in dollars the worst 2033 facility outcome Laura can accept, then sizing sleeve equity so that outcome holds in (say) 19 of 20 cases, instead of starting from expected return.

- Anchors: R-C37, R-C38, R-AN17, R-C60 ("understanding of her goals"). Quote: "an appropriate balance between pursuing growth and protecting the capital required for her goals" (case p.2 L72-74, VRF).
- Evidence: E11 (effectuation, VP).
 - The CFA standard allows "acceptable paths of risk" and treats volatility as irrelevant "beyond an absolute level of loss that would completely derail an investment portfolio" (E10, VP).
 - Her documented floor-first leap (B8a/B8b, VP there).
- Hypothesis (guess): yes. This is the Laura-specific version of the CFA loss-tolerance example (B4b Q15 proposes the CFA version; this adds the entrepreneur's version).
 - Illustration only (DER, E5): an affordable floor of about \$160k at p5 is consistent with ~60% sleeve equity; about \$170k with ~40%.
 - The number is the team's call. It is the "PM question" in grid 1.2.
 - Would change: **number** (sleeve equity derived from a stated affordable loss). **Sentence** (IPS risk tolerance).
 - Deliverables: IPS, FR. Domain: D6 (framing), D3 (the equity-to-floor mapping). Seed: 1.
 - North Star: a plan written in the decision rule of an entrepreneur ("decide what you can lose, then leap") is one she would recognise as hers. A "moderate risk tolerance" label is not.

Q7. (Seed 15) Is the bought 2031 floor Laura's pre-commitment in the effectuation sense ("Find people who'll commit to your idea")? And should the fundraising draft ask co-sponsors for commitments that sit on top of it ("the first ten years of operations and US\$X are already secured; we invite partners for the rest")?

- Anchors: R-C64, R-C66, R-C67, R-C72, R-AN30, R-S28. Quote: "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute." (case p.3 L110-112, VRF)
- Evidence: E11 "Crazy Quilt" (VP). Pride in Panels was built with the SF Public Library, a partner (B8a #14, VP).
 - Brief section 8 item 17: the seed-money effect is a mechanism, not a prediction.
 - B8b Q2 asks when she needed others to believe her. What is new here: the entrepreneur framework that names the mechanism, and the "ask structure" for the draft.
- Hypothesis (guess): yes, as one structural choice in the FR fundraising part: secured money first, then the range, then the invitation. It turns the pull quote's private conviction into a public, bought proof. The pull quote is unattributed (B8a/B8b), so link to the case, not to "Laura says".
- Would change: **sentence/structure** (order of the FR fundraising draft).
- Deliverables: FR. Domain: D5. Seed: 15.
- North Star: co-sponsors, and Laura, see that her own money moves first and is already in place. That is the strongest credibility signal a range can carry.

Q8. What does "financial flexibility as the project develops" buy for someone who builds in stages: (a) an overrun buffer (cash, needed within 1-3 years), (b) an option to expand once the residency proves itself (growth assets, 3-5 years) or (c) an option to pivot or relocate (liquid)? Should the held-back money's asset mix and horizon follow that answer?

- Anchors: R-C58, R-C84, R-AN19, R-C41, R-C85 ("financial flexibility Laura should preserve"). Quote: "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops." (case p.3 L102-103, VRF)
- Evidence (all VP via B8a):
 - "Always start small" (#26).
 - "All of it is way beyond our initial passion project." (#15)
 - "We definitely hope to offer more as the festival grows." (#16)

- "be OK with the fact that your significant projects will never be finished" (#31).
- "Experimented and shipped 10 features" (resume).
- Effectuation "Lemonade" (E11).
- The case: the ten years are "early operations" (R-C41); co-sponsors fund "any operating support beyond Laura's commitment" (R-C44).
- Hypothesis (guess): (b) first, (a) second. Her pattern is to prove something small, then scale it.
 - Implication (INT): part of the held-back money can stay in a balanced mix after 2033 rather than all in cash, with a stated overrun slice in short Treasuries.
 - Current thinking stops at "overrun buffer" (CLAUDE.md "project flexibility/overrun buffer").
 - Other files ask *how much* to hold back (B8b Q7, B4b Q23); none asks what it is *for*, which is what sets the horizon and the mix.
- Would change: **decision** (post-2033 asset mix of the held-back money). **Sentence** (FR flexibility definition).
- Deliverables: FR (principle in the IPS). Domain: D5 (with D6). Seed: null.
- North Star: "flexibility to grow what works" is a builder's word for flexibility. A contingency fund is a banker's.

Q9. Does Laura tolerate losses that only she bears (she helped launch an "Anti-Resume Project" to "normalize failure") differently from shortfalls others see (the co-sponsor range, the payments)? If so, should the post-2033 held-back money carry more market risk than the 2031-2033 range money?

- Anchors: R-C66, R-C71, R-C58, R-AN30. Quote: "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors." (case p.3 L114-115, VRF)
- Evidence: "To destigmatize alternate career paths and normalize failure, Laura helped launch the Anti-Resume Project" (Pulse Spikes; this is reporter text, VP via B8a 2f); "When I first posted the comic, I expected a way worse reaction" (OA, VP via B8a #8).
- Hypothesis (guess): yes. Her tolerance is high for her own visible failure and low for breaking a promise to partners.
 - So risk should be lowest on money others rely on (reserve, announced floor) and higher on money only she relies on (held-back money after 2033).
 - Tension with Q8 (a) overrun buffer. The team resolves it by splitting the held-back money by use.
 - Would change: **decision** (risk level of post-2033 held-back money). **Sentence** (IPS principle "risk sits only on money nobody else relies on").
- Deliverables: IPS, FR. Domain: D6. Seed: 32.
- North Star: it draws the line where she draws it, between her own risk and other people's trust.

Q10. Laura's ability to bear sleeve losses is highest in 2028-2029 and lowest in the year before the 2031 announcement. A -30% equity year in 2030 at 60% sleeve equity cuts the announced floor by about 18%. Should sleeve equity step down during 2029-2030, rather than stay flat until the 2031 lock?

- Anchors: R-C62, R-C63, R-C65, R-S24, R-I12, F-402. Quote: "demonstrates disciplined planning across Laura's changing time horizons and cash-flow needs" (SMAApply criterion 1, R-S24, VRF).
- Evidence:
 - E9 (-18%).
 - E6: over the 3-4 years to 2031, equities lag the locked 5.08% in 42-43% of JPM-model paths.
 - Investor.gov: "Those with a shorter time horizon may prefer to take on less risky or less volatile investments." (E14, VP)
 - The verified model holds 60% flat to 2031, then locks 80% (strategy_mc.py, VRF).
 - B4a Q16 proposes locking on *level* triggers. This question is about a *time* glide before the announcement.
- Hypothesis (guess): a small step-down (for example 60% → 40% in 2030) costs little median and protects the headline. But the announced range is set *after* the fact, so no promise breaks. The cost is a less "meaningful" signal, not lost credibility. The team should test it with D3 before adding any complexity; it may not earn its place.

- Would change: **decision** (sleeve glide path 2029-2031). **Sentence** (the IPS "planned changes" answer, R-I21).
- Deliverables: IPS, FR. Domain: D3. Seed: null.
- North Star: it gives the IPS "changing time horizons" a Laura-specific reason (the 2031 announcement) rather than a generic age-style glide path.

Q11. Over what horizon, and against what hurdle, should the sleeve be judged? The CFA standard says to set "a minimum time horizon for achievement of performance objectives". Should the hurdle be Laura's own riskless alternative (the surplus locked at about 5.08%, about \$204k in 2033) rather than the S&P 500?

- Anchors: R-C90, R-S26, R-I33 (FR evaluates implementation; R-S26: "evaluate funding reliability, the facility contribution, and financial flexibility under varying market outcomes"). Quote: "This evaluates how that strategy was implemented" (case p.4 L158, VRF).
- Evidence: E10 (VP). E3.
 - B7a Q11 asks which benchmark the FR should use for WInS. What is new here: a client-specific hurdle that exists only because the case's promise can be bought.
- Hypothesis (guess): yes.
 - Evaluation horizon: 2031 (the decision point), with yearly monitoring.
 - Hurdle: the locked alternative.
 - In the FR's WInS evaluation: the equity sleeve compared with what the same dollars would have done in the intermediate Treasury fund over the six weeks. The FR says openly that six weeks prove nothing either way.
- Would change: **decision** (FR benchmark). **Sentence** (IPS monitoring clause).
- Deliverables: IPS, FR. Domain: D8. Seed: null.
- North Star: "did the risk we took for you beat what you could have had for sure?" is the only performance question a client like Laura should care about.

Q12. Does Laura accept that her facility absorbs every rate move before January 2027 one for one (each 25bp fall means about \$7.3k more ladder cost, about \$9.9k less 2033 facility; -100bp about \$41k)? Should the IPS say so in one clause, so that the "two goals competing for the same money" is a stated choice?

- Anchors: R-C50, R-C57, R-C71, F-102, F-111, F-112. Quote: "After establishing the operating reserve in 2033, Laura must decide how much of the remaining portfolio she can responsibly contribute" (case p.3 L101-102, VRF).
- Evidence: E7 (DER). P(ladder cost > \$300k on 2027-01-01) ~24% (F-112, ASM).
 - B5a Q02 and B4a Q06 design the *rule*. What is new here: the cost stated in the client's own goal (facility dollars in 2033), which is how she would weigh it.
- Hypothesis (guess): yes. This is the only real competition between her two goals under lock-early. Naming it shows the priority rule (promise first; R-AN2) has a price, and that the firm told her the price.
- Would change: **sentence** (IPS priority clause). It supports the **decision** on the pre-2027 rate rule.
- Deliverables: IPS, FR. Domain: D1. Seed: null.
- North Star: telling a client the exact price of putting her promise first is a disclosure that earns her confidence (R-S25).

Q13. In Taiwan building-cost terms, what is the locked-equivalent surplus worth? \$204k in 2033 is about \$160k of 2026 construction cost at 3.54% a year, or about \$131k if the 2026 pace of 6.53% persisted. Should every co-sponsor figure carry a "today's-cost" equivalent next to the 2033 dollar figure?

- Anchors: R-C85, R-AN8, R-AN10, F-508, F-114. Quote: "the effect of inflation on portfolio projections and facility costs" (case p.4 L147, VRF).
- Evidence: E8 (DER; ASM rates from F-508).
 - B1b Q20 asks whether to tag figures "2033 dollars" vs "today's dollars". What is new here: the actual numbers, and the finding that the gap is 22-36% of the headline.

- Hypothesis (guess): yes, as a pair ("about US\$X in 2033, equal to about US\$Y of today's building costs"). This is how the case's inflation-on-facility-costs item gets answered without estimating the facility cost (R-C95).
- Would change: **number/sentence** (FR co-sponsor figures).
- Deliverables: FR. Domain: D4. Seed: null.
- North Star: co-sponsors plan in building costs, not in U.S. nominal dollars. Laura would not want to overstate by a quarter.

Q14. Once the payments are bought, which does Laura rank higher: a bigger 2033 facility contribution (a signal to co-sponsors) or more money held back? The case words the two differently: flexibility is her own recognition ("She recognizes..."); the signal is hedged ("may signal").

- Anchors: R-C58, R-C64, R-C60, R-C84. Quotes: "She recognizes that committing all remaining assets could limit her financial flexibility" (p.3 L102) vs "A meaningful personal contribution may signal that the residency is financially viable" (p.3 L110-111) (VRF).
- Evidence: word analysis (VRF).
 - Her stated values (INT from B8a #34-#38, VP there): "fit work around life", "avoid siloing", "don't want to pigeon-hole myself".
 - B8b Q7 and B4b Q23 size the share. What is new here: which goal ranks second, read from the case's own verbs.
- Hypothesis (guess): flexibility ranks slightly higher. The case gives it her voice ("She recognizes"); it gives the signal to others with a hedge. That argues for committing a clear minority of the post-reserve money as held back (the share is the team's call) and against "all but a token".
- Would change: **number** (facility share of the post-reserve surplus). **Sentence** (FR justification).
- Deliverables: FR (principle in the IPS). Domain: D6. Seed: 32.
- North Star: ranking her goals from her own words, not the team's taste, is "a thoughtful understanding of the client" (R-S25).

Q15. Should the Final Report include a short table: "what the profile does not tell us → what we assumed → what changes if we are wrong"? Rows would include: her outside resources; what the held-back money is for; her affordable loss; whether it may fund other ideas; the currency she thinks in; whether she runs the residency.

- Anchors: R-AN34, R-S25, R-C85, R-C56. Quote: "identify the assumptions supporting their recommendation" (case p.3 L98-99, VRF).
- Evidence: CFA 2010: "Individual investors may also require an assessment of their intellectual and emotional tolerance for potential losses associated with risks; for these assessments, interviews or questionnaires can be used." (VP; read in the same PDF as E10). Contacting Laura is banned (R-W16), so assumptions replace the interview.
- Hypothesis (guess): yes, 5-6 rows maximum. It shows the fact-find a real adviser would do, done honestly, without contacting her. The IPS gets one clause ("where the profile is silent we assume...").
- Would change: **sentence/table** (FR assumptions section).
- Deliverables: FR, IPS. Domain: D8 (with D9 for plain wording). Seed: 32.
- North Star: saying what you do not know about a client, and how the plan survives being wrong, is what makes her feel understood rather than analysed.

Q16. Should the 500-word IPS be organised on the IPS guide's own five client items ("goals, time horizons, liquidity needs, desired degree of funding certainty, and need for financial flexibility") rather than on the CFA grid? Legal, tax and unique circumstances would get one clause.

- Anchors: R-I6, R-I20, R-I21, R-I22. Quote: "It should reflect your client's goals, time horizons, liquidity needs, desired degree of funding certainty, and need for financial flexibility." (IPS p.1 L9-10, VRF)
- Evidence: INT. R-I6 is Wharton's own version of the objectives-and-constraints grid, with taxes and legal removed and certainty and flexibility added.
- Hypothesis (guess): yes. Using the reader's own checklist in its own order makes the IPS easy to score.

- Risk willingness vs ability sits inside "goals" and "funding certainty".
- One clause for out-of-scope items ("taxes and Taiwan law are outside this mandate", R-C97) shows awareness in about 10 words.
- Would change: **sentence/structure** (IPS outline).
- Deliverables: IPS. Domain: D9. Seed: null.
- North Star: judges read against their own list. Laura reads for her goals. One structure serves both.

Q17. Laura "has no shortage of ideas", and the case rules out withdrawals before 2033. Should the IPS present that rule as her own pre-commitment, and say that any new idea before 2033 is funded outside the portfolio (and that after 2033 the held-back money is the only portfolio source for it)?

- Anchors: R-C23, R-C34, R-C19. Quotes: "Laura has no shortage of ideas for the future." (p.2 L36-37); "she will neither add to nor withdraw from the portfolio before 2033." (p.2 L64-65) (VRF)
- Evidence: INT. Behavioural finance calls this kind of self-binding rule a commitment device (general knowledge, ASM; D6 to source).
- Hypothesis (guess): one clause, with low word cost and moderate value. It answers the risk a real adviser would see in a serial founder: the portfolio being raided for idea number 51.
- Would change: **sentence** (IPS governance).
- Deliverables: IPS. Domain: D6. Seed: null.
- North Star: it treats her many ideas as a strength the plan protects, not a risk it ignores.

Q18. Does the case's "capital required for her goals" grow in 2031? When the range's low end is announced, it becomes a public commitment. Should the IPS define protected capital as the reserve (from 2027) plus the announced floor (from 2031)?

- Anchors: R-C38, R-C66, R-C71, R-AN16. Quote: "protecting the capital required for her goals" (p.2 L73-74, VRF).
- Evidence: INT. The case's own logic: overpromising damages credibility (R-C66); the range must protect the payments (R-C71).
- Hypothesis (guess): yes. It gives one definition that explains both the 2027 ladder and the 2031 floor purchase, and why risk falls after 2031. The team's "bought, not forecast" rule (brief section 8 item 7) then follows from the case's own words.
- Would change: **sentence** (IPS definition of what is protected).
- Deliverables: IPS, FR. Domain: D8. Seed: null.
- North Star: one definition, read straight from her brief, carries the whole design. That is simplicity she would respect.

Tier 3: Final Report (Dec 4)

Q19. Why is the money with the longest horizon (the reserve, needed up to 2042) invested most safely, while the facility money (needed by 2031-2033) takes the risk? Which one-line explanation answers a reader who expects "longer horizon = more stocks" (as Investor.gov teaches)?

- Anchors: R-S24, R-C46, R-C47. Quote: "each payment is a fixed \$50,000 and is not adjusted for inflation." (p.3 L90, VRF)
- Evidence: E14 (VP). E6: even over 10 years, equities lag the locked rate in about 38% of model paths (DER).
 - B4b Q08 (Brunel) covers the practitioner side. What is new here: the public rule of thumb the reader brings, and the 38% number.
- Hypothesis (guess): a fixed nominal promise is matched, not grown. Horizon matters for *wants*, not for *bought* needs. One sentence in the FR, and possibly one clause in the IPS.
- Would change: **sentence** (FR, IPS).
- Deliverables: FR, IPS. Domain: D9. Seed: null.
- North Star: it anticipates the objection an educator would raise first.

Q20. If Laura runs the residency from 2033 ("the next chapter of her career"), her "continued business income", which the case lists as a possible outside source, may shrink exactly when the project needs flexibility. Should the held-back share assume smaller outside income after 2033?

- Anchors: R-C5, R-C44, R-AN24, R-C58. Quote: "who is planning the next chapter of her career" (p.1 L6, VRF); "continued business income" (p.3 L86, VRF).
- Evidence: INT only. Her 2025 "short break from writing to focus on teaching" (B8a 2f, VP there) shows her income mix changes over time. The case scenario is fictional (R-W4), so her real plans are not case facts.
- Hypothesis (guess): mention it as an assumption (Q15 row), not as a number. It nudges the held-back share up.
- Would change: **sentence** (FR assumption). Possibly a **number** (held-back share).
- Deliverables: FR. Domain: D6. Seed: 16.
- North Star: it thinks about her career *after* the plan's big date, which few teams will.

Q21. Which single table would let Laura see her whole client picture on one page: goal → required or desired return → risk she can bear → horizon → where the money sits? And is it the right opening exhibit for the FR's client section, in place of a narrative biography?

- Anchors: R-S25, R-S28 ("uses data effectively"), R-C60, R-C92. Quote: "Tailors the strategy to Laura's circumstances, priorities, risk considerations, and residency goals" (SMApply L51, VRF).
- Evidence: section 1 of this file is a first draft of the content (not the wording). Wharton's old guidance "Don't use precious space repeating known facts about your client" (B2b Q12, VP there).
- Hypothesis (guess): yes. A 4-5 row table (payments / 2031 floor / facility upside / held-back money), with a column for "stage when risk switches on or off" (Q1, Q10, Q18). It replaces a biography with understanding.
- Would change: **sentence/structure** (FR client section and the chart plan).
- Deliverables: FR. Domain: D9. Seed: 1.
- North Star: it answers "why us?" in one exhibit: the firm that turned her brief into a map of her money by goal and by date.

5. Leads for the specialists

- **D3 (quant):**
 - Rerun `strategy_mc.py` with sleeve bonds at the curve forwards (Q4). Add a "locked surplus" reference line (\$203,998) and report $P(\text{surplus} < \text{locked})$ (Q3).
 - Map equity weight to p5 floor so the team can pick equity from an affordable loss (Q6).
 - Test a 2029-2030 step-down (Q10).
 - Report the funded ratio by stage (Q1).
 - Formulas are in Appendix A.
- **D6 (client psychology):**
 - Read Blais & Weber (2006), *Judgment and Decision Making* 1, 33-47, cited on <https://www.dospert.org/>.
 - Try to open Heaton & Lucas (2000), JF 55(3) (UNVERIFIED LEAD), for household evidence.
 - Effectuation 101 (<https://effectuation.org/effectuation-101>) and the linked "Affordable Loss and the Plunge Decision" page.
 - Source "commitment device" (Q17) on a primary page before using it.
- **D5 (philanthropy):**
 - A reference-class number for cost overruns on small cultural or adaptive-reuse buildings (Flyvbjerg's databases). It would give Q8 (a) a size without estimating the facility cost.
 - "Crazy Quilt" framing for the fundraising order (Q7).
- **D8 (practice):**
 - CFA 2010 IPS PDF, sections 3b-3d (return requirement, risk paths, evaluation horizon, liquidity): <https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment->

policy-statement-individual-investors.pdf

- Use these for Q2, Q11 and Q15.
- **D4:** the construction-cost deflators in E8 depend on F-508. If D4 finds a rural or Yilan-relevant index (B8a Q10), update Q13.
- **D1:** state the pre-2027 rate rule's cost in 2033 facility dollars (Q12, E7).
- **D9:** use the IPS guide's R-I6 five-item order as the IPS skeleton (Q16). Use the one-page goal table as the FR client exhibit (Q21).
- **D10:** no new rule issue. Q1 remains subject to the Session Rules position limit (G ids in wins_week1_guardrails.md).

6. Sources (all accessed 2026-09-27 from this run's container)

- Official case, IPS guide, SMAApply page: repo paths in brief section 1 (VRF).
- `research/verified_2026-09-27/official_curve_pv.py` and `strategy_mc.py` (VRF; imported unchanged by Appendix A).
- CFA Institute, *Elements of an Investment Policy Statement for Individual Investors* (2010), PDF above (VP; four phrases confirmed verbatim with `fetch_text.py --grep`).
- Effectuation 101, <https://effectuation.org/effectuation-101> (VP; three phrases verbatim).
- DOSPERT, <https://www.dospert.org/> (VP; phrase verbatim).
- Chen, Miao & Wang, "Entrepreneurial Finance and Non-diversifiable Risk", NBER WP 14848, April 2009, <https://www.nber.org/papers/w14848> (VP; verbatim).
- NBER WP 20717 abstract (via the NBER site search API) (VP; abstract text).
- Investor.gov, "Asset Allocation", <https://www.investor.gov/introduction-investing/getting-started/asset-allocation> (VP; verbatim).
- Wikipedia, "Cost overrun", https://en.wikipedia.org/wiki/Cost_overrun (VP; general statement only).
- Laura's own words and record: through B8a (`phase_B/B8a_voice_of_laura.md`, section 2, each VP there with URL). I did not re-fetch them.
- Blocked or not found: Wiley (JF) 403; FINRA 403; Heaton & Lucas not located; WebSearch budget exhausted.

Appendix A: script used (scratch; formulas so D3 can reproduce)

The script is kept in the scratchpad because this agent writes only this file. It imports the two verified scripts unchanged. Core lines:

```
DF(y) = verified bootstrap discount factor to 1 Jan y (official 2026-09-25 curve)
PV27 = sum(50000*DF(y) for y in 2033..2042)/DF(2027) -> 292,264
dep28 = 150000*DF(2028)/DF(2027) -> 143,230
funded = (300000 + dep28)/PV27 -> 151.7%
break-even r: 300000(1+r)^6 + 150000(1+r)^5 = sum_{k=0..9} 50000/(1+r)^k -> 1.05%
locked surplus 2033 = (300000-PV27)*DF(2027)/DF(2033) + 150000*DF(2028)/DF(2033) -> 203,998
MC: strategy_mc.simulate("&quot;L&quot;", eq_w=w); P(surplus < 203,998); bond-at-forward case sets BD
= lognorm(5.08%, vol 3.48%)
P(equity < locked over T) = N(-(ln1.067 - ln1.0508) * sqrt(T) / 0.152)
```

What this teaches

A professional adviser does not start with "what should we buy?". The first step is filling in a grid about the client: what return each goal *needs*, how much risk the person *wants* and how much they *can afford*, when each dollar is needed, and what is special about them. For Laura the grid gives answers a generic plan misses.

- Her promise needs almost no return.
- Her ability to take risk switches on only when her second deposit arrives.
- The risk she takes buys little this year, so it has to be justified against the sure thing she could have instead.

- Her own entrepreneur's rule ("decide what you can afford to lose") is a better translation of "thoughtful risks" than any risk label.

The other lesson is honesty about gaps. Where the profile is silent, a good adviser says what was assumed and what changes if that assumption is wrong.

B9b Client Synthesis: Laura as a decision-maker choosing between firms

Agent B9b, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: "Client Synthesis (deep client understanding)". Starting angle: Laura as the person who reads competing proposals and picks one. She has a Wharton statistics degree, was a tech product manager who ran A/B tests, is an author-illustrator and professor, and builds community institutions. The question is how that person would judge a plan. The other B9 agent starts elsewhere.

This file is AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It contains questions, evidence, numbers and checklists. It contains **no text meant for submission**. The six students decide and write every deliverable in their own words, and must record AI use in the Final Report's Works Cited (R-W46). Privacy: I used only her public professional record. Her resume PDF and website also carry contact details, hobbies and family-related comics. I left all of that out on purpose. Nothing here suggests contacting her (R-W16).

Status labels follow brief section 3: **VP** = VERIFIED-PRIMARY (read and grep-checked by me, 2026-09-27), **VRF** = VERIFIED-REPO-FILE, **SU** = SNIPPET-UNVERIFIED, **ASM** = ASSUMPTION, **PU** = PARAPHRASE-UNVERIFIED, **INT** = INTERPRETATION (my reading, not a fact). Every "hypothesis" is a **guess**.

0. Summary (read this first)

1. Her own analytic work already labels its results by quality, and our plan should do the same.

In her data project "Rewriting Herstory" (lauragao.com) she publishes "The [unfinished] results" in a table whose last column is "Match Quality", with rows graded "Good", "Unsure" or "Bad" (VP). Her dance-music study ends with limitations and next steps ("Further probing ... would provide deeper analysis") and takes its definitions from the data source ("For the purposes of this study, I used Spotify's definitions") (VP). Someone who works like this will trust a plan that grades its own numbers: which are observed market prices, which are estimates, and which are assumptions. She will distrust a plan that presents one model output as fact. (Q9, Q10)

2. Her CV is a product-and-evidence CV, not a finance CV. Her resume lists:

- "Experimented and shipped 10 features" as a Twitter PM, in teams called "Growth ML" and "Abuse & Safety";
- "A/B testing" as a skill;
- as an Amazon data analyst, she "Investigated cause of 1.5M cases of transactions from unattributed ads";
- she "Beta-tested MVP with 300K+ users of economics platform, Marginal Revolution University";
- she ran "usability tests with nurses and doctors" with NIH researchers.

All VP. Four habits of mind follow (INT):

- compare against a **control**;
- **attribute** results to their causes;
- **test** with real users before launch;
- protect **guardrails** while optimising one goal.

Each gives a question below that most teams will not think to ask (Q3, Q7, Q11, Q12, Q14).

3. At today's yields her "required return" is below the Treasury yield. DERIVED from the case cash flows (VRF):

- The ten payments alone need only **1.05% a year** on her \$450k.
- Each extra \$50k of 2033 facility money adds about 1.05 points.
- Payments plus a \$200k facility need **5.25%**, about what Treasuries pay today (F-001, F-012).

A professional would open the client picture with this "return objective". It reframes the whole plan. Growth is not needed to *meet* her goals. It is optional upside for the facility, bought with risk. (Q2)

4. The "control arm" shows the growth sleeve's lift is modest. Scratch runs of the verified model (ASM inputs, see Q3) give the lock-early 2033 surplus p5/p50/p95 by sleeve equity weight:

Sleeve equity	Sleeve bonds at 4.0% (JPM, as in the verified model)	Sleeve bonds at 5.0% (ASM, near today's 5y)
0%	\$177k / \$195k / \$215k	\$183k / \$201k / \$222k
30%	\$174k / \$202k / \$235k	\$178k / \$206k / \$241k
60%	\$159k / \$207k / \$273k	\$161k / \$210k / \$277k

So 60% equity buys about **+\$9k at the median** and **+\$55k at the 95th percentile**, and costs about **-\$22k at the 5th percentile**, against an all-bond sleeve. A former A/B tester will ask "what is the lift over the control?", and this is the answer. It sharpens the open equity-weight decision (brief section 9) with the 0%-equity end, which the verified runs did not show (F-407 covers 50/60/70% only).

- The best answer to "why choose us" may be a job, not a return.** The job she hires a firm to do (INT): let her make public promises in 2031 that she never has to walk back. Her 2028 money comes from reputation-driven work (R-C27). The case warns that overpromising "could damage her credibility" (R-C66). The 2026-27 criterion asks for recommendations "that can earn her confidence" (R-S25). B2b found that Wharton moved the criterion from "win him/her over" to "earn her confidence". A plan built around that job reads as understanding, not analysis. (Q1)
- One seed-bank premise fails verification.** Seed 17 speaks of "learning to fall safely". Her second book's own page says the heroine, "Once dubbed the Queen of Balance as her school's top rock climber", "suffers an injury that sidelines her". The "Falling" in the title is falling in love. This is publisher copy about a fictional character, not Laura's words (VP). Any "fall safely" or "Queen of Balance" metaphor would be both a misreading and the gimmick she rejects ("No more gimmicks.", B8a #21). (Q22)
- Do not assume she is anti-AI.** She worked on "Growth ML" at Twitter. Her studio lists a client described as "AI-powered online learning software" (VP). How the index's AI concentration is described, and how the team records its own AI use, should be neutral and exact. It should not be written as a values statement on her behalf. (Q18)

Counts: **25 questions** (section 3), plus a client-picture table (section 2), 20 verified phrases from her own professional pages (section 4), and leads for the specialists (section 5).

1. Method and what is new versus B8a/B8b

- Read first:** the brief, v4 Parts 1-3, 5 and 6, CLAUDE.md, the case, and the Phase A registers. I also skimmed B2a/B2b, B4a/B4b, B7a/B7b and B8a/B8b so that I do not repeat them. B8a and B8b already cover her quotes on risk, gimmicks, purpose statements, staged institutions, late books and the pull quote. I cite their findings by number (e.g. "B8a #21") and do not re-ask their questions.
- New primary reading (all 2026-09-27, .venv/bin/python research/insight_v1/scripts/fetch_text.py URL --grep, each returned PHRASE FOUND VERBATIM: YES):**
 - her resume PDF (work history, skills);
 - four "Visual Data Stories" pages (/data-journalism, /rewriting-herstory, /the-evolution-of-dance-music, /philly-happy-hours);
 - /projects, /editorial, /hey, and the Kirby book page.

These show *how she analyses and presents evidence*. That is the question my starting angle poses, and B8a/B8b did not mine these pages.

- Web search:** the session's WebSearch budget was used up (200/200) before this agent started. So I found no new URLs by search. I read only pages linked from her own site, and I could not check two leads: her public statements on generative AI, and #PublishingPaidMe. Both are listed as open leads (section 5).
- Numbers:** two small derivations, both reproducible:
 - The required-return IRRs: scipy root-find on the case cash flows.
 - The control-arm table: runs `research/verified_2026-09-27/strategy_mc.py` unchanged. The only changes are the sleeve equity weight (0/30/60%) and, in one column, the sleeve-bond compound return (5.0%). The script is in the session scratchpad, not the repo. I was told to write only this file, so D3 should rebuild it under `research/insight_v1/scripts/`.
- Not re-asked (brief section 8):** lock-early vs growth-first, tiering, one curve, surplus-only risk and the other settled items. Q3 touches item 5 ("more sleeve equity buys range, not median"). The **new**

evidence is the 0% equity control and the 5% bond variant.

2. The professional client picture (CFA-style objectives and constraints)

This is the frame a top adviser fills in first. The CFA Asset Manager Code B.6.a asks managers to "Evaluate and understand the client's investment objectives, tolerance for risk, time horizon, liquidity needs, financial constraints, any unique circumstances" (via A3 SH-25, VP). The Rules page tells teams to operate by that Code (R-W29). The IPS guide's own definition names "risk tolerance" (R-I2). Each cell says what we know, its status, and what we still do not know (the gap becomes a question).

Element	What we know	Status	What we still do not understand (question)
Return objective	Payments alone need 1.05%/yr on \$450k; payments + \$100k facility 3.15%; + \$200k 5.25%; + \$250k 6.28%	Derived from R-C26, R-C27, R-C45 (VRF)	Is the facility a "need", a "want" or a "wish" for her? (Q2, Q25)
Risk willingness	"willing to take thoughtful risks"; "Although..." she wants a balance (R-C37, R-C38). Revealed pattern: she leapt only after a contract secured the base (B8a #4-#5)	VRF + VP via B8a	Is her risk budget already spent on her career and the residency? We cannot interview her (R-W16), so every inference must be labelled (Q4)
Risk ability	Payments: none needed once bought. Sleeve 2027-30: high (living costs outside, R-C33). Pledged floor after 2031: close to nil (R-C66). Post-2033: tied to project needs	VRF + INT	How big are her "financial resources outside the portfolio" (R-C33)? Unknown; state an assumption (Q6)
Time horizon	Five dated points: 2027, 2028, 2031, 2033, 2033-42 (R-C29; X-1). Multistage	VRF	Where does risk capacity step down, and does the IPS name each stage? (Q5)
Liquidity	None before 2033 (R-C34). \$50k on each 1 Jan 2033-42. Facility "after establishing the operating reserve in 2033" (R-C57)	VRF	Lump sum on 1 Jan 2033, or committed then and drawn in stages? (Q25)
Legal/regulatory	Taxes and Taiwan law out of scope (R-C97). CFA Code applies (R-W29). No-contact rule (R-W16)	VRF/VP	Should the IPS say in one clause what it deliberately does not cover? (Q8)
Unique circumstances	Public credibility is part of her income (BS-03). Self-employed, lumpy income (B8a #1). USD promise for a Taiwan project (R-AN10). Educator, "Professor of Comics" (VP). Data/PM background (VP)	Mixed	How does each one change a number or a sentence? (Q1, Q13, Q17, Q18)

What this frame says about the plan (INT): her *need* is small and can be bought. Her *want* (the facility) is where risk earns anything. Her *credibility* is the binding constraint after 2031. So the most "hers" plan fits one sentence (the team's own words, later): buy the need, take measured risk only for the want, and never announce what is not yet bought.

3. Questions (25), with reasoning

Format for each: question · anchor · hypothesis (a GUESS; evidence and status) · what would change and where · deliverables · domain · seed.

A. What she is buying, and the return objective

Q1. What "job" is Laura hiring an asset-management firm to do: grow her money, or let her make promises she never has to walk back?

- Anchors: R-C8 ("the investment strategy that Laura ultimately chooses"), R-C66 ("If Laura promises more than she can ultimately contribute, she could damage her credibility"), R-S25 ("presents recommendations that can earn her confidence").
- Hypothesis (guess): the second.
 - Her 2028 money comes from "publishing advances, speaking engagements, licensing" (R-C27, VRF). All of that depends on her reputation (BS-03).
 - B2b found (VP) that Wharton replaced "would win him/her over as a client" with "earn her confidence".

- A product manager thinks in "jobs to be done": what the customer is really trying to get done (INT; a standard product idea, not a case fact).
- What changes (sentence):
 - The central idea of the 50-word pitch (R-I17) and the first line of the IPS (R-I9) would name her promise, not the portfolio.
 - The FR opening would be written from the 2031 conversation backwards.
 - This also answers the North Star directly.
- Deliverables: IPS, FR · Domain: D6 · Seed: 1.

Q2. What is Laura's required return, and should the IPS say that her needs can be met at below today's Treasury yield?

- Anchors: R-C26, R-C27, R-C45, F-601 (cash flows); R-I2 (the IPS "outlines an investor's financial goals, risk tolerance").
- Hypothesis (guess, with numbers): yes. The internal rate of return (IRR: the single yearly return that turns the deposits into the withdrawals) on the case flows is:
 - 1.05% for the payments alone;
 - 2.10% with a \$50k facility in 2033;
 - 3.15% with \$100k;
 - 4.20% with \$150k;
 - 5.25% with \$200k;
 - 6.28% with \$250k.

Derived by scipy root-find; inputs VRF. The Treasury curve is 4.98-5.54% from 5 to 20 years (F-001, VP). So the payments plus roughly \$190-200k of facility money are reachable at close to Treasury yields. Equity risk pays only for amounts above that.

- What changes:
 - (number) A "required return" line in the FR client picture.
 - (sentence) The IPS risk/return sentence could say that growth is for the facility, not for the promise.
 - It also gives the team a principled top end for the 2031 range.
- Deliverables: IPS, FR · Domain: D3 · Seed: 26.

Q3. Should every recommendation be shown against a simple "control": the same plan with an all-Treasury sleeve? And is the lift from 60% sleeve equity worth its cost at the 5th percentile to Laura?

- Anchors: R-C38 ("an appropriate balance between pursuing growth and protecting the capital required for her goals"); F-407; brief section 9 (sleeve equity 50/60/70% open).
- New evidence:
 - The verified model (F-407) only compares 50/60/70% equity.
 - Adding 0% and 30% (scratch run, ASM inputs as in the model) gives the table in the summary: 60% vs 0% adds +\$9k at p50 and +\$55k at p95, and costs -\$22k at p5 (bond 4.0%); about the same at bond 5.0%.
 - 30% equity costs almost nothing at p5 (-\$5k) and adds +\$5k at p50 and +\$19k at p95.
 - Caveat: the "all-bond" sleeve here is an intermediate-Treasury fund with price risk. A sleeve of matched zero-coupon bonds would be even narrower.
- Hypothesis (guess):
 - A former A/B tester (resume: "A/B testing", "Experimented and shipped 10 features", VP) will look first for the control and the lift.
 - The honest case for 60% is the upside tail for the facility, not the median. The team may find 30-40% the better "thoughtful risk".
- What changes:
 - (decision) The sleeve equity weight; it also changes the WInS mix now (brief section 9 mirror).
 - (number) FR scenario table.
 - (sentence) The IPS names the trade-off as "upside for the facility vs a lower floor".
- Deliverables: WInS-now, IPS, FR · Domain: D3 · Seed: 26.

B. Risk willingness vs risk ability, horizons, constraints

Q4. How should the IPS state Laura's risk tolerance, given that we must infer it without ever speaking to her? And which evidence would change our inference?

- Anchors: R-C37/R-C38; R-AN17 (the case gives no risk-tolerance number); R-W16 (no contact); R-I2.
- Hypothesis (guess):
 - Willingness is **high for her career and the residency**. It is **moderate for this money**: her career leaps were taken only after a secured base (B8a #4-#5, VP). The words "Although ... balance" point the same way.
 - Ability is **zero for the payments, high for the sleeve until 2031**, and **low for whatever she announces in 2031**.
 - A statistics graduate will expect the inference to be *stated as an inference*: "we infer X from Y; if Z, we would change it". It should not be asserted as a trait.
- What changes:
 - (sentence) One IPS clause splitting willingness from ability.
 - (decision) The sleeve equity weight (with Q3).
 - (sentence) FR: "what would change our view" (e.g. if the 2028 deposit is late).
- Deliverables: IPS, FR · Domain: D6 · Seed: 18.

Q5. Which time-horizon stages should the IPS name, and at which boundary does her ability to take risk step down?

- Anchors: R-S24 ("disciplined planning across Laura's changing time horizons and cash-flow needs"); R-C29; R-I12 ("How will the portfolio's asset allocation and composition change as future funding needs approach").
- Hypothesis (guess): five stages:
 1. 2026 to Jan 2027: rate risk before the money arrives (F-111, F-112).
 2. 2027-2030: the sleeve grows; ability is high.
 3. 2031-2033: the announced floor is bought; ability is close to nil on that part.
 4. 2033-2042: the reserve pays itself out; no trading needed ("if at all", R-AN27).
 5. The flexibility pool after 2033: its horizon is the project's needs.

Most teams will write "long-term" once (INT).

- What changes: (sentence) the IPS's "planned changes" clause; (decision) the share of the sleeve locked in 2031, which brief section 9 lists as open.
- Deliverables: IPS, FR · Domain: D8 · Seed: null.

Q6. Is this portfolio all of Laura's investable wealth, or one goal-dedicated pool? What should the team assume and say?

- Anchor: R-C33 ("Laura's living expenses and short-term financial needs will be covered by income and financial resources outside the portfolio").
- Hypothesis (guess): one dedicated pool.
 - The case says outside "financial resources" exist but gives no size.
 - The honest move is to assume nothing about their size. Treat this pool's risk capacity on its own terms, and say so in one clause.
 - Do not use the unknown outside wealth to justify extra risk (INT). The case is fiction about a real person (R-W4), so her real finances must not be guessed.
- What changes: (sentence) an IPS assumption clause; (decision) it rules out arguments like "she can afford losses because she has other money".
- Deliverables: IPS, FR · Domain: D8 · Seed: null.

Q7. Should the IPS state one primary goal plus explicit guardrails (limits that are never breached), in plain words? For example: maximise the responsible facility contribution, subject to (1) all ten payments bought and (2) no 2031 announcement above what is already bought.

- Anchors: R-I10 ("What principles will guide your investment decisions?"); R-I28; R-C71 ("The proposed range must also protect the portfolio's ability to fund the ten-year operating commitment").

- Hypothesis (guess): yes.
 - "Primary metric plus guardrail metrics" is how product teams set up experiments (INT; general product practice).
 - Laura's Twitter team was "Abuse & Safety" (VP), a function built around not causing harm while shipping growth.
 - The structure is short enough for the 500 words, and a judge can read it as the CFA "objectives and constraints".
- What changes: (sentence) the order and shape of the IPS principles paragraph; (sentence) each TN reflection says which goal or guardrail the trade serves.
- Deliverables: TN, IPS · Domain: D9 · Seed: null.

Q8. Should the IPS include one "non-goals" clause, i.e. what the plan deliberately does not try to do?

- Examples of non-goals: beat the market, time interest rates, fund beyond 2042, estimate the facility cost.
- Anchors: R-C49 (after 2042 out of scope); R-C95 (no facility cost estimate); R-C97 (taxes, Taiwan law); R-I23 (focus on "strategy and decision-making framework ... rather than describing individual investments").
- Hypothesis (guess): yes, but in about 15 words.
 - Product specs routinely list non-goals (INT).
 - It shows the reader that scope was a choice, not an oversight.
 - It also protects against a rival's "but what about X" framing.
 - Risk: wasted words. Test it against the 500-word budget (R-AN39).
- What changes: (sentence) the IPS; (sentence) the FR scope line.
- Deliverables: IPS, FR · Domain: D10 · Seed: null.

C. What evidence she trusts, and what she rejects on sight

Q9. Should every key number in the Final Report carry an evidence grade: observed market price, historical estimate, or assumption, with an honest quality word? This is how her own "Match Quality" column works.

- Anchors: R-C56 ("identify the assumptions supporting their recommendation"); R-C85; R-S26 ("uses reasonable assumptions and projections"); R-S28 ("uses data effectively to support conclusions").
- Evidence (VP, lauragao.com/rewriting-herstory):
 - "The [unfinished] results are below."
 - A table with a final "Match Quality" column graded "Good", "Unsure" or "Bad".
 - The dance study defines terms from the data source: "For the purposes of this study, I used Spotify's definitions and values for different song attributes."
 - The dance study ends with limits: "Further probing of other song metrics' effects on danceability would provide deeper analysis."
- Hypothesis (guess): yes.
 - The team already uses status labels internally (brief section 3). A client-facing version is 3 grades in one column of the FR assumptions table.
 - It matches her own habit, and it is the opposite of false precision.
- What changes: (sentence and table) the FR assumptions table format; (sentence) the IPS can say in words that "certainty" rests on prices you can observe, not on a forecast.
- Deliverables: FR · Domain: D9 · Seed: 18.

Q10. What rounding rule should the deliverables use, so that no number looks more precise than the model behind it?

- Anchors: R-AN32 (statistics degree); R-C69 (confidence statement); F-405, F-407, brief section 8.6.
- Evidence (derived):
 - Changing the equity assumption from 6.7% to 5.0% moves the lock-early median from \$207k to \$200k (F-405).

- Changing the sleeve-bond assumption from 4% to 5% moves it from \$207k to \$210k (Q3 table).
- The Monte Carlo sampling error is tiny by comparison.
- So model uncertainty at the median is about $\pm \$7\text{-}10\text{k}$.
- Hypothesis (guess):
 - Dollar outcomes to the nearest \$10k, and ranges as round figures.
 - Probabilities to the nearest 5 points, with the model named.
 - Treasury-priced amounts, such as the ladder cost (\$292,264, bought at market), may keep more digits: they are prices, not forecasts.
 - A statistician rejects "\$207,413" on sight (INT, from SH-04).
- What changes: (number) every figure in the FR, the 2031 range and the co-sponsor draft; (sentence) one line in the FR stating the rounding rule.
- Deliverables: TN, FR · Domain: D3 · Seed: 18.

Q11. What can six weeks of WInS results honestly show a statistics graduate? Should the Trading Notes say that the P&L is noise, and point to a metric that is not?

- Anchors:
 - R-T11 ("The focus should be on the quality of your reasoning and the intended role of each decision").
 - R-W81 ("Gains and losses generated in WInS do not determine how your team is evaluated").
 - R-T20 ("supported, tested, or refined").
- Evidence (derived, ASM: JPM equity volatility 16.47%, 2026 realised volatility of the ladder value 7.24%, F-014):
 - One standard deviation of a 6-week return is about **5.6%** for equities.
 - It is about **2.5% (~\$7.6k on \$300k)** for a 65/35 mix.
 - It is about **\$7.2k** for the ladder value alone.
 - So a WInS gain or loss smaller than about \$15k (two standard deviations) is indistinguishable from chance.
- Hypothesis (guess):
 - Yes. The testable claim in six weeks is **hedge tracking**: does the Treasury holding move with the present value of the ten payments (about \$289 per 0.01%, F-103)? B4a Q20 proposes that log.
 - My addition is the reason a statistics-trained client would demand it: it is the only WInS claim with a signal-to-noise ratio above 1.
- What changes:
 - (sentence) TN reflections avoid "the trade worked" language.
 - (decision) Start the tracking log with the first trade (WInS-now).
- Deliverables: WInS-now, TN, FR · Domain: D3 · Seed: 18.

Q12. Should the Final Report attribute the WInS result to its causes (rate moves on the Treasury part, equity moves, commissions, cash drag), the way she once attributed 1.5M unexplained ad transactions?

- Anchors: R-C90 ("This evaluates how that strategy was implemented"); R-S26 ("integrates quantitative and qualitative analysis"); Guide p.5 "reasoning, assumptions, and tradeoffs" (R-new, VP per B7a/B7b).
- Evidence: resume, "Investigated cause of 1.5M cases of transactions from unattributed ads using SQL, R, and Excel" and "Created weekly reports for leadership of findings and data visualizations using Tableau" (VP).
- Hypothesis (guess):
 - Yes. A four-bar attribution chart is one of the few FR charts that "uses data effectively" without claiming skill from noise.
 - It also shows that the Treasury loss or gain was offset by the liability's change, as designed.
 - Data needed: daily holdings, trade log, the Treasury curve on the start and end dates, and commission count (\$25/\$10, F-607).
- What changes:
 - (decision) Record daily or weekly snapshots now; WInS closes Nov 6 (R-W58).
 - (FR chart) The FR chart list.

- Deliverables: WInS-now, FR · Domain: D3 · Seed: null.

Q13. Should her "wrong-way" income risk be shown as named scenarios (late, smaller, both) rather than as a modelled correlation between her income and markets? No data exists to estimate that correlation.

- Anchors: R-C27 ("She will contribute an additional \$150,000 at the beginning of 2028"); R-AN35 (the case says "will"); brief section 9 (wrong-way risk open); B8a Q08-Q09 (late books; political and budget exposure).
- Hypothesis (guess): scenarios.
 - Any correlation number for one person's creative income against the S&P 500 would be invented.
 - A statistician would see through a made-up coefficient (INT).
 - Three labelled team scenarios beyond the case are honest and simple:
 - deposit 12 months late;
 - deposit \$75k;
 - rates fall before January 2027 *and* the deposit is late (the joint tail, brief section 9).
- What changes: (decision) D3's modelling approach; (number) the FR stress table; (sentence) the assumption on "timing of cash flows" (R-C85).
- Deliverables: FR · Domain: D3 · Seed: 16.

D. How she would test and explain the plan to others

Q14. Before submitting, should the team "usability-test" the co-sponsor paragraph and the IPS's core rules with real non-finance readers, and report what changed, as she did with nurses and doctors?

- Non-finance readers here means, for example, school staff or parents. This involves no contact with Laura.
- Anchors: R-C72 ("draft part of Laura's fundraising materials"); R-S28 ("clearly and credibly to prospective co-sponsors"); R-S27 ("reflects meaningfully on the team's growth and response to challenges").
- Evidence (resume, VP):
 - "Collaborated with researchers at National Institute of Health for usability tests with nurses and doctors".
 - "Beta-tested MVP with 300K+ users".
 - Her editorial list includes NPR explainer comics, e.g. "There's a way to get healthier without even going to a gym. It's called NEAT" (VP).
 - She wrote "Recruiting for Tech for Non-Technical Students" (VP, title only).
- Hypothesis (guess):
 - Yes. A test takes about 30 minutes: 5 readers restate the range and the confidence in their own words, and the team counts the errors.
 - It is cheap, it is real process evidence for the Articulation criterion, and it catches jargon.
 - Few teams will do it (INT).
- What changes: (decision) add a test step before Dec 4; (sentence) the Articulation section reports the result, e.g. "3 of 5 misread X, so we changed Y".
- Deliverables: FR · Domain: D9 · Seed: 32.

Q15. Who exactly will read her 2031 fundraising material? Should the draft be built for 2-3 named reader types (her own habit of writing for personas) rather than for "co-sponsors" in general?

- Anchors: R-C64 ("A meaningful personal contribution may signal that the residency is financially viable"); R-C44 (co-sponsors, grants, collaborators); R-AN21.
- Evidence (VP, lauragao.com/philly-happy-hours): "we set out to build a guide for various types of students to select the best happy hour for any occasion", with sections such as "For the student drowning in debt". Her record of running Pride in Panels with a public-library partner, free and with mini-grants, is in B8a #14-#18 and B8b.
- Hypothesis (guess): three reader types are enough:

1. A public or cultural grant-maker, who needs proof of viability and a floor.
2. An institutional partner (e.g. a university or arts organisation), who needs to know the operations are funded.
3. A private donor, who needs to see the size of her own commitment ("skin in the game").

One paragraph can serve all three if it answers each one's first question. This is D5's research to confirm.

- What changes: (sentence) the content checklist for the FR fundraising draft.
- Deliverables: FR · Domain: D5 · Seed: null.

Q16. Should the Final Report include one simple visual of the promise that she, an illustrator, could redraw for co-sponsors? For example: ten blocks for the ten funded years, plus a bar for the facility range.

- Anchors: R-C72; R-S28 ("uses data effectively to support conclusions"); R-I55 (no graphics in the IPS, so the FR only).
- Evidence: she made "Interactive visualizations of career pathways" and uses Tableau (VP). "The Wuhan I Know" began as a short comic. Her view that comics "marry imagery with the power of words" is in B8a #25 (VP).
- Hypothesis (guess):
 - Yes, one visual only.
 - It must be simple enough to redraw by hand, and it must carry the core claim: bought vs estimated.
 - A decorative infographic would fail her "No more gimmicks." test (B8a #21).
- What changes: (FR chart) spec: data = ten payment dates marked "bought", plus the 2031 floor and the p50/p95 band from D3.
- Deliverables: FR · Domain: D9 · Seed: 32.

Q17. Should the co-sponsor wording be written so it survives translation into Mandarin? That means no idioms, "US\$" every time, an NT\$ reference with its date, and "confidence" defined in words.

- Anchors: R-AN10 (no currency named); R-C68 ("the dollar range"); BS-04 ("dollar" is ambiguous in Taiwan).
- Evidence: her resume lists "Languages: Mandarin Chinese" (VP). The residency is in Taiwan (R-C39). B8a #33 records her own care over translation choices in *Messy Roots* (VP).
- Hypothesis (guess):
 - Yes. Taiwanese partners may read her material in Chinese.
 - Idioms such as "skin in the game", "upside" or "downside" translate badly.
 - The cost is near zero.
 - The team should not translate anything itself; it only writes translatable English.
- What changes: (sentence) the FR fundraising draft style checklist.
- Deliverables: FR · Domain: D9 · Seed: null.

Q18. How should the team describe (a) the growth sleeve's AI and mega-cap concentration and (b) its own AI use, given that she has built ML and AI products and is not a known AI critic?

- Anchors: R-W46 ("how you use it must be recorded in your Works Cited pages"); R-S24 ("uses appropriate diversification"); seed 20.
- Evidence (VP): "Growth ML (2018-2019)" at Twitter; her studio client "Retainit, AI-powered online learning software". I could not search for her public statements on generative AI in art (search budget used up). That is an open lead (section 5).
- Hypothesis (guess):
 - Describe index concentration as a plain diversification fact, with the share from the fund's own holdings page. It should not be a values judgement made on her behalf.
 - Describe AI use exactly and modestly: what AI did, and what the students did.
 - Someone who shipped ML features knows how AI output fails. Overclaiming is the risk, not using AI.
- What changes: (sentence) the FR concentration sentence and the Works Cited AI statement; (decision) no "AI avoidance" tilt without her stated preference.

- Deliverables: FR · Domain: D2 · Seed: null.

E. Simplicity, failure and the evaluation of alternatives

Q19. What is the smallest set of rules (a "minimum viable plan") that still passes all five case tests? Can each extra rule show a measurable improvement?

- Anchors: R-C79 to R-C84 (the five tests); R-I30 ("clear, cohesive, and well-supported"); R-I23; design principle (brief).
- Hypothesis (guess):
 - Three rules may be enough: 1. Buy the ten payments when the money arrives (with the rates-fall variant). 2. The sleeve keeps a fixed mix, rebalanced by a band. 3. In 2031, announce only what is bought, plus a stated upside.
 - Everything else (TWD quote, Taiwan tilt, glide paths) must show a lift in a number, or go.
 - This is her "Always start small" (B8a #26) and last year's lesson in one test.
- What changes: (decision) the rule list in the IPS; (number) a "lift" line per extra rule in the FR.
- Deliverables: IPS, FR · Domain: D8 · Seed: null.

Q20. Does the plan show her, in one table, how each goal "fails softly": the worst realistic outcome for the payments, the 2031 announcement, the facility and flexibility, and who would notice?

- Anchors: R-C70 ("how favorable and unfavorable market outcomes could affect the amount she can provide"); R-C86; R-S26 ("under varying market outcomes").
- Evidence:
 - Her "Anti-Resume Project" to "normalize failure" and "33 job rejections" (B8a 2f, VP/reporter).
 - Her Twitter safety work (VP).
 - Brief section 8.8: under lock-early a missing 2028 deposit leaves the payments safe and the facility ~\$7-11k ("correct failure mode").
- Hypothesis (guess):
 - Yes, a four-row table.
 - Payments: fail only if the U.S. Treasury defaults.
 - Announcement: fails only if she announces more than is bought.
 - Facility: gets smaller, but never negative.
 - Flexibility: absorbs overruns.
 - Someone who normalises failure will trust a plan that shows its failures calmly, more than one that hides them (INT).
- What changes: (FR table); (sentence) a bad-case line in the co-sponsor draft.
- Deliverables: FR · Domain: D6 · Seed: null.

Q21. Which rejected alternatives should the Final Report show, with their numbers, as the team's "experiment log"? Candidates: growth-first, rolling short, equity-heavy, all-Treasury.

- Anchors: R-S27 ("clearly explains the team's research and decision-making process"); Guide p.5 "reasoning, assumptions, and tradeoffs" (R-new, VP per B7a); R-T9 ("aligned with, tested, or refined").
- Evidence: brief section 8.1-8.2 (growth-first misses 3-8%, tiering +\$20k per -100bp); Q3 table (all-Treasury control).
- Hypothesis (guess):
 - Show three rejected arms with one number each, and why they were rejected.
 - A product person reads a decision without its alternatives as untested (INT).
 - It is also the most honest evidence for the Articulation criterion.
- What changes: (FR section and chart) a small "options we tested" table; (sentence) one TN reflection can reference the test.
- Deliverables: TN, FR · Domain: D7 · Seed: null.

Q22. Which of Laura's own verified ideas can the strategy echo honestly? And which proposed echoes fail verification or turn into tokenism?

- Anchors: R-C21 (pull quote, unattributed per B8a/B8b); R-C10 (themes of her work); brief tokenism rule; seeds 15 and 17.
- Evidence (VP):
 - **Fails:** "fall safely" and "Queen of Balance". Her book page says the heroine is a "top rock climber" who "suffers an injury that sidelines her". "Falling" means falling in love. "Queen of Balance" is publisher copy about a fictional character.
 - **Fails as her words:** the pull quote (B8a Q01, B8b).
 - **Passes as a factual parallel, used sparingly:**
 - "If what you want doesn't exist, create it yourself" (B8a #11, PQ). Parallel: no iBonds-style fund exists for the 2038-2042 payments (F-206), so those rungs are built from individual Treasuries.
 - Her published "Match Quality" grades. Parallel: our evidence grades (Q9).
 - Her "floor-first" career leap (B8a #4-#5).
- Hypothesis (guess):
 - Echo at most one of her ideas in the FR, in the team's own words, with a citation.
 - Never echo in the IPS pitch, where it would read as flattery.
- What changes: (sentence) FR narrative; removes any book-title metaphor from all deliverables.
- Deliverables: IPS, FR · Domain: D6 · Seed: 15.

Q23. When she compares firms, would she trust one that says plainly which parts of the plan need almost no management (the bought payments) and which parts it actively manages (the sleeve)? And that charges or discloses fees accordingly?

- Anchors: R-C26 ("with an asset management firm"); R-W29 (CFA Code; F.4.d fee disclosure via SH-25); BS-01 (no fee modelled); B4a Q5 (fees on the reserve eat the \$7.7k headroom).
- Hypothesis (guess):
 - Yes. Saying "this part runs itself" is a costly honesty signal: the firm gives up a fee argument.
 - It also matches her CFPB internship and her financial-literacy teaching (B8a 2f, VP).
- What changes:
 - (sentence) An IPS clause on which parts are "set and hold" and which are "managed".
 - (number) The fee assumption in FR projections (A3: -\$17k median at 0.5%/yr).
- Deliverables: IPS, FR · Domain: D8 · Seed: 1.

Q24. How many sentences in the deliverables should describe Laura as a person (as opposed to her goals)? What may each rest on, so that she feels understood rather than analysed?

- Anchors: R-S25 ("demonstrates a thoughtful understanding of the client"); R-W4 ("While the client is real, the financial scenario is developed specifically for the competition"); privacy and tokenism rules (brief section 4).
- Hypothesis (guess):
 - Very few: perhaps 2-3 in the FR and 1 in the IPS.
 - Each one must link a *verified professional fact or the case's words* to a *decision*, for example "because her 2028 money depends on reputation, we ...".
 - Psychological readings of her identity, or claims about her real plans, would feel like being analysed. They also break the brief's rules (INT).
 - Understanding shows in how well the plan fits her, not in how much it says about her.
- What changes: (sentence) a checklist rule for all three deliverables: each Laura sentence = fact + source + decision.
- Deliverables: TN, IPS, FR · Domain: D10 · Seed: 32.

Q25. Should the facility contribution be framed as an amount committed in 2033 and paid as the project needs it, rather than a lump sum paid on 1 January 2033? Her own institutions scale in stages.

- Anchors:
 - R-C57 ("After establishing the operating reserve in 2033, Laura must decide how much of the remaining portfolio she can responsibly contribute").
 - R-C58 ("committing all remaining assets could limit her financial flexibility as the project develops").

- R-C84.
- Evidence: Pride in Panels grew in stages, e.g. "We definitely hope to offer more as the festival grows" (B8a #16, VP). B8b summary item 7.
- Hypothesis (guess):
 - A committed-but-drawn contribution keeps the undrawn part invested in short Treasuries. That preserves flexibility at little cost, fits her pattern, and stays inside the case, which fixes the *decision* in 2033, not the payment date.
 - Risk: it adds a moving part. Use it only if D5 finds that co-sponsors accept staged pledges.
- What changes: (decision) the FR facility recommendation's form; (sentence) the co-sponsor draft's verb ("commits ..., drawn as construction proceeds").
- Deliverables: FR · Domain: D5 · Seed: null.

4. Evidence register (new primary reading by B9b; all VP, grep-verified 2026-09-27)

Professional content only. Contact details, hobbies and family-related pages on the same site were read but not recorded.

#	Exact text	Source	Date of piece	What it shows (INT)
1	"Experimented and shipped 10 features that resulted in +4.5M DAU"	https://lauragao.com/s/Laura-Gao-Resume.pdf	undated, current	Experiment-driven decisions
2	"Product Management: JIRA, Confluence, A/B testing"	same	same	Control-vs-variant thinking
3	"Growth ML (2018-2019)"	same	same	Hands-on ML product work
4	"Abuse & Safety (2019-2020)" (as printed: "Abuse & Safety (2019-2020), Growth ML (2018-2019)")	same	same	Harm-limiting guardrails
5	"Investigated cause of 1.5M cases of transactions from unattributed ads using SQL, R, and Excel"	same	same	Attribution analysis
6	"Created weekly reports for leadership of findings and data visualizations using Tableau"	same	same	Regular reporting to decision-makers
7	"Beta-tested MVP with 300K+ users of economics platform, Marginal Revolution University"	same	same	Tests with real users; exposure to economics teaching
8	"Collaborated with researchers at National Institute of Health for usability tests with nurses and doctors"	same	same	Usability testing with non-specialists
9	"AI-powered online learning software"	same	same	Worked for an AI-product client
10	"Languages: Mandarin Chinese"	same	same	Can work with Taiwanese partners in Mandarin
11	"Bachelor of Science in Statistics and Information Decisions, Concentration in Product Design" (read on page; B8a also records it)	same	same	Statistics plus design
12	"The [unfinished] results are below." and the column "Match Quality" (rows "Good", "Unsure", "Bad")	https://lauragao.com/rewriting-herstory	undated (student-era project)	Publishes uncertain results with quality grades
13	"Using text analysis to explore powerful female counterparts of well-known men in history."	same	same	Data used to reframe a story
14	"For the purposes of this study, I used Spotify's definitions and values for different song attributes."	https://lauragao.com/the-evolution-of-dance-music	undated	Defines terms from the source
15	"However, tempo and speechiness display a diminishing returns relationship."	same	same	Comfortable with "sweet spot" findings
16	"Further probing of other song metrics' effects on danceability would provide deeper analysis."	same	same	States limits and next steps
17	"we set out to build a guide for various types of students to select the best happy hour for any occasion" and "For the student drowning in debt"	https://lauragao.com/philly-happy-hours	undated	Designs for reader personas

#	Exact text	Source	Date of piece	What it shows (INT)
18	"Only restaurants with Happy Hours on Thursday, Friday or Saturday have been included in this dataset."	same	same	Discloses data scope
19	"shapes the next generation of creators as a Professor of Comics at California College of the Arts"	https://lauragao.com/hey	current	Educator who assesses student work
20	"Once dubbed the Queen of Balance as her school's top rock climber, Kirby Tan suffers an injury that sidelines her for the rest of the season."	https://lauragao.com/kirbys-lessons-for-falling-in-love	current (book 2025)	Publisher copy about a fictional character; corrects seed 17's premise

Also seen (VP, titles only): "Recruiting for Tech for Non-Technical Students" and "There's a way to get healthier without even going to a gym. It's called NEAT" (NPR), both on <https://lauragao.com/editorial>. The same page describes The Sign.al: "Our bold interactive projects aim to help students discover their passions and live deliberately."

5. Leads for the specialists

• D3 (quant):

- Rebuild the control-arm runs (Q3) as `research/insight_v1/scripts/` code, with a docstring.
- Inputs: `strategy_mc.simulate("L", eq_w=w)` for w in $\{0, 0.3, 0.6\}$.
- Sleeve-bond compound 4.0% (JPM) and 5.0% (ASM).
- Add a zero-coupon-matched sleeve variant, where the sleeve buys 2033 STRIPS for the facility.
- Compute the required-return table (Q2).
- Compute the 6-week noise figures (Q11) with IEF/TLT realised volatility, if it can be fetched from iShares.
- Build the attribution template (Q12).
- Build scenario rows, not a correlation (Q13).

• D6 (client psychology):

- Write the willingness-vs-ability paragraph spec (Q4) using B8a #1-#10 plus section 2 here.
- Build the fail-softly table (Q20).
- Check that each Laura sentence meets "fact + source + decision" (Q24).

• D9 (communication):

- Usability-test protocol (Q14): 5 non-finance readers, restate-in-own-words scoring.
- Translation-safe style list (Q17).
- One-visual spec (Q16).
- Evidence-grade column (Q9).

• D5 (co-sponsors):

- Reader types for Taiwan arts funding (Q15).
- Whether funders accept staged founder pledges (Q25).
- Taiwan's arts-funding bodies were not researched here; their sites (e.g. the Ministry of Culture) were refused earlier in this run (brief section 2: artres.moc.gov.tw 403).

• D2: the concentration share for the chosen broad fund, from the issuer's holdings page (Q18).

• D13 (quotes): re-check rows 1-20 before any are quoted in a deliverable. The resume and the data pages are undated. Cite them as "lauragao.com, accessed [date]".

• Open leads I could not reach (search budget exhausted):

- Laura's public statements, if any, on generative AI in illustration.
- Whether she took part in 2020's #PublishingPaidMe advance-transparency campaign (would show how she talks about money in public).
- The Sign.al "Deconstructing Penn Pathways" page: <https://thesign.al/pages/deconstructing-penn-pathways/> (B8b: empty page).
- The Medium post "Your Freshman Summer, Demystified" (medium.com returned 403 to B8b).

6. Sources (all accessed 2026-09-27)

Source	URL	Status
Laura Gao resume (PDF)	https://lauragao.com/s/Laura-Gao-Resume.pdf	VP (grep)
Visual Data Stories index	https://lauragao.com/data-journalism	VP
Rewriting Herstory	https://lauragao.com/rewriting-herstory	VP
The Evolution of Dance Music	https://lauragao.com/the-evolution-of-dance-music	VP
Philly Happy Hours, Deconstructed	https://lauragao.com/philly-happy-hours	VP
Projects	https://lauragao.com/projects	VP
Editorial	https://lauragao.com/editorial	VP
About ("Hey")	https://lauragao.com/hey	VP
Kirby's Lessons for Falling (In Love) page	https://lauragao.com/kirbys-lessons-for-falling-in-love	VP (publisher copy)
Case, IPS guide, TN guide, SMAApply page	competition/official/2026_27/	VRF
Verified model	research/verified_2026-09-27/strategy_mc.py	VRF (script); my variants ASM
B8a, B8b, B2b, B4a, B4b, B7a, B7b	research/insight_v1/phase_B/	Cited for their findings; their status labels apply
Phase A registers	research/insight_v1/phase_A/	VRF

What this teaches

A professional adviser does not start with a portfolio. The first step is a client picture:

- what return she *needs* (here only about 1% a year for the payments);
- how much risk she is *willing* to take, and how much she *can* take (two different things);
- when she needs money, and what makes her situation unusual.

Reading how a client works also tells you what she will trust. Laura grades her own results "Good/Unsure/Bad", tests products with real users and compares every change against a control. So the plan most likely to earn her confidence:

- shows the simple alternative (all Treasuries) next to ours, with the difference in plain numbers;
- labels what is known, estimated or assumed;
- rounds to the precision the model deserves;
- says calmly how each goal would fail.

Understanding a client shows in how well the plan fits her, not in how much it says about her.

B10a Contrarian: attacking "lock early" from Laura's side

Agent B10a, insight_v1 run, Phase B (question generation). Written 2026-09-27. Lens: Contrarian (attack our own strategy). Starting angle: **Laura's side**. She "has been willing to take thoughtful risks" and wants "an appropriate balance between pursuing growth and protecting the capital required for her goals" (R-C37, R-C38). Would she call a plan that puts ~97% of her first deposit into Treasuries "balance"? Is there a design that keeps the ten payments just as certain but gives her more upside or more options? The other B10 agent starts elsewhere.

This file is AI-generated research (Claude Code) for Team Caplet, for brainstorming only. It holds questions, evidence, numbers and checklists. It has **no submission-ready prose**. The six students decide and write every deliverable in their own words, and must record AI use in the Final Report's Works Cited (R-W46). Laura appears only through the case and her public professional record. I left out every personal-life detail, including personal material on pages I opened. Nothing here suggests contacting her (R-W16).

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (I read it on the primary page on 2026-09-27; URL in section 6); **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **INT** = INTERPRETATION (my reading, never a fact); **DER** = derived from the verified simulation with the changes stated (script in Appendix A; its inputs carry their own labels). Every "hypothesis" below is a **guess**.

Terms used (defined once):

- **Ladder / lock**: buying Treasury bonds that pay out exactly on the ten payment dates, so the payments no longer depend on markets.
- **Sleeve / surplus**: the money left after the ladder. It pays for the facility contribution and flexibility.
- **Floor**: part of the sleeve moved into a Treasury maturing just before 2033. It becomes a known minimum facility amount.
- **Barbell**: holding only two very different things, safe bonds and stocks, with nothing in between. Here: a bought floor plus 100% stocks with the rest.
- **Ratchet**: locking part of a gain on several dates instead of one ("lock in good years").
- **p5 / p50 / p95**: the outcome that 5% / 50% / 95% of simulated paths fall below. p50 is the median.
- **Equity risk premium**: the extra yearly return stocks are expected to earn over safe government bonds.

0. Summary (read this first)

The attack **mostly fails, and the way it fails is itself the insight**. Three findings change sentences or a design choice. None of them reopens "lock early vs growth-first" (brief section 8.1).

1. **In 2026, taking risk buys Laura a wider range, not a higher middle**. I tested every Laura-friendly variant I could build (DER, Appendix A; payments stay funded in every case where the 2028 deposit arrives):

Design (JPM equities, sleeve bonds at 5.0%)	2033 surplus p5 / p50 / p95	vs current plan
Current plan (sleeve 60/40, lock 80% of sleeve in a 2-year Treasury in 2031)	\$161k / \$210k / \$277k	-
Sleeve 100% equity	\$137k / \$214k / \$332k	median +\$4k, p5 -\$24k
Delay the whole lock to Jan 2028 (use both deposits)	\$130k / \$214k / \$331k	median +\$4k, p5 -\$31k
Stage the 2033-36 rungs to Jan 2028	\$157k / \$213k / \$293k	median +\$3k, p5 -\$4k
Ratchet: lock the floor in thirds, 2029/30/31	\$171k / \$208k / \$259k	median -\$2k, p5 +\$10k
Barbell: lock 50% of the sleeve in Jan 2028 in a Treasury maturing before 2033, rest 100% equity	\$163k / \$210k / \$293k	median same, p5 +\$2k, p95 +\$16k, floor known 3 years earlier

Design (JPM equities, sleeve bonds at 5.0%)	2033 surplus p5 / p50 / p95	vs current plan
All-Treasury (sleeve 0% equity)	\$183k / \$201k / \$222k	median -\$9k, p5 +\$22k

The largest median gain from any extra risk is about **\$4k on a ~\$210k facility (about 2%)**. Laura's "growth" choice this year is really a choice about **spread**: how much p5 she gives up for how much p95.

2. **A second major house says U.S. stocks may not beat today's Treasuries at all.** Vanguard (22 July 2026, model run 30 June 2026) cut its 10-year expected return for U.S. equities to **4.2%-6.2%** a year (VP). Its midpoint, 5.2%, is about the same as the 10-year Treasury yield (5.17%, F-001). Using that midpoint, the sleeve's median is **\$201k / \$204k / \$204k** at 0% / 60% / 100% equity. Growth-first's median (\$205k) no longer beats lock-early's (\$204k), and its miss rate rises to 4.5% (DER). The brief's "more equity buys range, not median" (section 8.5) came from one house (JPM). **What is new is that a second house removes the median gain entirely.** That gives Laura an honest sentence: this year, her growth comes mostly from bond yields.
3. **The one design that beats the current plan without extra risk to the payments is a floor-first barbell.** It also matches Laura's documented pattern better than the current plan does. She leapt only after a contract secured her base: "So when she got the book deal with HarperCollins, she gave Twitter her notice." (Inverse/Input, 2022, VP). Her leap was then total, not 60%. INT: a bought facility floor plus a full-equity remainder is "base first, then leap". A 60/40 sleeve is a half-leap. Numbers: same median, slightly better p5, a higher p95, and a floor that is known from 2028, not 2031. The cost is a wider 2031 range, because stocks stay at risk until 2033. This needs D3's check with fat tails and historical crash sequences before anyone adopts it (Q4, Q17).
4. **Three Laura-side blind spots that change sentences:**
 - **"Balance" must be shown goal by goal.** The whole-portfolio equity share is ~1.5% in 2027, ~20% in 2028-30 and ~4% in 2031-32 (DER). Shown as one blended number, it reads as "no growth" (Q3).
 - **What is the firm's fee for?** A 1% fee on all assets cuts the median surplus from \$210k to **\$177k**. The same fee charged on the sleeve only gives **\$202k** (DER). A statistics graduate will ask why she pays a manager to hold a buy-and-hold ladder (Q1).
 - **Locking at the highest 10-year yield since 2007** (F-004) is a fact she can check, and it deserves a pre-commitment against regret: if rates rise 100bp after the lock, the ladder would have cost ~\$27k less (F-102) (Q16).
5. **The joint tail now has a size.** Suppose the short rungs have to wait until 2028 (because rates fell, or on purpose) and the 2028 deposit is **zero**. Then 24-28% of paths leave the payments short by **~\$8-10k** on average. With a \$75k deposit, no path is short (DER, Q7). That answers part of brief section 9's "UNANSWERED" joint tail.

Top five questions for the next phase: Q20 (how to choose between designs when the medians are equal), Q4 (barbell), Q2 (two houses), Q3 (balance by goal), Q14 (the all-Treasury rival).

1. Method

1. Read the brief, v4 Parts 1-3, 5 and 6, CLAUDE.md, the case, the Phase A registers, the WInS week-1 files, and the Phase B files that border this lens (B1a, B2a, B4a, B4b, B8a, B8b, B9a, B9b) so that I do not re-ask their questions. Where they got there first, I cite them and add only new numbers.
2. Built each design a smart rival or Laura herself might prefer, and ran it on the verified simulation's engine (`research/verified_2026-09-27/strategy_mc.py`: same seed, 200,000 paths, same JPM inputs). I changed one thing at a time and labelled each change (Appendix A). The baseline reproduces F-401/F-402 exactly (\$159k/\$207k/\$273k; floor \$127k/\$165k/\$217k).
3. Fixed one known inconsistency before comparing designs: the sleeve's bonds earn **5.0%**, not JPM's 4.0%. The ladder is priced at ~5.1% forwards, and IEF's yield to maturity is 5.17% (F-012, F-203; B9a raised this). This lifts the baseline to \$161k/\$210k/\$277k. All design comparisons in this file use this corrected baseline unless stated.
4. Web: the WebSearch budget for this session was already used up (200/200 calls). So I read known primary pages directly with `fetch_text.py` (Vanguard VCMM, Inverse, Russell) and grep-checked every

quote.

5. Filters for each question: (a) would the answer change a decision, a number or a sentence; (b) is it anchored in an R-id or F-id; (c) is it not already answered in brief section 8, or else is there new evidence (stated).

2. The questions (ranked; 19 kept, 6 parked in section 3)

Q20 (rank 1). When every design has about the same median, what rule should choose between them: the p5 Laura can accept, the p95 she hopes for, the floor co-sponsors can bank, or the simplest structure?

- Anchors: R-C38, R-C37, R-I11, R-S25; F-401, F-407.
- Official words: "she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals." (case p.2, VRF)
- Why it matters (DER, table in the summary): the medians of seven designs sit within \$201k-\$214k. Their p5 runs from \$130k to \$183k and their p95 from \$222k to \$332k. The team currently justifies "60% equity" as a middle setting. Given how flat the medians are, that is a **choice of spread**, and the case asks the team to recommend it ("recommend an appropriate balance").
- Hypothesis (guess): the rule that is most Laura's is "choose the highest bought floor that still leaves a real upside". Her documented pattern is a secured base, then a leap (Inverse 2022, VP). The co-sponsor task punishes a missed low end, not a missed high end (R-C66). B9a's "affordable loss" framing gives the p5 test; this question adds the tie-break: at equal medians, prefer the design with the higher known floor and fewer moving parts.
- Would change: **decision** (the selection rule written in the IPS: one sentence on how the balance was chosen); **sentence** (the IPS answer to R-I11 "How does your strategy balance growth, risk...").
- Deliverables: IPS, FR. Specialist: D6 (with D3 for the numbers). Seed: 29.
- North star: Laura sees her own decision rule (secure the base, then leap) turned into the plan's selection rule, not a generic "moderate" label.

Q4 (rank 2). Should the sleeve become a barbell: in January 2028 lock a facility floor in a Treasury maturing just before 2033, and hold the rest 100% in broad equity, instead of a rebalanced 60/40 sleeve plus an 80% lock in 2031?

- Anchors: R-C62 to R-C71, R-C38, F-402, F-206, F-001 (5-year 4.98%).
- Official words: "Rather than promising one exact amount, she wants to communicate a credible range for her potential contribution." (case p.3, VRF)
- Evidence (DER; JPM equities; sleeve bonds 5.0%; lock at the 5-year par yield 4.98%, ASM):

Share locked in Jan 2028	Floor (known 2028), median	2033 surplus p5 / p25 / p50 / p75 / p95
Current plan (for reference)	floor known only in 2031: \$167k median	\$161k / \$188k / \$210k / \$235k / \$277k
50%	~\$100k	\$163k / \$188k / \$210k / \$238k / \$293k
65%	~\$131k	\$175k / \$192k / \$208k / \$227k / \$265k
80%	~\$161k	\$186k / \$196k / \$205k / \$216k / \$238k

With Vanguard-midpoint equities (5.2%): current \$157k / \$204k / \$269k; barbell 50% \$159k / \$203k / \$280k (DER).

- Why the barbell can match the plan (INT): the current plan cuts its stock exposure sharply in 2031 (whole-portfolio equity ~4%, Q3). The barbell keeps stocks on less money for longer, with a hard bought floor underneath. This is buy-and-hold with a floor versus constant-mix rebalancing, a classic comparison in the asset-allocation literature. Perold & Sharpe's "Dynamic Strategies for Asset Allocation" (1988) is the usual reference; I did not open it (lead for D3).
- The cost: in 2031 the equity half still has two years of risk. Rough 2031 range (ASM: lognormal JPM, 2-year log sd ~21.5%): a low end of "floor + 75% of the 2031 equity value" is missed only if stocks fall more than 25% in two years, about 3% of paths. At medians that low end is roughly \$100k + 0.75 × ~\$96k ≈ \$172k, against the current plan's ~\$167k floor. **To be verified by D3; not yet simulated.**

- Hypothesis (guess): a barbell with 50-65% locked in 2028 weakly beats the current plan on p5 and p95 at the same median. It is also simpler to explain (two instruments, each with one job). It needs a 2031 rule for the equity part, and it must survive a fat-tailed or historical-sequence test (Q17).
- Would change: **decision** (sleeve structure and floor timing, before the IPS freezes on Nov 6); **number** (FR range tables); **WInS-now** (the sleeve's bond slot: Q11).
- Deliverables: WInS-now, TN, IPS, FR. Specialist: D3. Seed: 9.
- North star: a plan that mirrors how Laura actually took her biggest career risk (secure the base, then leap fully) is one she would recognise as hers. A 60/40 half-leap is what 6,000 other teams will write.

Q2 (rank 3). With Vanguard's June-2026 model giving U.S. equities 4.2%-6.2% a year (about today's 10-year Treasury yield) and JPM giving 6.7%, should the FR show both houses, and should the IPS say plainly that this year Laura's growth comes mostly from bond yields, with stocks adding range?

- Anchors: F-301, F-317, F-315, F-004, R-C77 ("reasonable return assumptions"), R-S26.
- Official words: "use reasonable return assumptions consistent with their strategy." (case p.4, VRF)
- New evidence (VP, Vanguard VCMM page dated 22 July 2026, read 2026-09-27): "our 10-year expected annualized return for U.S. equities declined from a range of 4.9%–6.9% to a range of 4.2%–6.2%". Developed ex-U.S. 4.5%–6.5%; emerging markets "2%–4%". Vanguard also warns that "valuations tend to be poor predictors of performance over the short or even intermediate term".
- DER with the Vanguard midpoint (5.2% compound, JPM volatility kept, ASM):
 - lock-early sleeve at 0/60/100% equity: medians **\$201k / \$204k / \$204k**; p5 \$183k / \$157k / \$130k;
 - growth-first: median **\$205k**, p5 \$5k, **miss 4.5%**; lock-early: \$157k / \$204k / \$269k.
 - Under JPM (bonds 5%): growth-first median \$238k vs lock-early \$210k, miss 2.7%.
- What is new versus brief section 8.1/8.5: those rest on JPM alone. The second house takes away growth-first's only advantage (its median) and removes any median reward for sleeve equity.
- Hypothesis (guess): show a two-house table in the FR, and state the returns as a range, not a point. The decision does not change, but the "why" gets stronger and more honest. It is a statistician's move: show that the conclusion survives the assumption that matters most.
- Would change: **number** (FR projection tables and assumptions list); **sentence** (IPS risk/return line).
- Deliverables: IPS, FR. Specialist: D2 (with D3). Seed: 26.
- North star: Laura, a statistics graduate, sees a plan that says what it does not know. Rivals quoting one 7-10% equity number will look naive next to it.

Q3 (rank 4). Should the IPS describe "balance" goal by goal (promise: no market risk; facility floor: none after it is bought; upside: full market risk), because the blended whole-portfolio equity share (~1.5% in 2027, ~20% in 2028-30, ~4% in 2031-32) would read to Laura as "no growth"?

- Anchors: R-C37, R-C38, R-I11, R-I12, F-409, R-AN45 ("funding purposes" as diversification).
- Official words: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance" (case p.2, VRF)
- Evidence (DER, medians): 2027 equity = 60% × \$7.7k of \$300k = **1.5%**. 2028: **20.4%** (matches F-409). 2031 after the 80% lock: sleeve ~\$187k, 20% of it at 60% equity, against a \$356k ladder = **4.1%**.
- Hypothesis (guess): stated per goal, the plan is "full protection where others cannot help, real risk where they can" (Q10), which reads as balance. As one blended percentage, it reads as a bond fund. The SMAApply page's "diversification across ... funding purposes" (R-AN45) is the official hook for per-goal language.
- Would change: **sentence** (IPS risk-tolerance and balance sentences; FR allocation chart: show three pots over time, not one blended equity line).
- Deliverables: IPS, FR, TN (reflections name the pot each trade serves). Specialist: D9 (with D6). Seed: 1.
- North star: Laura feels her "thoughtful risks" were heard and placed where they belong, rather than ignored.

Q1 (rank 5). What does Laura pay our firm for when about two-thirds of her money sits in a hold-to-maturity ladder, and should the IPS say that the fee applies to the sleeve only (or is a flat fee)?

- Anchors: R-C26 ("invest \$300,000 with an asset management firm"), R-W29 (CFA Asset Manager Code), R-C8, R-S25.
- Official words: "At the beginning of 2027, Laura plans to invest \$300,000 with an asset management firm." (case p.2, VRF)
- Evidence (DER; ASM: fee paid yearly out of the sleeve; ladder untouched): no fee median \$210k; **0.5%** on the sleeve only \$206k, on all assets \$193k; **1.0%** on the sleeve only \$202k, on all assets **\$177k (-16%)**. A3 found no fee assumption anywhere (BS-01) and that the Code requires fee disclosure (F.4.d, VP via A3).
- Hypothesis (guess): a firm that charges a percentage on a buy-and-hold Treasury ladder charges for work it does not do. A statistics graduate who ran A/B tests (B9b) will compare against the "control" of buying the ladder herself. An honest fee line (sleeve only, or flat) plus a list of what the firm actually does (the 2027 purchase rule, the floor rule, the 2031 statement, rebalancing, the co-sponsor numbers) answers "why us" better than any return claim.
- Would change: **sentence** (IPS or FR fee/governance line); **number** (FR surplus tables net of the stated fee).
- Deliverables: IPS, FR. Specialist: D8 (with D10 for the Code). Seed: 1.
- North star: she chooses the firm that is honest about what it earns and why. Almost no team will mention its own fee.

Q14 (rank 6). Would a rival that holds the whole sleeve in Treasuries (facility fully bought by 2028) beat us on "Investment Strategy", being simpler and more certain, or would judges mark it down for ignoring "pursuing growth"?

- Anchors: R-C38, R-S24, R-S25, R-C73 ("no single correct strategy"), F-409.
- Official words: "Teams may reach different conclusions about risk, asset allocation, liquidity, return expectations, funding confidence, the facility contribution, and the financial flexibility Laura should preserve." (case p.4, VRF)
- Evidence (DER): all-Treasury sleeve \$183k / \$201k / \$222k (p5/p50/p95), against our \$161k / \$210k / \$277k. Under the Vanguard midpoint the all-Treasury median (\$201k) is only \$3k below ours (\$204k). The rival could state an almost exact 2033 figure in 2027.
- Hypothesis (guess): the all-Treasury plan is the strongest "attack from safety". It loses on the case's own words: "pursuing growth" (R-C38) and "willing to take thoughtful risks" (R-C37). It also nearly removes the uncertainty the case asks teams to explain (R-C65, R-C70). Our answer must show *why* we keep some risk: the upside is where flexibility "as the project develops" comes from (Q9). "Because the case says growth" is not enough.
- Would change: **sentence** (FR "alternatives considered" section: name this rival and the reason we did not choose it); possibly a **decision** if the team finds the reason weak.
- Deliverables: FR (IPS one clause). Specialist: D7 (with D6). Seed: 29.
- North star: showing the rejected safe alternative, and why, proves our risk was a choice made for her.

Q6 (rank 7). If 80-100% of the sleeve is a bought floor in 2031, the co-sponsor "range" shrinks toward one number. Does that dodge the case's uncertainty task, and should the FR show how the range narrows from the 2027 view to the 2031 view to 2033?

- Anchors: R-C65, R-C67, R-C69, R-C70, R-AN9, R-S28, F-402.
- Official words: "In 2031, however, the value of her portfolio in 2033 remains uncertain." (case p.3, VRF)
- Evidence (DER, JPM, bonds 5%): floor 100% in 2031 gives surplus = floor (a point); floor 80% gives floor \$129k / \$167k / \$219k, with the surplus at least 25% above the floor in 58% of paths; floor 50% gives a \$104k median floor. From 2027, the 2033 outcome spans \$161k-\$277k (p5-p95).

- Hypothesis (guess): our structure *answers* the uncertainty by buying most of it away. A judge reading only the 2031 number could still think we skipped it. Showing the funnel (2027: wide; 2031: floor bought, narrow upside; 2033: known) makes "how favorable and unfavorable market outcomes could affect the amount" visible. It also fits the criterion's "investment uncertainty clearly and credibly" (R-S28).
- Would change: **sentence/chart** (a FR funnel chart; data: p5/p50/p95 by year, from Appendix A); **decision** (floor share: 80% vs a lower share that keeps a real range).
- Deliverables: FR. Specialist: D3 (with D9). Seed: 9.
- North star: co-sponsors see the uncertainty shrink on a schedule. That is credibility shown, not claimed.

Q9 (rank 8). Should the 2031 pledge be tied only to money already bought, with the equity upside labelled as Laura's flexibility money (never promised), so the range cannot be missed at the low end and flexibility grows in good markets?

- Anchors: R-C58, R-C66, R-C71, R-C84, R-AN19, R-I13.
- Official words: "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops." (case p.3, VRF)
- Hypothesis (guess): yes. The design then answers two case demands with one split. The bought floor is the credible pledge (overpromising becomes impossible at the low end). The unbought upside is the flexibility (R-C58), which is largest exactly when markets were kind. It fits her "Always start small" staging (B8a/B8b, VP there): the pledge is the first stage, and more can follow "as the project develops". The barbell (Q4) makes this split physical.
- Would change: **decision** (what the 2031 range's top end is made of; whether the top end is "may add" language); **sentence** (FR fundraising-draft checklist: verbs "set aside" for the floor, "may add" for upside; BS-05).
- Deliverables: FR (IPS one rule). Specialist: D5 (with D6). Seed: 11.
- North star: Laura never has to un-promise anything, and her upside becomes an option she controls. That is what an entrepreneur values.

Q5 (rank 9). Should the facility floor be built by a ratchet (lock a third of the target each January 2029, 2030, 2031) instead of on one date in 2031, to protect Laura against a crash in the year before she pitches co-sponsors?

- Anchors: R-C62, R-C65, R-C70, F-402, F-318; extends B4a Q16 (not modelled there).
- Official words: "They should explain how favorable and unfavorable market outcomes could affect the amount she can provide." (case p.3, VRF)
- New numbers (DER, JPM, bonds 5%, lock rates 4.96/4.94/4.81% ASM from F-001):
 - one-date 2031: surplus \$161k / \$210k / \$277k; floor \$129k / \$167k / \$219k;
 - thirds: surplus **\$171k / \$208k / \$259k**; floor **\$137k / \$165k / \$201k**;
 - after a bottom-decile 2030: floor median **\$142k (one date) vs \$156k (thirds)**.
- Hypothesis (guess): the ratchet buys +\$10k at p5 and +\$14k in the "crash just before the pitch" world, for -\$2k at the median and -\$18k at p95. Under a barbell (Q4) the question disappears, since the floor is bought in 2028. So decide Q4 first.
- Would change: **decision** (floor timing rule); **number** (FR floor distribution).
- Deliverables: FR (IPS one clause if adopted). Specialist: D3. Seed: 9.
- North star: it protects Laura's good luck as well as her promise, at the moment her credibility is on the line.

Q7 (rank 10). If the ladder costs more than \$300k on purchase day, is "buy the long rungs now, buy the 2033-36 rungs with the 2028 deposit" nearly riskless, and how big is the joint tail if the deposit then fails?

- Anchors: F-111, F-112, F-102, F-107, R-AN35, brief section 9 (joint tail "UNANSWERED").
- Official words: "She will contribute an additional \$150,000 at the beginning of 2028" (case p.2, VRF)

- New numbers (DER; ASM: 2027 yield change sd 71bp (F-014), duration of the 2033-36 rungs seen from 2028 5.2-6.4 years; stock-rate correlation 0 or 0.4):
 - deposit \$150k or \$75k: payments short in **0%** of paths;
 - deposit \$0: short in **24-28%** of paths, average shortfall **\$8-10k** (1.6-2.0% of the \$500k);
 - done deliberately when the ladder *is* affordable, staging adds only +\$3k at the median.
 - Hypothesis (guess): the current "longest rungs first, top up in 2028" rule is sound. The joint tail is small and can be stated in one line. The rule should be written as a funded-ratio rule, not a price cap (B2a Q4/Q5 made the wording point; this adds the size of the tail).
- Would change: **number** (FR risk register: joint tail ~\$8-10k in ~1 in 4 zero-deposit paths); **sentence** (IPS contingent rule).
- Deliverables: IPS, FR. Specialist: D3 (with D1). Seed: 4.
- North star: the plan names its one residual weakness and its size. Laura's statistics training rewards that.

Q8 (rank 11). Does any lock timing (for example buying the whole ladder with both deposits in January 2028) give Laura materially more upside?

- Anchors: F-101, F-106, F-110, R-C26, R-C27, brief section 8.1 (new variant only).
- Official words: "For long-term projections, teams should begin with Laura's \$300,000 investment in 2027, add the \$150,000 contribution in 2028" (case p.4, VRF)
- New numbers (DER, ASM as Q7, duration 9.9): a 2028 lock gives median **+\$4k**, p95 +\$54k, p5 **-\$31k**. With a \$75k deposit, payments are short in 0.6-1.6% of paths; with \$0, in 37-39% (average short \$24-29k).
- What is new: brief 8.1 compared growth-first to 2033 and partial locks with no deposit. This tests the one-year delay *with* the deposit. It still buys almost no median.
- Hypothesis (guess): no. The equity risk premium over 5% Treasuries is too thin to pay for a year's delay. Keep one FR sentence ("we tested waiting a year; it adds ~\$4k at the median and puts the payments at risk if the deposit fails").
- Would change: **sentence** (FR alternatives considered). Confirms the decision.
- Deliverables: FR. Specialist: D3. Seed: 4.
- North star: shows Laura we tried to find her more upside before recommending safety.

Q11 (rank 12). In WInS, should the "short Treasury in the growth money" slot be a defined-maturity fund ending December 2032 (the facility date) rather than a 1-3 year Treasury fund, so every holding has a job tied to a date?

- Anchors: F-206, R-AN45, R-C88, R-T (the guide's example note ties the bond to the commitment), G-ids in `wins_week1_guardrails.md` (position limit).
- Official words (Trading Notes guide example): "an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment" (VRF)
- Candidates (**PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading**; this season: listed in WInS and within the Session Rules position limit):
 - Primary: **IBTM**, iShares iBonds Dec 2032 Term Treasury ETF. Per S1 (VP there, 2026-09-24/25): 15 notes maturing Apr-Sep 2032, effective duration 5.05y, YTM 5.09%, 0.07% fee, \$555m, ~216k shares/day. **Not** on the 2025-26 list (historical only).
 - Alternate 1: a Treasury note or principal STRIP maturing in late 2032 (bonds are a permitted type, R-W; S1 lists CUSIPs; \$10 commission).
 - Alternate 2: keep **VGSH** (the current ticket; on the 2025-26 list, line 1113, VRF). Weaker date story.
 - Hypothesis (guess): if the team adopts a floor bought early (Q4) or keeps the 2031 floor as a dated bond, a Dec-2032 holding makes the WInS book show three dated jobs: payments (IEF/TLH), facility floor (Dec 2032), upside (VT). The equity weight stays at ~17-20%. Watch the overlap: IBTM also funds the Jan 2033 payment rung in S1's ladder map.
- Would change: **decision** (WInS-now: one security swap before notes are written); **sentence** (a TN candidate about the facility floor).

- Deliverables: WInS-now, TN. Specialist: D1 (with D10). Seed: null.
- North star: every holding answers "which of Laura's dates is this for?", which is the question she will ask.

Q16 (rank 13). Should the plan state that it locks at the highest 10-year Treasury yield since 2007, and pre-commit to no regret if rates rise after the lock?

- Anchors: F-004, F-003, F-102, F-007.
- Evidence: 10-year 5.18% on 2026-09-24, the highest close since 2007-07-06 (F-004, VP). If rates rise 100bp after the purchase, the same ladder would have cost \$264,890 instead of \$292,264 (F-102): a "regret" of ~\$27k that is invisible to the payments. The Fed's longer-run funds-rate median is 3.2% (F-007, VP).
- Hypothesis (guess): Laura's regret rule is "you're always going to regret not doing it" (Overachiever 2021, VP via B8a). Locking at a 19-year high fits it. Writing the no-regret rule down in advance stops a 2027 wobble if rates keep rising. One checkable fact beats a forecast.
- Would change: **sentence** (a TN reflection on the first hedge trade; the IPS "why now").
- Deliverables: TN, IPS. Specialist: D1 (with D6). Seed: 26.
- North star: gives Laura a fact she can check and a rule that protects her from second-guessing, not a rate forecast.

Q12 (rank 14). Should the WInS hedge be built in two dated steps under a rule written in advance (for example half now, half on a set date or if the 10-year falls below a stated level), so the Trading Notes rehearse Laura's January-2027 lock decision and show a strategy "tested or refined", without inventing a decision?

- Anchors: R-C88, R-AN40, R-AN43, F-111, F-112, R-W73.
- Official words: "This shows how selected investment decisions reflected, tested, or refined the team's developing strategy." (case p.4, VRF)
- Hypothesis (guess): the current ticket buys everything in one session. That is correct economically, but it gives three notes that say the same thing. Laura's real problem is rate risk *before* the lock (the ladder cost more than \$300k on 173 of 185 days in 2026, F-111), so a pre-written two-step rule is not a gimmick: it is the January rule in miniature. Risks: day-trading ban, position limit, and making a move for the sake of notes (B8a "No more gimmicks"). The test: would the team make the split if notes did not exist?
- Would change: **decision** (WInS-now order plan); **sentence** (TN note 1-2 content).
- Deliverables: WInS-now, TN. Specialist: D10 (with D7). Seed: null.
- North star: the notes show Laura's hardest real decision practised in public, not a one-click portfolio.

Q10 (rank 15). Should the IPS justify where the risk sits with the case's own asymmetry: outside money may fill a facility shortfall but never an operating shortfall?

- Anchors: R-C44, R-C48, R-AN6, R-I13, BS-17 (funders underfund operations).
- Official words: "Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement." (case p.3, VRF) and "Any remaining facility cost ... may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources." (case p.3, VRF)
- Hypothesis (guess): "risk sits only where others can help" is a one-clause, case-derived reason for the whole architecture. It turns "lock early" from a technique into a reading of Laura's situation. R-AN6 notes the asymmetry, but no file uses it as the justification for *where the risk sits*.
- Would change: **sentence** (IPS principle; pitch candidate content).
- Deliverables: IPS, FR. Specialist: D9 (with D5). Seed: 5.
- North star: the plan's logic comes from her case, not from a textbook.

Q25 (rank 16). Is part of "pursuing growth" growth of what the residency can raise, since a bought, credible floor may pull in co-sponsor money, and should the IPS say so without overclaiming?

- Anchors: R-C64, R-C38, brief section 8.17 (seed-money effect = mechanism, not prediction).
- Official words: "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project, and make potential co-sponsors more willing to contribute." (case p.3, VRF)
- Hypothesis (guess): yes, as a *mechanism* only (brief 8.17). A floor that cannot be missed is worth more to co-sponsors than an expected value that might be. So certainty is itself a growth tool for the project. The risk is that this becomes the fancy framing last year's lesson warns against. Keep it to one clause.
- Would change: **sentence** (IPS balance sentence; FR fundraising checklist).
- Deliverables: IPS, FR. Specialist: D5 (with D6). Seed: 1.
- North star: reframes Laura's "growth" as growth of her residency, which is what she actually wants to grow.

Q13 (rank 17). A pension at ~152% funded grows its surplus with no spending date. Laura's surplus has a date (2033) and a public promise (2031). Is the right professional analogue a pension "surplus glidepath" or a dated floor-plus-upside, and which would a judge recognise?

- Anchors: brief section 8 (pension logic accepted), F-117, B9a Q1 (152%), B4b (Russell 2026).
- Evidence (VP, Russell Investments, July 2026): "lock down the benefits promised, then grow surplus assets with a prudent process and disciplined understanding of risk."
- Hypothesis (guess): Russell's phrase fits the promise half. The pension analogue under-serves the surplus, because a pension's surplus has no deadline and no audience. Laura's does, which is exactly why a dated floor (Q4/Q5) is needed. One FR sentence on "where we depart from pension practice, and why" is a strong Client Knowledge signal.
- Would change: **sentence** (FR professional-practice paragraph).
- Deliverables: FR. Specialist: D8. Seed: 26.
- North star: shows the team knows the professional template and where Laura differs from it.

Q17 (rank 18). Do the barbell's and ratchet's advantages survive fat tails and real crash sequences (2000-02, 2008, 2022), or are they artefacts of the i.i.d. lognormal model?

- Anchors: brief section 11 (model gaps), F-316, F-318.
- Hypothesis (guess): the barbell's p5 should hold or improve (its floor is bought regardless of path). Its p95 depends on long bull runs. The 60/40 sleeve's rebalancing does better in choppy, mean-reverting markets. A block bootstrap on historical U.S. equity/Treasury returns would settle it.
- Would change: **decision** (adopt the barbell or not, Q4); **number** (FR tables).
- Deliverables: FR (IPS decision). Specialist: D3. Seed: null.
- North star: a statistics graduate trusts a design that was tried on history, not only on a smooth model.

Q18 (rank 19). Which single Laura-specific claim would make a judge who has read 50 "lock + growth sleeve" plans say ours is different, and can it be stated as one checkable fact?

- Anchors: R-S24 ("clear and creative investment thesis"), R-S25, B5-10 (rivals may converge), R-C8.
- Official words: "Your team hopes to develop the investment strategy that Laura ultimately chooses" (case p.1, VRF)
- Candidates (INT): (a) "the co-sponsor floor is bought in 2028, so the low end of the range cannot be missed" (needs Q4); (b) "we show two return houses and our conclusion survives both" (Q2); (c) "risk sits only where others can help" (Q10); (d) "our fee does not touch the promise" (Q1).
- Hypothesis (guess): (a) or (c). Each is checkable, comes from the case and matches her pattern. (b) and (d) are supporting evidence.
- Would change: **sentence** (pitch content checklist; IPS thesis line).
- Deliverables: IPS, FR. Specialist: D7. Seed: 29.
- North star: this is the North Star itself: one reason Laura can repeat to a co-sponsor.

3. Parked (logged, not carried forward)

- **Seed 5 (the 2033 ordering "before making the first operating payment or contributing to the facility").** Under lock-early the ladder *is* the reserve, so the order is met automatically. Brief 8.9 already reconciles "funded 2027, set aside 2033" in one sentence. No decision changes. Q10 uses the priority idea instead.
- **WlnS mirror at the literal 2027 book vs the post-2028 target.** B1a Q1 owns it. New evidence for B1a: the plan's whole-portfolio equity share is ~1.5% / ~20% / ~4% in 2027 / 2028-30 / 2031-32 (DER, Q3). So the WlnS ticket's ~20% VT matches only the 2028-30 window. That is fine if the notes say which year the book represents.
- **Is a small facility (~\$11k with no 2028 deposit) still "responsible"?** Brief 8.8 settled it ("correct failure mode").
- **Whether a broad index is "thoughtful risk" or "no view".** Folded into Q3 and Q20. Both houses rank developed ex-U.S. at or above U.S. (JPM EAFE 7.5% vs U.S. 6.7%, F-301/F-303; Vanguard 4.5%-6.5% vs 4.2%-6.2%, VP), which supports the global fund already in the ticket. No change.
- **Designers' intent from the case PDF date (2026-09-10).** B2a found it (the ladder cost > \$300k almost all year). Q7 adds the tail size; not re-asked.
- **Post-2033 use of flexibility money (option to expand vs buffer).** B9a Q8 and B4a Q17 own it; Q9 adds only the "unbought upside = flexibility" link.

4. Evidence found in this pass (with status)

#	Fact	Value	Status	Source
E1	Vanguard 10-year U.S. equity expected return	4.2%-6.2% a year (down from 4.9%-6.9%); VCMM run 30 June 2026; page dated 22 July 2026	VP	Vanguard VCMM page (section 6)
E2	Vanguard developed ex-U.S. / emerging	4.5%-6.5% / 2%-4%	VP	same
E3	Vanguard caveat	"valuations tend to be poor predictors of performance over the short or even intermediate term"	VP	same
E4	Laura's leap after a secured base	"So when she got the book deal with HarperCollins, she gave Twitter her notice." (reporter's account)	VP (re-grepped by me)	Inverse/Input, 2022-03-07
E5	Russell surplus glidepath	"lock down the benefits promised, then grow surplus assets with a prudent process and disciplined understanding of risk."	VP (re-grepped by me)	Russell Investments, July 2026
E6	Baseline reproduction	\$159k/\$207k/\$273k; floor \$127k/\$165k/\$217k	VRF (same as F-401/F-402)	Appendix A, section A
E7	Corrected baseline (sleeve bonds 5.0%)	\$161k/\$210k/\$277k; floor \$129k/\$167k/\$219k	DER	Appendix A, B/D
E8	Design table (summary item 1)	see summary	DER	Appendix A, B-K
E9	Vanguard-midpoint results	L sleeve 0/60/100%: medians \$201k/\$204k/\$204k; G median \$205k, miss 4.5%	DER (ASM: JPM vol kept)	Appendix A, C and G
E10	Ratchet vs one date	p5 \$171k vs \$161k; bad-2030 floor median \$156k vs \$142k	DER	Appendix A, E
E11	Joint tail (short rungs wait, deposit \$0)	24-28% of paths short, mean \$8-10k; \$0 short at a \$75k deposit	DER (ASM rate vol/ duration/ correlation)	Appendix A, F
E12	Lock delayed to 2028	median +\$4k, p5 -\$31k; 0.6-1.6% miss at \$75k, 37-39% at \$0	DER	Appendix A, I
E13	Fee drag	1% on all assets: median \$177k; on the sleeve only: \$202k	DER (ASM fee mechanics)	Appendix A, H
E14	Whole-portfolio equity share at medians	1.5% (2027), 20.4% (2028), 4.1% (2031)	DER	Appendix A, J
E15	Barbell	50% locked 2028: \$163k/\$210k/\$293k, floor ~\$100k known in 2028	DER (ASM 5y 4.98%)	Appendix A, K

#	Fact	Value	Status	Source
E16	IBTM facts	Dec 2032 iBonds; duration 5.05y; YTM 5.09%; \$555m; ~216k sh/day; not on the 2025-26 list	VP per S1 (not re-opened by me); list VRF	wins_now/S1_treasury_sleeve.md section 1c

Not reached: WebSearch (session budget spent, 200/200); Perold & Sharpe (1988) not opened; the JPM 2027 LTCMA is not yet out (F-315).

5. Leads for the specialists

• D3 (quant):

1. Add three designs to the verified model: barbell (lock X% of the sleeve in Jan 2028 to Dec 2032), ratchet (thirds 2029-31), and the current plan. Report p5/p25/p50/p75/p95 and the 2031 range + confidence for each (Q4, Q5, Q6).
2. Fix the sleeve-bond yield to ~5% (B9a flag; E7).
3. Run the Vanguard midpoint and range ends (4.2%/6.2%) next to JPM (Q2).
4. Replace i.i.d. lognormal draws with a historical block bootstrap (Q17).
5. Simulate the barbell's 2031 range rule (low end = floor + 75% of the equity value?) and its miss rate. Code base: Appendix A.

- **D2 (equity):** Vanguard VCMM page (E1-E3). Check whether Vanguard publishes a U.S. Treasury 10-year figure on the same run (the interactive table did not load as text). Check GMO/AQR/Research Affiliates 2026 numbers only if they change the "range, not median" conclusion.

- **D6 (client psychology):** the Inverse 2022 line (E4) plus B8a's regret quote (Overachiever 2021) are the Laura-side evidence for "secured base, then a full leap" (Q4, Q16, Q20). Do not use the surrounding personal material on the Inverse page (privacy rule).

- **D8 (practice):** fee practice for LDI/ladder mandates vs balanced mandates (Q1); Russell July 2026 (E5); where a dated surplus departs from pension practice (Q13); Perold & Sharpe 1988 for buy-and-hold vs constant-mix (Q4).

- **D1/D10 (rates, compliance):** IBTM availability in WInS, and the position limit when IBTM also serves the Jan 2033 payment rung (Q11). Also a two-step hedge rule that respects the day-trading ban (Q12).

- **D5 (co-sponsors):** does a floor "set aside since 2028" read as stronger than "set aside in 2031"? Wording tiers "set aside" vs "may add" (Q9; A3 BS-05).

- **D7 (competition):** whether past Top-50 IPSs used "safe bucket + growth bucket" wording (tests BS-10 and Q18).

6. Sources (all accessed 2026-09-27)

Source	URL / path	Status
Vanguard, "Vanguard Capital Markets Model® forecasts", 22 July 2026	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html	VP (phrases grep-verified 13:10 UTC)
Input/Inverse, feature on Laura Gao, 2022-03-07	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel	VP (one sentence grep-verified; personal content ignored)
Russell Investments, pension surplus article, July 2026	https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.html	VP (phrase grep-verified)
Overachiever Magazine interview, 2021-03-15 (regret quote)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	VP per B8a (not re-opened)
Case, IPS guide, TN guide, SMAApply page	competition/official/2026_27/	VRF
Case PDF metadata (created 2026-09-10)	Laura_Gao_2026_Client_Profile.pdf, read with pypdf	VRF
Verified model	research/verified_2026-09-27/strategy_mc.py	VRF
Phase A registers	research/insight_v1/phase_A/*.md	run files

Source	URL / path	Status
Phase B neighbours	B1a, B2a, B4a, B4b, B8a, B8b, B9a, B9b in <code>research/insight_v1/phase_B/</code>	run files
WInS week-1 files	<code>research/insight_v1/wins_now/</code> (S1, S3, S4)	run files
2025-26 ETF list (historical only)	<code>competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt</code> line 1113 (VGSH)	VRF

Appendix A. Script (run from the repo root; not saved as a separate file because this agent may write only this file)

Save as `research/insight_v1/scripts/B10a_contrarian_checks.py` if the main loop wants it kept. Run with `.venv/bin/python research/insight_v1/scripts/B10a_contrarian_checks.py` (about a minute).

```
&quot;&quot;&quot;B10a contrarian checks (scratch; code copied into phase_B/B10a_contrarian.md
appendix).
Inputs: same as research/verified_2026-09-27/strategy_mc.py (VERIFIED-REPO-FILE: ladder $292,264, 2y
4.81%,
JPM LTCMA 2026 US large cap 6.70/7.94/16.47, intermediate Treasuries 4.00/4.06/3.48, rho -0.01).
New inputs: Vanguard VCMM 30 Jun 2026 U.S. equities 10y 4.2-6.2% (VERIFIED-PRIMARY, midpoint 5.2%
used, ASSUMPTION
that JPM vol applies); sleeve bonds at 5.0% (ASSUMPTION ~ today's forward/yields F-012/F-203); par
yields 3y 4.94,
5y 4.98 (F-001); 10y yield vol 71bp/yr (F-014); rung costs F-107. Run: .venv/bin/python
B10a_checks.py
"""
import numpy as np
P, SEED = 200_000, 20260927
LAD = 292_264; LONG = 155_408; SHORT = 136_856 # F-107: 2037-42 and 2033-36 rungs at 2027-01-01
def lp(c, a):
    return np.log(1 + c), np.sqrt(2 * np.log((1 + a) / (1 + c)))
def draws(eq_c=0.067, eq_a=0.0794, bd_c=0.04, bd_a=0.0406, rho=-0.01, seed=SEED):
    rng = np.random.default_rng(seed)
    z1 = rng.standard_normal((P, 6)); z2 = rng.standard_normal((P, 6)); z3 = rng.standard_normal((P,
6))
    EQ = lp(eq_c, eq_a); BD = lp(bd_c, bd_a)
    zb = rho * z1 + np.sqrt(1 - rho**2) * z2
    return np.exp(EQ[0] + EQ[1] * z1) - 1, np.exp(BD[0] + BD[1] * zb) - 1, z1, z3
def pct(x, ps=(5, 25, 50, 75, 95)):
    return [int(round(np.percentile(x, p), -3) / 1000) for p in ps]
def lock_early(req, rbd, eq_w=0.6, floor_share=0.8, dep=150_000, y2=0.0481, ratchet=None,
start=None):
    s = np.full(P, 300_000.0 - LAD if start is None else start)
    locked = np.zeros(P)
    for y in range(6): # 2027..2032
        if y == 1: s = s + dep
        if ratchet is not None and y in ratchet: # lock share of CURRENT sleeve into
Treasury maturing 2033
            share, rate = ratchet[y]
            amt = share * s; s = s - amt; locked = locked + amt * (1 + rate) ** (6 - y)
        if ratchet is None and y == 4:
            amt = floor_share * s; s = s - amt; locked = locked + amt * (1 + y2) ** 2
        s = s * (1 + eq_w * req[:, y] + (1 - eq_w) * rbd[:, y])
    return locked + s, locked
req, rbd, z1, z3 = draws()
print("A. Baseline reproduce (JPM, bonds 4.0%):", pct(lock_early(req, rbd)[0]), "floor",
pct(lock_early(req, rbd)[1], (5, 50, 95)))
reqB, rbdB, _, _ = draws(bd_c=0.05, bd_a=0.0506)
print("B. Sleeve bonds at 5.0% (forward-consistent):&quot;)
for w in (0.0, 0.6, 1.0):
    print(f"    eq {w:.0%}: surplus p5/25/50/75/95 {pct(lock_early(reqB, rbdB, eq_w=w)[0])}")
reqV, rbdV, _, _ = draws(eq_c=0.052, eq_a=0.052 + (0.0794 - 0.067) * 1.0, bd_c=0.05, bd_a=0.0506)
print("C. Vanguard midpoint equities 5.2% compound, bonds 5.0%:")
for w in (0.0, 0.6, 1.0):
```

```

    print(f"    eq {w:.0%}: surplus {pct(lock_early(reqV, rbdV, eq_w=w)[0])}")
print("D. Floor share x sleeve equity (JPM equities, bonds 5.0%): surplus p5/25/50/75/95 | floor
p5/50/95 | 2031 range width p50 (surplus p95 - floor p5 proxy)")
for fs in (0.5, 0.8, 1.0):
    for w in (0.6, 1.0):
        sur, fl = lock_early(reqB, rbdB, eq_w=w, floor_share=fs)
        print(f"    floor {fs:.0%} eq {w:.0%}: {pct(sur)} | floor {pct(fl,(5,50,95))} | P(surplus&
gt;=1.25*floor) {np.mean(sur&gt;=1.25*fl):.2f}&quot;")
print("E. Ratchet vs one-date lock (both lock 80% of the sleeve by 2031, eq 60%, bonds 5.0%):")
one, fl1 = lock_early(reqB, rbdB)
# lock 1/3 of the target each Jan 2029, 2030, 2031 (shares of current sleeve that give 80%
cumulative if no returns)
rat = {2: (0.8 / 3 / 1.0, 0.0496), 3: ((0.8 / 3) / (1 - 0.8 / 3), 0.0494), 4: ((0.8 / 3) / (1 - 1.6
/ 3), 0.0481)}
two, fl2 = lock_early(reqB, rbdB, ratchet=rat)
print("    one-date 2031: surplus", pct(one), &quot;floor(2031)&quot;;, pct(fl1, (5, 50, 95)))
print("    thirds 2029/30/31: surplus", pct(two), &quot;floor(2031)&quot;;, pct(fl2, (5, 50, 95)))
# crash-in-2030 conditional: equity return in 2030 (y=3) below its 10th percentile
bad = req[:, 3] < np.percentile(req[:, 3], 10)
badB = reqB[:, 3] < np.percentile(reqB[:, 3], 10)
print("    given a bottom-decile 2030: one-date floor p50", int(np.median(fl1[badB]) / 1000), "k;
thirds floor p50", int(np.median(fl2[badB]) / 1000), "k")
print("F. Staged lock: 2027 buy only 2037-42 rungs; 2033-36 rungs bought Jan 2028 (deposit +
sleeve).")
D_SHORT = 5.2 # ASSUMPTION: approx duration of 2033-36 rungs seen from 2028 (5-8y zeros, PV-
weighted ~6.4y); test 6.4 too
for dur in (5.2, 6.4):
    for corr in (0.0, 0.4):
        for dep in (150_000, 75_000, 0):
            dy = 0.0071 * (corr * z1[:, 0] + np.sqrt(1 - corr**2) * z3[:, 0]) # 2027 yield change;
+corr: stocks down, yields down
            cost28 = SHORT * 1.045 * np.exp(-dur * dy)
            s = (300_000 - LONG) * (1 + 0.6 * reqB[:, 0] + 0.4 * rbdB[:, 0]) + dep
            short_fall = np.maximum(cost28 - s, 0)
            s = np.maximum(s - cost28, 0)
            for y in range(1, 6):
                if y == 4:
                    fl = 0.8 * s; s = s - fl
                s = s * (1 + 0.6 * reqB[:, y] + 0.4 * rbdB[:, y])
            sur = fl * 1.0481**2 + s
            base, _ = lock_early(reqB, rbdB, dep=dep)
            print(f"    dur {dur} corr {corr} dep {dep//1000}k: P(payments short) {np.mean(short_fall&
gt;0):.3f}, mean short ${short_fall[short_fall&gt;0].mean() if (short_fall&gt;0).any() else 0:,.0f};
surplus p5/50/95 {pct(sur,(5,50,95))} vs lock-all {pct(base,(5,50,95))}&quot;")
print("G. Growth-first vs lock-early median gap under JPM vs Vanguard equities (G = 75% eq to 2030,
then 60/40; reserve at flat 5.26% +/- 1pp)")
def growth_first(req, rbd, dep=150_000, seed=SEED):
    rng = np.random.default_rng(seed + 1)
    rate = 0.0526 + 0.01 * rng.standard_normal(P)
    res = sum(50_000 / (1 + rate) ** t for t in range(10))
    v = np.full(P, 300_000.0)
    for y, w in enumerate((0.75, 0.75, 0.75, 0.75, 0.6, 0.4)):
        if y == 1: v += dep
        v *= 1 + w * req[:, y] + (1 - w) * rbd[:, y]
    return v - res
for nm, (a, b) in [{"JPM eq, bonds 4%": (req, rbd), "JPM eq, bonds 5%": (reqB, rbdB), "Vanguard 5.2%
eq, bonds 5%": (reqV, rbdV)}.items()]:
    g = growth_first(a, b); l = lock_early(a, b)[0]
    print(f"    {nm}: G miss {np.mean(g&lt;0):.3f}; G p5/50/95 {pct(g,(5,50,95))}; L p5/50/95 {pct(l,
(5,50,95))}&quot;")
print("H. Fee drag: 0.5%/1.0% a year on the sleeve only vs on all assets (ladder fee paid from
sleeve)")
for fee in (0.005, 0.01):
    s = np.full(P, 300_000.0 - LAD); lad = LAD
    s2 = s.copy()
    for y in range(6):

```

```

if y == 1: s = s + 150_000; s2 = s2 + 150_000
if y == 4: fl = 0.8 * s; s = s - fl; fl2 = 0.8 * s2; s2 = s2 - fl2
r = 1 + 0.6 * reqB[:, y] + 0.4 * rbdB[:, y]
s = s * r * (1 - fee) # fee on sleeve only
s2 = s2 * r - fee * (s2 + lad + (fl2 if y >= 4 else 0)); lad *= 1.0508 # fee on all assets,
from sleeve
if y >= 4: fl = fl * 1.0481; fl2 = fl2 * 1.0481
print(f"    fee {fee:.1%}: sleeve-only median {int(np.median(fl+s)/1000)}k; all-assets median
{int(np.median(fl2+s2)/1000)}k")
print("I. Delay the whole lock to Jan 2028 (2027: all $300k at 60/40), ladder bought with both
deposits; dur 9.9, dy sd 71bp")
for corr in (0.0, 0.4):
    for dep in (150_000, 75_000, 0):
        dy = 0.0071 * (corr * z1[:, 0] + np.sqrt(1 - corr**2) * z3[:, 0])
        cost28 = 306_077 * np.exp(-9.9 * dy)
        s = 300_000 * (1 + 0.6 * reqB[:, 0] + 0.4 * rbdB[:, 0]) + dep
        short = np.maximum(cost28 - s, 0); s = np.maximum(s - cost28, 0)
        for y in range(1, 6):
            if y == 4: fl = 0.8 * s; s = s - fl
            s = s * (1 + 0.6 * reqB[:, y] + 0.4 * rbdB[:, y])
        sur = fl * 1.0481**2 + s
        base, _ = lock_early(reqB, rbdB, dep=dep)
        print(f"    corr {corr} dep {dep//1000}k: P(short) {np.mean(short>0):.3f} mean short
${short[short>0].mean()} if (short>0).any() else 0;,.0f}; surplus p5/50/95 {pct(sur,(5,50,95))} vs
lock-2027 {pct(base,(5,50,95))}&quot;)
print("J. Whole-portfolio equity share at medians: 2027, 2028-30, 2031-32")
s28 = (300_000 - LAD) * 1.05 + 150_000
print(f"    2027: {0.6*(300_000-LAD)/300_000:.1%}; 2028: {0.6*s28/(306_077+s28):.1%}; 2031: sleeve~
{s28*1.057**3:,.0f}, equity share {0.6*0.2*s28*1.057**3/(356_384+s28*1.057**3):.1%}&quot;)
print("K. Barbell: Jan 2028 lock share X of the sleeve in a Treasury maturing Jan 2033 (5y par
4.98%), rest 100% equity to 2033")
for eqset, (a, b) in {"JPM": (req, rbdB), "Vanguard mid": (reqV, rbdV)}.items():
    base, _ = lock_early(a, b)
    print(f"    [{eqset}] current plan (60/40, 80% floor 2031): p5/25/50/75/95 {pct(base)}")
    for X in (0.5, 0.65, 0.8):
        s28 = (300_000 - LAD) * (1 + 0.6 * a[:, 0] + 0.4 * b[:, 0]) + 150_000
        fl = X * s28 * 1.0498 ** 5
        e = (1 - X) * s28
        for y in range(1, 6): e = e * (1 + a[:, y])
        sur = fl + e
        print(f"    [{eqset}] X={X:.0%}: floor known in 2028 p50 {int(np.median(fl)/1000)}k; surplus
{pct(sur)}; P(below plan median 210k) {np.mean(sur<210_000):.2f}&quot;)

```

Printed results (2026-09-27 run, seed 20260927, 200,000 paths): the summary table, Q2, Q4, Q5, Q7, Q8, Q1 and Q3 numbers above are copied from this output unchanged.

What this teaches

1. **Attack your own plan with the client's words, not only with rival models.** "Is this balance?" is a sharper test than "is this optimal?". Here it showed that the plan was right, but that we had been *describing* it wrongly (one blended equity number instead of three jobs).
2. **When expected returns are close together, risk choices are about spread, not about the average.** In 2026 safe bonds pay about what stocks are expected to pay. Extra stocks mostly widen the range of outcomes. Say this out loud rather than implying that stocks add growth.
3. **Check an important assumption against a second independent source.** JPM and Vanguard disagree by about 1.5 points on U.S. stocks. A conclusion that survives both is one you can defend. One that needs the higher number is a bet.
4. **The simplest structure can be as good as the clever one.** A bought floor plus stocks (a barbell) did as well as a rebalanced mix plus a later lock, and it is easier to explain. Complexity has to earn its place, even inside a good plan.

B10b Contrarian: attack "lock early" from the case designers' and semifinal readers' side

Agent B10b, insight_v1 run, Phase B (question generation), written 2026-09-27. Lens: Contrarian (attack our own strategy). Starting angle: read the case the way its designers built it, with 2031 and 2033 as decision points and the operating reserve's "size and initial asset composition" asked for at the beginning of 2033. Then ask: does lock early answer the question Wharton asked, or step around it? Where would a smart rival look more tailored, creative or complete to a semifinal reader?

This file is AI-generated research for Team Caplet (Claude Code). It holds questions, evidence, numbers and checklists. It holds no text meant for submission. The six students decide and write every deliverable in their own words, and record AI use in the Final Report's Works Cited (R-W46). No finale material.

Status labels: VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED, ASSUMPTION (ASM), INTERPRETATION (INT), DERIVED (my arithmetic on labelled inputs).

0. Summary (read this first)

My verdict after attacking it: **lock early survives as the right architecture. But three of the reasons we give for it are weaker than we think, and in four places the plan could read as generic or evasive to a semifinal reader.**

- The designers wrote the case around a glide path, and our plan's equity path is a hump.** Wharton's own example note buys a Treasury fund to "begin preparing" for the commitment "as the residency funding date approaches" (R-T23, R-T25, VRF). The new 2026-27 Competition Guide asks "which principles will guide decisions as funding dates approach and payments are made" (VP, see section 5). Under lock early, whole-portfolio equity is about **1.5% on 1 Jan 2027, 20.4% after the 2028 deposit** (F-409) and about **4% after the 2031 floor lock** (DERIVED, script below). A reader expecting "less risk as the date nears" sees risk *rise* first. That is correct (risk capacity switches on with the 2028 deposit), but it must be explained in one clause or it reads as incoherent (Q2, Q16).
- NEW NUMBER: with the 2028 deposit present, "lock everything" is not what makes the payments certain.** I re-ran the verified model's inputs with partial locks (script `research/insight_v1/scripts/B10b_designer_checks.py`, ASM-based). With the full \$150k deposit, which the case says she "will" make (R-C27, R-AN35), locking **70%, 80%, 90% or 95%** of the ladder also gives **0.00% payment shortfall** in 200,000 paths, and a *slightly higher* median 2033 surplus (\$212.6k at 80% vs \$210.4k at 100%). Locking nothing gives 2.1% shortfall and +\$10.6k median. So the honest reason to lock 100% is the **facility range and the stressed deposit**, not payment certainty in the case's base. The p5 surplus falls from \$150k (100% lock) to \$128k (80%) and \$117k (70%). If the IPS or Final Report claims "anything less than a full lock fails", a statistics-trained reader can falsify it (Q4).
- The case's numbers were probably not set so that \$300k buys the promise.** The ladder cost more than \$300k on 173 of 185 trading days in 2026 (F-111). If the January 2027 curve looks like January-August 2026, "the first deposit buys the whole promise" disappears. A rate-robust headline survives: DERIVED (flat-curve ASM), the two deposits together still buy the full ladder unless rates fall about **420bp**; with a half-size 2028 deposit, about **250bp**. So "fully funded by January 2028 at the latest" is robust where "fully funded on day one" is not (Q3).
- The reserve answer can look like a dodge unless it answers the three things asked, in the case's words.** The case wants the reserve's size, initial composition and how composition changes (R-C52, R-C53). Lock early's honest answer is "it is the 2027 ladder, and it changes by itself". That is allowed ("if at all", R-AN27). It becomes an *answer* only with numbers: DERIVED (flat 5.35% ASM) the held ladder's duration rolls down from **about 4.1 years on 1 Jan 2033 to 0 in 2042**, with no trades. That is a glide path in maturity, not in stocks (Q1).
- The alternative most rivals will propose for the late payments, an equity "bucket" inside the reserve, costs 1.8-2.6 times more.** DERIVED (JPM large-cap inputs, lognormal ASM): to pay the 2040, 2041 or 2042 payment from equities held from 2033 with 95% probability needs about **\$59-62k** of stock per \$50k payment. At 99% it needs about **\$81k**. The Treasury price is about **\$31-35k**. One line of this in the Final Report shows we tested the obvious rival design and why it loses (Q9).

6. **The case's premise "In 2031 ... the value of her portfolio in 2033 remains uncertain" (R-C65) is made almost false by our own 2031 rule.** Under an 80% floor lock the 2031 range is roughly $\pm 5\%$ (B6a, B6b). The case also says teams "must recommend the dollar range" (R-C68). A reader could see "we give Laura a rule, and the range is nearly a point" as avoiding the task the designers built. The fix is to give both: a dollar range from today's view and the 2031 rule, and to say in one sentence why the range narrows (Q5).

7. **The biggest threat to what Laura's facility money buys is not markets; it is construction inflation.** DERIVED: Taiwan's construction cost index has risen about 3.5% a year since 2021 (F-508, F-509). Over 2026-2033 that is about **+27%**, against Taiwan CPI of about +15% at 2%. A two-year USD/TWD move has a standard deviation of about 6.8% (F-513). The case asks for assumptions on "the effect of inflation on ... facility costs" (R-C85). A USD-only, nominal range answers the market question and misses the one the case names (Q17).

8. **Where a rival beats us on paper:** a crisp facility-and-flexibility rule (ours is "mainly leftover"), a year-by-year allocation picture, a stated confidence number for the payments, and a larger headline facility number bought with hidden payment risk. The last one we should expose, not match (Q7, Q11, Q18).

Question count: **21** (section 3). By deadline: WInS-now/TN 3, IPS 8, FR 10. Most of them change a sentence or a chart. Q3, Q4 and Q7 could change a decision.

What is new versus Phase A and other Phase B files: the partial-lock frontier *with* the 2028 deposit (section 8 item 1 of the brief covered partial locks only with no deposit); the whole-portfolio equity path by year; the equity-bucket cost ratio; the reserve duration roll-down; the rate fall that breaks "fully funded by 2028"; the construction-inflation versus market-risk comparison; and a reading of the designers' glide-path language in the Trading Notes example and the Competition Guide.

1. Method

1. Read the brief, v4 Parts 1-3, 5 and 6, CLAUDE.md, the case, both guides, the SMAApply page, and the four Phase A files. Skimmed the summaries of all 18 Phase B files already written, so as not to repeat them (I name overlaps).
2. Read the 2026-27 Investment Competition Guide PDF myself (VP) and checked four quotes verbatim with `fetch_text.py --grep`.
3. Wrote down what a designer would expect from each case paragraph, then asked where lock early departs from that expectation, and whether the departure is (a) better and explained, (b) better but unexplained, or (c) weaker.
4. Tested four attacks with numbers, using the verified model's inputs (script in section 6).
5. Searched Wharton Global Youth's own article archive (WordPress search API) for teaching content on bond ladders, glide paths or liability matching that might show the designers' intended toolkit. Result: nothing relevant (VP negative result, section 7). My WebSearch budget for this session was already used up (the tool refused all calls), so a few leads below are marked for specialists to find.

2. The designer's reading of the case, paragraph by paragraph (INT unless labelled)

Case element (VRF)	What a designer probably pictured	What lock early does	Risk to us
"\$300,000 ... \$150,000" (R-C26, R-C27); \$450k < \$500k (R-AN13)	Money must grow to cover the promise; tension between growth and protection	Rates, not growth, close the gap; promise bought in 2027	"Solved in year 1" can look like the case was sidestepped, and it depends on September 2026 rates (F-111)
Example note: "begin preparing ... as the residency funding date approaches" (R-T23, R-T25)	A glide path: growth first, bonds added gradually	All at once, now; equity rises in 2028	Reads as the opposite of the model answer unless explained
"set aside ... at the beginning of 2033" (R-C50) + "size and initial asset composition" (R-C52) + change "if at all" (R-C53)	A reserve built in 2033 from a pooled portfolio, with a composition to choose	Relabel the 2027 ladder in 2033; no trades	Terse answer can look like a dodge

Case element (VRF)	What a designer probably pictured	What lock early does	Risk to us
"2031 is Year 5" (R-C29); range + confidence (R-C67 to R-C69); "remains uncertain" (R-C65)	A real forecasting problem two years out	Floor bought; range nearly a point	Looks like we avoided the uncertainty task
"no predetermined facility contribution"; "responsibly"; flexibility "as the project develops" (R-C57 to R-C60)	Teams choose a rule for how much to give and how much to keep	Facility = sleeve value; flexibility = leftover	Generic; no rule
"Teams may reach different conclusions about ... funding confidence" (R-C74)	Teams pick a confidence level	"Certain barring U.S. default"	No number; "how evaluated" may look thin (R-C55)
"the effect of inflation on portfolio projections and facility costs" (R-C85)	Real value of the facility money	USD nominal throughout	Misses a named assumption area
Guide: "how its components will support different needs" (VP)	Split portfolio by purpose	Ladder + sleeve	Strong fit; say it in these words

3. Questions

Format for each: question; anchors; hypothesis (a guess, with evidence); what it would change; deliverables; domain; seed; north-star link.

Tier 1: WInS now and Trading Notes (Oct 23)

Q15. Can three notes taken from a mostly-Treasury book show distinct roles, and do we already have a growth trade with a proper note?

- Anchors: Guide p.3 (R-new): a note should state the trade's "expected role in growth, liquidity, risk management, or future funding" (VP). R-T10 "Collectively, the decisions should demonstrate how your team used individual investments as part of a cohesive portfolio strategy" (VRF).
- Hypothesis (guess): if the book is about 80% hedge funds, the risk is three notes that all say "future funding". The four official role words give a ready test: aim for one "future funding" note, one "growth" note, one "risk management" note (a rule-driven rebalance or position-limit split). Overlap: B7b Q6 plans notes by "supported / tested / refined". This question plans them by *role*. Both can hold at once.
- Would change: which trades are placed in the next two weeks, and that each carries a role word in its note.
- Deliverables: WInS-now, TN. Domain: D9 (with D10 for rule checks). Seed: 30 (not in my list; null used).
- North star: shows Laura each dollar has a job she can name, not a pile of funds.

Q16. Wharton's example note says "begin preparing" and "reduce portfolio volatility". What single fact lets our hedge note explain "all at once, now" without contradicting the example?

- Anchors: R-T23 (VRF): "We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment." R-T25: "as the residency funding date approaches".
- Hypothesis (guess): the example encodes the gradual glide path most teams will copy. Our note needs one number that shows why waiting is the risky choice. Candidates: the promise costs about \$292k of her \$300k today (F-101); a 0.27% fall in rates uses up the headroom (F-104); and the ladder cost more than \$300k on 173 of 185 days this year (F-111). B1a Q2 already flagged the "volatility" wording. What is new here is the word "begin" (gradualism).
- Would change: the wording checklist for the hedge note and its reflection.
- Deliverables: TN, WInS-now. Domain: D9. Seed: null.
- North star: shows we did not copy the template; we chose against it for a reason tied to her money.

Q2. Our whole-portfolio equity share goes about 1.5% (2027) → 20% (2028) → 4% (2031). Will readers who expect a glide path read this hump as incoherent, and what one clause explains it?

- Anchors: R-I12 (VRF) "How will the portfolio's asset allocation and composition change as future funding needs approach and payments are made?"; Guide (VP) "which principles will guide decisions as funding dates approach and payments are made"; F-409; R-T25.

- Evidence: DERIVED in B10b_designer_checks.py: 2027 sleeve $\$7,736 \times 60\% \text{ equity} \div \$300\text{k} = 1.5\%$. 2028: 20.4% (F-409). 2031 after an 80% floor lock, with a rough median sleeve of \$190k (ASM) and the ladder at its forward value of \$356,384 (F-106): $0.6 \times 0.2 \times 190 \div 546 \approx 4.2\%$.
- Hypothesis (guess): the hump is right. Laura's ability to take risk exists only once the second deposit arrives (B9a: about 152% funded counting the 2028 money). It then falls as the facility date nears. That is a glide path for the *facility money*, and a zero-risk line for the *promise*. The clause must name both paths. A Final Report chart of "equity share by year, split promise/surplus" makes it obvious.
- Would change: an IPS sentence on how the allocation changes over time; a Final Report chart; a TN reflection if WInS shows about 20% equity while Laura's 2027 book is about 1.5% (already flagged in wins_now/; the new part is the whole path).
- Deliverables: IPS, FR, TN. Domain: D7 (with D3 for the chart data). Seed: 29.
- North star: a plan whose risk follows her cash flows, not a textbook age rule.

Tier 2: IPS (Nov 6)

Q3. Should the IPS headline promise "fully funded by January 2028 at the latest" instead of "the \$300,000 buys the promise", because the second claim is a September-2026 accident?

- Anchors: F-111 (ladder above \$300k on 173 of 185 trading days in 2026; \$316.5k on Jan 2); F-112 (about 24% chance it costs more than \$300k on 2027-01-01, ASM); R-C26, R-C27; R-AN13.
- Evidence: DERIVED (flat-curve, parallel-shift ASM, so a rough guide): the ten payments cost \$300k at a flat 5.09%, a fall of about 27bp. \$300k plus the 2028 deposit's present value (\$143,230, per B9a) covers the ladder down to a flat 1.16%, a fall of about 420bp. With a \$75k deposit the threshold is about 2.90%, a fall of about 250bp.
- Hypothesis (guess): the designers set round numbers before September's rate rise, when \$300k did *not* buy the promise. They probably expected both deposits and some growth to be needed. A headline that depends on today's curve can break before the money even arrives. A funded-ratio rule ("hedge as much as the money buys, finish with the 2028 deposit") keeps the same plan and survives any plausible January curve. Overlap: B2a Q4 (rate-conditional lock rule). New here: the 420bp/250bp break points and the proposal to move the *headline*, not just the rule.
- Would change: the elevator pitch and first IPS sentence (decision on what we promise), plus one Final Report sensitivity line.
- Deliverables: IPS, FR. Domain: D3 (verify with the zero curve, not a flat rate), D1. Seed: 4.
- North star: she gets a promise that does not depend on luck with the calendar.

Q4. With the 2028 deposit present, a 70-95% lock also shows no payment shortfall in the model. So what is our real reason to lock 100%, and are we overstating it anywhere?

- Anchors: R-C47 "high degree of certainty"; R-C74 "Teams may reach different conclusions about ... funding confidence"; R-AN35 ("will contribute"); F-401; F-410 (partial locks fail only in the no-deposit case).
- Evidence (DERIVED, B10b_designer_checks.py; ASM: 60/40 money outside the lock, JPM inputs, 2033 reserve at a flat $5.26\% \pm 1\text{pp}$, no 2031 floor):

Lock share	Shortfall (\$150k deposit)	Surplus p5 / p50 / p95	Shortfall (\$75k deposit)
100%	0.00%	\$150k / \$210k / \$299k	0.00%
90%	0.00%	\$139k / \$212k / \$318k	0.00%
80%	0.00%	\$128k / \$213k / \$337k	0.00%
70%	0.00%	\$117k / \$214k / \$357k	0.08%
50%	0.02%	\$93k / \$216k / \$396k	1.74%
0%	2.10%	\$34k / \$221k / \$496k	11.72%

- Hypothesis (guess): the full lock is still the best choice. It costs about \$2-3k of median against an 80% lock, and it buys \$22k more at p5, a much narrower facility range and immunity to a smaller deposit. But "anything less fails" is false in the case's own base. The true sentence is about the *price of certainty*: about \$11k of median facility money (about 5%) removes a roughly 2% chance of breaking the promise and lifts the bad-case facility money by about \$116k. Caveat: this model's 1pp rate spread for 2033 may understate how far rates can fall (F-406 widens it).

- Would change: the reasoning sentence in the IPS; the Final Report comparison table (add the frontier); and the "why not a partial lock" answer in the Final Report.
- Deliverables: IPS, FR. Domain: D3. Seed: 26.
- North star: a statistics graduate gets the true trade-off, priced, instead of a claim she could disprove.

Q12. Lock early settles the promise in January 2027. Does the IPS still show a full "decision-making framework" for the six years the designers built around 2031 and 2033?

- Anchors: R-C89 "formally establishes the strategy and the team's decision-making framework"; R-I23; R-AN11 (the timeline lists only event years); Guide (VP) "planned adjustments as funding dates approach".
- Hypothesis (guess): yes, if each decision point is written as trigger → action → record. Candidates: (1) January 2027 purchase if the ladder costs more than the money; (2) the 2028 deposit order of use; (3) sleeve rebalancing band; (4) the 2031 floor share; (5) the 2033 facility rule; (6) the flexibility rule; (7) what Laura is shown each year (funded ratio, BS-09). The council panel scored plain "static lock + sleeve" 6/10 on Creativity because it reads as "safe bucket plus growth bucket" (02_judge_panel.md, AI panel, low authority). The rules are what separate us.
- Would change: an IPS structure checklist (which rules must appear in 500 words). The Guide forbids redesign after the results, so every contingent rule has to be in the IPS (B1a Q8).
- Deliverables: IPS. Domain: D7, D8. Seed: 5.
- North star: she sees what we will do at each date she has marked, before it happens.

Q13. Will a first-screen reader see about 1.5% equity in 2027 and about 20% after 2028 as timid for a client who "has been willing to take thoughtful risks", and which single number reframes it?

- Anchors: R-C37, R-C38; R-S25 "risk considerations"; F-409; brief section 8 item 3 (risk measured on the surplus).
- New evidence: the 2027 figure (1.5%) is the one WInS literally represents (WInS = her \$300k, R-AN14), so it is the one readers see first. Readers include Wharton student evaluators before the asset managers (B7a).
- Hypothesis (guess): the reframing number is "60% of the money the promise does not need". Better still is one sentence on why the promise is not the place for her risk: career risk already sits in her income (B9a Q5, DOSPERT). Do not argue with total-portfolio percentages.
- Would change: an IPS sentence; the order of numbers in the Final Report.
- Deliverables: IPS, FR. Domain: D6. Seed: null.
- North star: honours her risk appetite where it cannot hurt her promise.

Q14. Criterion 1 says "uses appropriate diversification". Which argument will a fixed-income reader accept for a 2027 book that is about 97% one issuer (the U.S. Treasury)?

- Anchors: R-S24; R-W71 "diversification across asset classes, sectors, industries, market capitalizations, geographic regions, risk characteristics, and funding purposes" (VP); R-AN45.
- Hypothesis (guess): (a) diversification by *funding purpose* uses Wharton's own word list; (b) diversifying a matched hedge *adds* risk to a fixed-dollar promise; (c) the sleeve is broadly diversified. Say (a) and (b) in one clause. Do not add a foreign bond to "look diversified".
- Would change: one IPS clause; the Portfolio Analysis framing in the Final Report.
- Deliverables: IPS, FR. Domain: D1. Seed: null.
- North star: shows we understand why her promise should not be spread around.

Q20. The IPS guide names "liquidity" repeatedly. Where is liquidity in lock early, and does one clause cover it?

- Anchors: R-I6 ("liquidity needs"), R-I11, R-I21, R-I26; R-C33, R-C34 (no withdrawals before 2033); BS-07 (plan delay).
- Hypothesis (guess): liquidity need before 2033 is nil by the case. After 2033 the reserve pays itself out each January. U.S. Treasuries are among the most liquid securities if the residency is delayed. STRIPS (zero-coupon Treasuries) trade less actively than notes; D1 should check. One clause suffices. A rival's "cash bucket" answers a need Laura does not have.

- Would change: one IPS clause.
- Deliverables: IPS. Domain: D1. Seed: null.
- North star: no idle cash held for needs she has told us she does not have.

Q19. Is lock early now the textbook answer that AI tools hand every team, and what 2-3 Laura-specific decisions are missing from a generic liability-matching plan?

- Anchors: R-S24 "clear and creative investment thesis"; R-S28 "original thinking"; BS-10; R-W26 (AI allowed for brainstorming).
- Hypothesis (guess): at 5% yields, AI assistants given the case will often suggest a Treasury ladder plus a growth bucket plus a Monte Carlo "95%". A cheap test: paste the public case into two or three AI tools, record the architecture each proposes, and record this use in Works Cited. Our likely distinctive parts: the bought 2031 floor, a certainty definition that needs no model, the funded-ratio headline (Q3), and the price-of-certainty frontier (Q4). Overlap: B7b's field simulation. New here: an actual test, not an assumption.
- Would change: which ideas get the IPS's scarce words (decision).
- Deliverables: IPS, FR. Domain: D7. Seed: 29.
- North star: the answer to "why us" cannot be the part every rival also has.

Tier 3: Final Report (Dec 4)

Q1. Does "it is the 2027 ladder, and it changes by itself" answer the case's reserve question, or will it read as a dodge? What must the answer contain?

- Anchors: R-C50, R-C52 ("Teams must recommend its size and initial asset composition"), R-C53 ("if at all"), R-AN27, R-AN28; brief section 8 item 9.
- Evidence: DERIVED (flat 5.35% ASM; the official forward value is \$394,930, F-106): value before each payment and Macaulay duration (average time to the payments) run \$399.9k / 4.07y (2033), \$300.9k / 2.79y (2036), \$142.5k / 0.97y (2040), \$50k / 0y (2042).
- Hypothesis (guess): allowed and strong, *if* it names all three items: size (market value on 1 Jan 2033, plus face \$500k), composition (ten zero-coupon rungs, one per January), change (automatic maturity roll-down, no trading, shown as a chart). It also needs the one sentence "funded 2027, designated 2033" (brief 8.9). A bare "the ladder" answer risks being scored as not engaging.
- Would change: a Final Report checklist and a roll-down chart.
- Deliverables: FR. Domain: D1, D9. Seed: 5.
- North star: she can see exactly what sits in her reserve every year.

Q9. Should the Final Report show that the most common rival reserve design, an equity "bucket" for the late payments, costs 1.8-2.6 times the Treasury price?

- Anchors: R-C52, R-C53, R-C47; R-S26 "integrates quantitative and qualitative analysis".
- Evidence: DERIVED (B10b_designer_checks.py; JPM large cap 6.70% compound, 16.47% volatility; lognormal ASM):

Payment	Equity needed in 2033 at 95%	at 99%	Treasury price (flat 5.35% ASM)	Ratio
2040	\$61.5k	\$80.9k	\$34.7k	1.77x / 2.33x
2041	\$60.4k	\$80.9k	\$33.0k	1.83x / 2.46x
2042	\$59.1k	\$80.6k	\$31.3k	1.89x / 2.58x

- Hypothesis (guess): one line showing "we tested it, it costs nearly double for less certainty" beats a page defending the ladder. The figures are model outputs, not forecasts.
- Would change: one Final Report sentence or small table; it also strengthens the Articulation story (an alternative tested and rejected).
- Deliverables: FR. Domain: D3. Seed: 26.
- North star: she sees we checked the tempting option with her statistics toolkit.

Q10. Rivals who buy the reserve in 2033 at cautious rates will show reserves of about \$405-440k. Will a reader read our \$395k as "smaller, so riskier"?

- Anchors: R-C52; F-106 (\$394,930 on 2033-01-01); F-110 (\$405,391 at 5%, \$421,767 at 4%, \$439,305 at 3%); brief 8.9. New: the reader-comparison risk, not the choice of number.
- Hypothesis (guess): yes, for a first-screen reader. The fix is to show three numbers side by side once (cost \$292k in 2027, value \$395k in 2033, face \$500k) and the phrase "the interest rate was locked in 2027".
- Would change: one Final Report sentence or table.
- Deliverables: FR. Domain: D9. Seed: null.
- North star: her promise looks as safe as it is.

Q11. A certainty definition that needs no model can look as if we skipped "explain how they evaluated". What stress table and residual-risk list prove it, and should the residual U.S. default risk carry a market-based number?

- Anchors: R-C54, R-C55, R-C56; R-C82 "Addresses how investment uncertainty could affect both the operating commitment and the facility contribution"; R-S26 "evaluate funding reliability ... under varying market outcomes"; R-C74 "funding confidence".
- Hypothesis (guess): the evaluation should be a table in which the payment row does not move: rates ± 200 bp, equities -40% , 2028 deposit \$0 or late, TWD $\pm 12\%$. The residuals are listed (Treasury default, operations, the rate move before purchase, coupon reinvestment if STRIPS are unavailable). A market-implied default probability from U.S. credit default swap (CDS) prices would give a number. I could not read one: the page I tried loads its data by JavaScript (section 7). Named models give probabilities to two significant figures at most (brief 8.6).
- Would change: a Final Report table; possibly one IPS clause giving a confidence number.
- Deliverables: FR, IPS. Domain: D3, D1. Seed: 9.
- North star: a statistics graduate sees the certainty claim tested, not asserted.

Q5. The case says teams "must recommend the dollar range" and says the 2033 value "remains uncertain" in 2031. Our 2031 rule nearly removes that uncertainty. Do we give a dollar range today, a rule, or both, and how do we avoid the "one exact amount" the case rejects?

- Anchors: R-C65, R-C67 ("Rather than promising one exact amount"), R-C68 ("Teams must recommend the dollar range"), R-C69, R-C71, R-AN29 ("must").
- Evidence: 2031 floor p5/p50/p95 in today's view \$127k / \$165k / \$217k (F-402). In a 2031 view the band is about $\pm 4-5\%$ (B6b). At 80% lock the range is about US\$199-218k (B6a).
- Hypothesis (guess): both. Today's dollar range (with the model named) satisfies the "must". The 2031 rule shows how she will say it. One sentence explains why the range narrows: most of the uncertainty is gone by 2031, because the floor is bought. The width that remains should come from a *stated choice* (how much to keep flexible), not from markets. Overlap: B1a Q18, B6a Q7, B6b. New: the designer-premise framing and the literal "must".
- Would change: Final Report structure for the range section; the lock-share parameter (decision).
- Deliverables: FR. Domain: D5, D3. Seed: 9.
- North star: she can say one sentence to funders that stays true in 2033.

Q6. Under a floor + cap rule, what happens to good-market money, and does sending it to "flexibility" weaken the "meaningful contribution" signal the case describes?

- Anchors: R-C64 "A meaningful personal contribution may signal that the residency is financially viable"; R-C70 "explain how favorable and unfavorable market outcomes could affect the amount she can provide"; R-AN9.
- Hypothesis (guess): the case wants the favourable case to *affect the amount*. A hard cap makes the favourable case change only the leftover. A clean version: the range top is a promise ceiling, and upside above it is her choice in 2033 (an extra gift or flexibility), stated as such. Funders count only the unconditional part (B6a Q1), so the cap costs nothing with them.
- Would change: the favourable-outcome sentence in the fundraising draft checklist and in the facility rule.
- Deliverables: FR. Domain: D5. Seed: 9.
- North star: good luck is hers to direct, and she never has to walk a number back.

Q7. Our facility rule is "whatever the sleeve becomes" and flexibility is "mainly leftover". What crisp split rule would a rival state, and what should ours be anchored to?

- Anchors: R-C57, R-C58 ("committing all remaining assets could limit her financial flexibility as the project develops"), R-C60, R-C84, R-AN18, R-AN19, R-AN20 ("extent" may be a share or rule).
- Hypothesis (guess): a rule such as "contribute the bought floor plus a stated share of the rest; keep the remainder as project flexibility". Candidate anchors: the case's own words about the project developing; Laura's staged scaling (B9a Q8); goal tiers (needs / wants / wishes, B4b). The share is the team's decision. Without a rule, the "responsible" criterion is met by assertion only.
- Would change: a decision (the split), an IPS clause, a Final Report table.
- Deliverables: IPS, FR. Domain: D5, D6. Seed: 11.
- North star: she keeps room to act as the residency grows, which is how she has built everything.

Q8. Could part of the facility money be given in a second tranche after 2033, tied to project milestones, or does the case's 2033 framing rule that out?

- Anchors: R-C63 "how much she expects to contribute toward the facility in 2033"; R-C57; R-C58 "as the project develops"; R-C61 (no contingency fund needed).
- Hypothesis (guess): the co-sponsor range must describe the 2033 contribution, so a staged gift cannot be part of the range. It can be named as one possible use of the flexibility money. More than one sentence risks looking like a misread of the case.
- Would change: one Final Report sentence (or a decision not to mention it).
- Deliverables: FR. Domain: D5, D7. Seed: 11.
- North star: fits a founder who ships in stages, without bending the case.

Q17. Which is bigger for what Laura's facility money buys by 2033: market risk, USD/TWD risk, or Taiwan construction inflation? Does a USD-only nominal range miss the assumption the case names?

- Anchors: R-C85 "the effect of inflation on portfolio projections and facility costs"; R-AN8; R-AN10; F-508, F-509, F-513, F-501.
- Evidence (DERIVED): construction cost index about 3.5%/yr since 2021 → about +27% over 2026-2033 ($1.035^7 = 1.272$). Taiwan CPI at 2% → about +15%. Two-year USD/TWD standard deviation about 6.8% (since 2006). The 2031-view market band is about $\pm 5\%$ (B6b).
- Hypothesis (guess): construction inflation dominates. A fixed US\$ figure buys roughly a fifth less building in 2033 than today if that trend holds (ASM: trend continues). The range stays in US\$ (it protects the promise), but the Final Report needs one assumption line stating the purchasing-power effect. It does not need a facility-cost estimate (R-C95). Overlap: brief section 9 (TWD quote), B6b Q5. New: the ranking of the three risks.
- Would change: an assumption sentence and a possible small chart in the Final Report.
- Deliverables: FR. Domain: D4. Seed: null.
- North star: she is told what her gift will really buy, in the place she is building it.

Q18. If rivals show \$300k+ facility contributions from 8-10% return assumptions, should the Final Report show a same-assumptions comparison (ours vs growth-first vs partial lock) so a first-screen reader sees what those numbers cost in payment risk?

- Anchors: R-S26 "uses reasonable assumptions and projections"; R-C77 "reasonable return assumptions"; F-401; F-301; B7b (a 2025-26 semifinalist reported "a 70% chance of reaching the long-term goal"); B1b (last season's "9% annualized" habit).
- Hypothesis (guess): yes, as one chart: median and p5 facility money against the chance of breaking the promise, for 3-4 designs on the same JPM inputs (the Q4 frontier is the data). It turns "our number is smaller" into "our number is real".
- Would change: a Final Report chart (decision to include).
- Deliverables: FR. Domain: D3, D7. Seed: 29.
- North star: shows her the honest price of a bigger promise.

Q21. Lock early decides early, so where is the genuine "challenge and adaptation" story that the Articulation criterion scores?

- Anchors: R-S27 "reflects meaningfully on the team's growth and response to challenges"; Guide (VP) "You may identify decisions you would make differently"; R-C88 "reflected, tested, or refined".
- Hypothesis (guess): real candidates already exist. The team's original growth-first plan was dropped on evidence. The September rate rise changed whether \$300k buys the promise. A possible 25% position limit could force a three-fund hedge (G4). The WInS book was changed from the post-2028 mirror to the 2027 reality. Each needs a dated decision-log entry now. The council's AI panel scored the static version 6/10 on Articulation ("little process to narrate"; low authority).
- Would change: start a dated decision log this week; pick one Trading Note as "refined".
- Deliverables: WInS-now, TN, FR. Domain: D7. Seed: 31 (null used in the summary).
- North star: she hires people who changed their minds for good reasons.

4. Attacks I tested that failed (parked, with reasons)

- **"An equity bucket inside the reserve would honour her risk appetite."** Fails on cost (Q9: 1.8-2.6x).
- **"Wait until 2033 and buy the reserve then."** Already answered (brief 8.1). No new evidence.
- **"Tiering: roll 2037-42 short."** Already answered (brief 8.2).
- **"TIPS for the reserve."** Already answered (brief 8.10).
- **"More equity in the sleeve would lift the facility enough to matter."** Already answered (brief 8.5); B9b adds the 0% control. Not re-asked.
- **"WInS should hold 35% equity."** Already dropped in `wins_now/securities_and_allocation_v0.md` (no plan version holds it).

5. Evidence found (with status)

1. **2026-27 Investment Competition Guide** (5 pages), <https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf>, read 2026-09-27, VERIFIED-PRIMARY. Verbatim checks passed for:
 - "you will need to decide how to construct and diversify the portfolio, how its components will support different needs, how the portfolio's asset composition may change over time, and which principles will guide decisions as funding dates approach and payments are made." (p.2)
 - "the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding." (p.3)
 - "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." (p.5)
 - Also on p.2: "The goal is to develop a cohesive strategy in which your investment decisions work together to support the client's objectives across a range of possible market outcomes." First found by B1b/B7a/B7b. Not yet in `competition/official/2026_27/` (main loop's decision).
2. **Trading Notes example** (R-T23 to R-T25), VRF: "begin preparing", "as the residency funding date approaches".
3. **Model outputs** (DERIVED, ASM inputs as labelled): `research/insight_v1/scripts/B10b_designer_checks.py`, run 2026-09-27, runtime about 1 second, seed 20260927, 200,000 paths. Results in section 3 (Q2, Q4, Q9, Q1).
4. **Break-even rate falls** (DERIVED, flat-curve ASM, inline Python run 2026-09-27): cost \$300k at 5.09% (-27bp); \$371,615 at 2.90% (-246bp); \$443,230 at 1.16% (-420bp).

6. Script

`research/insight_v1/scripts/B10b_designer_checks.py`. Run from the repo root with `.venv/bin/python research/insight_v1/scripts/B10b_designer_checks.py`. The docstring lists inputs and status. Main limits: no 2031 floor in the frontier (so the 100% row's p5 is \$150k, not the verified \$159k); the 2033 reserve uses a flat 5.26% ± 1pp, which slightly overstates its cost (F-408); sleeve bonds earn JPM's 4.0%, which is conservative against today's yields (F-317; B9a Q4); i.i.d. lognormal returns; no fees.

7. Sources (all accessed 2026-09-27)

Source	URL / path	Status
Investment Competition Guide 2026-27	https://upenn.box.com/shared/static/b9kh53xila9czkzknk1xv5yegl3snlc1.pdf	VERIFIED-PRIMARY
Client case	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt	VERIFIED-REPO-FILE
IPS guide	competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt	VERIFIED-REPO-FILE
Trading Notes guide	competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt	VERIFIED-REPO-FILE
SMAApply criteria	competition/official/2026_27/SMAApply_Deliverables_Page_2026-09-27.md	VERIFIED-REPO-FILE
Phase A registers	research/insight_v1/phase_A/*.md (R-, F-, BS-, G- ids)	as labelled there
Verified model	research/verified_2026-09-27/strategy_mc.py	VERIFIED-REPO-FILE inputs
Wharton Global Youth article search (WordPress API)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/posts?search=... ("bond ladder", "glide path", "liability", "treasury", "target date")	VERIFIED-PRIMARY negative result: no teaching article on these topics found
U.S. 5-year CDS page	https://www.worldgovernmentbonds.com/cds-historical-data/united-states/5-years/	Not usable: values load by JavaScript, the text shows "-----"
WebSearch	tool	Refused: session search budget (200) already used
Council AI panel	research/council_2026-09-27/02_judge_panel.md	AI brainstorming, low authority

8. Leads for the specialists

- **D3 (quant):** re-run the Q4 frontier with (a) the 2031 floor included, (b) sleeve bonds at a current yield (about 5%), (c) the 2033 rate spread widened to 1.5-2pp, and (d) the 2028 deposit correlated with a bad 2027 equity year. Check whether 70-95% locks stay at 0% shortfall. Re-do the Q3 break-even with the zero curve (not a flat rate) and with the 2028 top-up bought at 2028 prices.
- **D3/D9:** build the chart data for Q2 (equity share by year, promise vs surplus), Q1 (reserve roll-down), Q18 (facility median and p5 against shortfall probability, 3-4 designs).
- **D1:** find a primary or official source for a market-implied U.S. default probability (for example a U.S. 5-year CDS level from a data vendor's static page or a central-bank publication). Check STRIPS liquidity (bid-ask) against on-the-run notes for the Q20 clause.
- **D4:** confirm the construction cost index trend (F-508) and whether DGBAS publishes a forecast. Compute the NT\$ purchasing power of a US\$ floor in 2033 under trend CCI and ± 1 standard-deviation FX moves (Q17).
- **D5:** a precedent for "floor promise, upside at the donor's discretion" wording (Q6), and whether staged gifts are common for founders of small residencies (Q8). Do not name organisations in deliverables (BS-15).
- **D7:** run the Q19 test (two or three AI tools, the public case text, record the architecture each proposes and log the use for Works Cited). Check whether any Wharton Global Youth teaching page covers liability matching; my API search found none.
- **D10:** confirm that a note naming a "role" word in WInS fits any character limit (G-ids; not published).

What this teaches

A good contrarian doesn't try to knock the plan down. It asks where a careful reader could fairly mark the plan down. Here the architecture held up under every attack: buying the promise with Treasuries beat each rival design we priced. Three of our *reasons* did not hold up as well. "Anything less than a full lock fails" is false when the second deposit arrives. "The \$300,000 buys the promise" depended on September's interest rates. And "the reserve doesn't change" answers the case's question only if we show how its maturities shorten each year. The lesson for investing is that being right is not enough when someone else checks your work: the argument has to be right too. The lesson about case writing is that the designers imagined a glide path (growth first, safety later), and a plan that departs from their picture has to say why, in their own words, before the reader has to ask.

Phase C filter log

Generated by `research/insight_v1/scripts/phase_C_survival.py` from the three skeptics' verdicts (roles: Wharton semifinal reader; already-answered checker; statistician/wrong-premise hunter).

Step	Count in	Dropped	Count out
Phase B questions (20 agents, 10 lenses)	472		472
Exact duplicates (normalised text)	472	0	472
Structural screen (no anchor or no stated change)	472	0	472
Librarian: semantic duplicates merged into clusters	472	224 merged away	248
Skeptic filters (so-what, anchor, 2-of-3 adversarial, simplicity)	248	200 parked	48

Why questions were parked (a question can fail several filters; kill reasons count each skeptic's kill):

- kill:already_answered: 318
- adversarial: 200
- so_what: 128
- kill:trivial: 95
- kill:no_so_what: 55
- kill:wrong_premise: 27
- kill:unanswerable: 26
- simplicity: 23
- anchor: 19
- kill:too_complex: 11

Survivors by domain tag: {'D5': 4, 'D8': 6, 'D1': 2, 'D10': 2, 'D6': 6, 'D9': 15, 'D7': 2, 'D3': 9, 'D2': 1, 'D4': 1}
Survivors by deliverable they affect: {'FR': 42, 'IPS': 34, 'TN': 15, 'WlnS-now': 13} Survivors with fewer than 3 votes (a skeptic call failed): 3

Parked questions are listed in `parked.json` with each skeptic's reason. Scope is semifinals only, so they are not turned into a Q&A bank.

What this teaches

Most questions that sound important fail at least one plain test: would the answer change anything we submit, is it tied to the case, has it already been answered, and can the insight be said in one sentence? Filtering hard before researching is how a small team spends its time on the few questions that move the result.

Phase C Librarian notes: merging the 472 Phase B questions

Summary: the 472 questions (20 Phase B agents, 10 lenses; exact duplicates already removed) collapse into **248 clusters**: 109 merged clusters that hold 333 questions, plus 139 singletons. Every input id appears in exactly one cluster; this was checked with Python (472 ids in, 472 ids out, no id repeated, none missing). No id was left unplaced. File: `research/insight_v1/phase_C/librarian_clusters.json` (list of `{source_ids, canonical_text, domain_tag}`, largest clusters first).

Method (ASSUMPTION: this is a judgement rule, not a formula): two questions were merged only when answering one would take the same research as answering the other and would change the same decision, number or sentence. Related questions that change different things stayed apart. For example, "what share of the 2031 sleeve to lock" (B6a-07/B6b-02/B10a-07), "lock progressively before 2031" (B4a-16/B10a-09/B9a-10) and "barbell in 2028" (B10a-02) are three clusters. So are "which chart" and "which words label the range ends". Each canonical question keeps the sharpest member's wording and its concrete anchor (for example \$300k, 173 of 185 days, 27bp, ~\$204k, 42-46% of paths).

Cluster sizes: 4 clusters of 7, 1 of 6, 6 of 5, 23 of 4, 27 of 3, 48 of 2, 139 of 1. Domain tags of the 248 clusters: D9 53, D5 30, D3 29, D7 25, D6 24, D8 23, D1 21, D10 21, D4 15, D2 7.

Biggest clusters (and how many separate lenses asked them)

Size	Lenses	Tag	What it is about
7	5 (B1,B2,B4,B7,B8)	D9	A fixed format for every WInS trade note: a one-line purpose, a role named in the Guide's own words, one dated cash flow of Laura's, one checkable number and one named risk
7	5 (B2,B3,B4,B8,B10)	D1	The rule if rates fall and the ladder costs more than \$300k in January 2027: buy what the deposit affords, longest rungs first, then top up from the 2028 deposit ("fully funded by 1 Jan 2028 at the latest")
7	5 (B1,B4,B8,B9,B10)	D5	An explicit rule or share for "financial flexibility" (the money kept uncommitted after the reserve), instead of "whatever is left"
7	3 (B1,B4,B7)	D7	Making the three Trading Notes "supported / tested / refined", with a pre-written rule that fires a real trade before Oct 23
6	5 (B1,B2,B3,B7,B9)	D10	Whether WInS should hold the literal 2027 book (about 97% ladder) or the post-2028 target mix, and the one sentence that joins WInS to the 16-year plan
5	4 (B2,B4,B7,B8)	D6	A pre-committed "bad-year rule" for Laura's emotions in a downturn
5	4 (B2,B4,B8,B9)	D5	Ordering the 2031 message: bought floor first, then a conditional stretch, then an invitation to partners
5	4 (B3,B5,B8,B9)	D2	AI exposure: no "AI hedge" tilt, plain disclosure of index concentration, any tilt left to Laura
5	3 (B1,B7,B10)	D8	Which "if X, then Y" rules the IPS must pre-declare, so the Final Report applies them instead of redesigning
5	2 (B3,B9)	D3	The correlation between the 2028 deposit and 2027 equity returns, or named scenarios instead
5	2 (B7,B10)	D7	What makes our lock-early plan different from the textbook answer that AI tools and hundreds of teams will give

Themes that many lenses asked on their own (a signal of importance)

These are the places where agents with different briefs (case anomalies, Wharton history, Laura's record, practitioner benchmarks, markets, co-sponsors, judging, behaviour, red team) reached the same question separately. Counting lenses rather than questions is fairer, because one agent can repeat itself.

- 1. The WInS book and the Trading Notes as evidence (tier 1, due now).** Five clusters, with 4-5 lenses each: the note format, supported/tested/refined, staged hedge trades, how our note differs from Wharton's Treasury-ETF example, and the "literal 2027 book" question. Taken together: WInS is the rehearsal of Laura's January 2027 purchase, and every note must show it.
- 2. Laura's risk tolerance, stated in her own terms.** Four clusters over B2, B3, B4, B7, B8, B9, B10: ability vs willingness; "floor first, then leap" from her career move; a low total equity share reading as

timid; a stated loss she can accept on the sleeve. A plain risk-tolerance sentence is the most-asked Laura-specific item.

3. **The 2031 co-sponsor range.** At least ten clusters: range today vs rule in 2031, confidence defined on the bottom, bought floor plus stretch, what funders count, the lock share, where money above the top goes, the chart, the late-2032 checkpoint. Every co-sponsor-facing lens (B1, B2, B4, B6, B8, B9, B10) came back to "the floor is the only promise".
4. **Pre-commitment rules in the 500-word IPS.** Contingency rules, governance, the rates-fall rule, the late-deposit sentence, the bad-year rule, no regret after the lock. The official "no redesign after results" line (Guide p.5) pushed five lenses to the same list.
5. **The 2028 deposit as the main uncertainty that is not market risk.** Correlation vs scenarios, how late, what label goes on "will", the joint tail with falling rates, and political rather than market drivers of her income. This is asked by B3, B5, B8, B9, B10.
6. **Refusing tokenism.** Identity-themed securities, book-title puns, "gimmicks", the swap test, Taiwan and Wuhan, the pull quote's attribution. Six lenses separately said to use her professional record, never her identity, as the reason.
7. **Fees and honest disclosure.** Fee on the sleeve only, what her CFPB and trading-desk background would check, AI-use record, banned "finance-bro" words, adviser-style disclosure instead of a pitch. Lenses B1, B2, B3, B4, B7, B8, B9, B10.
8. **Taiwan and currency risk.** TSMC and AI concentration in the sleeve, the single correlated crisis scenario, an NT\$ band vs a US\$ range, and facility-cost inflation. These questions are mostly from B3, B5 and B8 and are more spread out, as smaller clusters.

Lens-specific singletons are not weaker because they are singletons. Many are sharp technical checks that only one lens could raise (e.g. B5a-06 Nov-15 maturities as a payment-delay buffer, B1a-06 the WInS volume rule, B7b-14 the word counter). The ranking phase should judge them on value, not on repetition.

Judgement calls the next phase may want to revisit

- B2a-02 (narrative spine + WInS mirror) and B2a-11 (cash-flow calendar + note names cash flow) each have two parts; I placed them by their decision part (the WInS book; the note format).
- B8a-09 sits with "political vs market drivers of her income", not with the "media underweight" cluster, although it touches both.
- B10b-04 (equity "hump" incoherent) was merged with the "reads as timid" cluster because both are fixed by one goal-by-goal risk sentence.
- The historical-paths cluster (B6b-22, B10a-18, B5b-23) merges three tests of different outputs because they need the same backtest.

What this teaches

Many questions come down to a few decisions. 472 questions from 20 agents turned into 248 distinct ones, and about a dozen themes were asked independently by four or five different lenses. When people looking from different angles ask the same thing, that question is probably central, so it deserves research time first. Merging also forces precision: two questions are "the same" only if the same research would change the same decision. That test stops a librarian from blurring "how big is the floor" into "how do we word the floor", which need different work.

Part 3 - The research record, phase by phase · research/insight_v1/phase_D/D1_rates.md

D1 - Rates & fixed income: the dated rung in WInS (M044) and the January 2027 purchase rule (M004)

Agent D1 (Rates & Fixed-Income Analyst), run insight_v1, 2026-09-28. AI brainstorming output for the team: specifications, evidence, numbers and checklists. It is **not** text to submit. The team decides and writes every word. Scope (brief section 17): Trading Notes (TN, Oct 23) and IPS (Nov 6) only. Anything that is purely Final-Report material is one line in "Later (after Nov 9)". No relayed messages arrived during this work.

Status labels: VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SNIP), ASSUMPTION (ASM), DERIVED (computed by a script here from labelled inputs), INTERPRETATION (our reading; not a fact). Every security is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK** (brief sections 4 and 14).

Scripts (run from the repo root):

- `.venv/bin/python research/insight_v1/scripts/D1_wins_rung.py` (M044: bond prices, sizing, re-weighting, tracking)
- `.venv/bin/python research/insight_v1/scripts/D1_purchase_rule.py` (M004: rate-fall costs, the longest-first split, 2026 history, gap odds, buying all at once versus in stages, the joint tail)

Terms used below:

- **Ladder / rung:** Laura's real plan buys ten Treasury zero-coupon bonds (STRIPS), one per \$50,000 payment. Each is a "rung".
- **Coupon bond:** an ordinary Treasury that pays interest every six months and repays its face value at maturity.
- **Face value:** the amount a bond repays at maturity. **Accrued interest:** interest earned since the last coupon; the buyer pays it to the seller.
- **Duration:** roughly the % a bond's price moves when rates move 1 percentage point. **Cash-flow match:** the money arrives on the date it is needed. **Duration match:** the value moves like the payments' value, but no money is dated.
- **Gap:** how much more the ladder costs than the \$300,000 first deposit on 1 January 2027.
- **bp** = basis point = 0.01 percentage point. **Funded ratio** = money available ÷ cost of the ladder.

Summary: top findings (ranked by impact on reaching the semifinals and on making the plan hers)

1. **The IPS headline should be rate-conditional: "the whole first deposit buys the ladder on the day it arrives, longest-dated payments first. Any unbought part is the earliest payment(s), and it is the first use of the 2028 deposit, before any growth investment."** (M004, IPS; DERIVED numbers below)
 - Today the real Nov-15 STRIPS ladder costs \$294,387, so it fits under \$300k with \$5,613 to spare (VRF curve; DERIVED).
 - A 19bp fall before January creates a gap. The model odds of a gap are **24-31%** (exact-date zeros vs the real Nov-15 ladder; ASM zero-drift lognormal). Use these odds, **not** the 173/185-day count, which is history.
 - On the **worst curve of 2026** (Feb 27, 10-year at 3.97%) the gap is \$27,631. In 2028 dollars that is \$28,549, which is **19% of the \$150k deposit**.
 - Even then, only part of **one** payment waits: the 2033 payment. The gap exceeds a whole payment only after a ~150bp fall. The model odds of that are 0.01%. The gap exceeds the whole 2028 deposit only after a ~420bp fall (10-year under 1%).
 - 2. **Buy everything at once. Staging and yield triggers are rate bets with no reward (M004, IPS).** DERIVED, ASM:
 - Four quarterly tranches in 2027 leave the expected funded ratio almost unchanged (+0.2 points).
 - But they turn a known 1.019 into a **27% chance of ending under 1.00** (p5 0.966).
 - A "wait for a 5% margin" trigger does the same (26.5%; p5 0.912).

- This matches Guide p.5: the strategy "should not be rewritten simply because markets move". 3. **Longest-first is right, and now there is a number for it (M004, IPS)**. Suppose rates fall 100bp, a gap opens, and rates then fall another 100bp during 2027:
- longest-first: the waiting part (the 2033 payment) costs **+\$1.2k (+5%)** more;
- nearest-first: the waiting part (the 2042 payment) costs **+\$3.6-3.8k (+15%)** more.

Leaving the *shortest* payment unbought leaves the smallest, least rate-sensitive piece exposed (DERIVED).

4. **Name the joint tail plainly (M004, IPS rule; wording is the team's)**. Rates fall ~100bp before January **and** the 2028 deposit never arrives:

- about **\$31k of the \$50,000 January-2033 payment is unfunded** (6.3% of the \$500k promised; DERIVED);
- every other payment is bought;
- a \$75k deposit covers every rate fall down to the 2026 worst case;
- a deposit six months late costs only ~\$0.2-0.5k more if rates are unchanged.

The case says she "will" contribute (R-C27, R-AN35), so this is our own stress scenario and must be labelled as such.

5. **One dated Treasury "rung" in WInS is worth one \$10 trade, if WInS lists a suitable bond (M044, WInS-now + TN)**.

- It turns "we would buy a ladder" into a holding a fixed-income reader can check: a bond repaying \$50,000 on 15 Nov 2032, 47 days before the first payment.
- It is the parent bond of the real ladder's first STRIP (CUSIP 91282CFV8 → principal STRIP 912821KC8; VP via S1's MSPD table).
- The effect on hedge accuracy is tiny either way (DERIVED): +\$191 to -\$104 on a worst 50bp twist, against \$1,544 now.
- So the reason to buy it is the evidence it gives, not the hedge accuracy. It is a strong candidate to replace the VGSH buy as the third Trading Note.

6. **Which maturities WInS lists cannot be answered from public pages. The only public Stock-Trak bond list is from 2019 and has no Treasury maturing 2032-2035.**

- That list (stocktrak.com/available-bonds, dateModified 2019-09-06; VP for its content, **not** a 2026-27 WInS list) has 32 U.S. Treasuries.
- Among them are the **4.375% 15-Nov-2039** and **4.25% 15-Nov-2040** bonds, which are the parents of two of the real ladder's STRIPS.
- The 2026-27 WInS User Guide says the competition involves "a limited number of treasury/government bonds" (VP via A4).
- So the drop-down check is a tier-1 gate item. The fallback order is below. 7. **Small but useful: the case PDF's creation date (2026-09-10) is the day the real ladder cost \$299,990.**
- That is, almost exactly \$300,000 (DERIVED from the 2026-09-10 curve).
- INTERPRETATION only: it may be coincidence. Never claim it in a deliverable.
- What it does show: the \$300k "fit" is knife-edge, not comfortable, which is why the rate-conditional rule in finding 1 is needed.

Question M044 (tier 1: WInS now + Trading Notes Oct 23)

Question (canonical): Should the WInS book hold one or more individual Treasuries maturing near Laura's payment dates (Nov 2032-Nov 2041), to show cash-flow matching rather than only duration matching with constant-maturity ETFs, and which of those maturities does the WInS bond list actually offer?

One-sentence answer: Yes, one. If the WInS Bonds drop-down lists a Treasury maturing in the second half of a year from 2032 to 2041, buy one rung sized to one payment and fund it from IEF (not TLH). The fit barely changes, and it gives the TN set a checkable, dated piece of Laura's ladder. Which maturities are listed can only be seen when logged in, and the one public Stock-Trak list suggests the 2032-2035 notes may be missing.

Evidence

Claim	Source (accessed)	Status
Individual government bonds are a permitted type: "Treasury Bonds: Bonds available on WInS. The list includes Treasury bonds from the United States, ..." / "Any Government/Treasury Bonds from any exchange available on WInS"	https://wghsinvcomp.smaply.us/res/p/trading/ and /res/p/faqs/ (via phase_A/case_register.md R-W69, R-W89; read 2026-09-27)	VP
\$10 commission per Treasury bond trade; bond prices update once daily at the U.S. open; coupons every six months; bonds valued at the end	SMAApply FAQ (R-W91, R-W92)	VP
The platform lists "a limited number of treasury/government bonds"; the quote shows face value, last price, last coupon, accrued interest; "When you buy a bond in between coupon dates you will have to pay accrued interest to the seller"	2026-27 WInS User Guide p.7, p.11, https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf (re-read 2026-09-28; visible-text check by A4 in phase_A/wins_week1_guardrails.md)	VP. The PDF text layer also holds a hidden older line about "approved lists"; A4 found that it is not visible on the rendered page, so do not rely on it
Stock-Trak's public "Available Bonds" master list, dated 2019: 32 U.S. Treasuries, symbol format B-T- <coupon> - <ddmmyyyy> (e.g. B-T-4.375-15112039); maturities 2036-2044 include Feb/May/Aug/Nov bonds, among them 4.375% 15-Nov-2039 and 4.25% 15-Nov-2040; none maturing 2032-2035	https://www.stocktrak.com/available-bonds/ (linked from https://content.stocktrak.com/what-are-bonds/ : "All bonds, corporate and treasury, that we support are in one master list"); JSON-LD dateModified 2019-09-06; read 2026-09-28	VP for the page; relevance to 2026-27 WInS UNVERIFIED (generic, 7 years old)
Stock-Trak's own teaching assignment: "Stock Trak trades only a very limited number of bonds ... Treasuries are listed below the Corporates"	https://www.stocktrak.com/investments-bond-analysis-trading-project/ (read 2026-09-28)	VP (generic platform, not a season rule)
Candidate bonds exist and are strippable: 4.125% note 15-Nov-2032 (91282CFV8; principal STRIP 912821KC8); 4.375% bond 15-Nov-2039 (912810QD3; 912803DJ9); 4.25% bond 15-Nov-2040 (912810QL5; 912803DP5)	research/insight_v1/wins_now/S1_mspd_table5_2026-08-31_fixed_2032plus.csv (Treasury MSPD Table V, record date 2026-08-31, per S1)	VP per S1
IBTM (iShares iBonds Dec 2032 Term Treasury ETF): expense 0.07%; effective duration 5.05y; average YTM 5.03%; 15 holdings; net assets \$554.5m; close \$21.85; 30-day average volume 215,760 (all as of 2026-09-25/24); the funds "will terminate on or about October or December 15 of the year in each Fund's name"	https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf (read 2026-09-28; phrase verified verbatim)	VP. Not on the 2025-26 list (S1 table, VRF)
Constant-maturity funds never mature; three-quarters of TLH's bond weight matures after the last payment	wins_now/securities_and_allocation_v0.md §1 (74.7%), S4 finding 2	VRF (repo research)
2024-25 Session Rules screenshot showed a bond "Position Limit (Single Position) 100%"	2024 WInS guide p.6 via phase_A/wins_week1_guardrails.md	VP [PRIOR]: history only
S4 dropped an individual 2041 bond only as a position-limit fallback on complexity grounds; a single dated "demonstration rung" was never assessed	wins_now/S4_red_team.md finding 3	VRF

Numbers (D1_wins_rung.py; curve 2026-09-25 VRF; model prices ASM)

Bond (pays for)	Clean price	Accrued at 9/25	Yield	Model duration	Next coupon
4.125% 15-Nov-2032 (1 Jan 2033)	95.27	1.49	5.03%	5.27y	15 Nov 2026 (after the Nov 6 freeze)
4.375% 15-Nov-2039 (1 Jan 2040)	91.36	1.58	5.29%	9.64y	same
4.25% 15-Nov-2040 (1 Jan 2041)	89.31	1.54	5.34%	10.21y	same

Sizing and re-weighting. IEF/TLH are re-solved so that the whole hedge keeps the ticket's model duration (10.09y, which is 9.90 on issuer numbers). Tracking uses S4's forward-consistent method; the ETF-only worst 50bp twist is \$1,544 and the worst ± 100 bp is \$451.

Option	Rung	Rung \$ (share of \$300k)	IEF / TLH (share of \$300k)	Worst twist	Worst ± 100 bp
(iii) literal Jan-2027 book, face \$50,000	2032 note	\$48,378 (16.1%)	13.6% / 68.3%	\$1,735 (+191)	\$485

Option	Rung	Rung \$ (share of \$300k)	IEF / TLH (share of \$300k)	Worst twist	Worst $\pm 100\text{bp}$
	2039 bond	\$46,473 (15.5%)	28.1% / 54.5%	\$1,440 (-104)	\$435
	2040 bond	\$45,424 (15.1%)	29.9% / 52.9%	\$1,456 (-88)	\$438
	IBTM ~1,686 sh	\$36,841 (12.3%)	18.2% / 67.5%	\$1,710 (+166)	\$480
(ii) post-2028 mix scaled, face \$34,000	2032 note	\$32,897 (11.0%)	9.0% / 46.0%	\$1,737	\$485
	2039 bond	\$31,601 (10.5%)	18.8% / 36.6%	\$1,439	\$435
	IBTM ~1,147 sh	\$25,052 (8.4%)	12.2% / 45.5%	\$1,711	\$480

- **Hedge accuracy is not the argument.** Every change is under \$200, less than 0.1% of the hedge (DERIVED).
- **Coupon vs zero.** \$50,000 face of the 2032 note costs ~\$48.4k. It pays 13 coupons of \$1,031.25 (\$13,406) before its \$50,000 principal. The STRIP carved from the same bond costs ~\$37.3k and pays only the \$50,000 (DERIVED; S3). That coupon cash, and the need to reinvest it, is exactly why Laura's real ladder uses STRIPS (brief section 8 item 11). The rung note can make this point in one clause.
- **Accrued interest:** about \$750-800 on \$50k face, paid at purchase. No coupon arrives before the Nov 6 freeze (DERIVED).
- **Position limit:** the rung is 8-16% of the book, under any documented cap. **Watch-out:** a 2032 rung *raises* the TLH share (68% under (iii)), because the rest of the hedge must be longer. A 2039/2040 rung *lowers* it (53-55%). Under a cap below ~44%, the ticket's SPTL branch still applies to whatever TLH share remains.
- **Scaling (option ii):** the rung represents one payment at the same 0.685 scale as the rest of the book (\$34k face). A "\$50,000 = one payment" rung reads most naturally under option (iii). That is a small point in (iii)'s favour; it is not a reason to switch.

Implication (decision; WlnS-now + TN)

Decision rule for the WlnS gate session (tick in order; stop at the first "yes").

1. The Bonds drop-down lists a U.S. Treasury maturing **Jul-Dec 2032**. The 4.125% 15-Nov-2032 note (91282CFV8) is best: it pays for the first payment, the first year of the residency's operations.
 - Buy face \$50,000 under (iii), or ~\$34,000 under (ii).
 - Take the money out of **IEF**.
 - Re-solve IEF/TLH with `D1_wins_rung.py` on the trade date.
 - 2. Otherwise, it lists a Treasury maturing Jul-Dec of any year 2033-2041. The 15-Nov-2039 and 15-Nov-2040 bonds on Stock-Trak's old list are the likeliest.
 - Buy the same size, funded from IEF and TLH as the script says.
 - Bonus: its ~10-year duration eases the TLH position-limit problem.
 - 3. Otherwise, **IBTM** is listed as an ETF: buy ~\$37k under (iii) or ~\$25k under (ii). Alternate: IBTO (Dec 2033), which funds the 2034 payment.
2. Otherwise, **no rung**. Record "no suitable dated Treasury listed" in the decision log. Do not force it with a Feb/May bond far from a payment date.

Mechanics:

- The rung is bought **in the same session as the hedge**, so it adds one trade (\$10) and no extra re-weighting trades.
- If it is added later, the IEF trim is a genuine later decision and could itself be a note. Never buy and sell the same security on the same U.S. trading day.

Trading Note checklist for the rung (elements only; the team writes the words):

- which payment it stands for: 1 Jan 2033;
- its maturity date: 15 Nov 2032, 47 days before;
- its face: \$50,000, one payment (or "scaled like the rest of the book");
- that it is held to maturity, so later price moves do not change the \$50,000 it repays;

- that the other nine payments' rate sensitivity is carried by the ETFs (about 10 years together);
- the honest limit: it is a coupon bond; Laura's real ladder uses the zero-coupon STRIP cut from this same bond, because coupons would have to be reinvested;
- the rule: never sold because rates moved.

Never claim that the whole ladder is in WInS, that the ETFs "match" the payments, that the rung is "risk-free" in WInS value terms, or anything stronger than "certain in U.S. dollars barring a U.S. government default". Moody's cut the U.S. to Aa1 on 2025-05-16: SNIP (Wikipedia, read 2026-09-28; not checked on moodys.com).

Reflection (≤100 words) should contain: the difference between a dated cash flow and a duration stand-in; how that serves "a high degree of certainty" (case p.3 L91, VRF); and one thing the team learned, e.g. why coupons create reinvestment risk. No Laura quotes (D13 rule).

Add to the WInS gate checklist:

- screenshot the Bonds drop-down, list every U.S. Treasury maturing 2032-2042 (symbol, coupon, maturity);
- record the bond position limit;
- check whether the 2x-daily-volume rule shows a volume for bonds;
- record the accrued interest shown.

This upgrades ticket check 8 from "for the Final Report only" to "decides one order now".

- **Deadline tier:** 1. **Confidence:** medium. The logic is high-confidence; WInS availability is unknown.
- **Changes current strategy?** Yes, slightly. It adds one conditional order to the WInS ticket and a candidate third note in place of VGSH. It does not change the plan.
- **Criterion served most:** Portfolio Analysis (why each security belongs; quantitative + qualitative), then Investment Strategy.
- **What this teaches:** a fund can *behave* like a promise; only a bond with a date can *keep* one. Showing one real rung is simpler and more convincing than a longer explanation of durations.

Question M004 (tier 2: IPS Nov 6; one tier-1 by-product)

Question (canonical): If rates fall and the ladder costs more than \$300k on the January 2027 purchase date, should the IPS base rule be "buy as much as the first deposit affords, longest rungs first, and complete it from the 2028 deposit", with a stated shortfall limit and "fully funded by 1 January 2028 at the latest" as the headline? And should the purchase be made in full, in stages or on yield triggers, in a way Laura would recognise as hers?

One-sentence answer: Yes. Buy in full on arrival, longest-dated payments first. Any unbought part is always the earliest payment(s), completed as the first use of the 2028 deposit before any growth money. State the maximum dependence and the joint tail openly. Reject staging and triggers: they add a ~27% chance of being under-funded for +0.2 points of expected funding.

Evidence

Claim	Source (accessed)	Status
"All ten payments must be funded by the investment portfolio with a high degree of certainty." / "Teams may not rely on co-sponsors, grants, program fees, or other outside funding". The 2028 deposit is Laura's own contribution <i>to the portfolio</i> , so using it is not outside funding	case p.3 L91-92 (R-C47, R-C48); p.2 L43-45 (R-C26, R-C27)	VRF (reading of R-C48 is INTERPRETATION, low risk)
She "will contribute an additional \$150,000 at the beginning of 2028, using earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"; no other flows before 2033	case p.2 L43-45, L63-65 (R-C27, R-C34)	VRF
"A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move"	Competition Guide p.5 (2026_WGY_Investment_Competition_Guide.txt)	VRF
IPS "Focus on the strategy and decision-making framework ... rather than ... detailed financial calculations"; no citations or charts	2026_WGY_Investment_Policy-FINAL.txt L51; brief \$5	VRF

Claim	Source (accessed)	Status
Ladder cost above \$300k on 173 of 185 trading days of 2026 (exact-date); 176/185 for the real Nov-15 ladder	D1_purchase_rule.py [5] on scripts/data/D3/treasury_par_2026_raw.csv (treasury.gov snapshot; its 9/25 row equals the repo file)	VP inputs + DERIVED
Case PDF created 2026-09-10 15:19	phase_A/case_register.md §1.1	VRF
Practice anchor: Australia's defined-benefit standard sets a shortfall limit ("can be restored to a satisfactory financial position within one year") and a restoration period that "must not exceed three years"	APRA SPS 160, https://handbook.apra.gov.au/standard/sps-160 (both phrases verified verbatim 2026-09-28)	VP. Team knowledge only: the IPS bans citations, and the Final Report is out of scope for now
Laura's "I give myself a deadline for when I must post the art, finished or not." (Geeks OUT, 2022-05-11, about posting art) and "self-employed, with unstable income" (Overachiever, 2021-03-15)	phase_D/D13a_laura_quotes_verified.md rows B8b-Q31, B8b-Q18	VP. Not for TN/IPS (D13 rule); there is no verified Laura statement about investing, so any "hers" link is INTERPRETATION

Numbers (D1_purchase_rule.py; curve VRF/VP; method ASM, identical to the verified script)

[1]-[2] Cost at 2027-01-01 and the longest-first split (real Nov-15 STRIPS ladder; exact-date zeros in brackets)

Parallel fall before Jan 2027	Ladder cost	Gap	What waits	2028 \$ needed	% of \$150k
0 (today)	\$294,387 (\$292,264)	none (-\$5,613)	nothing	\$0	0%
-25bp	\$301,676 (\$299,596)	\$1,676 (none)	4% of the 2033 payment	\$1,751	1.2%
-50bp	\$309,169 (\$307,135)	\$9,169	24% of the 2033 payment	\$9,555	6.4%
-100bp	\$324,793 (\$322,864)	\$24,793	63% of the 2033 payment	\$25,711	17.1%
-150bp	\$341,313	\$41,313	all of 2033 + 2% of 2034	\$42,633	28.4%
2026 worst day (Feb 27)	\$327,631 (\$325,563)	\$27,631	~71% of the 2033 payment	\$28,549	19.0%

- **Break-evens:**

- a gap opens below -19bp (Nov-15 ladder) or -26bp (exact-date);
- the gap exceeds one whole payment below about -150bp;
- the gap exceeds the whole 2028 deposit below -421bp (10-year about 0.96%).

- **Gap odds on 2027-01-01** (ASM: zero-drift lognormal, 2026 realised volatility 7.16-7.24% a year, 0.27 years):

- $P(\text{gap}) = 24.3\%$ (exact-date) or 30.5% (Nov-15 ladder);
- p90 gap \$8.7k, p95 \$12.9k, p99 \$20.9k (Nov-15);
- $P(\text{gap} > \text{one whole payment}) = 0.01\%$.

- **[3] Longest-first versus nearest-first:** after a -100bp gap, a *further* -100bp during 2027 raises the Jan-2028 top-up by +\$1,265 (+5%) under longest-first, against +\$3,831 (+15%) under nearest-first.

- **[7] All at once versus staged versus trigger** (ASM: cost relative to T-bills is zero-drift; starting funded ratio 1.019):

Rule	Mean funded ratio	Chance under 1.00	p5
Buy all on the day	1.019	0% (once bought)	1.019
Four quarterly tranches	1.021	27.3%	0.966
Wait for a 5% margin, else buy at year-end	1.022	26.5%	0.912

The "all at once" row takes the purchase-day curve as given; the uncertainty *before* the purchase day is the gap odds above, and every rule shares it.

- **[8] Joint tail:**

- -50bp and no deposit: \$11,957 of the first \$50,000 payment unfunded (2.4% of \$500k);

- -100bp and no deposit: \$31,413 (6.3%);
- a \$75k deposit covers every case down to the 2026 worst;
- six months late (1 Jul 2028), with rates unchanged, the -100bp case needs +\$531.
- **[5] Case PDF date (2026-09-10):** the Nov-15 ladder cost **\$299,990** and the exact-date ladder \$297,879. See finding 7: INTERPRETATION only.

Implication (decision + one IPS rule; IPS, with one TN by-product)

IPS: rules to fix before Nov 6. These are the elements the team's own wording must contain; this is not drafted text.

1. **Buy on arrival, in full.** The first deposit buys the ladder on 1 January 2027. There is no staging and no waiting for better yields. That would be a rate forecast, and the Guide says strategy is not rewritten "because markets move".
 2. **Order: longest-dated payments first.** Reason in plain words: the piece left waiting is the smallest and least sensitive to rates, and it is the one needed soonest after the 2028 deposit arrives.
 3. **Completion before growth.** Any unbought part is the first use of the 2028 deposit, before any growth investment. If the deposit is late or arrives in instalments, completion comes first from each instalment.
 4. **Headline promise, rate-conditional:**
 - fully funded on 1 January 2027 if rates stay near today's;
 - otherwise fully funded on the day the second deposit arrives (1 January 2028 at the latest);
 - "fully funded" means the Treasuries are *bought*, not forecast.
 5. **Stated maximum dependence:** in words, not a table (the IPS bans detailed calculations). Content: "at most part of the first payment, and about one-fifth of the 2028 deposit even on 2026's most expensive day".
 6. **Joint tail, named once:** if rates fall sharply before January *and* the second deposit never comes, part of the first payment would be unfunded; every other payment is bought. This is the plan's known residual risk, and our own stress scenario (the case says "will").
 7. **If rates rise instead:** the few thousand dollars left over wait in Treasury bills until 2028, then join the growth money. This is already in the ticket (\$8).
- **For the elevator pitch** (≤50 words; the team's words): the one fact worth carrying is "the payments are bought first, with her own two deposits". Do **not** put "fits inside \$300,000" in the pitch: it was false on 176 of 185 days this year.
 - **TN by-product (tier 1, optional):** if the team writes a reflection on the hedge trade, it may say the hedge was sized to what the first deposit buys at today's prices, with the top-up rule stated. No probabilities in a note.
 - **Deadline tier: 2 (IPS). Confidence:** high on the direction (all at once, longest-first, completion first); medium on the odds (model ASMs).
 - **Changes current strategy?** Refines it without reversing it (brief §7 already says longest-first plus a 2028 top-up). Three changes:
 - the headline stops saying "fits inside \$300k";
 - "completion before growth" becomes an explicit rule;
 - staging and triggers are explicitly rejected in the IPS.
 - **Criterion served most:** Investment Strategy (a decision rule set before the surprise). Also Client Knowledge: Laura's 2028 income is her own career money, and she called self-employment "unstable income" (VP, but team knowledge only).
 - **Is it "hers"?** INTERPRETATION, with low risk if kept internal. A person who sets deadlines "finished or not" is well served by a rule that acts on the day the money arrives, not one that waits for a better price. Do not present this as her investment view (D13: no verified statement about investing exists).
 - **What this teaches:** when a promise is priced by today's market, "can we afford it?" has a date attached. Write the rule for the unlucky day before it happens. Leave the part most exposed to rates as small and as short as possible, and never wait for a better price on money that already has a job.

Already answered / mis-posed notes

- M004's member hypotheses asked for "FR sentence" and "re-run the surplus at the 2026-average curve" items. Those are Final-Report or D3-model work and are outside this scope (brief §17). Logged below.
- M044's "which maturities does WInS offer" cannot be answered from public sources. It is correctly a logged-in check (R-AN48), now with a symbol format and a fallback order.

Later (after Nov 9), one line each

- FR: say that part of today's surplus reflects unusually high September 2026 yields; D3 to re-run the surplus on the 2026-average curve.
- FR: joint-tail sensitivity line and the break-even rate falls (\$ table above).
- FR: optional practice analogy (APRA SPS 160 shortfall limit / restoration period) if it earns its place.

Sources

- Case and guides (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (p.2 L43-65, p.3 L88-92, p.4 L129); `2026_WGY_Investment_Competition_Guide.txt` p.5; `2026_WGY_Investment_Policy-FINAL.txt` L51.
- SMAApply Trading Details and FAQ (VP, 2026-09-27, via case_register R-W65-R-W92): <https://wghsinvcomp.smapply.us/res/p/trading/> ; <https://wghsinvcomp.smapply.us/res/p/faqs/>
- 2026-27 WInS User Guide (VP, re-read 2026-09-28): <https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf>
- Stock-Trak bond pages (VP content, generic, 2019; read 2026-09-28): <https://www.stocktrak.com/available-bonds/> ; <https://content.stocktrak.com/what-are-bonds/> ; <https://www.stocktrak.com/investments-bond-analysis-trading-project/>
- iShares IBTM (VP, read 2026-09-28): <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf>
- Treasury par curves 2026 (VP snapshot by D3): `research/insight_v1/scripts/data/D3/treasury_par_2026_raw.csv` ; `competition/official_market_data/daily-treasury-rates_2026-09.csv` (VRF).
- MSPD Table V (VP per S1): `research/insight_v1/wins_now/S1_mspd_table5_2026-08-31_fixed_2032plus.csv`.
- APRA SPS 160 (VP, 2026-09-28): <https://handbook.apra.gov.au/standard/sps-160>
- U.S. rating (SNIP, secondary, 2026-09-28): https://en.wikipedia.org/wiki/United_States_federal_government_credit-rating_downgrades
- Repo research (VRF): `wins_now/securities_and_allocation_v0.md`, `S1_treasury_sleeve.md`, `S3_final_report_ladder_and_reserve.md`, `S4_red_team.md`, `phase_A/fact_register.md` (F-101-F-117, F-206, F-605-F-608), `phase_A/wins_week1_guardrails.md`, `phase_D/D13a_laura_quotes_verified.md`.

What this teaches

1. **Behave versus keep.** Bond funds that roll forever can *move like* Laura's payments. Only a bond with a date can *pay* one. One visible rung explains the plan faster than a paragraph about duration.
2. **A promise priced by the market has a date on it.** \$300,000 bought the real Nov-15 ladder on only 9 of 185 trading days this year (12 for exact-date zeros). The honest plan says what happens on the other days, and makes the part that has to wait as small and as short as possible.
3. **Waiting is a forecast.** Buying in stages or on a trigger feels careful, but with a fixed promise it only adds a chance of falling short, for almost no expected gain.

Audit corrections (AX1b)

Auditor AX1b (rates cluster), 2026-09-28. The D1 text above is unchanged. Full audit: `research/insight_v1/phase_D/audit_rates_D1.md`. New check script: `.venv/bin/python research/insight_v1/scripts/AX1b_rates_audit.py`. **Reproduced:** both D1 scripts re-run and every number quoted above matches the output, except items 12-14 below. The 2026 curve snapshot is byte-identical to a fresh treasury.gov download (2026-09-28). Every quote was re-checked with the fetch helper and found verbatim. **Verdict: sound direction, pass with corrections.** Buy at once, longest-dated payments first, finish the ladder before any

growth money, and the conditional one-rung WinS order all survive. Nothing is blocking. Items 1-9 must be fixed before this reaches the strategy team.

Important

1. **The gap odds are understated for the real ladder** (summary 1, M004 [6]). 24.3% belongs to the idealised exact-date ladder. The real Nov-15 ladder has 30.5% on D1's inputs, and 30-35% across reasonable inputs:
 - 2026 volatility windows of 7.2-7.8%;
 - an "unchanged curve" drift, which adds ~\$1.0k to the January cost (+3.4 points).
 Model-free check: the 10-year yield fell by 19bp or more over 68 trading days in 32-36% of windows (FRED DGS10 since 1962 and since 1990; DERIVED, audit script [B]-[C]). Say "about 1 in 3" for the real ladder. Use "1 in 4" only for the exact-date ideal.
2. **"P(gap > one whole payment) = 0.01%" and rule 5's "at most part of the first payment" are thin-tail overclaims.**
 - Since 1990 the 10-year fell by 148bp or more over 68 trading days in 0.1% of windows (all in late 2008). Since 1962 it was 1.5%.
 - It fell by 100bp or more in 2.1% of windows since 1990 (the model says 0.4%), including 2008 and 2020.
 - A repeat of the worst 3-month fall since 1990 (-167bp, Sep-Dec 2008) leaves all of the 2033 payment and ~16% of 2034 waiting, which is \$48.6k (32%) of the 2028 deposit (DERIVED).
 - Fix: drop "0.01%" and "at most". State the 2026-worst figure as one day's figure, not a ceiling. Note that a \$75k deposit still covers the 2008-size case.
3. **"Fully funded ... 1 January 2028 at the latest" (M004 rule 4) overclaims.** It contradicts D1's own rule 3 (a late deposit or one paid in instalments) and the joint tail. Use "when the second deposit arrives (the case dates it to the start of 2028)".
4. **Guide p.5 is misapplied** (summary 2, rule 1). This is the same error AX2 found in D2 (C3). "Should not be rewritten simply because markets move" (Guide L156-157) sits in the post-IPS evaluation stage. A staging or trigger rule written into the IPS in advance would not breach it. The case against staging rests only on the numbers, so remove "matches Guide p.5" and "the Guide says".
5. **The 2019 Stock-Trak list says nothing about 2032-2035 maturities** (summary 6, M044 one-sentence answer).
 - No nominal Treasury maturing between Feb 2031 and Feb 2036 existed when that list was built. The 30-year bond was suspended from Feb 2002 to Feb 2006 (Wikipedia, SNIPPET-UNVERIFIED), and the list itself jumps from 15-Feb-2031 to 15-Feb-2036.
 - All 29 securities maturing 2032-2035 in S1's MSPD file are notes (VRF). They were issued from 2022 on (INTERPRETATION from the coupon and maturity pattern).
 - Nothing on the list was issued after Feb 2014, so the list is stale.
 - Fix: delete "suggests the 2032-2035 notes may be missing". The logged-in drop-down check stays tier 1.
6. **The IBTM fallback makes one TN checklist line false.** "Held to maturity, so later price moves do not change the \$50,000 it repays" does not hold for IBTM. iShares: the funds "do not seek to return any predetermined amount" and move to cash near termination (VP, [ishares.com/us/products/328944](https://www.ishares.com/us/products/328944), read 2026-09-28). If tier 3 is used, the note may say only "a fund that ends in December 2032". Notes are permanent.
7. **Third-note conflict with D12 and D10.** D1 makes the rung a "strong candidate to replace the VGSH buy". D12 gives VGSH the only "risk management" role word, so the three notes carry three distinct roles, and D10 lists note variety as a TN factor. Under (ii) the rung would be a second "future funding" note. Under (iii) there is no VGSH buy, so the rung is the natural extra note. Make the recommendation conditional on (ii) or (iii). The team decides.
8. **Rung rationale.** "The reason is the evidence it gives" reads as a trade made to show something. The ticket forbids that ("Do not create a decision just to get a note") and R-W88 discourages turnover. State the reason as implementation: the promise is dated, so where WinS lists a dated Treasury near a payment date, the book holds one, as the real ladder does. The evidence it gives is a by-product.

9. **The rung under a position cap below 44% is not computed.** `D1_wins_rung.py` solves only the no-cap case, but the ticket's gate (e) needs the branch picked in advance. A 2032 rung raises the long-fund need (TLH 68.3% under (iii)). Rule: if the cap is under 44%, buy the rung only if it is the 2039/2040 bond and the re-solve needs no new fund. Otherwise skip it. Re-run before the session.

Minor

10. The Stock-Trak page lists **72** Treasuries, not 32. Only 32 of them are still outstanding on 2026-09-28 (VP, re-read). Two symbols are mistyped: B-T-4.375-15052014 is the 15-May-2041 bond and B-T-5.25-15022028 is the 15-Nov-2028 bond. Search the drop-down by maturity, not by a built-up symbol. The linked page also says "There are no volume limit rules on Bonds" and "You can only use market orders" (VP; generic, UNVERIFIED for WInS). Add both to the gate check. Its "We only have US bonds" contradicts SMaply's six-country list, which confirms the page is out of date.
11. **Staging "+0.2 points"** comes from the zero-drift assumption (Jensen's inequality). Suppose instead the curve stays where it is and the waiting cash earns bill rates. Then staging *lowers* expected funding: tranches -0.21 points, trigger -0.35. P(under 1.00) rises to 34-35%, with p5 0.959/0.897 (audit [D]). Say "no reliable gain (+0.2 to -0.5 points) and a 27-35% chance of needing the 2028 deposit". The conclusion gets stronger.
12. Worst 2026 day: **68.7%** of the 2033 payment waits, not ~71%. The 2033 rung costs \$40,210 on the Feb-27 curve (audit [A]).
13. Coupon vs zero should compare prices on the same date. The STRIP 912821KC8 is ~\$36.8k on 2026-09-25 against the note at ~\$48.4k. \$37,253 is its 1 Jan 2027 forward value (audit [F]).
14. Accrued interest on \$50k face is **\$768-824** for settlement from 29 Sep to 9 Oct 2026, not "\$750-800" (audit [G]). WInS shows its own figure.
15. **The -421bp break-even is false precision.** At that shift the 1-month yield would be -0.17%. Say "a fall of more than 4 percentage points, the size of the largest 3-month fall on record (-410bp, Jul-Oct 1982)".
16. "Re-solve with `D1_wins_rung.py` on the trade date" does nothing as written. The script's curve and durations are fixed at 2026-09-25 (via `S1_hedge_weights.py`). Either update S1's inputs first, or use the hand formula with issuer durations on the trade date: IEF share of hedge = $((1-r) \cdot D_TLH + r \cdot D_rung - D_target) / (D_TLH - D_IEF)$.
17. **"The IPS bans detailed calculations" overstates.** IPS guide L51 says "Focus on ... rather than ... presenting detailed financial calculations". The explicit ban (L123) covers graphics, charts, images, attachments, links, footnotes and formal citations. "In words" is still the right advice.
18. "No probabilities in a note" is D1's own judgement, not an official rule. Label it as such.
19. **Near-final wording.** Summary 1's quoted headline, rule 5's quoted "content" and the pitch phrase are close to paste-ready. Turn them into lists of required elements. "Bought first, with her own two deposits" is also wrong in the ~2 in 3 paths where the first deposit alone buys the ladder. The element that holds in every path is "payments bought before any money takes stock-market risk".
20. **"Is it hers?"** The "finished or not" line is about posting art (D13 context rule). Drop the link to it; the rule stands on its numbers.
21. **Finding 7 (PDF creation date) is unearned.** The 19bp headroom and 176/185 days already show the fit is knife-edge. Remove it from the top-findings list; one line in the body at most.
22. **Wording.** "Today the real ladder costs \$294,387" should read "costs \$294,387 for purchase on 1 Jan 2027, priced from the 25 Sep curve". "What this teaches" #2 says \$300k "bought" the ladder on 9 of 185 days; say it "would have covered the January cost implied by that day's curve".
23. **Addition (supports D1): longest-first also leaves less of the promise unfunded if the deposit never comes.**
 - At -100bp: \$31.4k (2033) versus \$47.7k (2042) under nearest-first. At -50bp: \$12.0k versus \$19.0k (audit [E]).
 - Counterpoint for the team to weigh: under longest-first, any shortfall falls on the residency's first operating year.
24. **Cross-link (supports D1 under option (iii)).** AX2 flagged as blocking (D10 C1) that under option (iii) with no cap there are only two orders, while Oct 23 needs three executed trades. If WInS lists a suitable bond, the rung is a genuine third trade with a strategy reason, instead of or alongside AX2's BIL/SGOV leftover trade. This fits item 7: under (iii) the rung matters most.

D2: Equity & AI-concentration analyst: is sleeve equity paid enough in 2026? (question M030)

Agent D2, insight_v1 run, written 2026-09-27/28. AI-generated research for Team Caplet (brainstorming under the Wharton AI policy; record its use in Works Cited). None of this is text to submit: it gives numbers, evidence, a decision option and "what a strong sentence must contain" lists. The team decides and writes every word. The team has not yet approved the lock-early strategy. Scope follows brief section 17: WInS now, Trading Notes (Oct 23) and IPS (Nov 6) only. Final Report items appear as one-line "later (after Nov 9)" entries.

Script: `.venv/bin/python research/insight_v1/scripts/D2_equity_premium.py` (run from the repo root; the docstring lists every input with its status label).

Terms used:

- **Growth sleeve ("sleeve")**: the money the ten \$50,000 payments do not need (about \$7.7k in 2027 plus the \$150k deposit in 2028). It pays for the facility contribution and flexibility.
- **Equity premium**: how much more a year stocks are expected to earn than a Treasury held for the same period.
- **Compound (geometric) return**: the steady yearly rate that turns the start value into the end value. It is the number that drives the median outcome.
- **Hurdle rate**: what the same money earns with no stock risk, here a Treasury locked today for the sleeve's dates.
- **p5 / p50 / p95**: the outcome that 5% / 50% / 95% of simulated paths fall below. p50 is the median.
- **Forward P/E**: share price divided by analysts' expected earnings for the next 12 months. **CAPE** (Shiller P/E): price divided by the average of the last 10 years' inflation-adjusted earnings. An **earnings yield** is $1/P/E$.

Summary: top findings (ranked by impact on reaching the semifinals and on making the plan Laura's)

1. **The hurdle is known; the equity premium is not.** Money in the sleeve can lock about **5.2% a year** in Treasuries for its own dates (2028->2031: 5.20%; 2028->2033: 5.23%; forwards on the official 2026-09-25 curve, VERIFIED-REPO-FILE inputs, ASSUMPTION method). Against that, the expected extra return from stocks is **+1.5 to +1.8 points a year under J.P. Morgan** (U.S. large cap 6.70%, AC World 7.00%, data 2025-09-30), **about 0 under Vanguard's midpoint** (U.S. 4.2%-6.2%, a VT-like blend ~5.1%, June 30, 2026 run, VERIFIED-PRIMARY), and **about 0 on the forward earnings yield** ($1/19.2 = 5.21\%$, FactSet 25 Sep 2026, VERIFIED-PRIMARY). So across two houses and one yardstick the premium is **roughly -0.2 to +1.8 points a year; call it 0-2**. That confirms the Phase C guess.
2. **Moving sleeve equity from 60% to 50% costs almost nothing in the middle and protects the bottom.** Under all three return cases, 50% instead of 60% changes the 2033 median facility money by **-\$1.5k to +\$0.4k** (on ~\$200-210k), raises p5 by **\$6-7.5k**, raises the p5 of the 2031 bought floor by **\$5-6k**, and gives up **\$11-14k at p95** (model, ASSUMPTION-based). Going down to 40% repeats the pattern (median -\$0.6k to -\$3.2k vs 60%, p5 +\$12-15k). Recommendation (for team decision): **set the sleeve at 50% world stocks / 50% short Treasuries** and say why in one IPS sentence. Keep 60% only if the team explicitly values the p95 upside more than the floor it promises in 2031.
3. **Tier-1 consequence: decide the split before the first VT/VGSH order, not after.** At 50/50 the WInS option (ii) orders become **VT 17.0% (\$51,000, ~318 sh) / VGSH 16.0% (\$48,000, ~833 sh)** instead of VT 20.5% / VGSH 12.5% (PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK). Setting it once, up front, avoids a later "we cut stocks because of a forecast" trade that would contradict the Guide's warning against rewriting strategy "simply because markets move" (Guide p.5, VERIFIED-REPO-FILE). The hedge (66%) is unchanged.
4. **"Thoughtful risk" = risk that is paid.** This is the Laura link. The case says she "has been willing to take thoughtful risks" but wants "an appropriate balance between pursuing growth and protecting the

capital required for her goals" (case lines 70-73, VERIFIED-REPO-FILE). A plan that checks whether stock risk is paid before taking it, and takes it with exactly half of the free money, is a concrete reading of "thoughtful". It must be paired with the brief's goal-by-goal framing (promise: no market risk; free money: half in the world stock market), because a blended whole-portfolio share (17% in 2028 at 50/50) reads as timid on its own.

5. **Don't argue CAPE vs forward P/E: show why they disagree, then decide by robustness.** CAPE is 41.48, second only to Dec 1999's 44.19 (multpl.com, SECONDARY-READ); the forward P/E is 19.2, between its 10-year (19.0) and 5-year (19.8) averages (FactSet, VERIFIED-PRIMARY). They disagree because earnings are surging: CY2026 S&P 500 earnings +32.0%, with semiconductors +126% in Q3 and Q2 2027 growth expected at only +1.7% (FactSet 25 Sep 2026, VERIFIED-PRIMARY). Forward P/E looks normal only if AI-driven profits persist; CAPE looks extreme because it averages ten years of lower profits. Neither settles the premium, so the plan should not depend on either being right.
6. **What the broad fund really owns (AI in dollars).** At 50/50 the 2028 sleeve holds ~\$79k of VT; VT's top ten (21.7%, all AI-linked, a lower bound) is ~\$17k, **about 3.7% of Laura's whole portfolio**. The same money in an S&P 500 fund would carry ~\$32k in its 12-name AI basket (40.7%). So Laura's AI exposure is real but small, and a world fund halves it without any tilt (S2 figures, VERIFIED-PRIMARY issuer data; arithmetic ASSUMPTION inputs).
7. **Model hygiene before the IPS numbers are quoted anywhere.** `strategy_mc.py` gives sleeve bonds 4.00% (JPM, set when the 10-year was 4.16%, F-317), about 1 point below what short Treasuries pay today. That understates the no-stock alternative and overstates the reward for equity. The D2 script uses 4.8% (ASSUMPTION, anchored to VGSH's 4.55% SEC yield and 4.8-5.1% one-year forwards). With market-consistent bonds, even JPM's case gives 60% equity only +\$1.5k median over 50%.

M030 (tier 1 for the split decision; tier 2 for the IPS rule and sentence)

Canonical question: How many extra points a year does sleeve equity really expect over simply buying Treasuries in 2026 (JPM 6.7% vs Vanguard 4.2-6.2%; CAPE ~41 vs forward P/E 19.1; ~5.0% on a 6-year Treasury), and does that move the sleeve equity weight from 60% toward 40-50%?

Anchor: R-C37/R-C38 (case lines 69-74, "thoughtful risks", "appropriate balance"), R-C77 and Guide line 73 ("use reasonable return assumptions consistent with your strategy"), R-S26 (Portfolio Analysis), F-004, F-008, F-012, F-301, F-315, F-317, F-405, F-407. IPS guide line 46: "Explain how your team will approach risk and return" (VERIFIED-REPO-FILE).

Is it already answered? Partly. Brief section 8 item 5 ("more sleeve equity buys range, not median") rested on JPM alone, with sleeve bonds at a stale 4.00%. New here: a second house (Vanguard) and a market-consistent hurdle, which remove most of the median gain and turn the weight into a pure choice of spread. The question is well posed; one correction: the right hurdle is not "a 6-year Treasury today" (5.10% spot) but the forward rate for the sleeve's own dates, 2028->2031/2033 (5.20-5.23%), because the sleeve money only arrives in 2028.

One-sentence answer

Stocks are expected to beat a Treasury locked for the sleeve's own dates by only about 0-2 points a year (JPM +1.5 to +1.8, Vanguard and the forward earnings yield about 0), so cutting sleeve equity from 60% to 50% leaves the median facility money within about \$1.5k while raising the bad-case (p5) outcome by \$6-7.5k and the 2031 floor's p5 by \$5-6k: 50% is the better "thoughtful risk" setting, decided once, now.

Evidence

#	Claim	Source (access date)	Status
E1	Treasury par curve 2026-09-25: 1y 4.50, 2y 4.81, 3y 4.94, 5y 4.98, 7y 5.06, 10y 5.17	competition/official_market_data/daily-treasury-rates_2026-09.csv	VERIFIED-REPO-FILE
E2	Forward rates locked today: 2027->2033 5.14%, 2028->2031 5.20%, 2028->2033 5.23%, 6-year spot 5.10% (annual compounding)	D2 script section 1 (bootstrap method from official_curve_pv.py)	VERIFIED-REPO-FILE inputs; ASSUMPTION method

#	Claim	Source (access date)	Status
E3	JPM 2026 LTCMA: U.S. large cap 6.70% compound / 7.94% arith / 16.47% vol; AC World 7.00 / 8.28 / 16.78; AC World vs intermediate Treasuries correlation 0.00; data 2025-09-30	competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf p.2 (extracted 2026-09-27)	VERIFIED-REPO-FILE
E4	Vanguard, July 22, 2026: "our 10-year expected annualized return for U.S. equities declined from a range of 4.9%-6.9% to a range of 4.2%-6.2%"; developed ex-U.S. "4.5%-6.5%"; emerging markets "2%-4%"; forecasts are geometric; June 30, 2026 VCMM run	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html (read 2026-09-27)	VERIFIED-PRIMARY
E5	Vanguard on the same page: "valuations tend to be poor predictors of performance over the short or even intermediate term and should not serve as a primary reason for changing portfolio allocations"	same page	VERIFIED-PRIMARY
E6	10-year Treasury 4.44% on 2026-06-30 (Vanguard's run date) vs 5.17-5.18% now: bonds got ~0.7 points more attractive after Vanguard's run	FRED DGS10 CSV (read 2026-09-27); E1	VERIFIED-PRIMARY
E7	S&P 500 forward 12-month P/E 19.2 (5-yr avg 19.8, 10-yr avg 19.0; 20.4 on June 30); price +2.7% and forward EPS +8.9% since June 30; trailing P/E 25.8	FactSet Earnings Insight, 25 Sep 2026, https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_092526.pdf (read 2026-09-27)	VERIFIED-PRIMARY
E8	CY2026 earnings growth 32.0%; CY2027 15.4%; Q2 2027 1.7%; Q3 2026 IT sector +63.5%, semiconductors +126%, IT ex-semis +24.2%	same FactSet issue	VERIFIED-PRIMARY
E9	Shiller CAPE 41.48 (Sep 25, 2026); mean 17.42; max 44.19 (Dec 1999)	https://www.multpl.com/shiller-pe (read 2026-09-27)	SECONDARY-READ (aggregator; Shiller's own file not read)
E10	10-year TIPS real yield 2.83%; 10-year breakeven 2.34%	F-008, F-009 (phase_A/fact_register.md)	VERIFIED-PRIMARY
E11	VT top-10 21.7% (all AI-linked names, lower bound); S&P 500 top-10 38.8%, 12-name AI basket 40.7%; VT 37.7% non-U.S.	wins_now/S2_growth_sleeve.md sections 1-2; S3 ticket section 1 (issuer pages)	VERIFIED-PRIMARY (via S2/S3)
E12	VGSH SEC yield 4.55% [9/24]; VT \$160.03, VGSH \$57.59 closes [9/25]	wins_now/securities_and_allocation_v0.md section 1	VERIFIED-PRIMARY (via S3)
E13	Case: "willing to take thoughtful risks" ... "an appropriate balance between pursuing growth and protecting the capital required for her goals"	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt lines 69-74	VERIFIED-REPO-FILE
E14	Guide p.5: a long-term strategy "should not be rewritten simply because markets move or hindsight reveals a different outcome"	competition/official/2026_27/2026_WGY_Investment_Competition_Guide.txt line 156-157	VERIFIED-REPO-FILE

Numbers (D2 script; model outputs are ASSUMPTION-based)

Expected premium over the 2028->2033 Treasury hurdle (5.23%)

Source	Compound equity return	Premium, points a year
JPM AC World (VT-like)	7.00%	+1.77
JPM U.S. large cap	6.70%	+1.47
Vanguard U.S. low / mid / high	4.2 / 5.2 / 6.2%	-1.03 / -0.03 / +0.97
Vanguard VT-like blend low / mid / high (62.3% U.S.; non-U.S. 75% developed, 25% EM: ASSUMPTION)	4.08 / 5.08 / 6.08%	-1.15 / -0.15 / +0.85
Yardstick: forward earnings yield 1/19.2	5.21%	-0.02
Yardstick: CAPE earnings yield + breakeven (crude)	4.75%	-0.48

The yardsticks are not forecasts; they only show that valuation measures do not point to a large premium either.

2033 facility money and 2031 bought floor by sleeve equity weight (200,000 paths, seed 20260927; plan as in brief section 7; sleeve bonds 4.8%/2% vol; equity vol 16.78%; payments are funded by the ladder in every row)

Sleeve equity	JPM 7.00%: p5 / p50 / p95	Vanguard blend 5.08%: p5 / p50 / p95	Vanguard U.S. low 4.2%: p5 / p50 / p95	2031 floor p5 (JPM / Vg mid)
0%	\$189k / \$200k / \$211k	same	same	\$151k / \$151k
40%	\$173k / \$208k / \$252k	\$169k / \$202k / \$246k	\$167k / \$200k / \$243k	\$138k / \$135k
50%	\$167k / \$209k / \$265k	\$162k / \$203k / \$258k	\$160k / \$200k / \$254k	\$133k / \$130k
60% (current)	\$161k / \$211k / \$279k	\$155k / \$203k / \$270k	\$152k / \$200k / \$266k	\$129k / \$124k
70%	\$155k / \$212k / \$294k	\$148k / \$203k / \$282k	\$145k / \$199k / \$277k	\$124k / \$119k

Moving 50% -> 60%: median +\$1.5k (JPM), +\$0.2k (Vanguard mid), -\$0.4k (Vanguard low); p5 -\$6.2k / -\$7.1k / -\$7.5k; p95 +\$11-14k; 2031 floor p5 -\$4.9k to -\$5.9k. **Crash stress** (stocks -35% in 2030, the year before the floor is bought; JPM otherwise): floor median \$133k at 50% vs \$127k at 60%. Under JPM, 37% of paths at any weight from 40% to 70% end below the all-Treasury median (\$200k); under Vanguard, 46-47%. (That comparison is D3's M034; noted only.)

Whole-portfolio equity in 2028: 40% sleeve -> 13.7%; 50% -> 17.1%; 60% -> 20.5% (ladder ~\$305k + sleeve ~\$158k, S2 arithmetic, ASSUMPTION).

Implications

Deliverable	Type	Implication
WlnS now (tier 1)	decision + number	Before the first growth orders, the team picks the sleeve split once. D2 option: 50/50 -> option (ii) orders VT 17.0% (~\$51,000, ~318 sh) / VGSH 16.0% (~\$48,000, ~833 sh) , cash 1%, hedge 66% unchanged. Rebalancing band inside the sleeve VT 45-55% (replaces 55-65). Option (iii) (literal 2027 book) is unaffected. Securities: VT, alternates VTI+VXUS (62/38) or ITOT+IXUS; VGSH, alternates SHY or a WlnS-listed ~2-year Treasury note. PENDING WlnS AVAILABILITY + POSITION-LIMIT CHECK for every ticker. VT, VTI, VXUS, VGSH, SHY were on the 2025-26 list (historical only).
Trading Notes (tier 1)	sentence elements	If the VT or VGSH buy becomes a note, its reflection can carry the reason for the split. Elements only: (a) the growth money is what the payments do not need; (b) Treasuries for the same dates pay about 5%, known today; (c) two major forecasters disagree whether stocks will beat that (one says ~1.5-2 points more, one says about the same); (d) so half goes to world stocks for upside and half stays in short Treasuries for the amount Laura may promise co-sponsors; (e) the risk accepted: a fall shrinks the facility range, never the payments. Never say stocks are "overvalued" or predict a crash; never quote CAPE in a note (a note is ≤100 words and must be consistent with the IPS).
IPS (tier 2)	sentence + rule	"Rules to fix before Nov 6": (1) sleeve = 50% broad world stocks / 50% short Treasuries, band 45-55, rebalanced once a year (WlnS: weekly check); (2) the weight is set once on today's known Treasury yields and does not change with market forecasts or valuation readings afterwards (Guide p.5; Vanguard's own warning, E5); (3) returns are stated as a range across two published houses, not one point. What a strong IPS risk/return sentence must contain: the known Treasury rate for her dates (~5%); that expected extra reward for stocks is small this year (0-2 points); that stock risk is taken only with money the promise does not need; that this is how the firm reads "thoughtful risk". No citations or house names are needed in the IPS (formal citations are banned); "two major published forecasts" is enough.
Later (after Nov 9)	one-liners	FR: two-house projection table with returns as a range. FR: model sleeve bonds at market-consistent ~4.8-5.2%, not 4.00%. FR: one valuation sentence explaining CAPE vs forward P/E (earnings surge). FR: re-check whether JPM's 2027 LTCMA has appeared (F-315) and freeze one dated edition.

Deadline tier: 1 for the WlnS split (orders are imminent; the ticket's 60/40 is marked provisional); 2 for the IPS rule and sentence.

Confidence: medium. High that the premium is small and uncertain (two primary houses plus a primary yardstick agree on "small"); medium on 50% specifically, because the model is lognormal, i.i.d., fee-free and uses one volatility for every house; 40% is equally defensible on the same numbers, and 60% is defensible only on JPM's view.

Changes the current strategy? Yes, modestly: sleeve equity 60% -> 50% (brief section 7 and the wins_now ticket's VT 20.5% / VGSH 12.5% -> 17.0% / 16.0%; band 55-65 -> 45-55). The lock-early structure, the hedge and the 2031 method do not change.

Criterion served most: Investment Strategy (a clear rule, set once, traced to her words); second, Portfolio Analysis (quantitative and qualitative reason for each holding's size).

What this teaches: Before taking a risk, ask what the safe choice pays: when a Treasury locks 5% for your dates, stocks have to beat 5%, not 0%, and this year nobody can promise they will by much.

Devil's advocate (and answers)

- *"Laura took leaps; 50% is timid."* Her verified words praise taking the jump (D13a VERIFIED-PRIMARY; read the D13a context note before any use, and no Laura quotes in the Trading Notes or IPS), but the case asks for "thoughtful" risk and "balance". At 50% the p95 is still \$254-265k; the whole portfolio is 17% stocks only because 66% is a promise she has already made. Show risk goal by goal, not blended.
- *"Vanguard's 4.2-6.2% is from June; stocks got cheaper since (forward P/E 20.4 -> 19.2)."* True (E7), but Treasury yields rose ~0.7 points in the same period (E6), so the premium over bonds moved against stocks, not for them (forward earnings yield +0.3 points, 10-year yield +0.7).
- *"This is market timing."* It is a one-time strategic setting based on a known number (today's Treasury yields), not a forecast of a fall, and it is frozen afterwards. Changing it later because stocks rallied or fell would be timing.
- *"JPM is newer in format and official in the repo; use it alone."* JPM's data date is 2025-09-30, when the 10-year was 4.16% (F-010): its bond numbers are ~1 point stale. A 2027 edition may appear in October (F-315). Relying on one house whose inputs are a year old is the weaker position.
- *"The difference is only ~\$6k at p5; why bother?"* Because it is free: the median barely moves. When a change costs nothing in the middle and helps the promise to co-sponsors, it earns its (zero) complexity. If the team disagrees, keeping 60/40 is not wrong; it is a choice of spread and should be written as one.

Sources (all accessed 2026-09-27 unless stated)

- Vanguard, "Vanguard Capital Markets Model forecasts", July 22, 2026 (June 30, 2026 run): <https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html> (VERIFIED-PRIMARY, read with `fetch_text.py`).
- FactSet Earnings Insight, 25 Sep 2026 (also 18 and 11 Sep issues checked): https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_092526.pdf (VERIFIED-PRIMARY).
- multpl.com Shiller PE: <https://www.multpl.com/shiller-pe> (SECONDARY-READ).
- FRED DGS10/DGS5/DGS7/DFII10: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10,DGS5,DGS7,DFII10> (VERIFIED-PRIMARY).
- Repo files (VERIFIED-REPO-FILE): `competition/official_market_data/daily-treasury-rates_2026-09.csv`; `competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf` p.2; `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` lines 69-74, 130-132; `competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt` lines 20, 46, 68; `competition/official/2026_27/2026_WGY_Investment_Competition_Guide.txt` lines 73, 156-157.
- Prior run files: `research/insight_v1/wins_now/S2_growth_sleeve.md`, `securities_and_allocation_v0.md`; `research/insight_v1/phase_A/fact_register.md` (F-004, F-008-F-012, F-301, F-315, F-317, F-405, F-407); `research/insight_v1/phase_B/B5b_macro_2026.md`, `B10a_contrarian.md`; `research/insight_v1/phase_C/survivors.json` (M030).
- Script: `research/insight_v1/scripts/D2_equity_premium.py`.

What this teaches

1. **Compare risk with the safe alternative, not with zero.** A stock return of 6-7% sounds generous until you notice that a Treasury for the same dates already pays 5.2%. The reward for risk is the gap, and this year it is small.
2. **When experts disagree, pick the choice that is fine under all of them.** JPM and Vanguard differ by about 2 points a year on stocks. At 50% equity, Laura's median barely moves whichever is right, and her bad case is better.
3. **Two valuation numbers can both be true.** CAPE (41) and forward P/E (19) disagree because profits jumped 32% in a year, led by AI chips. Knowing why they disagree is more useful than picking a side.

4. **Decide once, then stop.** Setting the stock share from today's known bond yields is planning; changing it every time markets move is guessing. The competition guide rewards the first and warns against the second.

Audit corrections (AX2)

Auditor AX2 (cluster auditor: equity, Taiwan/FX, compliance), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: `research/insight_v1/phase_D/audit_markets_rules.md`. Check script: `research/insight_v1/scripts/AX2_audit_checks.py`.

Reproduction. `D2_equity_premium.py` re-run 2026-09-28: every hurdle rate, premium, percentile, crash-stress figure, equity share and share count quoted above reproduces exactly, except C9. **Sources re-opened 2026-09-28** (fetch helper, `--grep YES` unless stated): Vanguard VCMM page (U.S. 4.9%-6.9% -> 4.2%-6.2%; developed ex-U.S. 4.5%-6.5%; EM 2%-4%; "July 22, 2026"; "June 30, 2026, running"; "distribution of geometric returns"; the E5 sentence verbatim), VERIFIED-PRIMARY. FactSet Earnings Insight 25 Sep 2026 (forward P/E 19.2; 5-yr 19.8; 10-yr 19.0; 20.4 on June 30; price +2.7%, forward EPS +8.9%; CY2026 32.0% and CY2027 15.4% *projected*; Q2 2027 1.7%; IT 63.5%, semiconductors 126%, IT ex-semis 24.2%, all Q3 2026 *estimates*; trailing 25.8), VERIFIED-PRIMARY. FRED DGS10 4.16 (2025-09-30), 4.44 (2026-06-30), 5.18 (2026-09-24); T10YIE 2.34 (2026-09-25), VERIFIED-PRIMARY. multpl.com CAPE 41.48, mean 17.42, max 44.19 (Dec 1999), secondary. JPM PDF p.2: AC World 7.00/8.28/16.78, correlation with intermediate Treasuries 0.00, VERIFIED-REPO-FILE. Case L69-74; Guide L73 and L156-159, VERIFIED-REPO-FILE.

#	Severity	Issue	Correction
C1	important	"Known" / "lock" overclaim (summary 1, "Terms used", E2, TN element (b), IPS spec, devil's advocate, What this teaches 4). The 5.20-5.23% figures are forward rates <i>implied</i> by today's curve for money that arrives in January 2028. Laura cannot lock them now: the \$150k does not exist yet, and forwards or futures are derivatives, which WInS bans and which the plan should avoid too (SMAppl FAQ "Investments permitted (for BOTH contributions)", R-AN15 safe reading). The real rate is set when the money arrives.	Say "today's Treasury curve implies about 5.2% a year for her 2028-2033 dates (a market-implied hurdle, not a locked rate)". Never write "known" or "locked" about 2028 rates in a note or the IPS. E2 keeps its labels (VERIFIED-REPO-FILE inputs, ASSUMPTION method); the word "locked" is withdrawn.
C2	important	The forward earnings-yield yardstick is not like-for-like. The table adds 2.34% breakeven inflation to the CAPE earnings yield (to make it nominal), but compares the forward earnings yield (5.21%, roughly a <i>real</i> return) directly with a <i>nominal</i> Treasury rate.	Treated the same way: $5.21\% + 2.34\% = 7.55\%$, premium +2.3 points ; or 5.21% vs the 10-year TIPS real yield 2.83% (F-008) = +2.4 points real (AX2 script, section 2). So the yardsticks span about -0.5 (CAPE) to +2.3 (forward E/P), not "about 0". Corrected summary line: "central forecasts about 0 (Vanguard mid) to +1.8 (JPM); full published ranges and yardsticks about -1.2 to +2.4 points a year". "Small and uncertain" survives; "three sources agree on about 0" does not. The 50% option does not depend on it (it rests on the flat median across houses).
C3	important	Guide p.5 is misapplied. Guide L156-159 sits in the post-IPS "evaluation and reflection" stage. Before Nov 6 the official texts expect the strategy to evolve: "Although your strategy may continue to evolve before the IPS is submitted" (Guide L89-90); "Although your strategy may continue to evolve as you conduct additional research" (TN guide L18-19); notes show how decisions "tested, or refined" the strategy (TN guide L12; case L152-153).	A research-based change from 60/40 to 50/50 during trading is permitted and could itself be a strong note (the ticket, section 4, already lists "a change of the split after the team's research" as a genuine third-note source). "Decide before the first VT/VGSH order" is a <i>preference</i> (fewer trades, cleaner notes), not a rule. Delete "would contradict the Guide's warning". What must not happen: a change driven by a price move, or any change after Nov 6.
C4	important	The split is tier 1 only under WInS option (ii). Under option (iii) (D10's lean) there is no VT or VGSH order at all.	Route summary item 3 as "tier 1 if the team votes (ii); otherwise an IPS rule (tier 2) only".
C5	important	"A world fund halves AI exposure" overstates. The \$17k counts VT's top ten only and compares it with a 12-name S&P 500 basket. VT's non-U.S. part holds AI chip makers outside its top ten (S2 finding 4: TSMC, Samsung, SK hynix, ASML; "Going global does not remove AI exposure. It moves some of it to Asia").	Like-for-like: top-10 concentration falls from 38.8% to 21.7% (about -44%); total AI-linked exposure falls by less than that. Never write "halves AI exposure" in a VT note.

#	Severity	Issue	Correction
C6	important	TN element (d) conflicts with the plan's 2031 rule. "Half stays in short Treasuries for the amount Laura may promise co-sponsors" implies the 2031 promise is sized from the VGSH half. In the plan the 2031 amount is bought from a share of the <i>whole</i> sleeve (80% in the model, an open team parameter), stocks included. Notes are permanent.	Use the ticket's framing: VGSH is the low-risk part of the growth money that could later become part of a promised amount because it is already owned; no floor percentage and no "half = promise" link (ticket note 3; red-team finding 4).
C7	important	The frozen-weight rule rests on a reason that changes. IPS rule (2) freezes the weight, but the stated reason ("the expected extra reward for stocks is small this year") is a valuation reading that will change.	Base the IPS reason on what does not change: the growth money feeds a promise Laura makes in 2031, and under both published houses the median barely moves between 50% and 60% while the bad case improves. Otherwise the rule invites the re-timing D2 itself warns against.
C8	important	Contradiction with the WInS ticket on sleeve-bond returns. D2 models short Treasuries at 4.8% (market-consistent: VGSH SEC yield 4.55%; one-year forwards 4.8-5.1%). The ticket (section 8, "Model inputs to align") asks for "about 3.5-3.9%". Both cannot go into the model.	AX2 view: D2's market-based figure is the defensible one now (F-317: JPM bond inputs about 1 point stale); the ticket's line should be updated by the main loop. Also soften "about 1 point below what short Treasuries pay today" to "about 0.5-1 point" (4.00% vs 4.55% SEC yield and 4.8-5.1% forwards).
C9	minor	Sign error. "Going down to 40% ... median -\$0.6k to -\$3.2k vs 60%": script section 4 gives 40%→60% median +3.2k (JPM), +0.6k (Vanguard mid), -0.6k (Vanguard low).	40% vs 60%: median -\$3.2k to +\$0.6k ; p5 +\$12-15k (correct as written).
C10	minor	The "0% equity" row is not the no-stock alternative. It uses 4.8% with 2% volatility (p5-p95 \$189-211k). A sleeve invested at today's implied rates gives about \$204k with no spread (AX2 script, section 3; ASSUMPTION).	Against that comparator the share of paths below the Treasury-only outcome at 50-60% equity is 42-43% (JPM) and 51-52% (Vanguard mid) , not 37% and 46-47%. Under Vanguard's midpoint the 50-60% median (\$202.7-203.0k) is <i>below</i> the Treasury-only outcome. This strengthens D2's case; replace the quoted percentages.
C11	minor	"Free improvement" (summary as routed to the strategy team).	Say "near-zero median cost": 50% gives up \$11-14k at p95 and \$1.5k of median under JPM.
C12	minor	"Earnings jumped 32% in 2026" / "semiconductors +126% in Q3".	FactSet: "analysts are projecting earnings growth of 32.0%" for CY2026; the semiconductor figure is expected Q3 2026 growth. Say "projected" / "expected".
C13	minor	Rule (1) is silent on what happens after the 2031 floor is bought.	State whether the 50/50 applies to the un-locked remainder after 2031.
C14	minor	Labels and jargon. "SECONDARY-READ" is not a brief label; "SEC yield", "i.i.d.", "lognormal" are undefined.	Use SNIPPET-UNVERIFIED for multpl (secondary source read directly; Shiller's file not read). Define or drop the three terms.

Status after audit. Findings 2, 5 and 7 reproduce (7 amended by C8). Finding 1 downgraded from "confirmed" to "supported: the premium is small and uncertain, about -1.2 to +2.4 points a year across sources" (C1, C2). Finding 3 is a preference, not a rule (C3, C4). Finding 6 corrected (C5). Finding 4 stands as an INTERPRETATION of the case. The 50/50 option survives as a team choice; the team decides.

D3: Quant Modeler working file (simulation model, confidence statements, the 2031 range, stress scenarios)

Agent D3, insight_v1 run, Phase D. Written 2026-09-28. This is AI-generated research for Team Caplet. It holds numbers, evidence, checklists and "what a strong sentence must contain" specifications. It holds **no text to submit**: the six students decide and write every deliverable in their own words and record this AI use in the Final Report's Works Cited (Wharton AI policy, brief section 14). Laura appears only through the case and her public professional record.

Scope. Nine Phase C survivors: M034, M243, M039, M060, M028, M238, M053, M055, M059. They are worked in deadline order:

- tier 1 = WInS now / Trading Notes, 23 Oct;
- tier 2 = IPS, 6 Nov;
- tier 3 = Final Report, 4 Dec.

The extra mandate is the v2 model, `research/insight_v1/scripts/strategy_mc_v2.py`. Its full results are in `research/insight_v1/phase_D/D3_model_v2_results.md`.

Scripts (run from the repo root with `.venv/bin/python <script>`; each docstring lists inputs and status labels):

Script	Topic
<code>strategy_mc_v2.py</code>	model v2 and switches (a)-(g)
<code>D3_control_and_equity.py</code>	M034
<code>D3_funded_status_log.py</code>	M028
<code>D3_barbell.py</code>	M238
<code>D3_lock_frontier.py</code>	M243
<code>D3_range_rules.py</code>	M039, M060
<code>D3_joint_tail.py</code>	M053
<code>D3_ratchet.py</code>	M055
<code>D3_history_stress.py</code>	M059
<code>D3_data_snapshot.py</code>	data snapshots in <code>scripts/data/D3/</code>

All scripts are under `research/insight_v1/scripts/`.

Labels:

- VRF = VERIFIED-REPO-FILE.
- VP = VERIFIED-PRIMARY (read on the primary page or file, with its date).
- MODEL = an ASSUMPTION-based model output. It is not a forecast.
- ASSUMPTION = a modelling choice.

Model numbers are rounded to \$1k unless the table says otherwise.

Terms used:

- **Surplus**: the money left in 2033 after the ten payments are bought. It is the facility-plus-flexibility money.
- **Sleeve**: the invested money outside the payment ladder.
- **Floor**: the part of the sleeve moved into a 2-year Treasury in 2031, so it is certain by 2033.
- **p5/p50/p95**: 5 in 100 simulated futures end below p5; p50 is the median.
- **Riskless control**: the same money put in Treasuries maturing just before 2033, with no stocks at all.

Summary: top findings, ranked by impact on reaching the semifinals and on making the plan unmistakably Laura's

1. **The case's hardest sentence can be answered with a certainty instead of a model percentage (M039, M060).**

- The case asks for confidence that the 2033 contribution "will fall within that range" (p.3 L117-118, VRF).
- The bottom can be bought: the 2031 floor. The top can be a cap Laura sets herself, because the case says Laura "must decide how much" (p.3 L101, VRF). With both, "within the range" holds in 100% of modelled futures and in every fat-tail, low-return and historical test. The only exception is a U.S. default.
- The model is then needed for one number only: the chance of reaching the top (about 1 in 5 for a top at the 80th percentile).
- Without the cap, an "80% confident" statement lands within the range 62-93% of the time, depending on which model or history is used.
- Pair the cap with a share rule, "give up to about 90% of the post-reserve money". This keeps flexibility of \$15k/\$21k/\$30k (p5/p50/p95, central case). The case warns against committing everything (p.3 L102-103, VRF).
- Lock share: 80-90% of the 2031 sleeve. The median barely moves (\$209k at 50% to \$206k at 100%). 100% collapses the range into "one exact amount", which the case says Laura does not want (L115-116, VRF).

2. **Against the money Laura could lock for certain, stocks in the sleeve buy range, not a bigger expected facility (M034).**

- The riskless control is about \$204k by 2033, with no fee (MODEL).
- 60% sleeve equity adds about \$3k at the median under JPM. It adds about \$6k if sleeve bonds earn forward yields. It loses about \$4k under Vanguard's midpoint, and about \$2k after a 0.5% advisory fee on the sleeve.
- The plan ends below the control in 43-54% of futures.
- What equity buys is spread: on the same simulated path, p95 +\$70k against p5 -\$46k.
- The Investment Policy Statement (IPS) must justify equity as upside for the residency and flexibility, not as higher expected growth.
- WInS-now: fix the sleeve equity share (40-60% are all defensible) before the growth-fund (VT) order. WInS trade notes cannot be edited, and the note will record the share.
- 3. **The honest reason to lock all ten payments is that the certainty is bought, not forecast, and it is cheap (M243).**
- With the 2028 deposit present, 70-95% locks also never fail: in 200,000 model futures and in 93 historical windows.
- A full lock gives up only \$0.6-3.3k of median facility money under JPM, and nothing under Vanguard or lower 2033 rates.
- But a 90% lock fails in 13% of futures, and an 80% lock in 26%, if the deposit from a self-employed author does not come.
- The IPS must not say "anything less fails".
- The team should also say what it has been overstating: in about 30% of rate paths the real ladder costs more than \$300k in January 2027, and the first ~\$7k of the 2033 payment then waits for the 2028 deposit.
- 4. **The joint tail is small, and it is better stated in dollars than as a probability (M053).**
- Rates must first fall more than 19bp before January (about 30% of rate paths). Even then, the payments go short only if the 2028 deposit is almost entirely missing.
- When they do go short, the unfunded part of the 2033 payment averages \$9k (p95 \$24k; \$31k at a 100bp fall).
- A deposit of \$25k covers even a 100bp fall.
- The IPS needs one clause: the first use of the 2028 deposit is completing the ladder.
- "Longest payments first" cuts the price risk of the top-up about threefold compared with "nearest first".
- 5. **The WInS hedge demonstrably tracks the promise (M028).**
- In 2026, an IEF/TLH hedge valued from its actual holdings stayed within $\pm 0.42\%$ of the value of Laura's ten payments in 95% of six-week windows. That is about $\pm \$830$ on the \$198k WInS hedge.
- Over the same windows the payments' value itself swung by up to $\pm 2-3\%$ (MODEL; VP inputs).
- A five-number weekly log gives one checkable fact for a Trading Note reflection. The WInS-now to-do is small.
- 6. **Reject the barbell (M238).**

- The barbell means locking part of the growth money in January 2028 and holding the rest 100% in stocks.
- The 50% barbell matches the plan only by keeping half the facility money in stocks until 2033.
- In 2031 it can promise a bought bottom of only about \$101k (the plan: about \$165k).
- A 2031-32 crash after the announcement costs it about \$34k of median (the plan: \$8k).
- Larger barbells are just lower-equity points on the same frontier.
- Do not use "it mirrors her floor-first leap": D13 bars that trait. 7. **History check (M059).**
- A 2008-type year after the 2031 lock trims the facility money by about 5%.
- The same crashes before the lock cut it by 19-29%: 2022, 2008, 2000-02 and 1973-74. A 1929-31 replay cuts it by 48%.
- All ten payments stay bought in every case.
- The bell-curve model understates crash years: a -30% year happens in 0.3% of model years against 3.1% of S&P years. But its 10th percentile matches history's.
- Show one named historical row next to the model's p5 in the Final Report. 8. **No ratchet (M055).**
- Locking the floor in thirds (2029-31) raises the bad case (p5 +\$9k) but costs \$17k at p95.
- A plain ~42% equity sleeve reaches the same p5 within \$2k of median.
- A second IPS rule is not earned. 9. **The v2 model tells the team where to spend words.** See `D3_model_v2_results.md`.
- What matters: the 2028 deposit, fees charged on the ladder (a 1% fee on everything costs about \$32k of median), and which house's return forecast is used.
- What barely matters: fat tails; holding the WInS funds (VT/VGSH-like) instead of the model's assets.
- Central case: 2033 surplus \$152k/\$204k/\$277k (p5/p50/p95), against the verified \$159k/\$207k/\$273k.

Tier 1: WInS now and the Trading Notes (23 Oct)

M034. Judge the sleeve against Laura's riskless control, not a stock index

The question (Phase C): should every sleeve recommendation be judged against the whole surplus locked in Treasuries (about \$204k in 2033), given that 60% equity gives a median of about \$207-210k but a p5 of about \$159k? And would an all-Treasury rival beat us on simplicity?

One-sentence answer: Yes. Against the roughly \$204k Laura could lock without any fee, 60% sleeve equity adds only about +\$3k at the median under JPM (from -\$4k to +\$6k across the tested assumptions, and -\$2k after a 0.5% sleeve fee). The plan ends below the control in 43-54% of futures. So the equity buys a wider facility range, not a bigger expected facility, and it must be justified and benchmarked that way.

Numbers (`D3_control_and_equity.py`; MODEL unless labelled):

The riskless control. Two Treasury zeros maturing 1 January 2033 (F-012; F-106 method; VRF curve):

- the 2027 leftover: $\$7,736 \times 1.3512 = \$10,453$;
- the 2028 deposit: $\$150,000 \times 1.2903 = \$193,544$ on today's forwards;
- total **\$203,997**.

The 2028 part cannot be locked until the money arrives. With a yearly rate move of sd 0.71pp (F-014), the control is **\$193k/\$204k/\$215k** (p5/p50/p95).

Sleeve equity weight against the control (verified inputs; 80% floor locked in 2031):

Sleeve equity	2033 surplus p5/p50/p95	Median lift over control	Lift on the same path, p5 / p95	P(plan < control)
0%	\$177k/\$195k/\$215k	-\$9.3k	-\$31k / +\$14k	75%
20%	\$177k/\$199k/\$225k	-\$4.5k	-\$29k / +\$23k	61%
40%	\$170k/\$204k/\$247k	-\$0.5k	-\$36k / +\$44k	51%
50%	\$164k/\$205k/\$260k	+\$1.4k	-\$41k / +\$57k	48%
60% (current)	\$159k/\$207k/\$273k	+\$3.3k	-\$46k / +\$70k	46%
70%	\$154k/\$209k/\$287k	+\$5.0k	-\$51k / +\$84k	45%
100%	\$137k/\$214k/\$332k	+\$9.7k	-\$68k / +\$128k	43%

The median lift of 60% equity under other assumptions:

Assumption set	Median lift of 60% equity	P(plan < control)
Central case (STRIPS ladder, rate risk, AC World + short government bonds, fund fees)	+\$3.3k	45%
Sleeve bonds at the forward-implied 4.9%	+\$5.7k	43%
Vanguard midpoint 5.2% (VP range 4.2-6.2%)	-\$3.9k	54%
0.5% advisory fee on the sleeve (the control pays none)	-\$1.7k	51%

- Moving from 40% to 60% equity: p5 -\$8-10k, median +\$1.5-3.9k, p95 +\$22-26k, across the four sets.
- Whole-portfolio equity on 1 January 2028 (sleeve equity 40/50/60/70%): 13.6% / 17.0% / 20.4% / 23.8%. ASSUMPTION: medians; ladder at its forward value of \$306,077 (F-106).
- WInS option (ii) at a 66% hedge: VT/VGSH = 13.2/19.8% at 40% sleeve equity, 16.5/16.5% at 50%, 19.8/13.2% at 60%.

Evidence:

Claim	Source	Status
The case asks for "an appropriate balance between pursuing growth and protecting the capital required for her goals"	case p.2 L71-74	VRF
Laura "has been willing to take thoughtful risks"	case p.2 L69-71	VRF
JPM: U.S. large cap 6.70/7.94/16.47; AC World 7.00/8.28/16.78; intermediate Treasuries 4.00; short govt/credit 4.00/4.01/1.63	JPM LTCMA 2026 matrix p.2	VRF
JPM's bond assumption (4.0%) sits about 1pp below today's yields (IEF YTM 5.17%)	F-317, F-203	VRF/VP
Vanguard: U.S. equities 10-year expected return 4.2%-6.2% (run of 30 June 2026)	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html (accessed 2026-09-28)	VP
Her résumé lists "A/B testing" as a product-management skill (so a control-vs-plan comparison is a frame she uses professionally)	D13a B9b-Q02, https://lauragao.com/s/Laura-Gao-Resume.pdf	VP (D13a); use at most once, in the Final Report only

Mis-posed parts:

- The premise number is right: \$204.0k.
- The phrase "the whole surplus locked" is not: the 2028 money cannot be locked today. The control therefore carries about ±\$11k of 2028 rate risk.
- The "all-Treasury rival" is the riskless control itself: median \$204k (-\$3k against the 60% plan) and p5 \$193k (+\$34k). Keeping the sleeve but holding 0% equity is worse than both (median \$195k) because JPM's 4.0% bond assumption sits below the ~5.1% that locking earns. The rival's facility amount is known in 2028.
- On simplicity alone it would win. It loses on the case's "pursuing growth" and on giving co-sponsors a range. Name it in the Final Report's "alternatives considered".

Implication:

- *Decision (WInS-now)*: the team fixes the sleeve equity share before the VT order.
 - The evidence does not force a move from 60%.
 - 50% ("half of the free money") costs \$2k of median and \$13k of p95 and gains \$5k at p5.
 - The WInS growth note must state the share in the team's words. Notes are permanent (S3 ticket; A4).
- *Sentence (IPS)*: the "why any stocks" sentence must contain:
 1. the sure alternative exists;
 2. what equity adds (upside for the facility and flexibility);
 3. what it costs (a lower bad case);
 4. that the payments never depend on it.
- *Number (Final Report)*: one chart. A flat line for the riskless control (~\$204k) over the plan's p5-p95 fan, labelled "lift over the control".

- **Fee note:** if the fictional firm charges a fee, say what it is charged on. A fee on the sleeve alone (0.5%) erases the median lift. A fee on the ladder costs about \$11k of median per 0.5% (model v2 (c)).

Tier 1. Confidence: high on the direction; medium on each dollar figure. **Changes current strategy:** no. The structure stays; the justification and the benchmark change. **Criterion:** Portfolio Analysis ("reasonable assumptions and projections ... under varying market outcomes", R-S26).

What this teaches: a risky choice should always be compared with the best sure thing you could have had instead, not with a market index.

M028. A weekly funded-status log for the WInS hedge

The question: should the team log, from the first WInS trade, the value of Laura's ten payments on that day's curve, the ladder cost, the funded ratio and the WInS hedge value, so a Trading Note can report whether the IEF/TLH mix tracked the promise within X%?

One-sentence answer: Yes, weekly and light. In 2026 the IEF/TLH hedge moved within $\pm 0.42\%$ of the payments' value in 95% of six-week windows, while the payments' value itself swung $\pm 2-3\%$. So a weekly log gives a Trading Note one checkable, strategy-linked number. The skeptic is right that the log could be rebuilt later from public data, so it is useful but not urgent.

Numbers (D3_funded_status_log.py --backtest; 185 official curve dates, 2 Jan to 25 Sep 2026; MODEL on VP inputs):

Hedge (share of hedge)	Gap: hedge % change minus payments' % change over 30 trading days	95% of windows within	Worst
IEF/TLH 35.7/64.3	mean +0.07%, sd 0.18% (p5 -0.19%, p95 +0.42%)	$\pm \$827$ on a \$198k WInS hedge	+0.64%
IEF/TLT 62.1/37.9	sd 0.24% (p95 +0.54%)	$\pm \$1,076$	+0.88%

- **For scale:** the payments' value moved with sd 1.53% over six weeks. The 95th percentile move was \$5,723 on a \$198k book.
- **Whole period:** the payments' value (spot) fell from \$305,879 to \$288,924 (-5.5%). The IEF/TLH hedge fell 4.7%.
- **The same ladder's cost for January 2027** was \$318,655 (STRIPS) on 2 Jan 2026, against \$294,387 on 25 Sep. The funded ratio (\$300k ÷ cost) went from 0.94 to 1.02.
- **Since 1 Sep 2026:** the payments' value -3.18%; IEF/TLH -2.85% (gap +0.33% = +\$648).

The log specification (one line a week, same weekday; the team fills in column 5 from the WInS screen):

1. date and treasury.gov curve date;
2. the payments' value today (spot present value) and DV01, the dollars gained or lost per 0.01% move in rates (\$293/bp on 25 Sep);
3. the cost of the payments for 1 Jan 2027 (STRIPS) and the funded ratio (\$300k ÷ cost; 1.019 on 25 Sep);
4. the modelled hedge % change since the first trade;
5. the actual WInS hedge value and its % change since the first trade;
6. the gap between (5) and the payments' % change.

Tool: `.venv/bin/python research/insight_v1/scripts/D3_funded_status_log.py --date MM/DD/YYYY --since MM/DD/YYYY` after refreshing the curve with `D3_data_snapshot.py`.

Evidence:

Claim	Source	Status
A Trading Note should "capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding."	Guide p.3 L84-86 (2026_WGY_Investment_Competition_Guide.txt)	VRF
"A long-term investment strategy cannot be judged solely by what happens in the market over a few weeks."	Guide p.3	VRF
Reflections of ≤ 100 words must explain why, how the decision aligned with the strategy, and how it supported the client's goals, funding needs or risk considerations	TN guide L44-51	VRF
Fund holdings (IEF, TLH, TLT) as of 2026-09-24	wins_now/S1_fund_holdings_snapshot.csv	VP (S1)

Claim	Source	Status
Daily par curves 2026	treasury.gov CSV (snapshot data/ D3/treasury_par_2026_raw.csv, accessed 2026-09-28)	VP

Limits:

- The backtest measures curve-shape (twist) risk only. It uses fixed 24-Sep holdings (ASSUMPTION) and no fund price noise, premium/discount or fees.
- Real WInS values will add noise. Treat $\pm 0.4\%$ as a lower bound and expect perhaps $\pm 0.5\text{--}1\%$.

Implication:

- *Decision (WInS-now)*: start the weekly log with the first hedge trade (about 10 minutes a week).
- *Sentence (Trading Notes)*: a strong hedge-note reflection contains one measured pair of numbers: "the payments' value moved X%, our hedge moved Y%". Never claim "stable" or "reduced volatility" (S3 ticket).
- *Number (Final Report)*: a two-line chart over the six WInS weeks, the payments' value against the hedge value.

Tier 1. Confidence: medium-high (the model tracking is robust; real fund noise is unknown). **Changes current strategy:** no. **Criterion:** Portfolio Analysis ("demonstrates understanding and effective use of investment concepts and tools", R-S26).

What this teaches: judge a hedge against the thing it protects. A bond fund that lost 3% while the promise got 3% cheaper did its job.

M238. The barbell (lock a floor in January 2028, the rest 100% in stocks)

The question: should the sleeve become a barbell, instead of a rebalanced 60/40 sleeve plus an 80% lock in 2031?

One-sentence answer: No. The 50% barbell's equal median and better p95 come from keeping half the facility money in stocks until 2033. That weakens the 2031 message to co-sponsors (a bought bottom of about \$101k instead of about \$165k), and a 2031-32 crash after the announcement costs it about four times as much. Keep the 2031 lock, and keep VGSH rather than a Dec-2032 fund in WInS.

Numbers (D3_barbell.py; MODEL; the barbell locks at today's 5-year par yield of 4.98%, an ASSUMPTION as in B10a; the p80 top is described in M039):

Design	2033 surplus p1/p5/p50/p95	Floor known in	Median bought floor	Median p80 top	Floor ÷ top
Plan (60/40 sleeve, 80% lock in 2031)	\$144k/\$159k/\$207k/\$273k	2031	\$165k	\$212k	78%
Barbell 50%	\$150k/\$163k/\$210k/\$293k	2028	\$101k	\$232k	43%
Barbell 65%	\$166k/\$175k/\$208k/\$265k	2028	\$131k	\$223k	59%
Barbell 80%	\$181k/\$186k/\$205k/\$238k	2028	\$161k	\$214k	75%

- **A crash in 2031-32** (the bottom 10% of 2-year stock returns, after the 2031 announcement), median surplus:
 - plan \$199k (-\$8k), losing 28% of the announced range's width;
 - barbell 50% \$176k (-\$34k), losing 42% of the width.
- **With sleeve bonds at 5.0%** (this removes the JPM-4.0%-against-4.98%-lock gap): plan \$161k/\$210k/\$277k; barbell 50% \$163k/\$210k/\$293k (B10a's numbers reproduced). Barbell 80% is \$186k/\$205k/\$238k: the same "less stock" trade-off.
- **If rates fall to 4.0% by January 2028**, barbell 50% becomes \$159k/\$206k/\$288k (p5/p50/p95).
- **Historical 6-year windows** (93, rescaled to JPM's medians):

Design	Worst window	p10
Plan	\$105k (1929 start)	\$164k
Barbell 50%	\$138k	\$169k
Barbell 80%	\$176k	\$189k

The early lock helps in a Depression-type run. Against a crash in 2031-32 the plan's late lock is better.

- **Whole-portfolio equity on 1 January 2028:** plan 20.4%; barbell 50% 17.0%; barbell 80% 6.8%.

Evidence:

Claim	Source	Status
"Rather than promising one exact amount, she wants to communicate a credible range." A meaningful personal contribution "may signal that the residency is financially viable"	case p.3 L110-116	VRF
iShares Dec-2032 iBonds (IBTM) exists; no Dec 2037-2043 funds	F-206; S1	VP
The "floor-first leap" is NOT a verified Laura trait; do not use it	D13c summary item 2; brief section 16	VRF (D13)

Mis-posed parts:

- The question's north star ("mirrors how Laura took her biggest career risk") is barred by D13.
- Decide on numbers and on the 2031 message only.
- The question compares a 60/40 half-leap with a 100% leap. The real difference is when the floor is bought (2028 or 2031), and that is the same trade-off as the equity weight (M034) and the ratchet (M055).

Implication:

- *Decision (WInS-now and IPS):* no barbell. The WInS short-Treasury slot stays VGSH (S3 ticket), not IBTM.
- *Sentence (Final Report, alternatives considered):* the rejection reason must contain the 2031 bottom (\$101k against \$165k) and the crash-after-announcement exposure.

Tier 1 (the WInS sleeve bond slot) and 2 (the IPS freezes the structure). **Confidence:** medium-high.

Changes current strategy: no. **Criterion:** Investment Strategy ("disciplined planning across Laura's changing time horizons", R-S24).

What this teaches: two designs with the same median can differ in who carries the risk and when. Here the barbell moves risk to after the moment Laura has spoken to co-sponsors.

Tier 2: the IPS (6 Nov; the strategy freezes)

M243. The honest reason to lock 100% of the payments

The question: with the 2028 deposit present, a 70-95% lock also shows no payment shortfall. What is our real reason to lock 100%, and where are we overstating it?

One-sentence answer: A full lock is the only design whose certainty rests on prices visible today rather than a forecast. Partial locks also pass with the deposit, but they save the facility only \$0.6-3.3k of median under JPM (and nothing under Vanguard or lower 2033 rates). And they fail in 13-26% of futures if a self-employed author's 2028 deposit does not come.

Numbers (D3_lock_frontier.py; a lock share of every rung bought in January 2027; the rest invested 60/40; the unlocked payments bought in 2033 at 5.26% $\pm$ 1pp (the verified convention, about \$6k dear, F-408); no 2031 floor; MODEL). Each cell shows P(payments short) and the median surplus gap to the 100% lock:

Lock	Deposit \$150k (case)	Deposit \$75k	Deposit missing	Cut to \$0 in 25%, correlation 0.6	Vanguard 5.2%	Fat tails t4x1.09	2033 rates 4.0%
100%	0%; -	0%; -	0%; -	0%; -	0%; -	0%; -	0%; -
95%	0%; +\$0.6k	0%; +\$0.6k	2.3%; +\$0.5k	1.1%; -\$0.8k	0%; -\$0.5k	0%; +\$0.6k	0%; -\$0.5k
90%	0%; +\$1.1k	0%; +\$1.1k	12.8%; +\$1.0k	4.9%; -\$1.6k	0%; -\$1.0k	0%; +\$1.1k	0%; -\$1.0k
80%	0%; +\$2.2k	<0.01%; +\$2.2k	25.8%; +\$2.0k	9.0%; -\$3.4k	0%; -\$1.9k	0%; +\$2.3k	0%; -\$1.9k
70%	0%; +\$3.3k	0.08%; +\$3.2k	31.6%; +\$3.1k	10.7%; -\$5.2k	<0.01%; -\$2.9k	0.01%; +\$3.3k	<0.01%; -\$2.9k
0% (no lock)	2.1%; +\$10.6k	11.7%; +\$10.5k	41.0%; +\$10.2k	14.5%; -\$13.5k	3.6%; -\$9.9k	2.8%; +\$10.2k	3.6%; -\$9.8k

- A positive gap means the partial lock has the higher median. A negative gap means the full lock also has the higher median.
- Combined stress (Vanguard + \$75k + 2033 rates 4.5%): the 70% lock is short 0.27% of the time, 50% is short 4.1%, and the full lock has the higher median.
- History (D3_history_stress.py): 90% and 70% locks were short in **0 of 93** historical 6-year windows with the deposit.

Where we overstate "certain" (the part the team should say):

1. **"Never misses" is by construction, and holds only from the purchase date.** With the real ladder (\$294,387) the headroom is \$5.6k (19bp). In about 30% of rate paths the ladder costs more than \$300k in January 2027. The mean gap is \$7.0k (p95 over all paths \$12.5k), and the first part of the 2033 payment then waits for the 2028 deposit (model v2 (a); M053).
2. **The certainty is in nominal U.S. dollars.** At 2.5% inflation the 2042 payment buys \$33,681 in 2026 dollars (F-114, ASSUMPTION inflation).
3. **The residual risk is a U.S. Treasury default.** Broker mark-ups on the STRIPS are unknown; 0.25% moves the chance of costing more than \$300k from 30.2% to 32.6% (ASSUMPTION).

Evidence:

Claim	Source	Status
"All ten payments must be funded by the investment portfolio with a high degree of certainty." Teams must "define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty"	case p.3 L91, L97-99	VRF
"She will contribute an additional \$150,000" ("will")	case p.2 L43-45; R-AN35	VRF
Her own words describe self-employment as "unstable income" (career context, 2021)	D13a B8b-Q18 (Overachiever Magazine)	VP (D13a); Final Report only, in context
B10b's frontier (0.00% at 70-100%; 2.10% at 0%)	reproduced exactly by D3_lock_frontier.py	MODEL

Already answered / new: brief 8.1's partial-lock failure rates (35/26/14/3.4%) assumed no deposit. They are reproduced here in that case (26% at 80%, 13% at 90%, 2.3% at 95%; F-410's council numbers were not reproducible). What is new: the full lock is free under Vanguard or lower 2033 rates, and history shows no failures with the deposit.

Implication:

- *Sentence (IPS):* the reason-for-full-lock sentence must contain:
 1. certainty from prices, not a forecast;
 2. its price (a few thousand dollars of expected facility money, near zero under lower return forecasts);
 3. protection if the 2028 income is smaller or late;
 4. no claim that a partial lock would fail in the base case.
- *Number (Final Report):* the frontier table as a "price of certainty" chart, with P(short) by lock share and deposit scenario.

Tier 2. Confidence: high. **Changes current strategy:** no (the reason sentence changes). **Criterion:** Investment Strategy (a thesis that follows from the client's constraints, R-S24).

What this teaches: when two choices both pass the test, pick on what each one depends on. Buying the payments depends on today's prices; a partial lock depends on a forecast.

M039. Make "within the range" partly a policy: floor bought, top as a cap

The question: since Laura decides the 2033 contribution, should the range be partly a policy (contribute at most the top, keep any excess as flexibility), so the stated confidence is the chance of affording the bottom, and the top sits at a stated percentile?

One-sentence answer: Yes. With a bought bottom and a top that Laura commits never to exceed, "within the range" becomes a certainty in nominal U.S. dollars, barring a U.S. default, under every model and history tested. The model is then used only to say how likely the top is (about 1 in 5 for a p80 top). Add a share rule, "up to about 90% of the post-reserve money", so the cap does not leave her with no flexibility.

Numbers (D3_range_rules.py; strategy_mc_v2.py (f)/(g); MODEL):

Setup:

- The floor is 80% of the 2031 sleeve.
- The top = floor + unlocked money × the p-th percentile of its 2-year growth. The planning model's 2-year growth percentiles are p20 1.004, p50 1.119, p80 1.253, p90 1.331.

Contribution rule (verified base)	P(below)	P(within)	P(above)	P(top reached)	Contribution p5/p50/p95	Flexibility kept p5/p50/p95
Give all, no cap (p80 top)	0%	80%	20%	20%	\$159k/\$207k/\$273k	\$0/\$0/\$0
Give all, capped at the top	0%	100%	0%	20%	\$159k/\$207k/\$272k	\$0/\$0/\$6k (zero in 80% of futures)
Give 90%, no cap	0%	80%	20%	20%	\$143k/\$187k/\$246k	\$16k/\$21k/\$27k
Give 90%, capped	0%	100%	0%	20%	\$143k/\$186k/\$245k	\$16k/\$21k/\$30k
Give 80%, capped	0%	100%	0%	20%	\$127k/\$165k/\$217k	\$32k/\$42k/\$56k

- With a p90 top, P(top reached) falls to 10%.
- **Central case** (80% floor, p80 top, "give 90%, capped"): range \$146k-\$188k at the median; contribution \$136k/\$183k/\$248k; flexibility \$15k/\$21k/\$30k.
- **What Laura would announce in 2031** (80% floor, p80 top, give-all version; floor, most likely amount, top), by how 2027-2030 went:
 - bad (sleeve \$153k on 1 Jan 2031): \$134k / \$169k / \$173k;
 - middle (\$188k): \$165k / \$207k / \$212k;
 - good (\$232k): \$204k / \$256k / \$262k.
 - **Without a cap, the stated confidence depends on the model.** For an "80%" top, the share of outcomes within the range was:
 - 83.7% with fat tails (t4) and 83.2% under Vanguard;
 - 72.2% with stocks 3pp a year stronger;
 - 76.3% on history rescaled to JPM, and 61.9% on raw U.S. history (97 two-year periods).

For a "90%" top the spread is 81-97%. Every miss is on the upside.

Evidence:

Claim	Source	Status
"Laura must decide how much of the remaining portfolio she can responsibly contribute"	case p.3 L101-102	VRF
"state how confident they are that her 2033 contribution will fall within that range" (two ends, R-AN9)	case p.3 L117-118	VRF
"If Laura promises more than she can ultimately contribute, she could damage her credibility"	case p.3 L114-115	VRF
"committing all remaining assets could limit her financial flexibility as the project develops"	case p.3 L102-103	VRF
Brunel's goal-probability table: Needs 90-95; Wants 80-85; Wishes 65-75; Dreams 50-60	Das, Ostrov, Radhakrishnan & Srivastav, <i>Journal of Investment Management</i> 2018, Table 1, https://srdas.github.io/Papers/GBWM.pdf (grep-verified 2026-09-28)	VP
Her own results tables label her uncertainty " [unfinished]", graded Good/Unsure/Bad (a working habit of stating uncertainty)	D13a B9b-Q09, lauragao.com	VP (D13a); Final Report only

Hypotheses resolved: B1a-17/B1b-19 (with a cap, confidence = $P(\text{available} \geq \text{floor}) \approx 100\%$) and B4b-20 (without a cap, $P(\text{within})$ = the chosen percentile) are both right. They describe different rules, so report both numbers. A top at the p80-85 level would sit in Brunel's "wants" band (Das et al., VP). That is one practitioner framing, not a rule.

Implication:

- **Decision (IPS):** adopt a contribution rule with three parts:
 1. the floor is bought;
 2. the contribution is a share (~90%) of the post-reserve money;
 3. it never exceeds the announced top, and anything above stays as project flexibility.

- **Number (Final Report):** state the range with three figures (at least / most likely / at most) and one probability (reaching the top). The team writes the words.
- **Checklist for the co-sponsor passage** (Final Report; the team drafts it):
 - the bottom is already bought;
 - the ten operating payments are fully bought and separate;
 - the most likely amount;
 - the top and its chance;
 - that any excess stays with the project (flexibility), not withdrawn;
 - dollars stated as US\$ (R-AN10).

Tier 2 (IPS rule) and 3 (numbers). **Confidence:** high on the logic; medium on the percentile choice.

Changes current strategy: yes. It refines "an upside with a stated probability" (brief section 7) into a capped range with a share rule. **Criterion:** Creativity and Presentation ("communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors", R-S28).

What this teaches: when part of an outcome is a decision rather than a market result, a rule can remove uncertainty that no model can.

M060. What share of the 2031 sleeve to lock as the floor

The question: 80%, or toward 100%? A near-total lock costs only about \$1k of median but shrinks the range toward one number. Does that dodge the case's uncertainty task?

One-sentence answer: Lock 80-90%. The median hardly moves (\$209k at 50% to \$206k at 100%). The share sets how much of the announced range is already bought: an 80% lock gives a range about \$47k wide, a 90% lock about \$24k wide. A 100% lock turns the range into "one exact amount", which the case says Laura does not want. The 2027-view uncertainty is still shown in the Final Report.

Numbers (D3_range_rules.py section 1; MODEL; p80 top):

Lock share	2033 surplus p5/p50/p95	Bought floor p5/p50/p95	Median range width (p80 top)
50%	\$158k/\$209k/\$280k	\$79k/\$103k/\$135k	\$118k
60%	\$158k/\$208k/\$278k	\$95k/\$124k/\$162k	\$94k
70%	\$159k/\$208k/\$275k	\$111k/\$144k/\$190k	\$71k
80%	\$159k/\$207k/\$273k	\$127k/\$165k/\$217k	\$47k
90%	\$159k/\$207k/\$272k	\$143k/\$186k/\$244k	\$24k
100%	\$159k/\$206k/\$271k	\$159k/\$206k/\$271k	\$0

- **Central case with a 90% lock:** median range US\$183k-\$206k (give-all, capped). At 31.82 TWD/USD (F-510) that is NT\$5.82m-6.56m.
- **Exchange-rate caveat:** a ± 2 -standard-deviation two-year move in the Taiwan dollar exchange rate ($\pm 13.6\%$, using F-513's 6.8% two-year sd since 2006; ASSUMPTION that it repeats) moves the NT\$ value of the \$183k floor between NT\$5.03m and NT\$6.61m. In NT\$ terms, the exchange rate is then a bigger uncertainty than the markets.
- **2031 two-year yield** (80% floor; median floor):

Yield	Median floor
3.0%	\$159.2k
3.5%	\$160.8k
4.0%	\$162.4k
4.81%	\$164.9k

Evidence:

Claim	Source	Status
"Rather than promising one exact amount, she wants to communicate a credible range"	case p.3 L115-116	VRF
"In 2031, however, the value of her portfolio in 2033 remains uncertain."	case p.3 L112-113	VRF
The case never names a currency; the residency is in Taiwan	R-AN10	VRF

Claim	Source	Status
USD/TWD 31.82 (2026-09-18); two-year sd 6.8% since 2006	F-510, F-513 (FRED DEXTAUS)	VP inputs (derived)

Already answered / new: B6b-02's "100% costs about \$1k of median" is confirmed (\$207k → \$206k). New: the table of width against bought bottom; the 3-4% yield test (the floor is \$3-6k lower); and the finding that fixed-multiple tops are fragile (model v2 (f)).

Implication:

- *Number (IPS rule parameter; Final Report range):* the team picks 80% or 90%.
 - Choose 90% if it adopts the cap/share rule (M039): the bottom rises about \$20k and nothing is lost.
 - Keep 80% if it wants a visible upside tier (about \$45k).
- *Final Report chart:* a "funnel" of the facility money seen from 2027 (p5-p95 \$152k-\$277k, central) narrowing to the 2031 range. It shows the uncertainty shrinking on a schedule.

Tier 2. Confidence: medium-high. **Changes current strategy:** no. The parameter is an open item (brief section 9); the evidence favours 80-90%. **Criterion:** Creativity and Presentation (R-S28).

What this teaches: the width of a range is a choice about how much you buy now. It is not only a property of the market.

M053. The joint tail: rates fall before January 2027 AND the 2028 deposit is late or smaller

The question: how large is the joint tail, for example through a 2027 U.S. recession that both lowers yields and shrinks her publishing and speaking income?

One-sentence answer: Small, and best stated in conditional dollars.

- About 30% of rate paths need part of the 2028 deposit to finish the ladder (mean gap \$7k).
- The payments go short only if the deposit is also almost missing: on average \$9k of the 2033 payment (\$31k at a 100bp fall).
- Any deposit of about \$25k covers a 100bp fall.
- A recession link raises the chance of a shortfall when the deposit is missing (30% to 44-69%) but not its size.

Numbers (D3_joint_tail.py; STRIPS ladder; MODEL on VRF/VP inputs). Conditional dollars, no probabilities needed:

Rate move before 1 Jan 2027	Ladder cost	Gap over \$300k	Unbought payment face	Cost to finish in Jan 2028	Unfunded if deposit \$0 / \$10k / \$25k
0	\$294,387	\$0	\$0	\$0	0 / 0 / 0
-25bp	\$301,676	\$1,676	\$2,217 (2033)	\$1,751	\$2,217 / 0 / 0
-50bp	\$309,169	\$9,169	\$11,957 (2033)	\$9,555	\$11,957 / 0 / 0
-75bp	\$316,872	\$16,872	\$21,688 (2033)	\$17,540	\$21,688 / \$9,323 / 0
-100bp	\$324,793	\$24,793	\$31,413 (2033)	\$25,711	\$31,413 / \$19,196 / \$869
-150bp	\$341,313	\$41,313	\$50,886 (2033 and part of 2034)	\$42,633	\$50,886 / \$38,950 / \$21,047

Probabilities (the deposit scenarios are ASSUMPTION; the case says "will"):

Scenario	P(gap in Jan 2027)	P(payments short)	P(short, given the deposit is cut)	Short when short: mean / p95
Deposit arrives, or is late (2029)	30.2%	0%	-	-
Deposit \$10k	30.2%	8.2%	-	\$6.7k / \$18.7k
Deposit missing (markets independent)	30.2%	30.2%	-	\$9.1k / \$23.7k
Deposit missing, sd 0.48pp	34.4%	34.4%	-	\$12.5k / \$32.0k
Cut to \$0 in 25% of paths: independent	30.2%	7.7%	30.4%	\$9.1k / \$23.7k
Same, recession-linked (correlation 0.6; link 0.5)	30.1%	11.2%	44.4%	\$10.3k / \$25.9k

Scenario	P(gap in Jan 2027)	P(payments short)	P(short, given the deposit is cut)	Short when short: mean / p95
Same, extreme (0.9; 0.8)	30.0%	17.2%	69.0%	\$11.3k / \$26.6k

Rule check (-50bp gap; rates then move with sd 0.71pp during 2027):

- Longest-first leaves the 2033 payment for 2028. Its 2028 bill has a 5-95% range of \$9,040-\$10,108.
- Nearest-first leaves the 2042 payment. Its 2028 bill has a 5-95% range of \$8,137-\$11,244.
- So the plan's rule (longest first) carries about one-third of the top-up price risk. This settles brief section 10's "nearest-first vs longest-first" in favour of the current rule.

Evidence on the recession link:

Claim	Source	Status
In 12 S&P 500 years below -10% since 1928, the 10-year Treasury return was positive in 9 (not 1931, 1941, 2022)	Damodaran histretSP (snapshot data/D3/), accessed 2026-09-28	VP (dataset page)
Stock/10-year Treasury return correlation: 1928-2025 +0.02; 1966-99 +0.36; 2000-21 -0.65; 2022-25 +0.88	same	VP (derived)
The 10-year yield changed -124bp in 2002, -179bp in 2008, -99bp in 2020 and +236bp in 2022	FRED DGS10 (accessed 2026-09-28)	VP (derived)
The deposit comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"	case p.2 L44-45	VRF
No source was found for how publishing or speaking income moves in recessions	-	the gap stays UNVERIFIED; the size of the cut is an ASSUMPTION

Implication:

- *Sentence (IPS contingency clause)* must contain:
 1. if the ladder costs more than the first deposit, buy the latest payments first;
 2. the first use of the 2028 deposit is completing the ladder;
 3. no facility floor is announced until all ten payments are bought.
- *Number (Final Report risk register)*: one row, "rates fall before purchase AND the 2028 income is missing", given in conditional dollars (for example "\$12k of the first payment at a 50bp fall") and labelled beyond-case.

Tier 2. Confidence: high on the conditional dollars; low on the joint probability (the input cannot be known). **Changes current strategy:** no. It confirms longest-first and adds one clause. **Criterion:** Portfolio Analysis (funding reliability "under varying market outcomes", R-S26).

What this teaches: when a probability depends on something no one can know (will her income fall?), report "if X, then \$Y" instead of inventing a joint percentage.

M055. Lock the floor progressively (ratchet), or step equity down in 2029-30?

The question: should the floor be locked in stages (2029-31), by funded-status triggers, or by stepping sleeve equity down, given that a -30% year in 2030 cuts the announced floor by about 18%?

One-sentence answer: No. Locking in thirds lifts the bad-case floor but costs upside, and a single 2031 lock with about 42% equity gets the same bad case within \$2k of median. A dated step-down is fully matched by a constant ~51%. A second IPS rule is not earned.

Numbers (D3_ratchet.py; MODEL; the ratchet locks at today's 4y/3y/2y par yields of 4.96/4.94/4.81%, an ASSUMPTION as in B10a):

Design	Locked money for 2033 p5/p50/p95	2033 surplus p5/p50/p95	Locked money after a bottom-10% 2030
Plan: one date, 80% in 2031, 60% equity	\$127k/\$165k/\$217k	\$159k/\$207k/\$273k	\$141k
Thirds 2029/2030/2031	\$136k/\$164k/\$200k	\$169k/\$206k/\$256k	\$156k
Step-down 60/60/50/40 %, one date	\$131k/\$164k/\$207k	\$164k/\$205k/\$260k	\$148k
Constant 50% equity	\$131k/\$164k/\$206k	\$164k/\$205k/\$260k	\$144k
Constant 45%	\$134k/\$163k/\$201k	\$167k/\$205k/\$253k	\$145k

Design	Locked money for 2033 p5/p50/p95	2033 surplus p5/p50/p95	Locked money after a bottom-10% 2030
Constant 40%	\$136k/\$162k/\$197k	\$170k/\$204k/\$247k	\$146k

- **Dominance test** (same p5 of the 2033 surplus): a constant 42% equity sleeve gives \$169k/\$204k/\$250k against the ratchet's \$169k/\$206k/\$256k. The ratchet's edge is \$2k of median, inside model error. The step-down equals a constant 51% (\$164k/\$206k/\$261k).
- **Premise:** a stock year near -30% in 2030 cuts the plan's floor by 22% (median \$129k against \$165k). In the bell-curve model such a year is rare (0.3%); in history it happened in 3.1% of years (M059).

Evidence: the IPS guide asks for "planned changes in the portfolio as future funding needs approach" (IPS guide L46-47, VRF). The single 2031 lock and the 2033 run-down already answer it. B10a-09's numbers are reproduced (\$171k/\$208k/\$259k at 5% bonds; \$169k/\$206k/\$256k at 4%).

Implication:

- *Decision (IPS):* one dated lock (2031). If the team wants more protection in bad years, lower the sleeve equity instead (M034).
- *Sentence (Final Report, alternatives considered):* "ratchet tested and rejected", with the \$2k figure.

Tier 2. Confidence: medium-high. **Changes current strategy:** no. **Criterion:** Investment Strategy ("disciplined planning across Laura's changing time horizons", R-S24).

What this teaches: before adding a rule, check whether a simpler setting of an existing rule gives the same result. Here it does.

Tier 3: the Final Report (4 Dec)

M059. Check the sleeve against real historical paths

The question: should the 2031 floor, the 2033 contribution, and the barbell and ratchet claims be checked against real historical paths (after the Dec 1999 valuation peak, 2000-02, 2008, 2022), not only the i.i.d. lognormal model?

One-sentence answer: Yes, as one or two named rows in the Final Report.

- History hits only the money still invested. A crash after the 2031 lock trims the facility money by about 4-6%. The same crashes in 2028-30 cut it by 19-29%, and a 1929-31 replay by 48%. All ten payments stay bought.
- The lognormal model understates crash years (0.3% against 3.1%), but its 10th percentile matches history's.

Numbers (D3_history_stress.py; S&P 500 total return plus a bond proxy of 50% T-bill + 50% 10-year bond, ASSUMPTION; other years at JPM medians):

Episode (calendar years)	Placed in 2028-30 (before the lock): 2031 floor / 2033 money / vs model median	Placed in 2031-32 (after the lock)
1929-31 Depression	\$86k / \$107k / -48%	\$164k / \$195k / -6%
1973-74 oil shock	\$119k / \$149k / -28%	\$164k / \$194k / -6%
2000-02 dot-com (after the Dec 1999 valuation peak)	\$118k / \$148k / -29%	\$164k / \$199k / -4%
2008 crisis	\$128k / \$160k / -23%	\$164k / \$196k / -5%
2022 stocks and bonds both down	\$133k / \$167k / -19%	\$164k / \$198k / -5%

Model comparison: floor p5 \$127k; 2033 money p5 \$159k, median \$207k.

- **Crash frequency:** a stock year of -30% or worse happens in 0.29% of model years, 1.16% with fat tails (t4), and 3.1% of S&P years 1928-2025 (1931, 1937, 2008). For -20% or worse: model 2.9%, history 6.1%.
- **All 93 six-year windows 1928-2025, rescaled to JPM's medians:**

Design	Worst window	p10	Median
Plan	\$105k (1929 start)	\$164k	\$213k
Ratchet	\$124k	\$168k	\$212k

Design	Worst window	p10	Median
Barbell 50%	\$138k	\$169k	\$223k
Model (plan), for comparison	p1 \$144k	\$169k	\$207k

Raw history (not rescaled): plan worst \$111k, p10 \$175k, median \$227k.

Evidence:

Claim	Source	Status
Annual returns 1928-2025 (S&P 500 incl. dividends, 3-month T-bill, 10-year Treasury)	Damodaran histretSP, https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html (page dated January 2026; accessed 2026-09-28)	VP (dataset page)
"the VCMM may be underestimating extreme negative scenarios unobserved in the historical period"	Vanguard VCMM notes (accessed 2026-09-28)	VP
Teams "should also consider how favorable and unfavorable investment outcomes would affect their recommendations"	case p.4 L148-149	VRF

Already answered / new: the stats skeptic is right that the after-lock effect is arithmetic (20% unlocked × a known fall). The new findings are:

- the before-lock window is where history bites (-19% to -48%);
- the model understates crash years about tenfold;
- the barbell and ratchet help most in a Depression-type run.

Implication:

- *Number (Final Report):* one named-history row (2008 or 2000-02) next to the model's p5, with one line saying the payments are untouched.
- Keep the Depression replay as the honest worst case for the facility money only.

Tier 3. Confidence: medium-high (the bond proxy and calendar-year splitting are approximations).

Changes current strategy: no. **Criterion:** Portfolio Analysis (R-S26).

What this teaches: a model that averages over "typical" years can miss runs of bad years. History is the check that a real sequence would have survived.

The v2 model (extra mandate), in brief

Full tables are in `D3_model_v2_results.md`. The switches:

- (a) rate risk before the January 2027 purchase, using the real Nov-15 STRIPS ladder and the longest-first/top-up rule;
- (b) deposit scenarios: full, \$75k, late, missing, and correlated cuts;
- (c) fund expenses and advisory fees of 0/0.5/1.0% on the sleeve and/or the ladder;
- (d) AC World and short Treasuries (what WInS holds);
- (e) Student-t fat tails;
- (f) the 2031 range: P(within/below/above) by floor share and top rule;
- (g) contribution rules with the flexibility kept.

The verified base reproduces path by path (within \$1). The central case is \$152k/\$204k/\$277k (p5/p50/p95).

Sources (accessed 2026-09-28 unless noted)

- **Official, VRF:**
 - `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (lines cited)
 - `2026_WGY_Investment_Competition_Guide.txt` (p.3 L70-86)
 - `2026_WGY_Investment_Policy-FINAL.txt` (L45-52, L68-70)
 - `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L44-53)
- **Market data, VRF:**
 - `competition/official_market_data/daily-treasury-rates_2026-09.csv`
 - `JPM_LTCMA_2026_US_matrix_USD.pdf` p.2

- **Repo research** (lower authority; the numbers used here were reproduced):
 - `research/verified_2026-09-27/strategy_mc.py`, `official_curve_pv.py`
 - `research/insight_v1/phase_A/fact_register.md` (F-012, F-014, F-106, F-107, F-112, F-114, F-203, F-206, F-312, F-317, F-410, F-508, F-510, F-513)
 - `case_register.md` (R-AN9, R-AN10, R-AN35, R-S24-R-S28)
 - `wins_now/S1_treasury_sleeve.md`, `wins_now/securities_and_allocation_v0.md`
 - `phase_B/B10a_contrarian.md` and `B10b_designer_checks.py` (reproduced)
 - `phase_D/D13a_laura_quotes_verified.json`, `D13c_voice_map.md`
- **Primary web and data (VP):**
 - U.S. Treasury Daily Par Yield Curve 2026: https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&format=csv
 - FRED DGS10/DGS5/DGS2: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10>
 - Damodaran histretSP: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html
 - Vanguard VCMM forecasts (run of 30 June 2026): <https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html>
 - Das, Ostrov, Radhakrishnan & Srivastav (2018), "A New Approach to Goals-Based Wealth Management", *Journal of Investment Management*, Table 1: <https://srdas.github.io/Papers/GBWM.pdf>
- **Laura's words** (D13a VERIFIED-PRIMARY only; Final Report only, at most 1-2, in context; none in the Trading Notes or IPS):
 - B9b-Q02 (A/B testing, résumé);
 - B8b-Q18 ("unstable income", 2021);
 - B9b-Q09 ("[unfinished]" results).

What this teaches

1. **Buy what must be certain; model only what may vary.** The payments and the 2031 floor need no forecast because they are bought. Every probability in this file is about the money that is still invested.
2. **Compare every risk with the best sure alternative.** Against a sure ~\$204k, stocks add range, not much median. Saying so is more honest, and more convincing to a statistics-trained client, than a big expected-return number.
3. **Use rules where a model is weak.** A cap on the contribution turns "how confident are you?" from a model percentage into a promise Laura controls.
4. **Report conditional dollars when the probability cannot be known.** "If rates fall 50bp and the 2028 income vanishes, \$12k of the first payment is unfunded" is checkable. A joint percentage would be invented.
5. **Check models against history, and history against models.** The lognormal model gets the typical bad case right and misses runs of crashes. Named historical rows make the uncertainty concrete for co-sponsors.

Audit corrections (AX1)

Auditor AX1 (cluster auditor, rates and quant), 2026-09-28. D3's text above is unchanged; these are corrections to apply when the strategy team uses this file. Full audit: `research/insight_v1/phase_D/audit_rates_quant.md`. Checks: `.venv/bin/python research/insight_v1/scripts/AX1_quant_audit.py` (sections [1]-[8] below).

What holds. Every D3 script was re-run on 2026-09-28. Every number quoted in this file reproduces exactly. The v2 model matches the verified `strategy_mc.py` path by path (largest difference \$0.54, AX1 [1]). Every source AX1 re-opened says what D3 quotes: treasury.gov curve, FRED DGS10/DEXTAUS, Damodaran, Vanguard VCMM, Das et al. Table 1, the JPM matrix, and the case/guide lines.

Verdict: passes with corrections: 0 blocking, 7 important, 18 minor. The problems are about what the numbers mean, scope, and wording, not arithmetic.

#	Severity	Issue	Correction
C1	important	Scope (brief s17, 2026-09-28). The file is organised by tier 1-3 and never says which deliverable (TN or IPS) each item serves. It also contains Final Report material that s17 excludes: all of Tier 3 (M059); every "Number (Final Report)" chart spec (M034 fan, M028 two-line chart, M243 price-of-certainty chart, M060 funnel); the "alternatives considered" sentence specs (M238, M055); M039's "Checklist for the co-sponsor passage" (s17 names "fundraising excerpt drafts or checklists"); the 2031 dollar ranges and NT\$ figures (M039, M060); M053's risk-register row; and the three Laura-quote uses marked "Final Report only".	Keep, labelled by deliverable. TN: M034 (fix the sleeve equity share before the VT order; the VT note records it); M028 (optional, see C7). IPS, rules to fix before Nov 6: M243 (full lock plus the four reason elements); M053 (contingency clause: longest rungs first; the first use of the 2028 deposit completes the ladder; no facility floor is announced until all ten payments are bought); M060 (lock share 80-90% on one date in 2031); M039 (the method only, see C3); M055 (no ratchet); M238 (the lock timing, see C6); M034 (the "why any stocks" elements; fee note). Later list (after Nov 9), one line each: the charts, the alternatives-considered sentences, the co-sponsor checklist, the 2031 dollar/NT\$ ranges, M059's named-history rows, the Laura-quote uses.
C2	important	A rule result presented as a test result (summary 1, M039). "Within the range holds in 100% of modelled futures and in every fat-tail, low-return and historical test": with a cap, the contribution cannot exceed the top in <i>any</i> model, by definition, so the tests add nothing. "The only exception is a U.S. default" also leaves out three things: Laura overriding the cap, fees charged to the floor (v2 (c)), and a floor that was never bought (ladder unfinished, M053).	Say "inside the range by design (a rule Laura sets), not a model result". The informative number is P(top reached): about 1 in 5 at a p80 top, 1 in 10 at p90. Never state "100% confident". A statistics-trained reader would see it is by construction (SH-04; D5 finding 5; D5 audit C11 says the same).
C3	important	The recommended contribution rule gives up part of the bought floor for no gain (M039 decision). "Give ~90% of the post-reserve money, capped at the top" leaves 10% of the <i>bought</i> floor unpromised. D3's own table (g) row "Floor + half of the excess" produces the same contribution (\$143k/\$186k/\$245k) and the same flexibility (\$15-16k/\$21k/\$30k). But it announces the whole bought floor as the bottom: median \$165k vs \$148k; central case \$163k vs \$146k (AX1 [8]; MODEL). Uncapped, it also answers case L118 ("how favorable and unfavorable market outcomes could affect the amount she can provide", VRF), and it is D5's a/s rule.	Give the team one choice for the IPS method. (a) Bought floor + a share of the rest, uncapped. Confidence is stated on both sides: below the floor only on a U.S. default; above the top about 1 in 5 (model). (b) The same, capped, reported with P(top reached) only. Drop "give 90%" as the way to keep flexibility unless the team wants its bottom below what it has bought. Use one top percentile across files: D3 p80, D5 p85, D6 p90 (D5 audit C10).
C4	important	Median is not "expected" (summary 2, M034). "Stocks in the sleeve buy range, not a bigger expected facility" and "adds only about +\$3k" are medians. The <i>mean</i> (expected) lift of 60% sleeve equity over the all-Treasury benchmark is +\$6.5k (verified inputs), +\$7.0k (central), +\$9.4k (sleeve bonds 4.9%), -\$0.4k (Vanguard 5.2%) and +\$2.0k (0.5% sleeve fee) (AX1 [2]; MODEL). The +\$3.3k median also nets two separate effects. Equity adds +\$12.5-12.9k at the median over the same plan at 0% equity (JPM). But the plan's non-stock money is assumed to earn JPM's 4.0% while the benchmark locks at about 5.1% (F-317), which costs about -\$9k.	Say: under JPM, 60% sleeve equity adds a modest amount over locking everything in Treasuries (about \$3-6k at the median, \$7-9k on average), and nothing on average under Vanguard's lower forecast, while the spread widens a lot (same path: -\$46k / +\$70k). The four IPS sentence elements stand; element 2 becomes "a modest expected gain plus upside for the facility and flexibility".
C5	important	The "riskless control" is not riskless seen from today (M034; summary 2; What this teaches 2, "the best sure thing"). In D3's own central case, its 2033 value is \$170k/\$201k/\$233k (p5/p50/p95). That is because the Sep-Dec 2026 rate move changes both what is left after the ladder and the 2028 lock rate. Without that move it is \$190k/\$201k/\$213k; on verified inputs, \$193k/\$204k/\$215k (AX1 [3]). It becomes sure only on each lock date. Rate moves also change the benchmark but not the plan's sleeve bonds (v2 limit 1), so part of the same-path spread is rate noise, not stock risk.	Rename it "all-Treasury benchmark" and always give its range. In "Mis-posed parts", the rival's central-case p5 is \$170k (+\$18k against the plan's \$152k), not \$193k (+\$34k).

#	Severity	Issue	Correction
C6	important	The barbell is not dominated (M238; summary 6). "Larger barbells are just lower-equity points on the same frontier" is contradicted by D3's own model. At the same whole-portfolio equity in 2028 (6.8%), barbell 80% gives \$186k/\$205k/\$238k. The plan at 20% sleeve equity gives \$177k/\$199k/\$225k (JPM 4.0% bonds) or \$182k/\$205k/\$231k (5.0% bonds), so the barbell is better at every percentile. Barbell 50% (\$163k/\$210k/\$293k) beats the plan at 60% with 5.0% bonds (\$161k/\$210k/\$277k). In D3's own history table, the worst 6-year window is \$176k for barbell 80% against \$105k for the plan (AX1 [4]; MODEL; the barbell's January-2028 lock at 4.98% is an ASSUMPTION). Also, "\$101k against \$165k" compares a 50% lock with an 80% lock. At the same 80% share, the bought floors are \$161k (known in 2028) against \$165k (known in 2031).	Withdraw the frontier sentence and the \$101k-vs-\$165k comparison. Keep "no barbell" only as a judgement, on the two reasons that survive. (i) The barbell's unlocked money is 100% stocks through 2031-32, after Laura has spoken: in a bottom-decile 2031-32, the median falls \$34k at 50% and \$13k at 80%, against \$8k for the plan. (ii) The plan keeps more money growing in 2028-30 ("pursuing growth", case L72-73; whole-portfolio equity 20% against 7-17%). Add one genuine team choice to the IPS decision list: "Is the facility floor bought in January 2028 or January 2031?"
C7	important	"Demonstrably tracks" overstates a model backtest (summary 5, M028). The hedge is the 24-Sep holdings valued on each day's par curve, the same curve that values the payments. No actual fund prices are used. The 155 six-week windows overlap and come from one year (about 6 separate six-week periods).	Say: "In a 2026 holdings-based model backtest, an IEF/TLH mix moved within about $\pm 0.4\%$ of the payments' value over six-week windows, while that value moved $\pm 2-3\%$." If a Trading Note uses a tracking figure, label it as the team's own calculation (WinS screen vs curve model) and use one number only. Never frame it as a gain (Guide L79; TN guide L46-52 ask for reasoning). The log is optional (D3: "useful but not urgent"); skip it if nobody will keep it weekly.
C8	minor	"About 30% of rate paths" is a zero-drift model output. History supports it: the real Nov-15 STRIPS ladder cost $\leq$ \$300k on only 9 of 185 official curve dates in 2026 (first on 2026-09-10) (AX1 [5]). And the FRED 10-year yield fell ≥ 19 bp over 98 days in 35.8% of windows since 1990 (31.6% since 1962) (AX1 [6]). Separately, "the first $\sim$ \$7k of the 2033 payment then waits" mixes cost and face: \$7.0k is the mean January-2027 cost gap, while the unbought part of the 2033 payment averages \$9.1k of face (AX1 [7]; M053's own figure).	Keep "about 1 in 3" (AX1b made the same correction to D1). Add that the ladder cost more than \$300k on almost every day of 2026. Write "about \$9k of the 2033 payment (about \$7k at January-2027 prices)".
C9	minor	M053: "a recession link raises the chance ... but not its size". The mean shortfall also rises, \$9.1k $\rightarrow$ \$10.3k $\rightarrow$ \$11.3k (+13% / +24%). "Any deposit of about \$25k covers a 100bp fall" leaves \$869 unfunded.	"raises the chance a lot and the size a little"; "about \$26k".
C10	minor	M053 rule check: "This settles brief section 10's nearest-first vs longest-first". It settles the price-risk part only (the top-up bill range is about 3x narrower).	Say "supports", and name the other side in one line. Longest-first leaves part of the FIRST payment (2033) waiting. Nearest-first leaves part of the LAST (2042), with more face at risk (\$19.0k vs \$12.0k at -50bp) but nine more years to fix it (AX1 [9]). Longest-first still wins (less face and less price risk). AX1b checked the same rule for D1.
C11	minor	M055: at the same p5 (\$169k), the ratchet beats a constant 42% sleeve by \$2k at the median and \$6k at p95 (\$256k vs \$250k), so it is not dominated. Summary 8's "p5 +\$9k" is +\$10k (\$159k $\rightarrow$ \$169k).	The rejection stands on the brief's design principle: a second IPS rule for \$2-6k does not earn its place. Say that, not "a simpler setting gives the same result".
C12	minor	M059 / summary 7: "understates crash years about tenfold" rests on 3 of 98 historical years at a -30% cut-off. At -20% the gap is about 2x (2.9% vs 6.1%).	"The model has too few very bad years." Final Report material (C1).
C13	minor	M060 has three problems. (a) "The case says Laura does not want one exact amount" is an INTERPRETATION of L115-116; her stated reason is credibility under uncertainty. (b) "Choose 90% ... nothing is lost": with the cap/share rule the announced range narrows to about \$21k (central \$165k-\$186k), close to the single number D3 argues against. (c) "In NT\$ terms, the exchange rate is then a bigger uncertainty than the markets" is true only at a 90% lock: the range width is about \$24k against ± 2 sd FX of about $\pm$ \$25k. At 80% the market width (\$47k) is larger. The 6.8% two-year sd is the since-2006 window; since 1983 it is 9.4% (F-513; AX1 re-derived 6.74% / 9.41% from FRED DEXTAUS).	Label (a) INTERPRETATION. State the width in (b). Make (c) conditional on the lock share and name the window. NT\$ figures belong on the later list (C1).

#	Severity	Issue	Correction
C14	minor	M039: the 2031 range is lopsided by design. In every state the "most likely" figure sits only \$4-6k below the p80 top (\$169k/\$173k; \$207k/\$212k; \$256k/\$262k), while the floor is \$35-52k lower, because the floor assumes the unlocked money is lost entirely. D5 audit C6 found the same.	In the IPS method, call the bottom "the amount already bought", never a low-case forecast.
C15	minor	M039 checklist: "any excess stays with the project (flexibility), not withdrawn" is a second promise. If co-sponsors hear the excess goes to the project, they may count it, and the cap logic fails. The case frames flexibility as Laura's ("her financial flexibility as the project develops", L102-103, VRF).	"Laura decides the use of money above the top in 2033, under the stated rule." The checklist itself is later-list material (C1).
C16	minor	M243 frontier: the unlocked payments are bought in 2033 at the verified flat 5.26% $\pm$ 1pp convention, about \$6.4k dearer than forwards (F-408).	On forwards, a full lock costs about \$0.9-5.2k of median facility money for 95%-70% partial locks (ASSUMPTION arithmetic: (1 - lock share) $\times$ \$6.4k), not \$0.6-3.3k. The conclusion does not change.
C17	minor	M034's WInS weights "VT/VGSH 19.8/13.2% at 60%" treat the 1% cash float as outside the growth money. The ticket's 20.5/12.5% + 1% cash puts it inside (VT \$61,500 vs \$59,400). The model also invests the 2027 leftover 60/40 during 2027, while the ticket holds it in T-bills until 2028 (effect under \$1k).	Use the ticket's convention and recompute on the trade date: VT 20.5 / VGSH 12.5 at 60/40, or VT 17.0 / VGSH 16.0 at 50/50, with cash taken from the short-Treasury part (wins_now/securities_and_allocation_v1.md, which supersedes v0 and was written in parallel). Note the 2027 difference as a model simplification; v1 also keeps the leftover in T-bills.
C18	minor	M238: "The WInS short-Treasury slot stays VGSH, not IBTM" does not follow from the barbell test, because the barbell also moves the rest to 100% stocks. The bond part's instrument was never tested. D3's own benchmark runs suggest that locking the bond part to 2033 is worth about \$4-9k at the median: the 0%-equity plan trails the all-Treasury benchmark by \$9.3k (JPM 4.0% bonds) and \$3.9k (4.9% bonds) (AX1 [2]).	VGSH can stay for the ticket's own reasons (its short Treasuries become the 2031 note; it was on the 2025-26 list). Log "VGSH or a dated Dec-2032 Treasury fund for the growth money's bonds?" as an open team question. IBTM would be PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK . D1 separately proposes a dated WInS rung.
C19	minor	Laura's words (M034, M039 evidence rows). "A/B testing" in a résumé skills list is read as "a frame she uses professionally", and "[unfinished]" on a student-era project as "a working habit of stating uncertainty". Both over-read single words (a tokenism risk; D13 context rules). "More convincing to a statistics-trained client" (What this teaches 2) uses her degree (case L16, VRF) as a persuasion lever.	Downgrade both readings to INTERPRETATION. They are Final-Report-only and out of scope now. Drop the persuasion framing.
C20	minor	Wording: "Floor ... so it is certain by 2033" (Terms) and "answered with a certainty" (summary 1).	"certain in nominal US\$ if held to maturity, barring a U.S. Treasury default"; "a rule removes part of the uncertainty".
C21	minor	Submission-shaped templates: M028's "'the payments' value moved X%, our hedge moved Y%" and the quoted sentence in What this teaches 4 can be pasted as they are.	Restate as elements: which two numbers, over which dates, and whose calculation.
C22	minor	Jargon and false precision. Undefined here: i.i.d., lognormal, Student-t/nu, multivariate, dominance test, forward zero, bootstrap. Dollar-exact model outputs (\$203,997; $\pm$ \$827; 30.2%; 83.7%; \$33,681) are false precision for a deliverable.	Define each term or replace it ("bell-curve model", "fatter-tailed model"). In anything the team drafts from, round to \$1k and "about 1 in 3 / 1 in 5".
C23	minor	Cross-file: D2 recommends 50% sleeve equity (band 45-55) as an IPS rule; D3 says the evidence "does not force a move from 60%". Their numbers agree: 50% vs 60% changes the median by -\$1.5-2k, p5 by +\$5-6k and p95 by -\$11-14k.	Present one decision to the team with both files' numbers. The VT note records whichever share is chosen, decided before the VT order.
C24	minor	The "Vanguard" rows apply Vanguard's U.S. equity midpoint (5.2%) to a global fund, with JPM's volatility (a hybrid ASSUMPTION). Vanguard's own June-2026 ranges (developed ex-U.S. 4.5-6.5%, emerging 2-4%; VP, re-read 2026-09-28) give a VT-weighted midpoint of about 5.1% (D2's blend: 5.08%).	Label the rows "Vanguard U.S. midpoint + JPM volatility (ASSUMPTION)". The Vanguard case is, if anything, slightly generous to equity.
C25	minor	The real ladder cost (\$294,387) is a par-curve model price for the Nov-15 STRIPS (VRF inputs + ASSUMPTION method), not a STRIPS market quote. Real STRIPS yields differ from the fitted curve, so the 19bp headroom is approximate.	Say "about \$294k on today's curve (model price)". Re-price with actual STRIPS quotes on the purchase date (a 2027 action, not a deliverable item).

D3 model v2 results: what each new switch changes

Agent D3 (Quant Modeler), insight_v1 run, Phase D. Written 2026-09-28. This is AI-generated research for Team Caplet: numbers, tables and checklists. None of it is text to submit. The team decides and writes every deliverable in its own words, and records this AI use in the Final Report's Works Cited.

- **Model:** research/insight_v1/scripts/strategy_mc_v2.py. It extends the verified research/verified_2026-09-27/strategy_mc.py.
- **Run it** (from the repo root, about 25 seconds): .venv/bin/python research/insight_v1/scripts/strategy_mc_v2.py
- **Data snapshots** (research/insight_v1/scripts/data/D3/): written by D3_data_snapshot.py.
- **Question scripts:** D3_control_and_equity.py, D3_lock_frontier.py, D3_range_rules.py, D3_funded_status_log.py, D3_barbell.py, D3_joint_tail.py, D3_ratchet.py, D3_history_stress.py.
- **Status labels:**
 - **MODEL** means an ASSUMPTION-based model output, not a forecast.
 - **VRF** means VERIFIED-REPO-FILE.
 - **VP** means VERIFIED-PRIMARY.

Summary (read this first)

1. **The verified base still reproduces, path by path.** With every switch off, v2 matches the verified script within \$1 per path. The 2033 surplus (the money left after the payments are bought) is $p5/p25/p50/p75/p95 = \$159k/\$186k/\$207k/\$232k/\$273k$. The 2031 floor is $\$127k/\$165k/\$217k$, and growth-first misses 3.2% of the time. These are the same numbers as F-401/F-402.
2. **The D3 central case is a little lower and a little wider than the verified base.** The central case turns on:
 - the real Nov-15 STRIPS ladder (\$294,387);
 - rate moves between today and the January 2027 purchase;
 - what WInS actually holds (AC World equity plus short government bonds);
 - fund expense ratios.

It gives a 2033 surplus of **\$152k/\$204k/\$277k** (p5/p50/p95) and a 2031 floor of **\$121k/\$163k/\$220k** (MODEL). Against the verified base, p5 falls by \$7k and the median by \$3k.

3. **The payments are never short in any tested case where the 2028 deposit arrives.** This includes a \$75k deposit and a deposit one year late.
 - The ladder costs more than \$300,000 in January 2027 in about **30%** of rate paths. The mean gap is \$7k and the p95 gap is \$12.5k (MODEL; VP inputs).
 - Any deposit of about \$43k or more fills the gap, even after a 150bp fall in rates (about a 4-standard-deviation move at 2026 volatility).
 - Only a missing or near-zero deposit leaves part of the 2033 payment unfunded: \$9k on average when it happens, \$24k at p95. 4. **Robust numbers:**
 - Payments fully bought whenever the deposit arrives.
 - $P(\text{contribution below the bought floor}) = 0$.
 - A median facility amount of about \$200-210k in every asset, fat-tail and rate switch.
 - Fat tails move p1/p5/p95 by less than about \$7k.
 - Switching the model to what WInS holds moves the median by about +\$1k. 5. **Assumption-driven numbers:**
 - Anything that depends on the 2028 deposit. A \$75k deposit cuts the median by \$98k.
 - Advisory fees charged on the ladder: about -\$11k of median per 0.5% a year.
 - Which forecasting firm's return assumptions are used. Vanguard's midpoint gives about -\$6k of median.
 - The pre-purchase rate move: $P(\text{ladder costs more than } \$300k) = 24\text{-}34\%$.
 - The stated confidence of the range's top. An "80%" top lands 72-84% within the range across the model variants, and 62% on raw U.S. history. With a cap rule it is 100% by design. 6. **Three new**

facts the team should know:

- **A fee on the ladder is a claim on the promise money.** Charging 1% a year on everything costs about \$32k of median surplus. If the deposit is missing, the fees cannot be paid from the leftover money in 100% of paths.
- **The model understates crash years.** It gives a -30% stock year in 0.3% of years; S&P history shows 3.1% (1931, 1937, 2008). Even so, its 10th percentile matches history's.
- **Choosing the top of the range as a fixed multiple of the floor (for example 1.25x) is fragile.** It lands within the range 0%, 44% or 100% of the time depending on the floor share. Set the top from the unlocked money's own risk instead.

1. The model in one paragraph (plain English)

The model simulates 200,000 possible futures, each a set of yearly returns for stocks and bonds drawn from J.P. Morgan's published long-term assumptions. In each future:

- Laura's \$300,000 first buys ten zero-coupon Treasury bonds (Treasury bonds sold at a discount that pay one fixed amount on one date), one for each \$50,000 operating payment in 2033-2042.
- The leftover (the "sleeve"), plus the \$150,000 she adds in 2028, is invested in a stock/bond mix.
- In 2031 most of the sleeve moves into a 2-year Treasury note. That is the "floor" she can promise co-sponsors.
- In 2033 the floor plus the rest is the money for the facility and for flexibility.

Version 2 adds switches for what version 1 assumed away: rates moving before the January 2027 purchase, a smaller, late or missing 2028 deposit, fees, WInS-like funds, fat-tailed returns, rules for the 2031 range, and rules for the 2033 contribution. Every switch is off by default, so the default run is the verified model.

Terms:

- **p5/p50/p95:** 5 in 100 futures end below the p5 number; p50 is the median, the middle outcome.
- **Fat tails:** extreme years happen more often than a bell curve says.
- **Common random numbers:** every run uses the same random futures, so a difference between two runs is caused by the switch, not by luck.

2. Reproduction check (VRF inputs; MODEL outputs)

Quantity	Verified script	v2, all switches off
Ladder cost at 2027-01-01, exact Jan-1 dates	\$292,264	\$292,264 (computed: \$292,263.85)
2033 surplus p5/p25/p50/p75/p95	\$159k/\$186k/\$207k/\$232k/\$273k	identical (largest path difference \$0.54)
2031 floor p5/p50/p95	\$127k/\$165k/\$217k	identical
Growth-first miss rate; \$75k; \$0 deposit	3.2%; 13.7%; 40.8%	identical (exactly)
Equity 50/60/70%	\$164k/\$205k/\$260k; \$159k/\$207k/\$273k; \$154k/\$209k/\$287k	identical

The Nov-15 STRIPS ladder recomputes to **\$294,387**, the same as S1. The forward zero rate from 2027 to 2033 recomputes to 5.08% (F-012).

3. What each switch changes (one at a time, against the verified base; MODEL)

Changes are in \$k of the 2033 surplus. The last column is the chance that part of a payment is unbought.

Switch	Label	Δp5	Δp50	Δp95	P(payments short)
(a) Real Nov-15 STRIPS ladder, no rate move	VRF + method	-2.3	-3.0	-3.9	0%
(a) Rates move before purchase, sd 37bp (2026 volatility)	VP inputs / ASSUMPTION	-4.8	-0.6	+5.1	0%
(a) Both (the central-case setting)		-7.1	-3.6	+1.2	0%
(a) Both, sd 48bp (98-day moves since 1990)		-10.3	-3.8	+4.6	0%

Switch	Label	Δp5	Δp50	Δp95	P(payments short)
(b) Deposit \$75k	ASSUMPTION scenario	-75.5	-98.3	-129.6	0%
(b) Deposit late (arrives January 2029)	ASSUMPTION scenario	-1.0	-11.0	-26.8	0%
(b) Deposit missing	ASSUMPTION scenario	-151.3	-196.6	-258.6	0% without (a); 30% with (a)
(b) Deposit cut to \$0 in 25% of paths, correlation 0.6 with 2027 stocks	ASSUMPTION	-150.6	-13.9	-6.5	0% without (a); 7.6% with (a)
(c) Fund expenses 0.048% (60% VT 0.06% + 40% VGSH 0.03%)	VP	-0.3	-0.3	-0.4	0%
(c) Plus a 0.5% advisory fee on the sleeve	ASSUMPTION	-4.3	-5.3	-6.6	0%
(c) Plus a 1.0% advisory fee on the sleeve	ASSUMPTION	-8.3	-10.2	-12.8	0%
(c) Plus 0.5% on the sleeve and the ladder	ASSUMPTION	-14.6	-16.6	-19.1	0%
(c) Plus 1.0% on the sleeve and the ladder	ASSUMPTION	-28.6	-32.4	-37.3	0%
(d) JPM AC World equity instead of U.S. large cap	VRF	+0.3	+1.3	+3.0	0%
(d) AC World plus JPM short government/credit (the WInS mix)	VRF	-0.1	+1.1	+3.1	0%
(d) AC World plus short Treasuries at 3.5%	ASSUMPTION	-1.2	-0.2	+1.4	0%
(d) AC World plus short Treasuries at 4.9% (forward-implied)	ASSUMPTION	+1.9	+3.6	+6.2	0%
(d) Vanguard-midpoint U.S. equity return, 5.2%	VP range 4.2-6.2%	-4.7	-6.0	-7.8	0%
(e) Student-t, nu=4, same variance	ASSUMPTION	+2.0	-0.5	-3.0	0%
(e) Student-t, nu=4, scaled to the same 5% bad year	ASSUMPTION	-1.2	-0.3	+4.3	0%
2031 2-year yield 3.0% instead of 4.81%	ASSUMPTION	-4.4	-5.7	-7.4	0%

Reading the table. The deposit dwarfs everything else. Fees on the ladder come next, then the choice of forecasting firm and the 2031 yield. Fat tails and the choice of funds barely matter over six years. The central case (row 3 of the table, plus (d) WInS mix and fund expenses) is \$152k/\$204k/\$277k.

4. Switch by switch

(a) Rate risk between today and the January 2027 purchase

Rule modelled:

- If the ladder costs more than \$300,000, buy the longest payments first with the \$300,000.
- Buy the rest in January 2028 from the deposit, at January-2028 prices.

Rates: parallel moves with zero drift from today's forward prices (ASSUMPTION). The standard deviation is 0.37pp to January 2027: 71bp a year (F-014; recomputed from FRED DGS10 as 70.6bp) times $\sqrt{0.27}$. The sensitivity run uses 0.48pp, the standard deviation of 98-day moves in the 10-year yield since 1990 (FRED DGS10, VP inputs, derived). After the purchase, yields move with a standard deviation of 0.71pp a year; that only affects top-up prices.

Ladder	Pre-purchase sd	P(cost > \$300k)	Mean gap if > 0	p95 gap	P(short), deposit arrives	2033 surplus p5/p50/p95
Exact dates (\$292,264)	0	0%	-	-	0%	\$159k/\$207k/\$273k
Exact	0.37pp	23.8%	\$6,579	\$10,446	0%	\$154k/\$207k/\$278k
Exact	0.48pp	29.2%	\$9,195	\$16,095	0%	\$151k/\$206k/\$282k
STRIPS (\$294,387)	0	0%	-	-	0%	\$157k/\$204k/\$269k
STRIPS	0.37pp	30.2%	\$7,022	\$12,458	0%	\$152k/\$204k/\$275k
STRIPS	0.48pp	34.4%	\$9,690	\$18,070	0%	\$149k/\$203k/\$278k
STRIPS plus a 0.25% broker mark-up (ASSUMPTION, unsourced)	0.37pp	32.6%			0%	\$151k/\$203k/\$273k

Cost of the STRIPS ladder by rate move before purchase (VRF curve; same bootstrap method):

Move	Ladder cost
+25bp	\$287,297
0	\$294,387
-19bp	\$299,908 (the breakeven)
-25bp	\$301,676
-50bp	\$309,169
-75bp	\$316,872
-100bp	\$324,793
-150bp	\$341,313

Reading. The 24% in F-112 was for the exact-date ladder. With the real ladder the headroom is 19bp, not 27bp, so about 3 rate paths in 10 need part of the 2028 deposit before the promise is complete.

(b) 2028 deposit scenarios

Lock-early runs use the STRIPS ladder with the pre-purchase sd at 0.37pp. The growth-first ("G") runs use the verified convention.

Scenario	L: P(payments short)	L: mean short if short	L: 2033 surplus p5/p50/p95	G: miss
Full \$150k, 2028 (case)	0%	-	\$152k/\$204k/\$275k	3.2%
\$75k, 2028	0%	-	\$73k/\$105k/\$149k	13.7%
Late: \$150k in 2029	0%	-	\$150k/\$193k/\$249k	2.9%
Missing	30.2%	\$9,114 of payment face	\$0k/\$8k/\$33k	40.8%
Cut to \$75k in 25% of paths, correlation 0.6 with 2027 stocks	0%	-	\$87k/\$189k/\$268k	7.0%
Cut to \$0 in 25% of paths, correlation 0.6	7.6%	\$9,043	\$0k/\$189k/\$268k	15.4%
Same, plus 2026 rates linked to a coming 2027 recession (link 0.5)	11.2%	\$10,327	\$0k/\$191k/\$271k	15.4%
Extreme: correlation 0.9, link 0.8	17.2%	\$11,327	\$0k/\$194k/\$274k	16.9%

Correlation sensitivity (deposit cut to \$0 in 25% of paths):

Correlation with 2027 stocks	0.0	0.3	0.6	0.9
G's miss rate	12.6%	14.1%	15.4%	16.9%
L's shortfall (a (b)-only effect)	7.6% at every correlation			

L does not respond to this correlation alone. It responds only when the rate fall before January 2027 is also linked to the 2027 downturn (the "link" rows above).

Reading.

- The correlation is an ASSUMPTION. History gives a range: the stock/10-year-Treasury return correlation was +0.36 in 1966-99, -0.65 in 2000-21 and +0.88 in 2022-25 (Damodaran annual data, VP; `D3_joint_tail.py`).
- The case says the deposit "will" arrive (R-AN35), so every deposit row is a beyond-case stress. Label it that way.

(c) Fees

- Fund expenses are VP (issuer pages via the S3 ticket). Advisory fees are ASSUMPTION scenarios: the case names no fee and the firm is fictional.
- Fees on the ladder are paid out of the sleeve. A buy-and-hold zero-coupon ladder pays no income to cover a fee.

Fund expense	Advisory fee	Charged on	Surplus p5/p50/p95	Change in median
0	0	-	\$159k/\$207k/\$273k	-
0.048%	0	-	\$159k/\$207k/\$273k	-\$0.3k
0.048%	0.5%	sleeve	\$155k/\$202k/\$267k	-\$5.3k

Fund expense	Advisory fee	Charged on	Surplus p5/p50/p95	Change in median
0.048%	1.0%	sleeve	\$151k/\$197k/\$261k	-\$10.2k
0.048%	0.5%	sleeve + ladder	\$145k/\$191k/\$254k	-\$16.6k
0.048%	1.0%	sleeve + ladder	\$131k/\$175k/\$236k	-\$32.4k

- With the deposit arriving, fees never dent the bought 2031 floor (0% of paths).
- **Stress:** 1% on everything with the deposit missing. The fees exceed the leftover money in 100% of paths, by \$12.2k on average. They would have to come out of the payments' money.
- The riskless control (Treasuries Laura could buy herself) pays no fee and has a median of **\$204k**. A 0.5% sleeve fee takes away the whole median gain from equity (see M034 in `D3_quant.md`).

(d) Assets: what the sleeve holds

All rows are VRF JPM rows unless labelled.

Equity	Sleeve bonds	Surplus p5/p50/p95	2031 floor p5/p50/p95
U.S. large cap 6.70%	Intermediate Treasuries 4.00% (verified)	\$159k/\$207k/\$273k	\$127k/\$165k/\$217k
AC World 7.00%	Intermediate Treasuries	\$159k/\$209k/\$276k	\$127k/\$166k/\$219k
AC World	Short government/credit 4.00%, vol 1.63% (closest JPM row to VGSH)	\$159k/\$208k/\$276k	\$127k/\$166k/\$219k
AC World	Short Treasuries 3.5% (ASSUMPTION)	\$158k/\$207k/\$275k	\$126k/\$165k/\$218k
AC World	Short Treasuries 4.9%, forward-implied (ASSUMPTION)	\$161k/\$211k/\$279k	\$128k/\$167k/\$221k
Vanguard midpoint 5.2% (VP)	Intermediate Treasuries	\$155k/\$201k/\$265k	\$124k/\$161k/\$211k

- AC World correlations: 0.00 with intermediate Treasuries and 0.21 with short government/credit.
- U.S. large cap vs short government/credit: 0.16. All from the JPM matrix, p.2 (VRF).
- Vanguard source: U.S. equities "declined from a range of 4.9%–6.9% to a range of 4.2%–6.2%" (VCMM run of 30 June 2026; corporate.vanguard.com; VP, re-read 2026-09-28).

Reading. Moving the model to what WInS holds (VT-like and VGSH-like funds) changes the median by about +\$1k. This closes S3's red-team finding 10: the effect is below model noise.

(e) Fat tails: Student-t robustness

Setup:

- Multivariate t: stocks and bonds share the same extreme years.
- The median log return is unchanged. The draws are clipped at 8 standard deviations so the mean stays finite.
- The last row is scaled so its 5% bad year matches the bell-curve model's.

Tails	One-year stock p1/p5	Simulated arithmetic mean	2033 surplus p1/p5/p50/p95	Floor p5
Normal (verified)	-25.1% / -16.9%	7.94%	\$144k/\$159k/\$207k/\$273k	\$127k
t, nu=6	-27.8% / -16.2%	7.95%	\$142k/\$160k/\$207k/\$273k	\$128k
t, nu=4	-28.8% / -15.2%	7.92%	\$141k/\$161k/\$207k/\$270k	\$129k
t, nu=3	-28.4% / -13.2%	7.83%	\$142k/\$164k/\$207k/\$265k	\$131k
t, nu=4, x1.09	-31.4% / -17.0%	8.14%	\$137k/\$158k/\$207k/\$278k	\$126k

Reading. Fat tails that are random from year to year change the 2033 outcome by only a few \$k:

- six years of returns average out;
- only the sleeve is at risk;
- the payments are already bought.

What they do NOT capture is a run of bad years, and history shows such runs. For that test see `D3_history_stress.py` (M059 in `D3_quant.md`). Vanguard's own notes warn that its model "may be underestimating extreme negative scenarios unobserved in the historical period" (VP, 2026-09-28).

(f) The 2031 co-sponsor range

Setup:

- Laura buys a floor with share *a* of the sleeve on 1 January 2031.
- The top is set in 2031 as: floor + unlocked money × the p-th percentile of its 2-year growth under the planning model.
- The table assumes the contribution is the whole post-reserve surplus. Cap and share rules are in (g).

Floor share	Top rule	Median floor	Median top	Median width	P(within)	P(below)	P(above)
60%	p80	\$124k	\$218k	\$94k	80%	0%	20%
70%	p80	\$144k	\$215k	\$71k	80%	0%	20%
80%	p80	\$165k	\$212k	\$47k	80%	0%	20%
80%	p90	\$165k	\$215k	\$50k	90%	0%	10%
90%	p80	\$186k	\$209k	\$24k	80%	0%	20%
100%	any	\$206k	\$206k	\$0	100%	0%	0%

Two top rules to avoid:

- **"Top = the unlocked money keeps its 2031 value"** is within the range only 19% of the time, because that money usually grows.
- **"Top = 1.25 × floor"** is within 0% of the time at a 60-70% floor share, 44% at 80% and 100% at 90%. A top set as a fixed multiple of the floor is not tied to the money that is still at risk.

Is the stated confidence model-proof? The range is set with the planning model (80% floor, no cap). The outcomes come from:

Outcome model	P(within), p80 top	P(within), p90 top
Same model	80.0%	90.0%
t, nu=4	83.7%	92.1%
t, nu=4 x1.09	81.7%	90.5%
Vanguard midpoint	83.2%	92.0%
Stocks 1.7pp a year weaker	83.8%	92.4%
Stocks 3pp a year stronger	72.2%	84.8%
0.5% fee on the ladder	92.6%	96.8%
History 1928-2025, raw (97 two-year periods; S&P 500 + bond proxy)	61.9%	81.4%
History, rescaled to JPM's medians	76.3%	90.7%

- P(below the floor) = **0** in every row. The floor is bought.
- When the stated confidence is wrong, it is wrong on the upside: outcomes land above the top, which is good news for co-sponsors.

2031 two-year yield (M060 caveat), median floor at 80%:

Yield	Median floor	Change vs 4.81%
3.0%	\$159.2k	-\$5.7k
3.5%	\$160.8k	
4.0%	\$162.4k	-\$2.5k
4.81%	\$164.9k	-

(g) The 2033 contribution rule and the flexibility kept

Setting: 80% floor, top at p80, verified base.

Rule	Contribution p5/p50/p95	Flexibility kept p5/p50/p95	Median flexibility share	P(flexibility = 0)	P(within range)
Give everything	\$159k/\$207k/\$273k	\$0/\$0/\$0	0%	100%	80%
Give everything, capped at the top	\$159k/\$207k/\$272k	\$0/\$0/\$6k	0%	80%	100%
Give 90% of the surplus	\$143k/\$187k/\$246k	\$16k/\$21k/\$27k	10%	0%	80%
Give 80% of the surplus	\$127k/\$166k/\$219k	\$32k/\$41k/\$55k	20%	0%	80%
Give 90%, capped	\$143k/\$186k/\$245k	\$16k/\$21k/\$30k	10%	0%	100%

Rule	Contribution p5/p50/p95	Flexibility kept p5/p50/p95	Median flexibility share	P(flexibility = 0)	P(within range)
Give 80%, capped	\$127k/\$165k/\$217k	\$32k/\$42k/\$56k	20%	0%	100%
Floor + half of the excess above it	\$143k/\$186k/\$245k	\$15k/\$21k/\$30k	10%	0%	80%

The same rules in the central case (D3_range_rules.py section 6):

Floor share	Top	Rule	Median floor	Median top	P(within)	P(top reached)	Contribution p5/p50/p95	Flexibility p5/p50/p95
80%	p80	give 90%, capped	\$146k	\$188k	100%	20%	\$136k/\$183k/\$248k	\$15k/\$21k/\$30k
90%	p80	give 90%, capped	\$165k	\$186k	100%	20%	\$137k/\$183k/\$248k	\$15k/\$21k/\$28k
90%	p80	give everything, capped	\$183k	\$206k	100%	20%	\$152k/\$204k/\$275k	\$0/\$0/\$3k

Purchasing power (median, "give 80%" rule): \$166k nominal in 2033.

- In 2026 U.S. dollars at JPM's 2.5% inflation (F-312), that is **\$140k**.
- Deflated by the Taiwan construction-cost trend of 3.54% a year (F-508, 2021-2026 average), that is **\$130k**.

This is ASSUMPTION arithmetic. The case asks teams to explain inflation's effect on projections and facility costs.

5. Robust vs assumption-driven (what can be said with confidence)

Number or claim	Robust?	What drives it	How to present it
All ten payments bought in January 2027 at a known price	Robust (market prices, VRF)	U.S. Treasury credit only	"bought, not forecast"
The payments are complete once any deposit $\geq$ ~\$43k arrives	Robust across rate paths (the gap is \$41k at -150bp, a 4-standard-deviation move; finishing it in January 2028 costs about \$43k)	How far rates fall before January	Give the conditional dollar table (M053)
P(ladder costs more than \$300k in January 2027) $\approx$ 30% (24-34%)	Assumption-driven (rate volatility)	Rates in Sep-Dec 2026	"about 1 in 3 rate paths"; re-check on the purchase date
P(contribution below the bought floor) = 0	Robust (bought)	U.S. credit; no ladder fees charged to the floor	State it as certainty in nominal U.S. dollars
Median facility money $\approx$ \$200-210k	Fairly robust (\$197-211k across the (a), (d), (e) and 2031-yield switches)	Mostly bond yields, not stocks	"most likely about \$200k"
p5 of the facility money $\approx$ \$150-160k	Model-dependent (history's worst 6-year window: \$105k)	2027-2030 stock returns	Show one named historical row next to it (M059)
p95 $\approx$ \$270-280k	Model-dependent	Equity return assumption	Label as model
"80% / 90% confident the contribution lands in the range"	Assumption-driven (62-93% across models and history)	The top percentile and the return model	Round it ("about 8 / 9 in 10"), or use a cap so within = 100% and state only the chance of reaching the top
Anything involving the 2028 deposit being cut	Assumption-driven, beyond the case	An unknowable probability	Conditional dollars, not joint probabilities
Fee effects	Arithmetic, robust given the fee	The fee the firm charges	Say what the fee is charged on

6. What v2 still assumes away (known limits)

1. **Sleeve bond returns are independent of the modelled rate moves.** The shifts only price the ladder, top-ups and locks. JPM's 4.0% intermediate-Treasury return is also about 1pp below today's yields (F-317); the forward-implied switch in (d) tests this.
2. **Returns are the same distribution every year.** There is no momentum, no valuation regime and no mean reversion. History (M059) is the check.

3. **Only parallel moves of the whole curve before purchase.** A twist would change the ladder cost somewhat differently. The rungs sit between 6 and 15 years, so the error is small.
4. **The 2031 two-year yield is fixed** (sensitivity 3-4.81%). **The 2028 barbell lock rate is fixed** at 4.98% (sensitivity 4.0%).
5. **No fund price noise, bid/ask spread or premium/discount.** STRIPS broker mark-ups are unknown (0.25% tested).
6. **Money is in nominal U.S. dollars.** Taxes are out of scope (case). Taiwan-dollar exchange-rate risk is not in the model; F-513 gives a 2-year standard deviation of 6.8% since 2006.
7. **The historical bond proxy** (50% T-bill + 50% 10-year bond) is an ASSUMPTION.

7. Reproduce

From the repo root:

1. `.venv/bin/python research/insight_v1/scripts/D3_data_snapshot.py --offline` (re-parses the saved data)
2. `.venv/bin/python research/insight_v1/scripts/strategy_mc_v2.py` (the tables in sections 2-4)
3. The question scripts listed at the top (sections (f) and (g) detail: `D3_range_rules.py`)
 - Seed 20260927, 200,000 paths.
 - New random streams: `SeedSequence([20260927, k])` for $k = 1$ (rates), 2 (deposit), 3 (fat-tail mixing).

Sources (accessed 2026-09-28 unless noted)

- Repo, VRF:
 - `competition/official_market_data/daily-treasury-rates_2026-09.csv` (row 09/25/2026)
 - `competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf` p.2 (rows and correlations above)
 - `research/verified_2026-09-27/strategy_mc.py`, `official_curve_pv.py`
 - `research/insight_v1/wins_now/S1_treasury_sleeve.md` (STRIPS dates and costs)
 - `research/insight_v1/wins_now/securities_and_allocation_v0.md` (expense ratios)
 - `research/insight_v1/phase_A/fact_register.md` (F-012, F-014, F-106, F-112, F-312, F-317, F-508, F-513)
 - `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`
- U.S. Treasury Daily Par Yield Curve 2026 (VP): https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&format=csv
- FRED DGS2 / DGS5 / DGS10 (VP): <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10>
- Damodaran, "Historical Returns on Stocks, Bonds and Bills: 1928-2024" (page dated January 2026, rows to 2025; VP dataset page): https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html
- Vanguard, "Vanguard Capital Markets Model forecasts" (VCMM run of 30 June 2026; VP): <https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html>

What this teaches

- **A model is a machine for asking "what if", not for predicting.** The most useful output here is the tornado table in section 3. It shows which assumptions matter (the deposit, fees, the return forecast) and which do not (fat tails, the exact fund). That tells the team where to spend its words.
- **A number you have already bought needs no model.** The payments and the 2031 floor are certain in dollars because they are purchased. Everything the model says is about the money that is still invested.
- **An honest range is set from the money still at risk.** A rule of thumb like " $1.25 \times$ the floor" is not.

Audit corrections (AX1)

Auditor AX1 (cluster auditor, rates and quant), 2026-09-28. D3's text above is unchanged. Full audit: `research/insight_v1/phase_D/audit_rates_quant.md`. Checks: `.venv/bin/python research/insight_v1/scripts/AX1_quant_audit.py`. The companion list for `D3_quant.md` (C1-C25) applies here too wherever the same claims appear.

What holds. `strategy_mc_v2.py` was re-run and reproduces every table in sections 2-4. With all switches off it matches the verified model path by path (largest difference \$0.54 for lock-early, \$0.00 for growth-first; AX1 [1]). Inputs re-checked on 2026-09-28:

- JPM matrix rows and correlations (repo PDF p.2);
- the treasury.gov 09/25/2026 row (live = repo file);
- FRED DGS10: 70.6bp a year in 2026; 0.48pp sd of 98-day moves since 1990;
- Damodaran rows (2008, 2022, 2025);
- the Vanguard VCMM wording (both quotes verbatim).

Verdict: passes with corrections: 0 blocking, 3 important, 7 minor.

#	Severity	Issue	Correction
V1	important	Section 5, row 1 contradicts section 4(a). "All ten payments bought in January 2027 at a known price: Robust". But in about 30% of rate paths (24-34%, and 1 in 3 on history, AX1 [5]-[6]) the ladder costs more than \$300k. Part of the 2033 payment is then bought in January 2028, at a price not known today. The same slip appears in "What this teaches" 2 ("certain in dollars because they are purchased").	"All ten bought by January 2028 once the deposit arrives. All ten bought in January 2027 in about 2 of 3 rate paths." The robust statement is "bought, not forecast, from each purchase date; certain in nominal US\$ barring a U.S. Treasury default".
V2	important	Capped rules shown as "P(within) 100%" (summary 5; tables (g) and "the same rules in the central case"; section 5 row 8). That 100% is a rule, not a model result. The "Floor + half of the excess above it" row gives the same contribution and flexibility as "Give 90%, capped" but announces the whole bought floor as the bottom: median \$165k vs \$148k; central case \$163k vs \$146k (AX1 [8]).	Report capped rules only with P(top reached). Put "Floor + a share of the excess, uncapped" beside the capped rule as the team's choice, with its two-sided confidence (below the floor only on a U.S. default; above the top about 1 in 5). See D3_quant C2-C3 and D5 finding 5.
V3	important	Section 4(c): "The riskless control ... has a median of \$204k. A 0.5% sleeve fee takes away the whole median gain from equity." Seen from today the control is not riskless: central case \$170k/\$201k/\$233k (AX1 [3]). And the median gain is not the expected gain: the mean lift of 60% equity is +\$7.0k with no fee and +\$2.0k after a 0.5% sleeve fee (AX1 [2]).	"All-Treasury benchmark (sure only after each lock date)". "A 0.5% sleeve fee removes the median gain and most of the average gain." See D3_quant C4-C5.
V4	minor	Summary 3 and section 5 row 2: "any deposit of about \$43k ... even after a 150bp fall (about a 4-standard-deviation move) ... Robust". "4 standard deviations" is bell-curve language. The 10-year actually fell 174bp in 98 days (23 Sep to 30 Dec 2008). That fall needs about \$51k (gap \$49.6k; finishing in January 2028 \$51.0k, with no further fall in 2027). Falls of ≥ 150 bp over 98 days occurred in 0.11% of windows since 1990 and 1.43% since 1962 (AX1 [6]; FRED VP, derived).	"About \$43k covers a 150bp fall and about \$51k the worst 98-day fall since 1990, if rates do not fall further during 2027." Drop "robust".
V5	minor	Summary 5: "P(ladder costs more than \$300k) is 24-34%". 24% is the exact-date ladder, which Laura would not buy. The real ladder gives 30-34%, and history gives 35.8% for a ≥ 19 bp fall (since 1990). In 2026 the real ladder fit under \$300k on only 9 of 185 days (AX1 [5]-[6]).	"About 1 in 3" (AX1b made the same correction to D1).
V6	minor	Robustness wording. Summary 4 says "fat tails move p1/p5/p95 by less than about \$7k", but the t3 run moves p95 by -\$8k. Section 5 says "median $\approx$ \$200-210k (\$197-211k across the (a), (d), (e) and 2031-yield switches)", but the listed switches span \$201-211k; \$197k comes from the fee and Vanguard-central runs.	"About \$8k"; "\$201-211k across those switches; \$197k with a 1% sleeve fee".
V7	minor	Section 4(g) purchasing power uses the "give 80%" rule, while D3 recommends "give 90%, capped". It also deflates a US\$ amount by a NT\$ construction-cost index without the exchange rate (F-513).	Final Report material (brief s17): move to the later list. If kept, say which rule it uses and that USD/TWD is left out.
V8	minor	Section 6 limit 3: "the error is small" for parallel-only rate moves is not computed.	Say "not computed". For scale, S3/S4 found \$1.5-3.8k per 50bp twist for the WInS hedge (a different object).

#	Severity	Issue	Correction
V9	minor	Scope (brief s17). This is a model report, so it may stay as documentation, but it must say which deliverable it feeds. Its inputs to the IPS rules are (a) the payment-completion clause, (b) fees and what they are charged on, and (f)/(g) the method for the 2031 range and contribution. Its only TN input is via M034 (sleeve share) and M028. The dollar tables in (f)/(g) and section 5's "how to present it" column are Final-Report presentation.	Add a one-line "serves: IPS (rules); TN (via M034/M028)" header when the main loop merges. Move the presentation column and the dollar ranges to the later list.
V10	minor	Jargon: "tornado table" (What this teaches), "multivariate t", "clipped at 8 standard deviations", "common random numbers" (defined), "forward-implied".	Define "tornado table" (a list of switches sorted by how much each moves the result) and the others in one line each, or drop them.

Part 3 - The research record, phase by phase · research/insight_v1/phase_D/D4_taiwan_fx.md

D4 - Taiwan, FX and geopolitics: what Laura's facility money will actually buy (question M095)

Agent D4, run insight_v1, 2026-09-27. AI brainstorming output for the team: specifications, evidence and numbers, not submission-ready text. Status labels: VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SNIP), ASSUMPTION (ASM), DERIVED (computed by the script from the labelled inputs). Script: research/insight_v1/scripts/D4_facility_purchasing_power.py (run from the repo root; a full run on 2026-09-27 used live FRED copies and the live DGBAS construction-cost platform).

Terms used below:

- **NT\$** = New Taiwan dollar. **US\$** = U.S. dollar. "NT\$ per US\$" going **up** means the Taiwan dollar got weaker, so each US\$ buys **more** in Taiwan.
- **CCI** = Taiwan's official construction cost index (DGBAS), 2021 = 100.
- **Drift** = the expected, one-directional change (costs usually rise). **Spread** = the two-sided uncertainty around it (measured by the standard deviation, "sd", or the 5th-95th percentile band, "p5/p95").
- **Building power** = how much building Laura's money buys = US\$ amount x (NT\$ per US\$) / construction costs. In logarithms the three pieces simply add, so they can be compared side by side.

Summary: top findings (ranked by impact onreaching the semifinals and on making the plan hers)

1. **The answer depends on when you stand. The team has been comparing a 6-year cost drift with a 2-year FX band. Fix the horizon and the ranking flips (DERIVED).**

Vantage	Market (US\$ pool)	USD/TWD	Construction costs
Today -> 2033 (the Final Report's projection, 6.3 years)	largest spread: sd 0.16, p5/p95 -23%/+32%	sd 0.06 (since 2006) to 0.16 (since 1983)	largest expected effect: drift -11% to -20% building power; spread sd 0.11
2031 -> 2033 (what Laura tells co-sponsors)	smallest: sd 0.027, -3.8%/+5.1% (80% locked)	largest spread: sd 0.068, -11.6%/+11.4%	drift -4% to -7%; spread sd 0.058

- Variance shares (independent, ASM): 2026 view market 61% / FX 9% / construction 29%; **2031 view market 8% / FX 53% / construction 39%**.

- So: by the time Laura speaks in 2031, her own portfolio is the **least** uncertain part of what the gift will buy.

The locked floor did its job. What is left is the exchange rate and Taiwan building costs, which no US\$ portfolio can remove.

2. **Construction inflation is a drift, not a coin toss: say it as an assumption, with a number (case p.4 L147, R-C85).** A fixed US\$ amount buys about **11-12% less** building by January 2033 if Taiwan costs rise at CPI-like 1.9%/yr or the 2008-2026 CCI average (2.11%/yr), and about **20% less** at the post-2021 pace (3.54%/yr). The 2026 spike (+6.53% y/y) would mean -33% but should not be projected (F-509). This is the one assumption line the case explicitly asks for and that most teams will skip.
3. **The US\$ range stays binding; any NT\$ figure is "an illustration at a dated rate" with its own, wider band.** In the 2031 view the NT\$ band is about **-11%/+13%**, roughly 2.5-3 times the US\$ band (-3.8%/+5.1%). The bought US\$ floor is certain in US\$, but in NT\$ it would have come in *below* its dated NT\$ value in 51% of past 2-year windows since 2006, by more than 10% in 7%. B6b's guess is confirmed.
4. **Don't lock NT\$ in 2031; don't tilt to Taiwan (confirms the WInS ticket).** Locking NT\$ two years ahead costs about **5.9%** at today's rate gap (Taiwan 2y 1.66% is SNIP), roughly one FX standard deviation, and it would have paid off in only 23% of past 2-year windows. Keeping US\$ until contracts

are signed also keeps her flexibility. Adding Taiwan stocks (EWT) is not a currency hedge worth having (S2 already dropped it).

5. **Two natural cushions and one reinforcer (DERIVED, small samples).** (a) The Taiwan dollar tends to weaken when U.S. stocks fall (monthly correlation -0.38, 10 years). So in NT\$ terms a bad U.S. market is partly cushioned; the 2026-view combined sd falls from 0.209 to 0.189. (b) The only real Taiwan Strait crisis in the data (1995-96 missile tests) moved NT\$ per US\$ by at most +4.4%, and the August 2022 drills by about 0%. Both are inside normal 2-year noise. (c) Reinforcer: years with a weaker NT\$ have tended to be years of lower construction inflation (annual correlation -0.36, n=18). That widens the building-power band slightly. None of this covers a blockade or war, which is outside the data (ASM).

Does M095 change the current strategy? No change to the portfolio, the lock or the WInS ticket. It **adds** one FR assumption line (building-cost drift), one FR currency rule (US\$ binding, NT\$ dated), a quantitative reason for not converting to NT\$ in 2031, and one possible FR chart. **Deadline tier 3 (Final Report, Dec 4).**

M095. Which is largest for what the facility money buys by 2033: market risk, USD/TWD, or Taiwan construction inflation? Does any NT\$ figure need its own band or an "illustration at a dated rate" label?

Tier 3 (Final Report). Criterion served most: Client Knowledge and Objectives (it tells her what her gift will really buy where she is building it). Secondary: Creativity and Presentation (the case's own standard of communicating "investment uncertainty clearly and credibly to prospective co-sponsors").

Is the question well posed? Partly. The skeptics were right about two things. (1) The horizons must match: the original comparison put a 7-year cost drift next to 2-year FX and market bands (stats skeptic). (2) The *labelling* half (US\$ on every figure, NT\$ dated) was already specified in BS-04 (answered skeptic). The new work here is the **ranking on matched horizons, the size of the NT\$ band, and the drift-vs-spread split**. None of this existed before.

One-sentence answer

Seen from today, U.S. market risk is the biggest *uncertainty* and Taiwan construction inflation is the biggest *expected* loss of building power (about -11% to -20% by 2033). Seen from Laura's 2031 conversation, the locked floor has made the market the smallest factor, and the exchange rate (sd ~7%) and building costs dominate. So the range is quoted in US\$ as binding, with one purchasing-power assumption line, and any NT\$ figure is labelled an illustration at a dated rate, with a band about 2.5-3 times as wide.

Evidence

#	Claim	Source (access 2026-09-27)	Status
E1	Case requires assumptions on "the effect of inflation on portfolio projections and facility costs"	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt L146-148 (R-C85)	VRF
E2	Teams "are not expected to estimate that cost"	same file L164 (R-C95)	VRF
E3	"Teams must recommend the dollar range Laura should communicate"; no currency named anywhere	same file L117; R-AN10	VRF
E4	Residency is "in Taiwan"	same file L77	VRF
E5	2033 facility pool (lock-early, 60% equity sleeve, 80% floor): p5/p50/p95 \$159k/\$207k/\$273k	research/verified_2026-09-27/strategy_mc.py re-run	VRF model; lognormal returns ASM
E6	2031 view at the median 2031 sleeve (\$187.6k): 2033 p5/p50/p95 \$199.0k/\$206.9k/\$217.4k (-3.8%/+5.1%)	D4 script, B6b method	DERIVED (ASM: 2y yield 4.81% in 2031)
E7	USD/TWD 31.82 on 2026-09-18; daily series from 1983	https://fred.stlouisfed.org/graph/fredgraph.csv?id=DEXTAUS	VP (F-510)
E8	2-year moves since 2006: p5/p50/p95 -11.6%/-0.2%/+11.4%, sd 0.068; 6.3-year since 2006 -11.6%/-0.4%/+6.9%, sd 0.064; since 1983: 2y -17.5%/+14.5% (sd 0.096), 6.3y -31.7%/+28.4% (sd 0.165)	D4 script on E7 (monthly, overlapping windows)	DERIVED from VP

#	Claim	Source (access 2026-09-27)	Status
E9	CCI total reconstructed 2008-01 to 2026-08 (fit error ≤ 0.16 pts); latest 119.51; CAGR 2008-2026 2.11%/yr; since 2021 base 3.54%/yr; 2026 y/y +6.53%	DGBAS CCI platform https://www.stat.gov.tw/CCI/CCI_Site/CCIPrice/PartialPriceClassesList.aspx via D4 + A2 scripts	VP inputs, derived total (F-508 extended back to 2008)
E10	CCI calendar-year growth: 2009 -9.0%, 2015 -2.9%, 2021 +10.9%, 2022 +7.4%, 2024 +2.0%, 2025 +0.8%, 2026 (Jan-Aug avg) +4.4%	D4 script	DERIVED from VP
E11	CCI 2-year moves p5/p95 -5.0%/+16.4% (sd 0.058); 6.3-year -0.6%/+32.9% (median +6.1%, mean +13.7%, sd 0.113)	D4 script	DERIVED; only 18 years of data, so the 6.3y spread rests on ~3 independent periods
E12	Taiwan CPI forecasts 2027: 1.90% (DGBAS), 1.83% (CBC)	F-504, F-505 (eng.stat.gov.tw; cbc.gov.tw)	VP (Phase A)
E13	corr(monthly NT\$-per-US\$ change, S&P 500 return) = -0.38 (120 months to Sep 2026); annual -0.60 (n=10); a -10% S&P month ~ +1.4% NT\$ per US\$	FRED SP500 https://fred.stlouisfed.org/graph/fredgraph.csv?id=SP500 + E7	DERIVED from VP; FRED holds only 10 years of SP500
E14	Annual corr(CCI growth, NT\$-per-US\$ change) = -0.36 (2009-2026, n=18)	D4 script	DERIVED; weak evidence (small n)
E15	Third Taiwan Strait Crisis 21 July 1995 - 23 March 1996	https://en.wikipedia.org/wiki/Third_Taiwan_Strait_Crisis	SNIP (secondary source, read directly)
E16	NT\$ per US\$ 26.38 before the crisis, peak 27.55 (+4.4%), 27.31 at its end (+3.5%); Aug 2022 drills ~0.0%	D4 script on E7	DERIVED from VP; Aug 2022 dates (4-10 Aug) general knowledge, UNVERIFIED
E17	Taiwan 2-year government yield 1.66% (2026-08-10)	search snippet citing https://en.macromicro.me/charts/14586/taiwan-2-year-government-bond-yield	SNIP (page is script-rendered; not read)
E18	U.S. 2y par 4.81% (2026-09-25)	competition/official_market_data/ curve CSV	VRF
E19	BS-04: "US\$" on every figure, NT\$ reference with date and rate, one line on which currency is binding	research/insight_v1/phase_A/stakeholder_map.md L629-634	repo (Phase A)
E20	WinS ticket already drops EWT (Taiwan tilt = AI bet + tokenism; \$3.3k Taiwan exposure via VT)	research/insight_v1/wins_now/S2_growth_sleeve.md L43-48, L206-235	repo

Numbers and where they come from

A. Building power of a fixed US\$ in January 2033 (FX unchanged), from Aug 2026 costs (DERIVED from E9/E12):

Cost path (ASM choice)	Cost rise to Jan 2033	Building a fixed US\$ buys
Taiwan CPI forecast 1.9%/yr	+13%	-11%
CCI 2008-2026 average 2.11%/yr	+14%	-12%
CCI since 2021 base 3.54%/yr	+25%	-20%
2026 spike 6.53%/yr (do not project; F-509)	+49%	-33%
Suggested base: about 2-3.5%/yr, i.e. a fixed US\$ buys roughly 12-20% less building by 2033 (ASM: the		
18-year average and the post-2021 pace bracket it; CPI alone understates because wages rose 8.1% y/y, F-508).		

B. Spread on matched horizons (log sd; 90% bands ~ +/-1.645 sd; DERIVED):

	2026 -> 2033	2031 -> 2033
Market (US\$ pool)	0.164	0.027
FX, since 2006	0.064	0.068
FX, since 1983	0.165	0.096
FX, square-root-of-time from 10y daily vol 4.69%	0.118	0.066
Construction (CCI), since 2008	0.113	0.058
Combined (independent, FX since 2006)	0.209 (~-29%/+41%)	0.094 (~-14%/+17%)

	2026 -> 2033	2031 -> 2033
Combined with FX-equity corr -0.38	0.189	0.086
Caveat (stats skeptic, F-513): the FX answer depends heavily on the period. Since 2006 the 6-year spread is small		
(the central bank has smoothed the rate); since 1983 it includes the 1980s appreciation from ~40 to ~25 NT\$ per US\$		
and is as large as the market spread. State which period you use. The 2031-view ranking (FX >= construction >		
market) holds under every FX period.		

C. NT\$ illustration at the dated rate 31.82 (2026-09-18), FX only (DERIVED):

- Median 2033 pool US\$207k = NT\$6.60m; 2-year FX p5/p95 -> NT\$5.83m / NT\$7.35m.
- Median-state 2031 floor US\$165k = NT\$5.25m; 2-year FX p5/p95 -> NT\$4.64m / NT\$5.85m.
- 2031-view NT\$ band (market x historical 2-year FX, bootstrapped, independent): **-11.4%/+12.7%**, vs US\$ **-3.8%/+5.1%**.
- The bought US\$ floor, restated in NT\$, came in below its 2031 NT\$ value in 51% of past 2-year windows (since 2006), by >5% in 27%, by >10% in 7%.

D. Cost of locking NT\$ in 2031 (covered interest parity, ASM; E17 is SNIP): $((1.0166/1.0481)^2 - 1) = -5.9\%$ fewer NT\$ than spot. The realised median 2-year move since 2006 was -0.2%, and NT\$ strengthened by more than 5.9% in only 23% of windows. Locking costs about one FX sd for protection that would rarely have been needed. If the rate gap narrows by 2031, the cost falls. Re-check then.

Implications (decision / number / sentence) and deliverable

1. **FR, assumptions section: sentence (number).** A strong assumption line must contain: (a) the facility is paid in NT\$ and Laura's contribution is in US\$; (b) a named cost-growth assumption for Taiwan construction (the team's choice in the ~2-3.5%/yr band, citing DGBAS CCI with its date) and **why not CPI** (wages +8.1% y/y; costs ran ~3.5%/yr since 2021 but ~2.1%/yr since 2008); (c) what it means: a fixed US\$ amount buys roughly 12-20% less building by 2033; (d) that the team is **not** estimating the facility cost (case L164). No budget, no cost figure.
2. **FR, 2031 range / co-sponsor element: decision.** US\$ is the binding currency. Any NT\$ figure carries "at NT\$31.82 per US\$ on 18 Sep 2026 (FRED); illustrative only" and, if an NT\$ band is shown, it is the wider one (~+/-12% in the 2031 view). A strong sentence must say *which* currency is binding and *why the NT\$ number moves more* (the exchange rate, not her portfolio). Fits the message order already specified for M008 (floor bought, conditional stretch, invitation to co-fund).
3. **FR, uncertainty explanation: sentence.** The case asks how "favorable and unfavorable market outcomes" affect the amount (L117-118 area). Worth one line: by 2031 the market part is mostly settled (+/-4-5%). What still moves the *building* her gift pays for is the exchange rate and Taiwan costs, and co-sponsors budgeting in NT\$ face the same two. This is honest and specific to her project; a generic team will not have it.
4. **FR, flexibility rationale: sentence.** Keeping the post-floor leftover in US\$ until construction contracts are signed (no NT\$ lock in 2031) costs nothing now, avoids ~5.9% forward cost, and keeps her able to re-scope if costs or the project change. This supports the "flexibility as the project develops" reading (case p.3; brief section 7).
5. **FR, optional chart (Creativity and Presentation; the IPS bans charts).** "What moves the building her gift buys," two small grouped bars (2026 view vs 2031 view): market / FX / construction spread (sd) plus a separate marker for the construction drift. Data: table B plus table A. One chart, no more.
6. **IPS: none directly** (the IPS guide says the final range and projections are not expected). At most a clause in the 500 words that the promise and range are in US\$. **TN: none. WInS-now: none** (the EWT drop stands).
7. **Operating payments (adjacent, for the SH-12 erosion owner):** the same logic applies to the fixed \$50k a year in NT\$ terms. Wages (+8.1% y/y) matter more than construction for a residency's running costs. Route to whoever owns the erosion sentence; not re-worked here.

Taiwan Strait contingency (scope note)

- The case puts Taiwan legal and regulatory issues out of scope (case L169, VRF). A geopolitical blockade or conflict is outside every historical series used here (ASM). It would threaten the *project*, not the US\$ Treasury ladder that funds the ten payments.
- Plan consequences already in place: payments in US Treasuries (US-held); no Taiwan tilt (E20); leftover held as US\$ flexibility (implication 4). Nothing more is needed, and anything more would dress up a tail event as a forecast. One clause in the FR flexibility rationale is enough.
- Historical check (E16): the 1995-96 crisis moved the currency by at most +4.4%, within normal 2-year noise. UNVERIFIED general knowledge: the central bank intervened to stabilise it. Do not present this as evidence about a future crisis.

Confidence

- Ranking by vantage point: **high** (robust to FX period and to the CCI growth choice).
- Size of the construction drift (-11% to -20%): **medium** (18 years of CCI; the regime choice matters).
- NT\$ band (~+/-12% in 2031): **medium-high** since 2006; wider if the pre-2006 history is used.
- Hedge cost 5.9%: **low-medium** (Taiwan 2y yield is SNIP; the rate gap may change by 2031).
- Correlations: **low** (short samples).

Changes current strategy? No (adds FR content only). **Criterion:** Client Knowledge and Objectives (secondary: Creativity and Presentation; Portfolio Analysis for the matched-horizon method).

What this teaches (one line): compare risks over the same time span, and keep "what we expect to happen" (drift) separate from "how wrong we might be" (spread). A risk that is certain to erode value belongs in the assumptions, not the error band.

Sources (all accessed 2026-09-27)

Source	URL / path	Status
Official case	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (L77, L117, L146-148, L164, L169)	VRF
Verified MC model	research/verified_2026-09-27/strategy_mc.py	VRF (model inputs JPM LTCMA; ASM lognormal)
FRED DEXTAUS (Fed H.10 NT\$ per US\$)	https://fred.stlouisfed.org/graph/fredgraph.csv?id=DEXTAUS	VP
FRED SP500	https://fred.stlouisfed.org/graph/fredgraph.csv?id=SP500	VP
DGBAS construction cost index platform	https://www.stat.gov.tw/CCI/CCI_Site/CCIPrice/PartialPriceClassesList.aspx	VP inputs; total derived
DGBAS / CBC CPI forecasts	F-504, F-505 in research/insight_v1/phase_A/fact_register.md	VP (Phase A)
Third Taiwan Strait Crisis dates	https://en.wikipedia.org/wiki/Third_Taiwan_Strait_Crisis	SNIP (secondary)
Taiwan 2y yield 1.66% (2026-08-10)	https://en.macromicro.me/charts/14586/taiwan-2-year-government-bond-yield (via search snippet)	SNIP
Phase A/B/C files	research/insight_v1/phase_A/{fact_register, stakeholder_map, case_register}.md; phase_B/B6b_cosponsors.md (Q3, Q5); phase_C/survivors.json (M095)	repo
WInS growth sleeve (EWT drop)	research/insight_v1/wins_now/S2_growth_sleeve.md	repo
Script	research/insight_v1/scripts/D4_facility_purchasing_power.py (reuses A2_taiwan_cci.query)	this run

Blocked or not read: DGBAS attachment files on ws.dgbas.gov.tw (TLS failure, Phase A); Taipei Exchange / MacroMicro yield pages (script-rendered, no data in the page text).

What this teaches

- Laura's gift is decided in US dollars but spent in Taiwan dollars on Taiwan building costs. Three different things can shrink it, and they matter at different times: markets before 2031, the exchange rate and building costs after.

- Locking the floor in 2031 turns the biggest uncertainty (markets) into the smallest. That is what "protecting the commitment" looks like in numbers.
- Some risks are predictable drifts (costs usually rise). You state them as assumptions. Others are genuine unknowns (currencies). You state them as ranges and label the date and rate you used.
- A hedge has a price: locking NT\$ early costs about 6% today, roughly the size of the risk it removes. Sometimes the right decision is to disclose the risk rather than pay to remove it.

Audit corrections (AX2)

Auditor AX2 (cluster auditor: equity, Taiwan/FX, compliance), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: `research/insight_v1/phase_D/audit_markets_rules.md`. Check script: `research/insight_v1/scripts/AX2_audit_checks.py`.

Reproduction. `D4_facility_purchasing_power.py` re-run 2026-09-28 with live FRED (DEXTAUS 1983-10-03 to 2026-09-18, latest 31.82; SP500 10 years) and the live DGBAS CCI platform (latest 119.51; fit error 0.164 pts): every number in this file reproduces (market sd 0.164/0.027; FX sd 0.068/0.064/0.096/0.165; CCI sd 0.058/0.113; variance shares 61/9/29 and 8/53/39; drift -11%/-12%/-20%/-33%; correlations -0.38, -0.60, -0.36; crisis +4.4%/+0.2%; NT\$ band -11.4%/+12.7%; 51/27/7%; forward -5.9%; 23%). **Sources re-opened:** case L77, L114-120, L146-148, L164, L169 (VERIFIED-REPO-FILE); FRED and DGBAS through the live run (VERIFIED-PRIMARY inputs, derived outputs). Not re-opened: Wikipedia crisis dates (secondary) and the Taiwan 2-year yield 1.66% (SNIPPET-UNVERIFIED; the page is script-rendered).

#	Severity	Issue	Correction
C1	important	Scope. Brief section 17 (team leader, 2026-09-28, written after this file) limits the run to the Trading Notes and the IPS. Implications 1-5 are Final Report material, and implication 5 is a chart spec, which section 17 bans ("no ... visuals/chart specs").	Move implications 1-5 to one-line "later (after Nov 9)" entries. What serves the IPS now, as rules to fix before Nov 6 (method only, no figures): (a) the promise, the reserve and the 2031 range are stated in US\$, and any NT\$ figure is dated and illustrative; (b) facility money stays in US\$ Treasuries until the facility decision, with no currency conversion or hedge before then; (c) the IPS names the assumption that Taiwan building costs are expected to rise, so a fixed US\$ amount buys less (case L146-148). TN: none. WInS: none (EWT stays dropped).
C2	important	The 2031 ranking depends on the open floor share. "In 2031 the market is the smallest factor" holds because the model locks 80% of the growth money.	AX2 re-run (same method; AX2 script section 1): 2031-view market log sd at 60% sleeve equity = 0.027 (80% floor), 0.040 (70%), 0.054 (60%), 0.067 (50%); at 50% equity 0.023/0.034/0.045/0.056. FX is 0.068 and construction 0.058. So the market is the smallest factor only for floor shares of about 70% or more , and at 50% it ties FX. Change the confidence line from "high (robust ...)" to "high given a floor share of 70% or more ; the floor share is an open team parameter (ticket section 8)".
C3	important	The rules reason for not locking NT\$ is missing, and the 23% figure is a counterfactual. A two-year NT\$ forward is a derivative. WInS bans derivatives, and the SMAApply FAQ lists permitted investments "for BOTH contributions" (R-AN15; reading 1 is the brief's safe assumption), so a forward is out of bounds for the plan as well. The "paid off in only 23% of windows" figure applies <i>today's</i> 5.9% rate gap to every past window. The real gap varied (U.S. rates near zero in 2009-2015).	The conclusion "don't lock NT\$ in 2031" stands on the rules first and cost second. Say "5.9% fewer NT\$ than today's spot rate; that is a cost only if the exchange rate would otherwise not move (random-walk ASSUMPTION)". Label the 23% as "if today's rate gap had applied in every past window". Taiwan 2-year 1.66% stays SNIPPET-UNVERIFIED. Define covered interest parity: the forward rate is set by the gap between the two countries' interest rates.
C4	minor	The today-view ranking depends on the FX sample. With FX since 1983 the 6.3-year FX sd (0.165) ties the market (0.164).	Summary item 1: "largest spread, or tied with FX on the 1983 sample".
C5	minor	"51% of past 2-year windows below its dated NT\$ value" carries no information. For a zero-drift exchange rate it is a coin flip.	Lead with "more than 5% below in 27% of windows, more than 10% below in 7%".
C6	minor	Conflation ("That is what 'protecting the commitment' looks like"). The case's operating commitment (the ten payments) is protected by the ladder. The 2031 floor protects the credibility of the facility range (case L114-120).	Keep the two apart in every spec.

#	Severity	Issue	Correction
C7	minor	"A risk that is certain to erode value." CCI fell in 2009, 2013, 2015 and 2016 (script, calendar-year growth), and the 6.3-year p5 is -0.6%.	Say "expected to erode".
C8	minor	Implication 7 uses +8.1% as "wages". It is the CCI <i>construction</i> wage class (F-508), not wages in general.	Do not use it for the residency's running costs.
C9	minor	"Most teams will skip" (summary item 2) is unsupported.	Delete.
C10	minor	Jargon and labels. "log sd", "variance share", "bootstrapped" and "square-root-of-time" are undefined. "DERIVED" is not a brief label.	Define them or cut them. Map DERIVED to "ASSUMPTION method on VERIFIED-PRIMARY inputs".
C11	minor	Inputs use the 60% equity sleeve. D2 proposes 50%.	At 50% the 2031-view market sd falls from 0.027 to 0.023 (80% floor). The rankings do not change.

Status after audit. All numbers reproduce (script outputs on VERIFIED-PRIMARY inputs; model parts ASSUMPTION). Summary items 1 and 3 are conditional (C2, C4). Item 4's conclusion survives with a stronger, rules-based reason (C3). Items 2 and 5 stand with wording fixes (C7, C5). The deliverable routing changes: only the three IPS rules in C1 go forward now; everything else waits until after Nov 9.

D5 Philanthropy & Co-sponsor Analyst: the 2031 range, its confidence wording, kept flexibility, and the fundraising-excerpt checklist

Agent D5, insight_v1 run, Phase D. Written 2026-09-27. Questions: M003, M091, M107, M094 (from research/insight_v1/phase_C/survivors.json). AI-generated research for Team Caplet, for brainstorming only. This file contains **no submission-ready prose**: it gives numbers, evidence, decisions to take and "what a sentence must contain" checklists. The six students choose and write every word (Wharton AI policy: AI use must be recorded in Works Cited; students "must use their voice and words", R-W26/R-W49). No securities are recommended here (the 2031 two-year note is already in the wins_now ticket and never appears in WInS). Laura's public professional record only; nothing here suggests contacting her.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (read by me on the primary page on 2026-09-27 with `fetch_text.py --grep, phrase found verbatim`), **VRF** = VERIFIED-REPO-FILE, **SNIP** = SNIPPET-UNVERIFIED, **ASM** = ASSUMPTION, **INT** = my interpretation (not a fact). R-, F-, SH-, BS- ids point to the Phase A files.

Terms (defined once):

- **Floor**: the part of the growth money that is bought in January 2031 as a 2-year U.S. Treasury note maturing before the 2033 contribution date. Its 2033 value is known in 2031.
- **Lock share (a)**: the share of the growth money ("sleeve") bought as the floor in 2031.
- **Give-back share (s)**: the share of the *unlocked* remainder's 2033 value that is added to the facility gift; the other $(1 - s)$ stays uncommitted as financial flexibility.
- **Top of the range**: the gift the rule produces if the unlocked remainder grows at its q -th percentile ($q = 85\%$ here).
- **Percentile (p5, p50, p95)**: the value that 5%, 50% or 95% of modelled outcomes fall below. p50 = median.
- **Unconditional / conditional pledge**: a promise with no hurdle attached vs one owed only if a hurdle is cleared.

Script: research/insight_v1/scripts/D5_range_and_flexibility.py (run from the repo root; docstring lists inputs and labels). It uses the same random stream and inputs as the verified strategy_mc.py and reproduces its numbers exactly: 2031 floor at $a = 0.8$ p5/p50/p95 **\$127k/\$165k/\$217k** (F-402) and 2033 surplus **\$159k/\$207k/\$273k** (F-401). All model outputs below are **ASM-based** (JPM 2026 LTCMA inputs VRF; lognormal i.i.d. returns, 60/40 sleeve, no fees, 2031 two-year rate = today's 4.81%).

0. Top findings (ranked by impact on reaching the semifinals and on making the plan unmistakably Laura's)

1. **The current plan has a hidden contradiction, and one extra number fixes it (M003, M107; changes the strategy at parameter level)**. Brief section 7 says "promise a bought floor plus an upside" AND "keep leftover money mainly as project flexibility". But with the model's 80% lock, the leftover is only ~20% of the post-reserve money, and every dollar of it that is kept as flexibility is a dollar that cannot be the "upside" in the range. If all of it is kept, the range collapses to a single number (which the case steers away from, R-C67); if all of it is given, there is no flexibility (R-C58, R-C84). The fix is one plain rule with two parameters: **buy a share a of the growth money as the floor in 2031; in 2033 give the floor plus half ($s = 0.5$) of what the rest has grown to; keep the other half uncommitted**. Candidate default $a = 0.7$ (numbers in finding 2). This is the team's decision.
2. **The numbers for the candidate rule ($a = 0.7, s = 0.5, q = 0.85$), at the median 2031 growth money of \$188k (ASM model): range US\$144k - US\$180k; the bottom is bought, so $P(\text{below floor}) = 0$ barring U.S. default; $P(\text{above top}) = 15\%$ (model). Flexibility kept in 2033: **\$32k median (\$23k p5, \$45k p95), about 15% of the post-reserve money**. Each 10 points of lock share moves about \$20k into the counted floor and takes about \$10k each from flexibility and from the upside ($a = 0.6$: floor \$124k, flexibility 20%; $a = 0.8$: floor \$165k, flexibility 10%, range only 15% wide).**

3. **A professional co-funder counts only the bought floor as money (M091, confirmed with new primary checks).** Kresge counts "Written pledges or cash" and wants other sources "committed, imminent or backstopped" (VP); U.S. nonprofit accounting books a conditional promise only when "the barriers to entitlement are overcome" (FASB ASU 2018-08, VP); the seed-money experiments work through "an unconditional commitment" (Rondeau & List, VP). So the fundraising excerpt leads with the floor. **New: the midpoint is useless as a planning figure, in the conservative direction.** Given what is known in 2031, the gift lands below the range's midpoint in ~0% of modelled paths, and between **\$168k and \$184k** in 98% of them (p1-p95). The floor is a guarantee, not a forecast; the funder should plan on the floor and be told the likely figure separately.
4. **The range must be a 2031 rule, not dollar figures fixed today (M091, M107).** The case asks for "the dollar range Laura should communicate" (R-C69), but the range is communicated in 2031. A floor copied from today's median projection (\$144k) would be missed in **50%** of modelled paths; even today's p10-p90 band (\$143k-\$218k) would put the 2033 gift below its bottom in **10%** of paths, which is exactly the overpromise the case warns about (R-C66). The rule re-bases the floor on what she actually owns in 2031, so its floor is missed in 0%. The Final Report should show the rule, then the dollar range it gives at the median path, plus p10/p90 illustrations (**\$118k-\$147k** and **\$178k-\$223k**).
5. **Do not cap the gift at the top (M107).** If the gift is capped, P(within range) = 100% by construction, an empty confidence figure a statistics graduate would spot (SH-04). Uncapped, a good market visibly raises the amount, which the case requires ("how favorable ... outcomes could affect the amount", R-C70), and the excess is small (median **\$2k**, p95 **\$8k** above the top when it happens). The confidence statement becomes two-sided and honest: "below the floor: only if the U.S. Treasury defaults; above the top: about 1 in 7 (model)". Over-delivering never costs credibility; under-delivering does (R-C66).
6. **Minimum fundraising-excerpt checklist (M094): nine must-haves, six things to leave out** (section 4). New items not in the Phase A checklist: the floor cannot be handed over before 2033 (the case bans withdrawals, R-C34), so the form is "held in a Treasury note maturing before the contribution date"; the 2028-deposit risk is already resolved by 2031 and need not be mentioned; the likely figure is stated separately from the range; the date prepared and the 2033 confirmation date.
7. **Tier-2 consequence (IPS, Nov 6): the principle must be in the IPS, the numbers need not.** The IPS guide says the Final Report "should reflect the strategy established in your IPS" and the strategy "may not" be revised after Nov 6 (IPS guide p.2, VRF). A flexibility rule that first appears in the Final Report could read as a new strategy. So the IPS needs one line naming the principle (bought floor promised; a stated part of growth beyond it kept uncommitted for the project). Numbers and the range stay in the FR (the IPS guide says a "final facility-contribution range" is not expected).
8. **Brunel's "10%" is not a benchmark for this (M003, partly mis-posed).** On the primary page it is the example family's own wish ("The family also wants to reserve 10 percent of their total wealth ... for what they call 'opportunistic goals'", VP) and applies to total wealth, not to project money after a pledge. It cannot size Laura's flexibility. A better sanity anchor is verified and Laura-specific: Taiwan's construction cost index averaged **3.54%/yr** since 2021 (F-508, VP), which erodes a fixed US\$ pledge by **7.2% / 11.0% / 14.9%** over 2 / 3 / 4 years; the 1-sd two-year USD/TWD move is **6.8%** (derived). A ~15% flexibility share is the same order as three to four years of building-cost drift on her own pledge (ASM; a sanity check, not a contingency fund, which the case does not require, R-C61).

Whether Laura herself ranks flexibility above a bigger gift **cannot be tested** (contacting the client means disqualification, R-W16). The case gives both sides: she wants a "meaningful personal contribution" (R-C64) and she "recognizes that committing all remaining assets could limit her financial flexibility as the project develops" (R-C58). So the answer is a stated middle, with the reason written down as a team ASSUMPTION.

1. M003 (tier 2 for the IPS line; tier 3 for the numbers): how much of the post-reserve money stays uncommitted as flexibility?

Question (canonical): What explicit rule or share of the post-reserve surplus should stay uncommitted as "financial flexibility", rather than "whatever is left" or simply the unpledged upside, and does she rank it above a bigger facility contribution?

One-sentence answer: Keep flexibility by rule, not by leftovers: buy about 70% of the growth money as the 2031 floor, and in 2033 give the floor plus half of what the remaining 30% has grown to, which keeps

about 15% of the post-reserve money (about \$32k at the median, \$23k at p5) uncommitted for the project; whether Laura would rank this above a bigger gift cannot be tested, so it is a stated team assumption.

Evidence

Claim	Source (access 2026-09-27)	Status
"She recognizes that committing all remaining assets could limit her financial flexibility as the project develops."	Case p.3 (R-C58)	VRF
Test 5: "Preserves appropriate financial flexibility when determining the facility contribution."	Case p.4 (R-C84)	VRF
"Teams are not expected to determine the size or investment composition of a separate contingency fund or endowment."	Case p.3 (R-C61)	VRF
Living costs are covered outside the portfolio (so portfolio flexibility is project flexibility, not personal)	Case p.2 (R-C33)	VRF
Continued business income is one of the case's listed sources for remaining facility costs	Case p.3 (R-C44)	VRF
IPS: FR "should reflect the strategy established in your IPS"; strategy may not be revised after the deadline; a "final facility-contribution range" is not expected in the IPS	2026_WGY_Investment_Policy-FINAL.txt p.2 L68-77	VRF
Brunel's example family "also wants to reserve 10 percent of their total wealth (\$3.5 million) for what they call 'opportunistic goals'"	Brunel, CFA Institute Conference Proceedings Quarterly, March 2012, https://static1.squarespace.com/static/59e8d89d914e6b37450c946a/t/5c643e15104c7b43f7b84187/1550073366884/CFA+Goals+Based+WM_PDOC.pdf	VP (it is the client family's wish in a worked example, on total wealth: not a practitioner benchmark)
Taiwan construction cost index average 3.54%/yr since 2021; +6.53% y/y Aug 2026 is a one-year jump	fact_register F-508, F-509	VP (per A2)
Laura's verified lines about keeping options open ("avoid siloing yourself into an end-all-be-all career situation", Poets&Quants 2018; "I did 50 things...", Overachiever 2021) are about careers, not money	D13a table lines 98, 115	VP (context: career advice; use at most as character context in the FR, never as an investment rule; D13 limits the FR to 1-2 Laura quotes in total)

Numbers (script sections [1], [2], [8]; ASM-based model at the median 2031 growth money of \$188k)

Lock a	Give-back s	Floor (bought)	Top (q = 85%)	Range width	Kept flexibility p5/p50/p95	Share of post-reserve money (median)
0.6	0.5	\$124k	\$172k	39% of floor	\$30k/\$42k/\$60k	20%
0.7	0.5	\$144k	\$180k	25%	\$23k/\$32k/\$45k	15%
0.8	0.5	\$165k	\$189k	15%	\$15k/\$21k/\$30k	10%
0.8	0.0	\$165k	\$165k (a point)	0%	\$30k/\$42k/\$60k	20%
0.8	1.0	\$165k	\$213k	29%	\$0	0%
1.0	-	\$206k	\$206k (a point)	0%	\$0	0%

(Flexibility percentiles are across all 2027-2033 paths; the floor/top columns are at the median 2031 value.)

Implication

- **Decision (team):** choose a (lock share) and s (give-back share). Candidate default a = 0.7, s = 0.5. Alternates: a = 0.8 if the team weighs the co-sponsor signal most (floor +\$21k, flexibility falls to ~10%); a = 0.6 if it weighs flexibility most (~20%). Record the reason in the decision log (Articulation evidence).

- **IPS (Nov 6), one line must contain:** (1) the facility promise is only what has been bought two years ahead; (2) a stated part of any growth beyond it is kept uncommitted for the project's changes; (3) the operating payments are never touched. No dollar figures needed.
- **FR (Dec 4):** a small table of gift vs kept flexibility at p5/p50/p95 (a chart candidate: stacked bars "floor / added in 2033 / kept flexible" for p5, p50, p95); the reason for the share in one sentence tied to R-C58 ("as the project develops") and to building-cost drift (F-508), labelled as an assumption; one sentence that her living costs are outside the portfolio, so this is project flexibility.
- Deliverables: IPS, FR. Not TN (the TN guide excludes the facility contribution) and not WInS-now.

Deadline tier: 2 (IPS principle), then 3. **Confidence:** medium (the model reproduces verified outputs; the share itself is a judgement). **Changes current strategy:** yes, at parameter level: adds the give-back share s and moves the lock share from 0.8 to a candidate 0.7; resolves the brief section 7 tension. The wins_now ticket is unaffected (the 2031 share "never appears in a WInS note"). **Criterion:** Client Knowledge and Objectives (tailors to her "flexibility as the project develops"); also Portfolio Analysis (financial flexibility under varying outcomes). **What this teaches:** flexibility is only real if a rule reserves it before the money arrives; "whatever is left" is a hope, not a policy.

2. M091 (tier 3): does a co-funder count anything beyond the bought floor, and should the excerpt lead with the floor?

One-sentence answer: No: professional funders count written, unconditional money, so the bought floor is the headline planning figure, the upper part is information with a stated probability, and the midpoint should never be used because, given 2031 knowledge, the gift almost always lands well above it.

Evidence

Claim	Source (access 2026-09-27)	Status
Kresge counts as initial funds "Written pledges or cash from individuals, corporations, businesses and foundations ... All pledges must be paid within five years from your fundraising end date."	Kresge Foundation, <i>A Guide to the Challenge Grant</i> (updated 2011), https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf	VP (2011 document; whether Kresge still uses it is not checked)
Kresge: some sources "should be committed, imminent or backstopped"	same	VP
Kresge asks capital applicants: "Do you have a plan for sustaining support gained during the campaign after it concludes?"	same	VP
Conditional contributions are "unrecognized initially, that is, until the barriers to entitlement are overcome"	FASB ASU 2018-08, https://storage.fasb.org/ASU%202018-08.pdf	VP (U.S. nonprofit accounting; Taiwanese co-sponsors may follow other rules: UNVERIFIED)
958-605-55-51: a communication that "specified an intention to give" meant the charity "did not receive a promise to give"	same	VP (the example is a will that can be changed; applying it to Laura's upper figure is INT)
"A challenge gift is an unconditional commitment by a donor, or set of donors, to provide a given sum of money to the cause."	Rondeau & List, NBER w13728, https://www.nber.org/system/files/working_papers/w13728/w13728.pdf	VP
"challenge gifts positively influence contributions in the field, but matching gifts do not"	https://www.nber.org/papers/w13728 (abstract)	VP
Charities "maximize donations given by simply announcing the presence of a lead gift"	Huck & Rasul 2011 abstract, https://ideas.repec.org/a/eee/pubeco/v95y2011i5p351-362.html	VP
The case: she will "describe how much she expects to contribute"; she will "neither add to nor withdraw from the portfolio before 2033"	Case p.2-3 (R-C34, R-C63)	VRF

Numbers (script [3], [7] and the conditional check; ASM model)

- Given the median 2031 growth money and rule $a = 0.7$, $s = 0.5$: gift p1/p5/p50/p95 = **\$168k / \$170k / \$176k / \$184k**; floor \$144k; top \$180k; midpoint \$162k, below which the gift falls in ~0% of paths.
- A floor fixed today at the median (\$144k) is missed in 50% of paths; the 2031-rule floor in 0%.

- Kresge's five-year pledge window vs Laura's 2031 statement paid in 2033: two years, inside it (INT).

Implication

- **Decision:** the FR fundraising excerpt leads with the floor (US\$, form, date); the range follows; the confidence is split into P(at least the floor) and P(above the top). Also state the **likely figure** (the model median) as a separate, labelled number, because the range is deliberately lopsided (bottom guaranteed, outcomes cluster near the top). Do not present the midpoint.
- Wording choice for the team (no prose here): which verb goes with which tier (BS-05): the floor is "held/committed", the rest is "expected". Because of R-C34 she cannot hand money over in 2031, so do not imply the floor is already transferred or escrowed.
- New link worth one clause in the FR: Kresge's own word "backstopped" describes Laura's floor exactly (a pledge backed by a Treasury she owns) (INT).
- Deliverable: FR only. Tier 3.

Confidence: high on "lead with the floor" (three independent primary sources agree); medium on how Taiwanese public funders count pledges (not verified). **Changes current strategy:** no (it confirms brief 8.7 and the chair memo's "at least F; up to U"); it adds the "likely figure, not midpoint" point. The Phase C "answered" skeptic was right that the direction was known; what is new is the evidence and the lopsided-range number. **Criterion:** Creativity and Presentation ("communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors"). **What this teaches:** a number means what the reader's own books let it mean: a funder can only budget on money that is promised without conditions.

3. M107 (tier 3): where does good-market money above the top go, and what does it do to the confidence figure?

One-sentence answer: Let the rule run uncapped, so in the ~15% of paths where markets beat the top the gift exceeds it by a small amount (median \$2k, p95 \$8k) and the rest stays as flexibility; the confidence is then stated on both sides ("below the floor only if the U.S. defaults; above the top about 1 in 7"), whereas capping the gift would make "100% within range" true by construction and meaningless.

Evidence

Claim	Source	Status
"state how confident they are that her 2033 contribution will fall within that range" (two-sided)	Case p.3 (R-C69, R-AN9)	VRF
"explain how favorable and unfavorable market outcomes could affect the amount she can provide"	Case p.3 (R-C70)	VRF
"If Laura promises more than she can ultimately contribute, she could damage her credibility" (the risk is one-sided: overpromising)	Case p.3 (R-C66)	VRF
Statistician reader: a confidence that counts only the low end loses trust	stakeholder_map SH-04	VRF (Phase A judgement)
Brunel/Das goal-probability table: "Wants" 80-85%, "Needs" 90-95%	Das et al. 2018, https://srdas.github.io/Papers/GBWM.pdf , Table 1	VP (a practitioner scale, used here only to justify q = 85% as a "want", not a "need")

Numbers (script [4], [7]; ASM model; a = 0.7, s = 0.5, median 2031 value)

q (top percentile)	Top	P(below floor)	P(above top)
80%	\$179k	0 barring default	20%
85%	\$180k	0	15%
90%	\$182k	0	10%

When above the top: excess median \$2k, p95 \$8k; kept flexibility in those paths median \$38k. The choice of q barely moves the top (the unlocked part is small), so q is a wording choice more than a money choice.

Implication

- **Decision:** no cap. The top is "the most co-sponsors should plan on", not a limit on her generosity. The rule itself answers R-C70: bad markets move the gift toward the floor (never below it, never into the operating reserve); good markets move it to or slightly past the top, and grow the kept flexibility.
- **Sentence the FR must contain (checklist, not prose):** both tail probabilities with the model named; the direction of the only possible miss (upward); who decides about money above the top (Laura in 2033, under the stated rule).
- Deliverable: FR. Tier 3.

Confidence: high on "no cap / two-sided statement" (logic plus case wording); medium on $q = 85\%$ (a judgement). **Changes current strategy:** minor: replaces the chair memo's one-sided "up to U with about 90% model confidence" with the two-sided statement and fixes the destination of excess money. **Criterion:** Portfolio Analysis ("reasonable assumptions and projections to evaluate ... the facility contribution, and financial flexibility under varying market outcomes"). **What this teaches:** a confidence number is only informative if the event it describes can actually fail; design the rule first, then ask which way it can miss.

4. M094 (tier 3): the minimum checklist for the fundraising excerpt

One-sentence answer: Nine items make the excerpt credible to a finance reader and a non-specialist in one reading (floor, form, likely figure, range with two-sided confidence, good/bad markets, operations already funded, what is not promised, currency, date); leave out the model mechanics, facility-cost estimates, share-of-campaign figures, legal claims, personal details and any AI-drafted wording. (Partly already answered by stakeholder_map SH-05/BS-04 to BS-17; this merges and cuts it, and adds three new items.)

Must contain (each maps to a source)

1. **The floor in US\$, its form and maturity:** held in a U.S. Treasury note bought in January 2031 that matures before the 2033 contribution date; not transferable before 2033 (R-C34; new).
2. **The likely figure** (model median, labelled as a model number), separate from the range (M091; new).
3. **The range US\$ floor - US\$ top**, with the confidence on both sides: below the floor only on U.S. default; above the top about 1 in 7 (model named in one clause) (R-C69, R-AN9, SH-04).
4. **What good and bad markets do** in one line each: bad → toward the floor, never below it; good → at or slightly above the top, and the rest stays with the project as flexibility (R-C70, M107).
5. **Ten years of operating payments are already funded separately from her portfolio alone**, so co-sponsor money can go to the building (R-C48, BS-17; Kresge's "plan for sustaining support" question, VP).
6. **What is not promised:** facility cost overruns (her contribution does not rise beyond the stated range; scope adjusts), operating support beyond US\$50k a year, anything after 2042 (BS-16, R-C49; PKF 2019, VP: funders want to know "how the grantee would respond to any unexpected revenue shortfalls or cost overruns").
7. **Currency:** "US\$" on every figure; one NT\$ reference with date and rate (at the median floor of US\$144k: NT\$4.59m at 31.82 on 2026-09-18; NT\$4.30m-4.90m for a 1-sd two-year currency move of 6.8%) and one line that US\$ is binding (BS-04, F-510; script [6]).
8. **Date prepared and next confirmation** (2031 statement; confirmed at the 2033 contribution) (new; keeps corporate and public funders' calendars, SH-08).
9. **Sized from her portfolio, not as a share of a budget** (BS-06; R-C95 says teams need not estimate facility cost).

Leave out (and why)

- Monte Carlo mechanics, return assumptions, percentile tables → FR analysis section, not the excerpt (SH-09: non-specialist in one reading).
- The 2028-deposit risk → resolved before 2031; the floor already reflects it (new).
- Facility cost, budget, funding gap, campaign share → out of scope (R-C95).
- Legal enforceability claims → not the team's question (BS-05).
- Laura's personal details or quotes → privacy rule, and the excerpt is written as her material (D13 rules).

- Any AI-drafted wording → Wharton AI policy; the students write it.

Implication

- Sentence-level spec for the FR** (students write it). Length target: short enough to read in one pass; final format waits for the FR instructions released Nov 9 (R-C93). If space is tight, drop item 8, then item 9 (fold it into item 6).
- A good visual for the FR (not the excerpt): one horizontal bar per scenario (p5/p50/p95 of the 2031 growth money) split into "bought floor / added in 2033 / kept flexible".

Deadline tier: 3. **Confidence:** high (each item traces to the case or a primary source). **Changes current strategy:** no. **Criterion:** Creativity and Presentation (co-sponsor communication is named in its wording). **What this teaches:** credibility comes from saying exactly what is certain, what is likely and what is not promised, in the reader's own units.

5. Sources (all accessed 2026-09-27)

Official / repo (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (pp.2-4); `competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt` (p.2 L60-77); `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L20-21: co-sponsor draft not expected in TN); `competition/official/2026_27/SMApply_Deliverables_Page_2026-09-27.md` (L49-57 criteria); `research/insight_v1/phase_A/{case_register,fact_register,stakeholder_map}.md`; `research/insight_v1/phase_C/survivors.json`; `research/insight_v1/phase_D/D13a_laura_quotes_verified.md`; `research/insight_v1/wins_now/securities_and_allocation_v0.md`; `research/council_2026-09-27/01_chair_memo.md` s5; `research/verified_2026-09-27/strategy_mc.py`.

Web (VP, each phrase checked verbatim with `fetch_text.py --grep` on 2026-09-27):

Source	URL
Kresge Foundation, <i>A Guide to the Challenge Grant</i> (2011)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
FASB, ASU 2018-08 (Topic 958), June 2018	https://storage.fasb.org/ASU%202018-08.pdf
Rondeau & List, "Matching and Challenge Gifts to Charity", NBER w13728	https://www.nber.org/system/files/working_papers/w13728/w13728.pdf ; https://www.nber.org/papers/w13728
Huck & Rasul (2011), "Matched fundraising", J. Public Econ. 95(5-6), abstract	https://ideas.repec.org/a/eee/pubeco/v95y2011i5p351-362.html
PKF O'Connor Davies, "Best Practices: Due Diligence on Grantees", July 2019	https://www.pkfod.com/wp-content/uploads/2019/07/Best-Practices-Due-Diligence-on-Grantees-v4.pdf
Brunel, "Goals-Based Wealth Management in Practice", CFA Institute Conference Proceedings Quarterly, March 2012 (hosted copy)	https://static1.squarespace.com/static/59e8d89d914e6b37450c946a/t/5c643e15104c7b43f7b84187/1550073366884/CFA+Goals+Based+WM_PDOC.pdf
Das, Ostrov, Radhakrishnan & Srivastav (2018), <i>J. Investment Management</i> 16(3), Table 1	https://srdas.github.io/Papers/GBWM.pdf

Not verified by me: Taiwanese funders' pledge-counting rules (no primary source found; MOC pages refuse curl, per Phase A); whether Kresge still uses the 2011 guide.

What this teaches

A co-sponsor range is a promise design problem before it is a forecasting problem. Decide what is bought (the floor), what is shared (half of the growth beyond it) and what is kept (the other half), and the confidence statement writes itself: it can only miss upward. The same two numbers answer three case requirements at once: a credible range, favourable and unfavourable outcomes, and preserved flexibility. And the reader decides what a number means: a funder counts only money promised without conditions, so the plan should lead with exactly that.

Audit corrections (AY1)

Auditor AY1 (cluster auditor: client psychology and co-sponsors), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: `research/insight_v1/`

phase_D/audit_client_cosponsors.md. Check script: research/insight_v1/scripts/AY1_audit_checks.py (reuses this file's engine and random stream).

Reproduction. D5_range_and_flexibility.py re-run 2026-09-28: every number in sections 0-4 reproduces exactly (floor \$127k/\$165k/\$217k and surplus \$159k/\$207k/\$273k reconcile with F-401/F-402; the (a, s) table; \$118k-\$147k and \$178k-\$223k; 50% / 10%; P(above top) 15%, excess \$2k/\$8k; NT\$4.59m/4.30m/4.90m; 7.2/11.0/14.9%), except C4. **Sources re-opened 2026-09-28** (fetch helper, --grep found verbatim): Kresge guide ("Written pledges or cash", "All pledges must be paid within five years", "committed, imminent or backstopped", "plan for sustaining support"); FASB ASU 2018-08 ("barriers to entitlement are overcome"; 958-605-55-51 "did not receive a promise to give"); Rondeau & List ("an unconditional commitment by a donor"; abstract sentence); Huck & Rasul abstract; PKF 2019 ("how the grantee would respond to any unexpected revenue shortfalls or cost overruns"); Brunel 2012 ("reserve 10 percent of their total wealth"; "opportunistic goals"); Das et al. 2018 Table 1 (Wants/Fears 80-85, Needs/Nightmares 90-95). All VERIFIED-PRIMARY. Case lines R-C34, R-C58, R-C61, R-C66-R-C70 and IPS guide p.2 L68-77, VERIFIED-REPO-FILE.

#	Severity	Issue	Correction
C1	important	Scope (brief s17, 2026-09-28, written after this file). Sections 2-4 are Final Report material; the fundraising-excerpt checklist (section 4, M094) is named as out of scope ("fundraising excerpt drafts or checklists").	Keep for the IPS only the METHOD, as "IPS: rules to fix before Nov 6": (1) in 2031 buy a share a of the growth money as a Treasury note maturing before the 2033 contribution; that bought amount is the bottom of the range; (2) in 2033 the gift = the bought amount + a share s of what the rest has become, uncapped; the other (1 - s) stays uncommitted for the project; (3) the top = the bought amount + s x a stated percentile of the rest; confidence stated on both sides with the model named; (4) the operating payments are never touched. Move to the "later (after Nov 9)" list, one line each: the excerpt checklist (section 4), the NT\$ reference, the "likely figure" presentation, the p5/p50/p95 visual, the midpoint point, Kresge's pledge window.
C2	important	Withdrawn: "Kresge's own word 'backstopped' describes Laura's floor exactly" (s2 implication). Kresge's glossary: "Backstopping- Formal use of specific alternative resources to cover anticipated government funds, planned gifts or long-term financing. These alternative resources often are organizational money, which will be replaced once the expected funding is available." The "committed, imminent or backstopped" sentence applies only to "long-term financing, government funds (if a substantial amount), organizational funds or bequests", not to individual pledges (VERIFIED-PRIMARY, re-read).	Laura's floor is her own asset behind her own expected gift, not a backstop of someone else's funding. Drop the clause. Finding 3 should say: Kresge counts "Written pledges or cash" from individuals; it wants <i>certain other source types</i> (financing, government, organizational funds, bequests) committed, imminent or backstopped.
C3	important	"Guarantee" overclaim (finding 3 "The floor is a guarantee, not a forecast"; s2 "bottom guaranteed"; s3/s4 "never below it"). The floor is not a legal guarantee: she cannot transfer it before 2033 (R-C34), and the case says she will "describe how much she expects to contribute" (case L110, VERIFIED-REPO-FILE). By the FASB logic this file cites (958-605-55-51, an "intention to give" is not a promise to give), a co-sponsor could book even the floor only if Laura states it as a written, unconditional pledge, which is her legal and communication choice (out of scope).	Use "bought, not forecast", always with "in US\$, barring a U.S. Treasury default". Downgrade "a professional co-funder counts only the bought floor as money" from a finding to INT: "funders count unconditional written pledges; the bought floor is the only part Laura <i>could</i> state that way".
C4	minor	Arithmetic. "between \$168k and \$184k in 98% of them (p1-p95)": p1 to p95 spans 94% of paths by definition.	AY1 check [1]: 94.4% of paths fall in \$168k-\$184k; p1/p99 = \$167.8k/\$187.6k. Say "about 94% (p1-p95)" or "98% (p1-p99, \$168k-\$188k)".
C5	important	Cross-file contradiction with D6 on the lock share, resolved. D5 proposes a = 0.7; D6 says the behavioural evidence "supports 80-90% and argues against anything below 70%". D6's widths assume s = 1 (all unlocked money promised). The width depends on both: top / bottom $\approx 1 + s(1 - a)/a \times 1.21$ (q = 90%; AY1 check [2]).	D5's (a 0.7, s 0.5) gives 1.25-1.26x, narrower than D6's own base (a 0.8, s 1.0) at 1.29-1.30x. The comparable quantity is the share of the growth money that is promised but still at market risk in 2031, s(1 - a): 0.15 (D5) vs 0.20 (D6 base) vs 0.30 (a 0.7, s 1.0). D5's rule passes D6's width test. Present one team choice: "how much of the promised gift is still at market risk in 2031", not the lock share alone. (AX2's D4 C2 threshold of "70% locked" was also computed with s = 1.)

#	Severity	Issue	Correction
C6	minor	The range is lopsided by design (worth knowing before choosing wording). The lower half of US\$144k-US\$180k is reached only if the unlocked money loses about 36% in two years (AY1 [1]); the lowest ~65% of the width (floor to p1) is reached in under 1% of modelled paths. A reader who anchors on the midpoint is misled, and D6's evidence (Du et al.: "as precise as warranted") pulls the other way.	Not a rule change. For the IPS method, name the bottom as "the amount already bought", never as a low-case forecast. The "likely figure" vs a narrower stressed bottom is a Final-Report presentation choice (later list).
C7	minor	Model input. The floor assumes the 2031 two-year rate equals today's 4.81% (ASSUMPTION).	At 3.0% the floor is 3.4% lower; at 6.0% it is 2.3% higher (AY1 [4]). Small; another reason the IPS states the method and no dollar floor.
C8	minor	Tautology presented as evidence (finding 4). "A floor copied from today's median is missed in 50% of paths" and "the p10-p90 bottom is missed in 10%" are true by the definition of a median and a percentile.	Keep the conclusion (the range must be a 2031 rule); label the two numbers "by definition", not as findings.
C9	minor	Precision (finding 8). 7.2/11.0/14.9% are cost <i>increases</i> ; the pledge's buying power falls 6.7/9.9/13.0%. The index is in NT\$, so US\$ buying power also moves with USD/TWD.	Correct the wording if used; this is Final-Report material (later list).
C10	minor	Harmonise the top percentile. D5 uses q = 85% ("about 1 in 7 above the top"); D6 uses p90 ("about 9 in 10"); D8 rule 3 says "the top = [stated percentile] of the rest", which omits the give-back share s.	Pick one percentile for the IPS method. D8 rule 3 should read "top = bought amount + [share] x [stated percentile] of the rest" so that it matches D8 rule 4 and this file (flagged to the main loop; D8 not edited).
C11	minor	Cross-check with D3 model v2 (phase_D/D3_model_v2_results.md s4, written in parallel). D3 lists <i>capped</i> rules ("give 90%, capped": P(within) 100%) next to uncapped ones; D5 finding 5 says never cap. D3's "floor + half of the excess" row at an 80% floor share (\$143k/\$186k/\$245k) matches D5's a = 0.8, s = 0.5 row exactly.	Compatible if a capped rule is reported with D3's "P(top reached)" (about 20%), never with "P(within) = 100%". D5's preference for no cap stands as the simpler honest statement; the team chooses.

Status after audit. Finding 1 (hidden contradiction between "promise the upside" and "keep flexibility") stands: it is the most useful new result in the cluster. Findings 2 and 5 reproduce (5 stands). Finding 3: direction stands ("lead with what is bought"); the Kresge "backstopped" link is withdrawn and the "counts" claim is INT (C2, C3). Finding 4 stands with C8. Finding 6 is parked to the later list (C1). Finding 7 stands and is the part that goes forward. Finding 8 stands (Brunel is not a benchmark). Candidate a = 0.7 / s = 0.5 survives as a team choice and is compatible with D6 once the width is measured by s(1 - a) (C5).

D6: Client psychology and behavioural finance (M125, M018, M019, M228, M237, M197)

Agent D6 (Client Psychologist / Behavioural Finance), insight_v1 run, Phase D. Written 2026-09-27/28 (the session ran past midnight UTC). This is AI-generated research for Team Caplet. It gives specifications, evidence, numbers and checklists. **It contains no text to submit.** The six students decide everything and write every deliverable in their own words, and they record AI use in the Final Report's Works Cited (R-W46). Laura appears here only through the case and her public professional record. No personal-life material is recorded. Nothing here suggests contacting her (R-W16: contact means disqualification).

Script: `research/insight_v1/scripts/D6_behavioural_numbers.py` (run from the repo root with `.venv/bin/python`, about 3 seconds). It reproduces the verified base exactly: lock-early surplus in 2033 p5/p50/p95 \$159k/\$207k/\$273k, 2031 floor \$127k/\$165k/\$217k (F-401/F-402). All the figures below come from it unless another source is named.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (read on the primary page; URL and date in Sources); **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **DER** = derived by the script from labelled inputs; **INT** = my interpretation, never a fact about Laura.

Words used (defined once)

- **Promise money / the ladder:** Treasuries bought in January 2027 that pay the ten \$50,000 operating payments (2033-2042). About \$292k of the \$300k (F-101).
- **Growth money (the "sleeve"):** everything the payments do not need, roughly \$158k after the 2028 deposit. It pays for the facility contribution and for flexibility.
- **Floor:** part of the growth money moved into a 2-year Treasury in January 2031, which is already worth a known amount in 2033. The model locks 80% (an open parameter).
- **Risk need / risk-taking ability / behavioural loss tolerance:** the three parts of an investor's risk profile in CFA Institute's 2020 guide (VP). *Risk need* is how much risk a goal requires. *Ability* is how much loss the investor can bear without hurting her living standard or her goals. *Behavioural loss tolerance* is how she feels and acts when she loses money.
- **Willingness vs ability:** everyday names for loss tolerance vs ability.
- **Loss aversion:** people feel a loss more strongly than a gain of the same size. A common estimate is about 2 to 2.25 times as strongly (SNIP; Tversky & Kahneman 1992, not opened).
- **Myopic loss aversion:** loss aversion combined with checking results often (Benartzi & Thaler, VP).
- **Break-even effect / house-money effect:** after losses, people favour bets that could win back what they lost. After gains, they take more risk (Thaler & Johnson 1990, VP).
- **Reference point:** the number a person compares an outcome with to decide whether it feels like a gain or a loss.
- **Pre-commitment:** a rule agreed in calm times that decides what happens in a stressful moment.
- **Calibration:** stated confidence matches how often you turn out to be right.
- **p5 / p50 / p95:** in a simulation, the result that 5% / 50% / 95% of paths fall below. p50 is the median.

0. Summary: top findings, ranked by impact on reaching the semifinals and on making the plan hers

1. **Use a professional risk-profile framework to state Laura's risk tolerance (M019; IPS, tier 2).** The IPS must state a risk tolerance (IPS guide L3, VRF), and the case never gives one (R-AN17). CFA Institute's 2020 guide (VP) splits a risk profile into three parts: risk need, risk-taking ability and behavioural loss tolerance. Its reconciliation rules settle Laura's case without quoting her:
 - The promise money has no risk need: Treasuries cost \$292,264 of \$300,000 (F-101). Its ability to bear loss is nil, because "high degree of certainty" and "may not rely on co-sponsors" (case L91-92,

- VRF). CFA rule (VP): "Higher behavioral loss tolerance can be ignored when both the risk need and risk-taking ability are lower". So however willing she is, no market risk goes on the promise.
- The growth money has a low risk need: there is no facility target (case L104). Its ability to bear loss is high: living costs sit outside the portfolio and there are no withdrawals before 2033 (case L61-65, VRF). CFA rule (VP): "A lower risk need can be discounted when both risk-taking ability and behavioral loss tolerance are higher". So it holds stocks even though no target requires them.
 - Her *investment* loss tolerance is unknown. CFA measures it by interview or by past investing behaviour (VP), and we may not contact her. So the plan must not depend on her composure: every bad-year decision is pre-committed (finding 4). 2. **The job she hires the firm for is keeping every public number true, with growth second (M228; IPS pitch, tier 2).**
 - The case uses "growth" once (L73). It uses "certainty" 4 times, "credib-" 3 times and "confiden-" 3 times (DER word count, VRF).
 - None of the case's five "must" tests mentions growth (L134-145).
 - Wharton changed the client criterion from "would win him/her over" to "earn her confidence" (B2b, VP).
 - Credibility research (VP) says that once the outcome is known, people judge a source by *calibration*: was its confidence justified? Confidence alone does not earn credibility.
 - This mostly confirms stakeholder_map s7 ("promise first" as the central idea). What is new is the evidence base and a content spec for the pitch. 3. **A plain two-number rule picks the growth money's stock share (M237; IPS rule, FR table, tier 2).**
 - The medians hardly differ (\$201k-\$214k from 0% to 100% stock, JPM). So the *rule* decides the weight, and each candidate rule gives a different answer:
 - "max median" gives 100%;
 - "max p5" gives 0%;
 - a loss-averse client who compares against the all-Treasury alternative is nearly indifferent under JPM and prefers about 20% under Vanguard's lower forecast.
 - The rule that is plainest and closest to "protecting the capital" (case L73-74) is: **hold the most stock that still leaves about a 19-in-20 chance that the growth money ends 2033 with at least every dollar Laura put in (~\$158k).** It picks 60% under JPM's forecast and 50% under Vanguard's midpoint (DER). The pick does not change with heavier-tailed returns at the 1-in-20 level.
 - This confirms the provisional 60/40 and gives the IPS a checkable reason for it. 4. **Bad years are normal, and the plan is built so that none of them can touch the promise (M019 bad-year analysis; supports parked M007 and D8's M006).**
 - Under the current plan (DER, ASM rate model):
 - the growth money falls in about 27% of years (2028-2030);
 - at least one down year before 2033 happens in 62% of paths;
 - the whole statement shows a fall in about 1 year in 10, and at least once in half of all paths.
 - The rejected growth-first plan would show Laura a first annual statement (December 2027) with assets *below* the value of the ten payments in about 40% of paths.
 - Four documented traps (myopic loss aversion; break-even; house money; long-run regret of inaction) each get one pre-commitment (section 2.1, bad-year table). D8's rule set (M006) already holds the rules; D6 adds the evidence for them. 5. **Drop "floor-first leap" as Laura's trait (M018; TN/IPS/FR, tier 1).**
 - It rests on a reporter's sentence, and D13 ruled it out (brief s16).
 - The research also says career risk-taking does not predict investment risk-taking: risk attitudes are "highly domain-specific" (Weber, Blais & Betz 2002, VP). That is the case's own "Although" (R-AN16).
 - Base the IPS risk sentence on the case and the CFA framework (finding 1), not on her biography. 6. **Checklist for Trading Note reflections (M125; TN, tier 1).**
 - Each "how it serves Laura" line should echo one trait *stated in the case* (the checklist is in section 1.3).
 - Each line should pass the swap test: would it still be true for any other client? If so, it is generic.
 - This is mostly answered already (SH-01 to SH-04, D13c). What is new is the note-by-note mapping. 7. **How wide the 2031 range should be (M197; FR, tier 3).** The 2031 range runs from the bought floor to the p90 of the 2033 total, and its width is set almost entirely by the lock share:

lock share	top ÷ bottom
80%	1.30x
90%	1.13x
70%	1.52x
60%	1.81x

- Today's 2026 view (p5-p95) is 1.72x.
- Research on how people judge ranges (Du et al. 2011, Yaniv & Foster 1995, van der Bles et al. 2020; all VP) says people prefer the *narrowest range the evidence warrants* and pay little trust penalty for honest numeric ranges.
- So quote the 2031-view range, not today's spread. Keep the lock share at 70-90%: below 70% the range tops 1.5x; at 100% it becomes the "one exact amount" the case rejects.
- The exact cut-off where a range becomes "useless" is still an ASSUMPTION (no study gives one for co-sponsors).

1. TIER 1 (WInS now and Trading Notes, Oct 23)

1.1 M125: Which case-stated facts about how Laura thinks should every recommendation visibly mirror?

One-sentence answer: Every "how it serves Laura" line should echo one case-stated trait and pass the swap test:

- planning turns her ideas into reality;
- a statistics-trained reader;
- thoughtful risk within an "appropriate balance";
- a product manager's dated steps;
- a public voice whose credibility is an asset;
- flexibility "as the project develops".

This is largely already answered by SH-01 to SH-04 and the D13c checklist. The new part is the mapping to each Trading Note.

Evidence

claim	source	status
Criterion 2 rewards recommendations "that can earn her confidence" and tailoring to "Laura's circumstances, priorities, risk considerations, and residency goals"	competition/official/2026_27/SMApply_Deliverables_Page_2026-09-27.md (criteria block)	VRF
Case traits: "Thoughtful financial planning and long-term investing will play an important role in turning those ideas into reality" (L38-39); "Statistics & Information Decisions Management ... product manager" (L16-17); "willing to take thoughtful risks ... appropriate balance" (L69-74); "damage her credibility" (L114); "financial flexibility as the project develops" (L102-103)	Laura_Gao_2026_Client_Profile.txt	VRF
Past client Ladi Ayoola (2025): "not just about the numbers, but about understanding and connecting with people"	D13b (B2a-Q06), Wharton 2025 finale article	VP (via D13b)
The 2022 "never felt more 'seen'" line is a Wharton writer's paraphrase; only "seen" is the client's word	D13b (B2a-Q02)	VP as article text; PARAPHRASE-UNVERIFIED as the client's words
No Laura quotes in the Trading Notes or IPS	brief s16; D13c s5.3	binding rule
Swap test: keep a Laura-specific sentence only if it rests on her professional record or the case	D13c s5.1 (from B3b Q12)	rule
The TN reflection must explain "How the decision supported the client's goals, funding needs, or risk considerations" in ≤100 words	TN guide L46-52	VRF

Mapping to the three planned WlnS notes (securities_v0 ticket). Elements only; the team writes the words.

WlnS note	case trait it should echo (pick ONE; 100 words is tight)	swap-test check
Hedge (IEF/TLH)	Planning turns ideas into reality (L38-39): this trade sizes to a specific promise (~\$292k for ten \$50k payments), so the residency's operating money stops depending on markets	Fails if it says only "reduce volatility" (the guide's own example, which hundreds of teams will copy; M012)
Growth (VT)	Thoughtful risk within an appropriate balance (L69-74): stocks hold only money the promise does not need, so a fall shrinks the facility range, never the payments	Fails if it says "Laura likes risk" or "for growth" alone
Short Treasuries (VGSH)	Credibility (L114) or flexibility (L102-103): money that can later become an amount she can state to co-sponsors because it is already owned	Fails if it quotes a floor percentage (securities_v0 forbids it)

- Statistics degree (L16): use it at most once, in the FR or IPS where a probability is defined (securities_v0 s1 already spends it on VT). In a TN it is decoration.
- Product manager (L17): echoed by dated steps (2027 ladder, 2028 deposit, 2031 floor, 2033 reserve). That belongs in the IPS, not in a single TN.
- Past-client evidence ("seen"; "not just about the numbers") comes from finale events, which are outside this run's scope. Use it only as the reason for the checklist, never in a deliverable.

Implication: a sentence checklist for each TN reflection's "how it serves Laura" line (TN, Oct 23), reused for the IPS client paragraph. **Deadline tier:** 1. **Confidence:** high. **Changes strategy:** no. **Criterion:** Client Knowledge and Objectives.

What this teaches: a note feels tailored when the reader can point to the line in the case it answers. Adjectives about the client do not do that.

1.2 M018: Does Laura's "floor first, then leap" career story justify "lock the promise, then take risk with the surplus"?

One-sentence answer: No, the question is mis-posed. Do not use it.

- "Floor first" is a reporter's causal link: its "So" follows personal material, not a financial plan. D13 ruled that it must not be presented as her trait (brief s16).
- Research finds that risk-taking is "highly domain-specific", so a career leap says little about investment risk.
- The IPS risk sentence should rest on the case's "Although ... appropriate balance" and the CFA risk-profile rules (M019).
- Her two verified risk ideas can appear at most once, in the FR (D13c E2):
 - regret of *not* leaping;
 - letting down early backers is worse than failing yourself.

Evidence

claim	source	status
"So when she got the book deal with HarperCollins, she gave Twitter her notice." is the Input reporter's sentence (2022-03-07), and its "So" follows personal material	D13b line 254; D13c s3.4 and s7c	VP (reporter's words, not Laura's)
Binding: "Do NOT describe a 'floor-first leap' ... as Laura's trait"	brief s16	rule
Risk-taking "was highly domain-specific, i.e. not consistently risk-averse or consistently risk-seeking across all content domains" (financial investing is scored separately from social and other domains)	Weber, Blais & Betz 2002, <i>J. Behav. Decis. Making</i> 15(4):263-290, abstract on Princeton research portal	VP (abstract)
The case itself separates career risk from portfolio risk: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance..."	case L69-74; R-AN16	VRF
Her verified risk words: "you should always take the jump because you're always going to regret not doing it" (Overachiever 2021) and "letting other people down — people who were your earliest supporters and were the first to believe in you" (Daily Pennsylvanian 2016, student-era)	D13a/D13c (B1b-Q11, D13a-N01)	VP (via D13a; re-checked by D13c)

claim	source	status
Long-term regrets centre on failures to act; recent regrets on mistaken actions (Gilovich & Medvec 1995, as summarised by a 2023 replication, which found the interaction but a different exact pattern)	Richardson & Gilovich 2023, <i>R. Soc. Open Sci.</i> (PMC10282588)	VP (replication paper's summary; the 1995 original was not opened)

Numbers: none needed. The question fails on evidence, not on arithmetic.

Implication (decision, TN/IPS/FR):

- No WInS note, TN reflection or IPS sentence cites her career timeline or the book deal.
- The IPS risk sentence uses the case plus the three-part risk profile (M019).
- In the FR, at most one pair of her verified lines, dated and in context (D13c E2).

The honest link to her regret rule runs the other way. Long-run regret is about *not acting*. That is the reason the growth money holds real stock (M237) rather than the minimum.

Deadline tier: 1 (notes cannot be edited once saved). **Confidence:** high. **Changes strategy:** no (it changes the evidence and the wording only). **Criterion:** Client Knowledge and Objectives.

What this teaches: a person who takes big career risks may still want her savings handled carefully. Risk attitudes differ by domain, and the case's "Although" is signalling exactly that.

2. TIER 2 (IPS, Nov 6: the strategy freezes)

2.1 M019: How should the IPS state a risk tolerance the case never gives?

One-sentence answer: State it goal by goal in the three parts professionals use:

- **promise money:** risk need nil, ability nil, so no market risk whatever her appetite;
- **growth money:** risk need low, ability high, investment loss tolerance inferred and unknowable without an interview, so real but bounded stock exposure (50-60%, set by the M237 rule);
- **the announced 2031 floor:** ability nil once it is promised.

Name what would change the view, and pre-commit the bad-year decisions so that the plan does not depend on her composure.

Evidence

claim	source	status
IPS definition: "outlines an investor's financial goals, risk tolerance, and guidelines..." and helps "make disciplined decisions during changing market conditions"	IPS guide L3-5	VRF
The case states no risk tolerance, benchmark or return target	R-AN17 (word check)	VRF
Case risk wording: "Laura understands that investing involves uncertainty and periods of market volatility." (L68-69); "willing to take thoughtful risks ... appropriate balance between pursuing growth and protecting the capital required for her goals" (L69-74)	case	VRF
Ability inputs: living costs "covered by income and financial resources outside the portfolio"; no additions or withdrawals before 2033 except the two deposits (L61-65); payments "with a high degree of certainty" and teams "may not rely on co-sponsors..." (L91-92)	case	VRF
Three risk-profile factors: risk need, risk-taking ability, behavioural loss tolerance	CFA Institute, <i>Investment Risk Profiling: A Guide for Financial Advisors</i> (2020), Introduction and Figure 1	VP
"Higher behavioral loss tolerance can be ignored when both the risk need and risk-taking ability are lower"	same, "Reconciling and relating the IRP to a portfolio strategy"	VP
"A lower risk need can be discounted when both risk-taking ability and behavioral loss tolerance are higher."	same	VP
"Risk-taking ability sets an upper volatility bound ... Because risk-taking ability changes (especially as the goal time horizon shortens), this factor must be reassessed regularly."	same	VP
"Consequence of failure ... refers to the financial and emotional threats an investor faces if a goal is not achieved"	same, Risk Need section	VP

claim	source	status
"Risk composure refers to the likelihood that in a perceived or actual crisis, an investor will exhibit behavior fundamentally different from her past actions ... measured by evaluating an investor's past decisions and actions."	same	VP
"Where possible, the IPS should account for known liabilities to lend some quantitative basis to the risk tolerance assessment" and "More nuanced approaches may attempt to define multiple levels of risk associated with ... meeting a specific future financial goal"	CFA Institute, <i>Elements of an Investment Policy Statement for Individual Investors</i> (2010), s3c	VP
Risk as "the probability of failing to reach the threshold level in each mental account, and attitudes toward risk that vary by account"	Das, Markowitz, Scheid & Statman 2010, <i>JFQA</i> 45(2):311-334, abstract	VP
Contacting the client means disqualification	R-W16	VP (Phase A)
Team docs' "moderate to low capacity" conflates career income risk with portfolio capacity	B4b-05 (FCA FG11/05: capacity for loss is judged by the effect on "standard of living")	VP via B4b

Numbers (DER; current plan: growth money 60% stock, 80% locked in 2031; JPM inputs; ASM rate model)

measure	value
Promise: Treasury cost vs money available (risk need)	\$292,264 of \$300,000 (F-101): no return above Treasuries is needed
Growth money: chance a given year 2028-2030 is down	27%
Growth money: chance of at least one down year 2028-2032	62%
Growth money: worst single year 2028-2032, median path	about -\$3k (-2%)
Growth money: worst single year, bad case (p5)	about -\$21k (-12%) (thin-tailed model; 2008-type years were worse, D3/M059)
Ladder marked to market: chance a given year shows a fall	14% (p5 fall about -\$9k), but the payments' value falls by the same dollars, so they stay 100% covered
Whole statement: chance a given year 2027-2032 shows a fall	10%; at least one down year in 50% of paths
2033 growth money p5 / p50 / p95	\$159k / \$207k / \$273k (F-401)
For contrast, rejected growth-first plan: chance the Dec-2027 statement shows assets below the value of the ten payments	40% (before the 2028 deposit); 5.4% at some Jan 1 2028-2033 after it

Reading (INT):

- Ability on the growth money is high: a bad year costs facility dollars only, and none of her living costs depend on it.
- Ability on the promise is nil.
- Willingness for *investment* losses cannot be measured. CFA measures it by interview or by past investing behaviour (VP); we may not interview her, and her public record contains no investing behaviour (D13c V8: "We hold no verified statement by Laura about investing").

So the IPS must say the stock share is the team's inference. The design makes that inference low-stakes: if she turns out to be less composed than assumed, the worst action she could take (selling after a fall) can only shrink the facility range. It cannot touch the payments, because the ladder is not for sale under the plan's rules.

What a strong IPS risk passage must contain (spec, not text):

1. Risk tolerance stated **per goal**, not as one label ("moderate" is what most teams will write; SH-01).
2. For the payments: no risk *needed* (Treasuries already cost less than her deposit) and none *affordable* (the case demands high certainty with no outside help). So her appetite for risk is deliberately not used here.
3. For the growth money: she can bear losses (living costs are outside the portfolio; nothing is withdrawn before 2033), so it holds stocks even though no facility target requires them. The share is set by a stated rule (M237).
4. For the 2031 floor: once announced, it carries no risk.
5. That the investment-willingness part is inferred, with what would change it (list below).
6. Case words only (no Laura quotes, no citations: IPS guide L123).

What would change the inference (the FR's "what would change our view" line):

- A smaller or later 2028 deposit: ability falls before 2028 (already handled: the payments come first).
- A named facility target or date pressure (the case has none, L104): risk need rises, and the growth money should lock earlier rather than take more risk (CFA rule: need cannot exceed ability).
- Evidence that she prefers a larger *certain* facility to a chance at a bigger one: a lower stock share, down to the all-Treasury control.
- A lower equity outlook (for example the Vanguard midpoint, section 2.3): the same rule gives 50%, not 60%.
- Nothing she says about career risk: domain-specific (M018).

Bad-year behaviour and pre-commitments (what Laura will feel and do; my domain).

Each trap below is documented. Each pre-commitment is one rule. Several already exist for WInS (securities_v0 s5, parked M007). The 2031 ones are new.

moment	documented trap (source, status)	what she might feel or do (INT)	pre-commitment (spec)	already covered?
Rates rise; the ladder's statement value falls (14% of years)	Myopic loss aversion: investors "distinctly more sensitive to losses than to gains" who "evaluate their portfolios frequently" (Benartzi & Thaler, NBER w4369, VP)	"We lost \$9k on the safe part"	Report "10 of 10 payments covered" first and market value second; never sell the ladder after a rate rise	Partly: securities_v0 never-rule 1 (WInS); BS-09 (report funded status first)
Stocks fall in 2029-2030, before the 2031 pitch	Break-even effect: after losses, "outcomes which offer a chance to break even are especially attractive" (Thaler & Johnson 1990, VP)	Delay the 2031 lock "until it recovers", or add stock to catch up	The 2031 lock happens on its date (or by its pre-set ratchet, M055) whatever markets did; the stock share only rebalances within its band	Yes, in D8 (written in parallel): rule 3's trigger is the date 2031-01-01, and its governance line says rules change "never because markets moved" (phase_D/D8_practice.md, M006). New here: the documented reason (the break-even trap), which the FR can cite
Stocks rose strongly before 2031	House-money effect: "increased risk seeking in the presence of a prior gain" (same, VP)	Announce more than is bought, or keep more in stock	Only the bought amount is announced as the floor; any upside is labelled with its model probability	Partly: brief s8.7 (floor bought), BS-05 (verbs by tier)
Long run (2033 and after)	Long-run regrets centre on "failures to act" (Richardson & Gilovich 2023 summary, VP)	"We played it too safe"	The growth money's stock share is set by the M237 rule, never cut to the minimum to look safe	New link (D13c N1 raised the tension)
Any year	Contested size of the "behaviour gap": Morningstar estimates investors earned 7.0% vs funds' 8.2% a year (2015-2024), a 1.2-point gap; a 2026 FAJ study of the same sample finds poor timing costs "only 0.10% per year". Morningstar: allocation-fund investors kept "nearly 97%" of the funds' return	Trading on news	Rules above; yearly review on a fixed date. Claim them as cheap insurance , not as a proven 1.2%-a-year gain	New evidence; the size is disputed, so do not quote 1.2% as fact

The Guide supports this framing: "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." (Guide p.5 L156-157, VRF).

Implication:

- IPS: a risk passage built per goal (spec above). The 2031 lock's date trigger and "never because markets moved" already sit in D8's rule set (M006); D6 supplies the reason.
- FR: the what-would-change list; a one-row "bad year" illustration (growth money -12% in a p5 year means facility range smaller by about \$21k; payments unchanged).

Deadline tier: 2. **Confidence:** high for the framework and the structure; medium for the inferred share. **Changes strategy:** no. It confirms "risk only in the sleeve", and gives the evidence for D8's 2031 date trigger. **Criterion:** Client Knowledge and Objectives (also Investment Strategy).

What this teaches: "risk tolerance" is not one number. Professionals ask three questions: how much risk a goal *needs*, how much loss the investor *can* bear, and how she *feels* about losing. When the answers differ between goals, each pot of money gets its own answer.

2.2 M228: What job is Laura hiring an asset manager to do?

One-sentence answer: Both jobs, in a fixed order. First, make every number she will say in public true: ten years of operating money bought in 2027, and a floor bought in 2031. Second, grow what is left so that her facility contribution is as large as she can responsibly make it. The IPS pitch should name the first job as the central idea and the second as its purpose. This is largely already answered (stakeholder_map s7 "promise first"); what is new is the evidence base below.

Evidence

claim	source	status
Word count in the case: "growth" 1 (L73, "pursuing growth"); "certainty" 4 (L91, L98 twice, L136); "uncertain-" 3; "credib-" 3 (L114, L116, L143); "confiden-" 3; "responsib-" 4; "flexib-" 5	D6_behavioural_numbers.py [5] on the case text	DER from VRF
The five things the strategy must do (L134-145): support the payments with high certainty; a responsible facility contribution; address uncertainty; communicate "clearly and credibly"; preserve flexibility. None says "maximise growth"	case	VRF
"If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors."	case L114-115	VRF
Her 2028 money comes from reputation-driven work ("publishing advances, speaking engagements, licensing"), so credibility is part of her income	case L43-45; BS-03	VRF + INT (Phase A)
Criterion wording changed from "would win him/her over as a client" (2022-23, 2023-24) to "can earn her confidence" (2026-27)	B2b s0	VP (via B2b)
"whether a person is seen as credible ultimately depends on whether the person demonstrates good calibration. Credibility depends on whether sources were justified in believing what they believed."	Tenney, Spellman & MacCoun 2008, <i>JESP</i> (author PDF, in-press version)	VP
After an error, confident witnesses lose more credibility than unconfident ones	Tenney, MacCoun, Spellman & Hastie 2007, <i>Psychological Science</i> 18(1)	SNIP (abstract in search results; PubMed and SAGE blocked)
Communicating numeric uncertainty caused "only a small decrease in trust in numbers and trustworthiness of the source, and mostly for verbal uncertainty communication"	van der Bles et al. 2020, <i>PNAS</i> 117(14):7672-7683, abstract on the University of Groningen portal	VP

Reading (INT):

- In 2031 and 2033 co-sponsors will *see the outcome*. That is exactly the setting where calibration, not confidence, decides credibility.
- A firm whose job is "grow her money" will be judged on returns.
- A firm whose job is "make her promises keepable" is judged on whether the numbers held. The case scores the second (R-C66, R-S25, R-S28).
- The either/or in the question is false (stats skeptic): R-C38 asks for a balance. So the answer is "promise first, then growth", never "not growth".

What the 50-word pitch and the IPS's first line must contain (spec):

1. What is secured, and when: the ten payments are bought in 2027, before any risk is taken.
2. What the rest is for: growth for the facility contribution, with flexibility kept.
3. The credibility link: anything announced to co-sponsors in 2031 is already owned.

Exclude: return targets, fund names, "maximise", "guaranteed" without its qualifier, and any Laura quote.

Implication: the IPS pitch's central idea (IPS guide L16, "What is the central idea...") and the FR's opening. The FR could also open from the 2031 conversation and work backwards (B9b idea). That is a presentation

choice for the team.

Deadline tier: 2. **Confidence:** high for the direction; the "jobs to be done" label is optional. **Changes strategy:** no. **Criterion:** Client Knowledge and Objectives (and Creativity & Presentation for the pitch).

What this teaches: ask what a client could lose that she cannot get back. For Laura, money can be regrown but a broken public promise cannot. That tells you which job comes first.

2.3 M237: When the designs tie on the median, what rule should choose between them?

One-sentence answer: State a two-number safety-first rule: hold the most stock in the growth money that still leaves about a 19-in-20 chance that it ends 2033 with at least every dollar Laura put in (~\$158k). That picks **60%** stock under JPM's forecast and **50%** under Vanguard's lower midpoint, and it holds with heavier-tailed returns.

- Max-median and max-p5 rules give the extremes (100% and 0%).
- A loss-averse client who compares against the all-Treasury alternative is almost indifferent under JPM.

So the *reference point* decides, and the IPS should name the one it uses.

Evidence

claim	source	status
Medians barely move with the stock share; more stock buys range, not median	F-407; brief s8.5; B10a	VRF / DER
Risk defined as "the probability of failing to reach the threshold level in each mental account"	Das et al. 2010, <i>JFQA</i> , abstract	VP
People's risk-taking depends on prior gains and losses relative to a reference (house money / break-even)	Thaler & Johnson 1990, <i>Management Science</i> 36(6):643-660, abstract on EconPapers	VP
"Risk-taking ability sets an upper volatility bound to a portfolio recommendation."	CFA Institute 2020	VP
Vanguard (model run 2026-06-30): U.S. equities 4.2%-6.2% a year over 10 years	B10a s0 item 2	VP via B10a
JPM 2026 LTCMA: U.S. large cap 6.70% compound (7.94% arithmetic, 16.47% vol)	F-301	VRF
Case asks the team to "recommend an appropriate balance"	case L72-73	VRF

Numbers: design table. Case B inputs: JPM equities 6.70%; sleeve bonds 5.00%, following B10a's correction that today's Treasury yields are ~1pp above JPM's bond assumption (F-317; ASM). 80% of the growth money is locked in 2031. Riskless control (the whole growth money in Treasuries to 2033) ≈ \$203k (DER from F-012). Money put in = \$157,736.

stock share in growth money	2033 p5	p50	p95	P(ends below money put in)
0%	\$183k	\$201k	\$222k	0.0%
20%	\$182k	\$205k	\$231k	0.0%
40%	\$173k	\$208k	\$252k	0.6%
50%	\$167k	\$209k	\$264k	1.8%
60% (current)	\$161k	\$210k	\$277k	3.6%
70%	\$155k	\$211k	\$290k	5.9%
80%	\$149k	\$212k	\$303k	8.3%
100%	\$137k	\$214k	\$332k	12.9%

Case C uses Vanguard's midpoint (5.2%, JPM vol kept; ASM). Its medians are \$201k-\$204k for **every** share from 20% to 100%, and the p5 falls from \$180k to \$130k. P(below money put in) at 50/60/70% is 2.8/5.4/8.4%.

Which design each rule picks (DER):

rule	JPM (B)	Vanguard midpoint (C)
Maximise the median	100%	70% (a flat tie from 20% to 100%)
Maximise the bad case (p5)	0%	0%

rule	JPM (B)	Vanguard midpoint (C)
Most stock with P(below money put in) $\leq$ 5%	60%	50%
Same at $\leq$ 10%	80%	70%
Most stock with P(below \$160k) $\leq$ 5% (ASM threshold)	60%	50%
Same with \$170k	40%	20%
Loss-averse ($\lambda = 2.25$) vs the riskless control	flat: 40-100% all within about \$2k of each other; 0% worst	20%
Loss-averse ($\lambda = 2.25$) vs money put in	100%	80%
Fat-tail check (Student-t, 4 degrees of freedom, same volatility): P(below money put in) at 60% / 70%	3.2-3.8% / 4.9-5.5%: pick unchanged at 60%	not run

Reading (INT):

- Laura's "growth" choice this year is a choice about spread. The rule, not the forecast, decides the share.
- The reference point matters most:
 - Measured against what she put in, almost every design "gains", and a loss-averse reader wants lots of stock.
 - Measured against the all-Treasury alternative (which the FR should show, M034), roughly half of all paths "lose" at any stock share, because the median gain over the control is only about \$2-9k.
- The money-put-in rule is the simplest to explain and to check. Its two numbers are a threshold and a probability, the form a statistics graduate expects (Das et al.).
- It sits closest to the case's "protecting the capital" (INT: "capital required for her goals" mainly means the payments, so this is an extension, not the case's definition).
- It answers her long-run inaction regret: it does not set stocks to the minimum.
- **The threshold and the 1-in-20 are team choices (ASM).** The table shows what other choices would pick.

Implication:

- **Decision (IPS):** one rule sentence in the team's words. It must contain the threshold (money put in), the probability (about 19 in 20) and what it applies to (the growth money only).
- **Number:** growth-money stock share 60% if the team keeps JPM as its base; 50% if it adopts the more cautious house view. Do not quote the probability to one decimal in the IPS.
- **FR:** the design table (four rows is enough: 0/40/60/100%) with the all-Treasury control as a reference line. Chart: a p5-p50-p95 "whisker" per design, with the median band nearly flat (data: the table above).

Deadline tier: 2. **Confidence:** medium. The numbers are model properties; the rule is a judgement.

Changes strategy: no. It confirms the provisional 60/40 (WInS VT 20.5% / VGSH 12.5% unchanged) and gives it a stated reason. It would move to 50% only if the team adopts the Vanguard view. **Criterion:** Investment Strategy (also Portfolio Analysis).

What this teaches: when two choices have the same middle outcome, you are really choosing how wide the good and bad outcomes spread. Say what you are measuring against before you say what you chose.

3. TIER 3 (Final Report, Dec 4)

3.1 M197: How wide can the stated 2031 range be before non-specialists find it useless?

One-sentence answer: Quote the range Laura would state *in 2031*: from the bought floor to about the 90th percentile of her 2033 total. At an 80% lock share that is about 1.3x top-to-bottom (\$165k-\$215k at the median state), not today's 1.7x spread.

- Research says people prefer ranges "as precise as warranted by the information available, but not more precise", and honest numeric ranges cost little trust.
- Keep the lock share between 70% and 90%. Below 70% the range tops 1.5x; at 100% it becomes the "one exact amount" the case rejects.

- The exact point at which a range becomes "useless" is not established by any study we can verify (ASM).

Evidence

claim	source	status
Investors' preference for (im)precision "peaks for low levels of imprecision and diminishes when the range gets wider"; they "favor forecasts that are as precise as warranted by the information available, but not more precise"	Du, Budescu, Shelly & Omer 2011, <i>OBHDP</i> 114(2):179-189, abstract on RePEc	VP (both phrases verbatim)
"Coarse (imprecise) judgments are less informative than finely grained judgments; however, they are likely to be more accurate"; people "might accept errors in the interest of securing more informative judgments"	Yaniv & Foster 1995, <i>J. Exp. Psych.: General</i> 124(4):424-432, p.424 (author-hosted scan, read as an image)	VP
Numeric uncertainty caused only a small trust decrease, "mostly for verbal uncertainty communication"	van der Bles et al. 2020, <i>PNAS</i> , abstract	VP
Credibility depends on calibration once outcomes are known	Tenney, Spellman & MacCoun 2008	VP
Case: "Rather than promising one exact amount, she wants to communicate a credible range"; "state how confident they are that her 2033 contribution will fall within that range"	case L115-118	VRF
The confidence statement is two-sided: a range wide enough to be safe becomes useless; one narrow enough to be useful becomes risky	R-AN9	VRF
Du et al. studied earnings forecasts read by investors, not arts co-sponsors, and no numeric threshold appears in the abstract	stats skeptic note, M197	limit

Numbers (DER; stock 60% of the unlocked part; top = p90 of the 2033 total, given the 2031 state)

lock share in 2031	top ÷ bottom (p90 top)	(p95 top)	median ÷ floor
60%	1.81x	1.85x	1.68x
70%	1.52x	1.55x	1.44x
80% (model)	1.30x	1.32x	1.25x
90%	1.13x	1.14x	1.11x
100%	1.00x (one amount)	1.00x	1.00x

- With 100% stock in the unlocked part the ratios barely change: 1.34x at an 80% lock.
- The ratio does not depend on the sleeve's size in 2031, only on the lock share and the unlocked part's mix.
- In dollars (80% lock; 2033 US\$; model, not forecasts):
 - weak 2031 state (sleeve p10): floor \$134k, top \$175k;
 - median state: floor \$165k, top \$215k;
 - strong state (p90): floor \$204k, top \$265k.
- Today's 2026 view of the 2033 surplus (p5-p95): \$159k-\$273k, 1.72x.
- Confidence "within the range" at the 2031 view:
 - about 0% chance of falling below the floor (it is bought; residual risk U.S. default);
 - about 10% chance of landing above the top.

So it is "about 9 in 10", two-sided, with the model named (R-AN9, SH-04).

Reading (INT):

- Du et al.'s "as precise as warranted" is the key idea. In 2031 the information *warrants* a narrow range, because 80% of the money is already a bought Treasury and only two years of risk remain on the rest.
- Quoting today's 1.7x spread would be less precise than the information allows, and would read as "we don't know".
- A single number would be more precise than warranted, and the case forbids it.
- The Final Report must still give a dollar range now (R-C68). So present it as a rule plus the three illustrative 2031 states above (parked M040's answer), not as one 1.7x band.

Implication:

- **Number (FR):** range = bought floor to about the p90 of the 2033 total, about 1.3x at an 80% lock.

- **Decision input for D3's M060 (lock share):** behavioural evidence supports 80-90% and argues against anything below 70%. It cannot choose between 80% and 90%.
- **Chart (FR):** three bars (weak, median and strong 2031 states). Each shows the floor as a solid block and the upside to p90 as a lighter block, labelled "about 9 in 10 within". Data: the dollar rows above.

Deadline tier: 3. **Confidence:** medium (the ratios are model properties; the usefulness threshold is ASM).

Changes strategy: no. It supports the 80% lock share already in the model. **Criterion:** Creativity and Presentation ("communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors").

What this teaches: a range is useful when it is as narrow as your knowledge honestly allows. By 2031 the bought floor is what makes a narrow range honest.

4. Cross-question notes for the main loop

- **Consistency:**
 - The M237 rule gives 60% under the same JPM inputs that produce F-401. The WInS ticket (VT 20.5% / VGSH 12.5%) needs no change.
 - If D3's model v2 adopts sleeve bonds near 5%, the rule's pick is unchanged (case B).
 - If the team adopts Vanguard's view, the pick is 50%: VT about 17% / VGSH about 16% of the WInS book (ASM arithmetic on securities_v0's 33% growth money), for S3 to recompute.
- **The 2031 clause is already in D8's rule set:** a dated trigger (2031-01-01) plus a governance line saying rules change "never because markets moved" (phase_D/D8_practice.md, M006). D6 adds the behavioural evidence for why it matters (break-even after a fall; house money after a rise). No second clause is needed; the FR can cite the evidence when it reports that the rule "fired as written".
- **Quotes:** none of the six answers needs a Laura quote in the TN or IPS. The FR may use the D13c E2 pair once.
- **Parked items this supports:** M007 (bad-year rule), M033 (goal-by-goal risk to the first reader), M040 (range as a rule with an illustration).

5. Sources (accessed 2026-09-27/28 UTC)

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- Richardson & Gilovich (2023), replication of Gilovich & Medvec, *Royal Society Open Science* (PMC10282588): <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10282588/> (the 1995 original was not opened: its host failed TLS).
- Morningstar, *Mind the Gap 2025* (US): https://www.morningstar.com/content/cs-assets/v3/assets/blt9415ea4cc4157833/blt2c5c4d9171638c42/689b424311f3880edc4b4813/US_Mind_the_Gap_2025.pdf
- Fulkerson, Jordan, Riley & Yan (2026), "Bad Timing Does Not Cost Investors 15% of Their Funds' Returns", *FAJ*, abstract: <https://rpc.cfainstitute.org/research/financial-analysts-journal/2026/bad-timing-does-not-cost-investors-funds-returns>
- Tversky & Kahneman (1992), *J. Risk Uncertainty* 5:297-323: loss-aversion coefficient 2.25 is SNIP (the paper was not opened; the value is only a tested parameter here).

Repo files (VRF or labelled): competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (L16-17, L38-39, L43-45, L61-74, L91-92, L102-104, L114-118, L134-145); 2026_WGY_Investment_Policy-FINAL.txt (L3-5, L16, L20, L123); 2026_WGY_Trading_Notes_Analysis-FINAL.txt (L46-53); 2026_WGY_Investment_Competition_Guide.txt (L35, L70, L156-157, L182-183); SMAApply_Deliverables_Page_2026-09-27.md (criteria); research/insight_v1/phase_A/(case_register R-AN9/R-AN16/R-AN17; fact_register F-101, F-301, F-317, F-401-F-409; stakeholder_map SH-01 to SH-05, BS-03, BS-09, s7); phase_B/B10a_contrarian.md, B2b_why_laura_intent.md, B4b, B6b_cosponsors.md (Q18); phase_C/survivors.json, parked.json; phase_D/D13a*, D13b*, D13c_voice_map.md; wins_now/securities_and_allocation_v0.md; research/verified_2026-09-27/strategy_mc.py (engine reproduced). Script: research/insight_v1/scripts/D6_behavioural_numbers.py.

What this teaches

- **Ask three risk questions, not one.** How much risk does the goal need? How much loss can she bear? How will she feel when she loses? For Laura the answers differ between the promise and the growth money, so each gets its own answer. That is what "tailored" means.
- **A bold career does not mean a bold portfolio.** Risk attitudes differ by domain. The case's "Although" says the same thing in one word.
- **Plan for the bad year before it arrives.** A down year is more likely than not before 2033. The plan's strength is that every bad year has one meaning (a smaller facility range) and one pre-agreed response. That is easier to live with than a plan whose bad years raise the question "are the payments still safe?"
- **When the middle outcome ties, the comparison point decides.** Say what you measure against (the money she put in, or the all-Treasury alternative) before you pick a number.
- **Honest ranges earn trust.** People trust sources whose confidence proves justified. Give the narrowest range your bought floor supports, say how sure you are on both sides, and name the model.
- **Evidence can be contested.** The famous "1.2% a year behaviour gap" has a serious 2026 rebuttal. Use rules because they are cheap insurance, not because a disputed number says they pay.

Audit corrections (AY1)

Auditor AY1 (cluster auditor: client psychology and co-sponsors), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: research/insight_v1/phase_D/audit_client_cosponsors.md. Check script: research/insight_v1/scripts/AY1_audit_checks.py (reuses this file's engine and random stream).

Reproduction. D6_behavioural_numbers.py re-run 2026-09-28: every quoted number reproduces (base \$159k/\$207k/\$273k; the case B design table; every rule pick on the 10-point grid; 27% / 62% / -\$3k / -\$21k / 14% / -\$9k / 10% / 50%; 40% and 5.4% for growth-first; the M197 ratios and dollar states; the case word

counts, re-checked by grep on the case text). **Sources re-opened 2026-09-28** (fetch helper, --grep found verbatim unless stated): CFA Institute 2020 risk profiling (both reconciliation rules; "Risk-taking ability sets an upper volatility bound"; "measured by evaluating an investor's past decisions and actions"); CFA 2010 IPS paper ("account for known liabilities..."); Weber, Blais & Betz ("highly domain-specific"); Tenney, Spellman & MacCoun 2008 (found; the second sentence missed the grep only because the PDF writes "justified" with a ligature); Du et al. 2011 (both phrases); van der Bles et al. 2020; Thaler & Johnson 1990 (both phrases); Benartzi & Thaler w4369 (both phrases); Morningstar Mind the Gap 2025 (7.0% vs 8.2%, 1.2 points, "nearly 97%"); FAJ 2026 ("0.10% per year"); the 2022-24 criterion "would win him/her over as a client" (Wharton WordPress REST, page modified 2024-06-03) and the 2026-27 "earn her confidence" (SMAApply page L51). All VERIFIED-PRIMARY or VERIFIED-REPO-FILE. Yaniv & Foster 1995 (image-only scan) was not re-checked.

#	Severity	Issue	Correction
C1	important	"The rule picks 60% ... does not change with heavier-tailed returns" is a 10-point-grid artefact (finding 3, s2.3, s4). On a 5-point grid (AY1 [3], [3b]) the same rule picks: 60% (JPM, 4% bonds); 65% (JPM, 5% bonds, this file's own case B); 55% (Vanguard midpoint); 65% / 70% with fat tails (A / B); 55-60% with a 0.5% yearly fee; 50-55% with 1.0%. D6's own fat-tail line shows 70% at 4.9% (under 5%) in case B, so "pick unchanged at 60%" is wrong there. Also, a Student-t with the <i>same variance</i> makes the 5% tail lighter, so it is not a stress test.	Say: "the rule gives about 55-70% depending on the return, bond and fee assumptions; 60% sits inside that band, and so does D2's 50% under cautious inputs." The rule gives a checkable <i>reason</i> for a choice in the 50-70% band; it does not "confirm" 60%. IPS: state the rule and "about 60%" (or the team's pick), never a decimal. Historical crash sequences (D3) are the real stress test.
C2	important	Hedge-note trait row overclaims (s1.1 table; tier 1, notes are permanent). (a) "sizes to a specific promise (~\$292k)": under option (ii) the WInS hedge is \$198,000 (66% of \$300,000, the ladder's ~65% share of both deposits scaled; AY1 [5]); only under (iii) is the order about \$292k-294k. (b) "so the residency's operating money stops depending on markets": the funds never mature and move about 10% per 1-point rate change; the ticket already bans "match" and "stable".	(a) Say the hedge is set to the payments' rate sensitivity (about 10 years) and, under (ii), scaled to the share of the post-2028 portfolio the ladder takes; put the ~\$292k in the reflection as the real-plan cost. (b) Replace with "moves with the value of her payments when rates change".
C3	important	Internal inconsistency with M018 and the D13 rule. s1.2 says career risk-taking does not transfer to investment risk ("highly domain-specific"), then says of her regret line: "That is the reason the growth money holds real stock". Using her career-regret words as the reason for equity is the same domain transfer, and D13 bans writing her "investment philosophy" (brief s16).	The reason for holding stock is the case ("appropriate balance between pursuing growth and protecting the capital") plus the CFA risk-profile rules and the stated share rule. Regret research (Richardson & Gilovich) is context only. Brief s16 treats her regret line as a <i>tension the plan must answer</i> (why equity is only ~0-2% in 2027, ~20% in 2028-30, ~4% after the 2031 floor), not as a justification.
C4	important	IPS spec gaps (s2.2 pitch spec, s2.1 risk passage). Pitch item 1 "the ten payments are bought in 2027, before any risk is taken" holds only if the ladder costs no more than \$300k in January 2027 (it cost more on 173 of 185 trading days of 2026, brief s14; modelled ~24%, ASSUMPTION). The risk passage omits brief s16's second certainty gap.	Pitch/IPS spec item 1: "bought in January 2027; if prices have risen, the rest is completed from the 2028 deposit before any growth money is invested" (D8 rule 1, D12 rule 2). Add risk-passage item 7: "certain" means nominal US\$; a fixed \$50k buys less each year in Taiwan (named, not solved). Add item 8 (parked M033): state the whole-portfolio stock share by stage, not only the growth money's.
C5	important	M197 width guidance assumes all unlocked money is promised (s = 1). "Keep the lock share at 70-90%; below 70% the range tops 1.5x; behavioural evidence ... argues against anything below 70%" conflicts with D5's candidate a = 0.7, s = 0.5.	Width $\approx 1 + s(1 - a)/a \times 1.21$ at a p90 top (AY1 [2]). D5's (0.7, 0.5) gives 1.26x, narrower than this file's 80% base (1.30x). Restate the criterion as the width itself (about 1.3x or less) or the promised-but-at-risk share $s(1 - a)$ (about 0.20 or less), not the lock share. The evidence cannot choose between D5's (0.7, 0.5) and (0.8, 1.0).
C6	important	Scope (brief s17, 2026-09-28). s3.1 (M197 numbers, dollar states, bar chart), the FR bad-year illustration (s2.1), the FR design-table chart (s2.3) and "the FR could open from the 2031 conversation" (s2.2) are Final Report material.	Move to the "later (after Nov 9)" list. Keep for the IPS: the per-goal risk passage (with C4), the share rule (with C1), and the range METHOD (bottom = bought amount; top = a stated percentile given 2031 holdings; two-sided confidence, model named).

#	Severity	Issue	Correction
C7	minor	VGSH trait row ("money that can later become an amount she can state to co-sponsors because it is already owned").	Keep the ticket's "could later become <i>part of</i> a promised amount" (AX2 D2-C6): in the plan the 2031 amount is bought from the whole growth money, stocks included. Never "the floor". D12's role word "risk management" (the Guide's word, checkable with a number) is compatible; the echoed case trait fits "flexibility as the project develops" better than "credibility". Team chooses and records it.
C8	minor	"Risk need nil" (finding 1, s2.1 table).	Add "at today's prices": if the January 2027 price is above \$300k, the gap is a <i>funding</i> need met by the 2028 deposit, not a reason to take market risk. The conclusion (no market risk on the promise) stands.
C9	minor	"Report '10 of 10 payments covered' first" (bad-year table).	True only once the ladder is complete, which may be January 2028 (C4). Real-portfolio reporting is Final Report material.
C10	minor	Applicability labels. Tenney et al. study eyewitness credibility (a car-accident scenario); Du et al. study investors reading earnings forecasts.	Label their use for arts co-sponsors INT (the file already flags Du et al.).
C11	minor	Top percentile differs from D5 (p90 here, 85% in D5).	One percentile in the IPS method (see D5 C10).

Status after audit. Finding 1 (per-goal risk profile via the CFA framework) stands and is the strongest IPS contribution in the cluster. Finding 2 stands (word counts and criterion change re-verified) with C4. Finding 3 is downgraded from "confirms 60%" to "gives a stated reason for a 50-70% choice" (C1). Finding 4's numbers reproduce; its pre-commitments stand. Finding 5 stands, with C3. Finding 6 stands, with C2 and C7. Finding 7 is parked to the later list, and its lock-share guidance is restated (C5, C6).

D7 Wharton Intent Historian: M002 (how the three Trading Notes show a developing strategy) and M011 (what only our thesis can truthfully claim)

Agent D7, insight_v1 run, Phase D, written 2026-09-27. Lens: the client trend, Wharton's official framing, past case designs compared with this one, and what semifinal readers reward. This is AI-generated research for Team Caplet: evidence, numbers, specs and checklists. **None of it is text to submit.** The team decides and writes every word, including every WInS note, and records AI use in the Final Report's Works Cited (R-W46). Nothing here is for the finale.

Script: `research/insight_v1/scripts/D7_rule_trigger_odds.py` (run from the repo root: `.venv/bin/python research/insight_v1/scripts/D7_rule_trigger_odds.py`). Seed 20260927, 200,000 paths.

Terms used: **band** = a range around a target weight; the team trades only when a weight leaves it. **bp (basis point)** = 0.01 percentage point. **Hedge** = the Treasury funds whose value moves like the cost of Laura's ten payments. **Funded ratio** = hedge value divided by the value of the payments on the same day.

Summary of top findings (ranked by effect on reaching the semifinals and on making the plan Laura's)

- (M011, tier 2 IPS, strongest wording finding) The pitch must not claim "her ten payments are bought in January 2027".** That is false in roughly 1 in 4 to 1 in 3 modelled rate paths: the ladder costs more than \$300k if rates fall more than about 27bp before January (D7 script: ~23%; A2: 24.3%; S4: 30.7%; all ASSUMPTION-based). The claim only our plan can make truthfully *in every path* is a **rule**: *no dollar of Laura's money takes stock-market risk until all ten payments are bought* (with the 2028 deposit finishing the ladder first if rates fall). A rule is true by construction; a date is a forecast. The case's credibility test ("overpromising could damage her credibility", case p.3) applies to our own pitch first.
- (M002, tier 1 WInS-now/TN) No sensible rebalancing band will fire before Oct 23, so do not plan a "discipline trade".** With the ticket's weights, the chance that any band fires by Oct 20 is about 0% (55-65 growth band; whole-book equity +/-3 or +/-5 points) to 0.1-0.3% (a +/-2 to +/-3 point band). The hedge-duration band (9.65-10.15 years) drifts about 0.08 years at most in three weeks. Blueprint 01 section 7 ("Oct 1-16: at least one discipline/rebalance decision") should be dropped: planning it invites a staged trade, which R-W28 forbids and R-W88 discourages. Under lock-early the characteristic discipline is *not* trading, and a non-trade cannot be a Trading Note.
- (M002, tier 1) "Tested" does not need a new trade: it can live in the reflection of the hedge purchase.** The reflection (written ~Oct 20) may say how the decision was tested by a real market move. A 10-year yield move of 10bp or more between an Oct 1 fill and Oct 20 has a ~53% chance at the end date and ~83% at some close (realised 2026 volatility 4.45bp/day, VERIFIED-PRIMARY data, driftless ASSUMPTION). 10bp moves the payments' value about 1%. The test: did the hedge's % change track the payments' % change? This needs **data captured from the fill date** (WInS hedge values + treasury.gov curve on the same dates). The official wording is "supported, tested, *or* refined" in all four official texts: there is **no quota** of one note per type.
- (M002, tier 1) The only honest source of a "refined" note is a pre-announced planned adjustment or a real discovery.** Candidates, in order: (a) the Session Rules position limit forcing TLH to be split with SPTL (a real rule discovered at the first trade); (b) the growth split, which is genuinely undecided (60/40 provisional; CLAUDE.md calls the equity weight the weakest call) *if* the first growth note says so and names the date the team will decide; the Guide allows "planned adjustments" (p.5) and older Wharton guidance says of the strategy "Can you tweak it? Absolutely!" (VERIFIED-PRIMARY). If neither happens, three "supported" notes with honest "tested" reflections are fine.
- (M011, tier 2) Creativity this season is scored on the thesis's fit to a new kind of case, not on instruments.** Past cases (2021-22 to 2025-26) were growth or return targets with hobby hooks; this case is the first with a fixed liability larger than the deposits (111%), the first to use "certainty" and "co-

sponsor", and it has no hobbies at all (B2a/B1b, VERIFIED-PRIMARY past case pages; my re-check of the 2021-22 wording below). Three Laura-specific, checkable decisions carry the thesis: promise-before-growth (her 2028 money comes from career income she herself called "unstable", FR only); a 2031 floor that is bought, plus a stretch with a stated probability (the case's co-sponsor task); certainty defined as "bought at market prices, not forecast" (a statistics-trained client can check it). The premise that "AI tools will all say lock early" is **untested** and the decision does not depend on it.

6. **(M002, tier 1) Dated events in the window move rates but create no trade under our rules:** September jobs report Fri Oct 2 and September CPI Wed Oct 14 (both SNIPPET-UNVERIFIED; bls.gov refused access); the next FOMC meeting is **Oct 27-28, after the Oct 23 deadline** (VERIFIED-PRIMARY, federalreserve.gov). These are the moments to snapshot the hedge and the curve, not to trade.

M002 (tier 1: WInS-now and Trading Notes, Oct 23; also one IPS clause)

Question (canonical). Should the three Trading Notes be one "supported", one "tested" and one "refined" decision, and which pre-written rule (a WInS rebalancing band or rate trigger) and which dated market events between Sept 28 and Oct 20 can make a genuine rule-triggered trade happen before the Oct 23 deadline?

One-sentence answer. No quota applies and no sensible band will fire before Oct 23, so the team should choose three real trades (promise, growth, and either a real refinement or a third funding purpose), show "tested" inside the hedge note's reflection using a real rate move measured from data saved at the fill date, and never create a trade to fill a category.

Part-mis-posed. The question assumes a rule-triggered *trade* is the way to show "tested". Under lock-early the rules say "do not trade" after rate moves (ticket "never" rules 1-2), so the realistic rule-triggered event is a check with no trade. The better route to "tested" is the reflection, which the guide allows to be written after the fact (only the WInS note itself is frozen).

Evidence

Claim	Source	Status
The TN guide uses both "aligned with, tested, or refined" (instructions, L12) and "supported, tested, or refined" (L29-30); the Guide p.3 and the Infographic say "supported, tested, or refined"; the case says "reflected, tested, or refined" (L152). "Or" in every version: no one-of-each requirement	competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt L12, L29-30; 2026_WGY_Investment_Competition_Guide.txt L89; 2026_WGY_Competition_Infographic.txt L24-25; Laura_Gao_2026_Client_Profile.txt L152	VERIFIED-REPO-FILE
"A decision can still demonstrate thoughtful analysis and strategic thinking even if your team later sells the investment or changes its approach."	TN guide p.1 (R-T18)	VERIFIED-REPO-FILE
"A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move"	Guide p.5 L156	VERIFIED-REPO-FILE
"The competition does not require frequent or same-day trading." / "you should not be doing a lot of buying and selling of the stocks designated for the long-term portion of your portfolio"	SMApplly Trading Details and FAQ (R-W73, R-W88), read by Phase A 2026-09-27	VERIFIED-PRIMARY (via case_register)
Teams must not "fabricate analyses, invent teamwork and experiential stories"	Rules page (R-W28)	VERIFIED-PRIMARY (via case_register)
Trade notes cannot be edited; the three notes are quoted "exactly as it appears in WInS"	TN guide p.2 L44; Stock-Trak blog via phase_A/wins_week1_guardrails.md item 6	VERIFIED-REPO-FILE / VERIFIED-PRIMARY (vendor, not season-specific)
Older Wharton guidance on strategy: "Defining your strategy early is essential to doing well. Can you tweak it? Absolutely!" and "if you provide us with a team strategy that is buried somewhere in your experience, rather than clearly stated, defined and articulated up front, you will not be among the strongest teams."	https://globalyouth.wharton.upenn.edu/developing-strategy/ (undated, older season; accessed 2026-09-27, <code>fetch_text.py --grep YES</code>)	VERIFIED-PRIMARY (older-season page)

Claim	Source	Status
2026 FOMC meetings: "September 15-16", "October 27-28", "December 8-9"	https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm (accessed 2026-09-27)	VERIFIED-PRIMARY
Employment Situation for September 2026: Oct 2, 8:30 a.m. ET; CPI for September 2026: Oct 14, 8:30 a.m. ET	WebSearch snippet of bls.gov schedule pages (bls.gov returned HTTP 403 to the fetch helper)	SNIPPET-UNVERIFIED
Ticket plan: VT 60% / VGSH 40% of the growth money, band 55-65; hedge duration band 9.65-10.15y; duration refresh Oct 19-23; "Do not create a decision just to get a note"	research/insight_v1/wins_now/securities_and_allocation_v0.md sections 4-5	VERIFIED-REPO-FILE (a plan, not a fact)
Blueprint timeline: "Oct 1-16: at least one discipline/rebalance decision"	research/blueprints/01_trading_notes_blueprint.md section 7	VERIFIED-REPO-FILE (superseded by this finding)

Numbers (D7 script; all probabilities are ASSUMPTION-based model outputs, not forecasts)

Rule	Move needed	P(fires by Oct 20), weekly / daily checks
VT share of growth money, 55-65 (ticket)	VT vs VGSH -18.5% / +23.8%	0.00% / 0.00%
VT share of growth money, 57-63	-11.6% / +13.5%	0.12% / 0.15%
VT share of whole book 20.5%, +/-5 points	-28.9% / +32.7%	0.00% / 0.00%
VT share of whole book, +/-3 points	-17.7% / +19.1%	0.00% / 0.00%
VT share of whole book, +/-2 points	-12.0% / +12.6%	0.29% / 0.33%
Hedge duration band 9.65-10.15y	drift > 0.25y	worst case ~0.08y in 3 weeks (weight drift +/-0.03y for +/-50bp; issuer-duration drift ~-0.05y at the Jun-Sep pace) -> essentially 0

Inputs: realised 10-year daily change sd **4.45bp/day** (185 trading days of 2026, live treasury.gov CSV, VERIFIED-PRIMARY); equity vol 16.78% (JPM AC World, used as ASSUMPTION for short-run vol); correlation 0 (ASSUMPTION; JPM -0.01). Window: 19 U.S. trading days Sep 28-Oct 22; 13 days of market moves after an Oct 1 fill to Oct 20.

Rate moves available to a "tested" reflection (Oct 1 fill to Oct 20): |move| >= 5bp: 76% at end date; >= 10bp: 53% (83% touched at some close); >= 15bp: 35% (55%); >= 25bp: 12% (18%). Median |move| 10.8bp. A 10bp move changes the payments' value about 1.0% (~\$2,890 on the 2027 value; DV01 \$289/bp, VERIFIED via the verified script). Because the WInS hedge is scaled (\$198k of \$300k in option (ii)), compare **percentage** changes, never dollars.

Implication (decisions and specs; the team writes every word)

- **Decision (WInS-now + TN):** drop the planned "discipline/rebalance trade" from blueprint 01 section 7. Plan the three notes from trades that will happen anyway: (1) the hedge purchase, (2) the growth purchase, (3) either a real refinement (below) or the third funding purpose (VGSH, or SPTL if the position limit forces a split).
- **Decision (WInS-now, before the growth order):** if the team really has not settled the equity share of the growth money, the growth note should say the split is provisional and name the week the team will decide (e.g. "decide by Fri Oct 16 after our own research"). If the decision then changes the split, that one VT-against-VGSH trade is a genuine, pre-announced "refined" note (Guide p.5 "planned adjustments"). If the decision confirms 60/40, there is no trade and the log records "decided not to trade". **Never** buy at a split the team already considers wrong in order to "refine" it later: that is staging (R-W28).
- **Decision (WInS-now, data capture, owner needed):** on the fill date and on Oct 2, Oct 9, Oct 14, Oct 16 and Oct 20, save a screenshot of WInS position values (IEF, TLH/SPTL, VT, VGSH) and download that day's treasury.gov par curve. Then value the ten payments on each date with `research/verified_2026-09-27/official_curve_pv.py` (spot value). This feeds the "tested" reflection now and the Final Report's funded-ratio chart later (B7a Q12).

- **"Tested" reflection spec (hedge note, <=100 words, team's words):** must contain the rate move over the period (bp, from treasury.gov); the hedge's % change; the payments' % change on the same dates; what the rule told the team to do (hold); one sentence on what it means for Laura (the promise stayed funded while prices moved). If the move is under ~5bp, say the test was small rather than overstate it.
- **"Tested" reflection spec (growth note):** the growth money's % change and the fact that the payments were unaffected ("a fall shrinks the facility range, never the payments", ticket checklist 2).
- **IPS clause (tier 2):** one rebalancing sentence in the IPS (CFA IPS guidance says even a no-rebalance policy should be documented, per B4b, VERIFIED-PRIMARY by that agent). Content: bands exist for the growth money only; the hedge is never traded because rates moved.
- **Events calendar (check, never trade):** Oct 2 jobs report; Oct 14 CPI; Oct 12 Columbus Day (U.S. bond market closed while stocks trade, ASSUMPTION that WInS ETF prices still update); Oct 23 TN due; FOMC Oct 27-28 after the deadline (its effect can only appear in the IPS/FR, not in the notes).
- Deliverables: WInS-now, TN (and one IPS clause). Deadline tier: 1. Confidence: high on the band odds and on the "no quota" reading; medium on the value of a "tested" reflection (no public evidence of how readers score it).
- Changes current strategy? **No** to the strategy; **yes** to the TN plan (drop the planned rebalance; add a provisional-split sentence with a decision date to the growth note checklist; add a data-capture step to ticket step 1).
- Criterion served most: **Articulation of Competition Experience** ("clearly explains the team's research and decision-making process", R-S27), with Investment Strategy's "disciplined planning" (R-S24).
- What this teaches: a rule that almost never fires in three weeks is working as designed; you show discipline by measuring what the market did to your plan, not by inventing a trade.

M011 (tier 2: IPS pitch Nov 6; tier 3: Final Report)

Question (canonical). Since lock-early is now the textbook answer that AI tools and hundreds of teams will give, which 2-3 Laura-specific, checkable decisions make our "clear and creative thesis" different, and what one claim can only our strategy truthfully make in the 50-word pitch?

One-sentence answer. Lead the pitch with a rule, not a date: "no dollar takes stock-market risk until all ten payments are bought" is true in every rate path and is something growth-first and "balanced bucket" plans cannot say, and it is carried by three Laura-specific decisions (promise before growth, a bought 2031 floor with a stated stretch, certainty defined as bought rather than forecast).

Partly mis-posed. The premise ("AI tools and hundreds of teams will give lock-early") is **untested**. No public discussion of the Laura Gao case was found (two WebSearch queries, 2026-09-27, returned only official pages and third-party competition listings). The recommended claim does not depend on the premise: it is chosen because it is true, checkable and tied to Laura, so it holds whether rivals converge or not.

Evidence: what Wharton's framing and past cases say is "creative" this year

Claim	Source	Status
Criterion: "Presents a clear and creative investment thesis"	SMAApply deliverables page (R-S24), SMAApply_Deliverables_Page_2026-09-27.md	VERIFIED-REPO-FILE
Pitch: "Clearly communicate the central idea behind your approach and how it is designed to support your client's financial goals and future funding needs." (50 words max)	IPS guide p.1 L35-36 (R-I17)	VERIFIED-REPO-FILE
"There is no single correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions."	Infographic	VERIFIED-REPO-FILE
Case: overpromising could "damage her credibility and lose the confidence or participation of co-sponsors"	Case p.3 (R-AN30)	VERIFIED-REPO-FILE

Claim	Source	Status
Case: 2028 deposit comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"; she wants "an appropriate balance between pursuing growth and protecting the capital required for her goals"; degree in "Statistics & Information Decisions Management"	Case L44-45, L70-74, L16	VERIFIED-REPO-FILE
Laura on leaving her job: "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income." (Overachiever Magazine, 2021-03-15)	phase_D/D13a_laura_quotes_verified.md row B8b-Q18	VERIFIED-PRIMARY per D13a. Final Report only ; never in the TN or IPS (D13 rule)
2021-22 case (Nichole Jordan): a \$5,000 scholarship "for at least 10 years. Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends? Another option?"	https://globallyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/ (accessed 2026-09-27, --grep YES)	VERIFIED-PRIMARY
Case trend 2021-2026: payment stream was 50% of assets in 2021-22, 111% of deposits now; "certainty" 0 hits in past cases, 6 now; "co-sponsor" 0 before, 9 now; no hobby hooks this year; 2025-26 had a \$1.5m growth target and a values rule	phase_B/B2a_why_laura_intent.md summary 1-4; phase_B/B1b_case_anomalies.md section 1	VERIFIED-PRIMARY per those agents (past-case pages; 2025-26 text from a secondary copy, SNIPPET-UNVERIFIED)
"Your strategy should be unique to your team."	https://globallyouth.wharton.upenn.edu/developing-strategy/ (older season)	VERIFIED-PRIMARY
"Oftentimes, the best solutions are made up of simple, elegant ideas." (Melissa Ko Hahn, judge of the 2025 semifinal presentation videos)	https://globallyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/ (accessed 2026-09-27, --grep YES)	VERIFIED-PRIMARY (context: semifinal videos, not the written round)
"less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once." (DMV's Finest, 2022-23 champions, on their final report)	https://globallyouth.wharton.upenn.edu/news/wharton-investment-competition-raises-voices-from-the-2023-teams/ (--grep YES)	VERIFIED-PRIMARY
Distinctive elements already listed (model-free certainty; 2031 range as a rule; "buys range, not median")	research/council_2026-09-27/01_chair_memo.md section 7; phase_A/stakeholder_map.md BS-10, SH-18	VERIFIED-REPO-FILE (AI council brainstorming)

INTERPRETATION (D7): the designers moved the scoring weight from picking investments to reliability, flexibility and credibility. So a thesis is "creative" this year if it answers *when and how certainty is created, and how it is communicated to people who must believe her*, in a way a reader has not seen from other teams. Instruments are expected to be plain (Wharton's own example note buys an intermediate Treasury ETF, R-AN44).

Numbers: which candidate pitch claims are true in every path?

Candidate claim	True when	Source
"Her ten payments are bought in January 2027 with her first deposit"	Only if the ladder costs $\leq$ \$300k on the purchase date: false in ~23% (D7 one-factor), 24.3% (A2), 30.7% (S4) of modelled paths; the ladder cost more than \$300k on 173 of 185 trading days of 2026	D7 script [6]; brief section 14; ticket section 8 (ASSUMPTION-based)
"The payments are fully funded before any money is invested for growth" / "no dollar takes stock-market risk until all ten payments are bought"	Every path, because it is a rule (if rates fall first, the 2028 deposit completes the ladder before any growth buy). In the joint tail (rates fall and the 2028 deposit is missing) the rule still holds but some payments stay unfunded: the pitch must not add "guaranteed"	Brief section 7; ticket section 8; joint tail open (brief section 9)
"Her 2031 promise to co-sponsors is a floor already bought, plus an upside with a stated probability"	Every path by rule (the floor is whatever the bought note will pay); the stretch is a model number	Brief section 8 item 7; strategy_mc.py floor p5/p50/p95 \$127k/\$165k/\$217k
"95% certain"	Model-dependent; a statistics-trained reader can challenge it (brief section 8 item 6)	Avoid

The 2-3 Laura-specific, checkable decisions (content spec; the team chooses and words them)

1. **Promise before growth (lead idea).** Laura hook: her second deposit depends on career income (case L44-45; her own "unstable income", FR only). Checkable: WInS order sequence (hedge first, ticket step 1) and the 2027/2028 rule in the IPS. Truth test: passes in every path.
2. **A bought 2031 floor, plus a stretch with a stated probability.** Laura hook: the case's co-sponsor task and its credibility warning; criterion 5 scores exactly this communication (R-S28). Checkable in the FR. IPS gets only the principle (the guide says the final range is not expected in the IPS).
3. **Certainty = bought at market prices, not forecast** (nominal US\$, barring U.S. default). Laura hook: a statistics-trained client can check a price, not a model. Checkable by anyone with a Treasury price. Optional fourth for the FR only: report progress as a funded ratio (uses the M002 data capture).

Implication

- **Pitch content spec (IPS, tier 2; words are the team's; 50-word hard cap):** must contain (a) the central idea as a rule (promise bought before any growth risk); (b) what it does for Laura (payments certain in US\$; the growth money serves the facility contribution and flexibility); (c) optionally the 2031 idea (floor bought, upside stated honestly). Must **not** contain: a date-certain claim ("bought in January 2027"), "guaranteed"/"100%"/"95%", jargon (LDI, duration, DV01), any Laura quote, identity imagery or book-title puns (D13 rules), more than one number.
- **IPS first paragraph:** repeats the same central idea and states the if-rates-fall rule, because the Guide forbids redesign after results (p.5) and the chance of needing it is ~1 in 4 to 1 in 3.
- **FR (tier 3):** main chart for the lead idea = the funded ratio over the WInS period (data from M002) or the 2027-2033 "promise bought first, growth second" timeline; the one Laura quote allowed, if any, from D13a with outlet and year.
- **Optional AI comparison test (only if the team wants it; low priority):** after the team has drafted its own pitch (R-W51: "Did I come to my current conclusion before or after using generative AI tools?"), paste the public-facing parts of the case into 2-3 AI tools with one neutral prompt, record the architecture each suggests, and log tool, date and purpose in Works Cited (R-W46). It measures the premise; it does not change the recommended claim.
- Deliverables: IPS (pitch + first paragraph), FR; TN only as consistency (same words for the three jobs). Deadline tier: 2 (IPS Nov 6), FR parts tier 3. Confidence: high that the date-certain claim is unsafe and the rule claim is true; medium that it is *distinctive* (the field is unobserved).
- Changes current strategy? **No** to the strategy; **yes** to its wording: replace "bought in January 2027" with the rule wherever it is used as a headline (brief section 7 and chair memo sentences are internal; the pitch and IPS must use the rule).
- Criterion served most: **Investment Strategy** ("a clear and creative investment thesis", R-S24), with Client Knowledge ("recommendations that can earn her confidence", R-S25).
- What this teaches: the most distinctive claim is often the most honest one; a rule you control can be promised, a market price you do not control cannot.

Sources (accessed 2026-09-27 unless stated)

Primary web (VERIFIED-PRIMARY, read with `research/insight_v1/scripts/fetch_text.py`, quotes checked with `--grep`):

- Federal Reserve FOMC calendar: <https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>
- Treasury daily par yield curve 2026 CSV (via the D7 script): https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&format=csv
- 2021-22 case (Nichole Jordan): <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/>
- Developing a Strategy (older season): <https://globalyouth.wharton.upenn.edu/developing-strategy/>
- 2025 top-10 article (Hahn): <https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/>
- Tales from the 2023 teams (DMV's Finest): <https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-theses-from-the-2023-teams/>

Snippets and blocked (SNIPPET-UNVERIFIED / not used as fact):

- bls.gov schedule pages (HTTP 403 to the helper): https://www.bls.gov/schedule/2026/10_sched_list.htm , https://www.bls.gov/schedule/news_release/empisit.htm (dates from the WebSearch summary only).
- WebSearch for public discussion of the Laura Gao case (two queries): only official pages and third-party listings (e.g. <https://www.aralia.com/helpful-information/guide-to-the-wharton-global-high-school-investment-competition/> , <https://competemap.com/competitions/cmrw8ls8y0001vnne2cnjon1n>); nothing on case strategy.

Repo files (VERIFIED-REPO-FILE): `competition/official/2026_27/{Laura_Gao_2026_Client_Profile,2026_WGY_Trading_Notes_Analysis-FINAL,2026_WGY_Investment_Policy-FINAL,2026_WGY_Investment_Competition_Guide,2026_WGY_Competition_Infographic}.txt`, `SMAApply_Deliverables_Page_2026-09-27.md`; `research/insight_v1/context/brief.md`; `research/insight_v1/phase_A/{case_register, stakeholder_map, wins_week1_guardrails}.md`; `research/insight_v1/phase_B/{B7a_competition_meta, B2a_why_laura_intent, B1b_case_anomalies}.md`; `research/insight_v1/phase_C/survivors.json` (M002, M011); `research/insight_v1/phase_D/D13a_laura_quotes_verified.md`; `research/insight_v1/wins_now/securities_and_allocation_v0.md`; `research/blueprints/01_trading_notes_blueprint.md`; `research/council_2026-09-27/01_chair_memo.md`; `script research/insight_v1/scripts/D7_rule_trigger_odds.py`.

What this teaches

1. **Read the small words in the rules.** "Supported, tested, *or* refined" is a menu, not a checklist. Reading it as a quota would have pushed the team toward a staged trade, which the rules forbid.
2. **Measure before you plan.** A five-minute simulation shows a +/-5 point band has essentially no chance of firing in three weeks. Plans that depend on the market "doing something" by a deadline are wishes.
3. **Evidence has to be collected on the day.** A test of the hedge needs the hedge's value and the Treasury curve on the same dates; you cannot rebuild WInS values later.
4. **Say what you control.** "Bought in January 2027" depends on interest rates; "no stock risk until the payments are bought" depends only on the team. Laura's case warns that overpromising costs credibility; the pitch is the first place that warning applies.

Audit corrections (AY2)

Auditor AY2, 2026-09-28. The text above is left unchanged. Each item says what to correct, its severity (blocking / important / minor), and which deliverable it touches (TN or IPS). Evidence and the cross-file view are in `research/insight_v1/phase_D/audit_judges_practice_comms.md`. New check script: `research/insight_v1/scripts/AY2_audit_checks.py`.

Reproduced and confirmed (no change needed):

- **Script:** `D7_rule_trigger_odds.py` was re-run. Every quoted figure reproduces: 4.45bp/day; band odds; drift ~0.08y; 76/53/35/12%; 83% "touched"; ~23%.
- **Sources re-found verbatim with `fetch_text.py --grep`:** FOMC "October 27-28"; "Can you tweak it? Absolutely!"; "Your strategy should be unique to your team."; the 2021-22 case wording; DMV's Finest "less is more".
- **Repo lines as cited:** the TN guide, Guide p.5 L156-158, the Infographic and blueprint 01 L66.

Corrections

1. **[important, TN] Refinement candidate (a) is not a "refined" decision.**
 - Candidate (a) is the position-limit TLH/SPTL split. The `wins_now` ticket's gate boxes (c) and (e) require the Session Rules screenshot and the cap branch **before** the first order. The capped branch then places TLH and SPTL in the same first session.
 - Calling SPTL a "refinement" would describe a pre-planned purchase as a later change (R-W28 risk). SPTL also has the same job as TLH, so it adds no new funding purpose.
 - Keep (a) only if an order is actually rejected by a limit the team could not see beforehand, and the note says so. Candidate (b), the pre-announced provisional growth split with a decision week, remains the honest route to "refined".
2. **[important, TN] "Tested" must use a pre-registered window.** The window is the fill date to the Oct 20 close (or the last close before writing).

- The planning figure is the end-date chance of a move of 10bp or more (~53%).
- The "~83% touched at some close" figure is only reachable by picking the day with the biggest move, which is cherry-picking.
- Report small moves and poor tracking as they are. 3. **[important, TN + IPS] The pitch rule clashes with WInS option (ii).**
- The rule is "no dollar takes stock-market risk until all ten payments are bought". Under ticket option (ii), WInS buys VT (20.5%) on day one next to a hedge of about \$198k against a promise priced at about \$292k.
- This file's own TN plan (hedge note, growth note, VGSH note) assumes option (ii).
- Under (ii), the ticket's three-element scaling sentence must appear in the IPS, and a scaling clause in the hedge and growth notes. Otherwise a reader sees the pitch contradicted by the notes (case L161-162: "a clear and consistent investment strategy across all three").
- Under (iii), the rule is visibly true, but there is no growth note.
- This is a new input to gate box (b); it does not decide it. 4. **[important, IPS] Pitch spec (b), "payments certain in US\$", must be conditional.** Write "once bought, in US\$, barring a U.S. default". Otherwise it repeats the date-certain overclaim that finding 1 warns against.
- The joint tail also applies: rates fall before January 2027 **and** the 2028 deposit is smaller than the shortfall (about \$7k at -50bp, \$23k at -100bp; brief s6). 5. **[minor, TN] The data-capture step cannot be run as written.**
- `research/verified_2026-09-27/official_curve_pv.py` is hard-coded to the 2026-09-25 curve. It returns the 2027-01-01 forward value, not a spot value on other dates. Use `A2_curve_recheck.py (spot_pv / value_at)` or `D9_numbers.py [3]`.
- The treasury.gov yearly CSV keeps every date, so curves can be pulled later. Only the WInS position values need same-day screenshots.
- Two capture points (the fill date and the end date) are enough; the middle dates are optional.
- Check for IEF/TLH ex-dividend dates inside the window, or compare total returns. A monthly distribution (about 0.4%) is large next to a ~1% signal. 6. **[minor, TN] The two-decimal band odds (0.00% / 0.12% / 0.29%) are false precision from a thin-tailed model.**
- `AY2_audit_checks.py [2]` re-runs them under other assumptions (ASM):

Assumption	55-65 band	57-63 band / whole book +/-2
Fat tails (Student-t, 3 degrees of freedom)	~0.1%	~0.7%
Crisis-level 35% volatility	1-2%	11-15%

The conclusion is unchanged. Write "very unlikely (about 0-2% even in a crisis)". If a tighter band fires in a crash, that is a genuine rule-triggered note.

- [minor, IPS] The IPS clause "the hedge is never traded because rates moved" conflicts with the ticket's hedge duration band.** The ticket re-mixes IEF against TLH when the mix leaves 9.65-10.15y, and rate moves can cause that. Specify instead: the hedge is never sold or trimmed because rates moved; it is only re-mixed to keep its rate sensitivity near the payments'.
- [minor, TN] Wording of the "or" finding.** Summary finding 3 says "supported, tested, *or* refined" appears in all four official texts. In fact the TN guide L12 says "aligned with, tested, *or* refined" and the case L152 says "reflected, tested, *or* refined". The conclusion (every version says "*or*"; no quota) is confirmed.
- [minor, IPS] The case-trend claims need four fixes.**
 - (a) "certainty 6 now" counts "uncertainty" as well. The case has 4 "certainty" and 2 "uncertainty"; use D9's 4.
 - (b) The 2021-22 case also had a ten-year payment stream (a \$5,000 scholarship "for at least 10 years", re-verified), so "past cases were growth or return targets" is too broad. What is new is the size (face \$500k vs \$450k deposited) and the certainty/co-sponsor language.
 - (c) "111%" is face value. At today's prices the payments cost \$292k, 97% of the first deposit. Say "face value" so the figure does not imply the plan is underfunded.
 - (d) The 2025-26 part is SNIPPET-UNVERIFIED, so the summary's "VERIFIED-PRIMARY past case pages" label is mixed.
- [minor, IPS] The Hahn caveat is too narrow.** The page says she "served as a judge for the semifinals" and praised "your written reports and the videos" (re-verified; D13b agrees). Drop "not the

written round" and keep "past-season semifinal judge".

11. **[minor, TN] Label the R-W28 reading INT.** R-W28 bans lying about decisions and fabricating analyses. It does not name staged trades. "Staging, which R-W28 forbids" is a strong reading, not the rule's text: presenting such a trade as rule-driven would misstate the reasoning.
12. **[minor, scope] Move the Final Report items to the "later (after Nov 9)" list.** Brief s17 was added after this file was written. The items are the funded-ratio chart, "optional fourth for the FR only", the FR chart line under M011 and the Works Cited logging. The optional AI comparison test adds work without changing a decision: drop it unless the team asks for it.
13. **[minor, TN + IPS] Mark ready-made wording as illustrative.** Three phrases read as text to copy: the italic rule sentence (L23-24, L159-160) and the example note phrase "decide by Fri Oct 16 after our own research" (L121). Mark them as illustrative content; the team writes its own words (Wharton AI policy; brief s4).
14. **[minor, TN] "Oct 12 Columbus Day bond market closed" has no source here.** Label it ASSUMPTION. treasury.gov will show no curve for that date.
15. **[minor, TN] The "tested" reflection spec has to fit the official questions.** It lists five contents, but the TN guide asks each 100-word reflection to answer three questions: why; how it aligned; how it served the client's goals, funding needs or risk. The rate move and the two % changes (about 20-25 words) must sit inside those three answers, not replace them.

Verdict for D7: sound and useful. The "no quota" reading and "do not stage a trade" are confirmed. Before the TN plan is final, fix items 1-3; before the pitch spec is final, fix item 4.

D8 Professional-Practice Benchmark: our strategy against named industry frameworks

Agent D8, insight_v1 run, Phase D. Written 2026-09-27. Questions: M081 (tier 1), M006, M235, M109, M063, M026 (tier 2, with Final Report follow-ups). This is AI-generated research for Team Caplet, for brainstorming only. It holds **no submission-ready prose**: rule specifications, evidence, numbers and "a strong sentence must contain" checklists. The six students decide and write every word. Any framework named here that reaches the Final Report must go in its Works Cited, together with how AI was used (R-W46).

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (I opened the page on 2026-09-27 with `research/insight_v1/scripts/fetch_text.py` or `curl`), **VRF** = VERIFIED-REPO-FILE, **SNIP** = SNIPPET-UNVERIFIED, **ASM** = ASSUMPTION. R-, F-, BS-, SH-, X- ids point to `research/insight_v1/phase_A/`.

Numbers script: `.venv/bin/python research/insight_v1/scripts/D8_practice_numbers.py` (docstring lists inputs and labels; it reproduces the verified \$292,264 ladder cost and the \$394,930 reserve value first, as a check).

Terms used (plain English)

- **LDI (liability-driven investing)**: investing so the assets move like a known future bill. Pension funds do it.
- **Surplus**: the money above the market price of the promise. **Funded ratio**: assets divided by that price.
- **Glide path**: a pre-set plan for shifting from growth assets into matching assets. Pensions usually trigger each step when the funded ratio crosses a set level (a **trigger point**), not on a calendar date.
- **Goals-based wealth management (GBWM)**: one sub-portfolio per goal, each with its own horizon and its own acceptable chance of failure. Risk means "the chance of missing the goal".
- **Required return**: the yearly return a goal needs to be met. **Hurdle**: the return a riskier choice must beat.
- **Ability vs willingness to take risk**: ability (also called capacity for loss) is how big a fall the money can absorb without harming the goal; willingness is how much risk the person wants to take.
- **Rebalancing band**: a range around a target weight; you trade only when the weight leaves it.

0. Top findings (ranked by impact on reaching the semifinals and on making the plan unmistakably Laura's)

1. **The IPS needs four pre-declared rules, not three and not six (M006 + M235).** Professional IPS practice says it directly: the IPS offers "an objective course of action to be followed during periods of market disruption", and the process for changing it "should be clearly identified in advance" (CFA Institute 2010, VP). The Rules page binds teams to the Asset Manager Code, whose B.5.a says managers "Take only investment actions that are consistent with the stated objectives and constraints", and whose guidance says any flexibility "should be expressly understood and agreed to" beforehand (VP). The minimal set that covers all five case tests (R-C80-R-C84) is: **(1) promise first**: buy the ten payments with the first money; if they cost more than the money on hand, buy the latest-dated first and complete the rest from the 2028 deposit before anything else; **(2) growth mix with a yearly reset**; **(3) the 2031 range = what is already bought + a stated upside**; **(4) the 2033 order of use**: reserve, then a stated share for the facility, then flexibility. About 120-160 of the 500 words. Each extra rule must show a lift in a reported number or be dropped. The rebalancing rule passes that test: without it, the growth money's stock share wanders to 50-73% by 2031 (p5-p95), and about half of all paths leave a 55-65% band (script, ASM). Four professional precedents back the pattern: pension glide-path trigger points (BoFA, VP), APRA's "shortfall limit" and "restoration plan" (VP), Yale's two-year lagged spending rule (VP) and Russell's 2026 "surplus glidepath" (VP).
2. **Laura's needs can be met at today's Treasury yields; stocks are for upside, not need (M109).** Across both deposits the ten payments need only **1.05%/yr**. On the first deposit alone they need **5.09%/yr**, which is exactly what the bought ladder earns (the 27bp headroom). About **\$200k of 2033**

facility money needs 4.81%/yr on the growth money, below today's 5-year Treasury par yield (4.98%). A surplus held only in Treasuries at today's forward rates reaches ~\$204k in 2033. Stocks at 50/60/70% of the growth money add only about +\$9k/+\$11k/+\$12k in a middle-case projection (ASM: JPM 7.00% equity, bonds at the 5.23% forward). Yet at 60% the modelled 1-in-20 bad case is \$159k (F-407). The CFA IPS standard asks for exactly this statement, a "required ... growth rate ... to satisfy her future obligations" (VP). This does not change the architecture. It is strong devil's-advocate evidence for the open 50/60/70 decision, and it tells the team what the equity is *for*.

3. **Fees must be charged to the growth money, never to the bought payments (M026).** A typical U.S. adviser charges **1.00%** a year on accounts up to \$1M, and all-in costs run about **1.75%** up to \$500k (Kitces, reporting the Veres survey, 2017, VP). Taken *from the ladder*, a fee eats the headroom several times over: $0.15\% = 0.6x$, $0.5\% = 2.1x$, $1.0\% = 4.2x$ the \$7,736 headroom (script, ASM). Charged on the growth money only, a 1.0% fee costs the 2033 facility about \$10k. Charged on all assets but paid from the growth money, it costs about \$34k. The Code (F.4.d, VP) requires fee disclosure "in plain language", including gross- and net-of-fees returns. So: one IPS cost principle and one Final Report fee line with a sensitivity check.
4. **This is a goal-dedicated pool, and her ability to take risk differs by goal (M063).** The case puts her living costs "outside the portfolio" (R-C33). Under the Code's IPS guidance ("both the ability and willingness of the client to bear risk", VP) and the UK regulator's "capacity for loss" definition (VP), a loss on the surplus does not touch her standard of living. The payments, however, have **zero** capacity for loss, because the case bans outside money for them (R-C48). The IPS should say so and assume nothing about the size of her other wealth. It gives the risk-tolerance sentence a reason that belongs to her, without raising the equity weight.
5. **Chhabra's own words fix the team's bucket mapping and confirm "no tilts" in WInS now (M081).** In a 2016 interview (VP), Chhabra says an aspirational portfolio needs "a certain skill set" and "doesn't have to be an investment ... being really good at what you do, your passion—that could be part of your aspirational portfolio". Laura's career, books and the residency are her aspirational bucket. Her 2028 deposit already rides on them (R-AN31). The portfolio should hold only safety (the ladder) and market (broad funds). The team notes map "aspirational = flexibility reserve (stays invested for growth)". That puts the most risk on the money whose job is to absorb surprises. That is backwards, and a judge who knows the framework would notice.

Framework map (for the Final Report's strategy section; one table, cited there, not named in the IPS):

Framework (source, status)	What practitioners do	Our plan	Match?
Pension LDI, "surplus glidepath" (Russell, 2026-07, VP)	"lock down the benefits promised, then grow surplus assets"; "additional risk taken only on assets above a defined surplus threshold"	Ladder bought first; risk only in the growth money	Match
Pension glide-path triggers (BoFA, VP; NISA, VP)	De-risk "at set trigger points" chosen in advance	Rules 1-4 are triggers fixed on Nov 6	Match (rules are dated events plus one cost trigger, not funded-ratio levels, because the promise is already 100% hedged from day one)
APRA SPS 160 (VP)	Board-set "shortfall limit"; a "restoration plan" back to full funding within at most three years	Rule 1: if the ladder costs more than the cash, complete it from the 2028 deposit, first call on that money	Match in logic; our deadline is shorter (one year)
Yale spending rule (VP)	Spending set from the value "from two years ago", for predictable budgets	2031 range built from money already bought, applied in 2033	Match (ours is stricter: bought, not smoothed)
Goals-based WM (Brunel table in Das et al. 2018, VP; Chhabra 2015/2016, VP)	Needs at 90-95% success; aspirations in a separate risky bucket	Payments bought (above "needs"); no aspirational bucket in the portfolio	Match; deliberate difference explained in M081
Brunel: longer-dated needs may take more risk (via B4b, VP)	Time-segmented buckets	We hedge all ten payments now, including 2042	Deliberate difference: fixed nominal payments, no outside money (R-C48), duration already ~10y

Framework (source, status)	What practitioners do	Our plan	Match?
CFA IPS elements, individual (2010, VP)	Required return; ability and willingness; known liabilities; rebalancing policy "documented"; review process "identified in advance"; a spending calculus that includes fees	M109 / M063 / M006 / M026	Match once these items are added
CFA Asset Manager Code (VP; named in the Rules, R-W29)	Know the client (B.6.a); act within the stated mandate (B.5.a); plain fee disclosure (F.4.d)	Pre-set rules; fee principle	Match once the fee principle is added

1. M081 (tier 1: WInS now; then IPS and FR). Is Laura's aspirational bucket already filled by her career?

Question (canonical): Under Chhabra's risk-allocation buckets, is Laura's aspirational bucket already filled by her career, future books and the residency, so the portfolio should hold only safety (reserve) and market (broad sleeve) assets and refuse thematic, single-stock or country tilts?

One-sentence answer: Yes. Chhabra's own published words put skill-based bets ("being really good at what you do, your passion") in the aspirational bucket. Laura's creative business, her next books and the residency are that bucket, and her 2028 deposit already depends on them. So the portfolio holds only a safety part (the bought payments) and a market part (broad index funds): no Taiwan, AI, publishing or other themed tilt.

Is it already answered? The *decision* is. The wins_now ticket (section 1: "No leverage, inverse, thematic or single-country funds ... A Taiwan tilt would be tokenism") and S2_growth_sleeve.md already drop EWT and tilts. What is new: (a) primary-source wording replaces the SNIPPET used in Phase B; (b) the team's own notes mislabel the buckets, and that must be fixed before any teammate uses the framework in writing.

Evidence

Claim	Source	Status
"you should only have an aspirational portfolio if you have a certain skill set. Your education could be your aspirational portfolio. It doesn't have to be an investment. Specializing, being really good at what you do, your passion—that could be part of your aspirational portfolio."	Chhabra interview, WealthManagement.com, published 2016-05-13 (datePublished in page metadata): https://www.wealthmanagement.com/investment-news/q-a-with-ashvin-chhabra-you-re-investing-all-wrong (accessed 2026-09-27)	VP
"Everybody needs a safety portfolio. You should have a market portfolio because it's too expensive to have a safety portfolio. And then you may have an aspirational portfolio."	same	VP
"For those things you must do, the emphasis is going to be on lowering the risk of your not achieving these goals ... The goals you aspire to may require a somewhat higher level of risk." Three questions: safety / comfortable / aspire	CFA Institute Enterprising Investor, 2015-08-05: https://blogs.cfainstitute.org/investor/2015/08/05/ashvin-chhabra-on-wealth-management-the-value-proposition-lies-in-understanding-client-goals/ (accessed 2026-09-27)	VP
The 2005 paper: "risk allocation must precede asset allocation"; it includes "human capital" among the assets	Quoted by Princeton Info (secondary): https://princetoninfo.com/business/beat-the-market-consider-your-risks-and-asset-allocation/ (accessed 2026-09-27). Original paper: SSRN 403 / pm-research blocked	VP for the secondary page; the paper text itself is SNIP
Aspirational examples "owning a small business, or leveraging unique human capital ... entrepreneur"	Search summary only	SNIP (do not quote)
Team notes map "Aspirational risk bucket = flexibility reserve (stays invested for growth; dry powder)"	research/council_2026-09-27/team_notes_2026-09-27.md L15-18	VRF
Case: "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals."	case p.2 (R-C37, R-AN16)	VRF
The 2028 deposit comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures", all from her own work	case p.2 (R-C27, R-AN31)	VRF
Flexibility is for "as the project develops"	case p.3 (R-C58)	VRF

Numbers: \$150,000 of the \$450,000 in contributions (33%) comes from her own career (case, arithmetic). In a middle-case projection that share is 100% of the money above the ladder (F-409: the sleeve is ~34% of assets after 2028). So the facility money already rides on her human capital before any stock is bought.

Why the team mapping is backwards (devil's advocate): flexibility money exists to absorb surprises "as the project develops" (overruns, delays). In Chhabra's scheme it is closer to *safety* than to *aspiration*. Putting it in the riskiest bucket means the money for bad news is most likely to be down when bad news arrives. Corrected mapping: safety = the bought payments (and, from 2031, the bought facility floor); market = the growth money (broad index + short Treasuries); aspirational = outside the portfolio (her career, books, the residency itself).

Implication

- **WInS-now (decision, unchanged):** no tilts. Keep VT (or VTI+VXUS) and VGSH as in the ticket, PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK; alternates in the ticket's section 6. Nothing to trade differently.
- **TN (checklist):** the growth-note reflection can carry the reason in one clause: "one broad index because her own career already carries her concentrated bets". No framework name, no quote (D13 rules: no Laura quotes in TN/IPS).
- **IPS (sentence spec):** the risk-tolerance sentence must contain (1) her willingness ("thoughtful risks", the case's words), (2) where her big bets already are (career, and hence the 2028 deposit), (3) the consequence (the portfolio protects the promise and holds broad market risk only). No framework names (no citations allowed).
- **FR (sentence + table):** name Chhabra once, with the corrected mapping, cited to the 2016 interview and the 2015 CFA piece. Fix the team notes before anyone uses them.

Deadline tier: 1 (confirms WInS now; nothing to change). **Confidence:** high on the direction; medium on presenting the framework (the 2005 paper was not read directly). **Changes current strategy:** no.

Criterion: Client Knowledge and Objectives. **What this teaches:** a framework is only as good as its mapping. Put each bucket where its *job* is, not where its *name* sounds right.

2. M006 (tier 2: IPS; applied in FR). Which "if X, then Y" rules must the IPS pre-declare?

Question (canonical): Which contingent rules (rates fall before Jan 2027; 2028 deposit smaller or late; 2031 floor; 2033 reserve; facility share), each with trigger, action and record, must the IPS pre-declare before the Nov 6 freeze so the Final Report's numbers read as applications of the IPS rather than a redesign after results?

One-sentence answer: Four rules, each a trigger, an action and a record, about 120-160 words in total: (1) promise first (with the rates-fall and 2028-shortfall branches inside it), (2) the growth mix and its yearly reset, (3) the 2031 range = bought floor + stated upside, (4) the 2033 order of use (reserve, facility share, flexibility), plus one governance line (rules change only at a scheduled review, never because markets moved).

Evidence (professional practice)

Claim	Source	Status
"Perhaps most importantly, the IPS serves as a policy guide that can offer an objective course of action to be followed during periods of market disruption when emotional or instinctive responses might otherwise motivate less prudent actions."	CFA Institute, <i>Elements of an Investment Policy Statement for Individual Investors</i> , May 2010, p.1: https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf (accessed 2026-09-27)	VP
"A process for refreshing the IPS as investor circumstances and/or market conditions change should be clearly identified in advance." (element 2b)	same, p.6	VP
"Outside boundaries of acceptable variations from targets ... should be documented in the IPS ... If the policy is not to rebalance, this policy should be documented in the IPS." (element 4c)	same, p.17	VP

Claim	Source	Status
B.5.a "Take only investment actions that are consistent with the stated objectives and constraints of that portfolio or fund." Guidance: flexibility "is not improper but should be expressly understood and agreed to by Managers and clients."	CFA Institute <i>Asset Manager Code</i> , 2nd ed. (©2009, 2010), p.11-12: https://rpc.cfainstitute.org/sites/default/files/-/media/documents/code/amc/asset-manager-code-and-guidance-2nd-ed.pdf (accessed 2026-09-27)	VP
Rules page: "All teams should review the CFA Institute's Asset Manager Code and operate by these standards."	R-W29 via Phase A	VP (via A1)
Glide paths: "Meet fiduciary obligations without constant committee oversight by establishing a long-term framework for future de-risking at set trigger points."	BofA, "What You Need to Know About Glidepaths for Pension Plans": https://business.bofa.com/en-us/content/workplace-benefits/glidepaths-for-pension-plans.html (accessed 2026-09-27)	VP
"Fiduciaries must determine the appropriate trigger points for de-risking trades."	NISA, Dynamic LDI overview: https://www.nisa.com/perspectives/dldi/ (accessed 2026-09-27)	VP
Shortfall limit: the fund "can be restored to a satisfactory financial position within one year"; restoration plan within a period that "must not exceed three years"	APRA SPS 160: https://handbook.apra.gov.au/standard/sps-160 (accessed 2026-09-27)	VP
"we apply the targeted spending rate (5.25%) to the endowment's year-end value from two years ago"	Yale Provost: https://provost.yale.edu/how-exactly-does-spending-policy-work (accessed 2026-09-27)	VP
"lock down the benefits promised, then grow surplus assets with a prudent process"; "Any surplus strategy should preserve benefit security first, with additional risk taken only on assets above a defined surplus threshold."	Russell Investments, 2026-07: https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.html (accessed 2026-09-27)	VP
"A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." / "you may not redesign your strategy after observing the results"	competition/official/2026_27/2026_WGY_Investment_Competition_Guide.txt L156-158 (Guide p.5)	VRF
"Your team may not revise its investment strategy after the submission deadline."	IPS guide L73-74 (R-I35)	VRF
Laura's money arrives 2027-01-01, after the freeze; the ladder cost > \$300k on 173 of 185 trading days of 2026; P(cost > \$300k on 2027-01-01) ~24%	F-111 (VP inputs); F-112 (ASM)	as labelled

The rule specification (trigger / action / record). Parameters in [brackets] are team decisions; the words are the team's own.

#	Trigger	Action	Record (what the FR later shows "fired as written")	Case test covered
1 Promise first	Cash arrives (Jan 2027; Jan 2028)	Buy Treasuries maturing for the ten payments before anything else. If they cost more than the cash on hand: buy the latest-dated payments first; the 2028 deposit completes the rest before any money goes to growth. If the 2028 deposit is smaller or late: the promise is still completed first; only the facility range shrinks; the bought payments are never sold	Purchase-day cost vs cash (trigger = observed cost, not a model probability: stats skeptic, M006); funded ratio on each date	1 (certainty), 3 (uncertainty for the payments)
2 Growth mix	Once a year [or when outside the band]	Reset the growth money to [60/40] stocks/short Treasuries; band [55-65]	Yearly weights	3 (the risk held is the risk stated)
3 The 2031 range	2031-01-01	Buy [share] of the growth money as a 2-year Treasury maturing before 2033; that bought amount is the bottom of the range; the top = [stated percentile] of the rest; confidence stated	Floor bought; range; confidence	3, 4 (communicate)
4 2033 order of use	2033-01-01	Reserve = the ladder, held to maturity (nothing sold; this is the "no rebalancing" policy CFA 4c asks to be written down); facility = the bought floor + [share] of what is left; the rest kept as project flexibility	Reserve value, facility amount, flexibility kept	2 (responsible), 5 (flexibility)
G Governance	A scheduled yearly review	Rules change only at the review, for a change in Laura's circumstances, never because markets moved	Dated decision log (M021)	Guide p.5; Code B.5.a

Numbers: word budget ASM: 4 rules x 25-35 words + governance 10-15 words ≈ 110-155 of 500. The rates-fall branch matters in ~24% of modelled paths (F-112, ASM) and in 173 of 185 real 2026 trading days

(F-111, VP inputs).

Implication

- **IPS (decision):** adopt the four-rule structure. Fix every bracketed parameter by Nov 6. The IPS guide does not expect the final range or reserve numbers (L68-70), but a *rule* whose parameter is chosen after Nov 6 is a redesign. The IPS should name the parameter (for example "a fixed share of the growth money") even if the dollar outcome comes in the FR.
- **FR (sentence + table):** one table, "rule / what happened / what it produced", for the Articulation criterion. The Final Report applies the rules; it never adds one. Compare the four precedents in one line each at most.
- **WInS-now:** none new; the ticket's "never" rules and band are the WInS version of rules 1-2.
- **Open parameter left to the team:** latest-dated-first vs nearest-first (brief s9). Practice evidence (APRA) is about *having* a restoration rule, not its order. The ordering argument stays with D1/D3.

Deadline tier: 2. **Confidence:** high (the structure); medium (parameter values are team choices).

Changes current strategy: no (it formalises it). **Criterion:** Investment Strategy (with Articulation in the FR). **What this teaches:** a professional plan decides what it will do in bad times *before* they happen. That is the whole purpose of an IPS, and it is what protects a client from her adviser's hindsight.

3. M235 (tier 2: IPS). What is the smallest rule set that passes all five case tests?

Question (canonical): What is the smallest set of rules that still passes all five case tests, and can each extra rule show a measurable improvement?

One-sentence answer: Four rules (M006 table: promise first, growth mix, 2031 range, 2033 order of use). Three is too few, because without the 2033 order-of-use rule, tests 2 and 5 (responsible facility, flexibility) have no rule and the 2033 decision would be made after results. Every other candidate (a TWD quote, a Taiwan tilt, calendar glide paths, ratchets, extra bands) is either Final Report presentation or must show a lift in a reported number.

Is it mis-posed? Partly. The stats skeptic is right that "measurable lift" is false precision for wording choices. So the test applies to *rules* (things that move money), not to *presentation* (a second-currency quote is FR communication, not a rule). Merged with M006 in practice; kept separate here only for the lift test.

Test matrix (case p.4 tests, R-C80-R-C84, VRF)

Test	Rule 1 promise first	Rule 2 growth mix	Rule 3 2031 range	Rule 4 2033 order	Covered without the rule?
1 Payments with high certainty	Yes	-	-	(holds ladder to maturity)	No
2 Responsible facility contribution	-	-	(feeds)	Yes	No: dropping rule 4 fails test 2
3 Uncertainty for both goals	Payments	Facility risk as stated	Facility	-	Partly
4 Communicate clearly and credibly	-	-	Yes	-	No
5 Preserve flexibility	-	-	-	Yes	No: dropping rule 4 fails test 5

Lift of each candidate extra rule

Candidate	Lift (number)	Verdict
Rebalancing band (rule 2)	Without it, the growth money's stock share on 2031-01-01 is p5 50% / p50 62% / p95 73%, and 49% of paths sit outside 55-65% (script, ASM: JPM AC World 7.00% compound, 16.78% vol; bonds 3.9%, 2% vol; 100k paths)	Keep (one clause). It makes the risk held equal the risk stated (CFA 4c; Code B.5.a). Its <i>median</i> lift is ~0 (F-407: 50-70% moves the median by only \$2-4k)
Calendar glide path on the reserve	None: the ladder already shortens itself as rungs mature	Drop as a rule. Answer the IPS prompt ("composition change as payments approach") with rule 4's hold-to-maturity line

Candidate	Lift (number)	Verdict
Funded-status ratchet 2028-2031 (B4a Q16)	Not modelled; a lift of a few \$k at p5 would not earn its place	Drop unless D3 shows $\geq \sim \$10k$ at p5
Taiwan/AI tilt	Negative (M081)	Drop
TWD second quote of the range	Not a money rule	FR communication only
Yearly funded-ratio report	Not a money rule	FR presentation (Code F.2 clarity)

Implication: IPS decision: four rules plus one governance line. FR: a one-line "why no other rules" note in the Articulation section, which also answers last year's lesson (complexity must earn its place).

Deadline tier: 2. **Confidence:** medium-high. **Changes current strategy:** no. **Criterion:** Investment Strategy. **What this teaches:** a rule earns its place by covering a test nothing else covers, or by moving a number you report. Everything else is decoration.

4. M109 (tier 2: IPS; FR numbers). Required return per goal instead of one portfolio target?

Question (canonical): What is Laura's required return per goal (payments: the locked Treasury rate, only $\sim 1.05\%/yr$ across both deposits; facility: a stated hurdle), and should the IPS say her needs can be met below today's Treasury yield instead of setting one portfolio return target?

One-sentence answer: Yes. The payments need $1.05\%/yr$ across both deposits ($5.09\%/yr$ on the first deposit alone, which the bought ladder earns). About \$200k of facility money needs $4.81\%/yr$ on the growth money, below today's 5-year Treasury par yield of 4.98% . So stocks are there for facility money *above* roughly \$200k and for the case's "pursuing growth". The IPS should state a required return for the promise and a hoped-for return for the facility, not one portfolio target.

Evidence

Claim	Source	Status
IPS element 3b: "State the overall investment performance objective ... likely to incorporate descriptions of general funding needs"; example: "a required real growth rate of 4 percent to satisfy her future obligations"	CFA IPS elements 2010, p.9 (URL above)	VP
The case states no return target (R-AN17); an IPS "outlines an investor's financial goals, risk tolerance" (R-I2)	case; IPS guide L3	VRF
Case flows	F-601	VRF
Treasury par curve 2026-09-25: 5y 4.98% , 7y 5.06%	competition/official_market_data/daily-treasury-rates_2026-09.csv	VRF
JPM 2026 LTCMA AC World 7.00% compound; data as of 2025-09-30, when the 10y was $\sim 100bp$ lower than today, so its bond numbers look $\sim 1pp$ conservative	brief s6/s14	VRF / as labelled

Numbers (D8_practice_numbers.py; IRRs are exact arithmetic on VRF flows; projections ASM)

Goal	Required return
Ten payments, both deposits	$1.05\%/yr$
Ten payments, first deposit only	$5.09\%/yr$ (= the ladder's own yield; the 27bp headroom)
Payments + facility \$50k / \$100k / \$150k / \$165k / \$200k / \$250k	2.10 / 3.15 / 4.20 / 4.52 / 5.25 / $6.28\%/yr$ (whole portfolio)
Growth money only, for facility money of \$175k / \$200k / \$207k / \$225k / \$250k in 2033	2.08 / 4.81 / 5.53 / 7.29 / $9.55\%/yr$
Surplus held only in Treasuries at today's forwards (2027 leftover x 1.3513; 2028 deposit x 1.2903 = $5.23\%/yr$)	\$203,998 in 2033
Middle case, growth money 50/60/70% stocks (JPM 7.00%) + bonds at the 5.23% forward	\$212.9k / \$214.7k / \$216.5k (+ \$8.9k / + \$10.7k / + \$12.5k vs Treasury-only)
Verified model, 60% stocks (JPM bonds, lower than today's yields)	p5/p50/p95 \$159k / \$207k / \$273k (F-401, F-407)

Caveats (must travel with the numbers): the 5.23% forward for 2028-2033 is the market's break-even today, *not* lockable for the 2028 deposit (no derivatives; she does not have that money yet). The IRRs assume the 2028 deposit arrives in full (F-403/F-404 cover the shortfall cases). The verified model's median (\$207k) sits almost exactly at the Treasury-only figure because its bond inputs (JPM, 2025 data) are about 1pp below today's yields (brief s14). A consistent-input comparison is the +\$9-12k line above.

Devil's advocate (for the open 50/60/70 decision, brief s9): at today's yields, 60% stocks buy about +\$11k of middle-case facility money and cost about \$45k in a 1-in-20 bad case ($\approx$ \$204k Treasury-forward vs \$159k model p5; the two figures come from different input sets, so read this as an order of magnitude). This does **not** decide the weight. The case explicitly asks for "an appropriate balance between pursuing growth and protecting the capital" (R-C37), and some growth exposure answers that. But it is the strongest single argument for the low end of the band, and the team should be able to say it in their own words.

Implication

- **IPS (sentence spec):** the return-objective sentence must contain (1) the promise's required return = what the bought Treasuries earn (no number needed); (2) the facility's hoped-for return, with stocks named as the source of upside above what Treasuries alone would provide; (3) that no single portfolio return target is used, because the goals differ.
- **FR (number + chart):** a "required return per goal" line in the client picture. Chart candidate: a bar chart of the 2033 facility money under Treasury-only vs 50/60/70% stocks (median and p5), from one consistent model run (D3 to align the model's bond inputs with today's curve first).
- **Decision input:** sleeve equity weight (open); the 2031 range top.

Deadline tier: 2. **Confidence:** high on the IRRs; medium on the projections (ASM returns). **Changes current strategy:** not the architecture. It weakens the case for 60-70% stocks and should be put to the team as a devil's advocate point before Nov 6. **Criterion:** Client Knowledge and Objectives (with Investment Strategy). **What this teaches:** first ask what return each goal *needs*. Only then decide how much risk to take to earn more than that.

5. M063 (tier 2: IPS). One goal-dedicated pool, or all her wealth?

Question (canonical): Is this portfolio all of Laura's investable wealth or one goal-dedicated pool (the case says she has resources outside it), and should the IPS say so and let that raise her capacity to take risk with the surplus?

One-sentence answer: It is a goal-dedicated pool. The IPS should say so and split her ability to take risk by goal: none for the ten payments (the case forbids outside money for them), real but bounded for the surplus (no living cost depends on it). It should assume nothing about the size of her outside wealth, so this sharpens the risk-tolerance sentence without raising the stock weight.

Evidence

Claim	Source	Status
"Laura's living expenses and short-term financial needs will be covered by income and financial resources outside the portfolio."	case p.2 (R-C33)	VRF
"Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement."	case p.3 (R-C48)	VRF
An IPS should discuss "risk tolerances (both the ability and willingness of the client to bear risk), return objectives, time horizon, liquidity requirements, liabilities"; suitability is judged "in the context of the rest of the client's portfolio"	Asset Manager Code, B.6.a guidance, p.12	VP
"Where possible, the IPS should account for known liabilities to lend some quantitative basis to the risk tolerance assessment." Multiple risk levels: "avoiding financial catastrophe, maintaining a current standard of living, meeting a specific future financial goal, or developing significant further wealth"	CFA IPS elements 2010, element 3c, p.11	VP
"By 'capacity for loss' we refer to the customer's ability to absorb falls in the value of their investment. If any loss of capital would have a materially detrimental effect on their standard of living, this should be taken into account"	UK FSA FG11/05 <i>Assessing suitability</i> (2011), footnote 3: https://www.fca.org.uk/publication/finalised-guidance/fsa-fg11-05.pdf (accessed 2026-09-27)	VP
GBWM treats goals as separate "mental accounts"	Das et al. 2018, JOIM 16(3): https://srdas.github.io/Papers/GBWM.pdf (accessed 2026-09-27)	VP

Claim	Source	Status
"While the client is real, the financial scenario is developed specifically for the competition."	R-W4	VP (via A1)

Numbers: no new number is needed. F-407: stocks at 50/60/70% of the growth money give a median of \$205k/\$207k/\$209k and a p5 of \$164k/\$159k/\$154k (VRF script). Whatever the outside wealth, the choice moves the median by about $\pm$ \$2k. One limit on capacity: in 2027 the only surplus is \$7,736 (F-104), and the 2028 deposit depends on her career, which is the same risk as a bad year for publishing and speaking (wrong-way risk, R-AN31). So her capacity is lowest *before* the deposit arrives, and that is exactly when the plan holds almost no stocks (F-409).

Implication

- **IPS (sentence spec):** the risk-tolerance sentence must contain (1) the pool's single purpose (the residency goals; her living costs are met elsewhere); (2) ability by goal (none for the payments; bounded for the surplus); (3) willingness in the case's words ("thoughtful risks", "appropriate balance"); (4) no reliance on unstated outside wealth.
- **FR:** one line in the client picture; pair it with M081 (her big bets are outside the portfolio).
- **Kill:** any argument like "she can afford losses because she has other money". The case gives no size, and the scenario is fictional (R-W4).

Deadline tier: 2. **Confidence:** high. **Changes current strategy:** no. **Criterion:** Client Knowledge and Objectives. **What this teaches:** risk tolerance is not one number per person. It is one number per goal, and "ability" and "willingness" can point in different directions.

6. M026 (tier 2: IPS principle; FR numbers). What fee, and who pays it?

Question (canonical): What fee would a real adviser disclose for Laura's ~\$450k, and should the IPS say the reserve is held as directly owned Treasuries needing almost no management, with fees charged on or paid only from the growth sleeve so the ladder's ~\$7.7k headroom is never eroded?

One-sentence answer: A typical U.S. adviser would quote about 1.00% a year at this size (all-in about 1.75% with fund and platform costs, per the Veres survey reported by Kitces). Because a fee taken from the bought payments would exceed the \$7,736 headroom (0.5%/yr $\approx$ 2.1x it), the IPS should state one cost principle: the bought payments carry no ongoing fee and all costs are paid from the growth money. The Final Report should show projections net of a stated fee, with a sensitivity check.

Evidence

Claim	Source	Status
"the median advisory fee up to \$1M of assets under management really is 1%"; under \$250k the median is "almost 1.25%"; median all-in cost "1.75% for portfolios up to \$500k, 1.65% up to \$1M"; "the typical 1% AUM fee is really more of a 0.50% investment management fee, plus a 0.50% financial planning fee"	Kitces, "Financial Advisor Fees Comparison - All-In Costs For The Typical Financial Advisor?", 2017-07-31 (reporting Bob Veres' Inside Information survey of nearly 1,000 advisors): https://www.kitces.com/blog/independent-financial-advisor-fees-comparison-typical-aum-wealth-management-fee/ (accessed 2026-09-27)	VP for what Kitces reports; the survey is 2017 (dated; a current level is ASM)
F.4.d "Management fees and other investment costs charged to investors ..."; guidance: "At a minimum, Managers should provide clients with gross- and net-of-fees returns"; "Managers must not only use plain language ..."; "must disclose to prospective clients the average or expected expenses or fees clients are likely to incur"	Asset Manager Code, p.21	VP
A "spending calculus" that "reconciles investment return objectives, fees, taxes, inflation, and anticipated spending"; the example uses "fees of 1.2 percent"	CFA IPS elements 2010, p.10	VP
Case: "At the beginning of 2027, Laura plans to invest \$300,000 with an asset management firm." Fees are not mentioned; taxes are excluded	case p.2 (R-C26), p.4	VRF
Fund costs: IEF 0.15%; VT 0.06%; VGSH 0.03%	wins_now ticket s1 (issuer pages)	VP (via S3)
No projection in the repo includes a fee	BS-01	VRF

Numbers (D8_practice_numbers.py, ASM: growth money 60% stocks at 7.00% + 40% short Treasuries at 3.9% = 5.76%/yr; fee charged at the start of each year; ladder at forward values)

Fee	On growth money only	On all assets, paid from growth money	Taken from the ladder, 2027-42 (PV at 2027)
0	facility money 2033 \$209.3k	-	-
0.15% (IEF-type fund expense)	-	-	\$4.8k = 0.6x headroom
0.25%	-\$2.6k	-\$8.6k	-
0.50%	-\$5.2k	-\$17.1k	\$16.1k = 2.1x headroom
1.00%	-\$10.4k	-\$33.9k	\$32.3k = 4.2x headroom
Other checks: in 2027, a 1.0% fee on all assets is \$3,000 against a \$7,736 surplus, so it fits for one year. Product			
cost of the growth money with VT/VGSH: about 0.05%/yr. The STRIPS ladder has no ongoing fund fee.			

Implication

- **IPS (sentence spec, one clause):** it must contain (1) the bought payments are held directly to maturity, with no ongoing management charge; (2) every cost is paid from the growth money, so costs can never reduce the promise; (3) low cost as a principle. Do not state a fee number in the IPS.
- **FR (number):** state one base-case fee assumption and show the facility range net of it. Suggested base (ASM, team decides): 1.0% on the growth money plus fund costs ($\approx$ -\$10k); sensitivity: 1.0% on all assets paid from the growth money ($\approx$ -\$34k). Show gross and net per Code F.4.d. The 2031 range must be quoted net of fees, or it overpromises (R-C66).
- **WInS-now:** nothing new. The real WInS cost is \$25 per ETF trade (F-607), covered by the 1% cash float.

Deadline tier: 2 (IPS clause); numbers tier 3. **Confidence:** high on the principle; medium on the level (the fee survey is from 2017). **Changes current strategy:** no. It adds a principle and lowers FR facility figures by $\sim$ \$5-34k depending on the stated fee. **Criterion:** Portfolio Analysis (with Investment Strategy). **What this teaches:** a fee that looks small in percent can be large next to a thin margin of safety. Always compare costs with the headroom they eat, not with the whole balance.

7. Sources (all accessed 2026-09-27 with `fetch_text.py` or curl unless marked)

Source	URL / path	Status
CFA Institute, <i>Elements of an IPS for Individual Investors</i> (May 2010)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf	VP
CFA Institute, <i>Asset Manager Code</i> , 2nd ed. (©2009, 2010)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/code/amc/asset-manager-code-and-guidance-2nd-ed.pdf	VP
Kitces, Financial Advisor Fees Comparison (2017-07-31)	https://www.kitces.com/blog/independent-financial-advisor-fees-comparison-typical-aum-wealth-management-fee/	VP
Chhabra interview, WealthManagement.com (2016-05-13)	https://www.wealthmanagement.com/investment-news/q-a-with-ashvin-chhabra-you-re-investing-all-wrong	VP
CFA Institute Enterprising Investor on Chhabra (2015-08-05)	https://blogs.cfainstitute.org/investor/2015/08/05/ashvin-chhabra-on-wealth-management-the-value-proposition-lies-in-understanding-client-goals/	VP
Princeton Info on "Beyond Markowitz" (secondary)	https://princetoninfo.com/business/beat-the-market-consider-your-risks-and-asset-allocation/	VP (secondary)
Chhabra (2005), <i>Journal of Wealth Management</i> 7(4)	https://www.ssrn.com/abstract=925138	Blocked (SSRN 403); SNIP
Das, Ostrov, Radhakrishnan & Srivastav (2018), JOIM 16(3)	https://srdas.github.io/Papers/GBWM.pdf	VP
Russell Investments, pension surplus investing (2026-07)	https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.html	VP
BofA, glidepaths for pension plans	https://business.bofa.com/en-us/content/workplace-benefits/glidepaths-for-pension-plans.html	VP

Source	URL / path	Status
NISA, Dynamic LDI	https://www.nisa.com/perspectives/dldi/	VP
APRA SPS 160	https://handbook.apra.gov.au/standard/sps-160	VP
Yale Provost, spending policy	https://provost.yale.edu/how-exactly-does-spending-policy-work	VP
UK FSA FG11/05, <i>Assessing suitability</i>	https://www.fca.org.uk/publication/finalised-guidance/fsa-fg11-05.pdf	VP
Search results used only to find URLs (Kitces, Chhabra, glide paths)	WebSearch 2026-09-27	SNIP where not re-opened
Case, IPS guide, Competition Guide	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt; 2026_WGY_Investment_Policy-FINAL.txt; 2026_WGY_Investment_Competition_Guide.txt L156-158	VRF
Curve, verified model outputs	competition/official_market_data/daily-treasury-rates_2026-09.csv; research/insight_v1/phase_A/fact_register.md F-101-F-112, F-401-F-409	VRF
Prior agents reused (and re-verified here where marked VP)	research/insight_v1/phase_B/B4a_practice_benchmark.md, B4b_practice_benchmark.md; phase_A/stakeholder_map.md BS-01, BS-02, SH-25; wins_now/securities_and_allocation_v0.md	VRF

What this teaches

- 1. Professionals write the rules before the storm.** Every framework checked here (the CFA IPS standard, pension glide paths, Australia's restoration plans, Yale's spending rule) fixes its triggers in advance. Laura's plan earns trust the same way: four rules set on Nov 6 and applied, not invented, in December.
- 2. Ask what each goal needs before asking how much risk to take.** Her promise needs about 1% a year; about \$200k of facility money needs less than today's Treasury yield. Stocks are a choice for upside, and the team should say so plainly.
- 3. Risk tolerance is per goal.** No outside money may fund the payments, so they can bear no loss. The surplus can bear some, because her living costs do not depend on it.
- 4. Map frameworks by job, not by name.** Her career is her aspirational bucket; money kept for surprises belongs with safety.
- 5. Compare costs with the margin they eat.** 0.5% a year sounds small. Taken from the ladder, it is twice the ladder's safety margin.

Audit corrections (AY2)

Auditor AY2, 2026-09-28. The text above is left unchanged. Each item says what to correct, its severity (blocking / important / minor), and which deliverable it touches (TN or IPS). Evidence and the cross-file view are in `research/insight_v1/phase_D/audit_judges_practice_comms.md`. New check script: `research/insight_v1/scripts/AY2_audit_checks.py`.

Reproduced and confirmed (no change needed):

- **Script:** `D8_practice_numbers.py` was re-run. Every quoted figure reproduces: the \$292,264 and \$394,930 checks; the IRRs; \$203,998; +\$8.9k/+\$10.7k/+\$12.5k; the fee table; 0.6x/2.1x/4.2x; drift 50/62/73% with 49% outside the band.
- **Sources re-found verbatim with `fetch_text.py --grep:`**
 - CFA IPS 2010 (four quotes);
 - Asset Manager Code B.5.a, "expressly understood and agreed to", "gross- and net-of-fees returns", and B.6.a "ability and willingness" (found after a PDF line break);
 - Russell (three quotes, including "surplus glidepath");
 - Kitces ("really is 1%", "1.75% ... up to \$500k", "almost 1.25%");
 - Chhabra 2016 and CFA 2015;
 - APRA ("within one year", "must not exceed three years");
 - BofA, NISA, Yale, FCA FG11/05.
- **Team notes L18 mapping:** confirmed.

Corrections

1. **[important, IPS] Rule 1 overclaims in the joint tail.** Rule 1 says: "If the 2028 deposit is smaller or late ... only the facility range shrinks."
 - That is false when rates fall before January 2027 **and** the 2028 deposit is smaller than the shortfall (about \$7k at -50bp, \$23k at -100bp; brief s6). Some nearest-dated payments then stay unbought.
 - The frozen rule must say what happens in that case: all of the deposit goes to the unbought payments, no growth money is created, and the unfunded amount is stated. It must not claim that only the facility shrinks.
 - The case says she "will contribute" \$150k, so this is a stress beyond the case. A rule frozen on Nov 6 still must not overclaim.
2. **[important, IPS] Top finding 2 and the M109 answer overstate the headline.** The headline reads: "Laura's needs can be met at today's Treasury yields ... about \$200k needs 4.81%/yr, below today's 5-year Treasury (4.98%)."
- (a) Wrong horizon. The 5-year par yield covers 2026-2031, not 2028-2033. The matching figure is the 5.23% forward, which this file itself says is not lockable (the 2028 money arrives in 2028). If 5-year rates are 1 point lower in January 2028, a Treasury-only surplus gives about \$195k (AY2 arithmetic: $\$7,736 \times 1.3513 + \$150k \times 1.0423^5$).
- (b) Not a need. The facility has no required amount (case L104: "There is no predetermined facility contribution"). "\$200k" is an illustration.
- Correct to: the payments need what the bought ladder earns; about \$200k of facility money would need 4.81%/yr, below the 5.23% the curve implies for 2028-2033 today (not lockable). Keep "stocks are for upside" as a devil's-advocate argument, not a finding of fact.
3. **[upgrade, IPS] The devil's-advocate numbers hold on one input set.** `AY2_audit_checks.py` [1] re-ran them. Inputs (ASM): AC World 7.00% compound, 16.78% vol; bonds at the 5.23% forward, held fixed; 200k paths.
- 60% stocks: p5 \$155k (-\$49k vs Treasury-only \$204k), median +\$13k, p95 +\$104k.
- 50% and 70% stocks: p5 -\$41k / -\$56k; median +\$11k / +\$14k.
- Replace "+\$11k vs about \$45k (different input sets)" with "about +\$11-14k in the middle vs -\$41-56k in a bad case, one input set".
- Caveat: the bond rate is held fixed, so the uncertainty in the 2028 reinvestment rate is ignored.
4. **[minor, IPS] Present both sides of the equity weight.** This file calls M109 "the strongest single argument for the low end of the band". D13c, brief s16 and D9 M071 raise the opposite risk: the plan already looks very cautious for a client who "takes the jump", and the case asks for "pursuing growth". Both must reach the team together; neither decides the 50/60/70 weight.
2. **[minor, IPS] The fee comparison misreads the headroom.** "Taken from the ladder, a fee eats the headroom several times over: $0.15\% = 0.6x$ " is wrong for 0.15%, since 0.6x is less than the headroom. More basically, once the ladder is bought it has no spare money: any fee taken from it breaks full funding. For scale: 0.5%/yr over 2027-42 is about 2x the \$7,736 left over in 2027.
3. **[minor, IPS] Two different "middle cases" in one file.** M109 uses bonds at 5.23% (60% stocks: \$214.7k); M026 uses 3.9% (no fee: \$209.3k). Pick one, or label both, before any number travels.
4. **[minor, IPS] Rule 2 passes on governance, not on lift.** It fails this file's own "lift in a reported number" test: the median lift is about 0 (F-407), and weight drift is not an outcome number. It passes because the risk held then equals the risk stated (CFA element 4c; Code B.5.a). Say so.
5. **[upgrade, IPS] Cite the IPS guide as the main reason for rule 4.** The guide says the IPS "should also establish your strategic approach to preparing for the operating commitment, managing the operating reserve's asset composition over time, determining a responsible facility contribution, and preserving appropriate financial flexibility" (IPS guide L48-50, VRF). That is more direct than the test matrix.
6. **[minor, scope] Move the Final Report items to the "later (after Nov 9)" list, one line each.** Brief s17 was added after this file was written. The items are:
 - the framework-map table;
 - the "FR" bullets under M081, M006, M235 and M063;
 - the M109 FR chart candidate;
 - the M026 FR fee base case and sensitivity;
 - the Chhabra FR citation.

Keep now only the IPS rule specs, the IPS sentence specs and the TN clause content.

10. **[minor] The Yale quote is verbatim but trimmed.** Yale then "take[s] 20% of that amount and add[s] it to 80% of the total amount spent in the most recent fiscal year" (re-verified). So "spending set from the value from two years ago" misdescribes a smoothing rule, and the analogy to a bought floor is weak. It is Final Report only; park it.
11. **[minor] Two verbs are stronger than their sources.**
 - "Code F.4.d requires": the Code's guidance says Managers "should provide" gross- and net-of-fees returns "at a minimum". Use "recommends".
 - "binds teams to the Code": the Rules page says teams "should review ... and operate by these standards" (R-W29). "Binds" is strong.
12. **[minor, IPS] Word the governance line so it cannot read as permission to revise.** "Rules change only at a scheduled review" must not read as permission to revise the strategy after Nov 6. The IPS guide L73-74 says "may not revise its investment strategy after the submission deadline". Tie the review to Laura's circumstances from 2027 on, and never invoke it in the Final Report.
13. **[minor, IPS] The fee clause is the first thing to cut.** The case is silent on fees, so "no ongoing fee on the bought payments" is a team policy assumption, not a case fact. It is the first candidate to cut if the IPS runs over the word budget (budget table in the audit summary).
14. **[minor, TN] Mark the ready-made clause.** M081's TN clause "one broad index because her own career already carries her concentrated bets" (L134-135) reads as reflection text. Recast it as content (her concentrated bets are her career, which already drives the 2028 deposit) and let the team word it.
15. **[minor, IPS] Fix one citation.** M063 cites F-409 for "exactly when the plan holds almost no stocks", but F-409 covers the period after 2028 (17-24% stocks). The ~0% stocks in 2027 come from the ticket's real 2027 book (option iii).

Verdict for D8: strong on evidence (every practice quote verified) and on structure (four rules plus governance, now also backed by IPS guide L48-50). Fix items 1-2 before any rule or headline goes into the IPS spec. The devil's-advocate point is confirmed (item 3) but must travel with its counter-argument (item 4).

D9 Communication Analyst: can each idea be said plainly in the notes, the 50-word pitch and the 500-word IPS?

Agent D9, insight_v1 run, Phase D. Written 2026-09-28. Questions M001, M071, M143, M241, M031, M062, M226, M012, M036, M121, M191, M212 (`research/insight_v1/phase_C/survivors.json`), worked in deadline order.

This is AI-generated research for Team Caplet: specifications, evidence, numbers and checklists. **None of it is text to submit.** The six students choose and write every word of every WInS note, reflection, pitch and IPS, and record AI use in the Final Report's Works Cited (R-W46). Nothing here is for the finale. Laura appears only through the case and her public professional record. Nothing here suggests contacting her.

Scope (brief section 17). Only WInS trading (as the source of Trading Notes), the Trading Notes Analysis (TN, Oct 23) and the IPS (Nov 6) are in scope. Two questions (M031, and the table part of M212) are purely Final Report. They get one line each in "Later (after Nov 9)". The IPS-relevant part of M191 is kept as an IPS rule.

Scripts (run from the repo root; each docstring lists inputs and status labels):

- `.venv/bin/python research/insight_v1/scripts/D9_numbers.py` covers the vocabulary counts, the horizon odds, the price sensitivity with a live curve re-price, the trade-off costs and the independence odds.
- `.venv/bin/python research/insight_v1/scripts/D9_ips_page_fit.py [--draft file]` checks whether pitch + IPS fit on two pages.
- `.venv/bin/python research/insight_v1/scripts/D9_draft_checker.py --kind note|reflection|pitch|ips file` (or `--demo`) mechanically checks the team's own drafts. It counts words and flags them; it never rewrites anything.

Status labels: **VP** = VERIFIED-PRIMARY (read by D9 on the primary page, 2026-09-28, with `fetch_text.py --grep`, phrase found verbatim). **VRF** = VERIFIED-REPO-FILE. **DER** = derived by a D9 script from labelled inputs. **ASM** = ASSUMPTION. **SNIP** = SNIPPET-UNVERIFIED. **INT** = D9's interpretation (a judgement, not a fact). **CITED** = taken from another agent's file and not re-read by D9 (the file is named).

Summary: top findings, ranked by impact on reaching the semifinals and on making the plan unmistakably Laura's

1. (IPS compliance, tier 2; new) A full 50 + 500 words may not fit on two pages, and overflowing is an exclusion risk.

- Microsoft Word's "Double" spacing for Times New Roman 12 is 27.6 pt, because the font's line height is 1.15 em.
 - At that spacing, US Letter with 1-inch margins holds 23 lines a page, 46 on pages 2-3.
 - A full 50-word pitch plus 500-word IPS, laid out like Wharton's sample, needs about 44-46 lines. That leaves 0-2 spare. The largest IPS that still fits is about 525 words, and longer words shrink it.
 - A4 holds 50 lines (2-4 spare). Wharton's own sample uses a tighter 24 pt pitch, which would leave 8-10 spare.
 - The rule is "2-page maximum", and breaking a format rule means "will not be considered for semifinal selection" (IPS guide L82-83, L103-109, VRF).
 - Paper size is not specified. "Formatting choices not specified below are left to the team's discretion" (L101, VRF).
 - **Spec:** keep the IPS to about 470-490 words. Add no subtitle lines and no blank lines between paragraphs. Then check the exported PDF itself: the text must end on page 3. (DER, `D9_ips_page_fit.py`; line heights ASM.)
- ### 2. (TN, tier 1) Every WInS note needs a research element, and Wharton's own example has none.
- The Guide asks each note to capture "the supporting research or analysis" and its "role in growth, liquidity, risk management, or future funding" (Guide p.3 L84-86, VRF).

- Wharton's example note is 59 words and 413 characters. It has no number, no date and no source (DER, draft checker --demo).
- Hundreds of teams will copy its wording. The checker's clone test flags any draft that shares several three-word phrases with it.
- **Spec (M001): five elements, about 90 words or fewer.**
 1. The action, and what it is for.
 2. One Guide role word, plus the team's fixed label for that part of the portfolio.
 3. The dated Laura need it serves.
 4. One dated, checkable fact, with its source named in words.
 5. One named risk, and the rule that governs it.
- Agree this before the first order, because notes cannot be edited (R-AN43, G-list). 3. **(TN and IPS, tier 1) One certainty vocabulary, fixed before the first note (M062).** The official files never say "guarantee" or "safe" (0 uses in six documents, DER). The case gives each word a different job:
 - "certainty" belongs to the ten payments (case L91, L98, L136);
 - "confident" belongs to the 2033 contribution falling within the range (L117);
 - "confidence" and "credibility" belong to the people who must trust her: co-sponsors (L114-116), and in the criteria, Laura herself ("recommendations that can earn her confidence", SMApply L51);
 - "reliability" is Wharton's portfolio-level word ("funding reliability", IPS guide L20, L59).

One distinction is new. For Laura's real January 2027 bond ladder, "matched" and "locked" are true. For the WInS funds they are false: the funds never mature, so the notes must say "moves like" or "stands in for". Do not call the WInS Treasuries "the operating reserve". The case creates the reserve "at the beginning of 2033" (L94-95, VRF).

4. **(IPS, tier 2) Three questions (M241, M226 and M143) collapse into one principle, which**

saves words. Risk goes only where two things are true: the amount can flex, and others may help. It never goes into the fixed bill that only her portfolio may pay.

- The case supplies the asymmetry. Outside money may cover "any remaining facility cost", but "Teams may not rely on co-sponsors ... to meet this requirement" (L84-86 vs L91-92, VRF).
- The numbers answer the "longer horizon = more stocks" reflex, which Investor.gov states (VP). Under JPM's assumptions, stocks end below the Treasury rate that can be locked for the same date in about 4 paths in 10, at 6, 10 and 15 years alike (40%/39%/39%, DER, ASM lognormal).
- The reason is that today's locked rates (5.1-5.5%) sit only 1.2-1.6 points below JPM's 6.7% stock forecast, so the horizon does not rescue a fixed bill.
- The client benefit (M143) is the consequence: once the ladder is bought, the promise stops depending on her career income.
- Say that benefit conditionally ("once bought", not "in January 2027"). The ladder fits under \$300k in only about 69-77% of modelled paths (DER from A2/D7/S4, ASM). 5. **(TN, tier 1) The hedge note should lead with the promise, not with volatility (M012).**
- A 10bp move in rates changes the price of Laura's ten payments by about \$2,900.
- At 2026's realised rate volatility, the typical yearly swing is about \$20,000 (DER; DV01 VRF; annualisation ASM).
- So cash would look steady in WInS while the price of her promise moved. The Treasury funds are the low-risk choice *measured against the promise*, even though long bonds swing more in price than intermediate ones (TLT 3-year standard deviation 13.74% vs IEF 6.54%, F-201/F-202, VRF).
- **New wording point for gate box (b).** Option (iii), about 98% hedge (S3 ticket), can be described truthfully in one clause, because the hedge's size roughly equals the price of the payments.
- Option (ii) puts 66% in the hedge, about \$198k, against a promise that costs about \$292k. Every note then needs one extra clause saying WInS holds the after-2028 mix scaled to \$300k, or a reader sees a \$94k gap.
- That is a communication cost of (ii), not a reason to reject it. 6. **(IPS, tier 2) Numbers policy (M036): no tables, and three kinds of number.**
- Case facts are free, but repeating them wastes words. A past-season Wharton page says "Don't use precious space repeating known facts about your client and defining concepts" and "Excessive investing jargon doesn't necessarily dazzle the judges" (VP, retired 2023-24 page).
- Rule parameters are what make the frozen IPS testable later. Examples: the stock share of the growth money and its band, the 2031 lock share, and how the 2033 split is decided.

- Computed results are optional; use at most one, rounded and dated.
- Use no tables and no bullet lists of numbers. Tables are not named in the ban (L123), but the sample is plain paragraphs and the penalty is exclusion (INT). 7. **(IPS, tier 2) Name three trade-offs, each with what Laura gives up (M121).** The Guide p.5 and the Infographic both say strong teams explain "reasoning, assumptions, and tradeoffs" (VRF).
- **T1, certainty vs size.** She gives up about \$19k of median 2033 surplus and about \$267k of best-case (p95) upside. She gains +\$139k in the worst 5% of paths. It removes a 3.2% chance of missing a payment (13.7% if the 2028 deposit is halved).
- **T2, a 2031 promise vs more growth.** Each 10 points of lock share moves about \$20k into the bought floor (D5).
- **T3, buy all at once vs gradually.** A 100bp fall before purchase costs +\$30.6k; a 100bp rise saves \$27.4k. Waiting is a rate bet.
- In the IPS, state these without the numbers. The numbers wait for the Final Report. 8. **(IPS, tier 2) Laura in the IPS (M071).** Put her inside the rules as "because" clauses, not in a biography paragraph. Use case facts only. Every Laura clause must change a decision (the swap test: if the sentence still works with any client's name in it, cut it). Use no quotes (D13 rule; the IPS bans citations).
- As guidance only, this comes to roughly 40-60 of the 500 words, spread across the rules.
- The strongest case facts: the 2028 money is career income; the payments are fixed dollars that only her portfolio may pay; the co-sponsor promise is about credibility; her statistics degree (used once).
- 9. **(IPS, tier 2) The confidence method is an IPS rule; the numbers are Final Report (M191).**
- The IPCC note the question cites advises reporting both sides of a chance to avoid framing bias. It also says overwhelming evidence can be stated "as statements of fact without using uncertainty qualifiers" (VP). The question's reading (split the two tails) is a stretch, but the core idea survives.
- Applied here, the bottom of the 2031 range is already bought, so state it as a fact. Only the top needs a stated chance.
- The IPS fixes this method; D5 owns the numbers. 10. **(Tier 1) M212's traceability table is Final Report, but its raw material starts now.** Give the decision log one row per trade with the same five fields as the note template (finding 2), plus the provisional rule name the IPS will use. The Final Report table is then a copy-out job (later list).

North Star (INT). The plan becomes Laura's in words when every sentence answers her question: which of my dated needs does this serve, and what does it cost me? Generic teams write about the portfolio. Ours writes about her calendar.

Terms used in this file (plain English)

- **Hedge / "promise money":** the Treasury funds whose value moves like the price of Laura's ten \$50,000 payments (2033-2042).
 - **Ladder:** real Treasury bonds bought in January 2027, one maturing just before each payment. They are "matched" by date; WInS funds are not.
 - **Growth money ("sleeve"):** everything the payments do not need. It funds the facility contribution and flexibility.
 - **Basis point (bp):** 0.01 percentage point. **DV01:** dollars of value change per 1bp move in rates (\$289 for the ten payments).
 - **Floor (2031):** the part of the growth money bought in 2031 as a 2-year Treasury, so the bottom of the co-sponsor range is already owned.
 - **Percentile (p5/p50/p95):** in a simulation, the bad-case, middle and good-case results.
 - **Swap test (D13c):** replace "Laura" with any client's name. If the sentence still works, it is generic and should go.
-

TIER 1: WInS now and the Trading Notes (Oct 23)

M001: one note template from the first trade

Question. Should every WInS note open with a one-line purpose, use the Guide's role words and fixed team labels, tie itself to one dated cash flow, and carry one dated checkable fact and one named risk?

One-sentence answer. Yes, with one adjustment. Five elements, about 90 words or fewer, agreed before the first order. The research element is the one most teams (and Wharton's own example) will miss. The role word should come from the Guide's list and be paired with one fixed team label that never changes.

Evidence

Claim	Source	Status
A note should capture "the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding."	2026_WGY_Investment_Competition_Guide.txt p.3 L84-86	VRF
Notes are quoted "exactly as it appears in WInS"; "Yes, we will verify this."	TN guide L44, L53	VRF
Wharton's example is 59 words / 413 characters, has zero digits, no date and no source; it uses the role word "growth" and "reliable future cash flows"	D9_draft_checker.py --demo on TN guide L36-39	DER
Notes cannot be edited, only added to	2026-27 WInS User Guide via A4 (wins_week1_guardrails.md, R-AN43)	CITED (A4, VP there)
No note character limit is published	A4 section b4; ticket check 7	CITED
The reflection limit (100 words) applies to the reflection; the note is included separately	TN guide L42-46	VRF
Blueprint 01 section 3 lists action, role, reason now and risk; it lacks the "supporting research" element	research/blueprints/01_trading_notes_blueprint.md L30-36	VRF

The note spec (elements only; the team writes the words)

#	Element	What it must contain	What to avoid
1	Action + purpose	buy/sell, instrument type and ticker, rough share of the \$300,000, and in the same line what the position is for	starting with the market ("rates rose, so...")
2	Role	one Guide word (growth / liquidity / risk management / future funding) and the team's fixed label (table below)	switching labels between notes ("hedge", "safe bucket", "reserve", "ladder" read as four strategies, B7b-07)
3	Laura's dated need	one of: the ten \$50,000 payments 2033-2042; the facility contribution in 2033; the amount she can tell co-sponsors in 2031	generic "diversification" or "long-term growth"
4	One dated, checkable fact	e.g. the price of the ten payments at that date's Treasury prices (D9_numbers.py section [3] re-prices on the latest treasury.gov date), or a fund's duration from the issuer page with its "as of" date; source named in words, not a link	more than one number; undated numbers; forecasts presented as facts
5	Named risk + rule	the main risk accepted, and the pre-set rule that governs it (e.g. never sell the hedge after a rate rise)	"stable", "safe", "guaranteed" (M062)

- Length: aim for about 60-90 words and at most about 600 characters. (ASM: no limit is published. Paste the text into the WInS field before submitting and check that nothing is cut off.)
- Run `D9_draft_checker.py --kind note` on each draft. It flags overclaim words, missing role words, missing Laura anchors and closeness to Wharton's example.

Fixed labels (one per part of the portfolio; decide once, before trade 1)

Team label (INT: proposal, the team may rename)	Guide role word(s)	Laura need	Certainty word it may use (M062)
promise money (IEF, TLH / SPTL)	future funding; risk management	ten \$50,000 payments, 2033-2042	"moves like the price of her payments", "stands in for the bonds she buys in January 2027"
growth money (VT)	growth	facility contribution 2033; flexibility	"expected", never "will"
short-Treasury part of the growth money (VGSH)	liquidity; future funding	the amount she can promise co-sponsors in 2031	"already owned"
cash float	liquidity	commissions; no margin	none

Implication. *Sentence-level spec for every WInS note from the first trade (WInS-now → TN).* Update blueprint 01 section 3 by adding element 4 (research) and the fixed-label table. Tick gate box (d) in the wins_now ticket only once the note passes the checker and a second student has read it.

- Deadline tier 1. Confidence: high.
- Changes the strategy? No; this is a communication spec.
- Criterion: Articulation of Competition Experience ("clearly explains the team's research and decision-making process").

What this teaches. A good record of a decision says why, for whom, on what evidence and at what risk, and it says so before the outcome is known. Evidence written down after the result is not evidence.

M062: one certainty vocabulary before more notes are saved

Question. Should the team fix one certainty vocabulary and tie "certainty", "confidence" and "reliability" to one object each, so that no note says "guaranteed" or "safe"?

One-sentence answer. Yes. The official documents already assign each word to a different object, and never use "guarantee" or "safe". The team should copy that mapping and add one distinction the documents do not need: the difference between the WInS stand-in and Laura's real ladder.

Evidence (DER from D9_numbers.py [1], all six official files, VRF)

Word	Where the official files use it	Object
certainty (4 in the case, 1 in the IPS guide, 1 in the Guide)	"funded ... with a high degree of certainty" (case L91); "define what they consider a high degree of funding certainty" (L97-98); "Supports the ten-year operating commitment with a high degree of certainty" (L136); "desired degree of funding certainty" (IPS L10)	the ten payments
confident / confidence (3 in the case, 1 on SMAApply)	"state how confident they are that her 2033 contribution will fall within that range" (L117-118)	the 2033 facility range (a stated chance)
	"lose the confidence or participation of co-sponsors" (L114-115); "recommendations that can earn her confidence" (SMAApply L51)	people's trust (co-sponsors; Laura)
	"funding confidence" in the list of things teams "may reach different conclusions about" (L126-127)	ambiguous (INT: best read as the facility side, since the payments' level is fixed at "high")
reliability / reliable (1 case, 2 IPS guide, 2 TN guide, 4 Guide)	"funding reliability" (IPS L20, L59; TN L5; Guide L104, L144, L173); "reliable support for its early operations" (case L83)	the portfolio's ability to meet its commitments (Wharton's umbrella word)
credible / credibility (3 in the case, 1 on SMAApply)	"damage her credibility" (L114); "a credible range" (L116); "clearly and credibly" (L143)	Laura's standing with co-sponsors
guarantee / safe	0 uses in all six files	none; a note that uses them introduces words Wharton never used

The vocabulary (spec)

Object	Say	Never say	Why
Ten payments, real plan (from January 2027)	bought; locked in at market prices; matched by date; "certain in U.S. dollars, barring a U.S. government default" (chair memo section 4)	guaranteed; risk-free; 100%; "95% certain"	a model percentage for the payments would tell a statistics graduate we read the case carelessly (B1a-11)
Payments, WInS stand-in (now)	moves like the price of her payments; stands in for the ladder	matches; locked; mature when she pays; stable; reduces volatility	the funds never mature; three-quarters of TLH matures after 2042 (S3 ticket, red-team fix 2)
2033 facility range	expected; a stated chance with its model named; bottom "already set aside" once bought (BS-05)	promised (for the upside); guaranteed	the case wants a "credible range", and overpromising damages credibility (L114-116)
Whole portfolio	funding reliability (Wharton's word)	"safe portfolio"	umbrella term only
People	trust; credibility	"confidence" (keep that word for the range's stated chance)	avoids one word carrying two meanings in one sentence
WInS Treasuries	the Treasury funds that stand in for the bonds she buys in January 2027	"the operating reserve"	the case sets the reserve aside "At the beginning of 2033" (L94-95)

- Evidence that precise wording protects credibility when surprises occur: Jenkins, Harris & Lark (2019), cited by B6b as VP (abstract). CITED, not re-read by D9.

Implication. *Decision: a one-page certainty glossary (WInS-now, TN, IPS)*, built from the table above. It extends guardrail item 12, which lists strategy words but no certainty words. `D9_draft_checker.py` flags each of these words for a second look.

- Deadline tier 1. Confidence: high.
- Changes the strategy? No.
- Criterion: Creativity and Presentation ("communicates ... investment uncertainty clearly and credibly").

What this teaches. Careful readers treat different words as different claims. "Certain", "confident" and "reliable" are three promises of three strengths. Using each for exactly one thing is the cheapest way to show you read the brief as carefully as the person who wrote it.

M012: what the Treasury (hedge) note leads with

Question. Many teams will copy Wharton's example ("reduce portfolio volatility", "begin preparing"). What single fact and wording should our Treasury note lead with, so that long bonds read as reducing risk to Laura's promise?

One-sentence answer. Lead with the promise and its price. The position holds the value of Laura's ten fixed \$50,000 payments, which cost about \$X at [trade-date] Treasury prices. Name the risk that cash would leave open: each 0.1 percentage point move in rates changes that price by about \$2,900. Never claim the funds are "stable" or "reduce volatility".

Evidence and numbers

Claim	Number	Source	Status
Wharton's example purpose clause	"to reduce portfolio volatility and begin preparing for Laura's future operating commitment"	TN guide L36-37	VRF
Long Treasury funds are more volatile than intermediate ones	TLT 3y sd 13.74% vs IEF 6.54%	fact sheets 2026-06-30 (F-201, F-202)	VRF
Price of the ten payments for 2027-01-01	\$292,264 on the 2026-09-25 curve; re-priced by D9 from the live treasury.gov CSV (latest date 09/25/2026) and matching the verified script	<code>D9_numbers.py</code> [3]	DER (inputs VP)
Sensitivity	\$289 per bp → about \$2,890 per 10bp; \$7,736 of headroom = 27bp	brief s6; F-101/F-104	VRF inputs, DER
Typical yearly swing of that price	4.45bp/day realised in 2026 (D7) → about 71bp a year → about \$20,400 (1 sd)	<code>D9_numbers.py</code> [3]	DER; sqrt-time annualisation ASM
Three-week swing (a fill around Oct 1 to reflections around Oct 20)	about 17bp → about \$5,000	same	DER, ASM
The ladder cost more than \$300k on 173 of 185 trading days of 2026	Jan 2 \$316.5k; Feb 27 \$325.6k	F-111	VP inputs, DER
Hedge tracks the payments within about \$450 per 100bp (forward-consistent)	S3 ticket section 3	CITED	
"Stocks and Treasuries fell together in 2026" (question premise)	stocks were up about 13% YTD while IEF was down 4.20% YTD	B5b (stocks, CITED); IEF -4.20% YTD, F-203 (VP)	premise is loose: do not repeat it

Why this wording and not the example's (INT). For a client with a fixed-dollar bill, the risky position is the one whose value does *not* move with the bill. Cash keeps the WInS number still, while the price of her promise moves about \$20k in a typical year. The hedge's dollar value moves, but it moves together with the bill. That is one plain idea: "measured against her payments, not against the screen".

Option (ii) vs (iii): the words test (new; informs gate box (b), does not decide it)

	(ii) after-2028 mix scaled to \$300k	(iii) literal January 2027 book
Hedge size in WInS	66%, about \$198k (S3)	about 98%, about \$294k (S3)

	(ii) after-2028 mix scaled to \$300k	(iii) literal January 2027 book
Can the note say "sized to the price of her ten payments (about \$292k)" truthfully in one clause?	No. It needs a second clause (the promise's share once both deposits are in, about two-thirds), or a reader sees a \$94k gap	Yes, almost one-for-one
Words needed	about 15-20 more words in each hedge note	none extra

The S3 ticket's (ii) sentence already has the three contents. This finding only prices how many words it costs in a note. The team decides with gate box (b).

Checklist for the hedge note (adds to the S3 ticket's note checklist 1)

- [] The first clause says what the position is for (Laura's ten payments, 2033-2042).
- [] One dated number: that day's price of the payments (re-run `D9_numbers.py` [3] or `A2_curve_recheck.py` on the trade date).
- [] "Moves like / stands in for", never "matches"; no "stable", "safe" or "reduce volatility".
- [] One named risk and the rule: the value falls if rates rise, but so does the price of the payments, so the team never sells after a rise.
- [] Under option (ii), one clause explaining the scaled mix.

Implication. *Sentence-level spec for the hedge note (WInS-now → TN reflection 1).* Deadline tier 1. Confidence: high.

- Changes the strategy? No. It gives gate box (b) a communication input.
- Criterion: Portfolio Analysis ("understanding and effective use of investment concepts").

What this teaches. "Risky" always means risky *relative to something*. For someone saving toward a fixed bill, the right yardstick is the bill, not the screen.

M212 (tier 1 part): the decision log that later proves one strategy

Question. What single table lets a reader verify in one minute that the TN, IPS, WInS book and Final Report tell one strategy?

One-sentence answer. That table belongs to the Final Report and waits until after Nov 9. The part due now is to log every trade with the five M001 fields plus a provisional rule name, so the table later copies out rather than being reconstructed.

Evidence.

- "maintaining a clear and consistent investment strategy across all three" (case L161-162, VRF).
- "maintains consistency with the team's IPS" (SMAApply L49, VRF).
- "Strong submissions connect the client's goals, research, portfolio decisions, and final recommendations through one clear and cohesive investment strategy." (Infographic L40-41, VRF).
- Guardrails items 17-18 already require exact note copies and screenshots (VRF).

Decision-log columns (spec):

- date and time (ET);
- ticker, side, shares, fill price;
- note text exactly as WInS shows it;
- role word and team label;
- Laura need;
- fact and source;
- risk and rule;
- **provisional rule name** (e.g. "Rule 1: payments first"; the team names its rules once and reuses the names in the IPS);
- note candidate? (yes/no, and why).

Implication. *Decision: add the rule-name column now (WInS-now).* The table itself goes on the "Later" list.

- Deadline tier 1. Confidence: medium (the value depends on the team keeping the log).
- Changes the strategy? No.

- Criterion: Investment Strategy (consistency with the IPS).

What this teaches. Consistency is easier to record than to reconstruct. A column filled in on the day beats a paragraph written from memory.

TIER 2: the IPS (Nov 6; the strategy freezes)

M036: which numbers the words-only IPS carries, and whether tables are allowed

One-sentence answer. Use no tables, and carry only three kinds of number:

- case facts, used sparingly because they cost words;
- the rule parameters that make the frozen strategy testable later;
- at most one rounded, dated computed result.

It must also physically fit: about 470-490 IPS words on two pages, checked on the exported PDF.

Evidence

Claim	Source	Status
"Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted." Tables are not named	IPS guide L123	VRF
"Focus on the strategy and decision-making framework ... rather than describing individual investments or presenting detailed financial calculations."	IPS guide L51-52	VRF
Not expected: "final operating-reserve calculations, detailed portfolio projections, a final facility-contribution range"	IPS guide L68-69	VRF
"the IPS becomes the official record"; "may not revise its investment strategy after the submission deadline"; the Final Report "will evaluate how effectively it implemented that strategy"	IPS guide L73-77	VRF
"Formatting choices not specified below are left to the team's discretion. Keep the presentation simple"	IPS guide L101-102	VRF
Non-compliance: "will not be considered for semifinal selection"	IPS guide L82-83	VRF
"Excessive investing jargon doesn't necessarily dazzle the judges. Your report should reflect your team's voice." / "Don't use precious space repeating known facts about your client and defining concepts. However, it is okay to demonstrate that you understand your client beyond the case study."	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation (retired 2023-24 judging page, accessed 2026-09-28)	VP (past season, not 2026-27 rules)
Official PDFs are US Letter (612 x 792 pt); Wharton's sample IPS uses a 24 pt line pitch, a blank line before the IPS heading, and tab-indented paragraphs	pypdf on 2026_WGY_Investment_Policy-FINAL.pdf pp.5-6	VRF (measured)
Page fit (see table)	D9_ips_page_fit.py	DER; Word line height and Liberation-Serif = TNR widths are ASM

Page fit (50-word pitch + 500-word IPS, 1-inch margins; lines needed / available on pages 2-3)

Paper	Line height	Needed	Available	Spare	Max IPS words that fit
US Letter	27.6 pt (Word "Double", TNR 12)	44-46	46	0-2	about 525
A4	27.6 pt	46-48	50	2-4	about 550
US Letter	24 pt (Wharton's sample)	44-46	54	8-10	about 615
A4	24 pt	46-48	58	10-12	about 665

Proxy vocabulary averages 13.4 words per line on Letter. A text made only of long words (7+ characters) fits 9.0, so a draft heavy in words like "contribution" and "Treasury" needs more lines. Run the script with `--draft` on the real text, and in the end trust only the exported PDF.

Number policy (spec)

Kind	Examples	In the IPS?
Case facts	\$300,000 / \$150,000; ten \$50,000 payments 2033-2042; 2031	Only where a rule needs them; each costs words and adds nothing new

Kind	Examples	In the IPS?
Rule parameters (policy, not calculation)	stock share of the growth money and its band; share of the growth money bought as the 2031 floor; how the 2033 remainder is split between facility and flexibility (D5's give-back share); how often the plan is reviewed	Yes. These make the frozen IPS testable, and the Final Report is judged on implementing them
Computed results	"the ten payments cost about \$292k at September 2026 prices"	At most one, rounded and dated. Optional. Never "bought in January 2027" as a flat claim (D7 M011)
Model outputs	percentiles, miss rates, 4-in-10 odds, range dollars	No. These go to the Final Report

Format checklist (IPS):

- [] Headings exactly as in the sample.
- [] No blank line between IPS paragraphs.
- [] No tables and no bullet lists.
- [] No quotation marks around anyone's words.
- [] No URLs, footnotes or "(Source, year)".
- [] Title page on page 1 only; page break before page 2.
- [] Word count done with the same tool, with 5-10 words to spare.
- [] PDF of 5 MB or less; the text ends on page 3.

Implication. *Decision and number:* IPS target about 470-490 words; the numbers limited to the policy above; no tables (IPS).

- Deadline tier 2. Confidence: high on the policy. Medium on the exact line counts (ASM line heights), which is why the PDF check is the binding step.
- Changes the strategy? No.
- Criterion: Creativity and Presentation (and format compliance, a gate for everything else).

What this teaches. A limit is physical as well as numerical. "500 words" and "two pages" are two different constraints, and the tighter one depends on settings nobody told you about.

M143: name, as the client benefit, that the promise stops depending on her future earnings

One-sentence answer. Yes. It is the most Laura-specific benefit the plan can claim, and it fits in the pitch. It must be said as a consequence of a rule ("once the payments are bought"), not a date. In about 23-31% of modelled paths the ladder needs part of the 2028 deposit.

Evidence and numbers

Claim	Source	Status
The 2028 \$150,000 comes from "earnings from publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"	case L43-45	VRF
Laura called going self-employed a move to "unstable income" (2021 interview)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid (re-checked 2026-09-28)	VP. Final Report only: no quotes in the TN or IPS (D13 rule)
The \$150k is 33% of all the money she puts in	case L43-45	DER
P(ladder costs > \$300k on 2027-01-01): 24.3% (A2), 23% (D7), 30.7% (S4) → the promise is independent of 2028 income from January 2027 in about 69-77% of modelled paths, otherwise from January 2028	D9_numbers.py [5]	DER from ASM-based runs
The case says she "will contribute" the \$150k, so frame this as resilience, not as fixing a risk the case names	stats skeptic, R-AN35	CITED
Credibility is part of her income (reputation-driven advances and speaking)	stakeholder map BS-03	CITED (INT)

Content spec (pitch and IPS; elements only)

- The claim: after the payments are bought, neither markets nor her next contract can undo the ten-year promise.

- The condition, in plain words: bought with the first deposit if prices allow, otherwise finished with the second (the rule already in brief section 7).
- What still depends on her income: the size of the facility contribution, not the payments.
- Swap test: passes, because only a client with career-driven deposits gets this benefit.

Implication. *Sentence content for the 50-word pitch and the IPS opening (IPS); also the "why" of the hedge note (TN).*

- Deadline tier 2 (content can already shape the hedge note in tier 1). Confidence: high.
- Changes the strategy? No.
- Criterion: Client Knowledge and Objectives ("recommendations that can earn her confidence").

What this teaches. The best benefit statement names what the client no longer has to worry about. It stays honest by saying exactly when that becomes true.

M241: justify where risk sits with the case's own asymmetry

One-sentence answer. Yes. It is the case's own reason for the whole design: outside money may fill a facility shortfall but never Laura's ten payments. One IPS clause should say so in those terms. Note the wording nuance "never her ten payments", not "never operating costs". The case lets others fund "operating support beyond Laura's commitment".

Evidence

Claim	Source	Status
"Any remaining facility cost, and any operating support beyond Laura's commitment, may come from co-sponsors, grants, collaborators, program fees, continued business income, or other sources."	case L85-86	VRF
"All ten payments must be funded by the investment portfolio with a high degree of certainty. Teams may not rely on co-sponsors, grants, program fees, or other outside funding to meet this requirement."	case L91-92	VRF
The IPS guide asks exactly this: "How will your strategy protect the operating commitment while preserving flexibility for a responsible facility contribution?"	IPS guide L25-26	VRF
Already noted as an anomaly (R-AN6) and as "promise first" (BS-17); not yet used as the justification sentence	case_register, stakeholder_map	CITED

Merged principle (INT; with M226). Risk goes where the amount can flex and others may help. It never goes into the fixed bill that only her portfolio may pay. This single idea answers M241 (where risk sits), M226 (why the longest-dated money is the safest) and part of M143 (why the promise is independent). The IPS should state it **once**, then apply it in each rule. That saves 30-60 words against stating three separate reasons (INT).

Implication. *One IPS principle clause (IPS).* Too long for the pitch unless compressed to "risk only where others can help" (INT).

- Deadline tier 2. Confidence: high.
- Changes the strategy? No.
- Criterion: Investment Strategy ("a clear and creative investment thesis").

What this teaches. The strongest justification for a design is often already in the brief. Find the sentence where the rules treat two things differently, and build around that difference.

M226: why the longest-dated money is the safest (answering "longer horizon = more stocks")

One-sentence answer. A fixed-dollar bill is bought, not grown. Horizon earns risk only for money whose amount can flex. With today's Treasury rates close to JPM's stock forecast, stocks end below the locked amount in about 4 paths in 10 whether the horizon is 6, 10 or 15 years, so a longer wait does not rescue a fixed bill.

Evidence and numbers

Claim	Number	Source	Status
The reflex a reader brings	"Investors with a longer time horizon may feel comfortable taking on riskier or more volatile investments."	https://www.investor.gov/introduction-investing/getting-started/asset-allocation (accessed 2026-09-28)	VP
The bill is fixed	"each payment is a fixed \$50,000 and is not adjusted for inflation"	case L90	VRF
Locked rates 2027 → 2033/2037/2042	5.08% / 5.24% / 5.48%	brief s6 (forward zeros from the 2026-09-25 curve)	DER from VP
JPM U.S. large cap	6.70% compound, 7.94% arithmetic, 16.47% vol (data 2025-09-30)	brief s6	VRF
P(stocks end below the locked amount), U.S. large cap	40.3% (6y), 38.7% (10y), 38.5% (15y); simulation check 38.8% (10y, 200k paths, seed 20260927)	D9_numbers.py [2]	DER; lognormal i.i.d. ASM
Same, AC World (7.00%, 16.78%)	38.7% / 36.7% / 35.9%	same	DER, ASM
Stock premium over the locked rate	1.6 / 1.5 / 1.2 points a year	same	DER
Caveat: JPM's inputs date from when the 10y was 4.16%, about 100bp lower; the 2027 edition is not out	brief s14	VP there	

Why the odds barely fall with horizon (INT, plain English). The usual "time smooths out risk" story works when stocks are expected to beat the safe rate by a wide margin. Here the margin is only about 1.2-1.6 points a year, so the extra years add as much uncertainty as they add expected lead. Also, a single missed payment is a broken promise, not a bad average.

Content spec (IPS: one clause, no number; the 4-in-10 figure goes to the Final Report).

- The idea that the payments are a fixed-dollar bill, so they are bought.
- Horizon matters for the money whose size can change (the facility, flexibility).
- Say it with the M241 principle, not as a separate paragraph.

Implication. *One IPS clause (IPS).* The number is on the Final Report list.

- Deadline tier 2. Confidence: high on the principle. Medium on the exact odds (model ASM and JPM vintage).
- Changes the strategy? No.
- Criterion: Portfolio Analysis ("integrates quantitative and qualitative analysis").

What this teaches. Rules of thumb carry hidden conditions. "Long horizon means more stocks" assumes stocks are expected to win by a lot and that the goal can bend. Check both before applying it.

M071: how much of the IPS is about Laura, and on what each sentence may rest

One-sentence answer. Put Laura inside the rules as "because" clauses, roughly 40-60 of 500 words as guidance rather than a target, never as a biography paragraph. In the IPS, use only case facts that change a decision. Save beyond-the-case evidence, with its sources and at most one or two dated quotes, for the Final Report.

Evidence

Claim	Source	Status
"Always remember who your strategy is designed to serve." / "should demonstrate how your strategy is tailored to your client's financial goals, time horizons, and needs"	IPS guide L27, L45	VRF
"Don't use precious space repeating known facts about your client ... However, it is okay to demonstrate that you understand your client beyond the case study."	retired Wharton judging page (URL above)	VP (past season)
"sources do not need to be cited in the IPS. Relevant sources and supporting evidence must be included in the Final Report."	IPS guide L124-125	VRF
"While the client is real, the financial scenario is developed specifically for the competition."	R-W4	CITED (VP in Phase A)
No Laura quotes in TN or IPS; only D13a VERIFIED-PRIMARY lines may be her words; no "investment philosophy"	brief s16, D13c	CITED
Swap test	D13c section 8	CITED
The 30-50-word number is false precision; keep the principle	stats skeptic (M071)	CITED

Which facts earn a place (spec)

Fact	Status	Decision it changes	IPS?
The 2028 money is career income (advances, speaking, licensing)	VRF (case L43-45)	the payments are bought from the first deposit where possible; the facility absorbs a late or smaller deposit	Yes
Fixed nominal payments that only her portfolio may fund	VRF (L90-92)	risk sits only in money whose amount can flex (M241)	Yes
The 2031 promise must not overpromise; her credibility is at stake	VRF (L114-116)	the bottom of the range is bought, not forecast	Yes
Statistics degree	VRF (L16)	certainty defined as testable; every probability names its model	Once at most
"Willing to take thoughtful risks" plus "appropriate balance"	VRF (L70-74)	the growth money is real growth, not set at the minimum to look safe (D13c)	Yes, briefly
"unstable income" (her 2021 words)	VP	same as the first row	No (a quote): Final Report at most
Community/access priority; "earliest supporters" (2016, student-era)	VP (D13a)	2031 message order	No: Final Report at most
Identity, heritage, Taiwan as a personal reason for holdings	n/a	none (tokenism)	Never

Implication. Number and rule: about 40-60 Laura words across the rules; every Laura clause states a fact and the decision it changes (IPS).

- Deadline tier 2. Confidence: medium (the word range is judgement).
- Changes the strategy? No.
- Criterion: Client Knowledge and Objectives.

What this teaches. Understanding a client shows in how well the plan fits her, not in how much you say about her. One "because" per rule does more than a paragraph of biography.

M121: which three trade-offs the IPS names

One-sentence answer. Name three things Laura gives up, each in plain words:

- **T1:** some facility money in good markets, for ten payments that cannot miss;
- **T2:** some growth after 2031, for an amount she can promise co-sponsors;
- **T3:** the chance of cheaper bonds if rates rise, for not risking dearer ones if they fall (buying all at once).

State them without numbers in the IPS. The numbers wait for the Final Report.

Evidence and numbers (D9_numbers.py [4])

Trade-off	What she gives up	What she gets	Numbers	Status
T1 certainty vs size	about \$19k of median 2033 surplus (\$207k vs \$226k); about \$267k of p95 upside (\$273k vs \$540k)	never misses by construction vs 3.2% miss (13.7% if the 2028 deposit is \$75k; 40.8% if \$0); +\$139k in the worst 5% (\$159k vs \$20k)	brief s6 / F-401	DER from the verified run (ASM model)
T2 bought floor vs growth	each +10 points of 2031 lock share moves about \$20k into the floor and takes about \$10k each from the upside and from kept flexibility	a range bottom that is already owned	D5 finding 2	CITED (ASM model)
T3 all at once vs gradual	if rates rise 100bp before purchase the payments would have cost \$27,374 less	if rates fall 100bp they cost \$30,600 more; cost exceeded \$300k on 173 of 185 days in 2026	brief s6; F-111	VRF / DER

- **Source consistency warning.** D13c's re-run of the B10b script gives about \$11k at the median for a different comparison. The team should quote one source only (the verified `strategy_mc.py` run is the reference, brief s8) (DER).
- **Not a trade-off: inflation.** Say it as an assumption. The \$50,000 is fixed in dollars, so it buys less each year. That is the case's design, not a team choice. The IPS names it as an assumption; the Final Report sizes it (case L146-147 asks for the effect of inflation on "portfolio projections and facility costs").
- "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions." (Guide p.5 L183-184, VRF); "There is no single

correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions." (Infographic L42, VRF).

- D13c's candid clash (brief s16): whole-portfolio equity is about 0-2% in 2027, about 20% in 2028-30 and about 4% after 2031. T1 is the honest place to answer "is this too cautious for someone who takes the jump?" (INT).

Implication. *IPS checklist: three named trade-offs, each "gives up X to get Y" in words (IPS).*

- Deadline tier 2. Confidence: medium-high.
- Changes the strategy? No.
- Criterion: Investment Strategy (the Guide's "reasoning, assumptions, and tradeoffs").

What this teaches. Every choice has a price. Saying the price out loud is what makes a recommendation trustworthy, because the reader can see you looked at the other side.

M191 (IPS part): how the confidence in the 2031 range is stated

One-sentence answer. Partly mis-posed. The IPCC advice is about framing (report a chance both ways), not about splitting the two tails. But the useful rule survives and belongs in the IPS as method:

- the bottom of the range is money already bought in 2031, stated as fact;
- only the top carries a stated chance, with its model named.

D5 owns the numbers (Final Report).

Evidence

Claim	Source	Status
"Consider reciprocal statements to avoid value-laden interpretations (e.g., report chances both of dying and of surviving)." and "it may be appropriate to describe findings for which evidence and understanding are overwhelming as statements of fact without using uncertainty qualifiers."	IPCC AR5 Guidance Note, https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf (accessed 2026-09-28)	VP
"state how confident they are that her 2033 contribution will fall within that range" (two-sided)	case L117-118	VRF
With the floor bought in 2031, $P(\text{below}) = 0$ barring U.S. default; $P(\text{above top}) = 15\%$ at D5's candidate $q = 0.85$	D5 finding 2	CITED (ASM model)
A dollar range fixed today would be missed below in 10-50% of paths	D5 finding 4	CITED (ASM)

Implication. *IPS rule (method only):* "the range is set in 2031 from what is then owned; its bottom is bought; its top is a stated chance" (content, not wording). Put the numbers and the two-sided statement on the Final Report list.

- Deadline tier 2. Confidence: medium.
- Changes the strategy? No (it is brief section 7's rule, worded for testing).
- Criterion: Creativity and Presentation ("communicates ... investment uncertainty clearly and credibly to prospective co-sponsors").

What this teaches. When part of an answer is already certain, say it as a fact, and spend your probability words only where uncertainty really lives.

The pitch: which ideas fit in 50 words (spec, not text)

Candidate elements and a rough word cost (INT: estimated from typical phrasing lengths, not counted text):

Element	Why it earns pitch space	Approx. words	Keep?
Central idea: the ten payments bought first with U.S. Treasuries	the IPS guide asks for "the central idea" (L35-36)	12-15	Yes
Client benefit: the promise stops depending on markets or her future earnings, once bought (M143)	most Laura-specific claim; conditional wording	12-15	Yes
Growth purpose: the rest invested for growth toward the facility contribution and flexibility	covers "financial goals and future funding needs"	10-12	Yes
2031: a range whose bottom is already owned	the credibility theme; distinctive	8-12	If words allow

Element	Why it earns pitch space	Approx. words	Keep?
Case asymmetry (M241) / horizon logic (M226)	reasons, not the idea	15+	No: in the IPS body
Any computed number	IPS guide L51-52	3-6	No

Four elements at the low end already come to about 42-54 words. So a 50-word pitch holds three elements comfortably and a fourth only in very plain words. `D9_draft_checker.py --kind pitch` counts and flags.

TIER 3: Final Report (out of scope until Nov 9)

M031: the single co-sponsor chart

Answer. Final Report only; the IPS bans charts. Parked under brief section 17. The only action now is D7's point: capture WInS hedge values and the same-day treasury.gov price of the payments from the fill date, so later analysis uses data recorded at the time.

- Deadline tier 3. Confidence: n/a.
- Criterion: Creativity and Presentation.

Later (after Nov 9), one line each

- M031: one chart, ten payment blocks marked "bought" plus the 2031 floor bar, with a band for a weak, median and strong 2031 (D5 script data).
- M191: state P(below range) and P(above range) separately, with the model named (D5 numbers).
- M212: a six-row table: IPS rule → WInS trade → note as quoted → Final Report verdict, copied from the decision log.
- M226: the "about 4 in 10 even at 15 years" figure, with its model and JPM vintage stated.
- M121: the T1-T3 numbers, from one model source only.
- M071/M143: at most one or two dated Laura quotes in context (e.g. "unstable income", 2021), per D13a.

Sources (accessed 2026-09-28 unless noted)

Official, in the repo (VRF):

- `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (L16, L43-45, L70-74, L83-99, L102-103, L114-120, L126-127, L136-147, L161-162)
- `2026_WGY_Investment_Policy-FINAL.txt` (L10, L20, L25-27, L35-36, L45, L51-52, L59, L68-77, L82-83, L101-125) and its PDF (page size and sample layout)
- `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L5, L20-23, L36-53)
- `2026_WGY_Investment_Competition_Guide.txt` (p.3 L84-86; p.5 L183-184)
- `2026_WGY_Competition_Infographic.txt` (L40-42)
- `SMAApply_Deliverables_Page_2026-09-27.md` (L49-57)

Primary web (VP, re-read by D9 with `fetch_text.py --grep`):

- Investor.gov asset allocation page: <https://www.investor.gov/introduction-investing/getting-started/asset-allocation>
- IPCC AR5 Uncertainty Guidance Note: https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf
- Wharton retired judging page (REST): <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation>
- Overachiever Magazine interview (2021-03-15): <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>
- treasury.gov daily par yield curve 2026 CSV (live download inside `D9_numbers.py`; latest date 09/25/2026 at run time)

Market data (VRF): `competition/official_market_data/` (IEF/TLT fact sheets 2026-06-30; the curve CSV).

Other run files (CITED):

- `research/insight_v1/_context/brief.md` (s6, s7, s8, s14-s17)
- `phase_A/case_register.md` (R-AN6, R-AN43), `fact_register.md` (F-101, F-104, F-111, F-201-F-203, F-401, F-402, F-409), `stakeholder_map.md` (SH-16, BS-03, BS-05, BS-17), `wins_week1_guardrails.md` (section c)
- `wins_now/securities_and_allocation_v0.md`
- `phase_D/D5_cosponsors.md`, `D7_wharton_intent.md`, `D13a_laura_quotes_verified.md`, `D13c_voice_map.md`
- `research/blueprints/01_trading_notes_blueprint.md`, `research/council_2026-09-27/01_chair_memo.md` (sections 4, 7)

Not re-read by D9: Jenkins, Harris & Lark (2019), via B6b.

What this teaches

1. **Plain English is a discipline, not a tone.** Each idea must survive three squeezes: a 90-word note written before the outcome, a 50-word pitch, and a 500-word IPS that must also fit on two physical pages. An idea that cannot be said in one plain clause is usually not yet understood, or not yet one idea. Three of our questions turned out to be one principle said three ways.
2. **Words are claims.** "Certain", "confident", "reliable", "matched" and "safe" each promise something different. Choosing each word on purpose is part of the investment analysis, because it states exactly what risk remains.
3. **Measure risk against the goal.** For a client with a fixed bill, cash is the risky choice and long bonds are the careful one. The yardstick decides the answer.
4. **Write the record while it is cheap.** A dated fact and a named risk in each note, plus one decision-log column, cost minutes now and make every later deliverable consistent almost automatically.

Audit corrections (AY2)

Auditor AY2, 2026-09-28. The text above is left unchanged. Each item says what to correct, its severity (blocking / important / minor), and which deliverable it touches (TN or IPS). Evidence and the cross-file view are in `research/insight_v1/phase_D/audit_judges_practice_comms.md`.

Reproduced and confirmed (no change needed):

- **Scripts:** `D9_numbers.py`, `D9_ips_page_fit.py` and `D9_draft_checker.py --demo` were re-run. Every quoted figure reproduces:
 - vocabulary counts; 40.3/38.7/38.5% (simulation 38.8%);
 - \$2,890 per 10bp; about \$20.4k a year; T3 +\$30,600 / -\$27,374; 69-77%;
 - 44-46 of 46 lines; about 525 words;
 - 59 words, 413 characters, 0 digits.
 - **Sample PDF:** AY2 re-measured Wharton's sample with pypdf. It is US Letter at a 24.0 pt line pitch, with one blank line (48 pt) before the IPS heading.
- **Sources re-found verbatim:** Investor.gov; IPCC (two quotes); the retired Wharton judging page (two quotes); Overachiever "with unstable income".
- **Official lines as cited:** IPS guide L82, L101, L104, L116, L123-124; case L85 vs L91-92.
- **Extra support for M001:** the TN guide itself says notes "should document the research, analysis, and reasoning behind your investment decisions" (TN guide L7-8).

Corrections

1. **[important, TN] The VGSH row of the fixed-labels table overclaims.**
 - The row maps VGSH to "the amount she can promise co-sponsors in 2031" and gives "already owned" as its certainty word. This brings back the "floor proxy" overclaim the S4 red team removed (ticket, changes after red team, item 4).
 - In 2026-2030, VGSH is part of the growth money. The 2031 floor is bought only in 2031, from all of the growth money.
 - Use: role word "liquidity" (or "future funding"); label "short-Treasury part of the growth money"; Laura need "facility contribution and flexibility; the kind of asset a 2031 floor can later be built from" (the

ticket's "can later become"). No "already owned" before 2031.

- WInS notes are permanent, so this must be fixed before the first order.

2. **[important, IPS] The certainty vocabulary leaves out "uncertainty".**

- The official files use "uncertainty" 11 times (AY2 whole-word count): case 2, IPS guide 1, Guide 6, Infographic 1, SMAApply 1.
- Case test 3 says a strategy "Addresses how investment uncertainty could affect both the operating commitment and the facility contribution" (L137). The Creativity criterion asks for "investment uncertainty clearly and credibly".
- So the vocabulary must not imply that the payments carry no uncertainty. Add a row: "uncertainty" is the official word for what markets can do to both goals. For the payments it is limited to named residuals: rates before purchase, the 2028 deposit in the rates-fall branch, a U.S. default, and purchasing power in Taiwan.

3. **[important, IPS] The assumption line leaves out currency.** Brief s16, which is binding, requires both certainty gaps to be named plainly. One is that "certain" is in nominal U.S. dollars while the residency's costs are in Taiwan. Put the USD/TWD purchasing-power point into the same assumption clause as inflation. The words are the team's; D4 owns the numbers.

4. **[important, TN] The M012 one-sentence answer is a fill-in sentence for a permanent WInS note.** It reads: "The position holds the value of Laura's ten fixed \$50,000 payments, which cost about \$X at [trade-date] Treasury prices ..."

- WInS notes are quoted exactly and cannot be edited, so a template the team fills in becomes AI-written text in a submitted deliverable (Wharton AI policy; brief s4).
 - Recast it as the element list already in the checklist, and mark each quoted phrase "illustrative, do not copy".
 - The same applies to three phrases that are close to pitch- or IPS-ready: "measured against her payments, not against the screen" (L288), "risk only where others can help" (L483) and the M143 claim sentence (L445).
5. **[important, IPS] Nobody has added up the IPS elements across D7, D8 and D9.** This file owns the page-fit check.
- Together the three files ask for about 13-15 IPS elements. Their own estimates sum to roughly 310-475 words before connecting words, against this file's 470-490-word cap.
 - The ips_spec owner should set one budget (table in the audit summary, section 5). Merge the return objective into rule 2; merge risk tolerance with the M241 principle; fold inflation and currency into the certainty definition. Cut the fee clause first.
6. **[minor, IPS] "Locked" is the wrong word for 5.08/5.24/5.48% (L76-78, L497-510).** These are forward rates implied by the 2026-09-25 curve for a January 2027 purchase. Nothing is locked until her money arrives; D8 says this correctly for its 5.23%. Say "the rates today's curve implies for a January 2027 purchase". The 4-in-10 conclusion does not change in kind.

5. **[minor, IPS] T1's "+\$139k in the worst 5% of paths" misreads a percentile gap.** It, and "gives up \$267k of best-case upside", are differences between two strategies' percentiles, not gains in the same paths. Say "the bad case (5th percentile) is about \$139k higher". These numbers belong on the later list in any case.

6. **[minor, TN] Also fix blueprint 01's risk example.** When blueprint 01 section 3 is updated (M001 implication), remove its risk example "bond prices fall if rates rise, which does not matter for held-to-maturity matching" (blueprint L35). It is false for WInS funds, which never mature, and it contradicts this file's own vocabulary.

7. **[minor, TN + IPS] Some terms and vocabulary rows state "bought in January 2027" flatly.** Examples: L140 "Ladder: real Treasury bonds bought in January 2027"; L195 and L243 "stands in for the bonds she buys in January 2027". Add "starting January 2027; completed from the 2028 deposit if prices have risen", to match D7 M011 and this file's M143.

8. **[minor, TN] "Hundreds of teams will copy its wording" is D9's judgement.** Label it INT; there is no evidence about other teams' notes.

9. **[minor, IPS] Page fit: add a fallback.** (INT) The official sample is set at exactly 24 pt. So "Exactly 24 pt" line spacing is a defensible reading of "double-spaced" if a draft overflows under Word's "Double". The team decides, and the exported-PDF check stays binding. A common rule of thumb (about 250-300 double-spaced words a page) also puts 550 words plus headings right at two pages, which supports the 470-490 target.

10. **[minor, TN + IPS] Using the draft checker counts as AI use.** Running `D9_draft_checker.py` (an AI-built tool) on drafts must be logged in the team's AI-use record (R-W46), like any other AI help.
11. **[minor, TN + IPS] Pick one place for the statistics degree.** This file says "once at most" in the IPS, but the ticket uses it in the VT note and D7 uses it for the certainty definition. Across the TN and the IPS, pick one place (decoration/tokenism risk).
12. **[minor, TN + IPS] The option (ii) word cost (M012) is also a consistency cost.** D7's pitch rule ("no stock risk until all ten payments are bought") reads as contradicted by a day-one VT buy next to a \$198k hedge, unless the scaling clause appears (D7 correction 3). Give gate box (b) both inputs.

Verdict for D9: the strongest communication work in the cluster. The page-fit risk is real and confirmed on Wharton's own sample, and the note template, vocabulary and trade-off specs are well evidenced. Fix items 1-4 before the first WInS order and before the certainty glossary is adopted. Item 5 is the main cross-file task for whoever writes `ips_spec.md`.

D10 Compliance Officer: which book WInS shows (M005) and the IPS governance lines (M013)

Agent D10 (Compliance Officer), insight_v1 run. Written 2026-09-28 (the first official trading day). Scope follows brief sections 16-17: this file serves **WInS trading now + the Trading Notes (TN, Oct 23)** and **the IPS (Nov 6)** only. Final Report items are listed at the end as one-line "later" entries. Everything here is AI-generated research: specifications, rule checks, numbers and checklists. **None of it is text to submit.** The team decides, and writes every WInS note, reflection and IPS sentence in its own words (Wharton AI policy, quoted in section 4).

Status labels (brief section 3): VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED, ASSUMPTION, INTERPRETATION (my reading of an official text, not a fact).

Scripts (run from the repo root):

- `.venv/bin/python research/insight_v1/scripts/D10_wins_book_compliance.py` (M005: the three candidate books under each possible position limit; volume rule; equity-share consistency; position-limit drift).
- `.venv/bin/python research/insight_v1/scripts/D10_ips_page_fit.py` (M013: does 50 + 500 words fit on two double-spaced Times New Roman pages; the word budget left for governance).

Terms used: **hedge / promise money** = the Treasury funds standing in for the ten fixed \$50,000 payments (2033-2042). **Growth money / sleeve** = everything the payments do not need. **Funded ratio** = money set aside for the promise divided by the market price of the ten payments. **Position limit** = the most WInS lets the team hold in one security (shown only in the logged-in Portfolio Summary > Session Rules). **Book** = the list of holdings at one date.

Summary: top findings (ranked by impact on reaching the semifinals and on making the plan Laura's)

1. **The IPS page limit is the most likely way to lose a semifinal place, and 550 words barely fit** (tier 2, act now). At Times New Roman 12, double-spaced, 1-inch margins, the 50-word pitch plus a 500-word IPS fill about **92-105% of the two allowed pages on U.S. Letter** (script; ASSUMPTION-based layout model). Word's default 8pt space after each paragraph and more than about 7 paragraphs push it over. On A4 (an Australian default) it fits more easily (about 89-99%). A breach means "will not be considered for semifinal selection" (IPS guide L82-83, VRF). Checklist in M013.
2. **M005: no official rule decides which book WInS shows; the IPS sentence naming the date is required either way.** Both the post-2028 mix scaled to \$300k (ticket option ii) and the literal January-2027 book (option iii) pass every WInS rule, whatever the position limit turns out to be (script). What the rules do require is **consistency**: the frozen Nov-6 portfolio must "reflect the strategy established in your IPS" (IPS guide L75, VRF), and the Investment Strategy criterion scores "consistency with the team's IPS" (SMAApply, VRF). WInS equity differs from Laura's real January-2027 equity by **+20.5 points under (ii)** and **0 under (iii)** (script), so (ii) needs a longer explanation. **The "middle path" (about 3% growth slice) is mis-posed**: at today's prices the literal leftover is **\$4.4k-6.6k (1.5-2.2% of WInS)**, so **equity would be only 0.9-1.3%**, and it reverses the ticket's rule that the 2027 leftover waits in T-bills. Drop it. D10 leans to (iii) at medium-low confidence; (ii) is equally compliant if the three-element sentence is in the IPS.
3. **The book must be chosen before the first order, because trade notes are permanent** (tier 1). Notes cannot be edited, only added to (Stock-Trak, via A4), and must be consistent with the later IPS (TN guide L30-31, VRF). A later switch between books is allowed (notes "do not need to correspond to investments that remain", TN L25-26, VRF) but must be a real, logged decision, not a same-day reversal ("Day Trading: This is not permitted", 2026-27 User Guide, VP via A4).
4. **M013: the IPS needs four governance elements in about 30-45 words beyond D8's four money rules.** They are: (a) who decides: the rules decide; Laura herself makes only the two choices the case gives her; **no "our portfolio manager/advisor approves" wording**, because the real rules say advisors "may not make decisions" (Rules page, VP 2026-09-27); (b) review points at the case's own cash-flow dates plus a yearly check; (c) the event triggers **folded into one "promise first" priority**

rule rather than listed; (d) one rebalancing sentence that also states the deliberate **no-rebalancing between promise money and growth money** (CFA: "If the policy is not to rebalance, this policy should be documented in the IPS", VP). Cut the org chart, firm name or size, fees, reporting terms, the residency-delay trigger and citations.

5. **Why governance is a compliance issue, not decoration:** after Nov 6 "you may not redesign your strategy after observing the results" but "planned adjustments as funding dates approach" are allowed (Guide p.5, VRF). Only adjustments the IPS pre-authorises can appear later without looking like a post-freeze revision. A named trigger costs about 5-10 words; a missing one can cost the whole scenario analysis its legitimacy.
6. **A funded-ratio trigger "below 100%" can never fire once the ladder is bought** (it is cash-flow matched, so the ratio stays at 100% barring a U.S. default). It can only matter **before** the January 2027 purchase or while part of the promise waits for the 2028 deposit (the ladder cost more than \$300k on 173 of 185 trading days of 2026; about 24% modelled odds on 2027-01-01, F-111/F-112). So the real trigger is a yes/no test: "is the promise fully bought?"
7. **Deliverables of this D10 task:** this working file; two scripts (`D10_wins_book_compliance.py`, `D10_ips_page_fit.py`); and the structured answers for M005 and M013 returned to the run. No submission text, no commits.

1. M005 (tier 1: WInS now + TN; the sentence is tier 2: IPS)

Question (canonical, Phase C): Should the WInS book (exactly Laura's \$300k Year-1 deposit; the \$150k is never added) mirror her literal 2027 portfolio (about 97% Treasury ladder plus a small labelled surplus sleeve) rather than the post-2028 ~65/35 target mix, and which one IPS sentence says which book WInS represents?

Premise check. Two corrections before answering:

- "65/35" is hedge/sleeve, not bonds/equity; the post-2028 book holds about **20.4% equity** (F-409, VRF; ticket section 2). Option (i) (35% equity) is already dropped.
- "About 3% surplus sleeve" overstates the leftover. The ten payments cost \$292,264 (F-101) to \$294,387 (STRIPS ladder, ticket). After 4-7 commissions (\$25 each) and a \$1,000 cash float (ASSUMPTION, ticket band), the leftover is **\$4,438-6,636 = 1.5-2.2% of WInS**; at 60/40, **VT would be \$2.7k-4.0k (0.9-1.3%)** (script section 2).

One-sentence answer: No official document decides which book WInS shows, and both the literal January-2027 book and the scaled post-2028 mix pass every WInS rule, so the compliance requirement is only that the IPS names the date WInS represents and that the frozen portfolio can be produced by the IPS rules at that date; the "middle path" is mis-posed (about 1% equity, and it reverses the plan's 2027 leftover rule), and D10 leans to the literal January-2027 book because it is the only one that is true at a stated date without scaling.

1.1 What the official texts say (all re-read)

#	Claim	Source	Status
E1	"The WInS portfolio represents each team's implementation of its investment strategy during the competition. It does not determine Laura's actual portfolio value at the beginning of 2027."	Case p.4 (Laura_Gao_2026_Client_Profile.txt L129-130)	VRF
E2	"The \$300,000 is your team's WInS simulator balance. The additional \$150,000 contribution described in the Client Case Study will not be added to WInS."	https://wghsinvcomp.smaply.us/res/p/trading/ (read 2026-09-27/28, helper -- grep YES)	VP
E3	"There is no required sector allocation or minimum number of sectors. However, diversification remains an important investment principle." ... "funding purposes. The goal is to build an intentional mix of investments appropriate for your strategy, not simply to own a large number of securities."	same page	VP
E4	"Your portfolio and Final Report should reflect the strategy established in your IPS."	IPS guide L75; Guide p.2 L60-61	VRF
E5	Investment Strategy criterion: "uses appropriate diversification; and maintains consistency with the team's IPS."	SMAppl_Deliverables_Page_2026-09-27.md L49	VRF

#	Claim	Source	Status
E6	"The analysis should be consistent with the strategic approach your team is developing and will later articulate in its IPS." / "The Trading Notes you select do not need to correspond to investments that remain in your portfolio at the end of the competition."	TN guide L25-31	VRF
E7	Trading Note should capture "its expected role in growth, liquidity, risk management, or future funding." / WInS "is not the competition scorecard" / "your portfolio provides evidence of the decisions your team made"	Guide p.3 L70-86	VRF
E8	"Collectively, the decisions should demonstrate how your team used individual investments as part of a cohesive portfolio strategy."	TN guide L14-15	VRF
E9	"Focus on the strategy and decision-making framework that guide your portfolio rather than describing individual investments"	IPS guide L51-52	VRF
E10	"Day Trading: This is not permitted." / "Position Limit: This is how much of your portfolio you can invest in one single stock." (a 2024 screenshot shows a separate bond "Position Limit (Single Position) 100%")	2026-27 WInS User Guide p.6, via phase_A/wins_week1_guardrails.md	VP (via A4)
E11	Stock-Trak default "position limit is 25%"	edu.stocktrak.com/wharton/portfolio-faq/ (generic, 2023) via A4	VP (generic, not this season)
E12	Trade notes: "students cannot edit or delete their trade notes ... they can add more notes"	Stock-Trak blog 2017 via A4	VP (vendor, not season-specific)
E13	No document states which date's book WInS should show	phase_A/case_register.md X-7, R-AN14	VRF (A1's search)
E14	"Take only investment actions that are consistent with the stated objectives and constraints of that portfolio" (B.5.a); disclosures must be "truthful, accurate, complete, and understandable" (F.2)	CFA Institute <i>Asset Manager Code</i> , 2nd ed., via phase_A/stakeholder_map.md SH-25 and D8; the Rules page tells teams to "operate by these standards" (re-read 2026-09-27, --grep YES)	VP

1.2 Rule check of the three books (script sections 1 and 4)

All three use only ETFs and cash (permitted), no margin, no shorting, no derivatives. Every order is at most **0.2% of the "twice the current daily volume" cap** for the funds whose issuers publish volume (IEF, TLH, SPTL); Vanguard funds (VT, VGSH, VGIT, VGLT) publish no volume, so read it in WInS (ticket check 5). All securities remain **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK**.

Book	No limit	35% limit	25% limit	WInS equity
(ii) post-2028 mix scaled	IEF 23.6 / TLH 42.4 / VT 20.5 / VGSH 12.5 / cash 1; 4 trades, \$100	+ SPTL 8.4 (TLH 34); 5 trades, \$125; hedge ~10.2y	IEF 23.6 / TLH 24 / SPTL 18.4; 5 trades, \$125; hedge ~10.5y	20.5%
(iii) literal Jan-2027	IEF 35 / TLH 63 / cash 2; 2 trades, \$50	IEF 34 / TLH 34 / SPTL 30; 3 trades, \$75; hedge ~10.6y	IEF 24 / TLH 24 / SPTL 24 / VGIT 17.6 / VGLT 8.4; 5 trades, \$125; hedge 9.9y	0%
(mid) literal + growth slice	IEF 35.1 / TLH 63.1 / VT 0.9 / VGSH 0.6; 4 trades, \$100	5 trades, \$125	7 trades, \$175	0.9%

- **Continuous-limit risk** (ASSUMPTION that WInS re-checks the limit as prices move): under a 25% limit, a holding placed at 24% crosses 25% after rates fall about **101bp (ii, TLH)** or **118bp (iii, SPTL)**, or rise about **128bp (iii, IEF)** (first-order duration arithmetic, script section 4). The 10-year yield moved about 120bp from its February low to September in 2026 (F-003), so this is possible within six weeks but unlikely. If Session Rules show the limit is checked only when an order is placed, ignore it. Note (E10): the User Guide says the limit is per "single stock", and an older screenshot shows a separate 100% bond limit, so **read both lines** in Session Rules.
- None of the three breaks a rule. Rule compliance does not choose between them.

1.3 Consistency check (the binding requirement)

WInS equity minus the equity share of Laura's real portfolio at each plan date, in percentage points (script section 3; real shares: 0% in January 2027 under the T-bill rule, 20.4% in 2028-2030 (F-409), about 4% after

the 2031 floor (brief section 16), all ASSUMPTION-based):

Book	Jan 2027	2028-2030	After the 2031 floor
(ii)	+20.5	+0.1	+16.5
(iii)	0.0	-20.4	-4.0
(mid)	+0.9 (−0.1 if the growth-mix rule replaces the T-bill rule)	-19.5	-3.1

Reading (INTERPRETATION): every book matches exactly one date, so **the IPS must say which date WInS shows under any choice**. Under (iii) the date is the one Laura's real money first exists in the plan, so the explanation is short ("WInS is her first deposit on the day it is invested"). Under (ii), WInS holds about 20% stocks while the IPS's own first-year allocation is about 0%, so the sentence needs the three elements already specified in the ticket (section 2): the post-2028 mix; scaled to \$300,000; and that in January 2027 almost all of her real \$300,000 buys the ladder. Under Asset Manager Code F.2 (truthful, complete disclosure), a (ii) book should never be called "Laura's portfolio" in a note or the IPS without that scaling caveat.

1.4 The trade-offs the rules leave open (for the team's vote, ticket gate box b)

	(ii)	(iii)
Diversification criterion (E3, E5)	Three funding purposes visible in WInS (promise, growth, short Treasuries)	One funding purpose in WInS; diversification by purpose shown only in the IPS
TN variety (E7, E8)	Notes can show three roles: future funding, growth, liquidity	Notes show future funding / risk management only; the third note must come from a genuine later decision (duration refresh, position-limit response)
Consistency burden	Needs the three-element sentence (ASSUMPTION: ~30-40 words of 500)	Needs a short date sentence (ASSUMPTION: ~12-20 words)
What WInS says about Laura	"Her whole plan, in miniature"	"Promise first": what her real money does first; fits her verified concern about not letting down "people who were your earliest supporters" (D13a; use only as D13 allows, never in the TN or IPS)
Main risk	Reader sees ~20% stocks against a ~0% first-year plan and suspects inconsistency	Reader sees ~98% Treasuries for a client "willing to take thoughtful risks" and suspects timidity

D10 lean (medium-low confidence): (iii). Reasons: it is the only book true at a stated date without scaling; it needs the shortest reconciling sentence in a tight IPS (finding 1); it shows the plan's central priority in the WInS holdings themselves. **(ii) is equally compliant** if the three-element sentence is in the IPS and no note calls the holdings "Laura's portfolio". This is a team decision; record the vote and the reason in the decision log before the first order.

Drop the middle path. A 0.9-1.3% VT position (about \$2.7k-4.0k) would carry a \$25 commission and a whole trading note about a sliver that grows only in 2028. It also needs the ticket's 2027 leftover rule (T-bills; S4 finding 13) changed to "growth mix from day one". It adds a rule change and a weak note for no compliance gain.

1.5 Implication

- **WInS-now (decision, before the first order):** vote (ii) or (iii) using the table above; screenshot Session Rules (both position-limit lines); pick the cap branch from the ticket/script. The middle path is off the table.
- **TN (sentence spec, not text):** whichever book, each note's reasoning must be true of that book on its trade date. Under (ii), a note on VT or VGSH must say the holding represents the growth money after the second deposit. Under (iii), no note may imply WInS holds growth assets.
- **IPS (one sentence, tier 2), must contain:** (1) the date WInS represents (January 2027 for (iii)); "after both deposits, scaled to \$300,000" for (ii)); (2) that WInS holds stand-in funds that move like her payments, not the ladder itself; (3) for (ii) only: that in January 2027 almost all of her real \$300,000 buys the ladder. No tickers, no weights to decimals (E9).

Numbers and where they come from: leftover \$4,438-6,636 (1.5-2.2%); VT 0.9-1.3%; equity gap +20.5 / 0.0 pts; position counts 2-7; commissions \$50-175; worst order 0.19% of the volume cap; limit-drift 101-128bp (all from `D10_wins_book_compliance.py`, inputs VRF/VP as listed in its docstring).

Deadline tier: 1 (vote before the first order); the IPS sentence is tier 2. **Confidence:** high that both books comply and that the date sentence is required; medium-low on the lean to (iii). **Changes current strategy:** yes, slightly: it removes the middle path and moves the ticket's lean from (ii) to (iii); the lock-early strategy itself is unchanged. **Criterion served most:** Investment Strategy ("consistency with the team's IPS"; "appropriate diversification"). **What this teaches:** a simulator is a picture of a plan at one moment. A plan that changes over time can only be photographed at one date, so say which date the photo shows.

2. M013 (tier 2: IPS)

Question (canonical, Phase C): Which governance elements must the 500-word IPS carry (who decides what among automatic rules, the firm and Laura; dated reviews in 2028, 2031 and 2033; event triggers such as a funded ratio below 100%, a short or late deposit or a residency delay; one rebalancing rule), and which can be cut?

Status of the question: partly answered already. D8 (`phase_D/D8_practice.md`, M006/M235) set the four money rules (promise first; growth mix; 2031 range; 2033 order of use) at about 110-155 words plus one governance line. The "answered" skeptic killed M013 for that reason. What is new here: the compliance reasons (the advisor rule and the freeze rule), folding the triggers into rule 1, the funded-ratio point, the cut list, and the page-fit test.

One-sentence answer: On top of D8's four money rules, the IPS needs about 30-45 words of governance: the rules decide and Laura herself makes the two choices the case gives her (no "portfolio manager/advisor approves"); reviews at the case's own cash-flow dates plus a yearly check, where rules change only if her circumstances change and never because markets moved; event triggers folded into the "promise first" rule; and one rebalancing sentence that states the no-rebalancing policy between promise and growth money. Everything else (org chart, firm details, fees, reporting, residency-delay trigger, citations) is cut.

2.1 Evidence

#	Claim	Source	Status
G1	"Advisors may not make decisions on behalf of students or actively participate in team activities, trading decisions, or strategy development."	https://globallyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/ (read 2026-09-27, -- grep YES)	VP
G2	"Your portfolio manager (your team's teacher/advisor who makes the final investment decisions for your firm's portfolio)"	Case p.1 L4-5 (R-C3)	VRF
G3	The same line appears in the 2021-22 and 2022-23 cases, where the firm (WGAM) "currently manages a \$100,000,000 portfolio": it refers to the firm's own book, inherited wording	<code>phase_B/B1a_case_anomalies.md</code> E1 (Wharton archive)	VP via B1a (not re-checked by D10)
G4	"Laura must decide how much of the remaining portfolio she can responsibly contribute"; in 2031 "she will need to describe how much she expects to contribute"; "she wants to communicate a credible range"	Case p.3 L101-117	VRF
G5	"Investment Policy Statement. This formally establishes the strategy and the team's decision-making framework."	Case p.4 L155-156 (R-C89)	VRF
G6	"Once submitted, the IPS becomes the official record ... Your team may not revise its investment strategy after the submission deadline."	IPS guide L73-74	VRF
G7	"A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." / "you may not redesign your strategy after observing the results"	Guide p.5 L156-159	VRF
G8	IPS should explain "planned changes in the portfolio as future funding needs approach" and "How your strategy will guide portfolio construction and investment decisions as the client's needs change"	IPS guide L46-47, L61-62	VRF

#	Claim	Source	Status
G9	CFA <i>Elements of an IPS for Individual Investors</i> (2010): "2a. Specify who is responsible for determining investment policy, executing investment policy, and monitoring the results"; "It can reinforce the obligations of advisers to offer counsel and the obligations of principals to ultimately approve or disapprove of the policy." "2b. Describe the process for reviewing and updating the IPS." "A process for refreshing the IPS as investor circumstances and/or market conditions change should be clearly identified in advance." Example review "no less frequently than annually". "4c ... If the policy is not to rebalance, this policy should be documented in the IPS." "the IPS serves as a policy guide that can offer an objective course of action to be followed during periods of market disruption"	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf (read 2026-09-28 with the helper)	VP
G10	"Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted." "Submissions that do not meet the requirements below will not be considered for semifinal selection." TNR 12, double-spaced, 1-inch margins, pages 2-3 "2-page maximum", pitch max 50 words, IPS max 500 words	IPS guide L82-123	VRF
G11	Ladder cost > \$300k on 173 of 185 trading days of 2026; ~24% modelled chance on 2027-01-01	F-111 (VP inputs), F-112 (ASSUMPTION)	as labelled
G12	The case says the \$150,000 "will" be contributed	Case p.2 L43-45 (R-AN35)	VRF; a short or late deposit is the team's own stress scenario and must be labelled so
G13	Liberation Serif is metrically compatible with Times New Roman	https://en.wikipedia.org/wiki/Liberation_fonts (read 2026-09-28)	secondary source; used as ASSUMPTION

2.2 Keep / cut (each element traced to a source)

Element	Keep or cut	Why (source)	Word budget (ASSUMPTION)
Who decides	KEEP: the students' rules run the portfolio; Laura makes the two choices the case gives her (the 2031 range she announces, the 2033 facility amount), on the team's rule-based recommendation	G4 (the case gives those decisions to her); G9 2a (principal approves, adviser counsels); G1	10-15
"Our portfolio manager / advisor approves"	CUT	G1: in the real rules the advisor may not decide; G2/G3: the fiction's PM decides the <i>firm's</i> book and is inherited wording. Zero benefit, some risk of a reader thinking the advisor made strategy decisions	0
Review points	KEEP: at the case's cash-flow dates (Jan 2028 deposit, 2031 announcement, Jan 2033 reserve) plus a yearly check; rules change only for a change in Laura's circumstances, never because markets moved	G7, G8, G9 2b; D8 governance line	12-18
Event triggers	KEEP, but inside rule 1 (promise first) , not as a separate list: whenever the promise is not fully bought (ladder cost above the cash on hand; 2028 deposit short or late), the next money completes it before any growth	G6/G7 (a response must be pre-authorized to count as "planned"); G11 (binds in about 1 in 4 modelled paths); G12 (label the deposit scenario as the team's own)	0 extra (already in D8's rule 1)
"Funded ratio below 100%" as a trigger	Replace with the yes/no test "is the promise fully bought?"	Once the ladder is bought it is cash-flow matched, so the ratio stays at 100% barring default; the test only bites before the purchase or while the 2028 top-up is pending (G11)	0
Residency delay trigger	CUT	The case fixes 2033; a delay is outside the case (BS-07). A change in her circumstances is covered by the review line	0

Element	Keep or cut	Why (source)	Word budget (ASSUMPTION)
Rebalancing	KEEP one sentence: the growth mix is reset to its target once a year (band as the team sets it); promise money and growth money are never rebalanced against each other	G9 4c (a no-rebalancing policy must be written down); ticket section 5; D8 rule 2	10-15 (part may overlap D8's rule 2)
Record-keeping / decision log	Implied only (FR material)	Code D.4 via SH-25; FR is out of scope now	0
Org chart, firm name or size, fees, reporting frequency	CUT from the IPS	Not asked by the IPS guide (L53-65); fees belong to the FR projections (BS-01)	0
Naming CFA / citing sources	CUT	G10: "formal citations are not permitted"; naming a code in running text is not a formal citation (INTERPRETATION), but it spends words for no gain	0

Total governance ≈ 30-45 words, on top of D8's four money rules (110-155 words). That leaves about 300-360 words for the thesis, the client-specific reasoning (why this is Laura's plan), risk/return, liquidity and diversification, which the IPS guide asks for first (L53-65).

2.3 Page-fit test (script `D10_ips_page_fit.py`; ASSUMPTION-based layout model)

Liberation Serif widths (same as Times New Roman), Word-style "double" = 2 x single line (26.6pt; Word may use up to 27.6pt, which is tighter), word lengths sampled from Wharton's own guides, 400 random layouts per setting:

Paper	Paragraphs in the IPS	Space after paragraph	Share of the two pages used (median; p95)	Word slack at the median
Letter	5	0pt	92%; 101%	~52
Letter	7	0pt	94%; 101%	~38
Letter	7	8pt (Word default)	99%; 105%	~5
Letter	9	8pt	105%; 111%	~-31 (overflows)
A4	7	0pt	91%; 97%	~60
A4	9	8pt	99%; 107%	~7

Compliance checklist for the IPS file (tier 2; start testing with the first rough draft, not on Nov 6):

- [] Pitch at most 50 words and IPS at most 500 words **by the team's own count**; keep a margin (ASSUMPTION: aim for about 480) because Word and Google Docs count "U.S.", "\$50,000", hyphens and slashes differently.
- [] Space after paragraphs set to 0; at most about 7 IPS paragraphs; the two headings on one line each.
- [] Export to PDF and confirm the file is **exactly 3 pages** (title page + 2), at most 5 MB.
- [] Times New Roman 12 everywhere, including headings; double-spaced; 1-inch margins all sides.
- [] No tables, charts, images, links, footnotes, citations; no bullet graphics pasted as images.
- [] Title page: official team name exactly as on the roster; members as "First Name, Last Initial" exactly as on the roster; the **WinS username** (not the team name).
- [] Every sentence written by a student; AI use logged for the Works Cited later (section 4).

2.4 Implication

- **IPS (decision + word budget):** adopt the keep/cut table; spend about 30-45 words on governance; put the triggers inside the promise-first rule; state the no-rebalancing policy between promise and growth money in words.
- **IPS (format):** run the page-fit checklist on every draft; set paragraph spacing to 0 now.
- **TN:** none directly, except that no note should say "our portfolio manager decided" (A4 checklist item 13).

Numbers and where they come from: governance 30-45 words (ASSUMPTION budget); D8 rules 110-155 words (D8); page use 92-105% on Letter, 89-99% on A4 (script); rates-fall branch binds ~24% of modelled paths and 173/185 days of 2026 (F-111, F-112).

Deadline tier: 2. **Confidence:** high on the advisor/freeze reasoning and the cut list; medium on the page-fit numbers (a model of Word's layout, not the team's real file). **Changes current strategy:** no (it formalises the current rules; it only removes the funded-ratio and residency-delay triggers as separate items). **Criterion served most:** Investment Strategy (a clear framework consistent across deliverables). **What this teaches:** a policy is judged by what it says it will do before anything goes wrong. Rules written in advance let you act later without anyone asking whether you changed your mind after seeing the results.

3. Other compliance flags seen while working (outside M005/M013; one line each)

- **Today is the first trading day (Sep 28).** Nothing requires trading today ("does not require frequent or same-day trading", SMaply, VP); the ticket's five-box gate comes first.
- **"Required trading activity"** exists on the FAQ and Rules pages but is defined on no public page (FAQ re-read 2026-09-28, VP). Read the logged-in Trading Details page this week; the "first trade by Oct 10" claim is still UNVERIFIED (a 2023-24 rule per A4).
- **AI-use log:** "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages" (<https://globalyouth.wharton.upenn.edu/ai-policy/>, re-read 2026-09-28, VP). Start a dated log now (tool, date, what it was used for); the Works Cited itself is FR work.

4. Later (after Nov 9; Final Report) - one line each

- FR: Works Cited must record how AI was used (from the AI-use log).
- FR: school documentation needs an owner and an October request (BS-11; "Do not wait until the last minute", SMaply).
- FR: if (ii) was chosen, the evaluation of WInS must repeat that it was a scaled post-2028 mix.

Sources (access dates 2026-09-27/28)

- SMaply Trading Details: <https://wghsinvcomp.smaply.us/res/p/trading/> (VP; --grep checks run 2026-09-27/28).
- SMaply FAQs: <https://wghsinvcomp.smaply.us/res/p/faqs/> (VP).
- Wharton Rules & Roles: <https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/> (VP).
- Wharton AI policy: <https://globalyouth.wharton.upenn.edu/ai-policy/> (VP).
- CFA Institute, *Elements of an Investment Policy Statement for Individual Investors* (2010): <https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf> (VP, read 2026-09-28).
- CFA Institute, *Asset Manager Code*, 2nd ed. (via `phase_A/stakeholder_map.md` SH-25 and `phase_D/D8_practice.md`, VP there).
- Wikipedia, "Liberation fonts": https://en.wikipedia.org/wiki/Liberation_fonts (secondary, read 2026-09-28).
- Repo (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt`, `2026_WGY_Investment_Policy-FINAL.txt`, `2026_WGY_Trading_Notes_Analysis-FINAL.txt`, `2026_WGY_Investment_Competition_Guide.txt`, `SMaply_Deliverables_Page_2026-09-27.md`.
- Run files: `phase_A/case_register.md` (R-AN14, X-7, X-10, X-23), `phase_A/fact_register.md` (F-003, F-101, F-111, F-112, F-117, F-409), `phase_A/stakeholder_map.md` (BS-07, BS-08, BS-11, SH-25), `phase_A/wins_week1_guardrails.md` (A4), `wins_now/securities_and_allocation_v0.md`, `wins_now/S4_red_team.md` (finding 13), `phase_B/B1a_case_anomalies.md`, `phase_C/survivors.json` (M005, M013), `phase_D/D8_practice.md`, `phase_D/D13a_laura_quotes_verified.md`.
- Scripts: `research/insight_v1/scripts/D10_wins_book_compliance.py`, `research/insight_v1/scripts/D10_ips_page_fit.py`.

What this teaches

1. **Rules first, then taste.** Check what the rules force before debating what looks best. Here the rules force only one thing (say which date WInS shows), and the rest is a real choice the team can make and

defend.

2. **Check the premise with arithmetic.** "A 3% growth slice" sounded reasonable; the leftover is really 1.5-2.2%, so the stock position would be about 1%. A two-line calculation removed a whole option.
3. **A trigger that can never fire is not a safeguard.** Ask when each rule could actually bite. The funded ratio can only fall below 100% before the ladder is bought, so the real test is simply "is the promise fully bought?"
4. **Format is part of the strategy.** Two double-spaced pages hold about 550 words only if the layout is tight. Test the PDF early, because a page too many means the strategy is never read.

Audit corrections (AX2)

Auditor AX2 (cluster auditor: equity, Taiwan/FX, compliance), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: `research/insight_v1/phase_D/audit_markets_rules.md`. Check script: `research/insight_v1/scripts/AX2_audit_checks.py`.

Reproduction. Both D10 scripts were re-run on 2026-09-28. The three-book table, positions, commissions, leftover \$4,438-6,636, VT 0.9-1.3%, equity gaps (+20.5 / 0.0 / +0.9), limit-drift thresholds (101 / 118 / 128bp) and every page-fit row in section 2.3 reproduce exactly. The two exceptions are C5 and C6.

Sources re-opened 2026-09-28 (fetch helper `--grep`, all YES):

- SMAApply Trading Details: E2, E3, "does not require frequent or same-day trading", the 2x-volume rule and 200 trades.
- Rules & Roles: the advisor sentence (G1).
- CFA *Elements of an IPS* (2010): the 2a heading, "no less frequently than annually", 4c "If the policy is not to rebalance, this policy should be documented in the IPS", and "objective course of action".
- Wharton AI policy: the Works Cited sentence.
- SMAApply FAQ: "Investments permitted (for BOTH contributions)" (line 42 of the page text).

Repo texts re-read (VERIFIED-REPO-FILE): case L4-5, L101-120, L129-133; Guide L60-61, L70-90, L156-159; IPS guide L46-75, L82-123; TN guide L12-31, L42-53; SMAApply page L49. Not re-opened: the Asset Manager Code quotes (via SH-25/D8), B1a's 2021-23 case wording (G3), and the Stock-Trak blog and FAQ (via A4).

#	Severity	Issue	Correction
C1	blocking (for the (ii)/(iii) vote)	Option (iii) may not produce three executed trades by Oct 23. With no cap, (iii) has two orders (IEF, TLH). The ticket's duration refresh trades only if the hedge leaves 9.65-10.15 years, and a position-limit response happens only if a cap bites, so neither is guaranteed. The TN guide requires three notes from "trades your team executed in WInS" ("Yes, we will verify this", TN L42, L53). The ticket forbids creating a decision just to get a note.	The lean to (iii) is not decision-ready until the team names its genuine third trade in advance. The plan's own rule supplies one: the 2027 leftover "waits in T-bills" (ticket section 8), so buying BIL or SGOV for the ~2% leftover is a real, plan-consistent trade with a liquidity role (PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK; \$25 commission). State the cost as well: under (iii) all three notes are promise or liquidity notes, which is weaker on "a cohesive portfolio strategy" (TN L14-15). The ticket leans (ii) and D10 leans (iii); the team should see both sets of reasons side by side, including this one, before it votes (gate box b).
C2	important	"Pass every WInS rule, whatever the position limit" overclaims. The script tests only no cap, 35% and 25%. Below about 24%, option (iii) would need six or more Treasury funds, and the ticket's fallback is undefined (the script's VGIT/VGLT step does not check the cap). Volumes for the Vanguard funds (VT, VGSH, VGIT, VGLT) are unchecked, and availability is still pending.	Say "passes every published rule under no cap, a 35% cap or a 25% cap; Vanguard volumes and any cap under 25% are unchecked". Add a one-line rule for a cap under 25% before the first order (e.g. more funds at cap-1, re-run script section 1).
C3	important	The page-fit numbers are optimistic. The model compares continuous text height with two full pages, but a line cannot split across a page break (24 whole lines a page on Letter, not 24.4). Word's "Double" for Times New Roman 12 may also be 27.6pt; D10 mentions this but does not put a number on it.	AX2 re-run with D10's own functions, whole lines (AX2 script, section 4): Letter, 5 paragraphs, 0pt spacing: median 93.8% (p95 102.1%) at 26.58pt and 97.8% (p95 106.5%) at 27.6pt . Letter, 7 paragraphs, 0pt, 27.6pt: median 100.0%, so it does not reliably fit . D9's independent script agrees: at 27.6pt there are 0-2 spare lines on Letter (about 525 IPS words at most). Wharton's own sample uses a 24pt line and fits with 8-10 spare. The conclusion holds and is stronger: the layout is tight, so set spacing to 0 and test the exported PDF. Set the word target from the team's real PDF (D9's <code>--draft</code> option lays out the team's own words); "about 480 words" may be too many on Letter with Word defaults.

#	Severity	Issue	Correction
C4	minor	"The most likely way to lose a semifinal place" ranks risks without evidence.	Say "a real, cheap-to-avoid disqualification risk".
C5	minor	"Worst order 0.19% of the volume cap".	The script's maximum is 0.17% (SPTL, 3,724 shares, in the middle path at a 35% cap); for option (iii) it is 0.14%. These figures use 30-day or prior-day volume, not the day's current volume. The margin is so large that the conclusion stands.
C6	minor	A4 range "89-99%".	The script's A4 medians run 89.5-101.0% : 9 paragraphs with 8pt spacing and a 0.5in indent overflows.
C7	minor	The numbers assume the 60/40 sleeve.	If D2's 50/50 is adopted, (ii) becomes VT 17.0 / VGSH 16.0. The Jan-2027 equity gap is then +17.0 (not +20.5), and the real 2028-2030 equity share is 17.0% (F-409), not 20.4%.
C8	minor	The chance the ladder costs more than \$300k is quoted as 24% only.	Quote the range "about 24-31%" (F-112: 24%; ticket/S4: 30.7%), as brief section 16 does ("roughly 1 in 4 to 1 in 3").
C9	minor	Submission-ready prose. Section 1.3 gives a quotable sentence ("WInS is her first deposit on the day it is invested").	Replace it with elements: the date; stand-in funds; no scaling.
C10	minor	The "earliest supporters" line is student-era (Daily Pennsylvanian, 2016-01-28; D13a-N01). It is about entrepreneurship, not about a WInS book.	Label it student-era. Do not use a Laura quote to argue for a book choice (tokenism risk; D13 rules).
C11	minor	R-AN15 ("Investments permitted (for BOTH contributions)") is not discussed under M005.	Cite it as checked. It concerns which instruments are permitted, not which date WInS shows, so E13 stands.
C12	minor	"Cash-flow matched, ratio stays at 100%" is true only for the real STRIPS ladder.	The WInS funds are duration-matched stand-ins with a twist error of \$1.5-2.8k (ticket section 3). Say so wherever the funded-ratio point is applied to WInS.
C13	minor	"A missing one can cost the whole scenario analysis its legitimacy" overstates.	Say "a response not authorised in the IPS in advance may read as a post-freeze change (Guide L158)".

Status after audit. The M005 answer stands with C2: both books comply under the caps tested, one IPS sentence must name the date, and the middle path is dropped. The lean to (iii) is **not decision-ready** until C1 is attached. The M013 keep/cut table stands, and every quote behind it was re-verified. For the page fit, the direction is confirmed but the numbers are downgraded to "optimistic by about 1.5-6 points" (C3).

D12 Overflow Researcher (client side): M247, M234, M231

Agent D12, insight_v1 run, written 2026-09-28. AI-generated research for the team: evidence, numbers, specifications and checklists only. **Nothing here is text to submit.** The team decides and writes every word (Wharton AI policy, R-W26/R-W27; any AI use goes in the Works Cited, brief section 14).

Scope applied (brief section 17): only the Trading Notes (TN, due Oct 23) and the IPS (due Nov 6). Two of my three questions were tagged "Final Report" (M231, M234). I kept only the parts that serve TN or the IPS and moved the Final Report parts to the "Later (after Nov 9)" list at the end. No relayed user messages were received during this task.

Scripts (run from the repo root; each has a docstring listing inputs and status labels):

- `.venv/bin/python research/insight_v1/scripts/D12_note_roles.py` (M247)
- `.venv/bin/python research/insight_v1/scripts/D12_evidence_grades.py` (M231)
- `.venv/bin/python research/insight_v1/scripts/D12_reader_test.py` (M234)

Terms used: **role word** = one of the four jobs the Guide says a Trading Note should name ("growth, liquidity, risk management, or future funding", Guide p.3). **Hedge** = the Treasury funds standing in for the ten \$50,000 payments. **Growth money** = everything the payments do not need. **Duration** = roughly the % a bond fund falls if rates rise by 1 percentage point. **Paraphrase test** = a reader reads a passage and says in their own words what it means; the writer notes misreadings and fixes the text.

Summary: top findings, ranked by impact on reaching the semifinals and on making the plan hers

1. **NEW (tier 1, TN): WInS notes may be capped at 300 characters, about 43 words, and Wharton's own example note is 413 characters.** Stock-Trak's HowTheMarketWorks product says of its Trade Notes "we allow up to 300 characters" (VERIFIED-PRIMARY for that product, read 2026-09-28). WInS runs on Stock-Trak (edu.stocktrak.com/wharton). Whether the WInS 2026-27 note box uses the same cap is **UNVERIFIED**, and no WInS page states a limit. Wharton's example note is 59 words and 413 characters (script). A 300-character cut would lose its last 27%, including the sentence that links the trade to the plan. The 2026-27 WInS User Guide says the note is entered on the order-review screen: "Enter a Trade Note (these are important!) to discuss how the purchase fits into your overall strategy" (p.8, VERIFIED-PRIMARY). **Decision:** plan every note to fit in **300 characters**. On the first order, check the note box for a counter or cut-off and record what it shows. The ticket's note checklists (5-6 elements each) do not fit in about 43 words, so split them. Four elements go in the note. The rest goes in the 100-word reflection. (Details under M247.)
2. **(Tier 1, TN) Three distinct role words are available only if WInS holds growth money (option ii).** Option (ii) supports three role words: the hedge is *future funding*, VT is *growth*, and VGSH is *risk management* of the facility money (with liquidity). In a 20% stock fall, VGSH makes the growth money lose 12.4% instead of 20%. Option (iii), the literal January-2027 book, supports only *future funding* (plus cash as *liquidity*), so its three notes would show one role three times. The Guide says "or", so three roles are not required. But the TN guide's first sentence asks the notes to show the team's "approach to growth, risk, liquidity, funding reliability, financial flexibility". This is a TN reason to prefer (ii), and the team makes the final choice. As far as the repo shows, no trade has been placed yet: today is day 1 and the ticket's gate is still open.
3. **(Tier 2, IPS) Grading every number shows that the promise is "bought at market prices" except for four inputs nobody can observe yet. The IPS must cover those four with rules, not numbers.** The four are:
 - the ladder's price in January 2027;
 - whether the 2028 deposit arrives in full and on time, which matters for the payments only if that price is above \$300k;
 - the U.S. Treasury paying;
 - the modelled ~24% chance that the price is above \$300k (internal only).

Only 3 of the 19 key numbers are safe to state in the IPS without the words "we assume" or "history suggests": the case's fixed \$50,000 payments, the deposits, and the fact that the ladder cost more than \$300k on 173 of 185 trading days in 2026. Laura's own published results table grades every row "Good", "Unsure" or "Bad". She rated 8 of 14 rows below "Good" and published them under "The [unfinished] results are below." (VERIFIED-PRIMARY). That habit is the model. It is a private motivation for the team, not a line for the IPS.

4. **(Tier 2, IPS) A paraphrase test of the 50-word pitch and the IPS rule sentences is worth about 2 student-hours.** Use 6 unpaid readers who only restate the text: 3 without a finance background and 3 with one. The U.S. plain-language guide says "Conduct between 6 to 9 interviews". Its case study found readers who said a letter was clear yet misread one expert term (VERIFIED-PRIMARY). Six readers catch a misreading that half of all readers would make 98% of the time, and one that 1 in 5 would make 74% of the time. They often miss rare misreadings (47% at 1 in 10). Unpaid help from parents, industry professionals or other adults is allowed (R-W21). Paid "agents" are not, and readers must never supply wording (own-words rule).
5. **A caution that is easy to miss:** Wharton's example note uses "reduce portfolio volatility" and "greater stability". The wins_now ticket bans those words for our hedge, and the numbers support the ban. The hedge falls about 10% if rates rise 1 percentage point: -6.8% of the whole WInS book under (ii), -9.7% under (iii). It "reduces risk" only when measured against the payments, whose value falls about 10.2% in the same move. If a note uses "risk management" for the hedge, it must say "against the payments".

M247 (tier 1: WInS now + TN). Can three notes from a mostly-Treasury book show distinct official roles, and have we placed a growth trade?

Canonical question (survivors.json): "Can our three Trading Notes, drawn from a mostly-Treasury book, show distinct official roles (future funding, growth, risk management), and have we placed a growth trade with a role-worded note yet?" Skeptics: judge and stats kept it; "answered" voted kill as already answered (ticket s4, chair memo s6, guardrails C.12). What is new here: the 300-character finding, a role word for VGSH that no other note uses, the numeric role test, and the (ii)/(iii) consequence. The rest was already settled by the ticket and by D7 (which dropped the planned rebalance trade), and I do not repeat it.

One-sentence answer: Yes under option (ii): the hedge is *future funding*, VT is *growth* and VGSH is *risk management* of the facility money. Under option (iii) only *future funding* is honestly available. No growth trade has been placed yet (repo evidence). Every note must fit its role word, Laura's need and the risk accepted into about 300 characters until WInS shows otherwise.

Evidence

Claim	Source (access date)	Status
Note should state its "expected role in growth, liquidity, risk management, or future funding"	competition/official/2026_27/2026_WGY_Investment_Competition_Guide.txt L84-86 (Guide p.3)	VERIFIED-REPO-FILE
The TN deliverable should show the "approach to growth, risk, liquidity, funding reliability, financial flexibility, and future cash-flow needs"; the notes "Collectively" show "a cohesive portfolio strategy"	2026_WGY_Trading_Notes_Analysis-FINAL.txt p.1 (R-T10)	VERIFIED-REPO-FILE
Reflection explains how the decision "supported, tested, or refined" the strategy ("or": no quota)	TN guide p.1; Guide p.3	VERIFIED-REPO-FILE
"Enter a Trade Note (these are important!) to discuss how the purchase fits into your overall strategy." (order-review step, before Confirm)	https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf p.8 (read 2026-09-28)	VERIFIED-PRIMARY
"...their "Trade Notes" - a short sentence mentioning why they are placing this trade (almost like a tweet, but we allow up to 300 characters)."	https://www.howthemarketworks.com/updates/trade-notes-justify-trades/ (Stock-Trak, Inc. property; undated; read 2026-09-28, phrase found verbatim)	VERIFIED-PRIMARY for HowTheMarketWorks; UNVERIFIED for WInS 2026-27
Students "cannot edit or delete" notes but "can add more notes to the same trade (each note has its own timestamp...)"	https://www.stocktrak.com/new-feature-trade-notes/ (2017 blog; read 2026-09-28)	VERIFIED-PRIMARY (vendor, not season-specific)

Claim	Source (access date)	Status
Earlier files said "no character limit is published"	phase_A/wins_week1_guardrails.md L49, L97-98; scripts/D9_draft_checker.py docstring	VERIFIED-REPO-FILE (now superseded by the row above as a planning cap)
Example note: 59 words, 413 characters	D12_note_roles.py on TN guide L36-39	VERIFIED-REPO-FILE (count)
Planned weights (ii) IEF 23.6 / TLH 42.4 / VT 20.5 / VGSH 12.5 / cash 1; (iii) IEF 35 / TLH 63 / cash 2	wins_now/securities_and_allocation_v0.md ticket	provisional ticket (team has not voted)
No executed WlnS trade recorded in the repo (only decisions/demo/); ticket gate boxes (a)-(e) still unticked	repo, 2026-09-28	VERIFIED-REPO-FILE (absence); the team's WlnS account is not visible to D12

Numbers (script `D12_note_roles.py`)

Number	Value	From
Example note length	59 words / 413 characters; 27% would be cut at 300	TN guide text, counted
300 characters in words	about 43 words (7.0 characters per word, spaces included)	example note's own ratio (ASSUMPTION that team notes are similar)
Role test, rates +1 percentage point (first-order)	IEF -6.9%, TLH -11.6%, VGSH -1.9%, VT 0 (ASSUMPTION), payments' value -10.2%	issuer durations 6.86/11.59/1.9y; spot payment duration 10.16y (brief s14)
Role test, world stocks -20% (ASSUMPTION)	VT -\$12,300; the growth money (VT+VGSH, \$99,000) -12.4% instead of -20%; payments unaffected	ticket weights, option (ii)
Whole book, rates +1pp	(ii) -\$20,312 (-6.8%); (iii) -\$29,108 (-9.7%)	same
Role words the positions can honestly carry	(ii) 3 of 4 (future funding, growth, liquidity/risk management); (iii) 2 of 4 (future funding, liquidity)	same
Days left	19 U.S. weekdays from Mon Sep 28 to Thu Oct 22; 15 to Fri Oct 16 (leaves a week for reflections)	calendar; Columbus Day Oct 12 counted as an ETF trading day (ASSUMPTION)

Implication

- **Decision (WlnS-now, before the first order):** the 300-character rule. Draft each note to 300 characters or fewer (count with `D9_draft_checker.py --kind note`, which prints characters). On the first order, record in the decision log whether the note box shows a counter or cuts text off. If WlnS shows a larger limit, the team may use it, but shorter still reads better. **Do not plan on a second "added" note to finish a sentence.** It carries a later timestamp, and the TN guide asks for each note "exactly as it appears in WlnS". How Wharton treats added notes is UNVERIFIED; ask through Contact Us before relying on one.
- **Specification (TN): four elements must be in each ≤300-character note** (the team writes the words):
 1. the action and instrument type;
 2. **one role word** from the Guide's list;
 3. Laura's specific need it serves;
 4. the main risk accepted, or the rule that governs the position.

Everything else in the ticket's checklists moves to the reflection: the ~\$292k sizing, "closer to her dates", "stand-in for the January 2027 ladder", and the provisional 60/40 split with its decision date (D7).

- **Role word per planned note (option ii):**

Note	Role word	What the note must let a reader check	Never say
Hedge (TLH, or SPTL if the position limit forces a split)	future funding	Its value moves with the cost of the ten fixed payments	"match", "mature in her payment years", "stable", "reduce volatility" (ticket)
VT	growth	Only money the promise does not need; a fall shrinks the facility amount, never the payments	a return forecast
VGSH	risk management (of the facility money), with liquidity as the second word	Keeps part of the growth money out of stocks, so a 20% stock fall costs that money about 12% instead of 20%, and that part stays close to its value (-1.9% per +1pp)	a floor percentage; "the 2031 floor" (ticket)

This gives VGSH a role word the other two notes do not use. D6 (line 155) suggested "credibility or flexibility" for VGSH; "risk management" is the Guide's own word and can be checked with a number. The team chooses.

- **Decision input for gate box (b), (ii) vs (iii):** under (iii), all three candidate notes are hedge pieces (IEF, TLH, SPTL) with the same role word. The growth argument then lives only in the IPS. This is a real TN cost of (iii), but not a rule breach ("or" in the Guide). If the team chooses (iii), the notes must differ by what they show (supported / tested / refined, per D7), not by role.
- **Status check:** no growth trade has been placed, and none should be until gate boxes (a)-(e) are ticked. The ticket's target (fills by Fri Oct 2 ET) leaves 14 of the 19 weekdays for anything that must be tested or refined before Oct 23.

Deliverables: WInS-now, TN. Deadline tier 1. Confidence: **high** on the role mapping and the (iii) consequence; **medium** on the 300-character cap (the vendor states it for a sister product; WInS itself is unconfirmed). Changes current strategy? **Not the strategy; yes the ticket:** the note checklists split into note (4 elements, ≤300 characters) and reflection; VGSH gets the role word "risk management"; (iii) now has a named TN cost. Criterion served most: **Portfolio Analysis** ("explain why each security belongs"), then Investment Strategy (consistency).

What this teaches: a job title is only honest if the position behaves like it. A 1-point rate rise and a 20% stock fall hit each of the three funds very differently, so each note's role word can be checked with a number.

M231 (tier 2: IPS; the Final Report column is a later item). Should every key number carry an evidence grade, as her own "Match Quality" column does?

Canonical question: "Should every key number in the Final Report carry an evidence grade (observed price / historical estimate / assumption, plus a quality word), as her own 'Match Quality' column does?" All three skeptics kept it (priority 3). Under brief s17 the Final Report column is a later item. The part in scope: **what grading does to the IPS.**

One-sentence answer: Yes, in a light form. Grade every number now in the team's decision log. In the IPS, say the grade in words ("at today's prices", "history suggests", "we assume"). The exercise shows that the ten-payment promise rests on market prices except for four inputs nobody can observe yet, and the IPS must cover each of those with a rule.

Evidence

Claim	Source (access date)	Status
Her page: "The [unfinished] results are below." and a column headed "Match Quality" graded Good/Unsure/Bad; counts 6 Good, 5 Unsure, 3 Bad of 14; her "Initial takeaways" explain why weaker matches happened	https://lauragao.com/rewriting-herstory (undated; student-era per the page's "Wharton freshman" remark, ASSUMPTION), read 2026-09-28	VERIFIED-PRIMARY (also D13a B9b-Q09)
Case: teams "identify the assumptions supporting their recommendation" and "identify and explain their assumptions about investment performance, the timing of cash flows, outside funding, the effect of inflation..."	case p.3 L98-99 (R-C56), p.4 L146-148 (R-C85)	VERIFIED-REPO-FILE
Portfolio Analysis criterion: "uses reasonable assumptions and projections"	SMAApply page L53 (R-S26)	VERIFIED-REPO-FILE
IPS: "Focus on the strategy and decision-making framework ... rather than ... presenting detailed financial calculations"; no charts, footnotes or formal citations	IPS guide p.2, p.3 (R-I55)	VERIFIED-REPO-FILE
D13 rules: no Laura quotes in TN or IPS; the Rewriting Herstory line is safe only with context	brief s16; D13a/D13c	VERIFIED-REPO-FILE
Ladder cost more than \$300k on 173 of 185 trading days of 2026; P(> \$300k on 2027-01-01) ~24%	brief s14	VERIFIED (days, from treasury.gov curves) / ASSUMPTION (probability)

Numbers (script `d12_evidence_grades.py`; four grades: CASE, PRICED, HISTORY, ASSUMED)

- 19 key numbers registered. The **promise** rests on 2 CASE, 6 PRICED, 1 HISTORY and 4 ASSUMED inputs. The four ASSUMED inputs:
 1. the January 2027 ladder price (not yet observable);
 2. the ~24% chance that it is above \$300k (a model);
 3. the U.S. Treasury paying (the stated residual risk);
 4. the 2028 deposit arriving in full and on time. This one matters for the payments only if the January 2027 price is above \$300k. - The **facility outcome** rests on 1 CASE, 1 PRICED, 3 HISTORY and 4 ASSUMED inputs (JPM 7.00% AC World forecast, equity share, model percentiles, deposit).
- Only **3 of 19** can go into the IPS as plain facts: the fixed \$50,000 × 10 payments, the \$300k + \$150k deposits, and "the ladder cost more than \$300k on most trading days this year" (173 of 185). Every other number needs its grade said in words or should stay out. That fits the IPS guide's "no detailed calculations" and the stakeholder map's "IPS: at most one rounded number" (SH-16).
- Laura's own grades: 57% of her published rows were below "Good", and she published them.

Implication

- **IPS: rules to fix before Nov 6** (the content a strong IPS passage must contain; the team writes it):
 1. The certainty definition is in words and names its grade: bought at market prices at purchase, not forecast.
 2. **Rule for input 1:** if the ladder costs more than \$300k in January 2027, buy the longest-dated payments first and finish the ladder from the 2028 deposit before anything else. This is the current strategy (brief s7); grading makes it compulsory to state.
 3. **Rule for input 4:** if the 2028 deposit is late or short, the unfinished payments come first and the growth money waits. The facility amount absorbs the shortfall; the payments never do. This matches brief s8 item 8. State it as a rule.
 4. **Input 3** is named once as the only residual risk ("barring U.S. Treasury default", brief s7).
 5. Any forecast-based statement (growth, the 2031 range method) carries a word that marks it as assumed.
- **Decision log now (cheap, feeds TN reflections and the later FR):** add a "grade" column (CASE / PRICED / HISTORY / ASSUMED) to every number the team records. A TN reflection may use at most one number (blueprint 01 s4). Prefer a PRICED one, such as "about \$292k at today's Treasury prices", and say that it is a price.
- **Do not** mention Laura's Match Quality column in the IPS or the notes. It would be decoration and a citation the IPS bans. It stays as the team's private reason for the habit.

Deliverables: IPS (and the decision log feeding TN). Tier 2. Confidence: **high** on the dependency finding; **medium** that judges reward the grading as such (no public scoring evidence). Changes current strategy? **No**. It makes two existing rules (inputs 1 and 4) mandatory IPS content. Criterion: **Portfolio Analysis** ("reasonable assumptions", R-S26), then Client Knowledge.

What this teaches: sorting numbers into "bought", "history" and "assumed" shows exactly where a promise can still break. That is where a rule is needed.

M234 (tier 2: IPS; the co-sponsor paragraph test is a later item). Should the team usability-test its wording with about five non-finance readers?

Canonical question: "Before submitting, should the team usability-test the co-sponsor paragraph and the IPS rules with about five non-finance readers and report what changed, as she tested products with nurses and doctors?" The stats skeptic voted kill ("trivial"; n=5 is anecdotal). That criticism is right about statistics but not about the purpose: the test *finds misreadings*; it does not measure a rate. Under brief s17 the co-sponsor paragraph and the FR Articulation report are later items. In scope: the **50-word pitch and the IPS rule sentences**.

One-sentence answer: Yes. Before Nov 6, run one paraphrase test of the pitch and the IPS rules with 6 unpaid readers (3 without finance, 3 with). Readers only restate, never reword. Fix what two or more misread, retest with fresh readers, and log it for the Final Report later.

Evidence

Claim	Source (access date)	Status
"Conduct between 6 to 9 interviews"; ask readers "to tell you in their own words what that section means... Do not correct the participant"; a Veterans Benefits Administration test where readers "all said that it was clear" yet read the term "service-connected disability" in different ways	https://digital.gov/guides/plain-language/test/paraphrase-testing (U.S. plain-language guide), read 2026-09-28	VERIFIED-PRIMARY
Typical share of problems found per test user "is 31%"; "testing no more than 5 users"; five users find "85%"; with distinct groups, "3 users from each category if testing three or more groups", "3-4 users from each category if testing two groups"	https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/ (Jakob Nielsen, 2000-03-18), read 2026-09-28	VERIFIED-PRIMARY
Her resume: "Collaborated with researchers at National Institute of Health for usability tests with nurses and doctors"; "Beta-tested MVP with 300K+ users..."	https://lauragao.com/s/Laura-Gao-Resume.pdf (D13a B9b-Q04/Q05)	VERIFIED-PRIMARY (self-reported claims)
"Teams may seek guidance from a parent, industry professional, or other adult serving as a secondary advisor... However, the use of paid advisors, education consultants, or other agents is prohibited."	Rules & Roles (R-W21)	VERIFIED-PRIMARY via case_register
Students must use their own voice and words; no invented teamwork stories	R-W26-R-W28; brief s14	VERIFIED-PRIMARY via case_register
IPS pitch "Clearly communicate the central idea"; "Keep the presentation simple and focus on communicating the investment strategy clearly"	IPS guide p.1, p.3	VERIFIED-REPO-FILE
Semifinal readers fear jargon; who they are is not published	stakeholder_map SH-16	ASSUMPTION/judgement

Numbers (script `D12_reader_test.py`)

- Share of problems found (Nielsen-Landauer, $L = 31\%$): 3 readers 67%, 5 readers 84%, 6 readers 89%, 9 readers 96%.
- One specific misreading that a share p of readers would make, seen by at least 1 / at least 2 of 6 readers: $p = 50\%$: 98% / 89%; $p = 31\%$: 89% / 60%; $p = 20\%$: 74% / 34%; $p = 10\%$: 47% / 11%. So the test reliably catches the big misreadings and often misses rare ones. Report "k of 6 read X as Y" as a problem found, never as a percentage of people.
- Cost: about 60 reader-minutes, about 2 student-hours per round (ASSUMPTION: 10 minutes per reader, 2 students per session).

Implication

- IPS: test protocol (checklist, no drafted text):**
 - [] Window: after the TN are in (Oct 23). Round 1 about Oct 26-30 on the team's own draft; round 2 about Nov 2-3 with **fresh** readers on the revised text. Submit by Nov 6, 5:00 p.m. ET (5:00 p.m. EST = 9:00 a.m. Sat Nov 7 AEDT; U.S. clocks change Nov 1).
 - [] Readers: 6 unpaid adults or students **not** in any competing team. 3 with no finance background (like most co-sponsor readers; Laura's degree is in statistics, case p.1 L16, and her career is product work and art, not finance) and 3 with some finance background (closer to semifinal readers, who are unknown). The two groups follow Nielsen's guidance for distinct groups. The advisor may be one reader ("sounding board", R-W20).
 - [] Readers restate only. They never suggest words. The team does not correct them during the session.
 - [] Questions to ask (open, not yes/no): (1) "In your own words, what is this plan trying to do for Laura?" (2) "If stocks fell by a third next year, what happens to the \$50,000 payments?" (intended answer: nothing) (3) "If interest rates fall before January 2027, what does the team do?" (intended: longest payments first, the rest from the 2028 money) (4) "What could still stop a payment?" (5) "What will Laura be able to tell co-sponsors in 2031, and how sure will it be?" (the method, not numbers) (6) "What might confuse someone else?" (plain-language guide's indirect question)
 - [] Terms most likely to be misread (they carry expert meanings like "service-connected"): "fully funded", "high degree of certainty", "locked in", "hedge", "duration", "floor", "operating reserve", "nominal", "growth money/surplus". Check each one used in the draft.
 - [] Change rule: change a sentence if 2 or more readers misread it the same way, or if any reader gets question 2, 3 or 4 wrong (the promise's rules must be unmistakable).

- [] Log it without personal data: date, reader type (finance / non-finance), no names, which sentence was misread, what the team changed. This is true process evidence for the FR later (R-S27), and nothing may be invented (R-W28).
- **TN (optional, one line):** before an order, one person outside the note's author restates the note's role in one sentence. This sits inside the ticket's existing gate box (d) ("read by a second student").
- **Why this is Laura's way, internally only:** her resume records usability testing with the actual users. Do not write that link into the IPS; it would be decoration.

Deliverables: IPS (and a small TN gate step). Tier 2. Confidence: **high** that the method finds big misreadings cheaply and is allowed; **low-medium** on how much semifinal readers reward the clarity (not observable). Changes current strategy? **No** (a process step). Criterion: **Creativity and Presentation** (clear, credible communication, R-S28), then Articulation (R-S27, later).

What this teaches: a reader saying "it's clear" is not proof. Asking them to say it back in their own words is. Six people are enough to find the big misunderstandings, and the fixes matter more than the count.

Later (after Nov 9), one line each (Final Report; not worked here)

- M231: a grade column (CASE / PRICED / HISTORY / ASSUMED) in the FR assumptions table. Possibly one context-safe mention of her "Match Quality" habit (student-era, D13 rules).
- M234: paraphrase-test the co-sponsor fundraising paragraph with non-finance readers. Report "k of n misread X, so we changed Y" in the Articulation section.

Sources (all accessed 2026-09-28 unless noted)

- Stock-Trak / HowTheMarketWorks: <https://www.howthemarketworks.com/updates/trade-notes-justify-trades/> (undated; "300 characters" found verbatim); <https://www.stocktrak.com/new-feature-trade-notes/> (2017); <https://www.stocktrak.com/faq/what-are-trading-notes/> ; <https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf> (p.8). (<https://education.howthemarketworks.com/trade-notes-justify-trades/> returned HTTP 404.)
- Laura's public professional record: <https://lauragao.com/rewriting-herstory> ; <https://lauragao.com/s/Laura-Gao-Resume.pdf> (via D13a, 2026-09-27).
- Plain-language testing: <https://digital.gov/guides/plain-language/test/paraphrase-testing> ; <https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/> (2000-03-18).
- Repo (VERIFIED-REPO-FILE): `competition/official/2026_27/2026_WGY_Investment_Competition_Guide.txt` (L84-86), `2026_WGY_Trading_Notes_Analysis-FINAL.txt`, `2026_WGY_Investment_Policy-FINAL.txt`, `Laura_Gao_2026_Client_Profile.txt` (via R-C56, R-C85); `research/insight_v1/phase_A/{case_register,wins_week1_guardrails,stakeholder_map}.md`; `research/insight_v1/wins_now/securities_and_allocation_v0.md`; `research/insight_v1/phase_D/{D7_wharton_intent,D6_behavioural,D13a_laura_quotes_verified,D13c_voice_map}.md`; `research/insight_v1/phase_C/survivors.json`.
- Search used to find URLs (not evidence): WebSearch, 2026-09-28.

What this teaches

1. **Read the box before you write in it.** The best-researched note fails if the software cuts it at 300 characters. The small practical check (is there a limit?) comes before the clever content.
2. **A role is a behaviour, not a label.** "Future funding", "growth" and "risk management" are believable because each fund reacts differently to the same shock, and a reader can check that with one number.
3. **Say what kind of number it is.** A price you can check today, a pattern from history and a forecast deserve different words. Sorting them shows where the plan still needs a rule.
4. **Test understanding, not approval.** People say "clear" to be polite. Asking them to repeat the idea in their own words shows what they actually took away.

Audit corrections (AY1)

Auditor AY1 (cluster auditor: client psychology and co-sponsors), 2026-09-28. The text above is left unchanged; where it conflicts with these corrections, the corrections win. Full audit: `research/insight_v1/`

phase_D/audit_client_cosponsors.md. Check script: research/insight_v1/scripts/AY1_audit_checks.py.

Reproduction. D12_note_roles.py, D12_evidence_grades.py and D12_reader_test.py re-run 2026-09-28: every number reproduces (59 words / 413 characters, 27%, 43 words; -6.9/-11.6/-1.9/-10.2%; -\$20,312 / -\$29,108; 12.4%; 19 and 15 weekdays; 6/5/3 of 14; 67/84/89/96%; 98/89, 89/60, 74/34, 47/11%). The detection figures were also recomputed by hand (binomial). **Sources re-opened 2026-09-28** (fetch helper, --grep found verbatim unless stated): HowTheMarketWorks ("we allow up to 300 characters"); Stock-Trak 2017 blog ("cannot edit or delete", "can add more notes to the same trade"; it also says professors can make notes "Required"; no length limit stated); Stock-Trak FAQ "What are Trading Notes?" (no limit stated); 2026-27 WInS User Guide ("Enter a T trade Note (these are important!) ..." found with a PDF kerning space, on PDF page 10; no character limit anywhere in the guide); digital.gov paraphrase testing (all four phrases); Nielsen 2000 ("31%"; both group recommendations); lauragao.com/rewriting-herstory ("The [unfinished] results are below.", "Match Quality"; recount 6 Good / 5 Unsure / 3 Bad). Guide L84-86, TN guide L1-5 and L44, R-W20/R-W21, VERIFIED-REPO-FILE.

#	Severity	Issue	Correction
C1	important	The 4-element note spec drops two things the official Guide says a note should capture (M247 "Specification (TN)"). Guide L84-86: the note "should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding" (VERIFIED-REPO-FILE). D12's four elements (action and type; role word; Laura's need; risk or rule) omit "supporting research or analysis" and treat alignment only through "the rule". The action and ticker are already in the WInS trade record.	Note elements, in this order of priority: (1) the reasoning: Laura's specific need; (2) alignment with the strategy (which part of the plan this trade is); (3) one piece of supporting research or analysis a reader can check (for the hedge, e.g. "rate sensitivity about 10 years, like her payments"); (4) the role word. The risk accepted moves to the reflection if space is short. Notes are permanent: fix this in the TN pack before the first order.
C2	important	Evidence for the 300-character cap is weak and partly contrary (summary 1, M247). The only source is a sister product (HowTheMarketWorks, VERIFIED-PRIMARY for that product). No Stock-Trak or WInS page states any limit (AY1 re-check). Wharton's own 413-character example is counter-evidence (Wharton presumably wrote an example that fits its own box), not only a "27% would be cut" risk.	Keep "draft to 300 characters" as a cheap precaution, labelled "UNVERIFIED for WInS; evidence mixed". Verify at zero risk <i>before</i> the first real note: ask Stock-Trak support (the blog points to Live Chat) or Wharton via Contact Us; or type a long test text on the order-review screen and do not press Confirm, then clear it. Never cut C1's Guide elements to fit a limit that has not been seen.
C3	important	Option (iii) does not give three hedge trades (M247 "Decision input for gate box (b)"). SPTL is bought only if the Session Rules cap is under 44% (ticket s3). With no cap, (iii) is two orders (IEF, TLH).	Under (iii) with no cap, a third executed trade must come from a genuine later decision (for example the Oct 19-23 duration refresh, which trades only if the hedge is outside 9.65-10.15). So the TN cost of (iii) is larger than "one role three times": it may not yield three notes without a planned decision. AX2 made the same point as blocking for D10.
C4	important	IPS rules omit brief s16's second certainty gap (M231 "IPS: rules to fix before Nov 6").	Add rule/item 6: name in one clause that "certain" is in nominal US\$, and a fixed \$50,000 buys less each year in Taiwan (named, not solved; the case fixes the payments in nominal terms).
C5	minor	Citation. "p.8" for the Trade Note sentence.	PDF page 10 (pypdf check). Content verified.
C6	minor	False precision: "Only 3 of 19 key numbers are safe in the IPS". The 19 is this file's own register, so the count describes the list, not the plan. "173 of 185 trading days" is a computed count (bootstrapped curve), as of late September; it changes as 2026 goes on.	Say "only the case's fixed figures and one dated price fact". If the fact goes into the IPS, recompute and date it in the week of Nov 2, and use words ("on most trading days of 2026 so far"). The finding about four unobservable inputs stands.
C7	minor	"That habit is the model" (Match Quality column). The column grades a text-matching algorithm's outputs in a student project (VERIFIED-PRIMARY); treating it as her habit for financial numbers is interpretation.	Label INT. It stays private, as the file already says (no use in TN or IPS; no tokenism).
C8	minor	Assumption labels in M234. Nielsen-Landauer's L = 31% is for software-usability problems; using it for paraphrase misreadings is an analogy. The binomial detection figures assume independent readers.	Label both ASSUMPTION. The protocol stands (the digital.gov method itself is VERIFIED-PRIMARY).
C9	minor	Time zone. "5:00 p.m. EST = 9:00 a.m. Sat Nov 7 AEDT".	True for NSW/VIC/ACT/TAS; 8:00 a.m. AEST in Queensland (the team's state is not recorded, ticket s4).

#	Severity	Issue	Correction
C10	minor	VGSH wording across files. D12 "risk management"; ticket "a different funding purpose ... could later become an amount Laura can promise"; D6 "credibility or flexibility"; AX2 D2-C6 endorses the ticket.	Compatible if the note uses one Guide role word ("risk management" or "liquidity") and never implies VGSH is the 2031 floor (the floor is bought from the whole growth money). The 12.4%-vs-20% number is the ordinary effect of holding about 38% Treasuries in the growth money, not a special VGSH property; present it that way. Team chooses and records.

FYI for the main loop (not a D12 error). The 2026-27 WInS User Guide defines "Position Limit: This is how much of your portfolio you can invest in one single stock." (VERIFIED-PRIMARY, re-read). Whether it applies to ETFs is not stated; the ticket's capped-hedge branch already covers both cases, but the Session Rules screenshot should record whether ETFs are included.

Status after audit. Finding 1 downgraded from "WInS notes may be capped at 300 characters" to "a sister product caps at 300; WInS unverified; evidence mixed; check before the first note" (C2), and its element split is replaced by C1. Finding 2 stands with C3 and C10. Finding 3 stands (the four unobservable inputs are right), with C4 and C6. Finding 4 stands, with C8. Finding 5 stands (the hedge is "risk management" only against the payments).

AX1 audit: rates and quant (D3 model v2 and the nine D3 answers)

Agent AX1 (cluster auditor, rates and quant), insight_v1 run, 2026-09-28. This is an adversarial check of `research/insight_v1/phase_D/D3_quant.md` and `research/insight_v1/phase_D/D3_model_v2_results.md` before they reach the strategy team. It is AI-generated research (findings, numbers, checklists), **not text to submit**. The six students decide and write every word, and record this AI use in the Final Report's Works Cited (Wharton AI policy).

- **Scope:** brief section 17, Trading Notes (TN, Oct 23) and IPS (Nov 6) only.
- **Relayed messages:** none arrived during this audit.

What this task delivers

- This file.
- An "## Audit corrections (AX1)" section at the end of each D3 file: `D3_quant.md` has C1-C25 and `D3_model_v2_results.md` has V1-V10. D3's own text is unchanged.
- One check script: `.venv/bin/python research/insight_v1/scripts/AX1_quant_audit.py`. It takes about 15 seconds, imports D3's model code unchanged, and reproduces every number AX1 adds, in sections [1]-[9].

Nothing was committed.

Status labels (brief section 3):

- VP = VERIFIED-PRIMARY.
- VRF = VERIFIED-REPO-FILE.
- MODEL = an ASSUMPTION-based model output, not a forecast.
- ASSUMPTION = a modelling choice.
- INTERPRETATION = our reading, not a fact.

Terms used:

- **Sleeve / growth money:** the money outside the ten-payment ladder.
- **Floor:** the part of the sleeve moved into a 2-year Treasury in 2031.
- **p5/p50/p95:** 5 in 100 simulated futures end below p5. p50 is the median, the middle outcome.
- **Mean:** the average over all futures. It is what "expected" means.
- **All-Treasury benchmark:** D3's "riskless control". The surplus is locked in Treasuries maturing in 2033 instead of being invested.
- **Barbell:** lock part of the growth money in January 2028 and put the rest 100% in stocks.
- **Cap:** a rule that Laura never gives more than the announced top of the range.

Verdict (short)

D3 passes with corrections: 0 blocking, 8 distinct important issues (10 entries across the two files), 25 minor.

What holds:

- **Arithmetic:** all 10 D3 scripts reproduce every quoted number exactly. The v2 model matches the verified model path by path (largest difference \$0.54).
- **Sources:** every source AX1 re-opened says what D3 quotes.
- **Decisions that survive:**
 - lock all ten payments (M243);
 - the contingency clause: longest-dated payments first, and the 2028 deposit's first use completes the ladder (M053);
 - one dated floor lock in 2031, with no ratchet (M055; kept on simplicity grounds);
 - lock share 80-90% (M060);
 - judge the stocks against an all-Treasury benchmark, not a stock index (M034).

What changes:

1. The recommended 2031/2033 contribution rule ("give 90%, capped") promises less than is bought. D3's own table has a rule with the same money and flexibility that announces a bottom about \$17k higher.
2. "100% within the range" is a rule, not a test result.
3. "Stocks buy range, not a bigger expected facility" confuses median and mean. The expected gain under JPM is about \$7-9k, not \$3k.
4. The "riskless control" is not riskless seen from today.
5. The barbell is **not** dominated. When the floor is bought (2028 or 2031) is a genuine team choice.
6. "Demonstrably tracks" overstates a model backtest.
7. "All ten payments bought in January 2027: robust" contradicts D3's own 1-in-3 finding.
8. Much of D3 is Final Report material, which brief s17 moves to the later list.

1. Scripts re-run (2026-09-28, from the repo root)

Script	Result
strategy_mc_v2.py	Reproduces sections 2-4 exactly: the reproduction check, central case \$152k/\$204k/\$277k, the tornado rows, (a)-(g). Path-by-path match with research/verified_2026-09-27/strategy_mc.py: lock-early \$0.54 max difference, growth-first \$0.00 (AX1 [1])
D3_control_and_equity.py	All M034 tables and the 13.6/17.0/20.4/23.8% whole-portfolio equity shares reproduce
D3_lock_frontier.py	All M243 cells reproduce (e.g. deposit missing: 95/90/80/70% locks short 2.34/12.75/25.78/31.57%)
D3_range_rules.py	All M039/M060 tables reproduce (floor share 50-100%; bad/middle/good 2031 announcements; P(within) 61.9-92.6%)
D3_barbell.py	All M238 rows reproduce, including history (plan worst \$105k; barbell 80% \$176k)
D3_joint_tail.py	The conditional-dollar table, probabilities (30.2%; 8.2%; 30.17%; 44.4%/69.0%), rule check and Damodaran/FRED evidence all reproduce
D3_ratchet.py	Reproduces the premise (-22%), the designs and the dominance test (42% equity: \$169k/\$204k/\$250k; step-down = 51%)
D3_history_stress.py	Crash frequencies (0.29% / 1.16% / 3.1%), named episodes and window tables reproduce
D3_funded_status_log.py --backtest and --date 09/25/2026 --since 09/01/2026	±\$827 / 0.42%; payments' value -5.5% vs hedge -4.7%; \$288,924, 10.16y, \$293/bp; funded ratio 1.019; since 1 Sep -3.18% vs -2.85% (+\$648)
D3_data_snapshot.py --offline	Re-parses: Damodaran 98 years to 2025; 185 curve dates; FRED DGS2/5/10 to 2026-09-24
AX1_quant_audit.py (new)	[1] path match; [2] mean vs median equity lift; [3] benchmark range; [4] barbell vs the plan's equity frontier; [5] 2026 days the ladder fit under \$300k; [6] FRED 98-day fall base rates and the 2008 worst case; [7] gap in cost vs face dollars; [8] contribution rules at equal flexibility; [9] unbought face by buying order

2. Sources re-opened (the claims that matter most for decisions)

#	Claim in D3	Re-opened	Result
1	Par curve 2026-09-25: 2y 4.81, 5y 4.98, 10y 5.17, 20y 5.54	treasury.gov 2026 CSV, curl 2026-09-28	Confirmed (VP). The live 09/25/2026 row equals competition/official_market_data/daily-treasury-rates_2026-09.csv (VRF). No newer curve yet
2	10y rate vol 70.6bp a year in 2026; 0.48pp sd of 98-day moves since 1990	FRED DGS10 snapshot, recomputed; live spot checks (2008-12-31 2.25; 2026-09-24 5.18)	Confirmed (VP inputs, derived): 70.58bp; 0.477pp
3	10y yield: 2002 -124bp, 2008 -179bp (4.04 → 2.25), 2020 -99bp, 2022 +236bp	FRED DGS10	Confirmed (VP, derived)

#	Claim in D3	Re-opened	Result
4	USD/TWD 31.82 (2026-09-18); two-year sd 6.8% since 2006	FRED DEXTAUS, curl 2026-09-28, recomputed	Confirmed: 31.82; 6.74% since 2006, 9.41% since 1983 (VP, derived; F-513)
5	Damodaran histretSP "page dated January 2026"; 2008 S&P -36.55%, 10y bond +20.10%; 2022 -18.04% / -17.83%	https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html , fetch helper 2026-09-28	Confirmed (VP, dataset page). The page title still reads "1928-2024", but rows run to 2025 (2025 S&P +17.78%). The snapshot equals the live rows
6	Vanguard U.S. equities "declined from a range of 4.9%-6.9% to a range of 4.2%-6.2%" (run of 30 June 2026)	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html , --grep, 2026-09-28	Verbatim (VP). New context: developed ex-U.S. "4.5%-6.5%", emerging markets "2%-4%". So a VT-weighted Vanguard midpoint is about 5.1% (D2's blend: 5.08%). D3's 5.2% U.S. figure is slightly generous for a world fund
7	"the VCMM may be underestimating extreme negative scenarios unobserved in the historical period"	same page, --grep	Verbatim (VP)
8	Brunel goal-probability table: Needs 90-95, Wants 80-85, Wishes 65-75, Dreams 50-60	https://srdas.github.io/Papers/GBWM.pdf , fetch helper	Confirmed (VP): Table 1, <i>Journal of Investment Management</i> 16(3), 2018. It is one practitioner scale, not a rule
9	JPM 2026 LTCMA: U.S. large cap 6.70/7.94/16.47; AC World 7.00/8.28/16.78; intermediate Treasuries 4.00/4.06/3.48; short government/credit 4.00/4.01/1.63; correlations -0.01 / 0.00 / 0.16 / 0.21	competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf p.2 (pypdf)	Confirmed (VRF)
10	Case lines: "will contribute an additional \$150,000" (L43-45); "thoughtful risks ... appropriate balance" (L69-74); "high degree of certainty" and the definition task (L91, L97-99); "must decide how much" and "financial flexibility" (L101-103); range, confidence and "favorable and unfavorable market outcomes" (L110-120); WlnS gains not added (L129-133); statistics degree (L16)	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt	Confirmed (VRF). Note L118: the range must come with an explanation of how good and bad markets change "the amount she can provide". This matters for the cap (C2-C3)
11	IPS guide: "planned changes in the portfolio as future funding needs approach" (L46-47); "Focus on the strategy and decision-making framework" (L51-52); no final reserve, projections or range in the IPS (L68-70)	2026_WGY_Investment_Policy-FINAL.txt	Confirmed (VRF)
12	Guide p.3: "A long-term investment strategy cannot be judged solely by what happens in the market over a few weeks." (L79); Trading Note "should capture the reasoning behind the decision ..." (L84-86); TN guide reflection elements (L46-52)	2026_WGY_Investment_Competition_Guide.txt , 2026_WGY_Trading_Notes_Analysis-FINAL.txt	Confirmed (VRF)
13	Laura's words used: B9b-Q02 ("A/B testing", résumé), B8b-Q18 ("unstable income", 2021), B9b-Q09 ("[unfinished]")	phase_D/D13a_laura_quotes_verified.json	All three are VERIFIED-PRIMARY in D13a (AX1 did not re-fetch them). The <i>readings</i> D3 attaches to two of them are over-read (C19). All three are Final-Report-only, so out of scope now
14	IEF/TLH hedge weights 35.7/64.3	Ticket: issuer durations 6.86y / 11.59y (VP per S3)	Arithmetic confirmed: (11.59 - 9.90) / (11.59 - 6.86) = 35.7%. The iShares pages were not re-opened by AX1 (S3 re-read them on 2026-09-27)

Nothing had to be downgraded for a failed source check. Downgrades are about interpretation (section 3).

3. Does each implication follow from its evidence and the case? (the case wins)

Question	D3 conclusion	AX1 finding
M034, stocks vs benchmark	60% equity "buys range, not a bigger expected facility"; +\$3k median	Partly. The numbers are medians. The mean lift is +\$6.5-7.0k (JPM), +\$9.4k (bonds 4.9%) and -\$0.4k (Vanguard) (AX1 [2]). The +\$3.3k nets two effects. Equity adds +\$12.5-12.9k at the median over the same plan with 0% equity. The plan's non-stock money then trails locking by about \$9k, because JPM's bond assumption (4.0%) sits below today's ~5.1% lock yields (F-317). The benchmark is not riskless from today: central case \$170k/\$201k/\$233k (AX1 [3]). The direction stands (stocks add a modest gain and a lot of spread); the wording changes (C4-C5).
M028, WlnS hedge log	"Demonstrably tracks" within $\pm 0.42\%$	A model backtest: fixed holdings valued on the same curve as the payments, and 155 overlapping windows from one year. Keep it as "about $\pm 0.4\%$ in a model backtest". It is optional for a Trading Note and must not read as performance (C7).
M238, barbell	Reject: "larger barbells are just lower-equity points on the same frontier"	Does not follow. At the same 2028 equity share, barbell 80% beats the plan at every percentile, under both 4.0% and 5.0% bond assumptions. Its worst historical window is \$176k against the plan's \$105k (AX1 [4]). The "\$101k vs \$165k" comparison mixes lock shares (50% vs 80%). "No barbell" can stand only as a judgement: the barbell holds 100% stocks after Laura has spoken in 2031, and the plan keeps more growth in 2028-30. Lock timing becomes a team choice for the IPS (C6).
M243, full lock	The certainty is bought and cheap; partial locks fail only without the deposit	Follows. Minor point: the 2033 price convention (\$6.4k dear, F-408) understates the cost of certainty; on forwards it is about \$0.9-5.2k, not \$0.6-3.3k (C16). The "~30%" is also supported by history. The real ladder fit under \$300k on only 9 of 185 days in 2026, and 98-day 10-year falls of $\geq 19\text{bp}$ occurred in 35.8% of windows since 1990 (AX1 [5]-[6]) (C8).
M039, range as a policy	Adopt "floor bought; give ~90%; never above the top"	Logic right, rule choice wrong. "100% within" is by construction (C2). "Give 90%, capped" promises less than is bought. The uncapped "bought floor + half of the rest" gives the same contribution and flexibility with a bottom about \$17k higher (\$165k vs \$148k median; AX1 [8]). It also answers case L118 and matches D5 (C3).
M060, lock share	80-90%; 100% is the "one exact amount" the case rules out	Follows for 80-90%. "The case says Laura does not want one exact amount" is an INTERPRETATION: her reason is credibility. At 90% with the cap/share rule, the range is only about \$21k wide. The NT\$ claim holds only at 90% (C13).
M053, joint tail	Conditional dollars; longest-first "settles" the order	Follows , with small fixes. The shortfall size rises with the recession link (\$9.1k $\rightarrow$ \$11.3k). \$25k $\rightarrow$ about \$26k. "Settles" $\rightarrow$ "supports": longest-first leaves part of the <i>first</i> payment exposed, but less face (\$12.0k vs \$19.0k at -50bp; AX1 [9]) and less price risk (C9-C10).
M055, ratchet	Reject; a constant 42% "gets the same bad case"	Follows on simplicity grounds only. At the same p5, the ratchet is \$2k better at the median and \$6k better at p95, so it is not dominated (C11).
M059, history	Named rows; the model understates crash years "about tenfold"	Numbers reproduce. "Tenfold" rests on 3 bad years and becomes about 2x at a -20% cut-off (C12). Final Report material: later list.
v2 section 5	"All ten payments bought in January 2027 at a known price: Robust"	Contradicts v2 section 4(a). It holds in about 2 of 3 rate paths; otherwise completion waits for January 2028 (V1). "\$43k robust" becomes about \$51k at the 2008 worst 98-day fall of -174bp (V4).

4. Contradictions

Between specialists

1. D3 vs D5 on the cap.

- D3 M039 recommends a capped "give 90%" rule.
- D5 finding 5 says "Do not cap the gift at the top", because a 100% in-range figure is empty. D5 audit C11 also flagged this.
- AX1 adds a number that settles most of it. At equal flexibility, D5's uncapped "floor + share of the rest" rule announces a higher bought bottom (AX1 [8]).
- Resolution: one IPS method with two parameters (lock share; give-back share), uncapped by default. The capped variant is offered only with P(top reached).

2. D3 vs D2 on sleeve equity.

- D2 recommends 50% (band 45-55) as an IPS rule.
- D3 says 40-60% are all defensible, and the evidence "does not force a move from 60%".
- The two files' numbers agree (C23). One team decision is needed before the VT order.

3. Top percentile. D3 uses p80, D5 p85, D6 p90 (D5 audit C10). Pick one.

4. D3 vs D1 on the WlnS bond slot.

- D3 M238: "VGSH, not IBTM".
 - D1: a dated Nov-2032 Treasury rung could replace the VGSH note.
 - These are different roles (growth money's bonds vs the promise), but the untested question "VGSH or a dated bond for the growth money's bonds?" should go to the team once (C18).
5. **D3 vs D1 on the odds.** The two agree (D1 30.5%, D3 30.2% for the real ladder). AX1b and AX1 both say "about 1 in 3".

With brief section 14 (facts)

- No factual contradiction.
- D3 updates two brief figures, correctly:
 - P(ladder > \$300k) goes from ~24% (exact-date ladder) to ~30% (real ladder);
 - the fee effect is \$16.6k / \$32.4k (brief: ~\$17k / ~\$34k).
- D3's 20.4% equity share in 2028 matches brief s16 ("~20% in 2028-30").
- Brief s16 asks the plan to answer why whole-portfolio equity is so low. D3's corrected M034 wording (a modest expected gain, a lot of spread) is the honest input to that.

With the wins_now ticket

- **Which ticket:** checked against v0 (as assigned) and against `wins_now/securities_and_allocation_v1.md`, which another agent wrote in parallel and which now supersedes v0 for trading. v1 already adopts three things:
 - both splits (60/40: VT 20.5 / VGSH 12.5; 50/50: VT 17.0 / VGSH 16.0);
 - "about 1 in 3" for the ladder odds;
 - T-bills for the 2027 leftover.
 - **VT/VGSH weights** at 60%: D3 has 19.8/13.2% (cash outside the growth money); both tickets have 20.5/12.5% with 1% cash taken from the short-Treasury part (C17).
- **The 2027 leftover:** the model invests it 60/40, while the ticket holds it in T-bills until 2028 (effect under \$1k).
- **No ticket rule is contradicted.** D3 respects the ticket's "never claim 'stable'/'reduced volatility'" and "never say 'match'" rules. D3 uses "tracks" and "stand-in", not "matched".

5. Overclaims, jargon, false precision, complexity, privacy, tokenism, submission prose

- **Overclaims** (each corrected in the appended lists):
 - "100% of modelled futures ... in every test" (C2);
 - "not a bigger expected facility" (C4);
 - "riskless control" and "best sure thing" (C5);
 - "just lower-equity points on the same frontier" (C6);
 - "demonstrably tracks" (C7);
 - "Robust: all ten bought in January 2027" (V1);
 - "\$43k ... robust" (V4);
 - "settles" (C10);
 - "certain by 2033" without "nominal US\$, barring default" (C20);
 - "about tenfold" (C12).
- **"Guaranteed" and "matched" for funds:** none found. Good.
- **Jargon:**
 - undefined in `D3_quant.md`: i.i.d., lognormal, Student-t/nu, multivariate, dominance test, forward zero, bootstrap;
 - undefined in v2: "tornado table" (C22, V10).
- **False precision:** dollar-exact model outputs (\$203,997; ±\$827; 30.2%; 83.7%). Round in anything the team drafts from (C22).
- **Unearned complexity:**
 - the weekly six-column funded-status log is optional (C7);

- the M039 rule stacks four parameters (lock share, top percentile, give share, cap). D5's two-parameter uncapped rule delivers the same money (C3). The IPS has 500 words.
- **Privacy:** no problems found. No family, home or personal details; no student surnames.
- **Tokenism:** two over-read single words from Laura's public record ("A/B testing"; "[unfinished]"), and a persuasion use of her degree (C19). All are Final-Report-only and out of scope now.
- **Submission-shaped prose:** two fill-in sentences (C21). Otherwise D3 keeps to "must contain" specifications.

6. What goes forward to the strategy team (brief s17: TN or IPS only)

TN (Oct 23)

- **Before the VT order, fix the sleeve equity share** (D2: 50%; D3: 60% is defensible). The VT note records it. Notes cannot be edited.
- **Optional: a hedge-tracking number.** If the team keeps a weekly log, one reflection may cite one pair of numbers: the payments' value change (model, the team's calculation) and the WInS hedge change (screen). It must be labelled and must never be framed as a gain. PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK applies to IEF, TLH and VT as in the ticket.

IPS (Nov 6): rules to fix before the freeze

1. **Full lock.** Buy all ten payments on arrival of the first deposit. The reason: prices today, not a forecast. The cost: a few thousand dollars of median facility money under JPM, about zero under Vanguard. It protects against a smaller or late 2028 income. Never say a partial lock "fails" in the base case.
2. **Contingency clause.** Buy the longest-dated payments first. The first use of the 2028 deposit completes the ladder. No facility floor is announced until all ten payments are bought. About 1 in 3 rate paths need this (model and 2026 history).
3. **Lock timing and share.** One dated lock of 80-90% of the growth money into a 2-year Treasury in January 2031. **Open team choice (new):** January 2031 (plan) or January 2028 (barbell-type). Both are defensible (C6).
4. **Contribution rule (method, no dollar figures).** The bottom is the amount already bought. The gift is the bought floor plus a share of what the rest becomes. Choose uncapped (two-sided confidence) or capped (report only P(top reached)). Pick one top percentile. Laura decides the use of any money above the top (C2, C3, C14, C15).
5. **Stocks.** The "why any stocks" sentence must contain four things: that the all-Treasury alternative exists; a modest expected gain plus upside; a lower bad case; and that the payments never depend on stocks (C4). Say what any fee is charged on. A fee on the ladder is a claim on the promise money (v2 (c)).
6. **No ratchet.** One lock date only (C11).

Later (after Nov 9), one line each:

- M034 benchmark fan chart;
- M028 two-line chart;
- M243 price-of-certainty chart;
- M060 funnel chart and NT\$ range;
- M238/M055 alternatives-considered sentences;
- M039 co-sponsor checklist and three-figure 2031 ranges;
- M053 risk-register row;
- M059 named-history rows;
- v2 purchasing-power line;
- the three Laura-quote uses.

7. Corrections list (full wording in the appended sections)

D3_quant.md:

- **Important (7):**

- C1 scope;
- C2 "100% within" is a rule;
- C3 "give 90%, capped" promises less than is bought;
- C4 median is not expected;
- C5 the benchmark is not riskless;
- C6 the barbell is not dominated;
- C7 "demonstrably tracks" overstates a model backtest.
- **Minor (18):** C8-C25. They cover history support for the 1-in-3 odds, cost vs face, shortfall size, buying-order trade-off, ratchet not dominated, "tenfold", M060 interpretation and width, lopsided range, excess wording, 2033 price convention, ticket weights, VGSH vs dated bond, Laura-word readings, "certain" wording, fill-in sentences, jargon and precision, D2 harmonisation, Vanguard U.S.-only midpoint, and the STRIPS model price.

D3_model_v2_results.md:

- **Important (3):**
 - V1 "all ten bought in January 2027: robust" contradicts section 4(a);
 - V2 capped rules shown as "P(within) 100%";
 - V3 riskless control and median-only fee claim.
- **Minor (7):** V4 \$43k → about \$51k at the 2008 worst; V5 "about 1 in 3"; V6 robustness ranges; V7 purchasing power; V8 twist error not computed; V9 scope header; V10 jargon.

Sources (accessed 2026-09-28)

- **Official (VRF):**
 - [competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt](#) (L16, L43-45, L69-74, L88-120, L129-133, L146-149);
 - [2026_WGY_Investment_Policy-FINAL.txt](#) (L45-52, L68-70);
 - [2026_WGY_Investment_Competition_Guide.txt](#) (L17, L70, L79, L84);
 - [2026_WGY_Trading_Notes_Analysis-FINAL.txt](#) (L42-53).
- **Market data (VRF):**
 - [competition/official_market_data/daily-treasury-rates_2026-09.csv](#);
 - [JPM_LTCMA_2026_US_matrix_USD.pdf](#) p.2.
- **Primary web (VP):**
 - U.S. Treasury Daily Par Yield Curve 2026: https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv/2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&format=csv
 - FRED DGS10: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10>
 - FRED DEXTAUS: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DEXTAUS>
 - Damodaran histretSP: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html
 - Vanguard VCMM forecasts (run of 30 June 2026): <https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html>
 - Das, Ostrov, Radhakrishnan & Srivastav (2018), *Journal of Investment Management* 16(3), Table 1: <https://srdas.github.io/Papers/GBWM.pdf>
- **Repo research (lower authority):**
 - [research/verified_2026-09-27/strategy_mc.py](#);
 - [research/insight_v1/phase_A/fact_register.md](#) (F-012, F-014, F-106, F-112, F-317, F-408, F-508, F-513);
 - [wins_now/securities_and_allocation_v0.md](#);
 - [phase_D/D1_rates.md](#) and [audit_rates_D1.md](#);
 - [phase_D/D2_equity_ai.md](#), [D5_cosponsors.md](#), [D6_behavioural.md](#);
 - [phase_D/D13a_laura_quotes_verified.json](#).

What this teaches

1. **A number can be exactly right and still described wrongly.** Every D3 figure reproduced. The errors were in the words around them: "expected" for a median, "riskless" for something that is riskless only after it is bought, "robust" for something true in 2 of 3 futures.
2. **When a table says 100%, ask whether a rule made it 100%.** A cap makes "within the range" true by definition. The honest number is the one the rule does not control: the chance of reaching the top.
3. **To call an option "dominated", line the options up at the same risk.** Compared at the same stock exposure, the barbell was not worse. It was a different bet on *when* to take risk. That is a choice for the team, not a mistake to reject.
4. **Two rules that deliver the same money can promise different things.** "Give 90%, capped" and "floor plus half of the rest" leave Laura the same flexibility. Only one of them lets her announce everything she has already bought.
5. **Check a model's odds against history.** The model's "about 30%" chance that the ladder costs more than \$300k looked like a guess until 2026 itself showed the ladder above \$300k on 176 of 185 days.

AX1b audit: rates and fixed income (D1 rerun)

Agent AX1b (cluster auditor, rates), insight_v1 run, 2026-09-28. This is an adversarial check of research/insight_v1/phase_D/D1_rates.md (M044, the dated WInS rung, and M004, the January 2027 purchase rule) before it reaches the strategy team. It is AI-generated research: findings, numbers and checklists, not text to submit. The team decides and writes every word. Status labels follow brief section 3 (VP = VERIFIED-PRIMARY, VRF = VERIFIED-REPO-FILE, SNIP = SNIPPET-UNVERIFIED, ASM = ASSUMPTION, DERIVED = computed by a script from labelled inputs). Scope (brief section 17): Trading Notes (TN) and IPS only. No relayed messages arrived during this audit.

What this task delivers

- This file.
- An "## Audit corrections (AX1b)" section appended to the end of D1_rates.md, with 24 items. D1's own text is unchanged.
- One check script: .venv/bin/python research/insight_v1/scripts/AX1b_rates_audit.py. It imports D1's curve code unchanged and reproduces every number AX1b added.

Nothing was committed.

Terms: **gap** = how much more the ten-payment ladder costs than the \$300,000 first deposit on 1 Jan 2027. **Funded ratio** = money available ÷ cost of the ladder. **p5** = the level that 1 in 20 model outcomes fall below. **Drift** = the direction a model assumes prices move on average. **Fat tails** = big moves happen more often than a bell-curve model says.

Verdict (short)

D1 passes with corrections: 0 blocking, 9 important, 13 minor, plus 1 addition and 1 cross-link.

What holds:

- Both scripts reproduce exactly.
- Every web source re-opened says what D1 quotes.
- All four decisions survive: buy the whole ladder on arrival; buy the longest-dated payments first; finish the ladder before any growth money; buy one conditional dated rung in WInS.

What changes:

- **Headline odds:** the real ladder's chance of a gap is about **1 in 3**, not "24-31%".
- **Tail claims:** "0.01%" and the IPS wording "at most part of the first payment" are thin-tail overclaims.
- **Date promise:** "1 January 2028 at the latest" contradicts D1's own late-deposit rule.
- **Guide p.5 reason:** D1 uses it to reject staging, but it does not apply. This is the same error AX2 found in D2.
- **2019 Stock-Trak list:** it cannot tell us anything about 2032-2035 maturities. Those Treasuries did not exist yet.
- **IBTM fallback:** one Trading Note checklist line becomes false if IBTM replaces the bond.
- **Third note:** D1's suggestion to replace VGSH conflicts with D12 unless it is made conditional on option (ii) vs (iii).

1. Scripts re-run (2026-09-28)

Script	Result
D1_purchase_rule.py	Reproduces every quoted figure: ladder costs at each shift (exact-date and Nov-15); break-evens (-19/-26, -148/-153, -421/-423bp); 173/185 and 176/185 days; worst day (Feb 27) \$27,631 / \$28,549 / 19.0%; P(gap) 24.3% / 30.5%; p90/p95/p99 \$8.7k / \$12.9k / \$20.9k; longest-first vs nearest-first (+\$1,265 / +\$3,831); staging 1.0211 / 27.3% / 0.966 and trigger 1.0218 / 26.5% / 0.912; joint tail \$11,957 / \$31,413, +\$221 / +\$531 if late; 2026-09-10 \$299,990. Only the "71%" worst-day share is off (it is 68.7%). The script's curve check prints "09/25 row equals repo file: True".
D1_wins_rung.py	Reproduces every figure: bond prices, accrued interest, yields and durations; all 7 option rows; ETF-only \$1,544 / \$451; the +191 to -104 changes; 13 coupons of \$1,031.25; STRIP \$37,253. Caveat: the STRIP value is a 1 Jan 2027 forward, while the bond price is for 25 Sep 2026 (see minor item 13).

Script	Result
AX1b_rates_audit.py (new)	[A] worst-day share 68.7%; [B] odds sensitivity; [C] FRED base rates; [D] staging under a different drift; [E] unfunded face by buying order; [F] STRIP on the same date as the bond; [G] accrued interest by trade date; [H] short rates at -421bp.

Input check: AX1b downloaded the treasury.gov 2026 daily par-curve CSV again on 2026-09-28. After line-ending normalisation it is **byte-identical** to `scripts/data/D3/treasury_par_2026_raw.csv` (VP). The rows D1 relies on read:

- 02/27 10-year 3.97;
- 09/10 10-year 4.95;
- 09/25 10-year 5.17.

2. Sources re-opened (fetch helper, 2026-09-28)

#	Claim in D1	Source	Result
1	Stock-Trak master bond list; dated 2019; symbol format; 4.375% Nov-2039 and 4.25% Nov-2040 bonds; nothing maturing 2032-2035	https://www.stocktrak.com/available-bonds/ (JSON-LD datePublished 2017-02-21, dateModified 2019-09-06)	VP. But the list has 72 Treasuries, of which 32 are still outstanding on 2026-09-28. Two symbols are mistyped. It jumps from 15-Feb-2031 to 15-Feb-2036, and nothing on it was issued after Feb 2014
2	"All bonds, corporate and treasury, that we support are in one master list"	https://content.stocktrak.com/what-are-bonds/	VP verbatim. The same paragraph says "We only have US bonds", "You can only use market orders" and "There are no volume limit rules on Bonds" (generic; see minor item 10)
3	"Stock Trak trades only a very limited number of bonds ... Treasuries are listed below the Corporates"	https://www.stocktrak.com/investments-bond-analysis-trading-project/	VP verbatim. The page is dated 14 Dec 2016
4	"a limited number of treasury/government bonds"; "When you buy a bond in between coupon dates you will have to pay accrued interest to the seller"	2026-27 WInS User Guide PDF (edu.stocktrak.com)	VP verbatim. "Day Trading: This is not permitted." is also present, split as "Day T rading" in the text layer
5	"Bonds available on WInS. The list includes Treasury bonds from the United States, ..."	https://wghsinvcomp.smapply.us/res/p/trading/	VP verbatim
6	"Any Government/Treasury Bonds from any exchange available on WInS"; "\$10" per bond trade; "updated once daily at U.S. market open, and pay out every six months"	https://wghsinvcomp.smapply.us/res/p/faqs/	VP verbatim, all three
7	IBTM facts: 0.07% expense; 5.05y effective duration; 5.03% yield to maturity; 15 holdings; \$554.5m net assets; \$21.85 close; 215,760 average volume; "will terminate on or about October or December 15 ..."	https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf	VP. All figures are as of 2026-09-25. The same disclosure says the funds "do not seek to return any predetermined amount" (important item 6)
8	APRA SPS 160: "can be restored to a satisfactory financial position within one year"; "must not exceed three years"	https://handbook.apra.gov.au/standard/sps-160	VP verbatim (team knowledge only)
9	Case PDF created 2026-09-10 15:19	Case PDF metadata: CreationDate D: 20260910151951-04'00'	VRF
10	Case wording: "will contribute", "high degree of certainty", "may not rely"	<code>Laura_Gao_2026_Client_Profile.txt</code> L43-45, L91-92	VRF
11	Guide p.5 "should not be rewritten simply because markets move"	<code>2026_WGY_Investment_Competition_Guide.txt</code> L154-159	VRF. The line sits in the post-IPS "evaluation and reflection" stage (important item 4)
12	IPS guide: "Focus on ... rather than ... detailed financial calculations"	<code>2026_WGY_Investment_Policy-FINAL.txt</code> L51; the explicit ban is at L123	VRF. It is a focus instruction, not a ban (minor item 17)
13	CUSIPs 91282CFV8 → 912821KC8; 912810QD3 → 912803DJ9; 912810QL5 → 912803DP5	<code>wins_now/S1_mspd_table5_2026-08-31_fixed_2032plus.csv</code>	VRF (MSPD, per S1). Only \$26.1m of the 2032 note is stripped
14	Laura quotes B8b-Q31 ("finished or not") and B8b-Q18 ("unstable income")	<code>D13a_laura_quotes_verified.md</code> lines 100 and 127	VRF, VERIFIED-PRIMARY there. D1 keeps both out of TN/IPS (correct)

#	Claim in D1	Source	Result
15	Moody's cut the U.S. to Aa1 on 2025-05-16	Wikipedia (secondary)	Found. The status stays SNIP, as D1 labels it
16	(AX1b) The 30-year bond was suspended from 2002-02-18 to 2006-02-09	https://en.wikipedia.org/wiki/United_States_Treasury_security	Found verbatim; secondary, so SNIP. The Stock-Trak list's own jump from Feb-2031 to Feb-2036 agrees

3. Corrections (full wording in the appended section of D1_rates.md)

Important (fix before the strategy team uses D1)

1. **The gap odds understate for the real ladder.** Use about 1 in 3 (30-35%) for the Nov-15 ladder:
 - DERIVED from volatility windows of 7.2-7.8%;
 - an "unchanged curve" drift adds about \$1.0k to the January cost (+3.4 points).

Model-free cross-check: the 10-year yield fell by 19bp or more over 68 trading days in 32-36% of windows (FRED DGS10 since 1962 and since 1990). Caveat: 1982-2020 was a falling-rate era, which biases history toward falls.

2. **"0.01%" and "at most part of the first payment" are thin-tail overclaims.**
 - A 100bp-plus fall over three months happened in 2.1% of windows since 1990 (2008, 2020). The model gives 0.4%.
 - A 148bp-plus fall happened in 0.1% of windows since 1990 and 1.5% since 1962.
 - A repeat of late 2008 (-167bp) leaves all of the 2033 payment plus ~16% of 2034 waiting. That needs \$48.6k, about one-third of the \$150k deposit.
 - A \$75k deposit still covers it.
3. **"1 January 2028 at the latest" overclaims.** It contradicts D1's own late or instalment rule and the joint tail.
4. **Guide p.5 is misapplied** to reject staging. The case against staging rests on the numbers.
5. **The 2019 Stock-Trak list cannot tell us anything about 2032-2035 maturities.** Delete the inference. The logged-in check stays.
6. **IBTM fallback:** "repays \$50,000 / held to maturity" is false for an iBonds ETF.
7. **Third-note conflict with D12 and D10.** Make the recommendation conditional on option (ii) vs (iii).
8. **Rung rationale:** state it as implementation (the promise is dated), not as "evidence for the reader".
9. **The rung under a position cap below 44% is not computed.** Pre-decide: under a cap, only a 2039/2040 bond, or skip.

Minor (10-22):

- the bond-list count and symbol typos, plus the generic "no volume limit / market orders only" lines;
- staging "+0.2 points" is an artefact of the assumption (under an unchanged curve it is -0.2 to -0.5 points, with a 34-35% chance of ending under 1.00);
- the worst-day share is 68.7%;
- STRIP and note prices should be compared on the same date;
- accrued interest is \$768-824;
- -421bp is false precision;
- the script cannot "re-solve on the trade date" as written;
- the IPS "ban" is overstated;
- "no probabilities in a note" is a judgement, not a rule;
- three near-final quoted phrases, including the "her own two deposits" pitch fact, which is wrong in most paths;
- the "finished or not" over-reading;
- drop finding 7 (the PDF date);
- "today" and "bought" wording.

Addition (23): longest-first also leaves less face value unfunded if the deposit never comes:

- at -100bp, \$31.4k versus \$47.7k;
- at -50bp, \$12.0k versus \$19.0k.

Counterpoint: the shortfall then falls on the first operating year.

Cross-link (24): under option (iii), a listed rung is a genuine third executed trade. That answers AX2's blocking D10 C1.

4. Contradictions found

Between	What conflicts	Resolution (AX1b view)
D1 vs D12 / D10	D1: the rung replaces VGSH as the third note. D12: VGSH carries the only "risk management" role word, and D10 wants note variety	Conditional. Under (ii) keep VGSH and treat the rung as optional. Under (iii) the rung is the natural extra note (see cross-link 24). The team decides
D1 vs D8 rule 1	D8: "If the 2028 deposit is smaller or late: ... only the facility range shrinks"	D1 is right. If the deposit is missing, or smaller than the gap, part of the first payment stays unfunded. D8's wording needs the joint-tail exception. Main loop to fix; D8 is outside this cluster
D1 vs D10 (line 63), brief §14, F-112	"About 24%" odds are used for the real ladder	Use about 1 in 3 for the real Nov-15 ladder (item 1). The brief §16 wording "roughly 1 in 4 to 1 in 3" is acceptable
D1 vs S4 finding 9	30.5% vs 30.7%	Harmless: 7.16% (the ladder's own volatility) vs 7.24%. D1's figure is the internally consistent one
D1 vs the ticket (check 8)	The ticket says "Record ... the bond drop-down (for the Final Report only)"; D1 upgrades it to "decides one order now"	D1 is consistent with brief §17 (no Final Report work). The main loop should update ticket check 8
D1 vs brief §7	Brief: the ladder "fits inside \$300k at today's curve"	D1 is right to drop "fits inside \$300k" from the headline and pitch (history: 176 of 185 days above \$300k)
D1 vs D2 (AX2 C3)	Both misuse Guide p.5	Same fix: the Guide line is about not redesigning after results. Rules written in advance are allowed
D1 vs the ticket's hedge and leftover rules	None	IEF/TLH weights, the option (iii) 35.0/63.0 split and "leftover waits in T-bills" all match

5. Overclaims, jargon, privacy, tokenism, prose

- **Overclaims:** "0.01%", "at most", "at the latest", "matches Guide p.5", "no reward" and "suggests the 2032-2035 notes may be missing" (items 1-5, 11).
- **Checked and clean:** D1 correctly bans "match", "risk-free" and anything stronger than "certain in U.S. dollars barring a U.S. government default".
- **Jargon:** mostly defined. Not defined: "p5", "zero-drift lognormal" and "forward-consistent". This file defines the first two. Keep all three out of deliverables.
- **False precision:** -421bp; "0.01%"; accrued "\$750-800".
- **Unearned complexity:** finding 7 (the PDF date). The rung is earned under (iii) and optional under (ii).
- **Privacy:** clean. Both Laura lines are from her professional or public record and are kept out of TN/IPS.
- **Tokenism:** none.
- **Near-submission prose:** three quoted phrases (item 19). Convert them to elements.

6. What the strategy team should take from D1 (after corrections)

IPS: rules to fix before Nov 6 (elements, not wording):

- buy on arrival, in full;
- longest-dated payments first;
- finish the ladder before any growth money, including from each instalment if the deposit arrives late;
- the headline is conditional: fully bought in January 2027 if rates stay near today's (about 2 in 3), otherwise when the second deposit arrives;
- name the joint tail as our own stress case;
- name dependence on the 2028 deposit in words, as a typical and a bad case, never as "at most".

TN / WInS-now:

- read the Bonds drop-down (by maturity);
- if a Jul-Dec 2032 note is listed, or a 2033-2041 second-half maturity, one rung may be bought:
 - face \$50k under (iii), ~\$34k under (ii);

- funded from IEF;
- under a cap below 44%, only a 2039/2040 bond, or skip;
- IBTM is the last fallback, and its note must not say "repays \$50,000";
- every security is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK.

Sources (accessed 2026-09-28 unless stated)

- <https://www.stocktrak.com/available-bonds/> ; <https://content.stocktrak.com/what-are-bonds/> ; <https://www.stocktrak.com/investments-bond-analysis-trading-project/> (VP content; generic Stock-Trak, not a 2026-27 WInS rule)
- <https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf> (VP)
- <https://wghsinvcomp.smapply.us/res/p/trading/> ; <https://wghsinvcomp.smapply.us/res/p/faqs/> (VP)
- <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf> (VP)
- <https://handbook.apra.gov.au/standard/sps-160> (VP)
- <https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rates.csv> / 2026/all?type=daily_treasury_yield_curve&field_tdr_date_value=2026&page&_format=csv (VP; identical to the repo snapshot)
- FRED DGS10 daily 1962-2026: research/insight_v1/scripts/data/D3/fred_DGS10.csv (VP per D3; not re-downloaded)
- https://en.wikipedia.org/wiki/United_States_Treasury_security ; https://en.wikipedia.org/wiki/United_States_federal_government_credit-rating_downgrades (secondary, SNIP)
- Repo (VRF): case txt/PDF; Competition Guide p.5; IPS guide L51 and L123; [wins_now/securities_and_allocation_v0.md](#); [wins_now/S4_red_team.md](#); [wins_now/S1_mspd_table5_2026-08-31_fixed_2032plus.csv](#); [phase_A/case_register.md](#) (R-C27, R-C48, R-W28, R-W88, R-AN35, R-AN48); [phase_A/fact_register.md](#) (F-003, F-014, F-111, F-112); [phase_D/D8_practice.md](#), [D10_compliance.md](#), [D12_overflow_client.md](#), [D13a_laura_quotes_verified.md](#), [audit_markets_rules.md](#).

What this teaches

1. **A model's odds are only as good as its assumptions.** Two honest assumptions (forwards come true; the curve stays put) move the chance of a gap from 30% to 34%. History says about 1 in 3. Quote the range and the plain-English version, not one decimal.
2. **Bell curves hide crashes.** The model called a 150bp fall in three months a 1-in-10,000 event. It happened in late
3. A rule that says "at most" must survive the worst day in history, not just the worst day this year.
4. **Old evidence can be empty for a structural reason.** A 2019 list could not contain 2032 notes, because the U.S. had not issued them yet. Before reading meaning into an absence, ask whether the thing could have been there.
5. **A fund with a date is still a fund.** An iBonds ETF ends in a known month but does not promise a dollar amount. Only a bond repays a fixed amount on a fixed date. Words in a permanent Trading Note must match the instrument actually bought.

AX2 audit: equity (D2), Taiwan/FX (D4) and compliance (D10)

Agent AX2 (cluster auditor), insight_v1 run, 2026-09-28. This is an adversarial check of three specialist files before they reach the strategy team. It is AI-generated research: findings, numbers and checklists, not text to submit. The team decides and writes every word. Status labels follow brief section 3.

What this task delivers

- This file.
 - An "## Audit corrections (AX2)" section appended to the end of each of the three specialist files. Their own text is unchanged.
 - One check script, research/insight_v1/scripts/AX2_audit_checks.py, which reproduces every number AX2 added.
 - The corrections list and verdict returned to the run.
- Nothing was committed.

Files audited:

- research/insight_v1/phase_D/D2_equity_ai.md (M030, sleeve equity weight)
- research/insight_v1/phase_D/D4_taiwan_fx.md (M095, FX and building costs)
- research/insight_v1/phase_D/D10_compliance.md (M005 WInS book, M013 IPS governance)

Verdict (short)

All three files **pass with corrections**. Every script reproduces. Every web source I re-opened says what the specialists quote. None of the problems is a fabricated fact. They are overclaims, one inconsistent yardstick, one misapplied rule, scope drift, and one gap that could break the Trading Notes deliverable.

File	Verdict	What changes
D2	Pass with corrections (0 blocking, 8 important, 6 minor)	The 50/50 option survives as a team choice. The evidence line "three sources agree the premium is about 0" is withdrawn: across sources it is about -1.2 to +2.4 points a year, still small and uncertain. The "known/locked 5.2%" wording is withdrawn. "Decide once or break the Guide" is a preference, not a rule.
D4	Pass with corrections (0 blocking, 3 important, 8 minor)	Numbers are sound. Most of the file is Final Report material and is parked. Three IPS rules go forward (US\$ binding; no NT\$ conversion or hedge before the facility decision; building-cost drift named as an assumption). The "2031 ranking" holds only if at least ~70% of the growth money is locked in 2031.
D10	Pass with corrections (1 blocking, 2 important, 10 minor)	The M005 and M013 answers stand. The lean to option (iii) is not decision-ready : with no cap, (iii) gives only two trades, and three executed trades are needed for the Oct 23 notes. The page-fit numbers are optimistic; the IPS layout is tighter than D10 says.

1. Scripts re-run (2026-09-28)

Script	Result
D2_equity_premium.py	Reproduces exactly: hurdles 5.14/5.20/5.23/5.10%; all premiums; all percentile tables; crash stress; equity shares 13.7/17.1/20.5%; VT \$51,000 ~318 sh / VGSH \$48,000 ~833 sh. One text error: the 40%-vs-60% median has the wrong sign for one case (D2 C9).
D4_facility_purchasing_power.py (live FRED and DGBAS)	Reproduces exactly. DEXTAUS latest 31.82 (2026-09-18). CCI latest 119.51. Every sd, share, drift, correlation, band and hedge figure matches.
D10_wins_book_compliance.py	Reproduces, except that the worst volume use is 0.17%, not 0.19% (D10 C5).
D10_ips_page_fit.py	Every table row reproduces. The A4 range is 89.5-101.0%, not 89-99% (D10 C6). The model itself is optimistic (D10 C3).
AX2_audit_checks.py (new)	(1) 2031-view market spread by floor share. (2) D2 yardsticks like-for-like. (3) D2 sleeve vs a Treasury-only comparator. (4) Page fit with whole lines and a 27.6pt line.

Script	Result
D9_ips_page_fit.py (cross-check only)	At Word's 27.6pt double line: 0-2 spare lines on Letter (about 525 IPS words at most). At Wharton's own 24pt sample line: 8-10 spare. This agrees with AX2's re-run of the D10 model.

2. Sources re-opened (the most decision-relevant claims)

Read with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL --grep "..."` on 2026-09-28, unless the row says it is a repo file.

#	Claim (file)	Source	Result	Status
1	Vanguard U.S. equities 4.9%-6.9% -> 4.2%-6.2%; dev ex-U.S. 4.5%-6.5%; EM 2%-4%; July 22, 2026; June 30, 2026 run; geometric (D2 E4)	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html	All verbatim	VERIFIED-PRIMARY
2	"valuations tend to be poor predictors ... should not serve as a primary reason for changing portfolio allocations" (D2 E5)	same	Verbatim	VERIFIED-PRIMARY
3	Forward P/E 19.2; 5-yr 19.8; 10-yr 19.0; 20.4 on June 30; +2.7% price, +8.9% EPS; trailing 25.8 (D2 E7)	FactSet Earnings Insight 25 Sep 2026 PDF (URL in D2)	All present	VERIFIED-PRIMARY
4	CY2026 +32.0%, CY2027 +15.4%, Q2 2027 +1.7%, IT +63.5%, semis +126%, IT ex-semis +24.2% (D2 E8)	same	Present, but these are analyst projections and estimates , not outcomes	VERIFIED-PRIMARY (as projections)
5	10y 4.44% on 2026-06-30; 4.16% on 2025-09-30; 5.18% on 2026-09-24; Feb 27 low 3.97%; breakeven 2.34% (D2 E6, D10 F-003)	FRED DGS10, T10YIE CSV	All match	VERIFIED-PRIMARY
6	CAPE 41.48; mean 17.42; max 44.19 Dec 1999 (D2 E9)	https://www.multpl.com/shiller-pe	Matches	SNIPPET-UNVERIFIED (secondary read directly)
7	JPM AC World 7.00/8.28/16.78; correlation with intermediate Treasuries 0.00 (D2 E3)	competition/official_market_data/JPM_LTCMA_2026_US_matrix_USD.pdf p.2	Matches	VERIFIED-REPO-FILE
8	"willing to take thoughtful risks" / "appropriate balance" (D2 E13)	Case L69-74	Verbatim	VERIFIED-REPO-FILE
9	Guide p.5 "should not be rewritten simply because markets move" (D2 E14, D10 G7)	Guide L156-159	Verbatim, but it sits in the post-IPS evaluation stage . Guide L89-90 and TN guide L18-19 say the strategy "may continue to evolve" before the IPS	VERIFIED-REPO-FILE (D2 misapplies it)
10	Case asks for "the effect of inflation on portfolio projections and facility costs"; cost not to be estimated; "dollar range"; Taiwan legal out of scope (D4 E1-E4)	Case L146-148, L164, L117, L169	Verbatim	VERIFIED-REPO-FILE
11	"\$150,000 ... will not be added to WInS"; no sector minimum; "funding purposes"; "does not require frequent or same-day trading"; 2x volume; 200 trades (D10 E2, E3)	https://wghsinvcamp.smapply.us/res/p/trading/	All verbatim	VERIFIED-PRIMARY
12	"Advisors may not make decisions on behalf of students ..." (D10 G1)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/	Verbatim	VERIFIED-PRIMARY
13	CFA IPS elements: 2a; "no less frequently than annually"; 4c no-rebalance policy "should be documented in the IPS"; "objective course of action" (D10 G9)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf	All verbatim	VERIFIED-PRIMARY
14	AI use "must be recorded in your Works Cited pages" (D10 s.3)	https://globalyouth.wharton.upenn.edu/ai-policy/	Verbatim	VERIFIED-PRIMARY

#	Claim (file)	Source	Result	Status
15	IPS format: 2-page maximum; TNR 12; double-spaced; 1-inch margins; non-compliance "will not be considered for semifinal selection"; formatting not specified "left to the team's discretion" (D10 G10)	IPS guide L82-123 (L101 for discretion)	Verbatim	VERIFIED-REPO-FILE
16	Three notes "from trades your team executed in WInS" / "Yes, we will verify this"	TN guide L42, L53	Verbatim	VERIFIED-REPO-FILE
17	"Investments permitted (for BOTH contributions)" (R-AN15; used by AX2 against D4's NT\$ forward)	https://wghsinvcamp.smapply.us/res/p/faqs/ (line 42 of the text)	Present	VERIFIED-PRIMARY
18	USD/TWD 31.82; CCI 119.51; S&P monthly correlation -0.38 (D4)	Live FRED DEXTAUS/SP500 and DGBAS CCI through the D4 script	Match	VERIFIED-PRIMARY inputs, derived

Not re-opened, and still labelled as the specialists had them:

- Taiwan 2-year yield 1.66% (SNIPPET-UNVERIFIED; the page is script-rendered).
- Wikipedia dates for the 1995-96 crisis (secondary).
- The CFA Asset Manager Code quotes (via SH-25/D8).
- The 2021-23 case wording for the "portfolio manager" line (via B1a).
- The Stock-Trak blog and FAQ (via A4).

3. The corrections that matter most (full lists are appended to each file)

Blocking

1. **D10 C1: under option (iii), nothing guarantees a third executed trade before Oct 23.** With no cap there are two orders. The duration refresh and a position-limit response happen only if a band or cap is hit. The fix is to name a genuine third trade in advance. The plan's own rule gives one: the 2027 leftover "waits in T-bills", so buying BIL or SGOV for about 2% of the portfolio is a liquidity-role trade (PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK). The alternative is option (ii).

Important

2. **D2 C1: "the hurdle is known" / "can lock about 5.2%" is wrong as worded.** The 5.2% is implied by today's curve for money that arrives in 2028. It cannot be locked now: there is no money yet, and forwards are derivatives, which are banned. Call it a market-implied hurdle.
3. **D2 C2: the forward earnings-yield yardstick compares a real yield with a nominal rate.** D2 adds inflation to its CAPE yardstick but not to this one. Like-for-like it gives about +2.3 points, not about 0. Across sources the premium is about -1.2 to +2.4 points a year: still small and uncertain, but no longer "about 0".
4. **D2 C3: Guide p.5 does not forbid changing the split before Nov 6.** The official texts expect the strategy to evolve until the IPS. A research-based change could even be a strong note, and the ticket already says so.
5. **D2 C5 and C6 would matter in permanent notes.** First, "VT halves AI exposure" overstates: top-10 concentration falls about 44%, and VT holds Asian AI chip makers outside its top ten. Second, "half in short Treasuries for the amount Laura may promise" conflicts with the plan's 2031 rule, which buys the floor from a share of the whole sleeve.
6. **D2 C8 contradicts the ticket on the sleeve-bond return.** D2 uses 4.8% (market-consistent) and the ticket asks for 3.5-3.9%. The main loop must pick one; AX2 favours the market-based figure.
7. **D4 C1 (scope).** Implications 1-5 and the chart spec are Final Report material (brief section 17). Three IPS rules survive.
8. **D4 C2.** "By 2031 the market is the smallest factor" is true only if about 70% or more of the growth money is locked in 2031. That share is an open team parameter.
9. **D4 C3.** An NT\$ forward is a derivative, banned in WInS and (safe reading of R-AN15) in the plan. This is the first reason not to lock NT\$; the 5.9% cost is the second. The "23% of windows" figure is a counterfactual.

10. **D10 C2.** The claim that both books pass "whatever the position limit" was tested only for no cap, 35% and 25%. Vanguard volumes are unchecked.
11. **D10 C3.** The page fit is tighter than stated. With whole lines and Word's likely 27.6pt line, 550 words fill about 98-100% of two Letter pages even with 0pt spacing. D9 agrees. Test the real PDF early; the word target may need to be below 480.

4. Contradictions found

Between	Contradiction	Resolution (AX2 recommendation; the team decides)
D2 vs WInS ticket	Sleeve equity 50/50 (D2) vs 60/40 provisional (ticket)	A proposed change, not an error. Put it to the team as a choice of spread: median about equal, p5 +\$6-7.5k, p95 -\$11-14k at 50/50.
D2 vs WInS ticket	"Changing the split later contradicts the Guide" (D2) vs "a change of the split after the team's research" as a genuine note (ticket section 4)	The ticket is right (Guide L89-90; TN guide L18-19).
D2 vs WInS ticket	Sleeve-bond model return 4.8% (D2) vs "about 3.5-3.9%" (ticket section 8)	Use a market-consistent figure (VGSH SEC yield 4.55%, forwards 4.8-5.1%) with an ASSUMPTION label. The main loop updates the ticket line.
D2 vs D10	D2's tier-1 action (set the split before VT/VGSH orders) vs D10's lean to (iii) (no VT/VGSH orders)	The split is tier 1 only if (ii) is chosen; otherwise it is an IPS rule.
D10 vs WInS ticket	D10 leans (iii); the ticket (S3) leans (ii)	Both are compliant; this is a team vote (gate box b). The vote should see: (iii) is shorter to explain in the IPS but needs a named third trade (D10 C1) and shows no growth role in WInS. (ii) shows three roles but needs the three-element scaling sentence.
D10 numbers vs D2	D10's +20.5-point equity gap and 20.4% real equity assume 60/40	At 50/50: +17.0 points and 17.0% (F-409).
D10 vs D9	Page fit 92-105% (D10) vs "fits, spare 0-2 lines at 27.6pt; 8-10 at 24pt" (D9)	Same message once whole lines and 27.6pt are used: tight with Word defaults. Use one merged checklist, with D9's --draft tool on the team's own text.
D4 vs brief section 17	D4 routes almost everything to the Final Report, including a chart spec	Park it as "later (after Nov 9)"; three IPS rules go forward (D4 C1).
D4 vs brief section 14	None found. D4 uses CPI 2.04%, CCI 3.54%/yr since 2021 and the "3.2x is a one-year spike" reading.	No action needed.
D2/D10 vs brief section 16	D10 quotes 24% only for the ladder costing more than \$300k	Quote about 24-31% ("roughly 1 in 4 to 1 in 3").

5. Overclaims, jargon, privacy, tokenism and prose check

- **Overclaim words.** "Known" and "lock" for 2028 rates (D2); "halves" (D2); "free improvement" (D2 as routed); "certain to erode" (D4); "most likely way to lose" and "whatever the position limit" (D10); "cost the whole scenario analysis its legitimacy" (D10). All are corrected in the appended sections. No file calls the WInS funds a "match" for the payments: D10 correctly says "stand-in funds that move like her payments". D10 C12 adds that "cash-flow matched" applies to the real STRIPS ladder only.
- **False precision.** D4's variance shares (8/53/39%) and D10's page-use percentages rest on small samples or a layout model. Both are labelled, but they should be read as rough rankings. D4's "51% of windows" is a coin flip, not a finding.
- **Jargon.** D4: covered interest parity, log sd, variance share, bootstrapped. D2: SEC yield, i.i.d., lognormal. Define them or cut them.
- **Labels.** "SECONDARY-READ" (D2), "DERIVED" (D4) and "INTERPRETATION" (D10) are not brief labels. Map them to SNIPPET-UNVERIFIED or ASSUMPTION. INTERPRETATION is acceptable if it is defined in the file, as D10 does.
- **Privacy.** No problems. Only her public professional record is used; there is no personal data and no student surnames.
- **Tokenism.** D4 correctly keeps Taiwan as a case fact (where the building is), not an identity theme. D10 uses a student-era Laura quote ("earliest supporters", 2016) to argue for a WInS book. Label it and do not use it that way (D10 C10).

- **Submission-ready prose.** One quotable sentence in D10 section 1.3 (C9). D2's note elements are close to prose but acceptable as a checklist. Neither file drafts a note, a reflection or IPS text.

6. What survives and where it goes

Only Trading Notes (TN) and IPS items go forward (brief section 17).

- **WInS / TN (tier 1)**
 - Vote (ii) or (iii) first. If (iii), name the third trade: the T-bill leftover.
 - If (ii), choose the split before the VT and VGSH orders if the team can. A later, research-based change is allowed and can be a note.
 - Note wording must avoid: "known/locked" rates; "halves AI exposure"; "half = the promise"; "match". Use: "stand-in funds"; "the market-implied Treasury rate for her dates"; "a fall shrinks the facility range, never the payments".
- **IPS (tier 2): rules to fix before Nov 6**
 - Sleeve split, with its reason based on the 2031 promise and the flat median, not on "this year's premium".
 - One sentence naming the date WInS represents.
 - Governance in about 30-45 words (D10 keep/cut table, verified).
 - No rebalancing between promise money and growth money, stated in words (CFA 4c).
 - US\$ is the binding currency.
 - No NT\$ conversion or hedge before the facility decision.
 - Building-cost drift named as an assumption.
 - Format: space-after 0; test the PDF on the first draft; set the word target from the real PDF.
- **Later (after Nov 9), one line each:** D4 implications 1-5 (assumption line, co-sponsor currency sentence, uncertainty sentence, flexibility sentence, chart); D2's two-house projection table.

What this teaches

1. **Re-running the numbers is not the same as checking the argument.** Every script here reproduced to the decimal. The real errors were in the words around the numbers: a rate called "known" that is only implied, a yield compared with the wrong kind of rate, and a rule quoted from the wrong stage of the competition.
2. **Compare like with like.** An earnings yield is roughly a return after inflation. A Treasury yield is before inflation. Put them on the same basis before subtracting one from the other; here that moved the answer by more than 2 points.
3. **Read the sentence around the quote.** "Should not be rewritten simply because markets move" is real Wharton wording, but it applies after the IPS is submitted. Two pages earlier the same guide says the strategy may still evolve. Where a quote sits decides what it permits.
4. **Ask what a conclusion depends on.** "By 2031 the market is the smallest risk" is true because the plan locks 80% of the growth money. Change that open choice and the ranking changes too. A finding is only as robust as its least-settled input.
5. **Check deliverables, not just rules.** A WInS book can obey every trading rule and still leave the team one executed trade short of the three notes due on Oct 23. Compliance means being able to submit what is required.

AY1 audit: co-sponsors (D5), client psychology (D6) and client-side overflow (D12)

Agent AY1 (cluster auditor), insight_v1 run, 2026-09-28. This is an adversarial check of three specialist files before they reach the strategy team. It is AI-generated research: findings, numbers and checklists, not text to submit. The team decides and writes every word. Status labels follow brief section 3 (VERIFIED-PRIMARY = VP, VERIFIED-REPO-FILE = VRF, SNIPPET-UNVERIFIED, ASSUMPTION = ASM, INT = interpretation).

What this task delivers

- This file.
- An "## Audit corrections (AY1)" section appended to the end of each of the three specialist files. Their own text is unchanged; where it conflicts with the corrections, the corrections win.
- One check script, research/insight_v1/scripts/AY1_audit_checks.py (about 5 seconds; it re-uses the D5 and D6 engines and random streams, so every difference is in what is computed, not in the simulation).
- The corrections list and verdict returned to the run. Nothing was committed.

Files audited:

- research/insight_v1/phase_D/D5_cosponsors.md (M003, M091, M107, M094: 2031 range, flexibility, co-funders)
- research/insight_v1/phase_D/D6_behavioural.md (M125, M018, M019, M228, M237, M197: risk profile, bad years, stock share rule, range width)
- research/insight_v1/phase_D/D12_overflow_client.md (M247, M231, M234: note roles and length, evidence grades, reader test)

Scope applied (brief s17): only what serves the Trading Notes (TN, Oct 23) or the IPS (Nov 6). D5 and most of D6 were written before s17 (2026-09-28), so their Final Report parts are routed to the "later" list below, not audited in depth.

Verdict (short)

All three files **pass with corrections**. Every script reproduces. Every web source I re-opened says what the specialists quote (one link built on a source is withdrawn: D5's Kresge "backstopped" claim). No fabricated facts, no privacy or tokenism problem, and no submission-ready prose. The problems are one arithmetic slip, one misused source word, one rule claim that holds only on a coarse grid, two note-wording issues that matter because notes are permanent, and scope drift.

File	Verdict	What changes
D5	Pass with corrections (0 blocking, 4 important, 7 minor)	The key finding stands: promising the upside and keeping flexibility conflict unless a rule splits the post-2031 growth (give-back share s). The rule method goes to the IPS; the numbers, NT\$ figures and the excerpt checklist go to the later list. "Guarantee" becomes "bought, not forecast, barring U.S. default". Kresge's "backstopped" link is withdrawn. 98% becomes 94%.
D6	Pass with corrections (0 blocking, 6 important, 5 minor)	The per-goal risk profile (CFA framework) is the best IPS input in the cluster and stands. The stock-share rule gives about 55-70% depending on inputs, not "60%, robust to fat tails". The hedge-note row loses "sized to ~\$292k" (under option ii) and "stops depending on markets". Her career-regret line is context, not the reason for stocks. The pitch spec must allow for the ladder finishing from the 2028 deposit and must name the nominal-US\$ limit.
D12	Pass with corrections (0 blocking, 4 important, 6 minor)	The note-element spec is re-based on the Guide's own list (it had dropped "supporting research or analysis"). The 300-character cap is a precaution with mixed evidence; check it at zero risk before the first note. Option (iii) gives two trades, not three, without a cap. The IPS rule list gains the nominal-US\$ item.

1. Scripts re-run (2026-09-28)

Script	Result
D5_range_and_flexibility.py	Reproduces exactly (floor \$127k/\$165k/\$217k and surplus \$159k/\$207k/\$273k reconcile with F-401/F-402; every (a, s) row; \$118k-\$147k / \$178k-\$223k; 50% / 10%; P(above top) 15%; excess \$2k/\$8k; NT\$4.59m; 7.2/11.0/14.9%). One text slip: "\$168k-\$184k in 98% (p1-p95)" is 94% (D5 C4).
D6_behavioural_numbers.py	Reproduces exactly (design tables A/B/C; every rule pick on its 10-point grid; bad-year figures 27/62/14/10/50%, -\$3k/-21k/-9k; growth-first 40% / 5.4%; M197 ratios; word counts, re-checked by grep on the case).
D12_note_roles.py, D12_evidence_grades.py, D12_reader_test.py	Reproduce exactly; the binomial detection figures were also recomputed by hand.
AY1_audit_checks.py (new)	[1] D5 conditional gift percentiles; [2] range width as a function of lock share a AND give-back share s; [3] D6 rule on a 5-point grid (cases A, B, C; fat tails); [3b] same with a 0.5% / 1.0% yearly fee (ASM); [4] floor vs the 2031 two-year rate; [5] WlnS hedge dollars vs the \$292k ladder.

Key new numbers (AY1 script; all model properties under the specialists' labelled ASSUMPTIONS):

- D6's rule ("most stock with P(2033 growth money < money put in, \$157.7k) <= 5%") on a 5-point grid: JPM + 4% bonds **60%**; JPM + 5% bonds (D6's own case B) **65%**; Vanguard midpoint **55%**; fat tails **65% / 70%** (A / B); JPM with a 0.5% fee **55-60%**, with 1.0% **50-55%**.
- Range width top/bottom (p90 top): (a 0.7, s 0.5) **1.26x**; (0.8, 1.0) **1.30x**; (0.7, 1.0) **1.52x**; (0.6, 0.5) 1.40x. Width $\approx 1 + s(1 - a)/a \times 1.21$.
- D5 gift given the median 2031 state: p1/p5/p50/p95/p99 = \$167.8k / \$169.8k / \$175.8k / \$183.7k / \$187.6k; below the range midpoint in 0.00% of paths (the unlocked money would have to lose ~36% in two years).
- Floor vs the 2031 two-year rate: -3.4% at 3.0%, +2.3% at 6.0% (vs 4.81%).
- Option (ii) WlnS hedge \$198,000 = 68% of the \$292,264 ladder (the ladder is 64.9% of both deposits, scaled).

2. Sources re-opened (the most decision-relevant claims)

Read with `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL --grep "..."` on 2026-09-28 unless marked.

#	Claim (file)	Source	Result	Status
1	"we allow up to 300 characters" (D12)	https://www.howthemarketworks.com/updates/trade-notes-justify-trades/	Verbatim. It is a sister product	VP for that product; UNVERIFIED for WlnS
2	"Enter a Trade Note (these are important!) to discuss how the purchase fits into your overall strategy" (D12)	https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WlnS-Userguide.pdf	Present (PDF text reads "T rade"); PDF page 10 , not p.8; no character limit anywhere in the guide. Also: "Position Limit: This is how much of your portfolio you can invest in one single stock."	VP
3	Notes cannot be edited or deleted; more notes can be added, each timestamped (D12)	https://www.stocktrak.com/new-feature-trade-notes/	Verbatim; also "Required" notes option; no length limit	VP (vendor, 2017)
4	No note limit on Stock-Trak's FAQ	https://www.stocktrak.com/faq/what-are-trading-notes/	No limit stated	VP (absence)
5	"Conduct between 6 to 9 interviews"; "Do not correct the participant"; "service-connected disability"; "all said that it was clear" (D12)	https://digital.gov/guides/plain-language/test/paraphrase-testing	All verbatim	VP
6	L = "31%"; "3-4 users from each category if testing two groups" (D12)	https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/	Verbatim	VP
7	"The [unfinished] results are below."; "Match Quality"; 6 Good / 5 Unsure / 3 Bad (D12)	https://lauragao.com/rewriting-herstory	Verbatim; recount matches	VP (public professional page; student-era, ASM)

#	Claim (file)	Source	Result	Status
8	Kresge: "Written pledges or cash ..."; "committed, imminent or backstopped" (D5)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf	Verbatim, but the "backstopped" sentence covers only financing, government funds, organizational funds and bequests; glossary defines backstopping as the grantee's own alternative resources	VP; D5's "describes Laura's floor exactly" withdrawn
9	"barriers to entitlement are overcome"; "did not receive a promise to give" (D5)	https://storage.fasb.org/ASU%202018-08.pdf	Verbatim	VP
10	"A challenge gift is an unconditional commitment by a donor ..." (D5)	https://www.nber.org/system/files/working_papers/w13728/w13728.pdf (+ abstract page)	Verbatim	VP
11	Brunel: family "wants to reserve 10 percent of their total wealth"; Das 2018 Table 1 Wants 80-85 / Needs 90-95 (D5)	squarespace-hosted Brunel PDF (URL in D5); https://srdas.github.io/Papers/GBWM.pdf	Verbatim / table present	VP
12	CFA 2020 reconciliation rules; "upper volatility bound"; risk composure measured from past actions (D6)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/survey/investment-risk-profiling.pdf	All verbatim	VP
13	"highly domain-specific" (D6)	Princeton research portal (URL in D6)	Verbatim	VP (abstract)
14	Calibration drives credibility; "as precise as warranted"; "only a small decrease in trust" (D6)	Tenney et al. 2008 author PDF; RePEc Du et al.; RUG portal van der Bles et al.	Verbatim (Tenney's 2nd sentence has a PDF ligature)	VP; applicability to co-sponsors INT
15	House-money / break-even; myopic loss aversion; Morningstar 7.0% vs 8.2%, "nearly 97%"; FAJ "0.10% per year" (D6)	EconPapers; NBER w4369; Morningstar PDF; CFA FAJ page	All verbatim	VP
16	Old criterion "would win him/her over as a client" (D6 via B2b)	globalyouth.wharton.upenn.edu/WordPress/REST/slug/judging-and-evaluation/ (modified 2024-06-03)	Verbatim (not in the rendered page text; only via REST)	VP (historical)
17	Case word counts; five "must" tests; R-C34, R-C58, R-C61, R-C66-R-C70; Guide L84-86; TN guide L1-5, L44; IPS guide L60-77	repo official files	Match	VRF

Not re-checked: Yaniv & Foster 1995 (image-only scan); Tenney et al. 2007 (already SNIPPET-UNVERIFIED in D6).

3. Do the implications follow? (the case wins)

- **D5 finding 1 (flexibility needs a rule) follows** from case L102-107 ("committing all remaining assets could limit her financial flexibility"; no contingency fund required) and test 5 (L144). It is the most useful new result in the cluster. The give-back share s earns its place: $s = 0$ collapses the range to "one exact amount" (case L115 rejects it), $s = 1$ leaves no flexibility.
- **D5 finding 3 overreaches** (D5 C2, C3): the case says she will "describe how much she expects to contribute" (L110). An expectation is not a pledge, so no co-sponsor can "count" anything unless Laura chooses to pledge it. The sound part is the direction: lead with what is already bought.
- **D6 finding 1 follows** (CFA rules applied per goal; case L61-74, L91-92). **D6 finding 3 does not follow** as stated: the rule does not pin 60% (D6 C1). **D6's M018 reasoning** (career risk $\neq$ investment risk) is right, but the same file then uses her career-regret line as the reason for stocks (D6 C3).
- **D12 finding 2 follows** for option (ii); for (iii) the premise (three hedge trades) is wrong without a position cap (D12 C3). **D12 finding 1's element split conflicts** with the official Guide's own list of what a note should capture (D12 C1). **D12 finding 3 follows** and matches D8 rules 1 and brief s8 item 8.

4. Contradictions found (and how they resolve)

#	Between	Contradiction	Resolution
X1	D5 vs D6	D5 candidate lock share a = 0.7; D6 "supports 80-90%, argues against anything below 70%"	D6's widths assume s = 1. Measured the same way, D5's (0.7, 0.5) range is <i>narrower</i> (1.26x) than D6's (0.8, 1.0) at 1.30x. The comparable choice is the promised-but-at-risk share s(1 - a): 0.15 vs 0.20. Both pass D6's width test; the team picks. (AX2's D4 C2 "70% locked" threshold also assumed s = 1.)
X2	D5 vs D6	Top of range at the 85th percentile (D5) vs the 90th (D6)	One percentile in the IPS method; the money difference is small (top \$180k vs \$182k at the median state).
X3	D6 vs D2 (AX2 audit)	D6 "confirms 60%"; D2's 50/50 survives as a team option	The rule itself gives 50-70% across inputs (50% with JPM and a 1% fee). Both are inside; neither is "confirmed".
X4	D6 vs ticket	Hedge "sizes to ~\$292k"; ticket option (ii) hedge is 66% = \$198k	Scale wording (D6 C2).
X5	D12 vs ticket	(iii) "IEF, TLH, SPTL" notes; ticket buys SPTL only under a cap below 44%	Two trades without a cap (D12 C3; same as AX2's blocking finding on D10).
X6	D12 vs D6 vs ticket (and AX2 D2-C6)	VGSH: "risk management" / "credibility or flexibility" / "a different funding purpose"	Compatible: one Guide role word ("risk management" or "liquidity"), echo "flexibility as the project develops", never "the floor". Team records its choice.
X7	D12 vs Guide L84-86	Note spec omits "supporting research or analysis"	Guide wins (D12 C1).
X8	D6 / D12 vs brief s16	Pitch spec says the payments are "bought in 2027"; IPS rule lists omit the nominal-US\$ gap	Both certainty gaps must be named (D6 C4, D12 C4).
X9	D8 rule 3 vs D8 rule 4 / D5	Rule 3: "top = [stated percentile] of the rest" (all of it promised); rule 4 and D5 promise only a share s	Flag to the main loop: rule 3 should read "top = bought amount + [share] x [stated percentile] of the rest". D8 not edited (outside this cluster).
X10	D12 vs guardrails L49	"No character limit is published" vs "plan to 300"	Both true: no WInS limit is published; 300 is a sister-product figure used as a precaution.
X11	D5 vs D3 model v2 (written in parallel)	D5 "never cap the gift"; D3 lists capped rules with P(within) 100%	Report a capped rule with "P(top reached)" (about 20% in D3), never "P(within) 100%". D3's "floor + half of the excess" row equals D5's a 0.8 / s 0.5 row, so the engines agree. D3 also reports the model understates crash years (supports D6 C1).

Consistent with brief s14 and the ticket: \$292,264; 173 of 185 days; ~24% (ASM); spot duration 10.16y; IEF 6.86y / TLH 11.59y; the 66% hedge; "never" rules; no Laura quotes in TN or IPS; no "floor-first leap".

5. Overclaims, jargon, precision, complexity, privacy

- **Overclaims fixed:** "guarantee" / "bottom guaranteed" / "never below it" without "barring U.S. Treasury default" (D5 C3); "backstopped describes Laura's floor exactly" (D5 C2); "stops depending on markets" for bond funds (D6 C2); "confirms the provisional 60/40", "pick unchanged with fat tails" (D6 C1); "WInS notes may be capped at 300" stated as the headline (D12 C2). No file calls the funds "matched".
- **False precision:** "3 of 19" (D12 C6); "60%" from a 10-point grid (D6 C1); "98%" (D5 C4); tautological 50%/10% (D5 C8). Probabilities to one decimal are fine internally; D6 already bans decimals in the IPS.
- **Jargon to define or drop if any of it reaches the team pack:** "i.i.d.", "lognormal", "Student-t with 4 degrees of freedom", " λ (loss-aversion coefficient)", "calibration" (D6 defines the last). D5 and D12 define their terms.
- **Complexity:** D5's rule has three parameters (a, s, q). Each earns its place (a sets what is bought, s sets flexibility, q sets the top), and the whole rule fits in two plain sentences. D6's eight-rule comparison table is analysis, not IPS content: the IPS needs one rule.
- **Privacy and tokenism:** none found. Laura appears only through the case and her public professional record; D12's "Match Quality" and usability-testing links are kept private and labelled; no identity-based investment choice.
- **Submission-ready prose:** none. D5's quoted confidence line ("below the floor: only if ...; above the top: about 1 in 7") is the nearest thing to a sentence; treat it as a content spec, and the team writes its own.

6. The corrections (full wording in each file's appendix)

File	#	Severity	One line
D5	C1	important	Scope: keep the range/flexibility METHOD as IPS rules; move numbers, NT\$, excerpt checklist to "later"
D5	C2	important	Withdraw "Kresge's 'backstopped' describes Laura's floor exactly"; the phrase covers other source types
D5	C3	important	"Guarantee" → "bought, not forecast, in US\$, barring U.S. default"; "co-funders count the floor" is INT
D5	C5	important	Lock-share conflict with D6 resolved via $s(1 - a)$; D5's rule passes D6's width test
D5	C4, C6-C11	minor	98% → 94%; lopsided range; 2031 rate sensitivity; tautologies; cost vs buying power; harmonise q and D8 rule 3; capped rules in D3 need "P(top reached)"
D6	C1	important	Stock-share rule gives ~55-70%, not a robust 60%; fat-tail check is not a stress test
D6	C2	important	Hedge-note row: no "\$292k" as the WlnS size under (ii); no "stops depending on markets"
D6	C3	important	Career-regret line is not the reason for stocks (same domain transfer D6 rejects)
D6	C4	important	Pitch/IPS spec: payments may finish from the 2028 deposit; name the nominal-US\$ limit; whole-portfolio stock share
D6	C5	important	Width criterion must use $s(1 - a)$ or the width, not the lock share alone
D6	C6	important	Scope: M197 numbers and FR charts to "later"
D6	C7-C11	minor	VGSH wording; "risk need nil at today's prices"; "10 of 10 covered" timing; applicability labels; harmonise q
D12	C1	important	Note elements must follow Guide L84-86 (reasoning, alignment, supporting research, role)
D12	C2	important	300-character cap: mixed evidence; verify at zero risk before the first note
D12	C3	important	Option (iii) without a cap = two trades; a third note needs a planned genuine decision
D12	C4	important	Add the nominal-US\$ gap to the IPS rules
D12	C5-C10	minor	PDF page 10; "3 of 19" precision and dating of 173/185; Match Quality = INT; Nielsen = analogy; Queensland time; VGSH wording

7. What goes forward (the part that serves TN and the IPS)

TN pack (tier 1; notes are permanent):

- [] Before the first order: confirm the note-box limit at zero risk (Stock-Trak/Wharton support, or a long test text on the review screen without pressing Confirm). Draft to 300 characters until then.
- [] Each note carries the Guide's four elements: Laura's need (reasoning); which part of the plan it is (alignment); one checkable fact (supporting research); one role word.
- [] Hedge: role "future funding"; "moves with the value of her payments when rates change (about 10 years)"; the ~\$292k real-plan cost goes in the reflection; never "match", "mature in her years", "stable", "reduce volatility", "stops depending on markets".
- [] VT: role "growth"; only money the promise does not need; a fall shrinks the facility amount, never the payments.
- [] VGSH: role "risk management" or "liquidity"; flexibility as the project develops; never "the floor".
- [] If option (iii): plan the genuine third decision now (without a cap there are only two orders).
- [] Each "how it serves Laura" line passes the swap test (D6 s1.1). No Laura quotes; no career story as a reason.

IPS: rules to fix before Nov 6 (tier 2):

- Risk tolerance per goal (D6 finding 1, CFA 2020): payments need no market risk at today's prices and can bear none; growth money can bear losses (living costs outside; no withdrawals before 2033); her investment willingness is inferred, not observed; what would change the view.
- Two certainty gaps named plainly (brief s16): if the ladder costs more than \$300k in January 2027, the longest payments are bought first and the rest is completed from the 2028 deposit before any growth money is invested; a late or short 2028 deposit hits the facility amount, never the payments; "certain" means nominal US\$, barring a U.S. Treasury default, and a fixed \$50k buys less each year in Taiwan.
- Stock share of the growth money set by one stated rule (a threshold and a probability, e.g. "about 19 in 20 chance of ending 2033 with at least the money put in"), giving about 55-70% on current inputs; the team states its pick (e.g. 60%, or 50% on cautious inputs) and the whole-portfolio stock share by stage.

4. 2031 range method: buy a share a of the growth money as a Treasury note maturing before 2033 (the bottom = the amount bought); in 2033 give that plus a share s of what the rest has become, uncapped; keep the other (1 - s) for the project; top = the bought amount + s x a stated percentile of the rest; confidence stated on both sides with the model named; operating payments never touched. Choose (a, s) by "how much of the promised gift is still at market risk in 2031", s(1 - a) (candidates: D5 0.7/0.5; model 0.8/1.0).
5. Every forecast-based statement carries a word marking it as assumed; no dollar range or percentile in the IPS.
6. Paraphrase test of the 50-word pitch and rule sentences with 6 unpaid readers, two rounds (D12 M234), logged without personal data.

Later (after Nov 9), one line each: D5 fundraising-excerpt checklist (M094); NT\$ reference band; "likely figure" presentation and the lopsided-range wording; p5/p50/p95 floor/added/kept visual; Kresge pledge-window note; D6 M197 dollar states and bar chart; bad-year illustration; design-table chart; FR opening from the 2031 conversation; D12's own later items (grade column; co-sponsor paragraph paraphrase test).

Flags to the main loop (outside this cluster): D8 rule 3 wording should include the give-back share (X9); the ticket's VGSH note wording should match X6; the Session Rules screenshot should record whether the "one single stock" position limit covers ETFs.

Sources

All web sources in section 2 were accessed 2026-09-28 with `research/insight_v1/scripts/fetch_text.py` (plus curl on the Wharton WordPress REST endpoint <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation>). Repo files (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (L10-20, L40-50, L59-75, L86-120, L130-150); `2026_WGY_Investment_Competition_Guide.txt` (L80-90, L150-160); `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L1-53); `2026_WGY_Investment_Policy-FINAL.txt` (L60-80); `SMAApply_Deliverables_Page_2026-09-27.md` (L51); `research/insight_v1/phase_A/case_register.md` (R-W16, R-W20, R-W21); `wins_now/securities_and_allocation_v0.md`; `phase_D/D8_practice.md` (rule table); `phase_D/audit_markets_rules.md`; `phase_D/D13a_laura_quotes_verified.md`, `D13c_voice_map.md`. Scripts: `D5_range_and_flexibility.py`, `D6_behavioural_numbers.py`, `D12_*.py`, `AY1_audit_checks.py`.

What this teaches

- **Re-run the numbers on a finer grid before you call a result "robust".** A rule that "picks 60%" on 10-point steps picked 65% on 5-point steps with the same file's own inputs. Robustness is shown by moving the inputs, not asserted.
- **A quote can be exact and still be misused.** Kresge really wrote "backstopped", but for a different kind of money. Read the sentence around a word, and the glossary, before borrowing it.
- **Two experts can both be right and still disagree on paper.** D5 and D6 argued about the lock share because each held a different second number fixed. Name every parameter, and the disagreement often disappears.
- **The official guide beats a clever shortcut.** A 300-character plan that dropped "supporting research" would have saved space by cutting what Wharton asked for. Check the source text first, then optimise.
- **Promises need the right verbs.** "Bought" and "barring a U.S. default" are true; "guaranteed" and "stops depending on markets" are not. For a client whose public credibility is part of her income, the verb is part of the risk management.

AY2 cluster audit: Wharton intent (D7), practice benchmark (D8), communication (D9)

Agent AY2, insight_v1 run, Phase D audit. Written 2026-09-28. This is an AI-generated audit for Team Caplet. It checks three specialist files before they reach the strategy team. It holds no submission-ready prose. The team decides and writes every word of every note, reflection, pitch and IPS.

Relayed user question ("what would be delivered at the end of this task?"): this audit step delivers this summary, one "## Audit corrections (AY2)" section appended to the end of each of D7, D8 and D9, and one script, research/insight_v1/scripts/AY2_audit_checks.py. Nothing is committed.

Scope applied: brief section 17 (Trading Notes and IPS only). Every correction below says which deliverable it serves: **TN** (Trading Notes, Oct 23) or **IPS** (Nov 6).

Status labels as in brief section 3: VERIFIED-PRIMARY (VP), VERIFIED-REPO-FILE (VRF), SNIPPET-UNVERIFIED (SNIP), ASSUMPTION (ASM), INT (AY2's judgement).

1. Verdict

Pass with corrections. No blocking errors. Every number the three specialists quote reproduces from their scripts. Every decision-relevant quote I re-opened (30 checks) is verbatim on its source. The strategic direction of all three files holds. None of them changes the lock-early architecture.

The problems are in four areas:

- **Overclaims inside rule wording:**
 - D8 rule 1 says only the facility shrinks if the 2028 deposit is small, which is false in the joint tail.
 - D8's headline says "needs met at today's yields", but it compares the wrong horizon and treats the facility as a need.
 - D9 lets VGSH carry the 2031 promise and uses "already owned".
- **Cross-file contradictions:**
 - D7's pitch rule clashes with the WInS book under option (ii).
 - D7's "refined" candidate clashes with the ticket's gate.
 - D7's IPS hedge clause clashes with the ticket's duration band.
- **Missing official vocabulary:** "uncertainty" (11 official uses, and case test 3) and the currency gap that brief s16 makes binding.
- **Text that is too close to submission-ready:** mainly D9's fill-in sentence for a permanent WInS note, plus slogans in D7, D8 and D9.

There is also one collective risk: the three files together ask the 500-word IPS for 13-15 elements. That needs one word budget (section 5).

Confidence: high on the reproductions and the source checks. Medium on the word-budget estimate, which is built from the specialists' own guesses.

2. What was re-run and re-opened

Scripts (all re-run 2026-09-28 from the repo root; every quoted figure reproduced):

Script	Figures checked	Result
D7_rule_trigger_odds.py	4.45bp/day; band odds 0.00/0.12/0.29%; drift ~0.08y; 76/53/35/12% moves; 83% "touched"; P(ladder > \$300k) ~23%	Reproduced (live treasury.gov CSV, latest 09/25/2026)
D8_practice_numbers.py	\$292,264 and \$394,930 checks; IRRs 1.05/5.09%; \$203,998; +\$8.9k/+\$10.7k/+\$12.5k; fees -\$2.6k to -\$33.9k; 0.6x/2.1x/4.2x; drift p5/50/95 50/62/73%, 49% outside	Reproduced
D9_numbers.py	vocabulary counts; 40.3/38.7/38.5%; \$2,890 per 10bp; ~\$20.4k a year; T3 +\$30,600 / -\$27,374; 69-77%	Reproduced

Script	Figures checked	Result
D9_ips_page_fit.py	44-46 lines needed of 46 (Letter, 27.6pt); ~525 max IPS words	Reproduced (assumptions labelled in the script)
D9_draft_checker.py --demo	59 words, 413 characters, 0 digits, 57 shared three-word phrases	Reproduced
New: AY2_audit_checks.py	[1] consistent-input growth-money percentiles; [2] fat-tail and crisis-volatility band odds	See section 4

Also measured: Wharton's sample IPS PDF, with pypdf. It is US Letter at a 24.0 pt line pitch, with one blank line before the IPS heading (VRF). This confirms D9.

Sources re-opened with `fetch_text.py --grep`. All were served from the helper's cache of 2026-09-27/28 pages, which are the primary bytes fetched in this run. All returned **FOUND VERBATIM** unless noted.

#	Claim (file)	Source	Result
1	FOMC "October 27-28" (D7)	federalreserve.gov FOMC calendar	VP
2	"Can you tweak it? Absolutely!" / "Your strategy should be unique to your team." (D7)	globalyouth.wharton.upenn.edu/developing-strategy/	VP (older season)
3	"simple, elegant ideas" (D7)	Wharton 2025 top-10 article	VP. Speaker: Hahn, "a judge for the semifinals", who also praised the "written reports". D7's "videos only" caveat is too narrow
4	2021-22 case "for at least 10 years" (D7)	Wharton 2021-22 case page	VP. It also shows a past case with a 10-year payment stream
5	"less is more, and having an explainable strategy" (D7)	Wharton 2023 tales article	VP
6	CFA IPS 2010: "objective course of action ... market disruption"; "clearly identified in advance"; no-rebalance "should be documented"; "required real growth rate of 4 percent" (D8)	CFA IPS PDF	VP (4 of 4)
7	Asset Manager Code: B.5.a; "expressly understood and agreed to"; "gross- and net-of-fees returns"; B.6.a "ability and willingness" (D8)	CFA AMC PDF	VP. B.6.a first failed only because the PDF text layer has a line break ("risk")
8	Russell: "lock down the benefits promised ..."; "additional risk taken only on assets above a defined surplus threshold"; "surplus glidepath" (D8)	russellinvestments.com 2026-07	VP
9	Kitces: "really is 1%"; "1.75% for portfolios up to \$500k"; "almost 1.25%" (D8)	kitces.com 2017	VP (the survey is from 2017)
10	Chhabra 2016 "being really good at what you do, your passion"; CFA 2015 blog line (D8)	wealthmanagement.com; blogs.cfainstitute.org	VP
11	APRA SPS 160 "within one year" / "must not exceed three years"; BofA "at set trigger points"; NISA trigger points (D8)	apra.gov.au; bofa.com; nisa.com	VP
12	Yale "from two years ago" (D8)	provost.yale.edu	VP, but trimmed : it is the 20% leg of an 80/20 smoothing rule
13	FCA FG11/05 "capacity for loss" definition (D8)	fca.org.uk PDF	VP
14	Retired Wharton judging page: "Excessive investing jargon ..."; "precious space repeating known facts ..." (D9)	Wharton wp-json page	VP (past season)
15	IPCC: "reciprocal statements"; "statements of fact without using uncertainty qualifiers" (D9)	ipcc.ch AR5 note	VP
16	Investor.gov "longer time horizon may feel comfortable ..." (D9)	investor.gov	VP
17	Overachiever 2021 "with unstable income" (D7, D9)	overachievermagazine.com	VP. The full line opens "It's not an easy thing to say, ..." (D13a); it is Final Report only

Official files (VRF, line-checked):

- TN guide L7-8, L12, L29-30, L36-39, L44, L53;
- Guide L70, L85, L89, L156-158, L183;

- Infographic L24-25, L41-42;
- IPS guide L10, L20, L25, L35-36, L45-57, L59, L68-69, L73-74, L82, L101-104, L116, L123-124;
- case L16, L44, L63, L70-73, L85, L90-99, L114-119, L129-130, L136-143, L152, L161-162;
- SMAApply L49-57.

All say what the specialists say, with these exceptions:

- **TN guide L12 and case L152** use "aligned with, tested, or refined" and "reflected, tested, or refined". D7's summary says every official text reads "supported, tested, or refined".
- **"Certainty" counts:** the case has 4 "certainty" plus 2 "uncertainty". D7's "6" counts both words.

Not reproducible here:

- **D7's two WebSearch results** ("no public discussion of the case"): these are not decision-relevant and are left as D7 labelled them.
- **Word's 27.6 pt "Double" line height:** this cannot be tested without Word. It stays ASM, and the exported-PDF check stays binding.
- **A historical fat-tail check on real S&P 500 data:** the data download was not permitted in this session. It was replaced by a model sensitivity (section 4), labelled ASM.

3. Corrections, ranked (full list at the end of each file)

Rank	File	Severity	Deliverable	Correction (short)
1	D7 + D9 (and ticket)	important	TN + IPS	Under option (ii), D7's pitch rule "no stock-market risk until all ten payments are bought" is visibly contradicted by a day-one VT buy next to a ~\$198k hedge against a ~\$292k promise. D7's own TN plan assumes (ii). So under (ii), the ticket's three-element scaling sentence must appear in the IPS, and a scaling clause in the hedge and growth notes. Under (iii), the rule is visibly true but there is no growth note. This is a new input to gate box (b); it does not decide it.
2	D8	important	IPS	Rule 1's "only the facility range shrinks" is false in the joint tail: rates fall and the 2028 deposit is smaller than the shortfall (about \$7k at -50bp, \$23k at -100bp; brief s6). The frozen rule must say what then happens.
3	D9	important	TN	VGSH is mapped to "the amount she can promise co-sponsors in 2031", with "already owned" as its certainty word. That re-introduces the "floor proxy" overclaim S4 removed. Before 2031, VGSH is growth money; use the ticket's "can later become".
4	D7	important	TN	Refinement candidate (a), the TLH/SPTL cap split, is not a "refined" decision. The ticket's gate fixes the cap branch before the first order, so presenting SPTL as a refinement would misdescribe a planned buy.
5	D7	important	TN	"Tested" must use a pre-registered window (fill to Oct 20) and the end-date odds (~53%). The "83% touched at some close" figure only applies if the team picks the best date, which is cherry-picking.
6	D9	important	TN	The M012 answer gives a fill-in sentence for a permanent WInS note ("... cost about \$X at [trade-date] ..."). Recast it as elements, and mark quoted phrases "illustrative, do not copy".
7	D9	important	IPS	The vocabulary omits "uncertainty": 11 official uses, and case L137 asks how uncertainty affects both the payments and the facility. Name the payments' residual uncertainties rather than implying none.
8	D9	important	IPS	The assumption line names inflation but not currency. Brief s16 (binding) requires the nominal-USD vs Taiwan-costs gap to be named.
9	D8	important	IPS	"Needs met at today's yields" / "4.81% is below today's 5-year (4.98%)" compares the wrong horizon (2026-31 vs 2028-33). The facility has no required amount (case L104). Compare with the 5.23% forward instead, and say it is not lockable.
10	D7	important	IPS	Pitch spec (b), "payments certain in US\$", must be conditional: once bought, barring U.S. default.
11	D7/D8/D9	important	IPS	There is no joint word budget; see section 5.
12	D7	minor	TN	official_curve_pv.py cannot value other dates. Use A2_curve_recheck.py or D9_numbers.py [3]. Curves can be pulled later; only WInS screenshots are needed on the day. Check ex-dividend dates in the window.
13	D7	minor	IPS	"The hedge is never traded because rates moved" conflicts with the ticket's duration band. Say "never sold or trimmed; only re-mixed to keep its rate sensitivity".

Rank	File	Severity	Deliverable	Correction (short)
14	D7	minor	TN	Band odds to two decimals are false precision (section 4). The conclusion holds.
15	D8	minor	IPS	The fee clause is a team assumption the case never raises. Cut it first if words run short. "0.15% = 0.6x" does not "exceed" the headroom.
16	all	minor	TN + IPS	Scope (s17): D7's and D8's Final Report items move to the one-line "later" list. D9 already complies.
17	D7, D8, D9	minor	TN + IPS	Several ready-made phrases need marking as illustrative (listed per file).

4. Upgrades and downgrades from AY2's own checks

- **D8's devil's advocate: UPGRADED (confirmed on one input set).** D8 set a Treasury-forward middle case (\$204k) against a p5 from a different model (\$159k), and called the gap an order of magnitude. AY2_audit_checks.py [1] re-runs it with one input set. Inputs (ASM): AC World 7.00% compound, 16.78% vol; bonds at the 5.23% forward, deterministic; 200k paths.
 - At 60% stocks: p5 **\$155k (-\$49k)** against Treasury-only \$204k; median **+\$13k**; p95 +\$104k.
 - At 50% / 70%: p5 -\$41k / -\$56k; median +\$11k / +\$14k.
 - So the argument stands: stocks buy upside, not need. Read it as "about +\$11-14k in the middle vs -\$41-56k in a bad case".
 - Caveat: bond rates are held fixed, so the uncertainty in the 2028 reinvestment rate is ignored.
 - The opposite risk must travel with it: D13c/brief s16 say the plan already looks very cautious for a client who "takes the jump", and the case asks for "pursuing growth". Neither argument decides the 50/60/70 weight.
- **D7's band odds: conclusion CONFIRMED, precision DOWNGRADED.** AY2_audit_checks.py [2] re-runs D7's window under other assumptions.

Assumption (ASM)	55-65 band	57-63 band	Whole book +/-2 points
Fat tails (Student-t, 3 degrees of freedom)	0.1%	0.7%	0.7%
Crisis-level 35% volatility	1-2%	11-15%	11-15%

- "Do not plan a discipline trade" holds.
- Write "very unlikely (about 0-2% even in a crisis)", not "0.00%".
- If a tighter band does fire in a crash, that is a genuine rule-triggered note, not staging.

- **D8's four-rule structure: UPGRADED.** The IPS guide itself asks the IPS to "establish your strategic approach to preparing for the operating commitment, managing the operating reserve's asset composition over time, determining a responsible facility contribution, and preserving appropriate financial flexibility" (L48-50, VRF). That is a more direct reason for rule 4 than the test matrix.
- **D9's research element in every note: UPGRADED.** The TN guide says notes "should document the research, analysis, and reasoning behind your investment decisions" (L7-8, VRF). This is in addition to Guide p.3.
- **D9's page-fit risk: CONFIRMED on the sample.**
 - Wharton's own sample is set at exactly 24 pt.
 - (INT) "Exactly 24 pt" is therefore a defensible reading of "double-spaced" if a draft overflows under Word's "Double". The team decides; the exported PDF is the test.
 - A common rule of thumb, about 250-300 double-spaced words a page, also puts 550 words plus headings right at two pages (INT).
- **Downgrades:**
 - **D7's "past cases were growth/return targets":** the 2021-22 case also had a ten-year payment stream.
 - **D7's "111%":** this is face value. At today's prices the payments cost \$292k, 97% of the first deposit.
 - **D7's "certainty 6":** this count includes "uncertainty".
 - **D8's Yale analogy:** it rests on a trimmed quote.
 - **D8's "Code F.4.d requires":** the Code says "should provide".
 - **D9's "hundreds of teams will copy":** this is INT, not evidence.

5. One IPS word budget (cross-file; the ips_spec owner should fix it before drafting)

The IPS must fit about 470-490 words (D9, from the page-fit check). The three files each ask for "one clause", but nobody has added them up. The table uses the specialists' own estimates. It is INT, not counted text.

IPS element	Asked for by	Their estimate (words)	AY2 suggestion
Central idea + rates-fall branch (rule 1)	D7, D8	25-35	Keep. Add the joint-tail branch (correction 2)
Growth mix, band, yearly reset (rule 2)	D8, D7	25-35	Keep. Fold the return objective into its "because"
2031 range method (rule 3)	D8, D9 M191	25-35	Keep
2033 order of use / reserve composition (rule 4)	D8; IPS guide L48-50	25-35	Keep (required by the guide)
Governance line	D8	10-15	Keep. Word it so it cannot read as permission to revise after Nov 6
Where risk sits (M241/M226/M143 merged)	D9	~30	Merge with the risk-tolerance sentence
Return objective per goal	D8 M109	~25-30	Merge into rule 2
Risk tolerance by goal (ability/willingness)	D8 M063/M081	~30-40	Merge with "where risk sits"
Fee/cost principle	D8 M026	~20	Cut first if over budget (case silent on fees)
Certainty definition	brief s7; D9 M062	~20-25	Keep. Absorb inflation + currency here
Three trade-offs, "gives up X to get Y"	D9 M121	36-60	Keep at about 12 words each
Laura "because" clauses	D9 M071	40-60	Inside the rules, not extra
Inflation (+ currency) assumption	D9; brief s16	10-15	Inside the certainty definition
Option (ii) scaling sentence	ticket s2	25-30	Only if (ii) is chosen
"What / why" framing (IPS guide L53-57)	IPS guide	30-40	Opening lines
Sum before connecting words		about 310-475	Merging brings it to about 300-380, leaving room for plain connecting prose

6. Cross-specialist contradictions and tensions

#	Between	Issue	Resolution (decision owner)
C1	D7 M011 pitch rule vs ticket option (ii) vs D7 M002 TN plan	See rank 1	Gate box (b), team
C2	D7 refinement (a) vs ticket gate (c)+(e)	The cap branch is decided before trade 1, so it is not a refinement	Drop (a); keep (b), the pre-announced provisional split
C3	D7 IPS hedge clause vs ticket duration band	"Never traded" vs the IEF/TLH re-mix	Reword (rank 13)
C4	D9 VGSH label vs ticket red-team fix 4	"Already owned" 2031 promise vs "can later become"	Use the ticket's wording
C5	D8 M109 (low end of the band) vs D13c / D9 M071 ("not set at the minimum to look safe")	Opposite pressures on the equity weight	Show both sides to the team. Section 4 numbers apply to both
C6	Statistics degree used in three places (ticket VT note; D7 certainty definition; D9 "once at most")	Decoration risk	Pick one place across TN + IPS
C7	D9 "locked rates" vs D8 "forward, not lockable"	Vocabulary	Use D8's wording
C8	D7 "certainty 6" vs D9 "certainty 4"	Counting method	Use 4 (standalone word) and report "uncertainty" separately
C9	Blueprint 01 section 3 risk example ("does not matter for held-to-maturity matching") vs D9 vocabulary	False for WInS funds	Fix when blueprint 01 is updated

No contradiction was found with the brief section 14 facts (\$300k WInS cash, rules, P(ladder > \$300k) ~24%, durations, FOMC). All three files respect the D13 quote rules: no Laura quotes in the TN or IPS; "unstable income" is Final Report only and verified.

7. Privacy, tokenism, jargon, submission-ready prose

- **Privacy:** a scan of all three files for family, personal and student-data terms found nothing. Laura appears only through the case and her public professional record. No student surnames, emails or rankings.
- **Tokenism:** no identity- or Taiwan-based holdings; D8 and D9 both rule them out. The only risk is the statistics degree used as a hook in several places (C6).
- **Jargon:** D8's framework vocabulary (LDI, GBWM, glide path) is defined, and D8 keeps it out of the IPS. D7 bans "LDI, duration, DV01" from the pitch. D9's checker flags jargon.
- **Submission-ready prose (mark as illustrative):**
 - D7 L23-24 and L159-160 (the rule sentence in italics), and L121;
 - D8 L134-135 (a reflection clause);
 - D9 L266-269 (the note template), L288, L445 and L483.

None is a full drafted pitch, note or IPS paragraph. They are short enough to be copied straight into a 50-word pitch or a permanent note, which the AI policy forbids.

8. What goes to the strategy team (checklist)

- [] Gate box (b): decide (ii) vs (iii) with both new inputs: D9's word cost and D7's pitch-rule consistency (C1).
- [] TN plan: three real trades. "Refined" only via a pre-announced split decision. "Tested" on a pre-registered window, with WInS screenshots on the fill date and the end date.
- [] Label table for notes: VGSH as "short-Treasury part of the growth money" (C4); certainty words from D9 plus "uncertainty".
- [] IPS rules 1-4 plus governance, with the joint-tail branch in rule 1 and the reworded hedge clause.
- [] One word budget (section 5) before the first IPS draft.
- [] One place for the statistics degree.
- [] The Final Report items from D7 and D8 go to the one-line "later (after Nov 9)" list.

Sources (accessed 2026-09-28 via `research/insight_v1/scripts/fetch_text.py --grep`, helper cache of 2026-09-27/28)

- <https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm>
- <https://globalyouth.wharton.upenn.edu/developing-strategy/>
- <https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/>
- <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/>
- <https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-teams-from-the-2023-teams/>
- <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation>
- <https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-investors.pdf>
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- <https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.html>
- <https://www.kitces.com/blog/independent-financial-advisor-fees-comparison-typical-aum-wealth-management-fee/>
- <https://www.wealthmanagement.com/investment-news/q-a-with-ashvin-chhabra-you-re-investing-all-wrong>

- <https://blogs.cfainstitute.org/investor/2015/08/05/ashvin-chhabra-on-wealth-management-the-value-proposition-lies-in-understanding-client-goals/>
- <https://handbook.apra.gov.au/standard/sps-160> ; <https://business.bofa.com/en-us/content/workplace-benefits/glidepaths-for-pension-plans.html> ; <https://www.nisa.com/perspectives/dldi/>
- <https://provost.yale.edu/how-exactly-does-spending-policy-work>
- <https://www.fca.org.uk/publication/finalised-guidance/fsa-fg11-05.pdf>
- https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf
- <https://www.investor.gov/introduction-investing/getting-started/asset-allocation>
- <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>

Repo files (VRF):

- `competition/official/2026_27/` (the case; IPS, TN and competition guides; infographic; SMAApply page) and the IPS sample PDF;
- `research/insight_v1/_context/brief.md`;
- `phase_A/{case_register,fact_register}.md`;
- `phase_D/{D5_cosponsors,D13a_laura_quotes_verified,D13b_other_quotes_verified,D13c_voice_map}.md`;
- `wins_now/securities_and_allocation_v0.md`;
- `research/blueprints/01_trading_notes_blueprint.md`;
- `research/council_2026-09-27/team_notes_2026-09-27.md`;
- `research/verified_2026-09-27/official_curve_pv.py`.

Scripts:

- `research/insight_v1/scripts/{D7_rule_trigger_odds,D8_practice_numbers,D9_numbers,D9_ips_page_fit,D9_draft_checker}.py` (re-run);
- `research/insight_v1/scripts/AY2_audit_checks.py` (new).

What this teaches

1. **Re-running is cheap; re-reading is where errors hide.** Every number reproduced. The errors were in the words around the numbers: "only the facility shrinks", "locked" for a forward rate, "already owned" for money not yet promised.
2. **Two correct files can still contradict each other.** D7's pitch rule and the WInS option (ii) are each sensible alone. Together they tell a reader two different stories, so consistency has to be checked across files, not only within them.
3. **Compare like with like.** A middle case from one set of inputs against a bad case from another can mislead. Here the one-input re-run happened to confirm the point (about -\$49k at p5 for +\$13k in the middle). You only know that after doing it.
4. **Probabilities have a shape, not just a size.** "0.00%" from a thin-tailed model becomes 1-2% in a crisis, and 11-15% for tighter bands. Say "very unlikely", and know what would change it.
5. **A word limit is shared.** Five good one-clause ideas from three people can still overflow 500 words. Someone has to own the budget.

D13a: Laura Gao's own words, verified

Agent D13a (Quote Verifier for Laura's own words). All checks were made on 2026-09-27. The companion file `D13a_laura_quotes_verified.json` holds the same judgements, one object per id. The scripts are `research/insight_v1/scripts/D13a_laura_quotes_check.py` and `research/insight_v1/scripts/D13a_build_outputs.py`.

- The check script re-fetches every page and records whether each quote is found in `phase_D/_work/D13a_contexts.json`. That file holds no page text, because the text around a quote can contain the private material this check screens out.
- The build script writes the JSON and the table below from the recorded judgements.

Summary

- **Scope.** 104 Phase B ids: the 100 that D13b handed over as Laura's own words, plus the 4 copies of the case pull quote. I decided each speaker from the source page, not from the Phase B label. All 100 are Laura's. Each is one of: her answer in a Q&A; a direct quote an outlet marks "she says"; a turn labelled "Laura Gao:" in a podcast transcript; or a document she published herself (her website, resume, speaking guide, educators guide, a signed founder's note). Two are hers but are not statements: B8b-Q04 is a sample audience question in her speaking guide, and B8b-Q08 is a writing prompt for students.
- **Wording.** All 104 are verbatim (VERIFIED by `D13a_laura_quotes_check.py`, 106 of 106 found, including the 2 new quotes). The 4 pull-quote copies match the official case file.
- **Final status of the 104 Phase B ids:** 91 VERIFIED-PRIMARY, 6 PARAPHRASE-UNVERIFIED, 7 EXCLUDE-PRIVACY. Of the 6 PARAPHRASE-UNVERIFIED, 4 are the pull quote treated as Laura's words and 2 are podcast-transcript lines. The 104 ids hold 85 distinct quotes, because agents collected the same lines more than once; the table groups identical copies. I also found 2 new VERIFIED-PRIMARY quotes (D13a-N01, D13a-N02) while searching for the pull quote's source.
- **11 statuses changed against Phase B's claims:** 7 moved to EXCLUDE-PRIVACY, 2 moved down to PARAPHRASE-UNVERIFIED, and 2 moved up from SNIPPET-UNVERIFIED to VERIFIED-PRIMARY (B8b-Q40 and B8b-Q41, which the old fetch helper could not read).

Five findings to act on

1. **The case pull quote is not verified as Laura's words.** The case prints "The only person who needs to believe in something is yourself." as a quotation on her profile page, with a large opening quote mark. It names no speaker, although the 2021-22 case put a name line under its client quote. Eight web searches and 36 pages by or about Laura found no source. Cite it as the case's words, never as something she said (see "Pull-quote finding").
2. **One Phase B context changed the meaning of a quote.** B3b-Q09 "It's not published in Mandarin" was filed as a statement about the book's reach in China. On the page it answers a question about her family, so it is excluded under the privacy screen.
3. **Seven quotes fail the privacy screen** even though they are public and verbatim: B3b-Q09, B8b-Q23, B8b-Q24, B8b-Q34, B8a-Q20, B8a-Q22, B8a-Q23. They cover family, home and travel life, coming out, and her personal identity struggle.
4. **Podcast transcript lines are not safe to quote as her exact words.** The Maverick Show lines match only a transcript that has visible transcription errors and was produced three years after the episode. Two professional lines are held at PARAPHRASE-UNVERIFIED; the other three are excluded for privacy.
5. **Some verified lines are safe only with their context.** Each of the following breaks if trimmed:
 - "risk-adverse" (B8a-Q03) is about choosing a business major, not about investing.
 - "No more gimmicks." (B8b-Q14) is about fantasy drafts of a novel.
 - "diminishing returns" (B9b-Q12) is about dance songs.
 - "explain myself" (B8a-Q31) is about language choices in her memoir.
 - B8a-Q05 describes other people, not her.

Words used in this file

- **Verbatim:** the exact words appear on the page. Line breaks and curly versus straight quote marks are ignored.
- **Primary source:** where the words were first published, either by Laura herself or by the outlet that interviewed her.
- **Pull quote:** a short line printed large in a document's margin to catch the reader's eye.
- **Transcript:** a typed copy of spoken words. It is only as accurate as whoever, or whatever software, typed it.
- **VERIFIED-PRIMARY:** verbatim on the primary source, said or written by Laura, and the sentences around it keep its meaning.
- **PARAPHRASE-UNVERIFIED:** we cannot confirm the exact words, or cannot confirm that Laura said them. Do not put it in quotation marks as hers.
- **EXCLUDE-PRIVACY:** do not use, however accurate. It concerns family, a partner, relationships, health, home life, or something else outside her public professional work (brief section 4).

Shortlist: verified lines most likely to help the written deliverables

This is D13a's judgement (ASSUMPTION) about usefulness; the team decides. Every line here is VERIFIED-PRIMARY. They are evidence for the team's own sentences, not text to paste.

what it shows about Laura	id	her words (short)	source and date
Builds what is missing	B2b-Q07	"If what you want doesn't exist, create it yourself."	Poets&Quants, 2018-03-30 (student)
Takes career risks but sees the cost	B1b-Q11, B8a-Q07	"you should always take the jump..."; "...self-employed, with unstable income." (same answer)	Overachiever, 2021-03-15
Feels accountable to early backers (relevant to the 2031 co-sponsor range)	D13a-N01	"...letting other people down — people who were your earliest supporters and were the first to believe in you."	Daily Pennsylvanian, 2016-01-28 (student)
Labels her own uncertainty	B9b-Q09	"The [unfinished] results are below." (and she grades each match Good/Unsure/Bad)	her website, undated
Plans for the downside	B8a-Q13	"When I first posted the comic, I expected a way worse reaction"	Overachiever, 2021-03-15
Ships, then improves	B8b-Q31, B8b-Q32	"I give myself a deadline for when I must post the art, finished or not."	Geeks OUT, 2022-05-11
Will scrap work that is wrong	B8b-Q12, B8a-Q28	"I scrapped three completely different drafts..."	The Nerd Daily, 2025-03-08
Impact over metrics	B8b-Q21	"it's great having thousands of likes, but at the end of the day, it's just a number."	Overachiever, 2021-03-15
Community and access (relevant to the residency)	B8b-Q38, B8b-Q39, B3a-Q04	"Our No. 1 priority is to get as many diverse voices and stories in the room as possible..."; "Pro-bono visits can be offered on need-basis."	Local News Matters, 2026-02-12; speaking guide, 2024
Uses data to tell a story	B8a-Q19	"That [story] was something I could prove better with data"	Pulse Spikes, 2021-06-18

What a strong use of a Laura quote must contain:

- her exact words, in quotation marks, and nothing trimmed that changes the meaning;
- the outlet and the year;
- what she was talking about;
- a clear link to one decision in the plan.

Also say when a line comes from her student years (2016-2018). Never use a line about her identity, her family or her private life as a reason for an investment choice (brief section 4).

Full table: 104 Phase B ids and 2 new quotes (identical quotes share a row, ids joined by "=")

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B1b-Q11 = B8b-Q19 = B8a-Q08	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Answer to 'What pushed you to make the jump?' (from her tech job to full-time comics).	VERIFIED-PRIMARY	The same answer calls the move 'not an easy thing' and admits imposter syndrome, so it is a considered leap. It is about a career leap, not about investment risk: do not turn it into a portfolio weight.
B8a-Q09	"I feel like I've been jumping almost my whole life."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Opening of the same answer: friends call it a big jump; 'In college, I jumped majors almost every semester.'	VERIFIED-PRIMARY	About switching careers and majors.
B8b-Q20	"I did 50 things, and maybe I didn't like fully excel at each, but I got to try all of them and have so much fun with them."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same answer: how she hopes to look back on her life at 80, in contrast to 'I did this one thing and excelled at it for 50 years'.	VERIFIED-PRIMARY	Career and life outlook; no private detail. Spoken filler 'like' is in the original.
B8a-Q10	"I did 50 things, and maybe I didn't like fully excel at each, but I got to try all of them"	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same passage as B8b-Q20.	VERIFIED-PRIMARY	Ends before 'and have so much fun with them'; meaning kept. Prefer the full version (B8b-Q20).
B8b-Q18 = B8a-Q07	"It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same answer on leaving her tech job to be self-employed.	VERIFIED-PRIMARY	Her own words that self-employment means 'unstable income'; useful for why the 2028 deposit (advances, speaking, licensing) is uncertain. 'Health insurance' refers to job benefits, not her health.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q21	"it's great having thousands of likes, but at the end of the day, it's just a number. You never really understand the human connection behind it"	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Answer on handling attacks after 'The Wuhan I Know': why one reader's thank-you note meant more than metrics.	VERIFIED-PRIMARY	Sentence continues ', so seeing something tangible like that makes my whole week.'
B8a-Q11	"it's great having thousands of likes, but at the end of the day, it's just a number."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same passage as B8b-Q21.	VERIFIED-PRIMARY	First sentence only; meaning kept.
B8b-Q22	"I'd just be like, 'Hey, this is why you made this. This is who you're trying to help, and this is the positive effect you actually have.'"	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same answer: what she told herself when rereading a reader's supportive comment.	VERIFIED-PRIMARY	Use as the purpose of her work (who it helps). The surrounding text is about online racist abuse; leave that out.
B8a-Q12	"This is who you're trying to help, and this is the positive effect you actually have."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same passage as B8b-Q22.	VERIFIED-PRIMARY	Shorter form of B8b-Q22; meaning kept.
B8a-Q13	"When I first posted the comic, I expected a way worse reaction"	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Same answer: she expected abuse because her Twitter job was on the anti-abuse platform.	VERIFIED-PRIMARY	Continues ', just given what was going on at the time.' Shows she thought about the downside before publishing.
B8a-Q14	"Well, if people aren't gonna give it to us, I'll just make one."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Answer to 'What's coming up for you?': she pitched her next book as a happy queer love story 'just because there are barely any out there!'	VERIFIED-PRIMARY	A 'fill the gap yourself' line that matches B2b-Q07. It is about a book (her public work); cite it as a product decision, never as a reason for an investment choice.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q23	"I'm currently in Taiwan, Taipei."	Laura: her answer in a Q&A (interviewer Zoe Kim)	https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid	2021-03-15	Overachiever Magazine interview. Answer to 'Introduce yourself!'	EXCLUDE-PRIVACY	Where she lived in March 2021, in an answer that also gives personal details (home life). The case already places the residency in Taiwan; no quote is needed for that.
B2a-Q12	"After I graduated from Penn in 2018 with a degree in business analytics, I worked in tech as a product manager."	Laura: answer labelled 'LG:' in a Q&A by Braden Kelner	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	2022-04-06	Wharton Magazine Q&A. Answer to 'What do you personally get out of creating comics?': drawing while she worked as a product manager.	VERIFIED-PRIMARY	Her shorthand for her degree. The case says 'Statistics & Information Decisions Management'; her resume says 'Statistics and Information Decisions'; P&Q 2018 lists 'Economics with a concentration in Business Analytics'. Use the case wording for the degree. The sentence before it is private; do not quote it.
B2b-Q10 = B8b-Q25	"Soon enough, I had a comics diary that I could not only decompress into, but also take pride in how far I'd come when I reread it."	Laura: answer labelled 'LG:' in a Q&A by Braden Kelner	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	2022-04-06	Wharton Magazine Q&A. Same answer: a nightly 10-minute comic after work, kept as a 'comics diary'.	VERIFIED-PRIMARY	A small daily habit that added up to real progress.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q24	"Instead, I stand up to the beast by admitting I don't yet know the "right" answer, but whatever it is, it's not for anyone else to judge."	Laura: answer labelled 'LG:' in a Q&A by Braden Kelner	https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/	2022-04-06	Wharton Magazine Q&A. Answer to 'What is your favorite scene in the memoir, and why?' (the climax of Messy Roots).	EXCLUDE-PRIVACY	Verbatim and hers, but its subject is her personal identity struggle as told in her memoir. It sounds close to the case pull quote, which makes it tempting; using it to describe how she makes money decisions would turn her identity into decoration. Cite the book's public theme (identity, belonging) instead.
B2b-Q07 = B8a-Q01	"If what you want doesn't exist, create it yourself."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Answer to 'What is the biggest lesson you gained from studying business?'; she goes on to say she nearly transferred out of Wharton and created The Signal instead.	VERIFIED-PRIMARY	Her clearest short line on her entrepreneurial method. The same idea recurs in 2021 (B8a-Q14). Student-era (2018).
B8a-Q02	"However, I realized transferring was nothing more than running away from the problem."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Same answer: why she stayed at Wharton and founded The Signal and a design seminar.	VERIFIED-PRIMARY	The sentence before it contains family material; do not quote or paraphrase it. 'The problem' is the mismatch between her interests and the expected Wharton path; her response was to build something.
B2b-Q08 = B8a-Q06	"Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Her list of college activities in the profile.	VERIFIED-PRIMARY	First person ('I taught'), in a list she supplied. She taught financial literacy to high-school students; it says nothing about her own investing.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8a-Q03 = B8b-Q40	"However, this lack of structure and certainty can be off-putting for students who are easily influenced by those around them or are risk-adverse."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsfo.rundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Answer to 'What advice would you give to a student looking to major in a business-related field?'	VERIFIED-PRIMARY	'risk-adverse' is her spelling [sic]. It is about the vague, unstructured nature of a business major, NOT about investment risk; using it to describe her risk tolerance would trim it into a new meaning. Phase B's B8b-Q40 status (SNIPPET-UNVERIFIED) is now confirmed on the page.
B8a-Q04	"My advice is to avoid siloing yourself into an end-all-be-all career situation."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsfo.rundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Same answer; continues 'Instead, use college as your playground to explore as many interests as you can and stay open to new opportunities.'	VERIFIED-PRIMARY	Career advice to students (2018).
B8a-Q05	"each has juggled passion against practicality, social pressure against personal identity, and countless bouts of career pivoting"	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsfo.rundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Answer on her proudest achievement, The Sign.al: describing the people its articles featured.	VERIFIED-PRIMARY	Describes OTHER people (Sign.al interviewees), not Laura. Never present it as a self-description.
B8b-Q41	"Frankly, my first shock after arriving at Wharton was learning that "bankers" were not the people who exchanged checks for cash or chained pens to desks."	Laura: first-person questionnaire answer (profile by Jeff Schmitt; the closing nomination is by Lee Kramer, not her)	https://poetsandquantsfo.rundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/	2018-03-30	Poets&Quants 2018 Best & Brightest profile. Answer to 'What has surprised you most about majoring in business?'	VERIFIED-PRIMARY	A joke; low value. Phase B marked it SNIPPET-UNVERIFIED because the old helper dropped this page's text; now verbatim with the current helper and read on the page.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q12 = B8a-Q29	"I scrapped three completely different drafts before landing on this one."	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Answer to 'Did you face any challenges whilst writing and illustrating?' about her first fiction book, Kirby's Lessons for Falling (In Love).	VERIFIED-PRIMARY	Shows she will throw away finished work that is wrong.
B8b-Q13	"But in the process, I lost the heart of my writing—authentic, personal connections."	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Same answer: why she scrapped her early drafts after trying 'something wildly different'.	VERIFIED-PRIMARY	Em dash as on the page.
B8a-Q28	"So, I scrapped everything and went back to my roots. No more gimmicks."	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Same answer.	VERIFIED-PRIMARY	'Gimmicks' means the high-concept fantasy premises of the dropped drafts, not anything financial.
B8b-Q14	"No more gimmicks."	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Same answer.	VERIFIED-PRIMARY	Three words cut from a longer passage: quote B8a-Q28 instead. On its own it invites a meaning the page does not support (for example, 'she dislikes financial gimmicks').
B8b-Q15	"Every character is grappling with how to balance society's expectations with their own desires."	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Answer on her illustration process for Kirby's Lessons: describing the book's characters.	VERIFIED-PRIMARY	About her fictional characters, not herself. The lines just before it are personal; do not quote them.
B8b-Q16 = B8a-Q30	"I'm taking a short break from writing to focus on teaching my Writing Comics course at California College of the Arts!"	Laura: her answer in a Q&A by Elise Dumbleton	https://thenerdaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/	2025-03-08	The Nerd Daily Q&A. Answer on what comes next after Kirby (March 2025).	VERIFIED-PRIMARY	A dated plan. California College of the Arts is reported to close in 2027 (D13b items B3a-Q05, B3b-Q14), so this teaching role may end; check before relying on it.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q17 = B8a-Q31	"I wish I wrote about these nuances and tradeoffs made in an author's note at the end. But at the same time, I wish I didn't always have to explain myself."	Laura: answer labelled 'LG:' (questions by bookseller Shirley Mullin)	https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512	2022-02-07	ABA Indies Introduce Q&A (Bookweb). Answer to 'Is there anything in Messy Roots that you now wish you had not included?': how to show Wuhanese Mandarin to English readers.	VERIFIED-PRIMARY	'Tradeoffs' and 'explain myself' are about language choices as a bilingual author, not about how she wants decisions explained to her.
B8b-Q26 = B8a-Q15	"Now I get to fit work around life instead of the other way around, which is such a game-changer"	Laura: direct quote marked 'she says' in a feature by Annie Graham	https://www.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel	2022-03-07	Input (Inverse) feature. On the release of Messy Roots: her working life after leaving her tech job.	VERIFIED-PRIMARY	inputmag.com now redirects to inverse.com/input/...; the two copies are identical. The sentences around it describe her travels; do not use them.
B8b-Q30	"The quicker you get over your perfectionism, the faster you'll finish projects, get feedback, improve, and overcome imposter syndrome or "artist stage fright"."	Laura: her answer in a Q&A by Michele Kirichanskaya	https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/	2022-05-11	Geeks OUT interview. Answer to 'What advice might you have to give to other aspiring creators?', headed 'Post terrible work!'	VERIFIED-PRIMARY	The same sentences appear three weeks later in The Honey POP (a written answer she reused).
B8b-Q31	"I give myself a deadline for when I must post the art, finished or not."	Laura: her answer in a Q&A by Michele Kirichanskaya	https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/	2022-05-11	Geeks OUT interview. Same answer.	VERIFIED-PRIMARY	Also verbatim in The Honey POP (B8a-Q26); cite the earlier Geeks OUT piece.
B8b-Q32	"Ultimately my goal is to tell a story; I don't need to be perfect to be impactful."	Laura: her answer in a Q&A by Michele Kirichanskaya	https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/	2022-05-11	Geeks OUT interview. Same answer.	VERIFIED-PRIMARY	The answer ends: if she had kept the comic in drafts 'trying to get it perfect', she would never have published it and gotten the book deal. Also in The Honey POP (B8a-Q27).
B8a-Q25 = B8b-Q33	"Always start small and post that work, no matter how crappy it looks!"	Laura: her answer in a Q&A by Selina (The Honey POP)	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	2022-06-02	The Honey POP interview. Answer to 'What advice would you give to young artists looking to create their own webcomic?'	VERIFIED-PRIMARY	Opening line of the answer; the rest repeats her Geeks OUT answer word for word.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8a-Q26	"I give myself a deadline for when I must post the art, finished or not."	Laura: her answer in a Q&A by Selina (The Honey POP)	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	2022-06-02	The Honey POP interview. Same answer.	VERIFIED-PRIMARY	Duplicate of B8b-Q31 (Geeks OUT).
B8a-Q27	"Ultimately my goal is to tell a story; I don't need to be perfect to be impactful."	Laura: her answer in a Q&A by Selina (The Honey POP)	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	2022-06-02	The Honey POP interview. Same answer.	VERIFIED-PRIMARY	Duplicate of B8b-Q32 (Geeks OUT).
B8b-Q34	"Go at your own pace and always make sure to be in an environment in which you feel safe and supported by allies before you make any big reveals."	Laura: her answer in a Q&A by Selina (The Honey POP)	https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/	2022-06-02	The Honey POP interview. Answer to 'What's some advice you have for young LGBTQ+ readers who are still trying to find themselves?'	EXCLUDE-PRIVACY	Coming-out advice (sexuality, relationships), not her professional work. Turning it into a 'build a safe base before a big move' investment rule would use her identity as decoration.
B8b-Q35 = B8a-Q17	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	Laura: direct quote ('she says' / 'she reflects') in a profile by Ishita Shah	https://pulsespikes.org/story/laura-gao	2021-06-18	Pulse Spikes profile. Profile of her work: how writing and art go together for her.	VERIFIED-PRIMARY	
B8a-Q18	"I've always said that my main purpose is to tell a story"	Laura: direct quote ('she says' / 'she reflects') in a profile by Ishita Shah	https://pulsespikes.org/story/laura-gao	2021-06-18	Pulse Spikes profile. End of the profile, on her plans to keep writing and working with technology; continues 'It doesn't matter what medium I choose.'	VERIFIED-PRIMARY	
B8a-Q19	"That [story] was something I could prove better with data"	Laura: direct quote ('she says' / 'she reflects') in a profile by Ishita Shah	https://pulsespikes.org/story/laura-gao	2021-06-18	Pulse Spikes profile. On her data project Rewriting Herstory (matching famous men with lesser-known women).	VERIFIED-PRIMARY	'[story]' is Pulse Spikes' insertion.
B8b-Q36	"That [story] was something I could prove better with data,"	Laura: direct quote ('she says' / 'she reflects') in a profile by Ishita Shah	https://pulsespikes.org/story/laura-gao	2021-06-18	Pulse Spikes profile. Same passage as B8a-Q19.	VERIFIED-PRIMARY	Keeps the page's trailing comma; drop it when quoting.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q37	"All of it is way beyond our initial passion project. We're thrilled that the community really supported us and wanted us to continue to grow the festival,"	Laura: direct quote marked 'says Gao' (reporter JL Odom)	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	2026-02-12 (updated 2026-02-15)	Local News Matters (Bay City News) report. On the growth of Pride in Panels, the free San Francisco queer comics festival she co-organizes (second edition, 2026).	VERIFIED-PRIMARY	Trailing comma is the page's punctuation before 'says Gao'; drop it when quoting.
B8a-Q35	"All of it is way beyond our initial passion project."	Laura: direct quote marked 'says Gao' (reporter JL Odom)	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	2026-02-12 (updated 2026-02-15)	Local News Matters (Bay City News) report. Same passage as B8b-Q37.	VERIFIED-PRIMARY	First sentence only; meaning kept.
B8b-Q38	"Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on,"	Laura: direct quote marked 'says Gao' (reporter JL Odom)	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	2026-02-12 (updated 2026-02-15)	Local News Matters (Bay City News) report. On the festival's new mini grants to 20 comic creators ('says Gao about the grants').	VERIFIED-PRIMARY	The paper printed it with a leading ellipsis ('...Our No. 1 priority'), so the paper already cut words before it; keep the ellipsis to be exact. Trailing comma is page punctuation.
B8b-Q39 = B8a-Q36	"We definitely hope to offer more as the festival grows."	Laura: direct quote marked 'says Gao' (reporter JL Odom)	https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/	2026-02-12 (updated 2026-02-15)	Local News Matters (Bay City News) report. Same passage: the mini grants.	VERIFIED-PRIMARY	'We' means the festival organizers.
B8a-Q20	"Taiwan is like one of my top places to live in and to travel to all time"	Laura: turn labelled 'Laura Gao:' in the show's own transcript (host Matt Bowles); transcript has visible transcription errors and was not checked against the audio	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23; transcript PDF created 2025-10-22	The Maverick Show, episode 190, transcript PDF. Travel podcast: answer on how Taiwan was for her, including where she lived.	EXCLUDE-PRIVACY	About where she has lived and likes to travel (home life). Use the case for the Taiwan link. Verbatim only in the show's own transcript (speaker label 'Laura Gao:'). That transcript has visible transcription errors ('Yilan and Yulan', 'busing metropolis', 'coal side') and was made in October 2025 for a June 2022 episode; not checked against the audio. Phase B's 'AI-generated by Outcast AI' could not be confirmed on the PDF or the episode page.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8a-Q21	“Or there's also a little like some art residencies at the top of the mountains there.”	Laura: turn labelled 'Laura Gao:' in the show's own transcript (host Matt Bowles); transcript has visible transcription errors and was not checked against the audio	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23; transcript PDF created 2025-10-22	The Maverick Show, episode 190, transcript PDF. Lightning round, 'top three favorite travel destinations': recommending Yilan, Taiwan (monasteries, art residencies, tea farms).	PARAPHRASE-UNVERIFIED	Verbatim only in the show's own transcript (speaker label 'Laura Gao:'). That transcript has visible transcription errors ('Yilan and Yulan', 'busing metropolis', 'coal side') and was made in October 2025 for a June 2022 episode; not checked against the audio. Phase B's 'AI-generated by Outcast AI' could not be confirmed on the PDF or the episode page. A travel tip, not a statement of plans: do not infer a site or model for her residency from it.
B8a-Q22	“join a co living space even just for like a week or a month”	Laura: turn labelled 'Laura Gao:' in the show's own transcript (host Matt Bowles); transcript has visible transcription errors and was not checked against the audio	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23; transcript PDF created 2025-10-22	The Maverick Show, episode 190, transcript PDF. Answer to 'what is one travel hack?': where she makes local friends when travelling alone.	EXCLUDE-PRIVACY	A personal travel habit, not her professional work. Verbatim only in the show's own transcript (speaker label 'Laura Gao:'). That transcript has visible transcription errors ('Yilan and Yulan', 'busing metropolis', 'coal side') and was made in October 2025 for a June 2022 episode; not checked against the audio. Phase B's 'AI-generated by Outcast AI' could not be confirmed on the PDF or the episode page.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8a-Q23	"knowing that I don't have to own anything or be tied to any material possessions"	Laura: turn labelled 'Laura Gao:' in the show's own transcript (host Matt Bowles); transcript has visible transcription errors and was not checked against the audio	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23; transcript PDF created 2025-10-22	The Maverick Show, episode 190, transcript PDF. Answer to 'What impact do you think all of that travel has had on you?'	EXCLUDE-PRIVACY	Personal lifestyle (home life). Tempting as a clue to her attitude to money, but do not use it to set risk levels. Verbatim only in the show's own transcript (speaker label 'Laura Gao:'). That transcript has visible transcription errors ('Yilan and Yulan', 'busing metropolis', 'coal side') and was made in October 2025 for a June 2022 episode; not checked against the audio. Phase B's 'AI-generated by Outcast AI' could not be confirmed on the PDF or the episode page.
B8a-Q24	"So, I can't say no to this opportunity"	Laura: turn labelled 'Laura Gao:' in the show's own transcript (host Matt Bowles); transcript has visible transcription errors and was not checked against the audio	https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf	episode 2022-06-23; transcript PDF created 2025-10-22	The Maverick Show, episode 190, transcript PDF. On her future agent emailing her after the 2020 NPR interview, which led to the two-book HarperCollins deal.	PARAPHRASE-UNVERIFIED	Verbatim only in the show's own transcript (speaker label 'Laura Gao:'). That transcript has visible transcription errors ('Yilan and Yulan', 'busing metropolis', 'coal side') and was made in October 2025 for a June 2022 episode; not checked against the audio. Phase B's 'AI-generated by Outcast AI' could not be confirmed on the PDF or the episode page. For a quotable line on the career leap, use the Overachiever answer (B8b-Q18, B1b-Q11).

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8a-Q32	"I was a bit hesitant. I was like, I know this is kind of a sensitive topic. I don't know what I'll get from it"	Laura: direct quote ('Gao says/said') in a report by Aubri Juhasz	https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines	2020-04-04	NPR Goats and Soda report. On deciding in March 2020 to share 'The Wuhan I Know'.	VERIFIED-PRIMARY	The quote continues: 'But I got all kinds of responses that really warmed my heart.'
B8a-Q33	"It's disheartening to know that all they know are the bad parts of it"	Laura: direct quote ('Gao says/said') in a report by Aubri Juhasz	https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines	2020-04-04	NPR Goats and Soda report. On how people knew Wuhan only through the virus headlines.	VERIFIED-PRIMARY	Continues: 'About the virus and about the wet markets that people always want to bash on and point fingers at.'
B3b-Q09	"It's not published in Mandarin"	Laura: her answer in a Q&A by Malaka Gharib (edited for length and clarity)	https://www.npr.org/sections/goatsandsoda/2022/04/24/1093992912/the-pandemic-inspired-a-cartoonist-to-explore-their-wuhanese-roots-and-queer-ide	2022-04-24	NPR Goats and Soda Q&A. Answer to a question about her family members and the book.	EXCLUDE-PRIVACY	The question was about her family, and the sentence continues with a clause about them. Phase B's context ('whether Messy Roots reached China') trimmed it into a new meaning. If the market fact matters (no Mandarin edition as of April 2022), state it in your own words as a fact about the book, without quoting her.
B8a-Q34	"There wasn't much that was incorrect as much as there just wasn't any news about the overall picture of Wuhan outside of the coronavirus"	Laura: direct quote ('she told HuffPost') in a report by Brittany Wong	https://www.huffpost.com/entry/chinese-american-illustrator-life-in-wuhan_n_5ea9c739c5b63115cec2be5a	2020-04-29 (page: Apr 29, 2020, 08:07 PM EDT)	HuffPost report. On media coverage of Wuhan early in the pandemic.	VERIFIED-PRIMARY	Phase B dated it 2020-04-30 (the UTC date); the page shows Apr 29, 2020, 08:07 PM EDT.
B8a-Q37	"how long will this novelty factor last?"	Laura: answer labelled 'Gao:' in the student newspaper's Q&A	https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a	2016-01-28	The Daily Pennsylvanian Student Spotlight Q&A. Answer to 'What are your long-term and short-term goals?' for Draw Street Journal, her laptop-decal business, in its first week.	VERIFIED-PRIMARY	Her own rhetorical question; she answers it with a plan (marketing, a new design every month). Phase B's link to the case's 'small business' is plausible but an ASSUMPTION: the case does not name the business.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q02	“living proof that doodling on your homework can actually “get you somewhere” (take that, Mr. Dutton)!”	Laura's own site, third-person bio (self-published; quote as her website bio, not as speech)	https://lauragao.com/hey/	undated live page (accessed 2026-09-27)	her website bio ('Hey' page). First sentence.	VERIFIED-PRIMARY	Third-person bio on her own site. It names a private person (a teacher): if used, cut the parenthetical with an ellipsis, or use her first-person version from The Nerd Daily (2025-03-08): 'living proof that doodling on your Geometry homework can indeed get you somewhere in life!' (verified 2026-09-27). Low value for investment writing.
B8b-Q03	“Their work explores the hilariously messy intersections of queer immigrant stories”	Laura's own site, third-person bio (self-published; quote as her website bio, not as speech)	https://lauragao.com/hey/	undated live page (accessed 2026-09-27)	her website bio ('Hey' page). Second paragraph.	VERIFIED-PRIMARY	Her bio uses they/their; her site says she uses she/her and they/them (the case uses she/her). Cite as the theme of her public work only, never as a reason for an investment choice.
B9b-Q14	“shapes the next generation of creators as a Professor of Comics at California College of the Arts”	Laura's own site, third-person bio (self-published; quote as her website bio, not as speech)	https://lauragao.com/hey/	undated live page (accessed 2026-09-27)	her website bio ('Hey' page). The line on her teaching role.	VERIFIED-PRIMARY	Present tense on a live page; CCA is reported to close in 2027 (D13b items), so check before relying on it.
B8b-Q09	“Our bold interactive projects aim to help students discover their passions and live deliberately.”	Laura's own site, first person ('I co-founded')	https://lauragao.com/editorial	undated live page (accessed 2026-09-27)	her website, Editorial page. Describing The Sign.al, 'An “Anti-Culture Culture Club” I co-founded at the University of Pennsylvania'.	VERIFIED-PRIMARY	'Our' means The Sign.al club.
B8b-Q10	“Only after we realized EVERY student deserved to hear their words of wisdom did we decide to build something to cultivate it.”	Laura: founder's note signed 'Laura Gao, Founder of The Signal' (page linked from lauragao.com/editorial)	https://signaltemp.github.io/	c. 2022 (ASSUMPTION: 5 years after the January 2017 launch; the page's figures run to 2022)	The Sign.al 5-year anniversary page. Her founder's note: how the club began as alumni interviews.	VERIFIED-PRIMARY	'We' means she and her co-founders (named on the page; not recorded here).

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q11	"I couldn't be more proud of the current team keeping the legacy strong."	Laura: founder's note signed 'Laura Gao, Founder of The Signal' (page linked from lauragao.com/editorial)	https://signaltemp.github.io/	c. 2022 (ASSUMPTION: 5 years after the January 2017 launch; the page's figures run to 2022)	The Signal 5-year anniversary page. Last sentence of her founder's note.	VERIFIED-PRIMARY	She handed the venture to a team that kept it running.
B3a-Q02 = B3b-Q03	"Debuted on #9 on Indies Bestsellers List"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). LG Studios entry: her HarperCollins graphic novels.	VERIFIED-PRIMARY	Her own claim on her own resume (the line continues ', and featured on NPR and New York Times'); verified as her words, not independently. Her Wharton Magazine answer (2022) also mentions the Indie Bestseller List.
B3a-Q03	"Finalist for Harvey Award, Goodreads Choice Awards, and California Booksellers Award"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Same entry.	VERIFIED-PRIMARY	Self-reported. Her site bio says 'A Harvey and Goodreads Choice Award honoree'; her speaking guide says 'finalist'. Say finalist, not winner.
B3b-Q04	"Clients include: Retainit, AI-powered online learning software"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). LG Studios (her freelance studio, 2020 to present): first client listed.	VERIFIED-PRIMARY	A line break follows 'Clients include:' in the PDF; identical once whitespace is collapsed.
B9b-Q06	"AI-powered online learning software"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Her description of the client Retainit.	VERIFIED-PRIMARY	Fragment of B3b-Q04.
B9b-Q05	"Beta-tested MVP with 300K+ users of economics platform, Marginal Revolution University"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). LG Studios entry, under the client Retainit.	VERIFIED-PRIMARY	MVP = minimum viable product (a first working version). Self-reported.
B9b-Q04	"Collaborated with researchers at National Institute of Health for usability tests with nurses and doctors"	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). LG Studios client M3 (My Mask Movement), 3D-printed medical masks.	VERIFIED-PRIMARY	Quote as written ('National Institute of Health'; the agency is the National Institutes of Health).

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B9b-Q01	“Experimented and shipped 10 features that resulted in +4.5M DAU”	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Twitter Product Manager entry, 2018-2020.	VERIFIED-PRIMARY	DAU = daily active users. Self-reported; the line continues with examples (Breaking News updates, moderation tools).
B9b-Q07	“Growth ML (2018-2019)”	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Her Twitter teams ('Abuse & Safety (2019-2020), Growth ML (2018-2019)').	VERIFIED-PRIMARY	ML = machine learning.
B9b-Q03	“Investigated cause of 1.5M cases of transactions from unattributed ads using SQL, R, and Excel”	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Amazon Data Analyst entry, 2017 (Recommended Ads).	VERIFIED-PRIMARY	Self-reported.
B9b-Q02	“Product Management: JIRA, Confluence, A/B testing”	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Skills section.	VERIFIED-PRIMARY	A/B testing = trying two versions with real users to see which works better.
B9b-Q08	“Languages: Mandarin Chinese”	Laura's own resume (self-authored; claims are self-reported)	https://lauragao.com/s/Laura-Gao-Resume.pdf	PDF created 2024-10-24 (file metadata); live, accessed 2026-09-27	her resume (PDF on her site). Skills section.	VERIFIED-PRIMARY	A professional skill as listed; relevant only as a working language.
B3a-Q04	“Rates will differ based on travel, length of event, and books purchased. Pro-bono visits can be offered on need-basis.”	Laura's own promotional guide (PDF author field 'Laura Gao'; made in Canva)	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Publicity & Speaking Guide (PDF). Booking panel ('Email to book').	VERIFIED-PRIMARY	Her own promotional document. No fee amounts are published, so speaking income cannot be sized from it.
B8b-Q05 = B8a-Q38	“Pro-bono visits can be offered on need-basis.”	Laura's own promotional guide (PDF author field 'Laura Gao'; made in Canva)	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Publicity & Speaking Guide (PDF). Booking panel.	VERIFIED-PRIMARY	Second sentence of B3a-Q04.
B8b-Q04	“What motivated you to quit a secure job to become an artist-writer in the middle of a pandemic?”	Laura's own promotional guide (PDF author field 'Laura Gao'; made in Canva)	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Publicity & Speaking Guide (PDF). 'Popular topics and Q&A' page, under the talk 'Taking the Leap to Pursue Art Full-Time from a Traditional Tech Job'.	VERIFIED-PRIMARY	A sample audience question she lists, NOT a statement by her. Write 'her speaking guide lists the question ...'. It shows she frames her own story as leaving 'a secure job'.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B8b-Q06	“Tired of authors talking your students to sleep? Choose an interactive hands-on activity!”	Laura's own promotional guide (PDF author field 'Laura Gao'; made in Canva)	https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Publicity & Speaking Guide (PDF). Offer to schools, libraries and bookstores.	VERIFIED-PRIMARY	Promotional copy.
B8b-Q07	“Why must graphic novels be in competition with prose novels? How can kids benefit from both?”	Laura: guide with an author's note signed 'Laura Gao' (PDF author field 'Laura Gao')	https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Educators & Book Clubs Guide (PDF). Her signed author's note ('Teaching graphic novels'): what she asks educators who doubt graphic novels.	VERIFIED-PRIMARY	Phase B's date 'c.2022' is too early: the PDF was created 2024-03-24 and its bio says Kirby 'will be released in 2025'. A both-and habit of mind; do not stretch it into an asset-allocation claim.
B3b-Q13	“exercise students’ creativity in less daunting ways than traditional essay writing”	Laura: guide with an author's note signed 'Laura Gao' (PDF author field 'Laura Gao')	https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Educators & Book Clubs Guide (PDF). Same author's note: the activities she created for the discussion guide.	VERIFIED-PRIMARY	Fragment of 'Many of the activities I've created in the Discussion Guide below exercise students' creativity in less daunting ways than traditional essay writing.' Meaning kept.
B8b-Q08	“Write a letter to Future You.”	Laura: guide with an author's note signed 'Laura Gao' (PDF author field 'Laura Gao')	https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf	PDF created 2024-03-24 (file metadata)	her Messy Roots Educators & Book Clubs Guide (PDF). Discussion exercise for Chapter 10 ('Home is where the art is').	VERIFIED-PRIMARY	A writing prompt for students, not a statement about her own planning.
B9b-Q09	“The [unfinished] results are below.”	Laura's own site, first person ('I scraped')	https://lauragao.com/rewriting-herstory	undated (student-era project, ASSUMPTION)	her Rewriting Herstory project page. Text-analysis project matching famous men with women in their field.	VERIFIED-PRIMARY	The square brackets are hers. Her results table grades each match Good, Unsure or Bad: she labels her own uncertainty.
B9b-Q10	“For the purposes of this study, I used Spotify's definitions and values for different song attributes.”	Laura's own site, first person ('I decided to analyze')	https://lauragao.com/the-evolution-of-dance-music	undated (accessed 2026-09-27)	her 'Evolution of Dance Music' data study. Method section.	VERIFIED-PRIMARY	

id(s)	exact text	speaker check	URL	piece date	context	status	notes
B9b-Q11	"Further probing of other song metrics' effects on danceability would provide deeper analysis."	Laura's own site, first person ('I decided to analyze')	https://lauragao.com/the-evolution-of-dance-music	undated (accessed 2026-09-27)	her 'Evolution of Dance Music' data study. 'Takeaways' section.	VERIFIED-PRIMARY	
B9b-Q12	"However, tempo and speechiness display a diminishing returns relationship."	Laura's own site, first person ('I decided to analyze')	https://lauragao.com/the-evolution-of-dance-music	undated (accessed 2026-09-27)	her 'Evolution of Dance Music' data study. Findings on what makes a song danceable.	VERIFIED-PRIMARY	About songs, not finance. Do not present it as evidence that she thinks in investment terms.
B9b-Q13	"we set out to build a guide for various types of students to select the best happy hour for any occasion"	her own site, but written as 'we' (co-authors not named): a group project shown on her site	https://lauragao.com/philly-happy-hours	undated; Penn senior year per the text (c. 2017-18, ASSUMPTION)	'Philly Happy Hours, Deconstructed' project page. A Tableau data project on Philadelphia happy hours.	VERIFIED-PRIMARY	Written as 'we' (co-authors not named), so it is a group project shown on her site. Student-era and about bars: low value, and not a subject to feature in writing for a client.
B2a-Q13 = B2b-Q11 = B8b-Q01 = B8a-Q39	"The only person who needs to believe in something is yourself."	NOT confirmed as Laura's words: set as a pull quote with a large opening quote mark, no closing mark and no speaker line; no primary source found where she says it	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (lines 31-33; PDF p.1)	2026-09-10 (case PDF creation date)	Official case, Client Profile p.1 pull-quote box. In the right margin beside the paragraph on The Wuhan I Know, on the page titled 'MEET LAURA GAO / Storyteller, Entrepreneur, and Creative Visionary'.	PARAPHRASE-UNVERIFIED	Wording VERIFIED-REPO-FILE as the case's text (agrees with D13b). Attribution to Laura NOT verified: no name line (the 2021-22 case printed '— Nichole Jordan' under its client quote), and no public source found in 8 web searches and 36 pages by or about Laura (see the pull-quote finding). Cite it as the case's words (Client Profile p.1), never as something Laura said in an interview.

id(s)	exact text	speaker check	URL	piece date	context	status	notes
D13a-N01	"It's actually harder as an entrepreneur because not only are you letting yourself down, but more importantly, you're also letting other people down — people who were your earliest supporters and were the first to believe in you."	Laura: answer labelled 'Gao:' in the student newspaper's Q&A	https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a	2016-01-28	The Daily Pennsylvanian Student Spotlight Q&A. Answer to 'What were some major challenges you've had with Draw Street Journal?'	VERIFIED-PRIMARY	NEW (found by D13a; not in Phase B). The only place in 36 pages by or about Laura where she uses 'believe' about other people's support. Student-era (2016). The sentences before it describe a stressful day; do not quote them.
D13a-N02	"As girls, it is easy for us to put too much weight on other people's opinions. There's a social stigma that many women aren't mentally brave or prepared enough to explore the unknown alone. I don't think any of that matters though, if you are able to get past the hardest part: actually launching the venture."	Laura: answer labelled 'Gao:' in the student newspaper's Q&A	https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a	2016-01-28	The Daily Pennsylvanian Student Spotlight Q&A. Answer to 'Do you have any advice for your peers who also hope to start their own company?'	VERIFIED-PRIMARY	NEW (found by D13a). The closest verified match in her own words to the pull quote's idea: other people's doubts matter less than launching. Three consecutive sentences, quoted whole. The gender framing is hers; cite it as advice to new founders, not as a basis for any investment rule.

Pull-quote finding: "The only person who needs to believe in something is yourself."

What the case shows (VERIFIED-REPO-FILE: [competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt](#) lines 31-33, and the PDF page 1 rendered and viewed on 2026-09-27):

- The line sits in a pull-quote box in the right margin of page 1, beside the paragraph about The Wuhan I Know. The page is titled "MEET LAURA GAO / Storyteller, Entrepreneur, and Creative Visionary".
- It is set in large grey italics under a big gold opening quote mark. That mark is drawn as a graphic, which is why the text file does not show it. There is no closing mark and no name line.
- The only picture on the page is her headshot, which contains no text. Pages 2-4 have no pull quote.
- So the case presents the line as a quotation on Laura's page, and a reader will assume it is hers. But the case never says who said it.
- **Past cases named the speaker.** The 2021-22 case ends its client quote with "— Nichole Jordan" (VERIFIED-PRIMARY, <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/>, read 2026-09-27). D13b reports a name line under Connor Barwin's quote in the 2025-26 case. The 2026-27 case has none.

Search for a primary source where Laura says it: none found (2026-09-27).

- **Web searches (8).** None found the line in Laura's words:

- the exact phrase (only generic self-help pages on Medium, Pinterest, Facebook and blogs, none about Laura);
- "Laura Gao" with "believe in something";
- "Laura Gao" with "needs to believe";
- "only person who needs to believe" with Gao;
- the case's subtitle "Storyteller, Entrepreneur, and Creative Visionary" (no page but the case uses it);
- Wharton Global Youth with Laura Gao 2026 (no public Wharton article on her yet);
- Laura Gao talks with "believe in yourself";
- her book titles with "believe in".
- **Pages read and searched for "believ" (36 distinct pages or files with content):**
 - her website: 10 pages;
 - her resume, speaking guide and educators guide;
 - The Sign.al founder's page;
 - 15 interviews, profiles and news reports from 2016 to 2026 (all listed under Sources);
 - the podcast transcript and episode page;
 - the Playbook "Imagination Fund" spotlight, her CCA directory entry, the Commons Comics review, Wikipedia and Goodreads' quotes page;
 - the public SMaply client page, which only links to the same case PDF.

The phrase appears nowhere. The only place where she uses "believe" about other people's support is D13a-N01 (2016).

- **Could not check:**
 - archive.org answered "429 Too Many Requests" and reset connections;
 - thesign.al returned empty pages;
 - two lauragao.com addresses returned 404;
 - HarperCollins and bookseller pages refused access (403);
 - her posts on X and Instagram need a login and were not searched;
 - the full text of her books is not available.

The line could still come from a private conversation with Wharton, a talk, a social post or a book page.

Verdict

- **As Laura's words: PARAPHRASE-UNVERIFIED** (all 4 ids: B2a-Q13, B2b-Q11, B8b-Q01, B8a-Q39). As the case's words the wording is VERIFIED-REPO-FILE.
- **How to cite it:** as the case's framing of her outlook, for example "the case's profile of Laura highlights the line ..." (Client Profile, p.1). Do not write "Laura said", "in an interview" or "her motto".
- **Why this matters for the deliverables** (guidance, not text to submit). The case uses the line as her philosophy. The Phase A case register (anomaly R-AN30) sets it against the 2031 task. In 2031 co-sponsors must believe her dollar range, and case p.3 warns: "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors." (VERIFIED-REPO-FILE, case txt lines 114-115). Her own verified words hold both sides of that tension:
 - D13a-N02 (2016) says other people's doubts matter less than launching. That is the closest verified match to the pull quote.
 - D13a-N01 (2016) says letting down "people who were your earliest supporters and were the first to believe in you" is worse than letting yourself down.

A strong use pairs the case's line, cited as the case, with one of her own dated lines, and ties them to a promise she can keep, such as a floor already bought. Both 2016 lines are from her student years; say so.

- **A similar-sounding line to avoid:** B8b-Q24 ("...it's not for anyone else to judge"). It is verbatim and hers, but it is about her personal identity journey and is excluded under the privacy screen.

Laura-related quotes that are NOT her words (D13b's scope; listed so nobody quotes them as hers)

id(s)	text (short)	real speaker	D13a note
B2b-Q06 = B8b-Q27 = B8a-Q16 = B10a-Q03	"So when she got the book deal with HarperCollins, she gave Twitter her notice."	Input reporter Annie Graham, 2022-03-07	The "So" follows personal material in the paragraph before, so it is not evidence of a contract-first financial plan.
B8b-Q28	"She's hoping to finish in time for a 2024 release."	same reporter	The book came out in 2025 (Nerd Daily Q&A, 2025-03-08).
B2b-Q09	"Gao asked city officials what problem areas..."	Wharton Magazine writer Sara Albert, 2017-10-30	The same page has a direct Laura quote on that project: "They wanted to get an insider feel for what they should do to optimize the proposal," (VERIFIED-PRIMARY, https://magazine.wharton.upenn.edu/digital/whartons-pitch-to-amazon/ , read 2026-09-27).
B3a-Q01 = B3b-Q01, B3b-Q02, B8b-Q29	Rights Report lines on her HarperCollins deal	Publishers Weekly, July 2020 (trade press)	Not her voice.
B3b-Q12	"When her dynamic move at the Texas Youth Fall Invitational goes awry"	Publishers Weekly reviewer, 2024-12-12	About the fictional heroine.
B9b-Q15	"Once dubbed the Queen of Balance..."	third-person book description on her site (publisher-style copy; bookseller copies returned 403, so the publisher origin is not confirmed)	About the fictional heroine; not her voice.

Method

1. Loaded `phase_B/quotes_raw.json` (271 quotes) and `phase_D/D13_mechanical_check.json`. My scope was the 100 ids that D13b classed as Laura's own words, plus the 4 pull-quote ids.
2. Read every source page in full, not just the matching sentence. For each quote I checked the speaker, the piece date and the question being answered, and whether trimming changed the meaning. Dates come from page metadata where it exists (article dates; PDF creation dates for her resume and guides).
3. `D13a_laura_quotes_check.py` re-fetched every page with the current fetch helper. All 106 quotes were found verbatim (104 Phase B ids and 2 new). The 350 characters on each side of each quote, which I reviewed, were saved only outside the repo (option `--windows`), because they include private material from the source pages.
4. Applied the privacy screen (brief section 4 and this task) to every quote. Family, partner, relationships, health, home life and anything outside her public professional work were excluded even when public.
5. Pull quote: rendered case page 1 to see the layout. PyMuPDF was installed in the session scratchpad only, not in the project environment. I also ran 8 web searches and searched 36 pages by or about Laura (see above).
6. `D13a_build_outputs.py` records every judgement and writes the JSON and the table. It checks that grouped duplicates carry one judgement.

Sources (all accessed 2026-09-27)

Laura's words:

- Overachiever Magazine, 2021-03-15: <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>
- Wharton Magazine, 2022-04-06: <https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/>
- Poets&Quants, 2018-03-30: <https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/>
- The Nerd Daily, 2025-03-08: <https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/>
- ABA Bookweb, 2022-02-07: <https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512>
- Input/Inverse, 2022-03-07: <https://www.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel> (the Phase B links www.inputmag.com/... and nc.inverse.com/... serve the same text)
- Geeks OUT, 2022-05-11: <https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/>

- The Honey POP, 2022-06-02: <https://thehoneypop.com/2022/06/02/exclusive-interview-laura-gao-on-messy-roots-lgbtq-representation-and-more/>
- Pulse Spikes, 2021-06-18: <https://pulsespikes.org/story/laura-gao>
- Local News Matters, 2026-02-12: <https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/>
- The Maverick Show ep. 190, 2022-06-23: <https://www.themaverickshow.com/podcast/190-messy-roots-queer-identity-the-beauty-of-wuhan-and-writing-a-graphic-memoir-while-traveling-the-world-with-laura-gao/> and its transcript https://www.themaverickshow.com/wp-content/uploads/2025/10/190_Laura_Gao_Transcript.pdf
- NPR, 2020-04-04: <https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines>
- NPR, 2022-04-24: <https://www.npr.org/sections/goatsandsoda/2022/04/24/1093992912/the-pandemic-inspired-a-cartoonist-to-explore-their-wuhanese-roots-and-queer-ide>
- HuffPost, 2020-04-29: https://www.huffpost.com/entry/chinese-american-illustrator-life-in-wuhan_n_5ea9c739c5b63115cec2be5a
- The Daily Pennsylvanian, 2016-01-28: <https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a>
- Her site and documents: <https://lauragao.com/hey/>, <https://lauragao.com/editorial>, <https://lauragao.com/s/Laura-Gao-Resume.pdf>, <https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf>, <https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf>, <https://lauragao.com/rewriting-herstory>, <https://lauragao.com/the-evolution-of-dance-music>, <https://lauragao.com/philly-happy-hours>, <https://signaltemp.github.io/>

Also read for the pull-quote search:

- Laura's site: <https://lauragao.com/>, <https://lauragao.com/comics-and-stories/>, <https://lauragao.com/projects>, <https://lauragao.com/speaking-events>, <https://lauragao.com/kirbys-lessons-for-falling-in-love>
- Other pages about her: <https://www.playbook.com/blog/laura-gao-imagination-fund-recipient-spotlight/>, <https://portal.cca.edu/people/lauragao/>, <https://www.commonscomics.com/2022/messy-roots-by-laura-gao/>, https://en.wikipedia.org/wiki/Laura_Gao, https://www.goodreads.com/author/quotes/20531180.Laura_Gao, <https://magazine.wharton.upenn.edu/digital/whartons-pitch-to-amazon/>
- Competition pages: <https://wghsinvcomp.smapply.us/res/p/client/>, <https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/>
- Official case: `competition/official/2026_27/Laura_Gao_2026_Client_Profile.{txt,pdf}` (repo; SHA-256 in `manifest.yaml`).

What this teaches

- **A quote is three facts, not one:** the words, who said them, and what question they answered. The script got the words right for all 104 ids. Reading the page around them changed the status of 11 ids and attached a condition of use to about 20 more (for example "not about investing", "a question, not a statement", "do not quote the sentence before").
- **Short fragments are where meaning slips.** "risk-adverse", "No more gimmicks." and "It's not published in Mandarin" all mean something different once you read the sentence they came from.
- **A pull quote with no name is not a quote from the person.** Cite the document that printed it (the case), and look for the person's own words to stand beside it.
- **Public does not mean usable.** An interview can be published and accurate and still be about someone's family or private life. Writing about a real client means using only her professional record.
- **A transcript is a copy of speech, not the speech.** When the copy has obvious errors, treat its exact wording as unconfirmed until someone checks the audio.
- **Dates matter.** Several of her most quotable lines come from her student years (2016-2018). Saying so is more honest, and more convincing, than presenting them as what she thinks today.

D13b: verification of all non-Laura quotes (Phase B)

Access date for every check: 2026-09-27. Agent: D13b (Quote Verifier, "other" quotes). Companion file: [D13b_other_quotes_verified.json](#) (same rows, machine-readable).

Summary

- **Scope.** 171 of the 271 Phase B quotes are not Laura Gao's own words. They come from Wharton pages and guides, regulators, researchers, industry sources, news reporters, publishers, past clients, judges and past competitors. The case's unattributed pull quote is also in scope. The other 100 quotes are Laura's own words and belong to D13a.
- **Result.** All 171 have their wording confirmed verbatim on a primary page or official repo file (VERIFIED-PRIMARY). None is WRONG, and no wording needed repair beyond two small fixes (B5b-Q09, B2b-Q05). This status covers the wording and source only. Six quotes carry an attribution warning (below) that matters more than the wording.
- **The 21 quotes the mechanical check could not confirm** (11 NOT_FOUND, 6 NO_URL, 4 FETCH_FAILED) were all resolved as follows:
 - **Hidden or retired Wharton pages (5 + 2 Scribd).** B2b-Q01 to Q05 and the two Scribd quotes B1b-Q08/Q09 were read verbatim through Wharton's own WordPress REST API (`/wp-json/wp/v2/pages?slug=...`). The two Scribd quotes move up from SNIPPET-UNVERIFIED; cite the Wharton page, not Scribd.
 - **Blocked publishers (3).** PNAS and SAGE returned 403. The three abstracts (B6b-Q06/Q08/Q09) are verbatim in Europe PMC.
 - **SEC (1).** sec.gov returned 403. The phrase in B6b-Q18 is verbatim in the rule text on eCFR and Cornell LII, so it moves up from SNIPPET-UNVERIFIED.
 - **EUR-Lex (1).** EUR-Lex returned an empty bot-challenge page. B4a-Q09 is verbatim on legislation.gov.uk (Solvency II, Art. 101(3)).
 - **PDF line-break artefacts (2).** B4b-Q03 (CFA, "differ - ent") and B6b-Q02 (IPCC, "90- 95%") failed only because of line breaks in the PDF text.
 - **Wrong link type (1).** B7a-Q04 used a Box viewer link. The direct PDF link and the repo infographic file hold the text.
 - **Repo files (6).** The Trading Notes sample note (B5b-Q09, B10b-Q04) and the pull quote (B2a-Q13, B2b-Q11, B8b-Q01, B8a-Q39) are in the official case/guide text files.

Flags: past clients, judges and attribution (read before quoting any of these)

id	who	verdict
B2a-Q02	Past client Nichole Jordan (2022)	Not her words. Wharton's writer paraphrased her; only the word "seen" is hers. Say "Wharton reported..."
B2a-Q06	Past client Ladi Ayoola (2025 finale)	Confirmed direct quote.
B2a-Q08, B2b-Q02	Past client Connor Barwin (2025-26)	Confirmed. Wharton put an explicit name line under his pull quote.
B2a-Q04	Judge Eric Balchunas (2024 finale)	Not his words. The article paraphrases him (no quotation marks) and misspells his name as "Balchunus". Never write "Balchunas said".
B2a-Q03	Judge Andrea Vittorelli (2022 finale)	Confirmed. He was talking about finale slides and presentations.
B2a-Q05, B7a-Q13	Judge Zoe McCormick (2024 finale)	Confirmed. "and consider their total financial picture" is the writer's addition, outside the quote.
B2a-Q07	Judge Chirag Jain (2025 finale)	Confirmed.
B7a-Q12	Judge Vikas Keswani (2024 finale)	Confirmed. He is talking about analysis in finale presentations.
B2a-Q09	Semifinal judge Joshua Tam (2024)	Confirmed. Most relevant to our round.
B7a-Q06, B7b-Q06	Semifinal judge Melissa Ko Hahn (2025)	Confirmed. She talks about written reports; most relevant to our round.
B2a-Q13, B2b-Q11, B8b-Q01, B8a-Q39	Case pull quote "The only person who needs to believe in something is yourself."	Wording confirmed; speaker unknown. It sits in the "MEET LAURA GAO" sidebar (repo txt lines 31-33) with no attribution line. Past cases always named the speaker (" Nichole Jordan", " Connor Barwin"). An exact-phrase WebSearch found only generic self-help pages, and greps of 5 Laura interview pages found nothing. Cite it as the case's words, never as Laura's. D13a owns the final call.

No quote attributed to a past client or judge was invented. The two risks are the paraphrases (B2a-Q02, B2a-Q04) and the unattributed pull quote.

Other findings worth carrying forward

1. **The old "sector minimum = team size" rule was real, but in past seasons only.** The retired 2023-24 judging page (read via the REST API, B2b-Q03) says "portfolio is invested in at least as many sectors as there are team members". Alex Lamon's 2021 blog (B7a-Q16) repeats it. The 2026-27 rules have no such minimum (brief section 14). This explains where the third-party claim came from.
2. **Reporters' sentences about Laura are verbatim, but they are not her voice.** This covers B2b-Q06, B8a-Q16, B10a-Q03, B8b-Q27/28 (Input/Inverse), B2b-Q09 (Wharton Magazine), B3a-Q01/B3b-Q01/B3b-Q02/B8b-Q29 (Publishers Weekly) and B9b-Q15 (publisher copy). The Inverse paragraph around B2b-Q06 holds personal-life material. Do not reuse it (privacy rule).
3. **One context correction.** B7b-Q07 (KXAN): the Vandegrift team were 2026 Global Finalists ("headed to the global finals"), not only semifinalists. The "70% chance" line is the reporter describing the team's model; it says nothing about what judges reward.
4. **Truncation.** B5b-Q09 stops mid-sentence ("...and begin"). Use the full guide sentence (= B10b-Q04).
5. **Secondary sources where the primary document was not read.** Taipei Times on MSCI (B5b-Q06); Focus Taiwan on the U.S. Treasury FX report (B5b-Q07); Fortune on S&P (B5a-Q06); the ScienceDaily press release on Gneezy & Epley (B6b-Q11); EWC's summary of the Society of Authors survey (B3a-Q08). The wording is verified, but the underlying fact is second-hand. Say so when citing.
6. **Old-season Wharton text sitting on live pages.** This covers the 2021-22, 2023-24, 2024-25 and 2025-26 cases, the judging and deliverables pages, and the "Developing a Strategy" page. Old rules (approved stock list, mid-term report, WGAM audience) appear in them. Cite them as history, never as 2026-27 rules.
7. **Duplicates.** Several ids repeat the same sentence: B1a-Q03 = B1b-Q05; B1b-Q06 = B7b-Q05 = B7a-Q05; B3a-Q07 = B3b-Q10; B6a-Q08 = B6b-Q12; B4b-Q04 = B10a-Q04; the Guide p.3/p.5 lines; the pull quote x4; the Inverse sentence x3.

Source-type mix (171 quotes)

Exact counts (the JSON rows carry the fine-grained labels):

- official Wharton: 60 (2026-27 Guide/Infographic/rules pages plus past-season pages and news)
- official regulator, government, central bank or standard setter: 21 (Fed, U.S. Treasury, UK TPR, APRA, FCA, SEC/Investor.gov, FASB, IPCC, Taiwan county government, Solvency II, Yale, Vanderbilt)
- peer-reviewed or research working papers: 20
- industry: 46
- secondary (news, press releases, blogs, student work): 24

Method

1. Loaded [phase_B/quotes_raw.json](#) and [phase_D/D13_mechanical_check.json](#). Classified the speaker of each quote from its text and context. Quotes by Laura herself (100) went to D13a; everything else (171) is here.
2. Script [research/insight_v1/scripts/D13b_other_quotes_context.py](#) re-fetched every page with the fixed helper and saved a 350-character window around each quote ([phase_D/_work/D13b_contexts.json](#)). Every window was read to check speaker, context and trimming. Named speakers and dates were confirmed on the page (e.g. Vittorelli, Balchunas, McCormick, Hahn, Owens, all paper authors).

3. Re-tried all 21 non-VERBATIM quotes, using Wharton REST API, legislation.gov.uk, Europe PMC, eCFR, Cornell LII, the direct Box PDF and repo files.

4. Script `research/insight_v1/scripts/D13b_build_outputs.py` records the judgements and writes the JSON and the table below.

Status meaning here: VERIFIED-PRIMARY = the exact wording was read on the primary page or official repo file on 2026-09-27. For repo files this equals VERIFIED-REPO-FILE. "Status note" says where attribution is weaker than the wording.

Full table (171 rows)

id	exact text	speaker / source	source type	URL (where verified)
B1a-Q01	The members of your team hope to one day become portfolio managers who make the final investment decisions for WGAM's portfolio.	Wharton Global Youth, 2021-22 case study (client Nichole Jordan)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/
B1a-Q02	Starting in 2022, Nichole would like to provide a small scholarship of \$5,000 per year to a woman of color studying engineering at her alma mater, U.C. Davis. Nichole needs your help determining how best to fund this, as she wants to be able to offer the scholarship for at least 10 years. Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends?	Wharton Global Youth, 2021-22 case study (client Nichole Jordan)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/
B1a-Q03	Ladi believes it will cost about \$10,000 USD per housing unit; however, material costs and market dynamics are always changing.	Wharton Global Youth, 2024-25 case study (client Ladi Ayoola)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/
B1a-Q04	Barwin's case encouraged students to think beyond financial returns and consider how their strategies could support long-term community impact.	Wharton Global Youth news (Kara Dunn), 2026-03-20	official-Wharton	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition
B1a-Q05	7-year note auctions are usually announced in the second half of each month and generally auctioned a few business days later. They are issued on the last day of the month.	U.S. Treasury (TreasuryDirect)	official-government	https://www.treasurydirect.gov/auctions/general-auction-timing/
B1a-Q06	The iShares® iBonds® ETFs ("Funds") will terminate on or about October or December 15 of the year in each Fund's name.	BlackRock iShares product page (IBTR)	industry (issuer primary)	https://www.ishares.com/us/products/350028/ishares-ibonds-dec-2036-term-treasury-etf
B1b-Q01	You may identify decisions you would make differently, but you may not redesign your strategy after observing the results.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B1b-Q02	When your team executes a trade in WInS, the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf

id	exact text	speaker / source	source type	URL (where verified)
B1b-Q03	Your team will need to consider the client's goals, time horizons, required cash flows, funding commitments, liquidity needs, desired degree of funding certainty, risk tolerance, and communication with potential co-sponsors.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkzknk1xv5yegl3snlc1.pdf
B1b-Q04	There is no single correct strategy. Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions.	Wharton, 2026-27 Competition Infographic	official-Wharton (2026-27)	https://upenn.box.com/shared/static/l2p0l26svbbptmcizhmswyq5f7rbyp44.pdf
B1b-Q05	Ladi believes it will cost about \$10,000 USD per housing unit; however, material costs and market dynamics are always changing.	Wharton Global Youth, 2024-25 case study (client Ladi Ayoola)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2024-2025/
B1b-Q06	with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia	Wharton Global Youth news, 2026-01-27	official-Wharton	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investme
B1b-Q07	The Wharton Global Youth team is doing some reimagining of our signature competition	Wharton Global Youth news, 2025-05-01	official-Wharton	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-glob
B1b-Q08	Every dollar invested reflects MTWB's values — community-first, locally rooted, and sustainability-minded.	Wharton 2025-26 case study (client Connor Barwin)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study
B1b-Q09	Remember, your 10 weeks on WinS can only accurately assess the performance of short-term investments.	Wharton 2025-26 case study (client Connor Barwin)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study
B1b-Q10	above 25% (equivalent to three months) is better	Nonprofit Accounting Basics (industry guidance site)	industry	https://www.nonprofitaccountingbasics.org/internal-reporting/operating-reserve-ratio
B2a-Q01	Should she keep \$50,000 in cash? Should she invest in stocks that pay dividends? Another option?	Wharton Global Youth, 2021-22 case study (client Nichole Jordan)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/investment-competition/previous-winners/case-study-for-2021-2022/

id	exact text	speaker / source	source type	URL (where verified)
B2a-Q02	she had never felt more "seen" in her life	Wharton writer (Diana Drake) paraphrasing past client Nichole Jordan; only the word 'seen' is hers	official-Wharton	https://globalyouth.wharton.upenn.edu/news/top-teams-sail-to-success-in-the-2022-wharton-investment-competition-global-final
B2a-Q03	Cut slides, cut words, cut minutes	Judge Andrea Vittorelli (Wharton alum; global chairman, J.P. Morgan Insurance Investment Group), 2022 Global Finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/top-teams-sail-to-success-in-the-2022-wharton-investment-competition-global-final
B2a-Q04	to consider behavioral finance and their clients' emotions during a market downturn	Wharton writer paraphrasing judge Eric Balchunas (senior ETF analyst, Bloomberg Intelligence), 2024 finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024
B2a-Q05	take a look at this person holistically	Judge Zoe McCormick (senior investment manager, North American fixed income, abrdn), 2024 finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024
B2a-Q06	not just about the numbers, but about understanding and connecting with people	Past client Ladi Ayoola, 2025 Global Finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-glob
B2a-Q07	What distinguished managers was clear thoughtfulness and reasoning behind all the decisions that they put together.	Judge Chirag Jain (U.S. credit research analyst, Aberdeen Standard Investments), 2025 Global Finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/bam-investing-from-deerfield-academy-massachusetts-wins-the-2025-wharton-glob
B2a-Q08	approach every problem with confidence that you can figure it out, and humility to know that you don't have all the answers	Past client Connor Barwin (2025-26), pull quote with attribution	official-Wharton	https://globalyouth.wharton.upenn.edu/news/11-teams-advance-to-the-2026-wharton-global-high-school-investment-competition
B2a-Q09	The one thing that stood out to me was the ability to connect the client's objectives to an investment strategy.	Semifinal judge Joshua Tam (portfolio manager, Laurel Avenue Management), 2024	official-Wharton	https://globalyouth.wharton.upenn.edu/news/wharton-global-youth-hosts-five-virtual-events-to-announce-the-2024-investment-c
B2a-Q10	achieve specific financial goals and meet future cash-flow needs	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf

id	exact text	speaker / source	source type	URL (where verified)
B2a-Q11	his M.A. in Statistics and Ph.D. in Economics from the University of California at Berkeley	Wharton competition main page	official-Wharton (2026-27)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/
B2a-Q13	The only person who needs to believe in something is yourself.	Wharton 2026-27 case, p.1 sidebar pull quote (NO attribution line)	official-Wharton (2026-27, repo file)	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (repo)
B2b-Q01	Your team hopes to shape the winning investment strategy that Connor and the Make the World Better foundation ultimately choose to make a lasting community impact from an initial \$500,000 investment.	Wharton 2025-26 case study (hidden section of live page)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study
B2b-Q02	Wharton made me think more about the sustainability of a nonprofit, and the importance of building strong partnerships to keep the work going.	Past client Connor Barwin, pull quote in the 2025-26 case	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study
B2b-Q03	Client knowledge and objectives: strategy is tailored uniquely to the client's financial goals; would win him/her over as a client.	Wharton judging page (2023-24 version, retired)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation
B2b-Q04	Don't use precious space repeating known facts about your client and defining concepts. However, it is okay to demonstrate that you understand your client beyond the case study.	Wharton judging page (2023-24 version, retired)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation

id	exact text	speaker / source	source type	URL (where verified)
B2b-Q05	Final Report Audience: Your report should be directed to Wharton Global Asset Management's (WGAM) portfolio manager (your team's teacher/advisor).	Wharton deliverables page (2023-24 version, retired)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=deliverables
B2b-Q06	So when she got the book deal with HarperCollins, she gave Twitter her notice.	Input/Inverse reporter (not Laura's words)	secondary (news feature)	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel
B2b-Q09	Gao asked city officials what problem areas they needed help with, and these problem areas became the basis for the case competition's four subtopics	Wharton Magazine writer (not Laura's words), 2017-10-30	secondary (Wharton Magazine)	https://magazine.wharton.upenn.edu/digital/whartons-pitch-to-amazon/
B2b-Q11	The only person who needs to believe in something is yourself.	Wharton 2026-27 case, p.1 sidebar pull quote (NO attribution line)	official-Wharton (2026-27, repo file)	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (repo)
B3a-Q01	Donna Bray at HarperCollins/Balzer + Bray has acquired, at auction, world English rights to debut YA graphic memoir The Wuhan I Know, plus a second graphic work, by Laura Gao.	Publishers Weekly Rights Report, July 2020	secondary (trade press)	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.ht
B3a-Q05	Vanderbilt and the California College of the Arts have agreed for Vanderbilt to acquire the campus of CCA after the wind-down of CCA's operations in 2027	Vanderbilt University news release, 2026-01-13	official (institution press release)	https://news.vanderbilt.edu/2026/01/13/vanderbilt-university-to-establish-full-time-academic-campus-in-san-francisco/
B3a-Q06	The median author income for full-time authors from their books was \$10,000 in 2022, and their total median earnings from their book and other author-related income combined was \$20,000.	Authors Guild	industry (trade association)	https://authorsguild.org/news/key-takeaways-from-2023-author-income-survey/
B3a-Q07	The HarperCollins licensing deal provides for a \$5,000 fee per title, which will be split 50-50 between an author who chooses to participate and HarperCollins.	Authors Guild	industry (trade association)	https://authorsguild.org/news/harpercollins-ai-licensing-deal/
B3a-Q08	A quarter of illustrators (26%) and over a third of translators (36%) have already lost work due to generative AI.	European Writers' Council Society of Authors survey	secondary (summary of industry survey)	https://europeanwriterscouncil.eu/soa-survey-uk-ai-2024/
B3a-Q09	Book sales fell 1.8% in 2009, to \$23.86 billion, according to estimates released last week by the Association of American Publishers.	Publishers Weekly (Jim Milliot), 2010-04-12	secondary (trade press)	https://www.publishersweekly.com/pw/by-topic/industry-news/bookselling/article/42783-aap-book-sales-dipped-in-2009.html

id	exact text	speaker / source	source type	URL (where verified)
B3a-Q10	ICv2 estimates that sales of comics and graphic novels in the U.S. and Canada totaled \$2.2 billion in 2025, outstripping the highs of the COVID era.	ICv2 (Milton Griepp)	industry (market research)	https://icv2.com/articles/news/view/61573/icv2-white-paper-comics-graphic-novel-sales-reached-new-high-2025
B3a-Q11	The grant reinstatements come as a direct result of a November 21 federal court decision in a lawsuit brought against President Trump by 21 states.	American Library Association press release, 2025-12-03	industry (association press release)	https://www.ala.org/news/2025/12/ala-welcomes-reinstatement-all-federal-ims-grants-libraries
B3b-Q01	Donna Bray at HarperCollins/Balzer + Bray has acquired, at auction, world English rights to debut YA graphic memoir The Wuhan I Know, plus a second graphic work, by Laura Gao.	Publishers Weekly Rights Report, July 2020	secondary (trade press)	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.ht
B3b-Q02	The second graphic work is about a queer girl's many loves.	Publishers Weekly Rights Report, July 2020	secondary (trade press)	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.ht
B3b-Q05	Graphic novelists ranked second, earning a median book income of \$15,000 and a combined income of \$25,000 when including other author-related income.	Authors Guild	industry (trade association)	https://authorsguild.org/news/key-takeaways-from-2023-author-income-survey/
B3b-Q06	on signing the contract, three to six months after contract signing, on delivery of the manuscript, on acceptance of the manuscript, upon publication date, and one part 12 months after publication	Authors Guild	industry (trade association)	https://authorsguild.org/blog/everything-you-need-to-know-about-book-advances/
B3b-Q07	Total comics and graphic novel sales to consumers in the U.S. and Canada were around \$1.87 billion in 2023, down 7% from \$2.01 billion in 2022	ICv2 (Milton Griepp)	industry (market research)	https://icv2.com/print/article/57351
B3b-Q08	The ability to vary labor supply ex post induces the individual to assume greater risks in his investment portfolio ex ante.	Bodie, Merton & Samuelson, NBER WP 3954 (1992) abstract	research (working paper; later peer-reviewed JEDC 1992)	https://www.nber.org/papers/w3954
B3b-Q10	The HarperCollins licensing deal provides for a \$5,000 fee per title, which will be split 50-50 between an author who chooses to participate and HarperCollins.	Authors Guild	industry (trade association)	https://authorsguild.org/news/harpercollins-ai-licensing-deal/
B3b-Q11	On March 14, President Trump issued Executive Order 14238, which directed the elimination of the agency.	American Library Association press release, 2025-12-03	industry (association press release)	https://www.ala.org/news/2025/12/ala-welcomes-reinstatement-all-federal-ims-grants-libraries
B3b-Q12	When her dynamic move at the Texas Youth Fall Invitational goes awry	Publishers Weekly review of Kirby's Lessons	secondary (trade review)	https://www.publishersweekly.com/9780063067806
B3b-Q14	will close permanently in 2027	The Art Newspaper (Benjamin Sutton), 2026-01-13	secondary (news)	https://www.theartnewspaper.com/2026/01/13/california-college-arts-closing-vanderbilt-university-takeover
B4a-Q01	For schemes with no cash balance benefits, the duration is 10 years.	UK Pensions Regulator, DB funding code	official-regulator	https://www.thepensionsregulator.gov.uk/-/media/thepensionsregulator/files/import/pdf/db-funding-code-of-practice.ashx

id	exact text	speaker / source	source type	URL (where verified)
B4a-Q02	we expect schemes to target a minimum level of interest rate and inflation hedging of at least 90% on a low dependency funding basis.	UK Pensions Regulator, DB funding code	official-regulator	https://www.thepensionsregulator.gov.uk/-/media/thepensionsregulator/files/import/pdf/db-funding-code-of-practice.ashx
B4a-Q03	a 'shortfall limit' is the extent to which an RSE licensee considers that a fund can be in an unsatisfactory financial position with the RSE licensee still being able to reasonably expect that, because of corrections to temporary negative market fluctuations in the value of fund assets, the fund can be restored to a satisfactory financial position within one year.	APRA Prudential Standard SPS 160	official-regulator	https://handbook.apra.gov.au/standard/sps-160
B4a-Q04	the Government won't start any drawdowns from the Fund until at least 2032-33	Australian Treasurer media release	official-government	https://ministers.treasury.gov.au/ministers/jim-chalmers-2022/media-releases/future-future-fund
B4a-Q05	To determine the year's spending, we apply the targeted spending rate (5.25%) to the endowment's year-end value from two years ago.	Yale Office of the Provost	official (institution)	https://provost.yale.edu/how-exactly-does-spending-policy-work
B4a-Q06	The Verification Agent is responsible for independently confirming that the investments purchased for the Escrow Fund will be sufficient to fund the Debt Service payments on the Refunded Bonds	National Association of Bond Lawyers	industry (professional association)	https://www.nabl.org/bond-basics/verification-agent/
B4a-Q07	Hibernation strategies will burn through capital over time, squandering hard-won pension funding gains.	J.P. Morgan Asset Management, 'Rethinking the pension plan endgame' (Oct 2021)	industry	https://am.jpmorgan.com/content/dam/jpm-am-aem/global/en/institutional/insights/portfolio-insights/portfolio-strategy/Rethinking
B4a-Q08	taxable-bond and municipal-bond fund investors earned the smallest share of their funds' aggregate total returns, capturing only about half.	Morningstar, Mind the Gap 2025	industry	https://www.morningstar.com/content/cs-assets/v3/assets/blt9415ea4cc4157833/blt2c5c4d9171638c42/689b424311f3880edc4b
B4a-Q09	confidence level of 99,5 % over a one-year period.	EU Solvency II Directive 2009/138/EC, Art. 101(3)	official-legislation	https://www.legislation.gov.uk/eudr/2009/138/article/101
B4a-Q10	Because the investment return assumption for most plans also serves as the discount rate used to determine the present value of the plan's liabilities	NASRA issue brief	industry (association)	https://www.nasra.org/files/Issue%20Briefs/NASRAInvReturnAssumptBrief.pdf

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B4b-Q01	Perhaps most importantly, the IPS serves as a policy guide that can offer an objective course of action to be followed during periods of market disruption when emotional or instinctive responses might otherwise motivate less prudent actions.	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B4b-Q02	If the policy is not to rebalance, this policy should be documented in the IPS.	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B4b-Q03	Inappropriate use of different metrics to highlight or disguise certain risks should be avoided.	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B4b-Q04	lock down the benefits promised, then grow surplus assets with a prudent process and disciplined understanding of risk.	Russell Investments (Justin Owens), 2026-07-13	industry	https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.ht
B4b-Q05	Downgrades and defaults: The most material headwind facing hedge strategies is the asymmetric impact of downgrades and defaults on assets and liabilities.	J.P. Morgan Asset Management, 'Rethinking the pension plan endgame' (Oct 2021)	industry	https://am.jpmorgan.com/content/dam/jpm-am-aem/global/en/institutional/insights/portfolio-insights/portfolio-strategy/Rethinking
B4b-Q06	a random fluctuation in asset values can be turned into a permanent loss of capital.	Jean L.P. Brunel, CFA Institute Conference Proceedings Quarterly (Mar 2012)	industry (professional proceedings)	https://static1.squarespace.com/static/59e8d89d914e6b37450c946a/t/5c643e15104c7b43f7b84187/1550073366884/CFA+Goals
B4b-Q07	For those things you must do, the emphasis is going to be on lowering the risk of your not achieving these goals,	Ashvin Chhabra, quoted on CFA Institute Enterprising Investor blog	secondary (blog quoting practitioner)	https://blogs.cfainstitute.org/investor/2015/08/05/ashvin-chhabra-on-wealth-management-the-value-proposition-lies-in-understan
B4b-Q08	By 'capacity for loss' we refer to the customer's ability to absorb falls in the value of their investment.	UK FSA/FCA Finalised Guidance FG11/05 (Mar 2011)	official-regulator	https://www.fca.org.uk/publication/finalised-guidance/fsa-fg11-05.pdf
B4b-Q09	Ultimately, the key point is that outcomes, not probabilities, are what matter to clients	Kitces.com (Michael Kitces et al.)	industry (practitioner blog)	https://www.kitces.com/blog/monte-carlo-guardrails-probability-of-adjustment-success-client-communication-dynamic-retirement
B4b-Q10	However, we find that, using the same sample, poor timing by mutual fund investors costs them only 0.10% per year.	Fulkerson, Jordan, Riley & Yan, Financial Analysts Journal 82(3), 2026	peer-reviewed	https://rpc.cfainstitute.org/research/financial-analysts-journal/2026/bad-timing-does-not-cost-investors-funds-returns
B4b-Q11	achieve and balance the three objectives of: • maximising expected retirement income; • managing expected risks to the sustainability and stability of retirement income; and • having flexible access to expected funds during retirement.	APRA/ASIC information report, July 2023	official-regulator	https://www.apra.gov.au/sites/default/files/2023-07/Information%20report%20-%20Implementation%20of%20the%20retirement%20
B5b-Q01	Productivity growth is strong, and capital investment is robust.	FOMC statement, 2026-09-16	official-central bank	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm

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B5b-Q02	The forward 12-month P/E ratio for the S&P 500 is 19.1.	FactSet Earnings Insight	industry (data provider)	https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_0911
B5b-Q03	Incremental annual debt rose from 9% of capex in FY24 to 32% LTM by mid-2026	FactSet Insight article	industry (data provider)	https://insight.factset.com/hyperscalers-tap-external-financing-as-ai-capex-outruns-cash-flow
B5b-Q04	at a price that is effectively zero and as of the close of March 9, 2022	MSCI press release, March 2022	industry (index provider primary)	https://www.msci.com/eqb/pressreleases/archive/PR_Russia_Classification.pdf
B5b-Q05	Moody's Ratings (Moody's) has downgraded the Government of United States of America's (US) long-term issuer and senior unsecured ratings to Aa1 from Aaa	Moody's Ratings U.S. rating page	industry (rating agency primary)	https://www.moody's.com/web/en/us/about-us/usrating.html
B5b-Q06	Taiwan's weighting in the MSCI All Country World Index is set to rise to 3.13 percent from 3.08 percent, while its weighting in the MSCI Emerging Markets Index is to increase to 26.85 percent from 26.6 percent.	Taipei Times (Crystal Hsu), 2026-08-14	secondary (news)	https://www.taipeitimes.com/News/biz/archives/2026/08/14/2003862448
B5b-Q07	Taiwan's current account surplus grew to 19.6 percent of GDP in 2025, compared to 14.1 percent in 2024	CNA / Focus Taiwan	secondary (news agency)	https://focustaiwan.tw/business/202607240006
B5b-Q08	agreed to raise the monthly minimum wage by 4.745 percent, from NT\$29,500 (US\$927.41) to NT\$30,900	CNA / Focus Taiwan	secondary (news agency)	https://focustaiwan.tw/society/202609240022
B5b-Q09	We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin	Wharton 2026-27 Trading Notes Analysis guide, sample note (p.2)	official-Wharton (2026-27, repo file)	competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt (repo)
B5a-Q01	The Committee will deliver price stability.	FOMC statement, 2026-09-16	official-central bank	https://www.federalreserve.gov/newsevents/pressreleases/monetary20260916a.htm
B5a-Q02	Under historical excess cost growth in healthcare, this outer limit is likely reached within 20 years; there is a 25% chance of reaching it in 14 years.	Penn Wharton Budget Model (Smetters & He), 2026-06-04	research (university policy model, not peer-reviewed)	https://budgetmodel.wharton.upenn.edu/p/2026-06-02-when-does-federal-debt-reach-unsustainable-levels/
B5a-Q03	default of either explicit (Treasury) debt or implicit (pay-as-you-go) debt becomes a near certainty on a real (inflation-adjusted) basis	Penn Wharton Budget Model (Smetters & He), 2026-06-04	research (university policy model, not peer-reviewed)	https://budgetmodel.wharton.upenn.edu/p/2026-06-02-when-does-federal-debt-reach-unsustainable-levels/
B5a-Q04	The X-Date is likely to be reached in February 2028, when projected cash on hand and available extraordinary measures would be exhausted.	National Taxpayers Union	secondary (advocacy analysis)	https://www.ntu.org/publications/detail/projecting-the-debt-limit
B5a-Q05	relatively less likely to meet its debt obligations on time and in full than previously expected	Moody's Ratings U.S. rating page	industry (rating agency primary)	https://www.moody's.com/web/en/us/about-us/usrating.html
B5a-Q06	All three major ratings agencies have the US pegged one level below Triple-A with stable outlooks	Fortune, 2026-06-27	secondary (news)	https://fortune.com/2026/06/27/sp-global-us-sovereign-credit-rating-aa-stable-outlook-debt-deficits/

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B5a-Q07	Treasury anticipates maintaining nominal coupon and FRN auction sizes for at least the next several quarters.	U.S. Treasury Quarterly Refunding Statement, 2026-08-05	official-government	https://home.treasury.gov/news/press-releases/sb0590
B6a-Q01	Typically, we suggest at least 20 percent and sometimes as much as 50 percent of your private fundraising goal be raised before applying.	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6a-Q02	Written pledges or cash from individuals, corporations, businesses and foundations, including those resulting from direct appeals and unsolicited gifts. All pledges must be paid within five years from your fundraising end date.	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6a-Q03	Do you have a plan for sustaining support gained during the campaign after it concludes?	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6a-Q04	committed, imminent or backstopped	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6a-Q05	A challenge gift is an unconditional commitment by a donor, or set of donors, to provide a given sum of money to the cause.	Rondeau & List, NBER WP 13728 (2008)	research (working paper; later peer-reviewed)	https://www.nber.org/system/files/working_papers/w13728/w13728.pdf
B6a-Q06	We find evidence that challenge gifts positively influence contributions in the field, but matching gifts do not.	Rondeau & List, NBER WP 13728 (2008)	research (working paper; later peer-reviewed)	https://www.nber.org/papers/w13728
B6a-Q07	If charitable organizations can use lead gifts as they wish, our results show that they maximize donations given by simply announcing the presence of a lead gift.	Huck & Rasul, J. Public Economics 95(5) 2011, abstract (RePEc)	peer-reviewed	https://ideas.repec.org/a/eee/pubeco/v95y2011i5p351-362.html
B6a-Q08	larger match ratios (i.e., \$3:\$1 and \$2:\$1) relative to smaller match ratios (\$1:\$1) had no additional impact	Karlan & List, NBER WP 12338 abstract	research (working paper; published AER 2007)	https://www.nber.org/papers/w12338
B6a-Q09	A change in seed money from 10% to 67% of the campaign goal resulted in nearly a sixfold increase in contributions, while imposing a refund increased contributions by a more modest 20%.	List & Lucking-Reiley, abstract on RePEc (JPE 2002)	peer-reviewed (abstract)	https://ideas.repec.org/p/feb/natura/00301.html
B6a-Q10	specified an intention to give. The ability to modify a will at any time prior to death is well established; thus in 19X0 Church F did not receive a promise to give and did not recognize a contribution received.	FASB ASU 2018-08	official-standard setter	https://storage.fasb.org/ASU%202018-08.pdf
B6a-Q11	A challenge (matching) grant is a common form of a conditional promise to give.	FASB ASU 2018-08	official-standard setter	https://storage.fasb.org/ASU%202018-08.pdf

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B6a-Q12	The present value of the future cash flows (an "income approach" under ASC 820) is the valuation technique typically used when measuring the fair value of a contribution due more than one year from the date of the promise.	PwC Viewpoint NFP guide 7.3.2.2	industry (accounting firm)	https://viewpoint.pwc.com/dt/us/en/pwc/accounting_guides/not-for-profit-entities/Not-for-profit-entities/nfp07/7_3_unconditional.h
B6a-Q13	文化部為扶植國內藝術村營運發展，每年依據旨揭要點，評選補助國內藝術村單位	Lienchiang County Government notice of Taiwan Ministry of Culture subsidy call (2021)	official-government (Taiwan)	https://www.matsucc.gov.tw/%E6%96%87%E5%8C%96%E9%83%A8%E3%80%8C%E8%A3%9C%E5%8A%A9%E8%97%9D%E8%A1%93%E6%9D%91%E7%87
B6a-Q14	whether the sources of potential revenue are reliable	PKF O'Connor Davies, due-diligence guide (July 2019)	industry (accounting firm)	https://www.pkfod.com/wp-content/uploads/2019/07/Best-Practices-Due-Diligence-on-Grantees-v4.pdf
B6b-Q01	laypeople interpret IPCC statements as conveying probabilities closer to 50% than intended by the IPCC authors. We show that an alternative presentation format supplementing the verbal terms with numerical ranges increases the correspondence between the public's interpretations and the IPCC guidelines	Budescu, Por, Broomell & Smithson, Nature Climate Change 4 (2014)	peer-reviewed	https://www.nature.com/articles/nclimate2194
B6b-Q02	When there is sufficient information, it is preferable to specify the full probability distribution or a probability range (e.g., 90-95%) without using the terms in Table 1.	IPCC AR5 Guidance Note on uncertainties (2010)	official (intergovernmental body)	https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf
B6b-Q03	Consider reciprocal statements to avoid value-laden interpretations (e.g., report chances both of dying and of surviving).	IPCC AR5 Guidance Note on uncertainties (2010)	official (intergovernmental body)	https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf
B6b-Q04	experts need to specify the reference class, that is, the class of events to which a single-event probability refers.	Gigerenzer et al., Risk Analysis 25 (2005), abstract via Europe PMC	peer-reviewed	https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=TITLE%3A%2230%25%20chance%20of%20rain%22&form
B6b-Q05	When the upper limit of a range is described as a value that "can" occur (Experiment 5), outcomes both below and beyond this value were regarded as consistent with the forecast.	Teigen, Löhre & Hohle, Judgment and Decision Making 13(4)	peer-reviewed	https://www.cambridge.org/core/journals/judgment-and-decision-making/article/boundary-effect-perceived-post-hoc-accuracy-of-f
B6b-Q06	we observed only a small decrease in trust in numbers and trustworthiness of the source, and mostly for verbal uncertainty communication.	van der Bles, van der Linden, Freeman & Spiegelhalter, PNAS 2020	peer-reviewed	https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=DOI:%2210.1073/pnas.1913678117%22&resultType=core

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B6b-Q07	We consistently find that subsequent perceptions are least affected by the 'erroneous' prediction when it is expressed numerically, regardless of whether it is a point or range estimate.	Jenkins, Harris & Lark, J. Risk Research 22(5) 2019	peer-reviewed	https://ideas.repec.org/a/taf/jriskr/v22y2019i5p537-554.html
B6b-Q08	Advisors benefit from expressing themselves with confidence, but not from communicating false certainty.	Gaertig & Simmons (Wharton), Psychological Science 2018	peer-reviewed	https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=DOI:%2210.1177/0956797617739369%22&resultType=c
B6b-Q09	Our findings indicate that communicating uncertainty buffers against a loss of trust when evidence changes.	Dries, McDowell, Rebitschek & Leuker, Public Understanding of Science 2024	peer-reviewed	https://www.ebi.ac.uk/europepmc/webservices/rest/search?query=DOI:%2210.1177/09636625241228449%22&resultType=c
B6b-Q10	investors favor forecasts that are as precise as warranted by the information available, but not more precise.	Du, Budescu, Shelly & Omer, OBHDP 114(2) 2011	peer-reviewed	https://ideas.repec.org/a/eee/jobhdp/v114y2011i2p179-189.html
B6b-Q11	Exceeding a promise isn't viewed any more highly than keeping a promise, a study has demonstrated.	ScienceDaily reprint of Chicago Booth press release on Gneezy & Epley (2014)	secondary (press release)	https://www.sciencedaily.com/releases/2014/07/140717094556.htm
B6b-Q12	larger match ratios (i.e., \$3:\$1 and \$2:\$1) relative to smaller match ratios (\$1:\$1) had no additional impact.	Karlan & List, NBER WP 12338 abstract	research (working paper; published AER 2007)	https://www.nber.org/papers/w12338
B6b-Q13	Have a significant amount of your campaign pledges committed before the general public ever hears about the campaign.	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6b-Q14	If your plan includes any of the following sources of funds, they should be committed, imminent or backstopped	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6b-Q15	Written pledges or cash from individuals, corporations, businesses and foundations, including those resulting from direct appeals and unsolicited gifts.	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6b-Q16	Your project-cost estimates are firm and construction timetables have been established.	Kresge Foundation, A Guide to the Challenge Grant	industry (funder guidance)	https://kresge.org/sites/default/files/GuidetoChallengeGrant.pdf
B6b-Q17	The confidence interval around the median projected values is assumed to be symmetric and is based on root mean squared errors of various private and government forecasts made over the previous 20 years	FOMC Summary of Economic Projections, 2026-09-16	official-central bank	https://www.federalreserve.gov/monetarypolicy/fomcprojtabl20260916.htm

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B6b-Q18	criteria used and assumptions made	SEC Marketing Rule 206(4)-1 (17 CFR 275.206(4)-1)	official-regulator	https://www.ecfr.gov/api/renderer/v1/content/enhanced/current/title-17?part=275&section=275.206(4)-1
B7b-Q01	Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7b-Q02	A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7b-Q03	Do not treat Trading Notes as an afterthought. They provide a record of how your team put its strategy into practice.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7b-Q04	its expected role in growth, liquidity, risk management, or future funding	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7b-Q05	with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia	Wharton Global Youth news, 2026-01-27	official-Wharton	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investme
B7b-Q06	Overall, one of the tendencies of people who are just starting out is to fall into this trap of hiding behind fancy terms and formulas. I would encourage you to fight against that. Oftentimes, the best solutions are made up of simple, elegant ideas.	Semifinal judge Melissa Ko Hahn, 2025	official-Wharton	https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-compe

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B7b-Q07	Their model showed a 70% chance of reaching the long-term goal	KXAN Austin (Esmeralda Zamora) via Yahoo, 2026-03-22	secondary (local news)	https://www.yahoo.com/news/articles/female-vandergrift-team-advances-wharton-174003734.html
B7b-Q08	The Global Champion school will be permitted to fast-track one team to the semifinal round of the 2027-2028 competition.	Wharton competition Rules & Roles page	official-Wharton (2026-27)	https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/
B7b-Q09	less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once.	Brian Z, co-team leader, DMV's Finest (2022-23 champions)	official-Wharton	https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-teams-from-the-2023-teams/
B7a-Q01	Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7a-Q02	the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7a-Q03	You may identify decisions you would make differently, but you may not redesign your strategy after observing the results.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkyznk1xv5yegl3snlc1.pdf
B7a-Q04	Strong submissions connect the client's goals, research, portfolio decisions, and final recommendations through one clear and cohesive investment strategy.	Wharton, 2026-27 Competition Infographic	official-Wharton (2026-27)	https://upenn.box.com/shared/static/l2p0l26svbbptmcizhmswyq5f7rbyp44.pdf
B7a-Q05	with the help of our Wharton School student evaluators, as well as a professional review panel of asset managers from Aberdeen Investments in Philadelphia	Wharton Global Youth news, 2026-01-27	official-Wharton	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investme
B7a-Q06	A lot of work went into putting your written reports and the videos together, and it clearly came across	Semifinal judge Melissa Ko Hahn (Wharton MBA; managing member, Covepoint Capital Advisors), 2025	official-Wharton	https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-compe

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B7a-Q07	Still, there was this moment when we laid everything out in a rough draft of our final report and everything clicked — we really needed to cut things down. It also led us to reflect: less is more, and having an explainable strategy makes a lot more sense than going for so many different things at once.	Brian Z, co-team leader, DMV's Finest (2022-23 champions), Thomas Jefferson HS	official-Wharton	https://globalyouth.wharton.upenn.edu/news/wharton-investment-competition-teams-from-the-2023-teams/
B7a-Q08	*Note: if you provide us with a team strategy that is buried somewhere in your experience, rather than clearly stated, defined and articulated up front, you will not be among the strongest teams.	Wharton 'Developing a Strategy' page (older-season text)	official-Wharton (past season)	https://globalyouth.wharton.upenn.edu/developing-strategy/
B7a-Q09	They realize that the jargon and acronyms are just obfuscating the elegant simplicity of finance.	Prof. Michael R. Roberts (now the competition's Academic Director), interview 2022-05-31	official-Wharton (interview)	https://globalyouth.wharton.upenn.edu/news/prof-michael-roberts-wants-global-youth-students-to-discover-the-elegant-simplicity
B7a-Q10	assembled a group of 13 other Aberdeen professionals to read and evaluate the Region 3 reports	Wharton writer about Brianna Leporace (Aberdeen), 2018	official-Wharton	https://globalyouth.wharton.upenn.edu/news/kwhs-investment-competition-region-3-finalists/
B7a-Q11	provides evaluators from across the investment spectrum for the competition's semifinal round, reviewing more than 1,400 investment strategy proposals	abrdn press release (2023)	industry (competition partner primary)	https://www.aberdeenplc.com/en-gb/news/all-news/abrdn-enters-second-decade-supporting-wharton-investment-competition-for
B7a-Q12	In some cases, there was a lot of it. And in some cases, there was a little bit of it, but a lot of depth and accuracy. That is one of the things that in my mind differentiated better performances.	Judge Vikas Keswani (HPS Investment Partners), 2024 Global Finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024
B7a-Q13	When analyzing someone's investments, take a look at this person holistically	Judge Zoe McCormick (abrdn), 2024 Global Finale	official-Wharton	https://globalyouth.wharton.upenn.edu/news/spark-investments-bergen-county-academies-new-jersey-bring-the-heat-to-the-2024
B7a-Q14	The team's decision was proven with great success over the past few months.	Teen Ink: a 2023-24 team's published final report	secondary (student work)	https://www.teenink.com/opinion/all/article/1215809/Profitteering-Peninsula-Panthers-2023-2024-Wharton-Global-High-School-Inve
B7a-Q15	Teams that competed at least through the mid-term report submissions will receive a participation badge in February	Wharton Global Youth news, 2026-01-27	official-Wharton	https://globalyouth.wharton.upenn.edu/news/a-winning-season-thousands-of-teams-bigger-stakes-and-the-top-50-2026-investme
B7a-Q16	Students need to invest in as many industries as they have team members, but more is always better!	Alex Lamon, teacher-advisor (guest blog), 2021-09-08	official-Wharton blog (guest teacher, past season)	https://globalyouth.wharton.upenn.edu/essential-educator-blog/the-essential-educator-10-tips-for-teaching-the-wharton-investme

id	exact text	speaker / source	source type	URL (where verified)
B8b-Q01	The only person who needs to believe in something is yourself.	Wharton 2026-27 case, p.1 sidebar pull quote (NO attribution line)	official-Wharton (2026-27, repo file)	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (p.1 sidebar) (repo)
B8b-Q27	So when she got the book deal with HarperCollins, she gave Twitter her notice.	Input reporter (not Laura's words)	secondary (news feature)	https://www.inputmag.com/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel
B8b-Q28	She's hoping to finish in time for a 2024 release.	Input reporter (not Laura's words)	secondary (news feature)	https://www.inputmag.com/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel
B8b-Q29	Publication for the first book is planned for summer 2021	Publishers Weekly Rights Report, July 2020	secondary (trade press)	https://www.publishersweekly.com/pw/by-topic/childrens/childrens-book-news/article/83911-rights-report-week-of-july-20-2020.ht
B8a-Q16	So when she got the book deal with HarperCollins, she gave Twitter her notice.	Input/Inverse reporter (not Laura's words)	secondary (news feature)	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel
B8a-Q39	The only person who needs to believe in something is yourself.	Wharton 2026-27 case, p.1 sidebar pull quote (NO attribution line)	official-Wharton (2026-27, repo file)	competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (repo)
B9b-Q15	Once dubbed the Queen of Balance as her school's top rock climber, Kirby Tan suffers an injury that sidelines her for the rest of the season.	Publisher copy on lauragao.com (not Laura's own voice)	secondary (publisher copy)	https://lauragao.com/kirbys-lessons-for-falling-in-love
B9a-Q01	a required real growth rate of 4 percent	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B9a-Q02	establishing a minimum time horizon for achievement of performance objectives clarifies when action may be needed to resolve underperformance issues	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B9a-Q03	may be irrelevant beyond an absolute level of loss that would completely derail an investment portfolio	CFA Institute, Elements of an IPS for Individual Investors (2010)	industry (professional body)	https://rpc.cfainstitute.org/sites/default/files/-/media/documents/article/position-paper/investment-policy-statement-individual-inv
B9a-Q04	Instead of focusing on potential gains, decide what you're willing to lose.	Effectuation.org, Effectuation 101 (Sarasvathy framework)	industry/academic outreach	https://effectuation.org/effectuation-101
B9a-Q05	Find people who'll commit to your idea.	Effectuation.org, Effectuation 101 (Sarasvathy framework)	industry/academic outreach	https://effectuation.org/effectuation-101
B9a-Q06	financial decisions (separately for investing versus gambling), health/safety, recreational, ethical, and social decisions	DOSPERT scale site (Weber, Blais & Betz)	research (scale home page)	https://www.dospert.org/
B9a-Q07	Non-diversified entrepreneurs demand both systematic and idiosyncratic risk premium.	Chen, Miao & Wang, NBER WP 14848 (2009)	research (working paper; later peer-reviewed)	https://www.nber.org/papers/w14848
B9a-Q08	Those with a shorter time horizon may prefer to take on less risky or less volatile investments.	SEC Investor.gov	official-regulator	https://www.investor.gov/introduction-investing/getting-started/asset-allocation

id	exact text	speaker / source	source type	URL (where verified)
B10b-Q01	As you develop your strategy, you will need to decide how to construct and diversify the portfolio, how its components will support different needs, how the portfolio's asset composition may change over time, and which principles will guide decisions as funding dates approach and payments are made.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkzkn1xv5yegl3snlc1.pdf
B10b-Q02	A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkzkn1xv5yegl3snlc1.pdf
B10b-Q03	the Trading Note should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding.	Wharton, 2026-27 Investment Competition Guide	official-Wharton (2026-27)	https://upenn.box.com/shared/static/b9kh53xila9czkzkn1xv5yegl3snlc1.pdf
B10b-Q04	We are purchasing shares of an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment.	Wharton 2026-27 Trading Notes Analysis guide, sample note (p.2)	official-Wharton (2026-27, repo file)	competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt (p.2) (repo)
B10a-Q01	our 10-year expected annualized return for U.S. equities declined from a range of 4.9%-6.9% to a range of 4.2%-6.2%	Vanguard, VCMM return forecasts (2026-07-22)	industry	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html
B10a-Q02	valuations tend to be poor predictors of performance over the short or even intermediate term and should not serve as a primary reason for changing portfolio allocations	Vanguard, VCMM return forecasts (2026-07-22)	industry	https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html
B10a-Q03	So when she got the book deal with HarperCollins, she gave Twitter her notice.	Input/Inverse reporter (not Laura's words)	secondary (news feature)	https://nc.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel
B10a-Q04	lock down the benefits promised, then grow surplus assets with a prudent process and disciplined understanding of risk.	Russell Investments (Justin Owens), 2026-07-13	industry	https://russellinvestments.com/content/ri/us/en/insights/russell-research/2026/07/pension-surplus-investing-value-overfunding.ht

Sources used for re-tries (accessed 2026-09-27)

- Wharton WordPress REST API: <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=case-study> (and slug=judging-and-evaluation, slug=deliverables)
- Solvency II Art. 101: <https://www.legislation.gov.uk/eudr/2009/138/article/101>
- Europe PMC API (DOIs 10.1073/pnas.1913678117, 10.1177/0956797617739369, 10.1177/09636625241228449): <https://www.ebi.ac.uk/europepmc/webservices/rest/search>
- SEC Marketing Rule text: [https://www.ecfr.gov/api/renderer/v1/content/enhanced/current/title-17?part=275§ion=275.206\(4\)-1](https://www.ecfr.gov/api/renderer/v1/content/enhanced/current/title-17?part=275§ion=275.206(4)-1) and [https://www.law.cornell.edu/cfr/text/17/275.206\(4\)-1](https://www.law.cornell.edu/cfr/text/17/275.206(4)-1)
- Infographic direct PDF: <https://upenn.box.com/shared/static/l2p0l26svbptmcizhmswyq5f7rbyp44.pdf>
- Case and Trading Notes guide: competition/official/2026_27/ (repo, SHA-256 in manifest.yaml)

What this teaches

- A quote can be word-for-word correct and still be misattributed. The most common trap here was not wrong wording. It was a writer's paraphrase (Balchunas, Nichole Jordan) or a reporter's sentence being read as the person's own voice. Before you write "X said", check whether the words sit inside quotation marks next to X's name.

- "Not found" by a script often means "hidden", not "made up". Five Wharton quotes live in hidden or retired page sections that only the site's own API shows. Three abstracts sat behind publisher blocks but were open in Europe PMC. Two failed only because a PDF split a word across a line. Try the official back door before you call a quote wrong.
- Old rules can survive on live pages. The "one sector per team member" rule was genuine Wharton text in past seasons. That is exactly why it kept coming back as a claim for 2026-27, when it is no longer a rule. Always check the season of the page.
- An unnamed pull quote is the case's voice, not the client's, until a source proves otherwise. Writing "the case profile highlights..." costs nothing; putting words in Laura's mouth could cost credibility with the readers who know her.

D13c: Voice-of-Laura map (her verified words → values → strategy and communication)

Agent D13c (Voice-of-Laura mapper), insight_v1 run, Phase D. Written 2026-09-27. This is AI-generated research for Team Caplet. It holds specifications, evidence and checklists. It holds **no text to submit**: the six students write every deliverable in their own words and record AI use in the Final Report's Works Cited (R-W46). Laura appears here only through the case and her public professional record. Nothing here suggests contacting her (R-W16).

The gate for quotes. In sections 1-4 and 6, a line appears as Laura's words only if D13a marked it VERIFIED-PRIMARY (phase_D/D13a_laura_quotes_verified.json). All 50 D13a ids quoted or cited here were re-checked verbatim by D13c on fresh copies of their pages (script: research/insight_v1/scripts/D13c_voice_map_check.py, 50 of 50 pass). Everything else (the unattributed pull quote, transcript lines, private lines, reporter sentences, and 8 new lines D13c found) is kept apart in section 7.

Inputs read in full: the brief (sections 2-4 and 13-15 with care); the case (competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt); D13a (md and json); Phase B files B8a, B8b, B9a, B9b and B3b; the current allocation file wins_now/securities_and_allocation_v0.md (S3, v1); and the Phase C questions that touch her voice (M018, M049, M075, M100, M103, M126, M144, M216, M221, M243 and others).

Summary (read this first)

- Her words hold two risk ideas, and the plan must honour both.** She leaps: "you should always take the jump because you're always going to regret not doing it" (2021), after naming the cost, a move to "self-employed, with unstable income". And, as a student founder, she said letting down "people who were your earliest supporters and were the first to believe in you" is worse than letting yourself down (2016). INTERPRETATION: risk she carries herself is acceptable, even attractive, to her; risk that lands on people who trusted her is not. That split is the honest Laura-rooted reason for "buy the payments first, take risk only with the surplus". Neither "she is cautious" nor "she loves risk" fits her words.
- Correction to Phase B: "she leapt only after a contract secured her floor" is not in her words.** B8a, B8b, B9a, B9b and question M018 built on it. It rests on a reporter's sentence ("So when she got the book deal with HarperCollins, she gave Twitter her notice.", Input/Inverse 2022). D13a found, and D13c confirmed on a fresh copy, that its "So" follows personal material, not a financial plan. Stop describing "floor-first" as her trait.
- The candid clash.** On paper the plan is very cautious. Whole-portfolio equity is about 0-2% in 2027, about 20% in 2028-2030 and about 4% after the 2031 floor is bought (DER, B10b script re-run by D13c). A founder whose rule is "take the jump" could read that as the opposite of her. The plan can answer this, but only if the deliverables say it plainly:
 - the leap is the residency itself;
 - locking the payments costs about \$11k at the median and about \$200k at the 95th percentile of the 2033 surplus;
 - in return it removes a ~2% chance of not funding all ten payments (~12% if the 2028 deposit halves) (DER, B10b);
 - the growth money should not be set at the minimum just to look safe.
- Her working habits set the communication spec.**
 - Habits: a purpose statement before the work; a deadline, "finished or not"; three drafts scrapped; "prove better with data"; results labelled "[unfinished]".
 - Spec: purpose first; costs and gaps named; discarded drafts shown; every number graded and rounded; what the numbers secure stated before the numbers.
- Community.** Her newest words (Feb 2026) are about access: "...Our No. 1 priority is to get as many diverse voices and stories in the room as possible". And her festival grew after "the community really supported us". So "operations first" reads as hers. The 2031 message can follow the same order: her own part already bought; partners invited; more offered as support is proven (her festival phrase was "as the festival grows").
- What she would reject on sight:**
 - the case pull quote presented as "Laura said";

- book-title puns and "falling" metaphors;
 - her identity used as a reason for a holding;
 - a finance-prestige tone;
 - numbers more precise than the model behind them;
 - hidden costs;
 - a one-sided risk story.
5. **"Falling" has no usable line.** Her 2025 book's "falling" is fiction (a climber's fall, and falling in love). The only first-person "falling" line (Nerd Daily 2025) sits in an answer about her family, so it is excluded. Seed 17's "learning to fall safely" fails.
6. **Guardrails, in short:**
- Apply the swap test to every Laura sentence.
 - No Laura quotes in the Trading Notes or the IPS (the IPS bans "formal citations", R-I55).
 - At most 1-2 quotes in the Final Report, each dated and in context.
 - Label student-era lines (2016-2018) as such.
 - We hold no verified statement by her about investing, so never write her "investment philosophy".
7. **New for D13a to rule on:** 8 lines verified verbatim but not in D13a's register (section 7d). One corrects Phase B: "the Laura Gao I knew loved building apps, not excel models..." is her own first-person answer (Poets&Quants 2018), not the nominator's words.
8. **North Star link (INTERPRETATION).** The plan feels like hers if it:
- makes the leap certain for the people who will rely on it;
 - takes risk only where she alone bears the cost;
 - says its costs and gaps out loud;
 - sounds like a teacher, not a trading desk.

Words used in this file

- **Verified quote (VP):** D13a marked it VERIFIED-PRIMARY (her words, verbatim, meaning kept in context); D13c re-checked the wording on a fresh copy of the page on 2026-09-27.
 - **VRF:** VERIFIED-REPO-FILE, i.e. wording checked in an official file in this repo (the case, the guides).
 - **INTERPRETATION (INT):** D13c's reading of what her words show. Never a fact about her.
 - **DER:** derived by a script; its inputs carry their own labels. **ASM:** ASSUMPTION.
 - **Ladder:** Treasury bonds that mature on or just before each \$50,000 payment date, so each payment is already bought. The plan uses zero-coupon Treasuries (STRIPS), which pay one lump sum at maturity.
 - **Growth money (the "sleeve"):** money the ten payments do not need. It can take market risk.
 - **2031 floor:** an amount bought in a 2-year Treasury in 2031, so it can be promised to co-sponsors without a forecast.
 - **Percentile (p5, p50, p95):** in a simulation, p5 is the result that only 5% of paths fall below (a bad case), p50 the middle (median), p95 a good case.
 - **Regret rule:** choosing by asking which choice you would regret more later.
 - **Price of certainty:** what buying the payments up front gives up in good markets.
 - **Tokenism:** using a person's identity as decoration, or as a reason for a choice it does not justify.
 - **Student-era:** her words from 2016-2018, when she was a Penn student. Still hers, but older.
-

1. Values and decision habits her verified words show (all readings are INTERPRETATION)

Each value has 1-3 verified quotes (exact words, source, date, and what she was answering) and a note on how strong the evidence is. Case lines are the case's words (VRF), not hers.

V1. Risk: she leaps toward what she would regret not doing, and names the cost first

id	her exact words	source, date	what she was answering
B1b-Q11	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	Overachiever Magazine, 2021-03-15	"What pushed you to make the jump?" (tech job to full-time comics)
B8b-Q18	"It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""	same answer, 2021-03-15	same
B8a-Q32	"I was a bit hesitant. I was like, I know this is kind of a sensitive topic. I don't know what I'll get from it"	NPR Goats and Soda, 2020-04-04	deciding to share The Wuhan I Know

- Reading (INT): her risk rule is regret of *not* acting, applied to meaningful leaps. She says the downside in concrete words ("unstable income") and admits hesitation, then acts without knowing the outcome.
- Supporting lines (VP):
 - "I feel like I've been jumping almost my whole life." (B8a-Q09, 2021).
 - Her speaking guide lists the audience question "What motivated you to quit a secure job to become an artist-writer in the middle of a pandemic?" (B8b-Q04, guide PDF created 2024-03-24). This is a question she lists, not a statement, but it shows she presents her story publicly as leaving security.
- No verified line says she waits for a secured floor before leaping (see Summary item 2 and section 3.4).
- Case framing (VRF, R-C37/R-C38, p.2 L69-74): "Although she has been willing to take thoughtful risks throughout her entrepreneurial career, she wants her investment team to recommend an appropriate balance between pursuing growth and protecting the capital required for her goals."
- Strength: strong for "leaps, with the cost named" (two outlets, 2020-2021). Every one of these was a career or public risk that she carried herself; none is about investing.

V2. Failing people who believed in you is worse than failing yourself

id	her exact words	source, date	what she was answering
D13a-N01	"It's actually harder as an entrepreneur because not only are you letting yourself down, but more importantly, you're also letting other people down — people who were your earliest supporters and were the first to believe in you."	The Daily Pennsylvanian, 2016-01-28	"What were some major challenges you've had with Draw Street Journal?" (her student laptop-decal business)

- Reading (INT): she separates risk she carries from risk that falls on early backers, and ranks the second as worse. It is the only verified line where she uses "believe" about other people's support (D13a).
- The case makes the same point for 2031 (VRF, R-C66, p.3 L114-115): "If Laura promises more than she can ultimately contribute, she could damage her credibility and lose the confidence or participation of co-sponsors."
- Strength: moderate. It is one source from her student years, about a small business, not a residency. It matches the case, which is the stronger authority.

V3. Failure and imperfection: ship on a deadline, learn from feedback, scrap what is wrong

id	her exact words	source, date	what she was answering
B8b-Q31	"I give myself a deadline for when I must post the art, finished or not."	Geeks OUT, 2022-05-11 (also The Honey POP, 2022-06-02)	advice to aspiring creators ("Post terrible work!")
B8b-Q30	"The quicker you get over your perfectionism, the faster you'll finish projects, get feedback, improve, and overcome imposter syndrome or "artist stage fright"."	Geeks OUT, 2022-05-11	same answer
B8b-Q12	"I scrapped three completely different drafts before landing on this one."	The Nerd Daily, 2025-03-08	challenges writing her first fiction book

- Reading (INT): failure is a normal step in how she works. She sets deadlines, ships imperfect work, uses feedback, and throws away drafts that do not work. What she avoids is perfectionism that stops her shipping.
- Also VP:
 - "Ultimately my goal is to tell a story; I don't need to be perfect to be impactful." (B8b-Q32, Geeks OUT 2022-05-11, same answer);

- “Soon enough, I had a comics diary that I could not only decompress into, but also take pride in how far I’d come when I reread it.” (B2b-Q10, Wharton Magazine 2022-04-06, about a nightly 10-minute comic she drew while working in tech). A dated record of small steps is what a team decision log is (M021).
- Note: “helped launch the Anti-Resume Project ... to normalize failure” is a reporter’s description (Pulse Spikes), not her words (section 7c). Her own line on it is candidate D13c-C8 (section 7d).
- Strength: strong (three answers, 2022-2025, one repeated word for word in two outlets).

V4. “Falling”: no usable line in her own professional words

- Her 2025 book *Kirby’s Lessons for Falling (In Love)* is fiction. Its “falling” is a climber’s fall and falling in love. “Queen of Balance” is copy about the fictional heroine (B9b, VP there).
- The only first-person “falling” line in her interviews (The Nerd Daily, 2025-03-08) sits in an answer about her family and identity. It is **EXCLUDE-PRIVACY** (D13c; section 7b). This file does not reproduce it.
- So seed 17’s “learning to fall safely” has no basis in her words (B9b Q22 found the same). There is no value to map here; the only consequence is a guardrail (section 5).

V5. Build what is missing, with other people, and grow after support shows up

id	her exact words	source, date	what she was answering
B2b-Q07	“If what you want doesn’t exist, create it yourself.”	Poets&Quants, 2018-03-30 (student-era)	“What is the biggest lesson you gained from studying business?”
B8b-Q10	“Only after we realized EVERY student deserved to hear their words of wisdom did we decide to build something to cultivate it.”	The Sign.al 5-year page, founder’s note signed by her, c. 2022 (date ASM)	how her student club began
B8b-Q37	“All of it is way beyond our initial passion project. We’re thrilled that the community really supported us and wanted us to continue to grow the festival”	Local News Matters, 2026-02-12	growth of the free comics festival she co-organises (page’s trailing comma dropped)

- Reading (INT): her founder habit runs in stages: see a gap, check that the need is real, build it with others, then grow once the community backs it. The residency in the case fits this pattern (VRF, R-C18, p.1 L25-26: “she has combined artistic passion with strategic thinking and a willingness to pursue unconventional opportunities.”).
- Also VP: “Well, if people aren’t gonna give it to us, I’ll just make one.” (B8a-Q14, 2021, about her next book).
- Also VP: “I couldn’t be more proud of the current team keeping the legacy strong.” (B8b-Q11, the same founder’s note). She handed her club to a team that kept it running. INT: she values institutions that outlast their founder. At most one clause, if co-sponsors ask about life after 2042; funding beyond 2042 is out of scope (case p.3 L92-93).
- Strength: strong. The pattern spans 2018 to 2026 across a club, a book and a festival.

V6. Community and access: money serves getting people “in the room”

id	her exact words	source, date	what she was answering
B8b-Q38	“...Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on”	Local News Matters, 2026-02-12	the festival’s new mini grants to 20 creators (the paper prints the leading ellipsis)
B8b-Q39	“We definitely hope to offer more as the festival grows.”	same, 2026-02-12	same passage
B8b-Q05	“Pro-bono visits can be offered on need-basis.”	her Publicity & Speaking Guide (PDF created 2024-03-24)	the booking panel of her own guide

- Reading (INT): where she controls money, she spends it on access (grants, free entry, need-based pro-bono visits). The reporter adds that the festival is “remaining free for exhibitors and attendees” (fact, not her words).
- Limit: this says nothing certain about the residency’s fees. The case lists “program fees” as one possible outside source (R-C44). Any fee assumption must be labelled ASM.
- Strength: strong and recent (2024-2026).

V7. Authenticity: personal connection over clever concepts

id	her exact words	source, date	what she was answering
B8b-Q13	"But in the process, I lost the heart of my writing—authentic, personal connections."	The Nerd Daily, 2025-03-08	why she scrapped her early fiction drafts
B8a-Q28	"So, I scrapped everything and went back to my roots. No more gimmicks."	same answer	same

- Reading (INT): she judges her own work by whether it keeps "authentic, personal connections", and she drops high-concept framing that loses them.
- "Gimmicks" meant fantasy premises in her drafts (D13a). Applying the test to a portfolio document is our extension, and it should be called that. Never quote "No more gimmicks." on its own (B8b-Q14): cut down to three words, it invites meanings the page does not support.
- Strength: strong for her own craft (2025); interpretive when applied to us.

V8. Money and success: meaning over metrics, and freedom as the payoff

id	her exact words	source, date	what she was answering
B8b-Q21	"it's great having thousands of likes, but at the end of the day, it's just a number. You never really understand the human connection behind it"	Overachiever, 2021-03-15	why one reader's thank-you note mattered more than metrics
B8b-Q22	"I'd just be like, 'Hey, this is why you made this. This is who you're trying to help, and this is the positive effect you actually have.'"	same interview	what she told herself when she reread a reader's supportive comment
B8b-Q26	"Now I get to fit work around life instead of the other way around, which is such a game-changer"	Input (Inverse), 2022-03-07	her working life after leaving tech

- Reading (INT): she measures success by who is helped and by freedom over her time, not by the size of a number. When things go badly she anchors on purpose (B8b-Q22).
- Her few verified words about money:
 - she gave up a well-paid job with benefits for "unstable income" (V1);
 - she offers pro-bono visits (V6);
 - she once taught financial literacy (V12).
- **We hold no verified statement by Laura about investing, markets or wealth.** Any "her investment philosophy" sentence would be invented.
- Also: her "I did 50 things..." answer (B8b-Q20, quoted in full under V10) frames success as breadth of experience.
- Strength: strong for "meaning over metrics"; the evidence on money is thin by nature.

V9. Data inside a story; honest about what is unfinished

id	her exact words	source, date	what she was answering
B8b-Q35	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	Pulse Spikes, 2021-06-18	how writing and art go together for her
B8a-Q19	"That [story] was something I could prove better with data"	Pulse Spikes, 2021-06-18 (the bracket is the outlet's)	her data project Rewriting Herstory
B9b-Q09	"The [unfinished] results are below."	lauragao.com/rewriting-herstory (undated; student-era, ASM)	her own project page; the bracket is hers, and her table grades each match "Good", "Unsure" or "Bad"

- Reading (INT): data serves a stated purpose and proves a point inside a story. She labels her own results as unfinished and grades their quality.
- Also VP:
 - "I've always said that my main purpose is to tell a story" (B8a-Q18, Pulse Spikes 2021-06-18; D13a notes that she goes on to say the medium does not matter). Purpose is fixed; the tool can change;
 - she states limits and next steps: "Further probing of other song metrics' effects on danceability would provide deeper analysis." (B9b-Q11);

- she defines terms by their source: "For the purposes of this study, I used Spotify's definitions and values for different song attributes." (B9b-Q10);
- her résumé lists "Experimented and shipped 10 features that resulted in +4.5M DAU" (B9b-Q01) and "Product Management: JIRA, Confluence, A/B testing" (B9b-Q02). An A/B test compares a new version against a control.
- The case records a statistics degree (VRF, R-C12).
- Strength: strong (her own published data work plus her résumé). Dates are mostly undated or student-era.

V10. Keep options open

id	her exact words	source, date	what she was answering
B8a-Q04	"My advice is to avoid siloing yourself into an end-all-be-all career situation."	Poets&Quants, 2018-03-30 (student-era)	advice to a student choosing a business major
B8b-Q20	"I did 50 things, and maybe I didn't like fully excel at each, but I got to try all of them and have so much fun with them."	Overachiever, 2021-03-15	how she hopes to look back at 80

- Reading (INT): she resists locking herself into one path. The case gives this value its financial form (VRF, R-C58, p.3 L102-103): "She recognizes that committing all remaining assets could limit her financial flexibility as the project develops."
- Limit (D13a): these are career statements. Do not turn "50 things" into a diversification formula.
- Strength: moderate for her own words; strong once paired with the case.

V11. Healthy doubt about hype

id	her exact words	source, date	what she was answering
B8a-Q37	"how long will this novelty factor last?"	The Daily Pennsylvanian, 2016-01-28 (student-era)	her goals for her business in its first week; she answered her own question with a plan (D13a)

- Reading (INT): even about her own early success, she asked whether the excitement would last, and planned for when it faded.
- Strength: weak to moderate (one student-era line).

V12. She has taught money basics to teenagers, and she designs learning to be less daunting

id	her exact words	source, date	what she was answering
B2b-Q08	"Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia."	Poets&Quants, 2018-03-30 (student-era)	her list of college activities
B3b-Q13	"exercise students' creativity in less daunting ways than traditional essay writing"	her Educators & Book Clubs Guide (PDF created 2024-03-24)	her signed author's note on the activities she designed
B8b-Q06	"Tired of authors talking your students to sleep? Choose an interactive hands-on activity!"	her Publicity & Speaking Guide (2024)	promotional copy to schools

- Reading (INT): a team of high-school students is writing for someone who has taught financial literacy to high-school students. For her, plain explanation is a professional standard, not a courtesy.
- Related professional facts (VP, D13c check on Poets&Quants 2018, her questionnaire list of internships): "Trading Intern at Belvedere Trading in Chicago" and "Policy Analyst Intern at Consumer Financial Protection Bureau". These are facts, not value quotes. Her 2024 résumé does not list them (B8b).
- Strength: moderate. The facts are firm; the reading of her standards is inference.

V13. Tell the whole story, not only the frightening headline

id	her exact words	source, date	what she was answering
B8a-Q34	"There wasn't much that was incorrect as much as there just wasn't any news about the overall picture of Wuhan outside of the coronavirus"	HuffPost, 2020-04-29	media coverage early in the pandemic

id	her exact words	source, date	what she was answering
B8a-Q33	"It's disheartening to know that all they know are the bad parts of it"	NPR, 2020-04-04	how people knew the city only through virus headlines

- Reading (INT): she objects more to incomplete framing than to outright error. The case says her first public work was "Originally intended as a response to misinformation and anti-Asian racism" (VRF, p.1 L19-20).
- Limit: use this as a principle for how to present risk. Never put these lines, or Wuhan, next to Taiwan in a deliverable: that would join two different places (B3b Q08).
- Strength: moderate (two 2020 sources on one topic).

Decision habits at a glance (INTERPRETATION)

habit	her words (id)	where it shows up in our plan
decides a leap by asking what she would regret not doing	B1b-Q11	the residency is the leap; growth money is not set to the minimum
names the cost before leaping	B8b-Q18	the price of certainty is stated in dollars
expects the worst reaction before going public	B8a-Q13 "When I first posted the comic, I expected a way worse reaction" (2021)	bad-case numbers shown beside good ones
steadies herself by recalling who the work helps	B8b-Q22	a bad-year rule: the payments are already bought; the purpose is restated
writes a purpose statement for the work	B8b-Q35	the IPS opens with the purpose; each Trading Note names the holding's job
ships to a deadline, finished or not	B8b-Q31	purchase dates set by rule (no rate-timing); early WInS trades, then refinement
scraps drafts that lose the heart	B8b-Q12, B8b-Q13	show the discarded growth-first plan; cut gimmicks
builds once the need is proven, with a team	B8b-Q10, B8b-Q37	"proof before promise": the 2031 floor is bought, partners come on top
uses data as proof and labels unfinished results	B8a-Q19, B9b-Q09	evidence grades; every probability names its model
tests against a control	B9b-Q01, B9b-Q02	show the all-Treasury "control" next to the chosen mix

2. What each value means for the strategy and for how we talk to her

Format per value:

- **Strategy:** a decision the value supports (+) or challenges (–).
- **Earns trust / Feels inauthentic:** what earns her confidence, and what would feel inauthentic or "finance-bro" (show-off finance talk: bravado, jargon, chasing returns).
- **Effects:** the deliverable (TN = Trading Notes, Oct 23; IPS = Nov 6; FR = Final Report, Dec 4) and the scored criterion (R-S24 strategy, R-S25 client knowledge, R-S26 analysis, R-S27 articulation, R-S28 creativity and presentation).

Everything below is specification, not text to submit.

V1 Risk (leaps, with the cost named)

- Strategy:
 - (–) It challenges how cautious the plan looks. Whole-portfolio equity is about 0-2% (2027), about 20% (2028-2030) and about 4% after the 2031 floor (DER, B10b).
 - (+) It supports a real, not minimal, equity share in the growth money (open item: 50/60/70%; M075).
 - (+) It supports stating the price of certainty, so she can apply her own regret test (section 3.3, N1).
- Earns trust: for each big choice, one plain sentence of the form "we chose X; its price is Y".
- Feels inauthentic:
 - "a conservative portfolio for a cautious client";
 - "Laura has a high risk appetite, so we hold more stocks";
 - calling her "risk-averse" on the strength of B8a-Q03, which is about students choosing a business major;

- "safe" or "guaranteed" used as selling words.
- Affects: IPS (risk sentence: willingness vs ability, stated as the team's inference); FR (price-of-certainty table); TN (the growth note says what the equity is for). R-S25, R-S26.

V2 Failing others is worse than failing yourself

- Strategy:
 - (+) No market risk on money other people will rely on: the ten payments and, from 2031, the announced low end.
 - (+) Announce only what is already bought; the floor never falls.
 - (+) This is the Laura-rooted answer to M126 (residents rely on the payments) and M221 (risk sits only on money nobody else relies on).
- Earns trust: say, in her terms, which money others will rely on and show that it is owned before it is promised.
- Feels inauthentic:
 - leading with the top of the range;
 - "we expect to deliver \$X" with no floor;
 - a confidence number with no model named.
- Affects: IPS (principle: who relies on which money); FR (co-sponsor draft; the range = bought floor plus labelled upside); TN (the first Treasury note names the promise it stands for). R-S25, R-S28.

V3 Failure and iteration

- Strategy:
 - (+) Place the core WInS trades early and refine them later (M032).
 - (+) Keep a dated decision log.
 - (+) Show rejected options with one number each: growth-first, partial locks, all-Treasury (M108).
 - (–) It challenges any claim that the plan is final and perfect.
 - (–) It sits uneasily with the frozen IPS (section 3.3, N2).
- Earns trust: dated "first version, what we learned, what changed" (for example, the team's original ~75%-equity growth-first plan, dropped after testing; CLAUDE.md).
- Feels inauthentic: a polished success story with no mistakes; hindsight edits (WInS notes cannot be edited anyway, G-ids).
- Affects: TN (the guide allows a decision that "tested, or refined" the strategy, TN p.1 L12); FR Articulation. R-S27.

V4 "Falling"

- Strategy: none.
- Communication: never use "falling", "Queen of Balance" or "lessons for falling" frames (section 5). The case's own phrase "an appropriate balance" (R-C38) is fine; a book-derived "balance" metaphor is not.
- Affects: all three deliverables. R-S28.

V5 Build with others, in stages

- Strategy:
 - (+) The 2031 structure: floor bought, upside stated with its model, partners invited on top (M008).
 - (+) A facility contribution that can grow with results; committed in 2033 and possibly drawn in stages (M027, open).
 - (+) Keeping flexibility "as the project develops" (R-C58).
- Earns trust: her secured part presented as the first contribution that invites partners; more offered as support is proven.
- Feels inauthentic: casting her as a lone funder of "the dream"; promising growth before any proof.
- Affects: FR (co-sponsor draft; the form of the facility recommendation). R-S25, R-S28.

V6 Community and access

- Strategy:
 - (+) The case's own order, reserve before facility (R-C57), reads as her priority: the payments keep the doors open.

- (+) If a fee assumption is needed, label it ASM (M105). Do not state a fee policy for her.
- Earns trust: describe the payments by what they secure: "ten years of operating money".
- Feels inauthentic:
 - charity-speak about residents;
 - an invented budget or resident count (the case expects no business plan, R-C95);
 - lifting the festival's identity focus into a residency programme.
- Affects: FR (co-sponsor draft; assumptions table); IPS (one clause on why operations come first). R-S25.

V7 Authenticity

- Strategy (+): cut anything that exists only for show (M103):
 - a Taiwan equity tilt (EWT);
 - identity- or interest-themed funds;
 - branded framework names;
 - book puns;
 - model layers in the main text where "bought, not forecast" already answers the question.
- Earns trust: plain names for the parts of the plan ("the payments", "the growth money"); one idea per section.
- Feels inauthentic: "Messy Roots Portfolio"; "lessons for falling (in rates)"; comic-style layouts; AI images.
- Affects: all three. R-S28.

V8 Meaning over metrics; freedom

- Strategy:
 - (+) The headline measure is what is secured ("10 of 10 years of operating payments bought"), then dollars (M057).
 - (+) A real flexibility share after 2033 rather than everything into the building (R-C58, B8b-Q26).
- Earns trust: numbers translated into what they do (years funded; the facility range in 2033 US\$ and in today's building-cost terms).
- Feels inauthentic:
 - "maximise returns";
 - bragging about WInS gains or rank (the Guide p.3: WInS "is not the competition scorecard");
 - reporting only portfolio dollars.
- Affects: FR headline; TN ("reasoning ... not simply whether an investment gained or lost value", TN p.1 L15-16); IPS objective written as goals, not a return target. R-S25, R-S28.

V9 Data inside a story; honest uncertainty

- Strategy:
 - (+) Define "high degree of certainty" by what can be observed (payments bought at market prices), not by a model percentage (brief section 8 item 6).
 - (+) Show the control, an all-Treasury growth money, and the lift equity adds (B9b Q3).
 - (+) Grade each number: market price, estimate or assumption (M231).
 - (+) Round to the model's real precision (B9b Q10).
- Earns trust: every probability names its model; each chart proves one claim; unfinished parts are labelled.
- Feels inauthentic:
 - "95% confident" with no model named;
 - "\$207,413";
 - "200,000 simulated paths" in the main text;
 - a lock-early "0% miss" without the words "by construction" (brief section 11).
- Affects: FR (charts, assumptions table); IPS (certainty defined in words; no charts allowed, R-I55); TN (one checkable data point). R-S26, R-S28.

V10 Keep options open

- Strategy (+): keep part of the post-reserve surplus uncommitted (the share is open, M003); never "committing all remaining assets" (R-C58).

- Earns trust: say what the held-back money is for (an overrun buffer, or the chance to expand what works) and when it could be used.
- Feels inauthentic: a "rainy-day fund" with no purpose; the facility gift treated as the only goal.
- Affects: FR. R-S25.

V11 Doubt about hype

- Strategy (+): a broad index, not thematic or AI funds; state concentration plainly; no "AI hedge" story (M010).
- Earns trust: one plain concentration sentence using the fund's own published numbers.
- Feels inauthentic: trend-chasing trades; "the AI revolution"; buying what is hot in WInS.
- Affects: TN (growth note); FR. R-S24.

V12 Teacher of teenagers; plain words

- Strategy (+):
 - the fewest rules that do the job;
 - low-cost funds with costs disclosed (WInS commissions now; a fee assumption in the FR projections, BS-01; M052).
 - Reason: she taught financial literacy and interned at a consumer-protection agency and at a trading firm (V12 facts).
- Earns trust:
 - each term defined once;
 - a classmate could restate every rule;
 - costs named.
- Feels inauthentic: jargon in the main text ("alpha", "DV01", "convexity"); a sales tone; costs left out.
- Affects: all three. R-S25, R-S28.

V13 Whole story

- Strategy (+):
 - show both good and bad outcomes for every projection (the case asks for "favorable and unfavorable market outcomes", p.3 L118-119);
 - name Taiwan-related risks once and precisely (currency, construction costs, continuity), without caricature (M215).
- Earns trust: good case, middle, bad case, and what the plan does in each.
- Feels inauthentic: drama (war scenarios as spectacle), or silence about risks.
- Affects: FR (scenario table; co-sponsor draft). R-S26, R-S28.

Communication checklist (one page, drawn from sections 1-2)

Earns her confidence (R-S25 "recommendations that can earn her confidence"):

- [] The purpose comes first, in one sentence, before any instrument.
- [] Which money others rely on, and proof that it is bought before it is promised.
- [] The price of each big choice in one number, and the good-market give-up as well as the bad-market protection.
- [] The discarded drafts, with one number each and why they were dropped.
- [] Each number graded (market price / estimate / assumption), rounded, with its model named.
- [] What the numbers secure ("10 of 10 years") before portfolio dollars.
- [] The gaps in "certainty" named (section 3.3, N4).
- [] Costs named.
- [] Plain words; every term defined once.
- [] Her facts exactly right ("bestselling"; "finalist", not "winner", for the Harvey and Goodreads awards; D13a note on B3a-Q03).

Feels inauthentic or "finance-bro", as examples to avoid, with the reason:

- "beat the market", "alpha", "crush", "aggressive growth", "conviction picks": metric-chasing (V8, V12);
- "guaranteed", "risk-free", "95% sure" with no model: false precision (V9);
- WInS rank, P&L or "our winning trade": the Guide says WInS "is not the competition scorecard";
- book-title or plot puns: gimmicks (V7) and tokenism (section 5);

- "Laura says..." followed by the case pull quote: misattribution (section 4, R1);
- "Laura is risk-averse" or "Laura loves risk": each misreads her (V1, V2);
- walls of text and unexplained acronyms (V12).

3. Where our current strategy echoes her ideas honestly, and where it does not

3.1 The plan being compared (not yet approved by the team; CLAUDE.md, brief section 7, S3 allocation v1)

- **January 2027:** buy Treasuries (a STRIPS ladder) matching the ten \$50,000 payments, about \$292k-\$294k of the \$300k (brief section 6; S3). The small leftover waits in T-bills. If rates fall first, buy the longest-dated payments first and complete the ladder from the 2028 deposit.
- **January 2028:** the \$150k plus the leftover (about \$158k) becomes the growth money: 60% global equity (VT) and 40% short Treasuries (VGSH), provisional.
- **2031:** part of the growth money (80% in the model; an open parameter) moves into a 2-year Treasury. That is the floor Laura can promise co-sponsors, plus an upside with a stated probability.
- **2033:** the ladder is the operating reserve; the facility contribution comes from the rest; the leftover is kept mainly as project flexibility.
- **WInS:** either option (ii), 66% Treasury hedge (IEF/TLH), 20.5% VT, 12.5% VGSH and 1% cash, or option (iii), about 98% Treasuries (S3).

3.2 Honest echoes

#	plan element	her words	why the echo is honest	limit
E1	No market risk on money others will rely on (payments; the announced 2031 low end)	D13a-N01	Her words and the case (R-C66) point the same way	2016 student line about a small business
E2	Risk taken only where she alone bears the cost (the upside of the facility; post-2033 flexibility)	B1b-Q11, B8b-Q18 with D13a-N01	The risks she describes taking were career or public risks she carried herself; she ranks failing backers as worse. This is the true replacement for "floor-first"	Joins two sources; it is INT
E3	Proof before promise: floor bought in 2031, partners invited on top, more if markets allow	B8b-Q10, B8b-Q37, B8b-Q39	Her club and festival grew after the need and the support were shown	A club or festival is not a residency
E4	Operations first (reserve before facility)	B8b-Q38, B8b-Q05	Her record spends money on access; the order itself is the case's (R-C57)	Says nothing about fees
E5	"High certainty" = bought at market prices, not forecast; model numbers labelled	B9b-Q09, B8a-Q19	She labels her own results "[unfinished]" and uses data as proof	Honest only if the FR really grades its numbers
E6	The team scrapped its growth-first draft after testing	B8b-Q12	True event, not a metaphor (CLAUDE.md: growth-first "failed: 3-8% payment shortfall")	Only if Articulation tells it as it happened
E7	Buy on the date, by rule; no rate-timing	B8b-Q31	Same working habit: a deadline, "finished or not"	Her line is about posting art
E8	Keep flexibility; never commit everything	B8b-Q26, B8a-Q04	Consistent with her record	The case says it first (R-C58); cite the case
E9	No fund covers the 2038-2042 payments, so the ladder is built from individual Treasuries	B2b-Q07	A true fact: no iBonds Treasury fund exists for Dec 2037-2043 (brief section 14)	Trivial; keep it as a private design note, never as a slogan

3.3 Where it does not echo (candid), and how the plan could answer

These are options for the team and for Phase E, not decisions.

N1. The leap. Her risk words are "always take the jump". The plan:

- about 0-2% whole-portfolio equity in 2027;
- about 20% in 2028-2030 (at median values; more after strong markets);
- about 4% after the 2031 floor (DER, B10b re-run; ASM: sleeve at median values, 60/40 sleeve, 80% floor share);
- WInS option (iii): about 98% Treasuries.

S3 already names the weakness: "A reader may see about 98% Treasuries as timid for a client who 'has been willing to take thoughtful risks'". Unexplained, the plan reads as generic safety, which is the opposite of her voice.

What the numbers say (DER: the B10b frontier that question M243 cites; 2028 deposit \$150k; re-run by D13c 2026-09-27; ASM inputs: JPM returns, lognormal, no fees):

	2033 surplus p5	p50	p95	P(ten payments not fully funded)
Payments 100% locked in 2027	\$150k	\$210k	\$299k	0.00%
Nothing locked (60/40 until 2033)	\$34k	\$221k	\$496k	2.10% (11.72% if the 2028 deposit is \$75k)

So locking costs about \$11k at the median and about \$200k at the 95th percentile, and protects about \$116k at the 5th percentile.

Other labelled figures:

- B9b's scratch runs (ASM, not re-run by D13c): 60% equity in the growth money adds about +\$9k at p50 and +\$55k at p95 and costs about −\$22k at p5, against an all-Treasury growth money.
- B9a (DER, not re-run): locking the surplus as well would give about \$204k in 2033 with almost no market risk.

How the plan could answer:

- (a) **Say where the leap is** (IPS and FR). A strong sentence must contain:
 - the residency is the leap;
 - the money residents and partners will rely on is bought first;
 - the growth money takes a deliberate risk for a bigger facility;
 - what that risk can cost: the facility range, never the payments.
- (b) **Run her regret test both ways, with numbers** (FR, one small table). Rows: lock / no lock. Columns: strong markets / weak markets. Each cell says what she would regret. In strong markets locking gives up upside; in weak markets not locking could leave payments unfunded. That second regret falls on the people she said matter most (D13a-N01).
- (c) **Defend the growth-money equity weight in her terms** (IPS and FR). The median barely moves at 50/60/70% (\$205k/\$207k/\$209k, brief section 6), so the choice is about the tails.
 - Her regret rule argues against the minimum.
 - The case's "appropriate balance" (R-C38) argues against the maximum.
 - Never justify it with "she is cautious".
 - (d) **The 2031 floor share is open** (brief section 9). Her festival words ("We definitely hope to offer more as the festival grows.") fit a floor she can always keep, plus a labelled "more if markets allow".
 - A smaller floor share keeps more growth money invested until 2033: a lower announced number, but more upside.
 - The case wants "a meaningful personal contribution" (R-C64), so the trade-off needs pricing by D3/E1.
 - Her words do not pick the number.
 - (e) **WInS (ii) or (iii)**. Either way, one Trading Note reflection must answer "why so little equity?" in her terms: in 2027 almost all of her real \$300k buys the payments, and her ability to take risk starts with the 2028 money (B9a Q1).

Affects: IPS, FR, TN; R-S24, R-S25.

N2. Iteration versus a frozen plan.

- Her habit: ship, get feedback, revise, scrap (V3). A stronger line is waiting for D13a's ruling: "never be finished" (candidate D13c-C2).
- The plan: a ladder held to maturity, and an IPS that "becomes the official record" and cannot be revised after Nov 6. The Guide p.5 L156-157 (VRF) says a strategy "should not be rewritten simply because markets move or hindsight reveals a different outcome."
- How it could answer: fixed rules, with review points planned in advance:
 - the 2028 deposit;
 - the 2031 floor and announcement;

- the 2033 reserve and facility decision;
- yearly rebalancing of the growth money within bands.

The Guide itself allows "planned adjustments as funding dates approach". Show the refinement history in the TN and FR. Do not pretend the first version was right.

- Affects: IPS (planned-changes clause), TN, FR. R-S24, R-S27.

N3. "It's just a number" versus a number-heavy plan.

- The plan's working files speak in p5/p50/p95, "\$289 per basis point", "duration 9.90 years", "27bp headroom" and "200,000 paths".
- How it could answer:
 - lead with what the numbers secure;
 - keep mechanics in an appendix or in the team's notes;
 - round;
 - one chart per claim.
- She is also a data person (V9), so the answer is translation, not removing numbers.
- Affects: FR mostly; IPS wording. R-S28.

N4. Certainty language with holes in it. The definition "certain in nominal USD, barring U.S. Treasury default" is precise, but it is not the whole story. Two gaps a client who labels her own results "[unfinished]" would expect to see named:

- **Gap (i), the purchase price.** The lock is complete in January 2027 only if the ladder costs \$300k or less. In roughly 1 in 4 to 1 in 3 modelled rate paths it costs more (ASM-based: brief section 14; S3/S4). Part of the promise then waits on the 2028 deposit, which comes from income she herself called "unstable" (B8b-Q18). The joint tail, rates fall and the deposit is late or smaller, is still unanswered (brief section 9).
- **Gap (ii), nominal dollars.** The dollars are nominal US dollars. The residency pays costs in Taiwan, and a fixed \$50k buys less each year. Real erosion is also unanswered (brief section 9). Her No. 1 priority is people in the room (B8b-Q38); nominal certainty does not guarantee the same number of people.

How it could answer:

- one IPS clause and one FR box with the definition and both gaps in plain words;
- the rule if rates fall first;
- amounts in US\$ with a today's-cost equivalent.

Affects: IPS, FR. R-S26, R-S28.

N5. A self-sufficient fortress versus her partnership habit.

- The case requires her to fund the payments alone (R-C48), and the plan complies. But her institutions grew with partners (V5, V6).
- How it could answer: describe co-sponsors' role positively. They may fund the facility and extra operations (R-C44). Her secured part is the first commitment, and it invites theirs.
- Affects: FR (co-sponsor draft). R-S28.

N6. "50 things" versus one issuer. The 2027 book is almost entirely U.S. Treasuries. This is **not a real clash**: her line is about careers (D13a), so claim neither a clash nor an echo. If asked:

- the promise is concentrated on purpose, to match its currency and dates;
- diversification lives in the growth money (VT holds about 10,088 stocks, S3).

Affects: FR. R-S24.

N7. Creativity.

- Her record: "create it yourself" (B2b-Q07), and the case's "willingness to pursue unconventional opportunities" (R-C18).
- The plan: standard liability matching, which is what pension funds do. Say so.
- Its originality is fit, not invention, for example:
 - a floor bought so the 2031 range never has to be walked back;
 - the 2028 deposit treated as the moment her ability to take risk begins (B9a Q1).

- Do not call the plan "innovative".
- Affects: FR, IPS pitch. R-S24, R-S28.

3.4 Echoes proposed in Phase B that do not survive checking

Phase B echo	why it fails	use instead
"Floor-first leap" (B8a Q02, B8b Q3, B9a/B9b grids, M018)	Rests on a reporter's sentence (D13c-R1, section 7c); its "So" follows personal material (D13a; D13c re-read). Not her decision rule	E2 (risk she bears vs risk others bear)
"risk-adverse" as her profile (B8a-Q03)	About students choosing a business major	Do not describe her risk tolerance with it
"Both, not either-or" as an allocation rule (B8b-Q07)	About graphic vs prose novels in class (D13a)	The case's own "appropriate balance" (R-C38)
"Diminishing returns" (B9b-Q12)	About dance songs	Nothing
Wish for a trade-offs "author's note" as her rule for explaining decisions (B8b-Q17)	About translation choices; in the same answer she says "But at the same time, I wish I didn't always have to explain myself." (D13a)	Keep the trade-offs box short for readability, not "because she asked"
"Write a letter to Future You." as her planning habit (B8b-Q08)	A writing prompt for students	A plain pre-committed bad-year rule, no letter
Co-living, Yilan art residencies, Taiwan as a top place (Maverick Show)	Transcript lines: PARAPHRASE-UNVERIFIED or EXCLUDE-PRIVACY (D13a)	The case's residency description (R-C39, R-C40)
B3b-Q09 (a line on which languages her memoir appears in) used as a market fact	Answer to a family question: EXCLUDE-PRIVACY (D13a). Its text is not repeated here	State any market fact in our own words from a non-private source
Kirby's fall, "Queen of Balance" (B3b Q13, M144)	Fiction and publisher copy; the one first-person "fall" line is private	Nothing: no metaphor
The pull quote as "her motto"	Unattributed (D13a: 8 searches, 36 pages)	Cite it as the case's line (section 5)

4. What she would reject on sight (with evidence)

"Strength" rates how directly the evidence supports the claim. The team decides.

#	what	evidence	strength	where it bites
R1	The case pull quote presented as her words ("Laura said", "her motto")	D13a: no speaker line on the case page, and no source in 36 pages. Her first public work was "Originally intended as a response to misinformation" (case p.1, VRF)	strong	IPS pitch, FR
R2	Book-title puns and plot metaphors ("messy roots", "lessons for falling", "Queen of Balance", "fall safely")	B8b-Q13 and B8a-Q28 on her own drafts (extension, INT); the fiction and privacy findings (V4)	moderate, and costs nothing to follow	all
R3	Her identity as a reason for a holding (Taiwan or China tilt "for her heritage"; identity-themed funds; "returning to her roots")	Brief section 4 tokenism rule; the case gives no background-to-portfolio link and names no values screen (brief section 8 item 19); "Born in Wuhan, China, and raised in Texas" (case p.1): Taiwan is neither (B3b Q08); M045	strong	WInS now, TN, IPS, FR
R4	A finance-prestige tone (bravado, "beat the market", rank and P&L talk)	Her D13a-verified words: B8b-Q21 ("just a number") and B8b-Q41 (her "bankers" joke). Stronger lines await D13a's ruling (D13c-C3 "excel models" / "corporate coffee chats"; D13c-C4 "McKinsey or Goldman Sachs"). She is not anti-finance: Wharton degree, a trading internship and a consumer-protection internship (V12 facts). She would reject the tone, not finance	moderate	all
R5	False precision and unlabelled probabilities ("guaranteed", "95%" with no model, "\$207,413", "0% miss" without "by construction")	B9b-Q09 ("unfinished"), Good/Unsure/Bad grades); B9b-Q11 (limits stated); B9b-Q02 (A/B testing); statistics degree (R-C12)	moderate to strong	IPS, FR, TN
R6	Hidden costs or a sales tone	Consumer-protection internship and financial-literacy teaching (V12); the CFA Institute Asset Manager Code the Rules page points to (R-W29)	moderate (inference from her record)	IPS, FR

#	what	evidence	strength	where it bites
R7	A one-sided risk story (drama, or silence)	B8a-Q34, B8a-Q33; case p.1 on misinformation	moderate (a principle carried over)	FR
R8	A wall of text; talking down	B8b-Q06, B3b-Q13	weak to moderate (promotional copy), but easy to follow	FR
R9	AI-generated images or imitation comic art in the FR	No verified statement by her on AI (B3b, B9b). This is an ASSUMPTION from her profession. She has worked for an AI product client ("AI-powered online learning software", B9b-Q06), so never call her anti-AI	ASM	FR
R10	Guessing at her real life or finances (her teaching job's future, a writing break, where she lives), or treating the case as her real plan	"While the client is real, the financial scenario is developed specifically for the competition" (R-W4); brief section 4 privacy rule	strong (rule)	FR
R11	Overpromising: leading with the top of the range, or a range with no floor	Case R-C66; D13a-N01	strong	FR co-sponsor draft
R12	Hype trades (AI or thematic funds bought because they are hot)	B8a-Q37 (2016)	weak to moderate	WInS now, TN

5. Tokenism and privacy guardrails

5.1 Rules (checklist for every deliverable)

- [] **Swap test** (B3b Q12). Keep a Laura-specific sentence only if it rests on her professional record or the case: the statistics degree, income she called "unstable", institution-building, credibility with backers. Cut it if it only works because of her ethnicity, sexuality, gender or birthplace.
- [] **Themes, not reasons.** Identity, belonging, immigration and community may be named as what her books and public work are about, as context for whom the residency serves. They are never a reason for a security, a weight, a tilt or a screen.
- [] **No "falling", "Queen of Balance" or book-title metaphors** anywhere (V4, R2). The case's own "appropriate balance" (R-C38) is fine.
- [] **No invented philosophy.** We hold no verified statement by her on investing, markets or wealth (V8).
- [] **Pull quote.** Cite "The only person who needs to believe in something is yourself." as the case's line (Client Profile p.1). Never "Laura said", "in an interview" or "her motto" (D13a).
 - If it is used at all, pair it with one dated line of hers (D13a-N01, 2016, student-era) and tie both to a promise she can keep (a floor already bought).
 - The other near-match, D13a-N02 (2016), carries a gender framing that is hers. Leave it out of deliverables.
- [] **Date and context.** Every quote carries its outlet, year and topic. Say "as a student" for 2016-2018 lines.
- [] **No trimming into new meaning.** These lines are fragile: "risk-adverse" (B8a-Q03); "No more gimmicks." alone (B8b-Q14); "diminishing returns" (B9b-Q12); "explain myself" (B8b-Q17). B8a-Q05 describes other people, not her.
- [] **Pronoun:** "she", as the case uses. Her site lists she/her and they/them (D13a note on B8b-Q03).
- [] **Facts exactly as she states them:** "bestselling" (case); "finalist" for the Harvey and Goodreads awards (D13a note on B3a-Q03).
- [] **Real-life news stays out** (her college's future, a writing break, her home). Use industry-level facts only, to motivate labelled stress tests (B3b Q15).
- [] **Visuals:** no AI images and no comic-style pastiche (ASM, R9). Record AI use exactly (R-W46).
- [] **Never suggest contacting her** (R-W16: contact means disqualification).

5.2 Using her public work themes without decoration

allowed (spec)	not allowed
Her record as an institution-builder (a club, a festival) as evidence for how the 2031 message is ordered	Her identity as the reason Taiwan was chosen, or as the plan's theme
Her own word "unstable" for self-employed income, once, as her public statement, to justify why the payments are bought from the first deposit (M216)	Any guess at her actual finances, contracts or earnings
Her data habits (purpose, proof, "[unfinished]") as the reason every number is graded	Imitating her comics or data projects in the FR's design
One or two quotes in the FR, each tied to one decision	Quotes as decoration at the top of sections
The case's own description of her work's themes (R-C10: "identity, belonging, and the power of storytelling")	Linking those themes to an asset, a country weight or an ESG screen the case never names

5.3 Where a Laura quote may appear (spec; the cap is ASM)

- **TN:** none. There are 100 words for the reasoning behind a trade (TN p.2).
- **IPS:** none. "Formal citations are not permitted" (R-I55), and 500 words are scarce. Echo her principles in the team's own words (for example, the promise comes first; risk sits with money nobody else relies on).
- **FR:** at most 1-2, each with its outlet, year, what she was answering and the decision it supports. Suggested pair: D13a-N01 with the co-sponsor floor; B8b-Q12 with the growth-first draft the team scrapped.

5.4 Do-not-use list (ids and descriptions only; this file does not reproduce private text)

The full list is in section 7b: D13a's seven EXCLUDE-PRIVACY ids plus four more lines D13c flags.

6. The 15 strongest verified quotes for the team (all D13a VERIFIED-PRIMARY; re-checked by D13c)

#	id	exact words	source, date	why it matters (one line)
1	D13a-N01	"It's actually harder as an entrepreneur because not only are you letting yourself down, but more importantly, you're also letting other people down — people who were your earliest supporters and were the first to believe in you."	Daily Pennsylvanian, 2016-01-28 (student-era)	The only verified line on others' belief. It grounds "no risk on money others rely on" and "announce only what is bought"
2	B1b-Q11	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	Overachiever, 2021-03-15	Her own risk rule. The plan must answer it, or look timid (N1)
3	B8b-Q18	"It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""	Overachiever, 2021-03-15	She named her own income "unstable". This is why the 2028 deposit is uncertain and the promise is bought first. Use once; never speculate
4	B8b-Q38	"...Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on"	Local News Matters, 2026-02-12	Her newest stated priority is access, which makes "operations first" hers. It is about festival grants
5	B8b-Q37	"All of it is way beyond our initial passion project. We're thrilled that the community really supported us and wanted us to continue to grow the festival"	Local News Matters, 2026-02-12	Growth came after community support. This is the shape of the 2031 message
6	B8b-Q10	"Only after we realized EVERY student deserved to hear their words of wisdom did we decide to build something to cultivate it."	Sign.al founder's note, c. 2022 (date ASM)	Validate, then build: "proof before promise"
7	B2b-Q07	"If what you want doesn't exist, create it yourself."	Poets&Quants, 2018-03-30 (student-era)	Her founder method, and the residency itself. Many teams will quote it, so it is not distinctive
8	B8b-Q21	"it's great having thousands of likes, but at the end of the day, it's just a number. You never really understand the human connection behind it"	Overachiever, 2021-03-15	Report what numbers secure ("10 of 10 years"), not bare dollars
9	B8b-Q13	"But in the process, I lost the heart of my writing— authentic, personal connections."	The Nerd Daily, 2025-03-08	Her own authenticity test. Cut what is only for show. Pair with B8a-Q28, never "No more gimmicks." alone

#	id	exact words	source, date	why it matters (one line)
10	B8b-Q12	"I scrapped three completely different drafts before landing on this one."	The Nerd Daily, 2025-03-08	Showing the discarded growth-first draft is her own habit (Articulation)
11	B8b-Q31	"I give myself a deadline for when I must post the art, finished or not."	Geeks OUT, 2022-05-11	Purchase dates set by rule; early WInS trades refined later
12	B8b-Q35	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	Pulse Spikes, 2021-06-18	The IPS opens with a purpose; every Trading Note names the holding's job
13	B8a-Q19	"That [story] was something I could prove better with data"	Pulse Spikes, 2021-06-18	Each FR chart proves one claim; data serves the story
14	B9b-Q09	"The [unfinished] results are below."	lauragao.com/ rewriting-herstory (undated)	She labels unfinished results and grades them: grade our numbers and name the certainty gaps
15	B2b-Q08	"Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia."	Poets&Quants, 2018-03-30 (student-era)	She taught financial literacy to high-schoolers, so a high-school team must be plain and honest about costs

7. Separate labelled lists: lines NOT usable as her words (or not yet)

7a. PARAPHRASE-UNVERIFIED (never in quotation marks as hers)

id(s)	what	how it may be used
B2a-Q13 = B2b-Q11 = B8b-Q01 = B8a-Q39	The case pull quote "The only person who needs to believe in something is yourself."	As the case's words (VRF, Client Profile p.1; R-C21, R-AN30). Never as hers
B8a-Q21, B8a-Q24	Maverick Show transcript lines (a transcript with visible errors, not checked against the audio)	Not at all as quotes. For the career-leap idea use B8b-Q18 / B1b-Q11

7b. EXCLUDE-PRIVACY or do-not-use (never use, however accurate; text not reproduced)

- **D13a's seven EXCLUDE-PRIVACY ids:** B3b-Q09, B8b-Q23, B8b-Q24, B8b-Q34, B8a-Q20, B8a-Q22, B8a-Q23. Their topics are family, home and travel life, coming out, and her personal identity story.
- **Added by D13c (these were not in D13a's scope):**
 - The Nerd Daily 2025-03-08: the only first-person line about falling (and climbing back). It sits in an answer about her family and identity: EXCLUDE-PRIVACY.
 - Pulse Spikes 2021-06-18: the sentence about how her parents saw her student choices: EXCLUDE-PRIVACY. It sits directly before candidate C4.
 - Geeks OUT 2022-05-11: a line about where she lived and travelled (B8b V35): EXCLUDE-PRIVACY (home and travel life).
 - Mission Chronicle 2025 (B8b V44): a line about growing up between cultures. Identity context: do not use for finance (tokenism), as B8b itself advised.

7c. Not her words (reporter, publisher or case text that Phase B leaned on)

text (short)	real author	status and use
"So when she got the book deal with HarperCollins, she gave Twitter her notice." (D13c-R1)	Input/Inverse reporter, 2022-03-07	Verbatim on the page (D13c), but it is the reporter's words, and its "So" follows personal material. Not evidence of a floor-first rule
"Even before her comic went viral on Twitter, she'd made a tentative plan..."	same reporter	Reporter's account; do not present it as her plan
"remaining free for exhibitors and attendees" (D13c-R2)	Local News Matters reporter, 2026-02-12	Usable as a fact about the festival, not as her words
"helped launch the Anti-Resume Project ... normalize failure"; "her anti-resume includes receiving 33 job rejections"	Pulse Spikes reporter (2021); Daily Pennsylvanian reporter (2019-03-17)	Reporter paraphrase. The 2019 article says The Signal launched the project in 2019 and profiled her as an alumna. Cite as the reporters' account
"Once dubbed the Queen of Balance..."	book copy about the fictional heroine	Never about Laura
"Laura Gao believes stories have the power to change how people see the world."	the case (R-C9), in Wharton's house template (M136)	The case's framing (VRF). Never "Laura says"

text (short)	real author	status and use
"Laura Gao cares so passionately about making a difference..."	Wharton staff nominator on the Poets&Quants page	Not hers
B8a-Q05 "each has juggled passion against practicality..."	her words, but about other people	Never as a self-description

7d. D13c new finds: verbatim on fresh copies (2026-09-27), NOT in D13a's register

These need D13a's speaker, context and privacy ruling before anyone uses them as her words. D13c's reading is given for each.

id	exact words	source, date	why it matters	D13c caution
D13c-C1	"I think for me, that took a lot of fear out of it."	Overachiever, 2021-03-15 (same answer as B1b-Q11)	Her regret rule is how she manages fear	Quote only together with B1b-Q11 ("that" points back to it)
D13c-C2	"My biggest advice is to be OK with the fact that your significant projects will never be finished. It must constantly be sketched, erased, dropped in rain, ripped apart and sketched again."	Daily Pennsylvanian, 2016-01-28 (student-era; same answer as D13a-N02)	Iteration; the tension with a frozen IPS (N2)	Do not continue into the next sentence, which carries a gender framing
D13c-C3	"After all, the Laura Gao I knew loved building apps, not excel models; perfected her artwork, not her resume; and devoted herself to art galleries, not corporate coffee chats."	Poets&Quants, 2018-03-30 (student-era; her first-person answer)	Corrects Phase B, which credited it to the nominator. Her own contrast with finance-prestige culture (R4)	D13a's note on B8a-Q02 says "the sentence before it contains family material". On the page C3 is the sentence directly before B8a-Q02, and the family reference is in the sentence before C3, not in C3 itself. D13a to confirm; never quote or paraphrase that earlier sentence
D13c-C4	"And to my peers, who were like 'why aren't you going for McKinsey or Goldman Sachs instead of wasting your time doing art,' it seemed like a failure, too."	Pulse Spikes, 2021-06-18	Peer pressure toward prestige finance, told as "failure" (R4)	Its "And ... too" points back to the private sentence about her parents. D13a to rule whether it can stand alone. Until then, background only
D13c-C5	"There's a vibrant community of queer comic creators here in San Francisco, but why wasn't there any single event where we could celebrate that community and bring everyone together?"	Local News Matters, 2026-02-12	Build what is missing, in 2026 (V5)	Not needed: B2b-Q07 and B8a-Q14 carry the idea without naming an identity group
D13c-C6	"When we discuss COVID and how it connects with Chinese or Wuhanese people, we have to be mindful of how our perspective may negatively bias us against people from that region, and instead, focus on the full human story."	HuffPost, 2020-04-29	Complete framing (V13)	Principle only. Never quote it near Taiwan
D13c-C7	"mainly because I'm so interested in them and I don't want to pigeon-hole myself into one thing."	Pulse Spikes, 2021-06-18	Keeping options open (V10)	It starts mid-sentence: quote it with the outlet's preceding words "I have done a lot of different things,"
D13c-C8	"[I shared an anti-resume] mainly because I started off in business school, and then I ended up gravitating towards other things,"	Pulse Spikes, 2021-06-18 (the bracket is the outlet's)	Her own account of making failures public (V3)	The sentences after it are about her parents: never continue the quote

Two professional facts were also verified (D13c-F1, D13c-F2; section 1, V12): the Belvedere Trading and Consumer Financial Protection Bureau internships, as listed in 2018.

8. Method and verification record

1. Read the brief, the case and D13a in full. D13a decided each quote's status; D13c did not overrule any D13a verdict.

2. Read B8a, B8b, B9a, B9b and B3b in full. Took their value readings as hypotheses and kept only those that D13a-VP words support.
3. Re-read, on primary pages, the passages behind the two biggest Phase B claims:
 - the Input/Inverse "So when she got the book deal..." paragraph (confirming D13a);
 - the Overachiever "take the jump" answer.

Also read The Nerd Daily Q&A, Pulse Spikes, the Daily Pennsylvanian (2016 and 2019) and Poets&Quants for "falling", "failure" and finance lines. Private material on those pages was read but not recorded.

4. `research/insight_v1/scripts/D13c_voice_map_check.py` (run 2026-09-27, fresh copies via curl, default user agent):
 - 50 of 50 D13a ids quoted or cited here are D13a VERIFIED-PRIMARY and verbatim (a coverage check confirmed that every verified line quoted in full in this file is among them, or is a same-text copy of one);
 - 8 candidates (C1-C8), 2 facts (F1-F2), 2 reporter lines (R1-R2) and the ellipsis form of B8b-Q38 (P1) are verbatim;
 - 8 case lines are VRF.

The script prints no page text (privacy); fresh copies went only to the session scratchpad.

5. Re-ran `research/insight_v1/scripts/B10b_designer_checks.py` (seed 20260927). It reproduced the price-of-certainty frontier and the whole-portfolio equity path used in N1 (2027 1.5%, 2028 20.4%, 2031 4.2%). The 2027 figure is about 0% if the leftover waits in T-bills, as S3 plans.
6. Figures taken from other files without re-running them are labelled with their source: brief section 6 (lock-early and growth-first percentiles), B9a (the \$204k locked alternative) and B9b (the control runs).

9. Sources (all accessed 2026-09-27)

Laura's words (VP via D13a; wording re-checked by D13c):

- Overachiever Magazine, 2021-03-15: <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>
- The Daily Pennsylvanian, 2016-01-28: <https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a>
- Poets&Quants for Undergrads, 2018-03-30: <https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/>
- The Nerd Daily, 2025-03-08: <https://thenerddaily.com/laura-gao-kirbys-lessons-for-falling-in-love-interview/>
- Geeks OUT, 2022-05-11: <https://www.geeksout.org/2022/05/11/interview-with-creator-laura-gao/>
- Pulse Spikes, 2021-06-18: <https://pulsespikes.org/story/laura-gao>
- Local News Matters, 2026-02-12: <https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/>
- Input/Inverse, 2022-03-07: <https://www.inverse.com/input/culture/laura-gao-messy-roots-viral-tweet-comic-graphic-novel>
- NPR, 2020-04-04: <https://www.npr.org/sections/goatsandsoda/2020/04/04/823825436/the-wuhan-i-know-a-comic-about-the-city-behind-the-coronavirus-headlines>
- HuffPost, 2020-04-29: https://www.huffpost.com/entry/chinese-american-illustrator-life-in-wuhan_n_5ea9c739c5b63115cec2be5a
- Wharton Magazine, 2022-04-06: <https://magazine.wharton.upenn.edu/digital/a-wharton-grad-from-wuhans-exploration-of-identity/>
- BookWeb (ABA), 2022-02-07: <https://www.bookweb.org/news/indies-introduce-qa-laura-gao-1627512>
- The Signal 5-year page (c. 2022, ASM): <https://signaltemp.github.io/>
- Her site and documents: <https://lauragao.com/rewriting-herstory>, <https://lauragao.com/the-evolution-of-dance-music>, <https://lauragao.com/s/Laura-Gao-Resume.pdf> (PDF created 2024-10-24), <https://lauragao.com/s/Publicity-and-Speaking-Guide-Laura-Gao-e5n7.pdf> (PDF created 2024-03-24), <https://lauragao.com/s/Educators-Book-Clubs-Guide-Messy-Roots-klc2.pdf> (PDF created 2024-03-24)

Other pages read (not her words, or context only):

- The Daily Pennsylvanian, 2019-03-17: <https://www.thedp.com/article/2019/03/penn-anti-resume-ocr-recruiting-failure-signal>

Repo files (VRF or labelled):

- Case: competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt (p.1 L12, L13-14, L19-20, L25-26, L31-33; p.2 L69-74; p.3 L101-103, L110-120).
- IPS guide 2026_WGY_Investment_Policy-FINAL.txt L123 (R-I55).
- TN guide 2026_WGY_Trading_Notes_Analysis-FINAL.txt L12, L15-16.
- Competition Guide 2026_WGY_Investment_Competition_Guide.txt L70, L156-157.
- SMAApply_Deliverables_Page_2026-09-27.md L49-57 (R-S24 to R-S28).
- research/insight_v1/phase_D/D13a_laura_quotes_verified.{md,json}; phase_B/B8a, B8b, B9a, B9b, B3b; wins_now/securities_and_allocation_v0.md; phase_C/merged_questions.json; brief sections 6-9 and 14.
- Scripts: research/insight_v1/scripts/D13c_voice_map_check.py (new); research/insight_v1/scripts/B10b_designer_checks.py (re-run).

What this teaches

- **A quote is evidence of a value only when you know the question it answered.** "risk-adverse", "No more gimmicks." and "diminishing returns" all mean something narrower on the page than they seem to on their own.
- **Behaviour claims need the person's own words, not a reporter's "So".** The most-repeated Phase B insight ("she leaps only after securing a floor") turned out to rest on a journalist's link between two events. Her own words say something different, and more interesting: she jumps, she names the cost, and she treats letting down her backers as worse than failing herself.
- **A client can hold two risk ideas at once.** A good plan does not pick one of them; it shows which money each idea governs. Here that means payments others rely on are bought, and money only she depends on can take the leap.
- **Echoing a client honestly includes saying where you differ.** A plan that holds about 20% stocks in its riskiest years must explain itself to someone who tells herself to "take the jump". It should do so with numbers, not flattery.
- **Respect is accuracy, privacy and plain words.** The best-fitting "falling" line was private, so it stays out. The best tone for a former financial-literacy teacher is the one a classmate could follow.

E1 Strategy Architect: change proposals to the "lock early" strategy (for team approval)

Agent E1 (Strategy Architect), insight_v1 run, Phase E. Written 2026-09-28 (the first official WInS trading day). AI-generated research for Team Caplet: numbered change proposals, numbers, specifications and checklists. **None of it is text to submit.** The six students decide every change and write every WInS note, reflection, pitch sentence and IPS sentence in their own words; any AI use is recorded in the Final Report's Works Cited (Wharton AI policy, R-W46). Laura appears only through the case and her public professional record; nothing here suggests contacting her (R-W16).

Scope (brief section 17). Only WInS trading now (as the source of the Trading Notes), the Trading Notes Analysis (TN, Oct 23) and the IPS (Nov 6). Because the IPS freezes the strategy on Nov 6 and the Final Report must later apply it without redesign (IPS guide L73-75; Guide p.5 L156-159, VRF), the proposals fix the **decision rules** the Final Report will apply (reserve, 2031 range method, facility and flexibility, if-then rules), but give no Final Report content, no dollar range and no fundraising text. Purely Final Report items are one line each in section 5. **Relayed messages:** none arrived during this task.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY; **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION (a modelling or judgement input); **MODEL** = an ASSUMPTION-based model output, not a forecast; **INT** = E1's interpretation or judgement (an ASSUMPTION, never a fact about Laura). Ids: R- = case register, F- = fact register, SH-/BS- = stakeholder map, M- = Phase C questions (all under research/insight_v1/). Every security named is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK** (brief sections 4 and 14).

New script (this file): .venv/bin/python research/insight_v1/scripts/E1_split_and_slices.py (about 5 seconds; docstring lists inputs and labels). It re-uses D6's engine and random stream and prints: [1] the sleeve-split test across two forecasting houses, two bond assumptions and three fee levels; [2] which equity shares pass in every cell; [3] the whole-portfolio stock share by stage; [4] the 2031 "three slices" at 50/50 and 60/40.

Evidence read in full for this file: brief; phase_D/ D1-D12, D3_model_v2_results, D13a, D13c (sections 1-5), D13b flags, all four audit files and every "Audit corrections" appendix (corrections override the text above them); wins_now/securities_and_allocation_v0.md and S4_red_team.md; and, once they appeared in the repo during this task (checkpoint commit 32e31ac), phase_D/trading_now_brief.md and wins_now/securities_and_allocation_v1.md (agent T1). **The v1 ticket supersedes v0 for trading; where they differ, this file follows v1** ("the ticket" below means v1). T2_red_team.md does not exist in the repo on 2026-09-28. D3_quant.md and D3_model_v2_results.md carry no audit appendix yet (a D3 audit was in progress), so D3-only numbers below are pre-audit; the verified-base figures they reproduce (F-401, F-402) are unaffected. Also read: S2_growth_sleeve.md summary; phase_A/ case_register (R-C, R-I, R-T, R-S, R-AN), fact_register, stakeholder_map (summary, BS-01 to BS-20, section 7), wins_week1_guardrails (summary); phase_C/survivors.json and the parked items M007, M008, M015, M021, M033, M035, M040, M149; the council chair memo and round-2 referee ruling (history only); the official case, IPS guide, TN guide and Competition Guide text files.

Terms used (defined once):

- **Ladder:** ten zero-coupon U.S. Treasury bonds (STRIPS), one maturing shortly before each \$50,000 payment (2033-2042). **Rung:** one of those bonds.
- **Growth money (sleeve):** everything the ladder does not need (about \$158k after the 2028 deposit).
- **Floor (2031):** the part of the growth money bought in January 2031 as a Treasury maturing before 1 January 2033, so its 2033 value is known in 2031.
- **Funded ratio:** money available divided by the market price of the ten payments on the same day.
- **p5 / p50 / p95:** in a simulation, the value 5% / 50% / 95% of paths fall below (p50 = median).
- **bp:** basis point, 0.01 percentage point.
- **Option (ii) / (iii):** the two candidate WInS books in the ticket: (ii) the plan's mix after both deposits, scaled to \$300,000; (iii) Laura's literal January 2027 book.

0. Summary: the proposals ranked by deadline and impact

The architecture does not change: buy the ten payments with Treasuries first, take market risk only with money nobody else relies on, announce in 2031 only what is already bought (brief section 8 items 1-8 stand). What changes is that the frozen IPS must carry **rules, parameters and honest wording** that the current strategy (brief section 7) leaves loose, and that two tier-1 WInS decisions must be taken before the first order.

#	Change (one line)	Tier / deliverable	Confidence
P13	WInS book: vote (ii) or (iii) before the first order. E1 leans (ii) with a scaling sentence; (iii) is fully specified as the alternative, with a named third trade	1: WInS-now, TN, IPS	medium-low (judgement; both comply)
P5	Growth money 50% world stocks / 50% short Treasuries (was 60/40), chosen by one stated rule applied under both published forecasts	1 if (ii): WInS-now, TN; 2: IPS	medium
P14	Three-note plan per option; note elements from Guide L84-86; "tested" on a pre-registered window; no planned discipline trade; dated rung only where it is genuine	1: WInS-now, TN	high (method)
P2	Certainty = bought at market prices and held to maturity, in nominal US\$, barring a U.S. default, plus the two gaps named plainly	2: IPS; 1: TN vocabulary	high
P1	Central idea and pitch stated as a rule the firm controls, not a date ("bought in January 2027" and "fits inside \$300k" dropped)	2: IPS; 1: TN consistency	high
P3	January 2027 rule: buy in full on arrival, longest-dated first, finish from the 2028 deposit before any growth money; no staging or yield triggers	2: IPS	high (direction)
P4	2028 deposit rule (smaller, late, instalments) with the joint tail stated honestly (corrects D8 rule 1)	2: IPS	high
P7	2031 range method: three slices (bought / promised-at-risk / kept), uncapped, two-sided confidence; default 80/10/10, alternate 70/15/15	2: IPS	high (method), medium-low (numbers)
P8	Flexibility kept by rule, held in US\$ short Treasuries (safety, not growth); US\$ binding; no NT\$ conversion or hedge before the facility decision	2: IPS	medium-high
P9	Reserve = the ladder, held to maturity, self-liquidating; "bought 2027, designated 2033" reconciled	2: IPS	high
P6	Risk tolerance stated goal by goal (need / ability / behaviour), with where her big bets already are	2: IPS; 1: TN reflections	high
P10	Governance: who decides, review dates, bad-year pre-commitments, worded so it cannot read as permission to revise after Nov 6	2: IPS	high
P11	One consolidated if-then rule set (four rules + governance) replacing brief section 7's prose	2: IPS	high (structure)
P12	Fee principle only (no ongoing fee on the bought payments; all costs from growth money); first clause to cut	2: IPS	medium
P15	Dated decision log, evidence grades and AI-use log from today	1: WInS-now, TN	high
P16	IPS word budget, page-fit test on the exported PDF, and a paraphrase test with six readers	2: IPS	high

Section 2 (P17) lists what must **not** go into the IPS. Section 3 lists the conflicts E1 could not resolve, with both sides. Section 4 is the team's decision checklist in date order.

North Star link (INT). The plan becomes Laura's, not a generic liability match, through five checkable decisions, each tied to a case line: (1) the payments are bought before her career-income deposit can matter (case L43-45, R-C27); (2) nothing is announced to co-sponsors that is not already bought (L114-116, R-C66; her income is reputation-driven, BS-03); (3) certainty is defined by a price she can check, not a model (L16 statistics degree, R-AN32; L97-99); (4) stock risk sits only where she alone bears the cost (the case's "Although", L69-74, R-AN16); (5) flexibility for a project that will change is kept by rule, not by leftovers (L102-103, R-C58).

1. What stays the same (no change proposed)

- Lock the whole ladder; do not tier or roll the 2037-42 payments (brief section 8 items 1-2; F-108 +\$20k per -100bp) [VRF inputs, MODEL].
- STRIPS rather than coupon Treasuries for Laura's real ladder (coupon reinvestment risk ~\$5k, F-113) [ASM method].

- Risk measured on the growth money only; more equity "buys range, not median" (F-407; D3 M034) [MODEL].
- A single dated 2031 floor purchase; no ratchet, no barbell (D3 M055: a plain ~42% equity sleeve matches the ratchet's bad case within \$2k of median; D3 M238: a 50% barbell can promise only ~\$101k bought in 2031 vs ~\$165k) [MODEL].
- WInS hedge = IEF + TLH at duration weights (about 10 years), SPTL only if a Session Rules cap is under 44%; broad world equity (VT) and short Treasuries (VGSH); no Taiwan, AI, thematic, leveraged or single-country funds (ticket sections 1-3; S2; D8 M081) [VP issuer facts via S3/S4].
- The three "never" rules in WInS (never sell the hedge after a rate rise; never trim it after a rate fall; never day-trade) and the five-box gate before the first order (ticket) [VP rules via A4].
- No Laura quotes in the TN or the IPS; the case pull quote is the case's words, not hers; no "floor-first leap" (brief section 16; D13a; D13c) [VRF/VP].

2. Change proposals

Each proposal gives: **Change**; **Why** (case R-id or fact F-id, and the Laura link); **Numbers** (script and status); **Side effects and cost**; **Touches** (TN / IPS / WInS-now); **Complexity** (does it earn its place?); **Confidence**; **Team decides**.

P1. Central idea and the 50-word pitch: a rule the firm controls, not a date

Change.

- The central idea (IPS guide L16, L35-36, R-I9, R-I17) is stated as a rule: no dollar of Laura's money takes stock-market risk until all ten payments are bought (with the 2028 deposit finishing the ladder first if prices rise before January 2027). Replace every headline use of "the payments are bought in January 2027" and "the ladder fits inside \$300,000" (brief section 7) with this rule.
- **Pitch content spec (elements, not wording; 50-word hard cap, R-I16):**
 1. the rule above, in plain words (about 12-15 words);
 2. the client benefit, stated conditionally: once bought, the ten payments no longer depend on markets or on her future earnings (about 12-15 words; M143);
 3. what the rest is for: growth toward the facility contribution, with part kept for flexibility (about 10-12 words);
 4. optional, only if words allow: the 2031 range whose bottom is already bought (about 8-12 words).
- **Must not contain:** a date-certain claim; "fits inside \$300,000"; "guaranteed", "risk-free", "100%", "95%"; jargon (LDI, duration, DV01, STRIPS, liability matching); more than one number (preferably none); any Laura quote, the pull quote, identity imagery or book-title puns; "safe" (0 uses in the six official files, D9 M062).
- **IPS opening** (IPS guide L53-57: what, why): repeats the same rule and states the if-rates-fall branch once (P3), so the pitch and the IPS cannot drift apart.
- Every phrase in this file that describes the rule or a benefit is **content, not wording** (AY2 correction 13): the team writes its own sentences and does not copy these.

Why.

- The date claim is false in about 1 path in 3 for the real ladder (P3 numbers), and overpromising is the case's own credibility test (L114-115, R-C66) [VRF]. A rule is true in every path by construction (D7 M011) [INT].
- Laura link: her 2028 money comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (L44-45, R-C27, R-AN31) [VRF], so the benefit that only a client with career-driven deposits gets is that the promise stops depending on that income (D9 M143) [INT]. Swap test passes (D13c) [INT].
- The IPS guide asks for "the central idea behind your approach and how it is designed to support your client's financial goals and future funding needs" (L35-36) [VRF].

Numbers. Word costs: elements 1-3 about 34-42 words, a fourth only in very plain words (D9 pitch table) [INT]. P(ladder > \$300k on 2027-01-01) about 1 in 3 for the real Nov-15 ladder (P3) [MODEL]; this is why the date claim goes.

Side effects and cost. The pitch loses a vivid date. Under WInS option (ii) the rule is visibly contradicted by a day-one VT buy unless the IPS carries the scaling sentence (P13) (AY2 rank 1) [VRF]. **Touches.** IPS (pitch, opening); TN (every note must be consistent with it, TN L30-31). **Complexity.** Zero added words; it replaces a claim with a truer one. Earns its place. **Confidence.** High that the date claim is unsafe and the rule is true; medium that it is distinctive (rivals' submissions are unobserved, BS-10) [INT]. **Team decides.** Wording; whether element 4 fits.

P2. The certainty definition and its two honest gaps

Change. Replace "certain in nominal USD, barring U.S. Treasury default" (brief section 7) with a definition that has three parts and names two gaps. **Elements the IPS must contain (words are the team's):**

1. **Bought, not forecast:** each payment is matched by a Treasury that matures shortly before its date and is held to maturity (for Laura's real ladder; in WInS the funds only "move like" the payments, P14).
2. **How it is evaluated:** by price. At purchase the money covers the market price of the ten payments (funded ratio at least 1); no model percentage is used for the payments (case L97-99 asks how certainty "was evaluated", R-C54, R-C55) [VRF].
3. **The residual risk named:** certain in U.S. dollars, barring a U.S. Treasury default.
4. **Gap (i), the January 2027 price:** if rates fall before January, the ladder can cost more than \$300,000; the rule (P3) finishes it from the 2028 deposit, so until then part of the earliest payment depends on that deposit.
5. **Gap (ii), nominal US\$ vs Taiwan costs:** each payment is a fixed US\$50,000 "not adjusted for inflation" (L90, R-C46) and the residency spends in Taiwan, so the same payment buys less each year and moves with the exchange rate; named, not solved (support beyond her commitment may come from others, L85-86, R-C44) [VRF].
6. **Vocabulary:** "uncertainty" is the official word for what markets can do to both goals (11 uses across the official files; case test 3, L140-141) [VRF via AY2]. For the payments it is limited to the named residuals above; "certainty" belongs to the payments, "confident" to the 2033 range, "credibility" to Laura's standing (D9 M062) [VRF].
7. **Grade words:** every forecast-based statement carries a marker ("we assume", "history suggests"); prices carry "at today's prices" (D12 M231) [INT].

Why. The case requires teams to "define what they consider a high degree of funding certainty, explain how they evaluated that level of certainty, and identify the assumptions" (L97-99) [VRF]; a statistics-trained client (L16, R-AN32) would reject a model "95%" (brief section 8 item 6: no-lock miss rates 3.3-9.3% depending on rate-drift assumptions) [MODEL]. Brief section 16 makes both gaps binding [VRF]. D13c N4: a client who labels her own results "[unfinished]" would expect the holes named [INT; the line itself is VP but FR-only].

Numbers (IPS carries none of them; they justify the words):

- Gap (i): P3 below.
- Gap (ii): a fixed \$50,000 is worth \$42,063 (2033) and \$33,681 (2042) in 2026 dollars at JPM's 2.5% inflation (F-114) [ASM inflation]. Taiwan construction costs rose about 3.54% a year since 2021 and 2.11% a year 2008-2026 (F-508; D4) [VP inputs, derived]; a fixed US\$ amount buys about 11-20% less building by January 2033 on those paths (D4 table A) [ASM]. USD/TWD 31.82 on 2026-09-18; two-year moves sd 6.8% since 2006 (F-510, F-513) [VP inputs, derived].
- Residual credit: Moody's cut the U.S. to Aa1 on 2025-05-16 [SNIP, secondary; not checked on moodys.com].

Side effects and cost. About 45-60 IPS words (P16). Naming gap (i) admits that the promise is not complete on day one in about a third of paths; that is the honest price of not timing rates. **Touches.** IPS (definition); TN (the certainty vocabulary governs every note from the first trade). **Complexity.** Earns its place: it answers a "should" in the case directly and removes the one overclaim a statistics reader would catch. **Confidence.** High. **Team decides.** Wording; whether to mention the currency gap in the same clause as inflation (AY2 recommends it).

P3. The rule if rates fall before January 2027

Change. Fix in the IPS (elements, not wording):

1. **Buy in full on arrival** (1 January 2027). No staging, no waiting for a yield level.

2. **Longest-dated payments first** if the money does not cover all ten.
3. **Completion before growth:** any unbought part (always the earliest payment or payments) is the first use of the 2028 deposit, before any growth investment; if the deposit arrives in instalments, completion comes first from each.
4. **Headline, conditional:** fully bought in January 2027 if prices allow, otherwise when the second deposit arrives (the case dates it to the start of 2028). Do **not** write "by 1 January 2028 at the latest" (AX1b item 3).
5. **Dependence stated in words,** a typical case and a bad case, never "at most" (AX1b item 2).
6. **If rates rise instead:** the few thousand dollars left over wait in Treasury bills until 2028, then join the growth money (ticket section 8).
7. The reason against staging is the numbers, **not** Guide p.5 (that line governs the post-IPS stage; Guide L89-90 says the strategy may evolve before the IPS) (AX1b item 4; AX2 D2 C3) [VRF].

Why. The case requires "a high degree of certainty" (L91, R-C47) and forbids relying on outside money (L91-92, R-C48) [VRF]; the 2028 deposit is Laura's own contribution to the portfolio, so using it is not outside funding (D1) [INT, low risk]. A rule written before the surprise is what lets the Final Report apply rather than redesign (IPS guide L73-75; Guide p.5) [VRF].

Numbers (D1_purchase_rule.py, AX1b_rates_audit.py, strategy_mc_v2.py (a), D3_joint_tail.py):

- The real Nov-15 STRIPS ladder costs \$294,387 for purchase on 1 January 2027, priced from the 2026-09-25 curve; headroom \$5,613 (19bp) [VRF inputs, ASM method; D1, D3, S1].
- It would have cost more than \$300,000 on 176 of 185 trading days of 2026 so far (173 of 185 for the idealised exact-date ladder; F-111) [VP inputs, derived].
- $P(\text{cost} > \$300\text{k on } 2027-01-01)$: 30-34% for the real ladder, 24% for the exact-date ideal [MODEL, zero-drift, 2026 volatility]; model-free check: the 10-year yield fell 19bp or more over 68 trading days in 32-36% of windows since 1962/1990 [VP inputs (FRED DGS10), derived; AX1b]. Say "**about 1 in 3**" for the real ladder.
- Gap sizes: -50bp \$9,169; -100bp \$24,793 (top-up in January 2028 \$25,711, 17% of the deposit); the worst 2026 curve (Feb 27) \$27,631 (68.7% of the 2033 payment waits); a repeat of the worst 3-month fall since 1990 (-167bp, late 2008) leaves all of 2033 and ~16% of 2034 waiting, \$48.6k (32% of the deposit) [MODEL on VP inputs; D1, AX1b].
- Staging: four quarterly tranches or a "wait for a 5% margin" trigger give no reliable gain in the expected funded ratio (+0.2 to -0.5 points, depending on whether rates are assumed to follow today's forwards or stay put) and a 27-35% chance of ending below 1.00, i.e. of needing the 2028 deposit (tranche p5 0.96-0.97; trigger p5 0.90-0.91), against a known 1.02 if bought on the day [MODEL; D1 [7], AX1b item 11].
- Longest-first vs nearest-first: a further -100bp during 2027 adds +\$1,265 (+5%) to the top-up vs +\$3,831 (+15%); if the deposit never comes at -100bp, \$31.4k vs \$47.7k of payment face is left unbought [MODEL; D1, AX1b item 23].

Side effects and cost. In about a third of paths the ladder is complete only in January 2028, and the growth money starts smaller by the top-up (mean ~\$7k when a gap exists; D3 v2 (a)) [MODEL].

Counterpoint (AX1b item 23): under longest-first any shortfall falls on the residency's first operating year.

Touches. IPS (rule 1 of P11). TN: optional by-product only (the hedge reflection may say the hedge was set to what the first deposit buys at today's prices, with the top-up rule named). **Complexity.** One rule with branches, about 25-35 words (D8). Earns its place: it binds in about 1 path in 3. **Confidence.** High on the direction; medium on the odds (model assumptions; history agrees). **Team decides.** Wording only.

P4. The 2028 deposit rule (smaller, late, in instalments) and the joint tail

Change. Fix in the IPS (elements):

1. The 2028 deposit first completes any unbought payment (P3); only what is left becomes growth money.
2. **Smaller:** the payments are still completed first; the facility range shrinks.
3. **Late or in instalments:** completion comes first from each instalment; growth money starts when the ladder is complete; the facility range shrinks slightly.
4. **Joint tail (rates fall before January 2027 and the deposit is smaller than the gap):** all of the deposit goes to the unbought payments, no growth money is created, no 2031 floor is announced, and

the unfunded part of the earliest payment is stated. It is **not** true that "only the facility range shrinks" in this case; correct D8 rule 1 (AY2 rank 2; AX1b) [VRF].

5. **Labelled as the team's own stress scenario:** the case says she "will contribute" (L43-45, R-AN35) [VRF].

6. **No facility floor is announced until all ten payments are bought** (D3 M053) [INT].

Why. The deposit comes from income Laura herself described as unstable (VP, D13a B8b-Q18; FR only, never in TN or IPS) and book advances arrive in instalments (BS-13) [VP via A3]. The Portfolio Analysis criterion asks for "varying market outcomes" (R-S26) [VRF]. A frozen rule must not overclaim (IPS guide L73-74) [VRF].

Numbers (strategy_mc_v2.py (b), D3_joint_tail.py, D1_purchase_rule.py [8]):

- Deposit arrives in full, late (January 2029) or at \$75k: payments short in 0% of tested paths [MODEL]. Any deposit of about \$43k or more fills the gap even after a 150bp fall [MODEL].
- Late by one year: 2033 surplus median -\$11k; six months late at -100bp: +\$531 extra top-up cost [MODEL].
- \$75k deposit: 2033 surplus p5/p50/p95 \$73k/\$105k/\$149k (central case) [MODEL].
- Missing deposit: part of the 2033 payment unbought in about 30% of paths (the gap odds), mean \$9.1k, p95 \$23.7k; at -50bp \$11,957 and at -100bp \$31,413 of the first payment [MODEL]. A \$75k deposit covers every case down to the 2008-size fall (AX1b) [MODEL].
- Correlation of a cut deposit with a 2027 recession cannot be sourced (no data found on publishing/speaking income in recessions, D3 M053) [UNVERIFIED], so report conditional dollars, not a joint probability.

Side effects and cost. None to the structure; about 15-20 words inside rule 1. **Touches.** IPS.

Complexity. Folded into rule 1; earns its place (case-specific income source). **Confidence.** High. **Team decides.** Wording; whether the joint tail is named in the IPS or only in the Final Report (E1: name it once in the IPS, because a frozen rule must say what it does in its own worst case).

P5. Growth-money split: 50% world stocks / 50% short Treasuries, chosen by one stated rule

Change. Move the growth money from the provisional 60/40 (brief section 7; CLAUDE.md interview call 1, "weakest call") to **50/50**, rebalanced once a year within a **45-55%** band, and state the reason as **one rule applied under both published forecasts**: hold the most stock that still leaves about a 1-in-20 chance that the growth money ends 2033 with less than the money Laura put into it (~\$157.7k). The same 50/50 applies to the unlocked part of the growth money after the 2031 floor (answers D2 C13).

Why (this resolves D2 vs D6). D2 said 50/50 because the median is flat across forecasting houses; D6 said its rule "picks 60%"; AY1 showed the rule gives 55-70% depending on inputs. E1 ran D6's own rule under both houses, two bond assumptions and three fee levels: **50% is the largest share that passes under both houses' central forecasts with market-consistent bonds and a fee of up to 0.5%**; 60% passes only if J.P. Morgan's forecast is right and fees are low. So D6's rule and D2's two-house insight agree on 50% once the rule is applied to both houses [MODEL].

- Case: "willing to take thoughtful risks" and "an appropriate balance between pursuing growth and protecting the capital required for her goals" (L69-74, R-C37, R-C38) [VRF]. INT: risk that is checked against what the safe alternative pays, and sized by a stated rule, is a concrete reading of "thoughtful"; the rule always picks the *most* stock that passes its test, so it never lands at the minimum, which answers D13c's "not set at the minimum to look safe".
- The reason rests on things that do not change with this year's valuations (AX2 D2 C7): the growth money feeds a promise made in 2031, the median barely moves, the bad case improves.

Numbers (E1_split_and_slices.py [1]-[2]; all MODEL, i.i.d. lognormal, JPM 2026 LTCMA [VRF] and Vanguard VCMM 2026-06-30 [VP via D2/AX2]; sleeve bonds 4.0% [VRF, JPM] or 5.0% [ASM, market-consistent]; fees [ASM]):

Cell (house, bonds, fee)	Rule's pick	P(below money put in) at 50% / 60%	Median 50% / 60%	p5 50% / 60%
JPM AC World 7.00%, bonds 5.0%, no fee	65%	1.7% / 3.5%	\$210k / \$211k	\$168k / \$162k

Cell (house, bonds, fee)	Rule's pick	P(below money put in) at 50% / 60%	Median 50% / 60%	p5 50% / 60%
JPM AC World, bonds 5.0%, fee 0.5%	60%	2.4% / 4.5%	\$206k / \$208k	\$165k / \$159k
JPM U.S. 6.70%, bonds 4.0%, fee 1.0%	50%	4.5% / 6.9%	\$199k / \$201k	\$159k / \$154k
Vanguard U.S. midpoint 5.2%, bonds 5.0%, no fee	55%	2.8% / 5.4%	\$204k / \$204k	\$163k / \$157k
Vanguard midpoint, bonds 5.0%, fee 0.5%	50%	3.8% / 6.8%	\$200k / \$201k	\$160k / \$154k
Vanguard U.S. low end 4.2%, bonds 5.0%, no fee	50%	3.7% / 7.0%	\$200k / \$200k	\$161k / \$154k

- Robust test [MODEL]: pass in every cell of "both houses' central cases, bonds 5.0%, fee up to 0.5%" = **50%**; all JPM cells (both equity rows, both bond cases, all fees) = 50%; both central cases with no fee = 55%; every cell including Vanguard's low end = 40%.
- Moving 50% to 60%: median +\$0.2k to +\$2.1k; p5 -\$5k to -\$7k (E1 [1]); p95 +\$11-14k (D2) [MODEL]; 2031 floor p5 +\$5-6k at 50% (D2) [MODEL].
- Expected extra return of world stocks over the Treasury rate implied for her 2028-2033 dates (~5.2%, **implied, not lockable**): central forecasts about 0 (Vanguard midpoint) to +1.8 points (JPM); full published ranges and yardsticks about -1.2 to +2.4 points a year (D2 with AX2 C1-C2) [VP inputs, derived].
- WInS option (ii) orders at 50/50 (ticket v1 section 1): IEF 23.5% / TLH 42.5% / VT 17.0% (\$51,000, ~318 sh at \$160.03) / VGSH 16.0% (\$48,000, ~833 sh at \$57.59) / cash 1% (VT equals the plan's whole-portfolio stock share; the cash float comes out of the short-Treasury part); recompute on the trade date [VP prices 9/25; arithmetic].
- Whole-portfolio stock share (E1 [3], median path): 0% in 2027 (leftover in T-bills), **17.0%** after the 2028 deposit (20.4% at 60/40), **3.5%** after the 2031 floor at an 80% lock (5.2% at 70%) [MODEL].

Side effects and cost. Gives up \$11-14k at p95 and about \$0-2k of median [MODEL]. The whole portfolio looks even more cautious (17% stocks after 2028 instead of 20%), which sharpens D13c's "timid" tension; P6 answers it by stating risk goal by goal. Ticket v1 already carries both splits (VT 17.0 / VGSH 16.0, band 45-55 at 50/50); the team ticks one. D10's equity-gap figure under (ii) becomes +17.0 points (AX2 D10 C7).

Touches. WInS-now and TN only if option (ii) is chosen (AX2 D2 C4); otherwise IPS only (rule 2 of P11).

Complexity. Zero extra rules: it replaces an unexplained 60% with a stated rule of two numbers (a threshold and "about 1 in 20"), which the IPS needs anyway (IPS guide L46, "approach risk and return") [VRF]. Earns its place. **Confidence.** Medium. High that the median is flat and the choice is about spread; medium on 50% specifically (the fee and house assumptions decide between 50 and 60; the model is fee-free and thin-tailed by default). **Team decides.** 50/50 (passes under both houses) or 60/40 (passes only under JPM with low fees), written as a choice of spread either way. Decide **before** the VT/VGSH orders if (ii) (a preference, not a rule: a later research-based change is allowed before Nov 6 and could be a genuine "refined" note, AX2 D2 C3). If the team keeps 60/40, record why.

P6. Risk tolerance stated goal by goal

Change. The IPS states risk tolerance per goal, using the three parts professionals use (CFA Institute 2020: risk need, risk-taking ability, behavioural loss tolerance) [VP via D6/AY1]. **Elements:**

1. **Payments:** no risk *needed* at today's prices (the Treasuries cost less than her first deposit; if the January price is higher, that is a funding need met by the 2028 deposit, not a reason to take market risk, AY1 D6 C8) and none *affordable* (high certainty, no outside money, L91-92). So her appetite for risk is deliberately not used here.
2. **Growth money:** low need (no facility target, L104, R-C59), high ability (living costs outside the portfolio, L61-63, R-C33; no withdrawals before 2033, L63-65, R-C34; a loss shrinks only the facility range), so it holds real stock (P5) even though no target requires it.
3. **The 2031 floor:** once announced, it carries no market risk.
4. **Willingness** in the case's own words ("thoughtful risks", "appropriate balance"); her *investment* loss tolerance is the team's inference (it cannot be measured without contacting her, R-W16), so the bad-year decisions are pre-committed (P10).

5. **Where her big bets already are:** her career, books and the residency itself; the 2028 deposit already rides on them (R-AN31), so the portfolio holds only broad market risk, no tilts (D8 M081; Chhabra 2016 interview, VP).
6. Optional, one clause: the whole-portfolio stock share by stage (about none in 2027, about one-sixth after 2028, a few percent after 2031) so a first reader sees where the risk is and why (M033; AY1 D6 C4) [MODEL numbers in P5].
7. **Not** her career story as a reason (risk attitudes are "highly domain-specific", Weber, Blais & Betz 2002, VP via D6), not her regret line (AY1 D6 C3), not "risk-averse" (D13a context) [VP].

Why. An IPS "outlines an investor's financial goals, risk tolerance" (IPS guide L3, R-I2) and the case never states one (R-AN17) [VRF]. CFA reconciliation rules [VP]: "Higher behavioral loss tolerance can be ignored when both the risk need and risk-taking ability are lower" (payments) and "A lower risk need can be discounted when both risk-taking ability and behavioral loss tolerance are higher" (growth money). Laura link: the case's "Although" (R-AN16) separates her career risk-taking from what she wants for this money [VRF, INT].

Numbers (D6_behavioural_numbers.py) [MODEL]: under the plan the growth money falls in about 27% of years 2028-2030; at least one down year before 2033 in 62% of paths; the whole statement shows a fall in about 1 year in 10; the rejected growth-first plan would show assets below the value of the payments on the first December statement in about 40% of paths.

Side effects and cost. About 35-50 IPS words, merged with the "where risk sits" principle (D9 M241: outside money may cover the facility, never her ten payments, L85-86 vs L91-92) [VRF]. **Touches.** IPS; TN reflections (each "how it serves Laura" line echoes one case trait and passes the swap test, D6 M125).

Complexity. Replaces a one-word label ("moderate") with a structure the IPS guide asks for. Earns its place. **Confidence.** High. **Team decides.** Wording; whether to include element 6.

P7. The 2031 co-sponsor range method: three slices, bought bottom, uncapped, two-sided confidence

Change. Replace "a floor bought in a 2-year Treasury plus an upside with a stated probability" (brief section 7) with a method whose parameters are fixed by Nov 6 (method only; no dollar range in the IPS, IPS guide L68-69):

1. On 1 January 2031 split the growth money into **three slices**: a share **a** is bought as a Treasury maturing before 1 January 2033 (the bottom of the range); of the rest, a share **s** is promised as the stretch; the other (**1-s**) is kept for the project (P8).
 2. In 2033 the gift = the bought amount + $s \times$ what the rest has become, **uncapped**.
 3. The top = the bought amount + $s \times$ a stated high percentile of the rest's two-year growth (E1 default: the 90th, so "about 1 in 10" above the top).
 4. Confidence stated **both ways**: below the bottom only if the U.S. Treasury defaults; above the top about 1 in 10, with the model named (in the Final Report).
 5. The range is set in 2031 from what is then owned, never from a projection made today (M040).
 6. No range is announced until all ten payments are bought (P4); the operating payments are never touched (L119-120, R-C71).
- **E1 default: $a = 0.8$, $s = 0.5$ ("80/10/10": 80% bought, 10% promised but still at risk, 10% kept).** Alternate "70/15/15" ($a = 0.7$, $s = 0.5$, D5's candidate) if the team weights flexibility more.

Why. Case: "Rather than promising one exact amount, she wants to communicate a credible range" (L115-116, R-C67); "state how confident they are that her 2033 contribution will fall within that range" (L117-118, R-C69, two-sided per R-AN9); "explain how favorable and unfavorable market outcomes could affect the amount" (L118-119, R-C70); "committing all remaining assets could limit her financial flexibility" (L102-103, R-C58) [VRF]. D5's key finding: "promise the upside" and "keep flexibility" conflict unless one number (s) splits the rest [MODEL/INT, audited]. Laura link: her credibility with co-sponsors, and through it her reputation-driven income (BS-03), is at stake [VRF, INT].

- Why uncapped (D5 over D3): (i) one rule fewer; (ii) good markets visibly raise the gift, which R-C70 asks teams to explain; (iii) "100% within the range" by construction is an empty figure to a statistics reader (SH-04, AY1 X11); (iv) over-delivering never costs credibility, under-delivering does (R-C66) [INT].

- Why 80/10/10 as default: the highest bought bottom while still keeping flexibility by rule (R-C64 "meaningful personal contribution" vs R-C58); consistent with the verified model's 80% floor (F-402); behavioural evidence favours narrow honest ranges (D6 M197: "as precise as warranted", Du et al. 2011, VP); D4's finding that by 2031 markets are the smallest uncertainty needs about 70% or more locked (AX2 D4 C2) [MODEL]; the kept 10% is about three years of Taiwan building-cost drift at the post-2021 pace ($1.0354^3 = 1.11$) [ASM sanity check, not a contingency fund, R-C61].

Numbers (E1_split_and_slices.py [4]; 50/50 sleeve, JPM AC World, bonds 5.0%, no fee; MODEL):

Slices (bought / at-risk / kept)	Median bought bottom	Median top (p90)	Width	Gift p5/p50/p95	Kept p5/p50/p95
80/10/10 (default)	\$167k	\$192k	1.15x	\$151k / \$188k / \$238k	\$16k / \$21k / \$29k
70/15/15 (alternate)	\$146k	\$183k	1.26x	\$142k / \$178k / \$226k	\$24k / \$32k / \$43k
80/20/0 (give all, s = 1)	\$167k	\$216k	1.30x	\$168k / \$210k / \$266k	\$0 / \$0 / \$0
90/5/5	\$188k	\$200k	1.07x	\$159k / \$198k / \$251k	\$8k / \$11k / \$14k

- Width $\approx 1 + s(1-a)/a \times \sim 1.2$ (AY1 [2]); the comparable quantity across D5 and D6 is the share promised but still at risk, $s(1-a)$: 0.10 (default), 0.15 (alternate), 0.20 (D6/D3 base) [MODEL].
- Without a cap an "80%" top lands within the range 62-93% of the time across models and history; every miss is on the upside (D3 v2 (f)) [MODEL]. P(below the bottom) = 0 in every model and history test (bought) [MODEL].
- The bottom moves -3.4% if the 2031 two-year rate is 3.0% instead of 4.81%, +2.3% at 6.0% (AY1 [4]) [ASM rates].
- 90/5/5 is rejected: 1.07x is close to the "one exact amount" the case rejects (L115-116) [INT].

Side effects and cost. Three parameters (a, s, q). The range is lopsided by design: outcomes cluster near the top (D5/AY1 C6), so the Final Report must present a "likely figure", not the midpoint (later list). The 70/15/15 alternate lowers the bought bottom by about \$21k at the median. **Touches.** IPS (rule 3 of P11). Not TN (the TN guide excludes the facility and co-sponsor draft, L20-21); no floor share in any WInS note (ticket) [VRF]. **Complexity.** Each parameter moves money a reader cares about (bought / promised / kept); the rule fits two plain sentences (AY1). Earns its place, because without s the plan's "upside" and "flexibility" contradict each other. **Confidence.** High on the method; medium-low on the exact slices (a judgement between credibility and flexibility). **Team decides.** The slices (80/10/10 or 70/15/15), the top percentile (one value only; D5 used 85, D6 90, D3 80), cap or no cap (section 3, conflict C3).

P8. The flexibility rule and what the kept money holds

Change.

1. Flexibility is **reserved by rule** (the kept slice (1-s) of P7), not "whatever is left" (brief section 7).
2. The kept money is held in **US\$ short-term Treasuries / T-bills** from 1 January 2033 (the kept (1-s) part of the unlocked growth money moves there on that date), i.e. with safety, not "invested for growth". This corrects the team notes' mapping of flexibility to the aspirational bucket (D8 M081; team notes L15-18) [VRF].
3. **US\$ is the binding currency** for the promise, the reserve and the range; any NT\$ figure is dated and illustrative (D4 with AX2 C1) [INT].
4. **No NT\$ conversion or currency hedge before the facility decision:** a forward is a derivative, banned in WInS and, on the safe reading of "Investments permitted (for BOTH contributions)", in the plan (R-AN15, R-AN49; AX2 D4 C3) [VP rules; INT reading]; converting early would also be a currency bet.
5. The IPS **names the assumption** that Taiwan building costs are expected to rise, so a fixed US\$ gift buys less (case L146-147 asks for "the effect of inflation on portfolio projections and facility costs", R-C85) [VRF]; no numbers, no facility cost estimate (L164-165, R-C95).
6. Its purpose in words: changes "as the project develops" (L102-103), not Laura's living costs (outside, L61-63) and not a sized contingency fund (L106-107, R-C61) [VRF].

Why. Flexibility exists to absorb surprises; putting it in stocks means the money for bad news is most likely to be down when bad news arrives (D8 M081) [INT]. Laura link: the case's own "as the project develops" (R-C58, R-AN19) [VRF]. CLAUDE.md's provisional call 3 ("post-2033 leftover mainly as project flexibility/overrun buffer") is kept and made concrete [VRF].

Numbers. Kept money under the default 80/10/10: \$16k/\$21k/\$29k (p5/p50/p95), about 10% of the post-reserve money; 70/15/15: \$24k/\$32k/\$43k (E1 [4]) [MODEL]. Taiwan construction cost index +3.54%/yr since 2021 raises costs about 11% in three years (F-508; D5) [VP inputs, derived]. Locking NT\$ two years ahead would cost about 5.9% at today's rate gap (D4; Taiwan 2-year yield 1.66% is SNIP) [ASM]. **Side effects and cost.** Kept money earns short-Treasury rates, not equity returns (a small expected cost). About 15-25 IPS words (folded into the certainty/assumption clause and rule 4). **Touches.** IPS. **Complexity.** No new rule: it fixes the composition of money the plan already has. Earns its place. **Confidence.** Medium-high. **Team decides.** Whether the kept money is short Treasuries (E1) or stays 50/50 (the team's earlier "ventures second" idea); E1 sees no case line supporting growth for this money.

P9. The operating reserve: composition over time and which number is "the reserve"

Change. Fix in the IPS (elements):

1. The reserve **is the ladder**: bought from January 2027 (completed from the 2028 deposit if needed) and designated the operating reserve on 1 January 2033, before the first payment or any facility contribution (L94-95, R-C50, R-AN2, R-AN28) [VRF]. One clause reconciles "funded 2027, set aside 2033" (brief section 8 item 9).
2. **Composition over time**: held to maturity; each rung matures on 15 November of the year before its payment and waits about 47 days in Treasury bills; nothing is sold early; so the composition changes only as rungs mature (the case allows "if at all", L96, R-AN27) [VRF, ASM dates per S3].
3. **No rebalancing between the reserve and the growth money**, written down (CFA 2010 element 4c: "If the policy is not to rebalance, this policy should be documented in the IPS", VP via D8/D10).
4. **Which number**: the Final Report reports the reserve at market value on 1 January 2033 (~\$395k on today's forwards, F-106) alongside its cost and face value; the IPS names the method only.

Why. The IPS guide asks the IPS to "establish your strategic approach to ... managing the operating reserve's asset composition over time" (L48-50, R-I22) [VRF] and "How will the portfolio's asset allocation and composition change as future funding needs approach" (L22, R-I12) [VRF]. The honest answer is three dated steps (2028 completion and growth start; 2031 floor; 2033 reserve and facility), not a funded-ratio glide path: once bought, the ladder's funded ratio stays at 1 barring default, so a "below 100%" trigger can never fire (D10 finding 6; parked M149) [INT].

Numbers. Cost \$292,264 (exact dates) / \$294,387 (Nov-15 STRIPS) at 2027-01-01; forward value \$394,930 at 2033-01-01; face \$500,000 (F-101, F-106; D3) [VRF inputs, ASM method]. **Side effects and cost.** About 20-30 IPS words, shared with rule 4. **Touches.** IPS. **Complexity.** None added; answers a required element. Earns its place. **Confidence.** High. **Team decides.** Wording.

P10. Governance, review dates and bad-year pre-commitments

Change. About 30-45 IPS words (D10 M013), elements:

1. **Who decides**: the students' written rules run the portfolio; Laura makes the two choices the case gives her (the range she announces in 2031 and the 2033 facility amount), on the team's rule-based recommendation. **No** "our portfolio manager/advisor approves" (Rules page: "Advisors may not make decisions on behalf of students", VP; R-AN4).
2. **Review points** at the case's own dates: 1 January 2027 (purchase), 2028 (deposit, completion, growth start), 2031 (floor, announcement), 2033 (reserve, facility), plus a yearly check of the growth mix.
3. **What may change a rule**: a change in Laura's circumstances, never a market move; worded so it cannot be read as permission to revise the strategy after Nov 6 (IPS guide L73-74; AY2 D8 C12) [VRF].
4. **Bad-year pre-commitments** (folded into the rules, no separate clause; D6 bad-year table):
 - stocks fall: only the yearly reset inside the band; the ladder is never sold;
 - before 2031 after losses: the floor is bought on its date, not delayed "until it recovers" (break-even trap, Thaler & Johnson 1990, VP via D6);
 - after gains: only the bought amount is announced as the bottom (house-money trap, same source);

- the hedge in WInS is never sold or trimmed because rates moved; it is only re-mixed to keep its rate sensitivity near the payments' (AY2 D7 C7).

5. **Cut:** org chart, firm name or size, reporting frequency, residency-delay trigger, citations (D10 keep/cut).

Why. CFA 2010: the IPS offers "an objective course of action to be followed during periods of market disruption" and a review process "clearly identified in advance" [VP via D8/D10]. Guide p.5: "planned adjustments as funding dates approach" are allowed, redesign after results is not [VRF]. Laura link: "Laura understands that investing involves uncertainty and periods of market volatility" (L68-69, R-C36) [VRF]; her investment composure is unknown (P6), so the plan must not depend on it [INT].

Numbers. A down year for the growth money is more likely than not before 2033 (62% of paths) (D6) [MODEL]. **Side effects and cost.** 30-45 words. **Touches.** IPS; WInS-now (the ticket's "never" rules are the WInS version). **Complexity.** Earns its place: without it, Final Report actions could read as post-freeze changes (Guide L158) [VRF]. **Confidence.** High. **Team decides.** Wording.

P11. One consolidated if-then rule set (replaces brief section 7's prose)

Change. The IPS carries four money rules plus governance, each with a trigger, an action and what the Final Report will later show "fired as written" (D8 M006/M235, with every audit correction applied). Parameters in brackets must be fixed by Nov 6.

Rule	Trigger	Action	Covers (case test)
1. Promise first	Money arrives (Jan 2027; 2028 deposit, each instalment)	Buy the ten payments before anything else; longest-dated first if short; the 2028 deposit completes them before any growth; joint tail stated (P3, P4)	Test 1 certainty (L136); test 3 for the payments
2. Growth mix	Once a year, or when outside the band	Reset the growth money to [50/50] world stocks / short Treasuries, band [45-55]; same mix for the unlocked part after 2031 (P5)	Test 3 (the risk held is the risk stated)
3. The 2031 range	1 January 2031	Buy [$a = 0.8$] of the growth money as a Treasury maturing before 2033 (the bottom); top = bottom + [$s = 0.5$] × [90th] percentile of the rest; two-sided confidence; no announcement before the ladder is complete (P7)	Tests 3, 4 (L140-143)
4. 2033 order of use	1 January 2033	Reserve = the ladder, held to maturity; facility = bottom + [s] × what the rest became (uncapped); the other [$1-s$] kept in US\$ short Treasuries for the project (P8, P9)	Tests 2, 5 (L138, L145)
G. Governance	Case dates + yearly	Rules change only for a change in her circumstances, never because markets moved (P10)	IPS guide L73-74; Guide p.5

- **Three trade-offs** to name in words, each "gives up X to get Y", inside the rules' "because" clauses (D9 M121; Guide p.5 "reasoning, assumptions, and tradeoffs", VRF): T1 some facility money in strong markets for payments that cannot miss; T2 some growth after 2031 for an amount she can state in advance; T3 the chance of cheaper bonds if rates rise for never paying more because rates fell (buy on arrival). No numbers in the IPS.

Why. IPS guide L48-50 names exactly these four approaches (operating commitment, reserve composition, facility contribution, flexibility) [VRF]; without rule 4, tests 2 and 5 have no rule (D8 M235) [INT]. Rule 2 passes on governance (the risk held equals the risk stated), not on a lift in the median (AY2 D8 C7) [INT].

Numbers. 4 rules × 25-35 words + governance 10-15 ≈ 110-155 words (D8) [INT]. Without a rebalancing rule the growth money's stock share wanders to p5/p50/p95 50/62/73% by 2031 (D8 script) [MODEL]. **Side effects and cost.** About a quarter to a third of the 500 words. **Touches.** IPS. **Complexity.** Four rules is the smallest set that covers all five case tests; each extra candidate (ratchet, calendar glide path, TWD quote, tilts, funded-ratio trigger) failed the lift test (D8 M235; D3 M055) [MODEL/INT]. **Confidence.** High on structure; parameters as in P5/P7. **Team decides.** Every bracketed parameter, by Nov 6 (a rule whose parameter is chosen later is a redesign, D8).

P12. Fees: one cost principle, no number

Change. One IPS clause (elements): the bought payments are held directly to maturity with no ongoing charge; every cost is paid from the growth money, so costs can never reduce the promise; low-cost broad funds as a principle. No fee number in the IPS. **First clause to cut** if the IPS runs over its word budget (AY2 D8 C13).

Why. The case has Laura invest "with an asset management firm" (L43, R-C26) but names no fee [VRF]; no projection in the repo carried a fee (BS-01) [VRF]. The Asset Manager Code (which the Rules page tells

teams to "operate by", R-W29) recommends plain fee disclosure (F.4.d; "should provide" gross and net returns) [VP via D8, verb per AY2 C11]. A fee taken from the ladder breaks full funding (AY2 D8 C5) [INT]. **Numbers** (strategy_mc_v2.py (c); D8_practice_numbers.py) [MODEL, ASM fees]: fund expenses (~0.05%) cost about \$0.3k of median; a 0.5% advisory fee on the growth money -\$5.3k; 1.0% -\$10.2k; 1.0% on all assets paid from the growth money -\$32.4k; with the deposit missing, fees on everything exceed the leftover money in 100% of paths. A typical U.S. adviser at this size charges about 1.00% (Kitces reporting the 2017 Veres survey, VP; dated) [VP, ASM for 2026]. **Side effects and cost.** About 15-20 words. It is a team policy assumption, not a case fact. **Touches.** IPS. TN: none (WInS costs are \$25 per ETF trade, \$10 per bond trade, F-607 [VP]). **Complexity.** Low value per word; keep only if words allow. **Confidence.** Medium (principle high; its scoring value unknown). **Team decides.** Keep or cut.

P13. Which book WInS shows: vote (ii) or (iii) before the first order

Change. Take the vote now (ticket gate box (b)); no book is decision-ready without its conditions:

- **E1 lean: option (ii)** (the plan's mix after both deposits, scaled to \$300,000), **conditional on** (1) one IPS sentence carrying three elements (ticket v0 section 2 / v1 section 1): the post-2028 mix; scaled to \$300,000 with the Treasury part set to the payments' rate sensitivity (about 10 years); and the fact that in January 2027 almost all of her real \$300,000 buys the ladder; (2) each promise-money and growth note (or its reflection) marking the scaling ("after both deposits" / "after her 2028 deposit" as content, ticket v1 C2); (3) no note calling the holdings "Laura's portfolio" (Asset Manager Code F.2 truthful disclosure, via D10) [VP].
- **Alternative: option (iii)** (her literal January 2027 book, ~98% Treasuries), **conditional on** naming a genuine third executed trade in advance: the 2027 leftover in a T-bill fund (SGOV, alternates BIL then SHV, ~1.5%, bought in the same session as the hedge, the plan's own T-bill rule), or a dated Treasury rung if WInS lists one (P14); if neither is listed, (iii) cannot promise three executed trades, so choose (ii) (AX2 D10 C1, blocking; ticket v1 section 2) [VRF].
- **Decision test** (T1, adopted): choose (iii) if the team cannot state the (ii) scaling in its own words in about 30 IPS words. **Cap rule** (ticket v1 section 3): if Session Rules show a single-security cap below 25%, (iii) needs 5-6 Treasury funds, so switch to (ii) and re-vote, unless a separate higher bond limit and a listed Jul-Dec 2037-2041 Treasury let one bond carry the excess [MODEL/derived, T1 script].
- E1 and T1 reached the same lean independently from the same evidence (T1: "(ii), medium-low confidence").
- Correct the internal description (CLAUDE.md, brief section 7): "~65% Treasuries / ~35% equity" is hedge / growth money, not bonds / stocks; at 50/50 the book is hedge 66% / VT ~17% / VGSH ~16% / cash 1% (F-409; ticket section 2) [VRF].

Why (both sides; the rules decide neither).

- Neither book breaks a published rule under no cap, a 35% cap or a 25% cap; Vanguard volumes and caps below 25% are unchecked (D10 with AX2 C2) [VP rules; derived]. The binding requirement is consistency: "Your portfolio and Final Report should reflect the strategy established in your IPS" (IPS guide L75) [VRF].
- For (ii): the TN guide asks the notes to show the team's "approach to growth, risk, liquidity, funding reliability, financial flexibility, and future cash-flow needs" (L4-6) and a "cohesive portfolio strategy" (L14-15) [VRF]; (ii) gives three honest role words (future funding, growth, risk management/liquidity, D12) and "funding purposes" as a visible kind of diversification (R-AN45) [VP]; "The WInS portfolio represents each team's implementation of its investment strategy during the competition" (L129, R-C75) [VRF].
- For (iii): WInS starting cash equals her first deposit exactly and the \$150k "will not be added to WInS" (R-AN14, SMaply, VP), so (iii) is the only book true at a stated date without scaling; the pitch rule (P1) is visibly true in the holdings; its IPS sentence is shorter (~12-20 words vs ~25-30) (D10) [VP, INT].
- Against (ii): WInS shows ~17% stocks while her real 2027 money holds ~0% (equity gap +17.0 points at 50/50) and the \$198k hedge is 68% of the \$292k promise, so every hedge note needs a scaling clause (D9 M012; AY1 [5]) [derived].
- Against (iii): with no cap it gives two orders only; all notes are promise or liquidity notes (no growth decision is shown in WInS); ~98% Treasuries may read as timid for a client "willing to take thoughtful risks" (S3; D13c N1) [INT].

Numbers (ticket v1 sections 1-3, `T1_ticket_v1_numbers.py`; issuer durations IEF 6.86y, TLH 11.58y on 9/25, VP). (ii) at 50/50: IEF 23.5% / TLH 42.5% / VT 17.0% / VGSH 16.0% / cash 1%, 4 trades, ~\$100. (iii): IEF 34.9% / TLH 63.1% / SGOV ~1.5% (~\$4,500) / cash ~0.5%, 3 trades, \$75; the T-bill leg earns about \$18 in the WInS window against a \$25 commission, so its reason is the plan's rule, never WInS profit (T1, ASM) [VP prices; derived]. Leftover \$4,438-6,636 (1.5-2.2%) before commissions (D10). Securities and alternates (ticket v1 section 12, issuer pages re-read 2026-09-28, VP): VT (alternates VTI 62% + VXUS 38%, or ITOT + IXUS); VGSH (SHY, or a WInS-listed ~2-year Treasury note); TLH (IEF

- TLT at duration weights: 40.9/25.1 under (ii)); SPTL (VGLT); SGOV (BIL, SHV). On the 2025-26 list (history only, VRF): IEF, TLT, VT, VTI, VXUS, VGSH, SHY, BIL, SHV yes; TLH, SPTL, VGLT, VGIT, ITOT, IXUS, SGOV, IBTM no. All PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK. **Side effects and cost.** (ii) costs about 25-30 IPS words and 15-20 words per affected note; (iii) costs TN variety and needs a pre-named third trade. **Touches.** WInS-now (tier 1), TN, IPS (one sentence naming the date or state WInS represents, required under either option). **Complexity.** Either is one sentence of explanation; (ii) is one more trade type to explain. Earns its place either way because the sentence is required. **Confidence.** Medium-low on the lean (a judgement; D10 leaned (iii); S3, D12 and T1 lean (ii)). **Team decides.** (ii) or (iii), recorded with first names, date and reason in the decision log before the first order.

P14. The three Trading Notes plan (WInS-now and TN)

Change. Replace blueprint 01 section 7's planned "discipline/rebalance trade" (D7: no sensible band fires before Oct 23) with notes drawn only from trades that happen anyway. The slot table, note checklists and banned-word lists are already in ticket v1 sections 4-7 (T1); this proposal endorses them, with the points below. Also adopt v1's hedge re-mix band of **±0.25 years from the starting mix** (the old fixed 9.65-10.15 band clashed with the capped mixes at about 10.5 years; v1 change 5).

Plan under option (ii) (orders in one session, hedge first; target fills by Fri Oct 2 ET once the gate is passed):

1. **Hedge note** (TLH, or IEF; SPTL only if a cap forced it, and then it is *not* a "refined" decision, AY2 D7 C1): role word "future funding"; its "tested" evidence lives in the reflection.
2. **Growth note** (VT): role word "growth"; only money the promise does not need; a fall shrinks the facility range, never the payments; the split stated only if final (P5).
3. **Short-Treasury note** (VGSH): role word "risk management" (of the growth money) or "liquidity"; "short-Treasury part of the growth money"; never "the floor", never "already owned" before 2031 (AY2 D9 C1; AX2 D2 C6). Alternative third note only if genuine: a research-based change of the split announced in advance (the honest "refined" route, D7), or a band trade in a crash. A dated rung is optional under (ii) and not recommended (a second "future funding" note; complexity) (AX1b item 7).

Plan under option (iii):

1. Hedge note (TLH or IEF, as above; under (iii) it may carry the ~\$292-294k price of the payments in one clause, because the order is that size).
2. **The 2027 leftover in SGOV** (alternates BIL, SHV; "liquidity"), bought in the same session as the hedge, a plan rule, \$25 (ticket v1 section 2). One reflection under (iii) must answer "why so little equity": in January 2027 almost all of her real \$300,000 buys the payments, and room for stock risk starts with the 2028 money (D13c N1).
3. **Dated rung** if the WInS Bonds drop-down lists a U.S. Treasury maturing July-December of 2032-2041 (best: the 4.125% note of 15-Nov-2032, CUSIP 91282CFV8, parent of the ladder's first STRIP), face \$50,000, funded from IEF, one \$10 trade; the 2032 note raises TLH, so skip it under a cap below about 69%, and under a cap below 44% only a 2039/2040 bond with no new fund (D1 M044 with AX1b items 5-9; ticket v1 section 7) [VP via S1 MSPD; MODEL prices]. Its reason is implementation (the promise is dated), not display (AX1b item 8). Otherwise the third note is the IEF buy (weaker: same role). IBTM (Dec-2032 iBonds) is the last rung fallback and its note must not say "repays \$50,000" (iShares: the funds "do not seek to return any predetermined amount", VP via AX1b).

Note element spec (every note; the team writes the words). Use the ticket v1 section 5 core, which maps the Guide's four elements (reasoning, alignment, supporting research, role; Guide L84-86) [VRF] into about 300 characters: C1 one Guide role word plus the fixed team label (alignment); C2 Laura's dated need (reasoning; under (ii) fold the scaling in: "after both deposits" / "after her 2028 deposit"); C3 one dated research fact with its source named in words (prefer a price or issuer figure over a forecast); C4 the main

risk accepted or the rule that governs it. If a real limit forces a cut, keep C3 and drop C4 to the reflection (AY1 D12 C1: the Guide names research, not risk).

Reflection spec (≤100 words each; TN guide L46-52) [VRF]: answers the three official questions (why; alignment; how it served her goals, funding needs or risk). The "tested" evidence sits inside them (about 20-25 words): the 10-year rate move over the **pre-registered window** (fill date to the Oct 20 close), the hedge's % change and the payments' % change on the same dates, and what the rule said to do (hold). Report small moves or poor tracking as they are; never pick the best day (AY2 D7 C2). No Laura quotes (D13).

WInS-now checks (all but (f) are already in ticket v1 gate boxes (b)-(d) and section 10; tick before the first order):

- (f) split decided (P5) if option (ii) (v1 gate box (b) allows "provisional 60/40, decision by [date]"; E1 prefers a final decision now);
- (g) note-box limit checked at zero risk: type a long test text on the order-review screen without pressing Confirm, or ask Stock-Trak/Wharton; until seen, draft notes to 300 characters (a sister product's limit, VP for HowTheMarketWorks, UNVERIFIED for WInS; Wharton's own example is 413 characters) (AY1 D12 C2);
- (h) screenshot the Bonds drop-down, list Treasuries maturing 2032-2042, record the bond position limit and accrued interest (D1);
- (i) data capture: screenshot WInS position values on the fill date and on Oct 20; Treasury curves can be pulled later from treasury.gov; value the payments with `D3_funded_status_log.py`, `A2_curve_recheck.py` (spot) or `D9_numbers.py` [3], not `official_curve_pv.py` (fixed to 2026-09-25) (AY2 D7 C5); check IEF/TLH ex-dividend dates in the window.

Vocabulary for notes (D9 M062 with AY2 fixes): WInS Treasury funds "move like / stand in for" the payments; never "match", "mature in her payment years", "stable", "reduce volatility", "safe", "guaranteed", "risk-free"; never "known" or "locked" for 2028 rates (AX2 D2 C1); never "halves AI exposure" (AX2 D2 C5); never "the operating reserve" for WInS holdings (the case creates it in 2033, L94-95) [VRF].

Why. "Supported, tested, or refined" is a menu, not a quota (TN guide L12, L29-30; case L152) [VRF]. Staging a trade to fill a category risks misstating the reasoning (R-W28 reading, INT per AY2 D7 C11). Notes cannot be edited (Stock-Trak, VP via A4) and are quoted "exactly as it appears in WInS" (TN L44) [VRF]. Wharton's example note has no number, date or source (59 words, 413 characters; D9) [VRF, derived].

Numbers [MODEL]: a 10-year move of 10bp or more between an Oct 1 fill and Oct 20 has about a 53% chance at the end date (D7, realised 2026 volatility 4.45bp/day [VP inputs]); 10bp moves the payments' value about 1% (~\$2,890; DV01 \$289, F-103 [VRF]); in 2026 an IEF/TLH hedge stayed within ±0.42% of the payments' % change in 95% of six-week windows (a lower bound; expect ±0.5-1% with real fund noise, D3 M028) [MODEL on VP inputs]; band firing before Oct 23 is very unlikely (about 0-2% even in a crisis, AY2) [MODEL]. Role test for +1pp in rates: IEF -6.9%, TLH -11.6%, VGSH -1.9%, payments -10.2% (D12) [VP durations, derived]. **Side effects and cost.** One extra rung trade (\$10) under (iii) if listed; ~10 minutes a week of logging. **Touches.** WInS-now, TN. **Complexity.** Removes one planned trade; adds checks. Earns its place. **Confidence.** High on method; the rung depends on WInS listings (unknown until logged in). **Team decides.** Which three executed trades become the notes (after Oct 20); who writes and who second-reads each note.

P15. Decision log, evidence grades and AI-use log from today

Change. Start now (WInS-now): one row per trade and per "decided not to trade" moment, with columns: date/time (ET); ticker, side, shares, fill price; note text exactly as WInS shows it; role word and team label; Laura need; fact, source and **grade** (CASE / PRICED / HISTORY / ASSUMED, D12 M231); risk and rule; **provisional IPS rule name** (e.g. "Rule 1", D9 M212); note candidate? Also a dated AI-use log (tool, date, purpose) for the Works Cited (R-W46, VP). No student surnames or personal data (CLAUDE.md). **Why.** Consistency across the three deliverables is required (case L161-162, R-C92) and is cheaper to record than to reconstruct (D9) [VRF, INT]; Articulation needs true process evidence and forbids invented stories (R-S27, R-W28) [VRF/VP]; the AI policy requires the record [VP]. Parked M021. **Numbers.** None. **Side effects.** Minutes per trade. **Touches.** WInS-now, TN (feeds reflections), IPS (rule names). **Complexity.** Process only; earns its place. **Confidence.** High (value depends on the team keeping it). **Team decides.** The log owner.

P16. IPS format: word budget, page fit, paraphrase test

Change.

1. **Word budget** (owner: whoever writes `ips_spec.md`; AY2 section 5 merged with P1-P13) [INT]:

Element	Words (estimate)
Opening: central idea + what/why (P1)	30-40
Certainty definition + two gaps + inflation/currency assumption (P2, P8 items 3-5)	45-60
Rule 1 promise first, incl. rates-fall, deposit and joint-tail branches (P3, P4)	40-55
Risk by goal + where risk sits (P6)	35-50
Rule 2 growth mix + its reason (P5)	25-35
Rule 3 2031 range method (P7)	30-40
Rule 4 2033 order of use + reserve composition + flexibility (P8, P9)	35-45
Governance + review + no rebalancing between reserve and growth money (P10)	20-30
Three trade-offs, inside the rules' "because" clauses (P11)	0-30
WInS sentence ((ii) 25-30; (iii) 12-20) (P13)	12-30
Fee principle (P12; cut first)	0-20
Total before connecting words	about 270-435

2. **Page fit:** 50 + 500 words in Times New Roman 12, double-spaced, 1-inch margins leaves 0-2 spare lines on U.S. Letter at Word's "Double" (27.6 pt, about 525 IPS words at most) and fails to fit reliably with 7 paragraphs; Wharton's own sample is set at 24 pt (8-10 spare lines) (D9; AX2 D10 C3) [VRF measured, ASM line heights]. Spec: paragraph spacing 0, no blank lines, no tables or bullet lists, about 470-490 words or fewer, and the **exported PDF** must end on page 3; "Exactly 24 pt" is a defensible reading if a draft overflows (INT; the team decides). Non-compliance means "will not be considered for semifinal selection" (IPS guide L82-83, R-I39) [VRF].
3. **Paraphrase test** of the pitch and the rule sentences with 6 unpaid readers (3 without finance, 3 with), two rounds (about Oct 26-30 and Nov 2-3), readers restate only and never supply words; change a sentence if 2+ misread it or anyone gets "what happens to the payments if stocks fall by a third", "what if rates fall before January 2027" or "what could still stop a payment" wrong (D12 M234; digital.gov method VP). Unpaid adult help is allowed; paid help is not (R-W21) [VP].

Why. A breach of the page rule excludes the team regardless of strategy quality [VRF]. **Touches.** IPS. **Complexity.** Process; earns its place. **Confidence.** High that the layout is tight; medium on exact line counts. **Team decides.** Letter vs A4, the spacing reading, the reader list.

P17. What NOT to put in the IPS (checklist)

- Any dollar range, reserve calculation, percentile, miss rate, probability to a decimal or model output (IPS guide L68-69; D9 number policy) [VRF]. At most one rounded, dated price fact, graded in words.
- "Bought in January 2027" as a flat claim; "fits inside \$300,000"; "by 1 January 2028 at the latest"; "at most part of the first payment" (P1, P3).
- "Guaranteed", "risk-free", "100%", "95% certain", "safe", "matched" (for WInS funds), "locked" or "known" for 2028 rates, "only the facility shrinks" (joint tail).
- Charts, images, tables, footnotes, links, formal citations, framework or code names (LDI, GBWM, Chhabra, CFA, APRA, Kresge), house names for return forecasts ("two major published forecasts" is enough) (L123, R-I55) [VRF].
- Laura quotes; the pull quote as hers; "floor-first leap"; her career-regret line as the reason for stocks; "risk-averse"; identity, heritage or Taiwan as a reason for any holding; book-title or "falling"/"balance" puns (the case's own "appropriate balance" is fine) (D13 rules) [VRF].
- The statistics degree more than once across the TN and IPS together (pick one place; E1 suggests the certainty definition; ticket v1 has already removed it from the VT note checklist) (AY2 C6) [INT].
- "Our portfolio manager/advisor approves"; org chart; firm name or size; reporting frequency; a residency-delay trigger; a "funded ratio below 100%" trigger (P10, P9).
- Tickers, weights to decimals, CUSIPs, fund durations to two decimals ("about 10 years" is the honest precision, S4) (IPS guide L51-52) [VRF].
- A fee number; a TWD figure; any NT\$ conversion or hedge plan; an EWT or AI tilt.

- Guide p.5 quoted as a reason against staging or against changing the split before Nov 6 (AX2 D2 C3; AX1b item 4).
- WInS P&L, rank or "our winning trade" (Guide p.3: WInS "is not the competition scorecard") [VRF].
- The \$150k described as uncertain beyond one labelled stress-scenario clause (the case says "will", R-AN35).

3. Conflicts E1 could not fully resolve (both sides; the team decides)

#	Conflict	Side A	Side B	E1 view
C1	WInS book (ii) vs (iii)	(ii): S3, D12, T1 (three roles; shows growth; "funding purposes" diversification; TN guide L4-6, L14-15; copes better with a cap under 25%)	(iii): D10 (true at a stated date; shortest IPS sentence; pitch rule visibly true; R-AN14)	Lean (ii) at medium-low confidence; (iii) needs a pre-named third trade (AX2 blocking)
C2	Growth split 50/50 vs 60/40	60/40: D6 rule under JPM; D13c/D9 M071 (a "take the jump" client; do not look timid); CLAUDE.md provisional call	50/50: D2 (flat median across houses; better p5 and 2031 floor p5); D8 M109 (stocks are for upside, not need; -\$41-56k bad case vs +\$11-14k middle, AY2)	E1's test (P5) favours 50/50: it is the largest share passing D6's own rule under both houses with a modest fee. The conflict remains a judgement about how much weight JPM deserves
C3	Cap the 2033 gift at the top?	Cap (D3 M039): "within the range" becomes certain barring default; model needed only for P(top reached) (~1 in 5 at p80)	No cap (D5 M107, AY1): a capped "100% within" is empty for a statistics reader; uncapped answers R-C70 and misses only upward (excess median \$2k, p95 \$8k)	No cap (fewer rules, informative confidence). If the team caps, report "P(top reached)", never "P(within) = 100%" (AY1 X11)
C4	2031 slices	80/10/10 (verified model's 80% floor; D6 behavioural 80-90%; D4 needs ≥70% locked; higher bought bottom, R-C64)	70/15/15 (D5: more flexibility, ~15%, anchored to 3-4 years of building-cost drift; more growth to 2033)	Default 80/10/10; both pass the width test (1.15x vs 1.26x). A judgement between credibility and flexibility
C5	Top percentile q	80 (D3), 85 (D5, "about 1 in 7")	90 (D6, "about 1 in 10")	90 for the roundest honest statement; q barely moves the top because the at-risk slice is small (D5)
C6	Third note under (ii)	VGSH buy (D12: the only "risk management" role; ticket)	Dated rung (D1: a checkable dated cash flow)	VGSH under (ii); rung under (iii) (AX1b item 7)
C7	VGSH role word	"risk management" (D12, the Guide's word, checkable with a number)	"liquidity" / "a different funding purpose" (ticket); "flexibility" as the echoed trait (D6/AY1)	Compatible: one Guide role word, echo "as the project develops", never "the floor"
C8	Flexibility money after 2033	Short Treasuries (D8 M081: flexibility is closer to safety)	Stays invested for growth / "ventures second" (team notes; CLAUDE.md call 3 wording)	Short Treasuries; no case line supports growth for this money
C9	Longest-first ordering (values)	Longest-first (D1/D3: one-third of the top-up price risk; less face unfunded if the deposit never comes)	Its shortfall falls on the residency's first operating year (AX1b item 23)	Longest-first on the numbers; name the counterpoint in the team's own risk discussion
C10	Note length	Plan to 300 characters (D12, a sister product's limit; ticket v1 four-element core)	No WInS limit is published; Wharton's example is 413 characters (AY1 D12 C2)	Draft to 300 as a precaution and check the box at zero risk before the first note; if a real limit binds, keep the research fact and move the risk to the reflection
C11	Fee clause in the IPS	Keep a principle (D8; BS-01; Asset Manager Code)	Cut first if over budget (AY2; the case is silent)	Keep only if the PDF test leaves room
C12	Whole-portfolio stock share in the IPS	Show it by stage (M033; AY1 D6 C4) so a first reader sees the design	Leave it out; it draws attention to "timid" (INT)	Optional clause; the per-goal statement (P6) is the required part

Resolved during this synthesis (no longer conflicts): the gap odds ("about 1 in 3" for the real ladder; "1 in 4" only for the exact-date ideal, AX1b); Guide p.5 applies after the IPS, not before (AX2, AX1b); D8 rule 1's joint-tail wording (P4); D8 rule 3 must include the give-back share (AY1 X9; P7); the lock-share dispute between D5 and D6 is one number measured two ways, s(1-a) (AY1 X1); the page-fit estimates of D9 and D10 agree once whole lines and 27.6 pt are used (AX2 C3); the sleeve-bond model input is market-consistent (~4.8-5.0%), not the old ticket's 3.5-3.9% (AX2 C8; withdrawn in ticket v1 change 17); the WInS 50/50 weights are VT 17.0 / VGSH 16.0 / cash 1, not 16.5/16.5 (ticket v1 section 1).

4. What the team must decide, in date order (checklist)

Before the first WInS order (target this week; fills by Fri Oct 2 ET):

- [] Gate (a): vote on lock-early (first names, date, reason in the log).
- [] P13: option (ii) or (iii), with its conditions.
- [] P5: split 50/50 or 60/40 (only if (ii)).
- [] Gate (c)/(e): Session Rules screenshot (single-security limit; whether ETFs and bonds are covered) and the cap branch.
- [] P14 (g)/(h): note-box limit checked at zero risk; Bonds drop-down screenshot (rung decision under (iii)).
- [] Gate (d): each note drafted by one student to the P14 element spec, read by a second student.
- [] P15: decision log and AI-use log started; roles assigned (trader, note-writer, second reader, log owner).

By Oct 20-23 (Trading Notes): data capture on the fill date and Oct 20 (P14 (i)); choose three executed trades; reflections to the P14 spec; consistency check against the provisional IPS rule names.

By about Oct 26 (first IPS draft): P1-P12 parameters fixed: pitch elements; certainty definition wording; rule 1 branches; split and band; a, s, q and cap; flexibility composition; review points; fee clause in or out.

Oct 26-Nov 3: P16 paraphrase rounds and PDF page-fit test on every draft. **Nov 6, 5:00 p.m. ET:** submit; no "tidy-up" trades in the last days (ticket).

5. Later (after Nov 9), one line each (Final Report; not worked here)

- The 2031 range in dollars at the median and weak/strong 2031 states, "likely figure" vs midpoint (D5, D6, E1 [4]).
- The price-of-certainty and riskless-control comparison (~\$204k, D3 M034) and the rejected-alternatives table.
- Named historical rows next to the model's p5 (D3 M059).
- Fee assumption and net-of-fee projections (D8 M026, D3 v2 (c)).
- NT\$ reference with date and rate; building-power assumption line (D4).
- Fundraising-excerpt checklist and the operations-first lead (D5 M094; M008, M015).
- Reserve at market value on 1 Jan 2033 and the CUSIP-level ladder (S3 Final-Report file).
- At most one or two dated Laura quotes in context (D13a/D13c).
- The decision-log traceability table (D9 M212) and the paraphrase-test log (D12 M234).

Sources (accessed 2026-09-27/28; web sources were read by the specialists and re-opened by the auditors as cited)

Official, in the repo (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (L16, L38-45, L61-74, L83-120, L126-149, L152-169); `2026_WGY_Investment_Policy-FINAL.txt` (L3, L16-26, L35-36, L45-78, L82-125); `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L4-31, L36-53); `2026_WGY_Investment_Competition_Guide.txt` (L84-90, L154-159, L183-184); `SMAApply_Deliverables_Page_2026-09-27.md` (L49-57).

Market data (VRF): `competition/official_market_data/daily-treasury-rates_2026-09.csv`; `JPM_LTCMA_2026_US_matrix_USD.pdf` p.2.

Primary web (VP, as cited through the specialist files and audits): SMAApply Trading Details and FAQ (<https://wghsinvcamp.smaply.us/res/p/trading/>, [/res/p/faqs/](https://wghsinvcamp.smaply.us/res/p/faqs/)); Wharton Rules & Roles and AI policy (<https://globalyouth.wharton.upenn.edu/competitions/investment-competition/rules-roles/>, [/ai-policy/](https://globalyouth.wharton.upenn.edu/competitions/investment-competition/ai-policy/)); 2026-27 WInS User Guide (<https://edu.stocktrak.com/wharton/wp-content/uploads/sites/19/2026/09/2026-2027-WInS-Userguide.pdf>); Vanguard VCMM forecasts (<https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/vemo-return-forecasts.html>); FRED DGS10, DEXAUS; treasury.gov 2026 par-curve CSV; iShares IBTM page; CFA Institute *Investment Risk Profiling* (2020), *Elements of an IPS for Individual Investors* (2010), *Asset Manager Code* (2nd ed.); Weber, Blais & Betz (2002) abstract; Thaler & Johnson (1990) abstract; Du et al. (2011) abstract; Kitces (2017); Chhabra interview (WealthManagement.com, 2016); digital.gov paraphrase testing; HowTheMarketWorks trade-notes page. Full URLs are in the cited specialist files.

Secondary / snippet (SNIP): Moody's Aa1 downgrade date (Wikipedia); Taiwan 2-year yield 1.66%; multipl CAPE.

Research files (lower authority; numbers reproduced by their auditors): `research/insight_v1/phase_D/*.md` (D1-D12, D3 model v2, D13a-c, four audits), `phase_D/trading_now_brief.md` (T1); `wins_now/securities_and_allocation_v1.md` (the ticket, supersedes v0), `securities_and_allocation_v0.md`, `S2_growth_sleeve.md`, `S4_red_team.md`; `phase_A/{case_register,fact_register,stakeholder_map,wins_week1_guardrails}.md`; `phase_C/survivors.json`, `parked.json`; `research/council_2026-09-27/01_chair_memo.md`, `round2_referee_ruling.md` (history).

Scripts: `research/insight_v1/scripts/E1_split_and_slices.py` (new); re-used outputs of `T1_ticket_v1_numbers.py`, `D1_purchase_rule.py`, `D1_wins_rung.py`, `AX1b_rates_audit.py`, `strategy_mc_v2.py`, `D3_*.py`, `D2_equity_premium.py`, `D4_facility_purchasing_power.py`, `D5_range_and_flexibility.py`, `D6_behavioural_numbers.py`, `AY1_audit_checks.py`, `D7_rule_trigger_odds.py`, `D8_practice_numbers.py`, `D9_numbers.py`, `D9_ips_page_fit.py`, `D10_*.py`, `D12_*.py`.

What this teaches

1. **A strategy is mostly its rules, not its holdings.** The portfolio barely changes in these proposals. What changes is that every "we would probably..." becomes "if X happens, we do Y", decided before anyone knows whether X happens. That is what lets a plan frozen on Nov 6 still be applied honestly in December.
2. **When experts disagree, find the number they are each holding fixed.** D2 and D6 argued 50 against 60 because they trusted different forecasters; D5 and D6 argued about the lock share because each held a different give-back share. Running the same rule under both assumptions turned two arguments into one test.
3. **Say what you control.** "The payments are bought in January 2027" depends on interest rates; "nothing takes stock risk until the payments are bought" depends only on the team. Promises built on rules survive bad luck; promises built on dates do not.
4. **Honest gaps are part of the answer.** Naming the January price risk and the nominal-dollar gap costs a few words and removes the two objections a careful reader would raise first.
5. **Complexity must earn its place.** Each proposal says what it costs in words or trades. Anything that did not move a number a reader cares about (a ratchet, a barbell, a Taiwan tilt, a staged purchase) stayed out.

E2 Semifinal reader: scoring the E1 strategy as it would appear in three Trading Notes, a 50-word pitch and a 500-word IPS

Agent E2 (semifinal reader), insight_v1 run, Phase E. Written 2026-09-28. This is AI-generated research for Team Caplet: scores, reasons, numbers and checklists. **None of it is text to submit.** No pitch, IPS sentence, note or reflection is drafted here. Where a short phrase names an idea, it is a content label, not wording to copy. The six students decide every change and write every word themselves. Any AI use goes in the Final Report's Works Cited (Wharton AI policy, R-W46).

Serves: Trading Notes (TN, due Oct 23) and the IPS (due Nov 6) only (brief section 17). There is no Final Report content. Final Report items appear only as one-line "later" entries (section 6). **Relayed messages:** none arrived during this task.

Evidence read (audit corrections override the text above them):

- the brief;
- the official case, IPS guide, TN guide, Competition Guide, Infographic and SMAApply page;
- `phase_E/E1_change_proposals.md` (the object being scored);
- D1-D12, D3_model_v2_results, D13a, D13c, and all four audit files;
- every "Audit corrections" appendix. Note that the two D3 files now carry AX1 appendices, which E1 had not seen;
- `wins_now/securities_and_allocation_v1.md` (the ticket), `phase_D/trading_now_brief.md` and `S4_red_team.md`;
- Phase A (case_register R-AN list, stakeholder_map SH-14 to SH-20, fact_register);
- `phase_C/parked.json` items M007, M008, M015, M021, M033, M035, M040 and M149;
- the council chair memo and round-2 ruling (history only).

`T2_red_team.md` does not exist in the repo (checked with `find` on 2026-09-28), so the ticket v1 and `trading_now_brief` stand in for it. E1's numbers reproduce: `E1_split_and_slices.py` was re-run on 2026-09-28, and every figure in E1's P5 and P7 tables matches.

Status labels:

- **VP** = VERIFIED-PRIMARY, as carried by the cited file.
- **VRF** = VERIFIED-REPO-FILE.
- **SNIP** = SNIPPET-UNVERIFIED.
- **ASM** = ASSUMPTION.
- **MODEL** = an ASSUMPTION-based model output, not a forecast.
- **INT** = E2's judgement. **Every score in this file is INT.**

Terms:

- **Semifinal reader:** the person who scores the written deliverables and picks the Top 50. The task casts this person as a Wharton student evaluator or an asset manager. Who reads this season is not published (SH-16, ASM).
- **Fixed-income professional:** someone who manages bonds for a living.
- **Swap test:** put another client's name into a sentence. If it still works, the sentence is generic (D13c).
- **Two-minute test:** what a reader can say back after reading the pitch and the IPS's first paragraph once.
- **Ladder:** Treasury zero-coupon bonds, one for each \$50,000 payment.
- **Growth money:** everything the ladder does not need.
- **Option (ii):** WInS shows the plan's mix after both deposits, scaled to \$300k.
- **Option (iii):** WInS shows Laura's literal January-2027 book.

0. Summary

Verdict (INT). The substance is of Top-50 quality. The presentation, as E1 specifies it, is not yet.

- **The good news.** No idea in E1 is wrong in a way a reader would punish. The plan does three things most teams will not:
 - it defines certainty by a price she can check, not a model;
 - it names its own two gaps;
 - it promises co-sponsors only what is already bought.
 - **The problem.** E1 packs about **55 distinct ideas** into a 50-word pitch and a 500-word IPS (my count from P1-P13; section 3.2). That is about one new idea every 9 words. Last year's lesson was "strategy packaging was over-complex for the insight" (CLAUDE.md, VRF), and E1 repeats it at the level of wording rather than holdings.
- **One visible contradiction.** The recommended pitch rule is worded as a ban ("no stock risk until all ten payments are bought"). The recommended WInS book (option ii) buys world stocks on day one. A reader who meets the notes first sees the rule broken before the IPS explains the scaling.

Scores as specified by E1 (INT; the scale is defined in section 2):

Criterion (SMAApply wording, VRF)	Score	One-line reason
Investment Strategy	7	Excellent dated planning and pre-set rules. But "creative" is modest, diversification is never argued, and there is a consistency risk under (ii)
Client Knowledge and Objectives	7	Strongest Laura links (2028 career income, credibility) are real. But the pitch omits the residency, the plan looks timid, and the growth rule reads as loss-avoidance
Portfolio Analysis	7	Certainty by price, named residual risks, a pre-registered "tested" window. Three fixed-income slips; little visible data by design
Articulation of Competition Experience	5	Almost nothing in the TN shows challenge or change. The one true refinement story (growth-first dropped after testing) is unused
Creativity and Presentation	6	"Announce only what is bought" is original. But the IPS reads as a rulebook, is dense with jargon, and has no memorable line
Total	32/50	Competitive, but not a safe Top-50 read

After the ten fixes in section 4 (INT): about 8 / 8 / 8 / 6 / 8 = 38/50. Most of the gain comes from presentation and consistency, not from changing the strategy. Three fixes do change a rule or a parameter: F6 (the growth-share yardstick), F7 (floor timing and the growth money's bonds) and F4 (how the central idea is worded).

The ten fixes, ranked by score gained per hour:

1. **F1.** Cut the IPS to about 12 ideas and at most ~380 content words.
2. **F2.** Build the IPS on four dated decisions (2027, 2028, 2031, 2033), not "Rule 1-4 + G".
3. **F3.** Rebuild the pitch around a positive central idea, the residency, and "only promise what is owned".
4. **F4.** Word the central idea as a split of jobs, so it is true of both books, and put the scaling marker inside each (ii) note.
5. **F5.** Add the two IPS-guide items E1 never budgets: diversification and liquidity.
6. **F6.** Fix the growth-share rule's yardstick (money put in vs the safe alternative) and say what stocks are for.
7. **F7.** A fixed-income correction pack:
 - withdraw the \$101k/\$165k barbell figure;
 - add the "floor in 2028 or 2031?" choice;
 - give reasons for short bonds and for longest-first;
 - put "about" on the ladder price and name the model behind any odds.
8. **F8.** Use the true refinement history in one reflection.
9. **F9.** Give each note one checkable fact, and each reflection one extra job.
10. **F10.** A plain-language and authenticity pass: jargon table, paraphrase test aimed at the 2031 method, the team's own labels, and the statistics degree used at most once.

1. What the reader actually sees (reconstructed from E1's specification; elements, not text)

The reader has about 2,300 finished submissions in the pile. Last season, "Some 2,300 teams from 79 countries submitted final reports" (R-W99, VP via Phase A; R-AN60). The Top 50 is roughly the top 2% (arithmetic; INT that the same share applies this year).

Piece	What E1 puts there	What a reader takes away in about two minutes (INT)	Main risk
Note A: hedge (TLH/IEF)	Role "future funding"; the ten payments; a rate-sensitivity fact ("about 10 years"); the rule "never sold after a rate rise". Under (ii), "after both deposits"	"A Treasury ETF for the operating payments", the same trade as Wharton's own example (TN guide L36-39, VRF). It stands out only if the fact and the rule are there	Reads like the example (R-AN44). Under (ii), a careful reader sees a ~\$198k hedge against a ~\$292k promise (AY1 [5], derived)
Note B: VT	Role "growth"; only money the payments do not need; "after her 2028 deposit"; one fund fact or the split	"A world index fund for growth"	Generic unless it says whose money is at risk. Under (ii) it visibly breaks a ban-worded pitch rule
Note C: VGSH	"Risk management" or "liquidity"; the short-Treasury part of the growth money	"Short Treasuries to lower risk"	The weakest note. Its effect is the ordinary effect of not holding stocks (AY1 C10)
Reflections (≤100 words)	The three official answers, plus "tested" numbers, plus scaling, plus a case trait passing the swap test, plus one number	Crowded. Six jobs in 100 words reads mechanical	Each reflection tries to do everything (section 3.2)
Pitch (50 words)	The rule, the client benefit, what the rest is for, and optionally the 2031 bought bottom	"They buy the payments first and then invest the rest"	A prohibition as the central idea; no residency; the most distinctive idea (element 4) is optional
IPS (≈480 words)	Four money rules plus governance, certainty with two gaps, risk by goal, three trade-offs, the WinS sentence, a fee principle, currency and building costs, bad-year pre-commitments	Reads like a policy manual. The reader can restate the first rule, but not the 2031 method	Density, jargon and page overflow (section 3.2)

The two-minute test (four questions adapted from D12 M234; answers predicted INT):

1. What is bought first, and why? **Likely answered.**
2. What is the rest for? **Likely answered.**
3. What will Laura tell co-sponsors in 2031, and why will she never have to walk it back? **Likely missed.** E1 puts this in rule 3 as "a, s, q, uncapped". That is the plan's most original idea, and it is the least legible.
4. What could still stop a payment? **Partly answered.** E1 names gap (i), gap (ii), the joint tail, and late, smaller or instalment deposits in three different places. The reader loses the thread.

2. Scores per criterion, with reasons

Scale (ASM, used for all five criteria):

- **1-3:** missing or wrong.
- **4-5:** competent and generic, like a typical finished report.
- **6-7:** clearly above the field.
- **8:** stands out in a pile.
- **9-10:** among the few best a reader sees.

Two limits on the scores:

- A Top-50 read probably needs about 8 on the first three criteria (INT).
- The TN and IPS carry only part of what the Final Report will show, so Articulation and co-sponsor communication are capped here by format, not by quality.

2.1 Investment Strategy: 7/10

Official wording (SMAApply L49, VRF): "Presents a clear and creative investment thesis; demonstrates disciplined planning across Laura's changing time horizons and cash-flow needs; uses appropriate diversification; and maintains consistency with the team's IPS."

Earns points:

- **Disciplined planning across changing horizons** is the plan's best fit to any criterion. There is a dated action at each case date (2027 purchase; 2028 completion and growth; 2031 bought bottom; 2033 reserve and gift), each decided before Nov 6 (E1 P11, P3, P7, P9).
- The Guide says a strategy may include "planned adjustments as funding dates approach" (L156, VRF). E1's rules are exactly that.
- Pre-set if-then rules answer the IPS guide's "planned changes" and "operating reserve's asset composition over time" (L46-50, VRF).

Loses points:

- **Creativity is modest at the thesis level.** "Buy the promise, grow the surplus" is standard pension practice (D8 framework map; the Russell 2026 "surplus glidepath" wording is VP). D13c N7 already says: "Say so ... Do not call the plan 'innovative'."
 - The creative parts are narrower: a 2031 range whose bottom is bought, and a promise that stops depending on her career income. E1 buries the first in rule 3 and makes it optional in the pitch.
 - **"Appropriate diversification" is never argued.**
 - The IPS guide asks for "portfolio construction and diversification" (L46, VRF).
 - The 2027 book is ~97-98% one issuer (M035, parked high-priority; ticket section 2).
 - E1's word budget (P16) has no diversification line. The official "funding purposes" diversification (R-AN45, VP) appears only as a reason for option (ii) in P13.
 - **Consistency risk.** E1 recommends both a ban-worded central idea (P1) and option (ii) (P13). E1 itself says the rule is "visibly contradicted by a day-one VT buy unless the IPS carries the scaling sentence" (P1 side effects). But the notes are read first and are permanent (TN guide L44, VRF).
- **One withdrawn number.** E1 still cites the D3 comparison "\$101k vs ~\$165k" (E1 L97-98). AX1 C6 withdrew it: it compares a 50% lock with an 80% lock, and the barbell is not dominated (section 3.8, E-1).

What moves it to 8:

- F2 (dated spine);
- F3 (the bought-bottom idea in the pitch);
- F4 (no visible contradiction);
- F5 (a diversification sentence);
- F7 (the withdrawn figure removed; the floor-timing choice made).

2.2 Client Knowledge and Objectives: 7/10

Official wording (SMApply L51, VRF): "Tailors the strategy to Laura's circumstances, priorities, risk considerations, and residency goals; demonstrates a thoughtful understanding of the client; and presents recommendations that can earn her confidence."

Earns points:

- **The 2028 deposit is career income.** It comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (case L44-45, VRF). So buying the payments first makes her promise independent of that income. This passes the swap test, and it is the most Laura-specific benefit in the run (D9 M143; D7 M011).
- **Credibility is tied to her reputation-driven income** (BS-03, INT from VRF), not only to co-sponsors.
- **Risk is stated goal by goal** with the three-part CFA profile (P6; D6 finding 1, VP source). This beats the one-word "moderate" most teams will write (SH-01, INT).
- **Her two decisions are left to her.** Laura makes the choices the case gives her: the range she announces and the 2033 amount (P10). That is respectful and matches the case (L101, L110, VRF).

Loses points:

- **The residency is missing from the pitch spec.**
 - E1's pitch elements speak of "ten payments", "markets", "earnings" and "facility contribution" (P1).
 - The criterion names "residency goals". Laura's newest verified priority is people "in the room" (D13c V6, VP).
 - A reader wants to see what the money is for before how it is invested (D13c communication checklist: "purpose comes first").

- **The plan looks timid, and 50/50 sharpens it.** Whole-portfolio stocks are:
 - 0% in 2027;
 - 17.0% after 2028 (20.4% at 60/40);
 - 3.5% after the 2031 floor at an 80% lock (E1 [3], MODEL, reproduced).

The case's growth wish ("pursuing growth", L73, VRF) and her own verified "take the jump" line (D13c V1, VP) pull the other way. E1's answer, risk goal by goal, is right but costs words.

- **The share rule reads as capital protection, not "thoughtful risk".** The rule is "at most about 1-in-20 chance of ending below the money put in". E1's own "why" says the rule checks risk "against what the safe alternative pays" (P5). It does not (section 3.6, R3).
- **The \$150k is doubted too often.** The case says "She will contribute" (L43, VRF; R-AN35).
 - E1's P17 limits doubt to "one labelled stress-scenario clause".
 - Yet P2 (gap i), P3 (completion from 2028) and P4 (smaller; late or in instalments; joint tail) spend about 40-70 IPS words on it (P16 budget).
 - A reader may see a team that does not trust the client's own plan (INT).
- **Statistics-degree hook.** E1 spends it on the certainty definition (P17). AX1 C19 warns against using it as a "persuasion lever". Once, in passing, is the maximum (AY2 C6).

What moves it to 8: F3 (residency and purpose first), F6 (positive reason for the stocks), and F10 (one clause for the \$150k; the degree used once or not at all).

2.3 Portfolio Analysis: 7/10

Official wording (SMAApply L53, VRF): "Demonstrates understanding and effective use of investment concepts and tools; integrates quantitative and qualitative analysis; and uses reasonable assumptions and projections to evaluate funding reliability, the facility contribution, and financial flexibility under varying market outcomes."

Earns points:

- **Certainty by price, not model.** Each payment is "bought, not forecast", with the residual risk named (P2). The case asks teams to "define ... explain how they evaluated ... identify the assumptions" (L97-99, VRF).
- **The hedge is described by rate sensitivity** ("moves like", "about 10 years"), with banned words kept out (P14; ticket section 5).
- **The "tested" reflection** uses a pre-registered window (fill date to the Oct 20 close) and percentage changes (P14; AY2 C2). That is real use of a tool, visible in the TN.
- **Varying outcomes are handled by rules:** rates fall, the deposit is short, markets are good or bad (P3, P4, P7).
- **Forecast language is graded** in words (P2 item 7; D12 M231).

Loses points:

- **Little visible data, by design.** The IPS carries at most one number (P17), and a reflection one. This is correct for the format, but it caps the score before the Final Report.
- **Three fixed-income slips a professional would mark** (section 3.6):
 - short Treasuries for money with a known 2031/2033 use, with no reason given;
 - the growth rule measured against "money put in";
 - longest-first with no stated reason.
- **Model-based odds in the IPS** ("about 1 in 20", "about 1 in 10") need "under our assumptions". The IPS cannot cite a model (L123, VRF). D3 found the model "has too few very bad years" (AX1 C12, MODEL).

What moves it to 8: F6, F7, and F9 (one checkable number per note).

2.4 Articulation of Competition Experience: 5/10 (capped by what the TN and IPS can show)

Official wording (SMAApply L55, VRF): "Demonstrates teamwork, communication, and learning; clearly explains the team's research and decision-making process; and reflects meaningfully on the team's growth

and response to challenges."

Earns points:

- Notes written to an element specification before any order (ticket gate d).
- A dated decision log, including "decided not to trade" rows (P15).
- An honest "tested" window.

Loses points:

- **E1's TN plan makes three "supported" notes the likely outcome.** "Refined" happens only if a pre-announced split decision or an unforeseen limit occurs (P14). That is honest, but the reader sees no change of mind.
- **The strongest true story is unused.** The team's first plan (~75% equity, growth-first) was tested and dropped:
 - it left a 3-8% chance of a payment shortfall;
 - that chance was ~41-55% if the 2028 deposit is missing (CLAUDE.md; brief s8 item 1; VRF).

That is a genuine "refined" history (D13c E6: "True event, not a metaphor"). The TN guide asks how decisions "aligned with, tested, or refined" the strategy (L12, VRF). E1 never routes this story to a reflection.

- **No roles are assigned yet** (CLAUDE.md open item, VRF), so teamwork is invisible in the notes.
- **The research behind the plan is overwhelmingly AI-produced.** The risk is an IPS that reads as "artificial sophistication". Readers fear AI-written text (SH-16, ASM; R-W26, VP).

What moves it to 6: F8 (use the true story once), F10 (the team's own labels and voice). Most of this criterion is Final Report (instructions Nov 9; later list).

2.5 Creativity and Presentation: 6/10

Official wording (SMApply L57, VRF): "Presents a compelling, well-organized narrative in an authentic team voice; uses data effectively to support conclusions; demonstrates original thinking and meaningful reflection; and communicates Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors."

Earns points:

- **Original thinking.** In 2031 she announces only what is already bought, plus a stated chance of more (P7). This answers the case's credibility warning (L114-116, VRF).
- **Uncertainty communicated credibly.** The two named certainty gaps (P2) match calibration research: honest numeric uncertainty costs "only a small decrease in trust" (van der Bles et al. 2020, VP via D6).

Loses points:

- **The IPS is organised as a rulebook** (four rules, triggers, governance, pre-commitments, trade-offs), not as a narrative.
- **No single memorable sentence** survives the two-minute test (section 1).
- **Jargon density** (section 3.3). A past-season Wharton judging page says "Excessive investing jargon doesn't necessarily dazzle the judges" (VP, retired 2023-24 page, via D9). A 2025 semifinal judge praised "simple, elegant ideas" (VP via D7/AY2, past season).
- **Page-fit risk at E1's upper budget** (section 3.2). Non-compliance means "will not be considered for semifinal selection" (IPS guide L82-83, VRF).
- **The co-sponsor part of this criterion is mostly Final Report**, so it is capped here.

What moves it to 8: F1, F2, F3 and F10.

3. Attack sheet

3.1 What reads generic (fails the swap test or copies the field)

Item	Where in E1	Why it reads generic (INT)	What would make it Laura's
Hedge note as "a Treasury ETF for the operating payments"	P14 note A	It is Wharton's own example trade (TN guide L36-39, VRF). Many teams will copy it (R-AN44; S4 finding 7, INT)	Lead with the dated payments it stands for and one checkable rate-sensitivity fact. Add the never-sell rule. No "volatility"
VT note "world stocks for growth"	P14 note B	Any client, any team	Whose money is at risk: only money her payments do not need; a fall shrinks the facility, never the payments; after her 2028 deposit
VGSH note "short Treasuries to reduce risk"	P14 note C	Its effect is the ordinary effect of not holding stocks (AY1 C10)	A checkable number, e.g. a 20% stock fall costs the growth money ~10% at 50/50 (ticket section 5, ASM). Or replace it with a genuine third decision (F9)
"Yearly review", "low-cost broad funds", "rebalance within a band"	P10, P12, P5	Boilerplate in any IPS	Keep only what changes a decision: rules change for her circumstances, never because markets moved (P10 item 3)
"Risk is measured on the surplus"	P6, P11	True of every surplus-based plan	Tie it to her: the payments may never use outside money (L91-92, VRF), while the facility may (L85-86, VRF). That asymmetry is the case's own (D9 M241)
Pitch without the residency	P1	"Ten payments" could be any liability	Name what the payments secure (years of the residency's operations: "what the numbers secure", D13c V8)

3.2 What reads over-complex

1. About 55 distinct ideas are specified for the pitch plus the IPS (E2 count of the numbered elements in P1-P13, INT; many overlap). Grouped:

- the central rule and its benefit (2);
- certainty (3) and its gaps (2);
- the January rule (5);
- the 2028 deposit branches (4);
- the growth mix (4);
- risk by goal (6);
- the 2031 method (6);
- flexibility and currency (4);
- the reserve (3);
- governance (3) and bad-year pre-commitments (4);
- trade-offs (3);
- fees (1);
- the WInS sentence (3).

At about 480 words that is one idea per ~9 words. A plain-English target is about 12 ideas (F1 lists them).

2. The budget does not fit its own page limit.

- E1's P16 totals 270-435 words "before connecting words" (VRF, E1 table).
- The page-fit ceiling is about 470-490 IPS words. It may be lower on Letter paper with Word's 27.6 pt "Double" spacing (D9; AX2 D10 C3: "about 480 words may be too many").
- At E1's upper end only ~35-55 words remain to join the ideas into sentences. The IPS then either overflows or reads as a list (INT).
- The pitch has the same problem: elements 1-3 cost ~34-42 words, and adding element 4 reaches 42-54 against a 50-word cap (E1 P1; D9 pitch table, INT estimates).

3. The 2031 method has three parameters (a, s, q), is uncapped, and states confidence both ways. It is correct and it earns its place as money-moving logic (AY1 section 5). But a non-specialist cannot restate it in two minutes (D12's paraphrase questions 5 and 6). See F10 for how to simplify it without losing the rule.

4. Rule 1 carries five branches:

- buy on arrival;
- longest-first;

- completion before growth;
- instalments;
- the joint tail.

E1 budgets 40-55 words for them. The joint tail is also named in P2 gap (i). This is double coverage.

5. Each ≤100-word reflection is asked to carry:

- the three official answers;
- supported, tested or refined;
- the pre-registered "tested" numbers (~20-25 words);
- the scaling under (ii);
- a case trait passing the swap test;
- at most one number (ticket section 6; P14).

That is six jobs in 100 words.

6. Four bad-year pre-commitments and three trade-offs are listed as separate IPS content (P10 item 4; P11). Each is sound, but together they are about 40-60 words that restate the rules.

3.3 Jargon a reader would stumble on (plain replacement = concept, not wording)

Term	Where E1 would put it	Plain concept (the team words it)	Keep?
funded ratio (at least 1)	P2 item 2	the money covers today's price of all ten payments	Concept yes, term no
nominal (US\$)	P2, P8	fixed in dollars, not adjusted for rising prices	Once, defined
percentile / "90th" / p5	P7	"about 1 in 10 chance of more than the top"	Words only
give-back share <i>s</i> , lock share <i>a</i> , <i>q</i>	P7, P11	the part bought; the part of the rest added to the gift; the part kept	Letters never appear
uncapped	P7	if markets do better, she can give more	Concept yes
rate sensitivity / duration / "about 10 years"	P13, P14 notes	its price moves like the price of her payments when rates change	In notes, one short fact; in the IPS, the concept only
designated (the reserve) / self-liquidating	P9	bought in 2027, named the reserve in 2033; each bond pays out in its year	Concept yes
band 45-55	P5	reset once a year if stocks drift far from half	One number at most
STRIPS, LDI, CUSIP, DV01, glide path, twist	E1 bans these in the IPS (P17)	none	No. Keep the ban
"risk need / ability / behavioural loss tolerance"	P6	how much risk each goal needs, can bear, and how she might feel	Concepts yes; the CFA labels no (no framework names, P17)
"promise money", "growth money", "three slices", "short-Treasury part of the growth money"	E1 and ticket labels	the team's own labels	See F10: AI-coined labels in permanent notes are a voice risk (R-W26, VP)

3.4 What is unclear in two minutes

1. **"Longest-dated first."** A lay reader's instinct is to fund the nearest payment first (INT). Without a reason, it reads backwards. The reasons are short and on file:
 - the long payments carry most of the price risk, so the unbought piece is the smallest and least rate-sensitive;
 - the price risk of the waiting part is +5% vs +15% (D1; AX1b item 23, MODEL);
 - less face value is left unfunded if the deposit never comes (\$31.4k vs \$47.7k at -100bp, MODEL).

The honest counterpoint also belongs in one clause: the waiting piece is the residency's first year (AX1b item 23).

2. **The 2031 method** (section 3.2 point 3).

3. **Under (ii), why WInS holds 17% stocks** when the pitch says no stock risk before the payments are bought (3.5, row 1).
4. **Which number is the reserve.** It is bought in 2027 and named in 2033 (case L94-95 vs the plan). E1 budgets one reconciling clause (P9), which is right: a case-literal reader flags this first (R-AN28).
5. **Four different stock shares for one client:**
 - WInS (ii) 17%;
 - real 2027 0%;
 - 2028-30 ~17%;
 - after 2031 ~3.5%.

These appear if P6 element 6 is used alongside the WInS sentence. Pick one frame (F1).

3.5 Contradictions across the notes and the IPS

#	Claim in one place	Clashes with	Severity (INT)	Resolution
C1	Pitch/IPS: a ban on stock risk "until all ten payments are bought" (P1)	Option (ii) buys VT on day one; the notes are read before the IPS	High	F4: word the central idea as a split of jobs (true of both books), and put "after her 2028 deposit" inside the VT note
C2	IPS: the promise costs about \$292-294k	(ii) hedge is \$198k (66%)	Medium	"after both deposits" inside the hedge note (ticket C2); the price of the promise goes in the reflection
C3	The VT note states the split	The IPS rule 2 split; the frozen Nov 6 book must reflect the IPS (L75, VRF)	Medium	Decide the split before the VT order, or log a pre-announced research-based change and trade it before Nov 6 (AX2 D2 C3)
C4	The VGSH note ("risk management/liquidity")	The IPS 2031 bottom is bought from all of the growth money	Low if the banned words are kept	Never "floor", "already owned" or "the amount she can promise" in a note (ticket section 5)
C5	Notes: "moves like / stands in for"	The IPS certainty definition: "bought and held to maturity"	Low	The WInS sentence says the funds are stand-ins (P13; D10)
C6	E1 P17: the \$150k is doubted in at most one clause	E1 P2, P3 and P4 doubt it in three places	Medium	F10: one labelled stress clause; the rest lives in the rules' "if" branches without repeating "uncertain"
C7	E1 P5 "why": the rule checks risk against "what the safe alternative pays"	The rule's yardstick is the money put in (~\$157.7k), not the safe alternative	Medium (a stats-trained reader)	F6
C8	A reflection explains a split change "because we researched"	The governance line says rules never change "because markets moved"	Low	Allowed before Nov 6 (AX2 D2 C3). The reflection must name the research, not a price move

3.6 What a fixed-income professional would mark wrong (red pen)

#	Mark	Severity; deliverable	Evidence	Fix
R1	Short Treasuries (about 1.9-year duration) hold the bond half of money whose use dates are known (2031 purchase, 2033 gift). A professional asks why the rate is not locked to those dates. Rolling short bonds adds reinvestment risk that is never mentioned	Important; IPS (rule 2), TN (VGSH note)	D3 AX1 C18: locking the bond part to 2033 is worth about \$4-9k at the median (MODEL). VGSH duration 1.9y (VP via ticket)	F7(b): give one reason (the yearly reset and the 2031 purchase need money that holds its value), or make "short vs dated bonds for the growth money" a team choice before Nov 6
R2	The barbell rejection rests on a withdrawn comparison. "\$101k bought in 2031 vs ~\$165k" compares a 50% lock with an 80% lock	Important; IPS (rule 3 timing)	AX1 C6 on D3_quant.md (VRF): at the same 80% share, the bought floors are \$161k (known in 2028) vs \$165k (known in 2031). An 80% barbell beats a 20%-equity plan at every percentile (MODEL)	F7(a): withdraw it. Decide "is the facility floor bought in January 2028 or January 2031?" before Nov 6, with the surviving reasons (more money growing in 2028-30; the case's "pursuing growth"; a bad 2031-32 hits a barbell's all-stock remainder harder)

#	Mark	Severity; deliverable	Evidence	Fix
R3	The growth-share rule's yardstick is too low to test whether risk is paid. "At most ~1 in 20 below the money put in" uses a ~\$157.7k line. The same money in Treasuries at today's implied rates reaches about \$204k	Important; IPS (rule 2)	\$157,736 (brief s7 arithmetic, VRF); ~\$204k Treasury-only (D8, MODEL; implied, not lockable; seen from today it too varies, \$170k/\$201k/\$233k, AX1 C5). At 50/50 the rule's p5 is \$168k (E1 [1], MODEL). Paths ending below the Treasury-only outcome: 42-43% (JPM) and 51-52% (Vanguard mid) at 50-60% stocks (AX2 D2 C10, MODEL)	F6
R4	Longest-first is stated without its reason	Minor; IPS (rule 1)	D1; AX1b item 23 (MODEL)	F7(c): one "because" clause, plus the first-year counterpoint
R5	The ladder price is a model price from a fitted par curve, not a STRIPS quote, so the ~19bp headroom is approximate	Minor; IPS (at most one rounded price fact)	AX1 C25 (VRF inputs + ASM method)	"about", and re-price on the day (a 2027 action)
R6	"Risk management" for the hedge is true only measured against the payments. In absolute terms the whole (ii) book falls about 6.8% per +1pp	Minor; TN	D12 finding 5 (VP durations, derived)	Keep "future funding" as the hedge's role word (ticket), and say "against the payments" if "risk" appears
R7	Odds without a model. "about 1 in 20" (rule 2) and "about 1 in 10 above the top" (rule 3) are model properties. The model understates crash years	Minor; IPS	AX1 C12, D3 v2 (MODEL). The IPS bans citations (L123, VRF)	Mark them "under our return assumptions"; name the model in the Final Report (later list)
R8	VGSH's "12.4% instead of 20%" effect is presented as a special property	Minor; TN	AY1 C10	Present it as the ordinary effect of holding about half in Treasuries

Not marked (these hold up): the certainty definition with "barring a U.S. Treasury default" (P2); STRIPS for the real ladder (brief s8 item 11); no rolling of the 2037-42 payments (brief s8 item 2); "about 10 years" as the honest precision (S4); US\$ binding with no NT\$ forward (a derivative; AX2 D4 C3); no rebalancing between reserve and growth money, written down (CFA 2010 element 4c, VP via D8).

3.7 What fails "earn her confidence"

The criterion changed from "would win him/her over as a client" (2022-24) to "earn her confidence" (VP via AY1 #16; SMaply L51, VRF). Confidence is earned by calibration, not by volume (Tenney et al. 2008, VP via D6). Against that:

- **Failing items:**

- **Too many caveats.** At least six named risks in 480 words:
- gap (i);
- gap (ii);
- the joint tail;
- a late deposit;
- a smaller deposit;
- U.S. default;
- plus building costs and currency.

Honesty helps, but a document that is mostly about what could go wrong reads defensive (INT). Two gaps plus one residual, stated once, is enough for the IPS. The rest lives in the rules or the Final Report. - **The growth side is undersold.** E1 never tells her what she gains if markets are good, beyond "the range shrinks or grows". The case asks for "pursuing growth" (L73, VRF). Her verified risk words are about leaping (D13c V1, VP; never quoted in the TN or IPS). - **"Her big bets are her career."** As a diagnosis of her life, this can read presumptuous (INT). As a portfolio fact, it is fine: her 2028 deposit already rests on her books and speaking, so the portfolio does not add a second bet on the same things (D8 M081, with AY2 C14 recast as content). - **The statistics degree used as a lever** to make the certainty definition persuasive (AX1 C19; AY2 C6). - **The residency missing** from the first 50 words (2.2). - **Passing items (keep):** - announce only what is bought; - she makes the two decisions; - costs never touch the bought payments (P12, if words allow); - no quotes, identity hooks or puns (D13 rules).

3.8 Errors and stale items inside E1 (for the main loop)

#	Item	Evidence	Status
E-1	E1 L97-98 cites D3 M238 "a 50% barbell can promise only ~\$101k ... vs ~\$165k". Withdrawn by AX1 C6. E1 L97 also says the ratchet "matches ... within \$2k of median". AX1 C11: at the same p5 the ratchet is better by \$2k at the median and \$6k at p95. It is not dominated; the rejection stands only on the design principle	D3_quant.md Audit corrections (AX1) C6, C11 (VRF)	E1 predates the D3 audit (E1 L32-34 says D3 had "no audit appendix yet")
E-2	P5 "why" misdescribes the rule's yardstick (money put in vs the safe alternative)	E1 P5; D6 s2.3 (VRF)	3.6 R3
E-3	P17 limits the \$150k to one clause, but P2, P3 and P4 spend three	E1 P16/P17	3.5 C6
E-4	P16's upper budget plus connecting words exceeds the page-fit target	D9; AX2 D10 C3	3.2 point 2
E-5	D3-only numbers used by E1 are pre-audit. Example: "\$43k covers a 150bp fall" should add "about \$51k for the worst 98-day fall since 1990, if rates do not fall further in 2027"	AX1 V4 on D3_model_v2_results.md (VRF)	Minor
E-6	The IPS guide's explicit asks for "liquidity" and "portfolio construction and diversification" (L46, VRF) have no line in P16	IPS guide L45-47	F5
E-7	E1's section-4 decision checklist lacks AX1 C6's genuine team choice (floor bought in 2028 or 2031) and D3 C18's (short vs dated bonds for the growth money)	AX1 C6, C18 (VRF)	F7
E-8	T2_red_team.md is still absent (the task lists it as evidence)	find, 2026-09-28	Informational

4. The ten fixes that would most raise the score (ranked by score gained per hour; all INT)

Each fix says what to change, which deliverable (TN or IPS), which criteria move, and what it costs. Nothing here is wording. The students decide.

F1. Cut the IPS to about 12 ideas and at most ~380 content words (IPS; C&P +1, IS +0.5).

• Keep:

1. purpose and central idea;
2. certainty definition;
3. both gaps in one sentence;
4. the 2027 purchase rule with its one rates-fall branch;
5. 2028: finish the payments first, then the growth mix and its reason;
6. risk by goal, including why so little stock and where her big bets already are;
7. diversification and liquidity (F5);
8. the 2031 method;
9. 2033: reserve = the ladder, the gift, and flexibility held safely;
10. what can change a rule;
11. US\$ binding and the building-cost assumption;
12. the WInS sentence.

• Cut or fold:

- the fee clause (AY2 C13 already says cut first);
- the four bad-year pre-commitments as a list (fold them into "never because markets moved");
- the three trade-offs as separate sentences (at most one "gives up X to get Y" inside rules 1 and 3);
- P6 element 6 (stage stock shares);
- separate deposit branches (one stress clause);
- repeated certainty vocabulary.

- **Test:** the exported PDF ends on page 3 with at least 2 spare lines (D9/D10 checklists).

F2. Build the IPS on four dated decisions (IPS; IS +0.5, C&P +0.5, Client +0.5).

- Replace "Rule 1-4 + G, trigger/action" with a prose timeline: January 2027, January 2028, January 2031, January 2033.

- Each date gets one action and one "because" tied to Laura or the case. Branches become subordinate "if" clauses.
- This answers the IPS guide's "planned changes ... as future funding needs approach" (L46-47, VRF) in the order a reader expects.
- It fits her record of dated steps (a product manager, case L17, VRF; D6 M125, INT).
- The rules' content does not change. The Final Report can still show each rule "fired as written".

F3. Rebuild the pitch around a positive central idea (IPS; Client +0.5, C&P +0.5, IS +0.5).

- **Elements (at most three; zero numbers; the team words them):**
 1. what is secured first, named by what it secures for the residency (years of operation), not by instrument;
 2. why only Laura gets this benefit: once bought, the promise no longer depends on her career income or on markets (P1 element 2; D9 M143);
 3. what the rest does, merged with the credibility idea: it grows for her facility gift, and she will tell partners only the part already bought.
- **Drop:** the prohibition framing as the lead (see F4); "fits inside \$300k"; any date-certain claim (E1 P1 bans are right).
- **Test:** a reader can restate all three elements after one read (D12 protocol).

F4. Remove the (ii) contradiction at its root (TN and IPS; IS +0.5).

- Word the central idea as a split of jobs: the payments' money is bought and never takes stock risk; stock risk sits only in money the payments do not need.
- This is true of Laura's real plan and of the (ii) snapshot, where the scaled hedge is the bought promise and VT sits only in growth money. A time-worded ban is broken by a day-one VT buy.
- Keep the ordering (payments first) in the 2027/2028 rule, not in the pitch.
- Under (ii), the VT and hedge notes each carry their scaling marker **inside the 300 characters**, because a note is read before the IPS exists (ticket C2).
- **Judge lean on the book (INT):** (ii). It gives the only TN set with a growth decision, and the TN guide's first sentence asks for the "approach to growth" (L4-6, VRF).
 - Choose (iii) only if the team cannot state the scaling in its own words (E1's decision test).
 - (iii) with a dated rung is competitive; (iii) with only a ~\$4.5k T-bill note is weak to a reader. That note earns about \$18 against a \$25 commission (ticket section 2, ASM), so it reads as filler unless the leftover rule is stated.

F5. Add a diversification clause and a liquidity clause (IPS; IS +0.5, PA +0.25).

- **Diversification content:**
 - diversified by funding purpose (R-AN45, VP);
 - the Treasury concentration is deliberate, because the bill is in US\$ and dated;
 - the growth money holds one fund of about 10,000 companies worldwide (VT 10,088 stocks, VP via ticket).
- **Liquidity content:**
 - no withdrawals before 2033 (L63-65, VRF);
 - each year's payment cash arrives when its bond matures;
 - the money kept for the project sits in short Treasuries.
- About 25-35 words, paid for by F1's cuts. This answers M035 (parked, high priority) and the criterion's "uses appropriate diversification" (L49, VRF).

F6. Fix the growth-share rule's yardstick and say what stocks are for (IPS; TN if (ii); Client +0.5, PA +0.5).

- **Either** keep the loss-limit rule and describe it truthfully: a limit on the chance of losing her own money, not a test of whether stock risk is paid.
- **Or** add the comparison a statistics-trained client would expect: the growth mix against the all-Treasury alternative. Under JPM, 60% stocks add about \$7-9k on average, and nothing on average under Vanguard's midpoint (AX1 C4, MODEL).

- **Either way, add the positive reason.** The facility has no required size (L104, VRF), and she wants growth (L73). So stocks are there for a bigger gift in good markets, and a bad market costs facility money only.
- **50 vs 60:** E1's robust test picks 50%. The "earn her confidence" risk (timid) points up. The median barely moves (+\$0.2-2.1k, E1 [1], MODEL). A reader will accept either if the reason is stated as a choice of spread (AX2).
- **Judge lean (INT):** whichever the team picks, record why. Do not justify 50% by "safety" alone.

F7. The fixed-income correction pack (IPS; PA +0.5, IS +0.25).

- (a) Delete the \$101k/\$165k barbell figure (E-1). Put "facility floor bought in January 2028 or January 2031?" on the pre-Nov 6 decision list with the surviving reasons (R2). It is a rule the Final Report must later apply, so it must be fixed in the IPS (brief s17).
- (b) One clause on why the growth money's bonds are short (R1), or make "short vs dated" a pre-Nov 6 choice.
- (c) One "because" for longest-first, plus its first-year counterpoint (R4).
- (d) "About" on any ladder price (R5), and "under our assumptions" on any odds (R7).

F8. Put the true refinement story in one reflection (TN; Articulation +1).

- **Content:** the first plan (growth-first, ~75% equity) was tested against the case's certainty test and dropped; this trade is the refined plan in action (CLAUDE.md and brief s8 item 1, VRF numbers).
- Only one reflection carries it, in the team's words. It must be true as told (R-W28, VP).
- The (ii)/(iii) vote and the split decision are also real decisions with dates. Log them (P15) even if no note uses them.
- Assign roles before the first order (trader, note writer, second reader, log owner; E1 section 4), so the record shows six people.

F9. One checkable fact per note, one extra job per reflection (TN; PA +0.25, Client +0.25).

- **Note (≤300 characters until the box is checked):** in AY1 C1's order, Laura's dated need, then which part of the plan, one checkable fact, then the role word. The risk goes to the reflection if space is short.
 - **Hedge:** the payments it stands for, plus "about 10 years" rate sensitivity.
 - **VT:** only money the payments do not need, plus "after her 2028 deposit".
 - **VGSH:** its checkable number (R8 framing). Or replace it with the split-decision trade if that genuinely happens. Under (iii), use a listed dated rung.
 - **Reflections:** the three official answers, plus exactly one of the following:
 - tested numbers (hedge);
 - scaling (under ii);
 - why so little equity (under iii);
 - the refinement story (F8).

Assign which reflection carries which before Oct 19.

F10. Plain-language and authenticity pass (TN and IPS; C&P +0.5, Client +0.25).

- Apply the section 3.3 jargon table.
- Run D12's paraphrase test, with two extra target questions: restate the 2031 method, and say why longest-first.
- If 2 of 6 readers misread the 2031 method, present it by its effect (a bought bottom; part of the rest added; part kept) and let "about 1 in 10" carry the top. Fix q in the decision log before Nov 6, so no parameter is chosen after the freeze (D8).
- The team chooses its own labels for the parts of the portfolio in a meeting and logs the choice. AI-coined labels in permanent notes are a voice risk (R-W26, VP; SH-16).
- Use the statistics degree once or not at all (AY2 C6).
- The \$150k gets one labelled stress clause (E-3).

5. Checklists (tick before submitting)

Two-minute test (pitch + first IPS paragraph; 6 unpaid readers, D12 protocol):

- [] A reader can say what is bought first and what it secures for the residency.
- [] A reader can say why this matters for Laura in particular (her career income).
- [] A reader can say what Laura will tell partners in 2031 and why it will not be walked back.
- [] A reader can say what could still stop a payment (two gaps plus a U.S. default), without a list of six.
- [] No reader thinks the WInS book contradicts the pitch (ask them after showing the three notes).

TN trio (Oct 23):

- [] Three executed trades with three different jobs, or a stated reason why not (TN L42, L53, VRF).
- [] Every note passes the swap test and differs from Wharton's example by one checkable fact.
- [] Under (ii), the scaling marker is inside the hedge and VT notes.
- [] Exactly one reflection is "tested", on the pre-registered window; one carries the refinement story (F8); no reflection does all jobs.
- [] No banned words (ticket section 5 list; E1 P14 vocabulary). No floor share, reserve size or range.

Consistency (TN ↔ IPS ↔ frozen book on Nov 6):

- [] The central idea is worded as a split of jobs and matches every note (3.5 C1).
- [] The split in the VT note equals IPS rule 2, or a logged research-based change was traded before Nov 6 (C3).
- [] The IPS WInS sentence names the date or state the book shows (D10, VRF requirement).
- [] The same labels appear in the notes, the log and the IPS (D9 M001).

IPS: rules to fix before Nov 6 that E1 leaves open, from the reader's side (brief s17; the Final Report must apply them without redesign):

- [] The facility floor is bought in January 2028 or January 2031 (AX1 C6).
- [] Short vs dated Treasuries for the growth money's bond half (AX1 C18).
- [] The growth share (50 or 60) and the yardstick its reason uses (F6).
- [] a, s and q fixed in the log, cap or no cap (E1 C3-C5), stated in words in the IPS.
- [] The completion order (longest-first) with its reason (F7c).

6. Later (after Nov 9), one line each

- Articulation evidence beyond the notes (decision-log table, paraphrase-test log): Final Report.
- Co-sponsor communication (range in dollars, likely figure vs midpoint, fundraising excerpt): Final Report.
- The all-Treasury comparison and the named model behind every "1 in X": Final Report.
- Charts proving the "bought first" claim: Final Report (the IPS bans charts, L123).

Sources**Official files (VERIFIED-REPO-FILE; all under `competition/official/2026_27/`):**

- `Laura_Gao_2026_Client_Profile.txt`: L17, L43-45, L61-74, L85-107, L110-120, L129-133.
- `2026_WGY_Investment_Policy-FINAL.txt`: L3, L16-26, L35-36, L45-78, L82-83, L101-123.
- `2026_WGY_Trading_Notes_Analysis-FINAL.txt`: L4-31, L36-53.
- `2026_WGY_Investment_Competition_Guide.txt`: L70-90, L154-159, L183-184.
- `2026_WGY_Competition_Infographic.txt`: L40-42.
- `SMAApply_Deliverables_Page_2026-09-27.md`: L49-57.

Primary web sources, as carried and re-verified by the cited files (not re-opened by E2):

- Wharton retired judging page (jargon line), <https://globalyouth.wharton.upenn.edu/wp-json/wp/v2/pages?slug=judging-and-evaluation> (via D9/AY2).
- 2025 top-10 article (Hahn, "simple, elegant ideas"), <https://globalyouth.wharton.upenn.edu/news/lets-go-announcing-the-top-10-teams-advancing-to-the-wharton-investment-competitions-2025-global-finale/> (via D7/AY2).
- Rules & Roles and the AI policy (R-W26, R-W28; via Phase A and D10).
- SMAApply Trading Details, "funding purposes" (R-AN45; via Phase A and AX2).
- van der Bles et al. 2020 and Tenney et al. 2008 (via D6/AY1).

- Russell 2026 (via D8/AY2).
- Vanguard VT facts (via the ticket).

Research files (lower authority; audit corrections applied):

- `research/insight_v1/phase_E/E1_change_proposals.md`.
- `phase_D/ D1-D12, D3_model_v2_results, D3_quant (AX1 appendix), D13a, D13c, and the four audit files.`
- `phase_D/trading_now_brief.md`; `wins_now/securities_and_allocation_v1.md`; `wins_now/S4_red_team.md`.
- `phase_A/case_register.md` (R-AN28, R-AN35, R-AN44, R-AN45, R-AN60); `phase_A/stakeholder_map.md` (SH-16, SH-18, BS-03).
- `phase_C/parked.json` (M035, M033, M149).
- `research/council_2026-09-27/01_chair_memo.md` and `round2_referee_ruling.md` (history).
- `CLAUDE.md`.

Script re-run: `.venv/bin/python research/insight_v1/scripts/E1_split_and_slices.py` (2026-09-28; reproduces E1 [1]-[4]). No new script was written for this file. E2's own counts (ideas, words) are INT, from E1's text.

What this teaches

1. **A reader scores what reaches them, not what you know.** The run's research is deep and mostly correct. A semifinal reader sees 3 × 300 characters, 3 × 100 words, 50 words and 500 words. The skill is choosing which dozen ideas travel.
2. **Density is a risk, like any other.** Fifty-five good ideas in 500 words scores lower than twelve clear ones, because the reader cannot restate any of them. Complexity must earn its place in words as well as in holdings.
3. **Word a rule so it is true everywhere the reader will look.** A rule about timing (no stocks before a certain step) is true of Laura's plan but visibly false in a WInS book that buys stocks on day one. A rule about which money may carry stock risk is true in both.
4. **Check what a rule measures before you praise it.** A rule that limits the chance of ending below the money put in does not test whether stock risk is worth taking. For that, the yardstick is the safe alternative.
5. **Audits keep arriving.** E1 was written before the D3 audit and still carries a withdrawn number. Before anything is frozen, re-read the latest corrections, even for files you already know.

E3 Red team: "Laura Gao" reads E1's proposals as a client choosing between firms (a simulation)

Agent E3 (red team, client seat), insight_v1 run, Phase E. Written 2026-09-28. This is AI-generated research for Team Caplet: simulated reactions, evidence, numbers, specifications and checklists. **None of it is text to submit.** The six students decide every change and write every WInS note, reflection, pitch sentence and IPS sentence in their own words. AI use goes in the Final Report's Works Cited (Wharton AI policy, R-W46). Laura appears only through the case and her public professional record. Nothing here suggests contacting her (R-W16: contact means disqualification).

Relayed messages: none arrived during this task. **Scope (brief section 17):** Trading Notes (TN, Oct 23) and IPS (Nov 6) only. Every item says which of the two it serves. Where the IPS must fix a decision rule that the Final Report will later apply, it is listed under "IPS: rules to fix before Nov 6". Pure Final Report items are one line each in section 10. **Evidence read in full:** the brief; CLAUDE.md; the case, IPS guide, TN guide, Competition Guide and SMaply criteria; phase_E/E1_change_proposals.md; phase_D/ D1-D12, D3_model_v2_results, D13a, D13b flags, D13c, all four audit files and every "Audit corrections" appendix (corrections override the text above them); phase_D/trading_now_brief.md; wins_now/securities_and_allocation_v1.md (and v0, S1-S4 summaries); phase_A/ case_register (R-C, R-AN), fact_register, stakeholder_map, wins_week1_guardrails; phase_C/survivors.json and parked M007, M008, M015, M021, M033, M035, M040, M149; the council chair memo and round-2 ruling (history only). T2_red_team.md does not exist in the repo on 2026-09-28.

New script: .venv/bin/python research/insight_v1/scripts/E3_client_view_numbers.py (about 10 seconds, no network; docstring lists inputs and labels). It re-uses E1's and D6's engine and random stream and prints three client-facing numbers E1 does not: [1] what the payments cost as a share of her first deposit; [2] the whole-portfolio stock share year by year 2027-2032 and its six-year average; [3] how much of the likely 2033 gift is already bought when she speaks in 2031. E1's own script was re-run first and reproduces every figure E1 quotes.

How to read this file (important)

- **SC = the simulated client.** SC is a reader built only from (a) the official case (VERIFIED-REPO-FILE) and (b) lines that D13a marked VERIFIED-PRIMARY, used inside their stated context. SC is **not Laura**. Every reaction below is labelled **SIM**: a simulation, i.e. an INTERPRETATION of how a reader holding those stated wishes and public habits would likely react. SIM is never a fact about Laura and never her words.
- **Nothing in this file may be quoted, paraphrased or attributed to Laura in any deliverable.** Do not write "Laura would feel/say/prefer" in a note, reflection or the IPS. Only D13a VERIFIED-PRIMARY lines are her words, and brief section 16 keeps all of them out of the TN and the IPS.
- Reactions are written in the third person on purpose: first-person lines in quotation marks would look like quotes.
- Every phrase that describes content (for example "what the rule makes possible") is **content, not wording**.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY (re-checked on the primary page 2026-09-28 where marked); **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **MODEL** = an ASSUMPTION-based model output, not a forecast; **INT** = interpretation (a judgement); **SIM** = simulated client reaction (a kind of INT, always tied to an anchor). Ids: R- case register, F- fact register, SH-/BS- stakeholder map, M- Phase C, P- E1 proposals.

Terms used (defined once):

- **Ladder:** Treasury zero-coupon bonds, one maturing shortly before each \$50,000 payment (2033-2042).
- **Growth money:** everything the ladder does not need (about \$158k after the 2028 deposit).
- **Bought bottom (2031):** the part of the growth money bought in January 2031 as a Treasury maturing before 2033, so its 2033 dollar value is known when she speaks to co-sponsors.
- **Forced caution / chosen caution:** caution the case's own numbers impose (the payments cost almost all of the first deposit) versus caution the team chose (the stock share of the growth money, the 2031

lock share, where kept money sits).

- **p5 / p50 / p90:** in a simulation, the value 5% / 50% / 90% of paths fall below.
- **Swap test (D13c):** replace "Laura" with any client's name; if the sentence still works, it is generic.

0. Summary: the simulated client's verdict and the changes it asks for

SC verdict (SIM, anchored below). On substance, SC would very likely choose this firm. E1's design does the three things SC cares about most while still pursuing some growth (which an all-Treasury rival would not): (1) the residency's ten operating payments stop depending on markets and on her next contract once they are bought (case L43-45, L91-92, VRF); (2) she walks into 2031 meetings with a number she never has to walk back (case L114-116, VRF); and (3) it names the two things it cannot fix (the January 2027 price; what US\$ buy in Taiwan). SC's hesitation is not the architecture. It is that E1 is written from the trading desk outward: parameters, codes and behavioural traps come first, and what her money does for her residency comes second. A rival with the same architecture (BS-10) and plainer, warmer, more honest words would win her.

Ranked by impact on reaching the semifinals and on making the plan hers:

#	Finding (one line)	Deliverable	Confidence
1	Never call the 2031 stretch "promised". E1's summary table and P7 label the middle slice "promised-at-risk" / "promised but still at risk"; the case says she will describe what she "expects" (L110) and her "potential contribution" (L116), and D9's vocabulary bans "promised" for the upside. Rename the three parts by what each dollar does: bought / expected if markets allow / kept for the project	IPS (rule 3 wording); TN vocabulary	high
2	Separate forced caution from chosen caution in the IPS. The payments cost 97.4-98.1% of her first deposit at today's prices (E3 [1]), so ~0% stocks in 2027 is arithmetic, not a view of her temperament. Then state each chosen setting with its reason. Without this the plan reads as "cautious client"; with it, she can inspect every choice	IPS (risk passage); TN (the "why so little equity" or scaling line)	high
3	Say what the stocks are for, in her terms. Against an all-Treasury benchmark, stocks at 50% of the growth money add about \$1k at the median and end below it in about half of paths (D3 M034, MODEL). They are a chance at a larger gift and more project flexibility in good markets, paid for with a smaller gift (never the payments) in bad ones. E1 gives a model rule for the split but not its purpose	IPS (rule 2 "because"); TN (growth note/reflection under ii)	high
4	Keep the fee principle and extend it: costs never come from the payments or from the announced 2031 bottom . E1 ranks the fee clause "first to cut", yet E1's own 50% split rests on a fee assumption (with no fee, the same rule under both forecasts gives 55%, E1 [2], MODEL). Cutting it leaves the split's reason unstated and her cost question unanswered	IPS	high (logic); medium (word budget)
5	Purpose before prohibition in the pitch, and no tense slips. E1's pitch spec opens with "no dollar takes stock-market risk until...". SC would first look for what the plan makes possible. Nothing may read as already bought (in 2026 nothing is)	IPS (pitch)	medium-high
6	Keep behavioural diagnosis of her out of the TN and IPS ("loss tolerance", "inferred willingness", "house-money", "break-even trap", "composure"). The case says she "understands that investing involves uncertainty and periods of market volatility" (L68-69). Frame pre-commitments as the firm's own discipline and as protection of her public word	IPS (P6, P10); TN reflections	high
7	The 2031 method passes SC's "never overpromise" test once findings 1 and 4 are applied: at the median 2031 state, 89% of the likely 2033 gift is already bought when she speaks (80/10/10; 82% at 70/15/15), and the lower half of the range is reached in no modelled path (E3 [3], MODEL)	IPS (rule 3)	high (method); medium (numbers)
8	Option (ii) vs (iii) is a truth-at-a-date question for SC. Whichever book, run a zero-risk pre-trade check: two readers outside the team read the planned note drafts and say what Laura's real money holds in January 2027. If either says "stocks", the (ii) scaling is not carried by the notes	TN (before the first order)	medium
9	Name which year of the residency any January-2027 gap falls on. Longest-first puts it on the first operating year (2033); the IPS "because" must say why that is still the better rule (smallest, cheapest piece; first use of the next deposit; less of the promise unbought if the deposit never comes)	IPS (rule 1)	medium-high
10	Proportion: one clause per honest gap, and lead with what is secured. SC wants the plan to read like a launch, not a disclaimer (case L38-39, "turning those ideas into reality")	IPS	medium

Does the plan honour "thoughtful risks" AND "protecting the capital required for her goals"? (section 6) SIM: it honours "protecting" completely and "thoughtful" genuinely, but "risks" thinly: about

10% of her money sits in stocks on average over 2027-2032 (9.7% at 50/50 with an 80% lock; 11.7% at 60/40; E3 [2], MODEL). Most of that thinness is forced by the case's numbers. The part that is chosen is small, defensible and must be owned in words.

Could she speak to co-sponsors without ever overpromising? (section 7) SIM: yes in US dollars, barring a U.S. default, provided four conditions hold: the stretch is never called promised; costs never touch the announced bottom; no range is announced before all ten payments are bought (E1 P4, already); and US\$ is the binding currency (E1 P8, already).

1. What the simulation is allowed to stand on

1a. Case lines SC reads as her stated wishes (all VRF, [competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt](#))

Case line	What it tells SC	Ids
L5, L9-10: "a potential client"; "the investment strategy that Laura ultimately chooses"	She is comparing firms; the plan must earn her choice	R-C4, R-C8, R-AN5
L7-8: "seeking thoughtful financial planning"; L37-39: planning will play a role in "turning those ideas into reality"	She hires the firm to make ideas real, not to be told what she cannot do	R-C7, R-C24
L43-45: the \$150,000 comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures"	Her second deposit rides on her own career	R-C27, R-AN31
L61-65: living costs outside the portfolio; no other flows before 2033	The growth money can bear losses without touching her life	R-C33, R-C34
L68-74: "understands that investing involves uncertainty and periods of market volatility"; "Although she has been willing to take thoughtful risks ... an appropriate balance between pursuing growth and protecting the capital required for her goals"	She knows markets fall; she wants balance, not a choice of one side	R-C36-R-C38, R-AN16
L83: "reliable support for its early operations"	Operations are the reliability she is buying	R-C41
L85-86 vs L91-92: others may fund the facility and extra operations, never her ten payments	Where outside help may and may not go	R-AN6
L101-103: "Laura must decide how much ... she can responsibly contribute"; "committing all remaining assets could limit her financial flexibility as the project develops"	Her agency in 2033; flexibility is hers	R-C57, R-C58
L110-116: she will "describe how much she expects to contribute"; overpromising "could damage her credibility"; "a credible range for her potential contribution"	Expectation language, not pledge language	R-C63, R-C66, R-C67
L117-120: a dollar range, a confidence, favourable and unfavourable outcomes, protect the payments	What the 2031 method must produce	R-C68-R-C71, R-AN9
L146-148: assumptions on inflation, "portfolio projections and facility costs", flexibility	The currency and building-cost gap must be named	R-C85
L16: "Statistics & Information Decisions Management"; L19-20: first work "Originally intended as a response to misinformation"	A reader who checks numbers and dislikes incomplete stories	R-C12, R-AN32, R-AN33

1b. Her verified words SC may draw on (D13a VERIFIED-PRIMARY; all nine lines below re-found verbatim on the primary pages with `fetch_text.py --grep, 2026-09-28`). Research evidence only: none of these may appear in the TN or IPS (brief section 16)

Id	Exact words	Outlet, date, what she was answering	Caveat that travels with it
B8b-Q18	"It's not an easy thing to say, "let me give up my incredibly cushy, paying job with health insurance to be self-employed, with unstable income.""	Overachiever Magazine, 2021-03-15; leaving her tech job	A career move, not investing. Her own word for her income is "unstable". FR only, once
B1b-Q11	"But I always tell myself, you should always take the jump because you're always going to regret not doing it."	same answer, 2021-03-15	A career leap. Never a reason for a stock weight (D6 audit C3). It is the tension the plan must answer

Id	Exact words	Outlet, date, what she was answering	Caveat that travels with it
D13a-N01	"It's actually harder as an entrepreneur because not only are you letting yourself down, but more importantly, you're also letting other people down — people who were your earliest supporters and were the first to believe in you."	The Daily Pennsylvanian, 2016-01-28 (student-era); challenges of her student laptop-decal business	One source, student years, a small business, not a residency
B8a-Q13	"When I first posted the comic, I expected a way worse reaction"	Overachiever, 2021-03-15; posting The Wuhan I Know	About a public reaction, not money
B8b-Q35	"I can't think of many art pieces that I didn't also write some kind of purpose statement for."	Pulse Spikes, 2021-06-18; how writing and art go together	About her art; applying it to a policy document is INT
B8b-Q38 / B8b-Q39	"...Our No. 1 priority is to get as many diverse voices and stories in the room as possible, no matter the grounds that they start off on" / "We definitely hope to offer more as the festival grows."	Local News Matters, 2026-02-12; the festival's mini grants ("we" = organisers)	About festival grants; says nothing about residency fees
B2b-Q08	"Mentor for Tech It Out Philly and MoneyThink, two organizations in which I taught web development and financial literacy to high school students in West Philadelphia."	Poets&Quants, 2018-03-30 (student-era); her activities list	Says nothing about her own investing
B8b-Q21	"it's great having thousands of likes, but at the end of the day, it's just a number. You never really understand the human connection behind it"	Overachiever, 2021-03-15; one reader's thank-you note	Metrics vs meaning; not a view on finance

1c. Deliberately not used (and why)

- The case pull quote "The only person who needs to believe in something is yourself." is the case's words (VRF), not verified as hers (D13a: PARAPHRASE-UNVERIFIED as Laura's words). SC reads it as the case's framing only.
- "risk-adverse" (about choosing a business major), "No more gimmicks." on its own (about fantasy drafts), "explain myself" (about language choices), "diminishing returns" (about dance songs): context breaks (D13a).
- "[unfinished]" and "A/B testing": verified, but reading them as habits for financial numbers over-reads single words (D3 audit C19; D12 audit C7). Used here at most as INT, never as a lever.
- D13c candidates C1-C8: verbatim, but D13a has not ruled on them. Not used.
- The "floor-first leap": a reporter's sentence, barred by D13 (brief section 16).
- Privacy exclusions (ids only): B3b-Q09, B8b-Q23, B8b-Q24, B8b-Q34, B8a-Q20, B8a-Q22, B8a-Q23, plus the four D13c additions. There is **no verified statement by Laura about investing**: SC has no "investment philosophy" and none is invented here.

2. E1, proposal by proposal, from the client's chair

Reaction codes: **U** = feels understood; **A** = feels analysed (treated as a subject); **S** = feels sold to; **R** = a line she would reject on sight if it reached a deliverable; **N** = neutral (necessary, not felt).

E1 item	SC reaction (SIM)	Anchor	What would fix it (spec, not wording)	Serves
P1 pitch as a rule	U on honesty (a rule she controls cannot be falsified by rates); A/S on order (it opens with a prohibition, which reads as caution-first to a founder)	case L38-39, L69-74; B8b-Q35 (purpose statement, art context, INT)	Element order: (1) what the plan makes possible for the residency, (2) the rule, (3) what the rest is for. No word may imply anything is bought in 2026; the optional 2031 element must be future/conditional	IPS pitch
P2 certainty + two gaps	U : a definition she can check at a market price, plus the two holes named	case L97-99; L146-148	Keep; one clause per gap; graded words ("at today's prices", "we assume")	IPS; TN vocabulary
P3 January 2027 rule	U for "buy on arrival, no rate bets"; question : which operating year carries a gap	case L91-92; AX1b item 23	The "because" must say why the first operating year is the one left waiting (section 5, Q2)	IPS rule 1
P4 2028 deposit + joint tail	U : it treats her income honestly without doubting the case ("will", L43)	case L43-45; B8b-Q18 (FR only)	Keep to one clause, labelled the firm's own stress case	IPS rule 1

E1 item	SC reaction (SIM)	Anchor	What would fix it (spec, not wording)	Serves
P5 growth split 50/50 by a two-forecast rule	A/S : a model-selection argument, not a purpose; SC would ask what stocks are for and why her split depends on a fee nobody told her	case L72-74; E1 [2] (no fee → 55%), D3 M034	Purpose first (finding 3); state the fee assumption (finding 4); own the choice as "a steadier 2031 bottom" if 50/50 is kept	IPS rule 2; TN if (ii)
P6 risk by goal	U for "where her big bets already are" (her career, the residency); A if written as a scored assessment of her ("need low / ability high / willingness inferred")	case L69-71 ("throughout her entrepreneurial career"); R-AN17	Write each pot's job and what it can bear; say the payments carry no market risk however much appetite anyone has; no three-factor vocabulary	IPS; TN reflections
P7 2031 three slices	U on method (bought bottom, uncapped, two-sided); R on the word "promised" for the at-risk slice; A on codes (80/10/10, p90)	case L110, L114-116; D9 M062 vocabulary	Rename the parts by what each dollar does (finding 1); no codes in the IPS	IPS rule 3
P8 flexibility in US\$ T-bills; US\$ binding; no NT\$ hedge	U : flexibility kept by rule, and the firm noticed where the money is spent	case L102-103, L146-148	Keep	IPS rule 4 / certainty clause
P9 reserve = ladder	N/U : one sentence reconciling "funded 2027, designated 2033" answers the question she would ask	case L94-96	Keep	IPS
P10 governance, bad-year pre-commitments	U that she makes the 2031 and 2033 calls; A where the reason for pre-commitment is a trap she might fall into	case L68-69, L101	Frame as firm discipline and protection of her public word (finding 6)	IPS
P11 four rules + governance	N : the right skeleton	IPS guide L48-50	Keep; rename rule 3's parts	IPS
P12 fee principle, cut first	S (hidden cost) if cut; U if kept and extended	B2b-Q08 (student-era, INT); BS-01	Keep; extend to the announced bottom (finding 4)	IPS
P13 WInS (ii) vs (iii)	S risk under (ii) if any note or line lets a reader think her January 2027 money holds stocks; timid risk under (iii) unless the forced-vs-chosen answer is given	case L19-20 (misinformation origin, INT); R-AN14	Pre-trade two-reader check (finding 8); "two-thirds" must attach to her total money, never to the payments	TN; IPS WInS sentence
P14 three notes plan	U on "no staged trades" and honest "tested" windows; A if reflections become rate-tracking statistics	TN guide L15-16	One tracking clause at most; each reflection names her need, not her personality	TN
P15 decision log	N/U	TN guide L7-9	Keep	TN
P16 page fit, paraphrase test	U for the paraphrase test; N for the page-fit lawyering	IPS guide L82-83	Keep; add SC's questions (section 5) to the readers' list	IPS
P17 what not to put	U : it removes most of what SC would reject	brief s16	Add the items in section 4	IPS; TN

3. What makes her feel understood, and why each passes the swap test

Each item is SIM, tied to a case line; her verified words are supporting context only (Final Report at most).

- Once bought, the promise stops depending on her next contract** (E1 P1 element 2; D9 M143). Anchor: case L43-45 (VRF); her own word "unstable" for self-employed income (B8b-Q18, 2021, VP, FR only). Swap test: passes, because only a client whose second deposit is career income gets this benefit. SIM: the single strongest "understood" signal in E1. Condition: said conditionally ("once bought"), never as a date (E1 P1).
- Nothing is announced to co-sponsors that is not already bought when she speaks** (E1 P7 items 5-6; P4 item 6). Anchor: case L114-116 (VRF); D13a-N01 (2016, student-era, VP): she ranks letting

down early backers above letting herself down. Swap test: passes weakly on the case alone; strongly once BS-03 is added (her income is reputation-driven, so a walked-back number costs her twice, INT).

3. **Her risk-taking is located, not denied** (E1 P6 element 5): her concentrated bets are her career, her books and the residency, and the 2028 deposit already rides on them, so the portfolio holds only broad market risk. Anchor: case L69-71, L43-45 (VRF); D8 M081 (Chhabra 2016, VP). SIM: this is how the plan can honour "thoughtful risks" without pretending 17% stocks is bold.
4. **She keeps the two decisions the case gives her** (E1 P10: the range she announces in 2031; the facility amount in 2033). Anchor: case L101, L110 (VRF). SIM: a founder choosing a firm wants a firm that recommends, not one that decides for her.
5. **Operations come first, in the case's own order** (reserve, then facility, then flexibility). Anchor: case L83, L94-103 (VRF). Context only: her newest public priority is access for people "in the room" (B8b-Q38, 2026, festival grants, VP, INT link). SIM: the payments read as what keeps the residency running for its first decade, which fits that priority.
6. **The firm noticed where the money is spent** (E1 P2 gap ii, P8): US\$ is binding, Taiwan building costs are expected to rise, no NT\$ bet. Anchor: case L76-77, L146-148 (VRF). Swap test: passes (Taiwan is the case's location, not her identity).
7. **Every number says what kind of number it is** (E1 P2 element 7: price vs history vs assumption). Anchor: case L97-99 (VRF, "identify the assumptions"). SIM: a reader trained in statistics (case L16) trusts a graded number more than a precise one. INT only: her student-era project labels its own results "[unfinished]" (B9b-Q09); never cite that as a reason in a deliverable.
8. **She is not treated as a theme** (E1 P17; D13c R3): no Taiwan tilt, no identity-based holding, no book puns. SIM: the absence is felt.

4. What feels like being analysed or sold to, and what she would reject on sight

4a. Analysed (treated as a subject rather than a client) — fixes are specs, IPS and TN

Trigger in E1	Why SC reacts (SIM)	Anchor	Fix
Pre-commitments justified by "break-even trap", "house-money trap", "loss tolerance", "composure" (P6.4, P10.4, via D6)	Implies she may panic; the case says she already understands volatility	case L68-69 (VRF)	Keep the rules; give their reason as the firm's discipline and the protection of her public word. None of the trap vocabulary in the TN or IPS
E1 P6.4's point that her investment loss tolerance is the team's inference, if phrased as a finding about her	Reads as a diagnosis written about her, not to her	R-AN17 (case states no risk tolerance)	State it as a design property: the one place her appetite for risk cannot reach is the payments; the growth money's share is set by a stated rule
Her statistics degree as a hook (P2 "Why"; P17 "one place")	Flattery or a persuasion lever ("because you studied statistics, we...")	case L16 (VRF); D3 audit C19	Prefer zero explicit uses in TN + IPS; let the checkable definition show the respect. If used once, it must change a decision (every probability names its model), not compliment her
Codes: "80/10/10", "p90", "1-in-20", "45-55 band", "±0.25 years"	Trading-desk shorthand for her own money	D13c V12, R4 (INT)	Each parameter appears once, in plain words, next to what it does
"Thoughtful risk = risk that is paid" (D2 via P5) as a gloss on the case's phrase	The firm redefining her words	case L70 (VRF)	Present it as the firm's reading of the case, never as what she means

4b. Sold to (a pitch, not advice)

- **The fee clause cut first** (P12). SIM: a firm that does not say what it costs, or which pot pays, is selling. Anchor: B2b-Q08 (she taught financial literacy, student-era, VP, INT link); BS-01; D3 v2 (c): a 0.5% fee on the growth money costs about \$5k of median; 1% on everything about \$32k (MODEL, ASM fees).
- **A growth split whose stated reason is a model contest** (P5) rather than what it does for her residency.
- **Option (ii) WInS holdings described as "her portfolio"** (E1 P13 already bans this; keep the ban absolute).

4c. Reject on sight (checklist for every note, reflection, pitch and IPS draft)

- [] "Promised", "committed" or "pledged" for money that is still at market risk (the 2031 stretch).
- [] Any word implying the payments or the 2031 bottom are already bought (in 2026 nothing is); "bought in January 2027" or "fits inside \$300,000" as flat facts (E1 P17).
- [] Costs taken from the payments or from an announced bottom.
- [] A range reported as "100% within" because of a cap (E1 C3, D3 audit C2).
- [] A model percentage for the payments ("95% certain"); "guaranteed", "safe", "risk-free".
- [] An NT\$ figure presented as a commitment.
- [] Identity, heritage or Taiwan as a reason for a holding; book-title or "falling" puns; the pull quote as hers; the "floor-first leap"; her career-regret line as the reason for stocks.
- [] Behavioural-trap or risk-profile vocabulary applied to her.
- [] WInS funds called "the operating reserve", "matched" or "stable".
- [] The growth split justified by valuations ("stocks are expensive"): a reason that will change (AX2 on D2, C7).

5. The questions she would ask (SIM), and where the answer must already live

Each question is a SIM of what a client holding the case's wishes would ask a firm before choosing it. "Gap" means E1 does not yet give the IPS the content to answer it.

#	Question (content, not wording)	Where it must be answered	E1 status
Q1	What exactly is bought, when, and what happens if January 2027 prices are worse?	IPS rule 1	Answered (P3): buy on arrival, longest-dated first, finish from the 2028 deposit before any growth; about 1 in 3 modelled rate paths need part of the deposit (MODEL, AX1b)
Q2	If a gap opens, which year of the residency is exposed, and why that year?	IPS rule 1 "because"	Gap. Longest-first leaves the 2033 payment waiting. Reasons it still wins: the waiting piece is the smallest and least rate-sensitive, it is the first thing the next deposit buys (top-up price risk about one third of nearest-first), and if the deposit never came less of the promise would be unbought (\$31.4k vs \$47.7k at a 100bp fall; MODEL, AX1b item 23, D3 M053). The IPS needs the reason in its own "because"
Q3	If the middle outcome barely changes, what are the stocks for?	IPS rule 2 "because"; TN growth note (ii)	Gap. At 50% of the growth money, the median lift over an all-Treasury benchmark is about +\$1.4k and about 48% of paths end below it; on the same path the bad case is about \$41k lower and the good case about \$57k higher (D3 M034, MODEL; the benchmark itself is not riskless from today, D3 audit C5). So the stocks buy a chance at a larger gift and more flexibility in good markets
Q4	What do I pay, and from which money?	IPS cost clause	Gap/conflict (P12 cut first). See finding 4
Q5	What will I say in 2031, and is every word true in a bad market?	IPS rule 3	Answered in method (P7); needs the vocabulary fix (finding 1)
Q6	In a strong market, where does the extra go, and who decides?	IPS rules 3-4	Answered (uncapped: half to the gift, half kept, P7); add that money above the top is her decision in 2033 under the rule, so co-sponsors do not count it (D3 audit C15)
Q7	Is "certain" certain in Taiwan?	IPS certainty clause	Answered (P2 gap ii, P8): certain in US\$ barring a U.S. default; a fixed \$50k is worth about \$42k (2033) to \$34k (2042) in 2026 dollars at 2.5% inflation (F-114, ASM); a fixed US\$ buys about 11-20% less building by 2033 (D4, ASM on VP inputs)
Q8	Would you change my plan if markets crash in 2029?	IPS governance	Answered (P10): only her circumstances change a rule, never a market move. A down year for the growth money is more likely than not before 2033 (62% of paths at 60/40, D6, MODEL), and the case says she knows it (L68-69)
Q9	Why does WInS show what it shows?	IPS WInS sentence; TN notes	Answered if the date sentence is written (P13); see finding 8
Q10	What remains mine to decide?	IPS governance	Partly (P10). Content: the rules set the most she can responsibly announce; she decides whether and how to announce within it, and the use of kept money after 2033

#	Question (content, not wording)	Where it must be answered	E1 status
Q11	Why you, and not a firm that buys Treasuries with everything, or one that invests for growth and buys the payments in 2033?	IPS trade-offs (T1-T2 in P11)	Answered in principle. The all-Treasury rival is simpler and has a similar median but cannot "pursue growth" (L72-73) or offer co-sponsors any upside; growth-first misses a payment in 3.2% of modelled paths (13.7% if the 2028 deposit is \$75k) (F-401, F-403, MODEL). The IPS names both trade-offs without numbers
Q12	What happens after 2042, or if the residency is delayed?	Out of scope (case L92-93; R-AN34)	Later list (section 10)

Add Q2, Q3, Q4 and Q10 to the P16 paraphrase-test question list: if a reader cannot answer them from the draft IPS, the IPS is missing content, not only clarity (IPS).

6. Test 1: does the plan honour "thoughtful risks" AND "protecting the capital required for her goals"?

Numbers SC would look at (E3 script unless stated; MODEL on VRF/VP inputs, JPM AC World 7.00%, bonds 5.0%, no fee):

Stage (Jan 1)	Whole-portfolio stock share, 50/50, lock 0.8	60/40, lock 0.8	Stock share of the growth money only (50/50)
2027	0.0% (leftover waits in T-bills)	0.0%	n/a
2028-2030	17.0-17.2%	20.4-20.7%	50%
2031-2032 (after the bottom is bought)	3.5%	4.2%	10%
Six-year average	9.7%	11.7%	

At a 0.7 lock the six-year average is 10.3% (50/50) and 12.4% (60/40). In dollars, about \$79k of her ~\$158k growth money would be in world stocks in 2028 at 50/50 (D2, ASM).

Forced vs chosen (the distinction SC most needs):

Caution	Forced or chosen?	Evidence
~0% stocks in 2027	Forced. The ten payments cost \$292,264-\$294,387, i.e. 97.4-98.1% of the \$300,000 first deposit, on the 2026-09-25 curve	E3 [1]; F-101; S1 (VRF inputs, ASM method)
Payments bought first, all ten	Forced by the case's standard , cheap by the numbers: a full lock gives up about \$0.6-3.3k of median facility money against partial locks under JPM and nothing under Vanguard, and partial locks fail in 13-26% of paths if the 2028 deposit does not come	D3 M243 (MODEL; D3 audit C16: about \$0.9-5.2k on forwards)
Growth money at 50% (or 60%) stocks	Chosen. The rule picks 40-65% across the 24 tested cells of forecast, bonds and fee; 55% under both houses' central forecasts with no fee; 50% once a fee of up to 0.5% is assumed	E1 [1]-[2] (MODEL, reproduced)
80% (or 70%) of the growth money bought in 2031	Chosen , for her credibility, not for "capital required"	E1 P7; case L114-116
Kept money in US\$ T-bills after 2033	Chosen , for flexibility "as the project develops"	E1 P8; case L102-103

SC's reading (SIM):

- "Protecting the capital required for her goals": fully honoured. The capital her goals *require* is the payments (there is "no predetermined facility contribution", L104). They are bought, held to maturity, carry no fee, and the joint tail is named. SC would add that the 2031 lock protects something else she cares about, her word, and the IPS should say which of the two each protection serves (the D4 audit C6 makes the same distinction).
- "Thoughtful risks": honoured in kind (risk only where she alone bears the cost; set by a stated rule; not trimmed to the minimum) but thin in size. One dollar in ten in stocks on average is low for a client described through her career risk-taking. The honest defence is that most of it is forced, and that her real leap, the residency, is not in the portfolio at all. The IPS must make that visible, or the plan reads as a verdict on her nerve.
- **Where SC would call E1 too timid:**

1. *In words more than weights.* E1's budget gives up to about 60-80 of ~470 IPS words to the certainty definition with its two gaps plus the joint-tail clause (E1 P2, P4, P16 estimates; INT count), and opens the pitch with a prohibition. Honest, but the proportion and order make the plan sound defensive. Lead with what is secured and made possible; one clause per gap (ASM: about 15% of the IPS is a sensible ceiling).
2. *Stacked cautious filters on the split.* E1 picks 50% by requiring the rule to pass under the gloomier forecaster *and* a fee of up to 0.5% *and* market-consistent bonds. Each filter is reasonable; together they land near the low end of the 40-65% range the rule gives across E1's tested cells. If the team keeps 50/50, it should own it as a choice (a steadier 2031 bottom, simpler to explain), not present it as what "the rule" forces.
3. *Cost transparency treated as optional* (P12 cut first).

• **Where SC would call E1 too clever:**

1. Codes and parameters (80/10/10, a/s/q, p90, 1-in-20, "uncapped", "two-sided") where a sentence about what each dollar does would carry the same rule.
2. The (ii) "mix after both deposits, scaled to \$300,000" book: correct, but a model of her plan rather than her money on any day; it needs explaining in every note that touches it.
3. The CFA three-factor risk vocabulary if it reaches the IPS (E1 P17 already bans framework names; keep it out of the words too).
4. Reflections that turn into rate-tracking statistics (bp moves and % changes) instead of her need; one clause is enough.

- **Not too timid** (SC would accept on sight): the ladder bought first, the 2031 bottom bought before she speaks, the kept money in T-bills. Each protects something other people will rely on, which is where her student-era words put the higher stake (D13a-N01, VP, INT link, FR only).

7. Test 2: could she speak to co-sponsors in 2031 without ever overpromising?

The 2031 message at the median 2031 state (E3 [3]; 50/50 split, JPM AC World, bonds 5.0%, no fee, top = 90th percentile of the unlocked money's two-year growth; MODEL):

Parts (bought / expected if markets allow / kept)	Bought bottom	Likely 2033 gift	Top	Share of the likely gift already bought when she speaks	Gift lands in the lower half of the range	Kept for the project (median)
80/10/10 (E1 default)	\$167k	\$188k	\$192k	89%	needs a 35% two-year fall in the unlocked money: 0.0% of modelled paths	\$21k
70/15/15 (E1 alternate)	\$146k	\$178k	\$183k	82%	same threshold: 0.0% of modelled paths	\$32k

Across all paths (E1 [4], reproduced): 80/10/10 gift p5/p50/p95 \$151k/\$188k/\$238k; kept \$16k/\$21k/\$29k. The gift is at or above the bottom in every modelled path because the bottom is bought (residual risk: a U.S. Treasury default). Above the top: about 1 in 10 at the median state, by construction. The model is thin-tailed, so check history: D3's replays of five crash episodes placed after the 2031 purchase (1929-31, 1973-74, 2000-02, 2008, 2022) cut the facility money by only about 4-6% (D3 M059, 60/40 split, MODEL on VP data). In the worst row (1929-31) that is roughly an 18% fall in the unlocked money (INT arithmetic from D3's \$164k bottom and \$195k total), well short of the ~35% fall needed to reach the lower half.

SC's reading (SIM): the method lets her say three things that can all stay true: what is already bought, what she expects if markets allow, and what she keeps for the project. Almost nine-tenths of what she expects to give is owned when she speaks, so the range is narrow and honest ("as precise as warranted", Du et al. 2011, VP via D6; its applicability to arts co-sponsors is INT). **She can speak without overpromising in US\$, barring a U.S. default, if and only if:**

1. **The stretch is never "promised"** (finding 1). The case's verbs are "expects" (L110) and "potential contribution" (L116). A promise of at-risk money is exactly the overpromise the case warns against (L114-115).
2. **Costs never touch the announced bottom** (finding 4). A fee deducted from the bought 2031 Treasury would make the 2033 amount fall short of the announced bottom; D3's audit lists "fees charged to the floor" among the ways the bottom could be missed (D3 audit C2; D3 v2 (c), MODEL).

3. **No range before all ten payments are bought** (E1 P4 item 6, keep).

4. **US\$ is the binding currency; any NT\$ figure is dated and illustrative** (E1 P8, keep). In NT\$ terms nothing is certain: two-year USD/TWD moves have a standard deviation of about 6.8% since 2006 (F-513, VP inputs, derived).

Two residual discomforts SC would voice (SIM), neither a reason to change the method:

- The range is lopsided: the likely figure sits about \$4k under the top and about \$21k above the bottom. A co-sponsor who plans on the midpoint under-plans, which costs credibility nothing (FR presentation: later list).
- At 80/10/10 the kept money is thin (about \$21k median, roughly 11% of the likely gift). That is roughly two to three years of Taiwan building-cost drift at the post-2021 pace (3.54%/yr, F-508, VP inputs; ASM sanity check, not a contingency fund, which the case does not require, L106-107). SC would still lean 80/10/10 because the part others can bank on is larger (89% vs 82% bought), and the case lists "continued business income" among the project's other possible sources (L86). Team decides (E1 C4).

8. Changes E3 asks E6 to adopt (ranked; each is a spec, the team writes the words)

#	Change	Why (anchor)	Serves	Word cost	Conflict with E1?
E3-1	Rename rule 3's parts by what each dollar does: <i>bought</i> / <i>expected if markets allow</i> / <i>kept for the project</i> . Remove "promised" from every internal label so it cannot drift into drafts	case L110, L114-116; D9 M062	IPS rule 3; TN vocabulary	0	Yes: E1 P7 uses "promised but still at risk"
E3-2	IPS risk passage distinguishes forced caution (payments cost ~97-98% of the first deposit at today's prices; one graded price fact) from chosen settings (split, 2031 lock share, kept money), each with its reason	case L69-74; E3 [1]; E1 P17 allows one dated price fact	IPS; TN (iii) "why so little equity" or (ii) scaling	~10-15 (reuses the one allowed number)	Refines P6
E3-3	Rule 2's "because" states the purpose of stocks (a chance at a larger gift and more flexibility in good markets; a smaller gift, never the payments, in bad ones) before the rule that sets the share	case L72-74, L110-112; D3 M034	IPS rule 2; TN growth note (ii)	~10	Refines P5
E3-4	Keep the cost principle (do not cut first) and extend it: costs come only from money not yet promised to anyone, never from the payments or the announced 2031 bottom. State the fee assumption the split rests on, or re-state the split's reason without it (no fee, the rule gives ~55%)	E1 [2]; D3 audit C2; BS-01; B2b-Q08 (INT)	IPS	+3-5 on P12's 15-20	Yes: E1 P12 and AY2 rank it first to cut
E3-5	Pitch order: purpose (what the plan makes possible for the residency) → rule → what the rest is for; nothing reads as already bought	IPS guide L35-36; B8b-Q35 (INT)	IPS pitch	0	Refines P1 (order only)
E3-6	No behavioural diagnosis of her in the TN or IPS; pre-commitments framed as the firm's discipline and the protection of her public word	case L68-69; R-AN17	IPS (P6, P10); TN reflections	0	Refines P6/P10
E3-7	Rule 1's "because" names that any January-2027 gap waits on the first operating year and why that is still the better order	AX1b item 23; D3 M053	IPS rule 1	~8-12	Refines P3 (E1 kept the counterpoint to "the team's own risk discussion")
E3-8	Governance says what stays hers: whether and how to announce within the rule in 2031; the facility amount and the use of kept money in 2033; money above the top is her decision under the rule	case L101, L110; D3 audit C15	IPS governance	~8-10	Refines P10
E3-9	Statistics degree: zero explicit uses in TN + IPS preferred; if one, it must change a decision	D3 audit C19; AY2 C6	IPS; TN	saves ~6-10	Yes: E1 P17 suggests one use in the certainty definition
E3-10	Pre-trade two-reader check of the note drafts (what does her real money hold in January 2027?); the (ii) "about two-thirds" must attach to her total money after both deposits, never to the payments	case L161-162 (consistency); R-AN14	TN (before the first order)	0	New check; compatible with P13/P14

#	Change	Why (anchor)	Serves	Word cost	Conflict with E1?
E3-11	Proportion: one clause per honest gap; lead with what is secured; ~15% of IPS words as a ceiling for "what could still go wrong"	case L38-39	IPS	saves words	Refines P2/P4/P16

9. IPS: rules to fix before Nov 6 (client-seat content only; E1 P11 is the skeleton)

- **Rule 1 (promise first):** add the "because" for longest-first (E3-7); keep the joint tail to one labelled clause.
- **Rule 2 (growth mix):** purpose first (E3-3); the split and band fixed once, with the fee assumption it rests on stated or removed (E3-4); frozen against market moves (E1 P10).
- **Rule 3 (2031 range):** bought / expected if markets allow / kept (E3-1); set from what she owns in 2031; no range before the ladder is complete; uncapped with two-sided confidence and the model named (FR numbers later); costs never from the announced bottom (E3-4).
- **Rule 4 (2033 order of use):** reserve, then gift (bottom + the stated share of the rest), then kept money in US\$ short Treasuries; money above the top is her call under the rule (E3-8).
- **Governance:** who decides, including what stays hers (E3-8); rules change only for a change in her circumstances; no behavioural diagnosis (E3-6).
- **Certainty clause:** bought at market prices, held to maturity, in US\$, barring a U.S. default; gap (i) January price, gap (ii) Taiwan buying power, one clause each (E1 P2; E3-11).
- **Cost clause:** kept and extended (E3-4).
- **Risk passage:** forced vs chosen (E3-2); where her big bets already are (E1 P6.5).
- **Pitch:** purpose, rule, what the rest is for (E3-5).

TN checklist additions (before the first order and at reflection time):

- [] E3-10 two-reader check done and logged (no names, reader type only).
- [] No "promised", "committed" or "pledged" for any growth-money holding; no "two-thirds of the payments".
- [] Each reflection's "how it serves Laura" names her need (payments, facility, flexibility, timing of her deposits), not her personality; passes the swap test; no trap vocabulary; no degree.
- [] Under (iii): the "why so little equity" reflection gives the forced-vs-chosen answer (E3-2) in content.
- [] Under (iii), SIM: a dated Treasury rung repaying \$50,000 near the first payment date (if WInS lists one) is the most tangible single holding a non-specialist can point at; under (ii) its face is scaled (~\$34k), so it loses that tangibility (securities_v1 section 7).

10. Later (after Nov 9), one line each (Final Report; not worked here)

- The "likely figure" beside the range, not the midpoint (89% bought at the median 2031 state, E3 [3]).
- At most one or two dated Laura lines in context (candidates: D13a-N01 beside the bought bottom; B8b-Q18 beside the promise-first rule), each labelled with outlet, year and, for 2016, "student-era".
- The real-terms and Taiwan building-power sentence (F-114; D4 table A).
- One line on a delayed residency and on money after 2042 (BS-07; case L92-93).
- One line naming the operational controls behind "certain" (BS-18).
- The fee assumption and net-of-fee projections (D8 M026; D3 v2 (c)).

11. Conflicts E3 could not resolve (both sides; the team decides)

#	Conflict	Side A	Side B	E3 view (SIM/INT)
K1	Label for the 2031 middle slice	E1 summary table and P7 (and AY1's metric s(1-a)): "promised but still at risk"	D9 M062 vocabulary; case L110/L116: never "promised" for the upside	Rename (E3-1); the metric can stay internal

#	Conflict	Side A	Side B	E3 view (SIM/INT)
K2	Fee clause	E1 P12 / AY2: cut first (case silent on fees)	E1 P5's 50% rests on a fee assumption; SC's cost question; announced bottom exposed	Keep and extend (E3-4); if cut, re-state P5's reason
K3	WInS book	E1/T1/D12 lean (ii): three roles, a visible growth decision	D10 lean (iii): true at a stated date; SC values literal truth and a tangible \$50,000 rung	SC is close to neutral, slightly toward (iii) on credibility; the E3-10 check is the tie-breaker the team can run today
K4	Statistics degree	E1 P17 / D7: once, in the certainty definition	D3 audit C19 / SC: zero is safer	Prefer zero (E3-9)
K5	Pitch order	E1 P1: rule first (true in every path)	SC: purpose first, rule second	Reorder (E3-5); both elements stay
K6	Growth split	E1/T1/D2: 50/50 (passes under both houses with a fee $\leq 0.5\%$)	D6/D13c: 60/40 (JPM; do not look set at the minimum); E1 [2]: 55% with no fee under both houses	SC cares less about the number than about the purpose and an honest reason; whichever is chosen, own it as a choice of spread
K7	2031 parts	80/10/10: larger bought share (89% of the likely gift)	70/15/15: more kept flexibility (\$32k vs \$21k median)	SC leans 80/10/10 (others rely on the bought part); flexibility is thin but not required to be sized (L106-107)
K8	Where the longest-first counterpoint lives	E1 P3: in the team's own risk discussion	SC: in the IPS rule's "because"	IPS (E3-7), one clause

Sources (accessed 2026-09-28 unless noted)

Official, in the repo (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (L5-10, L16, L19-20, L25-26, L38-39, L43-45, L61-74, L76-77, L83-120, L123, L136-148, L169); `2026_WGY_Investment_Policy-FINAL.txt` (L3-5, L16-28, L35-36, L45-78, L82-83, L101-125); `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L3-31, L36-53); `2026_WGY_Investment_Competition_Guide.txt` (L70-94, L156-159, L181-186); `SMAApply_Deliverables_Page_2026-09-27.md` (criteria). Market data: `competition/official_market_data/daily-treasury-rates_2026-09.csv` (row 09/25/2026).

Laura's words (VP; D13a register, each re-found verbatim with `research/insight_v1/scripts/fetch_text.py --grep` on 2026-09-28):

- Overachiever Magazine, 2021-03-15 (B8b-Q18, B1b-Q11, B8a-Q13, B8b-Q21): <https://www.overachievermagazine.com/conversations/gskjklclyh93jwbb2809150ne9iyid>
- The Daily Pennsylvanian, 2016-01-28 (D13a-N01, student-era): <https://www.thedp.com/article/2016/01/draw-street-journal-q-and-a>
- Local News Matters, 2026-02-12 (B8b-Q38, B8b-Q39): <https://localnewsmatters.org/2026/02/12/sf-queer-comics-fest-expands-stays-free-in-second-year/>
- Pulse Spikes, 2021-06-18 (B8b-Q35): <https://pulsespikes.org/story/laura-gao>
- Poets&Quants for Undergrads, 2018-03-30 (B2b-Q08, student-era): <https://poetsandquantsforundergrads.com/students/2018-best-brightest-laura-gao-wharton-school/>
- B9b-Q09 ("[unfinished]", lauragao.com/rewriting-herstory) is cited from D13a (VP there) and used as INT only.

Research files (lower authority; numbers reproduced where stated): `research/insight_v1/phase_E/E1_change_proposals.md` (and `scripts/E1_split_and_slices.py`, re-run 2026-09-28, every figure reproduces); `phase_D/D1_rates.md` (+AX1b), `D2_equity_ai.md` (+AX2), `D3_quant.md` and `D3_model_v2_results.md` (+AX1), `D4_taiwan_fx.md` (+AX2), `D5_cosponsors.md`, `D6_behavioural.md`, `D12_overflow_client.md` (+AY1), `D7_wharton_intent.md`, `D8_practice.md`, `D9_communication.md` (+AY2), `D10_compliance.md` (+AX2), `D13a_laura_quotes_verified.md`, `D13c_voice_map.md`, the four audit files, `trading_now_brief.md`; `wins_now/securities_and_allocation_v1.md`, S1-S4; `phase_A/{case_register,fact_register,stakeholder_map,wins_week1_guardrails}.md`; `phase_C/survivors.json`, `parked.json`; `research/council_2026-09-27/01_chair_memo.md`, `round2_referee_ruling.md` (history).

Scripts: `research/insight_v1/scripts/E3_client_view_numbers.py` (new; reuses `E1_split_and_slices.py`, `D6_behavioural_numbers.py`, `A2_curve_recheck.py`).

What this teaches

1. **A client reads a plan from the outside in.** Analysts build from rules and parameters; a client first asks what her money will do for the thing she cares about, then what it costs, then what could go wrong. The same strategy can feel like advice or like a rulebook depending only on that order.
2. **Separate what the numbers force from what you chose.** When the payments cost 98% of the first deposit, holding almost no stocks in 2027 is arithmetic, not a judgement about the client. Saying so replaces an implied verdict on her temperament with a short list of choices she can inspect one by one.
3. **Verbs are part of risk management.** "Bought", "expected" and "kept" are three different promises. Calling at-risk money "promised" would recreate the one mistake the case warns about, even though the numbers underneath are sound.
4. **Understanding a client is not analysing her.** Behavioural research is useful for designing rules; applied to the client in her own document, it tells her she is being studied. Design the rules with it; explain them with her goals.
5. **Simulating a real person needs a fence.** Every reaction here rests on the case or on her verified words in their own context, and none of it may be put in her mouth. That fence is what makes the exercise honest rather than invented.

E4 Rival strategist: the strongest competing strategy for this case, and where the E1 plan loses to it

Agent E4 (red team, "the best strategist of a rival team that wants the same Top 50 place"), insight_v1 run, Phase E. Written 2026-09-28. This is AI-generated research for Team Caplet: a competing design, numbers, specifications and checklists. **None of it is text to submit.** No pitch, IPS sentence, WInS note or reflection is drafted here; every phrase that names an idea is a content label, not wording to copy. The six students decide every change and write every word. Any AI use goes in the Final Report's Works Cited (Wharton AI policy, R-W46). Laura appears only through the case and her public professional record; nothing here suggests contacting her (R-W16).

Scope (brief section 17). Only the Trading Notes (TN, Oct 23) and the IPS (Nov 6), plus WInS trading as the source of the notes. The rival strategy fixes the **decision rules** a Final Report would later apply (reserve, 2031 range method, facility and flexibility, if-then rules) but gives no Final Report content, no dollar range presented as a deliverable and no fundraising text. Final Report items are one line each in section 9. **Relayed messages:** none arrived during this task.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY; **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **MODEL** = an ASSUMPTION-based model output, not a forecast; **INT** = E4's interpretation or judgement (never a fact about Laura). Ids: R- = case register, F- = fact register, SH-/BS- = stakeholder map, M- = Phase C questions, P- = E1 proposals, F1-F10 = E2 fixes (all under research/insight_v1/). Every security named is **PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK** (brief sections 4 and 14).

New script: `.venv/bin/python research/insight_v1/scripts/E4_rival_numbers.py` (about 5 seconds, no network; the docstring lists every input with its label). It re-uses D6's engine and random stream (seed 20260927, 200,000 paths), **reproduces E1's P7 row exactly** (bottom \$167k, top \$192k, gift \$151k/\$188k/\$238k, kept \$16k/\$21k/\$29k) and then runs the rival on the same paths and on every historical 6-year window 1928-2025. Sections [1]-[7] of its output are cited below as E4 [n]. Cross-check: E1's average whole-portfolio stock share over 2027-2032 comes out at 9.7%, the same figure E3 reports from its own script (E3 [2]).

Evidence read in full (audit corrections override the text above them): the brief; CLAUDE.md; the case, IPS guide, TN guide, Competition Guide, Infographic and SMaply page; `phase_E/E1_change_proposals.md` (the plan being attacked), `phase_E/E2_judge.md`; D1-D12, `D3_model_v2_results.md`, `D3_quant.md`, D13a, D13c, all five audit files and every "Audit corrections" appendix (D1 AX1b, D2/D4/D10 AX2, D3 AX1, D5/D6/D12 AY1, D7/D8/D9 AY2); `phase_D/trading_now_brief.md` and `wins_now/securities_and_allocation_v1.md` (both as revised after T2), `wins_now/T2_red_team.md`, S1-S4; `phase_A/case_register` (R-C, R-W, R-AN, X), `fact_register`, `stakeholder_map`, `wins_week1_guardrails`; `phase_C/survivors.json` and parked M007, M008, M015, M021, M033, M035, M040, M149; the v4 prompt parts 2, 3, 5; the council chair memo and round-2 ruling (history only). `phase_E/E3_laura.md` appeared during this task; its summary was read and one figure is cross-checked above.

Terms used (defined once):

- **Ladder:** ten zero-coupon U.S. Treasury bonds (STRIPS), one maturing shortly before each \$50,000 payment (2033-2042). Identical in E1 and in the rival.
- **Growth money:** everything the ladder does not need (about \$158k on 1 January 2028).
- **Bottom (of the co-sponsor range):** the part of the facility gift that is already bought as a Treasury maturing before 1 January 2033, so its dollar value is known in advance. E1 buys it in January 2031; the rival in January 2028.
- **Barbell:** a split into two very different pieces with nothing in between: here a dated Treasury for the bottom and 100% world stocks for the rest.
- **Stock fund (rival):** the part of the growth money held in one broad world-stock fund from 2028 to 2033.
- **Cap:** a rule that the 2033 gift never exceeds the top announced in 2031; anything above stays with Laura.

- **Percent-years of stock:** the whole-portfolio stock share added up over 2028-2032 (a rough measure of how much stock risk a plan carries over time; divide by 6 for the 2027-2032 average).
- **p5 / p50 / p95:** in a simulation, the value 5% / 50% / 95% of paths fall below (p50 = median).

0. Summary

The rival in one paragraph (content, not wording). Buy each of Laura's promises on the day the money for it arrives, and put in stocks only the money nobody else relies on. January 2027: her first deposit buys the ten operating years (the same ladder and the same rates-fall rule as E1). January 2028: her second deposit, which the case says comes from "publishing advances, speaking engagements, licensing, and other entrepreneurial ventures" (case L44-45, VRF), first finishes any unbought payment, then buys the **building's minimum**: a Treasury maturing before 2033 that repays exactly the dollar amount of that deposit. What is left (about \$40k, about 9% of her money) goes into one world stock fund and is left alone until 2033. In 2031 nothing is traded: she tells co-sponsors a range whose bottom she has owned since 2028 and whose top is the bottom plus half of what the stock fund is worth that day; the top is reached if stocks are not lower two years later, which happened for U.S. stocks in 84% of two-year periods since 1928 (E4 [5], VP data, derived; 73% in the JPM world-stock model). In 2033 the ladder becomes the operating reserve, the building gets the minimum plus half the stock fund (never more than the announced top), and everything else is Laura's flexibility. (Content label for this file, not wording: "every dollar has a date and a job".) No forecast is needed for any number she will say in public except the one chance attached to the top, and that chance can be checked against a century of data.

Scoreboard, same model paths (E4 [2]-[5]; MODEL; JPM AC World 7.00%, growth-money bonds 5.0%, no fee, as E1 P7):

Measure	E1 (50/50 growth money, 80/10/10)	Rival default (bottom = the 2028 deposit)	Rival alternate (a = 2/3, same stock exposure as E1)
Facility + flexibility money in 2033, p5 / p50 / p95	\$168k / \$210k / \$266k	\$182k / \$207k / \$250k	\$176k / \$208k / \$265k
Bottom announced in 2031, p5 / p50 / p95	\$134k / \$167k / \$211k (depends on 2028-30 markets)	\$150k fixed, owned from Jan 2028	\$134k fixed, owned from Jan 2028
Bottom after a bad 2028-30 (worst 10% of 3-year stock returns): median (p5)	\$134k (\$119k)	\$150k (\$150k)	\$134k (\$134k)
Median top / width	\$192k / 1.15x	\$175k / 1.17x	\$167k / 1.24x
Gift p5 / p50 / p95	\$151k / \$188k / \$238k	\$165k / \$174k / \$188k	\$154k / \$166k / \$183k
Kept for the project p5 / p50 / p95	\$16k / \$21k / \$29k	\$16k / \$32k / \$66k	\$21k / \$42k / \$86k
What the top needs	a model percentile (p90); "about 1 in 10 above"	today's market value; reached in 73% (model) / 84% (history) of 2-year periods	same
Whole-portfolio stock share 2028 / 2031 (median)	17.0% / 3.5%	8.7% / 9.1%	11.4% / 11.9%
Stock percent-years 2028-32 (average 2027-32)	58 (9.7%)	45 (7.5%)	59 (9.8%)
History, every 6-year window 1928-2025 rescaled to JPM: total worst / p10 / median	\$120k / \$174k / \$214k	\$169k / \$185k / \$214k	\$159k / \$180k / \$218k
Rules the IPS must fix	growth mix 50/60 + band + yearly reset; a, s, q; cap or not; VGSH or dated bond (open); 2028 or 2031 (open)	the minimum rule (set by the case's own number); s = one half; cap	same, with a = 2/3

Where E1 loses to the rival (details and evidence in section 5):

1. E1's co-sponsor bottom is unknown until 2031 and falls with 2028-30 markets (to about \$134k in a bad decile, \$93k in a 1929-type window); the rival's is fixed and owned three years before she speaks.
2. E1's top rests on a model percentile that D3/AX1 found lands 81-97% "within" across models and history; the rival's top is a market value on the day, and its odds come from history, not a model.

3. E1 rejected the 2028 purchase on a comparison that AX1 has withdrawn (E2 E-1); at equal stock exposure the rival is better in the bad case and in every historical tail.
4. E1 holds the bond half of the growth money in rolling short Treasuries although its use-dates are known (E2 R1; AX1 C18); the rival dates it to December 2032.
5. E1's central idea is a prohibition that its own recommended WInS book breaks on day one (E2 C1); the rival's is a positive, dated rule that is true in both books.
6. E1's third Trading Note (VGSH) is its weakest (E2) and may not mention the co-sponsor promise; the rival's third note is the most Laura-specific holding in either plan and can name its job honestly.
7. E1 needs about 55 ideas in 550 words (E2 section 3.2); the rival's IPS needs about 10.
8. E1 cannot name anything co-sponsors can count on without a projection; the rival's bottom is set by a number the case itself gives (her 2028 deposit).

What E1 would need to win back (section 6): decide the open gate question now (ticket v1 gate box (b): "is the facility floor bought in January 2028 or January 2031?") and, if 2031 is kept, say in the IPS why the bottom may depend on 2028-30 markets; date the growth money's bond half (IBTM or a Jul-Dec 2032 Treasury, not VGSH); make the top checkable without a model; give the bottom a case-anchored meaning; lead the pitch with a positive dated idea; cut the IPS to about a dozen ideas.

Where the rival loses to E1 (section 7, honest): a lower median bottom (E1's is higher in 78% of paths) and a smaller median gift (\$174k vs \$188k); less stock over 2028-32 (45 vs 58 percent-years) and a lower p95; a cap that stops strong markets from visibly raising the gift; a 100% stock fund that is still at risk after Laura has spoken; and an instrument (IBTM or a 2032 note) whose WInS availability is unknown. The rival is not a free improvement; it is a different bet on **when** risk is taken and **what** is promised.

North Star link (INT). A client choosing between two firms that both buy her payments first would see the difference in 2031: one firm can tell her in 2028 the exact minimum she will be able to put on the table, sized to her own 2028 earnings; the other tells her it depends on the next three years.

1. The rival strategy (specification, not wording)

1.1 Design principle (one line each)

- **Each promise is bought the day its money arrives** (2027: operations; 2028: the building's minimum).
- **Only money nobody relies on takes stock risk, and it takes it fully** (one world stock fund, no bond dilution, no rebalancing, held to 2033).
- **Nothing is traded in 2031:** the range is read off what she owns (a bought minimum and a market value), so no forecast is announced.
- Every decision traces to a case line: L43-45 (two deposits, the second from her career), L91-92 (payments from the portfolio alone), L101-103 (flexibility), L110-120 (range, credibility, confidence), L94-96 (reserve), L69-74 (thoughtful risk, balance) (all VRF).

1.2 Four dated decisions (the IPS spine; E2 F2 asks for exactly this shape)

Date	Rule (trigger, action)	Why (case line; Laura link)	Status of the reason
1 Jan 2027	The first deposit buys the ten payments on arrival; if it falls short, longest-dated first; any unbought part is the first use of the next money; any leftover waits in T-bills	Same as E1 P3 (L91-92; the ladder cost more than \$300k on 176 of 185 days of 2026 and "about 1 in 3" rate paths, AX1b)	VRF case; VP inputs, MODEL odds
1 Jan 2028	The second deposit (each instalment if late) first completes the ladder; then buys a Treasury maturing before 1 Jan 2033 that repays the deposit's dollar amount (or all that is left, if less); the rest goes into one world stock fund	Her 2028 money is career income (L44-45); the case says "A meaningful personal contribution may signal that the residency is financially viable, demonstrate Laura's commitment to the project" (L110-112); the stock fund is money nobody relies on	VRF case; INT reading

Date	Rule (trigger, action)	Why (case line; Laura link)	Status of the reason
1 Jan 2031	Nothing is traded. Range = [the minimum, the minimum + one half of the stock fund's value that day]. Confidence stated two ways: below the bottom only on a U.S. Treasury default; the top reached if the stock fund is not lower two years later (history and model odds named)	"credible range", "how confident", "favorable and unfavorable" (L115-119); "In 2031, however, the value of her portfolio in 2033 remains uncertain" (L112-113)	VRF case; VP data (history), MODEL odds
1 Jan 2033	The ladder is designated the operating reserve, held to maturity (self-liquidating). Gift = minimum + one half of the stock fund, never above the announced top. Everything else (the other half, plus any gain above the top) is her flexibility, moved to short Treasuries; she decides its use	L94-96 ("if at all"), L101-107 (she decides; flexibility; no contingency fund required)	VRF case

- **Governance line (element):** the rules change only for a change in her circumstances, never because markets moved; the stock fund is never sold before 2033 to "protect" a gain or "wait for a recovery"; no rebalancing between the three parts, written down (CFA 2010 element 4c, VP via D8/D10); students decide, Laura makes the two choices the case gives her (range she announces; 2033 amount) (R-AN4; D10 M013, VP).
- **The parameters and their reasons:**
 - **Bottom = the 2028 deposit's dollar amount.** Not a free choice: the case supplies the number. It makes the minimum knowable in the IPS as a rule output, conditional only on the deposit arriving (the case says she "will", R-AN35) and on no extreme rate fall before January 2027 (E4 [7]: at -100bp the top-up of about \$26k comes out of the stock fund and the minimum stays whole; only at about -150bp, a late-2008-size move, does it dip to about \$147k; such 98-day falls occurred in about 0.1% of windows since 1990 and 1.4% since 1962, AX1 V4, VP data). At a lower January-2028 five-year rate the minimum stays the same and the stock fund shrinks (4.0%: about \$35k instead of \$40k, E4 [4], MODEL/ASM).
 - **s = one half** of the stock fund goes to the building (up to the top); the other half is kept. Reason (INT): by 2031 the uncertainties that dominate what the gift buys are the exchange rate and Taiwan building costs, not markets (D4 finding 1; true for lock shares of about 70% or more, AX2 D4 C2; the rival locks about 74%), so a real share is kept by rule for "as the project develops" (L102-103). Kept money is at least E1's at every percentile and larger at the median and p95 (E4 [2]).
 - **Cap at the announced top.** Makes the top a ceiling that needs no forecast. The informative confidence number is then the chance of reaching the top, never "100% within" (AX1 C2; D5 audit C11).
- **What the rival drops from E1 (and why it can):** the growth-money mix (50/50 vs 60/40), its 45-55 band and yearly reset, the percentile q, the uncapped give-back rule, VGSH, and the "bought in 2031 from a sleeve that moved for three years" dependency. None of these is needed when the only risky money is one stock fund held to a date.

1.3 Certainty, risk and currency (IPS content elements; the same honesty as E1 P2, applied to two promises)

- **Certainty definition (elements):** bought at market prices and held to maturity; evaluated by price, not a model; certain in nominal US\$ barring a U.S. Treasury default. Applies to the ten payments **and** to the building's minimum. The two gaps named once each: (i) the January-2027 price (the next money completes the ladder first); (ii) a fixed US\$ amount buys less each year in Taiwan (named, not solved) (brief section 16, binding).
- **Risk stated goal by goal (elements; CFA 2020 three-part profile via D6, VP; no framework name in the IPS):** operations: no risk needed, none affordable; building minimum: none once bought; stock fund: all of it, 100% stocks, sized so that even a total loss leaves every promise intact. This is a **maximum-loss statement**, not a probability: the most she can lose to markets is the stock fund, about 9% of her money in 2028 (E4 [6], MODEL arithmetic on VRF inputs). E1's equivalent rests on a model threshold ("about 1 in 20 below the money put in"), which E2 R3 says uses a weak yardstick.
- **Forced vs chosen caution (element; E3 finding 2):** the low stock share in 2027 is the case's arithmetic (the payments cost about 97-98% of the first deposit, F-101, S1); the chosen part is the minimum rule and s.

- **Currency and building costs (elements; AX2 D4 C1):** US\$ is binding for every promise; any NT\$ figure is dated and illustrative (Final Report); no NT\$ conversion or hedge before the facility decision (a forward is a derivative, banned in WInS and, on the safe reading of R-AN15, in the plan); the assumption that Taiwan building costs are expected to rise is named (case L146-147).
- **Diversification and liquidity (elements; E2 F5; R-AN45 "funding purposes", VP):** diversified by funding purpose and by date (a 2032 minimum, ten dated rungs 2032-2041, one undated stock fund of about 10,000 companies worldwide); the Treasury concentration is deliberate because the bills are in US\$ and dated; no withdrawals before 2033 (L63-65); each year's payment cash arrives when its bond matures.

1.4 Considered and rejected by the rival (one line each)

- **TIPS for the building's minimum:** protects U.S. CPI, not Taiwan building costs or the exchange rate, and makes the minimum's dollar amount unknown in 2028 (INT; brief section 8 item 10 for the payments).
- **A second lock in 2031 on top of the 2028 minimum:** a second rule for a narrower range; the design principle says no (AX1 C11 logic).
- **Uncapped top at a model percentile (E1/D5):** see section 7, X3; it is E1's better answer to "favorable outcomes".
- **Taiwan, AI or thematic tilts; EWT:** tokenism and concentration (D8 M081; S2; D13c R3).
- **Justifying the barbell by her career:** banned ("floor-first leap" is not her trait; career risk does not transfer, Weber et al. via D6; brief section 16). The rival's reasons are case lines only.

2. Numbers behind the scoreboard (all MODEL unless labelled; E4 script)

2.1 Same model, every design (E4 [2])

- Inputs: JPM 2026 LTCMA AC World 7.00% compound / 8.28% arithmetic (VRF); growth-money bonds 5.0% (ASM, market-consistent, as E1/D6 case B); rival bottom locked at today's 5-year par 4.98% (VRF curve; ASM that it holds in January 2028, the D3 barbell convention); E1 bottom at today's 2-year par 4.81% (ASM, as E1); no fees; i.i.d. lognormal returns (ASM).
- The grid (rival capped, $s = 1/2$ unless stated): $a = 0.50$ total \$164k/\$212k/\$297k, bottom \$101k; $a = 0.60$ total \$171k/\$210k/\$278k, bottom \$121k; $a = 2/3$ \$176k/\$208k/\$265k, bottom \$134k; default ($a \approx 0.744$) \$182k/\$207k/\$250k, bottom \$150k; $a = 0.80$ \$186k/\$206k/\$240k, bottom \$161k. Every row's total is at or above E1's p5 except $a = 0.50$, and the medians sit within \$4k of E1's \$210k.
- **s changes only the split between gift and kept money:** default rule at $s = 3/4$ gives gift \$172k/\$186k/\$207k and kept \$8k/\$19k/\$50k (E1-sized gift at the median, a thinner cushion in bad markets); at $s = 1$ the kept money is zero whenever stocks fall, which would breach case test 5 (L145), so $s = 1$ is out.
- **Share of E1's paths whose 2031 bottom is below the rival's:** 22% against the default (\$150k); 5% against $a = 2/3$ (\$134k); 40% against $a = 0.8$ (\$161k) (E4 [2]).
- **Share of the median 2033 gift already owned when she speaks in 2031:** rival default 86% (E4 [7]); E1 89% (E3 [3]). Both announce mostly bought money; the rival's bought part was fixed three years earlier.

2.2 Stress (E4 [3])

- **Bad 2028-2030** (worst 10% of three-year stock returns, before she speaks): E1 bottom median \$134k (p5 \$119k), gift \$151k, total \$168k. Rival default: bottom \$150k (p5 \$150k), gift \$165k, total \$186k.
- **Bad 2031-2032** (worst 10% of two-year returns, after she has spoken): E1 gift median \$185k (from \$188k), total \$203k; rival default gift \$169k (from \$174k), total \$189k. Both stay at or above the announced bottom in 100% of these paths (bought). The rival's stock fund carries more of the damage in dollars of total money (-\$18k vs -\$7k at the median), which is its main weakness (section 7, X4).

2.3 History, every 6-year window 1928-2025 placed in 2027-2032 (E4 [5]; Damodaran S&P 500 total return, VP dataset

page via D3 snapshot; bond proxy 50% bills + 50% 10-year, ASM as D3)

- **Raw history (93 windows):** total worst / p10 / median: E1 \$123k (1929 start) / \$180k / \$221k; rival default \$171k / \$190k / \$223k; rival a = 2/3 \$162k / \$187k / \$230k. Bottom worst: E1 \$95k; rival \$150k.
- **Rescaled to JPM's median returns:** E1 \$120k / \$174k / \$214k; rival default \$169k / \$185k / \$214k; rival a = 2/3 \$159k / \$180k / \$218k.
- **Named windows (rescaled):** 1929-34 E1 total \$120k (bottom \$93k) vs rival \$170k (bottom \$150k); 2000-05 E1 \$174k (bottom \$139k) vs rival \$186k; 2007-12 E1 \$179k (bottom \$142k) vs rival \$188k; 1973-78 E1 \$210k (bottom \$170k) vs rival \$194k (E1 wins when the crash is early and the recovery lands before 2031).
- **Top reached (the rival's only probability):** the S&P 500's two-year total return was zero or positive in 84% of 97 overlapping periods 1928-2025, 86% of non-overlapping ones, 89% since 1950 (E4 [5], VP data, derived); the JPM lognormal model gives 73% for world stocks (MODEL). History is U.S. only; VT is global (limit, labelled).

2.4 Sensitivities (E4 [4], [7])

- January-2028 five-year rate 4.00% / 4.98% / 5.23% (the 5.23% is the forward implied today, not lockable, AX2 C1): stock fund \$35k / \$40k / \$42k; total median \$199k / \$207k / \$209k; the minimum stays \$150k in all three.
- Deposit \$75k (beyond-case stress, labelled as the team's own, R-AN35): minimum \$75k by rule; total \$94k/\$109k/\$135k (E1 P4 reports \$73k/\$105k/\$149k for the same stress).
- Rates fall before January 2027: -50bp and -100bp leave the minimum at \$150k (the top-up comes out of the stock fund, which falls to about \$31k and \$15k); -150bp leaves about \$147k and no stock fund (E4 [7], VRF top-up table from D1/D3, ASM parallel shifts).

3. The rival's WInS book and Trading Notes plan (TN; tier 1)

3.1 Which book: the plan on 2 January 2028, scaled to \$300,000 (the rival's version of option (ii))

- **Why this date (INT):** it is the first day all three jobs exist (operations, the building's minimum, the stock fund). The rival's central idea is a split of jobs, not a ban on stocks before a date, so a day-one stock purchase in WInS does not contradict it (the fix E2 F4 asks E1 to make).
- **The consistency sentence the IPS must carry (elements; D10 1.5):** the date WInS shows (after both deposits); scaled to \$300,000; stand-in funds, not the ladder itself; in January 2027 almost all of her real \$300,000 buys the payments.

3.2 Orders (illustration at the 2026-09-25 closes; recompute every share count from the last close before

entering orders, T2 finding 1; E4 [7])

#	Holding	Weight	~Shares at 9/25	Job (Guide role word)	Primary / alternates	2025-26 list (history only, VRF)
1	IEF (7-10y Treasuries)	23.5%	~783	operations: future funding	IEF; VGIT then SPTI (ticket v1 s12)	IEF yes
2	TLH (10-20y Treasuries)	42.5%	~1,365	operations: future funding	TLH; IEF + TLT at duration weights; SPTL under a cap (ticket v1 s3)	TLH no, TLT yes
3	The building's minimum: a Jul-Dec 2032 U.S. Treasury (best: 4.125% note of 15-Nov-2032, CUSIP 91282CFV8) or IBTM (iShares iBonds Dec 2032 Term Treasury ETF)	24.3% (24.0% under a 25% cap)	note: about \$75k face; IBTM ~3,336	building minimum: future funding (and liquidity on the gift date)	1) the 2032 note (one \$10 trade); 2) IBTM (\$25); 3) if neither is listed, an undated stand-in of similar duration (IEI, 4.20y) labelled "moves like, not dated"	IBTM no; IEI no
4	VT (world stocks)	8.7%	~163	stock fund: growth	VT; VTI + VXUS (62/38); ITOT + IXUS	VT, VTI, VXUS yes

#	Holding	Weight	~Shares at 9/25	Job (Guide role word)	Primary / alternates	2025-26 list (history only, VRF)
	Cash float	about 1% (\$2,990 after 4 x \$25)		liquidity		

- Issuer facts (VP): IEF 6.86y, TLH 11.58y effective duration, both 0.15% (9/25, ticket v1); VT 0.06%, 10,088 stocks, 37.7% non-U.S. (8/31; ticket v1); **IBTM NAV \$21.83 (9/25), expense 0.07%, 30-day SEC yield 4.79% (9/24)** (re-read on <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf> on 2026-09-28 with the fetch helper); IBTM effective duration 5.05y, 15 notes maturing Apr-Sep 2032, net assets about \$555m, about 216k shares a day (9/24-9/25, VP via D1/S1). The page also says the funds "will terminate on or about October or December 15 of the year in each Fund's name" and "do not seek to return any predetermined amount" (both found verbatim, 2026-09-28). The 2032 note's model price (95.27 clean, about 1.5 accrued per 100 on 9/25) is D1's (ASM); WInS shows its own price.
- **Compliance:** every order is an ETF, a U.S. Treasury or cash (permitted, R-W67-R-W69, VP); no leverage, inverse, crypto-linked or thematic fund; each order is far below twice daily volume on published volumes (D10/AX2 C5; IBTM 3,336 shares vs about 216k a day); four or five trades; no same-day round trips (User Guide "Day Trading: This is not permitted", VP). Under a single-security cap: IBTM held at the cap minus one point; TLH split per the ticket's cap table. A separate bond limit, if shown, governs the 2032 note (read both Session Rules lines, ticket gate (c)).
- **What the book says at a glance (INT):** three jobs, three instruments. The building's minimum is dated (it ends in late 2032); the operations stand-ins (IEF/TLH) never mature but move like the dated payments (a listed 2032-2041 Treasury rung would add a second dated holding, ticket v1 s7); the stock fund is the smallest holding and the only one whose fall no promise's value mirrors.

3.3 The three notes (elements only; the team writes every word; ≤300 characters until the box is checked at zero

risk, ticket gate (c))

Note	C1 role + the team's own label	C2 Laura's dated need (with the scaling marker)	C3 one checkable fact (dated, source in words)	C4 the rule (or move to the reflection)	Never say
A. Building's minimum (the 2032 note or IBTM)	future funding (liquidity on the gift date)	the minimum of her 2033 building gift, sized to her 2028 deposit, "after her 2028 deposit"	the note's maturity (15 Nov 2032, 47 days before 1 Jan 2033) or IBTM's holdings (Apr-Sep 2032 notes) and end date (about Dec 15 2032) (iShares, dated)	bought the day its money arrives; never sold because rates moved	"repays \$X" for IBTM (AX1b item 6); "guaranteed", "risk-free", "locked 5.2%"; any range or 2031 top
B. Stock fund (VT)	growth	the building's upside and her flexibility, from the money left after the minimum, "after her 2028 deposit"	its share of the book (about 9%: all the stock the plan holds); or VT's stock count / top-ten share (Vanguard, dated)	held to 2033, never rebalanced or sold for market reasons; a fall shrinks the upside, never a promise	a return forecast; "halves AI exposure" (AX2 C5); "Laura likes risk"; the statistics degree
C. Operations (TLH, or IEF)	future funding	the ten \$50,000 payments 2033-2042, "after both deposits"	the security's own effective duration (TLH 11.58y or IEF 6.86y, iShares, dated); the mix of the two (about 9.9y at 23.5/42.5) is what moves about 10% per 1-point rate change, as the payments' value does (9.90y, VRF; the team's own calculation)	never sold after a rate rise; only re-mixed to keep about 10 years	match, matched, mature in her years, stable, reduce volatility, operating reserve

- **Reflections (≤100 words each; TN guide L46-52, VRF):** the three official answers, plus exactly one extra job each (E2 F9): note C carries the pre-registered **tested** window (fill date to the Oct 20 close: rate move in bp, hedge % change, payments' % change, labelled as the team's own calculation; ticket v1 s6); note A carries the **scaling** and the forced-vs-chosen caution line; note B carries the **refined**

story that is true for this team (the first plan, growth-first at about 75% equity, was tested and dropped: 3.2% miss, 40.8% if the 2028 deposit is missing; F-401, F-404, VRF; E2 F8). Never invent a refinement (R-W28).

- **Order of entry:** operations first, then the minimum, then VT, in one session (the time stamps show promise-first, D7). If a Jul-Dec 2032 note is listed, price it in the drop-down first (ticket v1 s7).
- **Distinctiveness check (INT):** note A differs from Wharton's own example (an intermediate Treasury ETF "to reduce portfolio volatility", TN guide L36-39, VRF) by a date, a person and a job; neither E1 note under (ii) can say what note A says, because E1's bottom is not bought until 2031 and its VGSH note is banned from naming it (ticket v1 s5).

3.4 Gate decisions the rival would take before the first order (ticket v1 gate (b) already lists the first two)

- [] Facility bottom bought in **January 2028** (not 2031): yes.
- [] Growth money's bond part: **dated to December 2032** (the 2032 note or IBTM), not VGSH.
- [] The minimum rule (= the 2028 deposit's amount) and s = one half, decided and logged with first names, date and reason, because note A records the rule permanently (TN guide L30-31: notes must be consistent with the later IPS).
- [] Bonds drop-down screenshot: every U.S. Treasury maturing Jul-Dec 2032 (and 2033-2041 for an optional rung).
- [] If neither a 2032 Treasury nor IBTM is listed: the undated stand-in is weak; the rival would then accept E1's (ii) book with the same notes' logic, or ask Wharton (Contact Us) before trading.

4. The rival's IPS (IPS; tier 2): elements, budget and rules to fix before Nov 6

4.1 Pitch content (≤50 words; zero numbers; no Laura quote, no pull quote, no puns; the team words it)

1. What the plan secures for the residency, and when: its ten operating years, then the building's minimum, each bought the day its money arrives.
 2. Whose money does what: the first deposit for operations; her own 2028 earnings for the building's minimum.
 3. What the rest is for: only money nobody relies on is in world stocks, for a larger gift and her flexibility, so in 2031 she tells partners only what she already owns (the fourth idea only if words allow; E2 F3 caps the pitch at three).
- Must not contain: a date-certain claim about the ladder; "fits inside \$300,000"; "guaranteed", "risk-free", "100%", "safe"; jargon (barbell, LDI, duration, STRIPS); identity imagery (D13c section 5).

4.2 The ideas the IPS carries (about 10; E2 F1 target about 12), with a rough word budget (INT estimates)

#	Idea (content)	Words
1	Purpose and central idea: each promise bought on arrival; stocks only where nobody relies on the money	30-40
2	2027 rule with its one rates-fall branch (longest-first, completed from the next money, one "because")	30-40
3	2028 rule: complete the ladder, buy the minimum equal to her deposit, one stock fund held to 2033	30-40
4	2031 rule: nothing traded; the bottom is owned; the top is today's value of half the stock fund; confidence both ways, history named in words	35-45
5	2033 rule: reserve = the ladder, held to maturity; gift up to the announced top; the rest hers, in short Treasuries	30-40
6	Certainty definition with the two gaps (price in January 2027; US\$ in Taiwan)	40-55
7	Risk by goal, as a maximum-loss statement; forced vs chosen caution	30-40
8	Diversification by purpose and date; liquidity by maturity dates	20-30
9	Governance: what may change a rule; no rebalancing between parts; who decides	20-30
10	The WInS date sentence (after both deposits, scaled, stand-ins)	20-30

#	Idea (content)	Words
	Total before connecting words	about 285-390

- Leaves room for connecting prose inside the 470-490-word page-fit target (D9; AX2 D10 C3). The exported PDF must end on page 3 (IPS guide L82-83, L103-109, VRF).
- Trade-offs to name inside the "because" clauses (Guide p.5 "reasoning, assumptions, and tradeoffs", VRF):** less stock in 2028-30 for a bottom that does not depend on those years; a top that cannot be exceeded for a top that needs no forecast; the chance of cheaper bonds after 2028 for never paying more because rates fell.

4.3 Rules the rival fixes before Nov 6 (so the Final Report applies, never redesigns; IPS guide L73-75, VRF)

- The four dated rules and the governance line (section 1.2), with: the minimum = the 2028 deposit's dollar amount (or what is left after completing the ladder, if less); s = one half; the cap at the announced top; the stock fund is one broad world fund, never rebalanced or sold before 2033; the kept money moves to short Treasuries on 1 Jan 2033.
- The certainty definition, the currency rules and the building-cost assumption (section 1.3).
- The confidence method for the top: history named in words (U.S. stocks, two-year periods since 1928) and the model named in the Final Report; never "100% within".
- No fee in the IPS; if the team keeps a cost clause, costs are paid from the stock fund, never from the ladder or the minimum (E3 finding 4; D8 M026).

5. Where the E1 plan loses to the rival (precise; each with evidence and the criterion it costs)

#	E1 weakness	Evidence	What the rival does instead	Criterion (SMAApply L49-57, VRF)
L1	The co-sponsor bottom is not known until 2031 and moves with 2028-30 markets. E1 bottom p5/p50/p95 \$134k/\$167k/\$211k; after a bad 2028-30, median \$134k (p5 \$119k); worst historical window \$93-95k (1929 start); 2000-05 \$139k	E4 [2], [3], [5] (MODEL; VP history); D3 M059 history table (before-lock crashes cut the facility money 19-48%)	A bottom equal to her 2028 deposit, owned from January 2028; worst historical window \$150k	Investment Strategy ("disciplined planning across Laura's changing time horizons"); Creativity & Presentation (co-sponsor credibility)
L2	The top is a model property. E1's top uses a percentile q of a model that "has too few very bad years" (AX1 C12); an 80-90% top lands within the range 62-97% of the time across models and history (D3 v2 (f)); E1 must also still choose q (E1 C5)	D3 v2 (f), AX1 C12, E2 R7	The top is a market value she can read on the day; its only probability comes from history (84% since 1928) and can be checked by a statistics graduate or a co-sponsor's accountant	Portfolio Analysis ("reasonable assumptions"); Client Knowledge (a statistics-trained client, R-AN32)
L3	E1's reason for not buying the bottom in 2028 has been withdrawn. E1 L97-98 cites "\$101k vs ~\$165k", which compares a 50% lock with an 80% lock; at the same 80% share the bottoms are \$161k (2028) vs \$165k (2031), and the barbell beats the plan at every percentile at equal stock exposure	AX1 C6 (VRF audit); E2 E-1, R2; ticket v1 gate (b) now lists "January 2028 or January 2031?" as open	At equal stock exposure (a = 2/3, 59 vs 58 percent-years) the rival's total is p5 +\$8k, median -\$2k, p95 -\$1k in the model, and better in every historical summary (worst \$159k vs \$120k; p10 \$180k vs \$174k; median \$218k vs \$214k, rescaled)	Investment Strategy; Portfolio Analysis
L4	Money with a known use-date sits in rolling short Treasuries (VGSH)	E2 R1; AX1 C18 (locking the bond part to 2033 is worth about \$4-9k at the median); T2 finding 5	The bottom is dated to December 2032, the month before the gift	Portfolio Analysis ("understanding and effective use of investment concepts")
L5	The central idea is a prohibition that E1's own WinS book breaks on day one (a time-worded ban next to a day-one VT buy)	E2 C1 and F4; AY2 D7 C3; E3 finding 5	A split-of-jobs idea that is true in the 2027 plan, the 2028 plan and the WinS book	Investment Strategy ("consistency with the team's IPS"); Creativity & Presentation

#	E1 weakness	Evidence	What the rival does instead	Criterion (SMAApply L49-57, VRF)
L6	E1's third note is the weakest in the set (VGSH: "the ordinary effect of not holding stocks", AY1 C10) and may not name the promise it prepares for (ticket v1 s5 bans "the 2031 floor", "already owned")	E2 section 1, 3.1; ticket v1 s5	Note A names a dated holding for the building's minimum, sized to her 2028 earnings; it is honest because that promise is bought in 2028 under the rule	Portfolio Analysis ("why each security belongs"); Client Knowledge
L7	Density. About 55 distinct ideas for the pitch plus IPS (one per ~9 words), the growth-mix debate (50 vs 60, band, reset), a, s, q, cap or no cap	E2 section 3.2	About 10 ideas; parameters: the minimum rule (set by a case number), one half, the cap	Creativity & Presentation ("a compelling, well-organized narrative"); last year's lesson (v4 Part 2)
L8	Nothing co-sponsors can count on can be named in the IPS without a projection; E1 P17 correctly bans dollar ranges	E1 P7, P17; IPS guide L68-69	The rule's output is a number the case already gives (her 2028 deposit), so the IPS can name what the building will at least receive as a rule, conditional on the deposit	Client Knowledge ("recommendations that can earn her confidence")
L9	The \$150k is doubted in three places (P2 gap (i), P3, P4 branches)	E2 C6, E-3	One clause: the minimum is sized to whatever arrives; a smaller or later deposit gives a smaller or later minimum; the payments come first	Client Knowledge (R-AN35 "will")
L10	E1's risk statement for the growth money is a model threshold with a weak yardstick ("about 1 in 20 below the money put in")	E2 R3, F6; AX1 C4 (median vs mean)	A maximum-loss statement that needs no model: at most the stock fund, about 9% of her money in 2028	Portfolio Analysis; Client Knowledge
L11	E1 cuts stocks to about 3.5% of her money in 2031 and holds almost none after she has spoken	E1 P5 [3]; D13c N1	The rival's stock fund stays invested to 2033 (about 9% rising); "pursuing growth" continues to the gift date (partial win only: see X2)	Client Knowledge ("thoughtful risks")

6. What E1 would need to win back (precise changes, in deadline order)

Tier 1, before the VGSH order (WInS-now and TN):

- Take the open gate decision explicitly** (ticket v1 gate (b); AX1 C6; T2 finding 5): "facility floor bought in January 2028 or January 2031?". If 2028: E1 becomes the rival's structure and should adopt section 3.2 row 3 and note A. If 2031: E1 must record why (the two surviving reasons, AX1 C6: more money growing in 2028-30 and less stock at risk after Laura has spoken) and accept L1 in writing.
- Replace VGSH with a dated December-2032 instrument** even if the bottom stays in 2031 (AX1 C18; E2 R1): the 2032 note or IBTM holds the growth money's bond half to a known date; the note then carries a checkable date instead of "the ordinary effect of not holding stocks".
- Reword the central idea as a split of jobs** before any note is saved (E2 F4), so no permanent note contradicts the IPS.

Tier 2, IPS by about Oct 26 (first draft):

- Make the top checkable without a model**, or at least name the history check next to q (D3 v2 (f): a p90 top lands within in 81% of raw history), and drop "promised" for the stretch (E3 finding 1).
- Give the bottom a case-anchored meaning** the IPS can name (for example, relating the bought share to her 2028 deposit), so co-sponsors hear a rule with a known output, not "a share of whatever the sleeve is worth in 2031".
- Cut to about a dozen ideas** (E2 F1): if the bottom moves to 2028, the growth-mix rule, its band and the 50/60 debate disappear (the stock fund is simply "all of what is left").
- Say E1's real advantages out loud** (section 7): a higher median bottom and gift, more stock in 2028-30, and an uncapped upside that visibly answers "favorable outcomes" (L118). If the team keeps E1, these are its answer to the rival, and they belong in the rules' "because" clauses.

7. Where the rival loses to E1 (honest), and what each side should steal

#	Rival weakness	Evidence	Size	What would fix it
X1	Lower median bottom and gift. E1's 2031 bottom exceeds \$150k in 78% of model paths (median \$167k); median gift \$188k vs \$174k	E4 [2]	-\$17k bottom, -\$14k gift at the median	Raise s to 3/4 (gift median \$186k; kept p5 falls to \$8k) or lock more (a = 0.8: bottom \$161k, less stock)
X2	Less stock over 2028-32 (45 vs 58 percent-years; 8.7% vs 17% of her money in 2028); lower p95 (-\$16k total, -\$50k gift)	E4 [2]	A real "pursuing growth" and "thoughtful risks" cost (L70-73; D13c N1)	The a = 2/3 alternate equalises stock exposure (59 percent-years) at the price of a \$134k bottom
X3	The cap: strong markets add to her flexibility, not to the gift; co-sponsors never see more than the top. D5 argued uncapped is the more honest answer to "how favorable ... outcomes could affect the amount" (L118)	D5 finding 5; AX1 C3	Excess above the top goes to Laura (her decision, AX1 C15)	Keep the cap but let Laura decide in 2033 to give more (a choice, not a promise)
X4	The stock fund is 100% stocks after she has spoken. A bad 2031-32 cuts the rival's total by \$18k at the median vs \$7k for E1; the gift stays within the range in both	E4 [3]; D3 M238	The bottom is untouched; the gift slides toward it	Accept as the price of a bottom fixed in 2028, or add a 2031 partial lock (a second rule; rejected on simplicity)
X5	Kept money sits in stocks until 2033 (D8 M081 says flexibility belongs with safety)	D8 M081	The kept money is still at least E1's at every percentile (\$16k/\$32k/\$66k vs \$16k/\$21k/\$29k)	It moves to short Treasuries on 1 Jan 2033 (both plans)
X6	WInS availability: no individual 2032 Treasury may be listed; IBTM is thin (about \$555m), not on the 2025-26 list, and "does not seek to return any predetermined amount"	D1; AX1b; iShares (VP)	Note A would lose its date if only an undated stand-in is available	Check the Bonds drop-down and search IBTM before the gate; otherwise ask Wharton
X7	The January-2028 five-year rate is unknown. At 4.0% the stock fund is about \$35k, not \$40k	E4 [4] (ASM)	Smaller upside; the minimum is unchanged	None needed; the rule adapts
X8	Novelty and explanation risk. A barbell can read as a "structured product"; the rival must never use the word or justify it by her career	D13c section 3.4; brief section 16	Word cost, not money	Describe it by its jobs and dates only

What the rival would steal from E1: P2 (certainty with two named gaps), P3/P4 (the 2027 rule, longest-first, the joint tail named once), P6 (risk by goal), P8 (US\$ binding; no NT\$ hedge), P10 (governance wording that cannot read as permission to revise), P15 (decision and AI logs), P16 (page-fit test and paraphrase test). E1 is right on all of these, and the rival adopts them unchanged. **What E1 should steal from the rival even if it keeps the 2031 bottom:** the dated December-2032 bond part (L4), the split-of-jobs central idea (L5), the maximum-loss framing of stock risk (L10), and the four-date IPS spine (E2 F2).

8. Open conflicts for the team (both sides; the team decides)

#	Conflict	For the rival	For E1	E4 view (INT)
C1	When is the facility bottom bought?	2028: fixed three years before she speaks; better bad case and history tails (L1, L3)	2031: more stock in 2028-30, higher median bottom and gift (X1, X2)	A genuine choice of when to take risk; the rival's version is easier to say and to check. Decide before the VGSH order
C2	Size of the bottom	Her 2028 deposit (a case number, no free parameter)	80% of a sleeve value (a share, output unknown until 2031)	The deposit rule is more tailored; a = 2/3 or 0.8 are the numeric alternates
C3	Top: capped market value or uncapped model percentile	No forecast; one historical probability	Good markets visibly raise the gift; D5's two-sided statement	Either is honest if the capped version reports only P(top reached)
C4	Stock exposure 2028-32	45 percent-years (default) or 59 (a = 2/3)	58	Pick the default for credibility, the alternate for growth; say which in one "because"
C5	WInS third note	Building minimum (dated, Laura-specific)	VGSH (risk management of growth money)	The rival's note is stronger if a dated instrument is listed

9. Later (after Nov 9), one line each (Final Report; not worked here)

- The co-sponsor range in dollars at median, weak and strong 2031 states, and the "share already owned" figure.
- The history table (named windows) next to the model's p5, for both designs.
- The fundraising-excerpt order (operations bought, minimum owned since 2028, the top and its historical odds, what is not promised, US\$ binding) as the team's own checklist.
- NT\$ illustration of the minimum at a dated rate, with the 2031-view exchange-rate band (D4).
- The rejected-alternatives table (growth-first, E1's 2031 lock if not chosen, uncapped percentile top, TIPS minimum).

Sources (accessed 2026-09-28 unless noted)

Official, in the repo (VRF): `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (L43-45, L61-74, L83-120, L126-149, L161-162); `2026_WGY_Investment_Policy-FINAL.txt` (L20-26, L45-52, L68-75, L82-83, L101-123); `2026_WGY_Trading_Notes_Analysis-FINAL.txt` (L11-15, L25-31, L36-53); `2026_WGY_Investment_Competition_Guide.txt` (L84-90, L154-159, L183-184); `SMAppl_Deliverables_Page_2026-09-27.md` (L49-57).

Market data (VRF): `competition/official_market_data/daily-treasury-rates_2026-09.csv` (row 09/25/2026: 2y 4.81, 5y 4.98); `JPM_LTCMA_2026_US_matrix_USD.pdf` p.2 (AC World 7.00/8.28/16.78).

Primary web (VP):

- iShares IBTM page, <https://www.ishares.com/us/products/328944/ishares-ibonds-dec-2032-term-treasury-etf> (read by E4 on 2026-09-28 with `research/insight_v1/scripts/fetch_text.py`; NAV, expense ratio, SEC yield and both quoted sentences found verbatim).
- Damodaran, Historical Returns on Stocks, Bonds and Bills, https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html (snapshot `research/insight_v1/scripts/data/D3/damodaran_histretSP_1928_2025.csv`, VP per D3 and AX1).
- SMAppl Trading Details and FAQ, <https://wghsinvcomp.smappl.us/res/p/trading/>, [/res/p/faqs/](https://wghsinvcomp.smappl.us/res/p/faqs/) (via `case_register` R-W56-R-W92); 2026-27 WInS User Guide (via A4, D1, AY1); Wharton Rules & Roles and AI policy (via Phase A); CFA Institute *Elements of an IPS* (2010) and *Investment Risk Profiling* (2020) (via D6, D8, D10). Not re-opened by E4.
- Issuer facts for IEF, TLH, TLT, VT, VTI, VXUS, IEI, VGIT, SPTI, SPTL as carried by `wins_now/securities_and_allocation_v1.md` and `wins_now/S1_treasury_sleeve.md` (VP there; not re-opened by E4).

Research files (lower authority; audit corrections applied): `research/insight_v1/phase_E/E1_change_proposals.md`, `E2_judge.md`, `E3_laura.md` (summary); `phase_D/ D1-D12, D3_model_v2_results, D3_quant` (AX1), `D13a, D13c, and audit_rates_quant.md, audit_rates_D1.md, audit_markets_rules.md, audit_client_cosponsors.md, audit_judges_practice_comms.md`; `phase_D/trading_now_brief.md`; `wins_now/securities_and_allocation_v1.md`, `T2_red_team.md`, `S1_treasury_sleeve.md`, `S2_growth_sleeve.md`, `S3_final_report_ladder_and_reserve.md`, `S4_red_team.md`; `phase_A/case_register.md` (R-AN4, R-AN6, R-AN9, R-AN14, R-AN15, R-AN20, R-AN32, R-AN35, R-AN45, R-AN52, R-W67-R-W92), `fact_register.md` (F-101-F-117, F-401-F-409), `stakeholder_map.md` (SH-04, SH-05, BS-03, BS-13, BS-17); `phase_C/parked.json` (M008, M015, M033, M035, M040, M149); `research/council_2026-09-27/01_chair_memo.md`, `round2_referee_ruling.md` (history).

Scripts: `research/insight_v1/scripts/E4_rival_numbers.py` (new); re-used engine `D6_behavioural_numbers.py`; reproduces `E1_split_and_slices.py` [4] row 1; top-up costs from `D1_purchase_rule.py` / `D3_joint_tail.py` tables.

What this teaches

1. **The same money can be promised at different times.** E1 and the rival buy the same payments (and, at the rival's $a = 2/3$ alternate, hold the same stock exposure); the difference is *when* the facility bottom is bought. Buying it in 2028 fixes it before three years of markets; buying it in 2031 lets those years raise or lower it. That timing is a strategy choice, not a detail.
2. **Compare designs at equal risk before calling one better.** At the same stock exposure, the dated barbell has a better bad case and better historical tails; at lower exposure it trades growth for certainty.

Line options up on the same risk first (AX1's lesson, applied).

3. **A number with no forecast in it is the most credible number you can say in public.** A bought minimum and a market value on the day need no model; only the chance of reaching the top does, and history can check it.
4. **Let the client's own facts set your parameters.** The case already gives a number that means something to Laura and to her partners (what her 2028 work earns); using it removes a free parameter and makes the plan visibly hers.
5. **A good rival shows you your own weak joints.** Most of what the rival wins comes from choices E1 left open (floor timing, a dated bond, a model-based top, a prohibition as the central idea). Closing those joints, whichever way the team decides, is what makes the plan hard to beat.

E5 Pre-mortem: "It is January 2027 and Team Caplet was not selected for the Top 50. Why?"

Agent E5 (pre-mortem analyst), insight_v1 run, Phase E. Written 2026-09-28 (the first official WInS trading day). AI-generated research for Team Caplet: causes, rough ratings, preventions, owners and dates. **None of it is text to submit.** It drafts no note, reflection, pitch or IPS sentence; where a short phrase names an idea it is a content label, not wording to copy. The six students decide every change and write every word; AI use goes in the Final Report's Works Cited (Wharton AI policy, R-W46).

Serves: the Trading Notes Analysis (**TN**, due Oct 23) and the IPS (**IPS**, due Nov 6) only, with WInS trading in scope only as the source of the notes (brief section 17). Rules the Final Report will later apply are listed under "IPS: rules to fix before Nov 6" (section 4). Pure Final Report items are one line each in section 7.

Relayed messages: none arrived during this task.

Status labels (brief section 3): **VP** = VERIFIED-PRIMARY; **VRF** = VERIFIED-REPO-FILE; **SNIP** = SNIPPET-UNVERIFIED; **ASM** = ASSUMPTION; **MODEL** = an ASSUMPTION-based model output, not a forecast; **DER** = derived by a script from labelled inputs; **INT** = E5's judgement. **Every likelihood and impact rating in this file is INT/ASM:** a rough ranking for deciding where to spend the team's hours, not a measured probability. Ids: R- case register, F- fact register, SH-/BS- stakeholder map, P- E1 proposals, F1-F10 E2 fixes, E3-n E3 changes (all under `research/insight_v1/`).

New script: `.venv/bin/python research/insight_v1/scripts/E5_premortem_checks.py` (under a second, no network; docstring lists inputs and labels). It prints [1] every deadline in Australian time, [2] U.S. trading hours in Australian time on the dates that matter, [3] the ranked risk register, [4] a robustness check of the ranking under two other impact weightings.

Evidence read (audit corrections override the text above them): the brief; CLAUDE.md; `phase_E/E1_change_proposals.md` (the plan being attacked), `E2_judge.md`, `E3_laura.md`; `wins_now/securities_and_allocation_v1.md` (the ticket), `phase_D/trading_now_brief.md`, `wins_now/T2_red_team.md`; `phase_D/D10_compliance.md` (+AX2), D7 (+AY2), D9 (+AY2), D12 (+AY1), the summaries and audit sections of D1-D6, D8, D3_model_v2_results, D13c sections 3-5, all five audit files; `phase_A/wins_week1_guardrails.md` (the G list), `case_register.md` (R-W, R-AN), `stakeholder_map.md` (SH-16 to SH-26, BS-01 to BS-20); the official case, IPS guide, TN guide, Competition Guide, Infographic and SMaply page; the council chair memo (history only).

Terms (defined once):

- **Pre-mortem:** imagine the project has already failed, then list the reasons. It finds risks that optimistic planning hides.
- **Likelihood (L):** rough chance the failure happens by its deadline if the team does nothing beyond what exists on 2026-09-28. Bands: VL ≈2%, L ≈6%, LM ≈11%, M ≈20%, MH ≈30%, H ≈40% (ASM).
- **Impact (I):** what the failure does to the chance of a Top-50 place. **Fatal** = excluded or disqualified; **Severe** = a whole deliverable or two criteria badly hurt, or an integrity question; **Major** = about 1-2 points on a criterion; **Moderate** = under a point; **Minor** = cosmetic (ASM).
- **Expected harm:** L × impact weight (Fatal 1.0, Severe 0.6, Major 0.3, Moderate 0.15), used only to rank.
- **Owner-type:** a role, not a named student: team leader, trader (the one student who places orders, plus a backup), note writer, second reader, log keeper (decision log and AI-use log), IPS lead writer, format checker, rates analyst, growth/risk analyst, client lead, admin liaison (roster and school paperwork, with the advisor for admin only). Roles follow CLAUDE.md's suggested list; the team assigns people.
- **Book (ii) / (iii):** the two candidate WInS books in the ticket: (ii) the plan's mix after both deposits, scaled to \$300,000; (iii) Laura's literal January 2027 book (about 98% Treasury funds).
- **Swap test:** put another client's name into a sentence; if it still works, the sentence is generic (D13c, INT).
- **AEDT / AEST:** Sydney summer time (UTC+11, from Sun 4 Oct 2026) / standard time (UTC+10). ET = U.S. Eastern.

0. Summary

0.1 The most likely failure story (INT, written from January 2027)

We never learned why: "Teams will not receive individual feedback" (R-W40, VP), the same silence as last season (CLAUDE.md, VRF). Looking back, the likeliest story is not a bad strategy. The substance was of Top-50 quality (E2 verdict, INT). The failure was in **ownership, packaging and process**:

1. The six students adopted a plan whose main calls they had deferred to an AI and never argued over (CLAUDE.md: "The team has not read the memos yet"; "the team deferred to the assistant's evidence-based calls", VRF). The notes, pitch and IPS carried AI-coined labels and a density no student would write, so the "authentic team voice" and "reflects meaningfully" parts of two criteria (SMApply criteria, VRF) scored low.
2. The IPS tried to carry about 55 ideas in about 480 words (E2 3.2, INT) and read as a rulebook, not a strategy a reader could restate.
3. Under book (ii) the day-one world-stock purchase appeared to contradict a pitch worded as a ban on stock risk (AY2 rank 1, VRF), and the scaling sentence that explained it was cut for space.
4. Nobody owned the PDF test, so the last draft's layout was never checked on Word's line height (D9/AX2, MODEL). This one is fatal, and cheap to prevent.
5. The work was squeezed into the last weekend. If any member sat NSW HSC exams (13 Oct to 5 Nov 2026, VP), that period covered both deadlines; the team's state and year levels are not recorded (UNVERIFIED whether it applies).

Against about 2,300 teams that finished last season (R-W99, VP), the Top 50 is roughly the top 2% (arithmetic; INT that the share is similar this year). At that margin, several "Major" losses together are enough to miss.

0.2 The top 10 causes (ranked by expected harm; E5 script [3]; ratings INT/ASM)

Rank	Id	Cause	Serves	L	Impact	The prevention in one line	Owner-type	By (Australian time unless ET)
1	PM-01	Strategy not owned by the students; voice reads AI-made	TN + IPS	H	Severe	Own-words gate before every vote; devil's-advocate pairs; blank-page drafting	team leader	before the gate vote (Wed Sep 30 AEST); IPS draft Mon Oct 26 AEDT
2	PM-02	IPS format breach (third page overflows; word cap by another counter; font; banned items)	IPS	M	Fatal	Fit on Letter at Word "Double"; ≤470 IPS words by two counters; PDF check by two students	format checker	every draft from Oct 26; final Wed Nov 4 AEDT
3	PM-03	Over-complex, jargon-dense pitch and IPS	IPS	H	Major	≤12 ideas; dated 2027-2028-2031-2033 spine; no codes; 6-reader paraphrase test	IPS lead writer	idea list Sat Oct 24; tests Oct 26-30 and Nov 2-3
4	PM-04	AI wording in permanent WInS notes or in reflections	TN	M	Severe	Notes drafted offline from elements only; text search of research/ for copied runs; no AI "polish"	note writer + second reader	before every order (first fills by Fri Oct 2 ET); reflections Oct 21-22 AEDT
5	PM-05	Team process breakdown (no roles, one login, holidays and exams, late drafting)	TN + IPS	H	Major	Roles and a shared exam/travel calendar this week; internal deadlines two days early	team leader	roles Tue Sep 29 AEST; calendar Wed Sep 30 AEST
6	PM-08	AI use not logged, or logged inaccurately	TN + IPS	M	Severe	Dated AI-use log from today, reconciled with the repo's commit history	log keeper	start Tue Sep 29 AEST; weekly
7	PM-06	Notes contradict the pitch/IPS (book date, scaling, split, labels)	TN + IPS	MH	Major	Central idea worded as a split of jobs; scaling marker inside (ii) notes; two-reader check; consistency table	log keeper + IPS lead writer	before the first order; checks Oct 22 and Nov 3 AEDT
8	PM-07	Plan reads timid or as analysing her; residency missing	TN + IPS	MH	Major	Purpose first in the pitch; forced vs chosen caution; stocks' purpose; one stress clause for the \$150k	client lead	pitch elements Sat Oct 24; draft check Fri Oct 30 AEDT

Rank	Id	Cause	Serves	L	Impact	The prevention in one line	Owner-type	By (Australian time unless ET)
9	PM-09	Generic thesis (fails the swap test; the hedge note is Wharton's example)	TN + IPS	MH	Major	Swap test on every note and IPS paragraph; lead with three Laura-specific decisions	client lead	each note; IPS Oct 26-Nov 3
10	PM-14	WinS rule breach (instrument, day trade, position cap, negative cash)	TN (WinS)	LM	Severe	One trader plus backup; A4 checklist read aloud; Session Rules screenshot; never a same-day sell	trader + second reader	Session Rules by Thu Oct 1 AEST; every order

- **Robustness (E5 script [4], DER on ASM weights):** PM-01 to PM-08 stay in the top 10 under all three weightings. Several ranks are ties at the same score (ranks 3-6, ranks 7-9), so read the order inside a tie as arbitrary. Rank 10 is a **tie** between PM-14 (rule breach), PM-15 (undefined activity minimum) and PM-16 (fewer than three executed trades). E5 puts PM-14 first because last year's team actually "traded assets outside the trading rules" (CLAUDE.md, VRF), a base-rate reason (INT). Under an exclusion-heavy weighting, PM-13 (missed deadline) and PM-18 (title-page mismatch) enter the top 10 in place of PM-09 and PM-14; under a quality-heavy weighting, PM-10 (overclaiming) enters in place of PM-14.
- **Cheap-and-fatal box (do these whatever the ranking; each costs under an hour, INT):** PM-02 PDF test, PM-13 deadline plan, PM-18 title page copied from the roster, PM-19 roster submitted early, PM-26 no contact with Laura, PM-27 no paid help, PM-28 membership kept at six.

0.3 What this file adds to E1-E3 (so the synthesis can merge it)

- It turns E2's scoring worries and E3's client worries into **owned, dated preventions**, and adds causes none of E1-E3 covers: ownership of the strategy by the students (PM-01), the accuracy of the AI-use record against a public repo (PM-08, PM-30), the Australian school calendar (PM-05), an undefined trading-activity rule (PM-15), unread SMAApply submission forms (PM-23), who can submit on SMAApply (PM-13), roster-to-title-page exactness (PM-18, PM-19), and shared-login discipline (PM-14).
- Its format stance is stricter than E1 P16: make the IPS fit under **Word's "Double" on U.S. Letter**, the strictest reading, so no interpretation of "double-spaced" or paper size is ever needed (INT). "Exactly 24 pt" (AY2, INT) stays a documented last resort, not a plan.

1. How the ratings were made (and their limits)

- **Likelihood** is E5's judgement of the chance, given the repo's state on 2026-09-28: no roles assigned, no trade placed, no vote taken, no IPS draft, the team "has not read the memos yet" (CLAUDE.md, VRF). It falls as preventions are done. It is not a model output (INT).
- **Impact** reads the official consequences literally where they exist: IPS format breaches mean "will not be considered for semifinal selection" (IPS guide L82-83, VRF); contact with the client, team size and paid help mean disqualification (R-W16, R-W17, R-W21, VP via A1/A4); missed deadlines lose eligibility ("To remain eligible for the Semifinals ... submit the competition deliverables by the published deadlines", Rules & Roles, VP via A4). Quality losses are scaled to E2's criterion scores (INT).
- **Which deliverables count for selection** is itself uncertain: the Rules page names "the strength of their Investment Policy Statement (IPS) and Final Reports" (R-W24, VP), while SMAApply says the TN, IPS and Final Report "are evaluated" (VRF). The safe reading is that all three are scored (R-AN41, INT). This file treats TN failures as real losses, and the IPS as the heavier deliverable.
- **Robustness:** the ranking was re-run with an exclusion-heavy and a quality-heavy set of impact weights (script [4]). The top 8 do not move out of the top 10 in any of the three runs. Treat ranks 9-18 as roughly equal.

2. The top 10 in detail

Each entry: what failed (January 2027 view); why it was plausible (evidence); ratings; **tripwires** (early warning signs to watch); **prevention** (specific steps); owner-type; dates. All dates are 2026.

PM-01. The strategy was not the students' own, and it showed (TN + IPS; L = H, Impact = Severe)

What failed. The key calls (lock early; book (ii) or (iii); the growth split; the 2031 slices; where kept money sits) were taken by accepting AI recommendations. The notes and IPS used AI-coined labels and research-file phrasing. Reflections described decisions the students could not explain in their own words. Readers scoring "an authentic team voice" (Creativity and Presentation) and "reflects meaningfully on the team's growth" (Articulation) marked the team down (SMAApply criteria, VRF). In the worst case a reader read it as AI-generated work "submitted as your own" (R-W26, VP).

Why it was plausible.

- The current state: "NOT YET DECIDED BY THE TEAM. The team has not read the memos yet"; "the team deferred to the assistant's evidence-based calls, provisionally" (CLAUDE.md, 2026-09-27, VRF).
- All team members are responsible for "Developing and implementing the team's investment strategy" (R-W15, VP via A1). Students "must use their voice and words" (R-W49, VP). Wharton's own self-check: "Did I come to my current conclusion before or after using generative AI tools?" (R-W51, VP).
- The research behind the plan is overwhelmingly AI-produced, with a risk of "artificial sophistication" (E2 2.4, INT; SH-16 "readers fear AI-written text", ASM).

Tripwires. A student cannot explain a rule without opening a file. A vote is logged with no student's own reason. Nobody argued the other side before a vote. Draft sentences use the research files' labels ("promise money", "three slices", "funded ratio").

Prevention.

1. **Own-words gate before every vote** (lock early, book, split; later the 2031 slices and kept money): each student writes 3-5 lines on the question with no AI file open, *then* reads the research, *then* votes. The log keeps both, dated. This is the order R-W51 asks about. Owner: team leader. By: the gate vote (target Wed Sep 30 AEST); IPS parameters by Mon Oct 26 AEDT.
2. **Devil's-advocate pairs:** for each of the five calls above, one student argues against it in the meeting; the log records why the team kept or changed it. Owner: team leader. By: the gate meeting; one IPS meeting about Oct 24-25.
3. **Explain-back check (internal learning, not finale preparation):** each student explains the central idea and the four dated decisions (2027, 2028, 2031, 2033) in two minutes without notes. Anyone who cannot, re-reads and retries. Owner: second reader. By: Sun Oct 25 AEDT, again Tue Nov 3 AEDT.
4. **Blank-page drafting:** the IPS and every reflection start from a blank page with research files closed; the research is used only afterwards to check facts and numbers. Owner: IPS lead writer. By: first IPS draft Mon Oct 26.
5. **The team names its own labels** for the parts of the portfolio in a meeting and logs that the proposals came from an AI source (T2 finding 8; E2 F10). Owner: note writer. By: before the first note.

PM-02. The IPS broke a format rule and was never read (IPS; L = M, Impact = Fatal)

What failed. The exported PDF spilled onto a fourth page, or the pitch counted 51 words in Wharton's counter, or a typed web address became a live link, or the file was set in a substitute font. "Submissions that do not meet the requirements below will not be considered for semifinal selection" (IPS guide L82-83, VRF).

Why it was plausible.

- Rules: title page 1 page; pages 2-3 "2-page maximum"; pitch "maximum 50 words"; IPS "maximum 500 words"; Times New Roman 12, double-spaced, 1-inch margins; PDF ≤5 MB; "Graphics, charts, images, attachments, external links, footnotes, and formal citations are not permitted" (IPS guide L84-123, VRF).
- At Word's "Double" line (27.6 pt for Times New Roman 12, ASM) on U.S. Letter, 50 + 500 words leave 0-2 spare lines; with 7 paragraphs and 0 pt paragraph spacing the text "does not reliably fit" (D9; AX2 on D10 C3; MODEL). Wharton's own sample is set at 24 pt (VRF, measured).
- E1's budget is 270-435 words before connecting words (P16, INT) and E2/E3 each add clauses (diversification, liquidity, forced vs chosen caution, cost principle). The pressure is upward.
- Word processors count "U.S.", "\$50,000", hyphens and slashes differently (D10 checklist, ASM).

Tripwires. Any draft above 470 IPS words. Any draft never exported to PDF. An addition after the last PDF check. Two students reporting different word counts.

Prevention (owner: format checker; a second student repeats every check independently):

1. Target **≤470 IPS words and ≤48 pitch words**, counted in Word **and** a second counter; if they disagree, use the higher count. By: every draft from Mon Oct 26 AEDT.
2. Lay out on **U.S. Letter at Word's "Double"**, paragraph spacing 0, no blank lines, at most 7 paragraphs, headings in Times New Roman 12. If it fits there it fits under any reading (INT). A4 is allowed by "Formatting choices not specified below are left to the team's discretion" (IPS guide L101, VRF; INT reading), but should not be relied on to make a long draft fit.
3. Export the PDF; open the exported file; confirm **exactly 3 pages**, the text ending on page 3 with at least 2 spare lines, 1-inch margins, file ≤5 MB, and the PDF's font list showing Times New Roman (a font substitution by Google Docs or LibreOffice is cheap to catch; whether a reader would treat it as a breach is ASM).
4. Search the PDF for "http", "www", a year inside brackets, and quotation marks: no links, no "(Source, year)" citations, no quotes of anyone's words; then look for tables, bullet lists and footnotes (D9 format checklist; D13 rules).
5. Freeze the words **Tue Nov 3 AEDT**; final PDF check **Wed Nov 4 AEDT**; nothing is added after it.

PM-03. The IPS was right but unreadable (IPS; L = H, Impact = Major)

What failed. Four rules, a governance clause, three 2031 slices with letters, bands, two named gaps, a joint tail, four bad-year pre-commitments and three trade-offs in about 480 words. The reader could restate the first rule but not the 2031 method, and no single sentence stuck. Last year's lesson, "strategy packaging was over-complex for the insight" (CLAUDE.md, VRF), repeated at the level of words.

Why it was plausible. E1 specifies about 55 distinct ideas for the pitch plus IPS, about one per 9 words (E2 3.2, INT). E2's two-minute test predicts the 2031 method is "likely missed" (INT). A past-season Wharton judging page: "Excessive investing jargon doesn't necessarily dazzle the judges" (VP, retired page, via D9/AY2). A 2025 semifinal judge praised "simple, elegant ideas" (VP via D7/AY2, past season).

Tripwires. More than about 12 distinct ideas in the draft. Any code or term from E2's jargon table (3.3): "80/10/10", "p90", "a/s/q", band numbers, "funded ratio", "duration", "STRIPS", "LDI", "DV01", "glide path". Headings of the form "Rule 1-4 + G" rather than dates. A non-finance reader cannot restate the 2031 method.

Prevention.

1. **Pick at most 12 ideas before drafting** (E2 F1's list is a starting menu); everything else goes to the team's "later" list. Owner: IPS lead writer. By: Sat Oct 24 AEDT (after the TN is in).
2. **Dated spine:** January 2027, January 2028, January 2031, January 2033, each with one action and one "because" tied to the case (E2 F2); branches become "if" clauses. Owner: IPS lead writer. By: first draft Mon Oct 26.
3. **One plain phrase per parameter**, used once; no letters or codes (E2 3.3; E3 4a). Owner: second reader. By: each draft.
4. **Paraphrase test** with 6 unpaid readers (3 without finance, 3 with), who restate and never supply words; change any sentence that 2 or more misread; target questions: the 2031 method, why longest-dated first, what could still stop a payment (D12 M234; E1 P16; digital.gov method VP via D12). Unpaid adult help is allowed; paid help is not (R-W21, VP). Owner: IPS lead writer. By: rounds Oct 26-30 and Nov 2-3.

PM-04. AI wording reached a permanent WInS note or a reflection (TN; L = M, Impact = Severe)

What failed. At a late-night gate session a note writer typed a phrase from the ticket, or lightly edited one, into the WInS note box. Notes cannot be edited and are quoted "exactly as it appears in WInS" in the TN, so the AI wording became part of a submitted deliverable. Reflections were then "tidied" in an AI chat.

Why it was plausible.

- "Include each Trading Note exactly as it appears in WInS" (TN guide L44, VRF); notes cannot be edited or deleted, only added to (Stock-Trak 2017 blog, VP vendor page via A4; not season-specific).
- The ticket and specialist files contained near paste-ready clauses (T2 finding 8; AY2 section 7, VRF); the ticket was since recast as elements, but several phrases remain short enough to copy (INT).

- Wharton's pre-baccalaureate AI guidance says "Don't use AI for personal reflection or opinion-based tasks" (R-W50, VP; written for another programme, a strong signal only, INT). Running the D9 draft checker counts as AI use (AY2 D9 C12, VRF).

Tripwires. A draft note with no paper or offline draft in the log. A draft note that shares any five-word run with a file in `research/`. A reflection drafted inside an AI chat.

Prevention.

1. Each note drafted by one student **offline, from the element list only** (role word, Laura's dated need, one dated checkable fact, the risk or rule; ticket section 5), with no research file open. Owner: note writer. By: before each order (first fills by Fri Oct 2 ET = 23:30 Fri Oct 2 AEST at the open).
2. **Copy check:** the second reader takes every run of five consecutive words in the draft and searches the `research/` folder for it (a plain text search, not an AI tool); any hit is rewritten by the note writer. Owner: second reader. By: before each order.
3. The draft is pasted into the decision log first, then into WInS; the saved note is copied back character for character with a screenshot (A4 checklist items 16-17). Owner: log keeper. By: each order.
4. **Reflections:** written by students without AI; any later AI use (for example checking a number) is logged. Owner: note writers. By: drafts Wed Oct 21, final Thu Oct 22 AEDT.

PM-05. The team's process broke down (TN + IPS; L = H, Impact = Major)

What failed. Six students, one shared login, no roles. The gate meeting slipped during the school holidays; the first fills came in week 2, shrinking the "tested" window; one student wrote every note at 1 a.m.; the IPS was drafted by one person over the last weekend while others had exams; nobody owned the PDF test; the second reader never saw the final version.

Why it was plausible.

- "No roles assigned yet" (CLAUDE.md, VRF). "All team members must play a contributing role" (R-W38, VP). "We strongly encourage teams to identify different roles for each team member" (SMAApply FAQ, VP via A4).
- NSW public-school spring holidays are Mon 28 Sep to Fri 9 Oct 2026, and Term 4 starts Tue 13 Oct (<https://education.nsw.gov.au/schooling/calendars/2026>, read 2026-09-28, VP). The whole WInS gate week falls in those holidays, and the roster deadline (Sat Oct 10, 08:00 AEDT) falls the morning after they end. The team's state and school type are not recorded, so this may not apply (UNVERIFIED for the team).
- The 2026 HSC written exams run from Tue 13 Oct to the morning of Thu 5 Nov (<https://www.nsw.gov.au/education-and-training/nesa/news/all/2026-hsc-timetables-released>, read 2026-09-28, `--grep` found verbatim, VP). That window covers the TN deadline and ends the day before the IPS deadline. It matters only if a member sits the HSC; year levels are not recorded (UNVERIFIED for the team).
- U.S. trading opens at 23:30 AEST before Oct 4 and 00:30-01:30 AEDT after it (E5 script [2], DER), so "watching the fill" means staying up.

Tripwires. The gate is not passed by Thu Oct 1 AEST. A week with no logged meeting. One student's name on more than half the log entries. No full IPS draft circulated by Mon Oct 26.

Prevention.

1. **Assign roles now:** trader (+ backup), note writer(s), second reader, log keeper, format checker, IPS lead writer, rates analyst, growth/risk analyst, client lead, admin liaison. Every student drafts at least one note or reflection and one IPS section. Owner: team leader. By: Tue Sep 29 AEST.
2. **Shared calendar** of each student's exams and travel; internal deadlines move around them. Owner: team leader. By: Wed Sep 30 AEST.
3. **Internal deadlines two days early:** gate passed Thu Oct 1 AEST; roster submitted Wed Oct 7 AEDT; TN submitted Thu Oct 22 AEDT; IPS first draft Mon Oct 26; words frozen Tue Nov 3; IPS submitted Thu Nov 5 AEDT. Owner: team leader.
4. **Weekly 30-minute check-in** with a written log line (who did what). Owner: log keeper. By: every Sunday.
5. **Time split:** after Oct 23, most hours go to the IPS (it may be the heavier deliverable for selection, R-W24 vs SMAApply; R-AN41). Owner: team leader.

PM-08. AI use was not recorded, or was recorded inaccurately (TN + IPS process; L = M, Impact = Severe)

What failed. In December the Works Cited said only "AI was used for brainstorming". The team's public repository shows a multi-agent research run that specified note elements, pitch elements and rule parameters. The record understated the use; or it omitted the draft checker and the fetch tool. An honest record made late looked worse than a clear dated one (INT).

Why it was plausible. "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages" (R-W46, VP) and "including how and where students used AI-generated information" (R-W49, VP). AI use not recorded is treated as misuse (A4 e8, INT). The repository is public on GitHub (git remote, VRF; CLAUDE.md "Repo is public", VRF), so anyone can compare the record with the files. The Works Cited itself is Final Report work (later); the log that feeds it starts now.

Tripwires. No AI-use log entry by Sunday Oct 4. Entries that say what the AI produced but not what the team decided or verified itself.

Prevention.

1. **Start the AI-use log today** with columns: date; tool; which student; purpose (brainstorm, fact-check, number check, draft check); what the AI produced (file path); what the team decided and whether before or after seeing the AI output (R-W51); what the team verified independently. Owner: log keeper. By: Tue Sep 29 AEST, then weekly.
2. **Backfill** the council, the blueprints, this research run (research/insight_v1/), the D9 draft checker and the fetch helper. Owner: log keeper. By: Sun Oct 4 AEDT.
3. **Reconcile** the log with the repository's commit history, which is itself a dated record. Owner: log keeper. By: Oct 23 (before the TN) and Nov 6 (before the IPS).

PM-06. The notes contradicted the pitch and the IPS (TN + IPS; L = MH, Impact = Major)

What failed. Under book (ii) the reader met the notes first: a world-stock fund bought on day one beside a hedge worth about two-thirds of the book, while the pitch was worded as a ban on stock risk until the payments are bought. The IPS sentence that explained the scaling had been cut for words. The growth note said one split and the IPS another, with no logged change, and the same holding had two labels. The reader saw two strategies.

Why it was plausible. AY2 rank 1 and E2 C1 flag exactly this (VRF). "Your portfolio and Final Report should reflect the strategy established in your IPS" (IPS guide L75, VRF); the Investment Strategy criterion "maintains consistency with the team's IPS" (VRF); "The analysis should be consistent with the strategic approach your team is developing and will later articulate in its IPS" (TN guide L30-31, VRF). Under (ii) WInS holds about 17.0 points more stock than Laura's real January 2027 money at 50/50, or 20.5 at 60/40 (D10 with AX2 C7, MODEL).

Tripwires. A pitch draft worded as a ban tied to time. No sentence in the IPS draft naming the date or state WInS shows. A (ii) hedge or growth note without its "after both deposits" / "after her 2028 deposit" marker. A split in the growth note that differs from the IPS rule. Two labels for one holding.

Prevention.

1. Word the central idea as a **split of jobs**, true of both books (E2 F4); keep the ordering (payments first) in the 2027/2028 rule. Owner: IPS lead writer. By: pitch elements Sat Oct 24.
2. Under (ii): the scaling marker **inside** each hedge and growth note, within the note-box limit; one IPS sentence with the elements: post-2028 mix, scaled to \$300,000, stand-in funds rather than the ladder, and that in January 2027 almost all of her real \$300,000 buys the ladder (D10 1.5; T2 16a; ticket section 5). Owner: note writer; IPS lead writer. By: before the first order; IPS draft.
3. **Two-reader check** before the first order: two readers outside the team read the note drafts and say what Laura's real money holds in January 2027; if either says "stocks", fix the drafts (E3-10). Owner: second reader. By: gate session.
4. **Consistency table** in the decision log (note → IPS rule name → label), checked before each submission. Owner: log keeper. By: Thu Oct 22 and Tue Nov 3 AEDT.
5. A split change before Nov 6 is allowed only as a logged, research-based decision with a trade, never because prices moved (AX2 D2 C3, VRF). Owner: growth/risk analyst. By: Fri Oct 16 ET if the split was

provisional (ticket calendar).

PM-07. The plan read as timid, or as a study of her rather than a plan for her (TN + IPS; L = MH, Impact = Major)

What failed. The reader saw almost no stocks (about 98% Treasury funds under (iii), about 17% stocks under (ii)) for a client who "has been willing to take thoughtful risks", no stated purpose for the stocks, the \$150,000 deposit doubted in three places, a long list of caveats, trap vocabulary about her ("loss tolerance", "house money"), and a pitch that never mentioned the residency. "Presents recommendations that can earn her confidence" (Client Knowledge, VRF) scored low.

Why it was plausible. Case L69-74 (VRF). The payments cost 97.4-98.1% of her first deposit at the Sept 25 curve, so little stock in 2027 is forced by prices, not chosen (E3 [1], MODEL on VRF inputs); the whole-portfolio stock share averages about 9.7% over 2027-2032 at 50/50 (E3 [2], MODEL). E2 2.2 finds the residency missing from the pitch spec, the \$150k doubted too often and the statistics degree used as a lever (INT). E3 findings 2, 3, 6 and 10 (SIM/INT).

Tripwires. No "residency" in the pitch draft. Any sentence about her temperament. "Risk-averse", "loss tolerance", "trap" or "composure" in a draft. More than about 15% of IPS words on what could go wrong (E3-11, ASM ceiling). The \$150k called uncertain more than once.

Prevention.

1. **Pitch order:** what the plan secures for the residency, then the rule, then what the rest is for (E3-5; E2 F3). Owner: client lead. By: Sat Oct 24.
2. **Forced vs chosen caution** stated in the risk passage, with at most one dated, rounded price fact (E3-2). Owner: client lead. By: first IPS draft Mon Oct 26.
3. **What the stocks are for** in the growth rule's "because" (E3-3; E2 F6). Owner: growth/risk analyst. By: Oct 26.
4. **One labelled stress clause** for the \$150,000; the case says she "will" contribute (case L43, R-AN35, VRF). Owner: IPS lead writer. By: Oct 26.
5. **No behavioural diagnosis** of her in any note, reflection or the IPS; pre-commitments framed as the firm's own discipline (E3-6). Owner: second reader. By: every draft.
6. Under (iii): one reflection answers "why so little equity" with the forced-vs-chosen content (ticket section 4). Owner: note writer. By: Thu Oct 22.

PM-09. The thesis read as generic (TN + IPS; L = MH, Impact = Major)

What failed. "Buy the payments with Treasuries, invest the rest" is standard pension practice, and many AI-assisted rivals reached it. The hedge note was Wharton's own example trade in different words. The Laura-specific reasons sat in the middle of rule 3.

Why it was plausible. Wharton's example note buys "an intermediate-term U.S. Treasury bond ETF ... for Laura's future operating commitment" (TN guide L36-39, VRF; R-AN44). Rivals using the same AI tools may reach the same architecture (BS-10, ASM). The surplus approach is named industry practice (Russell 2026, VP via D8/AY2). "Presents a clear and creative investment thesis" (Investment Strategy, VRF). D13c N7: do not call the plan "innovative" (INT).

Tripwires. A draft sentence that still works with another client's name. A hedge note with no dated need and no checkable fact. A pitch with no benefit only Laura gets.

Prevention.

1. **Swap test** on every note, reflection and IPS paragraph; failures are rewritten or cut (D13c, INT). Owner: client lead. By: each note; IPS drafts Oct 26-Nov 3.
2. Lead with the **three Laura-specific decisions**: the payments are bought before her career-income deposit can matter (case L43-45); nothing is announced to co-sponsors that is not already bought (L114-116); certainty is defined by a price she can check (L97-99) (E1 North Star; D7 M011; VRF case lines, INT selection). Owner: IPS lead writer. By: Oct 24 idea list.
3. **Hedge note:** her dated need, one checkable fact and the never-sell rule; no "reduce volatility" (ticket section 5; D9 M012). Owner: note writer. By: the first order.
4. Show originality through fit, never by claiming novelty. Owner: second reader. By: every draft.

PM-14. A trade broke a WInS rule (TN via WInS; L = LM, Impact = Severe)

What failed. Someone on the shared login bought a stock under \$5, a leveraged, inverse or crypto-linked fund, or an exchange-traded note; or sold a fund the same day to "fix" a mistake; or a Treasury fund crossed an unseen position limit after rates fell; or an order at the open pushed cash below zero. Last year's lesson repeated.

Why it was plausible.

- Last season the team "traded assets outside the trading rules" (CLAUDE.md, VRF).
- Permitted: cash, stocks $\geq \$5$, "Any Exchange-Traded Funds (ETFs) available on WInS", government/Treasury bonds on WInS; banned: margin, short selling, stock-secured debt, crypto, derivatives, "Anything other than the approved investments listed above" (SMAApply FAQ, VP via A4). "Day Trading: This is not permitted." (2026-27 WInS User Guide, VP via A4). Leveraged and inverse funds use derivatives internally (R-AN49, INT). The position limit is visible only in Session Rules; the Stock-Trak default is 25% (VP, generic page).
- "Each team shares one WInS account" (R-W14, VP): anyone can place an order.
- A holding placed one point under a 25% cap crosses it after about a 101-128bp rate move, or a 20-24% stock fall, or a 10-12% stock fall with a 50bp rally (D10; T2 [5]; MODEL, ASM that WInS re-checks the limit).
- Orders entered while the market is closed fill at the next open (SMAApply Trading Details, VP), so a thin cash float can be overrun (T2 finding 1, MODEL).

Tripwires. More than one person placing orders. An order that was not in the log before it was placed. Any holding within one point of the cap. The cash float under \$1,000.

Prevention.

1. **One trader and one backup** place every order; others never do (a team convention; the account itself stays shared, R-W14). Owner: team leader. By: Tue Sep 29 AEST.
2. The A4 "**before every trade**" **checklist** (guardrails section c) read aloud by the trader and ticked by a second student. Owner: trader + second reader. By: every order.
3. **Session Rules screenshot** (both position-limit lines, the day-trading tooltip, trades allowed) and the cap branch chosen in advance (ticket section 3). Owner: trader. By: Thu Oct 1 AEST, before the first order.
4. **Never sell the same security on the day it was bought**; an error is logged and lived with ("run with it", SMAApply FAQ, VP). Owner: trader. By: every order.
5. Recompute share counts from the last close; keep at least \$1,000 in cash; never add to a holding near the cap (ticket). Owner: trader. By: every order; weekly cap check by the log keeper.

3. The other 20 causes

Ratings INT/ASM; expected-harm ranks from E5 script [3]. "Owner" is a role, not a student.

Rank	Id	Cause (January 2027 view)	Serves	L	Impact	Evidence (labels)	Prevention	Owner-type	By
11	PM-15	A "required trading activity" minimum existed on a logged-in page and the low-trade plan (4-5 orders) missed it	TN (WInS)	LM	Severe	"Teams must meet the required trading activity and portfolio management guidelines" is undefined on public pages (R-W23, VP; D10 s3); "first trade by Oct 10" is a 2023-24 claim (SNIP, A4)	Read the logged-in Trading Details, Getting Started and weekly emails before the first order; if a minimum exists, ask Wharton (Contact Us) and meet it with plan-consistent trades logged with the rule as the reason; if none, log "none found" with a screenshot. Never stage a trade to fill a note category (T2 finding 13)	team leader	Thu Oct 1 AEST; weekly emails checked each Monday
12	PM-16	Only two usable executed trades by Oct 23 (under (iii) the T-bill leg vanished after a rate fall; an order was rejected; indecision delayed the fills)	TN	LM	Severe	"Select three (3) Trading Notes from trades your team executed in WInS"; "Yes, we will verify this" (TN L42, L53, VRF); (iii) T-bill leg under \$1,000 after a ~12bp fall, about 11% by Oct 2 and 19% by Oct 9 (T2 finding 1, MODEL)	Under (iii), apply the ticket's pre-set rule (leg under \$1,000 and no listed dated rung → revert to (ii)); market orders at the gate session; confirm "Filled" in Order History and a note on each trade	trader	fills by Fri Oct 2 ET; verified Mon Oct 5 AEDT; hard stop Fri Oct 9 ET
13	PM-10	Overclaiming words: "guaranteed", "safe", "risk-free", "95%", "bought in January 2027" or "fits inside \$300,000" as flat facts, "matched" or "operating reserve" for WInS funds, "locked"/"known" 2028 rates, "promised" for the 2031 stretch, "only the facility shrinks" (false in the joint tail), "innovative"	TN + IPS	H	Moderate (Major if in the pitch or a permanent note)	Ticket s5 and brief s2 banned lists; D7 M011 (date claim false in about 1 path in 3, MODEL); E3 finding 1 ("promised"); AY2 rank 2 (joint tail); the chair memo itself used "only the facility shrinks" (VRF, history)	A banned-word list pinned beside the note box and the IPS; the second reader searches every draft for each word; certainty words used only for their objects (D9 M062 with AY2 C2)	second reader	every note; IPS drafts Oct 26-Nov 3

Rank	Id	Cause (January 2027 view)	Serves	L	Impact	Evidence (labels)	Prevention	Owner-type	By
14	PM-11	A required IPS element was missing: the definition of "high degree of funding certainty" and how it was evaluated; diversification; liquidity; reserve composition over time; facility approach; flexibility; the inflation and facility-cost assumption; the "funded 2027, set aside 2033" reconciliation	IPS	M	Major	Case L94-99 and L146-148 (VRF); IPS guide L45-50 and L53-65 (VRF); E2 E-6: diversification and liquidity have no budget line in E1 (INT)	Checklist of every IPS-guide and case "should" item, ticked against the draft with the sentence number that covers it	IPS lead writer	Fri Oct 30 AEDT
15	PM-12	The IPS left decision rules open, so the Final Report had to choose parameters after the freeze and looked like a redesign	IPS	M	Major	"Your team may not revise its investment strategy after the submission deadline" (IPS guide L73-74, VRF); "you may not redesign your strategy after observing the results" (Guide p.5, VRF); open choices in E1 C1-C12, E2 5, T2 finding 5	Fix every parameter in section 4 below by vote, in the log, before the first full draft	team leader (vote); IPS lead writer	Mon Oct 26 AEDT
16	PM-13	A deadline was missed: the team planned to submit on Saturday morning, then an upload failed; or the only person with the SMAppl login was unavailable; or someone read "5 p.m." as Australian time for the lock	TN + IPS	L	Fatal	Deadlines 5:00 p.m. ET, "We will not grant any deadline extensions" (SMAppl, VRF). In Sydney: roster Sat Oct 10 08:00, TN Sat Oct 24 08:00, IPS Sat Nov 7 09:00 AEDT (one hour later because U.S. clocks change on Nov 1); Brisbane an hour earlier, Perth three hours earlier (E5 script [1], DER). Last U.S. session before the lock opens 01:30 and closes 08:00 Sat Nov 7 AEDT (script [2])	Internal submission two days early (Thu evening AEDT); never plan to use Saturday morning; confirm who can submit in SMAppl and that a second student can reach that person; keep the submitted PDF and a screenshot of the confirmation	team leader	TN Thu Oct 22; IPS Thu Nov 5 AEDT

Rank	Id	Cause (January 2027 view)	Serves	L	Impact	Evidence (labels)	Prevention	Owner-type	By
17	PM-18	IPS title page mismatch: team name not exactly as on the roster; names not "First Name, Last Initial" or not as on the roster; the team name given instead of the WInS username	IPS	L	Fatal	IPS guide L84-100 (VRF); "Your team username is NOT the same as your team name"; the username cannot be changed (SMAApply FAQ and User Guide, VP via A4)	Copy the team name and six names from the submitted roster (screenshot at submission) and the username from the WInS login screen, character for character; two students check the final PDF	admin liaison + format checker	capture Wed Oct 7; check Wed Nov 4 AEDT
18	PM-17	A wrong, stale or mislabelled number: 10y 4.22%, Taiwan CPI 2.4%, a \$295k ladder, \$500k WInS cash, "6,000 teams competed", "the rule picks 50% under both forecasts" (true only with a fee of 0.5% or more), the withdrawn \$101k/\$165k barbell figure, "111%" without "face value", "1 in 4" for the real ladder (about 1 in 3), a model ladder price shown as a quote, a "tested" figure computed with <code>official_curve_pv.py</code> (fixed to Sept 25) or distorted by a fund distribution	TN + IPS	MH	Moderate (Major if in a permanent note)	Brief s10 inconsistencies; T2 findings 2, 19; E2 E-1; AY2 D7 C5; D7 M011; ticket header (VRF)	Every number carries its date and kind (price, history, assumption) and comes from a primary source or a re-run script on the day; "tested" numbers from <code>A2_curve_recheck.py</code> or <code>D9_numbers.py</code> [3], with IEF/TLH ex-dividend dates checked; the list above pinned as "never use"; read the latest audit corrections before using any research number	rates analyst + second reader	each note; TN numbers Oct 21-22; the IPS's one number re-dated in the week of Nov 2
19	PM-22	The frozen Nov 6 book did not reflect the IPS: a tidy-up trade in the last days, a split change never traded, a drift outside the band, a final order that never filled	IPS	LM	Major	IPS guide L75 (VRF); ticket s8 "never tidy up in the last days"; orders after the close do not execute (A4 item 7, INT)	No discretionary trade after the words freeze; holdings compared with the IPS sentence about WInS on Nov 4; any rule-driven trade done by Fri Oct 30 ET	trader + IPS lead writer	freeze Tue Nov 3; check Wed Nov 4 AEDT
20	PM-23	The SMAApply submission forms had fields or format rules the team first saw on the deadline night	TN + IPS	LM	Major	"All deliverables must be submitted via your team's SurveyMonkey Apply account" (VRF); the TN form's contents are not in the repo (A4 b4: logged-in pages unread)	Open the TN and IPS forms in SMAApply; record fields, file types, limits and who can submit	team leader	Mon Oct 12 AEDT

Rank	Id	Cause (January 2027 view)	Serves	L	Impact	Evidence (labels)	Prevention	Owner-type	By
21	PM-20	Trading-note mechanics: a note cut off by an unseen limit, a typo, a note on the wrong order, a second "added" note to finish a sentence, curly quotes changing the "exact" text	TN	M	Moderate	300-character cap on a sister product (VP for that product, UNVERIFIED for WInS); Wharton's example is 413 characters (VRF count); added notes carry a later time stamp (Stock-Trak 2017, VP)	Zero-risk note-box test before the first real note (type about 450 characters on the review screen, do not confirm); draft to 300 characters until tested; paste plain text; check the saved note against the draft; ask Wharton whether an added note counts	trader + note writer	Thu Oct 1 AEST
22	PM-21	Reflections weak or non-compliant: over 100 words by Wharton's count; the three official questions not all answered; six jobs in 100 words; a "tested" window chosen after the fact; a trade staged to show "refined"	TN	M	Moderate	TN L46-52 (VRF); AY2 D7 C2 (pre-registered window); E2 3.2 point 5 and F9; R-W28 bans fabricated analyses (VP; staging reading is INT)	≤95 words by two counters; each reflection answers the three questions plus one extra job; WInS screenshots on the fill date and at the Oct 20 close (Wed Oct 21 07:00 AEDT); report small moves as they are; "refined" only if it really happened; the true "growth-first tested and dropped" history in one reflection if told truthfully (E2 F8)	note writers + log keeper	screenshots on the fill day and Oct 21; drafts Oct 21; final Oct 22 AEDT
23	PM-19	Roster missed or wrong; members lock and names must match the IPS	IPS	VL	Fatal	Roster due Oct 9 5:00 p.m. ET; unique email per student; "Once your official team roster is submitted, teams may not change their members" (SMAppl, VRF)	Each student confirms name spelling and email; submit early; screenshot the confirmation; full names, emails and birthdays never stored in the repo (CLAUDE.md, VRF)	admin liaison	Wed Oct 7 AEDT (deadline Sat Oct 10 08:00 AEDT)
24	PM-26	Someone on the team contacted Laura (a message, a comment on her posts, a question at an event, an email to her publisher about the case)	all	VL	Fatal (disqualification)	"Teams may not contact the Competition client. Teams that violate this rule will be disqualified." (R-W16, VP)	Team rule announced in writing: no interaction of any kind with Laura or her representatives until the competition ends; research uses only public pages	team leader	Tue Sep 29 AEST

Rank	Id	Cause (January 2027 view)	Serves	L	Impact	Evidence (labels)	Prevention	Owner-type	By
25	PM-27	Paid help, or a member enrolled in a non-Wharton course that "claims to teach" the competition; a paid tutor edited a draft	all	VL	Fatal	R-W21 and R-W28 (VP)	Each student confirms no such course or paid help; paraphrase readers unpaid and restate-only	team leader	Wed Sep 30 AEST
26	PM-28	Membership or team-leader rule breached (a member removed without approval; size outside 4-6; leader under 16 on Sep 28)	all	VL	Fatal	R-W11, R-W13, R-W17 (VP)	Keep six members; any change only with Wharton's prior approval; confirm the leader's eligibility before the roster	team leader	Wed Oct 7 AEDT
27	PM-24	A visible case misread: buying in 2027 not reconciled with "set aside at the beginning of 2033"; WInS gains mentioned as if they count; living costs taken from the portfolio; the facility treated as a dated liability	IPS	L	Major	Case L61-63, L94-95, L104, L132-133 (VRF); Guide p.3 "not the competition scorecard" (VRF); brief s8 item 19	One reconciling clause (bought 2027, designated the reserve on 1 Jan 2033; E1 P9); no WInS gain, loss or rank in any note, reflection or the IPS	client lead	Fri Oct 30 AEDT
28	PM-30	The public repository exposed the work: another team copied it, a reader found AI-written specifications that matched the submission, or student data leaked	TN + IPS	L	Major	Repo public (git remote; CLAUDE.md "Repo is public; the team leader accepted putting competition materials in it", VRF); "plagiarize existing strategies" is an ethics breach (R-W28, VP)	Team leader revisits the visibility decision until after Dec 4, or records why public stays; the PM-04 copy check covers phrasing; no student personal data in the repo	team leader	Fri Oct 2 AEST
29	PM-25	Laura misused: her quotes in the TN or IPS, the case pull quote attributed to her, an identity or Taiwan tilt, a book-title pun, a private detail, the statistics degree used as flattery	TN + IPS	LM	Moderate	Brief s16 D13 rules (VRF); D13c R1-R3, R10 (INT); E3 4c	Check every line against D13a and the brief's rules; statistics degree zero or one use (conflict K4)	client lead	every draft
30	PM-29	The advisor made or placed a decision, or a note or the IPS said "our portfolio manager approved" (the case's fictional framing)	TN + IPS	L	Moderate	"Advisors may not make decisions on behalf of students" (R-W19, VP); "Students should place trades on WInS, NOT the advisor" (SMAApply FAQ, VP); D10 M013 cut list	Advisor for admin only; no such wording in notes or the IPS; governance names who decides (E1 P10; E3-8)	team leader	now; IPS draft Oct 26

4. IPS: rules to fix before Nov 6 (from PM-12; the Final Report must later apply them without redesign)

Each is a vote recorded with first names, date and reason (brief s17; E1 P11; E2 section 5; E3 section 9). The content is in the cited files; this list only makes the deadline and owner explicit. **Owner: team leader runs the vote; IPS lead writer records the outcome in the draft. By: Mon Oct 26 AEDT, before the first full draft.**

1. **Completion rule:** buy on arrival; which payments first if money is short (longest-dated first, with its "because" and the first-operating-year counterpoint, E3-7); the 2028 deposit completes the ladder before any growth; one labelled joint-tail clause (E1 P3-P4).
2. **Growth mix:** 50/50 or 60/40, its band, a yearly reset, and a reason whose yardstick is stated (money put in, or the all-Treasury alternative; E2 F6; T2 finding 2 on the fee dependence).
3. **Growth money's bond half:** short Treasuries or dated to 2031/2033 (AX1 C18 via E2 R1).
4. **Floor timing:** bought in January 2028 or January 2031 (AX1 C6 via E2 R2).
5. **2031 range method:** the bought share, the share of the rest added to the gift, the top percentile, cap or no cap, two-sided confidence, no announcement before the ladder is complete; the parts named by what each dollar does (E1 P7; E3-1).
6. **2033 order of use:** the reserve is the ladder, held to maturity, designated on 1 Jan 2033; the gift; kept money in US\$ short Treasuries; US\$ binding; no NT\$ conversion or hedge before the facility decision (E1 P8-P9).
7. **Governance:** what stays Laura's decision; rules change only if her circumstances change, never because markets moved; no rebalancing between the reserve and the growth money (E1 P10; E3-8).
8. **Cost principle:** keep or cut (conflict C2 below), decided before words are counted.
9. **The WInS sentence:** the date or state the book shows, and stand-in funds rather than the ladder (D10 1.5).

5. Master calendar of preventions (Australian Eastern time unless marked ET; internal targets are ASM team rules)

When	What	Owner-type	Causes
Tue Sep 29 AEST	Roles assigned; one trader + backup; no-contact rule written down; AI-use log started	team leader; log keeper	PM-05, PM-14, PM-26, PM-08
Wed Sep 30 AEST	Shared exam/travel calendar; own-words notes and devil's-advocate pairs for the gate votes; paid-help confirmation	team leader	PM-05, PM-01, PM-27
Thu Oct 1 AEST	Gate: Session Rules screenshot (both limit lines), logged-in activity rules read, note-box test at zero risk, Bonds list, labels chosen by the team, book and split votes, note drafts copy-checked, two-reader check	trader; second reader; team leader	PM-14, PM-15, PM-20, PM-01, PM-04, PM-06
Fri Oct 2 ET (opens 23:30 AEST)	Target fills; screenshot positions (start of the "tested" window)	trader; log keeper	PM-16, PM-21
Fri Oct 2 AEST	Repo-visibility decision revisited and logged	team leader	PM-30
Sun Oct 4 AEDT	AI-use log backfilled (council, blueprints, this run, tools)	log keeper	PM-08
Mon Oct 5 AEDT	Every order "Filled" with a note; hard stop Fri Oct 9 ET	trader	PM-16
Wed Oct 7 AEDT	Roster submitted; names, team name and WInS username captured for the title page	admin liaison	PM-19, PM-18, PM-28
Mon Oct 12 AEDT	SMAApply TN and IPS forms opened and recorded; who can submit	team leader	PM-23, PM-13
Fri Oct 16 ET	Split decided if it was provisional (trade once or log "decided not to trade")	growth/risk analyst	PM-06, PM-12
Wed Oct 21, 07:00 AEDT	Oct 20 close: screenshot positions (end of the "tested" window); reflections drafted	log keeper; note writers	PM-21
Thu Oct 22 AEDT	TN consistency table, word counts, number check; TN submitted (deadline Sat Oct 24 08:00 AEDT)	log keeper; team leader	PM-06, PM-17, PM-21, PM-13
Sat Oct 24 AEDT	IPS idea list (≤ 12 ideas), pitch elements, dated spine	IPS lead writer; client lead	PM-03, PM-07, PM-09

When	What	Owner-type	Causes
Sun Oct 25 AEDT	Explain-back check, round 1	second reader	PM-01
Mon Oct 26 AEDT	Section 4 votes recorded; first full IPS draft from a blank page; first PDF test	team leader; IPS lead writer; format checker	PM-12, PM-01, PM-02
Oct 26-30	Paraphrase round 1; required-elements checklist (Fri Oct 30); case-misread check	IPS lead writer; client lead	PM-03, PM-11, PM-24
Fri Oct 30 ET	Any remaining rule-driven WInS trade completed	trader	PM-22
Nov 2-3	Paraphrase round 2; the IPS's one number re-dated; explain-back round 2; words frozen Tue Nov 3	IPS lead writer; rates analyst	PM-03, PM-17, PM-01
Wed Nov 4 AEDT	Final PDF check by two students; title page against the roster; holdings against the IPS sentence	format checker; admin liaison; trader	PM-02, PM-18, PM-22
Thu Nov 5 AEDT	IPS submitted (deadline Sat Nov 7 09:00 AEDT; the portfolio locks at the U.S. close, 08:00 AEDT the same morning)	team leader	PM-13

6. Conflicts the pre-mortem cannot settle (both sides; the team decides)

#	Conflict	Side A	Side B	Pre-mortem reading (INT)
C1	WInS book (ii) vs (iii)	(ii): three roles and a visible growth decision (E1, T1, D12, E2); lower PM-16 risk	(iii): true at a stated date, shortest IPS sentence (D10; E3 slightly toward (iii) on credibility)	(ii) carries PM-06 (contradiction) risk; (iii) carries PM-07 (timid) and PM-16 (only two decisions) risk. Neither is safer overall; each is safe only with its own prevention done before the first order
C2	Cost clause in the IPS	Cut first (E1 P12; AY2): lowers PM-02 page risk	Keep and extend (E3-4): a missing cost answer reads as selling, and the 50/50 reason rests on a fee assumption	Decide before the word count; if cut, the growth-mix reason must not depend on a fee
C3	Statistics degree	Once, in the certainty definition (E1 P17; D7)	Zero (E3-9; D3 audit C19)	Zero lowers PM-07 and PM-25 risk at no cost to substance
C4	"Double-spaced" reading	Word's "Double" (27.6 pt)	"Exactly 24 pt" matches Wharton's sample (AY2, INT)	Fit under the stricter reading; use 24 pt only as a documented last resort (PM-02)
C5	Paper size	U.S. Letter (Wharton's PDFs)	A4 (Australian default; more room; allowed by the discretion clause, INT)	Either is defensible; fitting on Letter removes the question
C6	Growth split	50/50 (E1, D2; better bad case)	60/40 (D6, D13c; less timid)	PM-07 pushes up, PM-17 warns the "50% under both forecasts" reason needs its fee; own the choice as a choice of spread (E2 F6)
C7	Semifinal basis	IPS and Final Reports only (Rules page, R-W24, VP)	All three deliverables evaluated (SMAApply, VRF)	Treat all three as scored, but weight hours toward the IPS after Oct 23 (PM-05)
C8	Public repository	Public, accepted by the team leader (CLAUDE.md)	Private until Dec 4 lowers PM-30	A team-leader decision; E5 only asks that it be revisited and logged

7. Later (after Nov 9), one line each (Final Report; not worked here)

- School documentation on letterhead: request through the advisor in October, as SMAApply says "Do not wait until the last minute" (VRF; BS-11); submitted with the Final Report on Dec 4.
- Works Cited with the AI-use record built from the PM-08 log (R-W46).
- Final Report format rules, released Nov 9: a fresh format pre-mortem then.
- The Final Report applies the section 4 rules "as written" (no parameter chosen after Nov 6).
- Deadline: Sat Dec 5 09:00 AEDT (E5 script [1]).

Sources

Official, in the repo (VERIFIED-REPO-FILE): [competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt](#) (L43-45, L61-63, L69-74, L91-99, L104, L114-116, L129-133, L146-148, L161-162); 2026_WGY_

Investment_Policy-FINAL.txt (L45-75, L82-125); 2026_WGY_Trading_Notes_Analysis-FINAL.txt (L11-31, L36-53); 2026_WGY_Investment_Competition_Guide.txt (L70, L89, L156-159, L183); 2026_WGY_Competition_Infographic.txt (L42); SMAApply_Deliverables_Page_2026-09-27.md (deliverables, roster fields, criteria).

Primary web, read by E5 on 2026-09-28 with `research/insight_v1/scripts/fetch_text.py` (VERIFIED-PRIMARY):

- NSW Department of Education, 2026 term dates: <https://education.nsw.gov.au/schooling/calendars/2026> ("Term 4: Tuesday 13 October to Thursday 17 December"; "Spring: Monday 28 September to Friday 9 October").
- NSW Government / NESA, "2026 HSC Timetables released": <https://www.nsw.gov.au/education-and-training/nesa/news/all/2026-hsc-timetables-released> (--grep "13 October" found verbatim: "starting with English Paper 1 on Tuesday 13 October and finishing with Physics in the morning of Thursday 5 November").
- Found first through WebSearch (2026-09-28); the snippets were then confirmed on the pages above.

Primary web, as carried by the cited files (VP there; not re-opened by E5): Wharton Rules & Roles, FAQ and AI policy (R-W11 to R-W51 via `phase_A/case_register.md` and A4); SMAApply Trading Details and FAQ (via A4, D10, AX2); 2026-27 WInS User Guide and Stock-Trak pages (via A4, D12); retired Wharton judging page and 2025 top-10 article (via D9, D7, AY2); digital.gov paraphrase testing (via D12); Russell 2026 (via D8/AY2).

Research files (lower authority; audit corrections applied): `research/insight_v1/phase_E/E1_change_proposals.md`, `E2_judge.md`, `E3_laura.md`; `wins_now/securities_and_allocation_v1.md`, `T2_red_team.md`; `phase_D/trading_now_brief.md`, `D10_compliance.md` (+AX2), `D7_wharton_intent.md` (+AY2), `D9_communication.md` (+AY2), `D12_overflow_client.md` (+AY1), `D13c_voice_map.md`, `audit_markets_rules.md`, `audit_judges_practice_comms.md`; `phase_A/wins_week1_guardrails.md`, `case_register.md`, `stakeholder_map.md`; `CLAUDE.md`; `research/council_2026-09-27/01_chair_memo.md` (history).

Scripts: `research/insight_v1/scripts/E5_premortem_checks.py` (new; deadline clock via Python zoneinfo, ranking from the ASM ratings above). Figures quoted from other scripts are as reproduced by their auditors (T2, AX2, AY1, AY2, E2, E3).

What this teaches

1. **Most projects fail at the joints, not in the engine.** The strategy survived every audit; the likely failures are in who owns it, how densely it is written, whether two documents tell the same story, and whether a PDF has three pages. A pre-mortem looks at the joints on purpose.
2. **Rank by likelihood times impact, then do the cheap fatal ones anyway.** A missed deadline or a fourth page is unlikely, but it ends everything and costs an hour to prevent. Low-probability, high-cost risks with cheap fixes are the best deals in risk management, in markets as well as in competitions.
3. **Every risk needs an owner and a date.** "The team should check the PDF" means nobody will. A named role and a day turn a worry into a task.
4. **Watch for tripwires, not just outcomes.** You cannot see a lost semifinal place until January, and Wharton gives no feedback. You can see today whether anyone can explain the 2031 rule without notes, or whether a draft is 510 words. Early signs are the only feedback you will get.
5. **Honesty about how you worked is part of the work.** An AI-use record written as you go costs minutes; one reconstructed later, against a public record anyone can read, costs credibility. That is the same lesson the case teaches about Laura's 2031 promise: say only what you can stand behind.

Phase G verification of the six insight_v1 outputs

Summary (5 lines)

1. I checked six files adversarially against `phase_E/E6_final_spec.md` and the output of `scripts/E6_final_checks.py` (re-run 2026-09-28, exit 0).
2. I found 19 problems: 1 blocking, 7 important and 11 minor. I fixed 18 in place with minimal edits. One minor item was left because it is consistent with the spec's illustration.
3. The blocking problem was privacy. A parked question in `open_questions.md` quoted a D13a EXCLUDE-PRIVACY line and called it Laura's advice. That text is now withheld.
4. The important problems were: rates-fall figures in `strategy_changes.md` that the spec had already corrected; Laura's word "unstable" in the IPS B2 element list; an unqualified "lock-early misses none"; a lock-early design mixed with REC in one comparison; and three files naming tickers without the PENDING label.
5. After the fixes, every number matches E6 [1]-[8] or the spec. The deadlines, the (ii)R book, REC vs ALT, the three notes and rules R1-R10 agree across files. Every Laura line presented as hers is D13a VERIFIED-PRIMARY.

Serves: Trading Notes (Oct 23) and IPS (Nov 6), as a quality check on the files the team will read. This file is AI-generated verification. It contains no deliverable text.

Status labels: as in brief s3 (VP, VRF, SNIP, ASM, MODEL, INT). Severity: **blocking** = breaks a hard rule (privacy, quote, submission text); **important** = a wrong or overclaimed number, a cross-file contradiction or a missing mandatory label; **minor** = precision, wording or labelling.

1. Problems found

#	File	Problem	Severity	Fixed
1	open_questions.md	Parked item M217 quoted a D13a EXCLUDE-PRIVACY line (B8b-Q34, a personal-life advice line) and called it "Laura's identity-context advice"	blocking	yes: the quote text is withheld and the entry points to the id and the ruling
2	strategy_changes.md	I9 gave the rates-fall stock fund as "\$23k / \$7k / \$0, minimum \$137k" (only the "5-year unchanged" branch). It also said "the spec's R2 row still prints E4 [7]'s \$31k / \$15k / \$147k", which is false: the spec now prints E6 [8]'s ranges	important	yes: now \$20-23k (-50bp), \$1-7k (-100bp), and at -150bp no fund and a minimum of \$127-137k, as in E6 [8] and the spec R2
3	ips_spec.md	Section (e), item 4, is the list of what B2 must contain. It carried Laura's own word "unstable" inside the gap (i) element, so it read as IPS content. Brief s16 bans Laura quotes in the IPS	important	yes: gap (i) now cites the case (career income, L43-45). Her word is marked "background only, never in the IPS or the notes" (B8b-Q18)
4	why_us.md	R1 said "lock-early misses none" next to growth-first's miss rate if the deposit is \$0. That is a certainty claim stronger than the evidence. The joint tail (rates fall before January 2027 with no deposit) leaves \$11,957 or \$31,413 of the 2033 payment unfunded (E6 R2)	important	yes: the claim is now scoped to the model's 9/25 curve, and the joint tail is stated
5	why_us.md	R1 compared growth-first with lock-early "\$207k / \$273k". Those numbers come from <code>strategy_mc.py</code> , which models the council's 60%-equity design, not REC. REC's p95 on the E6 basis is \$250k	important	yes: the source design is labelled and REC's \$207k/\$250k (E6 [2]) is given
6	why_us.md	Names IEF, TLH, IBTM and VT with no PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK label	important	yes: a label line and a pointer to the alternates
7	strategy_changes.md	Names IEF, TLH, TLT, IBTM, VGIT, SPTL, VGSH, VT and EWT with no PENDING label	important	yes: a label line and a pointer to the alternates
8	open_questions.md	Names 12 tickers (e.g. IEF, TLH, IBTM, SGOV) with no PENDING label	important	yes: a label line and a pointer to the alternates
9	ips_spec.md	The B6 evidence for VT said "PENDING WInS AVAILABILITY CHECK" (no position-limit part, no alternates). R10 did not point to alternates	minor	yes: the full label, VT alternates (VTI + VXUS, ITOT + IXUS), and an alternates pointer in R10

#	File	Problem	Severity	Fixed
10	securities_and_allocation.md	The summary was one paragraph, not the required 5-line summary	minor	yes: 5 numbered lines with the same content
11	why_us.md	The reasons were named R1-R8, which collide with the spec's decision rules R1-R10 (e.g. "makes R3, R6 and R8 visible")	minor	yes: a one-line note that R1-R8 here are reasons, not rules
12	ips_spec.md	Word budget said content "about 390" and total "about 455-460". The ten blocks sum to 385 and the spec says 385 / 455	minor	yes
13	ips_spec.md	Section (d) gave the stock fund "a six-year horizon to 2033". The fund is bought in January 2028 and held to January 2033: five years	minor	yes
14	ips_spec.md	The Vanguard-type return input was labelled "VP via the cited files". E6 [4] labels the 5.08% blend an ASSUMPTION hybrid	minor	yes: "5.08% compound, an ASM hybrid built on VP inputs, E6 [4]"
15	why_us.md	"we trade about \$14k of median gift". The spec says \$174k against about \$186-188k, which is \$12-14k	minor	yes
16	strategy_changes.md	Insight 4 quoted "takes the jump", an altered form of D13a B1b-Q11 ("you should always take the jump"). Brief s3 says never polish a quote	minor	yes: the exact wording is restored
17	open_questions.md	Summary item 3 and the "spec inconsistency" note said the R2 issue was fixed in three files. strategy_changes.md was still wrong (item 2)	minor	yes: the note now lists all four files and the corrected ranges
18	trading_notes_pack.md	Section 3 (position-limit branches) named SPTL, VGLT, VGIT and IBTM with no label in that section	minor	yes: a label line
19	strategy_changes.md	T3 says "4 trades, \$100". That is true on the IBTM route; on the 2032-note route it is 3 x \$25 + \$10 = \$85	minor	no: it matches the spec's IBTM illustration, and the spec's "4-6 trades, \$50-150" covers both

Observations (not problems, left unchanged):

- why_us.md says "No blind multi-evaluator tournament was run". During this check, phase_E/tournament/ (firm_A-F, fairness_log) was being written, with no verdict yet. The statement is true now. If a verdict lands, update why_us.md limit 5 and section 0.
- open_questions.md M020 mentions the pronouns listed on her public professional site. They are not in the D13a EXCLUDE-PRIVACY list, and the case itself uses "she". I judged this public-professional and left it.
- open_questions.md M103 ("would Laura call 'gimmicks'") is verbatim parked-question text in a log. It is not a recommendation. D13a limits "No more gimmicks." to its drafts context, and ips_spec.md f3 already warns about it.
- Near-sentence content copied from the spec's pitch element ("each deposit buys a promise when it arrives") appears in ips_spec.md, why_us.md and strategy_changes.md. So do short vocabulary phrases in quotation marks ("moves like a Treasury maturing in late 2032"). All come from the spec and are labelled as content or vocabulary, not drafted text. The five-word copy check (trading_notes_pack s8) will catch any verbatim reuse.
- The D13a JSON holds 106 entries: 93 VERIFIED-PRIMARY, 7 EXCLUDE-PRIVACY and 6 PARAPHRASE-UNVERIFIED. Brief s16 says 91 / 7 / 6. The difference is the later N-series additions (e.g. D13a-N01). It does not affect the six files.

2. Checks run

Ground truth. I ran `.venv/bin/python research/insight_v1/scripts/E6_final_checks.py` on 2026-09-28 (exit 0). Sections [1]-[8] were used for every model number: the REC and ALT totals, bottoms, gifts and kept money; P(top); stock shares; the 53% / -\$23k / +\$32k same-path figures; \$157,736 and 1.73%; 97.4% and 98.1%; the stress, Vanguard-input and history rows; the fees; the (ii)R and 25%-cap books; and the rates-fall rows.

Check 1: numbers. Every dollar figure and percentage in the six files was compared with E6 [1]-[8] or the spec (R1-R10, s5.2-5.3, s5.10, s6.1, s7). Numbers outside E6 were traced to their cited sources:

- 78% ALT bottom above \$150k: E4 X1.
- About \$1k and 48% versus all-Treasury: E3 Q3 / D3 M034.

- \$292 per bp and "up to about 600 characters": ticket v1.
- \$35k at a 4.0% five-year rate: E4 [4] / X7.
- 3.2%, 13.7% and 40.8%, and \$207k / \$273k: CLAUDE.md, `strategy_mc.py`.

I recomputed with Python:

- **Share counts:** all 15 at the 9/25 closes, with no mismatch.
- **(ii)R cash:** \$3,090 before and \$2,990 after fees.
- **Hedge mix and re-solves:** the IEF share is 0.356. The re-solves (VGIT 16.6 / TLH 49.4; SPTI 16.3 / 49.7; IEF 40.9 / TLT 25.1) and the 20%-cap book (about 9.9y) all match.
- **Other figures:**
 - the \$150k minimum costs \$118,047 at 4.98%;
 - the note costs \$72,570 at 95.27 + 1.49;
 - Nov 15 to Jan 1 is 47 days;
 - the IPS blocks sum to 385 words.

The known-and-fixed items were confirmed: the rates-fall ranges \$20-23k / \$1-7k / minimum \$127-137k (now also in `strategy_changes.md`), and the \$5,613 = 19bp headroom (every file uses it; \$7.7k / 27bp appears only as CLAUDE.md's superseded exact-date figure).

Check 2: cross-file agreement.

- **Deadlines:** I converted them with `zoneinfo`:
 - Oct 9, 5 pm ET = Sat Oct 10, 08:00 AEDT;
 - Oct 23, 5 pm ET = Sat Oct 24, 08:00 AEDT;
 - Nov 6, 5 pm ET = Sat Nov 7, 09:00 AEDT;
 - the Oct 20 close = Wed Oct 21, 07:00 AEDT;
 - the Oct 2 open = 23:30 AEST.

All files agree.

- **Plan terms:** book (ii)R, REC vs ALT, and the three notes (A = TLH, tested; B = the minimum, refined only if logged, otherwise supported; C = VT, supported, with the growth-first story) are the same in all files.
- **Rules:** R1-R10 in `ips_spec.md` match spec s4.

Check 3: no drafted text. I read every file in full. There are element lists, checklists and banned-word lists only. There are no notes, reflections, pitch sentences or IPS sentences.

Check 4: no Final Report content. I grepped for "Final Report" and "FR". Every hit is a pointer (Works Cited, "numbers go to the Final Report", "the Final Report states the fee assumption"), and each matches the spec's rules. Final Report items appear only in the `open_questions.md` s5 "Later" list, which matches spec s10 plus two one-line additions (M014, Works Cited).

Check 5: privacy and quotes.

- **Privacy:** I grepped the six files for privacy terms (family, home, Taipei, co-living, memoir, pronoun, coming out and others) and for D13-restricted phrases ("unstable", "take the jump", "earliest supporters", "gimmick", "risk-adverse", the pull quote, "floor-first", "investment philosophy"). I also read the full `parked.json` text of M017, M020, M072, M145, M217 and M221.
- **Quote ids:** I resolved all 20 Laura ids cited in the six files against `phase_D/D13a_laura_quotes_verified.json`: D13a-N01, B1b-Q11, B8b-Q18/B8a-Q07, B8b-Q38, B8b-Q37, B8b-Q10, B8b-Q21, B8b-Q13, B8b-Q12, B8b-Q31, B8b-Q35, B8a-Q19, B9b-Q09, B8a-Q13, B2b-Q08, B3a-Q04, B8b-Q39, B8a-Q03 and B8b-Q14. All are VERIFIED-PRIMARY, and the quoted fragments are verbatim. No Laura quote is recommended for the notes or the IPS after fix 3.

Check 6: ticker labels. I counted the labels and tickers in each file. After fixes 6-9 and 18, every file that names a ticker carries the full PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK label and points to same-type alternates. `securities_and_allocation.md` s1-s2 and s6 list them.

Check 7: certainty words. I grepped for "guaranteed", "100%", "matched" / "match", "risk-free", "riskless" and "misses none". Remaining uses are banned-word lists, composition statements ("100% stocks"), MODEL-labelled results that hold by construction (the gift is at or above the bottom in 100% of paths), or "matched" for the real plan's held-to-maturity Treasuries (spec B2(a)). None applies to a fund.

Check 8: file hygiene.

- **Required parts:** each file has status labels, a 5-line summary (`securities_and_allocation.md` after fix 10), a "Serves:" line naming TN and/or IPS, and a closing "What this teaches".
- **Scripts:** the scripts the files tell the team to run exist and accept the stated flags (`D9_draft_checker.py --kind pitch|ips|note|reflection`, `D9_ips_page_fit.py --draft`, `A2_curve_recheck.py`, `D9_numbers.py`).

Nothing was committed.

What this teaches

A final check is worth most where files copy numbers from each other. The one stale figure survived because a note said "fixed in three files" when a fourth also needed the fix. The privacy slip hid in a truncated log line that nobody meant as advice. Checking against the script's own output, and not against another file's copy of it, is what caught both.

Appendices

The brief every agent worked from, the surviving questions, and the index of scripts and data.

1. Appendix A - The shared agent brief
2. Appendix B - Questions: statistics and the 48 survivors
3. Appendix C - Scripts and data index

Shared agent brief: insight_v1 run (read this FIRST, fully)

RELAYED MESSAGES: if a message from the user is relayed to you while you work (for example a question about what the run will deliver, or a scope change), do NOT stop or replace your assigned task. Note it in one line in your file, apply any scope rule it states (the current scope is brief section 17: Trading Notes and IPS only, no Final Report work), and complete your assignment. The main loop answers the user.

Run started 2026-09-27 (today). Team Caplet (six Australian high-school students: Ray (team leader), Ahaan, Darren, Harry, Eric, Young) in the 2026-27 Wharton Global High School Investment Competition. This run executes the locked prompt `research/ultracode_prompt_v4_deep_strategy.md`. Read that file's PART 1 (context), PART 2 (lessons from last year), PART 3 (mission), PART 5 (rules) and PART 6 (seed questions) in addition to this brief. Also read `CLAUDE.md` (project memory). The official case wins over every other source.

NORTH STAR: "Why would Laura choose our firm's strategy over 6,000 others?" Every output must answer to it. Skill and a correct strategy are necessary but not sufficient; the winning strategy also reflects who she is, what she values and how she thinks, in ways she would recognise as hers.

DESIGN PRINCIPLE (last year's main failure): complexity must earn its place. Best insight, highest value, least complexity. Compliance with this year's WInS rules comes before any strategy idea. Every decision must trace back to Laura or the case.

1. Where things are (all paths relative to repo root /home/user/caplet-investment)

Official (authority order: case > guides > criteria page):

- Client case: `competition/official/2026_27/Laura_Gao_2026_Client_Profile.txt` (4 pages; PDF alongside)
- IPS guide: `competition/official/2026_27/2026_WGY_Investment_Policy-FINAL.txt` (6 pages; PDF alongside)
- Trading Notes guide: `competition/official/2026_27/2026_WGY_Trading_Notes_Analysis-FINAL.txt` (2 pages)
- Deliverables + evaluation criteria: `competition/official/2026_27/SMApply_Deliverables_Page_2026-09-27.md`
- Manifest (SHA-256 verified 2026-09-27): `competition/official/2026_27/manifest.yaml` Market data (team downloads, primary): `competition/official_market_data/` (Treasury par curve Sept 2026 CSV, IEF and TLT fact sheets 2026-06-30, JPM 2026 LTCMA USD matrix PDF). Historical only (NOT this year's list): `competition/historical/2025_26/25-26-WGHIC-Approved-ETF-List.txt`. Verified scripts: `research/verified_2026-09-27/official_curve_pv.py`, `strategy_mc.py` (both re-run 2026-09-27 and reproduce: \$292,264; lock-early surplus p5/p50/p95 \$159k/\$207k/\$273k; growth-first 3.2% miss). Prior AI council work (brainstorming, lower authority, many numbers superseded): `research/council_2026-09-27/` (start with `01_chair_memo.md`, `round2_referee_ruling.md`), `research/blueprints/01_trading_notes_blueprint.md`, `research/team_materials/review_why_laura_and_public_profile.md`. Team's two client documents (NOT in the repo, never copy their text into the repo; read-only):
- `/tmp/claude-0/-home-user-caplet-investment/4ef462a5-52dd-5480-b22c-ecbc4432d179/scratchpad/team_docs/team_client_profile.md`
- `/tmp/claude-0/-home-user-caplet-investment/4ef462a5-52dd-5480-b22c-ecbc4432d179/scratchpad/team_docs/team_why_wharton_chose_laura.md` These are team research from public sources; many claims are unverified (e.g. CCA-Vanderbilt acquisition, festival figures). Treat them as leads to verify, never as facts. Where they conflict with the case, the case wins. Python: use `/home/user/caplet-investment/.venv/bin/python` (numpy, scipy, pandas, matplotlib, pypdf installed). Run scripts from the repo root. Save any new script under `research/insight_v1/scripts/` with a docstring that lists inputs (with status labels) and how to run it.

2. Source access (checked 2026-09-27 11:35 UTC; updated 12:40 UTC)

IMPORTANT: the WebFetch tool is BLOCKED (EGRESS_BLOCKED) for most hosts, but curl through the proxy WORKS. To read a page use the helper (from the repo root): `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL` (prints visible text; handles PDFs) `.venv/bin/python research/insight_v1/scripts/fetch_text.py URL --grep "exact phrase"` (verifies a quote verbatim) or plain `curl -sSL URL`. (Helper fixed 2026-09-27 ~13:50 UTC: an HTML-parsing bug hid text on some pages, e.g. Poets&Quants; a 'NOT FOUND' from an earlier run is not evidence a quote is absent.) WebSearch (load with ToolSearch "select:WebSearch") works for finding URLs; then open the page with the helper before calling anything VERIFIED-PRIMARY. Reachable with curl: home.treasury.gov, fred.stlouisfed.org, www.ishares.com, am.jpmorgan.com, globalyouth.wharton.upenn.edu, magazine.wharton.upenn.edu, lauragao.com, en.wikipedia.org, poetsandquantsforundergrads.com, wghsinvcomp.smaply.us (public pages only), www.federalreserve.gov, nuvoices.com, legacy.diversebooks.org, www.bookweb.org, thenerddaily.com, www.apra.gov.au. Refused by the site (403): www.futurefund.gov.au, afpglobal.org, candid.org help pages, artres.moc.gov.tw, SSRN; ws.dgbas.gov.tw data files fail TLS. If a source is blocked or refuses, say so; never guess its content.

3. Status labels (use on EVERY factual claim and number)

- VERIFIED-PRIMARY: you read it on the primary source page/file yourself (give URL + access date or repo path).
- VERIFIED-REPO-FILE: from an official or primary file in this repo (give path + page/line).
- SNIPPET-UNVERIFIED: from a search result snippet or secondary source you could not confirm on the primary page.
- ASSUMPTION: a modelling or judgement input we chose; say why.
- PARAPHRASE-UNVERIFIED: wording attributed to a person that you could not confirm verbatim on a primary source. Quotes: exact wording only, from the primary source, with URL, date of the piece, and context. Never invent, trim into new meaning, or polish a quote.

4. Hard rules (v4 Part 5 + CLAUDE.md)

- SCOPE = SEMIFINALS ONLY (team leader, 2026-09-27). The Top 50 is chosen on the written deliverables (Trading Notes Oct 23, IPS Nov 6, Final Report Dec 4). Do NOT produce anything for the finale round: no finale pitch prep, no judge Q&A or defence question banks. Questions that fail the filters are simply "parked" (logged), not turned into a Q&A bank. "Judges" below means the semifinal readers of the written deliverables.
- Official materials and verified data only; never invent facts; label assumptions.
- NO submission-ready prose: no drafted elevator pitch, IPS paragraphs, trading notes, reflections, report text or fundraising paragraphs. Produce specifications, evidence, numbers, checklists and "what a strong sentence must contain" guidance that the team turns into its own writing. Wharton AI policy: AI for brainstorming; AI-generated text may not be submitted as students' work; AI use must be credited.
- Privacy: Laura's PUBLIC PROFESSIONAL record only. Do not record family members, partner, birth name, home/address, personal-life details or anything not bearing on her professional/public work, even if a source or the team docs contain it. Never suggest contacting her. No student personal data (no surnames, emails, birthdays).
- Identity: her books and public work are about identity, belonging, immigration and community; you may cite these themes as her public work. Never use identity as decoration or as a reason to pick investments (tokenism).
- Securities and allocations ARE allowed (team leader, 2026-09-27, superseding the earlier "no tickers" rule): you may recommend specific securities (ETF tickers, Treasury maturities/STRIPS) and portfolio weights, both for the WInS portfolio and for Laura's long-term portfolio. This year's approved list, starting cash and WInS rules are still UNKNOWN; the team will check approval later. So every security recommendation must: (a) carry the label "PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading"; (b) give a primary pick plus 1-2 alternates of the same instrument type, so a substitute exists if a pick is not approved; (c) say whether it was on the 2025-26 approved list (historical only: [competition/historical/2025_26/](#)); (d) cite key facts (expense ratio, duration or maturity, holdings/index, liquidity) from the issuer's primary page with access date; (e) trace to the strategy and to Laura or the case. Prefer low-cost, liquid, broad, transparent funds; no leverage, inverse or thematic bets unless the case clearly justifies them.

- Write only under `research/insight_v1/` (your assigned path). Do not edit official files, configs, CLAUDE.md, src/ or tests/. Do not commit or push (the main loop does that).
- Plain English for high-school students who want to learn; define each technical term the first time it appears; end every file you write with a short "What this teaches" note.
- Deadline priority: tier 1 = WInS trading now + Trading Notes (Oct 23); tier 2 = IPS (Nov 6, trading ends and the strategy freezes); tier 3 = Final Report (Dec 4; instructions released Nov 9).

5. The case in brief (verify against the case file; the case wins)

- \$300,000 in at start of 2027 (Year 1); \$150,000 at start of 2028 from publishing advances, speaking, licensing, other ventures. No other additions/withdrawals before 2033. Living costs outside the portfolio. Year = calendar year - 2026; all flows at the beginning of the year.
- Ten fixed \$50,000 payments (NOT inflation-adjusted) at the start of each year 2033-2042 to the residency's operating expenses; funded by the portfolio alone with "a high degree of certainty"; teams may not rely on co-sponsors, grants, fees etc. for them.
- Start of 2033, BEFORE the first payment or any facility contribution: set aside an "operating reserve". Teams recommend its size and initial asset composition, how composition changes as payments approach/are made; define "high degree of funding certainty", explain how it was evaluated, and state assumptions.
- Facility contribution: no preset amount; recommend and justify what she can responsibly contribute after the reserve; "committing all remaining assets could limit her financial flexibility as the project develops". Teams need not size a separate contingency fund/endowment, estimate facility cost, budget, funding gap or business plan.
- 2031 (two years before): she approaches co-sponsors and must describe how much she expects to contribute in 2033, as a credible dollar RANGE plus the team's confidence that the 2033 contribution falls in it; explain favourable and unfavourable outcomes; the range "must also protect the portfolio's ability to fund the ten-year operating commitment". Overpromising damages credibility. Teams draft part of her fundraising materials.
- Co-sponsors etc. may fund the facility and operating support beyond her commitment, but not her commitment.
- She "has been willing to take thoughtful risks" and wants "an appropriate balance between pursuing growth and protecting the capital required for her goals".
- WInS portfolio = implementation of the strategy during the competition; WInS gains/losses NOT added to projections.
- Teams should explain assumptions about investment performance, timing of cash flows, outside funding, the effect of inflation on portfolio projections AND FACILITY COSTS, and financial flexibility.
- Taxes, Taiwan legal/regulatory issues and funding beyond 2042 are out of scope.
- Framing fiction: "You are a team of young analysts at an asset management company. Your portfolio manager (your team's teacher/advisor who makes the final investment decisions for your firm's portfolio)..." Real rules: advisors may not make decisions or trade; the team's advisor handles admin only.
- Case quote: "The only person who needs to believe in something is yourself."

Deliverables (official): Trading Notes Analysis Oct 23 (3 notes quoted exactly from executed WInS trades - "Yes, we will verify this" - each with a reflection of 100 words or fewer: why, fit with strategy, how it serves Laura's goals, funding needs or risk; focus on reasoning, not gains; the note trades need not still be held; must be consistent with the later IPS; not expected: reserve size, projections, facility amount, co-sponsor draft). The guide's own example note buys "an intermediate-term U.S. Treasury bond ETF to reduce portfolio volatility and begin preparing for Laura's future operating commitment". IPS Nov 6: title page (official team name, members "First Name, Last Initial", WInS username) + 50-word elevator pitch + 500-word IPS on pages 2-3 (2-page maximum), Times New Roman 12, double-spaced, 1-inch margins, PDF ≤5 MB; no graphics, charts, images, attachments, external links, footnotes or formal citations; non-compliance = "will not be considered for semifinal selection"; IPS "becomes the official record"; "may not revise its investment strategy after the submission deadline"; not expected in IPS: final reserve calculations, detailed projections, final facility range, co-sponsor draft; focus on "strategy and decision-making framework ...

rather than describing individual investments or presenting detailed financial calculations". Final Report Dec 4 (instructions Nov 9, "beginning of Week 7"). Evaluators consider the three deliverables together. Evaluation criteria (verbatim headings; no weights published): Investment Strategy; Client Knowledge and Objectives; Portfolio Analysis; Articulation of Competition Experience; Creativity and Presentation (the last includes "an authentic team voice" and communicating "Laura's potential facility contribution and investment uncertainty clearly and credibly to prospective co-sponsors").

6. Verified numbers (reproduced 2026-09-27)

- Treasury par curve 2026-09-25 (treasury.gov CSV in repo): 1m 4.04, 3m 4.24, 6m 4.33, 1y 4.50, 2y 4.81, 3y 4.94, 5y 4.98, 7y 5.06, 10y 5.17, 20y 5.54, 30y 5.49. The 10y rose ~38bp during September 2026 (4.79 on 9/1).
- Ten payments valued at 2027-01-01 on that curve: \$292,264; -25bp \$299,596; -50bp \$307,135; -100bp \$322,864; +50bp \$278,198; +100bp \$264,890. DV01 \$289/bp; duration 9.90y; headroom under \$300k: \$7,736 (27bp). Forward zeros 2027-2033/2037/2042: 5.08%/5.24%/5.48%. Same payments valued at 2033-01-01 on forwards: ~\$394.9k.
- IEF effective duration 6.95y, TLT 15.31y, 0.15% expense each (fact sheets 2026-06-30) -> 64.8% IEF / 35.2% TLT duration match (duration match only; NOT a cash-flow match).
- JPM 2026 LTCMA (data 2025-09-30): U.S. large cap 6.70% compound (7.94% arithmetic, 16.47% vol); AC World 7.00%; EAFE 7.50%; U.S. intermediate Treasuries 4.00%; long Treasuries 4.90%; cash 3.10%; inflation 2.50%; correlation U.S. large cap vs intermediate Treasuries -0.01.
- strategy_mc.py (200k paths, seed 20260927): lock-early never misses (BY CONSTRUCTION); 2033 surplus p5/p50/p95 \$159k/\$207k/\$273k; 2031 floor (80% of sleeve in 2y at 4.81%) p5/p50/p95 \$127k/\$165k/\$217k. Growth-first 3.2% miss (13.7% if 2028 deposit \$75k; 40.8% if \$0); surplus \$20k/\$226k/\$540k. Sleeve equity 50/60/70%: median \$205k/\$207k/\$209k, p5 \$164k/\$159k/\$154k. Equities 5.0% compound: L median \$200k; G miss 5.5%.
- UNVERIFIED (snippets): Fed hike to 3.75-4.00% on 2026-09-16; Taiwan construction costs +6.54% y/y (Aug 2026); Taiwan CPI Aug 2026 2.04% (another snippet says 2.4%); USD/TWD ~31.8 (FRED mirror; 10y daily vol 4.7%); last year's list is historical only; "sector minimum = team size", "200-trade cap", "first trade by Oct 10" are third-party claims only; 2025-26 starting cash \$500k (this season's unknown).

7. Current strategy ("lock early"; NOT yet approved by the team)

Buy Treasuries matching the ten payments in January 2027 (fits inside \$300k at today's curve; if rates fall first, buy the longest-dated payments first and top up the rest from the 2028 deposit); take market risk only with the surplus ("growth sleeve", ~60% equity); in 2031 promise co-sponsors a floor already bought in a 2-year Treasury plus an upside with a stated probability; "high certainty" = fully funded at market prices, cash-flow/duration matched, residual risk U.S. default ("certain in nominal USD, barring U.S. Treasury default"); after 2033 keep leftover money mainly as project flexibility. In WInS, mirror the post-2028 target (~65% Treasuries in an IEF/TLT duration mix, ~35% broad equity).

8. Already answered convincingly by prior work (do NOT re-ask as open questions; you may challenge them only

with NEW evidence, and say what is new)

1. Lock early beats growth-first / wait-until-2033 (G shortfall 3-8%, 13.7-17% at a \$75k 2028 deposit, ~41-55% at \$0; partial locks 50/80/90/95% still fail 35/26/14/3.4% with no 2028 deposit).
2. Rolling 2037-42 payments short ("tiering") is rejected (+\$20k cost per -100bp, +\$43k per -200bp).
3. Risk is measured on the surplus; the equity weight applies to the sleeve only.
4. Price off one bootstrapped zero curve: ladder \$292,264 (not \$295k/\$297k/\$333k).
5. More sleeve equity buys range, not median (+\$0-10k median vs -\$11-47k at p5).
6. Certainty is market-consistent/model-free (3-part definition); a model "95%" is fragile (no-lock 3.3-9.3% spread comes from rate-drift assumptions).
7. The 2031 floor must be bought (2y Treasury), not a model percentile; a missing 2028 deposit means "aspiration only".

8. Missing 2028 deposit under lock-early: payments safe, facility ~\$7-11k ("correct failure mode").
9. Which number is the reserve (cost ~\$292k in 2027 / market value ~\$395k on 2033-01-01 / face \$500k): report market value on 1/1/2033; reconcile "funded 2027, set aside 2033" in one sentence.
10. TIPS are not the core hedge (liability is nominal).
11. STRIPS (zero-coupon) beat coupon Treasuries for matching (coupon reinvestment risk ~\$5k).
12. iBonds Dec-20XX Treasury ETFs cover 2033-37 payments only; nothing found for 2038-42 (snippets).
13. IEF/TLT ~64.8/35.2 duration match; twist gap ~\$3-4k cannot be closed with two funds.
14. JPM 6.7% is compound (7.94% arithmetic).
15. Taiwan construction costs ran ~3.2x CPI (snippets): don't inflate facility cost at CPI; TWD risk cannot be hedged inside a USD promise: disclose, quote in two currencies.
16. Diagnosed bugs: actuary DF(0) bug; Vasicek drift bias; team's "\$626k" implies 5.99%.
17. List & Lucking-Reiley (2002) seed-money effect: a mechanism, not a prediction for institutions.
18. Judging is not on P&L; Top 50 chosen on written deliverables; three deliverables judged together.
19. Case-reading errors in team docs already corrected: facility is not a dated fixed liability; no income-gap cash buffer (living costs outside); residency is a creative residency in Taiwan, not her home; no ESG preference is named in the case; teaching income is not part of the case.

9. Open or disputed (good territory for questions and research)

- Sleeve equity weight 50/60/70% (is ~20-24% total equity after 2028 Laura's "thoughtful risk"?).
- 2031 rule parameters (share of sleeve locked as floor; USD-only vs also TWD quote; convert part of floor to TWD?).
- WInS mirror: post-2028 ~65/35 vs literal 2027 book (~99% bonds); the mirror implies ~35% total equity while the plan implies ~20-24% (inconsistency).
- WInS rules unknown: approved list, capital, trade limits, sector minimum and whether broad ETFs count, whether Treasury/defined-maturity/STRIPS funds are allowed.
- Rule if rates fall before January 2027 (P(cost > \$300k) estimated 23-38%, ASSUMPTION-based): long-rungs-first vs nearest-first vs staged purchase.
- Joint tail: rates fall AND the 2028 deposit is missing/cut (up to ~\$23-33k of the 2033 payment unfunded) - UNANSWERED.
- Real erosion of the fixed \$50k operating budget (inflation/construction) - UNANSWERED; must be said aloud.
- Case interpretation: does "set aside at the beginning of 2033" allow economic funding in 2027? Is "high certainty" a number or "bought, not forecast"?
- Probability and correlation of a smaller 2028 deposit with a bad 2027 market (wrong-way risk).
- Stock-rate correlation relevant to the 2033 reserve cost.
- EWT/Taiwan tilt keep or drop; human-capital underweight of publishing/media; how to present AI exposure (her choice).
- Coupon-reinvestment rule if STRIPS unavailable; facility "scope clause" if building costs exceed X.
- Post-2033 flexibility: how much to keep after the facility contribution.
- Rebalancing bands; which three trades become Notes.

10. Known cross-file inconsistencies (resolve against official/primary files; the later verified value wins)

Ladder cost \$295k (chair memo) vs \$292,264 (verified); Taiwan CPI 2.4% vs 2.04%; rung order nearest-first (round 1) vs longest-first (round 2, current); 20y yield 5.45/5.51 vs 5.54 (official); TWD vol "5-7%" vs 4.7% computed; coupon arithmetic \$20k/\$6.7k vs \$15.5k/\$5.2k; Fed hike "verified in search" vs UNVERIFIED; chair memo's "Final Report (due Nov 9)" is wrong (instructions Nov 9, report Dec 4); IEF duration 6.95 (fact sheet) vs "~7.5" (blog); winners.md lesson "≥95% MC + floor" contradicts the council; 2025-26 field "6,300+ registered" vs "nearly 6,000 competed".

11. Known gaps in the verified model (strategy_mc.py)

Lock-early "0% miss" is by construction (assumes purchase at today's curve and a perfect cash-flow match); no 2028 top-up logic; no rate risk between Sept 2026 and the Jan 2027 purchase (27bp cushion); 2028 deposit independent of markets (no wrong-way risk); 2033 rate independent of bond returns; i.i.d. lognormal returns (no fat tails), no fees; growth sleeve uses U.S. large cap only; flat 5.26% used for the growth-first 2033 reserve (~\$401k) vs ~\$395k on forwards (slightly overstates G's shortfall).

12. Output hygiene for every agent

Write your detailed file at the exact path you are given. Structure: short summary first; then the body; every claim labelled; sources listed with URL + access date; a "What this teaches" note at the end. Keep jargon out or define it. Return the structured summary you are asked for (it feeds the next phase).

13. PRIORITY: understand the client deeply (team leader, 2026-09-27) and how Wharton judges

The single most important thing in this run is a DEEP understanding of Laura Gao: her goals, constraints, risk tolerance (willingness vs ability), time horizons, values and way of thinking, researched BEYOND what the case states (her public professional record, her books, her own words), and a strategy tailored to her rather than a generic portfolio. Every agent should ask: "does this make the plan more hers?"

How Wharton judges (the official 2026-27 wording is in SMAApply_Deliverables_Page_2026-09-27.md and wins; the team leader's summary below also draws on past-season Wharton teacher guides found on Scribd, a secondary source, SNIPPET-UNVERIFIED for 2026-27):

- Investment Strategy: a clear, creative investment thesis; primarily long-term with appropriate shorter-term thinking; portfolio decisions consistently follow that thesis; appropriate diversification.
- Client Knowledge & Objectives: understand the client deeply; research beyond what is given; tailor the strategy to her goals, constraints, risk tolerance and time horizon rather than a generic portfolio.
- Portfolio Analysis: genuine understanding of the investments; quantitative AND qualitative analysis; explain why each security belongs and how it fits the strategy.
- Articulation of Competition Experience: explain the decision-making process, not just the final portfolio; show teamwork, communication, challenges, adaptations and learning.
- Creativity & Presentation: a compelling narrative; effective graphs/visuals rather than walls of text (Final Report only: the IPS bans charts); an authentic team voice rather than artificial sophistication; clear, persuasive communication. Implications for every output: tie each recommendation to one of these five areas; prefer insights that show understanding of Laura over technical flourishes; where an idea would be best shown as a chart in the Final Report, say which chart and what data it needs.

14. PHASE A RESULTS (2026-09-27; these SUPERSEDE earlier sections where they conflict)

Files: `research/insight_v1/phase_A/case_register.md` (324 R-ids: R-C case, R-I IPS guide, R-T trading-notes guide, R-S SMAApply page, R-W public Wharton web pages, R-AN anomalies), `fact_register.md` (F-ids), `stakeholder_map.md` (SH/BS ids), `wins_week1_guardrails.md` (G ids). Anchor your work to these ids. WInS 2026-27 rules are now VERIFIED-PRIMARY (public SMAApply "Trading Details" and FAQ pages, read 2026-09-27; R-W56-R-W92, F-605-F-608, G1-G4): \$300,000 virtual starting cash (= Laura's Year-1 deposit; the \$150k is not added); trading 2026-09-28 to 2026-11-06, then the portfolio is frozen; up to 200 trades; no trade above 2x a security's daily volume; permitted: cash, stocks priced $\geq$ \$5, "Any Exchange-Traded Funds (ETFs) available on WInS", "Any Government/Treasury Bonds from any exchange available on WInS" (so INDIVIDUAL Treasury bonds are a permitted type); banned: margin, short selling, stock-secured debt, crypto, derivatives, anything else; NO sector minimum (the third-party "sector minimum = team size" claim is FALSE); commissions \$25 per stock/ETF trade and \$10 per Treasury bond trade; bond prices update daily, coupons semiannual; no separate approved ETF list this year (the 2025-26 list is history only). 2026-27 WInS User Guide: "Day Trading: This is not permitted."; a per-security "Position Limit" exists but its value is shown only on the logged-in Portfolio Summary > Session Rules page (the Stock-Trak default is 25% per security - UNVERIFIED for this season); trade notes cannot be edited, only added to. The SMAApply FAQ says "Investments permitted (for BOTH contributions)" (meaning unclear). Remaining approval check for any security: is it actually listed in WInS, and does it fit the Session Rules position limit? So recommendations are now "PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK". Semifinal basis: the Rules & Roles page

says Top 50 are chosen on the IPS and Final Reports; SMAApply and the case say all three deliverables are evaluated - treat all three as scored. AI policy: "If you use AI to assist you in any way during the competition, how you use it must be recorded in your Works Cited pages"; students "must use their voice and words". The Rules page tells teams to operate by the CFA Institute Asset Manager Code. Contacting the client = disqualification. "While the client is real, the financial scenario is developed specifically for the competition." Field: 2025-26 had 6,300+ registered teams but ~2,300 (2,339) submitted final reports. Updated numbers (VERIFIED-PRIMARY unless noted): FOMC 2026-09-16 raised to 3.75-4.00% (12-0), IORB 3.90%, SEP medians fed funds 4.1 end-2026, longer run 3.2, PCE inflation 3.7% in 2026. IEF duration 6.86y (YTM 5.17%), TLT 14.88y (YTM 5.54%) as of 2026-09-24 -> duration match to the 9.90y liability (valued at 2027-01-01) = 62.2% IEF / 37.8% TLT; to the SPOT duration today (10.16y; spot PV \$288,924) = 58.9% / 41.1% (use spot duration for hedging the WInS book today). The same ten-payment ladder cost MORE than \$300k on 173 of 185 trading days of 2026 (\$316.5k on Jan 2; \$325.6k on Feb 27; \$313.0k on Jun 30); it has been <= \$300k only since 2026-09-10; P(cost > \$300k on 2027-01-01) ~24% (ASSUMPTION: zero-drift lognormal, 2026 realised vol 7.24%). Value of the payments on 2031-01-01 at forwards \$356,384; on 2033-01-01 \$394,930. Taiwan CPI Aug 2026 +2.04% y/y (the 2.4% figure is wrong); DGBAS forecasts 2.07%/1.90% (2026/2027), CBC 2.03%/1.83%; CBC discount rate 2% (held 2026-09-17); Taiwan overnight call rate 0.812%. Taiwan construction cost index 119.50 (Aug 2026, 2021=100), +6.53% y/y, but flat in 2024-25 and ~3.54%/yr since 2021 (the "3.2x CPI" ratio is a one-year spike). USD/TWD 31.82 (FRED, 2026-09-18); vol of daily log changes 1y 4.03%, 3y 5.64%, 5y 5.25%, 10y 4.69%; 2-year realised moves sd 6.8% since 2006, 9.4% since 1983. iShares iBonds Treasury funds exist for Dec 2026-2036, 2044-46, 2054-56, NONE for Dec 2037-2043 (only the Jan 2033-2037 payments can be matched by them). JPM 2026 LTCMA also: EM equity 7.80% compound (9.74% arith, 20.93% vol); long Treasuries 4.90/5.69/13.02; TIPS 4.30; U.S. Aggregate 4.80; large cap vs long Treasuries correlation 0.02; data date 2025-09-30 when the 10y was 4.16% (~100bp below today) - bond assumptions look ~1pp conservative vs today's yields; the 2027 edition is not out yet. Blind spots found (stakeholder map): no management-fee assumption anywhere (~\$17k lower median surplus at 0.5%/yr, ~\$34k at 1.0%; ASSUMPTION arithmetic); the CFA Asset Manager Code unused; credibility is part of Laura's human capital (her 2028 income is reputation-driven); funders underfund operations, so her covering ten years of operations lets co-sponsor money go to the building; the case never names a currency ("dollar range") - write US\$ explicitly; book advances arrive in instalments, so the 2028 deposit can be LATE as well as smaller; school documentation has no owner yet. The case pull quote "The only person who needs to believe in something is yourself." has no attribution in the PDF text layer: verify before presenting it as Laura's words.

15. NEW OFFICIAL DOCUMENTS (found by B7a, filed 2026-09-27)

competition/official/2026_27/2026_WGY_Investment_Competition_Guide.{pdf,txt} (5 pages) and 2026_WGY_Competition_Infographic.{pdf,txt} (1 page), from the public SMAApply guide page; SHA-256 in manifest.yaml. Key lines (quote from the .txt files; cite as "Guide p.N"): "Judges want to understand not only what your team decided to do, but also the reasoning, assumptions, and tradeoffs behind those decisions." (p.5); "The competition recognizes thoughtful strategy, research and analysis, client understanding, disciplined decisions, management of risk and uncertainty, communication, and creativity." (p.5); "Your team will need to consider the client's goals, time horizons, required cash flows, funding commitments, liquidity needs, desired degree of funding certainty, risk tolerance, and communication with potential co-sponsors." (p.2); "It is up to your team to determine what additional research and analysis you need" (p.2); "The goal is to develop a cohesive strategy in which your investment decisions work together to support the client's objectives across a range of possible market outcomes." (p.2); the Trading Note "should capture the reasoning behind the decision, including its alignment with your strategy, the supporting research or analysis, and its expected role in growth, liquidity, risk management, or future funding." (p.3); WInS "is not the competition scorecard" and teams are not evaluated on ranking, number of trades, outperformance or making money (p.3); "A long-term strategy may include planned adjustments as funding dates approach, but it should not be rewritten simply because markets move or hindsight reveals a different outcome." (p.5). Infographic: "Strategy guides every decision."; "Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions." Phase B produced 472 questions (20 agents) and 271 Laura quotes (259 marked VERIFIED-PRIMARY by the agents; D13 will re-verify every one).

16. PHASE C AND D13 RESULTS (2026-09-27/28; supersede earlier sections where they conflict)

Phase C: 472 questions -> 248 merged -> 48 survivors (phase_C/survivors.json, filter_log.md); 200 parked (phase_C/parked.json, mostly judged "already answered" by Phase A, the wins_now ticket or the council). High-priority parked items that were killed as already answered (their answers must still reach the final spec): M008 (2031 message order: floor bought, conditional stretch, invitation to co-fund), M007 (bad-year rule), M015 (fundraising leads with the fully pre-funded ten years of operations), M021 (dated decision log from today), M149 (funded-status glide path wording), M033 (whole-portfolio equity share visible to a first reader), M035 (diversification shown to a fixed-income reader), M040 (range as a rule vs a forecast). D13 quote verification (phase_D/D13a_laura_quotes_verified.md/.json, D13b_other_quotes_verified.md/.json, D13c_voice_map.md): Laura quotes 91 VERIFIED-PRIMARY, 6 PARAPHRASE-UNVERIFIED, 7 EXCLUDE-PRIVACY; all 171 non-Laura quotes verified (2 wording corrections; 6 attribution warnings, e.g. Wharton paraphrases of a judge and a past client). RULES FROM D13 (binding for all later phases):

- Only D13a VERIFIED-PRIMARY lines may be presented as Laura's words, with outlet, year and context; label student-era lines; several verified lines are safe only in context (e.g. "risk-adverse" is about choosing a business major, not investment risk; "No more gimmicks." is about fantasy drafts; "explain myself" is about language choices) - read the D13a notes before using any line.
- The case pull quote "The only person who needs to believe in something is yourself." is VERIFIED-REPO-FILE as the CASE's text but NOT verifiable as Laura's words: cite it as the case's words only.
- Do NOT describe a "floor-first leap" (quit only after the book deal secured a floor) as Laura's trait: it rests on a reporter's sentence next to personal material, not her words.
- There is no verified statement by Laura about investing: never write her "investment philosophy". No "falling safely" / book-title puns; the case's own "appropriate balance" is fine.
- No Laura quotes in the Trading Notes or the IPS (the IPS bans formal citations); at most 1-2 in the Final Report.
- Her verified words carry two risk ideas: regret-driven leaps ("you should always take the jump because you're always going to regret not doing it") and not letting down "people who were your earliest supporters and were the first to believe in you" (student-era, Daily Pennsylvanian 2016). The second supports "promise first"; the first means the plan must honestly answer why whole-portfolio equity is only ~0-2% in 2027, ~20% in 2028-30 and ~4% after the 2031 floor.
- Two certainty gaps must be named plainly: the ladder may cost more than \$300k in January 2027 (roughly 1 in 4 to 1 in 3 modelled rate paths), so part of the promise can wait on 2028 income she called "unstable"; and "certain" is in nominal USD while the residency's costs are in Taiwan and a fixed \$50k buys less each year.

17. SCOPE: NO FINAL REPORT WORK YET (team leader, 2026-09-28; supersedes earlier mentions of the Final Report)

Work only for (1) WInS trading now, (2) the Trading Notes Analysis (Oct 23) and (3) the IPS (Nov 6). Do NOT produce Final Report material: no Final Report narrative, chapter plans, visuals/chart specs, Works Cited plans, fundraising excerpt drafts or checklists, final operating-reserve calculations, detailed projections, or final 2031 range numbers presented as Final Report content. BUT the IPS freezes the strategy on Nov 6 and the Final Report must later apply it without redesign (Guide p.5), so the STRATEGY must already contain the decision RULES the Final Report will apply: how the operating reserve is funded and changes; how the 2031 co-sponsor range is determined (the method, not final figures); how the facility contribution and financial flexibility are decided; which "if X then Y" rules apply. Put such items under "IPS: rules to fix before Nov 6". Anything purely Final-Report goes to a short "later (after Nov 9)" list in open_questions.md, one line each. Tightened the same day (team leader): "We only need items for trading notes and IPS." WInS trading is in scope only as the source of the Trading Notes (notes must quote executed WInS trades). Every output must say which of the two deliverables it serves (TN or IPS); drop anything that serves neither. The two primary outputs of this run are trading_notes_pack.md (which trades to make and when, what each WInS note and each <=100-word reflection must contain, checklists only) and ips_spec.md (what the 50-word pitch and the 500-word IPS must contain, the decision rules to fix before Nov 6, format compliance) - specifications, never drafted text.

Appendices · generated from the run's files

Appendix B - Questions: statistics and the 48 survivors

Serves: background for the Trading Notes and IPS. Generated from the run's JSON files (phase_B/questions.json, phase_C/merged_questions.json, survivors.json, parked.json, phase_B/quotes_raw.json).

B.1 How many questions and quotes

- Phase B produced **472 questions** from 20 agents and **271 quotes** (verified in Phase D13).
- The librarian merged them into **248 distinct questions**; three skeptics judged each one.
- 48 survived** all four filters and went to the specialists; **200 were parked** (listed one line each in *Open questions*, Part 1).

Lens	Questions
B1 Case anomalies	57
B2 Why Laura / designer intent	41
B3 Laura's world and income	41
B4 Practitioner benchmark	50
B5 2026 macro and markets	47
B6 Co-sponsors	43
B7 Competition and judges	54
B8 Voice of Laura	53
B9 Client synthesis	46
B10 Contrarian	40

B.2 The 48 questions that survived (ranked by the skeptics' priority, 1-5)

#	Question (merged)	Specialist	Priority	Deliverables
M003	What explicit rule or share of the post-reserve surplus should stay uncommitted as 'financial flexibility' (anchored to Laura's stated wish to keep options open and try many ventures, and checked against benchmarks such as Brunel's 10%), rather than 'whatever is left' or simply the unpledged upside, and does she rank it above a bigger facility contribution?	D5	5.0	FR, IPS
M006	Which contingent 'if X, then Y' rules (rates fall before Jan 2027; 2028 deposit smaller or late; 2031 floor; 2033 reserve; facility share), each with trigger, action and record, must the IPS pre-declare before the Nov 6 freeze so the Final Report's numbers read as applications of the IPS rather than a redesign after results?	D8	5.0	FR, IPS
M004	If rates fall and the ladder costs more than \$300k on the January 2027 purchase date (it did on 173 of 185 trading days in 2026), should the IPS base rule be 'buy as much as the first deposit affords, longest rungs first, and complete it from the 2028 deposit' with a stated shortfall limit and 'fully funded by 1 January 2028 at the latest' as the headline, and should the purchase be made in full, in stages or on yield triggers in a way Laura would recognise as hers?	D1	4.67	FR, IPS, TN, WInS-now
M005	Should the WInS book (exactly Laura's \$300k Year-1 deposit; the \$150k is never added) mirror her literal 2027 portfolio (about 97% Treasury ladder plus a small labelled surplus sleeve, since her ability to take risk only switches on in 2028) rather than the post-2028 ~65/35 target mix, and which one IPS sentence says which book WInS represents?	D10	4.33	FR, IPS, TN, WInS-now
M018	Does Laura's documented career 'thoughtful risk' (leaving a steady tech job only after the HarperCollins book deal was signed; her talk 'Taking the Leap to Pursue Art Full-Time from a Traditional Tech Job') show 'floor first, then leap', and should the IPS use it as evidence for 'lock the promise, then take risk with the surplus' instead of 'high appetite, so more equity'?	D6	4.33	FR, IPS, TN
M237	When every tested design has about the same median 2033 surplus, what rule should choose between them: the p5 Laura can accept, the p95 she hopes for, the floor co-sponsors can bank, or the simplest structure?	D6	4.33	FR, IPS

#	Question (merged)	Specialist	Priority	Deliverables
M001	Should every WInS trade note from now on open with a one-line purpose, name its role in the Guide's own words (growth, liquidity, risk management, future funding) using the same fixed labels the IPS and Final Report will use, link it to one dated cash flow of Laura's, and carry one dated, checkable data point and one named risk?	D9	4.0	FR, IPS, TN, WInS-now
M071	How many of the IPS's 500 words should describe Laura as a person rather than her goals, which beyond-the-case facts earn a place, and what may each sentence rest on so she feels understood rather than analysed?	D9	4.0	FR, IPS, TN
M002	Should the three Trading Notes be one 'supported', one 'tested' and one 'refined' decision, and which pre-written rule (a WInS rebalancing band or rate trigger) and which dated market events between Sept 28 and Oct 20 can make a genuine rule-triggered trade happen before the Oct 23 deadline?	D7	3.67	IPS, TN, WInS-now
M011	Since lock-early is now the textbook answer that AI tools and hundreds of teams will give, which 2-3 Laura-specific, checkable decisions make our 'clear and creative thesis' different, and what one claim can only our strategy truthfully make in the 50-word pitch?	D7	3.67	FR, IPS, TN
M019	Should the IPS state Laura's risk tolerance, which the case never states, in two parts, ability (low for the promise, high capacity for loss on the sleeve) and willingness (inferred without speaking to her, so ~60% sleeve equity is set by willingness and the 2031 promise), and which evidence would change that inference?	D6	3.67	FR, IPS
M026	What fee would a real adviser disclose for Laura's ~\$450k, and should the IPS say the reserve is held as directly owned Treasuries needing almost no management, with fees charged on or paid only from the growth sleeve so the ladder's ~\$7.7k headroom is never eroded?	D8	3.67	FR, IPS
M034	Should every sleeve recommendation be judged against Laura's riskless control (the whole surplus locked at ~5.08%, ~\$204k in 2033) rather than the S&P 500, given 60% sleeve equity gives median ~\$207-210k but p5 ~\$159-162k and ends below \$204k in 42-46% of paths, and would an all-Treasury rival beat us on simplicity?	D3	3.67	FR, IPS, WInS-now
M109	What is Laura's required return per goal (payments: the locked Treasury rate, only ~1.05%/yr across both deposits; facility: a stated hurdle), and should the IPS say her needs can be met below today's Treasury yield instead of setting one portfolio return target?	D8	3.67	FR, IPS
M143	Should the IPS name, as the client benefit, that from the Treasury purchase onward her ten-year promise stops depending on her own future earnings (book deals, speaking, licensing)?	D9	3.67	FR, IPS, TN
M241	Should the IPS justify where risk sits with the case's own asymmetry, that outside money may fill a facility shortfall but never an operating shortfall?	D9	3.67	FR, IPS
M030	How many extra points a year does sleeve equity really expect over simply buying Treasuries in 2026 (JPM 6.7% vs Vanguard 4.2-6.2%; CAPE ~41 vs forward P/E 19.1; ~5.0% on a 6-year Treasury), and does that move the sleeve equity weight from 60% toward 40-50%?	D2	3.5	FR, IPS
M243	With the 2028 deposit present (the case says she 'will' make it), a 70-95% lock also shows no payment shortfall in the model; what is our real reason to lock 100%, and where are we overstating it?	D3	3.5	FR, IPS
M031	Which single Final Report chart shows the promise and the 2031 range most honestly (ten 'bought' blocks plus a floor bar with a narrow ~±5% whisker for three 2031 starting states, versus a fan chart of the 2026-view \$159k-\$273k band), in a form Laura, an illustrator and data-story maker, could redraw for co-sponsors?	D9	3.33	FR
M039	Since Laura decides the 2033 contribution herself, should 'will fall within that range' be partly a policy outcome (contribute at most the top, keep any excess as flexibility), so the stated confidence is the probability of affording the bottom and the top is set at a stated percentile?	D3	3.33	FR, IPS
M044	Should the WInS book hold one or more individual Treasuries maturing near Laura's payment dates (Nov 2032-Nov 2041) to show cash-flow matching rather than only duration matching with constant-maturity ETFs, and which of those maturities does the WInS bond list actually offer?	D1	3.33	TN, WInS-now
M060	What share of the 2031 sleeve should be locked as the floor (80%, or toward 100%, which costs only about \$1k of median 2033 value but shrinks the range toward one number), and does a near-total lock dodge the case's uncertainty task while funders reward stretch gifts?	D3	3.33	FR, IPS

#	Question (merged)	Specialist	Priority	Deliverables
M062	Before more un-editable WInS notes are saved, should the team fix one certainty vocabulary (bought/locked, model-expected, possible) and tie each of the case's three words (certainty, confidence, reliability) to one object, so no note says 'guaranteed' or 'safe' in a way later deliverables must qualify?	D9	3.33	FR, IPS, TN, WInS-now
M226	Why is the longest-horizon money (reserve, to 2042) invested most safely while the facility money (needed by 2031-33) takes the risk, and which one line answers a reader who expects 'longer horizon = more stocks'?	D9	3.33	FR, IPS
M228	What job is Laura hiring an asset-management firm to do: grow her money, or let her make public promises in 2031 that she never has to walk back?	D6	3.33	FR, IPS
M012	Since hundreds of teams will clone Wharton's example note (a Treasury ETF bought to 'begin preparing' and 'reduce portfolio volatility') and stocks and Treasuries fell together in 2026, what single fact and wording should our Treasury note lead with ('matches her ten payments', bought all at once, now) so long-duration bonds read as reducing risk to Laura's promise rather than to portfolio volatility?	D9	3.0	TN, WInS-now
M028	Should the team log from the first WInS trade the daily value of Laura's ten payments on that day's curve (~\$289 per bp), the ladder cost, the funded ratio and the WInS hedge value, so a Trading Note can report whether the IEF/TLT mix tracked the promise within X%?	D3	3.0	FR, TN, WInS-now
M036	With charts, citations and 'detailed financial calculations' excluded, which few numbers and rules must the 500-word IPS carry in words to be testable, and are small tables allowed?	D9	3.0	IPS
M063	Is this portfolio all of Laura's investable wealth or one goal-dedicated pool (the case says she has resources outside it), and should the IPS say so and let that raise her capacity to take risk with the surplus?	D8	3.0	FR, IPS
M091	Would a professional co-funder count any part of Laura's range as money raised beyond the written, bought floor, and should the fundraising part therefore lead with the floor rather than the range or its midpoint?	D5	3.0	FR
M107	Under a floor-plus-cap range rule, where does good-market money go if the 2033 portfolio lands above the top of the 2031 range, and does sending it to flexibility undercut the 'meaningful contribution' signal and the case's demand to show how favourable outcomes affect the amount?	D5	3.0	FR
M121	Which three tradeoffs should the IPS name explicitly, since the Guide, the Infographic and the FAQ all say strong teams explain 'reasoning, assumptions, and tradeoffs'?	D9	3.0	IPS
M125	Past clients said what won them over (feeling 'seen', plans 'not just about the numbers'); with no personal hooks in Laura's case, which case-stated facts about how she thinks should every recommendation visibly mirror so she feels understood rather than analysed?	D6	3.0	FR, IPS, TN
M191	Should the confidence statement give the chance of missing the range on each side separately (below the floor versus above the top), following the IPCC advice to give reciprocal statements?	D9	3.0	FR
M212	What single table would let a reader verify in one minute that our TN, IPS, WInS book and FR tell one strategy (IPS rule, WInS trade, note quoted, FR evaluation)?	D9	3.0	FR, WInS-now
M231	Should every key number in the Final Report carry an evidence grade (observed price / historical estimate / assumption, plus a quality word), as her own 'Match Quality' column does?	D9	3.0	FR
M238	Should the sleeve become a barbell: in January 2028 lock a facility floor in a Treasury maturing just before 2033 and hold the rest 100% in broad equity, instead of a rebalanced 60/40 sleeve plus an 80% lock in 2031?	D3	3.0	FR, IPS, TN, WInS-now
M247	Can our three Trading Notes, drawn from a mostly-Treasury book, show distinct official roles (future funding, growth, risk management), and have we placed a growth trade with a role-worded note yet?	D9	3.0	TN, WInS-now
M013	Which governance elements must the 500-word IPS carry (who decides what among automatic rules, the firm and Laura; dated reviews in 2028, 2031 and 2033; event triggers such as a funded ratio below 100%, a short or late deposit or a residency delay; one rebalancing rule), and which can be cut for a nameless, sizeless firm?	D10	2.67	FR, IPS
M053	How large is the joint tail in which rates fall before January 2027 (ladder above \$300k, long rungs bought first) and the 2028 deposit is then late or smaller, for example through a 2027 U.S. recession that both lowers yields and shrinks her publishing and speaking income?	D3	2.67	FR, IPS

#	Question (merged)	Specialist	Priority	Deliverables
M081	Under Chhabra's risk-allocation buckets, is Laura's aspirational bucket already filled by her career, future books and the residency, so the portfolio should hold only safety (reserve) and market (broad sleeve) assets and refuse thematic, single-stock or country tilts?	D8	2.67	FR, IPS, WInS-now
M094	What minimum checklist must a funder-grade founder contribution statement in Laura's fundraising part meet to be credible to a finance reader and a non-specialist at once, and what can be dropped to keep it short?	D5	2.67	FR
M095	Which is largest for what Laura's facility money buys by 2033: market risk, USD/TWD risk (2-year p5/p95 -12.1%/+10.6% since 2006 against a ~-4%/+5% market band) or Taiwan construction inflation, and does any NT\$ figure need its own band or an 'illustration at a dated rate' label?	D4	2.67	FR
M234	Before submitting, should the team usability-test the co-sponsor paragraph and the IPS rules with about five non-finance readers and report what changed, as she tested products with nurses and doctors?	D9	2.67	FR
M235	What is the smallest set of rules that still passes all five case tests, and can each extra rule show a measurable improvement?	D8	2.67	FR, IPS
M055	Should the facility floor be locked progressively before 2031 (a ratchet each January 2029-2031, funded-status triggers, or stepping sleeve equity down in 2029-30) rather than on one date, given that a -30% equity year in 2030 at 60% sleeve equity cuts the announced floor by ~18%?	D3	2.33	FR, IPS
M059	Should the sleeve results (the 2031 floor, the 2031-33 band, the 2033 facility contribution, and the barbell and ratchet advantages) be checked against real historical paths (after the Dec 1999 CAPE peak, 2000-02, 2008, 2022, and actual two-year outcomes of a sleeve-like mix) rather than only the i.i.d. lognormal model?	D3	2.33	FR, IPS
M197	How wide can the stated range be before non-specialists find it useless, given that the preference for range forecasts peaks at low imprecision and falls as ranges widen?	D6	2.33	FR

What this teaches

Most questions that sound important fail a plain test. Filtering hard before researching is how a small team spends its time on the few that matter.

Appendices · generated from the run's files

Appendix C - Scripts and data index

Serves: checking any number in this record. Every model number can be re-run from the repository root with `.venv/bin/python <script>`. The final strategy's numbers come from `insight_v1/scripts/E6_final_checks.py`; the verified base case from `verified_2026-09-27/strategy_mc.py` and `official_curve_pv.py`.

C.1 Python scripts (61)

Script	What it does (from its docstring)
<code>insight_v1/scripts/A2_curve_recheck.py</code>	<code>A2_curve_recheck.py</code> - independent re-check of every Treasury-curve number the team may quote. What it does (plain English): rebuilds the verified ladder valuation (research/verified_2026-09-27/official_curve_pv.py, same method, re-implemented here so it can run on any date), then answers the follow-
<code>insight_v1/scripts/A2_jpm_extract.py</code>	<code>A2_jpm_extract.py</code> - read the J.P. Morgan 2026 LTCMA U.S. dollar matrix straight from the PDF in the repo. What it does (plain English): opens the PDF with pypdf, finds each asset-class row, and prints the four numbers JPM gives (compound return 2026, arithmetic return 2026, annualised volatility, co
<code>insight_v1/scripts/A2_taiwan_cci.py</code>	<code>A2_taiwan_cci.py</code> - Taiwan construction cost index (CCI, 營造工程物價指數) from the official DGBAS platform. Why: the facility will be built in Taiwan, so building-cost inflation matters for the co-sponsor range. The headline "+6.54% y/y in August 2026" was only a news snippet, and the official download file
<code>insight_v1/scripts/A2_twd_vol.py</code>	<code>A2_twd_vol.py</code> - USD/TWD level and volatility from the primary FRED series DEXTAUS. What it does (plain English): downloads the official daily exchange rate "Taiwan dollars per one U.S. dollar" (Federal Reserve H.10 noon buying rate, FRED series DEXTAUS), then measures how much it moves. "Volatility"
<code>insight_v1/scripts/AX1_quant_audit.py</code>	<code>AX1_quant_audit.py</code> - AX1 (cluster auditor, rates and quant) checks on <code>D3_quant.md</code> and <code>D3_model_v2_results.md</code> . What it does (plain English): re-uses the D3 model code unchanged (<code>strategy_mc_v2.py</code> , <code>D3_funded_status_log.py</code>) and the saved D3 data snapshots to reproduce every number the AX1 audit adds. l
<code>insight_v1/scripts/AX1b_rates_audit.py</code>	<code>AX1b_rates_audit.py</code> - independent checks on <code>D1_rates.md</code> (auditor AX1b, insight_v1 run, 2026-09-28). PROVISIONAL audit research, not team-approved. It reuses D1's own curve code (imported, unchanged) and adds checks that D1 did not run. What it checks (plain English): [A] Worst 2026 day (Feb 27): wha
<code>insight_v1/scripts/AX2_audit_checks.py</code>	<code>AX2_audit_checks.py</code> - reproducible checks behind the AX2 audit corrections to D2, D4 and D10. Agent AX2 (cluster auditor: equity, Taiwan/FX, compliance), insight_v1 run, 2026-09-28. Audit file: research/insight_v1/phase_D/audit_markets_rules.md What it does (plain English): 1. D4 (correction C2): ho
<code>insight_v1/scripts/AY1_audit_checks.py</code>	<code>AY1</code> (cluster auditor: client psychology and co-sponsors) - checks behind the corrections to D5, D6 and D12. How to run (repo root): <code>.venv/bin/python research/insight_v1/scripts/AY1_audit_checks.py</code> (a few seconds) It re-uses the specialists' own engines (same random stream, same inputs), so every dif
<code>insight_v1/scripts/AY2_audit_checks.py</code>	<code>AY2_audit_checks.py</code> - audit checks behind the AY2 corrections to D7 (band odds) and D8 (devil's advocate numbers). Agent AY2 (cluster auditor: Wharton intent, practice benchmark, communication), insight_v1 run, 2026-09-28. What it does (plain English): [1] D8 M109 used a Treasury-forward middle case
<code>insight_v1/scripts/B10b_designer_checks.py</code>	<code>B10b</code> contrarian checks: attack "lock early" from the case designers' side with numbers. What it computes (all ASSUMPTION-based model outputs, not forecasts): 1. "Price of certainty" frontier: lock a fraction f of the ten-payment ladder in January 2027, invest the rest of the money 60/40 (JPM large c
<code>insight_v1/scripts/D10_ips_page_fit.py</code>	<code>D10_ips_page_fit.py</code> - does a 50-word pitch + 500-word IPS fit on the two allowed pages? (question M013, word budget) Agent D10 (Compliance Officer), insight_v1 run, 2026-09-28. What it does (plain English): The IPS rules allow pages 2-3 only ("2-page maximum") for a 50-word elevator pitch plus a 500
<code>insight_v1/scripts/D10_wins_book_compliance.py</code>	<code>D10_wins_book_compliance.py</code> - rule checks for the three candidate WInS books (question M005). Agent D10 (Compliance Officer), insight_v1 run, 2026-09-28. What it does (plain English): 1. Builds the three candidate WInS books at \$300,000: (ii) the post-2028 mix scaled to \$300k (ticket option ii), (ii
<code>insight_v1/scripts/D12_evidence_grades.py</code>	<code>D12</code> (Overflow Researcher, client side), question M231: grade each key number by the kind of evidence behind it. What it does 1. Counts the grades in Laura's own published "Match Quality" column (Rewriting Herstory page) as the precedent. 2. Lists the key numbers the team may use in the Trading Notes
<code>insight_v1/scripts/D12_note_roles.py</code>	<code>D12</code> (Overflow Researcher, client side), question M247: can three Trading Notes show distinct official roles? What it computes 1. Length of Wharton's own example Trading Note (words, characters) against the 300-character Trade Note limit that Stock-Trak's HowTheMarketWorks product states, and a chara

Script	What it does (from its docstring)
insight_v1/scripts/D12_reader_test.py	D12 (Overflow Researcher, client side), question M234: how many outside readers does a restate-it test need? What it computes 1. Nielsen-Landauer problem-discovery curve: share of problems found with n readers, $1 - (1 - L)^n$, for $L = 31\%$. 2. For one specific misreading that a share p of all readers
insight_v1/scripts/D13_mechanical_quote_check.py	D13 step 1 (mechanical): re-fetch every Phase B quote's source page and test whether the exact text appears verbatim (after normalising whitespace and curly quotes). No judgement; agents review context and failures after. Run from the repo root: .venv/bin/python research/insight_v1/scripts/D13_mecha
insight_v1/scripts/D13a_build_outputs.py	D13a step 2: build the verified list of Laura Gao's own words (Phase B quotes whose speaker is, or is presented as, Laura Gao) plus the case pull quote, as JSON and as a Markdown file. Inputs (status labels): - research/insight_v1/phase_B/quotes_raw.json (Phase B claims; exact text taken from here)
insight_v1/scripts/D13a_laura_quotes_check.py	D13a step 1: re-check every Phase B quote whose speaker is (or is presented as) Laura Gao against its source page with the fixed fetch helper, and save the text around each quote so speaker, context and trimming can be judged. Inputs (status labels): - research/insight_v1/phase_B/quotes_raw.json (Ph
insight_v1/scripts/D13b_build_outputs.py	D13b: build the verified table of all Phase B quotes whose speaker is NOT Laura Gao. Inputs (status labels): - research/insight_v1/phase_B/quotes_raw.json (Phase B claims; to be checked) - research/insight_v1/phase_D/D13_mechanical_check.json (script re-fetch: VERBATIM / NOT_FOUND / NO_URL / FETCH_F
insight_v1/scripts/D13b_other_quotes_context.py	D13b helper: for every Phase B quote whose speaker is NOT Laura Gao, re-fetch the source page (fixed helper, cached) and print the text around the quote so a human/agent can check speaker and context (not trimmed into a new meaning). Inputs (status labels): - research/insight_v1/phase_B/quotes_raw.j
insight_v1/scripts/D13c_voice_map_check.py	D13c (Voice-of-Laura mapper): re-check every Laura quote that D13c_voice_map.md uses, plus the new candidate lines D13c found, against a FRESH copy of each source page, and check the case lines the map quotes. Why: the map may use a line as Laura's words only if D13a marked it VERIFIED-PRIMARY. This
insight_v1/scripts/D1_purchase_rule.py	D1_purchase_rule.py - the January 2027 ladder purchase rule if rates fall first (question M004). Agent D1 (Rates & Fixed-Income Analyst), insight_v1 run, 2026-09-28. PROVISIONAL research, not team-approved. What it answers (plain English): [1] What the ten-payment ladder costs on 2027-01-01 after pa
insight_v1/scripts/D1_wins_rung.py	D1_wins_rung.py - one dated "demonstration rung" in the WInS hedge (question M044). Agent D1 (Rates & Fixed-Income Analyst), insight_v1 run, 2026-09-28. PROVISIONAL research, not team-approved. Every security here is PENDING WInS AVAILABILITY + POSITION-LIMIT CHECK. What it computes (plain English):
insight_v1/scripts/D2_equity_premium.py	D2 (Equity & AI-Concentration Analyst), question M030: how much extra return does the growth sleeve's equity really expect over simply buying Treasuries for the same horizon, and does that move sleeve equity from 60% to 40-50%? Run from the repo root: .venv/bin/python research/insight_v1/scripts/D2_
insight_v1/scripts/D3_barbell.py	D3_barbell.py - M238: should the sleeve be a "barbell" (lock a facility floor in January 2028 in a Treasury that matures just before 2033, hold the rest 100% in broad equity) instead of a rebalanced 60/40 sleeve plus an 80% lock in 2031? What it computes (model outputs, not forecasts; common random
insight_v1/scripts/D3_control_and_equity.py	D3_control_and_equity.py - M034: judge the sleeve's equity against Laura's riskless control, not against a stock index. Question (phase_C survivors M034): should every sleeve recommendation be judged against the money Laura could lock for certain (the whole surplus in Treasuries maturing just before
insight_v1/scripts/D3_data_snapshot.py	D3_data_snapshot.py - download and freeze the public data series used by the D3 (Quant Modeler) scripts. What it does (plain English): fetches three public data sets through the environment's proxy (curl), parses them, and writes small CSV snapshots under research/insight_v1/scripts/data/D3/ so ever
insight_v1/scripts/D3_funded_status_log.py	D3_funded_status_log.py - M028: a weekly "funded status" log for the WInS Treasury hedge, and a 2026 backtest of how closely an IEF/TLH (or IEF/TLT) hedge tracked the value of Laura's ten payments. Question (phase_C survivors M028): should the team log, from the first WInS trade, the value of Laura'
insight_v1/scripts/D3_history_stress.py	D3_history_stress.py - M059: check the plan (and the barbell / ratchet / partial-lock alternatives) against real historical return sequences, not only the i.i.d. lognormal model. What it computes: 1. Crash-year frequency: model vs history (how often a stock year is -30% or worse). 2. Named episodes
insight_v1/scripts/D3_joint_tail.py	D3_joint_tail.py - M053: how large is the joint tail in which rates fall before the January 2027 purchase (the ladder costs more than \$300,000, so the longest payments are bought first) AND the 2028 deposit is late, smaller or missing? What it computes: 1. A conditional dollar table (no probabilities
insight_v1/scripts/D3_lock_frontier.py	D3_lock_frontier.py - M243: what is the honest reason to lock 100% of the payments in January 2027? Question (phase_C survivors M243): with the 2028 deposit present, a 70-95% lock also shows no payment shortfall in the model; what is our real reason to lock 100%, and where are we overstating it? Wha

Script	What it does (from its docstring)
insight_v1/scripts/D3_range_rules.py	D3_range_rules.py - M039 and M060: the 2031 co-sponsor range (floor share, top rule, cap rule) and how confident Laura can honestly be that her 2033 contribution lands inside it. Questions (phase_C survivors): - M060: what share of the 2031 sleeve should be locked as the floor (80%, or towards 100%)
insight_v1/scripts/D3_ratchet.py	D3_ratchet.py - M055: should the facility floor be locked progressively (a third each January 2029-2031) or by stepping sleeve equity down in 2029-30, rather than on one date in 2031? What it computes (model outputs, not forecasts; common random numbers): 1. The premise: how much a -30% stock year i
insight_v1/scripts/D4_facility_purchasing_power.py	D4_facility_purchasing_power.py - which risk moves "how much building Laura's facility money buys" the most by 2033: U.S. market risk, the USD/TWD exchange rate, or Taiwan construction-cost inflation? (Phase D question M095.) Plain English: Laura's facility contribution is decided and promised in US
insight_v1/scripts/D5_range_and_flexibility.py	D5 (Philanthropy & Co-sponsor Analyst): the 2031 co-sponsor range, its confidence wording, and how much of the 2033 post-reserve money stays uncommitted as "financial flexibility" (questions M003, M091, M107, M094). How to run (repo root): .venv/bin/python research/insight_v1/scripts/D5_range_and_fl
insight_v1/scripts/D6_behavioural_numbers.py	D6 (Client Psychologist / Behavioural Finance) numbers for questions M237, M019, M197 and M228. What it computes (every output is a MODEL PROPERTY under the stated assumptions, not a forecast): [1] Reproduces the verified base (strategy_mc.py): lock-early surplus 2033 p5/p50/p95 at sleeve equity 60%
insight_v1/scripts/D7_rule_trigger_odds.py	D7_rule_trigger_odds.py - can a pre-written WInS rule genuinely fire before the Oct 23 Trading Notes deadline? Agent D7 (Wharton Intent Historian), insight_v1 run, 2026-09-27. Question M002 (and one truthfulness check for M011). What it does (plain English): 1. Measures how much the 10-year Treasury
insight_v1/scripts/D8_practice_numbers.py	D8 Professional-Practice Benchmark: numbers for M109 (required return per goal), M026 (fees) and M006/M235 (does a rebalancing rule move anything measurable?). Run from the repo root: .venv/bin/python research/insight_v1/scripts/D8_practice_numbers.py Inputs and status labels - Case cash flows: +\$
insight_v1/scripts/D9_draft_checker.py	D9_draft_checker.py - a mechanical check for the team's OWN drafts of WInS notes, TN reflections, the pitch and the IPS. It never writes or rewrites text. It counts words and characters and flags words that the D9 communication spec (research/insight_v1/phase_D/D9_communication.md) says to avoid o
insight_v1/scripts/D9_ips_page_fit.py	D9_ips_page_fit.py - Will a 50-word pitch plus a 500-word IPS fit on the two allowed pages? Question (D9, IPS format compliance; serves M036): the IPS guide allows "Pages 2 and 3: Investment Strategy, 2-page maximum" holding the 50-word pitch and the 500-word IPS, in Times New Roman 12, double-space
insight_v1/scripts/D9_numbers.py	D9_numbers.py - every number the D9 Communication Analyst file quotes, recomputed in one place. Sections (run all; nothing is random except where stated): [1] M062 How the official documents use certainty / confidence / reliability words (counts + the object each word is attached to). Inputs: the si
insight_v1/scripts/E1_split_and_slices.py	E1 (Strategy Architect) - two checks behind the change proposals in phase_E/E1_change_proposals.md. How to run (repo root, about 20 seconds): .venv/bin/python research/insight_v1/scripts/E1_split_and_slices.py It re-uses D6's engine (research/insight_v1/scripts/D6_behavioural_numbers.py: same random
insight_v1/scripts/E3_client_view_numbers.py	E3 (Laura-as-client red team) - three client-facing numbers behind phase_E/E3_laura.md. How to run (repo root, about 10 seconds, no network needed): .venv/bin/python research/insight_v1/scripts/E3_client_view_numbers.py It re-uses E1's module (research/insight_v1/scripts/E1_split_and_slices.py), whi
insight_v1/scripts/E4_rival_numbers.py	E4 (rival strategist) - numbers behind research/insight_v1/phase_E/E4_rival.md. How to run (repo root, about 20 seconds): .venv/bin/python research/insight_v1/scripts/E4_rival_numbers.py WHAT IT COMPARES (every output is a MODEL PROPERTY under the stated assumptions, not a forecast) E1 = the plan in
insight_v1/scripts/E5_premortem_checks.py	E5 (pre-mortem analyst) - deadline clock in Australian time and the risk ranking behind phase_E/E5_premortem.md. How to run (repo root, under a second, no network needed): .venv/bin/python research/insight_v1/scripts/E5_premortem_checks.py It prints: [1] Every 2026-27 deliverable deadline (5:00 p.m.
insight_v1/scripts/E6_final_checks.py	E6 (Chief Strategist) - numbers behind research/insight_v1/phase_E/E6_final_spec.md and E6_our_strategy_one_page.md. How to run (repo root, about 20 seconds, no network): .venv/bin/python research/insight_v1/scripts/E6_final_checks.py WHAT IT DOES (every model output is a MODEL PROPERTY under the st
insight_v1/scripts/E7_tournament_field_numbers.py	Model numbers for the six tournament firm summaries (research/insight_v1/phase_E/tournament/firm_*.md). Inputs (labels as in brief s3): - Engine, returns, reserve pricing, seed and path count: research/verified_2026-09-27/strategy_mc.py (VRF inputs: JPM 2026 LTCMA U.S. large cap 6.70% compound / 16.

Script	What it does (from its docstring)
insight_v1/scripts/S2_growth_sleeve.py	S2 growth-sleeve analysis (insight_v1, wins_now). AI-generated research code for Team Caplet. What it does 1. Concentration of the S&P 500 today (SPYM full holdings, 2026-09-24): top-1, top-10, and an "AI-linked mega-cap" basket (definition below is an ASSUMPTION, chosen to be transparent, not exhaust
insight_v1/scripts/S3_allocation_numbers.py	S3_allocation_numbers.py - numbers behind research/insight_v1/wins_now/securities_and_allocation_v0.md (agent S3). What it does (plain English): 1. Rebuilds the official 2026-09-25 Treasury zero curve exactly as research/verified_2026-09-27/official_curve_pv.py does, and checks the \$292,264 / 9.90y
insight_v1/scripts/S4_red_team_checks.py	S4 red-team checks for research/insight_v1/wins_now/securities_and_allocation_v0.md (agent S3's WInS plan). PROVISIONAL week-1 research (2026-09-27). Not approved by the team. Every ticker is PENDING APPROVAL CHECK: confirm on this year's WInS approved list/rules before trading (this season: WInS av
insight_v1/scripts/T1_ticket_v1_numbers.py	T1_ticket_v1_numbers.py - numbers behind research/insight_v1/wins_now/securities_and_allocation_v1.md and research/insight_v1/phase_D/trading_now_brief.md (agent T1, Trading-Now editor, insight_v1 run, 2026-09-28). PROVISIONAL research for Team Caplet. Not approved by the team. Every ticker is PENDI
insight_v1/scripts/T2_red_team_checks.py	T2_red_team_checks.py - arithmetic behind research/insight_v1/wins_now/T2_red_team.md (agent T2, adversarial red team of the WInS ticket v1 and the trading-now brief, insight_v1 run, 2026-09-28). PROVISIONAL research for Team Caplet, not approved by the team. Every ticker is PENDING WInS AVAILABILIT
insight_v1/scripts/build_merged_questions.py	Phase C main-loop step: build merged questions from the Librarian's clusters. Every source question id must be in exactly one cluster; merged fields are the union/concatenation of member fields. Run from repo root: .venv/bin/python research/insight_v1/scripts/build_merged_questions.py Writes researc
insight_v1/scripts/build_record_pdf.py	Build the complete insight_v1 research record as one PDF (main-loop utility; no model inputs). Run from the repo root: .venv/bin/python research/insight_v1/scripts/build_record_pdf.py Needs: markdown, reportlab, pypdf (in .venv) and the Playwright headless Chromium at /opt/pw-browsers. Output: resea
insight_v1/scripts/collect_phase_B.py	Collect Phase B question-generation results from the workflow journals into one file, give every question an id, and do the exact-duplicate step of Phase C (normalised text match). Main-loop utility; no model inputs. Run from the repo root: .venv/bin/python research/insight_v1/scripts/collect_phase_
insight_v1/scripts/fetch_text.py	Fetch a web page through the environment's egress proxy and print it as plain text. Why: in this environment the WebFetch tool is blocked for many hosts, while curl through the proxy works. Usage (from the repo root): .venv/bin/python research/insight_v1/scripts/fetch_text.py URL [--grep "exact phra
insight_v1/scripts/phase_C_survival.py	Phase C main-loop step: combine the three skeptics' verdicts per merged question and apply the four filters. A merged question SURVIVES only if all hold (v4 Part 4, Phase C): 1. so-what: a majority of skeptics say its answer would change a decision, number or sentence; 2. anchor: a majority say it i
insight_v1/scripts/route_phase_D.py	Route Phase C survivors to Phase D specialists (main-loop utility). Each specialist takes at most CAP questions (highest skeptic priority first); the rest go to an overflow researcher (D11 for the markets side D1-D4/D10, D12 for the client side D5-D9). Prints four workflow argument objects. Run from
insight_v1/scripts/strategy_mc_v2.py	strategy_mc_v2.py - Laura Gao lock-early model, version 2 (D3 Quant Modeler, insight_v1 run, 2026-09-28). WHAT IT IS (one paragraph, plain English) Each of 200,000 simulated "futures" draws yearly returns for stocks and bonds from J.P. Morgan's published long-term assumptions. Laura's \$300,000 first
verified_2026-09-27/official_curve_pv.py	Cost of Laura's ten \$50k payments (Jan 1 2033..2042), valued at 2027-01-01, from the OFFICIAL Treasury par yield curve of 2026-09-25 (home.treasury.gov CSV uploaded by the team). Method: par yields as semiannual bond-equivalent yields, linear interpolation onto a 0.5y grid, bootstrap discount factor
verified_2026-09-27/strategy_mc.py	Monte Carlo comparison of two strategies for Laura Gao, 2027-2033, using VERIFIED inputs where available. VERIFIED inputs (files in competition/official_market_data/): - Cost at 2027-01-01 of the ten \$50k payments on the official 2026-09-25 Treasury curve: \$292,264 (research/verified_2026-09-27/offi

C.2 Data files (29)

File	Size
phase_B/questions.json	583 KB
phase_B/quotes_raw.json	148 KB
phase_C/librarian_clusters.json	79 KB
phase_C/merged_questions.json	454 KB
phase_C/parked.json	870 KB
phase_C/questions_deduped.json	583 KB
phase_C/survivors.json	263 KB
phase_D/D13_mechanical_check.json	86 KB

File	Size
phase_D/D13a_laura_quotes_verified.json	103 KB
phase_D/D13b_other_quotes_verified.json	141 KB
scripts/data/D3/damodaran_histretSP_1928_2025.csv	5 KB
scripts/data/D3/fred_DGS10.csv	262 KB
scripts/data/D3/fred_DGS2.csv	204 KB
scripts/data/D3/fred_DGS5.csv	262 KB
scripts/data/D3/treasury_par_2026_raw.csv	14 KB
scripts/data/S2/spym_holdings.csv	16 KB
scripts/data/S2/xb_holdings.csv	1 KB
scripts/data/S2/xlc_holdings.csv	1 KB
scripts/data/S2/xle_holdings.csv	0 KB
scripts/data/S2/xf_holdings.csv	2 KB
scripts/data/S2/xli_holdings.csv	2 KB
scripts/data/S2/xlk_holdings.csv	2 KB
scripts/data/S2/xlp_holdings.csv	1 KB
scripts/data/S2/xlre_holdings.csv	1 KB
scripts/data/S2/xlu_holdings.csv	1 KB
scripts/data/S2/xlv_holdings.csv	2 KB
scripts/data/S2/xly_holdings.csv	1 KB
wins_now/S1_fund_holdings_snapshot.csv	62 KB
wins_now/S1_mspd_table5_2026-08-31_fixed_2032plus.csv	11 KB

What this teaches

A number you cannot re-run is an opinion. Keeping every script beside the claim it supports is what lets a reader trust the result.